# ALGEBRAIC LOCI OF HODGE CLASSES

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Abstract. We prove that in every family of abelian varieties of Weil type, for every CM field, dimension and discriminant, the Weil classes are algebraic on a dense set that contains an explicit point of complex multiplication and is either the whole family or meagre, and that the real multiplication of a K3 surface behaves in the same way on its square; that the Hodge conjecture is equivalent to two statements, one on abelian varieties and one on the classes reached from them by correspondences, each of them the existence of an interior point in such a locus; that every rational Hodge class is algebraic on the very general complete intersection of diagonal quadrics of Vandermonde type, on every power of the K3 surfaces with complex multiplication by $\mathbb{Q}(\sqrt{-5})$ or $\mathbb{Q}(\sqrt{-7})$ of the Fricke families, and on the squares of the K3 surfaces with real multiplication by $\sqrt{6}$ and by the cubic fields $\mathbb{Q}(\zeta_7)^+$ and $\mathbb{Q}(\zeta_9)^+$ constructed here; and we carry each of the two statements to a first open instance, where one object, one class or one correspondence decides it.

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This paper joins and replaces two earlier ones by the authors: *Algebraic loci of Weil classes on abelian varieties*, written with Priyabrata Mandal (archive [BMB26]), and *Dense algebraic loci of Hodge classes*, whose earlier version is the Hexagon preprint 2610.00315v2, `https://hexagonmath.org/2610.00315v2` (archive [BB26code]). Every result of both is included with its proof, and what was open there is taken further in Part 6. The preprint *Relative secant cycles and Hodge classes* [BB26] contains errors and incomplete proofs and is replaced by the present paper.

Every proof is given in the text. Programs that repeat much of the finite arithmetic, and a Lean 4 certificate of part of it, are archived as described in the statement on data availability at the end of the paper; no statement rests on them.

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## 1. Introduction

Let $X$ be a smooth complex projective variety. The Hodge conjecture asserts that the cycle class map

$$\mathrm{cl}^p_X \colon \mathrm{CH}^p(X)_{\mathbb{Q}} \longrightarrow \mathrm{Hdg}^p(X) \coloneqq H^{2p}(X,\mathbb{Q}) \cap H^{p,p}(X)$$

is surjective for every $p$. For $p = 1$ this is the theorem of Lefschetz on $(1,1)$ classes.[1] Beyond degree two the best understood Hodge classes are those of abelian varieties, and the first of them that divisors do not produce are the Weil classes. Weil exhibited them on abelian varieties with an action of an imaginary quadratic field whose two eigenspaces on the tangent space have the same dimension, showed that they are Hodge classes outside the subring generated by divisor classes, and proposed them as test cases for the Hodge conjecture [Wei77]. The construction works for every CM field $F$ in place of the quadratic one. On a general abelian variety of Weil type the Hodge ring is generated by divisor classes together with the Weil classes, a flat $F$-line of classes of middle degree, so for these varieties the Hodge conjecture is the algebraicity of the Weil classes.

Markman proved that the Weil classes are algebraic on the abelian sixfolds of split Weil type for every imaginary quadratic field, and, through a product argument of Schoen [Sch98, Proposition 10], on every abelian fourfold of Weil type [Mar25a, Mar25b].[2] With the description of the Hodge rings of abelian varieties of dimension at most five by Moonen and Zarhin [MZ99], this gives the Hodge conjecture for abelian fourfolds and fivefolds. His sheaves are built from secant sheaves on an abelian variety of half the dimension, and they extend along the family because they satisfy an equivariant form of semiregularity. This paper is about the set of members of a family on which a flat Hodge class is algebraic, which we call its algebraic locus. For the Weil classes we determine it for every CM field and every dimension at once; we then show that the whole Hodge conjecture is two statements about such loci, one on abelian varieties and one on the classes that correspondences carry from them, prove each on the families where it can be proved, and take each to the first place where it is open and build there.

1.1. **The main theorem.** Fix a CM field $F$ of degree $2m$, an integer $n \geq 2$ and a discriminant $\delta$. The abelian varieties of $(F,n,\delta)$-Weil type are the members of one family over a connected period domain $\mathcal{D}_F$ of dimension $mn^2$, a product of $m$ copies of the bounded symmetric domain of $\mathrm{SU}(n,n)$, and their Weil classes form a flat local system of rank $2m$ over it (Setup 19.1, Proposition 19.3). We call the set

$$\Sigma_F = \{\, s \in \mathcal{D}_F : \text{the Weil classes of } A_s \text{ are algebraic} \,\}$$

[1] The rational coefficients are essential. The theorem of Lefschetz holds integrally in degree two, but the integral analogue fails in higher degree: Atiyah and Hirzebruch found torsion classes of type $(p,p)$ that are not classes of cycles, and Kollár found non-torsion ones. Every statement in this paper concerns $\mathbb{Q}$-coefficients.

[2] In the normalisation of Definition 15.22 the discriminant is the class of $(-1)^n \det H$ in $\mathbb{Q}^{\times}_{>0}/\mathrm{N}_{K/\mathbb{Q}}(K^{\times})$. Markman uses the class of $\det H$ itself, in which the same sixfolds have discriminant $(-1)^3 = -1$.

the *algebraic locus* of the family. A property holds for the *very general member* if it holds outside a countable union of proper closed analytic subsets of $\mathcal{D}_F$.[3]

**Theorem 1.1** (Main theorem)**.** *For every CM field $F$, every $n \geq 2$ and every discriminant $\delta$, the algebraic locus $\Sigma_F$ of the Weil classes has the following properties.*

(i) *It contains an explicit point $s_0$ at which $A_{s_0}$ is isogenous to a product of abelian varieties with complex multiplication by $F$, and at which the Weil classes are polynomials with rational coefficients in divisor classes.*

(ii) *It is dense in $\mathcal{D}_F$.*[4]

(iii) *It is the preimage of a countable union of Zariski closed subsets of the Shimura variety of the family, and either $\Sigma_F = \mathcal{D}_F$ or $\Sigma_F$ is meagre.*

(iv) *The following are equivalent: $\Sigma_F$ is closed; $\Sigma_F$ has an interior point; the Weil classes are algebraic on every member of the family; the Hodge conjecture holds for the very general member of the family.*

For $n = 1$ the Weil classes have degree two and are algebraic by the theorem of Lefschetz, and all four statements hold trivially. The proof of Theorem 1.1 is completed in Section 19, after the imaginary quadratic case has been treated in full in Part 2. Figure 2 draws the picture it describes: a connected domain, a dense set built from one explicit point, and a decomposition of that set into countably many closed algebraic strata.

The theorem carries the Hodge conjecture for the very general abelian variety of Weil type, for every CM field, every dimension and every discriminant, to a single topological statement about one explicit set: the dense set $\Sigma_F$, which is a countable union of closed algebraic subsets of a connected Shimura variety, has to be closed. The rest of the paper is organised around that statement. Part 1 shows why the classical constructions do not produce Weil classes and fixes the explicit form of the classes; Part 2 proves the theorem for imaginary quadratic fields; Part 3 determines what closing $\Sigma_F$ requires of a single object at a single point; Part 4 completes the proof for every CM field and computes the requirement exactly for quartic fields; Part 5 places the Weil classes inside the whole conjecture; and Part 6 builds at the two places where the conjecture is first open (Figure 1).

The paper also proves the Hodge conjecture itself on a family of varieties of every dimension. Let $X \subset \mathbb{P}^N$ be a complete intersection of $c$ diagonal hypersurfaces $\sum_i a_{ki} x_i^d = 0$ of degree $d$. It is *of Vandermonde type* if $a_{ki} = \mu_i \sum_{l=1}^{c} g_{kl} \lambda_i^{l-1}$ for distinct numbers $\lambda_0, \dots, \lambda_N$, nonzero numbers $\mu_i$ and an invertible matrix $(g_{kl})$; for $c \geq 2$ this says that the columns of $(a_{ki})$ are distinct points of a rational normal curve, and every smooth complete intersection of two diagonal hypersurfaces of the same degree is of this type. Such an $X$ is isomorphic to a variety $X_{d,N-c}(\lambda)$ that depends only on $\lambda$ (Theorem 20.55), and *very general* refers to $\lambda$. For $d = 2$ we call $X$ a *Vandermonde quadric intersection*.

**Theorem 1.2** (Diagonal complete intersections)**.** *Every rational Hodge class is algebraic on the very general Vandermonde quadric intersection, in every dimension and in every*

[3] A subset of $\mathcal{D}_F$ is *meagre* if it is contained in a countable union of closed subsets with empty interior. Since $\mathcal{D}_F$ is a complex manifold, the Baire category theorem shows that a meagre set has dense complement, so a statement that holds off a meagre set holds on a dense set of members; the very general members are those off a particular meagre set, a countable union of proper analytic subsets.

[4] Density comes from the Hecke orbit of $s_0$: the group $G_F(\mathbb{Q})$ of rational points of the unitary group of the hermitian form acts on $\mathcal{D}_F$ through isogenies that respect the $F$-action and the polarisation up to a scalar, so it preserves the algebraicity of the Weil classes, and the closure of $G_F(\mathbb{Q})$ contains the identity component of $G_F(\mathbb{R})$, which acts transitively on $\mathcal{D}_F$; the proof uses the rational isotropic vectors of the hermitian form, which exist by the Hasse principle.

*projective space. The same holds for the very general complete intersection of Vandermonde type of diagonal cubics or quartics of dimension at most seven, or of diagonal sextics of dimension at most five, and for every smooth complete intersection of two diagonal cubics, of two diagonal quartics or of two diagonal sextics in* $\mathbb{P}^6$.

The first two statements are Theorem 20.66(iii) and the third is Corollary 20.56(ii). The variety $X$ is a quotient of a power of a generalised Fermat curve, so its Hodge classes come from abelian subvarieties of the Jacobian of that curve. For quadrics the monodromy of the curve along a chain of vanishing cycles is the full symplectic group on every eigenspace, so every Hodge class comes from powers of polarisations and no theorem on abelian varieties of higher dimension is needed. For cubics, quartics and sextics we compute the monodromy by an induction on the number of branch points (Proposition 20.64), and the classes it leaves are divisor classes, Weil classes of abelian fourfolds of Weil type, and Weil classes of abelian sixfolds of split Weil type, all algebraic by the theorems of Lefschetz and Markman [Mar25a, Mar25b]. In odd dimension every Hodge class is a multiple of a power of the hyperplane class. So the bound seven for cubics and quartics is the point where abelian varieties of dimension eight appear, and the bound five for sextics comes from abelian sixfolds of non-split Weil type, which occur in dimension six.

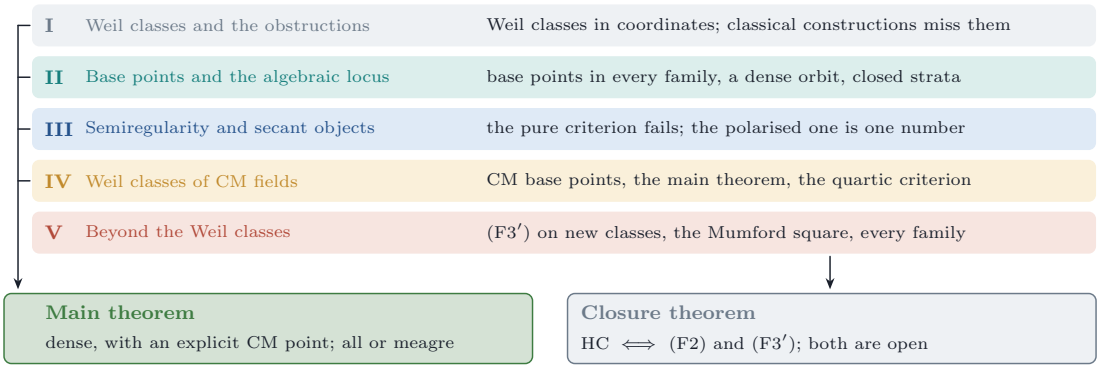



FIGURE 1. The first five parts of the paper. Parts I to IV lead to Theorem 1.1; Part V places the Weil classes inside the conjecture, which by the closure theorem of Section 27 is equivalent to the two statements (F2) and (F3′); Part VI builds at the first open instance of each and ends where Section 26 says.

1.2. **The proof.** The first step is a base point. In Theorem 19.5 the member $A_{s_0}$ is isogenous to $B_\Phi^n \times B_{\bar{\Phi}}^n$, where $B_\Phi$ is an abelian $m$-fold with complex multiplication by $F$ of a CM type determined by the polarisation, and the Weil classes there are written out as polynomials with rational coefficients in $F$-balanced divisor classes. For an imaginary quadratic field $K$ there is a second base point in every family, with quaternionic multiplication: whenever an algebra $(-d, b)_{\mathbb{Q}}$ with $b > 0$ acts compatibly with $K$, the Weil line is spanned by the real and imaginary parts of $\Theta^n$ for an explicit $(1,1)$ class $\Theta$, and the discriminant of the family is the class of $b^n$, which can be any discriminant (Theorem 15.16, Theorem 15.25, Theorem 15.27). Products of abelian surfaces of Weil type give a third, since the Weil class is multiplicative: $\omega(A_1)\,\omega(A_2) = \omega(A_1 \times A_2)$ for the $K$-valued Weil class $\omega$ (Theorem 15.7, Theorem 15.10). None of these base points uses anything beyond the theorem of Lefschetz on $(1,1)$ classes.

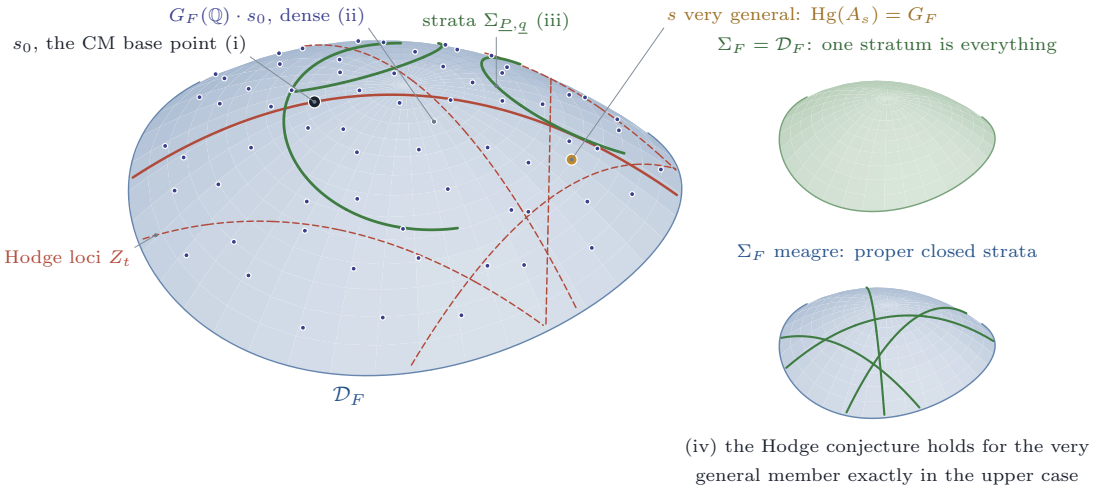



FIGURE 2. The algebraic locus of Theorem 1.1, drawn over a real slice of the period domain $\mathcal{D}_F$. Left: the explicit CM base point $s_0$ of part (i); its Hecke orbit $G_F(\mathbb{Q}) \cdot s_0$, which is dense by part (ii); three of the strata $\Sigma_{\underline{P},\underline{q}}$ of part (iii), each closed and algebraic, through points of the orbit; the Hodge loci $Z_t$, dashed, one of them through $s_0$, off which the Hodge group of the member is all of $G_F$ (Lemma 19.4); and a very general point $s$. Right: the two cases of the dichotomy of part (iii). By part (iv) the Hodge conjecture holds for the very general member exactly in the upper case, where one stratum is the whole domain.

The second step moves the base point. An element of $G_F(\mathbb{Q})$, where $G_F = \mathrm{Res}_{F_0/\mathbb{Q}} \mathrm{SU}(V, H)$ and $F_0$ is the totally real subfield, induces after clearing denominators an isogeny between two members and fixes the Weil classes, so $\Sigma_F$ is a union of $G_F(\mathbb{Q})$-orbits (Lemma 16.6). The hermitian form is isotropic by the Hasse principle, the rational points of the unipotent radicals of the parabolic subgroups are dense in their real points, and these real points generate the identity component of $G_F(\mathbb{R})$; so every orbit is dense (Proposition 16.7, Theorem 19.7(ii)).

The third step is properness. The relative Hilbert scheme of the universal family, for one tuple of Hilbert polynomials, is proper over the Shimura variety,[5] and the class of a flat family of cycles is locally constant, so the set of points at which a Weil class is represented by cycles with prescribed Hilbert data is Zariski closed; $\Sigma_F$ is the countable union of these strata (Theorem 16.4). The Baire category theorem then gives the dichotomy of part (iii) (Corollary 16.5). Part (iv) needs one more fact, that the very general member has Hodge group $G_F$; we prove it directly, from the simplicity of the real factors $\mathrm{SU}(n, n)$ (Lemma 19.4, Corollary 19.8).

For an imaginary quadratic field everything can be written in coordinates. On the standard model of Section 4, built from a $K$-vector space of dimension $2n$ with a hermitian

[5]By Grothendieck's construction the Hilbert scheme of a projective morphism, for a fixed Hilbert polynomial, is projective over the base; here the base is the Shimura variety of the family, after a level structure that makes the universal family exist, and the morphism is the universal abelian scheme.

form of signature $(n,n)$, the Weil line has the integral basis

$$\omega_1 = \sum_{|T| \text{ even}} (-1)^{|T|/2} d^{n-|T|/2}\, w^{(T)}, \qquad \omega_2 = \sum_{|T| \text{ odd}} (-1)^{(|T|-1)/2} d^{n-(|T|+1)/2}\, w^{(T)},$$

where $T$ runs over the subsets of $\{1,\dots,2n\}$ and $w^{(T)}$ is the monomial taking $y_j$ for $j \in T$ and $x_j$ otherwise (Theorem 4.9). A member with Hodge group $\mathrm{SU}(V,H)$ has exactly three Hodge classes in degree $2n$ and one in every other degree (Theorem 4.14), so the Hodge conjecture for it, in every degree at once, is the algebraicity of the one class $\omega_1$, with $2^{2n-1}$ integer coefficients (Corollary 4.15). By Deligne's theorem [Del82] that class is absolutely Hodge at every point,[6] and the Hodge conjecture for the family is equivalent to its algebraicity at every point of one connected domain (Corollary 15.35).

1.3. **Closing the algebraic locus.** By Theorem 1.1(iv) the Hodge conjecture for the very general member of a Weil family is the closedness of $\Sigma_F$, and by (iii) it is enough to show that one stratum has an interior point. Two statements would do this, and the paper determines what each of them asks.

The first bounds the Hilbert data. If at every point of the orbit of the base point the Weil class is represented by a cycle whose Hilbert data comes from one finite set, then finitely many strata cover a dense set and one of them is everything (Theorem 16.9). Read on an arbitrary representative this bound is equivalent to its conclusion (Theorem 16.17); read on the transported cycle, the one representative that can be named, it fails, because the transport multiplies the degree of a component by $c^{2n}$ and divides it only by the order of a stabiliser (Theorem 16.15). So the bound restates the problem rather than reducing it. The bound is needed only on a small set: by the theorem of Pila, Shankar and Tsimerman on the André–Oort conjecture [PST21],[7] it suffices along one sequence of CM points that leaves every proper special subvariety, and such a sequence exists inside the orbit (Theorem 19.14). Every member at which this paper exhibits algebraic Weil classes has endomorphisms beyond the field, so it lies in the locus of members with smaller Hodge group. That locus contains the base point, is dense, is stable under $G_F(\mathbb{Q})$ and is a countable union of closed algebraic strata, exactly as $\Sigma_F$ is, and it is meagre (Proposition 19.12, Corollary 19.13). The properties listed in Theorem 1.1 therefore do not decide the dichotomy by themselves, and a proof that $\Sigma_F$ is everything has to reach a member with full Hodge group.

The second is semiregularity. By the theorem of Buchweitz and Flenner, in the form of Pridham for perfect complexes [BF03, Pri24],[8] a perfect complex at one base point whose Chern character is a nonzero multiple of a Weil class plus a polynomial in the polarisation, and whose semiregularity map is injective, extends along the whole family and makes $\Sigma_F$ everything (Theorem 16.29, Theorem 19.7(v)). The paper computes the injectivity

[6] A class is absolutely Hodge if its image under every automorphism $\sigma$ of $\mathbb{C}$, through the comparison of Betti cohomology with algebraic de Rham cohomology of the conjugate variety $X^\sigma$, is again a Hodge class. Algebraic classes have this property, so it is a necessary condition for algebraicity; Deligne proved it for every Hodge class on an abelian variety.

[7] In the form used here: an irreducible closed subvariety of a Shimura variety of abelian type that contains a Zariski dense set of CM points is a special subvariety. Hence a sequence of CM points that leaves every proper special subvariety is Zariski dense, and so is every infinite subsequence of it.

[8] The semiregularity map of a perfect complex $E$ on $X$ sends $\mathrm{Ext}^2(E,E)$ to $\bigoplus_q H^{q+2}(X,\Omega_X^q)$ by the trace of the product with powers of the Atiyah class of $E$; it carries the obstruction to deforming $E$ along a first order deformation of $X$ to the obstruction to keeping $\mathrm{ch}(E)$ of Hodge type. If it is injective, $E$ deforms, formally and then along the whole family, in every direction in which $\mathrm{ch}(E)$ stays a Hodge class. Bloch introduced it for local complete intersection subvarieties.

condition exactly. The annihilator of the Weil class in $H^{0,2}$ is carried by the polarisation isomorphically onto the tangent space of the family (Lemma 9.6, Theorem 9.17), so no object whose Chern character lies on the Weil line and on which $H^2(\mathcal{O}_A)$ acts faithfully is semiregular (Theorem 9.7). When the Chern character is exactly a Weil class the criterion is never met, for any CM field and any $n \geq 2$: a semiregular object would extend to the very general Weil torus of the field, a complex torus that is not algebraic, on which every coherent sheaf has vanishing Chern character strictly between degree zero and the top degree (Proposition 17.4, Theorem 17.5, Theorem 19.19). With powers of the polarisation added the criterion becomes one number: for a Chern character $N\omega + \sum_k c_k \theta^k$ and $n \geq 3$ every object has $\dim \operatorname{Ext}^2(E,E) \geq (4+\rho)n^2 - 2n$, where $\rho \leq 3$ is the rank of the Hankel matrix of the numbers $k!\, c_k$, and an object attaining this bound meets the criterion (Theorem 17.9).[9] A flatness condition on the deformations of the object suffices in place of injectivity (Proposition 17.15). By descent and scalar extension the statement is needed only for the split families of the CM fields of degree at least four (Proposition 20.13), and for one family it is equivalent to the Lefschetz standard conjecture for the total space of the family over a curve through a base point (Corollary 20.31).

For a quartic CM field and $n = 2$ we compute the rank exactly for every shape. It takes fifteen values; the least, 80, is taken only by the pure shape, which the Weil tori exclude, so an object meeting the criterion has $\dim \operatorname{Ext}^2(E,E) \geq 88$ and a polynomial part that is exponential or linear at each real place (Theorem 19.29, Corollary 19.30). The Hodge locus of every such character is determined: it is the Weil tori of the field only for the characters of $\theta^4$ shape, which are therefore excluded at rank 88, and the polarised family itself for every other character of that rank (Proposition 19.32, Corollary 19.33).

1.4. **Secant objects.** The objects that a secant construction supplies are examined in Section 18. On the split member $X \times \widehat{X}$ of Section 14 the secant plane is written down in every dimension, and a secant sheaf has rank $a$, $c_1 = bt$, $c_2 = \frac{ad+b^2}{2}t^2$ and Bogomolov discriminant $\mathrm{N}_{K/\mathbb{Q}}(b + a\sqrt{-d})\, t^2$ (Theorem 14.5, Theorem 14.13). No object whose Chern character is an effective combination of exponentials of divisor classes contributes to the Weil line, by a positivity argument with the norm form (Theorem 18.5), and no object with a line bundle summand off the polarisation line meets the criterion, so no sum of line bundles does in any dimension (Theorem 9.20, Corollary 18.21). Convolutions of line bundles, their iterated cones, are cut down by the maps between the pieces. Fewer than $2n$ pieces with nondegenerate differences never meet the criterion (Theorem 18.26). On $E_0^{2n}$, $E_0 = \mathbb{C}/\mathbb{Z}[i]$, the signed Chern characters of $2 \cdot 4^{n-1}$ line bundles add up to a real Weil class (Proposition 18.27), and with three multiples of the polarisation no convolution of these pieces meets the criterion at $n = 3$ (Theorem 18.35), nor at $n = 4$ when the shifts of the pieces lie in six consecutive values (Corollary 18.48). The weakened criterion of [Mar25b, Question 11.4] reduces, for the transform of $\mathcal{F} \boxtimes \mathcal{F}$, to the single identity $\operatorname{At}(\mathcal{F})^2 = -d\Theta^2 \cdot \mathrm{id}$ for the sheaf $\mathcal{F}$ on the fourfold (Theorem 18.56); a local computation at an isolated double point of the support decides it in the negative for Markman's candidate in dimension eight (Theorem 18.61), and at a quartic CM field a curve on which a complex is locally free is tangent to at most one of the two eigenplanes of the real multiplication, which decides Markman's explicit pair [Mar25c, Example 11.2.7] in the same way (Theorem 19.43, Corollary 19.45).

[9] That is, $\rho$ is the rank of the $(2n-1) \times 3$ matrix with entries $\mu_{i+j}$, $0 \leq j \leq 2n-2$, $0 \leq i \leq 2$, where $\mu_k = k!\, c_k$ (Setup 17.8). A vector in its kernel is a linear recurrence of order at most two among the $\mu_k$, and $\rho$ does not change when the character is multiplied by $e^{t\theta}$.

The supports are constrained as well. A Cohen–Macaulay support carries no local obstruction, while two smooth branches of codimension two meeting transversally at a point carry one (Corollary 18.77). For a smooth support, a surface $S$ in the fourfold, Grothendieck–Riemann–Roch fixes every numerical invariant, and the Bogomolov–Miyaoka–Yau inequality gives $d \leq b^2$ for the twist $b\Theta$, which leaves only $d \in \{1, 3, 5, 7\}$ for the twist $3\Theta$ of Markman's candidate (Theorem 18.67, Corollary 18.68). Complete intersections are excluded for every $b$ and $d$ (Theorem 18.81), two-term complexes of powers of the polarisation at every rank, twist and discriminant (Theorem 18.89), and a Hilbert–Burch resolution of a secant support has rank at least two (Proposition 18.86); the secant class is an integer combination of classes of line bundles exactly when $d \equiv 15$ or $23 \pmod{24}$ (Theorem 18.83).

1.5. **The classical constructions.** Part 1 explains why two classical lines of attack do not produce Weil classes, and the explanations are different in kind. The first imports an object from a projective space, restricts it to the dual abelian variety and transports it back by the Fourier–Mukai transform. Its Chern character stays in the subring generated by the polarisation (Theorem 7.3), because the cohomological transform carries powers of the dual polarisation to rational multiples of powers of the polarisation in closed form (Theorem 6.5); and a construction equivariant for the field lands in the isotypic component of the norm character, which the Weil line misses, since it sits at the two ends of the grading (Theorem 5.6). The second is induction on dimension through hyperplane sections, and the restriction map kills every primitive class of middle degree (Theorem 10.3); below the middle degree the return trip is the Lefschetz standard conjecture (Theorem 10.8). The Kuga–Satake construction, which changes the variety rather than cutting it, reaches the Weil line only in dimension two (Proposition 11.11).[10] A rational class of the Weil line has a $K^\times$-orbit spanning a plane, so any construction reaching it must be bilinear in a pair of inputs exchanged by $\mathrm{Gal}(K/\mathbb{Q})$ (Proposition 13.1), and Markman's construction is of that kind.

1.6. **Weil classes and the Hodge conjecture.** Part 5 places the Weil classes inside the conjecture. Two statements carry all of it. The first, (F2), asks that every Hodge class on an abelian variety outside the subring generated by divisor classes and Weil classes of quotients be algebraic; multiplying by an exceptional class on the square of a Mumford fourfold moves every Hodge class of every abelian variety outside that subring, so (F2) is equivalent to the Hodge conjecture for all abelian varieties (Proposition 21.49). The second, (F3′), asks that every Hodge class on every smooth projective variety be algebraic modulo the images of Hodge classes of abelian varieties under algebraic correspondences (Definition 20.34). The Hodge conjecture is equivalent to (F2) and (F3′) together (Proposition 20.35, Theorem 27.1).

We prove (F3′) for every variety whose cohomology is reached from abelian varieties by correspondences, a class closed under products, surjective images, blow-ups, projective bundles and Hilbert schemes of points on surfaces, which contains every Fermat hypersurface and every K3 surface whose Kuga–Satake correspondence is algebraic (Proposition 20.35, Proposition 20.42); for every variety of dimension at most three, and on a variety of dimension $k$ in the degrees 0, 2, $2k-2$ and $2k$; for every fourfold and fivefold whose zero-cycles are supported in dimension at most three, among them every uniruled fourfold

[10] The construction attaches to a polarised Hodge structure $T$ of weight two with $h^{2,0} = 1$ an abelian variety $A$ with $H^1(A, \mathbb{Q}) = C^+(T)$, the even Clifford algebra, and an embedding of $T$ into $\mathrm{End}(H^1(A, \mathbb{Q}))$ by Hodge structures, which is a Hodge class on the product of a variety carrying $T$ with $A \times A$; the Kuga–Satake Hodge conjecture asks that this class be algebraic.

and every rationally connected fivefold (Proposition 20.43, Proposition 20.44); for every fourfold and fivefold dominated by a variety of that class (Proposition 20.46); for every smooth ample divisor of odd dimension in a variety of that class, and in even dimension outside the middle degree and for a very general member (Proposition 20.47); and, by a toric compactification whose boundary is again of the same kind, for every hypersurface of simplex type, so for every smooth Delsarte hypersurface of every dimension and every cyclic cover branched along one, whose Hodge conjecture then follows from that of Fermat varieties of one degree (Theorem 20.52, Corollary 20.53). A complete intersection of diagonal hypersurfaces whose coefficients are of Vandermonde type, which holds for every smooth complete intersection of two of them, is a quotient of a power of a generalised Fermat curve, so it satisfies (F3′), and its Hodge conjecture reduces to abelian varieties attached to the characters of the curve; for two diagonal hypersurfaces of degree 3, 4 or 6 in $\mathbb{P}^6$ these have dimension at most five, and the Hodge conjecture holds, the exceptional classes being Weil classes of abelian fourfolds of Weil type (Theorem 20.55, Corollary 20.56). At a very general configuration the monodromy of the curve leaves only the Hodge classes that come from polarisations and from Weil classes; we compute that monodromy for cyclic covers of degree three, four and six by an induction on the number of branch points (Proposition 20.64), so the Hodge conjecture holds for the very general such intersection of quadrics in every dimension, of cubics and quartics up to dimension seven, and of sextics up to dimension five, in every projective space (Theorem 20.66). On the powers, Hilbert schemes and moduli spaces of a K3 surface whose transcendental field is a CM field, and on the fourfolds of $\mathrm{K3}^{[2]}$ type with field $\mathbb{Q}$ or a CM field, the Hodge conjecture itself holds (Proposition 20.39, Proposition 20.40).

The first instance of (F2) is a pair of classes on the square of a Mumford fourfold, and Section 21 studies it in detail. The two classes are the real multiplication by a totally real cubic field on the primitive part of $H^2$, and they are algebraic once the Kuga–Satake class of one K3 surface of Picard number thirteen is algebraic (Proposition 21.2, Theorem 21.4); they are algebraic at every CM point (Theorem 20.29); they are rigid, their Hodge locus being the compact Mumford curve (Theorem 21.21); and they are represented by algebraic cycles at every closed point of every reduction of the curve modulo a prime, so that their algebraicity on every member is a bound on the Hilbert data of those cycles (Proposition 21.42, Theorem 21.43). For an abelian scheme over a curve whose invariant classes are algebraic the Lefschetz standard conjecture holds on the total space (Theorem 20.21), and over a compact Shimura curve the total space of a Mumford family satisfies the Hodge conjecture (Proposition 20.26).

The structure that Theorem 1.1 finds in one family is present in every family that carries (F2) or (F3′) (Section 22). Every Hodge class of every abelian variety lies in an abelian scheme with a member of CM type, and (F2) is equivalent to the statement that in every such scheme the algebraic locus of a flat Hodge class that is algebraic at a CM member has an interior point (Proposition 22.4, Theorem 22.5); the base points of the Weil families are the seeds, through André's theorem on the Hodge classes of abelian varieties of CM type. (F3′) is equivalent to the absence of Hodge classes off the part of the cohomology reached from abelian varieties by algebraic correspondences (Proposition 22.7). On the squares of K3 surfaces with real multiplication, where (F3′) is first open, the algebraic locus of the real multiplication contains every CM point, is dense, is stable under rational Hodge isometries, and is either the whole family or meagre (Theorem 22.9). In each case the property still missing is the interior point (Theorem 22.10).

The last part of the paper takes the two statements at the first places where they are open and builds. At (F3′), the real multiplication of a K3 surface by $\sqrt{6}$, by $\mathbb{Q}(\sqrt{-5})$ and $\mathbb{Q}(\sqrt{-7})$, and by the cubic fields $\mathbb{Q}(\zeta_7)^+$ and $\mathbb{Q}(\zeta_9)^+$ is made algebraic on whole families, by chains of symplectic quotients, by Fricke isogenies and by dihedral correspondences (Theorem 23.5, Theorem 23.9, Theorem 23.14), and the surfaces of the Mumford fourfolds of the two cyclic fields of conductor 7 and 9, where the locus of Theorem 22.9 was open, are settled; Proposition 23.15 says that the same correspondences reach no other cubic field. At (F2), the object the polarised criterion asks for is sought at the point of complex multiplication $E_0^6$ and in the families through it: the product and direct-sum constructions, the convolutions of line bundles and of abelian threefolds, and the supports of a complex meeting the criterion are worked out, each ending at a precise statement of what fails (Theorem 24.3, Theorem 24.23, Theorem 24.33, Theorem 24.27), and the two exceptional classes of a Mumford fourfold are shown to be algebraic exactly when one Hodge class on the product of its square with an abelian sixfold is (Theorem 24.45). Section 25 records what specialisation, degeneration, blow-ups, dominant maps, derived equivalences and residues do to the two open statements: they move them and do not weaken them. Where the paper stops is stated in Section 26.

Finally, the implications of the paper and of the literature form a finite rule set, and its consequences can be determined by hand. They give the minimal sets of statements from which the conjecture follows (Theorem 20.70, Remark 21.50): (F3′) together with the Lefschetz standard conjecture, with the variational Hodge conjecture for algebraic classes, or with (F2), besides two sets that are each equivalent to the conjecture.[11] Section 27 collects the results in one theorem.

Every proof is written to be checked by hand. Where an argument rests on a finite calculation, the calculation is given where it is used, as an explicit identity, a determinant or a case analysis. The archives [BMB26] and [BB26code] hold programs that repeat much of this arithmetic and a Lean 4 certificate of part of it; no statement of the paper rests on them.

## Part 1. Weil classes and the obstructions

### 2. Conventions

All varieties are over $\mathbb{C}$ and all cohomology is singular cohomology with the indicated coefficients. For a smooth projective variety $X$ we write $H^{p,q}(X)$ for the Hodge components of $H^{p+q}(X,\mathbb{C})$, and

$$\mathrm{Hdg}^p(X) = H^{2p}(X,\mathbb{Q}) \cap H^{p,p}(X)$$

for the space of rational Hodge classes of type $(p,p)$. A class is *algebraic* if it lies in the image of the cycle class map $\mathrm{cl}^p_X\colon \mathrm{CH}^p(X)_{\mathbb{Q}} \to \mathrm{Hdg}^p(X)$. The algebraic classes form a subring of $H^*(X,\mathbb{Q})$, because the cycle class map carries intersection products to cup products.[12]

[11] The Lefschetz standard conjecture asks that the inverse of the hard Lefschetz isomorphism $L^{\dim X-k}\colon H^k(X,\mathbb{Q}) \to H^{2\dim X-k}(X,\mathbb{Q})$ be induced by an algebraic correspondence. The variational Hodge conjecture asks that a flat section of Hodge classes over the base of a smooth projective family that is algebraic at one fibre be algebraic at every fibre.

[12] The intersection product on $\mathrm{CH}^*(X)_{\mathbb{Q}}$ is defined because $X$ is smooth, and its compatibility with the cycle class map is proved in [Ful98, Chapter 19]. The integral form of the question, with $\mathbb{Z}$ in place of $\mathbb{Q}$, is a different problem and is not considered in this paper.

An abelian variety $A$ of dimension $g$ has $H^1(A,\mathbb{Q})$ of dimension $2g$ and a canonical isomorphism

$$H^k(A,\mathbb{Q}) \;\cong\; \bigwedge^k H^1(A,\mathbb{Q}) \tag{1}$$

of Hodge structures, with the cup product on the left corresponding to the wedge product on the right. We use (1) without further comment; it is what makes every computation below finite.[13] The dual abelian variety is $A^\vee = \mathrm{Pic}^0(A)$, and $H^1(A^\vee,\mathbb{Q}) \cong H^1(A,\mathbb{Q})^\vee$.

A polarisation of $A$ is a class $\theta \in \mathrm{NS}(A)\otimes\mathbb{Q}$ that is ample; equivalently, an alternating form $E$ on $H_1(A,\mathbb{Q})$ satisfying the Riemann relations. The polarisation is principal when $E$ is unimodular on $H_1(A,\mathbb{Z})$, and then $\theta^g/g!$ is the class of a point. We write $\theta'$ for the polarisation on $A^\vee$ induced by $\theta$ through the isogeny $\phi_\theta\colon A \to A^\vee$; when $\theta$ is principal, $\phi_\theta$ is an isomorphism and $\phi_\theta^*\theta' = \theta$.

For an endomorphism $f$ of $A$ we write $f^*$ for the induced map on cohomology. When a field $K$ acts on $A$ through $\iota\colon K \hookrightarrow \mathrm{End}^0(A) = \mathrm{End}(A)\otimes\mathbb{Q}$, each $\tau \in K$ acts on $H^*(A,\mathbb{Q})$ through $\iota(\tau)$, and we speak of the action of $\tau$ when $\iota$ is understood. For nonzero $\tau$ in the order $\iota^{-1}(\mathrm{End}(A))$ the element $\iota(\tau)$ is an isogeny; every element of $K$ is a rational multiple of an element of this order, so the general case follows by $\mathbb{Q}$-linearity, and no statement below depends on the distinction.

The Chern character of a bounded complex $\mathcal{E}^\bullet$ of locally free sheaves is $\mathrm{ch}(\mathcal{E}^\bullet) = \sum_i (-1)^i \mathrm{ch}(\mathcal{E}^i)$. On a smooth projective variety every bounded complex of coherent sheaves is quasi-isomorphic to such a complex, and the resulting class depends only on the object of $D^b$. We write $\mathrm{ch}_k$ for the component in $H^{2k}$. On an abelian variety the tangent bundle is trivial, and so is the Todd class. Grothendieck–Riemann–Roch therefore reads $\mathrm{ch}(f_!x) = f_*\,\mathrm{ch}(x)$ for a morphism $f$ between abelian varieties, and $\mathrm{ch}\colon K_0(A)_\mathbb{Q} \to \mathrm{CH}^*(A)_\mathbb{Q}$ is an isomorphism of rings [Ful98, Chapter 15]; both facts are used without further mention. For the inclusion of a smooth subvariety, whose normal bundle need not be trivial, the Todd class of the normal bundle enters, and it is written out where it matters.

For the Fourier–Mukai transform we follow [Muk81]. The Poincaré bundle $\mathcal{P}$ on $A\times A^\vee$ is normalised so that $\mathcal{P}|_{A\times\{0\}}$ and $\mathcal{P}|_{\{0\}\times A^\vee}$ are trivial. With $p\colon A\times A^\vee \to A$ and $q\colon A\times A^\vee \to A^\vee$ the projections, the transform is

$$\Phi_\mathcal{P}\colon D^b(A^\vee) \to D^b(A), \qquad \Phi_\mathcal{P}(\mathcal{G}) = Rp_*(Lq^*\mathcal{G} \otimes^L \mathcal{P}).$$

On cohomology we write the induced map with the same symbol,

$$\Phi_\mathcal{P}\colon H^*(A^\vee,\mathbb{Q}) \longrightarrow H^*(A,\mathbb{Q}), \qquad \Phi_\mathcal{P}(x) = p_*(q^*x\cdot \mathrm{ch}(\mathcal{P})),$$

and the Grothendieck–Riemann–Roch theorem gives $\mathrm{ch}(\Phi_\mathcal{P}(\mathcal{G})) = \Phi_\mathcal{P}(\mathrm{ch}(\mathcal{G}))$ because the Todd class of an abelian variety is trivial; see Proposition 6.1.

For an imaginary quadratic field $K$ we write $\mathrm{N} = \mathrm{N}_{K/\mathbb{Q}}$ for the norm and $\overline{\tau}$ for the conjugate of $\tau \in K$. Characters of $K^\times$ are written multiplicatively, and $\mathrm{N}^n$ denotes $\tau \mapsto \mathrm{N}(\tau)^n$.

Finally, we say that a construction is *of ambient type* if it has the shape of the hypothesis of Theorem 7.3: a morphism $i\colon A^\vee \to \mathbb{P}^{N-1}$ defined by a linear system, an object $\mathcal{F}$ of $D^b(\mathbb{P}^{N-1})$, and the class $\mathrm{ch}_k(\Phi_\mathcal{P}(Li^*\mathcal{F}))$. Nothing is assumed about $\mathcal{F}$: it may be the structure sheaf of a secant variety, of a join, of a determinantal locus or of any other subscheme, and it may be twisted by any line bundle from the ambient space.

[13] Topologically $A = V/\Lambda$ is a real torus of dimension $2g$, so $H^*(A,\mathbb{Z}) = \bigwedge^*\mathrm{Hom}(\Lambda,\mathbb{Z})$ by the Künneth formula for $(S^1)^{2g}$, with the cup product corresponding to the wedge product. The Hodge decomposition of $H^k$ is the $k$-th exterior power of $H^1 = H^{1,0}\oplus H^{0,1}$, because the translation-invariant forms $dz_I \wedge d\overline{z}_J$ are harmonic for the flat metric and represent every class.

## 3. Abelian varieties of Weil type

We fix the notion of Weil type, construct the Weil line, prove that it consists of Hodge classes, and record the two facts about it that the obstruction theorems use: its dimension, and its position relative to the subring generated by divisor classes. The standard references are [BL04] for abelian varieties and [Voi02] for Hodge theory. Apart from the Riemann relations, the Lefschetz theorem on $(1,1)$ classes and the description of the Néron–Severi group by symmetric endomorphisms, every result of the section is proved here or in Section 5.

3.1. **Weil type.** Let $K = \mathbb{Q}(\sqrt{-d})$ be an imaginary quadratic field, with $d$ a squarefree positive integer, and let $^-$ denote the nontrivial element of $\mathrm{Gal}(K/\mathbb{Q})$.

**Definition 3.1** (Weil type)**.** Let $A$ be a complex abelian variety, let $\iota\colon K \hookrightarrow \mathrm{End}^0(A)$ be an embedding, and let $\eta$ be a polarisation class on $A$. The triple $(A,\iota,\eta)$ is of $(K,-1,n)$-*Weil type* if

(i) $\dim A = 2n$;
(ii) the endomorphism $\iota(\sqrt{-d})$ acts on the tangent space $H^0(A,\Omega^1_A)^\vee \cong H^{1,0}(A)^\vee$ with the two eigenvalues $\pm\sqrt{-d}$ each occurring with multiplicity $n$;
(iii) the Rosati involution attached to $\eta$ restricts on $\iota(K) \subset \mathrm{End}^0(A)$ to complex conjugation, that is $\iota(\tau)^\dagger = \iota(\overline{\tau})$ for all $\tau \in K$.

Condition (ii) is the balanced signature condition. It separates Weil type from the unbalanced case, in which the eigenvalues $\pm\sqrt{-d}$ occur with different multiplicities, and it is the condition under which the Weil line consists of Hodge classes (Remark 4.10). Condition (iii) says that the polarisation is compatible with the $K$-action, and the middle entry $-1$ of $(K,-1,n)$ records the sign in this compatibility: the Rosati involution restricts to conjugation on $\iota(K)$, not to the identity. By Lemma C.4 the sign is forced once $\iota(K)$ is stable under the Rosati involution. Some authors suppress the sign and speak of $K$-Weil type of dimension $2n$.

**Lemma 3.2** ($H^1$ is a $K$-space)**.** *Let $(A,\iota,\eta)$ be of $(K,-1,n)$-Weil type and put $V = H^1(A,\mathbb{Q})$. Then $\iota$ makes $V$ a $K$-vector space of dimension $2n$, and $\dim_{\mathbb{Q}} V = 4n$.*

*Proof.* The algebra $\mathrm{End}^0(A)$ acts $\mathbb{Q}$-linearly on $V$ by pullback. Pullback reverses the order of composition, but $K$ is commutative, so $\tau \mapsto \iota(\tau)^*$ is a unital ring homomorphism $K \to \mathrm{End}_{\mathbb{Q}}(V)$. Since $K$ is a field, $V$ becomes a $K$-vector space, and since $\dim_{\mathbb{Q}} V = 2\dim A = 4n$, we get $\dim_K V = 4n/[K:\mathbb{Q}] = 2n$. ∎

*Notation* 3.3. Write $V_{\mathbb{C}} = V \otimes_{\mathbb{Q}} \mathbb{C}$. Since $\iota(\sqrt{-d})^2 = -d$ on $V$, the operator $\iota(\sqrt{-d}) \otimes 1$ on $V_{\mathbb{C}}$ has the two eigenvalues $\pm\sqrt{-d}$, and

$$V_{\mathbb{C}} = V_+ \oplus V_-, \qquad V_\pm = \ker\bigl(\iota(\sqrt{-d}) \mp \sqrt{-d}\bigr), \tag{2}$$

with $\dim_{\mathbb{C}} V_\pm = 2n$ each, because $V$ is free of rank $2n$ over $K$ and $K \otimes_{\mathbb{Q}} \mathbb{C} \cong \mathbb{C} \times \mathbb{C}$. Complex conjugation on $V_{\mathbb{C}}$, coming from the real structure $V \otimes \mathbb{R}$, interchanges $V_+$ and $V_-$. For general $\tau = a + b\sqrt{-d} \in K$ the operator $\iota(\tau)$ acts on $V_+$ by the scalar $\tau$ and on $V_-$ by $\overline{\tau}$.

*Remark* 3.4. The two scalars in Notation 3.3 are different functions of $\tau$. For $\tau \notin \mathbb{Q}$ the operator $\iota(\tau)$ is $\mathbb{Q}$-linear on $V$ but not a scalar; it becomes a scalar on each piece only after extension to $\mathbb{C}$ and splitting. This is the source of the characters studied in Section 5. A class defined over $\mathbb{Q}$ can be an eigenvector of every $\iota(\tau)$ only with an eigenvalue in $\mathbb{Q}$; the function $\mathrm{N}(\tau) = \tau\overline{\tau}$ is $\mathbb{Q}$-valued, and $\tau^{2n}$ is not (Remark 5.8).

3.2. **The Hodge structure.** Write the Hodge decomposition as $V_{\mathbb{C}} = V^{1,0} \oplus V^{0,1}$, with $V^{1,0} = H^{1,0}(A)$ and $V^{0,1} = H^{0,1}(A)$. The $K$-action preserves it, because $\iota(\tau)$ is induced by an endomorphism of $A$ and is therefore a morphism of Hodge structures. The eigenspaces of an operator that preserves both summands are the direct sums of its eigenspaces on the two summands, so each of $V_+$ and $V_-$ splits: with $V_\pm^{p,q} = V_\pm \cap V^{p,q}$,

$$V_\pm = V_\pm^{1,0} \oplus V_\pm^{0,1}. \tag{3}$$

**Lemma 3.5** (Balanced signature)**.** *For $(A, \iota, \eta)$ of $(K, -1, n)$-Weil type,*

$$\dim_{\mathbb{C}} V_+^{1,0} = \dim_{\mathbb{C}} V_-^{1,0} = n, \qquad \dim_{\mathbb{C}} V_+^{0,1} = \dim_{\mathbb{C}} V_-^{0,1} = n.$$

*Proof.* By Definition 3.1(ii), $\iota(\sqrt{-d})$ acts on the tangent space with each eigenvalue of multiplicity $n$. The tangent space is dual to $H^{1,0}(A) = V^{1,0}$, and the action of $\iota(\sqrt{-d})$ on $V^{1,0}$ by pullback is the transpose of its action on the tangent space, so it has the same eigenvalues with the same multiplicities; that is, $\dim V_+^{1,0} = \dim V_-^{1,0} = n$. Since $\dim_{\mathbb{C}} V_\pm = 2n$ by Notation 3.3 and $V_\pm = V_\pm^{1,0} \oplus V_\pm^{0,1}$, the remaining two dimensions are $n$ each. ∎

3.3. **The Weil line.**

**Definition 3.6** (Weil line)**.** Let $(A, \iota, \eta)$ be of $(K, -1, n)$-Weil type and $V = H^1(A, \mathbb{Q})$. The *Weil line* is

$$W(A) \;=\; \Big(\textstyle\bigwedge_{\mathbb{C}}^{2n} V_+ \oplus \bigwedge_{\mathbb{C}}^{2n} V_-\Big) \cap H^{2n}(A, \mathbb{Q}),$$

the intersection being taken inside $H^{2n}(A, \mathbb{C}) = \bigwedge_{\mathbb{C}}^{2n} V_{\mathbb{C}}$.

**Proposition 3.7** (Structure of the Weil line)**.** *Let $(A, \iota, \eta)$ be of $(K, -1, n)$-Weil type. Then*

(i) *$W(A)$ is a $\mathbb{Q}$-subspace of $H^{2n}(A, \mathbb{Q})$ of dimension 2, stable under $\iota(K)$, and free of rank one as a $K$-module;*
(ii) *$W(A) \subset H^{n,n}(A)$, so every element of $W(A)$ is a Hodge class;*
(iii) *$W(A)$ is canonically isomorphic to $\bigwedge_K^{2n} V$ as a $K$-module.*

*Proof.* (i) Write $W_{\mathbb{C}} = \bigwedge^{2n} V_+ \oplus \bigwedge^{2n} V_-$. Each summand is one-dimensional over $\mathbb{C}$ because $\dim_{\mathbb{C}} V_\pm = 2n$, so $\dim_{\mathbb{C}} W_{\mathbb{C}} = 2$. The splitting (2) is the eigenspace decomposition of the operator $\iota(\sqrt{-d})$, which is defined over $\mathbb{Q}$, so every automorphism of $\mathbb{C}$, acting on $V_{\mathbb{C}}$ through the second factor, either preserves $V_+$ and $V_-$ or interchanges them, according to its restriction to $K$; complex conjugation interchanges them. Hence $W_{\mathbb{C}}$ is stable under $\mathrm{Aut}(\mathbb{C})$, and by Galois descent it is defined over $\mathbb{Q}$: the space $W(A) = W_{\mathbb{C}} \cap H^{2n}(A, \mathbb{Q})$ has $\dim_{\mathbb{Q}} W(A) = 2$ and $W(A) \otimes \mathbb{C} = W_{\mathbb{C}}$. It is stable under $\iota(K)$ because $\iota(\tau)$ preserves each of $V_\pm$, hence each summand. A two-dimensional $\mathbb{Q}$-space with a faithful $K$-action is free of rank one over $K$.

(ii) By (3) and Lemma 3.5, the line $\bigwedge^{2n} V_+$ equals $\bigwedge^n V_+^{1,0} \otimes \bigwedge^n V_+^{0,1}$, which is of Hodge bidegree $(n, n)$. The same computation applies to $\bigwedge^{2n} V_-$. Hence $W_{\mathbb{C}} \subset H^{n,n}(A)$ and so $W(A) \subset H^{2n}(A, \mathbb{Q}) \cap H^{n,n}(A) = \mathrm{Hdg}^n(A)$.

(iii) Since $K \otimes_{\mathbb{Q}} \mathbb{C} \cong \mathbb{C} \times \mathbb{C}$, with the two factors corresponding to the two embeddings of $K$, we have $V \otimes_{\mathbb{Q}} \mathbb{C} = V \otimes_K (K \otimes_{\mathbb{Q}} \mathbb{C}) = V_+ \oplus V_-$, and exterior powers over $K$ commute with this base change:

$$\Big(\textstyle\bigwedge_K^{2n} V\Big) \otimes_{\mathbb{Q}} \mathbb{C} \;\cong\; \bigwedge_{\mathbb{C}}^{2n} V_+ \oplus \bigwedge_{\mathbb{C}}^{2n} V_- \;=\; W_{\mathbb{C}} \;\subset\; H^{2n}(A, \mathbb{C}).$$

Every map in this chain is compatible with the action of $\mathrm{Aut}(\mathbb{C})$ through the second factor, which on the right permutes $V_+$ and $V_-$ as in (i). By Galois descent the composite is induced by an injective $\mathbb{Q}$-linear map $\bigwedge_K^{2n} V \to H^{2n}(A,\mathbb{Q})$, which commutes with $\iota(K)$ because each map in the chain does. Its image is a $\mathbb{Q}$-subspace of $W_{\mathbb{C}} \cap H^{2n}(A,\mathbb{Q}) = W(A)$ of dimension two, hence all of $W(A)$. ∎

*Remark* 3.8. Some authors call a generator of $W(A)$ *the* Weil class and speak of a one-dimensional space; the difference is the usual one between a $K$-line and its underlying $\mathbb{Q}$-plane. Nothing below depends on the convention, since the obstruction theorems concern the whole of $W(A)$.

3.4. **The Weil line and the divisor subring.** The Weil line is not accounted for by divisor classes. We record the statement in the form in which it is used and derive it from the character computation of Section 5.

**Definition 3.9.** The *divisor subring* $D^*(A) \subset H^*(A,\mathbb{Q})$ is the $\mathbb{Q}$-subalgebra generated by $\mathrm{NS}(A) \otimes \mathbb{Q} \subset H^2(A,\mathbb{Q})$.

Every element of $D^*(A)$ is algebraic, by the Lefschetz theorem on $(1,1)$ classes and the fact that the algebraic classes form a subring. The Hodge conjecture for $A$ in degree $n$ is therefore equivalent to the algebraicity of a complement of $D^{2n}(A)$ in $\mathrm{Hdg}^n(A)$. For a very general member of a Weil family the Weil line is such a complement, by Theorem 4.14; at special members the divisor subring can grow and meet the Weil line (Remark 20.5).

**Proposition 3.10.** *Let $(A,\iota,\eta)$ be of $(K,-1,n)$-Weil type with $\mathrm{End}^0(A) = \iota(K)$, and suppose $n \geq 1$. Then*

$$W(A) \cap D^{2n}(A) = 0.$$

*Proof.* By Proposition 3.7(i) and Theorem 5.6, the space $W(A) \otimes \mathbb{C}$ is the sum of the $K^\times$-eigenspaces for the characters $\tau \mapsto \tau^{2n}$ and $\tau \mapsto \overline{\tau}^{2n}$. The space $\mathrm{NS}(A) \otimes \mathbb{Q}$ is isomorphic to the space of elements of $\mathrm{End}^0(A)$ fixed by the Rosati involution [BL04, Section 5.2]. Since $\mathrm{End}^0(A) = \iota(K)$ and the Rosati involution restricts to conjugation on $\iota(K)$, the fixed elements are those of $\iota(\mathbb{Q})$, so $\mathrm{NS}(A) \otimes \mathbb{Q} = \mathbb{Q}\,\eta$. By Lemma 5.2 the class $\eta$ is a $K^\times$-eigenclass for the norm character N, so every element of $D^{2n}(A) = \mathbb{Q}\,\eta^n$ is an eigenclass for $\mathrm{N}^n$. By Lemma 5.4 the characters $\tau \mapsto \tau^{2n}$ and $\tau \mapsto \overline{\tau}^{2n}$ both differ from $\tau \mapsto \mathrm{N}(\tau)^n$. Eigenvectors for pairwise distinct characters are linearly independent, so $W(A) \otimes \mathbb{C}$ meets the $\mathrm{N}^n$-eigenspace, which contains $D^{2n}(A)$, only in zero. ∎

The mechanism behind the proof is that of Remark 5.8: $K^\times$ can act on a rational class through a character only if the character is $\mathbb{Q}$-valued, and the $\mathbb{Q}$-valued characters occurring in the cohomology are the powers of the norm. The Weil line has rank one over $K$, not over $\mathbb{Q}$, and as a $K$-line of exterior degree $2n$ it carries the $2n$-th power of $\tau$ itself. Theorem 5.6 and Lemmas 5.2 and 5.4 are proved in Section 5, and nothing there uses Proposition 3.10, so the forward reference is not circular.

## 4. Weil classes in coordinates

Section 3 defines the Weil line as an intersection inside $H^{2n}(A,\mathbb{C})$. That description is convenient for proofs but exhibits no single class, and in this section we write the Weil classes out. We build an abelian variety of $(K,-1,n)$-Weil type from a $2n$-dimensional $K$-vector space with a hermitian form, write two integral generators $\omega_1, \omega_2$ of the Weil line as sums of $2^{2n-1}$ monomials each, and display them in full for $n = 2$ and in a table for $n = 3$. The coordinates then give two things that the abstract description does not: an

elementary second proof that the Weil line misses the powers of the polarisation, and a complete count of the Hodge classes in every degree, which reduces the Hodge conjecture for a general member to the algebraicity of the single class $\omega_1$.

### 4.1. **The standard model.**

**Setup 4.1** (The model)**.** Fix a squarefree positive integer $d$, let $K = \mathbb{Q}(\sqrt{-d})$, and fix the embedding $K \hookrightarrow \mathbb{C}$ carrying $\sqrt{-d}$ to $\delta := i\sqrt{d}$. Fix $n \geq 1$ and let

$$V = K^{2n}, \qquad u_1, \dots, u_{2n} \text{ the standard } K\text{-basis},$$

regarded as a $\mathbb{Q}$-vector space of dimension $4n$ with the basis

$$x_j := u_j, \qquad y_j := \sqrt{-d}\, u_j, \qquad 1 \leq j \leq 2n.$$

Let $S\colon V \to V$ be multiplication by $\sqrt{-d}$, so that in this basis

$$Sx_j = y_j, \qquad Sy_j = -d\, x_j. \tag{4}$$

Let $\epsilon_j = -1$ for $1 \leq j \leq n$ and $\epsilon_j = +1$ for $n < j \leq 2n$, and let $Q$ be the alternating $\mathbb{Q}$-bilinear form on $V$ with

$$Q(x_j, y_j) = \epsilon_j, \qquad Q(x_j, x_k) = Q(y_j, y_k) = 0, \qquad Q(x_j, y_k) = 0 \ \ (j \neq k). \tag{5}$$

Finally let $\eta \in \bigwedge^2 V$ be the class with the same coefficients,

$$\eta = \sum_{j=1}^{2n} \epsilon_j\, x_j \wedge y_j. \tag{6}$$

Over $\mathbb{C}$ the operator $S$ has the two eigenvalues $\pm\delta$, and its eigenvectors can be written down directly.

**Lemma 4.2** (Eigenbasis)**.** *For $1 \leq j \leq 2n$ and $s \in \{+, -\}$ put*

$$u_j^s \;=\; \tfrac{1}{2}\Big(x_j - s\,\tfrac{\delta}{d}\, y_j\Big) \;\in V_{\mathbb{C}}. \tag{7}$$

*Then $Su_j^s = s\,\delta\, u_j^s$, the vectors $u_1^+, \dots, u_{2n}^+$ form a basis of $V_+$ and $u_1^-, \dots, u_{2n}^-$ a basis of $V_-$, and complex conjugation on $V_{\mathbb{C}}$ interchanges $u_j^+$ and $u_j^-$.*

*Proof.* Using (4) and $\delta^2 = -d$,

$$Su_j^s = \tfrac{1}{2}(y_j + s\,\tfrac{\delta}{d}\, d\, x_j) = \tfrac{s\delta}{2}(x_j + \tfrac{1}{s\delta} y_j) = \tfrac{s\delta}{2}(x_j - s\tfrac{\delta}{d} y_j) = s\,\delta\, u_j^s,$$

where the third equality uses $1/\delta = -\delta/d$. The change of basis $(x_j, y_j) \mapsto (u_j^+, u_j^-)$ is block diagonal with $2 \times 2$ blocks of determinant $\delta/(2d) \neq 0$, so the $4n$ vectors $u_j^\pm$ are linearly independent. Since $V_+$ and $V_-$ have dimension $2n$ each by Notation 3.3, the $u_j^+$ form a basis of $V_+$ and the $u_j^-$ a basis of $V_-$. Conjugation fixes $x_j$ and $y_j$ and sends $\delta$ to $-\delta$, which interchanges the two formulas. ∎

**Definition 4.3** (The complex structure)**.** Define a Hodge structure of weight one on $V$ by

$$V^{1,0} = \langle u_1^+, \dots, u_n^+,\, u_{n+1}^-, \dots, u_{2n}^- \rangle_{\mathbb{C}}, \qquad V^{0,1} = \overline{V^{1,0}}. \tag{8}$$

The first $n$ basis directions contribute to $V^{1,0}$ through their $+$-eigenvector and the last $n$ through their $-$-eigenvector, so each of $V_\pm$ meets $V^{1,0}$ in a subspace of dimension $n$. The signs $\epsilon_j$ of (5) are matched to this choice and are not free: the proof of Theorem 4.4 shows that the second Riemann relation holds for them and fails as soon as one of them is reversed (see also Remark 4.6). Reversing all of them replaces $Q$ by $-Q$, and the construction then goes through with $V^{1,0}$ and $V^{0,1}$ interchanged.

**Theorem 4.4** (The model is of Weil type)**.** *With the data of Setup 4.1 and Definition 4.3, the pair $(V, V^{1,0})$ is a polarised rational Hodge structure of weight one with polarising form $Q$, so $V = H^1(A, \mathbb{Q})$ for an abelian variety $A$ of dimension $2n$; the multiplication $S$ makes $K$ act on $A$ up to isogeny; $\eta$ is the class of the polarisation; and $(A, \iota, \eta)$ is of $(K, -1, n)$-Weil type in the sense of Definition 3.1.*

*Proof.* We verify the two Riemann bilinear relations, then the three conditions of Definition 3.1.

*The form vanishes on $V^{1,0}$.* By (5) the form $Q$ is block diagonal for the decomposition of $V$ into the planes $\langle x_j, y_j\rangle$, so $Q(u_j^s, u_k^t) = 0$ whenever $j \neq k$. Within one block, $Q(u_j^s, u_j^s) = 0$ because $Q$ is alternating, and

$$Q(u_j^+, u_j^-) = \tfrac{1}{4}\, Q(x_j - \tfrac{\delta}{d} y_j,\ x_j + \tfrac{\delta}{d} y_j) = \tfrac{1}{4} \cdot \tfrac{2\delta}{d}\, Q(x_j, y_j) = \frac{\delta\, \epsilon_j}{2d}. \tag{9}$$

The basis (8) of $V^{1,0}$ contains, for each $j$, exactly one of $u_j^+, u_j^-$. Hence any two of its members either lie in different blocks or are equal, and in both cases $Q$ vanishes on them. So $Q(V^{1,0}, V^{1,0}) = 0$.

*Positivity.* Let $0 \neq v = \sum_j a_j v_j \in V^{1,0}$, where $v_j$ is $u_j^+$ for $j \leq n$ and $u_j^-$ for $j > n$. Then $\overline{v} = \sum_j \overline{a_j}\, \overline{v_j}$ with $\overline{v_j}$ the other eigenvector in the same block, so again only the diagonal terms survive:

$$i\, Q(v, \overline{v}) = \sum_{j \leq n} i\, |a_j|^2\, Q(u_j^+, u_j^-) + \sum_{j > n} i\, |a_j|^2\, Q(u_j^-, u_j^+).$$

By (9) and $\delta = i\sqrt{d}$ the first kind of term is $i \cdot \delta\epsilon_j/(2d) = -\epsilon_j/(2\sqrt{d})$ and the second is $+\epsilon_j/(2\sqrt{d})$. With $\epsilon_j = -1$ for $j \leq n$ and $+1$ for $j > n$ both are $+1/(2\sqrt{d})$, so $i\, Q(v, \overline{v}) = \|a\|^2/(2\sqrt{d}) > 0$. This is the second Riemann relation. For the complex structure (8) it forces the signature of $\epsilon$ to be $(n, n)$, since reversing any single $\epsilon_j$ makes the corresponding term negative.

*Weil type.* Condition (i) is $\dim A = \frac{1}{2} \dim_{\mathbb{Q}} V = 2n$. For (ii), $V^{1,0}$ is spanned by the $u_j^+$ with $j \leq n$ and the $u_j^-$ with $j > n$, so $\dim V_+^{1,0} = \dim V_-^{1,0} = n$; by duality, as in the proof of Lemma 3.5, this is the balanced eigenvalue condition on the tangent space. For (iii), write $\tau = a + b\sqrt{-d}$, so that $\iota(\tau)$ acts by

$$x_j \longmapsto a\, x_j + b\, y_j, \qquad y_j \longmapsto -bd\, x_j + a\, y_j,$$

and $\iota(\overline{\tau})$ by the same formulas with $b$ replaced by $-b$. Since $Q$ is block diagonal and $\iota(\tau)$ preserves each block, it suffices to compare $Q(\iota(\tau)v, w)$ with $Q(v, \iota(\overline{\tau})w)$ for basis vectors $v, w$ of one block, where

$$\begin{aligned} Q(\iota(\tau)x_j, y_j) &= a\epsilon_j = Q(x_j, \iota(\overline{\tau})y_j), \\ Q(\iota(\tau)x_j, x_j) &= -b\epsilon_j = Q(x_j, \iota(\overline{\tau})x_j), \\ Q(\iota(\tau)y_j, y_j) &= -bd\, \epsilon_j = Q(y_j, \iota(\overline{\tau})y_j), \end{aligned}$$

and the remaining pair $(y_j, x_j)$ follows from the first line with $\tau$ replaced by $\overline{\tau}$ and the antisymmetry of $Q$. So $\iota(\tau)^\dagger = \iota(\overline{\tau})$. Equivalently, in the form used later, $\iota(\tau)^* \eta = \mathrm{N}(\tau)\, \eta$: by (7), $x_j \wedge y_j = \frac{2d}{\delta}\, u_j^+ \wedge u_j^-$, and $\iota(\tau)$ multiplies $u_j^+ \wedge u_j^-$ by $\tau\overline{\tau}$.

Finally, $\eta$ is of type $(1, 1)$, because each $x_j \wedge y_j$ is a multiple of $u_j^+ \wedge u_j^-$ and in each block one of $u_j^+, u_j^-$ lies in $V^{1,0}$ and the other in $V^{0,1}$. Since in addition $\eta^{2n} \neq 0$, the class $\eta$ is a polarisation class. ∎

**Lemma 4.5** (The hermitian form)**.** *On the model of Setup 4.1 there is a unique $K$-hermitian form $H$ on $V$, linear in the first variable, with*

$$Q(v,w)=\tfrac{1}{2}\operatorname{Tr}_{K/\mathbb{Q}}(\delta^{-1}H(v,w)),\qquad \text{namely}\qquad H(v,w)=Q(\sqrt{-d}\,v,\,w)+\sqrt{-d}\,Q(v,w), \tag{10}$$

*and in the basis $u_1,\dots,u_{2n}$ it is $H=\operatorname{diag}(-\epsilon_1,\dots,-\epsilon_{2n})$, of signature $(n,n)$.*

*Proof.* $K$-linearity in the first variable is the computation

$$H(\sqrt{-d}v,w)=Q(-dv,w)+\sqrt{-d}\,Q(\sqrt{-d}v,w)=\sqrt{-d}\,H(v,w),$$

and the hermitian symmetry $\overline{H(w,v)}=H(v,w)$ follows from $Q(v,\sqrt{-d}\,w)=-Q(\sqrt{-d}\,v,w)$, which is condition (iii) of Definition 3.1 for $\tau=\sqrt{-d}$. Multiplying by $\delta^{-1}$ and taking the trace kills the first summand of (10), since $Q$ is $\mathbb{Q}$-valued and $\operatorname{Tr}_{K/\mathbb{Q}}(\delta^{-1})=0$, and doubles the second; this gives $Q$. For uniqueness, if $H'$ is a second such form and $z=H(v,w)-H'(v,w)$, then $\operatorname{Tr}_{K/\mathbb{Q}}(\delta^{-1}z)=0$, and replacing $v$ by $\sqrt{-d}\,v$ gives $\operatorname{Tr}_{K/\mathbb{Q}}(z)=0$; together these force $z=0$. Finally $H(u_j,u_j)=Q(y_j,x_j)=-\epsilon_j$ and $H(u_j,u_k)=0$ for $j\neq k$, so the signature is the number of indices with $\epsilon_j=-1$ against the number with $\epsilon_j=+1$, that is $(n,n)$. ∎

*Remark* 4.6. The two signatures that have appeared express the same condition. The hermitian form $H$ has signature $(n,n)$ by Lemma 4.5, and the $K$-action on the tangent space has eigenvalue multiplicities $(n,n)$ by Definition 3.1(ii). The positivity computation in the proof of Theorem 4.4 passes from one to the other: the sign $\epsilon_j$ decides both which eigenvector of the $j$-th block is holomorphic and the sign of $H$ on that block.

*Remark* 4.7. Taking the lattice $\mathcal{O}^{2n}$ for an order $\mathcal{O}\subset K$ and scaling $Q$ to be integral on it produces an abelian variety and not only a Hodge structure. Nothing below depends on the lattice, so we work with the rational Hodge structure throughout. Varying (8) over the period domain of the hermitian form gives the Weil family, of dimension $n^2$ inside the $n(2n+1)$-dimensional moduli of principally polarised abelian $2n$-folds (Proposition B.8).

## 4.2. **The two generators.**

*Notation* 4.8. For $T\subseteq\{1,\dots,2n\}$ write

$$w^{(T)}=w_1^{(T)}\wedge w_2^{(T)}\wedge\cdots\wedge w_{2n}^{(T)},\qquad w_j^{(T)}=\begin{cases}y_j, & j\in T,\\ x_j, & j\notin T.\end{cases}$$

The $2^{2n}$ elements $w^{(T)}$ are part of the monomial basis of $\bigwedge^{2n}V$, which has dimension $\binom{4n}{2n}$; they are the monomials that contain exactly one of $x_j,y_j$ for each $j$.

**Theorem 4.9** (The Weil line in coordinates)**.** *Let $(A,\iota,\eta)$ be the model of Theorem 4.4 and let*

$$\omega_1=\sum_{\substack{T\subseteq\{1,\dots,2n\}\\ |T|\text{ even}}}(-1)^{|T|/2}\,d^{\,n-|T|/2}\,w^{(T)},\qquad \omega_2=\sum_{\substack{T\subseteq\{1,\dots,2n\}\\ |T|\text{ odd}}}(-1)^{(|T|-1)/2}\,d^{\,n-(|T|+1)/2}\,w^{(T)}. \tag{11}$$

*Then*

(i) *$\omega_1$ and $\omega_2$ have integer coefficients and each has exactly $2^{2n-1}$ nonzero terms;*
(ii) *$\omega_1,\omega_2$ is a $\mathbb{Q}$-basis of the Weil line $W(A)$ of Definition 3.6;*
(iii) *writing $\omega_\pm=u_1^\pm\wedge\cdots\wedge u_{2n}^\pm$ for the generators of $\bigwedge^{2n}V_\pm$,*

$$\omega_+ +\omega_-=\frac{\omega_1}{2^{2n-1}d^{\,n}},\qquad \frac{\omega_+-\omega_-}{\delta}=-\frac{\omega_2}{2^{2n-1}d^{\,n}};$$

(iv) *$\iota(\tau)^*\omega_+ = \tau^{2n}\omega_+$ and $\iota(\tau)^*\omega_- = \overline{\tau}^{\,2n}\omega_-$ for all $\tau \in K^\times$, so on the plane $\langle \omega_1, \omega_2 \rangle$ the element $\iota(\tau)$ acts by the $\mathbb{Q}$-rational matrix of multiplication by $\tau^{2n}$ on $K$;*

(v) *$\omega_1$ and $\omega_2$ are of Hodge type $(n, n)$, hence are Hodge classes.*

*Proof.* (iii) and (i). Expanding (7) in the product defining $\omega_s$ and collecting the monomials $w^{(T)}$,

$$\omega_s = \frac{1}{2^{2n}} \sum_T \Big(-s\,\frac{\delta}{d}\Big)^{|T|} w^{(T)}, \tag{12}$$

because the factor from index $j$ contributes $x_j$ with coefficient $1/2$ and $y_j$ with coefficient $-s\delta/(2d)$, and the monomials $w^{(T)}$ appear in the order $j = 1, \dots, 2n$ with no reordering sign. Therefore

$$\omega_+ + \omega_- = \frac{1}{2^{2n-1}} \sum_{|T| \text{ even}} \Big(\frac{\delta}{d}\Big)^{|T|} w^{(T)}, \qquad \frac{\omega_+ - \omega_-}{\delta} = -\frac{1}{2^{2n-1}} \sum_{|T| \text{ odd}} \frac{\delta^{|T|-1}}{d^{|T|}}\, w^{(T)},$$

since the terms with $|T|$ odd cancel in the first sum and those with $|T|$ even in the second. Now $\delta^{2m} = (-d)^m$, so for $|T| = 2m$ the coefficient in the first sum is $(-1)^m d^{-m}$, and for $|T| = 2m+1$ the coefficient in the second is $-(-1)^m d^{-m-1}$. Multiplying by $2^{2n-1} d^n$ clears denominators, since $m \le n$ in the first case and $m + 1 \le n$ in the second, and gives (11). The term counts are $\sum_m \binom{2n}{2m} = \sum_m \binom{2n}{2m+1} = 2^{2n-1}$.

(ii). By (iii), $\omega_1$ and $\omega_2$ lie in $\bigwedge^{2n} V_+ \oplus \bigwedge^{2n} V_-$. They are fixed by complex conjugation, which interchanges $\omega_+$ and $\omega_-$ by Lemma 4.2 and sends $\delta$ to $-\delta$, and by (i) they have rational coefficients. Hence they lie in $(\bigwedge^{2n} V_+ \oplus \bigwedge^{2n} V_-) \cap H^{2n}(A, \mathbb{Q})$, which is $W(A)$ by Definition 3.6 and has dimension two by Proposition 3.7(i). They are linearly independent over $\mathbb{C}$ since $\omega_+, \omega_-$ are, so they form a basis of $W(A)$.

(iv). The first two identities are immediate, since $\iota(\tau)$ multiplies each of the $2n$ factors $u_j^+$ by $\tau$ and each $u_j^-$ by $\overline{\tau}$. For the last clause, by (iii) the space $W(A) \otimes \mathbb{C}$ is $\mathbb{C}\omega_+ \oplus \mathbb{C}\omega_-$, on which $\iota(\tau)$ acts by $\tau^{2n}$ and $\overline{\tau}^{\,2n}$; so under the isomorphism $W(A) \cong K$ of Proposition 3.7(iii) the action of $\iota(\tau)$ is multiplication by $\tau^{2n}$, whose matrix in the basis $\omega_1, \omega_2$ has rational entries.

(v). By (8) the monomial $\omega_+ = u_1^+ \wedge \cdots \wedge u_{2n}^+$ has $n$ factors in $V^{1,0}$, namely those with $j \le n$, and $n$ factors in $V^{0,1}$; so $\omega_+$ is of type $(n, n)$, and likewise $\omega_-$. By (iii), so are $\omega_1$ and $\omega_2$. ∎

*Remark* 4.10 (Balanced signature). The balanced condition enters the proof of Theorem 4.9 only in part (v). If (8) is replaced by a complex structure with $\dim V_+^{1,0} = p$ and $\dim V_-^{1,0} = 2n - p$, everything else being kept, then $\omega_+$ has $p$ holomorphic factors and is of type $(p, 2n-p)$, so $W(A)$ is a rational sub-Hodge structure of $H^{2n}(A, \mathbb{Q})$ of type $\{(p, 2n-p), (2n-p, p)\}$, and it contains a nonzero Hodge class if and only if $p = n$. The balanced signature in Definition 3.1(ii) is therefore the condition under which the Weil line contains Hodge classes at all. Proposition C.7 proves this for every abelian variety with a Rosati-stable action of $K$.

4.3. **The case $n = 2$.** Here $\dim A = 4$, $\dim_{\mathbb{Q}} V = 8$, $H^4(A, \mathbb{Q})$ has dimension $\binom{8}{4} = 70$, and each generator has $2^3 = 8$ terms. Writing the monomials in the order $j = 1, 2, 3, 4$ as

in Notation 4.8, (11) reads

$$\omega_1 = d^2\, x_1x_2x_3x_4 \;-\; d\sum_{|T|=2} w^{(T)} \;+\; y_1y_2y_3y_4, \tag{13}$$

$$\omega_2 = d\sum_{|T|=1} w^{(T)} \;-\; \sum_{|T|=3} w^{(T)}, \tag{14}$$

with the wedge signs omitted. For $d = 1$, written out in full,

$$\begin{aligned}\omega_1 &= x_1x_2x_3x_4 - y_1y_2x_3x_4 - y_1x_2y_3x_4 - y_1x_2x_3y_4 - x_1y_2y_3x_4 - x_1y_2x_3y_4 - x_1x_2y_3y_4\\ &\qquad + y_1y_2y_3y_4,\\ \omega_2 &= y_1x_2x_3x_4 + x_1y_2x_3x_4 + x_1x_2y_3x_4 + x_1x_2x_3y_4\\ &\qquad - y_1y_2y_3x_4 - y_1y_2x_3y_4 - y_1x_2y_3y_4 - x_1y_2y_3y_4.\end{aligned}$$

The square of the polarisation is $\eta^2 = 2\sum_{j<k} \epsilon_j\epsilon_k\, x_jy_jx_ky_k$, a sum of six monomials, none of which is among the sixteen above; Proposition 4.11 below shows that this separation holds in every dimension. Figure 3 arranges the sixteen monomials $w^{(T)}$ on the vertices of the 4-cube.

4.4. **The case $n = 3$.** Here $\dim A = 6$, $\dim_{\mathbb{Q}} V = 12$, $H^6(A, \mathbb{Q})$ has dimension $\binom{12}{6} = 924$, and each generator has $2^5 = 32$ terms. The coefficients depend only on $|T|$, so the two classes are recorded completely by Table 1.

| $\omega_1$ | | | | $\omega_2$ | | | |
|---|---|---|---|---|---|---|---|
| $\lvert T\rvert$ | terms | coefficient | at $d=2$ | $\lvert T\rvert$ | terms | coefficient | at $d=2$ |
| 0 | 1 | $+d^3$ | $+8$ | 1 | 6 | $+d^2$ | $+4$ |
| 2 | 15 | $-d^2$ | $-4$ | 3 | 20 | $-d$ | $-2$ |
| 4 | 15 | $+d$ | $+2$ | 5 | 6 | $+1$ | $+1$ |
| 6 | 1 | $-1$ | $-1$ | | | | |
| | 32 | | | | 32 | | |

TABLE 1. The Weil classes at $n = 3$. Each row gives the number of index sets $T$ of that size and the common coefficient of the corresponding monomials $w^{(T)}$ in (11). The two classes together involve all 64 monomials containing exactly one of $x_j, y_j$ for each $j$, out of the 924 monomials of $H^6$.

The pattern has a closed description. By Theorem 4.9(iii) and (7),

$$\omega_1 + \delta\,\omega_2 = 2^{2n}d^n\,\omega_- = d^{-n}\prod_{j=1}^{2n}(d\,x_j + \delta\,y_j), \tag{15}$$

the product being the wedge product taken in the order $j = 1, \dots, 2n$. The proof of Theorem 4.9 uses of the basis only that the $u_j$ form a $K$-basis of $V$, so (11) and (15) define generators of the Weil line, and hold, for every $K$-basis $u_1, \dots, u_{2n}$ of $H^1(A, \mathbb{Q})$ of an abelian variety of $(K, -1, n)$-Weil type, with $x_j = u_j$ and $y_j = \sqrt{-d}\,u_j$. For $n = 3$ and $d = 1$ the coefficients in Table 1 are $+1, -1, +1, -1$ for $\omega_1$ and $+1, -1, +1$ for $\omega_2$, and the two classes are the real and imaginary parts of $\prod_{j=1}^{6}(x_j + \sqrt{-1}\,y_j)$.

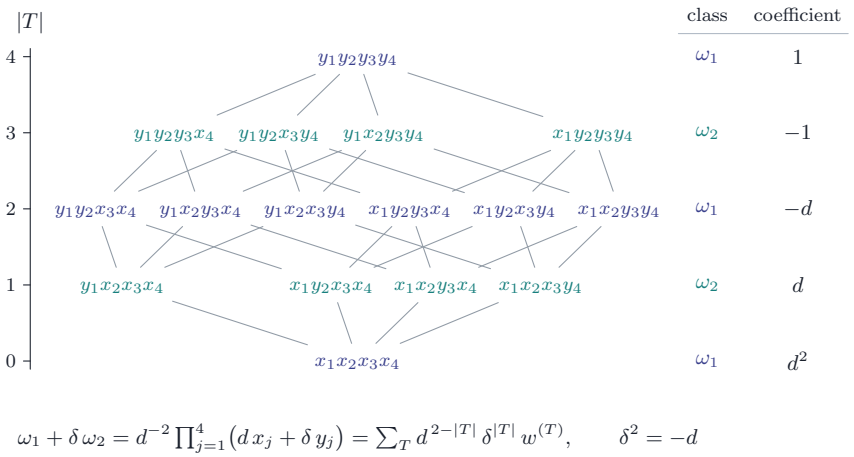



FIGURE 3. The $2^{2n} = 16$ monomials $w^{(T)}$ of Notation 4.8 at $n = 2$, as the vertices of the 4-cube, projected so that the height of $w^{(T)}$ is $|T|$. An edge joins two monomials whose index sets differ in one element, that is, two monomials related by a single substitution $x_j \mapsto y_j$. The two colours are the bipartition by the parity of $|T|$: the eight even vertices carry $\omega_1$ and the eight odd ones carry $\omega_2$. The coefficient of $w^{(T)}$ depends only on $|T|$ and is given by (13) and (14).

4.5. **Disjoint supports.** Proposition 3.10 is deduced from the character computation of Section 5. In the model the Weil line can be separated from the powers of the polarisation without any representation theory: the Weil classes and the powers of $\eta$ are supported on disjoint sets of monomials.

**Proposition 4.11** (Disjoint supports)**.** *In the model of Theorem 4.4, every monomial occurring in $\omega_1$ or in $\omega_2$ contains exactly one of $x_j, y_j$ for every $j \in \{1, \dots, 2n\}$, and every monomial occurring in $\eta^k$ contains either both or neither, for every $j$ and every $k \geq 0$. Consequently*

$$W(A) \cap \mathbb{Q}\,\eta^n = 0,$$

*and more generally $W(A)$ meets the span of the monomials occurring in powers of $\eta$ in zero.*

*Proof.* The first assertion holds by (11) and Notation 4.8. For the second, $\eta$ is the combination (6) of the $2n$ two-forms $x_j \wedge y_j$, which pairwise commute and square to zero, so

$$\eta^k = k! \sum_{\substack{S \subseteq \{1,\dots,2n\} \\ |S| = k}} \Big(\prod_{j \in S} \epsilon_j\Big) \bigwedge_{j \in S} (x_j \wedge y_j),$$

and the monomial indexed by $S$ contains both $x_j$ and $y_j$ for $j \in S$ and neither for $j \notin S$. A monomial cannot satisfy both patterns for a given $j$, and since $2n \geq 1$ there is at least one index at which to test, so the two families of monomials are disjoint. Since $\omega_1, \omega_2$ span $W(A)$ by Theorem 4.9(ii), a nonzero element of $W(A)$ has a nonzero coefficient on some $w^{(T)}$. As the monomials form a basis of $\bigwedge^{2n} V$ and $w^{(T)}$ occurs in no power of $\eta$, such an element does not lie in the span of the monomials occurring in powers of $\eta$, and in particular not in $\mathbb{Q}\,\eta^n$. ∎

*Remark* 4.12. The two proofs that the Weil line misses the powers of the polarisation differ in scope. Proposition 4.11 concerns one model and one basis, and it is short because the basis makes the $K$-action block diagonal. The character proof is basis free, applies to every abelian variety of Weil type, and identifies the obstruction as a difference of characters. The first can be checked by hand; the second is the argument that Theorem 5.6 extends.

4.6. **The Hodge ring of a general member.** It remains to count the Hodge classes, which measures how much of the Hodge conjecture for $A$ is at stake. The Hodge classes of an abelian variety are the invariants of its Hodge group, the special Mumford–Tate group $\mathrm{Hg}(A) \subseteq \mathrm{SL}(V)$ [Del82]: $\mathrm{Hdg}^k(A) = H^{2k}(A,\mathbb{Q})^{\mathrm{Hg}(A)}$.

**Lemma 4.13** (The Hodge group)**.** *Let $(A,\iota,\eta)$ be of $(K,-1,n)$-Weil type and let $H$ be the $K$-hermitian form of Lemma 4.5. Then*

$$\mathrm{Hg}(A) \subseteq \mathrm{SU}(V,H).$$

*Proof.* The group $\mathrm{Hg}(A)$ is a $\mathbb{Q}$-algebraic subgroup of $\mathrm{SL}(V)$ which fixes every Hodge class in every tensor construction on $V$. It fixes $\eta \in \bigwedge^2 V$, hence preserves $Q$; and it commutes with $\iota(K)$, because each $\iota(\tau)$ is a Hodge class in $V \otimes V^\vee$. A $\mathbb{Q}$-linear automorphism of $V$ commuting with the $K$-action and preserving $Q$ preserves $H$, by the formula (10) expressing $H$ through $Q$ and the $K$-action. Hence $\mathrm{Hg}(A) \subseteq \mathrm{U}(V,H)$. Now $W(A)$ consists of Hodge classes by Proposition 3.7(ii), so $\mathrm{Hg}(A)$ fixes $W(A)\otimes\mathbb{C}$ pointwise, and in particular it fixes the line $\bigwedge^{2n} V_+$ pointwise. An element $g \in \mathrm{U}(V,H)$ acts on $\bigwedge^{2n} V_+$ by $\det_K(g)$, so every $g \in \mathrm{Hg}(A)$ has $\det_K(g) = 1$, which is the definition of $\mathrm{SU}(V,H)$. ∎

For a very general member of the Weil family, that is, for every member whose point in the period domain $\mathcal{D}_{n,n}$ of Proposition B.8 lies outside a certain countable union of proper closed analytic subsets, the containment of Lemma 4.13 is an equality; this is Weil's theorem [Wei77], in the form given by Moonen and Zarhin [MZ99]. The restriction to very general members cannot be replaced by a Zariski open condition, since the locus where the endomorphism algebra jumps is a countable union of such subsets, by Theorem B.6, and is dense in the classical topology. For such a member the Hodge classes can be counted in every degree.

**Theorem 4.14** (The Hodge ring of a general member)**.** *Let $(A,\iota,\eta)$ be of $(K,-1,n)$-Weil type with $\mathrm{Hg}(A) = \mathrm{SU}(V,H)$. Then for every $k$ with $0 \le k \le 2n$,*

$$\dim_{\mathbb{Q}} \mathrm{Hdg}^k(A) = \begin{cases} 3, & k = n, \\ 1, & k \neq n, \end{cases} \qquad \text{and} \qquad \mathrm{Hdg}^n(A) = \mathbb{Q}\,\eta^n \oplus W(A).$$

*Proof.* Complexify. The group $\mathrm{SU}(V,H)\otimes\mathbb{C}$ is $\mathrm{SL}(V_+) \cong \mathrm{SL}_{2n}(\mathbb{C})$, acting on $V_+$ tautologically and on $V_-$ through the isomorphism $V_- \cong V_+^\vee$ induced by $H$. By Proposition 5.1,

$$H^{2k}(A,\mathbb{C}) = \bigoplus_{r+s=2k} \bigwedge^r V_+ \otimes \bigwedge^s V_+^\vee,$$

so it suffices to find the invariants in each summand. An invariant is a weight-zero vector annihilated by every raising operator. Fix a basis of $V_+$, take the diagonal torus of $\mathrm{SL}(V_+)$ as maximal torus, and for $T,U \subseteq \{1,\dots,2n\}$ let $v_T \in \bigwedge^{|T|} V_+$ and $v_U^\vee \in \bigwedge^{|U|} V_+^\vee$ be the corresponding monomials in the basis and its dual. In the weight lattice of $\mathrm{SL}_{2n}$ the weight of $v_T \otimes v_U^\vee$ is $\mathbf{1}_T - \mathbf{1}_U$ modulo $(1,1,\dots,1)$; since the entries of $\mathbf{1}_T - \mathbf{1}_U$ lie in $\{-1,0,1\}$, the only multiples of $(1,\dots,1)$ it can equal are $0$ and $\pm(1,\dots,1)$, giving

$$T = U \quad (\text{so } r = s), \qquad (T,U) = (\{1,\dots,2n\},\emptyset), \qquad (T,U) = (\emptyset,\{1,\dots,2n\}).$$

The last two occur only for $(r,s)=(2n,0)$ and $(0,2n)$, each with a one-dimensional weight-zero space consisting of the invariant lines $\bigwedge^{2n} V_+$ and $\bigwedge^{2n} V_-$, which are SL-invariant because SL has trivial determinant. For $r=s$ the weight-zero space has dimension $\binom{2n}{r}$ and its invariant subspace is the line spanned by the image of the identity under $\mathrm{End}(\bigwedge^r V_+) \cong \bigwedge^r V_+ \otimes \bigwedge^r V_+^\vee$, which is one dimensional because $\bigwedge^r V_+$ is an irreducible $\mathrm{SL}_{2n}$-module and Schur's lemma applies.

Collecting the contributions: for $k \neq n$ the only one is $(r,s)=(k,k)$, of dimension one and spanned by $\eta^k$, which is a nonzero invariant lying in that summand by Lemma 5.2. For $k=n$ there are three contributions, $(n,n)$, $(2n,0)$ and $(0,2n)$, of total dimension three; the first is spanned by $\eta^n$ and the last two span $W(A)\otimes\mathbb{C}$ by Lemma 5.3. Taking invariants commutes with extension of scalars from $\mathbb{Q}$ to $\mathbb{C}$, so the dimensions over $\mathbb{Q}$ are the same. ∎

Figure 4 displays these dimensions for $n=2,3,4$.

**Corollary 4.15** (Reduction to one class)**.** *Let $(A,\iota,\eta)$ be as in Theorem 4.14. The Hodge conjecture holds for $A$ in every degree if and only if the single class $\omega_1$ of (11) is algebraic.*

*Proof.* For $k \neq n$ the Hodge classes are the multiples of $\eta^k$, which are algebraic, being powers of a divisor class. For $k=n$ they are spanned by $\eta^n$, $\omega_1$ and $\omega_2$ by Theorem 4.14. Finally, $\omega_2$ is algebraic as soon as $\omega_1$ is. By Theorem 4.9(iv) the plane $W(A)$ is a free $K$-module of rank one generated by $\omega_1$. For $\tau \in K^\times$ some positive integer multiple $m\,\iota(\tau)$ lies in $\mathrm{End}(A)$, and on $H^{2n}(A,\mathbb{Q})$ the operator $\iota(\tau)^*$ is $m^{-2n}$ times the pullback by $m\,\iota(\tau)$, so it carries algebraic classes to algebraic classes. Any $\tau$ with $\tau^{2n} \notin \mathbb{Q}$, which exists by the argument of Lemma 5.4, moves $\omega_1$ to a class with a nonzero $\omega_2$ component, and so $\omega_2$ is a rational combination of the algebraic classes $\omega_1$ and $\iota(\tau)^*\omega_1$. ∎

*Remark* 4.16. Corollary 4.15 explains why the Weil line is the standard test case. On a general abelian variety of Weil type the Hodge conjecture concerns one explicitly written class with $2^{2n-1}$ integer coefficients, and no space of classes of unknown structure. The obstruction theorems of this paper constrain the ways in which that class can be produced, and Section 15 exhibits in every Weil family a member on which the whole Weil line consists of polynomials in divisor classes.

## 5. The character grading and the position of the Weil line

This section proves the character obstruction, Theorem 5.6, which is used in the proof of Proposition 3.10. The group $K^\times$ acts on $H^{2n}(A,\mathbb{Q})$ through the endomorphisms $\iota(\tau)$. This action grades the cohomology, and the Weil line and the power $\eta^n$ of the polarisation lie in different graded pieces; once the grading is written down, the obstruction follows in a few lines. The one point that needs care is the character through which $K^\times$ acts on the Weil line. It is $\tau \mapsto \tau^{2n}$, together with its conjugate, and not a power of the norm, even though the Weil line is a $K$-line and multiplication by $\tau$ on a $K$-line has determinant $\mathrm{N}(\tau)$ (Remark 5.9).

5.1. **The grading.** Throughout this section $(A,\iota,\eta)$ is of $(K,-1,n)$-Weil type, $V = H^1(A,\mathbb{Q})$, and $V_\mathbb{C} = V_+ \oplus V_-$ is the splitting of Notation 3.3.

**Proposition 5.1** (Character grading)**.** *For every $k \geq 0$,*

$$H^k(A,\mathbb{C}) \;=\; \bigoplus_{r+s=k} \Big( \textstyle\bigwedge^r V_+ \otimes \bigwedge^s V_- \Big), \tag{16}$$

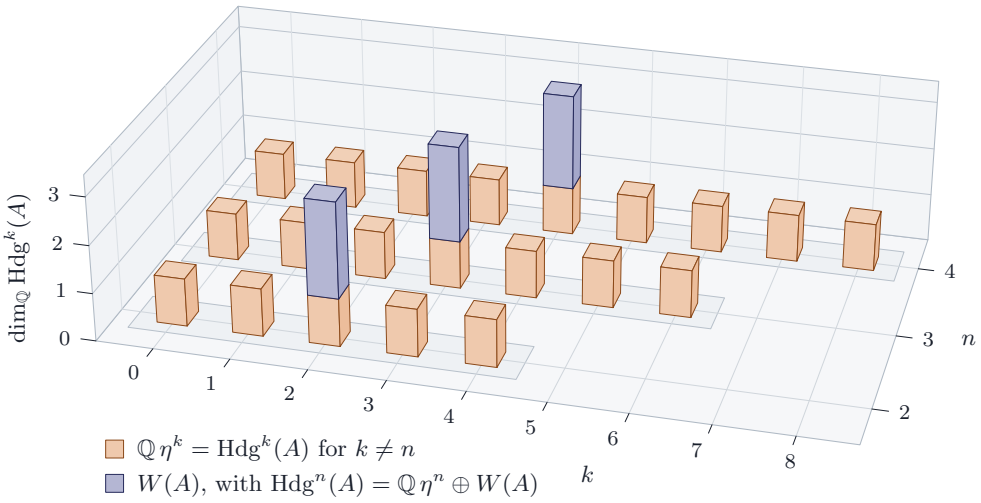



FIGURE 4. The dimensions of $\mathrm{Hdg}^k(A)$ for a general abelian variety of $(K, -1, n)$-Weil type, $0 \le k \le 2n$, at $n = 2, 3, 4$, as computed in Theorem 4.14. Every degree carries the line $\mathbb{Q}\,\eta^k$; the middle degree $k = n$ carries in addition the two dimensions of the Weil line $W(A)$, and nothing else occurs. The profile is the same for every $n$.

*and $\iota(\tau)^*$ acts on the summand indexed by $(r, s)$ by the scalar $\tau^r \overline{\tau}^s$. The summands are the isotypic components for the action of $K^\times$, and the characters $\tau \mapsto \tau^r \overline{\tau}^s$ for distinct pairs $(r, s)$ with $r + s = k$ are pairwise distinct.*

*Proof.* By (1), $H^k(A, \mathbb{C}) = \bigwedge^k V_{\mathbb{C}}$, and the exterior power of a direct sum decomposes as $\bigwedge^k (V_+ \oplus V_-) = \bigoplus_{r+s=k} \bigwedge^r V_+ \otimes \bigwedge^s V_-$, which is (16). By Notation 3.3, $\iota(\tau)^*$ acts on $V_+$ by $\tau$ and on $V_-$ by $\overline{\tau}$. Since it is an algebra homomorphism for the wedge product, it acts on $\bigwedge^r V_+ \otimes \bigwedge^s V_-$ by $\tau^r \overline{\tau}^s$.

For the last statement, suppose that $\tau^r \overline{\tau}^s = \tau^{r'} \overline{\tau}^{s'}$ for all $\tau \in K^\times$, where $r + s = r' + s' = k$. Then $s - s' = r' - r$, and dividing gives $(\tau/\overline{\tau})^{r-r'} = 1$ for every $\tau \in K^\times$. If $r \neq r'$, the image of the map $c(\tau) = \tau/\overline{\tau}$ therefore lies in the group of $(r - r')$-th roots of unity of $K$, which is finite: the group of all roots of unity of the imaginary quadratic field $K$ has order 2, 4 or 6. But $c$ has infinite image. Two elements have the same image exactly when their ratio lies in $\mathbb{Q}^\times$, and the elements $m + \sqrt{-d}$ with $m \in \mathbb{Z}$ are pairwise non-proportional over $\mathbb{Q}$. Hence $r = r'$, and so $s = s'$. Since the characters of the summands are pairwise distinct, the summands are the isotypic components.

The distinctness is that of characters of the torus $\mathrm{Res}_{K/\mathbb{Q}}\,\mathbb{G}_m$, not of their values at a given $\tau$: the values at $\tau$ for $(r, s)$ and $(r', s')$ agree if and only if $\tau/\overline{\tau}$ is a root of unity of order dividing $r - r'$, and this can happen (Remark 5.5). ∎

**Lemma 5.2** (The polarisation sits in the middle)**.** *Let $(A, \iota, \eta)$ be of $(K, -1, n)$-Weil type. Then*

$$\iota(\tau)^* \eta = \mathrm{N}_{K/\mathbb{Q}}(\tau)\,\eta \qquad \text{for all } \tau \in K^\times,$$

*so $\eta$ lies in the summand $(r, s) = (1, 1)$ of (16) and $\eta^n$ lies in the summand $(r, s) = (n, n)$, on which $K^\times$ acts by $\mathrm{N}^n$.*

*Proof.* Let $E$ be the alternating form on $H_1(A,\mathbb{Q})$ attached to $\eta$. By condition (iii) of Definition 3.1, the Rosati involution $\dagger$ of $\eta$ satisfies $\iota(\tau)^\dagger = \iota(\overline{\tau})$, so by the defining property of the Rosati involution

$$E(\iota(\tau)x,\, y) = E(x,\, \iota(\tau)^\dagger y) = E(x,\, \iota(\overline{\tau})y) \qquad \text{for all } x, y.$$

Therefore

$$\begin{aligned}(\iota(\tau)^* E)(x,y) &= E(\iota(\tau)x, \iota(\tau)y) = E(x, \iota(\overline{\tau})\iota(\tau)y)\\ &= E(x, \iota(\mathrm{N}(\tau))y) = \mathrm{N}(\tau)\, E(x,y),\end{aligned}$$

using $\overline{\tau}\tau = \mathrm{N}(\tau) \in \mathbb{Q}$ and the $\mathbb{Q}$-linearity of $\iota$. Under the identification of $H^2(A,\mathbb{Q}) = \bigwedge^2 H^1(A,\mathbb{Q})$ with the alternating forms on $H_1(A,\mathbb{Q})$, which is compatible with pullback, $\eta$ corresponds to $E$, so this is the displayed identity. Since $\mathrm{N}(\tau) = \tau\overline{\tau}$, Proposition 5.1 identifies the isotypic component of N in $H^2(A,\mathbb{C})$ with the summand $(r,s) = (1,1)$, and $\eta$ lies in it. Because $\iota(\tau)^*$ is a ring homomorphism, $\iota(\tau)^*\eta^n = \mathrm{N}(\tau)^n\eta^n$, and in the same way this places $\eta^n$ in the summand $(n,n)$. ∎

**Lemma 5.3** (The Weil line sits at the ends)**.** *Let $(A,\iota,\eta)$ be of $(K,-1,n)$-Weil type. Then*

$$W(A)\otimes_{\mathbb{Q}}\mathbb{C} = \left(\bigwedge^{2n} V_+\right) \oplus \left(\bigwedge^{2n} V_-\right),$$

*the summands $(r,s) = (2n,0)$ and $(0,2n)$ of (16) in degree $2n$, on which $K^\times$ acts by $\tau \mapsto \tau^{2n}$ and $\tau \mapsto \overline{\tau}^{2n}$.*

*Proof.* This is Definition 3.6 together with Proposition 3.7(i), which says that $W(A)$ is the $\mathbb{Q}$-structure on that pair of lines; the characters are read off from Proposition 5.1 with $(r,s) = (2n,0)$ and $(0,2n)$. ∎

**Lemma 5.4** (The two characters differ)**.** *For every $n \geq 1$ the characters $\tau \mapsto \tau^{2n}$ and $\tau \mapsto \mathrm{N}_{K/\mathbb{Q}}(\tau)^n = (\tau\overline{\tau})^n$ of $K^\times$ are distinct, as are $\tau \mapsto \overline{\tau}^{2n}$ and $\mathrm{N}^n$.*

*Proof.* This is the case $(r,s) = (2n,0)$, $(r',s') = (n,n)$ of the last assertion of Proposition 5.1; we write the argument out in this case. For $\tau \neq 0$ the identity $\tau^{2n} = (\tau\overline{\tau})^n$ is equivalent to $(\tau/\overline{\tau})^n = 1$, so the two characters agree if and only if $(\tau/\overline{\tau})^n = 1$ for every $\tau \in K^\times$.

Consider the map $c\colon K^\times \to K^\times$, $c(\tau) = \tau/\overline{\tau}$. Two elements have the same image if and only if their ratio is fixed by conjugation, that is, lies in $\mathbb{Q}^\times$. The elements $m + \sqrt{-d}$ with $m \in \mathbb{Z}$ are pairwise non-proportional over $\mathbb{Q}$, so $c$ takes infinitely many values.

On the other hand, $\{\zeta \in K : \zeta^n = 1\}$ is contained in the group $\mu(K)$ of roots of unity of $K$, and for an imaginary quadratic field $\mu(K)$ is finite: it has order 4 for $d = 1$, order 6 for $d = 3$, and order 2 otherwise. Since the image of $c$ is infinite, it is not contained in this finite set, so some $\tau \in K^\times$ has $(\tau/\overline{\tau})^n \neq 1$, and the characters differ. The second pair is obtained from the first by conjugation. ∎

*Remark* 5.5 (Explicit witnesses). The proof does not exhibit a $\tau$ at which the two characters differ, and the natural candidate $\tau = 1 + \sqrt{-d}$ is not always one. For $d = 1$ its ratio $\tau/\overline{\tau}$ is $\sqrt{-1}$, a primitive fourth root of unity, so the two characters agree at this $\tau$ whenever $4 \mid n$; for $d = 3$ the ratio is $(-1+\sqrt{-3})/2$, a primitive cube root of unity, and they agree whenever $3 \mid n$. The candidate $\tau = 3 + \sqrt{-d}$ fails as well, at $d = 3$ with $6 \mid n$, where the ratio is $(1+\sqrt{-3})/2$, a primitive sixth root of unity. In each case the characters agree at $\tau$ exactly when the order of the ratio divides $n$. The element $\tau = 2 + \sqrt{-d}$ is a witness for every $d$ and every $n$. Its ratio is $(4-d+4\sqrt{-d})/(4+d)$, which is not $\pm 1$, and for $d \notin \{1,3\}$ these are the only roots of unity in $K$; for $d = 1$ and $d = 3$ the ratios $(3+4\sqrt{-1})/5$ and

$(1+4\sqrt{-3})/7$ are not roots of unity either, since their traces $6/5$ and $2/7$ are not integers. The same holds for $4+\sqrt{-d}$ and $3+2\sqrt{-d}$. The finiteness argument in the proof covers all fields at once and needs no witness. Figure 5 shows the powers of these ratios for $d=1$ and $d=3$.

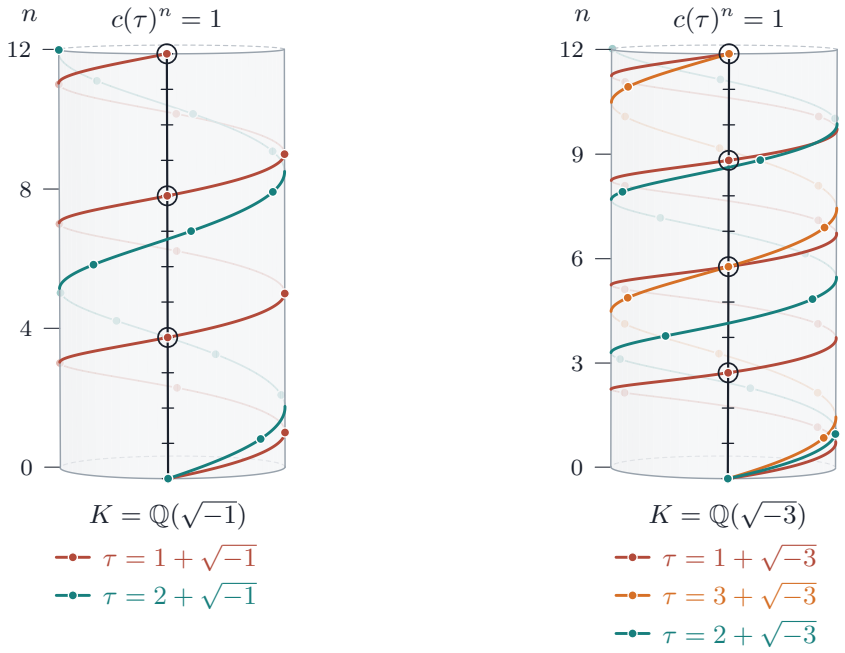



FIGURE 5. Remark 5.5. The values $c(\tau)^n = (\tau/\overline{\tau})^n$, $0 \le n \le 12$, on the unit circle, stacked over $n$ as a helix; the characters $\tau^{2n}$ and $\mathrm{N}^n$ agree at $\tau$ exactly where the helix meets the vertical line $c^n = 1$ (circled). Left, $K = \mathbb{Q}(\sqrt{-1})$: the candidate $1+\sqrt{-1}$ returns every fourth step, and $2+\sqrt{-1}$ never. Right, $K = \mathbb{Q}(\sqrt{-3})$: $1+\sqrt{-3}$ returns every third step, $3+\sqrt{-3}$ every sixth, and $2+\sqrt{-3}$ never.

## 5.2. **The obstruction.**

**Theorem 5.6** (Character obstruction)**.** *Let $(A,\iota,\eta)$ be of $(K,-1,n)$-Weil type and write $H^1(A,\mathbb{Q})\otimes\mathbb{C} = V_+ \oplus V_-$ for the decomposition into eigenspaces of $\iota(\sqrt{-d})$. Then*

$$H^{2n}(A,\mathbb{Q})\otimes\mathbb{C} = \bigoplus_{r+s=2n} \bigwedge^r V_+ \otimes \bigwedge^s V_-,$$

*with $\iota(\tau)$ acting on the $(r,s)$ summand by $\tau^r\overline{\tau}^s$. The Weil line $W(A)$ spans the two extreme summands $(r,s)=(2n,0)$ and $(0,2n)$, on which $K^\times$ acts by $\tau^{2n}$ and $\overline{\tau}^{2n}$, while $\eta^n$ lies in the middle summand $(r,s)=(n,n)$, on which $K^\times$ acts by the norm character $\mathrm{N}_{K/\mathbb{Q}}(\tau)^n$. In particular, for every $n \ge 1$,*

$$W(A) \cap H^{2n}(A,\mathbb{Q})^{\mathrm{N}^n} = 0,$$

*so a class that is a $K^\times$-eigenclass for $\mathrm{N}^n$ has zero component along the Weil line.*

The theorem shows that the obstruction is structural. A construction that is equivariant for the $K$-action returns an eigenclass, so it must choose a character in advance, and the norm character is the one that the polarisation already occupies. The Weil line sits at the two ends of the $K^\times$-grading, as far from the middle as the grading allows; Figure 6 draws the grading and marks the two positions.

*Proof of Theorem 5.6.* The grading (16) and the action on its summands are given by Proposition 5.1, the position of $W(A)$ by Lemma 5.3, and that of $\eta^n$ by Lemma 5.2. For the last assertion, write $H^{2n}(A,\mathbb{Q})^{\mathrm{N}^n}$ for the space of classes $x$ with $\iota(\tau)^*x = \mathrm{N}(\tau)^n x$ for all $\tau \in K^\times$. After extension of scalars to $\mathbb{C}$ this is the isotypic component of the character $\mathrm{N}^n$, which by Proposition 5.1 is the single summand $(r,s) = (n,n)$. By Lemma 5.4 the characters $\tau^{2n}$ and $\overline{\tau}^{2n}$ differ from $\mathrm{N}^n$, so by Lemma 5.3 the space $W(A) \otimes \mathbb{C}$ meets the summand $(n,n)$ in zero. Intersecting with $H^{2n}(A,\mathbb{Q})$ gives $W(A) \cap H^{2n}(A,\mathbb{Q})^{\mathrm{N}^n} = 0$. ∎

**Corollary 5.7** (Equivariant classes)**.** *Let $x \in H^{2n}(A,\mathbb{Q})$ satisfy $\iota(\tau)^*x = \mathrm{N}(\tau)^n x$ for all $\tau \in K^\times$. Then the component of $x$ along $W(A)$, with respect to any $K^\times$-stable complement of $W(A)$, is zero. In particular, $x$ is a Weil class only if $x = 0$.*

*Proof.* Let $C$ be a $K^\times$-stable complement of $W(A)$ in $H^{2n}(A,\mathbb{Q})$ and let $p$ be the projection onto $W(A)$ along $C$. Since both summands are $K^\times$-stable, $p$ commutes with every $\iota(\tau)^*$, so $\iota(\tau)^*p(x) = \mathrm{N}(\tau)^n p(x)$ for all $\tau \in K^\times$. Hence $p(x)$ lies in $W(A) \cap H^{2n}(A,\mathbb{Q})^{\mathrm{N}^n}$, which is zero by Theorem 5.6. If $x$ is a Weil class, then $x = p(x) = 0$. ∎

Corollary 5.7 is the form of the obstruction that applies to constructions. A construction that is natural in the $K$-action produces an eigenclass, whose character is determined by how the input transforms. The only character available to a rational eigenclass in $H^{2n}(A,\mathbb{Q})$ is $\mathrm{N}^n$ (Remark 5.8). By Lemma 5.2 this character is already occupied by the polarisation, and by Lemma 5.4 it is not a character of the Weil line.

*Remark* 5.8 (Rational eigenclasses)*.* To land on the Weil line, a construction must produce classes on which $K^\times$ acts through $\tau \mapsto \tau^{2n}$ or $\tau \mapsto \overline{\tau}^{2n}$, and neither character is $\mathbb{Q}$-valued. This rules out rational eigenclasses. Indeed, let $0 \neq x \in H^k(A,\mathbb{Q})$ satisfy $\iota(\tau)^*x = \chi(\tau)\,x$ for all $\tau \in K^\times$. Then $\chi(\tau) \in \mathbb{Q}$, because $\iota(\tau)^*$ preserves $H^k(A,\mathbb{Q})$. By Proposition 5.1, $x$ lies in a single summand $(r,s)$ and $\chi(\tau) = \tau^r\overline{\tau}^s$. Since $\chi$ is $\mathbb{Q}$-valued, it equals its conjugate $\tau \mapsto \tau^s\overline{\tau}^r$, and the distinctness of the characters in Proposition 5.1 gives $r = s$. So every rational $K^\times$-eigenclass in $H^{2n}(A,\mathbb{Q})$ is an eigenclass for $\mathrm{N}^n$, and Corollary 5.7 applies to it. A construction that reaches the Weil line must therefore produce a class that is not a $K^\times$-eigenclass. A nonzero element of $W(A)$ has nonzero components in both summands $(2n,0)$ and $(0,2n)$, since complex conjugation interchanges them, and its $K^\times$-orbit spans the $\mathbb{Q}$-plane $W(A)$. No rational eigenvalue condition isolates the Weil line: such conditions cut out only the diagonal summands $(r,r)$, and the Weil line lies at the two ends of the grading.

*Remark* 5.9 (Determinant and character)*.* A natural chain of statements leads to the opposite conclusion, that $K^\times$ acts on the Weil line through a power of the norm. The Weil line is $\bigwedge_K^{2n} V$, a free $K$-module of rank one; such a module is two-dimensional over $\mathbb{Q}$; and multiplication by $\tau$ on it is a $\mathbb{Q}$-linear map of determinant $\mathrm{N}(\tau)$. Each statement is true, but the last concerns the $K$-module structure and not the action of $\iota(\tau)$. By Lemma 5.3, $\iota(\tau)$ acts on the Weil line as multiplication by $\tau^{2n}$, with determinant $\mathrm{N}(\tau)^{2n}$, and a space on which $K^\times$ acts through multiplication by $\tau^{2n}$ is not one on which it acts by the scalar $\mathrm{N}(\tau)^n$. The first is a $K$-line and the second a $\mathbb{Q}$-eigenspace, and they would agree only if $\tau^n = \overline{\tau}^n$ for all $\tau \in K^\times$, which Lemma 5.4 excludes. Figure 6 draws the distinction.

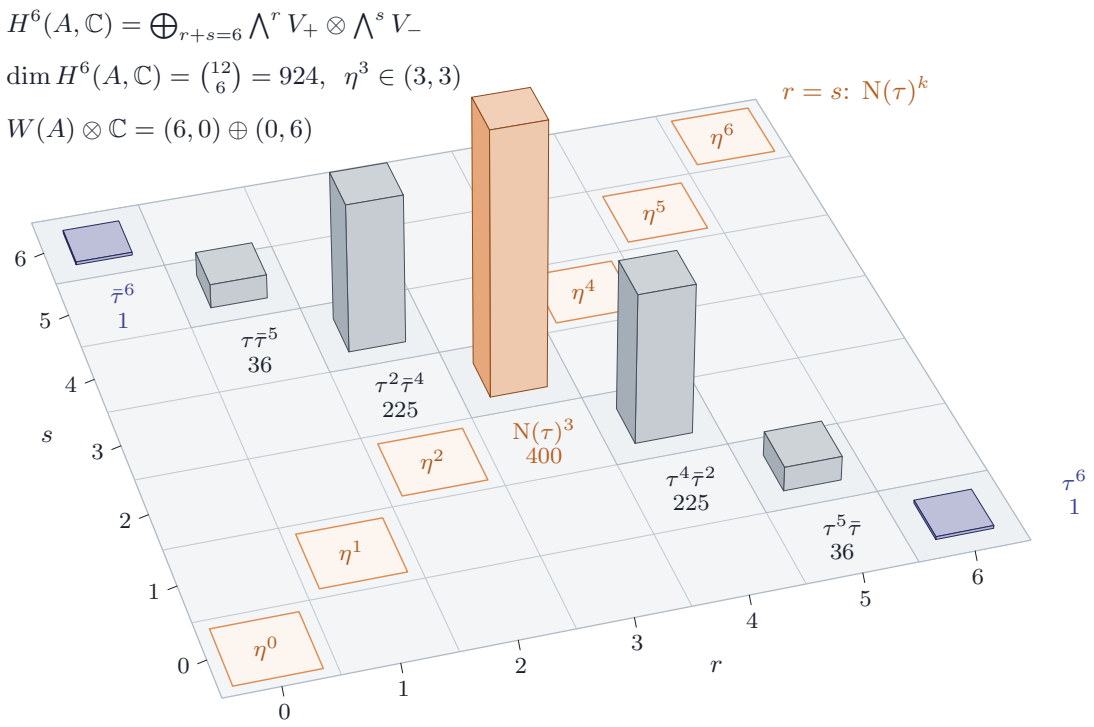



FIGURE 6. The $K^\times$-grading at $n = 3$ on the grid of pairs $(r, s)$, $0 \le r, s \le 6$. The summands $\bigwedge^r V_+ \otimes \bigwedge^s V_-$ of $H^6(A,\mathbb{C})$ lie on the antidiagonal $r + s = 6$ and are drawn at their dimensions $\binom{6}{r}\binom{6}{s}$, which sum to $\binom{12}{6} = 924$; on the summand $(r, s)$ the group $K^\times$ acts by $\tau^r\bar\tau^s$. The two end cells $(6, 0)$ and $(0, 6)$, with characters $\tau^6$ and $\bar\tau^6$, span the Weil line; the middle cell $(3, 3)$, with character $\mathrm{N}(\tau)^3$, contains $\eta^3$. The outlined cells on the diagonal $r = s$ are the summands on which $K^\times$ acts through a power of the norm, and they carry the powers $\eta^k$. That the end cells and the middle cell differ is the content of Theorem 5.6.

## 6. The cohomological Fourier–Mukai transform

With the character obstruction proved, we turn to the divisor obstruction of Theorem 7.3. Its one nontrivial ingredient is a computation: the cohomological Fourier–Mukai transform sends each power of the polarisation class to a rational multiple of a power of the polarisation class on the other side. We prove this in closed form. The computation takes place in a cohomology ring which, together with the classes involved, depends up to isomorphism only on the dimension, so one calculation covers every principally polarised abelian variety at once. Since the Todd class of an abelian variety is trivial, the passage from sheaves to cohomology is a formal step.

### 6.1. Grothendieck–Riemann–Roch on an abelian variety.

**Proposition 6.1.** *Let $A$ be an abelian variety of dimension $g$ and $\mathcal{G}$ an object of $D^b(A^\vee)$. Then*

$$\mathrm{ch}(\Phi_{\mathcal{P}}(\mathcal{G})) = \Phi_{\mathcal{P}}(\mathrm{ch}(\mathcal{G})),$$

*where the transform on the right is the cohomological one,* $\Phi_{\mathcal{P}}(x) = p_*\big(q^*x \cdot \mathrm{ch}(\mathcal{P})\big)$.

*Proof.* Grothendieck–Riemann–Roch for the projection $p\colon A \times A^\vee \to A$, a proper morphism of smooth varieties, reads

$$\mathrm{ch}\big(Rp_*\mathcal{K}\big) \cdot \mathrm{td}(T_A) = p_*\big(\mathrm{ch}(\mathcal{K}) \cdot \mathrm{td}(T_{A\times A^\vee})\big)$$

for $\mathcal{K} \in D^b(A \times A^\vee)$. The tangent bundle of an abelian variety is trivial, so $\mathrm{td}(T_A) = 1$ and $\mathrm{td}(T_{A\times A^\vee}) = 1$. Take $\mathcal{K} = Lq^*\mathcal{G} \otimes^L \mathcal{P}$, so that $Rp_*\mathcal{K} = \Phi_{\mathcal{P}}(\mathcal{G})$; the multiplicativity of the Chern character and $\mathrm{ch}(Lq^*\mathcal{G}) = q^*\,\mathrm{ch}(\mathcal{G})$ give the statement. ∎

### 6.2. **The universal model.**

*Notation* 6.2. Let $A$ be a principally polarised abelian variety of dimension $g$ with polarisation class $\theta$, and put $V = H^1(A, \mathbb{Q})$, a $\mathbb{Q}$-vector space of dimension $2g$ carrying the symplectic form attached to $\theta$. Choose a symplectic basis $e_1, \dots, e_{2g}$ of $V$ for this form, so that

$$\theta = \sum_{j=1}^{g} e_j \wedge e_{g+j}. \tag{17}$$

The principal polarisation identifies $A^\vee$ with $A$ and $H^1(A^\vee, \mathbb{Q})$ with the dual $V^\vee$; let $f_1, \dots, f_{2g}$ be the basis dual to $e_1, \dots, e_{2g}$, so that

$$\theta' = \sum_{j=1}^{g} f_j \wedge f_{g+j}. \tag{18}$$

Then $H^*(A \times A^\vee, \mathbb{Q}) = \bigwedge^*(V \oplus V^\vee)$.

**Lemma 6.3** (The Poincaré class)**.** *In the model of Notation 6.2,*

$$\ell := c_1(\mathcal{P}) = \sum_{i=1}^{2g} e_i \wedge f_i \;\in\; V \otimes V^\vee \subset \bigwedge^2(V \oplus V^\vee), \tag{19}$$

*the canonical element of* $V \otimes V^\vee = \mathrm{Hom}(V, V)$ *corresponding to the identity, and* $\mathrm{ch}(\mathcal{P}) = e^\ell$.

*Proof.* The Poincaré bundle is normalised to be trivial on $A \times \{0\}$ and on $\{0\} \times A^\vee$. The components of $c_1(\mathcal{P})$ in $\bigwedge^2 V$ and in $\bigwedge^2 V^\vee$ are its restrictions to these two subvarieties, so they vanish, and $c_1(\mathcal{P})$ lies in the middle summand $V \otimes V^\vee$ of $\bigwedge^2(V \oplus V^\vee) = \bigwedge^2 V \oplus (V \otimes V^\vee) \oplus \bigwedge^2 V^\vee$. The defining property of $\mathcal{P}$ is that its restriction to $A \times \{\lambda\}$ is the line bundle $\lambda$; on cohomology this says that the induced map $H^1(A^\vee, \mathbb{Q}) \to H^1(A, \mathbb{Q})^\vee$ obtained by contracting $c_1(\mathcal{P})$ is the canonical identification. Equivalently, the element of $\mathrm{Hom}(V, V)$ determined by $c_1(\mathcal{P})$ is the identity, which is (19). Since $\mathcal{P}$ is a line bundle, $\mathrm{ch}(\mathcal{P}) = e^{c_1(\mathcal{P})}$. ∎

*Remark* 6.4. Every principally polarised abelian variety of dimension $g$ gives, up to isomorphism, the same data in Notation 6.2: a symplectic space of dimension $2g$ over $\mathbb{Q}$, its dual, the symplectic form, and the canonical element. The constants computed below therefore depend only on $g$ and $k$, and not on $A$.

6.3. **The closed form.**

**Theorem 6.5** (Transform of the polarisation powers)**.** *Let $A$ be a principally polarised abelian variety of dimension $g$. Then for every $0 \le k \le g$,*

$$\Phi_{\mathcal{P}}(\theta'^{\,k}) \;=\; (-1)^{\,g(g+1)/2\,+\,k}\,\frac{k!}{(g-k)!}\,\theta^{\,g-k}, \tag{20}$$

*and $\Phi_{\mathcal{P}}(\theta'^{\,k}) = 0$ for $k > g$. In particular $\Phi_{\mathcal{P}}(\mathbb{Q}[\theta']) = \mathbb{Q}[\theta]$.*

*Proof.* Work in the model of Notation 6.2. By definition $\Phi_{\mathcal{P}}(x) = p_*(q^*x \cdot e^{\ell})$, and $p_*$ extracts the coefficient of the top exterior form $f_1 \wedge \cdots \wedge f_{2g}$ in the $V^\vee$ variables: it sends $y \wedge f_1 \wedge \cdots \wedge f_{2g}$ to $y$ for $y \in \bigwedge^* V$, and kills every class of lower degree in the $V^\vee$ variables.

The classes $e_i \wedge f_i$ have even degree, so they commute, and they have square zero. Hence for every $m$,

$$\frac{\ell^m}{m!} = \sum_{\substack{S \subseteq \{1,\dots,2g\} \\ |S| = m}} \prod_{i \in S} e_i \wedge f_i, \tag{21}$$

the product being taken in increasing order of $i$. Likewise, from (18),

$$\frac{\theta'^{\,k}}{k!} = \sum_{\substack{T \subseteq \{1,\dots,g\} \\ |T| = k}} \prod_{j \in T} f_j \wedge f_{g+j}, \qquad \frac{\theta^{\,g-k}}{(g-k)!} = \sum_{\substack{U \subseteq \{1,\dots,g\} \\ |U| = g-k}} \prod_{j \in U} e_j \wedge e_{g+j}. \tag{22}$$

Fix $k$. In the product $\theta'^{\,k} \cdot e^{\ell}$, the terms that survive $p_*$ are those of total degree $2g$ in the $f$ variables. A term of (22) indexed by $T$ carries the $2k$ variables $\{f_j, f_{g+j} : j \in T\}$, and a term of (21) indexed by $S$ carries the $|S|$ variables $\{f_i : i \in S\}$. For the product to be nonzero the two index sets must be disjoint, and for the total $f$-degree to be $2g$ we need

$$S = \{1, \dots, 2g\} \setminus (T \cup (g+T)), \qquad |S| = 2g - 2k.$$

So $S$ is determined by $T$, only the term $\ell^{2g-2k}/(2g-2k)!$ of $e^{\ell}$ contributes, and the surviving terms are in bijection with the $k$-subsets $T$ of $\{1, \dots, g\}$. Writing $U = \{1, \dots, g\} \setminus T$ we have $S = U \cup (g+U)$, so the $V$-part of the term indexed by $T$ is $\prod_{i \in U \cup (g+U)} e_i$, which up to sign is the term of (22) indexed by $U$. The bijection $T \leftrightarrow U$ is the one between the terms of $\theta'^{\,k}/k!$ and those of $\theta^{\,g-k}/(g-k)!$.

It remains to compute the sign $\sigma$ with which each term contributes, and to check that it is the same for all $T$. The sign is that of the rearrangement of the ordered word

$$\Big(\prod_{j \in T} f_j f_{g+j}\Big) \cdot \Big(\prod_{i \in S} e_i f_i\Big)$$

into the target word $\big(\prod_{j \in U} e_j e_{g+j}\big) \cdot (f_1 \cdots f_{2g})$, which uses the same generators. Relabelling the $g$ symplectic pairs by a permutation $\pi$ of $\{1, \dots, g\}$ is an automorphism of the exterior algebra. It carries the word for $T$ to the word for $\pi(T)$, because the blocks $f_j f_{g+j}$ and $e_i f_i$ have degree two and may be reordered freely, and it carries the target word for $U$ to the target word for $\pi(U)$, because reordering $f_{\pi(1)} \cdots f_{\pi(g)} f_{g+\pi(1)} \cdots f_{g+\pi(g)}$ costs the square of the sign of $\pi$. Since the permutations of $\{1, \dots, g\}$ act transitively on the $k$-subsets, $\sigma = \sigma(g,k)$, and we evaluate it at $T = \{1, \dots, k\}$. Put $r = g-k$. Moving the $2r$ variables $e_i$ of the second factor to its front costs $(-1)^{r(2r-1)} = (-1)^r$, moving them further past the $2k$ variables of the first factor costs nothing, and arranging them as $\prod_{j \in U} e_j e_{g+j}$ costs $(-1)^{r(r-1)/2}$. The $f$ variables are left in the order $f_1 f_{g+1} \cdots f_k f_{g+k}\, f_{k+1} \cdots f_g\, f_{g+k+1} \cdots f_{2g}$, and bringing them

to $f_1 \cdots f_{2g}$ costs $(-1)^{k(k-1)/2+rk}$. As $r + r(r-1)/2 + k(k-1)/2 + rk = g(g+1)/2 - k$, this gives

$$\sigma(g,k) = (-1)^{\,g(g+1)/2\,+\,k}. \tag{23}$$

Assembling, and accounting for the factorials in (22),

$$\Phi_{\mathcal{P}}(\theta'^{\,k}) = k! \cdot \sigma(g,k) \cdot \frac{\theta^{\,g-k}}{(g-k)!},$$

which is (20). For $k > g$ we have $\theta'^{\,k} = 0$, because $\theta'^{\,k}$ lies in $\bigwedge^{2k} V^\vee$, which vanishes for $2k > 2g$. The last assertion follows because the constants in (20) are nonzero for every $0 \le k \le g$. ∎

*Remark* 6.6 (The sign). The sign (23) depends only on $g$ and $k$, by the transitivity argument in the proof, and the count at $T = \{1, \dots, k\}$ fixes it. For even $g = 2n$, the case of Weil type, $g(g+1)/2 = n(2n+1)$ has the parity of $n$, and (23) reads $\sigma(2n,k) = (-1)^{n+k}$.

**Corollary 6.7** (Stability of the polarisation subring)**.** *Let $A$ be a principally polarised abelian variety and let $x \in H^*(A^\vee, \mathbb{Q})$ lie in the subring $\mathbb{Q}[\theta']$ generated by $\theta'$. Then $\Phi_{\mathcal{P}}(x) \in \mathbb{Q}[\theta]$.*

*Proof.* The ring $\mathbb{Q}[\theta']$ is spanned over $\mathbb{Q}$ by the powers $\theta'^{\,k}$, Theorem 6.5 sends each of them to a rational multiple of a power of $\theta$, and $\Phi_{\mathcal{P}}$ is linear. ∎

*Remark* 6.8 (Non-principal polarisations). If $\theta$ is a polarisation of degree $D^2$ rather than a principal one, the isogeny $\phi_\theta \colon A \to A^\vee$ has degree $D^2$, and $\phi_\theta^* \theta' = D\theta$ for a suitable normalisation of $\theta'$. The argument of Theorem 6.5 goes through with $\theta$ replaced by $D^{-1}\phi_\theta^*\theta'$, and the constants change by powers of $D$, all of which are nonzero rationals. The statements below concern only membership in $\mathbb{Q}[\theta]$, not the values of the constants, so we state them for principal polarisations and use them in general.

## 7. The divisor obstruction

The first way of reaching a Weil class is to import an object from a projective space. One restricts an object $\mathcal{F}$ of $D^b(\mathbb{P}^{N-1})$ to the dual abelian variety through a morphism $i \colon A^\vee \to \mathbb{P}^{N-1}$ and carries the result back to $A$ by the Fourier–Mukai transform $\Phi_{\mathcal{P}}$. Secant varieties of $A^\vee$ are the usual choice of $\mathcal{F}$, because a dimension count makes the codimension come out right, and this is the construction of the preprint cited on the title page. This section shows that no construction of this shape produces a Weil class. The proof needs, besides the closed form of Section 6, two facts about a morphism to projective space: the Chern character of a restricted object is a polynomial in the pulled-back hyperplane class, and this class is the first Chern class of the line bundle defining the morphism. When the Néron–Severi group of $A^\vee$ has rank one, the transform therefore lands in $\mathbb{Q}[\theta]$. We then give two proofs of Corollary 7.6, one through Proposition 3.10 and one through the characters of Section 5, and isolate the equivariance hypothesis on which the second depends.

### 7.1. Statement and proof.

**Lemma 7.1** (Restriction from an ambient projective space)**.** *Let $Y$ be a smooth projective variety, let $i \colon Y \to \mathbb{P}^{N-1}$ be a morphism defined by a linear system, let $h \in H^2(\mathbb{P}^{N-1}, \mathbb{Q})$ be the hyperplane class, and let $\mathcal{F} \in D^b(\mathbb{P}^{N-1})$. Then*

$$\operatorname{ch}(Li^*\mathcal{F}) \in \mathbb{Q}\big[\, i^*h \,\big] \subset H^*(Y, \mathbb{Q}).$$

*Proof.* The Chern character commutes with derived pullback: $\mathrm{ch}(Li^*\mathcal{F}) = i^*\,\mathrm{ch}(\mathcal{F})$. For a locally free sheaf this follows from the naturality of Chern classes. It extends to $D^b(\mathbb{P}^{N-1})$ because the Chern character is additive on distinguished triangles and, by Beilinson's theorem, every object of $D^b(\mathbb{P}^{N-1})$ is quasi-isomorphic to a finite complex whose terms are direct sums of line bundles $\mathcal{O}(-m)$ with $0 \le m \le N-1$. All that is needed is that the classes of line bundles generate $K_0(\mathbb{P}^{N-1})$, which also follows from the Koszul resolutions of the structure sheaves of linear subspaces. The cohomology ring of $\mathbb{P}^{N-1}$ is $\mathbb{Q}[h]/(h^N)$, so $\mathrm{ch}(\mathcal{F})$ is a polynomial in $h$ with rational coefficients, and applying the ring homomorphism $i^*$ gives a polynomial in $i^*h$. ∎

Remark 20.3 records which other ambient varieties behave in the same way.

**Lemma 7.2** (The hyperplane class of a linear system)**.** *Let $A$ be an abelian variety and $i\colon A^\vee \to \mathbb{P}^{N-1}$ a morphism defined by a linear system $|M|$ with $M \in \mathrm{Pic}(A^\vee)$. Then $i^*h = c_1(M) \in \mathrm{NS}(A^\vee) \otimes \mathbb{Q}$.*

*Proof.* A morphism to $\mathbb{P}^{N-1}$ defined by the linear system $|M|$, or by a subsystem of it, satisfies $i^*\mathcal{O}(1) \cong M$ by construction. Hence $i^*h = c_1(i^*\mathcal{O}(1)) = c_1(M)$. ∎

**Theorem 7.3** (Divisor obstruction)**.** *Let $A$ be a principally polarised complex abelian variety of dimension $g$ with $\mathrm{NS}(A^\vee) \otimes \mathbb{Q} = \mathbb{Q} \cdot \theta'$, let $i\colon A^\vee \to \mathbb{P}^{N-1}$ be any morphism defined by a linear system, and let $\mathcal{F}$ be any object of $D^b(\mathbb{P}^{N-1})$. Put $\mathcal{E} = \Phi_{\mathcal{P}}(Li^*\mathcal{F})$. Then*

$$\mathrm{ch}_k(\mathcal{E}) \in \mathbb{Q} \cdot \theta^k \qquad \text{for every } k \ge 0.$$

*Proof.* By Lemma 7.1, $\mathrm{ch}(Li^*\mathcal{F}) \in \mathbb{Q}[i^*h]$, and by Lemma 7.2, $i^*h = c_1(M)$ for the line bundle $M$ defining $i$. Since $\mathrm{NS}(A^\vee) \otimes \mathbb{Q} = \mathbb{Q}\theta'$, we have $c_1(M) = m\theta'$ for some $m \in \mathbb{Q}$, and therefore $\mathrm{ch}(Li^*\mathcal{F}) \in \mathbb{Q}[\theta']$. By Proposition 6.1, $\mathrm{ch}(\mathcal{E})$ is the image of $\mathrm{ch}(Li^*\mathcal{F})$ under the cohomological transform $\Phi_{\mathcal{P}}$, which carries $\mathbb{Q}[\theta']$ into $\mathbb{Q}[\theta]$ by Corollary 6.7. Hence $\mathrm{ch}(\mathcal{E}) \in \mathbb{Q}[\theta]$, and since $\mathbb{Q}[\theta] \cap H^{2k}(A, \mathbb{Q}) = \mathbb{Q}\theta^k$, taking the component in $H^{2k}(A, \mathbb{Q})$ gives the statement. ∎

*Remark* 7.4 (The hypothesis on the Néron–Severi group)*.* The hypothesis $\mathrm{NS}(A^\vee) \otimes \mathbb{Q} = \mathbb{Q}\theta'$ holds at the very general point of any positive-dimensional family of abelian varieties of Weil type with $\mathrm{End}^0(A) = \iota(K)$, which is the case in which the Weil classes lie outside the divisor subring (Proposition 3.10); see Lemma 7.5. Without it, the conclusion is only that $\mathrm{ch}_k(\mathcal{E})$ lies in the image under $\Phi_{\mathcal{P}}$ of the divisor subring of $A^\vee$. Theorem 5.6 makes no hypothesis on NS, which is one reason for giving two proofs of Corollary 7.6.

**Lemma 7.5.** *Let $\mathcal{A} \to S$ be the universal family of abelian varieties of $(K, -1, n)$-Weil type over the corresponding moduli space, with $n \ge 2$. At a very general point $s \in S$ one has $\mathrm{End}^0(A_s) = \iota(K)$ and $\mathrm{NS}(A_s) \otimes \mathbb{Q} = \mathbb{Q}\eta_s$, of dimension one.*

*Proof.* The locus in $S$ where $\mathrm{End}^0$ is strictly larger than $\iota(K)$ is a countable union of proper closed subvarieties: each additional endomorphism imposes a nontrivial algebraic condition on the period point, and there are only countably many candidates, the endomorphisms of the lattice $H_1(A_s, \mathbb{Z})$. So $\mathrm{End}^0(A_s) = \iota(K)$ at a very general point $s$, and there the Hodge group is the full special unitary group $\mathrm{SU}(V, H)$ of the $K$-hermitian form $H$ attached to the polarisation (Lemma 4.5), by Weil's computation of the Hodge group of a general abelian variety of Weil type [Wei77]. The space $\mathrm{Hdg}^1(A_s) = \mathrm{NS}(A_s) \otimes \mathbb{Q}$ is the space of Hodge-group invariants in $H^2(A_s, \mathbb{Q})$. By Proposition 5.1 in degree two, $H^2 \otimes \mathbb{C} = \bigwedge^2 V_+ \oplus (V_+ \otimes V_-) \oplus \bigwedge^2 V_-$. The group $\mathrm{SU}(V, H)$ acts on $V_+ \otimes V_- \cong V_+ \otimes V_+^\vee$ with a one-dimensional space of invariants, spanned by the class of $H$, and on $\bigwedge^2 V_\pm$ with

no invariants, since $2n \geq 4$ and $\bigwedge^2$ of the standard representation of $\mathrm{SL}_{2n}$ has no invariant vector unless $2n = 2$. Hence the invariants are one-dimensional, and they are spanned by $\eta_s$. ∎

7.2. **Consequences for Weil classes.** Since a Weil class is by construction not in the subring generated by divisor classes, the theorem excludes Weil classes at once.

**Corollary 7.6.** *With notation as in Theorem 7.3, suppose in addition that $g = 2n$ and that $(A, \iota, \eta)$ is of $(K, -1, n)$-Weil type. Then $\mathrm{ch}_n(\mathcal{E})$ is a Weil class only if $\mathrm{ch}_n(\mathcal{E}) = 0$.*

The first proof passes through Proposition 3.10. The second replaces that proposition by the characters of Section 5 and explains in terms of representations why the first had to succeed: the $K^\times$-isotypic decomposition of $H^{2n}(A, \mathbb{Q})$ separates the two classes in question (Figure 7).

*Proof of Corollary 7.6.* We work under the hypothesis of Theorem 7.3. Let $g = 2n$ and let $(A, \iota, \eta)$ be of $(K, -1, n)$-Weil type with $\mathrm{NS}(A) \otimes \mathbb{Q} = \mathbb{Q}\eta$. Then $\theta$ is a nonzero multiple of $\eta$, and Theorem 7.3 gives $\mathrm{ch}_n(\mathcal{E}) \in \mathbb{Q} \cdot \theta^n = \mathbb{Q} \cdot \eta^n \subset D^{2n}(A)$. The proof of Proposition 3.10 uses the hypothesis $\mathrm{End}^0(A) = \iota(K)$ only to obtain $\mathrm{NS}(A) \otimes \mathbb{Q} = \mathbb{Q}\eta$, so it applies here and gives $W(A) \cap D^{2n}(A) = 0$. Hence if $\mathrm{ch}_n(\mathcal{E})$ is a Weil class, it lies in $W(A) \cap D^{2n}(A) = 0$. ∎

*Second proof of Corollary 7.6, via characters.* By Lemma 5.2 the class $\eta^n$ is a $K^\times$-eigenclass for the character $\mathrm{N}^n$, and hence so is every nonzero element of $\mathbb{Q}\eta^n$. As in the first proof, $\mathrm{ch}_n(\mathcal{E}) \in \mathbb{Q}\eta^n$ by Theorem 7.3, so $\mathrm{ch}_n(\mathcal{E})$ satisfies $\iota(\tau)^* \mathrm{ch}_n(\mathcal{E}) = \mathrm{N}(\tau)^n \mathrm{ch}_n(\mathcal{E})$ for all $\tau \in K^\times$. By Corollary 5.7 it is a Weil class only if it vanishes. ∎

The second proof uses Theorem 7.3, and with it the hypothesis on NS, only to know that $\mathrm{ch}_n(\mathcal{E})$ is an eigenclass for $\mathrm{N}^n$; the rest of the argument uses nothing but this equivariance. We record the resulting statement, which does not mention the ambient space at all.

**Theorem 7.7** (Equivariance suffices)**.** *Let $(A, \iota, \eta)$ be of $(K, -1, n)$-Weil type and let $\mathcal{E} \in D^b(A)$ be an object such that for every $\tau$ in the order $\iota^{-1}(\mathrm{End}(A)) \subset K$ one has*

$$\iota(\tau)^* \mathrm{ch}_n(\mathcal{E}) = \mathrm{N}_{K/\mathbb{Q}}(\tau)^n \ \mathrm{ch}_n(\mathcal{E}).$$

*Then $\mathrm{ch}_n(\mathcal{E})$ is a Weil class only if $\mathrm{ch}_n(\mathcal{E}) = 0$.*

*Proof.* The hypothesis says that $\mathrm{ch}_n(\mathcal{E})$ is an eigenclass for $\mathrm{N}^n$ under the nonzero elements of the order, and it is then one under all of $K^\times$. Indeed, since the order spans $K$ over $\mathbb{Q}$, every $\tau \in K^\times$ can be written as $\sigma/m$ with $\sigma$ in the order and $m$ a nonzero integer. The endomorphism $\iota(m)$ is multiplication by $m$, which acts on $H^{2n}(A, \mathbb{Q})$ as $m^{2n} = \mathrm{N}(m)^n$. Hence $\iota(\tau)^* \mathrm{ch}_n(\mathcal{E}) = \mathrm{N}(\sigma)^n m^{-2n} \mathrm{ch}_n(\mathcal{E}) = \mathrm{N}(\tau)^n \mathrm{ch}_n(\mathcal{E})$, and Corollary 5.7 applies. ∎

*Remark* 7.8 (Comparison of the hypotheses). Theorem 7.3 requires the object to be restricted from an ambient projective space and imposes a condition on $\mathrm{NS}(A^\vee)$; Theorem 7.7 requires only that the resulting class be equivariant for the norm character. The second hypothesis is weaker, and it holds for any construction whose input is $K$-invariant. This includes the secant constructions, since the secant varieties of a $K$-invariant subvariety are $K$-invariant. We state both because the first can be checked by inspection of the construction, while the second requires an equivariance computation.

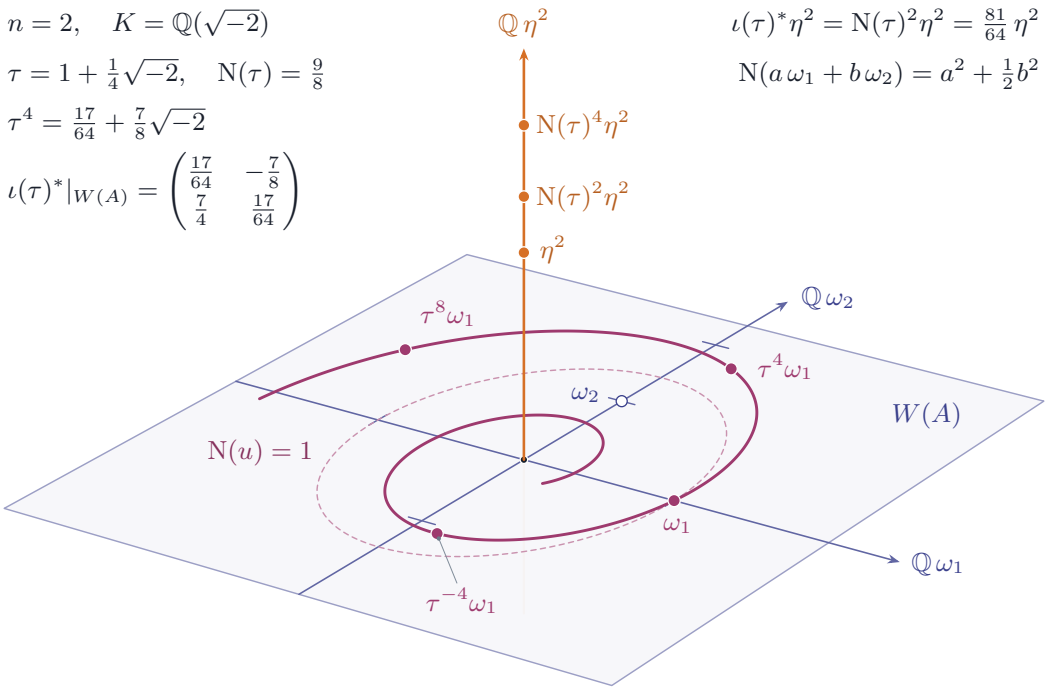



FIGURE 7. The linear algebra behind Theorem 7.3, drawn inside $H^{2n}(A, \mathbb{Q})$. The Weil line $W(A)$ is a plane, free of rank one over $K$ and of rank two over $\mathbb{Q}$, and $K^\times$ acts on it through $\tau^{2n}$, so the orbit of any nonzero class is a spiral that spans the plane and lies on no line in it. The transverse axis carries $\eta^n$, on which $K^\times$ acts through the norm. A construction returning a single eigenclass returns a line, and by Proposition 13.1 no line inside $W(A)$ is stable.

## 8. Secant numerology

The obstructions of Theorems 5.6 and 7.3 concern where a class can lie in cohomology, and neither of them uses a count of dimensions. A secant construction is also subject to such a count, and this section shows that the count obstructs nothing. It asks for a polarisation on $A^\vee$ whose Euler characteristic lies below the values at which the theorem of Lefschetz guarantees that a linear system is free of base points or very ample. Those values are sufficient conditions and not necessary ones, and already on abelian surfaces there are base-point-free and very ample systems below them.

### 8.1. **The dimension count.**

**Setup 8.1** (The secant numerology)**.** Let $A$ be of $(K, -1, n)$-Weil type, so that $\dim A = \dim A^\vee = 2n$. Suppose that $A^\vee$ is embedded in $\mathbb{P}^{N-1}$ by a complete linear system $|M|$ with $M$ ample. An ample line bundle on an abelian variety has no higher cohomology, so $N = h^0(M) = \chi(M)$. The $n$-th secant variety of a nondegenerate $2n$-fold has expected dimension

$$\dim \mathrm{Sec}^n(A^\vee) = n(2n+1) - 1 = 2n^2 + n - 1,$$

and for a construction that intersects $\mathrm{Sec}^n(A^\vee)$ with $A^\vee$ in the expected dimension inside $\mathbb{P}^{N-1}$,

$$\dim\bigl(\mathrm{Sec}^n(A^\vee)\cap A^\vee\bigr) = (2n^2+n-1)+2n-(N-1) = 2n^2+3n-N.$$

Requiring this dimension to be $n$, so that the construction produces a class of the required codimension, forces

$$N = \chi(M) = 2n^2+2n. \tag{24}$$

**Lemma 8.2.** *For every $n \geq 2$,*

$$2n^2+2n < 2^{2n} < 3^{2n},$$

*and the ratio $2^{2n}/(2n^2+2n)$ tends to infinity.*

*Proof.* The right inequality is clear. Since $2^{2n} = 4^n$, put $q(n) = 4^n/(2n^2+2n)$, so that $q(2) = 16/12 > 1$. The ratio of consecutive values is

$$\frac{q(n+1)}{q(n)} = \frac{4(2n^2+2n)}{2(n+1)^2+2(n+1)} = \frac{4n}{n+2},$$

which is at least 2 for $n \geq 2$. Hence $q(n) \geq 2^{n-2}q(2) > 1$ for every $n \geq 2$, and $q(n) \to \infty$. ∎

8.2. **The Lefschetz thresholds.** The theorem of Lefschetz says that for $M$ ample on an abelian variety, $M^{\otimes 2}$ is free of base points and $M^{\otimes 3}$ is very ample [BL04, Section 4.5]. If $M = P^{\otimes 2}$ with $P$ ample on an abelian variety of dimension $m$, then $\chi(M) = 2^m\chi(P) \geq 2^m$, since $\chi(L) = (L^m)/m!$ for every line bundle $L$ by Riemann–Roch, and similarly $\chi \geq 3^m$ in the very ample case. Read as thresholds, these bounds would say that there is no base-point-free system below $2^m$ and no very ample system below $3^m$. Both readings turn a sufficient condition into a necessary one, and both fail already for abelian surfaces.

**Proposition 8.3** (The thresholds are not necessary)**.** *Let $S$ be a general complex abelian surface, so $m = 2$.*

(i) *A polarisation of type $(1,3)$ on $S$ has $\chi = 3 < 4 = 2^2$ and its complete linear system is free of base points; the associated morphism $S \to \mathbb{P}^2$ is a six-sheeted covering.*

(ii) *A polarisation of type $(1,5)$ on $S$ has $\chi = 5 < 9 = 3^2$ and is very ample; the image is a Horrocks–Mumford abelian surface of degree 10 in $\mathbb{P}^4$* [HM73].

*In particular neither $2^m$ nor $3^m$ is a lower bound for the Euler characteristic of a base-point-free, respectively very ample, ample line bundle on an abelian variety of dimension $m$.*

*Proof.* By Riemann–Roch on an abelian surface, a polarisation $L$ of type $(1,d)$ has $\chi(L) = (L^2)/2 = d$. The remaining assertions belong to the classification of $(1,d)$-polarised abelian surfaces; see Gross and Popescu [GP98, §1.2], where the base locus of $|L|$ is determined for each $d$: it is empty for $d \geq 3$, consists of four points for $d = 2$, and $|L|$ is very ample for $d \geq 5$ on a general surface, the case $d = 5$ being the Horrocks–Mumford embedding of [HM73]. In that case the image has degree $(L^2) = 10$. For $d = 3$ the morphism $S \to \mathbb{P}^2$ defined by $|L|$ contracts no curve, because $L$ is ample, so it is finite and surjective, of degree $(L^2) = 6$. ∎

**Corollary 8.4.** *The count of Setup 8.1 leads to no contradiction. The requirement $\chi(M) = 2n^2+2n$ of (24) is compatible with $M$ being very ample for all $n$ for which a polarisation of that Euler characteristic and suitable type exists on a $2n$-dimensional abelian variety. At $n = 2$ the requirement $\chi = 12$ on an abelian fourfold is met by polarisations of*

*type* $(1,1,2,6)$ *and* $(1,1,1,12)$*, and nothing in the numerology excludes very ampleness for either.*

*Remark* 8.5 (The threshold reading)*.* The supplementary material of the preprint cited on the title page asserts that the polarisation required by (24) admits no base-point-free linear system, on the basis of a table of the values $2n^2+2n$, $2^{2n}$ and $3^{2n}$. The table is correct, and Lemma 8.2 reproduces it, but by Proposition 8.3 the inference drawn from it does not hold, and the claim is withdrawn. The obstructions of Theorems 5.6 and 7.3 are of a different nature: they bound where a class can lie in a decomposition of cohomology, and no choice of polarisation, ambient space or object affects them.

## 9. The semiregularity target

Markman's construction at $n=2$ deforms a sheaf across a family and controls the obstruction by a semiregularity argument in the sense of Bloch and of Buchweitz and Flenner [Blo72, BF03]. As for the secant construction of Section 8, one may ask whether a dimension count blocks this route for $n \geq 4$. We compute the target of the semiregularity map and show that it grows too fast for the count to impose any such constraint, so that neither route meets a numerical obstruction. The constraint that does exist is structural. In Section 9.2 we show that the annihilator of the Weil line in $H^{0,2}(A)$ has dimension $n^2$, and that it prevents every object whose Chern character is concentrated on the Weil line from being semiregular as soon as $H^2(\mathcal{O}_A)$ acts on the object faithfully. In Section 9.3 we identify this annihilator with the tangent space to the Weil family.

### 9.1. The dimension of the target.

**Lemma 9.1** (Hodge numbers of an abelian variety)**.** *For an abelian variety $A$ of dimension $g$ one has $h^{p,q}(A) = \binom{g}{p}\binom{g}{q}$.*

*Proof.* By (1) and the compatibility of the Hodge decomposition with the exterior algebra structure, $H^{p,q}(A) = \bigwedge^p H^{1,0} \otimes \bigwedge^q H^{0,1}$, and $\dim H^{1,0} = \dim H^{0,1} = g$. ∎

**Proposition 9.2** (The semiregularity target)**.** *Let $A$ be an abelian variety of dimension $g = 2n$. The target of the semiregularity map, namely $\bigoplus_q H^{q+2}(A, \Omega_A^q)$, has dimension*

$$\sum_{q \geq 0} h^{q,q+2}(A) = \binom{4n}{2n-2}.$$

*Proof.* By Lemma 9.1 with $g = 2n$,

$$\sum_{q \geq 0} h^{q,q+2} = \sum_{q \geq 0} \binom{2n}{q}\binom{2n}{q+2} = \sum_{q \geq 0} \binom{2n}{q}\binom{2n}{2n-q-2} = \binom{4n}{2n-2},$$

the last step being the Vandermonde convolution. ∎

**Corollary 9.3** (The dimension count)**.** *The target dimension $\binom{4n}{2n-2}$ takes the values $28, 495, 8008, 125970, 1961256, 30421755$ for $n = 2, \dots, 7$, growing by a factor exceeding ten at each step. Any source whose dimension grows polynomially in $n$, for instance the quadratic quantity $\binom{2n}{2} = n(2n-1)$, which takes the values $6, 15, 28, 45, 66, 91$ over the same range, is eventually negligible against it. In particular, for large $n$ a comparison of the two dimensions cannot obstruct the existence of a semiregular sheaf; the inequality it produces points the other way.*

*Remark* 9.4 (Consistency at $n = 2$). At $n = 2$ the target has dimension 28, and the sheaf produced by Markman has $\dim \mathrm{Ext}^2 = 12$, well below it, which is consistent with the construction succeeding there. Nothing in the numbers changes character between $n = 2$ and $n = 4$, so if the construction cannot be carried out for larger $n$, the reason is not this comparison.

*Remark* 9.5 (The remaining requirement). Corollary 9.3 does not show that the semiregularity route works. It removes one apparent obstruction and leaves the actual requirement in place, namely the construction of a sheaf on $A$ whose Chern character has a nonzero component along the Weil line. By Theorem 5.6 such a sheaf cannot be $K$-equivariant with the norm character, so it cannot be produced by any functorial construction from $K$-invariant data on $A^\vee$; Markman's sheaf at $n = 2$ is not produced in this way. In this sense the problem for $n \geq 3$ is one of breaking equivariance, and the dimensions play no role in it.

9.2. **The annihilator of the Weil line and objects of rank zero.** By Remark 9.5, what remains is to construct a sheaf whose Chern character has a component along the Weil line. The most economical way to arrange this is to make the Chern character lie *entirely* on the Weil line, that is, to take an object of rank zero with every $\mathrm{ch}_k$ vanishing except $\mathrm{ch}_n$, as the construction of Section 14 does. We show that this choice is incompatible with semiregularity whenever $H^2(\mathcal{O}_A)$ acts faithfully on the object, for a reason that depends only on the position of the Chern character and not on how the object is built.

We first recall the semiregularity map, on which the rest of this subsection turns. For a coherent sheaf $\mathcal{E}$ on a smooth projective variety $A$ the semiregularity map is defined through the Atiyah class $\mathrm{At}(\mathcal{E}) \in \mathrm{Ext}^1(\mathcal{E}, \mathcal{E} \otimes \Omega^1_A)$: a class $\xi \in \mathrm{Ext}^2(\mathcal{E}, \mathcal{E})$ is sent to the trace of $\xi \cdot \exp(\mathrm{At}(\mathcal{E}))$, and the component of that trace in $H^{q+2}(A, \Omega^q_A) = H^{q,q+2}(A)$ is the $q$-th coordinate of $\sigma_{\mathcal{E}}(\xi)$. Bloch introduced the map for local complete intersections [Blo72], and Buchweitz and Flenner extended it to arbitrary coherent sheaves and perfect complexes [BF03]. If $A$ moves in a family $\mathcal{A} \to S$, the obstruction to extending $\mathcal{E}$ to first order along $v \in T_s S$ is a class $\mathrm{ob}_v(\mathcal{E}) \in \mathrm{Ext}^2(\mathcal{E}, \mathcal{E})$, and the theorem of Buchweitz and Flenner identifies $\sigma_{\mathcal{E}}(\mathrm{ob}_v(\mathcal{E}))$ with the derivative of $\mathrm{ch}(\mathcal{E})$ in the direction $v$, projected off the diagonal of the Hodge decomposition. Hence if $\mathrm{ch}(\mathcal{E})$ stays of Hodge type along $S$, then $\sigma_{\mathcal{E}}(\mathrm{ob}_v(\mathcal{E})) = 0$, and if $\sigma_{\mathcal{E}}$ is also injective, then $\mathrm{ob}_v(\mathcal{E}) = 0$ and $\mathcal{E}$ deforms. Injectivity thus converts a statement about the Chern character into a statement about the sheaf, and it is the one hypothesis in the argument that sees more than $\mathrm{ch}(\mathcal{E})$.

Two special cases are all that is needed below. For a line bundle $L$ one has $\mathrm{Ext}^2(L, L) = H^2(\mathcal{O}_A) = H^{0,2}(A)$, and $\sigma_L$ is multiplication by $e^{c_1(L)}$. For a direct sum, $\mathrm{Ext}^2$ is the sum of the $\mathrm{Ext}^2(\mathcal{E}_i, \mathcal{E}_j)$, and $\sigma$ acts on each diagonal block $H^2(\mathcal{O}_A) \cdot \mathrm{id}_{\mathcal{E}_i}$ by multiplication by the Chern character of that summand. Both formulae follow at once from the definition by the trace.

**Lemma 9.6** (The annihilator of the Weil line in $H^{0,2}$)**.** *Let $A$ be of $(K, -1, n)$-Weil type and let $\omega \in W(A) \otimes \mathbb{R}$ be any nonzero real element of the Weil line. Then the multiplication map*

$$H^{0,2}(A) \longrightarrow H^{n,n+2}(A), \qquad z \longmapsto z \cdot \omega,$$

*has kernel of dimension exactly $n^2$, spanned by the products $u \wedge v$ with $u \in V^{0,1}_+$ and $v \in V^{0,1}_-$. In particular the kernel is the same subspace for every $\omega$ on the line.*

*Proof.* Write $\omega = a\,\alpha_+ + b\,\alpha_-$ with $\alpha_\pm$ generating $\bigwedge^{2n} V_\pm$, which is possible because $W(A) \otimes \mathbb{C} = \bigwedge^{2n} V_+ \oplus \bigwedge^{2n} V_-$ (see the proof of Proposition 3.7). Complex conjugation

interchanges $V_+$ and $V_-$, so $\overline{\alpha}_+ = \mu\,\alpha_-$ for some $\mu \neq 0$, and comparing $\omega$ with $\overline{\omega} = \omega$ gives $b = \overline{a}\,\mu$. Since $\omega \neq 0$, the coefficients $a$ and $b$ are therefore both nonzero. As $H^{0,1} = V_+^{0,1} \oplus V_-^{0,1}$ with each summand of dimension $n$, the space $H^{0,2} = \bigwedge^2 H^{0,1}$ decomposes as

$$H^{0,2} = \bigwedge^2 V_+^{0,1} \;\oplus\; \left(V_+^{0,1} \otimes V_-^{0,1}\right) \;\oplus\; \bigwedge^2 V_-^{0,1},$$

of dimensions $\binom{n}{2}$, $n^2$ and $\binom{n}{2}$. For $u \in V_+^{0,1}$ and $v \in V_-^{0,1}$ the mixed product $u \wedge v$ satisfies $u \wedge v \wedge \alpha_+ = 0$, because $u \wedge \alpha_+$ lies in $\bigwedge^{2n+1} V_+ = 0$, and $u \wedge v \wedge \alpha_- = 0$ for the same reason with $v$. So the middle summand lies in the kernel.

Conversely, let $z = z_{++} + z_{+-} + z_{--}$ be in the kernel. For the same reason $z_{++} \wedge \alpha_+$, $z_{--} \wedge \alpha_-$ and $z_{+-} \wedge \omega$ vanish, so $z \cdot \omega = a\, z_{--} \wedge \alpha_+ + b\, z_{++} \wedge \alpha_-$. If $n = 1$ the outer summands are zero and there is nothing to prove. If $n \geq 2$ the two terms lie in the distinct summands $\bigwedge^{2n} V_+ \otimes \bigwedge^2 V_-$ and $\bigwedge^2 V_+ \otimes \bigwedge^{2n} V_-$ of $\bigwedge^{2n+2} V_{\mathbb{C}} = \bigwedge^{2n+2}(V_+ \oplus V_-)$, so each of them vanishes. Wedging with a generator of $\bigwedge^{2n} V_+$ is injective on $\bigwedge^2 V_-$, and wedging with a generator of $\bigwedge^{2n} V_-$ is injective on $\bigwedge^2 V_+$. Since $a \neq 0$ we get $z_{--} = 0$, and since $b \neq 0$ we get $z_{++} = 0$, so the kernel is the middle summand. ∎

The reality of $\omega$ is needed: for the non-real element $\omega = \alpha_+$ of $W(A) \otimes \mathbb{C}$ the same computation gives the kernel $V_+^{0,1} \otimes V_-^{0,1} \oplus \bigwedge^2 V_+^{0,1}$, of dimension $n^2 + \binom{n}{2}$. Figure 9 draws the mixed block that forms the kernel.

**Theorem 9.7** (The rank-zero obstruction)**.** *Let $A$ be of $(K, -1, n)$-Weil type. Let $\mathcal{E}_1$ and $\mathcal{E}_2$ be coherent sheaves on $A$ such that*

$$\operatorname{ch}(\mathcal{E}_1) - \operatorname{ch}(\mathcal{E}_2) = c\,\omega, \qquad c \neq 0, \quad \omega \in W(A),$$

*such that $\operatorname{Ext}^2(\mathcal{E}_1, \mathcal{E}_2) = \operatorname{Ext}^2(\mathcal{E}_2, \mathcal{E}_1) = 0$, and such that the map $H^2(\mathcal{O}_A) \to \operatorname{Ext}^2(\mathcal{E}_1, \mathcal{E}_1)$, $z \mapsto z \cdot \mathrm{id}_{\mathcal{E}_1}$, is injective, as it is whenever $\operatorname{rk} \mathcal{E}_1 \neq 0$. Put $\mathcal{E} = \mathcal{E}_1 \oplus \mathcal{E}_2$. Then the semiregularity map*

$$\sigma_{\mathcal{E}} \colon \;\; \operatorname{Ext}^2(\mathcal{E}, \mathcal{E}) \longrightarrow \bigoplus_{q \geq 0} H^{q,q+2}(A)$$

*has kernel of dimension at least $n^2$. In particular $\mathcal{E}$ is not semiregular, for any $n \geq 1$.*

*Proof.* The vanishing of the off-diagonal $\operatorname{Ext}^2$ groups gives $\operatorname{Ext}^2(\mathcal{E}, \mathcal{E}) = \operatorname{Ext}^2(\mathcal{E}_1, \mathcal{E}_1) \oplus \operatorname{Ext}^2(\mathcal{E}_2, \mathcal{E}_2)$, and each summand receives $H^2(\mathcal{O}_A)$ through the $\mathcal{O}_A$-module structure, $z \mapsto z \cdot \mathrm{id}_{\mathcal{E}_i}$. For $z \in H^2(\mathcal{O}_A) = H^{0,2}(A)$ consider the element $\xi_z = z \cdot \mathrm{id}_{\mathcal{E}_1} - z \cdot \mathrm{id}_{\mathcal{E}_2}$. Its component in $\operatorname{Ext}^2(\mathcal{E}_1, \mathcal{E}_1)$ is $z \cdot \mathrm{id}_{\mathcal{E}_1}$, so by hypothesis $z \mapsto \xi_z$ is injective. The semiregularity map is compatible with direct sums, and on $H^{0,2} \cdot \mathrm{id}_{\mathcal{E}_i}$ it is multiplication by the Chern character of $\mathcal{E}_i$, so

$$\sigma_{\mathcal{E}}(\xi_z) = z \cdot \operatorname{ch}(\mathcal{E}_1) - z \cdot \operatorname{ch}(\mathcal{E}_2) = c\, z \cdot \omega.$$

If $\omega = 0$ this vanishes for every $z$, and $\dim H^{0,2} = n(2n-1) \geq n^2$. Otherwise $\omega$, being rational, is a nonzero real element of the Weil line, and by Lemma 9.6 the product vanishes for every $z$ in an $n^2$-dimensional subspace of $H^{0,2}$. The corresponding $\xi_z$ are linearly independent, and hence $\dim \ker \sigma_{\mathcal{E}} \geq n^2$.

It remains to justify the claim in the statement that the hypothesis on $\mathcal{E}_1$ holds in positive rank. For a coherent sheaf $\mathcal{F}$ the map $H^2(\mathcal{O}_A) \to \operatorname{Ext}^2(\mathcal{F}, \mathcal{F})$, $z \mapsto z \cdot \mathrm{id}_{\mathcal{F}}$, is the edge map of the local-to-global spectral sequence, and $\operatorname{tr}(z \cdot \mathrm{id}_{\mathcal{F}}) = \operatorname{rk}(\mathcal{F})\, z$. When $\operatorname{rk} \mathcal{F} \neq 0$ the trace therefore splits the map up to the factor $\operatorname{rk} \mathcal{F}$, and the map is injective; this applies in particular to a direct sum of line bundles. For a sheaf of rank zero the map can vanish: for a skyscraper sheaf it factors through $H^2$ of the structure sheaf of a point, which is zero. The direct sum $\mathcal{E}_1 \oplus \mathcal{E}_2$ is used only to produce an element of $\operatorname{Ext}^2$ whose image under $\sigma$ is $z \cdot (\operatorname{ch} \mathcal{E}_1 - \operatorname{ch} \mathcal{E}_2)$ rather than $z \cdot (\operatorname{ch} \mathcal{E}_1 + \operatorname{ch} \mathcal{E}_2)$. ∎

**Corollary 9.8** (The construction of Section 14)**.** *Let $L_1, \dots, L_m$ be line bundles on $A$ with $\sum_i c_1(L_i) = 0$ such that the virtual class $\sum_i([L_i] - [L_i^{-1}])$ has Chern character on the Weil line and every difference $\varepsilon\, c_1(L_i) - \varepsilon' c_1(L_j)$ with $\varepsilon, \varepsilon' \in \{\pm 1\}$ and $(\varepsilon, i) \neq (\varepsilon', j)$ is nondegenerate of index different from 2. Then the sheaf $\bigoplus_i (L_i \oplus L_i^{-1})$ has semiregularity kernel of dimension at least $n^2$. At $n = 3$ the explicit object of Section 14 has kernel of dimension exactly $15 = n^2 + n(n-1)$, of which the $n^2 = 9$ dimensions on the odd side are the ones supplied by Theorem 9.7; the remaining $n(n-1) = 6$ come from the even side and are specific to the line-bundle structure.*

*Proof.* Take $\mathcal{E}_1 = \bigoplus L_i$ and $\mathcal{E}_2 = \bigoplus L_i^{-1}$. Then $\operatorname{Ext}^2(\mathcal{E}_1, \mathcal{E}_2) = \bigoplus_{i,j} H^2(L_i^{-1} \otimes L_j^{-1})$ and $\operatorname{Ext}^2(\mathcal{E}_2, \mathcal{E}_1) = \bigoplus_{i,j} H^2(L_i \otimes L_j)$, and the first Chern classes $\mp(c_1(L_i) + c_1(L_j))$ occurring here, for all $i$ and $j$, are differences of the kind in the hypothesis, with $\varepsilon' = -\varepsilon$. A nondegenerate line bundle has cohomology only in the degree equal to its index, so the index hypothesis gives the required $\operatorname{Ext}^2$ vanishing. The sheaf $\mathcal{E}_1$ has rank $m \neq 0$, and the Chern character hypothesis is that of Theorem 9.7. Since $e^{\lambda}$ and $e^{-\lambda}$ combine into $\cosh\lambda$ and $\sinh\lambda$, which occupy disjoint degrees, the map splits by parity, and the odd part is the annihilator of Lemma 9.6 tensored with the sign vector. At $n = 3$ the dimensions are read off from the proof of Proposition 18.14, where the signs are absorbed in the same way: the kernel consists of the vectors $(u_i^{-k} z)_i$ with $z$ in the mixed block $V_+^{0,1} \otimes V_-^{0,1}$ for $k = 3$, which change sign under $u_i \mapsto -u_i$, and with $z$ in $\bigwedge^2 V_+^{0,1}$ or $\bigwedge^2 V_-^{0,1}$ for $k = 2$ or $k = 4$, which do not. ∎

*Remark* 9.9 (Uniformity over the configurations). By Corollary 9.8 the kernel has dimension at least $n^2$ for every configuration of line bundles $L_i^{\pm 1}$ whose virtual Chern character lies on the Weil line and whose differences are nondegenerate of index different from 2. The deficiency is therefore not special to the six line bundles of Proposition 18.14: Theorem 9.7 produces it from the vanishing of the rank alone.

Theorem 9.7 excludes the economical choice described at the start of this subsection. An object on which $H^2(\mathcal{O}_A)$ acts faithfully and which realises a Weil class through semiregularity cannot have its Chern character concentrated on the Weil line. It must carry a nonzero rank, or nonzero Chern character in other degrees, so that $z \mapsto z \cdot \operatorname{ch}(\mathcal{E})$ is injective on $H^{0,2}$; the degree-two component of that product is $\operatorname{rk}(\mathcal{E}) \cdot z$, and rank zero is the case in which it vanishes. Markman's sheaves have nonzero rank, and this is essential to his method: the class he shows to be deformation-invariant is the normalised character $\exp(-c_1(\mathcal{E})/\operatorname{rk}\mathcal{E})\operatorname{ch}(\mathcal{E})$, which divides by the rank, and the objects at $n = 3$ are reflexive sheaves [Mar25a, Mar25b]. This narrows the fourth ingredient of Remark 14.16: it is a single sheaf of positive rank whose Chern character has a Weil component alongside other components, rather than a difference of two sheaves whose classes cancel outside the Weil line.

Theorem 9.7 is stated for a direct sum because there the element $\xi_z$ is explicit. The mechanism applies to any perfect complex on which $H^2(\mathcal{O}_A)$ acts faithfully, and in particular to the object used in Section 12.

**Proposition 9.10** (Complexes with concentrated Chern character)**.** *Let $A$ be of $(K, -1, n)$-Weil type and let $E$ be a nonzero perfect complex on $A$ with $\operatorname{ch}_k(E) = 0$ for $k \neq n$ and $\operatorname{ch}_n(E) = N\omega$ for some $N \neq 0$ and some nonzero $\omega \in W(A)$. If the natural map*

$$H^2(\mathcal{O}_A) \longrightarrow \operatorname{Ext}^2(E, E), \qquad z \longmapsto z \cdot \mathrm{id}_E, \tag{25}$$

*is injective, then* $\dim\ker\sigma_E \geq n^2$*, so* $E$ *is not semiregular. The map* (25) *is injective whenever* $E$ *is represented by a virtual sum of line bundles, in particular at every base point of Corollary 15.4, Theorem 15.25 and Theorem 15.10.*

*Proof.* For any perfect complex the semiregularity map satisfies $\sigma_E(z\cdot\mathrm{id}_E) = z\cdot\mathrm{ch}(E)$, since $\mathrm{tr}(z\cdot\mathrm{id}_E\circ\exp\mathrm{At}(E)) = z\cdot\mathrm{tr}\exp\mathrm{At}(E) = z\cdot\mathrm{ch}(E)$. Hence for $z$ in the annihilator of $\omega$ in $H^{0,2}(A)$,

$$\sigma_E(z\cdot\mathrm{id}_E) = N\,z\cdot\omega = 0,$$

and that annihilator has dimension $n^2$ by Lemma 9.6. If (25) is injective, the classes $z\cdot\mathrm{id}_E$ for $z$ in a basis of the annihilator are linearly independent, which gives the bound.

For the last sentence, write the class of $E$ in $K_0(A)$ as $\sum_i \varepsilon_i[L_i]$ with $\varepsilon_i = \pm 1$ and $L_i$ line bundles, and realise it by the direct sum of the $L_i$ with $\varepsilon_i = 1$ and of the shifts $L_i[1]$ of those with $\varepsilon_i = -1$. The map (25) is then $z \mapsto (z,\dots,z)$ into $\bigoplus_i \mathrm{Ext}^2(L_i,L_i) = \bigoplus_i H^2(\mathcal{O}_A)$, which is injective. At a split member $\omega_1$ is a polynomial in $\beta,\widehat{\beta},\ell$ by Theorem 15.2(iii), at a quaternionic member it is a polynomial in divisor classes by Theorem 15.25, and at a product it is one by Theorem 15.10; in each case the Chern character is a ring expression in classes of line bundles, so such a representation exists. ∎

**Proposition 9.11** (The class forces the size of the object)**.** *Let* $A$ *be of* $(K,-1,n)$*-Weil type, let* $\omega$ *be a nonzero element of the Weil line and let* $E$ *be a perfect complex on* $A$ *with*

$$\mathrm{ch}(E) = r + N\omega, \qquad r \in H^0(A,\mathbb{Q}),\ N \neq 0,$$

*and no other component. Then*

$$\chi(E,E) \;=\; (-1)^n N^2 \int_A \omega^2,$$

*independently of* $r$*, and consequently*

$$\sum_i \dim \mathrm{Ext}^i(E,E) \;\geq\; N^2\Big|\int_A \omega^2\Big| \;>\; 0.$$

*Proof.* The Todd class of an abelian variety is 1, so Riemann–Roch for the Ext algebra reads $\chi(E,E) = \int_A \mathrm{ch}(E)^\vee\,\mathrm{ch}(E)$ with $\mathrm{ch}_k^\vee = (-1)^k\,\mathrm{ch}_k$. Expanding,

$$\mathrm{ch}(E)^\vee\,\mathrm{ch}(E) = r^2 + (1+(-1)^n)rN\omega + (-1)^n N^2\omega^2,$$

and of the three terms only the last has degree $4n$, so only it survives the integral; the rank enters solely through $r^2$, which sits in degree zero. For the inequality, $\chi(E,E) = \sum_i (-1)^i \dim\mathrm{Ext}^i(E,E)$, so the total dimension is at least $|\chi(E,E)|$. Finally $\int_A \omega^2 \neq 0$. Write $\omega = \alpha_+ + \alpha_-$ with $\alpha_\pm \in \bigwedge^{2n} V_\pm$, both nonzero by the proof of Lemma 9.6. Each $\alpha_\pm$ is a top form on its own eigenspace and squares to zero, so $\omega^2 = 2\alpha_+\alpha_-$, which is a nonzero multiple of the fundamental class. ∎

*Remark* 9.12 (The constant). The bound concerns the whole Ext algebra and does not depend on the rank, so increasing the rank does not weaken it. The constant $|\int\omega^2|$ depends on the normalisation of $\omega$; measured against $\eta$ it picks up factors of $d$ and of the polarisation type, and no dimension-free growth rate can be read off from it. The bound uses only that the Weil class is not isotropic. Together with Proposition 9.10 it constrains an object carrying the class in two ways. If the Chern character is concentrated on the Weil line, the semiregularity map has a kernel of dimension at least $n^2$ unless $H^2(\mathcal{O}_A)$ acts on the object degenerately; and whatever the rank, the Ext algebra has dimension at least $N^2|\int\omega^2|$.

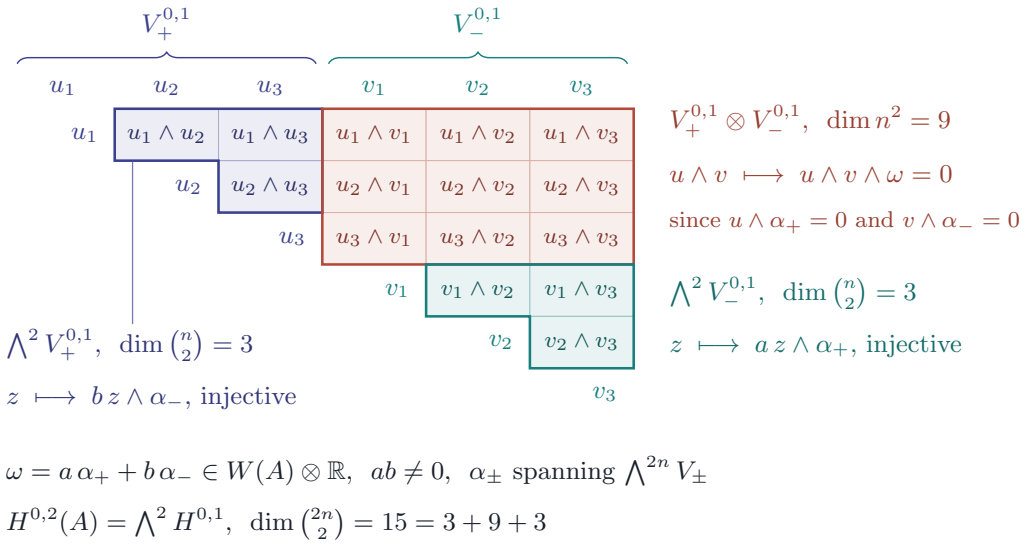



FIGURE 8. The structure behind Lemma 9.6, drawn at $n = 3$. $H^{0,2}(A) = \bigwedge^2 H^{0,1}$ splits into three blocks by how many factors come from $V_+^{0,1}$ and how many from $V_-^{0,1}$. The two outer blocks, of dimension $\binom{n}{2} = 3$ each, are carried injectively by multiplication with the Weil class. The middle block, of dimension $n^2 = 9$ and spanned by the mixed products, goes to zero, because each such product repeats a factor of $\alpha_+$ and a factor of $\alpha_-$. Its dimension $n^2$ is the deficit in Proposition 9.10.

*Remark* 9.13 (Consequences for the semiregularity reduction). Theorem 16.29 proves $\Sigma = \mathcal{D}_{n,n}$, that is, the algebraicity of the Weil class on the whole family, under the hypothesis that $\sigma_E$ is injective for the complex $E$ of Proposition 16.27, whose Chern character is $N\omega_1$ in degree $n$ and zero elsewhere. This complex satisfies the hypotheses of Proposition 9.10 at every base point of Corollary 15.4, Theorem 15.25 and Theorem 15.10, so there the hypothesis of Theorem 16.29 fails for it, by $n^2$ dimensions.

This does not make Theorem 16.29 vacuous, for two reasons. First, the hypothesis concerns a chosen perfect complex representing a class in $K_0(A)_{\mathbb{Q}}$, and distinct representatives of one class have distinct $\mathrm{Ext}^2$ groups; what is ruled out is the representative that the construction supplies. Second, Proposition 9.10 needs (25) to be injective, which is a real restriction for an object of rank zero: it fails, for instance, for a sheaf supported at a point. A representative escaping Proposition 9.10 must therefore receive $H^2(\mathcal{O}_A)$ non-injectively, which for a rank-zero object means that its support is degenerate. Of the two reductions of Section 12, the degree bound of Theorem 16.9 is untouched by anything here, and the semiregularity reduction of Theorem 16.29 is unavailable for the complexes that the constructions at these base points supply.

9.3. **The annihilator and the tangent space to the family.** The number $n^2$ of Lemma 9.6 is also the dimension of the Weil family. We show in this subsection that the polarisation identifies the annihilator with the tangent space to the family, canonically and in every dimension. Two consequences follow. First, the deficiency of the semiregularity map at an object of rank zero cannot be removed by a better choice of object, since it is

the family itself, seen inside $H^{0,2}$. Second, the first-order Hodge locus of the Weil line is the Weil family, so the directions that the semiregularity map cannot see are those along which the class stays Hodge and along which it would have to be carried.

Write $V_\pm$ for the two eigenspaces of $K$ on $H^1(A,\mathbb{C})$, each of dimension $2n$, and

$$V_\pm^{1,0} = V_\pm \cap H^{1,0}, \qquad V_\pm^{0,1} = V_\pm \cap H^{0,1},$$

each of dimension $n$ by Lemma 3.5. Let $E$ denote the polarisation form on $H^1(A,\mathbb{Q})$, extended to $H^1(A,\mathbb{C})$. We use throughout that $E$ vanishes on $V_+ \times V_+$ and on $V_- \times V_-$ and pairs $V_+$ with $V_-$ perfectly, which is condition (iii) of Definition 3.1 read on the eigenspaces, and that $E$ vanishes on $H^{1,0}\times H^{1,0}$ and on $H^{0,1}\times H^{0,1}$, which is the first Riemann bilinear relation. For the first statement, let $x, y \in V_+$. Then $E(\sqrt{-d}\,x, y) = \sqrt{-d}\,E(x,y)$ from the eigenvalue, and $E(\sqrt{-d}\,x, y) = -E(x, \sqrt{-d}\,y) = -\sqrt{-d}\,E(x,y)$ from Definition 3.1(iii), so $E(x,y)=0$. The same holds on $V_-$, and the nondegeneracy of $E$ then forces the pairing between $V_+$ and $V_-$ to be perfect. Together, the two vanishing statements say that of the four blocks of $E$ on $(V_+^{1,0}\oplus V_+^{0,1}) \times (V_-^{1,0}\oplus V_-^{0,1})$ only $V_+^{1,0}\times V_-^{0,1}$ and $V_+^{0,1}\times V_-^{1,0}$ are nonzero, and each of these two is a perfect pairing of $n$-dimensional spaces. This block structure is all that the proof of Theorem 9.17 uses of $E$.

A first-order deformation of $A$ is a class in $H^1(A, T_A)$, which for an abelian variety is canonically $\operatorname{Hom}(H^{1,0}, H^{0,1})$, since the tangent bundle is trivial and $H^{1,0}$ is the dual of its space of sections. Such a $v$ is extended to $H^1(A,\mathbb{C})$ by zero on $H^{0,1}$ and acts on $H^k(A,\mathbb{C}) = \bigwedge^k H^1(A,\mathbb{C})$ as the derivation it generates; we write $v \lrcorner \xi$ for that action. A Hodge class $\xi$ of type $(p,p)$ remains of type $(p,p)$ to first order along $v$ exactly when $v\lrcorner\xi = 0$. We call $v$ *polarised* when $E(vx,y)+E(x,vy)=0$ for all $x,y\in H^{1,0}$, which is the condition that the deformation preserve the polarisation. The polarised $v$ form a space of dimension $n(2n+1) = \dim \mathcal{A}_{2n}$.

**Proposition 9.14** (The first-order Hodge locus of the Weil line)**.** *Let $A$ be of $(K,-1,n)$-Weil type and let $v$ be a polarised first-order deformation of $A$. The following are equivalent.*

(i) *$v\lrcorner\omega = 0$ for every $\omega$ in the Weil line.*
(ii) *$v$ commutes with the action of $K$.*
(iii) *$v(V_+^{1,0}) \subseteq V_+^{0,1}$ and $v(V_-^{1,0}) \subseteq V_-^{0,1}$.*

*The space of such $v$ has dimension $n^2$, and under Proposition B.8 it is the tangent space at $[A]$ to the Weil family $\mathcal{D}_{n,n}$. Each of the two rational generators of the Weil line already cuts it out by itself.*

*Proof.* The equivalence of (ii) and (iii) is immediate. Since $K$ acts on $V_\pm$ by the two scalars $\pm\sqrt{-d}$, commuting with $K$ means preserving $V_+$ and $V_-$. The extension of $v$ kills $H^{0,1}$ and takes values in $H^{0,1}$, so preserving $V_\pm$ amounts to $v(V_\pm^{1,0}) \subseteq V_\pm \cap H^{0,1} = V_\pm^{0,1}$.

For (i) and (iii), let $\alpha_+$ generate $\bigwedge^{2n} V_+$ and write $\alpha_+ = e_1\wedge\cdots\wedge e_{2n}$ for a basis $e_1,\dots,e_{2n}$ of $V_+$ adapted to $V_+ = V_+^{1,0}\oplus V_+^{0,1}$. As a derivation,

$$v\lrcorner\alpha_+ = \sum_{j=1}^{2n} e_1\wedge\cdots\wedge v(e_j)\wedge\cdots\wedge e_{2n}.$$

Split $v(e_j)$ along $H^1(A,\mathbb{C}) = V_+\oplus V_-$. The $V_+$ components contribute $\operatorname{tr}(\pi_+\circ v|_{V_+})\,\alpha_+$, and that trace vanishes: the map $\pi_+\circ v|_{V_+}$ kills $V_+^{0,1}$ and carries $V_+^{1,0}$ into $V_+^{0,1}$, so in the adapted basis its matrix has zero diagonal blocks. Writing $v(e_j)_- = \sum_k c_{jk} f_k$ for a basis

$f_1, \dots, f_{2n}$ of $V_-$, the $V_-$ components contribute

$$\sum_{j,k} \pm c_{jk}\, e_1 \wedge \cdots \wedge \widehat{e_j} \wedge \cdots \wedge e_{2n} \wedge f_k,$$

and the $4n^2$ monomials appearing here are distinct members of the standard basis of $\bigwedge^{2n} H^1(A, \mathbb{C})$, hence linearly independent. So $v \lrcorner \alpha_+ = 0$ if and only if every $c_{jk}$ vanishes, which is $v(V_+) \subseteq V_+$. The same argument on $\alpha_-$ gives $v \lrcorner \alpha_- = 0$ if and only if $v(V_-) \subseteq V_-$. The Weil line spans $\bigwedge^{2n} V_+ \oplus \bigwedge^{2n} V_-$ over $\mathbb{C}$ by Proposition 3.7, and each of the two rational generators of Theorem 4.9 has both components nonzero, so a single generator already forces both conditions. This is (i) if and only if (iii).

For the dimension, let $v$ satisfy (iii), and write $v_+$ and $v_-$ for its restrictions to $V_+^{1,0}$ and $V_-^{1,0}$. If $x$ and $y$ both lie in $V_+^{1,0}$, then $v_+x$ and $y$ both lie in $V_+$, so both terms of the polarisation condition vanish, and likewise if both lie in $V_-^{1,0}$. For $x \in V_+^{1,0}$ and $y \in V_-^{1,0}$ the condition reads $E(v_+x, y) = -E(x, v_-y)$. The left side is determined by $v_+$, and the pairing between $V_+^{1,0}$ and $V_-^{0,1}$ is perfect, so for each $y$ there is exactly one $v_-y \in V_-^{0,1}$ satisfying this for all $x$, and it depends linearly on $y$. Hence $v_-$ is determined by $v_+$, which may be chosen freely, and the polarised deformations satisfying (iii) form a space of dimension $\dim \mathrm{Hom}(V_+^{1,0}, V_+^{0,1}) = n^2$. By Proposition B.8 the family $\mathcal{D}_{n,n}$ is an open subset of the Grassmannian of $n$-planes in $V_+$, whose tangent space at $V_+^{1,0}$ is $\mathrm{Hom}(V_+^{1,0}, V_+/V_+^{1,0}) = \mathrm{Hom}(V_+^{1,0}, V_+^{0,1})$, and the identification just made is that isomorphism. ∎

**Proposition 9.15** (Infinitesimal rigidity of divisor classes)**.** *Keep the notation above, put $P = V_+^{1,0}$ and $P' = V_-^{1,0}$ as in the proof of Proposition B.8, so that $\overline{P} = V_-^{0,1}$ and $\overline{P}' = V_+^{0,1}$, and write a class $\delta \in H^{1,1}$ in the four blocks*

$$a \text{ on } P \otimes \overline{P}, \quad b \text{ on } P' \otimes \overline{P}', \quad c \text{ on } P \otimes \overline{P}', \quad e \text{ on } P' \otimes \overline{P},$$

*the first two carrying the norm character of $K^\times$ and the last two the characters $\tau^2$ and $\overline{\tau}^2$; reality makes $a$ and $b$ anti-hermitian and ties $e$ to $-\overline{c}^T$. Identify the tangent space $\mathrm{Hom}(P, \overline{P}')$ to $\mathcal{D}_{n,n}$ with the $n \times n$ matrices $V$ as in Theorem 9.17, and write* alt *for the alternating part of a matrix, viewed in $\bigwedge^2$. Then contraction with a tangent vector is*

$$\Phi_\delta(V) \;=\; \big( Va + b^T V, \;\; \mathrm{alt}(Vc), \;\; \mathrm{alt}(-V^T e) \big), \tag{26}$$

*with values in $\overline{P}' \otimes \overline{P} \oplus \bigwedge^2 \overline{P}' \oplus \bigwedge^2 \overline{P} = H^{0,2}$, and $\ker \Phi_\delta$ is the tangent space to the locus where $\delta$ stays of type $(1,1)$. Consequently:*

(i) *$\Phi_\delta = 0$ if and only if $\delta$ is a real multiple of $\eta$. The rigid classes form a real line, which is the infinitesimal form of the statement that $\mathrm{NS}(A)_{\mathbb{Q}}$ has rank one at the general member.*

(ii) *For the norm-character class with block $a$ and $b = c = e = 0$,*

$$\dim \ker \Phi_\delta \;=\; n\,(n - \operatorname{rank} a),$$

*so such a class cuts a locus of codimension $n \operatorname{rank} a$, at least $n$.*

(iii) *A general class has $\ker \Phi_\delta = 0$: to first order it rigidifies the family to a point.*

*Proof.* For (26), a polarised deformation commuting with $K$ sends $P$ into $\overline{P}'$ and $P'$ into $\overline{P}$, and the polarisation condition $E(vx, y) + E(x, vy) = 0$ forces the second matrix to be $-V^T$, as in the proof of Proposition 9.14. Applying $v$ to the $(1,0)$ slot of each block and collecting terms in $H^{0,2}$ gives the three displayed components; the sign in the first comes from $\overline{P} \wedge \overline{P}' = -\overline{P}' \wedge \overline{P}$.

(i) If $\Phi_\delta = 0$ then $\operatorname{alt}(Vc) = 0$ for every $V$, which for $n \geq 2$ forces $c = 0$, and likewise $e = 0$. The first component gives $Va = -b^T V$ for every $V$; taking $V$ to run over the matrix units shows $a$ is a scalar $\lambda$ and $b^T = -\lambda$. Anti-hermitian scalars are the purely imaginary ones, so the solutions form a real line. It contains $\eta$, whose blocks are $a = i$, $b = -i$, and the two therefore agree.

(ii) With $b = c = e = 0$ the condition is $Va = 0$, that is, the rows of $V$ annihilate the column space of $a$; the solution space has dimension $n(n - \operatorname{rank} a)$.

(iii) By (ii), the class with $a$ of rank $n$ (for instance $a = i$) and $b = c = e = 0$, which is real since $a$ is anti-hermitian, has $\ker \Phi_\delta = 0$. Since $\Phi_\delta$ depends linearly on $\delta$, the classes with $\ker \Phi_\delta \neq 0$ are cut out by the vanishing of the maximal minors of $\Phi_\delta$ and form a proper real algebraic subset of the real classes of type $(1,1)$, which gives (iii). ∎

*Remark* 9.16 (The divisorial supply). Proposition 9.15(ii) explains why the divisorial supply of Theorem 15.16 cannot be enlarged by a better choice of class. Any norm-character class other than a multiple of $\eta$ has $\operatorname{rank} a \geq 1$ and so cuts a locus of codimension at least $n$; that locus is proper, and by (i) only the real multiples of $\eta$ cut nothing. Each further class therefore yields a subvariety and never an open set, which is Proposition 15.18 with the codimension made explicit.

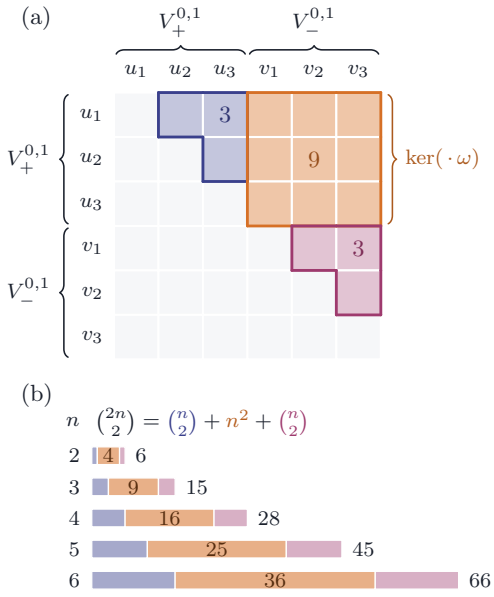



FIGURE 9. The geometry behind Lemma 9.6, drawn schematically for the real shadow of the hermitian form. The ruled surface is the null cone of $H$; the two transverse planes are the conjugate isotropic subspaces $V_+$ and $V_-$, which conjugation interchanges, so that neither is defined over $\mathbb{Q}$ and only their sum is. The segments join a vector of one to its conjugate in the other, and the strip they sweep stands for the mixed block $V_+^{0,1} \otimes V_-^{0,1}$, of dimension $n^2$, which multiplication by the Weil class sends to zero and which is the deficit in Proposition 9.10.

**Theorem 9.17** (The annihilator is the tangent space)**.** *Let $A$ be of $(K,-1,n)$-Weil type and let $\omega$ be a rational generator of the Weil line. The polarisation induces an isomorphism*

$$T_{[A]}\mathcal{D}_{n,n} \xrightarrow{\sim} \operatorname{Ann}_{H^{0,2}(A)}(\omega),$$

*canonical in the sense that it depends on no choice of basis and on no choice of $\omega$ on the line. Explicitly, if $p_1,\dots,p_n$ is a basis of $V_+^{1,0}$ and $q_1,\dots,q_n$ the basis of $V_-^{0,1}$ dual to it under $E$, then $\varphi \in \operatorname{Hom}(V_+^{1,0},V_+^{0,1})$ corresponds to*

$$\sum_{a=1}^{n} q_a \wedge \varphi(p_a) \;\in\; V_-^{0,1}\otimes V_+^{0,1} \;\subset\; \bigwedge^2 H^{0,1} = H^{0,2}(A).$$

*Proof.* The form $E$ pairs $V_+$ with $V_-$ perfectly and vanishes on $V_+^{1,0}\times V_-^{1,0}$, the latter because both spaces lie in $H^{1,0}$. Hence $E$ restricted to $V_+^{1,0}\times V_-$ kills $V_-^{1,0}$ and descends to a pairing

$$V_+^{1,0}\times V_-^{0,1}\longrightarrow \mathbb{C}.$$

This pairing is perfect: both spaces have dimension $n$, and a vector of $V_+^{1,0}$ orthogonal to $V_-^{0,1}$ is orthogonal to all of $V_-$, hence zero. It gives a canonical isomorphism $(V_+^{1,0})^*\cong V_-^{0,1}$, and therefore

$$T_{[A]}\mathcal{D}_{n,n} = \operatorname{Hom}(V_+^{1,0},V_+^{0,1}) = (V_+^{1,0})^*\otimes V_+^{0,1} \;\cong\; V_-^{0,1}\otimes V_+^{0,1},$$

the first equality being the tangent space computation at the end of the proof of Proposition 9.14. Since $H^{0,1}=V_+^{0,1}\oplus V_-^{0,1}$, the map $q\otimes u\mapsto q\wedge u$ identifies $V_-^{0,1}\otimes V_+^{0,1}$ with the middle summand of $\bigwedge^2 H^{0,1}$, and that summand is the annihilator by Lemma 9.6. The displayed formula is the composite of these isomorphisms written in a dual pair of bases, so it is independent of the basis chosen; and Lemma 9.6 gives the same subspace for every $\omega$ on the line, so the isomorphism does not depend on $\omega$ either. ∎

**Corollary 9.18** (The kernel contains the tangent space to the family)**.** *Let $A$ be of $(K,-1,n)$-Weil type and let $E$ be a nonzero perfect complex on $A$ satisfying the hypotheses of Proposition 9.10, that is, with Chern character concentrated on the Weil line and with (25) injective. Then $\ker\sigma_E$ contains a subspace canonically isomorphic to $T_{[A]}\mathcal{D}_{n,n}$, namely the image of the composite*

$$T_{[A]}\mathcal{D}_{n,n} \;\cong\; \operatorname{Ann}_{H^{0,2}}(\omega) \;\hookrightarrow\; H^2(\mathcal{O}_A) \xrightarrow{z\mapsto z\cdot\mathrm{id}_E} \operatorname{Ext}^2(E,E).$$

*By Proposition 9.14 that subspace is also the tangent space to the first-order Hodge locus of the class $\mathrm{ch}_n(E)$.*

*Proof.* Immediate from Theorem 9.17, Proposition 9.10 and Proposition 9.14. ∎

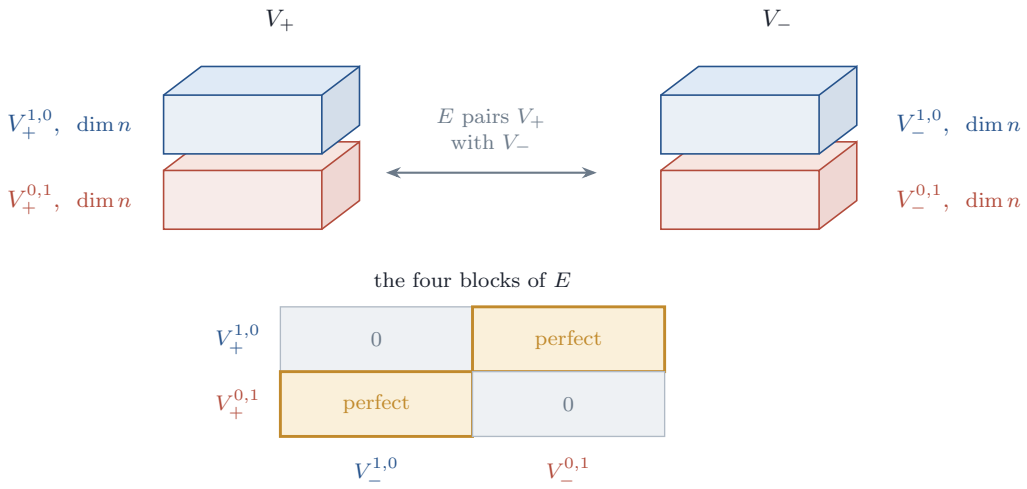
**Theorem 9.19** (The weakened criterion is unavailable for concentrated Chern characters)**.** *Let $A$ be of $(K,-1,n)$-Weil type and let $E$ be a nonzero perfect complex on $A$ satisfying the hypotheses of Proposition 9.10. Let*

$$\mathrm{ev}_E\colon HH^2(A)\longrightarrow \operatorname{Ext}^2(E,E)$$

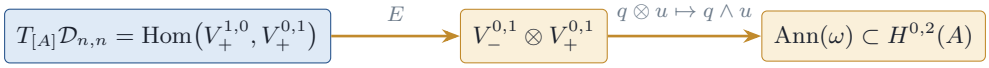
*be the evaluation on $E$ of the natural transformations $\mathrm{id}\to\mathrm{id}[2]$ of $D^b(A)$, as in* [Mar25b, Section 11.4]*. Then*

$$\mathrm{ev}_E(H^2(\mathcal{O}_A))\cap\ker\sigma_E = \Delta := \{\, z\cdot\mathrm{id}_E \;:\; z\in \operatorname{Ann}_{H^{0,2}}(\omega)\,\},$$

$K$ splits $H^1(A,\mathbb{C})$ into two halves, and each half splits by Hodge type.




The two zeros are the first Riemann relation, $E = 0$ on $H^{1,0} \times H^{1,0}$ and on $H^{0,1} \times H^{0,1}$.
The upper right block is the one that moves the tangent space.




Both spaces have dimension $n^2$. The composite depends on no basis and on no choice of $\omega$ on the line.

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$n = 2$: $\dim T = \dim \mathrm{Ann} = 4$ $\quad n = 3$: 9 $\quad n = 4$: 16
checked exactly over $\mathbb{Q}(i)$

FIGURE 10. Theorem 9.17. The polarisation has only two nonzero blocks on $V_+ \times V_-$: it pairs $V_+^{1,0}$ with $V_-^{0,1}$ and $V_+^{0,1}$ with $V_-^{1,0}$, each perfectly, and the other two blocks vanish by the first Riemann relation. The upper right block turns the tangent space $\mathrm{Hom}(V_+^{1,0}, V_+^{0,1})$ to the Weil family into $V_-^{0,1} \otimes V_+^{0,1}$, which is the middle block of $\bigwedge^2 H^{0,1}$ drawn in Figure 8 and the annihilator of the Weil line. The two $n^2$-dimensional spaces are therefore the same subspace of $H^{0,2}$, and the directions that the semiregularity map cannot see are the directions of the family along which the class stays Hodge.

*a subspace of dimension $n^2$. In particular $\sigma_E$ is not injective on the image of $\mathrm{ev}_E$, so $E$ satisfies neither the hypothesis of* [Mar25b, Lemma 11.3] *nor the weakened hypothesis of* [Mar25b, Question 11.4].

*Proof.* The Hochschild cohomology $HH^*(A) = \mathrm{Ext}^*_{A\times A}(\mathcal{O}_\Delta, \mathcal{O}_\Delta)$ is the algebra of natural transformations of the identity functor, and the Hochschild–Kostant–Rosenberg isomorphism [Cal05, BF08] writes $HH^2(A) \cong H^2(\mathcal{O}_A) \oplus H^1(A, T_A) \oplus H^0(A, \bigwedge^2 T_A)$, with no correction from the Todd class because $T_A$ is trivial. The first summand is the image of $H^2(\mathcal{O}_A) = \mathrm{Ext}^2(\mathcal{O}_A, \mathcal{O}_A)$ under pullback to $A \times A$, and the natural transformation it defines sends a complex $F$ to the composite $F = F \otimes \mathcal{O}_A \xrightarrow{\mathrm{id}\otimes z} F \otimes \mathcal{O}_A[2] = F[2]$, that is, to $z \cdot \mathrm{id}_F$. This is the statement that the action of $HH^*(A)$ on $\mathrm{Ext}^*(F, F)$ restricts on $H^*(\mathcal{O}_A)$ to the $\mathcal{O}_A$-module structure. Concretely, a class $z \in H^2(\mathcal{O}_A)$ is represented by a 2-extension of $\mathcal{O}_A$ by itself; its pullback $\pi_1^* z$ to $A \times A$ is a 2-extension of $\mathcal{O}_{A\times A}$ by itself, and tensoring with $\mathcal{O}_\Delta$ gives a 2-extension of $\mathcal{O}_\Delta$ by itself, which is the element of $HH^2(A)$ in question. The Fourier–Mukai transform with kernel $\mathcal{O}_\Delta$ is the identity, and the transform with kernel this extension is the extension of the identity by itself twisted by $z$, so its value on $F$ is $z \cdot \mathrm{id}_F$. Markman uses this identification when he writes the left vertical arrow of the diagram in the proof of [Mar25b, Lemma 11.3] as the HKR isomorphism followed by contraction with $\mathrm{ch}(E)$. On the summand $H^2(\mathcal{O}_A)$ contraction is the cup product $z \mapsto z \cdot \mathrm{ch}(E)$, which is $\sigma_E(z \cdot \mathrm{id}_E)$ by the computation in the proof of Theorem 9.7.

Hence $\mathrm{ev}_E(z) = z{\cdot}\mathrm{id}_E$ for $z \in H^2(\mathcal{O}_A)$, and $\sigma_E(z{\cdot}\mathrm{id}_E) = z{\cdot}\mathrm{ch}(E) = N\, z{\cdot}\omega$ by the proof of Proposition 9.10, which vanishes exactly for $z \in \mathrm{Ann}_{H^{0,2}}(\omega)$. So $\mathrm{ev}_E(H^2(\mathcal{O}_A)) \cap \ker \sigma_E = \Delta$, and $\dim \Delta = \dim \mathrm{Ann}_{H^{0,2}}(\omega) = n^2$ by Lemma 9.6 and the injectivity of (25). Since $\Delta \neq 0$, the restriction of $\sigma_E$ to the image of $\mathrm{ev}_E$ has a nonzero kernel. The hypothesis of [Mar25b, Lemma 11.3] is that $\ker \mathrm{ev}_E$ equals the kernel of contraction with $\mathrm{ch}(E)$ and that $\mathrm{ev}_E$ is surjective; the two conditions together would make $\sigma_E$ injective on all of $\mathrm{Ext}^2(E, E)$, contradicting Proposition 9.10. The weakened hypothesis of [Mar25b, Question 11.4] is injectivity of $\sigma_E$ on the image, which we have just shown to fail. ∎

**Theorem 9.20** (A line bundle summand off the polarisation line obstructs)**.** *Let $A$ be of $(K, -1, n)$-Weil type with $n \geq 2$ and let $T = T_{[A]}\mathcal{D}_{n,n}$. Let $E$ be a perfect complex on $A$ with a direct summand $L[p]$, $L$ a line bundle, and suppose that $\mathrm{ch}(E)$ remains of Hodge type to first order along $\mathcal{D}_{n,n}$, that is $v \lrcorner \mathrm{ch}(E) = 0$ for every $v \in T$. If $c_1(L)$ is not a real multiple of $\eta$, then there is $v \in T$ with*

$$\mathrm{ev}_E(v) \neq 0 \qquad \text{and} \qquad \sigma_E(\mathrm{ev}_E(v)) = 0.$$

*Consequently $E$ is not semiregular, does not satisfy the weakened hypothesis of* [Mar25b, Question 11.4]*, and is obstructed to first order along $v$. The same holds for a twisted complex, with $c_1(L)$ replaced by the twisted first Chern class of the summand, which is $c_1(L)$ shifted by the class common to all summands.*

*Proof.* The class $c_1(L)$ is a real class of type $(1,1)$ and not a real multiple of $\eta$, so by Proposition 9.15(i) the contraction $\Phi_{c_1(L)} \colon T \to H^{0,2}(A)$ is not zero: there is $v \in T$ with $v \lrcorner c_1(L) \neq 0$. On the summand $H^1(A, T_A)$ of $HH^2(A)$ the evaluation map is contraction with the Atiyah class, $\mathrm{ev}_E(v) = v \lrcorner \mathrm{At}(E)$. This is the top row of the diagram of Buchweitz and Flenner reproduced as [Mar25b, (2.1)]; its identification with the Hochschild evaluation is in [BF08], and Markman uses it in the proof of [Mar25b, Lemma 11.3]. The Atiyah class of a direct sum is the direct sum of the Atiyah classes, and that of $L[p]$ is $c_1(L)$, so the component of $\mathrm{ev}_E(v)$ in the summand $\mathrm{Ext}^2(L[p], L[p]) = H^{0,2}(A)$ of $\mathrm{Ext}^2(E, E)$ is $v \lrcorner c_1(L) \neq 0$. On the other hand the same diagram gives $\sigma_E(\mathrm{ev}_E(v)) = v \lrcorner \mathrm{ch}(E)$, which

vanishes by hypothesis. So $\mathrm{ev}_E(v)$ is a nonzero element of $\ker\sigma_E \cap \operatorname{im}\mathrm{ev}_E$; this is the failure of both hypotheses, and $v \lrcorner \mathrm{At}(E) \neq 0$ is the statement that $E$ does not lift to a first-order deformation of the pair along $v$, as recalled in [Mar25b, Section 2]. Twisting by a class $B$ replaces the Atiyah class of every rank one summand by its shift by $B$ and $\mathrm{ch}(E)$ by the twisted Chern character, and nothing else in the argument changes. ∎

*Remark* 9.21 (Scope of the kernel computation). The semiregularity criterion converts a statement about the Chern character into a statement about the object by requiring $\sigma_E$ to be injective. Corollary 9.18 says that for an object whose Chern character is concentrated on the Weil line this requirement fails, and that the failure is measured by the tangent space to the family across which the class is to be transported, in every dimension and with no hypothesis beyond (25). This concerns the size and the position of the kernel. By itself it does not say that the obstruction classes are nonzero, nor that the Weil class fails to deform. For the explicit object of Proposition 18.14 the obstruction classes are computed in Proposition 18.16: they are nonzero in every direction of the family normal to the split locus, they lie in $\ker\sigma_E$, and they meet the subspace of Corollary 9.18 only in zero. The Weil class deforms along the family, by Proposition 9.14, while that object does not.

The same computation bears on a weaker criterion, which it settles only in part. Markman asks in [Mar25b, Question 11.4] whether the semiregularity theorem survives when $\sigma_E$ is required to be injective only on the image of the evaluation map $\mathrm{ev}_E$ rather than on all of $\mathrm{Ext}^2(E,E)$, and records in his Section 12 that an affirmative answer would carry the method to higher dimensions and to fields of larger degree. By Theorem 9.19 the classes of Corollary 9.18 lie in that image, so for concentrated Chern characters the weakened hypothesis fails as well, in every dimension and by the same $n^2$ dimensions. By Theorem 9.20 it fails for every object with a line bundle summand off the polarisation line, whatever the rank and at every member of the family, and by Corollary 18.21 this excludes every sum of line bundles. For the explicit split object of Proposition 18.14 the failure is complete: Proposition 18.18 shows that every one of the fifteen dimensions of $\ker\sigma_E$ is the value of $\mathrm{ev}_E$ on a Hochschild class that already preserves $\mathrm{ch}(E)$ to first order.

The next question is whether an object of positive rank can satisfy the weakened hypothesis without satisfying the original one. Markman's own sheaves are of positive rank and are not of the concentrated shape (Remark 9.9); for them he avoids the question by descending to a quotient on which $\sigma_E$ is injective on an invariant subspace containing the image of $\mathrm{ev}_E$ [Mar25b, Proposition 11.5]. In his Section 12 he proposes a rank one complex on a principally polarised abelian fourfold, built from $(d+9)/2$ translates of the theta divisor, as a candidate for the weakened criterion in dimension eight. Theorem 18.61 shows that the candidate fails, for every discriminant, at every isolated double point of its support.

Voisin's account of the subject [Voi26] makes the same point qualitatively, for the variational Hodge conjecture in general: semiregular objects are hard to construct and too restricted to carry a general argument. Corollary 9.18 gives this observation a quantitative form for one family of classes, in which the restriction amounts to $n^2$ dimensions and those dimensions are the tangent directions of the family. Totaro's account [Tot26] describes Markman's method as the construction of enough semiregular sheaves to prove the conjecture for abelian fourfolds and fivefolds, and records that it needed the semiregularity theorem for twisted sheaves, which Pridham supplied [Pri24].

## 10. The hyperplane obstruction

This section sets the constructions on abelian varieties aside and turns to the other classical approach: induction on the dimension of the variety through hyperplane sections. One places the variety in a family of hyperplane sections, uses the algebraicity of the Hodge locus to control where a class stays of type $(p,p)$, appeals to the conjecture in one dimension lower, and transports the resulting cycles back up. Each ingredient of such an argument is a theorem: the Lefschetz hyperplane theorem, hard Lefschetz, the algebraicity of the Hodge locus [CDK95], and the properness of relative Hilbert schemes. The question is whether they compose into an induction. They do not, for an elementary reason that applies to every argument of this shape, however the family of hypersurfaces is chosen.

Throughout, $X$ is a smooth complex projective variety of dimension $d$, $j\colon Y \hookrightarrow X$ is a smooth hypersurface in the linear system of an ample divisor, and $L = c_1(\mathcal{O}_X(Y)) \in H^2(X,\mathbb{Q})$.

### 10.1. Two classical inputs.

**Theorem 10.1** (Lefschetz hyperplane theorem)**.** *The restriction $j^*\colon H^k(X,\mathbb{Q}) \to H^k(Y,\mathbb{Q})$ is an isomorphism for $k \le d-2$ and is injective for $k = d-1$.*

This is the topological form of the theorem. The standard proof, due to Andreotti and Frankel, is by Morse theory on the smooth affine variety $X \smallsetminus Y$; see [Voi03, Chapter 1].

**Lemma 10.2** (Gysin and restriction)**.** *For every $k$ and every $x \in H^k(X,\mathbb{Q})$,*

$$j_*j^*x = x \cup L.$$

*Moreover $j_*\colon H^k(Y,\mathbb{Q}) \to H^{k+2}(X,\mathbb{Q})$ is injective for $k \ge d$.*

*Proof.* The first identity is the projection formula together with $j_*1 = [Y] = L$. For the second, Poincaré duality on $X$ and on $Y$ identifies $j_*\colon H^k(Y) \to H^{k+2}(X)$ with the dual of

$$j^*\colon H^{2d-k-2}(X) \longrightarrow H^{2(d-1)-k}(Y) = H^{2d-k-2}(Y),$$

so $j_*$ is injective on $H^k(Y)$ if and only if this $j^*$ is surjective. By Theorem 10.1 that $j^*$ is surjective whenever $2d-k-2 \le d-2$, that is whenever $k \ge d$. ∎

### 10.2. The middle degree.

**Theorem 10.3** (Hyperplane obstruction)**.** *Let $d = 2p$ and let $\alpha \in H^{2p}(X,\mathbb{Q})$ be primitive. Then $j^*\alpha = 0$ for every smooth ample hypersurface $j\colon Y \hookrightarrow X$.*

The primitive middle degree is where the Hodge conjecture is hard, and it is where the restriction map returns nothing. Figure 11 draws the Lefschetz decomposition of an abelian fourfold and the part of it that restriction annihilates. The proof takes three lines.

*Proof of Theorem 10.3.* Let $d = 2p$ and let $\alpha \in H^{2p}(X,\mathbb{Q})$ be primitive. A class of degree $k \le d$ is primitive when $L^{d-k+1}$ annihilates it, so in the middle degree primitivity means $L\alpha = 0$, that is, $H^d_{\mathrm{prim}}(X,\mathbb{Q}) = \ker\bigl(L\colon H^d(X,\mathbb{Q}) \to H^{d+2}(X,\mathbb{Q})\bigr)$. By Lemma 10.2,

$$j_*\bigl(j^*\alpha\bigr) = \alpha \cup L = 0.$$

The class $j^*\alpha$ lies in $H^d(Y,\mathbb{Q})$, on which $j_*$ is injective by the case $k = d$ of Lemma 10.2. Hence $j^*\alpha = 0$. ∎

**Corollary 10.4** (Blindness in the middle degree)**.** *Let $d = 2p$ and let $\alpha, \beta \in H^{2p}(X, \mathbb{Q})$ be primitive Hodge classes. Then $j^*\alpha = j^*\beta = 0$ for every smooth ample hypersurface $Y$. In particular no property of the restricted classes distinguishes $\alpha$ from $\beta$, and no statement about Hodge classes on hypersurfaces of $X$, however strong, carries any information about the primitive middle cohomology of $X$.*

*Remark* 10.5 (The size of what is lost)*.* The primitive middle cohomology is not a small correction term. When $X$ is a smooth hypersurface of degree $m$ in $\mathbb{P}^{d+1}$, the dimension of $H^d_{\mathrm{prim}}(X, \mathbb{Q})$ grows like $m^{d+1}$; and when $d = 2p$, the Hodge conjecture for $X$ in degree $2p$ is the statement that the Hodge classes in $H^d_{\mathrm{prim}}(X, \mathbb{Q})$ are algebraic, since the remaining Hodge classes of degree $2p$ are the multiples of $L^p$. By Corollary 10.4, all of this cohomology is invisible to hyperplane restriction.

10.3. **Below the middle degree.** Below the middle degree the restriction map is injective, and in degrees at most $d-2$ it is an isomorphism, so a class can be transported downwards. The difficulty moves to the return trip.

**Definition 10.6** (Lefschetz standard conjecture)**.** For a smooth projective variety $X$ of dimension $d$ with ample class $L$, hard Lefschetz gives isomorphisms $L^{d-k}\colon H^k(X, \mathbb{Q}) \to H^{2d-k}(X, \mathbb{Q})$, and makes $H^*(X, \mathbb{Q})$ a representation of $\mathfrak{sl}_2$ in which $L$ raises the degree by 2 and $H$ acts on $H^k$ by $k - d$. Let $\Lambda$ denote the unique endomorphism of degree $-2$ with $[L, \Lambda] = H$. The Lefschetz standard conjecture $B(X)$ asserts that $\Lambda$ is induced by an algebraic cycle on $X \times X$; this is equivalent to Kleiman's formulations [Kle68, Kle94].

**Lemma 10.7** ($\Lambda$ undoes $L$ on primitive classes)**.** *Let $\alpha \in H^k(X, \mathbb{Q})$ be primitive with $k \le d$. Then $\Lambda L\alpha = (d - k)\,\alpha$.*

*Proof.* This follows from the relation $[L, \Lambda] = H$, where $H$ acts on $H^k$ by the scalar $k - d$, and from $\Lambda\alpha = 0$, which holds because the primitive classes of degree at most $d$ are the lowest weight vectors of the $\mathfrak{sl}_2$-representation. Explicitly, $\Lambda L\alpha = L\Lambda\alpha - [L, \Lambda]\alpha = 0 - (k - d)\alpha = (d - k)\alpha$. ∎

**Theorem 10.8** (The return trip)**.** *Let $2p \le d - 2$, so that $j^*\colon H^{2p}(X, \mathbb{Q}) \to H^{2p}(Y, \mathbb{Q})$ is an isomorphism. If $j^*\alpha$ is algebraic on $Y$, then $L\alpha$ is algebraic on $X$. The implication "$L\alpha$ algebraic $\Rightarrow$ $\alpha$ algebraic", asked of all smooth projective $X$ and all Hodge classes $\alpha$, is equivalent to the Lefschetz standard conjecture.*

*Proof of Theorem 10.8.* Let $2p \le d - 2$, so $j^*\colon H^{2p}(X, \mathbb{Q}) \to H^{2p}(Y, \mathbb{Q})$ is an isomorphism by Theorem 10.1. If $j^*\alpha = \mathrm{cl}(W)$ for some $W \in \mathrm{CH}^p(Y)_{\mathbb{Q}}$, then $j_*W \in \mathrm{CH}^{p+1}(X)_{\mathbb{Q}}$ and

$$\mathrm{cl}(j_*W) = j_*j^*\alpha = L\alpha$$

by Lemma 10.2 and the compatibility of the cycle class map with proper push-forward, so $L\alpha$ is algebraic.

For the second statement, suppose $B(X)$ holds and $\alpha$ is primitive of degree $2p < d$ with $L\alpha$ algebraic. Then $\Lambda$ is induced by an algebraic correspondence, hence carries algebraic classes to algebraic classes, and $\alpha = (d-2p)^{-1}\Lambda(L\alpha)$ by Lemma 10.7, so $\alpha$ is algebraic. The same holds without primitivity. Write $P^m$ for the primitive part of $H^m(X, \mathbb{Q})$. Iterating the relation $[L, \Lambda] = H$ as in the proof of Lemma 10.7 shows that $\Lambda L$ acts on the summand $L^iP^{2p-2i}$ of the Lefschetz decomposition of $H^{2p}(X, \mathbb{Q})$ by the scalar $(i + 1)(d - 2p + i)$, which is nonzero because $2p < d$. So $\Lambda L$ is invertible on $H^{2p}(X, \mathbb{Q})$, its inverse there is a polynomial in $\Lambda L$ with rational coefficients, and $\alpha = (\Lambda L)^{-1}\Lambda(L\alpha)$ is algebraic.

Conversely, suppose that for every smooth projective variety and every Hodge class $\alpha$ on it the algebraicity of $L\alpha$ implies that of $\alpha$. Applied repeatedly, this shows that on every smooth projective variety of dimension $e$ the hard Lefschetz isomorphisms $L^{e-k}\colon H^k \to H^{2e-k}$, $k \le e$, carry the algebraic classes of degree $k$ onto those of degree $2e-k$, since the preimage of an algebraic class under $L^{e-k}$ is a Hodge class. This is Kleiman's conjecture $A$ for every smooth projective variety, and $A$ for $X \times X$ implies $B(X)$ [Kle68, Kle94]. ■

Together the two theorems give the following.

**Corollary 10.9** (No dimensional induction through hyperplane sections)**.** *Let $X$ be smooth projective of dimension $d$ and let $\alpha \in \mathrm{Hdg}^p(X)$. An argument that deduces the algebraicity of $\alpha$ from the Hodge conjecture for smooth ample hypersurfaces of $X$, using only $j^*$ and $j_*$, returns no information when $2p = d$ and $\alpha$ is primitive, and otherwise establishes the algebraicity of $L\alpha$ rather than of $\alpha$.*

*Proof of Corollary 10.9.* Suppose the argument uses only $j^*$ and $j_*$ to relate classes on $X$ to classes on smooth ample hypersurfaces $Y \subset X$. First let $2p = d$ with $\alpha$ primitive. Then $j^*\alpha = 0$ by Theorem 10.3. A cycle on $Y$ of codimension $p-1$ has its class in $H^{d-2}(Y, \mathbb{Q})$, which is $j^*H^{d-2}(X, \mathbb{Q})$ by Theorem 10.1, so by Lemma 10.2 its image under $j_*$ lies in $L \cdot H^{d-2}(X, \mathbb{Q})$. By the Lefschetz decomposition this subspace meets the primitive part of $H^d(X, \mathbb{Q})$ only in 0, so $j_*$ produces $\alpha$ only when $\alpha = 0$, and no conclusion about $\alpha$ is available. In the remaining cases the best that $j_*j^*$ produces is $L\alpha$, and Theorem 10.8 identifies the passage from $L\alpha$ to $\alpha$ with the Lefschetz standard conjecture. ■

10.4. **Two gaps in inductive arguments.** In practice an induction of this shape breaks at one of two steps, each of which looks harmless.

*Remark* 10.10 (The dimension bookkeeping). A common formulation places $X$ in a Lefschetz pencil by choosing a generic pencil $\{X_t\}_{t \in \mathbb{P}^1}$ of hyperplane sections *of* $X$ and then setting $X = X_{t_0}$ for some $t_0$. The two requirements are incompatible: the members of a pencil of hyperplane sections of $X$ have dimension $d-1$, so none of them is $X$. An induction that reads $\dim X_t = d-1 < d$ in its inductive step, and $X_{t_0} = X$ when it transports the resulting cycle back, is using both readings of the same symbol.

Neither way of resolving the ambiguity rescues the argument. If $X$ is required to be a member of the pencil, then the pencil lives in an ambient variety of dimension $d+1$ and every member has dimension $d$; the inductive parameter does not decrease and the induction is circular. If instead $\dim X_t = d-1$, then the class $\alpha$ lives on $X$ and not on the members, and the only available transports are $j^*$ and $j_*$. This is the situation of Theorems 10.3 and 10.8, and Corollary 10.9 applies.

*Remark* 10.11 (Pointwise existence is not a family). The second gap appears in arguments that produce cycles fibrewise and then assemble them. Suppose an inductive hypothesis furnishes, for each $t$ in the base $S$ of a family, a cycle $Z_t$ with a prescribed class. It does not follow that the Hilbert polynomial $t \mapsto P_{Z_t}$ is constructible, or semicontinuous, or takes finitely many values. Those are theorems about a flat family, and the input here is a bare existence statement at each point with no coherence between points. Without constructibility there is no section of a relative Hilbert scheme over a dense open subset, and therefore nothing for the valuative criterion of properness to extend. The step looks safe because the relative Hilbert scheme of each fixed Hilbert polynomial is proper over $S$; but properness extends a section that exists over a punctured curve, and supplies no section where none is given.

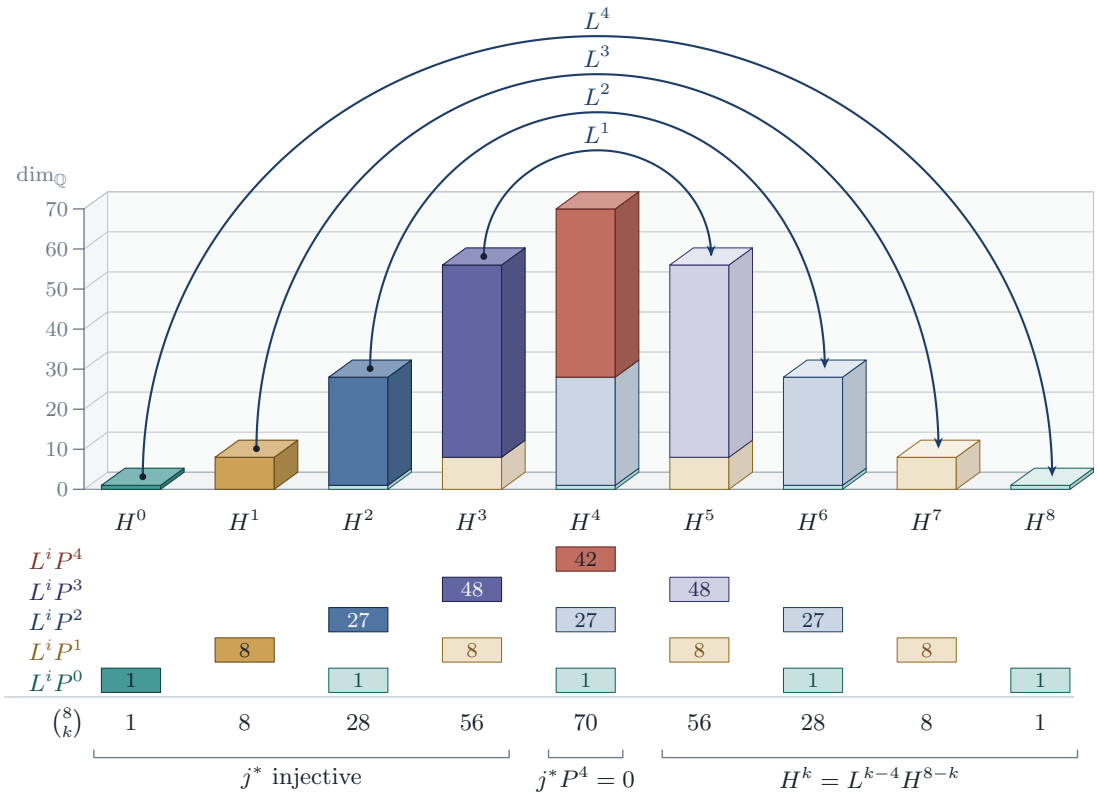



FIGURE 11. The Betti numbers $\binom{8}{k}$ of an abelian fourfold, drawn as prisms and divided according to the Lefschetz decomposition $H^k = \bigoplus_i L^i P^{k-2i}$: the primitive part $P^k$ is drawn dark, and each image $L^i P^{k-2i}$ is drawn light in the colour of $P^{k-2i}$. The arcs are the hard Lefschetz isomorphisms $L^{4-k}\colon H^k \to H^{8-k}$. Below the middle degree the restriction $j^*$ to a smooth ample hypersurface is injective and a class descends, but the return trip $j_*j^*$ multiplies by $L$, and recovering the class is the Lefschetz standard conjecture; above the middle degree a class is carried to the lower half by an arc. The primitive middle cohomology $P^4$, of dimension $70 - 28 = 42$, is annihilated by $j^*$ (Theorem 10.3), and neither an arc nor a restriction reaches it.

## 11. The Kuga–Satake construction

Theorem 10.3 blocks induction on dimension through hyperplane sections in the primitive middle degree. One classical construction does reach a middle degree Hodge class without any such induction, and we set it out in full, both because it shows what a successful route looks like and because its boundary is numerical and can be located. It is the construction of Kuga and Satake [KS67], which builds on Satake's Clifford algebra families and their Riemann forms [Sat66]. We follow the form given by Deligne [Del72], to whom the Hodge-theoretic properties that make it useful are due, in particular the embedding of Proposition 11.7. After building it we show that Theorem 10.3 has no bearing on it. For abelian varieties of Weil type with Hodge group $\mathrm{SU}(V,H)$ we then determine when it applies to a piece of $H^2$: exactly when $n=1$, where the Weil line consists of divisor classes and there is nothing to prove, or $n=2$ with trivial discriminant, where the Weil line lies in $H^4$.

### 11.1. The construction.

**Setup 11.1** (K3 type)**.** Let $T$ be a $\mathbb{Q}$-vector space of dimension $b$ with a nondegenerate symmetric form $q$, and let $T$ carry a Hodge structure of weight two, $T_{\mathbb{C}} = T^{2,0} \oplus T^{1,1} \oplus T^{0,2}$, polarised by $q$, with

$$\dim_{\mathbb{C}} T^{2,0} = 1.$$

Such a structure is said to be *of K3 type*. Write $T^{2,0} = \mathbb{C}\sigma$. The Hodge–Riemann relations give $q(\sigma,\sigma) = 0$ and $q(\sigma,\overline{\sigma}) > 0$, and make $q$ negative definite on the real points of $T^{1,1}$; so the real plane $P = \langle \Re\sigma, \Im\sigma \rangle \subset T_{\mathbb{R}}$ is positive definite and $q$ has signature $(2, b-2)$ on $T_{\mathbb{R}}$. Since $q(\Re\sigma, \Im\sigma) = 0$ and $q(\Re\sigma) = q(\Im\sigma)$, we may normalise $\sigma$ so that $e_1 = \Re\sigma$ and $e_2 = \Im\sigma$ are orthonormal.

**Definition 11.2** (Clifford algebra)**.** $C(T,q)$ is the quotient of the tensor algebra on $T$ by the ideal generated by $v \otimes v - q(v)$ for $v \in T$. It is $\mathbb{Z}/2$-graded, $C = C^+ \oplus C^-$, by the parity of the word length.

**Lemma 11.3.** *For an orthogonal basis $f_1, \dots, f_b$ of $T$ the products $f_S$ over subsets $S \subseteq \{1,\dots,b\}$, taken in increasing order, form a basis of $C(T,q)$. Hence $\dim_{\mathbb{Q}} C = 2^b$ and $\dim_{\mathbb{Q}} C^{\pm} = 2^{b-1}$, and $C^- \cdot C^+ \cdot C^- \subseteq C^+$.*

*Proof.* The relations $f_i^2 = q(f_i)$ and $f_i f_j = -f_j f_i$ for $i \neq j$, the latter obtained by expanding $(f_i + f_j)^2 = q(f_i) + q(f_j)$, let one rewrite any word in the $f_i$ as a scalar times an increasing product, so the $f_S$ span; they are independent by the standard filtration argument comparing $C$ with $\bigwedge^* T$. The dimension counts are $\sum_k \binom{b}{k} = 2^b$ and $\sum_{k \text{ even}} \binom{b}{k} = 2^{b-1}$. The last inclusion holds because the grading is multiplicative and odd plus even plus odd is even. ∎

**Lemma 11.4** (The complex structure)**.** *With $e_1, e_2$ as in Setup 11.1, put $J = e_1 e_2 \in C^+ \otimes_{\mathbb{Q}} \mathbb{R}$. Then $J^2 = -1$, and left multiplication by $J$ is an $\mathbb{R}$-linear operator on $C^+ \otimes_{\mathbb{Q}} \mathbb{R}$ whose square is $-\mathrm{id}$.*

*Proof.* $J^2 = e_1e_2e_1e_2 = -e_1e_1e_2e_2 = -q(e_1)q(e_2) = -1$, since the orthogonal vectors $e_1$ and $e_2$ anticommute and $e_i^2 = q(e_i) = 1$. Left multiplication by an element of $C^+$ preserves $C^+$, and the square of left multiplication by $J$ is left multiplication by $J^2 = -1$. ∎

Figure 12 draws the positive plane $P$, the null cone of $q$ and the period domain in the smallest case $b = 3$.

**Definition 11.5** (Kuga–Satake variety)**.** Fix a lattice $C^+_{\mathbb{Z}} \subset C^+$ stable under multiplication. The complex torus

$$\mathrm{KS}(T) = \big(C^+ \otimes_{\mathbb{Q}} \mathbb{R},\, J\big) / \, C^+_{\mathbb{Z}}$$

has dimension $\frac{1}{2}\dim_{\mathbb{Q}} C^+ = 2^{b-2}$. Satake's construction of a Riemann form on $C^+$ compatible with $J$ [Sat66] makes it an abelian variety, and by construction

$$H^1(\mathrm{KS}(T), \mathbb{Q}) = C^+(T)$$

as a Hodge structure of weight one, the Hodge decomposition being the eigenspace decomposition of left multiplication by $J$.

**Example 11.6.** For a projective K3 surface, $T$ is the transcendental lattice inside $H^2(X, \mathbb{Q})$; taking all of $H^2$ with $b = 22$ gives $\dim \mathrm{KS} = 2^{20} = 1{,}048{,}576$. At the other end, $b = 3$ gives $\dim \mathrm{KS} = 2$, an abelian surface, and $b = 6$ gives $\dim \mathrm{KS} = 16$; these are the two cases that occur below, for $n = 1$ and for $n = 2$.

### 11.2. **The transport of Hodge classes.**

**Proposition 11.7** (The embedding)**.** *Fix $v_0 \in T$ with $q(v_0) \neq 0$. The map*

$$\Psi : T \longrightarrow \mathrm{End}_{\mathbb{Q}}(C^+), \qquad \Psi(v)(x) = v\, x\, v_0,$$

*is injective and is a morphism of Hodge structures $T(1) \to \mathrm{End}\big(H^1(\mathrm{KS}(T), \mathbb{Q})\big)$, both sides having weight zero. Consequently a Hodge class in $T$, that is a rational class of type $(1,1)$, is carried to a Hodge class of type $(0,0)$ in $\mathrm{End}(H^1(\mathrm{KS}(T), \mathbb{Q}))$, that is to an element of $\mathrm{End}^0(\mathrm{KS}(T))$.*

*Proof.* The map lands in $\mathrm{End}(C^+)$ by the last part of Lemma 11.3, since $v, v_0 \in T \subset C^-$. It is injective because $\Psi(v)(1) = v\, v_0$ and $v_0$ is invertible in $C(T, q)$, with $v_0^{-1} = v_0/q(v_0)$, so $\Psi(v) = 0$ forces $v = 0$. That it respects the Hodge structures is Deligne's computation [Del72, §4]: the Hodge structure on $\mathrm{End}(C^+)$ is induced by $J$ acting on both sides, and $\Psi$ intertwines the weight-two structure on $T$, twisted once, with it. The last sentence follows from the definition of a morphism of Hodge structures and the Lefschetz theorem on $(1,1)$ classes, in the form which says that for an abelian variety $B$ a Hodge class of type $(0,0)$ in $\mathrm{End}(H^1(B, \mathbb{Q}))$ is an element of $\mathrm{End}^0(B)$. ∎

*Remark* 11.8 (Algebraicity of the transport). An element of $\mathrm{End}^0(\mathrm{KS}(T))$ is a rational multiple of an endomorphism, whose graph is an algebraic cycle on $\mathrm{KS}(T) \times \mathrm{KS}(T)$, so the image of a Hodge class under $\Psi$ is algebraic. Carrying that conclusion back to the variety $X$ from which $T$ came requires the correspondence $\Psi$ itself to be induced by an algebraic cycle on $X \times \mathrm{KS}(T) \times \mathrm{KS}(T)$. What is proved in general is that $\Psi$ is absolute Hodge, by Deligne's theorem for abelian varieties together with the analysis of the Kuga–Satake correspondence, and that it is motivated in André's sense [And96a]; van Geemen [vG00] surveys the cases in which algebraicity has been established. The Kuga–Satake construction therefore converts the Hodge conjecture for $T$ into the algebraicity of one specific correspondence.

### 11.3. **Independence from the hyperplane obstruction.**

**Proposition 11.9.** *Theorems 10.3 and 10.8 do not constrain the Kuga–Satake route.*

*Proof.* Both theorems are statements about the pair of maps $j^*$ and $j_*$ attached to the inclusion $j : Y \hookrightarrow X$ of a smooth member of an ample linear system, and both proofs use two properties of that pair which a general correspondence does not have: the Lefschetz hyperplane theorem, which makes $j^*$ an isomorphism in degrees at most $d-2$ and, through Lemma 10.2, forces the vanishing of Theorem 10.3 in the primitive middle degree; and the

identity $j_*j^* = \cdot\cup L$ of Lemma 10.2, which makes the return trip multiply by the polarisation instead of recovering the class. The Kuga–Satake correspondence is a cohomology class on a triple product of varieties of unrelated dimensions, $\dim \mathrm{KS}(T) = 2^{b-2}$ against $\dim X$, and it has neither property; in particular the composite of $\Psi$ with its transpose is a nonzero multiple of the identity of $T$, not a power of a polarisation of $X$. So neither of the two properties on which Theorem 10.3 rests holds for $\Psi$. ∎

Proposition 11.9 shows what an escape from Theorem 10.3 must look like: a correspondence whose composite with its transpose is not the Lefschetz operator. Section 13 makes the analogous identification for Theorems 5.6 and 7.3.

11.4. **Abelian varieties of Weil type.** The limitation of the Kuga–Satake construction is the hypothesis $\dim T^{2,0} = 1$. For abelian varieties of Weil type the rational sub-Hodge structures of $H^2$, and hence the possible values of $\dim T^{2,0}$, can be listed completely.

**Proposition 11.10** (The pieces of $H^2$)**.** *Let $(A, \iota, \eta)$ be of $(K, -1, n)$-Weil type. In the character grading of Proposition 5.1,*

$$H^2(A, \mathbb{C}) = \textstyle\bigwedge^2 V_+ \;\oplus\; (V_+ \otimes V_-) \;\oplus\; \bigwedge^2 V_-, \tag{27}$$

*the middle summand is the isotypic component of the norm character and is defined over $\mathbb{Q}$, and the outer two are interchanged by conjugation so their sum is defined over $\mathbb{Q}$. The Hodge numbers are*

$$(h^{2,0}, h^{1,1}, h^{0,2}) = \begin{cases} \left(\binom{n}{2},\, n^2,\, \binom{n}{2}\right) & \text{on } \bigwedge^2 V_\pm, \\ (n^2,\, 2n^2,\, n^2) & \text{on } V_+ \otimes V_-, \end{cases}$$

*and the total is $2\binom{2n}{2} + 4n^2 = \binom{4n}{2} = \dim H^2(A, \mathbb{Q})$.*

*Proof.* The splitting is (16) at $k = 2$. The characters are $\tau^2$, $\tau\overline{\tau} = \mathrm{N}(\tau)$ and $\overline{\tau}^2$, so the middle summand is the norm-isotypic component, and it is self-conjugate, hence defined over $\mathbb{Q}$; the outer two are exchanged. For the Hodge numbers use (3) and Lemma 3.5: $V_\pm^{1,0}$ and $V_\pm^{0,1}$ have dimension $n$ each, so $(\bigwedge^2 V_+)^{2,0} = \bigwedge^2 V_+^{1,0}$ has dimension $\binom{n}{2}$ and $(\bigwedge^2 V_+)^{1,1} = V_+^{1,0} \otimes V_+^{0,1}$ has dimension $n^2$, while $(V_+ \otimes V_-)^{2,0} = V_+^{1,0} \otimes V_-^{1,0}$ has dimension $n^2$ and the $(1,1)$ part has dimension $2n^2$. The total is $2(2\binom{n}{2} + n^2) + 4n^2 = 2(2n^2 - n) + 4n^2 = 8n^2 - 2n = \binom{4n}{2}$. ∎

**Proposition 11.11** (Sub-Hodge structures of K3 type)**.** *Let $(A, \iota, \eta)$ be of $(K, -1, n)$-Weil type with $\mathrm{Hg}(A) = \mathrm{SU}(V, H)$. Then $H^2(A, \mathbb{Q})$ is the direct sum of the three rational sub-Hodge structures*

$$\mathbb{Q}\,\eta, \qquad (V_+ \otimes V_-)_{\mathbb{Q}} \ominus \mathbb{Q}\eta, \qquad R = \left(\textstyle\bigwedge^2 V_+ \oplus \bigwedge^2 V_-\right)_{\mathbb{Q}},$$

*with $h^{2,0}$ equal to $0$, $n^2$ and $n(n-1)$ respectively. The first two are irreducible, and so is $R$ when $n \geq 3$. When $n = 2$, the structure $R$ is irreducible if the discriminant of $(A, \iota, \eta)$ in the sense of Definition 15.22 is nontrivial, and it is the direct sum of two isomorphic sub-Hodge structures of K3 type, each of dimension six, if the discriminant is trivial. Consequently $H^2(A, \mathbb{Q})$ contains a rational sub-Hodge structure of K3 type if and only if either $n = 1$, or $n = 2$ and the discriminant is trivial.*

*Proof.* A rational sub-Hodge structure $W \subseteq H^2(A, \mathbb{Q})$ is stable under $\mathrm{Hg}(A)$: the orthogonal projector onto $W$ for the polarisation is a Hodge class in $\mathrm{End}(H^2)$, hence is fixed by $\mathrm{Hg}(A)$, hence $W$ is a $\mathrm{Hg}(A)$-submodule. Conversely a $\mathrm{Hg}(A)$-submodule is a sub-Hodge

structure, because the Hodge decomposition is the eigenspace decomposition of an element of $\mathrm{Hg}(A)(\mathbb{R})$. So the rational sub-Hodge structures are the $\mathbb{Q}$-subrepresentations of $\mathrm{SU}(V,H)$. For the same reason the Hodge numbers of a sub-Hodge structure depend only on its complexification as a representation of $\mathrm{Hg}(A)_{\mathbb{C}}$.

After complexification $\mathrm{SU}(V,H)$ becomes $\mathrm{SL}(V_+)$, acting on $V_+$ tautologically and on $V_- \cong V_+^\vee$ dually. In (27) the outer summands $\bigwedge^2 V_+$ and $\bigwedge^2 V_- \cong \bigwedge^2 V_+^\vee$ are irreducible, and the middle summand is $V_+ \otimes V_+^\vee = \mathrm{End}(V_+) = \mathfrak{sl}(V_+) \oplus \mathbb{C}$, the sum of the adjoint representation and the trivial one. Conjugation fixes the two pieces of the middle summand and exchanges the outer two, so the three listed subspaces are rational sub-Hodge structures, and the first two, having irreducible complexifications, are irreducible. The trivial summand is spanned by $\eta$ by Lemma 5.2, and has $h^{2,0} = 0$ since $\eta$ is of type $(1,1)$. The adjoint summand has $h^{2,0} = n^2$, since removing the one-dimensional trivial piece from $V_+ \otimes V_-$ removes a $(1,1)$ class only. The structure $R$ has $h^{2,0} = 2\binom{n}{2} = n(n-1)$ by Proposition 11.10.

The complexification of a rational subrepresentation of $R$ is a subrepresentation of $\bigwedge^2 V_+ \oplus \bigwedge^2 V_-$ stable under conjugation. For $n \geq 3$ the two summands are not isomorphic, because $\bigwedge^2 V_+^\vee \cong \bigwedge^{2n-2} V_+$ and $2n-2 \neq 2$. The subrepresentations are then 0, the two summands and the whole; conjugation exchanges the summands, so only 0 and the whole are stable under it, and $R$ is irreducible. For $n = 1$ both summands are the trivial representation, and $R$ is the plane of Hodge classes $W(A) \subset H^2(A,\mathbb{Q})$.

Let $n = 2$. Then $\bigwedge^2 V_+^\vee \cong \bigwedge^2 V_+ \otimes (\bigwedge^4 V_+)^\vee \cong \bigwedge^2 V_+$ as representations of $\mathrm{SL}(V_+)$, so $\mathrm{End}_{\mathrm{Hg}(A)}(R) \otimes \mathbb{C} \cong M_2(\mathbb{C})$ and $\mathrm{End}_{\mathrm{Hg}(A)}(R)$ is a quaternion algebra over $\mathbb{Q}$. To identify it, put $V = H^1(A,\mathbb{Q})$, a $K$-vector space of dimension four by Lemma 3.2. The natural map $\bigwedge^2_{\mathbb{Q}} V \to \bigwedge^2_K V$ is $\mathrm{SU}(V,H)$-equivariant, and its kernel has complexification $V_+ \otimes V_-$, so it restricts to an isomorphism of $R$ onto $\bigwedge^2_K V$; in particular $K \subseteq \mathrm{End}_{\mathrm{Hg}(A)}(R)$. The wedge product $B\colon \bigwedge^2_K V \times \bigwedge^2_K V \to \bigwedge^4_K V \cong K$ is a symmetric $K$-bilinear form, invariant under $\mathrm{SU}(V,H)$ because $\mathrm{SU}(V,H)$ acts trivially on $\bigwedge^4_K V$. Let $H_2$ be the hermitian form induced by $H$ on $\bigwedge^2_K V$, and let $\varphi$ be the $K$-antilinear map defined by $B(x,y) = H_2(y,\varphi(x))$. Then $\varphi$ commutes with $\mathrm{SU}(V,H)$, so $\varphi^2$ is a $K$-linear endomorphism commuting with $\mathrm{SU}(V,H)$; it is a scalar by Schur's lemma, and a rational one because it commutes with $\varphi$. Choose a $K$-basis $e_1,\dots,e_4$ of $V$ orthogonal for $H$, with $H(e_i,e_i) = h_i \in \mathbb{Q}^\times$, and identify $\bigwedge^4_K V$ with $K$ through $e_1 \wedge e_2 \wedge e_3 \wedge e_4$. Then $\varphi(e_1 \wedge e_2) = (h_3h_4)^{-1} e_3 \wedge e_4$ and $\varphi(e_3 \wedge e_4) = (h_1h_2)^{-1} e_1 \wedge e_2$, so $\varphi^2 = (\det H)^{-1}$. Hence $\mathrm{End}_{\mathrm{Hg}(A)}(R) = K \oplus K\varphi$ is the quaternion algebra $(-d, \det H)_{\mathbb{Q}}$. It is split if and only if $\det H \in \mathrm{N}_{K/\mathbb{Q}}(K^\times)$, that is, since $(-1)^2 \det H = \det H$, if and only if the discriminant is trivial. If it is not split it is a division algebra, and $R$ is irreducible. If it is split, $R = R_1 \oplus R_2$ with $R_1 \cong R_2$ of dimension six, and the complexification of each $R_i$ is isomorphic to $\bigwedge^2 V_+$ as a representation of $\mathrm{SL}(V_+)$, so by Proposition 11.10 each $R_i$ has Hodge numbers $(1,4,1)$. Each $R_i$ lies in the primitive part of $H^2(A,\mathbb{Q})$, which is the $\mathrm{Hg}(A)$-stable complement of $\mathbb{Q}\eta$, and is polarised by the restriction of the Hodge–Riemann form; so $R_i$ is of K3 type.

Finally, a rational sub-Hodge structure of $H^2(A,\mathbb{Q})$ is a direct sum of irreducible ones, each isomorphic to $\mathbb{Q}\eta$, to the adjoint structure, to $R$, or, for $n = 2$ and trivial discriminant, to $R_1$; and $h^{2,0}$ is additive. A structure of K3 type has $h^{2,0} = 1$. For $n = 1$ the adjoint structure has $h^{2,0} = 1$. For $n \geq 3$ the nonzero values $n^2$ and $n(n-1)$ both exceed 1. For $n = 2$ the values are 0, 4, and 2 for $R$ or 1 for $R_1$, so a structure of K3 type exists if and only if the discriminant is trivial. ∎

**Corollary 11.12** (Where the construction applies)**.** *Let $(A,\iota,\eta)$ be of $(K,-1,n)$-Weil type with $\mathrm{Hg}(A) = \mathrm{SU}(V,H)$. The Kuga–Satake construction applies to a rational sub-Hodge*

*structure of $H^2(A,\mathbb{Q})$ if and only if either $n=1$, or $n=2$ and the discriminant is trivial. For $n=1$ the Weil line lies in $H^2(A,\mathbb{Q})$ and consists of divisor classes, hence of algebraic classes by the Lefschetz theorem on $(1,1)$ classes, so nothing needs transporting. For $n=2$ the Weil line lies in $H^4(A,\mathbb{Q})$, outside the structures to which the construction applies.*

*Proof.* The first clause is Proposition 11.11 with Setup 11.1. For $n=1$ we have $2n=2$, so $W(A)\subseteq H^2(A,\mathbb{Q})\cap H^{1,1}(A)$, which is $\mathrm{NS}(A)\otimes\mathbb{Q}$ by the Lefschetz theorem on $(1,1)$ classes. For $n=2$ the Weil line lies in degree $2n=4$. ∎

*Remark* 11.13 (Higher weight). For $n\geq 2$ the Weil line sits in $H^{2n}(A,\mathbb{Q})$ with $2n\geq 4$, and the Clifford construction has no known counterpart in weight greater than two. It turns a quadratic form into an algebra; the polarisation of a Hodge structure of weight $w$ is symmetric only when $w$ is even, and even then there is no reason for its Clifford algebra to carry the right complex structure. One can instead apply the construction to a sub-Hodge structure of $H^2$ and try to recover the degree $2n$ class from the resulting abelian variety. Proposition 11.11 rules this out for $n\geq 3$, and for $n=2$ at nontrivial discriminant. At $n=2$ and trivial discriminant the two structures of K3 type exist, but the Hodge conjecture for abelian fourfolds of Weil type is in any case a theorem of Markman [Mar25a, Mar25b], proved at trivial discriminant also by Floccari and Fu [FF26]. What remains conceivable for $n\geq 3$ is a construction of Kuga–Satake shape for the weight $2n$ structure directly, producing an auxiliary variety on which the Weil line becomes a divisor class or an endomorphism. None is known, and Theorem 5.6 does not obstruct one: such a construction is a correspondence and need not produce a $K^\times$-eigenclass. This is the gap of Remark 20.4, seen from the other side.

## 12. Reduction to the middle degree

The two preceding sections dealt with restriction to hypersurfaces and with the one classical correspondence that avoids it. Here we combine the first of them with the Lefschetz decomposition to locate the difficulty of the Hodge conjecture in general. The result, Theorem 12.2, is an equivalence in which neither side is easier than the other. Its use is to isolate the primitive classes of middle degree as the one place where a new method is needed, and by Theorem 10.3 restriction to hyperplane sections sees nothing there.

### 12.1. **Reduction to primitive classes.**

**Proposition 12.1** (Lefschetz decomposition of Hodge classes)**.** *Let $X$ be smooth projective of dimension $d$ with ample class $L$. Every $\alpha\in\mathrm{Hdg}^p(X)$ can be written*

$$\alpha=\sum_{r\geq\max(0,\,2p-d)} L^r\beta_r,\qquad \beta_r\in\mathrm{Hdg}^{p-r}(X)\cap H^{2p-2r}_{\mathrm{prim}}(X,\mathbb{Q}),$$

*and the $\beta_r$ are uniquely determined. Consequently the Hodge conjecture holds for $X$ if and only if every primitive rational Hodge class on $X$ is algebraic.*

*Proof.* The Lefschetz decomposition $H^k(X,\mathbb{Q})=\bigoplus_r L^r H^{k-2r}_{\mathrm{prim}}(X,\mathbb{Q})$ is a decomposition of rational Hodge structures, because $L$ is a rational class of type $(1,1)$ and the primitive parts are kernels of morphisms of Hodge structures. The rationality of $L$ is essential here: for a Kähler class that is not a real multiple of a rational one the primitive subspaces are in general not rational, and a rational Hodge class need not decompose into rational primitive pieces. A Hodge class $\alpha$ therefore has Hodge classes $L^r\beta_r$ as its components. By hard Lefschetz, $L^r$ is injective on $H^{2p-2r}_{\mathrm{prim}}(X,\mathbb{Q})$ for $r\geq\max(0,2p-d)$, and it is a morphism

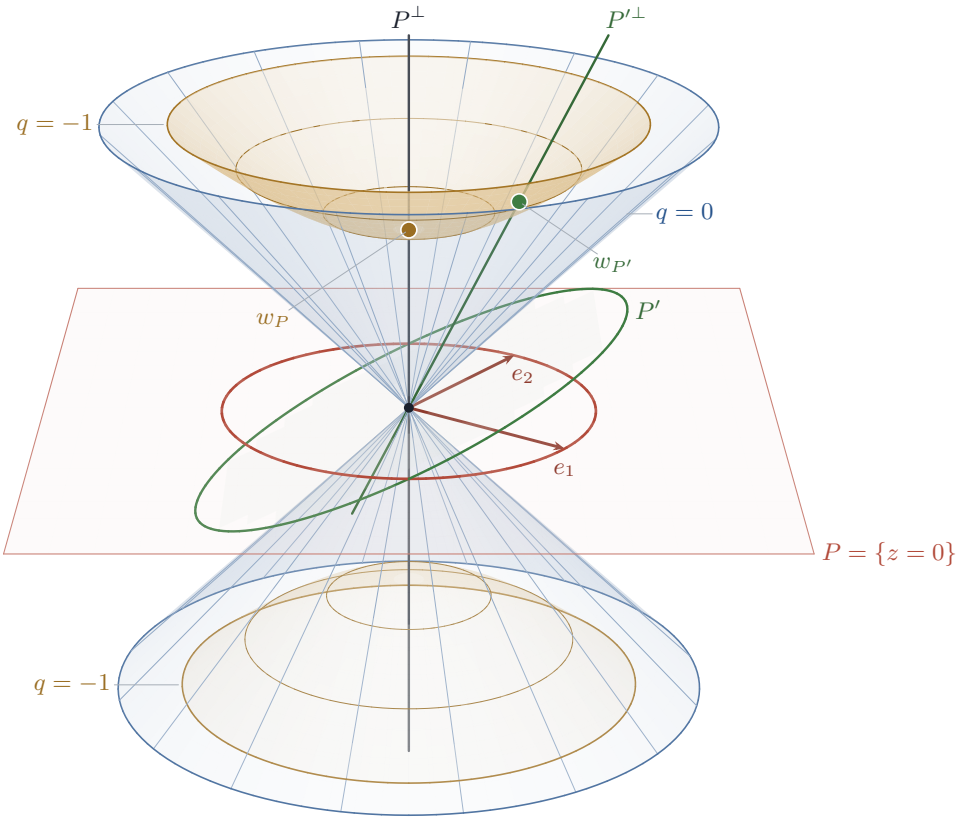



FIGURE 12. The geometry from which the Kuga–Satake construction starts, in the smallest case $b = 3$. The quadratic form is $q = x^2 + y^2 - z^2$, of signature $(2, 1)$, with its null cone $q = 0$; the horizontal plane $P = \{z = 0\}$ is positive definite and meets the cone only at the origin. The construction needs such a plane, and Setup 11.1 supplies one: $P = \langle \Re\sigma, \Im\sigma \rangle$ for the period $\sigma$ spanning $T^{2,0}$. An orthonormal basis $e_1, e_2$ of $P$ gives $J = e_1 e_2$ with $J^2 = -1$ by Lemma 11.4, and $J$ is the complex structure of $\mathrm{KS}(T)$. A positive definite plane $P$ is determined by the negative line $P^\perp$, which meets the upper sheet of the hyperboloid $q = -1$ in one point $w_P$; that sheet is the period domain, and a second plane $P'$ with its point $w_{P'}$ is drawn for comparison. The hypothesis behind all of this is $\dim T^{2,0} = 1$, and by Proposition 11.11 an abelian variety of Weil type with Hodge group $\mathrm{SU}(V, H)$ supplies it in $H^2$ only when $n = 1$, or $n = 2$ at trivial discriminant.

of Hodge structures of bidegree $(r,r)$, so each $\beta_r$ is uniquely determined and is itself a Hodge class. For the last statement, if each primitive $\beta_r$ is algebraic then so is $L^r\beta_r$, since $L$ is algebraic and algebraic classes form a subring, and hence so is $\alpha$; the converse holds because primitive Hodge classes are Hodge classes. ∎

12.2. **The equivalence.**

**Theorem 12.2.** *The following are equivalent.*

(i) *The Hodge conjecture holds for every smooth complex projective variety.*
(ii) *The Lefschetz standard conjecture $B(X)$ holds for every smooth complex projective variety $X$, and every primitive rational Hodge class of middle degree on every smooth complex projective variety of even dimension is algebraic.*

*Proof.* (i) implies (ii). The second half of (ii) is a special case of (i). For $B(X)$, note that $\Lambda$ is a morphism of Hodge structures of bidegree $(-1,-1)$, hence is given by a Hodge class on $X\times X$ through the standard correspondence between morphisms of Hodge structures and Hodge classes on the product, and (i) applied to $X\times X$ makes that class algebraic. This is Kleiman's argument [Kle68, §2].

(ii) implies (i). We argue by induction on $d=\dim X$. For $d\le 1$ the Hodge conjecture is classical. Let $d\ge 2$ and assume the conjecture for all smooth projective varieties of dimension less than $d$. By Bertini there is a smooth hypersurface $j\colon Y\hookrightarrow X$ in $|mH|$ for an ample $H$ and $m\gg 0$; put $L=c_1(\mathcal{O}_X(Y))$, and take primitivity and the operator $\Lambda$ with respect to this $L$, which is harmless because $B(X)$ does not depend on the choice of ample class [Kle68, Kle94]. By Proposition 12.1 it suffices to prove that a primitive Hodge class $\alpha\in H^{2p}(X,\mathbb{Q})$ is algebraic and, by hard Lefschetz, primitive classes occur only in degrees at most $d$, so $2p\le d$.

If $2p=d$, then $\alpha$ is a primitive Hodge class of middle degree on a variety of even dimension, so it is algebraic by the second half of (ii).

If $2p<d$, the class $j^*\alpha$ is a rational Hodge class on $Y$ because $j^*$ is a morphism of Hodge structures. As $\dim Y=d-1<d$, the inductive hypothesis makes $j^*\alpha$ algebraic, say $j^*\alpha=\mathrm{cl}(W)$ with $W\in \mathrm{CH}^p(Y)_{\mathbb{Q}}$. Then $\mathrm{cl}(j_*W)=j_*j^*\alpha=L\alpha$ by Lemma 10.2, so $L\alpha$ is algebraic. By $B(X)$ the operator $\Lambda$ carries algebraic classes to algebraic classes, and by Lemma 10.7, $\alpha=(d-2p)^{-1}\Lambda(L\alpha)$, which is therefore algebraic; the scalar is defined because $d-2p\ge 1$. ∎

*Remark* 12.3 (The content of the equivalence). $B(X)$ is itself a consequence of (i), so Theorem 12.2 does not reduce the problem to a smaller one; it locates the difficulty. The induction in the proof that (ii) implies (i) runs through every degree except the middle one, where it rests entirely on the hypothesis. By Theorem 10.3, the step it uses below the middle degree, restriction to a hypersurface, returns zero on primitive classes of middle degree, so no refinement of that step extends the induction to that degree.

**Corollary 12.4** (Primitive classes of middle degree)**.** *Any proof of the Hodge conjecture must, at some point, produce an algebraic cycle representing a primitive Hodge class of middle degree on a variety of even dimension, by a method that does not factor through restriction to hyperplane sections of that variety.*

*Proof.* By Theorem 12.2, the Hodge conjecture includes the algebraicity of the primitive Hodge classes of middle degree on varieties of even dimension, and by Corollary 10.4 every such class restricts to zero on every smooth ample hypersurface. ∎

*Remark* 12.5 (Methods that do not factor through hyperplane sections). Corollary 12.4 does not say that the conjecture is out of reach, and the settled cases come from known methods outside its scope. The Kuga–Satake construction (Section 11) replaces a weight-two Hodge structure of K3 type by the $H^1$ of an abelian variety; the class it produces is of type $(1,1)$ on a product, hence algebraic by Lefschetz, and it is not a restriction to a subvariety. Degenerations and limits of Hodge structures move the variety instead of cutting it. Correspondences from other varieties, of which the Kuga–Satake construction is one instance and Markman's construction [Mar25a, Mar25b] another, are also outside the scope. Arguments through the coniveau filtration, which show that a class supported on a proper closed subvariety is controlled by that subvariety, are compatible with everything here: they lower the dimension legitimately, because the class is supported on the smaller variety, whereas a primitive class of coniveau zero is supported on no proper closed subvariety.

## 13. Equivariance and Markman's construction

We return to abelian varieties of Weil type. By Theorem 5.6, a $K^\times$-eigenclass for the norm character has no component along the Weil line, and we ask what the contrapositive demands of a construction that does reach the Weil line. The demand does not concern the size of the object, the ambient geometry or any dimension count; it concerns the way the object transforms under $K^\times$. We state it in a concrete form and then show how Markman's construction meets it.

### 13.1. **The orbit criterion.**

**Proposition 13.1** (Weil classes generate a plane)**.** *Let $(A,\iota,\eta)$ be of $(K,-1,n)$-Weil type and let $\alpha \in W(A)$ be nonzero. Then*

$$\operatorname{span}_{\mathbb{Q}}\{\iota(\tau)^*\alpha \,:\, \tau \in K^\times\} \;=\; W(A),$$

*a two-dimensional $\mathbb{Q}$-vector space, and as a representation of $K^\times$ it is irreducible over $\mathbb{Q}$.*

*Proof.* By Proposition 3.7(i) the space $W(A)$ is free of rank one over $K$, and $\iota(\tau)^*$ acts on it as multiplication by the scalar $\tau^{2n} \in K$. Hence the span in question is $S\cdot\alpha$, where $S = \operatorname{span}_{\mathbb{Q}}\{\tau^{2n} : \tau \in K^\times\} \subseteq K$. Taking $\tau = 1$ shows $1 \in S$. By the argument in the proof of Lemma 5.4 the map $c(\tau) = \tau/\overline{\tau}$ has infinite image while $\mu(K)$ is finite, so there is $\tau_0$ with $(\tau_0/\overline{\tau_0})^{2n} \neq 1$, that is with $\tau_0^{2n} \neq \overline{\tau_0}^{2n}$, that is with $\tau_0^{2n} \notin \mathbb{Q}$. Then $S \supseteq \mathbb{Q} + \mathbb{Q}\tau_0^{2n} = K$, so $S = K$ and the span is $K\alpha = W(A)$.

For irreducibility, a proper nonzero $K^\times$-stable $\mathbb{Q}$-subspace would be a line $\mathbb{Q}\alpha'$, on which $K^\times$ would act by a $\mathbb{Q}$-valued character. Since $\iota(\tau)^*\alpha' = \tau^{2n}\alpha'$ and $W(A)$ is free over $K$, this forces $\tau^{2n} \in \mathbb{Q}$ for all $\tau$, which we have just excluded. ∎

**Corollary 13.2** (The orbit criterion)**.** *Let $\mathcal{E}$ be an object of $D^b(A)$, or $Z$ a cycle in $\mathrm{CH}^n(A)_{\mathbb{Q}}$, whose class $\mathrm{ch}_n(\mathcal{E})$, respectively $\mathrm{cl}(Z)$, is a nonzero Weil class. Then there is no function $\chi\colon K^\times \to \mathbb{Q}$ with*

$$\iota(\tau)^*\,\mathrm{ch}_n(\mathcal{E}) = \chi(\tau)\;\mathrm{ch}_n(\mathcal{E}) \qquad \text{for all } \tau \in K^\times.$$

*Equivalently, the $K^\times$-orbit of the class spans a plane and not a line.*

*Proof.* Such a $\chi$ would make $\mathbb{Q}\cdot\mathrm{ch}_n(\mathcal{E})$ a $K^\times$-stable line inside $W(A)$, contradicting Proposition 13.1. ∎

Corollary 13.2 says in what sense a construction that reaches the Weil line must break equivariance. Suppose that a construction assigns to the data $(A,\iota,\eta)$ an object $\mathcal{E}(A,\iota,\eta)$ naturally, in the sense that an isomorphism of the data induces an isomorphism of the

objects. For $\tau \neq 0$ in the order of $K$ the endomorphism $\iota(\tau)$ is an isogeny of $A$ commuting with $\iota$ and multiplying $\eta$ by $\mathrm{N}(\tau)$, and one expects naturality to give $\iota(\tau)^*\mathcal{E} \cong \mathcal{E}$ up to a normalisation imposed by the change of polarisation, whose effect on $\mathrm{ch}_n$ is multiplication by a rational number $\chi(\tau)$, a power of the norm. Whenever that is the case, Corollary 13.2 shows that $\mathrm{ch}_n(\mathcal{E})$ is not a nonzero Weil class. A construction that reaches the Weil line must therefore depend on a choice that $\mathrm{Gal}(K/\mathbb{Q})$ moves. The rest of this section describes such a choice.

13.2. **The two conjugate isotropic subspaces.** The choice that $\mathrm{Gal}(K/\mathbb{Q})$ moves is already visible in Section 5. The splitting $V_{\mathbb{C}} = V_+ \oplus V_-$ of (2) is defined over $K$ and not over $\mathbb{Q}$, since conjugation interchanges the two summands. Hence $\bigwedge^{2n} V_+$ and $\bigwedge^{2n} V_-$ are each defined over $K$, and only their sum is rational. A class in $W(A)$ is of the form $w + \overline{w}$ with $w \in \bigwedge^{2n} V_+$, and we call $\bigwedge^{2n} V_+$ and $\bigwedge^{2n} V_-$ the two conjugate halves.

**Definition 13.3** (Conjugate pair data)**.** By *conjugate pair data* for $(A, \iota, \eta)$ we mean a choice of one of the two summands $V_\pm$, equivalently of one of the two embeddings $K \hookrightarrow \mathbb{C}$, equivalently of one of the two points of a $\mathrm{Gal}(K/\mathbb{Q})$-orbit of size two attached to the $K$-structure.

**Proposition 13.4** (Corestriction produces the plane)**.** *Let $w \in \bigwedge^{2n} V_+$ be nonzero. Then $w + \overline{w}$ and $\sqrt{-d}\,(w - \overline{w})$ lie in $W(A)$ and span it over $\mathbb{Q}$. In particular, the rational classes in the Weil line are the $\mathrm{Gal}(K/\mathbb{Q})$-corestrictions of the classes in one conjugate half, and any construction producing such a class must produce, in effect, a pair of conjugate inputs and combine them.*

*Proof.* Conjugation interchanges $w$ and $\overline{w}$ and sends $\sqrt{-d}$ to $-\sqrt{-d}$, so both elements are fixed by it and are therefore rational. Both lie in $\bigwedge^{2n} V_+ \oplus \bigwedge^{2n} V_-$ by construction, hence in $W(A)$ by Definition 3.6. They are linearly independent over $\mathbb{Q}$ because $w$ and $\overline{w}$ are linearly independent over $\mathbb{C}$: a relation $q_1(w + \overline{w}) + q_2\sqrt{-d}\,(w - \overline{w}) = 0$ with $q_1, q_2 \in \mathbb{Q}$ forces $q_1 + q_2\sqrt{-d} = 0$, that is $q_1 = q_2 = 0$. Since $W(A)$ has dimension two by Proposition 3.7(i), they span it. ∎

A construction reaching the Weil line therefore has to be, in a suitable sense, bilinear in the two conjugate halves: it takes one input attached to $V_+$ and one attached to $V_-$ and pairs them. A construction fed a single $K$-invariant input cannot do this, by Corollary 13.2.

13.3. **Markman's construction.** Markman's construction [Mar25a, Mar25b] produces Weil classes, and it meets the criterion in the manner just described. We set out only the structure relevant here; the details are in [Mar25b].

The construction does not begin on $A$ or on $A^\vee$. It begins on an abelian $n$-fold $X$, that is, on a variety of half the dimension of $A$. Inside $H^{\mathrm{ev}}(X, \mathbb{Q})$ one selects a plane $P$ spanned by Hodge classes, and one requires that $P$ meet the spinor variety of even pure spinors in two points. Those two points are conjugate over the imaginary quadratic field $K$; they correspond to two maximal isotropic subspaces $W_1$ and $W_2$ that are interchanged by $\mathrm{Gal}(K/\mathbb{Q})$. This is the conjugate pair data of Definition 13.3, realised concretely.

To the two points one attaches *secant sheaves* on $X$, coherent sheaves whose Chern characters are rational points of $P$. In the worked case in which $X$ is the Jacobian of a curve of genus three, the secant sheaf is $\mathcal{I}_{\cup C_i}(\Theta)$, the twist by the theta bundle of the ideal sheaf of a union of translates of the Abel–Jacobi image of the curve. Call the two sheaves used $\mathcal{F}_1$ and $\mathcal{F}_2$.

The abelian variety of Weil type is then $A = X \times \widehat{X}$, of dimension $2n$, with its $K$-action and polarisation inherited from $P$. The object that carries the Weil class is

$$E = \Phi(\mathcal{F}_1^\vee \boxtimes \mathcal{F}_2), \tag{28}$$

where $\Phi\colon D^b(X \times X) \to D^b(X \times \widehat{X})$ is Orlov's derived equivalence. One then shows that the characteristic class $\kappa(E)$ remains of Hodge type under every deformation of $X \times \widehat{X}$ as a polarised abelian $2n$-fold of Weil type, which is what makes the class algebraic on the whole family. The conclusion in [Mar25b] is the algebraicity of the Weil classes for abelian sixfolds of Weil type of trivial discriminant in the normalisation of Definition 15.22 (discriminant $-1$ in Markman's normalisation), and, through Schoen's product argument [Sch98, Proposition 10], for all abelian fourfolds of Weil type, whence the Hodge conjecture for abelian fourfolds.

**Proposition 13.5** (Which hypotheses are escaped)**.** *The object* (28) *satisfies none of the hypotheses of Theorem 7.3 and does not satisfy the hypothesis of Theorem 7.7. Specifically:*

(i) The source is not a projective space. *The input $\mathcal{F}_1^\vee \boxtimes \mathcal{F}_2$ lives on $X \times X$, an abelian variety. The proof of Theorem 7.3 used that $H^*(\mathbb{P}^{N-1}, \mathbb{Q})$ is generated in degree two, so that* ch *of any object is a polynomial in the hyperplane class. The cohomology of $X \times X$ is not generated in degree two: it contains Künneth classes of every degree, and* $\mathrm{ch}(\mathcal{F}_1^\vee \boxtimes \mathcal{F}_2)$ *is not confined to any divisor subring.*

(ii) The transform is not $\Phi_{\mathcal{P}}$ on $A \times A^\vee$. *The relevant equivalence is between $D^b(X \times X)$ and $D^b(X \times \widehat{X})$, and Theorem 6.5, which computes the transform with Poincaré kernel on powers of the polarisation of $A$, says nothing about it.*

(iii) The construction is not $K$-equivariant. *The two tensor factors of* (28) *carry the two conjugate points of $P \cap$ (spinor variety), one each. Interchanging them is the action of the nontrivial element of* $\mathrm{Gal}(K/\mathbb{Q})$*, and it exchanges $E$ with a different object. Hence* $\mathrm{ch}_n(E)$ *is not a $K^\times$-eigenclass for the norm, and Corollary 13.2 is satisfied rather than violated.*

*Proof.* Items (i) and (ii) are a comparison of hypotheses and need no argument beyond reading (28). For (iii), if $\mathrm{ch}_n(E)$ were an eigenclass for the norm, then by Corollary 5.7 it would have zero component along the Weil line, whereas Markman shows that this component is nonzero [Mar25b]. ∎

The three hypotheses of Proposition 13.5 are related. A construction that takes its input from projective space is normally equivariant for whatever acts on the input, and an equivariant construction normally returns a single eigenclass, so the three tend to hold or fail together.

*Remark* 13.6 (Where secant geometry must be placed)*.* Secant geometry can produce Weil classes: Markman's construction is a secant construction. The obstructions bear on its placement. Secant geometry placed on the dual $A^\vee$ of the $2n$-fold inside an ambient projective space, and transported by the Poincaré kernel, has three defects at once, by Proposition 13.5: its input lies where the cohomology is generated by a divisor, it uses the transform to which Theorem 6.5 applies, and it feeds the construction $K$-invariant data, since a secant variety of a $K$-invariant subvariety is $K$-invariant. Placing the secant geometry on an abelian $n$-fold removes all three, and it does so because halving the dimension makes room for the two conjugate inputs.

13.4. **Beyond the known range.** The algebraicity of the Weil classes is known for $n = 2$, for every discriminant, and for $n = 3$ with trivial discriminant; see [Mar25b] and, for a different route at $n = 2$ and trivial discriminant, [FF26]. We state what a construction of the same shape would have to supply for $n \geq 4$.

**Question 13.7.** Fix $n \geq 4$ and an imaginary quadratic field $K$. Does there exist an abelian $n$-fold $X$, a plane $P \subset H^{\mathrm{ev}}(X, \mathbb{Q})$ spanned by Hodge classes meeting the spinor variety in two points conjugate over $K$, secant sheaves $\mathcal{F}_1, \mathcal{F}_2$ on $X$ realising those two points, and a deformation argument showing that $\kappa(\Phi(\mathcal{F}_1^\vee \boxtimes \mathcal{F}_2))$ remains of Hodge type across the moduli of polarised abelian $2n$-folds of Weil type with multiplication by $K$?

Each of the four ingredients is a concrete demand. By Corollary 13.2, any answer must be bilinear in a pair of conjugate inputs, since that is the only way a rational class can acquire a $K^\times$-orbit spanning a plane. Section 14 supplies the first three ingredients explicitly, for every $n \geq 2$ and every imaginary quadratic field, with $X$ an arbitrary polarised abelian $n$-fold. The fourth, the deformation argument, is isolated in Remark 14.16, which says exactly what it requires.

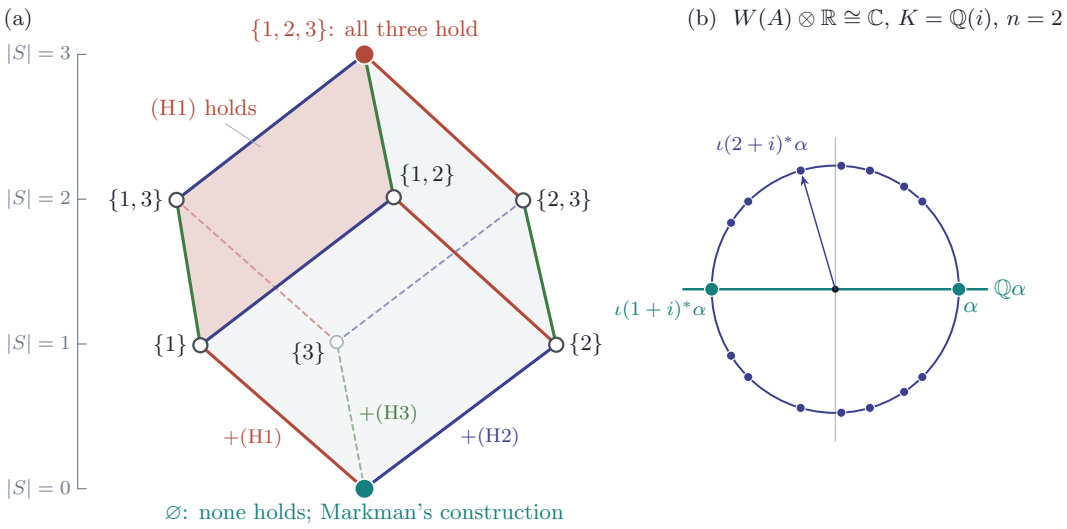



FIGURE 13. (a) The eight positions a construction can occupy, drawn as the Boolean lattice of the subsets $S$ of three hypotheses: (H1) the Chern character of the input lies in the subring generated by divisor classes; (H2) the output is a $K^\times$-eigenclass; (H3) the output spans a single line rather than a plane. (H1) and (H2) are the hypotheses negated in (i) and (iii) of Proposition 13.5, and (H3) is what Corollary 13.2 excludes for a nonzero Weil class. At the top vertex all three hold, and both Theorem 7.3 and Theorem 5.6 apply; on the shaded face (H1) holds, and Theorem 7.3 alone already applies. Markman's construction sits at the bottom vertex, where none of the three holds. (b) The orbit of a Weil class $\alpha$ under $K^\times$ for $K = \mathbb{Q}(i)$ and $n = 2$, in $W(A) \otimes \mathbb{R} \cong \mathbb{C}$ and normalised to the unit circle. Since $\iota(\tau)^*$ acts through $\tau^4$, the class $\iota(1+i)^*\alpha = -4\alpha$ stays on the line $\mathbb{Q}\alpha$, while $\iota(2+i)^*\alpha$, with $(2+i)^4 = -7 + 24i$, does not; the orbit spans the plane, as Proposition 13.1 asserts.

# Part 2. Base points and the algebraic locus

## 14. An explicit secant plane in every dimension

Question 13.7 asks for four things: an abelian $n$-fold $X$, a plane $P \subset H^{\mathrm{ev}}(X,\mathbb{Q})$ of Hodge classes meeting the spinor variety in two points conjugate over $K$, a pair of secant sheaves realising the plane, and a deformation argument. This section supplies the first three, for every $n \geq 2$ and every imaginary quadratic field $K$. The first two are written in closed form (Theorems 14.2 and 14.5). The third is first reduced to an explicit equation on a Chern character (Theorem 14.7) and then solved, in every dimension, by sheaves of positive rank whose classes are combinations of the powers of a single line bundle (Theorem 14.13). The fourth ingredient is the deformation argument, and Remark 14.16 says exactly what it requires.

The framework is Markman's [Mar25a, Mar25b]: the Weil structure on $X \times \widehat{X}$ obtained by embedding $K$ into $\mathrm{End}^0(X \times \widehat{X})$, the notion of a $K$-secant sheaf, whose Chern character lies in a plane meeting the spinor variety at two conjugate points, and the deformation step that completes the argument at $n = 3$ are all his. What we add is the explicit form of the first three ingredients, uniformly in $n$ and $K$. The closed form rests on two facts: the two conjugate isotropic subspaces demanded by Corollary 13.2 are the graphs of multiples of a single element of the Néron–Severi group, and the pure spinor of the graph of a two-form is the exponential of that form.

### 14.1. The Weil structure.

**Setup 14.1.** Let $X$ be a complex abelian variety of dimension $n$ with $H_1(X,\mathbb{Q}) = U$ of dimension $2n$ and complex structure $J$ on $U \otimes \mathbb{R}$. Let $d$ be a squarefree positive integer, $s = \sqrt{-d}$ and $K = \mathbb{Q}(s)$. Let $\beta \in \mathrm{NS}(X) \otimes \mathbb{Q}$ be nondegenerate, regarded as an alternating form on $U$, so that the Riemann relation $J^{\top}\beta J = \beta$ holds, and write $\phi_\beta \colon X \to \widehat{X}$ for the associated isogeny. Put

$$A = X \times \widehat{X}, \qquad \dim A = 2n, \qquad H_1(A,\mathbb{Q}) = U \oplus U^{\vee},$$

and let

$$M = \begin{pmatrix} 0 & -\beta^{-1} \\ d\,\beta & 0 \end{pmatrix} \in \mathrm{End}^0(A), \tag{29}$$

acting on $H_1(A,\mathbb{Q}) = U \oplus U^{\vee}$ by $(u,\phi) \mapsto (-\beta^{-1}\phi,\ d\,\beta u)$.

**Theorem 14.2** (An explicit Weil structure)**.** *In Setup 14.1:*

(i) *$M^2 = -d$, so $\iota(s) = M$ defines an embedding $\iota \colon K \hookrightarrow \mathrm{End}^0(A)$;*
(ii) *$M$ commutes with the complex structure of $A$, and for an alternating form $\beta$ on $U$ this is equivalent to the Riemann relation, that is to $\beta \in \mathrm{NS}(X) \otimes \mathbb{Q}$;*
(iii) *$M$ acts on $H^{1,0}(A)$ with eigenvalues $s$ and $-s$, each of multiplicity $n$, so $(A,\iota,\eta)$ is of $(K,-1,n)$-Weil type for a suitable polarisation $\eta$;*
(iv) *the eigenspaces $W_{\pm} \subset H_1(A,\mathbb{C})$ of $M$ for $\pm s$ are maximal isotropic for the canonical pairing on $U \oplus U^{\vee}$, are interchanged by complex conjugation, and are the graphs of the two-forms $\mp s\beta$.*

*Proof.* (i) Block multiplication gives $M^2 = \mathrm{diag}(-\beta^{-1}\,d\beta,\ -d\beta\,\beta^{-1}) = -d \cdot \mathrm{id}$.

(ii) The complex structure of $A = X \times \widehat{X}$ is $J_A = \mathrm{diag}(J, -J^{\top})$. The relation $MJ_A = J_A M$ amounts to the two identities $d\beta J = -dJ^{\top}\beta$ and $-\beta^{-1}(-J^{\top}) = J(-\beta^{-1})$, and each is equivalent to $\beta J = -J^{\top}\beta$. This is the Riemann relation: from $J^{\top}\beta J = \beta$ and

$J^2 = -1$ one gets $J^\top\beta = \beta J^{-1} = -\beta J$, as required. Conversely $\beta J = -J^\top\beta$ gives $J^\top\beta J = -\beta JJ = \beta$.

(iii) $M$ is block anti-diagonal and $H^{1,0}(A) = H^{1,0}(X) \oplus H^{1,0}(\widehat{X})$ with both summands of dimension $n$, and $M$ exchanges them. If $M(x,y) = \lambda(x,y)$ then $-\beta^{-1}y = \lambda x$ and $d\beta x = \lambda y$, so $y = d\beta x/\lambda$ and, substituting, $-dx/\lambda = \lambda x$, that is $\lambda^2 = -d$. For each of the two values of $\lambda$ the assignment $x \mapsto (x, d\beta x/\lambda)$ is an isomorphism from $H^{1,0}(X)$ onto the $\lambda$-eigenspace, which is therefore of dimension $n$. That is the balanced signature of Definition 3.1(ii). For (iii) of that definition, let $\eta_0$ be any polarisation of $A$ and put $\eta(x,y) = \eta_0(x,y) + d^{-1}\eta_0(Mx, My)$. A positive integer multiple $mM$ of $M$ is an isogeny, so the second term is the positive multiple $(dm^2)^{-1}(mM)^*\eta_0$ of a polarisation, and $\eta$ is a polarisation. Since $M^2 = -d$, $\eta(Mx,My) = \eta_0(Mx,My) + d\,\eta_0(x,y) = d\,\eta(x,y)$, and as $M^{-1} = -M/d$ this gives $\eta(Mx,y) = -\eta(x,My)$: the Rosati involution of $\eta$ sends $M$ to $-M$, that is $\iota(\tau)^\dagger = \iota(\overline{\tau})$. When $\beta$ is a polarisation an explicit choice of $\eta$ is recorded in Proposition 15.19(i).

(iv) The same computation identifies $W_\lambda = \{(x, d\beta x/\lambda)\}$, the graph of the form $d\beta/\lambda$, and $d\beta/\lambda = d\beta/(\pm s) = \mp s\beta$ since $d/s = -s$. The canonical pairing on $U \oplus U^\vee$ is $\langle (x,\xi),(x',\xi')\rangle = \xi'(x) + \xi(x')$, and on the graph it gives $d\beta(x',x)/\lambda + d\beta(x,x')/\lambda = 0$ because $\beta$ is alternating; so $W_\lambda$ is isotropic, and it has dimension $2n$, half of $\dim(U\oplus U^\vee) = 4n$, hence is maximal. Conjugation sends $s$ to $-s$ and so interchanges the two. ∎

The placement of the factor $d$ in (29) is a convention, fixed once and for all. The endomorphism $\left(\begin{smallmatrix} 0 & -d\beta^{-1} \\ \beta & 0 \end{smallmatrix}\right)$ also squares to $-d$, commutes with the complex structure and defines a Weil structure on the same $A$, but it is a different embedding of $K$: its eigenspaces are the graphs of $\mp s\beta/d$ rather than of $\mp s\beta$, its polarisation is $\beta + d\widehat{\beta}$ rather than $d\beta + \widehat{\beta}$, and the class in $\bigwedge^2 V_+ \oplus \bigwedge^2 V_-$ it produces is $\beta - d\widehat{\beta}$ rather than $\gamma = d\beta - \widehat{\beta}$. The convention of (29) is the one whose action on cohomology is $x \mapsto -Bx$, $\xi \mapsto dB^{-1}\xi$ in the proof of Theorem 15.2, and every formula below refers to it. For $d = 1$ the two conventions coincide.

*Remark* 14.3. Theorem 14.2 says that abelian $2n$-folds of Weil type of the split form $X \times \widehat{X}$ exist for every $n$, every imaginary quadratic $K$ and every polarised abelian $n$-fold $X$, and that the Weil structure is written down by one element of $\mathrm{NS}(X)$. These are not general members of the moduli of Weil type. The Weil classes on them are of interest because Markman's deformation argument, where it can be carried out, propagates algebraicity from them to the general member; see Remark 14.16.

14.2. **The secant plane.** Write $t = \beta \in \mathrm{NS}(X) \otimes \mathbb{Q}$, so that by Theorem 14.2(iv) the two conjugate maximal isotropic subspaces are the graphs of $\mp s\,t$; we use a separate letter because everything below holds for any nondegenerate $t \in \mathrm{NS}(X) \otimes \mathbb{Q}$. The pure spinor attached to the graph of a two-form $\omega$ is $e^\omega$, so the two conjugate points of the even spinor variety determined by the Weil structure are $[e^{-st}]$ and $[e^{st}]$.

**Definition 14.4** (The secant plane)**.** With $t$ and $d$ as above, put

$$u_t = \sum_{j\ge 0} \frac{(-d)^j}{(2j)!}\, t^{2j}, \qquad v_t = \sum_{j\ge 0} \frac{(-d)^j}{(2j+1)!}\, t^{2j+1}, \tag{30}$$

both sums being finite because $t^k = 0$ for $k > n$, and let $P_t = \mathbb{Q}u_t \oplus \mathbb{Q}v_t \subset H^{\mathrm{ev}}(X,\mathbb{Q})$.

**Theorem 14.5** (The plane is an explicit rational secant)**.** *With notation as above:*

(i) *$u_t$ and $v_t$ are rational Hodge classes, and $e^{\pm st} = u_t \pm s\,v_t$;*

(ii) *$P_t$ is a two-dimensional $\mathbb{Q}$-subspace of $H^{\mathrm{ev}}(X,\mathbb{Q})$ spanned by Hodge classes, and $\mathbb{P}(P_t)$ is a line defined over $\mathbb{Q}$ passing through the two conjugate points $[e^{\pm st}]$ of the even spinor variety;*

(iii) *that line is not contained in the spinor variety, so it meets it in exactly those two points.*

*Proof.* (i) Each $t^k$ is a rational Hodge class because $t \in \mathrm{NS}(X)\otimes\mathbb{Q}$ is of type $(1,1)$ and the algebraic classes form a subring. Separating $e^{st} = \sum_k s^k t^k/k!$ into even and odd powers of $s$ and using $s^{2j} = (-d)^j$, $s^{2j+1} = (-d)^j s$ gives $e^{st} = u_t + s v_t$, and conjugating gives $e^{-st} = u_t - s v_t$.

(ii) The two elements are linearly independent over $\mathbb{Q}$ because $u_t$ has nonzero component in $H^0$ and $v_t$ does not, while $v_t \neq 0$ since $t \neq 0$. From (i), both $e^{st}$ and $e^{-st}$ lie in $P_t \otimes K$, so $\mathbb{P}(P_t)$ passes through the two points.

(iii) The even spinor variety is cut out by quadrics, by Cartan's description of pure spinors, so a line either lies inside it or meets it in a subscheme of length at most two. The point $[u_t]$ of $\mathbb{P}(P_t)$ is not a pure spinor: a pure spinor with nonzero component in $H^0$ is a scalar multiple of $e^{\omega}$ for a two-form $\omega$, and $u_t$ has zero component in $H^2$ while its component in $H^4$ is $-\frac{d}{2}t^2 \neq 0$, whereas $e^{\omega}$ with vanishing $H^2$ part is 1. So the line is not contained in the spinor variety. The two points of (ii) are distinct, since $u_t + s v_t$ and $u_t - s v_t$ have the same component 1 in $H^0$ and $v_t \neq 0$; hence they exhaust the intersection. ∎

Figure 16 draws the case $n = 2$, in which the line $\mathbb{P}(P_t)$ meets the spinor quadric in the two conjugate points $[e^{\pm st}]$.

**Example 14.6.** For $n = 4$ and $d = 1$, $u_t = 1 - \frac{1}{2}t^2 + \frac{1}{24}t^4$ and $v_t = t - \frac{1}{6}t^3$. For $n = 4$ and $d = 3$, $u_t = 1 - \frac{3}{2}t^2 + \frac{3}{8}t^4$ and $v_t = t - \frac{1}{2}t^3$. All four classes are rational and lie in the subring $\mathbb{Q}[t]$.

14.3. **The Chern character equation.** The next theorem computes the invariants that a sheaf with Chern character in the secant plane must have. On the side of sheaves, the demands of Question 13.7 reduce to this one equation.

**Theorem 14.7** (The invariants of a secant sheaf)**.** *Let $X$, $t$, $d$ be as above and let $\mathcal{F}$ be a coherent sheaf on $X$ with $\mathrm{ch}(\mathcal{F}) \in P_t$, say $\mathrm{ch}(\mathcal{F}) = a\,u_t + b\,v_t$ with $a, b \in \mathbb{Q}$. Then*

$$\mathrm{rk}\,\mathcal{F} = a, \qquad c_1(\mathcal{F}) = b\,t, \qquad \mathrm{ch}_2(\mathcal{F}) = -\tfrac{ad}{2}\,t^2, \qquad c_2(\mathcal{F}) = \tfrac{ad+b^2}{2}\,t^2,$$

*and more generally $\mathrm{ch}_k(\mathcal{F})$ is the degree $k$ term of (30) scaled by $a$ or $b$ according to the parity of $k$. The Bogomolov discriminant is*

$$\Delta(\mathcal{F}) = 2a\,c_2(\mathcal{F}) - (a-1)c_1(\mathcal{F})^2 = \mathrm{N}_{K/\mathbb{Q}}(b + as)\cdot t^2. \tag{31}$$

*In particular, if $t$ is ample and $\mathcal{F} \neq 0$, then $\Delta(\mathcal{F})\cdot H^{n-2} > 0$ for every ample $H$, so the Bogomolov inequality places no obstruction on the existence of $\mathcal{F}$.*

*Proof.* The first four identities are read off from (30) degree by degree, using $\mathrm{ch}_2 = \frac{1}{2}(c_1^2 - 2c_2)$ to convert $\mathrm{ch}_2$ into $c_2$. For (31),

$$2a\cdot\tfrac{ad+b^2}{2}t^2 - (a-1)b^2t^2 = (a^2d + ab^2 - ab^2 + b^2)t^2 = (a^2d + b^2)t^2,$$

and $\mathrm{N}_{K/\mathbb{Q}}(b+as) = (b+as)(b-as) = b^2 + a^2 d$. If $\mathcal{F} \neq 0$ then $\mathrm{ch}(\mathcal{F}) \neq 0$, so $(a,b) \neq (0,0)$ and the norm is positive; and $t^2 H^{n-2} > 0$ for $t$ and $H$ ample. This gives the last assertion. ∎

*Remark* 14.8 (The norm form as the discriminant). Identity (31) says that the discriminant of the sheaf is the norm form of $K$ evaluated at the coordinates $(a, b)$ of its Chern character in the plane. The two conjugate points of the spinor variety are the two isotropic directions of this norm form after extension of scalars to $K$, and no nonzero rational point of the plane is isotropic. The Chern character of a secant sheaf is a rational point of the line $\mathbb{P}(P_t)$, and so never an isotropic one, while the Weil structure is carried by the two isotropic points of that line, which are defined over $K$. In this sense the conjugate pair of Corollary 13.2 is built into the invariants of the sheaf.

**Corollary 14.9** (The case of an abelian surface)**.** *Let $n = 2$, so $X$ is an abelian surface and $A = X \times \widehat{X}$ an abelian fourfold of Weil type, and let $t$ be ample. A nonzero sheaf with $\mathrm{ch}(\mathcal{F}) \in P_t$ has Mukai vector $v(\mathcal{F}) = (a,\, bt,\, -\frac{ad}{2}t^2)$ and Mukai self-pairing*

$$\langle v(\mathcal{F}), v(\mathcal{F})\rangle = \mathrm{N}_{K/\mathbb{Q}}(b + as) \cdot t^2 > 0.$$

*Proof.* On an abelian surface the Mukai pairing of $v = (r, c, e)$ with itself is $c^2 - 2re$, which here is $b^2t^2 - 2a(-\frac{ad}{2}t^2) = (b^2 + a^2d)t^2$. It is positive because $(a, b) \neq (0, 0)$ and $t^2 > 0$. ∎

Corollary 14.9 is consistent with the case $n = 2$ being settled: on an abelian surface, positivity of the self-pairing of a primitive Mukai vector of positive rank is sufficient for the moduli spaces of stable sheaves with that vector to be non-empty.

14.4. **The genus three Jacobian.** At $n = 3$ the equation of Theorem 14.7 determines a secant sheaf of rank one almost completely, and it reproduces the shape of the object Markman uses. The computation serves as a check on the framework of this section.

**Proposition 14.10** (Rank one at $n = 3$)**.** *Let $X$ be the Jacobian of a smooth curve of genus three, let $\Theta$ be the theta divisor and $C \subset X$ the Abel–Jacobi curve, so that $\Theta^2/2 = [C]$. Take $t = \Theta$ and $a = 1$ in Theorem 14.7. Then a rank one torsion-free sheaf $\mathcal{F}$ with $\mathrm{ch}(\mathcal{F}) \in P_\Theta$ and $c_1(\mathcal{F}) = b\Theta$ is of the form $\mathcal{F} = \mathcal{I}_Z(b\Theta)$ with*

$$[Z] = \frac{b^2 + d}{2}\,\Theta^2 = (b^2 + d)\,[C].$$

*If $Z$ is a disjoint union of $m$ translates of $C$, then $b = 1$ and*

$$m = d + 1.$$

*Proof.* A rank one torsion-free sheaf on a smooth projective variety is the tensor product of its double dual, which is a line bundle, with the ideal sheaf of a subscheme of codimension at least two. With $c_1 = b\Theta$ this gives $\mathcal{F} = \mathcal{I}_Z(b\Theta)$ for such a subscheme $Z \subset X$, and $\mathrm{ch}(\mathcal{I}_Z(b\Theta)) = e^{b\Theta}(1 - \mathrm{ch}(\mathcal{O}_Z))$. Comparing the components in $H^4$ with Theorem 14.7 at $a = 1$, which gives $\mathrm{ch}_2 = -\frac{d}{2}\Theta^2$, yields $\frac{b^2}{2}\Theta^2 - [Z] = -\frac{d}{2}\Theta^2$, that is $[Z] = \frac{b^2+d}{2}\Theta^2$. The Poincaré formula $\Theta^{g-1}/(g-1)! = [C]$ at $g = 3$ reads $\Theta^2/2 = [C]$, whence $[Z] = (b^2 + d)[C]$.

Suppose now that $Z$ is a disjoint union of $m$ translates of $C$. Translates of $C$ are homologous to $C$, so $m = b^2 + d$. Each translate is a smooth curve of genus three and $X$ has trivial tangent bundle, so Grothendieck–Riemann–Roch gives $\mathrm{ch}(\mathcal{O}_Z) = m[C] - 2m[\mathrm{pt}]$. Using $\Theta^3 = 6[\mathrm{pt}]$ and $\Theta \cdot [C] = 3[\mathrm{pt}]$, the component of $\mathrm{ch}(\mathcal{I}_Z(b\Theta))$ in $H^6$ is $(b^3 - 3bm + 2m)[\mathrm{pt}]$, and Theorem 14.7 requires it to equal the degree three term $-bd\,\Theta^3/3! = -bd\,[\mathrm{pt}]$. Substituting $m = b^2 + d$ gives $-2(b-1)(b^2+d) = 0$, so $b = 1$ and $m = d + 1$. ∎

*Remark* 14.11. The sheaf used in [Mar25b] on a genus three Jacobian is $\mathcal{I}_{\cup C_i}(\Theta)$, the twist by the theta bundle of the ideal sheaf of a union of translates of the Abel–Jacobi curve.

| $d$ | 1 | 2 | 3 | 5 | 7 | 11 |
|---|---|---|---|---|---|---|
| $m = d+1$ | 2 | 3 | 4 | 6 | 8 | 12 |

TABLE 2. The number $m = d + 1$ of translates of the Abel–Jacobi curve in Proposition 14.10, as a function of the field $K = \mathbb{Q}(\sqrt{-d})$.

This is the shape that Proposition 14.10 forces, including the twist by $\Theta$ itself, with the number of translates determined by the field.

14.5. **Existence of secant sheaves.** Theorem 14.7 describes a secant sheaf but does not produce one. We show that such sheaves exist on every polarised abelian variety, in every dimension and for every imaginary quadratic field, and that their classes can be assembled from the powers of a single line bundle.

Two observations do the work. The first is that the Chern character wanted is a rational point of a plane, and a plane has no preferred scale: a positive integer multiple $Mp$ of a rational point $p$ of $P_t$ is again a rational point of $P_t$, and since the Hodge conjecture concerns classes with rational coefficients, replacing $p$ by $Mp$ costs nothing. The second is that, once denominators may be cleared, the Chern characters of the powers of one line bundle span the subring $\mathbb{Q}[t]$ in which the secant plane lies, because the matrix expressing them in the basis $t^k/k!$ is a Vandermonde matrix on distinct nodes. What remains is the passage from a class in $K_0$ to a sheaf, and for that the only hypothesis needed is that the rank be positive.

**Lemma 14.12** (Positive rank is enough)**.** *Let $Y$ be a smooth projective variety over a field and let $\alpha \in K_0(Y)$ satisfy $\operatorname{rk}\alpha > 0$. Then $\alpha = [\mathcal{F}]$ for some coherent sheaf $\mathcal{F}$ on $Y$.*

*Proof.* Write $m = \dim Y$ and $a = \operatorname{rk}\alpha \geq 1$, and let $F^cK_0(Y)$ be the topological filtration, the subgroup generated by the classes $[\mathcal{O}_Z]$ of closed integral subschemes $Z \subseteq Y$ of codimension at least $c$. Since $Y$ is smooth, every coherent sheaf has a finite locally free resolution, so $K_0(Y) = K_0'(Y)$ and the filtration is exhaustive with $F^{m+1} = 0$. The class of a coherent sheaf supported in codimension at least $c$ lies in $F^c$ and is congruent modulo $F^{c+1}$ to $\sum_i \ell_i[\mathcal{O}_{Z_i}]$, where $\sum_i \ell_i Z_i$ is its cycle in codimension $c$; and $F^1$ is the kernel of the rank, while $F^1/F^2$ is $\operatorname{Pic}(Y)$ through the first Chern class [Ful98, §15.1 and Ex. 15.1.5]. We construct coherent sheaves $\mathcal{F}_c$ of rank $a$ with $\alpha - [\mathcal{F}_c] \in F^c$, by induction on $c$.

Start at $c = 2$. Let $L$ be a line bundle with $c_1(L) = c_1(\alpha)$ and set $\mathcal{F}_2 = L \oplus \mathcal{O}_Y^{\oplus(a-1)}$. Then $\alpha - [\mathcal{F}_2]$ has rank zero and first Chern class zero, hence lies in $F^2$.

For the inductive step let $\delta = \alpha - [\mathcal{F}_c] \in F^c$ with $c \geq 2$. Modulo $F^{c+1}$ the class $\delta$ is a finite sum $\sum_j m_j[\mathcal{O}_{Z_j}]$ with $m_j \in \mathbb{Z}$ and $\operatorname{codim} Z_j = c$. The terms with $m_j > 0$ are removed by replacing $\mathcal{F}_c$ with $\mathcal{F}_c \oplus \bigoplus_{m_j>0} \mathcal{O}_{Z_j}^{\oplus m_j}$, which changes neither the rank nor the codimension of the remaining difference. For a term with $m_j < 0$ we must subtract, and here the rank is used. The restriction $\mathcal{F}_c \otimes \mathcal{O}_{Z_j}$ is a quotient of $\mathcal{F}_c$ and has rank at least $a \geq 1$ at the generic point of the integral scheme $Z_j$, by semicontinuity of the fibre dimension. Dividing $\mathcal{F}_c \otimes \mathcal{O}_{Z_j}$ by a subsheaf of generic corank one gives a surjection $\mathcal{F}_c \twoheadrightarrow \mathcal{Q}$ with $\mathcal{Q}$ a coherent $\mathcal{O}_{Z_j}$-module of rank one. Then $[\mathcal{Q}] \in F^c$ with associated cycle $[Z_j]$, so $[\mathcal{Q}] - [\mathcal{O}_{Z_j}] \in F^{c+1}$. The kernel of that surjection is a coherent sheaf, still of rank $a$ because $\mathcal{Q}$ is supported in positive codimension, and its class is $[\mathcal{F}_c] - [\mathcal{Q}]$. Performing this $|m_j|$ times for each $j$ with $m_j < 0$ produces $\mathcal{F}_{c+1}$ with $\alpha - [\mathcal{F}_{c+1}] \in F^{c+1}$ and $\operatorname{rk}\mathcal{F}_{c+1} = a$. At $c = m + 1$ the difference lies in $F^{m+1} = 0$, so $\alpha = [\mathcal{F}_{m+1}]$. ∎

**Theorem 14.13** (Secant sheaves exist in every dimension)**.** *Let $X$ be an abelian variety of dimension $n$, let $t \in \mathrm{NS}(X) \otimes \mathbb{Q}$ be a polarisation and let $d > 0$. Let $p$ be a rational point of the secant plane $P_t$ whose $u_t$ coordinate is positive. Then there are a positive integer $M$, an integer $D > 0$ with $Dt \in \mathrm{NS}(X)$, integers $n_0, \dots, n_n$ and a coherent sheaf $\mathcal{F}$ on $X$ such that, setting*

$$\alpha = \sum_{j=0}^{n} n_j \,[\mathcal{O}_X(jDt)] \in K_0(X), \tag{32}$$

*one has $\alpha = [\mathcal{F}]$ in $K_0(X)$, $\mathrm{ch}(\mathcal{F}) = M\,p$ and $\mathrm{rk}\,\mathcal{F} = M\,a > 0$, where $a$ is the $u_t$ coordinate of $p$. The integers $M$ and $n_j$ depend only on $n$, $d$, $D$ and the coordinates of $p$, and are obtained by inverting one Vandermonde matrix. Moreover $Mp$ is again a rational point of $P_t$; applying the statement to two linearly independent rational points of $P_t$ produces a pair of coherent sheaves whose Chern characters span $P_t$.*

*Proof.* If the conclusion holds for $Np$, where $N$ is a positive integer, with multiple $M'$, then it holds for $p$ with $M = NM'$, and the hypothesis on $p$ is unchanged by the scaling. Clearing the denominators of the coordinates of $p$, we may therefore assume $p = au_t + bv_t$ with $a, b \in \mathbb{Z}$ and $a > 0$. Choose $D > 0$ with $Dt \in \mathrm{NS}(X)$, so that $\mathcal{O}_X(jDt)$ is a line bundle for every $j \in \mathbb{Z}$.

Since $t$ is a polarisation, $t^n \neq 0$ and $t^{n+1} = 0$ in $H^{\mathrm{ev}}(X, \mathbb{Q})$, so the $n+1$ classes $t^k/k!$ with $0 \le k \le n$ are a basis of the subring $\mathbb{Q}[t]$. In that basis

$$p = \sum_{k=0}^{n} c_k \,\frac{t^k}{k!}, \qquad c_k = \begin{cases} a(-d)^{k/2}, & k \text{ even}, \\ b(-d)^{(k-1)/2}, & k \text{ odd}, \end{cases}$$

by (30), and all the $c_k$ are integers. On the other side,

$$\mathrm{ch}(\mathcal{O}_X(jDt)) = e^{jDt} = \sum_{k=0}^{n} (jD)^k \,\frac{t^k}{k!},$$

so in the same basis the $n+1$ vectors $\mathrm{ch}(\mathcal{O}_X(jDt))$ for $j = 0, \dots, n$ are the columns of the Vandermonde matrix $\big((jD)^k\big)_{0 \le j,k \le n}$ on the $n+1$ distinct nodes $0, D, 2D, \dots, nD$. Its determinant is $\prod_{0 \le i < j \le n} (jD - iD) = D^{n(n+1)/2} \prod_{0 \le i < j \le n} (j - i) \neq 0$, so the system

$$\sum_{j=0}^{n} x_j \,(jD)^k = c_k \qquad (0 \le k \le n)$$

has a unique solution $x \in \mathbb{Q}^{n+1}$. Let $M$ be the least common multiple of the denominators of the $x_j$ and put $n_j = Mx_j \in \mathbb{Z}$. With $\alpha$ as in (32),

$$\mathrm{ch}(\alpha) = \sum_j n_j e^{jDt} = \sum_{k=0}^{n} \Big( \sum_j Mx_j (jD)^k \Big) \frac{t^k}{k!} = M \sum_{k=0}^{n} c_k \frac{t^k}{k!} = Mp.$$

In particular $\mathrm{rk}\,\alpha = \mathrm{ch}_0(\alpha) = Mc_0 = Ma > 0$, so Lemma 14.12 gives a coherent sheaf $\mathcal{F}$ with $[\mathcal{F}] = \alpha$. The Chern character is a function of the class in $K_0(X)$, whence $\mathrm{ch}(\mathcal{F}) = Mp$ and $\mathrm{rk}\,\mathcal{F} = Ma$. Finally $Mp = (Ma)u_t + (Mb)v_t$ is a rational point of $P_t$, and two linearly independent rational points stay linearly independent after scaling, which gives the last assertion. ∎

Figure 15 draws the theorem at $n = 2$ and $d = 2$, with the points of the secant plane that integer combinations of line bundles reach.

*Remark* 14.14 (The factorials and the multiple $M$). The integer $M$ is not an artefact of the proof. The lattice spanned by the Chern characters of the powers of a single line bundle is spanned equally well by the classes $(e^t-1)^k$ for $0 \le k \le n$, because the $(n+1)$st finite difference of a polynomial of degree $n$ vanishes; and $(e^t-1)^k = t^k + \cdots$ has leading term $t^k$, not $t^k/k!$. The secant plane, by contrast, is spanned by classes whose coordinates in the basis $t^k/k!$ are integers. The index between the two lattices can be computed exactly. Write $j^k = \sum_{i \le k} S(k,i)\, i! \binom{j}{i}$ with the Stirling numbers $S(k,i)$ of the second kind. As $j$ runs over $0, \dots, n$, the vectors $\left(\binom{j}{i}\right)_{0 \le i \le n}$ form a unitriangular basis of $\mathbb{Z}^{n+1}$, so the vectors $(j^k)_{0 \le k \le n}$, which are the coordinates of $\operatorname{ch} \mathcal{O}_X(jt)$ in the basis $t^k/k!$, span the image of $\mathbb{Z}^{n+1}$ under the lower triangular matrix $T = \left(S(k,i)\, i!\right)_{k,i}$, whose diagonal is $0!, 1!, \dots, n!$. The index is therefore $\prod_{k \le n} k!$. The inverse of $T$ is $\left(s(k,i)/k!\right)_{k,i}$, with the Stirling numbers $s(k,i)$ of the first kind, because $\binom{j}{k} = \sum_i s(k,i) j^i / k!$; so $n!\, T^{-1}$ is integral, and for every integral point $p$ of the secant plane the least admissible $M$ divides $n!$. It is often larger than 1. At $n=4$, $d=3$, $a=b=1$ the least multiple is $M=6$, a divisor of $4!$, and the witness is

$$\alpha = -23\,[\mathcal{O}_X] + 66\,[\mathcal{O}_X(t)] - 54\,[\mathcal{O}_X(2t)] + 20\,[\mathcal{O}_X(3t)] - 3\,[\mathcal{O}_X(4t)],$$

of rank $-23+66-54+20-3 = 6 = Ma$; in degree one its coefficient is $66-108+60-12 = 6 = Mb$, and in degrees two to four the sums $\sum_j n_j j^k$ are $-18$, $-18$ and $54$, which are $6c_2$, $6c_3$ and $6c_4$. The solution of the Vandermonde system is unique and the $n_j$ have no common factor, 23 being prime to 6, so $M=6$ is the least multiple, and the freedom to rescale is needed. Figure 14 shows the least multiple in further cases.

For a principal polarisation $\theta$ the classes $\theta^k/k!$ are integral: in a symplectic basis $x_1, \dots, x_g, y_1, \dots, y_g$ of $H^1(X,\mathbb{Z})$, $\theta = \sum_i x_i y_i$, the products $x_i y_i$ commute and square to zero, and so $\theta^k/k! = \sum_{|I|=k} \prod_{i \in I} x_i y_i$. This is Poincaré's formula in a form that does not need a curve, and it is why $t^k/k!$ is the natural integral basis in which to take the coordinates $c_k$.

### 14.6. **The three ingredients.**

**Theorem 14.15** (Three of the four ingredients)**.** *For every $n \ge 2$, every squarefree $d > 0$ and every polarised abelian $n$-fold $X$ carrying a nondegenerate $\beta \in \mathrm{NS}(X) \otimes \mathbb{Q}$, the first three of the four ingredients required by Question 13.7 exist and are explicit:*

- (a) *the abelian $2n$-fold of Weil type is $A = X \times \widehat{X}$ with the $K$-action (29), by Theorem 14.2;*
- (b) *the plane is $P_t$ of Definition 14.4 with $t = \beta$, it is spanned by Hodge classes and it meets the spinor variety in exactly the two conjugate points $[e^{\pm st}]$, by Theorem 14.5;*
- (c) *a pair of coherent sheaves $\mathcal{F}_1, \mathcal{F}_2$ on $X$ whose Chern characters are linearly independent rational points of $P_t$ exists, by Theorem 14.13, and each may be taken to have the class of an explicit integer combination of powers of one line bundle.*

*The invariants of any such sheaf are forced by the explicit system of Theorem 14.7, which imposes $n-1$ linear conditions on a Chern character already constrained to $\mathbb{Q}[t]$, and, when $\beta$ is ample, are unobstructed numerically by (31).*

*Proof.* Items (a), (b) and (c) are the cited theorems. For (c), the proof of Theorem 14.13 uses of $t$ only that $t^n \neq 0$ and that a positive multiple of $t$ is integral; both hold for $t = \beta$, the first because $\beta$ is nondegenerate. For the count in the last sentence, a Chern character

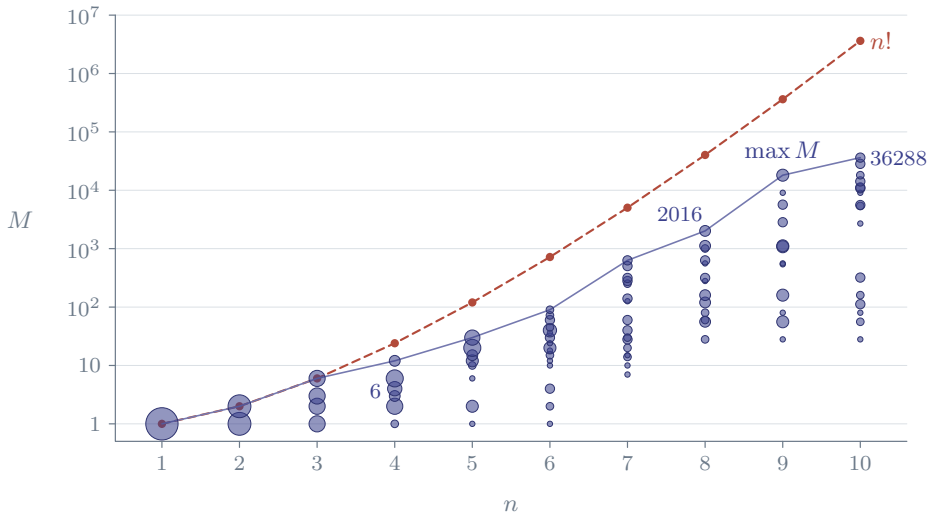



FIGURE 14. Remark 14.14. The least multiple $M$ of Theorem 14.13 for $1 \le n \le 10$, over the seventy-two cases $d \in \{1,2,3,5,7,11\}$, $a \in \{1,2,3\}$, $b \in \{\pm 1, \pm 2\}$ for each $n$, read off from the solution of the Vandermonde system, on a logarithmic scale; the area of a disc counts the cases with that value. The solid line joins the largest values and the dashed one is $n!$, which every $M$ divides by the remark: the quotient of $\mathbb{Z}^{n+1}$ by the lattice of the characters $(1, j, \dots, j^n)$ of the powers of a line bundle is dual to the integer-valued polynomials of degree at most $n$, spanned by the $\binom{x}{k}$, whose coefficients have denominators dividing $n!$.

in the subring $\mathbb{Q}[t] \subset H^{\mathrm{ev}}(X, \mathbb{Q})$ is a polynomial of degree at most $n$ in $t$, so lies in a space of dimension $n+1$, while $P_t$ has dimension two, and $(n+1) - 2 = n - 1$. ∎

*Remark* 14.16 (The deformation step). The fourth ingredient of Question 13.7 is untouched by this section, and it is the hardest. Given the pair of secant sheaves $\mathcal{F}_1, \mathcal{F}_2$ on $X$ supplied by Theorem 14.13, one must still show that the characteristic class of $\Phi(\mathcal{F}_1^\vee \boxtimes \mathcal{F}_2)$ remains of Hodge type under every deformation of $X \times \widehat{X}$ as a polarised abelian $2n$-fold of Weil type, and that the object deforms with the variety, as a twisted sheaf to first order. Markman carries out this step at $n = 3$ [Mar25b], through a semiregularity analysis. Our contribution to it is negative: Theorem 18.56 identifies the Hochschild classes that the weakened form of that analysis must kill for a secant object, and Theorem 18.61 shows that the candidate Markman proposes for $n = 4$ does not kill them. Without the step one obtains algebraicity only on the split locus $X \times \widehat{X}$, where the Weil classes are of much less interest, since the general abelian $2n$-fold of Weil type is simple and is not of that form.

Accordingly, nothing in this section proves the Hodge conjecture for abelian $2n$-folds of Weil type at any $n \ge 4$. What is established is that three of the four ingredients are written down for every $n$, the third with an explicit witness in line bundles, and that the fourth is isolated.

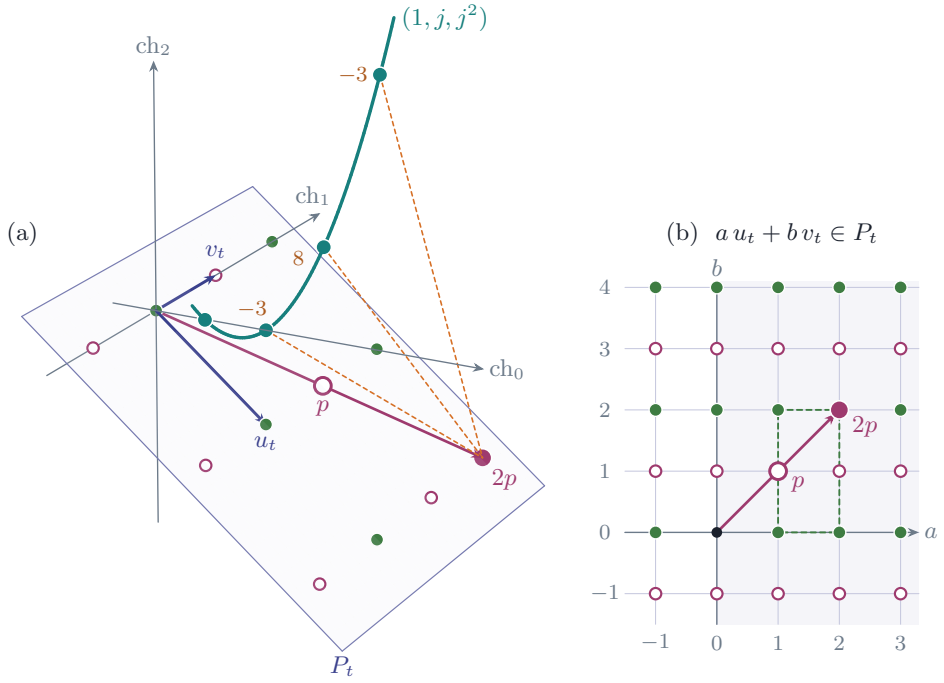



FIGURE 15. Theorem 14.13 at $n = 2$ and $d = 2$. (a) The coordinates $(\mathrm{ch}_0, \mathrm{ch}_1, \mathrm{ch}_2)$ in the basis $1$, $t$, $t^2/2$. The Chern characters of the powers of one line bundle lie on the moment curve $j \mapsto (1, j, j^2)$, and the Vandermonde determinant says that any three of its points are affinely independent. The integral points of $P_t = \mathbb{Q}u_t \oplus \mathbb{Q}v_t$ split by the criterion $\mathrm{ch}_2 \equiv \mathrm{ch}_1 \bmod 2$ into those reached by integer combinations of line bundles, drawn filled, and those missed, drawn open. The point $p = u_t + v_t$ is missed and $2p$ is reached, by the witness $-3[\mathcal{O}_X] + 8[\mathcal{O}_X(t)] - 3[\mathcal{O}_X(2t)]$, whose three terms are joined to $2p$ by dashed segments. (b) The same points in the coordinates $(a, b)$ of $a\,u_t + b\,v_t$. Here $\mathrm{ch}_1 = b$ and $\mathrm{ch}_2 = -2a$, so the criterion reads $b \equiv 0 \bmod 2$; the shaded half-plane $a > 0$ is the range of Theorem 14.13.

## 15. Base points in every Weil family

The secant construction of Section 14 lives on the split variety $X \times \widehat{X}$ and stops short of the deformation step. In this section we study the split member and the other special members of a Weil family through the Weil classes themselves. On the split member the Weil classes are polynomials in three divisor classes (Theorem 15.2); on products of abelian surfaces of Weil type they are polynomials in divisor classes (Theorem 15.10); and on abelian varieties with quaternionic multiplication extending $K$, and on their products, they are polynomials in two divisor classes, which yields an algebraic base point in every Weil family, of every discriminant (Theorem 15.27). What remains is the propagation of one flat class over one connected period domain, formulated in Sections 15.3 and 15.7.

15.1. **The split member in closed form.** We begin with the split member, where a single formula, valid for every $n$ and every $d$, writes the Weil classes as polynomials in three divisor classes. The split locus is thin. A general abelian $2n$-fold of Weil type is simple,

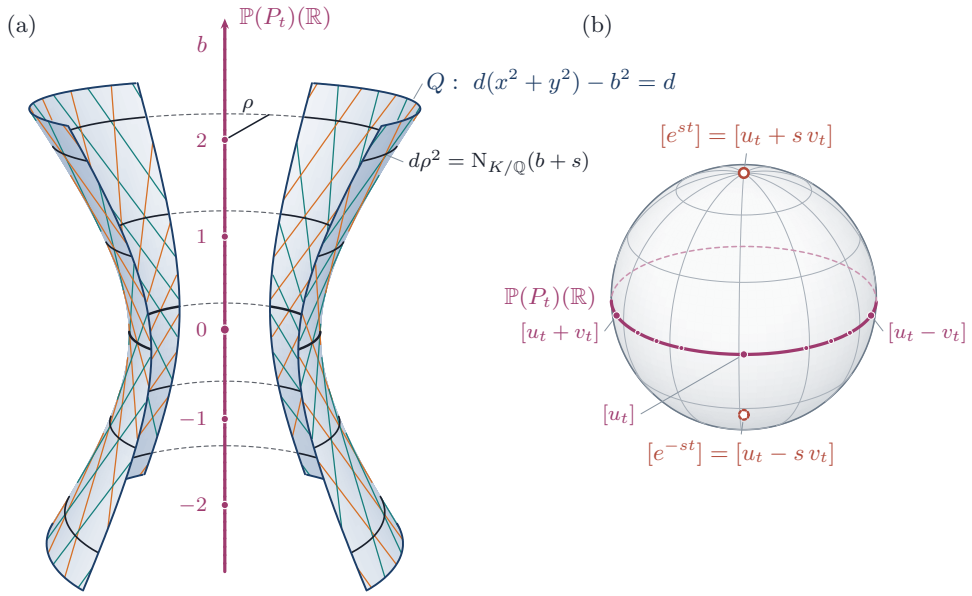



FIGURE 16. The geometry of Theorem 14.5 in the first case, $n=2$. The even spinor variety is then the smooth six-dimensional quadric of classes isotropic for the Mukai pairing in $\mathbb{P}(H^{\mathrm{ev}}(X,\mathbb{C}))\cong\mathbb{P}^7$, and the figure shows a real two-dimensional section of it, a quadric surface with its two rulings. The line $\mathbb{P}(P_t)$ of Definition 14.4 is defined over $\mathbb{Q}$ and meets the quadric only in the two points $[e^{\pm st}]$, which are conjugate over $K$ and not individually rational; they are drawn as real points. This is the configuration the Weil line demands: a class defined over $\mathbb{Q}$ cannot single out one of the two points, so the construction must carry the pair, and the pair spans a plane rather than a line. Proposition 13.1 is the same statement on the cohomological side, and Figure 13 places it among the hypotheses a construction can satisfy.

hence not isogenous to a product, and for $n\geq 2$ the members $X\times\widehat{X}$ form a subvariety of $\mathcal{D}_{n,n}$ of positive codimension (Proposition 15.19(iii)). Corollary 15.4 is therefore a base case, and Question 15.13 is a question about propagation.

*Notation* 15.1. On $A=X\times\widehat{X}$ write

$$\beta\in H^2(A,\mathbb{Q}),\qquad \widehat{\beta}\in H^2(A,\mathbb{Q}),\qquad \ell=c_1(\mathcal{P})\in H^2(A,\mathbb{Q})$$

for the pullback of the polarisation of $X$, the pullback of the dual polarisation of $\widehat{X}$, and the class of the Poincaré bundle. All three are divisor classes, so every polynomial in them is algebraic. Put

$$\gamma=d\,\beta-\widehat{\beta}. \tag{33}$$

**Theorem 15.2** (The Weil line of the split member)**.** *Let $A=X\times\widehat{X}$ carry the $K$-action (29) determined by a principal $\beta$, and let $V_{\pm}\subset H^1(A,\mathbb{C})$ be the eigenspaces of $\iota(\sqrt{-d})$. Then*

(i) *$\gamma\mp\sqrt{-d}\,\ell$ lies in $\bigwedge^2 V_{\pm}$, so it is an eigenclass for the character $\tau\mapsto\tau^2$ on $H^2(A,\mathbb{C})$ respectively for its conjugate;*

(ii) *the generators of* $\bigwedge^{2n} V_\pm$ *are*

$$\omega_\pm = \frac{(-1)^{n(n-1)/2}}{n!\,d^{\,n}}\bigl(\gamma \mp \sqrt{-d}\,\ell\bigr)^n; \tag{34}$$

(iii) *consequently the Weil line of* $A$ *is*

$$W(A) = \Bigl\langle\ \Re\bigl(\gamma - \sqrt{-d}\,\ell\bigr)^n,\ \ \Im\bigl(\gamma - \sqrt{-d}\,\ell\bigr)^n\ \Bigr\rangle_{\mathbb{Q}},$$

*and every Weil class on* $A$ *is a polynomial with rational coefficients in the three divisor classes* $\beta, \widehat{\beta}, \ell$*, hence is algebraic.*

*Proof.* Write $U = H^1(X,\mathbb{Q})$ with basis $x_1,\dots,x_{2n}$ chosen so that $\beta = \sum_{j\le n} x_j \wedge x_{n+j}$, let $\xi_1,\dots,\xi_{2n}$ be the dual basis of $H^1(\widehat{X},\mathbb{Q}) = U^\vee$, and let $B\colon U \to U^\vee$ be the isomorphism induced by $\beta$, so that $Bx_j = \xi_{n+j}$ and $Bx_{n+j} = -\xi_j$. The endomorphism (29) acts on homology, and its transpose $M^*$ acts on $H^1(A,\mathbb{Q}) = U \oplus U^\vee$ by $u \mapsto -Bu$ and $\phi \mapsto d\,B^{-1}\phi$. Indeed, in the basis $u_1,\dots,u_{2n}$ of $H_1(X,\mathbb{Q})$ dual to the $x_i$, the map $\beta\colon H_1(X) \to H^1(X)$ sends $u_j \mapsto x_{n+j}$ and $u_{n+j} \mapsto -x_j$, so (29) sends $u_j \mapsto d x_{n+j}$, $u_{n+j} \mapsto -d x_j$, $x_{n+j} \mapsto -u_j$ and $x_j \mapsto u_{n+j}$, where $x_i$ is regarded as an element of $H_1(\widehat{X},\mathbb{Q}) = H^1(X,\mathbb{Q})$ and $\xi_i$ is its dual class; pairing with the dual basis gives the stated action. The square of $M^*$ is $-d$; so, writing $\delta = \sqrt{-d}$,

$$V_\pm = \Bigl\{\, u \pm \tfrac{\delta}{d}\,Bu \ :\ u \in U_{\mathbb{C}} \Bigr\}, \tag{35}$$

since $M^*(u \pm \frac{\delta}{d}Bu) = -Bu \pm \delta u = \pm\delta(u \pm \frac{\delta}{d}Bu)$.

(i) Take the sum over $j \le n$ of the wedges of the two eigenvectors attached to $x_j$ and to $x_{n+j}$:

$$\sum_{j=1}^{n}\Bigl(x_j \pm \tfrac{\delta}{d}Bx_j\Bigr) \wedge \Bigl(x_{n+j} \pm \tfrac{\delta}{d}Bx_{n+j}\Bigr) = \beta \pm \tfrac{\delta}{d}\Sigma - \tfrac{1}{d}\widehat{\beta},$$

using $\delta^2 = -d$, where $\Sigma = \sum_j (x_j \wedge Bx_{n+j} + Bx_j \wedge x_{n+j})$ and $\sum_j Bx_j \wedge Bx_{n+j} = \sum_j \xi_{n+j} \wedge (-\xi_j) = \widehat{\beta}$. Now

$$\Sigma = \sum_{j=1}^{n}(x_j \wedge (-\xi_j) + \xi_{n+j} \wedge x_{n+j}) = -\sum_{i=1}^{2n} x_i \wedge \xi_i = -\ell.$$

So the displayed sum is $\frac{1}{d}(d\beta - \widehat{\beta} \mp \delta\ell) = \frac{1}{d}(\gamma \mp \delta\ell)$. It is a sum of wedges of pairs of vectors of $V_\pm$, hence lies in $\bigwedge^2 V_\pm$, which proves (i); the statement about characters is Proposition 5.1 at $(r,s) = (2,0)$ and $(0,2)$.

(ii) By (i) the $n$-th power $(\gamma \mp \delta\ell)^n$ lies in $\bigwedge^{2n} V_\pm$, which is a line by Proposition 3.7. It is nonzero: expanding $\gamma \mp \delta\ell = d\beta - \widehat{\beta} \mp \delta\ell$, the only term of the $n$-th power lying in $\bigwedge^{2n} U$ is $d^n\beta^n$, and $\beta^n \neq 0$ because $\beta$ is nondegenerate; the remaining terms all involve at least one $\xi$, so no cancellation can occur. Hence $(\gamma \mp \delta\ell)^n$ spans $\bigwedge^{2n} V_\pm$ and is a scalar multiple of $\omega_\pm = \bigwedge_{i=1}^{2n}(x_i \pm \frac{\delta}{d}Bx_i)$. To find the scalar, compare the components in $\bigwedge^{2n} U$: that of $\omega_\pm$ is $x_1 \wedge \cdots \wedge x_{2n}$, while

$$d^n\beta^n = d^n\,n!\bigwedge_{j=1}^{n}(x_j \wedge x_{n+j}) = d^n\,n!\,(-1)^{n(n-1)/2}\,x_1 \wedge \cdots \wedge x_{2n},$$

the sign being that of the permutation carrying $(1, n+1, 2, n+2, \dots, n, 2n)$ to $(1,2,\dots,2n)$. Inverting gives (34).

(iii) The two displayed classes are the real and imaginary parts of $(\gamma - \delta\ell)^n$, which up to the nonzero rational factor in (34) are $\omega_- + \omega_+$ and $(\omega_- - \omega_+)/\delta$; by Proposition 3.7(i)

these span $W(A)$ over $\mathbb{Q}$. Expanding $(\gamma - \delta\ell)^n$ by the binomial theorem and separating even from odd powers of $\delta$ writes each of them as a rational polynomial in $\beta, \widehat{\beta}, \ell$, because $\delta^{2m} = (-d)^m$ and $\delta^{2m+1} = (-d)^m\delta$. Products of divisor classes are algebraic, so both are algebraic. ∎

**Example 15.3.** The $2n$ classes $x_1, \dots, x_{2n}$ form a $K$-basis of $H^1(A, \mathbb{Q})$, on which $\sqrt{-d}$ acts by $x \mapsto -Bx$, so (15) applies to them; with (34) it gives the integral generators of Theorem 4.9 for this basis in closed form,

$$\omega_1 + \sqrt{-d}\,\omega_2 = \frac{(-1)^{n(n-1)/2}}{n!}\,(\gamma + \sqrt{-d}\,\ell)^n.$$

At $n = 2$ this reads

$$\omega_1 = \tfrac{1}{2}(d\,\ell^2 - \gamma^2), \qquad \omega_2 = -\gamma\,\ell,$$

and at $n = 3$,

$$\omega_1 = \tfrac{1}{2}\,d\,\gamma\,\ell^2 - \tfrac{1}{6}\gamma^3, \qquad \omega_2 = \tfrac{1}{6}\,d\,\ell^3 - \tfrac{1}{2}\gamma^2\ell,$$

with $\gamma = d\beta - \widehat{\beta}$ throughout.

**Corollary 15.4** (Algebraicity on the split locus)**.** *For every $n \geq 1$, every squarefree $d > 0$ and every principally polarised abelian $n$-fold $X$, the Weil classes of $A = X \times \widehat{X}$ with the $K$-action (29) are algebraic, and an algebraic representative is given by (34) as an explicit rational combination of intersections of three divisors.*

*Proof.* This is Theorem 15.2(iii), together with the Lefschetz theorem on $(1,1)$ classes, which makes $\beta, \widehat{\beta}, \ell$ algebraic, and the fact that the algebraic classes form a subring of $H^*(A, \mathbb{Q})$. ∎

*Remark* 15.5 (Consistency with the character obstruction). Corollary 15.4 does not contradict Theorem 5.6. At the split point $\mathrm{NS}(A) \otimes \mathbb{Q}$ contains the three independent classes $\beta, \widehat{\beta}, \ell$, so the hypothesis $\mathrm{NS}(A) \otimes \mathbb{Q} = \mathbb{Q}\eta$ of Proposition 3.10 fails, and this is the situation of Remark 20.5. What Theorem 5.6 forbids is a $K^\times$-eigenclass for the norm character meeting the Weil line, and $\gamma \mp \sqrt{-d}\,\ell$ is an eigenclass for $\tau^2$ and for $\overline{\tau}^2$, not for the norm. The formula (34) is the first place in the paper where a class in the relevant eigenspace is exhibited, and such a class exists because the split member has enough divisors to build one.

15.2. **Products of abelian surfaces.** A second source of members with enough divisors is the product, an operation that the Weil line respects. Throughout this subsection $K$ is fixed and all Weil structures are taken with respect to it.

**Lemma 15.6** (Products of Weil type)**.** *Let $(A_1, \iota_1, \eta_1)$ be of $(K, -1, n_1)$-Weil type and $(A_2, \iota_2, \eta_2)$ of $(K, -1, n_2)$-Weil type. Then $A = A_1 \times A_2$, with the diagonal action $\iota = \iota_1 \times \iota_2$ and the product polarisation $\eta = p_1^*\eta_1 + p_2^*\eta_2$, is of $(K, -1, n_1 + n_2)$-Weil type.*

*Proof.* Write $V_i = H^1(A_i, \mathbb{Q})$, so $V = H^1(A, \mathbb{Q}) = V_1 \oplus V_2$ as $K$-vector spaces and $\dim_K V = 2n_1 + 2n_2$. The Rosati involution of the product polarisation is the product of the two Rosati involutions, so it restricts on $K$ to conjugation, which is the sign $-1$ of Definition 3.1. For the multiplicities, the Hodge decomposition of $V$ is the direct sum of those of $V_1$ and $V_2$, and the eigenspace decomposition is likewise $V_\pm = V_{1,\pm} \oplus V_{2,\pm}$. Hence

$$\dim_{\mathbb{C}} V_+^{1,0} = \dim_{\mathbb{C}} V_{1,+}^{1,0} + \dim_{\mathbb{C}} V_{2,+}^{1,0} = n_1 + n_2,$$

and the same for the other three, which is Lemma 3.5 for $n = n_1 + n_2$. ∎

For an abelian variety $A$ of $(K,-1,n)$-Weil type and a $K$-basis of $H^1(A,\mathbb{Q})$, let $\omega_1(A),\omega_2(A)$ be the generators of the Weil line given by (11) in that basis, and write

$$\omega(A)=\omega_1(A)+\sqrt{-d}\,\omega_2(A)\ \in\ K\otimes_{\mathbb{Q}}H^{2n}(A,\mathbb{Q})$$

for the $K$-valued Weil class; by (15) its image in $H^{2n}(A,\mathbb{C})$ is $d^{-n}\prod_j(d\,x_j+\delta\,y_j)$. On a product we use the $K$-basis obtained by concatenating bases of the factors.

The Weil line is the top exterior power over $K$, and a top exterior power of a direct sum is the tensor product of the top exterior powers. This is the content of the next theorem; what has to be checked is its effect on the rational generators, which are defined through the real structure rather than over $K$.

**Theorem 15.7** (The Weil class is multiplicative)**.** *With the notation of Lemma 15.6, and cup product on $H^*(A,\mathbb{Q})$ taken after pullback along the two projections,*

$$\begin{aligned}\omega_1(A)&=\omega_1(A_1)\,\omega_1(A_2)-d\,\omega_2(A_1)\,\omega_2(A_2),\\ \omega_2(A)&=\omega_1(A_1)\,\omega_2(A_2)+\omega_2(A_1)\,\omega_1(A_2),\end{aligned}\tag{36}$$

*that is $\omega(A_1)\,\omega(A_2)=\omega(A_1\times A_2)$ for the $K$-valued Weil class. For $k$ factors, $\prod_j\omega(A_j)=\omega(A_1\times\cdots\times A_k)$.*

*Proof.* Let $u_1,\dots,u_{2n_1}$ and $u_{2n_1+1},\dots,u_{2n}$ be the $K$-bases of $H^1(A_1,\mathbb{Q})$ and $H^1(A_2,\mathbb{Q})$, so that together they form the $K$-basis of $H^1(A,\mathbb{Q})$, and put $x_j=u_j$ and $y_j=\sqrt{-d}\,u_j$ as in (15). By (15),

$$\begin{aligned}\omega(A)&=d^{-n}\prod_{j=1}^{2n}(d\,x_j+\delta\,y_j)\\ &=\Big(d^{-n_1}\prod_{j\le 2n_1}(d\,x_j+\delta\,y_j)\Big)\Big(d^{-n_2}\prod_{j>2n_1}(d\,x_j+\delta\,y_j)\Big)=\omega(A_1)\,\omega(A_2)\end{aligned}$$

in $H^{2n}(A,\mathbb{C})$, the products being taken in the order of the basis, so that no sign arises. Expanding $\big(\omega_1(A_1)+\delta\omega_2(A_1)\big)\big(\omega_1(A_2)+\delta\omega_2(A_2)\big)$ with $\delta^2=-d$, and comparing rational parts and $\delta$-parts, which is legitimate because the classes $\omega_j(A_i)$ are rational and $1,\delta$ are linearly independent over $\mathbb{Q}$, gives (36). The statement for $k$ factors follows by induction. ∎

Figure 17 shows the multiplicativity and the loci it reaches.

**Corollary 15.8** (Algebraicity propagates along products)**.** *If the Weil classes of $A_1$ and of $A_2$ are algebraic, then so are those of $A_1\times A_2$, and an algebraic representative is the rational combination of external products given by (36).*

*Proof.* Algebraic classes are closed under pullback along the projections and under cup product, and (36) expresses $\omega_1(A)$ and $\omega_2(A)$ in those operations applied to the $\omega_j(A_i)$. ∎

In dimension two there is nothing to prove.

**Lemma 15.9** (Dimension two)**.** *Let $A_0$ be of $(K,-1,1)$-Weil type, so $\dim A_0=2$. Then $W(A_0)\subset H^2(A_0,\mathbb{Q})$ and both generators are algebraic.*

*Proof.* The Weil classes lie in $H^{2n}=H^2$ and are of type $(1,1)$ by Proposition 3.7(ii), so they are algebraic by the Lefschetz theorem on $(1,1)$ classes. No hypothesis on $A_0$ beyond the Weil structure is used. ∎

**Theorem 15.10** (An algebraic Weil class in every dimension)**.** *Let $n \geq 1$, let $d > 0$ be squarefree, and let $A_1, \dots, A_n$ be abelian surfaces of $(K, -1, 1)$-Weil type. Then $A = A_1 \times \cdots \times A_n$ is of $(K, -1, n)$-Weil type and its Weil classes are algebraic; explicitly, by Theorem 15.7 and Lemma 15.9 they are the rational part and the coefficient of $\sqrt{-d}$ in $\prod_{j=1}^{n}(\omega_1(A_j) + \sqrt{-d}\,\omega_2(A_j))$, a polynomial with integer coefficients in $2n$ divisor classes.*

*Proof.* Lemma 15.6 gives the Weil structure, Lemma 15.9 gives algebraicity of each factor's classes, and Corollary 15.8 applied $n-1$ times gives the conclusion; the displayed form is Theorem 15.7 for $k = n$ factors. Each $\omega_j(A_i)$ is a divisor class by Lemma 15.9, so the product is a polynomial in $2n$ of them. ∎

*Remark* 15.11 (The product locus). Theorem 15.10 is unconditional and holds for every $n$; like Corollary 15.4 and Theorem 15.27, it produces an algebraic Weil class in every dimension using nothing beyond the Lefschetz theorem on $(1,1)$ classes. The product construction supplies a second family of base points, defined for every $n$ and every $d$, at which the class is written down explicitly. It also shows that the algebraicity locus is closed under products, so that a settled case $n_0$ propagates to every multiple of $n_0$, and settled cases propagate to their sums. Combined with Markman's theorem at $n = 2$ [Mar25a, Mar25b], this gives, for every $n$, algebraicity on the locus of products of abelian fourfolds and surfaces of Weil type, which has dimension $2n$ when $n$ is even.

The reach of these results is limited. The family of polarised abelian $2n$-folds of $(K, -1, n)$-Weil type has dimension $n^2$, and the locus of products of $n$ surfaces has dimension $n$, so for $n \geq 2$ it is a proper closed subvariety, smaller than the quaternionic locus of Theorem 15.25, whose dimension is $n(n+1)/2$. A general member of the family is simple, hence not isogenous to a product, and neither Theorem 15.10 nor Corollary 15.4 applies to it. The general member lies outside all of these loci, and by Theorem 15.14 what joins them to it is the propagation statement.

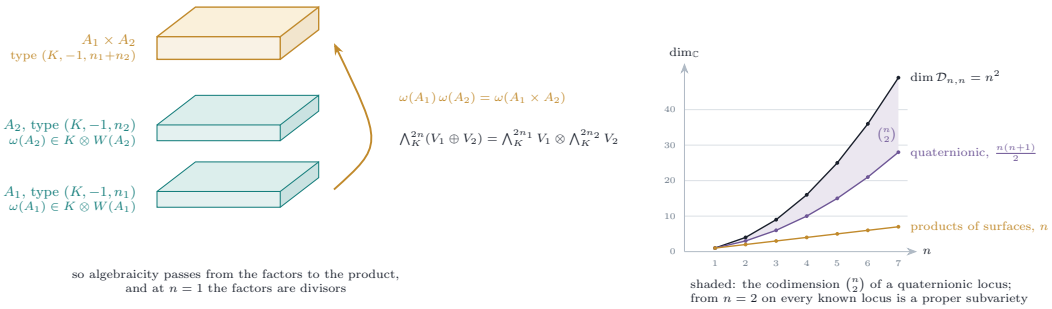



FIGURE 17. Theorem 15.7 and its reach. Left: the Weil line of each factor is a $K$-line, the Weil line of the product is the tensor product of the two over $K$, and the $K$-valued Weil class is multiplicative. Algebraicity therefore passes from the factors to the product, and at $n = 1$ the factors are divisor classes, so a product of $n$ abelian surfaces of Weil type carries an explicit algebraic Weil class in every dimension. Right: the dimensions of the loci on which the class is known to be algebraic, against the dimension $n^2$ of the family. Both the quaternionic locus and the product locus are proper subvarieties once $n \geq 2$, and a general member lies on neither.

15.3. **The residual statement.** The results above allow the remaining question to be stated in one sentence, without auxiliary hypotheses.

**Setup 15.12** (The family)**.** Fix $n \geq 2$ and a squarefree $d > 0$. Let $\mathcal{D}_{n,n}$ be the period domain of Proposition B.8 and let

$$\pi \colon \mathcal{A} \longrightarrow \mathcal{D}_{n,n}$$

be the family of polarised abelian $2n$-folds of $(K, -1, n)$-Weil type it parametrises, with $\omega$ the flat section of $R^{2n}\pi_*\mathbb{Q}$ given fibrewise by the class $\omega_1$ of (11). Let $\mathcal{D}^{\mathrm{spl}} \subset \mathcal{D}_{n,n}$ be the locus of split members $X \times \widehat{X}$ carrying the action (29).

**Question 15.13** (The residual statement)**.** Is $\omega_s$ algebraic for every $s \in \mathcal{D}_{n,n}$?

**Theorem 15.14** (The reduction)**.** *Fix $n \geq 2$ and a squarefree $d > 0$, and keep the notation of Setup 15.12.*

(a) *If Question 15.13 has a positive answer, then the Hodge conjecture holds, in every degree, for every abelian $2n$-fold of $(K, -1, n)$-Weil type with $\mathrm{Hg}(A) = \mathrm{SU}(V, H)$.*

(b) *Conversely, if the Hodge conjecture holds for every abelian $2n$-fold of $(K, -1, n)$-Weil type, then Question 15.13 has a positive answer.*

(c) *$\mathcal{D}^{\mathrm{spl}}$ is nonempty, $\omega_s$ is algebraic at every $s \in \mathcal{D}^{\mathrm{spl}}$ with the explicit representative (34), and $\mathcal{D}_{n,n}$ is connected.*

*Thus Question 15.13 is the variational Hodge conjecture for one flat class on one connected family; it is necessary as well as sufficient, and its base case is supplied. Theorem 15.27 supplies a base case in every family, for every discriminant, so nonemptiness of the base locus is never the obstacle.*

*Proof.* (a) Let $A$ be of $(K, -1, n)$-Weil type with $\mathrm{Hg}(A) = \mathrm{SU}(V, H)$. By Proposition B.8 its Hodge structure is a point $s$ of $\mathcal{D}_{n,n}$, so $A$ is a fibre of $\pi$, and the Weil line of $A$ is generated over $K$ by $\omega_s$. A positive answer makes $\omega_s$ algebraic, hence makes the whole Weil line algebraic by the argument of Corollary 4.15. By Theorem 4.14 the Weil line together with the powers of $\eta$ exhausts $\mathrm{Hdg}^*(A)$, and the powers of $\eta$ are algebraic, so every Hodge class on $A$ is algebraic.

(b) For every $s$ the class $\omega_s$ is a Hodge class on the fibre $A_s$, by Proposition 3.7(ii), and $A_s$ is of $(K, -1, n)$-Weil type by construction. So the Hodge conjecture for $A_s$ gives its algebraicity.

(c) $\mathcal{D}^{\mathrm{spl}}$ is nonempty because the construction (29) applies to any principally polarised abelian $n$-fold and produces a point of $\mathcal{D}_{n,n}$ by Theorem 14.2; algebraicity there is Corollary 15.4; and $\mathcal{D}_{n,n}$ is connected because it is a bounded symmetric domain, by Proposition B.8. ∎

*Remark* 15.15 (The target). The preceding sections remove three parts of the problem. Theorem 4.14 settles how many classes are at issue: one, in one degree, on each fibre. Theorem 7.3 and Theorem 5.6 exclude two families of candidate constructions. Theorem 15.2 supplies the base case: the class is algebraic somewhere in the family, with a representative that can be written on a single line.

The next step is propagation, which is Grothendieck's variational Hodge conjecture for the pair $(\pi, \omega)$ of Setup 15.12. Theorem 15.14 shows that for the Hodge conjecture on abelian varieties of Weil type it is necessary as well as sufficient, so every route to the conjecture meets an equivalent difficulty. Section 16.1 analyses the propagation; there the passage from a formal deformation to an algebraic one, usually the harder half, is supplied by properness of the Hilbert scheme. What survives is either an infinitesimal statement at

a single point or a bound on the complexity of representing cycles along one orbit, and the second is equivalent to its conclusion (Theorem 16.17).

15.4. **The divisorial criterion.** Theorem 15.2 is a computation on one member of the family, but the mechanism behind it is not special to that member. Isolating it gives a criterion that can be tested at any point; we then determine where it holds.

**Theorem 15.16** (The divisorial criterion)**.** *Let $(A, \iota, \eta)$ be of $(K, -1, n)$-Weil type and suppose there exists $\Theta \in \bigwedge^2 V_-$ with*

(i) *$\Theta$ nondegenerate as an alternating form on $V_-^\vee$, equivalently $\Theta^n \neq 0$;*
(ii) *$\Theta$ of Hodge type $(1,1)$;*
(iii) *$\Theta + \overline{\Theta}$ rational, that is in $H^2(A, \mathbb{Q})$.*

*Then the Weil classes of $A$ are algebraic. Explicitly, writing $g = \Theta + \overline{\Theta}$ and $\lambda = (\Theta - \overline{\Theta})/\sqrt{-d}$, both $g$ and $\lambda$ are divisor classes and*

$$W(A) = \left\langle \, \Re(g + \sqrt{-d}\,\lambda)^n, \;\; \Im(g + \sqrt{-d}\,\lambda)^n \, \right\rangle_{\mathbb{Q}}, \tag{37}$$

*so every Weil class is a rational polynomial in two divisor classes.*

*Proof.* Conjugation interchanges $\bigwedge^2 V_-$ and $\bigwedge^2 V_+$, so $\overline{\Theta} \in \bigwedge^2 V_+$, and both $g$ and $\lambda$ are fixed by conjugation, hence real. Rationality of $g$ is (iii); that of $\lambda$ comes from the action of $K^\times$. Fix $\tau = a + b\sqrt{-d} \in K^\times$ with $ab \neq 0$. By Proposition 5.1 the operator $\iota(\tau)^*$ acts on $\bigwedge^2 V_+$ by $\tau^2 = a^2 - db^2 + 2ab\sqrt{-d}$ and on $\bigwedge^2 V_-$ by $\overline{\tau}^2$, so

$$J := \frac{\iota(\tau)^* - (a^2 - db^2)}{2ab}$$

is a $\mathbb{Q}$-rational endomorphism of $H^2(A, \mathbb{Q})$ preserving the space $R$ of Proposition 11.11 and acting on $\bigwedge^2 V_\pm$ by $\pm\sqrt{-d}$. Applying it to $g = \Theta + \overline{\Theta}$ gives $J(g) = -\sqrt{-d}\,\Theta + \sqrt{-d}\,\overline{\Theta} = -\sqrt{-d} \cdot \sqrt{-d}\,\lambda = d\,\lambda$, so $\lambda = J(g)/d$ is rational. Both $g$ and $\lambda$ are of type $(1,1)$ by (ii) and conjugation symmetry, so both lie in $\mathrm{NS}(A) \otimes \mathbb{Q}$ by the Lefschetz theorem on $(1,1)$ classes, and both are algebraic.

By (i) the form $\Theta$ on the $2n$-dimensional space $V_-$ has nonzero Pfaffian, so $\Theta^n$ is a nonzero element of the line $\bigwedge^{2n} V_-$, and it therefore spans it; conjugating, $\overline{\Theta}^n$ spans $\bigwedge^{2n} V_+$. Hence $\Theta^n + \overline{\Theta}^n$ and $(\Theta^n - \overline{\Theta}^n)/\sqrt{-d}$ span $W(A)$ over $\mathbb{Q}$, by Proposition 3.7(i). Finally $2^n\Theta^n = (g + \sqrt{-d}\lambda)^n$, and expanding by the binomial theorem with $(\sqrt{-d})^{2m} = (-d)^m$ writes the real and imaginary parts as rational polynomials in $g$ and $\lambda$. Products of divisor classes are algebraic. ∎

**Corollary 15.17.** *The split member of Theorem 15.2 satisfies the criterion, with $\Theta = \gamma - \sqrt{-d}\,\ell$ and $\gamma = d\beta - \widehat{\beta}$; the two divisor classes of (37) are $2\gamma$ and $-2\ell$.*

*Proof.* Theorem 15.2(i) places $\Theta = \gamma - \sqrt{-d}\,\ell$ in $\bigwedge^2 V_+$ and its conjugate in $\bigwedge^2 V_-$. The hypotheses and the conclusion of Theorem 15.16 are symmetric under replacing $\Theta$ by $\overline{\Theta}$, which interchanges $V_+$ and $V_-$, fixes $g$ and changes the sign of $\lambda$, so the criterion may be applied to $\Theta$. The classes $\gamma$ and $\ell$ are divisor classes, hence of type $(1,1)$, so $\Theta$ is of type $(1,1)$ and $\Theta + \overline{\Theta} = 2\gamma$ is rational. Nondegeneracy is the nonvanishing of $\Theta^n$, proved in Theorem 15.2(ii). ∎

The base case is therefore one instance of a general criterion. The next two results determine how far the criterion reaches, and both limit it.

**Proposition 15.18** (The criterion fails at a general point)**.** *Let $(A,\iota,\eta)$ be of $(K,-1,n)$-Weil type with $n\geq 2$ and $\mathrm{Hg}(A)=\mathrm{SU}(V,H)$. Put*

$$R=\Big(\textstyle\bigwedge^2 V_+\oplus\bigwedge^2 V_-\Big)\cap H^2(A,\mathbb{Q}).$$

*Then $R\cap\mathrm{NS}(A)_{\mathbb{Q}}=0$, and no $\Theta$ as in Theorem 15.16 exists.*

*Proof.* By Proposition 11.11 the space $R$ is a rational sub-Hodge structure of $H^2(A,\mathbb{Q})$, and the intersection $R\cap\mathrm{NS}(A)_{\mathbb{Q}}$ is a rational sub-Hodge structure of $R$ of pure type $(1,1)$. The complexification of a nonzero rational sub-Hodge structure of $R$ is a nonzero $\mathrm{Hg}(A)_{\mathbb{C}}$-subrepresentation of $\bigwedge^2 V_+\oplus\bigwedge^2 V_-$, so it contains a constituent isomorphic to $\bigwedge^2 V_+$ or to $\bigwedge^2 V_-$; by the proof of Proposition 11.11, Hodge numbers depend only on the representation, and by Proposition 11.10 each of $\bigwedge^2 V_\pm$ has $h^{2,0}+h^{0,2}=2\binom{n}{2}=n(n-1)$, which is positive for $n\geq 2$. So no nonzero rational sub-Hodge structure of $R$ is of pure type $(1,1)$, and $R\cap\mathrm{NS}(A)_{\mathbb{Q}}=0$. The argument covers the case $n=2$ with trivial discriminant, where $R$ is reducible. A $\Theta$ as in Theorem 15.16 would give the nonzero element $\Theta+\overline{\Theta}$ of $R\cap\mathrm{NS}(A)_{\mathbb{Q}}$. ∎

**Proposition 15.19** (The split locus)**.** *Fix $n\geq 1$ and a squarefree $d>0$, and let $A=X\times\widehat{X}$ carry the $K$-action (29) determined by a nondegenerate $\beta\in\mathrm{NS}(X)\otimes\mathbb{Q}$ of elementary divisors $(d_1,\dots,d_n)$.*

(i) *Let $B$ be the Gram matrix of $\beta$ in a basis of $U$, and let $\widehat{\beta}$ be the form on $U^\vee$ with Gram matrix $-B^{-1}$ in the dual basis, which for principal $\beta$ is the class of Notation 15.1. The polarisation of $A$ is $\eta=\beta+\widehat{\beta}/d$, a positive rational multiple of $d\beta+\widehat{\beta}$; its Gram matrix on $H_1(A,\mathbb{Q})=U\oplus U^\vee$ is the block form $Q=B\oplus(-B^{-1}/d)$, and $M$ is anti-self-adjoint for it.*

(ii) *In the $K$-basis $u_1,\dots,u_{2n}$ of $H_1(A,\mathbb{Q})$ the hermitian form of Lemma 4.5 has Gram matrix $\sqrt{-d}\,B$, so*

$$\det H=(-d)^n\det B=(-d)^n(d_1\cdots d_n)^2,$$

*and $(-1)^n\det H=d^n(d_1\cdots d_n)^2$ is a norm from $K$. Every split member therefore lies in the Weil family of trivial discriminant, whatever $\beta$ is chosen.*

(iii) *The split members form a locus of dimension $n(n+1)/2$ inside $\mathcal{D}_{n,n}$, of codimension $n(n-1)/2$, which is zero only for $n=1$ and is a divisor for $n=2$.*

(iv) *For $X$ with $\mathrm{Hg}(X)=\mathrm{Sp}(H^1(X,\mathbb{Q}))$ one has $\mathrm{NS}(A)_{\mathbb{Q}}=\langle\beta,\widehat{\beta},\ell\rangle$, of rank three, and in degree $2n$ the monomials in those three classes span the whole space of Hodge classes of $A$.*

*Proof.* (i) In the chosen basis of $U$ and the dual basis of $U^\vee$ the map $\beta\colon U\to U^\vee$ has matrix $B^{\top}=-B$ and its inverse has matrix $-B^{-1}$, so (29) becomes $M=\left(\begin{smallmatrix}0 & B^{-1}\\ -dB & 0\end{smallmatrix}\right)$, with $M^2=-d$ and $M^{\top}=\left(\begin{smallmatrix}0 & dB\\ -B^{-1} & 0\end{smallmatrix}\right)$, using $B^{\top}=-B$ and $(B^{-1})^{\top}=-B^{-1}$. For a block form $Q=\left(\begin{smallmatrix}P & \Lambda\\ -\Lambda^{\top} & S\end{smallmatrix}\right)$ the Rosati relation $M^{\top}Q+QM=0$ reads $B\Lambda^{\top}+\Lambda B=0$, $dBS+PB^{-1}=0$, $B^{-1}P+dSB=0$ and $B^{-1}\Lambda+\Lambda^{\top}B^{-1}=0$. Taking $\Lambda=0$ and $P=B$ forces $S=-B^{-1}/d$ from the second relation, and the third then reads $-I+I=0$. So $Q=B\oplus(-B^{-1}/d)$, which is the Gram matrix of $\beta+\widehat{\beta}/d$. For principal $\beta$ one may take $B=\left(\begin{smallmatrix}0 & I\\ -I & 0\end{smallmatrix}\right)$, so that $-B^{-1}=B$ and $\widehat{\beta}=\sum_j\xi_j\wedge\xi_{n+j}$ as in Notation 15.1.

(ii) The set $u_1,\dots,u_{2n}$ is a $K$-basis because $MU=d\beta(U)=U^\vee$, and $H(u_i,u_j)=Q(Mu_i,u_j)+\sqrt{-d}\,Q(u_i,u_j)$ by (10); the first term vanishes since $Mu_i\in U^\vee$, $u_j\in U$ and

$Q$ has no cross block by (i), and the second is $\sqrt{-d}\,\beta_{ij}$. Hence the Gram matrix is $\sqrt{-d}\,B$ and $\det H = (\sqrt{-d})^{2n}\det B = (-d)^n \det B$. The determinant of an alternating matrix is the square of its Pfaffian, here $(d_1\cdots d_n)^2$. Now $d = \mathrm{N}_{K/\mathbb{Q}}(\sqrt{-d})$ is a norm and every square is a norm, and the norms form a subgroup of $\mathbb{Q}^\times$, so $d^n(d_1\cdots d_n)^2$ is a norm.

(iii) A split member is determined by $X$ together with $\beta$; the moduli space of polarised abelian $n$-folds has dimension $\dim \mathcal{A}_n = n(n+1)/2$, and the type of $\beta$ is discrete. By Proposition B.8 the ambient domain has dimension $n^2$, and $n^2 - n(n+1)/2 = n(n-1)/2$.

(iv) The Hodge classes of $A = X \times \widehat{X}$ are the invariants of $\mathrm{Hg}(A) = \mathrm{Hg}(X) = \mathrm{Sp}(U)$ acting on $\bigwedge^*(U \oplus U^\vee)$. In degree two the invariants are the three lines spanned by $\beta$, $\widehat{\beta}$ and $\ell$, one from each Künneth summand $\bigwedge^2 U$, $U \otimes U^\vee$ and $\bigwedge^2 U^\vee$. In higher degree, identify $U^\vee$ with $U$ through $\beta$, so that $\bigwedge^*(U \oplus U^\vee) \cong \bigwedge^*(U \otimes \mathbb{Q}^2)$ as representations of $\mathrm{Sp}(U)$. By the first fundamental theorem of invariant theory for the symplectic group, the invariants of $\mathrm{Sp}(U)$ in $U^{\otimes 2m}$ are spanned by the tensors attached to the pairings of the $2m$ factors, each a product of copies of the invariant element of $U \otimes U$ dual to $\beta$. The projection of $(U \otimes \mathbb{Q}^2)^{\otimes 2m}$ onto $\bigwedge^{2m}(U \otimes \mathbb{Q}^2)$ is equivariant and surjective, hence surjective on invariants since $\mathrm{Sp}(U)$ is reductive, and it carries such a tensor, tensored with vectors of $\mathbb{Q}^2$, to a product of invariants of degree two. So the invariants are generated by $\beta$, $\widehat{\beta}$ and $\ell$, and in degree $2n$ the monomials $\beta^i\widehat{\beta}^j\ell^k$ with $i+j+k=n$ span the Hodge classes. ∎

*Remark* 15.20 (Two residual questions). The Weil families for a fixed $K$ and a fixed $n$ are classified by the discriminant of the hermitian form, an element of the infinite group $\mathbb{Q}^\times_{>0}/\mathrm{N}_{K/\mathbb{Q}}(K^\times)$. By Proposition 15.19(ii) the split construction of Section 15.1 produces a base point in only one of them, the family of trivial discriminant; products of abelian surfaces of Weil type (Theorem 15.10, Lemma 15.26) reach every discriminant, but only on a locus of dimension $n$. Read together with Theorem 15.14, this divides Question 15.13 into two questions of different kinds.

(R1) *Trivial discriminant.* A base point exists, with the explicit cycle (34), and what is missing is propagation off a locus of codimension $n(n-1)/2$. At $n=2$ that locus is a divisor and the propagation is known, by Markman [Mar25a, Mar25b] and by Floccari and Fu [FF26]; combined with Theorem 15.14(a) this recovers the Hodge conjecture for abelian fourfolds of Weil type of trivial discriminant with full Hodge group. At $n=3$ it is settled as well, by Markman's theorem on the sixfolds of trivial discriminant [Mar25a, Mar25b], for every $K$. For $n \geq 4$ it is the propagation statement, and by Proposition 20.13 it is then the only question in the imaginary quadratic case.

(R2) *Nontrivial discriminant.* The split construction never lands there, by Proposition 15.19(ii), so it supplies no base point; and by Proposition 15.18 no divisorial argument can supply one at a general member. A base point must come from a special member at which $R$ acquires Hodge classes.

Question (R2) is settled in Section 15.5: the special members required are built from abelian varieties with quaternionic multiplication extending $K$ and their products, they exist in every family, and on them Theorem 15.16 applies. What then remains is the propagation question of (R1), posed in every family, each of which has a base point.

*Remark* 15.21 (Codimension of the split locus). A component of the Noether–Lefschetz locus of a weight two variation has codimension at most $h^{2,0}$, here $n(n-1)$. The split locus has codimension $n(n-1)/2$ by Proposition 15.19(iii), half of that, so it is a component of half the expected codimension. This reflects Proposition 15.19(iv): at a split point

$R$ acquires two independent Hodge classes at once, namely $\gamma$ and $\ell$, and each of them costs $h^{2,0}/2$ conditions rather than $h^{2,0}$. Whether there are components of the expected codimension carrying only one class, and whether the criterion of Theorem 15.16 can be met on them, is not needed below: question (R2) is answered in Section 15.5 through quaternionic multiplication.

15.5. **A base point in every family.** By Proposition 15.19(ii) the split construction reaches a single discriminant class. Theorem 15.16 does not depend on the origin of $\Theta$, and in this subsection $\Theta$ comes from an endomorphism rather than from a pair of divisors chosen by hand. This source reaches every class, and it settles question (R2) of Remark 15.20.

**Definition 15.22** (Discriminant)**.** For $(A,\iota,\eta)$ of $(K,-1,n)$-Weil type with hermitian form $H$ as in Lemma 4.5, the *discriminant* is the class of $(-1)^n \det H$ in $\mathbb{Q}^\times_{>0}/\mathrm{N}_{K/\mathbb{Q}}(K^\times)$. The sign makes the representative positive, because $H$ has signature $(n,n)$; and the class is well defined because changing the $K$-basis multiplies $\det H$ by a norm. Hermitian forms of rank $2n$ over $K$ with signature $(n,n)$ are classified by this class, so it labels the Weil families.

The classes that enter Theorem 15.16 have a description by endomorphisms: a class in $R\cap \mathrm{NS}(A)_{\mathbb{Q}}$ is the same thing as a symmetric endomorphism that conjugates $K$.

**Lemma 15.23** (Classes in $R$ and endomorphisms)**.** *Let $(A,\iota,\eta)$ be of $(K,-1,n)$-Weil type. The map $\mu\mapsto g_\mu$, $g_\mu(x,y)=\eta(\mu x,y)$, is a bijection between*

(a) *the elements $\mu\in \mathrm{End}^0(A)$ with $\mu^\dagger=\mu$ and $\mu\,\iota(\tau)=\iota(\overline{\tau})\,\mu$ for all $\tau\in K$, and*
(b) *the elements of $R\cap\mathrm{NS}(A)_{\mathbb{Q}}$.*

*Under it, $\mu$ invertible corresponds to $g_\mu$ nondegenerate, hence to a nondegenerate $\Theta$. If such a $\mu\neq 0$ exists and the centraliser of $\iota(K)$ in $\mathrm{End}^0(A)$ is $\iota(K)$, then $\mu^2\in\mathbb{Q}^\times$ and $\iota(K)+\iota(K)\mu$ is a quaternion algebra $(-d,\mu^2)_{\mathbb{Q}}$ inside $\mathrm{End}^0(A)$.*

*Proof.* A class in $\mathrm{NS}(A)_{\mathbb{Q}}$ is a symmetric homomorphism $A\to A^\vee$ up to isogeny, that is $\eta\mu$ with $\mu^\dagger=\mu$. The condition $g_\mu(\iota(\tau)x,y)=g_\mu(x,\iota(\tau)y)$, which says that $g_\mu$ is $K$-bilinear and hence lies in $R=\bigwedge^2_K V$, reads $\eta(\mu\iota(\tau)x,y)=\eta(\mu x,\iota(\tau)y)=\eta(\iota(\tau)^\dagger\mu x,y)=\eta(\iota(\overline{\tau})\mu x,y)$, using Definition 3.1(iii), so it is the condition $\mu\iota(\tau)=\iota(\overline{\tau})\mu$. For the last sentence, $\mu^2$ commutes with $\iota(K)$, hence lies in $\iota(K)$ by hypothesis, and $\mu\mu^2\mu^{-1}=\mu^2$ forces it to be fixed by conjugation, so $\mu^2\in\mathbb{Q}^\times$. Then $\iota(\sqrt{-d})$ and $\mu$ generate the quaternion algebra with the stated invariants. ∎

The criterion therefore asks for quaternionic multiplication extending $K$. Abelian varieties with this structure are easy to construct, and their invariants can be computed.

**Setup 15.24** (The quaternionic model)**.** Let $b\in\mathbb{Q}_{>0}$ and let $B=(-d,b)_{\mathbb{Q}}$ be the quaternion algebra with basis $1,i,j,ij$ and $i^2=-d$, $j^2=b$, $ij=-ji$. Let $V=B^n$ as a left $B$-module, of dimension $4n$ over $\mathbb{Q}$, let $K=\mathbb{Q}(i)\subset B$ act through $i$, and fix $a_1,\dots,a_n\in\mathbb{Q}^\times$ and

$$E(x,y)=\sum_{k=1}^{n} a_k\,\mathrm{trd}\big(\overline{x_k}\,i\,y_k\big), \tag{38}$$

where trd is the reduced trace and $\bar{}$ the canonical involution of $B$.

**Theorem 15.25** (Weil type from quaternionic multiplication)**.** *With the data of Setup 15.24:*

(i) *$E$ is alternating and nondegenerate, and its Rosati involution is $\beta \mapsto i^{-1}\overline{\beta} i$, which restricts to complex conjugation on $K$ and fixes $j$; it is positive, the form* $\operatorname{trd}(\beta\beta^{\dagger})$ *being diagonal with entries $2, 2d, 2b, 2db$ on the basis;*

(ii) *in the $K$-basis $e_1, e_1 j, \dots, e_n, e_n j$ of $V$ the hermitian form of Lemma 4.5 is*

$$H = \operatorname{diag}(2a_1 d,\ -2a_1 bd,\ \dots,\ 2a_n d,\ -2a_n bd),$$

*of signature $(n, n)$;*

(iii) *there is a complex structure on $V_{\mathbb{R}}$ commuting with $B$ and polarised by $E$, and for every such the resulting abelian variety $A$ is of $(K, -1, n)$-Weil type, the balanced signature being forced rather than imposed;*

(iv) *the discriminant of $A$ is the class of $b^n$;*

(v) *$\mu =$ left multiplication by $j$ satisfies Lemma 15.23(a), so $g = g_\mu$ lies in $R \cap \mathrm{NS}(A)_{\mathbb{Q}}$ and is nondegenerate, the divisorial criterion Theorem 15.16 applies, and the Weil classes of $A$ are algebraic.*

*Proof.* (i) $\operatorname{trd}(z) = \operatorname{trd}(\overline{z})$ and $\overline{\overline{x}\overline{i}y} = \overline{y}\,\overline{\overline{i}}\,x = -\overline{y}ix$ give $E(y,x) = -E(x,y)$. Nondegeneracy is the nondegeneracy of the reduced trace pairing. For the Rosati, $E(\beta x, y) = E(x, \beta^{\dagger} y)$ for all $x, y$ reads $\overline{\beta} i = i\beta^{\dagger}$, that is $\beta^{\dagger} = i^{-1}\overline{\beta} i$. Then $i^{\dagger} = -i$ and, since $(-j)i = ij$, also $j^{\dagger} = i^{-1}(ij) = j$. Positivity is the displayed diagonal, which is positive because $d > 0$ and $b > 0$.

(ii) $V = Ke_k \oplus Ke_k j$ over all $k$, since $B = K \oplus Kj$. By (10), $H(x,y) = E(ix,y) + \sqrt{-d}\,E(x,y)$. On one factor, $E(ix,y) = a\operatorname{trd}(\overline{ix}\,i\,y) = -a\operatorname{trd}(\overline{x}i^2 y) = ad\operatorname{trd}(\overline{x}y)$, so

$$H(x,y) = a\big[d\operatorname{trd}(\overline{x}y) + \sqrt{-d}\operatorname{trd}(\overline{x}iy)\big].$$

Now $\operatorname{trd}(1) = 2$, $\operatorname{trd}(i) = \operatorname{trd}(j) = \operatorname{trd}(ij) = 0$, $\overline{j}j = -b$ and $\overline{j}ij = ib$, which give $H(1,1) = 2ad$, $H(j,j) = -2abd$ and $H(1,j) = 0$. Since $b > 0$ the two entries of each block have opposite signs, so the signature is $(n,n)$.

(iii) A complex structure commuting with $B$ and compatible with $E$ is the same as a subspace $V^{1,0} \subset V_{\mathbb{C}}$, stable under $B$, with $V_{\mathbb{C}} = V^{1,0} \oplus \overline{V^{1,0}}$ and the two Riemann relations. Being $K$-stable it splits as $P \oplus P'$ with $P \subset V_+$ and $P' \subset V_-$; left multiplication by $j$ is $\sigma$-semilinear, so it carries $V_+$ to $V_-$, and $B$-stability forces $P' = jP$. Conversely, for any $n$-dimensional $H$-negative definite $P \subset V_+$ the subspace $P \oplus jP$ is $B$-stable and satisfies the Riemann relations, by the computation in the proof of Theorem 4.4 applied blockwise; such $P$ exists because $H$ has signature $(n,n)$ by (ii). Since $j$ is invertible and carries $P$ isomorphically to $P'$, one gets $\dim P = \dim P' = n$, which is Definition 3.1(ii). The general theory of abelian varieties with prescribed endomorphism structure is in [BL04, Ch. 9].

(iv) By (ii), $\det H = \prod_k (2a_k d)(-2a_k bd) = (-1)^n b^n \big(2^n d^n \prod_k a_k\big)^2$, so $(-1)^n \det H$ is $b^n$ times a square, and squares are norms.

(v) Left multiplication by $j$ satisfies $\mu^{\dagger} = \mu$ by (i) and $\mu\,\iota(\tau) = \iota(\overline{\tau})\,\mu$ because $ji = -ij$; it is invertible because $\mu^2 = b \neq 0$. Lemma 15.23 puts $g_\mu$ in $R \cap \mathrm{NS}(A)_{\mathbb{Q}}$ and makes it nondegenerate, and Theorem 15.16 applies to the component $\Theta$ of $g_\mu$ in $\bigwedge^2 V_-$. ∎

**Lemma 15.26** (Products and discriminants)**.** *Let $(A_1, \iota_1, \eta_1)$ and $(A_2, \iota_2, \eta_2)$ be of $(K, -1, n_1)$- and $(K, -1, n_2)$-Weil type. Then $A_1 \times A_2$ with the diagonal $K$-action and the product polarisation is of $(K, -1, n_1 + n_2)$-Weil type, its hermitian form is the orthogonal sum, its discriminant is the product of the two discriminants, and if both factors satisfy the divisorial criterion then so does the product.*

*Proof.* The first assertion is Lemma 15.6. Signatures and hermitian forms add over an orthogonal sum, so the signature is $(n_1 + n_2, n_1 + n_2)$ and $\det H = \det H_1 \det H_2$, whence

$(-1)^{n_1+n_2}\det H$ is the product of the two normalised determinants. For the criterion, $V_- = V_{1,-} \oplus V_{2,-}$, and $\Theta = \Theta_1 + \Theta_2$ lies in $\bigwedge^2 V_-$, is of type $(1,1)$, has $\Theta + \overline{\Theta}$ rational, and is nondegenerate on the direct sum because each summand is nondegenerate on its factor. ∎

**Theorem 15.27** (A base point in every Weil family)**.** *Let $K = \mathbb{Q}(\sqrt{-d})$, let $n \geq 1$, and let $\delta$ be any class in $\mathbb{Q}^\times_{>0}/\mathrm{N}_{K/\mathbb{Q}}(K^\times)$. Then there is an abelian $2n$-fold $A$ of $(K,-1,n)$-Weil type with discriminant $\delta$ whose Weil classes are algebraic, with an explicit representative as a rational polynomial in two divisor classes.*

*Proof.* Choose $b \in \mathbb{Q}_{>0}$ representing $\delta$.

If $n$ is odd, take the quaternionic model of Setup 15.24 with this $b$. Its discriminant is $b^n = b \cdot (b^{(n-1)/2})^2$, which is $\delta$, and its Weil classes are algebraic by Theorem 15.25(v).

If $n$ is even, then $b^n$ is a square and the model alone reaches only the trivial class. Take instead $A = A_1 \times A_2$ with $A_1$ the quaternionic model at $n_1 = 1$ and parameter $b$, and $A_2$ the quaternionic model at $n_2 = n-1$ and parameter $d$. Both factors satisfy the criterion, so the product does by Lemma 15.26; and the discriminant is $b^1 \cdot d^{n-1}$, which is $\delta$ because $d = \mathrm{N}_{K/\mathbb{Q}}(\sqrt{-d})$ is a norm. The dimension is $2(1+(n-1)) = 2n$.

In both cases the representative is (37) for the class $g_\mu$ of Theorem 15.25(v), or, in the second case, for the sum of the two such classes. ∎

**Corollary 15.28** (Reduction to propagation)**.** *Every Weil family contains a member at which the Weil classes are algebraic. Consequently Question 15.13 reduces to propagation alone: the Hodge conjecture for abelian $2n$-folds of Weil type, for every $n$ and every discriminant, holds if and only if the class $\omega$ of Setup 15.12 propagates from the nonempty locus where it is known to be algebraic to the rest of its connected family.*

*Proof.* The first sentence is Theorem 15.27 applied to each class, together with the fact that the families are indexed by those classes (Definition 15.22). The second is Theorem 15.14, the base locus being nonempty in every family by the first. ∎

*Remark* 15.29 (Two features of the residual statement)*.* By Theorem 15.14 and Corollary 15.28, the residual statement is the variational Hodge conjecture for one flat class on one connected family, with a base point in every family, and it is necessary as well as sufficient. Two of its features show which tools cannot settle it.

The base locus is thin. By Proposition 15.18 the divisorial criterion fails at every point with full Hodge group, so the locus where it holds is contained in the Noether–Lefschetz locus of $R$, a countable union of proper closed algebraic subvarieties of $\mathcal{D}_{n,n}$ by Theorem B.6. The divisorial method cannot enlarge the base locus beyond special points: every point it produces is again special.

The propagation is not formal. By Remark B.11 its infinitesimal form would follow from semiregularity, in the sense of Bloch, of the cycle at a base point, and that cycle is explicit in every family, being a rational polynomial in two divisor classes. Its semiregularity is a computation in the normal bundle that we do not carry out; for the perfect complexes of Proposition 16.27, whose Chern character is concentrated in the degree of the Weil class, semiregularity fails (Proposition 9.10, Theorem 17.5). The other half of the difficulty, that semiregularity gives deformations only in the formal category, is removed in Section 16.1: once the algebraic locus is known to be a countable union of closed subvarieties, a formal deformation at one point is enough.

The secant route of Section 14 needs, at $n \geq 4$, an object on $X$ whose Chern character lies on the secant plane and which also supplies the deformation step of Remark 14.16. No such object is known, but its numerical invariants raise no obstruction.

**Proposition 15.30** (Numerical admissibility of the secant plane)**.** *Let $t$ be a principal polarisation on $X$ and let* $\mathrm{ch} = au+bv$ *be a rational point of the secant plane of Theorem 14.5, with $a, b \in \mathbb{Z}$. Then*

(i) *every* $\mathrm{ch}_k$ *is an integral multiple of the integral class $t^k/k!$, the multiple being $\pm ad^j$ in even degree $k = 2j$ and $\pm bd^j$ in odd degree $k = 2j+1$;*

(ii) *for all integers $a$, $b$ and every positive integer $d$, the Chern classes obtained from* ch *by Newton's identities are integral in every degree: $c_k = \gamma_k\, t^k/k!$ with $\gamma_0 = 1$, $\gamma_1 = b$ and*

$$\gamma_{k+1} \;=\; b\,\gamma_k + d\,k\,(a-k+1)\,\gamma_{k-1} \qquad (k \geq 1);$$

(iii) *the discriminant is $\Delta = 2rc_2 - (r-1)c_1^2 = (a^2d + b^2)t^2 = \mathrm{N}_{K/\mathbb{Q}}(b + a\sqrt{-d})\, t^2$, which is positive, so Bogomolov's inequality never binds.*

*Proof.* (i) is the definition of $u$ and $v$: the coefficient of $t^k$ is $a(-d)^j/(2j)!$ or $b(-d)^j/(2j+1)!$, and multiplying by $k!$ clears the factorial and leaves $\pm ad^j$ or $\pm bd^j$. For (ii), write the total Chern class as a power series $c(t) = \sum_k \gamma_k t^k/k!$ in the variable $t$. Newton's identities in generating-function form say $\log c = \sum_{k\geq 1}(-1)^{k-1}(k-1)!\,\mathrm{ch}_k$, and with the coefficients of (i) the two halves of this sum are recognised series:

$$\log c(t) \;=\; -a\sum_{j\geq 1}\frac{(-d)^j t^{2j}}{2j} + b\sum_{j\geq 0}\frac{(-d)^j t^{2j+1}}{2j+1} \;=\; \frac{a}{2}\log(1+dt^2) + \frac{b}{\sqrt{d}}\arctan(\sqrt{d}\,t).$$

Differentiating, $(1+dt^2)\,c'(t) = (b+adt)\,c(t)$. Comparing the coefficients of $t^k/k!$ on the two sides gives $\gamma_{k+1} + d\,k(k-1)\gamma_{k-1} = b\gamma_k + ad\,k\,\gamma_{k-1}$, which is the stated recursion. Its coefficients are integers and it starts from $\gamma_0 = 1$, $\gamma_1 = b$, so every $\gamma_k$ is an integer, and $c_k = \gamma_k\, t^k/k!$ is an integral class because $t^k/k!$ is. (iii) is the computation recorded in Theorem 14.7, and the norm form of an imaginary quadratic field is positive definite. ∎

*Remark* 15.31 (Numerical consistency of the secant object). Proposition 15.30 excludes a purely arithmetic obstruction. If the object required by the construction at $n \geq 4$ were forbidden by its numerical invariants, the secant route would fail on arithmetic grounds alone. It is not: the Chern character is integral, the Chern classes are integral in every degree by Proposition 15.30(ii), the discriminant is positive, and neither Riemann–Roch nor Bogomolov's inequality stands in the way. Together with Proposition 9.11, which bounds the size of the object from below without excluding it, and the parameter count of Remark 14.11, which fixes the rank one shape at $n = 3$, every count available to us is consistent with the existence of the object, and none of them produces it. What is missing at $n \geq 4$ is the object itself.

15.6. **The quaternionic locus as a Lagrangian Grassmannian.** Theorem 15.25 produces points of the family carrying an admissible endomorphism $\psi$. The locus of all such points has dimension $n(n+1)/2$, by a Morita argument recalled in Remark 16.26. Here we identify the locus itself, and the identification shows where the residual difficulty lies.

**Proposition 15.32** (The quaternionic locus)**.** *Keep Setup 15.24 and let $\psi$ be $K$-semilinear, invertible, Rosati-symmetric, with $\psi^2 = b$. For $x, y \in V_+$ put*

$$\omega_\psi(x,y) \;=\; E(x,\psi y).$$

(i) *$\omega_\psi$ is a nondegenerate alternating form on $V_+$, and it is alternating precisely because $\psi$ is Rosati-symmetric.*

(ii) *A point $P = V_+^{1,0}$ of $\mathcal{D}_{n,n}$ lies on the locus where $\psi$ is a morphism of Hodge structures if and only if $P$ is isotropic for $\omega_\psi$.*

(iii) *Since $\dim_{\mathbb{C}} V_+ = 2n$ and $\dim_{\mathbb{C}} P = n$, such a $P$ is Lagrangian, so the locus is the open subset of $\mathrm{LG}(n, V_+, \omega_\psi)$ cut out by negative definiteness of $H$. Its dimension is $n(n+1)/2$ and its codimension in $\mathcal{D}_{n,n}$ is $n(n-1)/2$.*

*Proof.* (i) $\psi$ carries $V_+$ to $V_-$, since it is semilinear, and $E$ pairs $V_+$ with $V_-$ perfectly, so $\omega_\psi$ is defined and nondegenerate. Rosati symmetry is the identity $E(\psi x, y) = E(x, \psi y)$; with $E$ alternating it gives $\omega_\psi(y, x) = E(y, \psi x) = -E(\psi x, y) = -E(x, \psi y) = -\omega_\psi(x, y)$.

(ii) Suppose $\psi$ preserves the Hodge structure. Being semilinear it carries $V_+^{1,0}$ into $V_-^{1,0}$, so $\psi P \subseteq V^{1,0}$, and $E$ vanishes on $V^{1,0}$ by the first Riemann relation; hence $E(P, \psi P) = 0$, which is the isotropy of $P$. Conversely, if $P$ is isotropic then $\psi P$ lies in the $E$-annihilator of $P$ inside $V_-$. That annihilator has dimension $n$, because $E$ pairs $V_+$ with $V_-$ perfectly, and it contains $V_-^{1,0}$, which also has dimension $n$ and is $E$-orthogonal to $P$ since both lie in $V^{1,0}$. The two therefore agree, so $\psi P = V_-^{1,0}$ and $\psi$ preserves $V^{1,0}$.

(iii) A Lagrangian Grassmannian $\mathrm{LG}(n, 2n)$ has dimension $n(n+1)/2$, its tangent space at a Lagrangian $P$ being the symmetric forms on $P$; negative definiteness is an open condition and does not change the dimension. Subtracting from $\dim \mathcal{D}_{n,n} = n^2$ gives $n(n-1)/2$. ∎

Figure 18 draws these loci as a pencil of leaves.

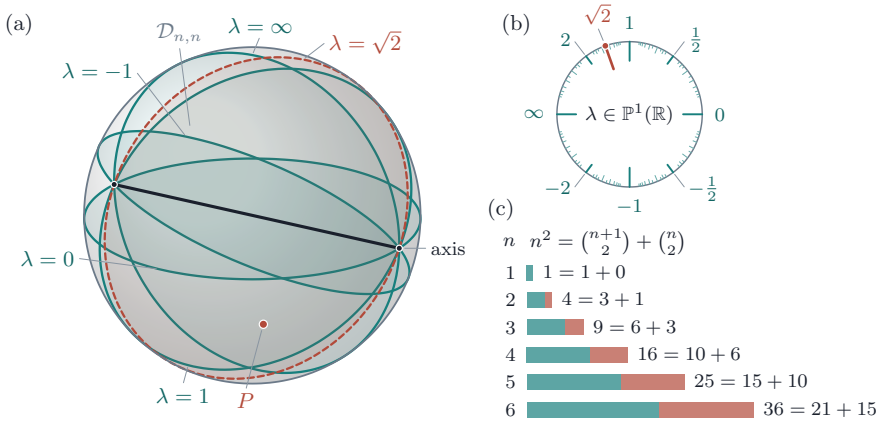



FIGURE 18. The quaternionic loci of Proposition 15.32, drawn as a pencil. Each admissible $\psi$ gives one leaf, an open piece of $\mathrm{LG}(n, V_+, \omega_\psi)$ of dimension $n(n+1)/2$. As $\psi$ varies over the reals the leaves through a common axis form a continuous family and sweep out $\mathcal{D}_{n,n}$. The two heavy leaves are those with $\psi$ rational, and only such leaves are quaternionic loci of abelian varieties. The drawing is faithful to the continuity of the family and to the discreteness of its rational members, but not to dimensions: in the family the leaves have dimension $n(n+1)/2$ inside $n^2$, hence codimension $n(n-1)/2$, which vanishes only at $n = 1$ (at $n = 2$ the leaves are divisors, Remark 16.26), and $\psi$ ranges over a real variety rather than an interval.

*Remark* 15.33 (Real and rational quaternionic loci). Fix a point $P$ of $\mathcal{D}_{n,n}$, and let $\psi_0$ be left multiplication by $j$ at a point $P_0$ given by Theorem 15.25(iii). The group $\mathrm{SU}(V,H)(\mathbb{R})$ acts transitively on the bounded symmetric domain $\mathcal{D}_{n,n}$ of Proposition B.8, so $P = gP_0$ for some real $g$; since $g$ is $K$-linear and preserves $H$, hence $E$, the map $\psi = g\psi_0 g^{-1}$ is $K$-semilinear, invertible and Rosati-symmetric, satisfies $\psi^2 = b$, and preserves the Hodge structure at $P$. It satisfies every condition of Proposition 15.32 except rationality, and $P$ lies on its locus. So the quaternionic loci, taken over the *real* $\psi$, cover $\mathcal{D}_{n,n}$; they cover it with room to spare, since each has dimension $n(n+1)/2$ and they are parametrised by a positive-dimensional family. What prevents the quaternionic loci of abelian varieties from covering it is that $\psi$ must lie in $\mathrm{End}^0(A)$, hence be rational, and the rational $\psi$ form a countable set.

This is the dichotomy of Corollary 16.5 in its simplest form. The obstruction is the countability of a set of parameters inside an uncountable one, and it involves neither deformation theory nor positivity. What is needed is a mechanism that produces, from one rational $\psi$, a family of cycles representing $\omega$ over an open subset of $\mathcal{D}_{n,n}$ rather than over the locus of $\psi$ alone; by Corollary 16.5, such a family would answer Question 15.13 in the affirmative.

15.7. **Absolutely Hodge classes and the exact form.** The algebraicity of a Hodge class is usually approached through an intermediate property. A rational class of type $(p,p)$ on a smooth projective variety $X$ over $\mathbb{C}$ is *absolutely Hodge* if, for every automorphism $\sigma$ of $\mathbb{C}$, the conjugate under $\sigma$ of its de Rham component is the de Rham component of a rational class of type $(p,p)$ on the conjugate variety $X^\sigma$. The implications

$$\text{algebraic} \implies \text{absolutely Hodge} \implies \text{Hodge} \tag{39}$$

are both elementary. The Hodge conjecture asserts the converse of their composite. The converse of the second is the weaker statement, and Deligne proved it for abelian varieties, so on abelian varieties the Hodge conjecture is the converse of the first.

**Theorem 15.34** (Deligne [Del82])**.** *Every Hodge class on a complex abelian variety is absolutely Hodge. Moreover,* Principle B*: if $\pi\colon \mathcal{X} \to S$ is smooth projective over a connected smooth base, and $\alpha$ is a flat section of $R^{2p}\pi_*\mathbb{Q}(p)$ that is Hodge at every point of $S$ and absolutely Hodge at one point, then it is absolutely Hodge at every point.*

Together with Theorem 15.14 and Theorem 15.27, this locates the remaining problem.

**Corollary 15.35** (Reduction to an absolutely Hodge class)**.** *Let $\pi\colon \mathcal{A} \to \mathcal{D}_{n,n}$ and $\omega$ be as in Setup 15.12. Then $\omega_s$ is absolutely Hodge for every $s$, and the following are equivalent:*

(i) *the Hodge conjecture holds, in every degree, for every abelian $2n$-fold of $(K,-1,n)$-Weil type with $\mathrm{Hg}(A) = \mathrm{SU}(V,H)$;*

(ii) *the absolutely Hodge class $\omega_s$ is algebraic for every $s$.*

*This holds for every $K$, every $n$ and every discriminant.*

*Proof.* The first assertion is Theorem 15.34. It also follows from Principle B together with Theorem 15.27, which supplies a point of $\mathcal{D}_{n,n}$ at which $\omega$ is algebraic and hence absolutely Hodge, and with the connectedness of $\mathcal{D}_{n,n}$ from Proposition B.8.

That (ii) implies (i) is Theorem 15.14(a). Conversely, the Hodge group of $A_s$ is $\mathrm{SU}(V,H)$ for $s$ outside a countable union of proper closed analytic subsets of $\mathcal{D}_{n,n}$, by Weil's theorem [Wei77] recalled before Theorem 4.14. Under (i) the class $\omega_s$ is algebraic at all such $s$, so the set $\Sigma$ of points at which it is algebraic is not meagre, and Corollary 16.5 gives $\Sigma = \mathcal{D}_{n,n}$, which is (ii). ∎

## 16. The algebraic locus and the propagation criteria

The results of Section 15 are pointwise: a class, a criterion and a cycle at a point, and in Corollary 15.35 an implication to be proved at every point. This section studies the set of points at which the Weil class is algebraic. Properness of the Hilbert scheme makes it a countable union of closed subvarieties (Theorem 16.4), and rational isogenies make it dense (Proposition 16.7); together these turn the propagation question into a dichotomy (Corollary 16.5). Two criteria for propagation follow, a bound along one orbit (Theorem 16.9) and a rank at one point (Theorem 16.29). We treat the bound first, together with the transport of cycles along isogenies, which decides one of its two readings, and with the limits of representatives built from products of divisors; the rest of the section is devoted to the rank.

16.1. **The algebraic locus.** Question 15.13 is a question about the set

$$\Sigma \;=\; \{\, s \in \mathcal{D}_{n,n} \;:\; \omega_s \text{ is algebraic on } A_s \,\}.$$

This set is restricted in shape, and the restriction changes the form of the remaining problem: instead of a class to construct at every point, one is left with a bound to prove along one orbit, or a rank to compute at one point. The reason is that the Hilbert scheme is proper, so each piece of $\Sigma$ cut out by bounded data is Zariski closed. The mechanism is standard: the locus where a fixed Hodge class becomes algebraic is a countable union of closed subvarieties, the companion of the statement that the Hodge locus itself is one (see [Voi03, Chapter 5] and, in the form used in Theorem B.6, [CDK95]). What is particular to the present situation is that the class is determined by Theorem 4.14 and written out by Theorem 4.9, that its algebraic locus is nonempty in every family by Theorem 15.27, and that the group acting on the family is known.

**Setup 16.1** (The quotient and the universal family)**.** Keep Setup 15.12. Fix an integer $N \geq 3$ and let $\Gamma$ be the subgroup of $\mathrm{SU}(V,H)(\mathbb{Q})$ preserving the lattice $L \subset V$ used to build the family and acting trivially on $L/NL$. Write

$$p\colon \mathcal{D}_{n,n} \longrightarrow \mathcal{S} = \Gamma\backslash\mathcal{D}_{n,n}.$$

Since $N \geq 3$ the group $\Gamma$ is torsion-free, so it acts freely on $\mathcal{D}_{n,n}$, and by Baily and Borel [BB66] the quotient $\mathcal{S}$ is a smooth quasi-projective complex variety; it is irreducible because $\mathcal{D}_{n,n}$ is connected. The level structure rigidifies, so $\mathcal{S}$ carries a universal family

$$f\colon \mathcal{A} \longrightarrow \mathcal{S},$$

and the relative polarisation $3\eta$ is relatively very ample, which fixes a relative projective embedding and hence a notion of degree in the fibres. Write $G = \mathrm{SU}(V,H)$, an algebraic group over $\mathbb{Q}$, so that $G(\mathbb{R})$ acts transitively on $\mathcal{D}_{n,n}$ and $\Gamma \subset G(\mathbb{Q})$.

We record two facts that are used repeatedly below.

**Lemma 16.2** (The section descends)**.** *The section $\omega$ of Setup 15.12 is $\Gamma$-invariant, hence descends to a flat section of $R^{2n}f_*\mathbb{Q}$ over $\mathcal{S}$ that is of type $(n,n)$ at every point.*

*Proof.* The Weil line is $W = \bigwedge_K^{2n} V$, and an element of $\mathrm{SU}(V,H)$ is by definition a $K$-linear automorphism of $V$ of $K$-determinant 1. Its action on the top $K$-exterior power is multiplication by that determinant, so $\Gamma$ acts trivially on $W$ and in particular fixes $\omega$. Flatness is then immediate, and the Hodge type at every point is Proposition 3.7(ii). ∎

**Lemma 16.3** (One component is enough)**.** *Let $(A, \iota, \eta)$ be of $(K, -1, n)$-Weil type with $\iota(\mathcal{O}_K) \subseteq \mathrm{End}(A)$, and let $W \subset H^{2n}(A, \mathbb{Q})$ be its Weil line. If one nonzero element of $W$ is algebraic, every element of $W$ is algebraic.*

*Proof.* Among the elements $\tau_m = m + \sqrt{-d}$ of $\mathcal{O}_K$ with $1 \le m \le 7$ the ratios $\tau_m/\overline{\tau}_m$ are pairwise distinct, since $\tau_m \overline{\tau}_{m'}$ is rational only when $m = m'$. The group of roots of unity in an imaginary quadratic field has order 2, 4 or 6, so some $m$ in that range has $\tau = \tau_m$ with $\tau/\overline{\tau}$ not a root of unity, and then $\tau^{2n} \neq \overline{\tau}^{2n}$, that is $\tau^{2n} \notin \mathbb{Q}$, for every $n \ge 1$. A range is needed, since the witness $m = 1$ fails for two fields: for $d = 1$ the ratio attached to $1 + \sqrt{-1}$ is a primitive fourth root of unity, and for $d = 3$ the ratio attached to $1 + \sqrt{-3}$ is a primitive cube root of unity, so $(1 + \sqrt{-d})^{2n}$ is rational when $d = 1$ and $n$ is even, and when $d = 3$ and $3 \mid n$.

Since $\tau \in \mathcal{O}_K$, the element $\iota(\tau)$ is an endomorphism of $A$, so $\iota(\tau)^*$ on $H^{2n}(A, \mathbb{Q})$ is induced by the graph correspondence and therefore carries algebraic classes to algebraic classes. On $H^1$ it is multiplication by $\tau$, so on $\bigwedge_K^{2n} V$ it is multiplication by $\tau^{2n}$. As $\tau^{2n} \notin \mathbb{Q}$ and $W$ is a one-dimensional $K$-vector space, the pair $w$, $\tau^{2n} w$ is a $\mathbb{Q}$-basis of $W$ for every $w \neq 0$. Explicitly, fix a $K$-basis $w_0$ of $W$ and write $\tau^{2n} = a + b\sqrt{-d}$, so that $b \neq 0$, and $w = (w_1 + w_2\sqrt{-d})\, w_0$; the determinant of the pair $w, \tau^{2n} w$ in the $\mathbb{Q}$-basis $w_0, \sqrt{-d}\, w_0$ is $b(w_1^2 + d w_2^2)$, which is nonzero when $w \neq 0$. If $w$ is algebraic then so is $\iota(\tau)^* w = \tau^{2n} w$, and the two together span. ∎

Lemma 16.3 writes out the last step of the proof of Corollary 4.15, where the witness $\tau$ has to be chosen.

**Theorem 16.4** (The algebraic locus is a countable union of closed subvarieties)**.** *Keep Setup 16.1. For a finite tuple $\underline{P} = (P_1, \dots, P_m)$ of Hilbert polynomials of degree $n$ and a tuple $\underline{q} \in \mathbb{Q}^m$ put*

$$\mathcal{H}_{\underline{P}} \;=\; \mathrm{Hilb}^{P_1}(\mathcal{A}/\mathcal{S}) \times_{\mathcal{S}} \cdots \times_{\mathcal{S}} \mathrm{Hilb}^{P_m}(\mathcal{A}/\mathcal{S}),$$

$$\Sigma_{\underline{P},\underline{q}} \;=\; \mathrm{pr}\Big( \{\, \underline{Z} \in \mathcal{H}_{\underline{P}} \;:\; \textstyle\sum_i q_i [Z_i] = \omega \,\} \Big) \;\subseteq\; \mathcal{S}.$$

*Then each $\Sigma_{\underline{P},\underline{q}}$ is Zariski closed in $\mathcal{S}$, and*

$$p^{-1}\Big( \bigcup_{\underline{P},\underline{q}} \Sigma_{\underline{P},\underline{q}} \Big) \;=\; \Sigma,$$

*the union running over a countable index set.*

*Proof.* Each $\mathrm{Hilb}^{P_i}(\mathcal{A}/\mathcal{S})$ is projective over $\mathcal{S}$ by Grothendieck's construction of the Hilbert scheme [Gro61], so the fibre product $\mathcal{H}_{\underline{P}}$ is projective over $\mathcal{S}$, and being of finite type it has finitely many connected components.

Fix a connected component $T$, replace it by its reduction, which changes neither the underlying space nor any cohomology group, and let $\mathcal{Z}_i \subset \mathcal{A} \times_{\mathcal{S}} T$ be the pullback of the $i$-th universal subscheme. It is flat over $T$ and its fibres have Hilbert polynomial $P_i$ of degree $n$, so the associated $n$-cycles, the top-dimensional components counted with their multiplicities, are the fibres of a cycle on $\mathcal{A} \times_{\mathcal{S}} T$ whose class restricts on the fibre over $t$ to $[Z_{i,t}]$. Restriction of a global class to the fibres of a topologically locally trivial family is a flat section of the local system, so $t \mapsto \sum_i q_i [Z_{i,t}]$ is flat on $T$; and $\omega$ is flat on $\mathcal{S}$ by Lemma 16.2, hence flat after pullback. Two flat sections of a local system over a connected base agree everywhere as soon as they agree at one point, so the locus where they agree is a union of connected components of $\mathcal{H}_{\underline{P}}$. That locus is therefore itself projective over $\mathcal{S}$, and the image of a proper morphism is Zariski closed.

For the equality of sets, let $s \in \Sigma$. Then $\omega_s = \sum_i q_i[Z_i]$ for finitely many closed integral subschemes $Z_i \subset A_s$ of dimension $n$ and rationals $q_i$; taking $P_i$ to be the Hilbert polynomial of $Z_i$ puts $p(s)$ in $\Sigma_{\underline{P},\underline{q}}$. The converse holds because the class of the $n$-cycle of a closed subscheme of dimension $n$ is algebraic. The index set is countable, since a Hilbert polynomial is determined by its finitely many integer coordinates in the binomial basis, and $\mathbb{Q}$ is countable. ∎

The bound on the Hilbert polynomials cannot be dropped: without it one takes the image of an infinite disjoint union, and an infinite union of closed sets need be neither closed nor constructible, and the dichotomy below is what can be said of $\Sigma$ itself. Tuples are needed because a rational cycle is a rational combination of several integral ones, which may have different Hilbert polynomials. The relative Chow variety of effective $n$-cycles of bounded degree would serve equally well, with the same proof, since only properness over the base and local constancy of the cycle class are used.

**Corollary 16.5** (The dichotomy)**.** *Exactly one of the following holds.*

(a) $\Sigma = \mathcal{D}_{n,n}$*, and the Hodge conjecture holds in every degree for every abelian* $2n$*-fold of* $(K, -1, n)$*-Weil type with* $\mathrm{Hg}(A) = \mathrm{SU}(V, H)$ *in the family.*

(b) $\Sigma$ *has empty interior in the analytic topology. Its complement is then a dense* $G_\delta$*, so* $\omega_s$ *fails to be algebraic for every* $s$ *outside a meagre set, and the Hodge conjecture fails.*

*In particular, if* $\omega_s$ *is algebraic for all* $s$ *in some nonempty open subset of* $\mathcal{D}_{n,n}$*, then it is algebraic for all* $s$*.*

*Proof.* Suppose $\Sigma$ contains a nonempty open set; then so does its image in $\mathcal{S}$, and we may pick a nonempty connected open $U \subseteq \mathcal{S}$ contained in that image. By Theorem 16.4 the countably many sets $\Sigma_{\underline{P},\underline{q}} \cap U$ are closed in $U$ and cover it. A connected complex manifold is a Baire space, so one of them has nonempty interior in $U$. A Zariski closed subset of the irreducible variety $\mathcal{S}$ either is all of $\mathcal{S}$ or is a proper closed subvariety, and a proper closed subvariety of a connected complex manifold has empty interior. Hence that $\Sigma_{\underline{P},\underline{q}}$ is all of $\mathcal{S}$, so $\Sigma = \mathcal{D}_{n,n}$ and (a) holds by Theorem 15.14(a). If $\Sigma$ has empty interior, its complement is the intersection of the countably many dense open sets $\mathcal{D}_{n,n} \setminus p^{-1}(\Sigma_{\underline{P},\underline{q}})$, and Baire again gives that this intersection is dense. At each of its points $\omega_s$ is a Hodge class by Lemma 16.2 and is not algebraic, so the Hodge conjecture fails there. ∎

Figure 19 draws the two cases.

**Lemma 16.6** (Invariance under rational isogenies)**.** $\Sigma$ *is stable under the action of* $G(\mathbb{Q})$ *on* $\mathcal{D}_{n,n}$*.*

*Proof.* A point of $\mathcal{D}_{n,n}$ is a complex structure $J$ on $V_\mathbb{R}$ compatible with $H$, and $g \in G(\mathbb{Q})$ carries $J$ to $gJg^{-1}$. The map $g$ is then $\mathbb{Q}$-rational and $\mathbb{C}$-linear from $(V_\mathbb{R}, J)$ to $(V_\mathbb{R}, gJg^{-1})$, so after clearing denominators it induces an isogeny $A_J \to A_{gJ}$, whose graph is an algebraic correspondence. Algebraicity of a rational cohomology class is preserved by such a correspondence and by its transpose. Finally $g$ lies in $\mathrm{SU}(V, H)$, so by the computation in Lemma 16.2 it fixes the Weil line pointwise; the correspondence therefore carries $\omega_J$ to a nonzero rational multiple of $\omega_{gJ}$, and $\omega_{gJ}$ is algebraic when $\omega_J$ is. ∎

Two density statements should be distinguished. For a class that does not remain of Hodge type over the whole family, the locus where it is of Hodge type is a countable union of proper closed subvarieties, each algebraic by Theorem B.6, and density there concerns the distribution of special points. For the class $\omega$ of Setup 15.12 the Hodge locus is the

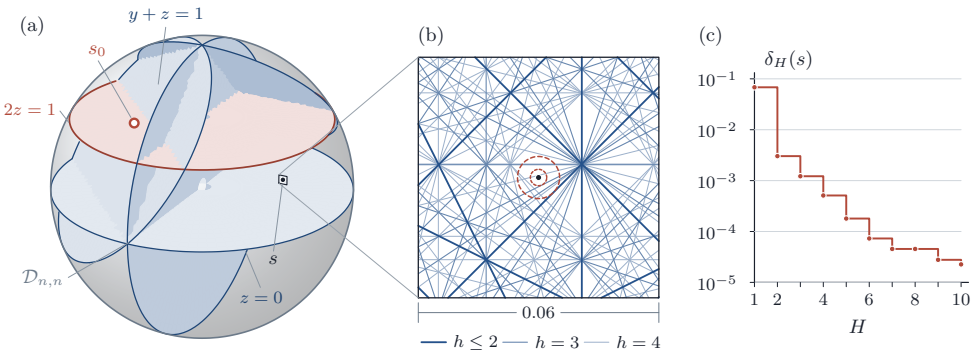



FIGURE 19. The dichotomy of Corollary 16.5. Each stratum $\Sigma_{\underline{P},\underline{q}}$ is Zariski closed by Theorem 16.4, and there are countably many of them; the base point $s_0$ of Theorem 15.27 lies on one. By Proposition 16.7 the union of the strata is dense, so every neighbourhood of a point $s$ lying off all of them meets infinitely many strata. The conjecture for the family is the assertion that no such $s$ exists, and by Corollary 16.5 it is enough that one stratum has nonempty interior.

whole family by Proposition 3.7(ii), so Theorem B.6 gives no information, and the set that carries information is the algebraic locus. The next proposition concerns that set, and its proof uses no Hodge theory: the algebraic locus is a union of orbits of a group whose rational points are dense in its real points.

**Proposition 16.7** (The algebraic locus is dense)**.** *Let $n \ge 2$. Then $\Sigma$ is dense in $\mathcal{D}_{n,n}$ in the analytic topology, and its image in $\mathcal{S}$ is dense.*

*Proof.* The hermitian form $H$ has rank $2n \ge 4$ over $K$ and signature $(n,n)$, so it is isotropic over $\mathbb{R}$; over every $\mathbb{Q}_p$ a hermitian form of rank at least three over a quadratic extension is isotropic; and hermitian forms over an imaginary quadratic field satisfy the Hasse principle [Sch85, Ch. 10, §6]. So $H$ is isotropic over $\mathbb{Q}$ and $G$ is $\mathbb{Q}$-isotropic. The group $G = \mathrm{SU}(V,H)$ is simply connected and almost $\mathbb{Q}$-simple, so by Platonov's theorem $G(\mathbb{Q})$ is generated by the rational points of the unipotent radicals of its proper parabolic $\mathbb{Q}$-subgroups [PR94, Thm. 7.6 and §7.2]. For a unipotent group $U$ over $\mathbb{Q}$ the exponential identifies $U(\mathbb{Q})$ with a $\mathbb{Q}$-vector space inside the corresponding $\mathbb{R}$-vector space $U(\mathbb{R})$, so $U(\mathbb{Q})$ is dense in $U(\mathbb{R})$. The closure of $G(\mathbb{Q})$ in $G(\mathbb{R})$ therefore contains $U(\mathbb{R})$ for every such $U$, and those generate the identity component $G(\mathbb{R})^+$, which acts transitively on $\mathcal{D}_{n,n}$. Hence every $G(\mathbb{Q})$-orbit in $\mathcal{D}_{n,n}$ is dense.

By Theorem 15.27 the set $\Sigma$ is nonempty, and by Lemma 16.6 it is $G(\mathbb{Q})$-stable, so it contains a dense orbit. The projection $p$ is continuous and surjective, so the image is dense. ∎

In particular, if case (b) of Corollary 16.5 occurs, then $\Sigma$ and its complement are both dense, and the two sets are distinguished by Baire category and not by position.

**Corollary 16.8** (A Baire category criterion)**.** *Let $n \ge 2$. The Hodge conjecture holds for every abelian $2n$-fold of $(K,-1,n)$-Weil type with $\mathrm{Hg}(A) = \mathrm{SU}(V,H)$ in the family of Setup 15.12 if and only if $\Sigma$ is non-meagre in $\mathcal{D}_{n,n}$, and if and only if $\Sigma$ has nonempty interior.*

*Proof.* If $\Sigma = \mathcal{D}_{n,n}$, both conditions hold, and so does the conjecture, by Theorem 15.14(a). Otherwise case (b) of Corollary 16.5 holds, so $\Sigma$ has empty interior; it is then the countable union of the closed sets $p^{-1}(\Sigma_{\underline{P},\underline{q}})$, each with empty interior, and so it is meagre. The points $s$ with $\mathrm{Hg}(A_s) = \mathrm{SU}(V,H)$ form the complement of a countable union of proper closed analytic subsets of $\mathcal{D}_{n,n}$, by Weil's theorem [Wei77] recalled before Theorem 4.14, so by Baire they are not all contained in the meagre set $\Sigma$. At a point of this kind outside $\Sigma$ the Hodge conjecture fails for $A_s$. ∎

16.2. **The first criterion: a bound along one orbit.** The proof of Proposition 16.7 produces the dense algebraic locus by transporting one cycle along rational isogenies, and an isogeny multiplies degrees. The dense set so obtained is therefore stratified by unbounded data, and this is what prevents Corollary 16.5 from applying to it. A bound on the data along one orbit removes the difficulty.

**Theorem 16.9** (A bound along the orbit closes the family)**.** *Let $n \geq 2$ and let $s_0 \in \Sigma$ be one of the base points supplied by Theorem 15.27. Suppose that there is a finite set $\mathcal{T}$ of pairs $(\underline{P},\underline{q})$ such that for every $g \in G(\mathbb{Q})$ the class $\omega_{g\cdot s_0}$ is represented by some cycle whose Hilbert data lies in $\mathcal{T}$. Then $\Sigma = \mathcal{D}_{n,n}$, and the Hodge conjecture holds in every degree for every abelian $2n$-fold of $(K,-1,n)$-Weil type with $\mathrm{Hg}(A) = \mathrm{SU}(V,H)$ in that family.*

*Proof.* The orbit $G(\mathbb{Q})\cdot s_0$ is dense by the proof of Proposition 16.7, and by hypothesis it is the union of the finitely many sets $G(\mathbb{Q})\cdot s_0 \cap p^{-1}(\Sigma_{\underline{P},\underline{q}})$ with $(\underline{P},\underline{q}) \in \mathcal{T}$. The union of the finitely many closed sets $p^{-1}(\Sigma_{\underline{P},\underline{q}})$, $(\underline{P},\underline{q}) \in \mathcal{T}$, therefore contains a dense set, and so it is all of $\mathcal{D}_{n,n}$; in particular $\Sigma = \mathcal{D}_{n,n}$. In fact one of these sets is already all of $\mathcal{S}$: each $\Sigma_{\underline{P},\underline{q}}$ is Zariski closed in the irreducible variety $\mathcal{S}$, hence either equal to $\mathcal{S}$ or nowhere dense, and a finite union of nowhere dense sets is nowhere dense. The Hodge conjecture for the family follows by Theorem 15.14(a). ∎

*Remark* 16.10 (Choice of the representing cycle). We call the hypothesis of Theorem 16.9 the bounded criterion (P1); Remark 17.7 lists it beside the rank criterion (P2) of Theorem 16.29. It quantifies over representing cycles at each point of the orbit and asks only that, at each point, one of them have bounded complexity. A stronger hypothesis would fix the cycle, taking at $g\cdot s_0$ the transport of the cycle at $s_0$ along the isogeny of Lemma 16.6. Theorem 16.15 below shows that this stronger form is never satisfied by a base cycle with a component of finite stabiliser, since the transported data is then unbounded. The proof above uses only the weaker hypothesis, which is what the covering argument needs.

The hypothesis of Theorem 16.9 can be sharpened by locating the part of the Hilbert data that is at issue. The leading coefficient is not.

**Proposition 16.11** (The leading coefficient is constant on the orbit)**.** *Let $s_0 \in \Sigma$ and $g \in G(\mathbb{Q})$, and let $\omega_s$ denote the Weil class at $s$ and $\eta$ the polarisation class. Then*

$$\int_{A_{g\cdot s_0}} \omega_{g\cdot s_0}\cdot\frac{\eta^n}{n!} \;=\; \int_{A_{s_0}} \omega_{s_0}\cdot\frac{\eta^n}{n!},$$

*so the degree of the transported class with respect to the polarisation, which is the leading coefficient of the Hilbert polynomial of any cycle representing it, is the same at every point of the orbit.*

*Proof.* By Lemma 16.6 the correspondence transporting the class is induced by a $\mathbb{Q}$-rational map $g \in \mathrm{SU}(V,H)$. It preserves the hermitian form and hence the polarisation class, it fixes the Weil line pointwise by Lemma 16.2, and its determinant over $\mathbb{Q}$ is $\mathrm{N}_{K/\mathbb{Q}}(\det_K g) = 1$,

so it acts trivially on the top exterior power of $V^\vee$, in which both integrals are evaluated. The two integrals are therefore equal. The identification of the number with the leading coefficient is Hirzebruch–Riemann–Roch on $A$, whose Todd class is 1 (for the embedding by $3\eta$ of Setup 16.1 the leading coefficient is $3^n$ times the number); the remaining coefficients involve the structure sheaf of the cycle and not only its class. ∎

*Remark* 16.12 (The source of unboundedness). Write a representing cycle at a point of the orbit, the transported cycle or any other, as a difference $\xi^+ - \xi^-$ of effective cycles. By Proposition 16.11 the difference of their degrees is constant along the orbit, and by the boundedness of effective cycles of bounded degree in the fibres of the polarised family $f\colon \mathcal{A} \to \mathcal{S}$, if $\deg \xi^+$ were bounded along the orbit then only finitely many tuples $(\underline{P}, \underline{q})$ could occur and Theorem 16.9 would apply. So the hypothesis of that theorem is equivalent to a bound on the effective part alone.

This bound is not automatic, and the mechanism that threatens it can be seen directly. The transport at $g$ is by the isogeny obtained from $g$ by clearing denominators; if $g$ has denominator $m$ on the reference lattice, which has rank $4n$, the isogeny has degree $m^{4n}$ (Lemma 16.13 below), and pushing a cycle forward along it multiplies its multiplicities and degrees by powers of $m$ (Lemma 16.14). Elements of $G(\mathbb{Q})$ of bounded denominator form a discrete subset of $G(\mathbb{R})$, so along a dense orbit the denominators, and with them the isogeny degrees, are unbounded. What Theorem 16.9 asks is that the cycles nevertheless simplify: that at each point of the orbit the class $\omega$, whose degree has not changed, admits an effective representation no more complicated than before. For the transported cycle itself the answer is negative as soon as a component of the base cycle has finite stabiliser (Theorem 16.15 below), and for a cycle chosen at each point the bound is equivalent to the conclusion (Theorem 16.17). The present remark shows only that the question reduces from the whole Hilbert datum to the degree of the effective part, and that the part of the datum determined by cohomology is constant.

16.3. **Transport along isogenies.** The mechanism of Remark 16.12 can be made explicit, and doing so settles one of the two readings of (P1): the reading in which the cycle at $s_0$ is transported along the orbit rather than chosen afresh at each point. Write $\Lambda$ for the reference lattice, of rank $4n$ over $\mathbb{Z}$, and $E$ for the alternating form of the polarisation, so that a point $s$ of $\mathcal{D}_{n,n}$ is a complex structure on $\Lambda \otimes \mathbb{R}$ and $A_s = (\Lambda \otimes \mathbb{R})/\Lambda$. Throughout, $\dim A_s = 2n$, and a component of a cycle representing $\omega_s$ has dimension $n$.

**Lemma 16.13** (The transport in coordinates)**.** *Let $g \in G(\mathbb{Q})$ and let $c = c(g)$ be the least positive integer with $cg\Lambda \subseteq \Lambda$. Then $\varphi_g := cg$ induces an isogeny $A_{s_0} \to A_{g\cdot s_0}$ with*

(i) $\deg \varphi_g = c^{4n}$;

(ii) $\varphi_g^* \eta_{g\cdot s_0} = c^2 \eta_{s_0}$ *and* $\varphi_g^* \omega_{g\cdot s_0} = c^{2n} \omega_{s_0}$;

(iii) *the transported cycle representing $\omega_{g\cdot s_0}$ is $c^{-2n}\, \varphi_{g*}(Z)$, where $Z$ represents $\omega_{s_0}$.*

*Moreover* $\sup_{g \in G(\mathbb{Q})} c(g) = \infty$.

*Proof.* (i) The index of $cg\Lambda$ in $\Lambda$ is $|\det_{\mathbb{Q}}(cg)| = c^{4n} |\det_{\mathbb{Q}} g|$, and $\det_{\mathbb{Q}} g = \mathrm{N}_{K/\mathbb{Q}}(\det_K g) = 1$ for $g \in \mathrm{SU}(V, H)$.

(ii) $g$ preserves $E$, so $(cg)^* E = c^2 E$, which is the first identity. On $H^{2n} = \bigwedge^{2n}(\Lambda^\vee \otimes \mathbb{Q})$ the map $\varphi_g^*$ is $c^{2n} \bigwedge^{2n} g^\vee$, and $g \in \mathrm{SU}(V, H)$ fixes the Weil line pointwise by Lemma 16.2.

(iii) $\varphi_{g*}\varphi_g^* = \deg \varphi_g$, so $\omega_{g\cdot s_0} = (\deg \varphi_g)^{-1} \varphi_{g*}\varphi_g^* \omega_{g\cdot s_0} = c^{-4n}\, c^{2n} \varphi_{g*} \omega_{s_0} = c^{-2n} \varphi_{g*} \omega_{s_0}$ by (i) and (ii).

For the last sentence, suppose that $c(g) \le C$ for every $g \in G(\mathbb{Q})$. Then $(C!)\, g\Lambda \subseteq \Lambda$ for every $g$, so $G(\mathbb{Q})$ lies in the discrete subset $(C!)^{-1} \mathrm{End}_{\mathbb{Z}}(\Lambda)$ of $\mathrm{End}(\Lambda \otimes \mathbb{R})$. But by

the proof of Proposition 16.7 the closure of $G(\mathbb{Q})$ in $G(\mathbb{R})$ contains $G(\mathbb{R})^+$, a Lie group of positive dimension, which is a contradiction. ∎

**Lemma 16.14** (Push-forward along an isogeny)**.** *Let $\varphi\colon A \to A'$ be an isogeny of abelian varieties and $Z \subseteq A$ an integral closed subvariety. Put $G(Z) = \{x \in A : Z + x = Z\}$, a closed algebraic subgroup. Then*

$$\varphi_*[Z] = m\,[\varphi(Z)], \qquad m = \bigl|\ker\varphi \cap G(Z)\bigr|.$$

*If moreover $\dim Z = n$ and $\varphi^*\eta' = c^2\eta$, then*

$$\deg_{\eta'}\varphi(Z) = \frac{c^{2n}}{m}\,\deg_{\eta} Z.$$

*Proof.* The map $\varphi|_Z\colon Z \to \varphi(Z)$ is finite and surjective, and its degree $m$ is the number of points of a general fibre. For $z \in Z$ the fibre through $z$ is $Z \cap (z + \ker\varphi)$, which as a set of translations is $\{t \in \ker\varphi : z+t \in Z\}$. For a fixed $t$ the locus $\{z \in Z : z+t \in Z\} = Z \cap (Z-t)$ is all of $Z$ when $t \in G(Z)$ and a proper closed subset otherwise; since $\ker\varphi$ is finite, a general $z$ avoids the finitely many proper subsets, and there $\{t \in \ker\varphi : z + t \in Z\} = \ker\varphi \cap G(Z)$. This is the first identity, and $\varphi_*[Z] = m[\varphi(Z)]$ by the definition of the push-forward of cycles.

For the degree, the projection formula gives

$$m\deg_{\eta'}\varphi(Z) = \int \varphi_*[Z]\cdot\frac{\eta'^n}{n!} = \int [Z]\cdot\frac{(\varphi^*\eta')^n}{n!} = c^{2n}\int [Z]\cdot\frac{\eta^n}{n!} = c^{2n}\deg_{\eta} Z.$$

∎

For example, for multiplication by $c$ on an abelian variety of dimension $2n$, an abelian subvariety $Z$ of dimension $n$ has $G(Z) = Z$, so $m = |Z[c]| = c^{2n}$ and $\deg\varphi(Z) = \deg Z$, while a $Z$ whose stabiliser meets $A[c]$ trivially has $m = 1$ and $\deg\varphi(Z) = c^{2n}\deg Z$.

**Theorem 16.15** (The transported cycle)**.** *Let $s_0 \in \Sigma$ and let $Z = \sum_i q_i Z_i$, with $q_i \neq 0$ and the $Z_i$ integral of dimension $n$, represent $\omega_{s_0}$.*

(i) *For $g \in G(\mathbb{Q})$ the transported cycle is $\sum_i c(g)^{-2n} m_i(g)\, q_i\,[\varphi_g(Z_i)]$ with $m_i(g) = |\ker\varphi_g \cap G(Z_i)|$, and the $i$-th component has degree $c(g)^{2n}\deg_\eta(Z_i)/m_i(g)$.*

(ii) *If some $Z_i$ has finite stabiliser $G(Z_i)$, then $\deg\varphi_g(Z_i) \geq c(g)^{2n}\deg_\eta(Z_i)/|G(Z_i)|$, which is unbounded on $G(\mathbb{Q})$. The Hilbert data of the transported cycle then takes infinitely many values, and the hypothesis of Theorem 16.9 in the form that fixes the transported cycle is not satisfied by $Z$.*

(iii) *Consequently, a base cycle for which the transported data is bounded must have every component of positive-dimensional stabiliser, that is, every $Z_i$ must be stable under translation by a positive-dimensional abelian subvariety of $A_{s_0}$.*

*Proof.* (i) is Lemma 16.13(iii) and Lemma 16.14.

(ii) $\ker\varphi_g \cap G(Z_i) \subseteq G(Z_i)$, so $m_i(g) \leq |G(Z_i)|$, a constant; the displayed bound is (i), and $c(g)$ is unbounded by Lemma 16.13. Distinct degrees of the components give distinct Hilbert polynomials, so the data takes infinitely many values.

(iii) is the contrapositive of (ii), together with the fact that an algebraic subgroup of positive dimension in an abelian variety has an abelian subvariety of positive dimension as its identity component. ∎

*Remark* 16.16 (The two readings of (P1))*.* Three points should be kept apart.

First, (P1) has two readings, and Theorem 16.15 settles one of them. Read with the cycle fixed at $s_0$ and transported, it fails for every base cycle written down in this paper:

those cycles are intersections of divisors, and an intersection of $n$ divisors on an abelian $2n$-fold has finite stabiliser unless the divisors are pulled back from a quotient, which the ones here are not. Read with the cycle chosen afresh at each point of the orbit, which is the form of the hypothesis of Theorem 16.9, it is not affected by Theorem 16.15.

Second, the difference between the two readings is the content of Lemma 16.14. The class $\omega_{g\cdot s_0}$ has constant degree along the orbit, by Proposition 16.11; what grows is the complexity of the particular representative that the transport produces, because the isogeny spreads each component out by the factor $c^{2n}/m_i$. When a component of the base cycle has finite stabiliser, the transport does not produce a representative of bounded complexity, whether or not one exists.

Third, an approach to (P1) must either produce a base cycle supported on subvarieties with positive-dimensional stabiliser, a strong requirement for a Weil class, or abandon the transport and bound instead the minimal complexity of a representative, which is a statement about the Chow variety of the family and not about any one cycle. Neither is achieved here.

The bound on the minimal complexity of a representative is the hypothesis of Theorem 16.9, and it is the weakest hypothesis under which the covering argument runs. The next theorem shows that it is equivalent to the conclusion. Write $\mathbf{W}(K,n,\delta)$ for the statement that every abelian $2n$-fold of $(K,-1,n)$-Weil type with discriminant $\delta$ (Definition 15.22) has algebraic Weil classes, $\delta$ being the discriminant of the family of Setup 15.12.

**Theorem 16.17** (Equivalence of the bounded criterion)**.** *Let $n \geq 2$, let $s_0 \in \Sigma$ and keep Setup 15.12. The following are equivalent.*

(a) *There is a finite set $\mathcal{T}$ of pairs $(\underline{P},\underline{q})$ such that at every point of the orbit $G(\mathbb{Q})\cdot s_0$ the class $\omega$ is represented by a cycle whose Hilbert data lies in $\mathcal{T}$; that is,* (P1) *holds at $s_0$.*

(b) *There is a single pair $(\underline{P},\underline{q})$ such that at every point of $\mathcal{D}_{n,n}$, not merely of the orbit, the class $\omega$ is represented by a cycle with that datum.*

(c) $\Sigma = \mathcal{D}_{n,n}$.

(d) $\mathbf{W}(K,n,\delta)$ *holds, that is, the Weil classes are algebraic on every member of the family.*

*In particular Theorem 16.9 is an equivalence, and* (P1) *is a reformulation of the conjecture for the family and not a reduction of it. The same holds for every CM field, with Theorem 19.7(iv) in place of Theorem 16.9.*

*Proof.* (a) implies (c) is Theorem 16.9. (b) implies (a) is immediate, the orbit being contained in $\mathcal{D}_{n,n}$ and $\mathcal{T}$ being the one pair. (d) implies (c) trivially, and (c) implies (d) because the algebraicity of $\omega_s$ makes the whole Weil line of $A_s$ algebraic, by the argument of Corollary 4.15, written out in Lemma 16.3.

It remains to deduce (b) from (c). If $\Sigma = \mathcal{D}_{n,n}$ then $\Sigma$ has nonempty interior, and the proof of Corollary 16.5 shows that some $\Sigma_{\underline{P},\underline{q}}$ is then all of $\mathcal{S}$: the countably many sets $\Sigma_{\underline{P},\underline{q}} \cap U$ are closed in a nonempty connected open $U \subseteq \mathcal{S}$ and cover it, so by Baire one of them has nonempty interior in $U$, and a proper Zariski closed subvariety of the irreducible variety $\mathcal{S}$ has empty interior. By the definition of $\Sigma_{\underline{P},\underline{q}}$ in Theorem 16.4 as the image of a proper morphism, every $s \in \mathcal{S}$ then carries a tuple $\underline{Z}$ of subschemes with Hilbert polynomials $\underline{P}$ and $\sum_i q_i[Z_i] = \omega$, which is (b).

For a CM field the same argument applies without change, using Theorem 19.7(iii) in place of Theorem 16.4 and Corollary 16.5. ∎

*Remark* 16.18 (Status of the first criterion). By Theorem 16.17, (P1) is a restatement of the conjecture for the family and not a route to it. Theorem 16.9 is a valid implication, and its covering argument is the only place in the paper where density of the rational orbit is converted into a global statement; but the implication has a converse, so verifying its hypothesis for a family is as hard as proving the conclusion for that family. The boundedness is nevertheless a concrete finiteness statement about the Chow variety of the universal family, and a reformulation of $\mathbf{W}(K,n,\delta)$ that mentions neither Hodge theory nor deformation theory.

The two propagation criteria are therefore not on the same footing. The rank criterion (P2) of Theorem 16.29 is a sufficient condition with no evident converse: injectivity of one semiregularity map at one point implies the conjecture for the whole family, while $\Sigma = \mathcal{D}_{n,n}$ carries no information about the deformation theory of any particular perfect complex. In its pure form, with Chern character a Weil class, it is never satisfied (Theorem 17.5); the polarised criterion (P2′) of Remark 17.6, in which the Chern character may also carry powers of the polarisation, is the form studied in Section 17.

16.4. **Products of divisors.** The intersection numbers needed below are Pfaffians. Let $E$ be a nondegenerate alternating form on a real vector space $V$ of dimension $2g$, let $\eta$ be its class, with the orientation for which $e = \int \eta^g > 0$, and for an $E$-self-adjoint endomorphism $\phi$ write $x_\phi(u,v) = E(\phi u, v)$, an alternating form.

**Lemma 16.19** (Pfaffians of self-adjoint classes). *For real $s$, $\int (s\eta + x_\phi)^g = e\, p_\phi(s)$, where $p_\phi$ is the monic polynomial of degree $g$ with $p_\phi^2 = \det(s+\phi)$. In particular:*

(i) $2g \int x_\phi \eta^{g-1} = e \operatorname{tr}\phi$, *and if* $\operatorname{tr}\phi = 0$ *then* $2g(g-1)\int x_\phi^2 \eta^{g-2} = -e \operatorname{tr}(\phi^2)$;

(ii) *if* $\operatorname{tr}\phi = 0$ *and* $\phi^2 = \beta$ *is a nonzero real scalar, then $g$ is even and* $\int x_\phi^g = e\,(-\beta)^{g/2}$.

*Proof.* In a basis with Gram matrix $G$ for $E$, the class $s\eta + x_\phi$ has Gram matrix $(s+\phi)^T G$, of determinant $\det G \det(s+\phi)$. Its Pfaffian divided by $\operatorname{Pf}(G)$ is therefore a polynomial in $s$, monic of degree $g$, whose square is $\det(s+\phi)$; and $\int \omega^g/e$ is $\operatorname{Pf}(\omega)/\operatorname{Pf}(G)$ for every alternating form $\omega$. This is the identity. Write $p_\phi(s) = s^g + p_1 s^{g-1} + p_2 s^{g-2} + \cdots$ and compare with $\det(s+\phi) = s^{2g} + \operatorname{tr}(\phi)\, s^{2g-1} + \frac{1}{2}\big((\operatorname{tr}\phi)^2 - \operatorname{tr}(\phi^2)\big)s^{2g-2} + \cdots$: then $2p_1 = \operatorname{tr}\phi$ and $2p_2 + p_1^2 = \frac{1}{2}\big((\operatorname{tr}\phi)^2 - \operatorname{tr}(\phi^2)\big)$, while the binomial expansion of $\int (s\eta + x_\phi)^g$ gives $g\int x_\phi \eta^{g-1} = e\,p_1$ and $\binom{g}{2}\int x_\phi^2 \eta^{g-2} = e\,p_2$; this is (i). Under the hypothesis of (ii) the eigenvalues of $\phi$ are $\pm\sqrt{\beta}$, each with multiplicity $g$ because the trace vanishes, so $\det(s+\phi) = (s^2-\beta)^g$; this is the square of a polynomial only for $g$ even, and then $p_\phi(s) = (s^2-\beta)^{g/2}$ and $\int x_\phi^g = e\,p_\phi(0)$. ∎

At the quaternionic points of Setup 15.24 the mechanism of Remark 16.12 can be seen explicitly. Take $n = 2$ and the integral lattice $\Lambda_b = \mathcal{O}_b^2$, $\mathcal{O}_b = \mathbb{Z}\langle 1, i, j, ij\rangle$, for $b \in \mathbb{Z}_{>0}$. Put

$$g_b(x,y) = E_b(jx,y), \qquad \lambda_b(x,y) = -\tfrac{1}{d}\, g_b(ix,y), \qquad Z_b = g_b^2 - d\lambda_b^2, \qquad Z_b' = 2g_b\lambda_b,$$

so that $Z_b + \sqrt{-d}\, Z_b' = (g_b + \sqrt{-d}\,\lambda_b)^2$ and $Z_b, Z_b'$ span the Weil plane. Write $x \in \mathcal{O}_b$ as $x = u + vj$ with $u, v \in \mathbb{Z}[i]$, where $i^2 = -d$ as in Setup 15.24, so that $\mathbb{Z}[i] = \mathbb{Z}[\sqrt{-d}]$; the pair $(\Lambda_b, i)$ is then the same $\mathbb{Z}[i]$-module $\mathbb{Z}[i]^4$ for every $b$, and we use this identification throughout.

**Proposition 16.20** (Divisibility of the quaternionic Weil cycle). *With these data, for every $b \in \mathbb{Z}_{>0}$:*

(i) $E_b = E_u + b\,E_v$, $g_b = b\,g_1$ *and* $\lambda_b = b\,\lambda_1$, *where* $E_u, E_v, g_1, \lambda_1$ *are integral on* $\mathbb{Z}[i]^4$ *and do not depend on* $b$*; in particular* $g_1, \lambda_1 \in \mathrm{NS}(A_b)$*;*

(ii) $Z_b = b^2 Z_1$ *and* $Z_b' = b^2 Z_1'$ *in* $\mathrm{CH}^2(A_b)$ *modulo algebraically trivial cycles, for every choice of line bundles representing* $g_b, \lambda_b$*; in cohomology the equalities hold in* $\wedge^4\Lambda^*$*;*

(iii) $\int \eta_b^4 = b^2 \int \eta_1^4$, $\int g_b^2\eta_b^2 = -\frac{b}{3}\int \eta_b^4$ *and* $\int Z_b^2 = \frac{4b^2}{3}\int \eta_b^4$*, so that* $(\int g_b^2\eta_b^2)^2 = \frac{1}{12}\int Z_b^2 \int \eta_b^4$*;*

(iv) *for* $\alpha \in K^\times$ *the element* $\alpha j$ *generates the same algebra at the same point, and replacing* $j$ *by* $\alpha j$ *multiplies the two ratios in* (iii) *by* $N(\alpha)$ *and* $N(\alpha)^2$*.*

*Proof.* For $x = u + vj$ one has $\overline{x} = \overline{u} - vj$, and the relations $ju = \overline{u}j$, $j^2 = b$ give $\overline{x}\,i\,y \equiv \overline{u}\,i\,u' + b\,v\,i\,\overline{v'}$ modulo $Kj$. Since trd vanishes on $Kj$ and restricts to $\mathrm{tr}_{K/\mathbb{Q}}$ on $K$,

$$E_b(x,y) = \textstyle\sum_k a_k\big(\mathrm{tr}(i\,\overline{u_k}u_k') + b\,\mathrm{tr}(i\,v_k\overline{v_k'})\big).$$

From $jx = b\overline{v} + \overline{u}j$ one reads off $g_b(x,y) = b\sum_k a_k\big(\mathrm{tr}(i\,v_k u_k') + \mathrm{tr}(i\,\overline{u_k}\overline{v_k'})\big)$, which is $b\,g_1$, and $\lambda_b = b\lambda_1$ follows. The traces take integral values on $\mathbb{Z}[i]$, and $g_1 = b^{-1}g_b$ and $\lambda_1 = b^{-1}\lambda_b$ are rational multiples of Hodge classes, so they are integral Hodge classes and lie in $\mathrm{NS}(A_b)$ by the Lefschetz theorem on $(1,1)$-classes; this proves (i). Then $g_b + \sqrt{-d}\,\lambda_b = b(g_1 + \sqrt{-d}\,\lambda_1)$ and (ii) holds in cohomology. Choose $L, M$ with $c_1(L) = g_1$ and $c_1(M) = \lambda_1$. The cycle $c_1(L^b)^2 - d\,c_1(M^b)^2$ equals $b^2(c_1(L)^2 - d\,c_1(M)^2)$ in $\mathrm{CH}^2(A_b)$, and another choice changes both sides by algebraically trivial cycles.

For (iii), $E_u$ and $E_v$ live on complementary halves $U, V'$ of $\Lambda \otimes \mathbb{Q}$ of rank four, so $\eta_u^3 = \eta_v^3 = 0$ and $\int \eta_b^4 = 6b^2 \int \eta_u^2\eta_v^2$. The form $g_b$ is $x_\phi$ for $\phi = j$, which is $E_b$-self-adjoint by Theorem 15.25(i), anticommutes with $i$, so has trace 0, and has $j^2 = b$; hence Lemma 16.19(i), with $g = 4$, gives $\int g_b^2\eta_b^2 = -\frac{1}{24}\mathrm{tr}(j^2)\int \eta_b^4 = -\frac{8b}{24}\int \eta_b^4$. For real $a, c$ the class $a g_b + c\lambda_b$ is $x_\phi$ for $\phi = j(a - \frac{c}{d}\,i)$, again self-adjoint of trace 0, and $\phi^2 = (a + \frac{c}{d}i)\,j^2\,(a - \frac{c}{d}i) = b\,(a^2 + c^2/d)$. So Lemma 16.19(ii) gives $\int (a g_b + c\lambda_b)^4 = b^2(a^2 + c^2/d)^2 \int \eta_b^4$, and comparing coefficients, $\int g_b^4 = b^2 \int \eta_b^4$, $\int g_b^2\lambda_b^2 = \frac{b^2}{3d}\int \eta_b^4$ and $\int \lambda_b^4 = \frac{b^2}{d^2}\int \eta_b^4$. Hence $\int Z_b^2 = \int g_b^4 - 2d\int g_b^2\lambda_b^2 + d^2\int \lambda_b^4 = \frac{4b^2}{3}\int \eta_b^4$.

For (iv), $(\alpha j)^2 = N(\alpha)b$, and $g_{\alpha j} + \sqrt{-d}\,\lambda_{\alpha j}$ is $\alpha$ or $\overline{\alpha}$ times $g_j + \sqrt{-d}\,\lambda_j$. ∎

*Remark* 16.21 (Growth of the polarisation). At these points no multiple growing with $b$ enters when the Weil class is written in integral terms: by Proposition 16.20(ii) the cycle $Z_b$ is $b^2 Z_1$, with $Z_1$ one integral algebraic cycle built from the two fixed divisor classes $g_1, \lambda_1$. The factor that the orbit argument would otherwise have to absorb into the coefficients $\underline{q}$ of Theorem 16.9 is therefore absent. What grows is the polarisation: $\int \eta_b^4 = b^2 \int \eta_1^4$, so the points $A_b$ carry polarisations of unbounded degree, and an isogeny carrying $\eta_b$ to the polarisation of one fixed family has degree $\int \eta_b^4 / \int \eta^4$, which grows like $b^2$. This is the unbounded isogeny degree of Remark 16.12, located at an explicit family of points, and the integrality of $Z_1$ does not affect it.

The base points of Theorem 15.27, and every algebraicity statement for Weil classes proved in this paper, represent $\omega$ through products of divisor classes. The next result shows that a representation of that kind can hold with bounded degrees only on a nowhere dense set, so the bound asked for in Theorem 16.9 has to come from cycles that are not products of divisors. For a class $x \in H^2(A_s, \mathbb{R})$ put

$$t(x) = \frac{\int x\,\eta^{2n-1}}{\int \eta^{2n}}, \qquad I(x) = t(x)^2 \int \eta^{2n} - \int (x - t(x)\eta)^2\,\eta^{2n-2}.$$

Since $\mathcal{D}_{n,n}$ is contractible, the local system $R^2\pi_*\mathbb{Z}$ is trivial; write $H^2_{\mathbb{Z}}$ for its fibre, identified with $H^2(A_s,\mathbb{Z})$ at every $s$. Both $t$ and $I$ are polynomial functions on $H^2_{\mathbb{Z}}$ and do not depend on $s$.

**Lemma 16.22** (The Hodge norm of a divisor)**.** *Let $h_s$ be the Hodge norm on $H^2(A_s,\mathbb{R})$ defined by the polarisation, normalised by $h_s(\eta)=\int\eta^{2n}$. Then*

(i) *$h_s(x)=I(x)$ for every real class $x$ of type $(1,1)$ at $s$;*

(ii) *if $D$ is an effective divisor on $A_s$ then $I([D])\le 2\bigl(\int[D]\eta^{2n-1}\bigr)^2/\int\eta^{2n}$.*

*Proof.* Write $x=t\eta+x_0$ with $t=t(x)$, so that $x_0$ is primitive. The Lefschetz decomposition is orthogonal for $h_s$, and on primitive classes $h_s(x_0)=-\int x_0\wedge C_sx_0\wedge\eta^{2n-2}$, where $C_s$ is the Weil operator, positive by the Hodge–Riemann relations. For $x_0$ of type $(1,1)$, $C_sx_0=x_0$, which gives (i). An effective divisor on an abelian variety is nef, since a translate of it meets any given curve properly, so $\int[D]^2\eta^{2n-2}\ge 0$. Expanding, $\int[D]^2\eta^{2n-2}=t^2\int\eta^{2n}+\int x_0^2\eta^{2n-2}$, hence $-\int x_0^2\eta^{2n-2}\le t^2\int\eta^{2n}$ and $I([D])\le 2t^2\int\eta^{2n}$, which is (ii). ∎

**Proposition 16.23** (Divisor products of bounded degree)**.** *In Setup 15.12, fix $C>0$ and let $\Sigma_C\subset\mathcal{D}_{n,n}$ be the set of points $s$ at which $\omega_s$ is a $\mathbb{Q}$-linear combination of classes $[D_1]\cdots[D_n]$ with every $D_i$ an effective divisor on $A_s$ of degree $\int[D_i]\eta^{2n-1}\le C$. Then $\Sigma_C$ is contained in a locally finite union of proper closed analytic subsets of $\mathcal{D}_{n,n}$. In particular $\Sigma_C$ is nowhere dense, and on any dense subset of $\mathcal{D}_{n,n}$ at which $\omega$ is a combination of products of effective divisors the degrees of those divisors are unbounded on every nonempty open set.*

*Proof.* For $x\in H^2_{\mathbb{Z}}$ let $\mathrm{Hdg}(x)\subset\mathcal{D}_{n,n}$ be the closed analytic set where $x$ is of type $(1,1)$. If $x\notin\mathbb{Q}\eta$ it is proper, because $\mathrm{NS}(A_s)\otimes\mathbb{Q}=\mathbb{Q}\eta$ at a very general point by Lemma 7.5. Let $s\in\Sigma_C$. Not every $[D_i]$ in the representation lies in $\mathbb{Q}\eta$, since $\mathbb{Q}\eta^n$ and the Weil line carry different characters by Lemma 5.4. So $s\in\mathrm{Hdg}(x)$ for some $x$ in

$$X_C=\bigl\{x\in H^2_{\mathbb{Z}}\setminus\mathbb{Q}\eta\ :\ I(x)\le 2C^2/\textstyle\int\eta^{2n}\bigr\},$$

by Lemma 16.22(ii). Now let $U\subset\mathcal{D}_{n,n}$ be open with compact closure and $s_0\in U$. The Hodge norm depends continuously on the point, so $h_{s_0}\le c_U\,h_s$ for some $c_U$ and all $s\in U$. If $\mathrm{Hdg}(x)$ meets $U$ at $s$ and $x\in X_C$, then by Lemma 16.22(i) $h_{s_0}(x)\le c_Uh_s(x)=c_UI(x)\le 2c_UC^2/\int\eta^{2n}$, and only finitely many integral classes satisfy this. So the sets $\mathrm{Hdg}(x)$, $x\in X_C$, form a locally finite family of proper closed analytic subsets. Their union is closed with empty interior in the connected manifold $\mathcal{D}_{n,n}$, and it contains $\Sigma_C$. ∎

*Remark* 16.24 (Divisorial representatives). The loci on which this paper proves $\omega$ algebraic are the split, quaternionic and product loci and their translates under $G(\mathbb{Q})$, and on each of them $\omega$ is a rational polynomial in divisor classes, since an isogeny carries $\mathrm{NS}\otimes\mathbb{Q}$ isomorphically. Proposition 16.23 shows that no bounded family of such cycles covers a dense set, so the finiteness in Theorem 16.9 cannot be obtained by bounding the divisors that produce $\omega$ at the points where it is known to be algebraic. It would have to come from representatives of another kind, such as the transported cycles of Lemma 16.6, which are push-forwards along isogenies rather than products of divisors on the target; and by Theorem 16.15 their data is unbounded as soon as a component of the base cycle has finite stabiliser. The unbounded isogeny degree of Remark 16.12 and Remark 16.21 is thus unavoidable for divisorial constructions. The argument uses nothing about $\omega$ beyond the fact that it is not a multiple of $\eta^n$, and nothing about the family beyond the generic Picard number, so it applies equally to any variational argument that rests on products of divisors.

At $n=2$ the phenomenon can be exhibited on one fixed polarised lattice. Take $\Lambda=\mathcal{O}^2$ with $\mathcal{O}=\mathbb{Z}\langle 1,i,j,ij\rangle$, $i^2=-d$, $j^2=1$, and $E$ as in (38) with $a_1=a_2=1$. The lattice and the polarisation are fixed, and all the loci below lie in the one family $\mathcal{D}_{2,2}$ that they define. Let $R=\{x:\ x(iu,v)=x(u,iv)\}\subset H^2(\Lambda,\mathbb{Q})$ be the space of $K$-bilinear classes, where the divisor classes of the quaternionic points live, and for $x\in R$ let $\phi_x=E^{-1}x$, so that $x(u,v)=E(\phi_x u,v)$. Put $e=\int\eta^4$ and, for $k\in\mathbb{Z}$,

$$C_k=\begin{pmatrix}1 & k\\ k & -1\end{pmatrix},\qquad \phi_k(u_1,u_2)=\tfrac{1}{2d}\big(j(u_1+ku_2),\,j(ku_1-u_2)\big),\qquad x_k(u,v)=E(\phi_k u,v),$$

so that $\phi_k$ is left multiplication by the matrix $(2d)^{-1}C_k\,j$ and $x_k=x_0+k(x_1-x_0)$ is a pencil.

**Proposition 16.25** (A pencil of quaternionic loci)**.** *In this setting, for every $d$ and every $k\in\mathbb{Z}$:*

(i) *$e=384\,d^4$; $R$ has dimension 12, every $x\in R$ has $t(x)=0$ and $I(x)=\frac{1}{24}\operatorname{tr}(\phi_x^2)\,e$, and the form $I$ has signature $(8,4)$ on $R$;*

(ii) *$x_k$ is a primitive integral class in $R$, and $\phi_{x_k}^2=\beta(k)$ with $\beta(k)=(1+k^2)/(4d^2)$;*

(iii) *the set $\operatorname{Hdg}(x_k)$ is a quaternionic locus with algebra $(-d,\beta(k))\cong(-d,1+k^2)$, nonempty of dimension 3, and loci with different values of $\beta$ are not related by any automorphism of $(\Lambda,E,i)$;*

(iv) *at a very general point of $\operatorname{Hdg}(x_k)$ one has $\mathrm{NS}=\mathbb{Z}\,\frac{1}{2d}\eta\oplus\mathbb{Z}x_k\oplus\mathbb{Z}\,\iota x_k$, where $\iota x=x(i\cdot,\cdot)$, and the minimum of $I$ on $\mathrm{NS}\setminus\mathbb{Q}\eta$ is $\mu_k=32\,d^2(1+k^2)$.*

*Consequently, at a very general point of $\operatorname{Hdg}(x_k)$ every effective divisor $D$ whose class is not a multiple of $\eta$ has $\int[D]\eta^3\ge 32\sqrt{6}\,d^3\sqrt{1+k^2}$, which grows linearly in $|k|$.*

*Proof.* Two facts about $\mathcal{O}$ are used throughout. The reduced trace form $\operatorname{trd}(wz)$ is diagonal on $1,i,j,ij$ with entries $2,-2d,2,2d$, so the dual lattice $\mathcal{O}^\sharp=\{w\in B:\operatorname{trd}(w\mathcal{O})\subseteq\mathbb{Z}\}$ is $\mathbb{Z}\frac12\oplus\mathbb{Z}\frac{i}{2d}\oplus\mathbb{Z}\frac{j}{2}\oplus\mathbb{Z}\frac{ij}{2d}$. And a form $y(u,v)=\sum_{k,l}\operatorname{trd}(\overline{u_l}\,w_{lk}\,v_k)$ on $\mathcal{O}^2$ with $w_{lk}\in B$ is integral if and only if every $w_{lk}$ lies in $\mathcal{O}^\sharp$, because $\operatorname{trd}(\overline{u}wv)=\operatorname{trd}(wv\overline{u})$ and the products $v\overline{u}$ with $u,v\in\mathcal{O}$ span $\mathcal{O}$.

(i) The Gram matrix of $\operatorname{trd}(\overline{u}iv)$ on $1,i,j,ij$ pairs 1 with $i$ and $j$ with $ij$, with entries $\mp 2d$ and $\pm 2d$ and zeros elsewhere, so $E$ has elementary divisors $2d$ four times on $\Lambda$ and $e=4!\cdot(2d)^4=384\,d^4$. The involution $\sigma(x)=d^{-1}x(i\cdot,i\cdot)$ of $\bigwedge^2V^*$ has $R$ as its $(-1)$-eigenspace, since $x(iu,v)=x(u,iv)$ is equivalent to $x(iu,iv)=-d\,x(u,v)$. Over $\mathbb{C}$, $V_{\mathbb{C}}=W\oplus\overline{W}$ with $W,\overline{W}$ the eigenspaces of $i$ for $\pm\sqrt{-d}$, each of dimension 4, and $\sigma$ is $-1$ on $\bigwedge^2W^*\oplus\bigwedge^2\overline{W}^*$ and $+1$ on $W^*\otimes\overline{W}^*$; so $\dim R=6+6=12$. For $x\in R$ the map $\phi_x$ is $E$-self-adjoint, as $x$ and $E$ are alternating, and anticommutes with $i$, as $i$ is $E$-skew-adjoint by Theorem 15.25(i): $E(\phi_x iu,v)=x(u,iv)=E(\phi_x u,iv)=-E(i\phi_x u,v)$. Hence $\operatorname{tr}\phi_x=-\operatorname{tr}(i\phi_x i^{-1})=-\operatorname{tr}\phi_x=0$, and Lemma 16.19(i), with $g=4$, gives $t(x)=0$ and $I(x)=-\int x^2\eta^2=\frac{1}{24}\operatorname{tr}(\phi_x^2)e$.

For the signature regard $V_{\mathbb{R}}$ as a complex vector space through $i/\sqrt{d}$, with the hermitian form $H$ of Lemma 4.5, which has signature $(2,2)$ by Theorem 15.25(ii), and an $H$-orthogonal basis in which $H=\operatorname{diag}(h_1,\dots,h_4)$. The maps $\phi_x$, $x\in R\otimes\mathbb{R}$, are the $E$-self-adjoint antilinear maps. Writing $\phi(v)=A\overline{v}$, self-adjointness for $E=d^{-1/2}\operatorname{Im}H$ says exactly that $S=h\overline{A}$ is antisymmetric, and then $\operatorname{tr}_{\mathbb{R}}(\phi^2)=2\operatorname{Re}\operatorname{tr}(A\overline{A})=-4\sum_{k<l}|S_{kl}|^2/(h_kh_l)$. Four of the six pairs $k<l$ have $h_kh_l<0$ and two have $h_kh_l>0$, and each $S_{kl}$ is a complex coordinate, so $I$ has signature $(8,4)$.

(ii) $\phi_k$ anticommutes with $i$ because $ji=-ij$ and the entries of $C_k$ are rational. It is $E$-self-adjoint because $C_k$ is symmetric and left multiplication by $j$ is self-adjoint for

$\mathrm{trd}(\overline{u}iv)$ on each factor, by Theorem 15.25(i); so $x_k \in R$. Since the entries of $C_k$ commute with $j$, $\phi_k^2 = (2d)^{-2}C_k^2 j^2 = (1+k^2)/(4d^2)$. In the form of the first paragraph $x_k$ has $w_{lk} = -(2d)^{-1}c_{kl}\, j\, i = (2d)^{-1}c_{kl}\, ij$, with $c_{kl}$ the entries of $C_k$, which lie in $\mathcal{O}^\sharp$; so $x_k$ is integral, and $x_k((1,0),(ij,0)) = (2d)^{-1}\mathrm{trd}(ij\cdot ij) = 1$, so it is primitive.

(iii) Write $\phi = \phi_k$ and $\beta = \beta(k)$. The map $\phi$ is $E$-self-adjoint, anticommutes with $i$, and $\phi^2 = \beta > 0$, so $V_{\mathbb{R}} = V_+ \oplus V_-$ with $V_\pm$ the $\pm\sqrt{\beta}$ eigenspaces, $E$-orthogonal to each other, and $i$ exchanges them. A complex structure $J$ on $V_+$ compatible with $E|_{V_+}$, extended to $V_- = iV_+$ by $J(iw) = iJw$, commutes with $i$ and $\phi$ and is compatible with $E$, and $i$ has eigenvalues $\pm\sqrt{-d}$ each with multiplicity 2 on $V^{1,0}$; so these $J$ are points of $\mathcal{D}_{2,2}$ at which $x_k$ is of type $(1,1)$, and they form a copy of the Siegel space of $V_+$, of dimension 3, in agreement with Proposition 15.32. The algebra generated by $i$ and $\phi$ is $(-d,\beta)$, and $(-d,\beta) \cong (-d, 1+k^2)$ because the two differ by the square $(2d)^2$. An automorphism $\gamma$ of $(\Lambda, E, i)$ sends $\phi_x$ to $\gamma\phi_x\gamma^{-1}$ and so preserves $\mathrm{tr}(\phi_x^2) = 8\beta$.

For the first half of (iv), let $S_k \subset \mathcal{D}_{2,2}$ be the Siegel space just described and $\mathcal{C}_k \subset \mathrm{Sp}(V_{\mathbb{R}}, E)$ the centraliser of $i$ and $\phi$. An element of $\mathcal{C}_k$ preserves $V_+$ and is determined by its restriction there, because it commutes with $i$, so $\mathcal{C}_k \cong \mathrm{Sp}(V_+)$, acting on $V_{\mathbb{R}} = V_+ \oplus iV_+$ as $g \oplus igi^{-1}$. A real class is of type $(1,1)$ at $J$ exactly when it is invariant under $J$. The points of $S_k$ form one orbit $\{gJ_0g^{-1} : g \in \mathcal{C}_k\}$ of a fixed point $J_0$. Let $z$ be a real 2-form invariant under every point of $S_k$, and write $\mathfrak{c} = \mathrm{Lie}(\mathcal{C}_k) \cong \mathfrak{sp}(V_+)$, $\mathfrak{k}$ for the centraliser of $J_0$ in $\mathfrak{c}$ and $\mathfrak{p} = [\mathfrak{c}, J_0]$. For $X \in \mathfrak{c}$ the curve $g_t = \exp(tX)$ gives $(g_tJ_0g_t^{-1})^*z = z$ for all $t$, and the derivative at $t=0$ says that $[X, J_0]$ annihilates $z$, so $\mathfrak{p}$ annihilates $z$. The annihilator of $z$ in $\mathfrak{c}$ is a Lie subalgebra, so it contains $\mathfrak{p} + [\mathfrak{p},\mathfrak{p}]$. Now $\mathfrak{c} = \mathfrak{k} \oplus \mathfrak{p}$ with $[\mathfrak{k},\mathfrak{p}] \subseteq \mathfrak{p}$ and $[\mathfrak{p},\mathfrak{p}] \subseteq \mathfrak{k}$, so $\mathfrak{p} + [\mathfrak{p},\mathfrak{p}]$ is an ideal of $\mathfrak{c}$; it is nonzero, and it is therefore all of $\mathfrak{c}$, because $\mathfrak{c} \cong \mathfrak{sp}_4(\mathbb{R})$ is simple: its complexification $\mathfrak{sp}_4(\mathbb{C})$ is simple, with irreducible root system $C_2$, and an ideal of a real form complexifies to an ideal. Hence $z$ is annihilated by $\mathfrak{c}$ and, since $\mathrm{Sp}(V_+)$ is connected, invariant under all of $\mathcal{C}_k$. So the classes of type $(1,1)$ at every point of $S_k$ are the invariants of $\mathrm{Sp}(V_+)$ in $\bigwedge^2(V_+ \otimes \mathbb{R}^2) = (\bigwedge^2 V_+ \otimes S^2\mathbb{R}^2) \oplus (S^2V_+ \otimes \bigwedge^2 \mathbb{R}^2)$, a space of dimension 3 spanned by $\eta$, $x_k$ and $\iota x_k$. Every other rational class $z$ is of type $(1,1)$ only on a proper closed analytic subset of the connected manifold $S_k$, and a very general point avoids these countably many subsets.

For the lattice, $\iota x_k$ is the form of $\phi_k i = -i\phi_k$, so every class in $\mathbb{Q}\eta + \mathbb{Q}x_k + \mathbb{Q}\iota x_k$ is $y = s\eta + E(\phi_k\lambda\cdot,\cdot)$ with $s \in \mathbb{Q}$ and $\lambda \in K$ acting by left multiplication. In the form of the first paragraph, $y$ has $w_{lk} = \delta_{lk}\, s\, i + (2d)^{-1}c_{kl}\overline{\lambda}\, ij$. The first summand lies in $K$ and the second in $Kj$, and $\mathcal{O}^\sharp$ is the direct sum of its parts in $K$ and in $Kj$, so $y$ is integral if and only if $s \in (2d)^{-1}\mathbb{Z}$ and $(2d)^{-1}c_{kl}\overline{\lambda}\, ij \in \mathcal{O}^\sharp$ for all $k,l$. Writing $\overline{\lambda} = p - q\, i$, one has $\overline{\lambda}\, ij = p\, ij + qd\, j$, and since $c_{11} = 1$ and all $c_{kl}$ are integers this holds exactly when $p, q \in \mathbb{Z}$, that is $\lambda \in \mathbb{Z}[i]$. This is the stated description of NS. Now $t(\eta) = 1$ and $t$ vanishes on $R$, so $I(s\eta + z) = s^2 e + I(z)$ for $z \in R$, and $(\phi_k\lambda)^2 = \phi_k^2\overline{\lambda}\lambda = N(\lambda)\beta(k)$, so by (i)

$$I\big(s\eta + E(\phi_k\lambda\cdot,\cdot)\big) = s^2 e + \tfrac{8}{24}N(\lambda)\beta(k)e = s^2 e + 32\, d^2(1+k^2)N(\lambda).$$

A class off $\mathbb{Q}\eta$ has $\lambda \neq 0$, hence $N(\lambda) \geq 1$, and $\lambda = 1$, $s = 0$ gives $x_k$; so $\mu_k = 32\, d^2(1+k^2)$. The final assertion follows from Lemma 16.22(ii): an effective $D$ with $[D] \notin \mathbb{Q}\eta$ has $(\int [D]\eta^3)^2 \geq e\,\mu_k/2 = 6144\, d^6(1+k^2)$. ∎

Each locus of the pencil carries algebraic Weil classes, by the divisorial criterion, since $x_k$ is nondegenerate, and on the $k$-th of them the divisors that produce those classes have degree at least of order $|k|$. This is Proposition 16.23 seen on explicit quaternionic loci in one fixed polarised lattice, and Figure 20 illustrates it.

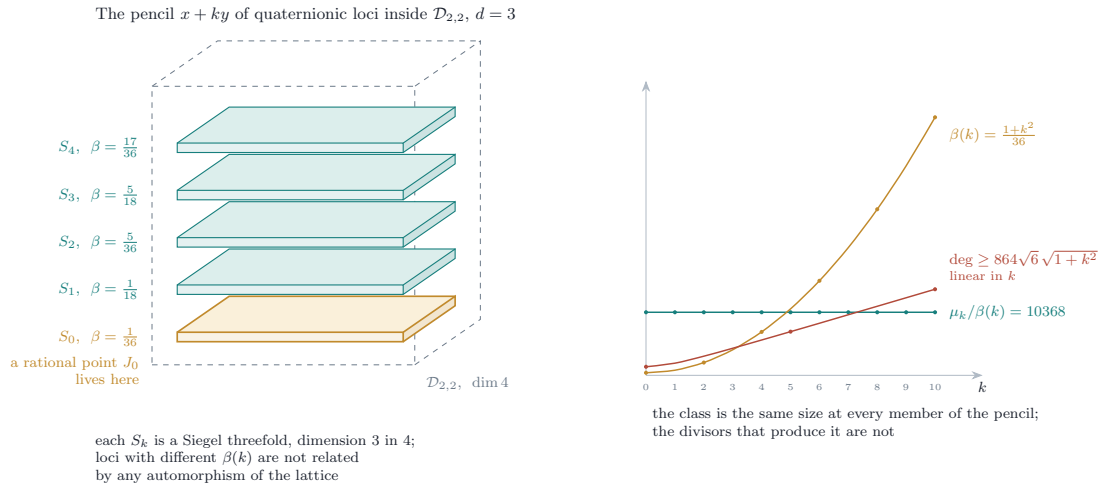



FIGURE 20. Proposition 16.25 at $d = 3$. Left: the members $S_k$ of the pencil, each a Siegel threefold inside the four-dimensional family, with the member $k = 0$, where $\beta = \frac{1}{36}$ is a square and a rational point $J_0$ can be written down, drawn apart. Right: along the pencil the ratio $\mu_k/\beta(k)$ is the constant $10368 = \int \eta^4/3$, so $\mu_k$ grows like $k^2$ and the least degree of a divisor not proportional to $\eta$ grows linearly in $k$. The Weil class is a flat class of the one fixed lattice and does not change; the divisors that produce it must.

*Remark* 16.26 (The case $n = 2$). Let $n = 2$, so $\dim \mathcal{S} = n^2 = 4$. For every $n$ the quaternionic locus of Setup 15.24 has dimension $n(n+1)/2$, by a Morita argument. The quaternion algebra $B = (-d, b)$ contains $K$ and acts on $V$, and $B \otimes \mathbb{R} \cong M_2(\mathbb{R})$ because $b > 0$, so Morita equivalence writes $V_{\mathbb{R}}$ as $W \otimes_{\mathbb{R}} \mathbb{R}^2$ with $W$ symplectic of dimension $2n$. The complex structures commuting with $B$ and compatible with $E$ are then the complex structures on $W$ compatible with its symplectic form, that is the points of the Siegel space $\mathfrak{H}_n$, of dimension $n(n+1)/2$, and the codimension in $\mathcal{D}_{n,n}$ is $n^2 - n(n+1)/2 = n(n-1)/2$. At $n = 2$ the quaternionic locus is therefore the symmetric space of $\mathrm{Sp}(4, \mathbb{R})$ inside $\mathcal{D}_{2,2}$, of dimension 3, so it is a divisor. Its $G(\mathbb{Q})$-translates are infinitely many pairwise distinct irreducible closed subvarieties of $\mathcal{S}$ of dimension 3, since a finite union of proper closed analytic subsets cannot be dense. If infinitely many of them carry cycles with one and the same data $(\underline{P}, \underline{q})$, then the closed set $\Sigma_{\underline{P},\underline{q}}$ has an irreducible component containing infinitely many distinct irreducible threefolds, so that component has dimension at least 4 and therefore equals $\mathcal{S}$. No density is needed at $n = 2$, only infinitude. Combined with the theorem of Moonen and Zarhin [MZ99] that every Hodge class on an abelian fourfold lies in the subalgebra generated by divisor classes and Weil classes, the conclusion would be the Hodge conjecture for all abelian fourfolds of Weil type, hence for all abelian fourfolds. That conclusion is a theorem of Markman [Mar25a, Corollary 1.6.1], proved by a different route, so at $n = 2$ the argument above would give a second proof; its interest lies in $n \geq 3$.

16.5. **The second criterion: a rank at one point.** The other way from a dense algebraic locus to an open one is to make the cycle move, and the classical tool for that is semiregularity. The theorem below adds nothing to the theory of semiregularity itself. Its content is that, after Theorem 4.14, Theorem 15.27 and Theorem 16.4, the semiregularity

of a single object at a single point implies the Hodge conjecture for the whole family, with no further propagation argument. We begin with the object: it exists, and its Chern character is concentrated in one degree and is a flat Hodge class on the whole family.

**Proposition 16.27** (The $K$-theory class at a base point)**.** *Let $s_0 \in \Sigma$ be a base point of Theorem 15.27 and write $A_0 = A_{s_0}$. There are an integer $N \geq 1$ and a perfect complex $E$ on $A_0$ with*

$$\mathrm{ch}_k(E) = 0 \quad (k \neq n), \qquad \mathrm{ch}_n(E) = N\,\omega_1.$$

*The tuple $v = \mathrm{ch}(E)$ extends to a flat section of $\bigoplus_k R^{2k} f_* \mathbb{Q}$ over $\mathcal{S}$ which is of type $(k,k)$ at every point.*

*Proof.* By Theorem 15.27 there is $\xi \in \mathrm{CH}^n(A_0)_{\mathbb{Q}}$ with $\mathrm{cl}(\xi) = \omega_1$. For a smooth quasi-projective variety over a field the Chern character $\mathrm{ch}\colon K_0(A_0)_{\mathbb{Q}} \to \mathrm{CH}^*(A_0)_{\mathbb{Q}}$ is a ring isomorphism [Ful98, Ex. 15.2.16(b) and Cor. 18.3.2], so there is a unique $u \in K_0(A_0)_{\mathbb{Q}}$ with $\mathrm{ch}(u) = \xi$, and its Chern character has no component outside degree $n$. Choose $N \geq 1$ such that $Nu$ is the image of a class in $K_0(A_0)$; that class is a difference of classes of vector bundles, hence the class of a perfect complex $E$, and $\mathrm{ch}(E) = N\xi$. Passing to cohomology gives the displayed identities.

For the last sentence, $\mathrm{ch}(E)$ has a single nonzero component, equal to $N\omega_1$, and $\omega$ is a flat section of type $(n,n)$ everywhere by Lemma 16.2. ∎

**Theorem 16.28** (Buchweitz and Flenner [BF03]; Pridham [Pri24])**.** *Let $X$ be a smooth projective complex variety, $E$ a perfect complex on $X$, and $X \hookrightarrow \mathcal{X} \to \operatorname{Spec} R$ a deformation of $X$ over an Artin local $\mathbb{C}$-algebra along which $\mathrm{ch}(E)$ stays of Hodge type. If the semiregularity map*

$$\sigma_E\colon \operatorname{Ext}^2_X(E,E) \longrightarrow \bigoplus_{q \geq 0} H^{q+2}(X, \Omega^q_X), \qquad \sigma_E(\zeta) = \sum_q \tfrac{1}{q!}\operatorname{tr}\big(\zeta \circ \mathrm{At}(E)^q\big),$$

*is injective, then $E$ extends to a perfect complex on $\mathcal{X}$.*

In this generality, for perfect complexes and for every obstruction annihilated by $\sigma_E$ rather than only the curvilinear ones, the theorem is proved by Pridham [Pri24], who identifies $\sigma_E$ with the tangent map of a generalised Abel–Jacobi map on the derived moduli of perfect complexes; for coherent sheaves see also [BLM23]. Perry [Per26] extends it to noncommutative varieties with group actions, which covers twisted derived categories and answers a question of Markman.

**Theorem 16.29** (Semiregularity at one point closes the family)**.** *Let $n \geq 2$, let $s_0$ and $E$ be as in Proposition 16.27, and suppose the semiregularity map $\sigma_E$ of Theorem 16.28 is injective. Then $\Sigma = \mathcal{D}_{n,n}$, and the Hodge conjecture holds in every degree for every abelian $2n$-fold of $(K,-1,n)$-Weil type with $\mathrm{Hg}(A) = \mathrm{SU}(V,H)$ in that family.*

*Proof.* Let $\mathfrak{M}$ be the stack of perfect complexes with vanishing negative self-extensions on the fibres of $f\colon \mathcal{A} \to \mathcal{S}$. It is algebraic and locally of finite presentation over $\mathcal{S}$ [Lie06], and at a point $[E]$ over $s_0$ its deformation theory relative to $\mathcal{S}$ has tangent space $\operatorname{Ext}^1(E,E)$ and obstruction space $\operatorname{Ext}^2(E,E)$. If $\operatorname{Ext}^{<0}(E,E) \neq 0$, which that stack excludes,[14] use instead the analytic Kuranishi germ of pairs from the proof of Theorem 17.5, which has the same tangent and obstruction spaces; the argument below then gives a nonempty analytic open set in place of a Zariski open one, which is all that Corollary 16.5 needs.

[14] Lieblich's stack parametrises the complexes he calls universally gluable, those with $\operatorname{Ext}^{<0}(E,E) = 0$; without that condition a complex can have automorphisms of negative degree, and the natural moduli object is the derived stack of Toën and Vaquié.

By Proposition 16.27 the Chern character of $E$ stays of Hodge type along every direction in $\mathcal{S}$, because it is a flat section of Hodge classes on the whole base. So Theorem 16.28 applies to every infinitesimal deformation of $A_0$ inside the family, and injectivity of $\sigma_E$ kills every obstruction. The morphism $\mathfrak{M} \to \mathcal{S}$ is therefore formally smooth at $[E]$, and being locally of finite presentation it is smooth on an open neighbourhood of $[E]$, which we take connected. Smooth morphisms are open, so the image of that neighbourhood is a nonempty Zariski open subset $U \subseteq \mathcal{S}$.

For $s \in U$ pick a point of the neighbourhood over $s$, that is, a perfect complex $E_s$ on $A_s$. Chern characters are locally constant in families, so $\mathrm{ch}(E_s)$ is the parallel transport of $\mathrm{ch}(E)$: it is $N\omega_1$ in degree $n$ and zero elsewhere. Chern characters of perfect complexes are algebraic, so $N\omega_1$, and hence $\omega_1$, is algebraic on $A_s$; the rest of the Weil line follows by Lemma 16.3. So $p^{-1}(U) \subseteq \Sigma$, which therefore contains a nonempty open set, and Corollary 16.5 gives $\Sigma = \mathcal{D}_{n,n}$. Apply Theorem 15.14(a). ∎

*Remark* 16.30 (The hypothesis is not met by the object above). Injectivity of $\sigma_E$ is automatic for some objects and not for others, and its degree-zero component already shows both. For a line bundle $L$ one has $\mathrm{Ext}^2(L,L) = H^2(\mathcal{O}_X)$ and $\sigma_0 = \mathrm{tr}$ is the identity on it, so $\sigma_L$ is injective for every line bundle on every variety, and Theorem 16.28 returns the classical fact that a divisor class deforms wherever it stays of type $(1,1)$. The object of Proposition 16.27 has rank zero, so its trace component is not an isomorphism, and in general not injective, the source having no reason to be as small as $h^2(\mathcal{O}_{A_0}) = \binom{2n}{2}$. The hypothesis of Theorem 16.29 therefore rests on the components with $q \geq 1$, which is where semiregularity stops being formal and becomes a computation.

For the object of Proposition 16.27 the hypothesis fails: by Proposition 9.10 it has $\dim \ker \sigma_E \geq n^2$ at every base point constructed in this paper (see also Remark 9.13). The hypothesis concerns a choice of perfect complex representing a class in $K_0$, and Proposition 9.10 does not reach every representative. Theorem 16.31 and the results after it describe what a representative satisfying the hypothesis would have to look like, and Theorem 17.5 shows that there is none. The proof of Theorem 16.29 applies word for word to the polarised criterion (P2′) of Remark 17.6, and Remark 17.13 records which of the results below carry over to it.

16.6. **Consequences of the rank condition.** The hypothesis of Theorem 16.29 is a condition on a rank, which cannot be inspected until the object is written down. It nevertheless has consequences that hold for every representative at once, because the semiregularity map composed with the evaluation map of Hochschild cohomology is contraction into the Chern character (Setup 18.54), and the Chern character here is a Weil class. The annihilator of a Weil class in $HT^2$ is large, and injectivity forces the evaluation map to vanish on the whole of it. We compute the annihilator and draw the consequences. They are the exact analogue for (P2) of what Theorem 18.61 and Theorem 18.89 do for the secant route in Section 18, and they rest on the same local products.

**Theorem 16.31** (Consequences of injectivity)**.** *Let $A$ be an abelian $2n$-fold of $(K,-1,n)$-Weil type, $n \geq 2$, and write $H^1(A,\mathbb{C}) = V_+ \oplus V_-$ for the eigenspaces of $K$, $V_\pm^{1,0}$, $V_\pm^{0,1}$ for their Hodge components and $T_\pm \subset H^0(T_A) = T_0A$ for the eigenspaces of $K$ on the tangent space, each of dimension $n$. Let $\omega = a\alpha_+ + b\alpha_-$ be a class on the Weil line with $ab \neq 0$, for instance any nonzero rational one, and let $E$ be a nonzero perfect complex on $A$ with $\mathrm{ch}(E) = N\omega$, $N \neq 0$, and no other component.*

(i) *The annihilator of* $\operatorname{ch}(E)$ *in* $HT^2(A)$ *is*

$$\operatorname{Ann}_{HT^2}(\operatorname{ch} E) = (V_+^{0,1} \otimes V_-^{0,1}) \ \oplus \ \operatorname{Hom}_K(H^{1,0}, H^{0,1}) \ \oplus \ (T_+ \otimes T_-),$$

*of dimension* $n^2 + 2n^2 + n^2 = 4n^2$*: the mixed block of* $H^{0,2}$*, which is the annihilator of Lemma 9.6; the* $K$*-linear first order deformations of* $A$*, which contain the tangent space of the polarised Weil family; and the mixed bivectors.*

(ii) *If* $\sigma_E$ *is injective, then* $\operatorname{ev}_E$ *vanishes on this space. Explicitly:* $z \cdot \mathrm{id}_E = 0$ *for every* $z \in V_+^{0,1} \otimes V_-^{0,1}$*;* $E$ *extends to first order along every* $K$*-linear deformation of* $A$*; and the translation classes satisfy* $A_\theta A_{\theta'} = 0$ *in* $\operatorname{Ext}^2(E,E)$ *for every* $\theta \in T_+$ *and* $\theta' \in T_-$*.*

(iii) *Under the same hypothesis the algebra map* $H^*(\mathcal{O}_A) \to \operatorname{Ext}^*(E,E)$*,* $z \mapsto z \cdot \mathrm{id}_E$*, kills the ideal generated by the mixed block, which is* $\bigoplus_{a,b\geq 1} \bigwedge^a V_+^{0,1} \otimes \bigwedge^b V_-^{0,1}$*; so the image of* $H^{0,k}(A)$ *has dimension at most* $2\binom{n}{k}$ *for* $1 \leq k \leq n$ *and is zero for* $k > n$*, and every endomorphism* $f$ *of* $E$ *has* $\operatorname{tr}(f) = 0$ *in* $H^0(\mathcal{O}_A) = \mathbb{C}$*.*

(iv) *Under the same hypothesis, if at some point* $p$ *of its support* $E$ *is locally isomorphic to* $i_*F$ *with* $F$ *locally free of positive rank on a smooth closed subvariety* $Z \ni p$ *of codimension at least two, then* $T_+ \subseteq T_pZ$ *or* $T_- \subseteq T_pZ$*.*

*Proof.* (i) Write $H^*(A,\mathbb{C}) = \bigwedge^*(H^{1,0} \oplus H^{0,1})$ and $\alpha_\pm$ for a generator of $\bigwedge^{2n} V_\pm = \bigwedge^n V_\pm^{1,0} \otimes \bigwedge^n V_\pm^{0,1}$. The three summands of $HT^2$ act as follows: $z \in H^{0,2} = \bigwedge^2 H^{0,1}$ by wedge product; $v \in H^1(T_A) = \operatorname{Hom}(H^{1,0}, H^{0,1})$ as the derivation that replaces one $(1,0)$ factor by its image; $\pi \in H^0(\bigwedge^2 T_A) = \bigwedge^2 (H^{1,0})^\vee$ by double contraction of $(1,0)$ factors. For $z$, the computation is Lemma 9.6. For $v$, decompose $\operatorname{Hom}(H^{1,0}, H^{0,1})$ into its four blocks by eigenspaces. An element of the block $\operatorname{Hom}(V_+^{1,0}, V_+^{0,1})$ replaces a factor of $\bigwedge^n V_+^{1,0}$ in $\alpha_+$ by an element of $V_+^{0,1}$, which is already exhausted in $\alpha_+$, so it kills $\alpha_+$; it acts trivially on $\alpha_-$, which has no $V_+^{1,0}$ factor. Likewise $\operatorname{Hom}(V_-^{1,0}, V_-^{0,1})$ kills $\omega$. A nonzero element of the block $\operatorname{Hom}(V_+^{1,0}, V_-^{0,1})$ carries $\alpha_+$ to a nonzero element of $\bigwedge^{n-1} V_+^{1,0} \otimes V_-^{0,1} \otimes \bigwedge^n V_+^{0,1}$ and kills $\alpha_-$, and a nonzero element of $\operatorname{Hom}(V_-^{1,0}, V_+^{0,1})$ carries $\alpha_-$ to a nonzero element of a different summand of $H^{n-1,n+1}$ and kills $\alpha_+$. Since $ab \neq 0$ and the two summands are independent, $v$ kills $\omega$ if and only if its two mixed blocks vanish, that is, if and only if $v$ is $K$-linear. The space of $K$-linear maps has dimension $2n^2$, and it contains the $n^2$-dimensional tangent space of the polarised family, which is its symmetric part. For $\pi$, a mixed bivector $\theta \wedge \theta'$ with $\theta \in T_+ = (V_+^{1,0})^\vee$ and $\theta' \in T_- = (V_-^{1,0})^\vee$ kills $\alpha_+$ because $\theta'$ kills every factor of $\bigwedge^n V_+^{1,0}$, and kills $\alpha_-$ symmetrically; a nonzero bivector in $\bigwedge^2 T_+$ carries $\alpha_+$ to a nonzero element of $\bigwedge^{n-2} V_+^{1,0} \otimes \bigwedge^n V_+^{0,1}$ and kills $\alpha_-$, and symmetrically for $\bigwedge^2 T_-$, the two images lying in different summands; so the annihilator in $H^0(\bigwedge^2 T_A)$ is $T_+ \otimes T_-$. Since the images of the three summands of $HT^2$ lie in the three different Hodge components $H^{n,n+2}$, $H^{n-1,n+1}$ and $H^{n-2,n}$ of $H^*(A,\mathbb{C})$, the annihilator is the direct sum of the three annihilators computed, of dimensions $n^2$, $2n^2$ and $n^2$.

(ii) By the commutative square $\sigma_E \circ \operatorname{ev}_E = (\,\cdot\,) \lrcorner \operatorname{ch}(E)$ of Setup 18.54, $\sigma_E$ vanishes on $\operatorname{ev}_E(\operatorname{Ann}_{HT^2}(\operatorname{ch} E))$; if $\sigma_E$ is injective, $\operatorname{ev}_E$ vanishes there. The three explicit statements are (54) on the three summands of (i): on $H^{0,2}$ the evaluation map is $z \mapsto z \cdot \mathrm{id}_E$; on $H^1(T_A)$ it is the obstruction map $v \mapsto v \lrcorner \operatorname{At}(E)$, whose vanishing on $v$ is the extension of $E$ to first order along $v$ [Tod09]; and on a bivector $\theta \wedge \theta'$ it is $\frac{1}{2}(\theta \wedge \theta') \lrcorner \operatorname{At}(E)^2 = A_\theta A_{\theta'}$, as in the proof of Theorem 18.61.

(iii) The map $z \mapsto z \cdot \mathrm{id}_E$ is a homomorphism of graded algebras from $H^*(\mathcal{O}_A) = \bigwedge^* H^{0,1}$ to the graded centre of the Yoneda algebra, so its kernel is an ideal, and by (ii) it contains the mixed block $V_+^{0,1} \otimes V_-^{0,1}$. The ideal generated by the mixed block in $\bigwedge^*(V_+^{0,1} \oplus V_-^{0,1}) = \bigwedge^* V_+^{0,1} \otimes \bigwedge^* V_-^{0,1}$ is the sum of the summands $\bigwedge^a V_+^{0,1} \otimes \bigwedge^b V_-^{0,1}$ with $a, b \geq 1$, since each such summand consists of multiples of the mixed block, while the remaining summands, with $a = 0$ or $b = 0$, meet the ideal in zero; the quotient in degree $k \geq 1$ is $\bigwedge^k V_+^{0,1} \oplus \bigwedge^k V_-^{0,1}$, of dimension $2\binom{n}{k}$, and zero for $k > n$. In particular the top class of $H^{2n}(\mathcal{O}_A)$, which is mixed for $n \geq 1$, acts by zero. Serre duality on $A$, whose canonical bundle is trivial, is the perfect pairing $\mathrm{Ext}^{2n}(E, E) \times \mathrm{Hom}(E, E) \to H^{2n}(\mathcal{O}_A) \cong \mathbb{C}$, $(\xi, f) \mapsto \mathrm{tr}(\xi \circ f)$; for $\xi = w \cdot \mathrm{id}_E$ with $w$ the top class this is $w \cdot \mathrm{tr}(f)$, where $\mathrm{tr}(f) \in H^0(\mathcal{O}_A) = \mathbb{C}$, so $w \cdot \mathrm{id}_E = 0$ forces $\mathrm{tr}(f) = 0$ for every $f$.

(iv) Let $\mathcal{O}$ be the local ring of $A$ at $p$ and choose coordinates with $Z = \{x_1 = \cdots = x_c = 0\}$, $c \geq 2$. Locally $E \cong (\mathcal{O}/J)^{\oplus r}$ with $J = (x_1, \ldots, x_c)$, a regular sequence, so $\mathrm{Ext}^*_{\mathcal{O}}(\mathcal{O}/J, \mathcal{O}/J) = \bigwedge^*_{\mathcal{O}/J} N$ as a graded algebra, where $N = \mathrm{Hom}(J/J^2, \mathcal{O}/J)$ is the normal module with basis the classes $\partial_1, \ldots, \partial_c$ dual to $x_1, \ldots, x_c$; the $\mathrm{Ext}^1$ is computed on the Koszul resolution exactly as in the proof of Lemma 18.60, and the product is the wedge because the Yoneda algebra of a Koszul complex is the exterior algebra on the dual of the generators. For $E$ of rank $r$ the local Yoneda algebra is $\bigwedge^* N \otimes M_r(\mathcal{O}/J)$. By Lemma 18.59 the germ of $A_\theta$ at $p$ is the jet class along $\theta$; on the Koszul complex the jet class of a derivation $D$ is the chain map $-D(d)$, and for $D = \theta = \sum_i c^i \partial_i$ its component in $\mathrm{Ext}^1 = N \otimes M_r(\mathcal{O}/J)$ is $\sum_{i \leq c} (c^i \bmod J)\, \partial_i \otimes \mathrm{id}$, the class of $\theta$ modulo $T_Z$. Multiplicativity of the germ map then gives the germ of $A_\theta A_{\theta'}$ as $(\theta \bmod T_Z) \wedge (\theta' \bmod T_Z) \otimes \mathrm{id}$ in $\bigwedge^2 N \otimes M_r(\mathcal{O}/J)$, whose value at $p$ is $\bar\theta(p) \wedge \bar\theta'(p) \in \bigwedge^2 (T_pA/T_pZ)$. If $A_\theta A_{\theta'} = 0$ for all $\theta \in T_+$, $\theta' \in T_-$, then $\bar a \wedge \bar b = 0$ for all $\bar a$ in the image $I_+$ of $T_+$ and $\bar b$ in the image $I_-$ of $T_-$ in $N_p = T_pA/T_pZ$, and $I_+ + I_- = N_p$ since $T_+ \oplus T_- = T_pA$. If both $I_\pm$ were nonzero, every nonzero $\bar b \in I_-$ would be proportional to every nonzero $\bar a \in I_+$, so $I_+ = I_-$ would be a line and $N_p = I_+ + I_-$ of dimension one, against $c \geq 2$. Hence $I_+ = 0$ or $I_- = 0$, which is the claim. ∎

**Corollary 16.32** (Locally free components are excluded)**.** *Let $A$ be as in Theorem 16.31 and suppose that no proper abelian subvariety $B \subset A$ has $T_0B \supseteq T_+$ or $T_0B \supseteq T_-$; equivalently, that $A$ has no nonzero $K$-stable abelian subvariety on which $K$ acts through a single embedding. This holds whenever $A$ is simple, at every split member $X \times \widehat{X}$ with $X$ simple of endomorphism algebra $\mathbb{Q}$, and at a general point of the quaternionic locus of Setup 15.24. Let $E$ be a perfect complex on $A$ with $\mathrm{ch}(E) = N\omega$ as above and $\sigma_E$ injective. Then no irreducible component $Z_0$ of the support of $E$ of codimension at least two has a dense open subset along which $E$ is a locally free sheaf of positive rank on $Z_0$. In particular $E$ is not a sheaf, and along every component of its support of codimension at least two it has, generically, either at least two nonvanishing cohomology sheaves or a cohomology sheaf that is not locally free on the reduced support.*

*Proof.* Suppose that $Z_0$ is such a component, with $U \subseteq Z_0^{\mathrm{sm}}$ dense open and $E|_U$ a locally free sheaf of positive rank on $U$. By Theorem 16.31(iv), $U = U_+ \cup U_-$ with $U_\pm = \{p \in U : T_\pm \subseteq T_pZ_0\}$, both closed in $U$; as $U$ is irreducible, $U = U_+$ say. Then every constant vector field $\theta \in T_+$ is tangent to $Z_0$ along $U$, hence along $Z_0^{\mathrm{sm}}$. The set $S = \{(p, t) \in Z_0 \times \mathbb{C} : p + \exp(t\theta) \in Z_0\}$ is a closed analytic subset of the irreducible space $Z_0 \times \mathbb{C}$, and it contains an open neighbourhood of $Z_0^{\mathrm{sm}} \times \{0\}$, because the local flow of the restriction of $\theta$ to the submanifold $Z_0^{\mathrm{sm}}$ exists on such a neighbourhood; so $S = Z_0 \times \mathbb{C}$, and $Z_0 + \exp(t\theta) \subseteq Z_0$, hence $= Z_0$, for every $t$. Let $G = \{x \in A : Z_0 + x = Z_0\}$, a closed algebraic subgroup of $A$, and $B$ its identity component, an abelian subvariety. The

connected set $\exp(T_+)$ contains 0 and lies in $G$, hence in $B$, so $T_+ \subseteq T_0B$. By hypothesis $B = A$, so $Z_0 = A$, against the codimension. The last sentence of the statement is the negation of the condition just excluded. If $E$ were a sheaf, its support would have codimension at least $n$, because the Chern character of a nonzero sheaf begins in the codimension of its support, with the support cycle, whose multiplicities are positive; this is excluded by Theorem 16.39(iii) below, whose proof uses nothing from the present one beyond the translation argument above.

For the equivalence in the hypothesis, let $\sqrt{-d}$ act on $A$ through an order of $K$. If an abelian subvariety $B \neq A$ has $T_0B \supseteq T_+$, then $H_1(B,\mathbb{Q}) \cap \sqrt{-d}\,H_1(B,\mathbb{Q})$ is a $K$-stable sub-Hodge structure, hence $H_1(C,\mathbb{Q})$ for a $K$-stable abelian subvariety $C \subseteq B$, and $T_0C = T_0B \cap \sqrt{-d}\,T_0B$ still contains $T_+$, on which $\sqrt{-d}$ acts by a scalar. The $K$-stable complement $C'$ of $C$ for the polarisation is nonzero, since $C \subseteq B \neq A$, and $T_0C'$ is a $K$-stable complement of $T_0C \supseteq T_+$, so $T_0C' \subseteq T_-$ and $K$ acts on $C'$ through a single embedding. Conversely, if a nonzero $K$-stable $C' \neq A$ has $T_0C' \subseteq T_-$, its $K$-stable complement $B$ is proper with $T_0B \supseteq T_+$. For the three cases: a simple $A$ has no nonzero proper abelian subvariety; for $X \times \widehat{X}$ with $X$ simple and $\mathrm{End}^0(X) = \mathbb{Q}$ the sub-Hodge structures of $H_1(X) \otimes \mathbb{Q}^2$ are $H_1(X) \otimes W$ with $W \subseteq \mathbb{Q}^2$, and since $K$ acts on $\mathbb{Q}^2$ through a quadratic field, $W$ is $K$-stable only for $W = 0$ or $\mathbb{Q}^2$; and at a general quaternionic member $A$ is either simple or, when the quaternion algebra of Setup 15.24 splits, isogenous to $C \times C$ with $C$ a general principally polarised abelian $n$-fold, whose sub-Hodge structures are again $H_1(C) \otimes W$ with the same conclusion. ∎

16.7. **The Hochschild action and the two branches.** Theorem 16.31 computes the annihilator summand by summand and reads off what the vanishing of the evaluation map on it means for $E$. That description hides a structure which is easily brought out and gives more: the annihilator is the mixed part of a splitting of $HH^1(A)$ into two halves, one for each of the two conjugate pieces $\alpha_+$ and $\alpha_-$ of the Weil class, and the same splitting describes the annihilator in every degree. The results of this subsection concern the Hochschild cohomology of $A$ alone, and their consequences for $E$ follow formally.

**Setup 16.33** (Hochschild cohomology in these terms)**.** For an abelian variety $A$ the Hochschild–Kostant–Rosenberg isomorphism identifies

$$HH^*(A) \;\cong\; HT^*(A) = \bigoplus_{p,q} H^p\big(\textstyle\bigwedge^q T_A\big) = \textstyle\bigwedge^*\big(H^{0,1}(A) \oplus H^0(T_A)\big),$$

as graded rings, the grading being $p+q$: the Todd class of $A$ is trivial, so the twist by $\mathrm{td}^{1/2}$ in [Cal05], [CRV12] is the identity and the plain identification is already multiplicative. Write $HH^1(A) = H^{0,1}(A) \oplus H^0(T_A)$, of dimension $4n$, so that $HH^*(A)$ is the exterior algebra on it, of dimension $2^{4n}$. It acts on $H^*(A,\mathbb{C}) = \bigwedge^*(H^{1,0} \oplus H^{0,1})$ by wedging with the classes of $H^{0,1}$ and contracting the classes of $H^{1,0}$ against $H^0(T_A)$; the two families of operators involve disjoint generators, so they pairwise anticommute and the action is an action of the exterior algebra. In degree two this is the operator $x\lrcorner$ of Setup 18.54, and $\mathrm{ev}_E\colon HH^*(A) \to \mathrm{Ext}^*(E,E)$ is a homomorphism of graded rings into the graded centre, with $\sigma_E \circ \mathrm{ev}_E = (\,\cdot\,)\lrcorner\,\mathrm{ch}(E)$ [BF08, Section 6], [Cal05], [CRV12].

**Theorem 16.34** (The annihilator is the mixed part of a splitting)**.** *Let $A$ be of $(K,-1,n)$-Weil type with $n \geq 2$, and let $\omega = a\alpha_+ + b\alpha_-$ lie on the Weil line with $ab \neq 0$. Put*

$$P = \mathrm{Ann}_{HH^1}(\alpha_+) = V_+^{0,1} \oplus T_-, \qquad Q = \mathrm{Ann}_{HH^1}(\alpha_-) = V_-^{0,1} \oplus T_+.$$

(i) $\dim P = \dim Q = 2n$ *and* $HH^1(A) = P \oplus Q$*; in particular* $\mathrm{Ann}_{HH^1}(\omega) = 0$.

(ii) $\mathrm{Ann}_{HH^2}(\omega) = P \wedge Q$, *of dimension* $4n^2$. *This is the space computed in Theorem 16.31(i), written as the mixed part of the splitting.*

(iii) *Contraction with* $\alpha_-$ *is injective on* $\bigwedge^* P$, *and contraction with* $\alpha_+$ *is injective on* $\bigwedge^* Q$; *the two images meet in exactly one line, in degree* $2n$.

(iv) *For every* $k \geq 1$,

$$\mathrm{Ann}_{HH^k}(\omega) = \Big( \bigoplus_{\substack{i+j=k \\ i,j\geq 1}} \textstyle\bigwedge^i P \wedge \bigwedge^j Q \Big) \;\oplus\; \epsilon_k,$$

*where* $\epsilon_k = 0$ *for* $k \neq 2n$ *and* $\epsilon_{2n}$ *is the line of* (iii). *So* $\dim \mathrm{Ann}_{HH^k}(\omega) = \binom{4n}{k} - 2\binom{2n}{k} + \delta_{k,2n}$ *for* $k \geq 1$, *while* $\mathrm{Ann}_{HH^0}(\omega) = 0$, *and the annihilator is the ideal generated by its degree-two part together with that one line.*

*Proof.* Choose bases $x_1, \dots, x_n$ of $V_+^{1,0}$ and $y_1, \dots, y_n$ of $V_-^{1,0}$, and bases $\overline{x}_i$ of $V_+^{0,1}$ and $\overline{y}_j$ of $V_-^{0,1}$ (the bar is notation only: complex conjugation carries $V_+^{1,0}$ to $V_-^{0,1}$), so that $\alpha_+ = x_1 \cdots x_n \overline{x}_1 \cdots \overline{x}_n$ and $\alpha_- = y_1 \cdots y_n \overline{y}_1 \cdots \overline{y}_n$, and write $\partial_{x_i}, \partial_{y_j}$ for the dual basis of $H^0(T_A) = T_+ \oplus T_-$.

(i) Of the $4n$ basis operators, $\overline{x}_i \wedge$ kills $\alpha_+$ because $\overline{x}_i$ already occurs in it, and $\partial_{y_j} \lrcorner$ kills $\alpha_+$ because no $y$ occurs in it; so $P \supseteq V_+^{0,1} \oplus T_-$. Conversely, the classes $\overline{y}_j \wedge \alpha_+$ are linearly independent, as are the classes $\partial_{x_i} \lrcorner \alpha_+$, and the two families lie in different summands of $H^*$, so no nonzero combination of the operators $\overline{y}_j \wedge$ and $\partial_{x_i} \lrcorner$ kills $\alpha_+$: $P = V_+^{0,1} \oplus T_-$, of dimension $2n$. The computation for $Q$ is the same with the roles of the two eigenspaces exchanged, and $P \cap Q = 0$ with $\dim P + \dim Q = 4n$, so $HH^1 = P \oplus Q$. A class of degree one that kills $\omega$ kills $\alpha_+$ and $\alpha_-$ separately, the two images lying in different summands and $ab \neq 0$, so it lies in $P \cap Q = 0$.

(iii) A monomial $\overline{x}_I \partial_{y_J} \in \bigwedge^* P$ sends $\alpha_-$ to $\pm\, \overline{x}_I\, y_{[n]\setminus J}\, \overline{y}_{[n]}$, and these are linearly independent for distinct pairs $(I, J)$, which is the injectivity; the statement for $Q$ is symmetric. For the intersection, the image of $\bigwedge^* P$ is spanned by monomials with no factor $x$ and all $n$ factors $\overline{y}$, and the image of $\bigwedge^* Q$, by the same computation, by monomials with no factor $y$ and all $n$ factors $\overline{x}$. Both images are spanned by monomials, so their intersection is spanned by the monomials common to both, and the only such monomial is $\overline{x}_{[n]}\overline{y}_{[n]}$. It is the image of $\pi_P = \overline{x}_{[n]}\partial_{y_{[n]}}$ and of $\pi_Q = \overline{y}_{[n]}\partial_{x_{[n]}}$, both of degree $2n$. So the intersection is the single line spanned by that class.

(ii) and (iv) Decompose $\bigwedge^k HH^1 = \bigoplus_{i+j=k} \bigwedge^i P \wedge \bigwedge^j Q$. A summand with $i, j \geq 1$ annihilates $\alpha_+$, because it has a factor in $P$ and the action is an action of the algebra, and annihilates $\alpha_-$ for the same reason with $Q$; so it lies in the annihilator of $\omega$. The sum of these summands over all $k$ is the ideal generated by $P \wedge Q$, since each of their monomials has, after reordering, a factor $p \wedge q$ with $p \in P$ and $q \in Q$. On the remaining two summands, $\xi_P \in \bigwedge^k P$ has $\xi_P \lrcorner \omega = b\, \xi_P \lrcorner \alpha_-$ and $\xi_Q \in \bigwedge^k Q$ has $\xi_Q \lrcorner \omega = a\, \xi_Q \lrcorner \alpha_+$, so a combination $\xi_P + \xi_Q$ annihilates $\omega$ if and only if $b\, \xi_P \lrcorner \alpha_- = -a\, \xi_Q \lrcorner \alpha_+$. By (iii) both contractions are injective and their images meet only in degree $2n$, in one line; so this happens only for $k = 2n$, and then only on the line spanned by $b^{-1}\pi_P - a^{-1}\pi_Q$, with $\pi_P$ and $\pi_Q$ scaled so that $\pi_P \lrcorner \alpha_- = \pi_Q \lrcorner \alpha_+$. In degree zero the annihilator is zero because $\omega \neq 0$. Taking $k = 2$ and $n \geq 2$, so that $2 \neq 2n$, gives (ii), the dimension being $\dim P \cdot \dim Q = 4n^2$; the general count is $\binom{4n}{k} - \binom{2n}{k} - \binom{2n}{k}$ plus the exceptional line. ∎

**Corollary 16.35** (Consequences for the Hochschild action)**.** *Let* $A$ *and* $\omega$ *be as above and let* $E$ *be a nonzero perfect complex on* $A$ *with* $\mathrm{ch}(E) = N\omega$, $N \neq 0$.

(i) *Unconditionally,* $\dim \mathrm{Ext}^2(E,E) \geq 2n(2n-1)$.

(ii) *If* $\sigma_E$ *is injective then* $\mathrm{ev}_E$ *vanishes on* $P \wedge Q$*, hence on the ideal it generates, and the ring homomorphism* $\mathrm{ev}_E \colon HH^*(A) \to \mathrm{Ext}^*(E,E)$ *factors through*

$$R \;=\; \textstyle\bigwedge^* P \;\oplus_{\mathbb{C}}\; \bigwedge^* Q, \qquad \dim_{\mathbb{C}} R = 2^{2n+1} - 1,$$

*the two exterior algebras glued along their scalars, inside* $HH^*(A)$ *of dimension* $2^{4n}$*. At* $n = 2$ *that is* $31$ *out of* $256$*.*

(iii) *Granting the compatibility* $\sigma_E \circ \mathrm{ev}_E = (\,\cdot\,) \lrcorner \mathrm{ch}(E)$ *in degree* $k$ [Cal05], [CRV12]*, the induced map on* $R$ *is injective away from the line of Theorem 16.34(iii), so*

$$\dim \mathrm{Ext}^k(E,E) \;\geq\; 2\binom{2n}{k} \quad (1 \leq k < 2n), \qquad \sum_{k \geq 1} \dim \mathrm{Ext}^k(E,E) \;\geq\; 2^{2n+1} - 3.$$

*Proof.* (i) By Setup 16.33, $\ker \mathrm{ev}_E \subseteq \mathrm{Ann}_{HH^2}(\mathrm{ch}\, E)$ in degree two, since $\sigma_E(\mathrm{ev}_E(\xi)) = \xi \lrcorner \mathrm{ch}(E)$ and the right side is nonzero off the annihilator. Hence $\dim \mathrm{Ext}^2(E,E) \geq \dim HH^2(A) - \dim \mathrm{Ann}_{HH^2}(\omega) = \binom{4n}{2} - 4n^2 = 2n(2n-1)$ by Theorem 16.34(ii). No hypothesis on $\sigma_E$ is used.

(ii) $\mathrm{ev}_E$ vanishes on the annihilator by Theorem 16.31(ii), its kernel is an ideal because it is a ring map, and the ideal generated by $P \wedge Q$ in $\bigwedge^*(P \oplus Q) = \bigwedge^* P \otimes \bigwedge^* Q$ is the sum of the summands with $i, j \geq 1$, whose complement is $\bigwedge^* P \oplus_{\mathbb{C}} \bigwedge^* Q$ of dimension $2 \cdot 2^{2n} - 1$.

(iii) With the compatibility in degree $k$, the argument of (i) gives $\ker \mathrm{ev}_E \subseteq \mathrm{Ann}_{HH^k}(\omega)$, and by Theorem 16.34(iv) that annihilator meets $\bigwedge^k P \oplus \bigwedge^k Q$ only in the exceptional line. So the map induced on $R$ has kernel contained in that one line and its image has dimension at least $\dim R - 1 = 2^{2n+1} - 2$. One of those dimensions sits in degree zero, where the image is $\mathbb{C} \cdot \mathrm{id}_E$, nonzero because $E$ is; the remaining $2^{2n+1} - 3$ lie in positive degrees, which is the second inequality. For $1 \leq k < 2n$ the exceptional line has no component in degree $k$, so $\bigwedge^k P \oplus \bigwedge^k Q$, of dimension $2\binom{2n}{k}$, maps injectively, which is the first. ∎

*Remark* 16.36 (The two branches)*.* The ring $R$ of Corollary 16.35(ii) is the ring of functions on two odd affine spaces of dimension $2n$ crossing at the origin. In these terms, if $E$ satisfies the criterion, the deformations of $D^b(A)$ along which its Hochschild action survives form two branches, $P$ and $Q$, which meet only at the trivial deformation and are the annihilators of the two conjugate pieces $\alpha_+$ and $\alpha_-$ of the Weil class. The criterion says that $E$ sees the two halves of the class separately and their interaction not at all.

The splitting $HH^1 = P \oplus Q$ is not the splitting into eigenspaces of $K$, which are $V_+^{0,1} \oplus T_+$ and $V_-^{0,1} \oplus T_-$; $P$ and $Q$ cut across them. It is the splitting into the annihilators of $\alpha_+$ and of $\alpha_-$, and it exists because each of the two pieces of the Weil class is a top form on an eigenspace. Nor is the conclusion a contradiction in itself. $R$ is a perfectly good ring, and the bound of (iii) is consistent with the value of $\chi(E,E)$ computed in Proposition 9.11, since $\sum_k (-1)^k 2\binom{2n}{k} = 0$. The corollary says how large such an $E$ must be and how little of the Hochschild cohomology it may see, but nothing in the Hochschild computation forbids an $E$ of the required shape. The exclusion comes from a different direction, deformation to non-algebraic Weil tori, in Theorem 17.5.

16.8. **The numerical form of the criterion and the support.** The same square gives two further results. The first converts the rank of $\sigma_E$ into a single dimension, the form in which a candidate can be tested. The second locates the support of $E$, and it excludes the first place one would look: a cycle representing the Weil class.

**Theorem 16.37** (The criterion as a dimension)**.** *Let* $A$*,* $\omega$ *and* $E$ *be as in Corollary 16.35, so that* $\mathrm{ch}(E) = N\omega$ *with* $N \neq 0$ *and no other component.*

(i) *If* $\dim \mathrm{Ext}^2(E,E) = 2n(2n-1)$*, the least value that Corollary 16.35(i) allows, then* $\mathrm{ev}_E\colon HH^2(A) \to \mathrm{Ext}^2(E,E)$ *is surjective with kernel exactly* $P \wedge Q$*, the map* $\sigma_E$ *is injective, and every class of* $\mathrm{Ext}^2(E,E)$ *is a sum of products* $\mathrm{ev}_E(p)\,\mathrm{ev}_E(p') + \mathrm{ev}_E(q)\,\mathrm{ev}_E(q')$ *with* $p, p' \in P$ *and* $q, q' \in Q$*, hence lies in the graded centre of* $\mathrm{Ext}^*(E,E)$*.*

(ii) *Conversely, if* $\sigma_E$ *is injective, then* $\mathrm{ev}_E(HH^2(A))$ *has dimension exactly* $2n(2n-1)$*, and* $\sigma_E$ *carries it isomorphically onto* $HH^2(A) \lrcorner \omega$*.*

(iii) *Consequently, if at one base point* $s_0$ *of a family some perfect complex* $E$ *with* $\mathrm{ch}(E) = N\omega_1$ *has* $\dim \mathrm{Ext}^2(E,E) = 2n(2n-1)$*, then* $\Sigma = \mathcal{D}_{n,n}$*, and the Hodge conjecture holds in every degree for every member of that family with* $\mathrm{Hg}(A) = \mathrm{SU}(V,H)$*.*

*Proof.* (i) The square of Setup 16.33 gives $\ker \mathrm{ev}_E \subseteq \mathrm{Ann}_{HH^2}(\omega) = P \wedge Q$ in degree two, the equality being Theorem 16.34(ii). Hence $\operatorname{rank} \mathrm{ev}_E \geq \binom{4n}{2} - 4n^2 = 2n(2n-1) = \dim \mathrm{Ext}^2(E,E)$, so $\mathrm{ev}_E$ is onto and its kernel, of dimension $4n^2$ and contained in $P \wedge Q$, is all of $P \wedge Q$. Every class of $\mathrm{Ext}^2(E,E)$ is then $\mathrm{ev}_E(\xi)$ for some $\xi$, and $\sigma_E(\mathrm{ev}_E(\xi)) = N\,\xi \lrcorner \omega$ vanishes only for $\xi \in P \wedge Q = \ker \mathrm{ev}_E$; so $\sigma_E$ is injective. Since $HH^2(A) = \bigwedge^2 P \oplus (P \wedge Q) \oplus \bigwedge^2 Q$ and $\mathrm{ev}_E$ is a ring map into the graded centre, the image is $\mathrm{ev}_E(\bigwedge^2 P) + \mathrm{ev}_E(\bigwedge^2 Q)$, which is the description of the classes.

(ii) By Theorem 16.31(ii) the kernel of $\mathrm{ev}_E$ in degree two contains $P \wedge Q$, and by the argument of (i) it is contained in it, so it equals $P \wedge Q$ and the image has dimension $2n(2n-1)$. The composite $\sigma_E \circ \mathrm{ev}_E$ is contraction into $\mathrm{ch}(E) = N\omega$, whose kernel is exactly $P \wedge Q = \ker \mathrm{ev}_E$, so $\sigma_E$ maps the image isomorphically onto $HH^2(A) \lrcorner \omega$.

(iii) By (i) the map $\sigma_E$ is injective, and Theorem 16.29 applies. ∎

Theorem 16.37(iii) is a sufficient condition for $\mathbf{W}(K,n,\delta)$ that mentions no map at all: one object, at one point, and one number. At $n=2$ the number is 12 and at $n=3$ it is 30. By Theorem 17.5 no complex attains it, but the first two parts of the theorem hold for the polarised criterion (P2′) with the number $r(\gamma)$ of Theorem 17.9(iv) in place of $2n(2n-1)$, and that is the form in which the polarised criterion can be tested.

The dimension of $\mathrm{Ext}^2(E,E)$ is computable in a way the rank of $\sigma_E$ is not. For a complex built from sheaves whose cohomology is known, the local-to-global spectral sequence $E_2^{p,q} = H^p(A, \mathcal{E}xt^q(E,E)) \Rightarrow \mathrm{Ext}^{p+q}(E,E)$ reduces the count to the cohomology of the local $\mathcal{E}xt$ sheaves, and on an abelian variety the cohomology of a sheaf supported on an abelian subvariety, or of a homogeneous bundle, is a question of linear algebra. The rank of $\sigma_E$, by contrast, requires the trace of the exponential of the Atiyah class on every class of $\mathrm{Ext}^2$, a computation with the object and not only with its invariants. Part (ii) says that, on the central part of $\mathrm{Ext}^2$, nothing is lost by reading the criterion this way. Whether an injective $\sigma_E$ forces the whole of $\mathrm{Ext}^2(E,E)$ to be central is, by Serre duality, a statement about $\mathrm{Ext}^{2n-2}$, which the square in degree two does not reach; nothing below uses it.

The second result rests on a local fact about complexes supported at a point, which we state and prove first.

**Lemma 16.38** (Two jet classes on a complex of finite length)**.** *Let* $R = \mathbb{C}[[x_1, \dots, x_c]]$ *with* $c \geq 2$*, let* $M$ *be a bounded complex of finitely generated free* $R$*-modules whose cohomology has finite length, and put* $\chi(M) = \sum_i (-1)^i \ell(H^i(M))$*. For* $a \in \mathbb{C}^c$ *let* $J(a) \in \mathrm{Ext}^1_R(M,M)$ *be the jet class of* $M$ *along the constant vector field* $\sum_i a_i \partial_i$*, represented by the chain map* $-\sum_i a_i \partial_i(d)$ *as in the proof of Theorem 16.31(iv). If* $\chi(M) \neq 0$ *and* $a, b$ *are linearly independent, then* $J(a)J(b) \neq 0$ *in* $\mathrm{Ext}^2_R(M,M)$*.*

*Proof.* Let $T$ be an abelian variety of dimension $c$ and $t \in T$ a point. Linear coordinates on the universal cover identify the completed local ring $\widehat{\mathcal{O}}_{T,t}$ with $R$ in such a way that the translation invariant vector fields become the constant fields $\sum_i a_i \partial_i$: the exponential map from $T_t T = \mathbb{C}^c$ is a local biholomorphism carrying the constant vector fields to the translation invariant ones, and its germ at the origin identifies $\widehat{\mathcal{O}}_{T,t}$ with $\mathbb{C}[[x_1, \dots, x_c]]$ compatibly with derivations. A complex of $R$-modules with finite length cohomology is the same thing as a complex of $\mathcal{O}_{T,t}$-modules with finite length cohomology, since modules of finite length over either ring are modules over $\mathcal{O}_{T,t}/\mathfrak{m}^N$ for $N$ large, so $M$ defines a perfect complex $M_T$ on $T$ supported at $t$. Its local $\mathcal{E}xt$ sheaves are skyscrapers at $t$, so the local-to-global spectral sequence gives $\operatorname{Ext}^*_T(M_T, M_T) = \operatorname{Ext}^*_R(M, M)$, and by Lemma 18.59 the translation classes $A_a$, $A_b$ of $M_T$ restrict to $J(a)$, $J(b)$. Apply the square of Setup 18.54 on $T$ to the bivector $a \wedge b$:

$$\sigma_{M_T}(A_a A_b) = (a \wedge b) \lrcorner \operatorname{ch}(M_T).$$

A sheaf of length $\ell$ supported at a point has Chern character $\ell \cdot [\mathrm{pt}]$, and Chern characters are additive over the cohomology of a complex, so $\operatorname{ch}(M_T) = \chi(M)\,[\mathrm{pt}]$. The class of a point is a nonzero multiple of $dz_1 \cdots dz_c\, d\bar{z}_1 \cdots d\bar{z}_c$, and contracting it with $a \wedge b$ removes two holomorphic factors and leaves a nonzero class of type $(c-2, c)$, because $a$ and $b$ are independent. So the right side is nonzero when $\chi(M) \neq 0$, hence so is $A_a A_b$, and so is its restriction $J(a)J(b)$, the restriction map being an isomorphism. ∎

The hypothesis on $\chi(M)$ cannot be dropped. The Euler characteristic is the only additive invariant here: the Grothendieck group of complexes with finite length cohomology over a regular local ring is $\mathbb{Z}$, generated by the residue field, since every module of finite length has a composition series whose factors are all isomorphic to it. The lemma says that the product of two independent jet classes detects this generator, and on a complex of Euler characteristic zero the product can vanish. Let $K$ be the Koszul resolution of the residue field $k$ of $\mathbb{Q}[x_1, x_2]$, on which the jet class of a derivation $D$ is the chain map $-D(d_K)$, and let $u = \partial_2(d_K)\partial_1(d_K) \colon K \to K[2]$ represent $J(\partial_2)J(\partial_1)$. The entries of $d_K$ are $\pm x_1, \pm x_2$, so $u$ has constant entries. Let $C$ be the cone of $u$, with $C^m = K^{m+1} \oplus K^{m+2}$ and $d_C(a, b) = (-d_K a,\ u(a) + d_K b)$. As $u$ induces zero on cohomology, $C$ has two cohomology modules, each isomorphic to $k$, so $\chi(C) = 0$. Since $\partial_i(u) = 0$, the matrix $\partial_i(d_C)$ is block diagonal, and the product $J(\partial_2)J(\partial_1)$ on $C$ is $(a, b) \mapsto (u(a), u(b))$, which equals $d_C H + H d_C$ for $H(a, b) = (b, 0)$: it is null homotopic, the gluing has killed it. On the split complex $k \oplus k[1]$, whose Euler characteristic is also zero, the same product is $u \oplus u[1]$, which is not null homotopic because $u \neq 0$ in $\operatorname{Ext}^2(k, k)$ by the lemma applied to $k$.

**Theorem 16.39** (The support of an object meeting the criterion)**.** *Let $A$ be an abelian $2n$-fold of $(K, -1, n)$-Weil type with $n \geq 2$, let $\omega = a\alpha_+ + b\alpha_-$ with $ab \neq 0$, and let $E$ be a perfect complex on $A$ with $\operatorname{ch}(E) = N\omega$, $N \neq 0$, no other component, and $\sigma_E$ injective. For an irreducible component $Z$ of the support of $E$ write*

$$m_Z(E) = \sum_i (-1)^i\, \ell_{\mathcal{O}_{A,Z}}(\mathcal{H}^i(E)_Z)$$

*for the multiplicity of $Z$ in the cycle of $E$.*

(i) *If $Z$ has codimension at least two and $m_Z(E) \neq 0$, then $Z$ is a union of cosets of an abelian subvariety $B \subsetneq A$ with $T_0 B \supseteq T_+$ or $T_0 B \supseteq T_-$.*

(ii) *If $A$ has no proper abelian subvariety $B$ with $T_0 B \supseteq T_+$ or $T_0 B \supseteq T_-$, which by Corollary 16.32 holds when $A$ is simple, at every split member $X \times \widehat{X}$ with $X$ simple*

*and* $\mathrm{End}^0(X) = \mathbb{Q}$*, and at a general point of the quaternionic locus, then* $m_Z(E) = 0$ *for every component* $Z$ *of codimension at least two, and the cycle of* $E$ *is a divisor.*

(iii) *At every member of the family, whatever its endomorphisms, the support of* $E$ *has an irreducible component of codimension less than* $n$*.*

*In particular no perfect complex whose support has codimension at least* $n$*, at any member of any Weil family, satisfies the criterion of Theorem 16.29. This includes every complex assembled from the structure sheaves of the components of a cycle representing* $\omega$*, the base cycle of Theorem 15.27 among them.*

*Proof.* (i) Let $c \geq 2$ be the codimension of $Z$ and let $U \subseteq Z$ be the dense open set of smooth points of $Z_{\mathrm{red}}$ that lie on no other component of the support. Fix $\theta \in T_+$ and $\theta' \in T_-$; by Theorem 16.31(ii), $A_\theta A_{\theta'} = 0$ in $\mathrm{Ext}^2(E,E)$. Suppose that at some $p \in U$ the images $\bar\theta, \bar\theta'$ of $\theta, \theta'$ in $N_p = T_pA/T_pZ$ are linearly independent, and complete them by images of constant fields $\nu_3, \dots, \nu_c$ to a basis of $N_p$. In linear coordinates on the universal cover the constant fields are coordinate derivations, and the germ $S$ at $p$ of the translate of the span of $\theta, \theta', \nu_3, \dots, \nu_c$ is a smooth germ of dimension $c$, transverse to $Z$ at $p$, to which $\theta$ and $\theta'$ are tangent. Let $j\colon S \to A$ be the inclusion and $M = Lj^*E$ on $\widehat{\mathcal{O}}_{S,p} \cong \mathbb{C}[[x_1, \dots, x_c]]$. Derived restriction is a ring map $\mathrm{Ext}^*(E,E) \to \mathrm{Ext}^*(M,M)$, and it carries the germ of $A_\theta$ to the jet class of $M$ along $\theta|_S$, because the Atiyah class is functorial for pullback and $\theta$ is tangent to $S$ [BF03, Section 3]. So $J(\theta)J(\theta') = 0$ on $M$. Near $p$ the support of $E$ is $Z$, and $Z \cap S = \{p\}$, so $M$ has finite length cohomology; its Euler characteristic is the intersection multiplicity of the cycle of $E$ with $S$ at $p$, which, $S$ being transverse to $Z$ at a smooth point lying on no other component, is $m_Z(E)$, by Serre's formula for intersection multiplicities applied to each cohomology sheaf and summed with signs [Ser00, Chapter V].[15] Since $\theta$ and $\theta'$ are independent directions on $S$, Lemma 16.38 applied to $M$ gives $m_Z(E) = 0$.

So if $m_Z(E) \neq 0$, then at every $p \in U$ and for all $\theta \in T_+$, $\theta' \in T_-$ the images satisfy $\bar\theta \wedge \bar\theta' = 0$ in $\bigwedge^2 N_p$. As in the proof of Theorem 16.31(iv), the images $I_\pm$ of $T_\pm$ in $N_p$ span $N_p$, which has dimension $c \geq 2$, so one of them is zero: $T_+ \subseteq T_pZ$ or $T_- \subseteq T_pZ$. The two conditions cut out closed subsets of the irreducible set $U$ whose union is $U$, so one of them, say the first, holds on all of $U$, and the argument of Corollary 16.32 shows that $Z$ is stable under translation by the identity component $B$ of its stabiliser, an abelian subvariety with $T_0B \supseteq T_+$. Since $Z \neq A$, $B \neq A$.

(ii) follows from (i), since no such $B$ exists; if $A$ itself is a component of the support, its multiplicity is the rank of $E$, which is zero, so the cycle of $E$ is a divisor.

(iii) Suppose every component of the support has codimension at least $n$. Then $\mathrm{ch}_k(E) = 0$ for $k < n$ and $\mathrm{ch}_n(E) = \sum_Z m_Z(E)[Z]$, the sum over the components of codimension exactly $n$; this is Riemann–Roch without denominators in its simplest case [Ful98, Chapter 15]. By (i), each $Z$ with $m_Z(E) \neq 0$ is $\pi^{-1}(\pi(Z))$ for the quotient $\pi\colon A \to A/B$ by an abelian subvariety $B$ with $T_0B \supseteq T_+$ or $T_0B \supseteq T_-$, so $[Z] = \pi^*[\pi(Z)]$ lies in the subring of $H^*(A,\mathbb{C})$ generated by $\pi^*H^1(A/B,\mathbb{C})$, the space of classes of degree one that vanish on $T_0B$ and on its conjugate. When $T_0B \supseteq T_+$, a $(1,0)$-form vanishing

[15] Serre's formula expresses $\sum_i (-1)^i \ell(\mathrm{Tor}_i(F, \mathcal{O}_S))$, for a module $F$ and a smooth germ $S$ meeting its support properly at $p$, as a sum over the components $Z'$ of the support of $\ell(F_{Z'})$ times the intersection multiplicity of $Z'$ and $S$ at $p$. For $S$ transverse to a component smooth at $p$ the intersection multiplicity is one. For a complex the Tor spectral sequence gives the same formula with alternating sums over the cohomology, which define $m_Z(E)$.

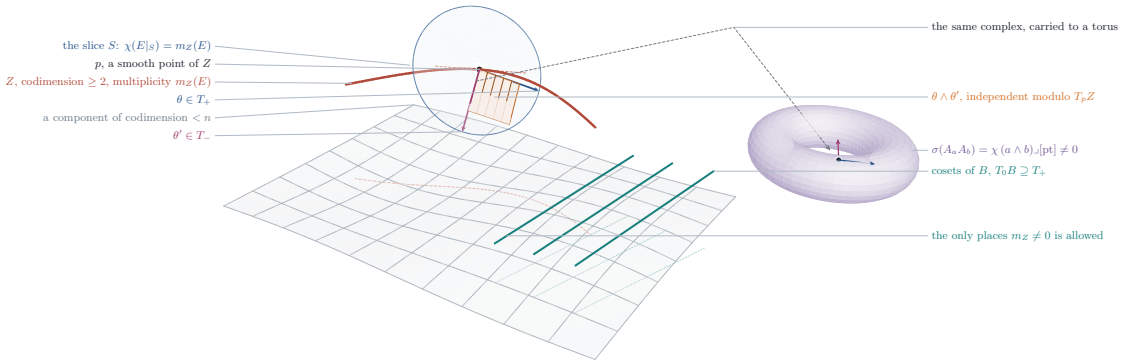



FIGURE 21. The proof of Theorem 16.39. The grey sheet stands for a component of the support of codimension less than $n$, which the theorem forces to exist; the clay curve $Z$ for a component of codimension at least two. At a smooth point $p$ of $Z$ the directions $\theta \in T_+$ and $\theta' \in T_-$ span the slice $S$, transverse to $Z$, on which $E$ restricts to a complex of finite length with Euler characteristic $m_Z(E)$; the amber parallelogram is the bivector $\theta \wedge \theta'$, independent modulo $T_pZ$. Carried to a torus of the same dimension, the product of the two jet classes is seen by the semiregularity map as $m_Z(E)$ times the class of a point contracted with the bivector, which is not zero, so $m_Z(E) = 0$ or the bivector degenerates. The teal lines are cosets of an abelian subvariety $B$ tangent to $T_+$, the only components of codimension at least two on which a nonzero multiplicity survives. The scene is schematic in the dimensions and exact in the incidences.

on $T_+$ lies in $V_-^{1,0}$, and a $(0,1)$-form vanishing on the conjugate of $T_+$ lies in $V_+^{0,1}$, because complex conjugation exchanges the two eigenvalues of $K$;[16] so the subring lies in $\bigwedge^*(V_-^{1,0} \oplus V_+^{0,1})$. When $T_0B \supseteq T_-$ it lies in $\bigwedge^*(V_+^{1,0} \oplus V_-^{0,1})$. Choose bases of the four pieces. Both exterior algebras are spanned by monomials in these bases, so their sum is too, and $\alpha_+$, a single monomial in $n$ generators of $V_+^{1,0}$ and $n$ of $V_+^{0,1}$, is not among the monomials spanning it; nor is $\alpha_-$. As $ab \neq 0$, the class $N\omega$ has nonzero coordinates on both and is not in the sum, which contradicts $\mathrm{ch}_n(E) = N\omega$. So some component has codimension less than $n$. The last assertion follows, since a complex assembled from structure sheaves of the components of a cycle of codimension $n$ is supported on that cycle. ∎

Figure 21 draws the slice and the bivector of the proof.

16.9. **Indecomposability and endomorphisms.** The numerical criterion of Theorem 16.37 fixes one number, $\dim \mathrm{Ext}^2(E,E)$. Together with Serre duality, the lower bounds of Corollary 16.35 in the other degrees and the Hodge–Riemann relations for the Weil class, it fixes a good deal more: the object that carries the class can be taken indecomposable, and it has at least three linearly independent endomorphisms at $n = 2$ and, granting the compatibility of Corollary 16.35(iii), at $n = 3$, the first case in which the criterion is needed at a nontrivial discriminant.

[16]The field $K$ acts on $H_1(A,\mathbb{Q})$ by rational endomorphisms, so on the complexification its action commutes with complex conjugation, which carries the eigenspace of $\sqrt{-d}$ for one embedding of $K$ to that for the other. On the tangent space the conjugate of $T_+$ is the eigenspace in $T^{0,1}$ for the second embedding; it pairs with $V_-^{0,1}$, not with $V_+^{0,1}$.

**Theorem 16.40** (The shape of an object meeting the criterion)**.** *Let $A$, $\omega$ and $E$ be as in Theorem 16.37, so that* $\mathrm{ch}(E) = N\omega$ *with* $N \neq 0$ *and no other component, and suppose that* $\dim \mathrm{Ext}^2(E,E) = 2n(2n-1)$*. Write* $e_k = \dim \mathrm{Ext}^k(E,E)$*.*

(i) $\chi(E,E) = \sum_k (-1)^k e_k = (-1)^n N^2 \int_A \omega^2$*, and this number is positive.*

(ii) $E \cong E_0 \oplus E'$ *with* $E_0$ *indecomposable,* $\mathrm{ch}(E_0) = \mathrm{ch}(E)$*,* $\dim \mathrm{Ext}^2(E_0,E_0) = 2n(2n-1)$*,* $\mathrm{ch}(E') = 0$ *and* $\mathrm{Ext}^2(E_0,E') = \mathrm{Ext}^2(E',E_0) = \mathrm{Ext}^2(E',E') = 0$*. So $E_0$ meets the criterion as well.*

(iii) *If $E$ is indecomposable, then* $\mathrm{End}(E) = \mathbb{C}\cdot \mathrm{id} \oplus J$ *with $J$ the nilpotent radical, every $f \in J$ annihilates* $\mathrm{Ext}^2(E,E)$ *and* $\mathrm{Ext}^{2n-2}(E,E)$ *under the Yoneda product, and* $\mathrm{ev}_E(\bigwedge^{2n} P) = \mathrm{ev}_E(\bigwedge^{2n} Q)$ *is a single line in* $\mathrm{Ext}^{2n}(E,E)$*. If moreover $e_1 = 4n$, then $J$ annihilates* $\mathrm{Ext}^1(E,E)$ *as well.*

(iv) *If $n = 2$ or $n = 3$, then*

$$\dim \mathrm{End}(E) \;\geq\; 2 + \tfrac{1}{2}\,\chi(E,E) \;>\; 2,$$

*so $E$ has at least three linearly independent endomorphisms, and at $n = 3$ also $e_1 \leq e_0 + 9$. In particular no simple object meets the criterion: no stable sheaf, no complex with* $\mathrm{End} = \mathbb{C}$*, and no image of one under an autoequivalence of $D^b(A)$; and an indecomposable object that meets it has a nonzero nilpotent endomorphism. The smallest profiles $(e_0,\dots,e_{2n})$ that the constraints allow are $(3,8,12,8,3)$ at $n = 2$, with $\chi = 2$, and $(3,12,30,40,30,12,3)$ and $(3,12,30,41,30,12,3)$ at $n = 3$, with $\chi = 2$ and $\chi = 1$.*

*Parts (i) and (ii), and part (iv) at $n = 2$, use the compatibility $\sigma_E \circ \mathrm{ev}_E = (\,\cdot\,)\lrcorner\,\mathrm{ch}(E)$ of Setup 16.33 in degree two only, as Theorem 16.37 does. Part (iii), and part (iv) at $n = 3$, use it in the further degrees in which Corollary 16.35(iii) grants it: $2n-2$ and $2n$ for (iii), and three for (iv).*

*Proof.* (i) The Todd class of $A$ is trivial, so the Riemann–Roch theorem gives $\chi(E,E) = \int_A \mathrm{ch}(E)^\vee \mathrm{ch}(E)$, and $\mathrm{ch}(E)^\vee = (-1)^n N\omega$ since $\omega$ has degree $2n$. The polarisation pairs $V_+$ only with $V_-$ (see Figure 10), so its class $\theta$ lies in $V_+ \otimes V_-$ and $\theta \wedge \alpha_\pm = 0$, because $\alpha_\pm$ spans the top exterior power of $V_\pm$. So $\omega$ is a nonzero rational primitive class of type $(n,n)$ in the middle degree, and the Hodge–Riemann relations give $(-1)^{n(2n-1)} \int_A \omega^2 > 0$, where $(-1)^{n(2n-1)} = (-1)^n$.

(ii) The category $D^b(A)$ has finite dimensional morphism spaces and split idempotents, so $E = \bigoplus_i E_i$ with each $E_i$ indecomposable. The evaluation map and the semiregularity map are compatible with direct sums, and by Theorem 16.37(i) the kernel of $\mathrm{ev}_E$ in degree two is $P \wedge Q$; so $\mathrm{ev}_{E_i}(P \wedge Q) = 0$ and, composing with $\sigma_{E_i}$, $(P \wedge Q)\lrcorner\,\mathrm{ch}(E_i) = 0$ for every $i$. We claim that the only classes of Hodge type killed by $P \wedge Q$ are those of $\mathbb{C}\alpha_+ \oplus \mathbb{C}\alpha_-$. Write $H^*(A,\mathbb{C}) = \bigwedge^*(A_+ \oplus B_+ \oplus A_- \oplus B_-)$ with $A_\pm = V_\pm^{1,0}$ and $B_\pm = V_\pm^{0,1}$, and grade it by the four degrees $(i,j,k,l)$. As in Setup 16.33, $P$ acts by wedging with $B_+$ and contracting $A_-$, and $Q$ by wedging with $B_-$ and contracting $A_+$, so $P \wedge Q$ is spanned by four kinds of operators, each of which shifts the four degrees by a fixed amount: wedging with $b_+ \wedge b_-$ kills a component of degree $(i,j,k,l)$ for all choices exactly when $j = n$ or $l = n$; wedging with $b_+$ after contracting $a_+$, exactly when $i = 0$ or $j = n$; contracting $a_-$ and wedging with $b_-$, exactly when $k = 0$ or $l = n$; and contracting $a_-$ and $a_+$, exactly when $i = 0$ or $k = 0$. Each of these is a statement about a tensor product of maps that are injective unless the degree in question is extreme. A class of type $(p,p)$ has $i+k = j+l = p$ in every component. If $j = n$, then $l = n$ would force $i = k = n$, against the fourth condition, so $l < n$ and the third condition gives $k = 0$; hence $i = n + l$, so $i = n$ and $l = 0$: the component lies

in $\bigwedge^n A_+ \otimes \bigwedge^n B_+ = \mathbb{C}\alpha_+$. If $j < n$, the first two conditions give $l = n$ and $i = 0$, hence $k = n$ and $j = 0$: the component lies in $\mathbb{C}\alpha_-$. This proves the claim. Since $\mathrm{ch}(E_i)$ is a rational class of Hodge type killed by $P \wedge Q$, it lies on the rational Weil plane, and when it is nonzero Corollary 16.35(i) gives $\dim \mathrm{Ext}^2(E_i, E_i) \geq 2n(2n-1)$, a nonzero rational Weil class having both coefficients $a$ and $b$ nonzero. As $\mathrm{Ext}^2(E,E) = \bigoplus_{i,j} \mathrm{Ext}^2(E_i, E_j)$ has dimension $2n(2n-1)$ and $\sum_i \mathrm{ch}(E_i) = N\omega \neq 0$, exactly one summand $E_0$ has nonzero Chern character and every other block of $\mathrm{Ext}^2$ vanishes.

(iii) An indecomposable object of a category with finite dimensional morphism spaces and split idempotents has a local endomorphism ring, so $\mathrm{End}(E) = \mathbb{C} \cdot \mathrm{id} \oplus J$ with $J$ nilpotent. By Theorem 16.37(i), $\mathrm{ev}_E$ maps $\bigwedge^2 P \oplus \bigwedge^2 Q$ isomorphically onto $\mathrm{Ext}^2(E,E)$. By Serre duality $e_{2n-2} = e_2$, and contraction with $\omega$ is injective on $\bigwedge^{2n-2} P \oplus \bigwedge^{2n-2} Q$ by Theorem 16.34(iv), since $2n - 2 \neq 2n$; so, with the compatibility in degree $2n-2$, $\mathrm{ev}_E$ maps that space isomorphically onto $\mathrm{Ext}^{2n-2}(E,E)$. In degree $2n$ the semiregularity map is the trace, since $H^{q+2n}(\Omega_A^q) = 0$ for $q > 0$, so $\mathrm{tr}\,\mathrm{ev}_E(\xi)$ is the $H^{0,2n}$ component of $\xi \lrcorner \mathrm{ch}(E)$; for the generators $\pi_P$, $\pi_Q$ of $\bigwedge^{2n} P$ and $\bigwedge^{2n} Q$ it is $Nc_P$ and $Nc_Q$ with $c_P c_Q \neq 0$. Indeed $\pi_P$ wedges with a basis of $B_+$ and contracts a basis of $A_-$, so it kills $\alpha_+$, whose factor $\bigwedge^n B_+$ is already full, and carries $\alpha_-$ to $\pm$ the generator $\bigwedge^n B_+ \wedge \bigwedge^n B_-$ of $H^{0,2n}$; so $c_P = \pm b$ and likewise $c_Q = \pm a$, both nonzero. For $f \in \mathrm{End}(E)$ and $x \in \bigwedge^2 P \oplus \bigwedge^2 Q$ write $f \cdot \mathrm{ev}_E(x) = \mathrm{ev}_E(\varphi_f(x))$; then $f \mapsto \varphi_f$ is a homomorphism of algebras. Let $x \in \bigwedge^2 P$ and $y \in \bigwedge^{2n-2} P$, and write $x \wedge y = \beta(x,y)\pi_P$. Since $\mathrm{ev}_E$ is a ring map that kills the ideal generated by $P \wedge Q$,

$$\beta(x,y)\,\mathrm{tr}(f\,\mathrm{ev}_E(\pi_P)) = \mathrm{tr}(f\,\mathrm{ev}_E(x)\,\mathrm{ev}_E(y)) = \mathrm{tr}\,\mathrm{ev}_E(\varphi_f(x) \wedge y) = Nc_P\,\beta(\varphi_f(x)_P, y),$$

and the pairing $\beta$ is perfect, so $\varphi_f(x)_P = \mu_f x$ with $\mu_f = \mathrm{tr}(f\,\mathrm{ev}_E(\pi_P))/Nc_P$. The same computation with $y \in \bigwedge^{2n-2} Q$, where $x \wedge y$ lies in the ideal generated by $P \wedge Q$, gives $\varphi_f(x)_Q = 0$. So $f$ acts on $\mathrm{ev}_E(\bigwedge^2 P)$ by a scalar $\mu_f$, and in the same way on $\mathrm{ev}_E(\bigwedge^2 Q)$ by a scalar $\nu_f$; $\mu$ and $\nu$ are characters of $\mathrm{End}(E)$ and vanish on $J$. So $J$ kills $\mathrm{Ext}^2(E,E)$, on both sides because $\mathrm{ev}_E$ lands in the graded centre, and the same argument with the roles of the two degrees exchanged shows that it kills $\mathrm{Ext}^{2n-2}(E,E)$. In particular $\mathrm{tr}(f\,\mathrm{ev}_E(\pi_P)) = \mu_f Nc_P = 0$ for $f \in J$, so $\mathrm{ev}_E(\pi_P)$ and, likewise, $\mathrm{ev}_E(\pi_Q)$ lie in the annihilator of $J$ under the perfect pairing of $\mathrm{End}(E)$ with $\mathrm{Ext}^{2n}(E,E)$. That annihilator is a line, $J$ having codimension one, and both elements are nonzero, their traces being $Nc_P$ and $Nc_Q$; so they span the same line. Finally, if $e_1 = 4n$ then $\mathrm{Ext}^1(E,E) = \mathrm{ev}_E(HH^1)$, because $\mathrm{ev}_E$ is injective on $HH^1$, as shown in (iv). For $f \in J$ and $h \in HH^1$ write $f\,\mathrm{ev}_E(h) = \mathrm{ev}_E(h')$; then $\mathrm{ev}_E(h' \wedge h'') = f\,\mathrm{ev}_E(h \wedge h'') = 0$ for every $h'' \in HH^1$, so $h' \wedge HH^1 \subseteq P \wedge Q$, which forces $h' = 0$ since $\dim P = \dim Q \geq 2$.

(iv) By Serre duality $e_k = e_{2n-k}$. The map $\mathrm{ev}_E$ is injective on $HH^1$ using degree two alone: if $\mathrm{ev}_E(h) = 0$ then $\mathrm{ev}_E(h \wedge h') = 0$ for every $h'$, so $h \wedge HH^1 \subseteq \mathrm{Ann}_{HH^2}(\omega) = P \wedge Q$, and $h = 0$ as in (iii). So $e_1 \geq 4n$, and Corollary 16.35(iii) gives $e_3 \geq 2\binom{6}{3} = 40$ at $n = 3$. At $n = 2$ the Euler characteristic is $2e_0 - 2e_1 + 12$, and at $n = 3$ it is $2e_0 - 2e_1 + 60 - e_3$. So $2e_0 = \chi + 2e_1 - 12 \geq \chi + 4$ at $n = 2$, and $2e_0 = \chi + 2e_1 + e_3 - 60 \geq \chi + 4$ at $n = 3$, where also $e_1 = e_0 + 30 - \frac{1}{2}(\chi + e_3) \leq e_0 + 9$. With (i) this gives the bound. An autoequivalence preserves End, and a stable sheaf is simple. An indecomposable object with $\dim \mathrm{End} \geq 3$ has $J \neq 0$. For the smallest profiles take $e_0 = 3$. At $n = 2$, $\chi$ is even and positive, so $2e_1 = 18 - \chi \leq 16$, and $e_1 \geq 8$ forces $e_1 = 8$ and $\chi = 2$. At $n = 3$, $2e_1 + e_3 = 66 - \chi \leq 65$ with $e_1 \geq 12$ and $e_3 \geq 40$ forces $e_1 = 12$ and $e_3 \in \{40, 41\}$, with $\chi = 2$ and $\chi = 1$. ∎

*Remark* 16.41 (Objects excluded by the shape). Theorem 16.40 gives necessary conditions only, and their effect is to exclude the objects one would construct first. Line bundles, semi-homogeneous bundles, skyscraper sheaves, stable sheaves and their Fourier–Mukai transforms are simple, and direct sums of them are excluded by (ii) together with Theorem 20.32. At $n = 3$, granting the compatibility of Corollary 16.35(iii), the shape allowed is an indecomposable object with a local endomorphism algebra of dimension at least three whose radical kills $\mathrm{Ext}^2$ and $\mathrm{Ext}^4$ and pairs to zero with the line $\mathrm{ev}_E(\bigwedge^6 P)$. The smallest conceivable one has $\mathrm{Hom}(E,E)$ of dimension three, $\mathrm{Ext}^1(E,E)$ equal to the image of $HH^1(A)$, of dimension twelve and killed by the radical, and $\mathrm{Ext}^3(E,E)$ of dimension forty or forty-one. Iterated self-extensions and thickenings are the natural objects with nilpotent endomorphisms, but an extension of a simple object $F$ by itself has $\mathrm{ch} = 2\,\mathrm{ch}(F)$, so $F$ would already have to carry the class, and the question returns one level down. For $n \geq 4$ write $e_k = 2\binom{2n}{k} + \delta_k$ for $1 \leq k \leq 2n-1$, with $\delta_k \geq 0$ and $\delta_2 = \delta_{2n-2} = 0$. Since $\sum_{k=1}^{2n-1}(-1)^k\binom{2n}{k} = -2$, the Euler characteristic is $2e_0 - 4 + \sum_k (-1)^k\delta_k$, and an excess $\delta_k$ in an even degree $4 \leq k \leq 2n-4$ can make it positive with $e_0 = 1$; so the bound on End disappears. By Theorem 17.5 none of these shapes is realised.

## Part 3. Semiregularity and secant objects

### 17. The semiregularity criterion in its pure and polarised forms

17.1. **Weil tori and the pure form of the criterion.** The object $E$ of Section 16 has Chern character exactly $N\omega$. For such an object, injectivity of $\sigma_E$ would make $E$ extend along every $K$-linear deformation of $A$ (Theorem 16.31(ii)). These deformations reach non-algebraic tori, on which the Weil class is a Hodge class but not a Chern class of any coherent sheaf. This settles the pure form of the criterion, negatively, in every dimension.

**Setup 17.1** (Weil tori)**.** Let $A$ be an abelian $2n$-fold of $(K,-1,n)$-Weil type, $V = H^1(A,\mathbb{Q})$ with its action of $K$, and $V\otimes\mathbb{C} = V_+ \oplus V_-$ the two eigenspaces, each of dimension $2n$. For complementary $n$-dimensional subspaces $X, Y \subset V_+$ the subspace $X \oplus \overline{Y}$ of $V\otimes\mathbb{C}$ is the space of holomorphic 1-forms of a complex structure on the real torus underlying $A$; call the resulting complex torus $T_{X,Y}$. On $T_{X,Y}$ the field $K$ acts with signature $(n,n)$, so $T_{X,Y}$ is a Weil torus in the sense of [Voi02b], and every complex structure on that real torus commuting with $K$ and of signature $(n,n)$ arises in this way. The pairs $(X,Y)$ form a connected complex manifold $S$ of dimension $2n^2$, containing the point of $A$; the polarised Weil family is the submanifold of dimension $n^2$ on which the polarisation of $A$ is of type $(1,1)$, and the tangent space of $S$ is the space $\mathrm{Hom}_K(H^{1,0}, H^{0,1})$ of $K$-linear first order deformations in Theorem 16.31(i). Since $\alpha_+$ spans $\bigwedge^{2n} V_+ = \bigwedge^n X \otimes \bigwedge^n Y$ and $X$ is holomorphic while $Y$ is antiholomorphic, the Weil classes are of type $(n,n)$ on every $T_{X,Y}$. We say that a property holds for a very general member of $S$ if it holds outside a countable union of proper analytic subsets.

**Lemma 17.2** (The Hodge classes of a very general Weil torus)**.** *For a very general member $T$ of $S$, $\mathrm{Hdg}^k(T) = 0$ for $1 \leq k \leq 2n-1$, $k \neq n$, and $\mathrm{Hdg}^n(T)$ is the rational Weil plane.*

*Proof.* There are countably many rational classes, and the Hodge locus of each is an analytic subset of the connected manifold $S$; so a rational class that is of Hodge type at a very general member is of Hodge type on all of $S$. Let $G = \mathrm{Res}_{K/\mathbb{Q}}\mathrm{GL}_{2n}$ be the group of $K$-linear automorphisms of $V$. Its real points act transitively on $S$, since $\mathrm{GL}(V_+)$ acts transitively on pairs of complementary subspaces. The Mumford–Tate group $M$ of a very general member lies in $G$, since $K$ acts by endomorphisms of every member, and it

contains the Mumford–Tate group of every member, whose Hodge classes include those of the very general one. So the real Lie algebra of $M$ contains the complex structure $J_s$ of every member, which is the derivative of its Hodge cocharacter, and therefore contains the $\operatorname{Ad} G(\mathbb{R})$-span of one $J_s$. This span is an ideal of $\mathfrak{g}_{\mathbb{R}} = \mathfrak{gl}_{2n}(\mathbb{C})$ containing the non-central element $J_s$, which acts on $V_+$ by $i$ on $X$ and by $-i$ on $Y$; hence it contains $\mathfrak{sl}_{2n}(\mathbb{C})$, and $M$ contains $G^{\mathrm{der}} = \operatorname{Res}_{K/\mathbb{Q}} \mathrm{SL}_{2n}$. Over $\mathbb{C}$ this is $\mathrm{SL}(V_+) \times \mathrm{SL}(V_-)$ acting on $\bigwedge^*(V_+ \oplus V_-) = \bigwedge^* V_+ \otimes \bigwedge^* V_-$, whose invariants are spanned by 1, $\alpha_+$, $\alpha_-$ and $\alpha_+\alpha_-$. Hodge classes are invariant under $M$, which gives the statement. ∎

**Lemma 17.3** (A very general Weil torus has no subvarieties)**.** *A very general member $T$ of $S$ has no analytic subsets other than finite sets and $T$ itself.*

*Proof.* Let $Z \subset T$ be an irreducible analytic subset of codimension $c$ with $0 < c < 2n$. For every Kähler class $\kappa$ the class $[Z]$ satisfies $\int_T [Z]\,\kappa^{2n-c} = \int_Z \kappa^{2n-c} > 0$, so $[Z]$ is a nonzero Hodge class of degree $2c$. By Lemma 17.2 this forces $c = n$ and $[Z]$ into the Weil plane. Take for $\kappa_0$ the class of a positive definite hermitian form that is block diagonal for the splitting $H^{1,0} = X \oplus \overline{Y}$. Its blocks lie in $X \otimes \overline{X}$ and $\overline{Y} \otimes Y$, both inside $V_+ \otimes V_-$, so every power of $\kappa_0$ has a factor from $V_+$ and a factor from $V_-$ in each term, and $\alpha_\pm \wedge \kappa_0^n = 0$ because $\alpha_\pm$ spans the top exterior power of $V_\pm$. So $\int_T [Z]\,\kappa_0^n = 0$, which is impossible. ∎

**Proposition 17.4** (Coherent sheaves on a very general Weil torus)**.** *Let $n \geq 2$ and let $T$ be a very general member of $S$. Every coherent analytic sheaf $F$ on $T$ has $\mathrm{ch}_k(F) = 0$ in $H^{2k}(T, \mathbb{Q})$ for $1 \leq k \leq 2n-1$. For $n = 2$ this is Voisin's theorem that every coherent sheaf on a general Weil torus of dimension four has $c_2(F) = 0$* [Voi02b, Theorem 10]*, whose proof we follow.*

*Proof.* Chern classes of coherent sheaves on $T$ exist in rational Deligne cohomology, extend those of bundles and are additive [Gri10], so they are Hodge classes. By Lemma 17.3 the torsion of $F$, the locus where the torsion-free quotient is not locally free, and the cokernel of the map from that quotient to its reflexive hull are all supported on finite sets, and a sheaf supported on a finite set has Chern character in the top degree only. So it suffices to treat a reflexive sheaf $G$. Every coherent sheaf on $T$ has $c_1 = 0$, since $\mathrm{Hdg}^1(T) = 0$ (Lemma 17.2); so every torsion-free sheaf has slope zero for every Kähler class, and a torsion-free sheaf is stable for one Kähler class exactly when it has no saturated subsheaf of intermediate rank, which does not depend on the class. A Jordan–Hölder filtration of $G$ therefore has stable torsion-free factors $Q_j$, stable for every Kähler class, and $\mathrm{ch}(G) = \sum_j \mathrm{ch}(Q_j)$. By the theorem of Bando and Siu [BS94] the stable reflexive sheaf $Q_j^{**}$ carries an admissible Hermite–Einstein metric for every Kähler class $\kappa$ and satisfies $\int_T (2r c_2 - (r-1)c_1^2)(Q_j^{**})\,\kappa^{2n-2} \geq 0$, with equality exactly when $Q_j^{**}$ is a projectively flat vector bundle. Here $c_1 = 0$ and $c_2(Q_j^{**}) \in \mathrm{Hdg}^2(T)$. For $n \geq 3$ that group is zero; for $n = 2$ it is the Weil plane, and then $\int_T c_2(Q_j^{**})\,\kappa_0^2 = 0$ for the class $\kappa_0$ of the proof of Lemma 17.3. In either case equality holds for some $\kappa$, so $Q_j^{**}$ is a projectively flat bundle with $c_1 = 0$, hence flat, and its rational Chern classes of positive degree vanish. Since $Q_j$ and $Q_j^{**}$ differ only on a finite set, $\mathrm{ch}_k(Q_j) = 0$ for $1 \leq k \leq 2n-1$, and the same holds for $G$ and for $F$. ∎

**Theorem 17.5** (The pure form of the criterion is never met)**.** *Let $n \geq 2$, let $A$ be an abelian $2n$-fold of $(K, -1, n)$-Weil type, $\omega$ a nonzero rational Weil class, and $E$ a perfect complex on $A$ with $\mathrm{ch}(E) = N\omega$, $N \neq 0$, and no other component. Then $\sigma_E$ is not injective. In particular $\dim \mathrm{Ext}^2(E, E) > 2n(2n-1)$, the hypotheses of Theorem 16.29 and Theorem 16.37(iii) are satisfied by no object at any member of any Weil family of an*

*imaginary quadratic field, and the propagation statement* (P2) *of Remark 17.7, in the form stated there, is false.*

*Proof.* Suppose that $\sigma_E$ is injective. Since $A$ is projective, $E$ is isomorphic to a bounded complex $E^\bullet$ of holomorphic vector bundles. The class $N\omega$ is of Hodge type on every member of $S$ (Setup 17.1), so by the theorem of Buchweitz and Flenner in Pridham's form (Theorem 16.28, [BF03, Pri24]), every obstruction to extending $E$ along an Artinian deformation of $A$ inside $S$ vanishes, exactly as in the proof of Theorem 16.29. The deformations of the complex structure inside $S$ together with those of the holomorphic structures and differentials of $E^\bullet$ are governed by a differential graded Lie algebra of Dolbeault type with finite dimensional cohomology, and Kuranishi's method [Kur65], in the form of [GM88, Sections 2–3], gives an analytic germ $(M, m_0)$ with a morphism $\pi\colon (M, m_0) \to (S, s_0)$ and a complex of holomorphic vector bundles on the pulled back family of tori deforming $E^\bullet$, whose completion is a hull of the formal deformation functor. The vanishing of the obstructions says that $\pi$ is formally smooth, hence smooth, so its image contains a neighbourhood $U$ of $s_0$. Thus every member $T_s$ with $s \in U$ carries a bounded complex of holomorphic vector bundles $E_s^\bullet$ whose Chern character is the parallel transport of $N\omega$. Choose $s \in U$ very general. By Proposition 17.4, $\mathrm{ch}_n(E_s) = \sum_i (-1)^i \mathrm{ch}_n(E_s^i) = 0$, while it is the transport of $N\omega$, which is not zero. This contradiction proves that $\sigma_E$ is not injective. The argument uses only that $\mathrm{ch}(E)$ is of Hodge type on every member of $S$, so it applies verbatim when $\mathrm{ch}(E) - N\omega$ has components in degrees 0 and $4n$ as well. The remaining statements follow, since $\dim \mathrm{Ext}^2(E, E) \geq 2n(2n-1)$ by Corollary 16.35(i), and equality forces $\sigma_E$ to be injective by Theorem 16.37(i). ∎

*Remark* 17.6 (The polarised criterion). Theorem 17.5 turns Theorem 16.31(ii) from a constraint into a contradiction: an object that extends along every $K$-linear deformation of $A$ reaches non-algebraic tori, where the Weil class is a Hodge class but not a Chern class. The results that describe such an object, Theorem 16.31, Corollary 16.35(ii)–(iii), Theorem 16.37, Theorem 16.39, Theorem 20.32 and Theorem 16.40, remain true as implications, but they are vacuous: no object satisfies their hypotheses. They record what the pure form asks of an object, and Remark 17.13 says which of them extend to the polarised form defined next.

The obstruction does not reach a Chern character with a component that is of Hodge type on the polarised family and not beyond it, unless a twist removes that component. Call (P2′) the statement that at one base point of each Weil family some perfect complex $E$ with

$$\mathrm{ch}(E) = N\omega_1 + \sum_k c_k\, \theta^k, \qquad N \neq 0,\ c_k \in \mathbb{Q},$$

$\theta$ the polarisation, has injective semiregularity map. The proof of Theorem 16.29 applies to it word for word, with the deformation taken over the polarised family, along which $\mathrm{ch}(E)$ stays of Hodge type: at a member $A_s$ it produces $\mathrm{ch}_n(E_s) = N\omega_1 + c_n\theta^n$ algebraic, and $\theta^n$ is algebraic, so $\omega_1$ is. The form above is already the most general one allowed by the subring generated by $\theta$ and the Weil classes: a Weil class is primitive, so $\theta\omega_1 = 0$ and that subring is spanned by the powers of $\theta$ and the Weil plane. Theorem 17.5 excludes directly a Chern character that remains of Hodge type on all of $S$, which by Lemma 17.2 is a multiple of the Weil class plus terms in degrees 0 and $4n$, and after a line bundle twist the characters $c\, e^{t\theta} + N\omega_1 + c'\theta^{2n}$ with $t\theta$ integral (Theorem 17.9(v)). For every other polarised character with some $c_k \neq 0$, $1 \leq k \leq 2n-1$, the component of the Hodge locus of $\mathrm{ch}(E)$ through $A$ is the polarised family itself, whose members are abelian varieties, so nothing

forces $E$ onto non-algebraic tori and the argument cannot be run on $E$ (Proposition 17.11); the one exception is a ratio at $n = 2$ at which the locus is five-dimensional (Remark 17.12). The semiregular objects in Markman's proofs for abelian fourfolds and for the sixfolds of trivial discriminant [Mar25a, Mar25b] are not of the pure form: their Chern characters are not multiples of a Weil class, and they are deformed only along families on which those characters stay of Hodge type, and nothing below depends on whether they have exactly the form above. From here on, in Corollary 20.71, Theorem 20.70 and the frontier of Section 20, (P2) is read in this polarised form. The annihilator computations of Section 16 assume that $\mathrm{ch}(E)$ is a pure Weil class; the next subsection carries them out for a Chern character with components in every even degree (Theorem 17.9, Proposition 17.10, Proposition 17.11).

The same obstruction does not reach the Mumford target of Section 21.2. The exceptional classes there are rigid: their Hodge locus among all complex tori is the Mumford curve (Theorem 21.21, Proposition 21.24), so a semiregular complex with Chern character $N\omega$ on the square of a Mumford fourfold would only have to deform along the curve, whose members are all algebraic.

*Remark* 17.7 (Two finite statements). After Theorem 16.9 and Theorem 16.29 the Hodge conjecture for abelian varieties of Weil type follows from either of the following, each of which is a statement about explicit finite data and neither of which mentions a general point. The first is settled by Theorem 16.17, and we record it because it is the form in which the question first presents itself. The second is false as stated here (Theorem 17.5), and its polarised form, (P2′) of Remark 17.6, is computed in closed form below.

(P1) *A bound.* At every point of the orbit $G(\mathbb{Q}) \cdot s_0$ the class $\omega$ is represented by a cycle whose Hilbert data comes from one finite set. Theorem 16.15 shows that the transport of the base cycle does not satisfy this, for any cycle whose components have finite stabiliser, and Theorem 16.17 shows that, read on an arbitrary representative, the statement is equivalent to $\mathbf{W}(K, n, \delta)$ itself. It is therefore a reformulation of the conjecture for the family, and we record it as such rather than as a criterion.

(P2) *A rank.* The semiregularity map of a perfect complex with the Chern character of Proposition 16.27 at one base point is injective. In this form the statement is false (Theorem 17.5); its polarised form, in Remark 17.6, allows the Chern character to carry powers of the polarisation, and what follows records what the pure form asks. The source of the map is $\mathrm{Ext}^2(E, E)$ for a single object, and its target has dimension $\sum_q h^{q,q+2}(A_0) = \binom{4n}{2n-2}$, computed in Proposition 9.2. So the target is large, which is the direction that helps, and the question is the rank of the map. Full rank would force the following (Theorem 16.31): the evaluation map of Hochschild cohomology must vanish on the $4n^2$-dimensional annihilator of the Weil class, so $E$ must be killed by the mixed block of $H^{0,2}$, must extend along every $K$-linear deformation, must have all products of transverse translation classes zero, and must have no endomorphism of nonzero trace; and at a base point with no abelian subvariety tangent to an eigenspace it can be a locally free sheaf on no smooth germ of its support of codimension at least two (Corollary 16.32). The representative that the construction supplies, a virtual sum of line bundles, fails at the first of these (Proposition 9.10), and any representative that is a sheaf fails at the last. Theorem 16.34 identifies that annihilator as the mixed part of a splitting $HH^1 = P \oplus Q$ into the annihilators of the two conjugate pieces of the class, and Corollary 16.35 draws the consequence: the whole Hochschild action of $E$ factors through a ring of dimension $2^{2n+1} - 1$ inside one of dimension $2^{4n}$, while $\dim \mathrm{Ext}^2(E, E) \geq 2n(2n-1)$ for every representative, semiregular or not. Theorem 16.37 turns the rank condition into that number:

equality suffices. Theorem 16.39 locates the support: at every member it has a component of codimension less than $n$, so no complex carried by a cycle representing $\omega$ can serve, and at a general member every component of codimension at least two has multiplicity zero. Theorem 16.40 fixes its shape: an indecomposable summand carries the class and meets the criterion alone, and at $n = 2$, and at $n = 3$ granting the compatibility of Corollary 16.35(iii), it has at least three independent endomorphisms, so no simple object serves.

Neither statement is the variational Hodge conjecture: (P1) says nothing about deformations, and (P2) is an infinitesimal statement at one point, the global propagation being supplied by Theorem 16.4, which is what Remark B.11 identifies as missing. Nor is either implied by the results of Section 15.5: the base points and the cycles are constructed there, and (P1) and (P2) are questions about those same cycles that the construction does not answer. In the forms stated here both are settled: (P1) is equivalent to its conclusion, and (P2) is false. Its polarised form in Remark 17.6 is the criterion that the rest of this section computes.

17.2. **The polarised criterion in closed form.** The annihilator computations of Section 16 assume that $\mathrm{ch}(E)$ is a pure Weil class, as Remark 17.6 notes. This subsection carries them out for a Chern character with components in every even degree. In degree two the answer is complete and holds for every $n$; in the other degrees it is a closed formula, with one correction term in the middle degree. The same computation gives the Hodge locus of the character, and with it the list of the obstructions of this paper that apply to polarised characters (Proposition 17.11, Remark 17.13). None of this constructs an object or decides whether one exists. It gives, for each shape of Chern character, the number that $\dim \mathrm{Ext}^2(E,E)$ has to reach and the exact kernel that $\mathrm{ev}_E$ must then have, which is the form in which a candidate can be tested.

**Setup 17.8** (Polarised characters)**.** Let $A$ be an abelian $2n$-fold of $(K,-1,n)$-Weil type, $n \geq 2$, with polarisation $\theta$, and let $\omega = u\alpha_+ + \overline{u}\,\alpha_-$ be a nonzero real class on the Weil line, in the notation of Theorem 16.31. For complex numbers $c_0, \dots, c_{2n}$ put

$$p(\theta) = \sum_{k=0}^{2n} c_k \theta^k, \qquad \gamma = \omega + p(\theta), \qquad \mu_m = m!\, c_m,$$

and let $\rho \in \{0,1,2,3\}$ be the rank of the $(2n-1) \times 3$ Hankel matrix $H = [\mu_{j+i}]$, $0 \leq j \leq 2n-2$, $0 \leq i \leq 2$. Write $r(\gamma)$ for the rank of contraction $HT^2(A) \to H^*(A,\mathbb{C})$, $\xi \mapsto \xi \lrcorner \gamma$. For $a < n \leq b$, in the basis of Theorem 16.34 adapted to an eigenbasis $e_j$ of $V_+$ and $f_j$ of $V_-$ with $e_j, f_j$ of type $(1,0)$ for $j < n$ and $j \geq n$ respectively, write $z_{ab} = w(e_b)w(f_a) \in V_+^{0,1} \otimes V_-^{0,1}$, $v^1_{ab} = w(e_b)c(e_a)$ and $v^2_{ab} = w(f_a)c(f_b)$ in $H^1(T_A)$, and $\pi_{ab} = c(e_a)c(f_b) \in T_+ \otimes T_-$, where $w$ is wedge product with a $(0,1)$ class and $c$ contraction of a $(1,0)$ class.

The matrix $H$ is a catalecticant of the moment sequence $\mu_m = m!\, c_m$. A nonzero vector $(x,\lambda,w)$ in its kernel is a linear recurrence $x\mu_j + \lambda\mu_{j+1} + w\mu_{j+2} = 0$ of order at most two, so $\rho \leq 2$ says that the moments satisfy one, and the shapes of Theorem 17.9(ii) are its solutions, read off from the roots of $wT^2 + \lambda T + x$. For a Chern character, $\gamma = N\omega_1 + \sum_k c_k\theta^k$ with rational $c_k$, and $u$ is determined by $N\omega_1$. Since $\alpha_\pm^2 = 0$, $\int_A \omega^2 = 2|u|^2 \int_A \alpha_+\alpha_-$; scaling the eigenbases so that $\int_A \alpha_+\alpha_- = (-1)^n \int_A \theta^{2n}/(2n)!$, which is possible because the sign of $\int_A \alpha_+\alpha_-$ is $(-1)^n$ by the Hodge–Riemann relations, one has $|u|^2 = (-1)^n \frac{(2n)!}{2} \int_A \omega^2 / \int_A \theta^{2n}$. The rank $\rho$ does not change when $\gamma$ is multiplied by $e^{t\theta}$, which acts on the moments by a

binomial transform, and $\omega e^{t\theta} = \omega$ because a Weil class is primitive; so $\rho$ is an invariant of $\gamma$ up to twist.

**Theorem 17.9** (The polarised criterion as a number)**.** *In Setup 17.8:*

(i) *For $n \geq 3$,*

$$\operatorname{Ann}_{HT^2}(\gamma) = \bigoplus_{a<n\leq b} \{x\, z_{ab} + y_1 v^1_{ab} + y_2 v^2_{ab} + w\, \pi_{ab} \ : \ (x,\ i(y_1 - y_2),\ w) \in \ker H\},$$

*so that $\dim \operatorname{Ann}_{HT^2}(\gamma) = n^2(4-\rho)$ and $r(\gamma) = (4+\rho)n^2 - 2n$, whatever $u \neq 0$ is.*

(ii) *$\rho = 3$ for a general $p$. If $\rho = 3$, and either $n \geq 3$ or $n = 2$ and $e = 0$ in the notation of (iii), then $\operatorname{Ann}_{HT^2}(\gamma)$ is exactly the tangent space $T_{[A]}\mathcal{D}_{n,n}$ of the polarised Weil family, spanned by the $n^2$ vectors $v^1_{ab} + v^2_{ab}$, inside $H^1(T_A)$, with no component in $H^{0,2}$ or in $H^0(\bigwedge^2 T_A)$; and $r(\gamma) = \binom{4n}{2} - n^2 = 7n^2 - 2n$. Moreover $\rho \leq 2$ exactly when $p$ is one of $Ae^{s\theta} + Be^{t\theta}$, $(A + B\theta)e^{t\theta}$, $Ae^{t\theta} + B\theta^{2n}$, $A\theta^{2n-1} + B\theta^{2n}$, and $\rho \leq 1$ exactly when $p \in \mathbb{C}e^{t\theta} \cup \mathbb{C}\theta^{2n}$; the pure case $\rho = 0$ gives back $r = 2n(2n-1)$ of Corollary 16.35(i).*

(iii) *For $n = 2$, $\dim \operatorname{Ann}_{HT^2}(\gamma) = 4(4-\rho) + e$ with $e = \dim\ker((HK)^2 - |u|^2) \leq 2$ and $K$ the antidiagonal matrix with entries $1, -2, 1$; so $r(\gamma) \in \{12, 16, 18, 20, 22, 23, 24\}$, and $e > 0$ exactly when $X(X-I)^2 = 4J^2$, where $X = |u|^2$ and $I$, $J$ are the two invariants of the binary quartic $\sum_m \binom{4}{m}\mu_m x^{4-m}y^m$.*

(iv) *Let $E$ be a perfect complex on $A$ with $\operatorname{ch}(E) = \gamma$ and $N \neq 0$. Then $\dim \operatorname{Ext}^2(E,E) \geq r(\gamma)$ with no hypothesis on $\sigma_E$. If equality holds, then $\operatorname{ev}_E$ is onto with kernel $\operatorname{Ann}_{HT^2}(\gamma)$ and $\sigma_E$ is injective, so that (P2′) holds at that point and, by Remark 17.6 and Theorem 16.29, the Hodge conjecture holds for every member of the family with full Hodge group. Conversely, if $\sigma_E$ is injective then $\ker \operatorname{ev}_E = \operatorname{Ann}_{HT^2}(\gamma)$.*

(v) *If $p \in \mathbb{C}e^{t\theta} + \mathbb{C}\theta^{2n}$ with $t\theta$ the class of a line bundle $L$, a set of shapes with $\rho \leq 2$ that contains $\mathbb{C}\theta^{2n}$, then no perfect complex $E$ with $\operatorname{ch}(E) = \gamma$ has injective $\sigma_E$.*

*Proof.* (i) Let the torus $(\mathbb{C}^\times)^{2n}$ act by $e_j \mapsto \tau_j e_j$, $f_j \mapsto \tau_j^{-1} f_j$. It preserves the Hodge decomposition and $\theta$, and scales $\alpha_\pm$ by $(\prod_j \tau_j)^{\pm 1}$; contraction is equivariant. A weight of $HT^2$ has at most two nonzero coordinates, so for $n \geq 3$ two such weights never differ by a nonzero multiple of $(1, \dots, 1)$, and the images of one weight component in the $\theta$-part and in the two $\alpha$-parts land in distinct weight spaces. Hence $\operatorname{Ann}(\gamma) = \operatorname{Ann}(p(\theta)) \cap \operatorname{Ann}(\alpha_+) \cap \operatorname{Ann}(\alpha_-)$, and the last two meet in $P \wedge Q$ by Theorem 16.34(ii). The space $P \wedge Q$ is the sum over $a < n \leq b$ of the four-dimensional weight spaces spanned by $z_{ab}, v^1_{ab}, v^2_{ab}, \pi_{ab}$, and $\operatorname{Ann}(p(\theta))$ is torus-stable, so the computation splits pair by pair. Put $m = e_b \wedge f_a$ and $\theta' = \theta - t_a - t_b$, where $t_j = is_j e_j \wedge f_j$ with $s_j = \pm 1$ the sign that makes $\theta = \sum_j t_j$ positive. Since the $t_j$ are even with $t_j^2 = 0$, direct evaluation gives

$$z \lrcorner \theta^k = m\, \theta'^k, \quad v^1 \lrcorner \theta^k = ik\, m\, \theta'^{k-1}, \quad v^2 \lrcorner \theta^k = -ik\, m\, \theta'^{k-1}, \quad \pi \lrcorner \theta^k = k(k-1)\, m\, \theta'^{k-2},$$

so for $\xi = xz + y_1 v^1 + y_2 v^2 + w\pi$ and $\lambda = i(y_1 - y_2)$,

$$\xi \lrcorner p(\theta) = m \wedge \sum_{j=0}^{2n-2} \frac{\theta'^j}{j!}(x\mu_j + \lambda\mu_{j+1} + w\mu_{j+2}).$$

The classes $m \wedge \theta'^j$, $0 \leq j \leq 2n-2$, are nonzero and of distinct degrees, so $\xi$ kills $p(\theta)$ exactly when $(x, \lambda, w) \in \ker H$. The map $(x, y_1, y_2, w) \mapsto (x, \lambda, w)$ is onto with kernel spanned by $v^1 + v^2$, which gives $1 + (3 - \rho)$ dimensions for each of the $n^2$ pairs. (ii) The vector $v^1_{ab} + v^2_{ab}$ is $K$-linear and kills $\theta$, and these $n^2$ vectors span the tangent space of the

polarised family by Proposition 9.14; when $\rho = 3$ they are all of the annihilator. If $\rho \le 2$, a nonzero $(x, \lambda, w) \in \ker H$ turns the equations $x\mu_j + \lambda\mu_{j+1} + w\mu_{j+2} = 0$ into a linear recurrence of full rank on $\mu_0, \dots, \mu_{2n}$, whose two-dimensional space of solutions is spanned by the listed shapes according to the roots of $wT^2 + \lambda T + x$; $\rho \le 1$ is the intersection of two such spaces. (iii) At $n = 2$ the argument of (i) fails for exactly one pair of weights, $\epsilon_2 + \epsilon_3$ in $\bigwedge^2 P$ and $-\epsilon_0 - \epsilon_1$ in $\bigwedge^2 Q$, whose difference is $(1, 1, 1, 1)$; these give one coupled block of size eight, whose kernel is isomorphic to $\ker(X - T)$ with $T$ conjugate on its image to $(HK)^2$, and the characteristic polynomial of $HK$ is $y^3 - Iy - 2J$, which is traceless, so $e \le 2$. (iv) is the argument of Theorem 16.37(i)–(ii) with $\mathrm{Ann}_{HT^2}(\gamma)$ in place of $P \wedge Q$: $\sigma_E \circ \mathrm{ev}_E$ is contraction into $\gamma$, so $\ker \mathrm{ev}_E \subseteq \mathrm{Ann}_{HT^2}(\gamma)$. (v) If $\gamma - \omega$ has components in degrees 0 and $4n$ only, the proof of Theorem 17.5 applies verbatim. If $p = c\,e^{t\theta} + c'\theta^{2n}$ with $t\theta = c_1(L)$, then $\mathrm{ch}(E \otimes L^{-1}) = \gamma e^{-t\theta} = c + \omega + c'\theta^{2n}$, because $\omega\theta = 0$ and $\theta^{2n+1} = 0$, and $\sigma_{E\otimes L^{-1}} = e^{-c_1(L)} \cdot \sigma_E$ is injective exactly when $\sigma_E$ is, so the first case applies to $E \otimes L^{-1}$. ∎

For a general shape, then, the polarised criterion asks of one complex that $\dim \mathrm{Ext}^2(E, E)$ be 24 at $n = 2$, 57 at $n = 3$ and 104 at $n = 4$, against 12, 30 and 56 for the pure form, and injectivity then forces nothing beyond first-order extension along the polarised family, which is what the criterion is for. The larger number comes from the components $c_k\theta^k$: every direction of $HT^2$ other than the $n^2$ directions of the family moves $\gamma$ off the Hodge classes, so an object meeting the bound has $\mathrm{ev}_E$ injective on a complement of $T_{[A]}\mathcal{D}_{n,n}$. Figure 22 plots the number against $n$ for the four Hankel ranks.

**Proposition 17.10** (The Hochschild profile of a polarised character)**.** *In Setup 17.8, let $\rho_k$ be the rank of contraction $HT^k(A) \to H^*(A, \mathbb{C})$ into $\gamma$, put $C_l = l!\,c_l$, let $r'$ be the rank of the $(n+1) \times (n+1)$ Hankel matrix $H' = [C_{p+q}]$, and $d = \dim\ker((K'H')^2 - |u|^2)$ with $K' = \mathrm{antidiag}((-1)^a\binom{n}{a})$.*

(i) *$\rho_0 = \rho_{2n} = 1$, $\rho_k = 0$ for $k > 2n$, and for $1 \le k \le 2n-1$, $k \ne n$,*

$$\rho_k = 2\binom{2n}{k} + M_k \min(k+1,\, 2n+1-k,\, r'), \qquad M_k = \sum_{\substack{i+j=k \\ 1\le i,j\le n-1}} \binom{n}{i}\binom{n}{j};$$

*in the middle degree the same formula holds with $d$ subtracted. In particular $\rho_k = \rho_{2n-k}$, and the general profiles are $(1, 8, 24, 8, 1)$ at $n = 2$ and $(1, 12, 57, 112, 57, 12, 1)$ at $n = 3$.*

(ii) *$\chi(E, E) = V\,Q(c) + W$ for a perfect complex with $\mathrm{ch}(E) = \gamma$, where $V = \int_A \theta^{2n} > 0$, $W = (-1)^n \int_A \omega^2 > 0$ and $Q(c) = \sum_k (-1)^k c_k c_{2n-k}$; and $\chi(E, E)$ is even.*

(iii) *Granting the compatibility of Corollary 16.35(iii) in every degree, $\dim \mathrm{Ext}^k(E, E) \ge \rho_k$. At $n = 2$, at the points with $d = 1$ one has $\rho_2 = 23$, which is odd, so $\dim \mathrm{Ext}^2(E, E) = r(\gamma)$ cannot hold there, since $\chi(E, E)$ is even and Serre duality gives $\chi(E, E) \equiv \dim \mathrm{Ext}^2(E, E) \bmod 2$.*

*Proof.* (i) Split $HT^1 = Q \oplus P$ as in Theorem 16.34, so that $HT^k = \bigoplus_{i+j=k} \bigwedge^i Q \otimes \bigwedge^j P$ and $Q$, $P$ act on the two tensor factors of $H^*(A) = H_+ \otimes H_-$ separately. A mixed monomial kills $\omega$, and on $p(\theta)$ the block of mixed monomials with fixed index sets $I$, $J$ has entries $C_{s+t+m+m'}$ depending only on $s + t$ and $m + m'$, so its rank is that of a catalecticant of the binary form $\sum_k c_k x^k y^{2n-k}/(2n-k)!$, which is $\min(k+1, 2n+1-k, r')$ by the classical Hilbert function of an apolar ideal. The pure sources $\bigwedge^k Q$ and $\bigwedge^k P$ kill $\alpha_-$ and $\alpha_+$ respectively and contract the other class injectively (Theorem 16.34(iii)). To see where

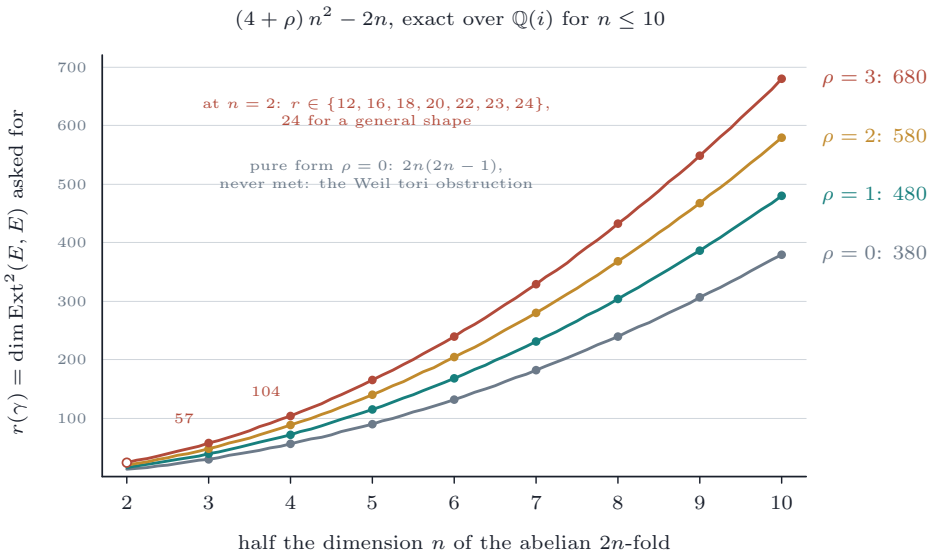



FIGURE 22. The dimension $r(\gamma) = (4+\rho)n^2 - 2n$ that $\mathrm{Ext}^2(E,E)$ must have for a complex with Chern character $\gamma = N\omega_1 + \sum_k c_k\theta^k$ to meet the polarised criterion, for the four Hankel ranks $\rho$ of Theorem 17.9; the dots mark the values at $n = 2, \dots, 10$, and at $n = 10$ the space $HT^2$ has dimension 780. The pure form, $\rho = 0$, is the bottom curve $2n(2n-1)$, which no complex attains (Theorem 17.5); a general shape lies on the top curve $7n^2 - 2n$. At $n = 2$ the weight argument fails for one pair of weights and seven values occur.

their images go, use the torus of the proof of Theorem 17.9(i) and put $I_Q = \{j < n\}$, $I_P = \{j \ge n\}$. The basis vectors $w(f_j), c(e_j)$ of $Q$ have weight $-\epsilon_j$ for $j \in I_Q$, the vectors $w(e_j), c(f_j)$ of $P$ have weight $\epsilon_j$ for $j \in I_P$, $\theta$ has weight 0 and $\alpha_\pm$ has weight $\pm\sum_j \epsilon_j$. If $\xi \in \bigwedge^k Q$ is a monomial, $k < 2n$, and another monomial $\eta$ of $HT^k$ sends $p(\theta)$ or $\alpha_-$ to the weight of $\xi \lrcorner \alpha_+$, which is $+1$ at every $j \in I_P$, then it is not $\alpha_-$, which would need two factors from $P$ at every $j \in I_P$; so $\eta$ has one factor from $P$ for each $j \in I_P$, $\xi$ has weight at most $-1$ at every $j \in I_Q$, and $k \ge n$. For $k = n$ this forces $\xi$ and $\eta$ to lie in the spans $D_Q \subset \bigwedge^n Q$ and $D_P \subset \bigwedge^n P$ of the $2^n$ monomials with one factor for each index. The same holds with $P$ and $Q$ exchanged. A mixed monomial kills $\omega$, so for $k < n$ the matrix of contraction is block triangular, the pure blocks injective onto targets of their own, and $\rho_k$ is $2\binom{2n}{k}$ plus the rank of the mixed blocks; for $k > n$ the formula follows from Lemma 21.33, which also gives $\rho_k = \rho_{2n-k}$.

For $k = n$, a pure monomial outside $D_Q \cup D_P$ repeats an index, so it kills 1 and $t_j$ at that index and hence $p(\theta)$, and keeps a target of its own; no mixed monomial reaches the weights $\sum_{j\in I_P}\epsilon_j$ and $-\sum_{j\in I_Q}\epsilon_j$ of the images of $D_Q \oplus D_P$, since it would need $n$ factors from one side. So $\rho_n$ is the value of the formula minus the dimension of the kernel of the block $\beta\colon D_Q \oplus D_P \to T_1 \oplus T_2$, $T_1$ and $T_2$ being the targets of those two weights. For $R \subseteq I_Q$ and $J \subseteq I_P$ let $X_R = \prod_{j\in I_Q} z_j$ with $z_j = w(f_j)$ for $j \in R$ and $c(e_j)$ otherwise, and $Y_J = \prod_{j\in I_P} z_j$ with $z_j = c(f_j)$ for $j \in J$ and $w(e_j)$ otherwise, the factor of largest index acting first, and write $\alpha_+ = a\, e_0 \cdots e_{2n-1}$, $\alpha_- = b\, f_0 \cdots f_{2n-1}$. Since $\theta^{2n}/(2n)! = \prod_j t_j = \prod_j s_j\, e_0 \cdots e_{2n-1} f_0 \cdots f_{2n-1}$, the normalisation of Setup 17.8 gives

$ab=(-1)^n\prod_j s_j$. As the $t_j$ are even with $t_j^2=0$, $p(\theta)=\sum_S \mu_{|S|}\prod_{j\in S}t_j$. Each operator acts on one factor of $H^*(A)=\bigotimes_j \bigwedge^*\langle e_j,f_j\rangle$ with the Koszul sign of the factors before it, and in each of the four products below the parities of those factors do not depend on $R$ or $J$; the signs are $+1$ on $p(\theta)$, $\epsilon_1=(-1)^{n(n-1)/2}$ for $X_R$ on $\alpha_+$ and $\epsilon_2=(-1)^{n(3n-1)/2}$ for $Y_J$ on $\alpha_-$. With $\tau_R=e_n\cdots e_{2n-1}\prod_{j\in R}t_j$ and $\tau'_{R'}=f_0\cdots f_{n-1}\prod_{j\in R'}t_j$, and using $c(f_j)t_j=-is_je_j$, $c(e_j)t_j=is_jf_j$ and $e_jf_j=-is_jt_j$, one finds

$$X_R\lrcorner\,\alpha_+=\epsilon_1 a\prod_{j\in R}(is_j)\,\tau_R,\qquad Y_J\lrcorner\,p(\theta)=\sum_R\mu_{|J|+|R|}\prod_{j\in J}(-is_j)\,\tau_R,$$
$$Y_J\lrcorner\,\alpha_-=\epsilon_2 b\prod_{j\in I_P\setminus J}(-is_j)\,\tau'_{I_P\setminus J},\qquad X_R\lrcorner\,p(\theta)=\sum_{R'}\mu_{n-|R|+|R'|}\prod_{j\in I_Q\setminus R}(is_j)\,\tau'_{R'}.$$

For $x=\sum_R x_R X_R$ and $y=\sum_J y_J Y_J$ put $\tilde x_R=\prod_{j\in R}(is_j)\,x_R$ and $\tilde y_J=\prod_{j\in J}(-is_j)\,y_J$, and let $\Pi_P=\prod_{j\in I_P}(-is_j)$, $\Pi_Q=\prod_{j\in I_Q}(is_j)$. Then $\beta(x,y)=0$ says

$$\epsilon_1 a\,u\,\tilde x_R=-\sum_J\mu_{|J|+|R|}\,\tilde y_J,\qquad \epsilon_2 b\,\overline{u}\,\Pi_P(-1)^{|J|}\tilde y_J=-\Pi_Q\sum_R\mu_{2n-|R|-|J|}(-1)^{|R|}\tilde x_R$$

for all $R$ and $J$. So $\tilde x_R=g_{|R|}$ and $(-1)^{|J|}\tilde y_J=h_{|J|}$ depend only on $|R|$ and $|J|$, and with $D=\operatorname{diag}((-1)^c\binom{n}{c})$ and the exchange matrix $J_0$, so that $DJ_0=K'$ and $J_0DJ_0=(-1)^nD$, the conditions read $\epsilon_1 a\,u\,g=-H'Dh$ and $\epsilon_2 b\,\overline{u}\,\Pi_P h=-\Pi_Q J_0H'J_0Dg$. Eliminating $h$ gives $(H'K')^2g=\lambda g$ with $\lambda=(-1)^n\epsilon_1\epsilon_2\,ab\,|u|^2\,\Pi_P/\Pi_Q$, and $\epsilon_1\epsilon_2=(-1)^n$, $ab=(-1)^n\prod_j s_j$, $\Pi_P/\Pi_Q=(-1)^n\prod_j s_j$, so $\lambda=|u|^2$. Conversely an eigenvector $g$ determines $h$, $x$ and $y$. Hence $\dim\ker\beta=\dim\ker((H'K')^2-|u|^2)=d$, as $H'K'$ and $K'H'$ are conjugate.

(ii) $\operatorname{ch}(E)^\vee$ changes the sign of the components of odd codegree, $\omega\theta=0$ kills the cross terms, and Hodge–Riemann gives $W>0$; parity holds because the Chern character of a perfect complex on a complex torus is integral, and the intersection form on $\bigwedge^{2n}H^1(A,\mathbb{Z})$ pairs each monomial only with its complement, so $\int x^2$ is even. (iii) is the argument of Corollary 16.35(iii). ∎

The profile also shows what the polarised criterion leaves unconstrained. For the pure form, Theorem 16.40(iv) derives $\dim\operatorname{End}(E)\ge 3$ at $n=2$ and $n=3$ from $\chi(E,E)>0$. For a polarised character, $\chi(E,E)=VQ(c)+W$ can have either sign, and when it is small enough the Euler characteristic forces nothing. At $n=2$, in the model of Setup 4.1 with $d=1$, where $\int\eta^4=24$ and $\int\omega_1^2=8$ (see the proof of Proposition 17.11), the character $\omega_1+p(\eta)$ with $c=(2,1,0,1,-1)$ has $Q(c)=2c_0c_4-2c_1c_3+c_2^2=-6$, so $\chi(E,E)=-144+8=-136$, and a simple object is not excluded.

**Proposition 17.11** (The Hodge locus of a polarised character)**.** (i) *Let $T$ be a complex torus, let $x$ be a real class of type $(k,k)$ on $T$, and for $v\in\operatorname{Hom}(H^{1,0},H^{0,1})=H^1(T_T)$ let $J_v$ be the complex structure on the underlying real torus with $H^{1,0}_v=(1+v)H^{1,0}$. Then $x$ is of type $(k,k)$ for $J_v$ if and only if $v\lrcorner x=0$. So in this chart the Hodge locus of $x$ is the linear space $\operatorname{Ann}_{H^1(T_T)}(x)$, and it coincides with its first-order locus.*

(ii) *Let $J$ be any complex structure on the real torus underlying $A$. If $\theta^k$ is of type $(k,k)$ for $J$ for some $1\le k\le 2n-1$, then $\theta$ is of type $(1,1)$ for $J$.*

(iii) *In Setup 17.8 with $n\ge 3$, the germ at $A$ of the Hodge locus of $\gamma$ among all complex structures is the polarised Weil family $\mathcal{D}_{n,n}$, of dimension $n^2$, if $c_k\neq 0$ for some $1\le k\le 2n-1$, and the family of all $K$-linear deformations, of dimension $2n^2$,*

*otherwise. In the first case the connected component of the Hodge locus through $A$ is $\mathcal{D}_{n,n}$, and all its members are abelian varieties polarised by $\theta$.*

(iv) *For $n = 2$ the same holds, with one exception: if $c_1 = c_3 = 0$ and $c_2^2 \int_A \theta^4 = \frac{3}{4}\int_A \omega^2$, the germ has dimension 5. In the model of Setup 4.1 this is the character $N\omega_1 \pm \frac{d}{2}N\eta^2 + c_0 + c_4\eta^4$; with rational data the ratio is attained exactly in the families of trivial discriminant.*

*Proof.* (i) Write $D_v$ for the derivation of $\bigwedge^* H^1$ extending $v$. Since $v^2 = 0$ on $H^1$, $\bigwedge^*(1+v) = \exp(D_v)$, so the Hodge filtration of $J_v$ is $F_v^p = \exp(D_v)F^p$. Now $\exp(-D_v)x = \sum_m (-1)^m D_v^m x/m!$ with $D_v^m x$ of type $(k-m, k+m)$, so $x \in F_v^k$ exactly when $D_v x = v \lrcorner x = 0$; and a real class lies in $F_v^k \cap \overline{F_v^k} = H_v^{k,k}$ as soon as it lies in $F_v^k$. (ii) Let $t \mapsto t_J$ be the circle acting on $H^{p,q}$ by $t^{p-q}$. It acts on $\bigwedge^* H^1(A,\mathbb{R})$ by algebra automorphisms and fixes $\theta^k$, so the path $t \mapsto t_J\theta$ lies in $\{y \in \bigwedge^2 H^1(A,\mathbb{R}) : y^k = \theta^k\}$. At a nondegenerate $y$ the differential $h \mapsto k\, y^{k-1}h$ of $y \mapsto y^k$ is injective on $\bigwedge^2$ when $2 + (k-1) \le 2n$, by the linear hard Lefschetz theorem for the symplectic form $y$ on a space of dimension $4n$: for a nondegenerate $y \in \bigwedge^2 U^\vee$ on a real space $U$ of dimension $2m$, multiplication by $y^j$ is injective from $\bigwedge^p$ to $\bigwedge^{p+2j}$ whenever $p + j \le m$, because $y$ generates an $\mathfrak{sl}_2$-action on $\bigwedge^* U^\vee$ in which $\bigwedge^p$ has weight $p - m$; here $m = 2n$, $p = 2$ and $j = k-1$. So near $\theta$ that set is discrete, the path is constant, and $\theta$ is of type $(1,1)$. The bound $k \le 2n-1$ is sharp, since $\theta^{2n}$ is a multiple of the volume form for every complex structure. (iii) By (i), applied to each homogeneous component, the Hodge locus of $\gamma$ in the chart is $\mathrm{Ann}_{H^1(T_A)}(\gamma)$, the part of $\mathrm{Ann}_{HT^2}(\gamma)$ in $H^1(T_A)$. In the description of Theorem 17.9(i) that part is $x = w = 0$ with $(0,\lambda,0) \in \ker H$; the vector $(0,1,0)$ lies in $\ker H$ exactly when $\mu_1 = \cdots = \mu_{2n-1} = 0$, so the part is spanned by the $n^2$ vectors $v_{ab}^1 + v_{ab}^2$, which span $T_{[A]}\mathcal{D}_{n,n}$, when some $c_k$ with $1 \le k \le 2n-1$ is nonzero, and by all $v_{ab}^1, v_{ab}^2$, the $K$-linear deformations, otherwise. A germ of submanifold of dimension $n^2$ inside a linear space of the same dimension is that space. On $\mathcal{D}_{n,n}$ the class $\theta$ stays of type $(1,1)$ and nondegenerate, so the signature of the hermitian form it defines is constant and every member is polarised by $\theta$. The same argument at every member shows that $\mathcal{D}_{n,n}$ is open in the Hodge locus; it is closed in the space of complex structures, since $K$-linearity is a closed condition and a limit of positive hermitian forms that stays nondegenerate stays positive; and it is connected. (iv) At $n = 2$ the weight argument of Theorem 17.9(i) fails for one pair of weights. The other weights contribute as in (iii): the $K$-linear deformations, of dimension 8, when $c_1 = c_2 = c_3 = 0$, and the tangent space of the polarised family, of dimension 4, otherwise. The remaining pair of weights couples $\xi' \in \mathrm{Hom}(V_+^{1,0}, V_-^{0,1})$ with $\xi'' \in \mathrm{Hom}(V_-^{1,0}, V_+^{0,1})$. Choose bases $x_{\pm,k}$ of $V_\pm^{1,0}$ and $y_{\pm,k}$ of $V_\pm^{0,1}$, $k = 1,2$, with $\theta = \sum_k (x_{+,k}y_{-,k} + x_{-,k}y_{+,k})$, put $\beta_\pm = x_{\pm,1}x_{\pm,2}y_{\pm,1}y_{\pm,2}$, a generator of $\bigwedge^4 V_\pm$, and write $\omega = u_+\beta_+ + u_-\beta_-$. The component of $(\xi' + \xi'')\lrcorner\gamma$ of degree 3 in $V_-$ and 1 in $V_+$ is $u_-\,\xi''\lrcorner\beta_- + c_2\,\xi'\lrcorner\theta^2$, because $c_1 = c_3 = 0$ and $\xi'\lrcorner\theta^4 = 4\theta^3(\xi'\lrcorner\theta) = 0$; the component of degree 3 in $V_+$ is the same with $+$ and $-$ exchanged. Here $\xi'\lrcorner\theta$ lies in the line spanned by $y_{-,1}y_{-,2}$, so $\xi'\lrcorner\theta^2 = 2\theta\,(\xi'\lrcorner\theta)$ is a multiple of $\theta\, y_{-,1}y_{-,2}$. The map $\xi'' \mapsto \xi''\lrcorner\beta_-$ is injective, and $\theta\, y_{-,1}y_{-,2} = \xi_0''\lrcorner\beta_-$ for $\xi_0''(x_{-,1}) = -y_{+,2}$, $\xi_0''(x_{-,2}) = y_{+,1}$, which has $\xi_0''\lrcorner\theta = 2y_{+,1}y_{+,2}$. With $\xi_0'$ defined in the same way, a solution has $\xi' = a\xi_0'$ and $\xi'' = b\xi_0''$ with $u_+a + 4c_2b = 0$ and $4c_2a + u_-b = 0$. So the annihilator meets this pair in a line when $16c_2^2 = u_+u_-$ and in zero otherwise; since $\theta^4 = 24\,\beta_+\beta_-$ and $\omega^2 = 2u_+u_-\,\beta_+\beta_-$, the condition is $c_2^2 \int \theta^4 = \frac{3}{4}\int \omega^2$. In the model, where $\int \theta^4$ and $\int \omega^2$ are 24 and $2|u|^2$ times the same volume, the ratio reads $|u| = 4|c_2|$. In the model of Setup 4.1, $\eta^4 = 4!\prod_j \epsilon_j x_j y_j$ gives $\int \eta^4 = 24$; in $\omega_1^2$ only

the products $w^{(T)}w^{(T^c)}$ survive, and each of the eight with $|T|$ even contributes $d^2$, so $\int\omega_1^2 = 8d^2$; likewise $\int\omega_2^2 = 8d$, and $\omega_1\omega_2 = 0$ because $T$ and $T^c$ have the same parity. This gives $c_2 = \pm dN/2$. For a general rational Weil class $\omega = x\omega_1 + y\omega_2$ and hermitian form $\mathrm{diag}(-a_1,-a_2,a_3,a_4)$ one has $\int\omega^2 = 8d\,\mathrm{N}_{K/\mathbb{Q}}(y+x\sqrt{-d})$ and $\int\theta^4 = 24\,a_1a_2a_3a_4$, so the ratio asks $(c_2/N)^2 = d\,\mathrm{N}_{K/\mathbb{Q}}(y+x\sqrt{-d})/(4a_1a_2a_3a_4)$, a rational square exactly when $a_1a_2a_3a_4$ is a norm from $K$. ∎

The argument of Theorem 17.5 therefore does not extend to the polarised form. A semiregular complex deforms along the Hodge locus of its Chern character. For the pure form that locus is the whole $2n^2$-dimensional family of $K$-linear tori, whose very general member is not algebraic; for a polarised character with some $c_k \neq 0$, $1 \le k \le 2n-1$, it is the polarised family itself, all of whose members are abelian varieties, and the argument through Weil tori has nothing to act on. This concerns the argument run on $E$ itself: an autoequivalence can still remove a component, and a line bundle twist is how the argument reaches the polarised characters of Theorem 17.9(v). The only other case it can reach is the exceptional ratio of (iv), where the Hodge locus leaves the polarised family.

*Remark* 17.12 (The exceptional ratio at $n = 2$). At $n = 2$, with $c_1 = c_3 = 0$ and $c_2^2\int\theta^4 = \frac{3}{4}\int\omega^2$, put $\gamma_2 = \omega + c_2\theta^2$, and let $\mathfrak{g}$ be its stabiliser in $\mathfrak{gl}(H^1(A,\mathbb{R}))$. In the notation of the proof of Proposition 17.11(iv), put $(e_1,\dots,e_4) = (x_{+,1},x_{+,2},y_{+,1},y_{+,2})$ and $(f_1,\dots,f_4) = (y_{-,1},y_{-,2},-x_{-,1},-x_{-,2})$; then $\theta = \sum_j e_jf_j$, $\omega = u_+e_{1234} + u_-f_{1234}$ with $e_{1234} = e_1\wedge\cdots\wedge e_4$, and the ratio is $16c_2^2 = u_+u_-$. Passing to the basis $se_j$, $s^{-1}f_j$ with $s^8 = u_+/u_-$ fixes $\theta$ and makes the two coefficients of $\omega$ equal, and passing from $f_j$ to $\sqrt{-1}f_j$, if necessary, changes the sign of $\theta^2$ and fixes $f_{1234}$. So over $\mathbb{C}$ the class $\gamma_2$ is a nonzero multiple of

$$\tfrac{1}{2}(e_{1234}+f_{1234}) - \tfrac{1}{8}\Big(\textstyle\sum_j e_jf_j\Big)^2,$$

which is the Cayley form $\frac{1}{2}\omega_0^2 + \Re\Omega_0$ of $\mathbb{C}^4$, with $\omega_0 = \frac{\sqrt{-1}}{2}\sum dz_j\wedge d\bar z_j$ and $\Omega_0 = dz_1\wedge\cdots\wedge dz_4$, written in the basis $e_j = dz_j$, $f_j = d\bar z_j$ [HL82]. The stabiliser of the Cayley form in $\mathrm{GL}(8,\mathbb{R})$ is $\mathrm{Spin}(7)$, acting on $\mathbb{R}^8$ by its spin representation, and its invariants in $\bigwedge^*\mathbb{R}^8$ are spanned by 1, the form and the volume form [Bry87]. A stabiliser in $\mathfrak{gl}$ is cut out by linear equations, so it commutes with extension of scalars, and $\mathfrak{g}\otimes\mathbb{C} \cong \mathfrak{so}(7,\mathbb{C})$ acts on $H^1(A,\mathbb{C})$ by the spin representation, irreducibly, with invariants in $\bigwedge^* H^1$ spanned by 1, $\gamma_2$ and the class of a point. At an ordinary ratio the stabiliser is the Lie algebra $\mathfrak{su}(2,2)$ of the Hodge group, of dimension 15, which also fixes $\gamma_2$ here.

The real form is $\mathfrak{so}(4,3)$. The spin representation of $\mathfrak{so}(7,\mathbb{C})$ carries an invariant symmetric form, unique up to a scalar by irreducibility, so $\mathfrak{g}$ preserves a real symmetric form $B$ on $H^1(A,\mathbb{R})$. On $H^1(A,\mathbb{C}) = V_+\oplus V_-$ the subalgebra $\mathfrak{su}(2,2)\otimes\mathbb{C} = \mathfrak{sl}(V_+)$ acts on $V_+$ and dually on $V_-$, and its only invariant bilinear forms pair $V_+$ with $V_-$; so $B$ is a multiple of the real part of the hermitian form of $\theta$, of signature $(4,4)$, and $\mathfrak{g}\subseteq\mathfrak{so}(4,4)$. The real forms of $\mathfrak{so}(7,\mathbb{C})$ are the $\mathfrak{so}(p,q)$ with $p+q=7$, and the maximal compact subalgebra $\mathfrak{so}(p)\oplus\mathfrak{so}(q)$ lies in a maximal compact subalgebra of $\mathfrak{so}(4,4)$, which is $\mathfrak{so}(4)\oplus\mathfrak{so}(4) \cong \mathfrak{su}(2)^4$. A simple subalgebra of $\mathfrak{su}(2)^4$ projects injectively to a factor, so it has dimension 3; hence $p,q\le 4$, and $\{p,q\} = \{4,3\}$. So $\gamma_2$ is the invariant 4-form of $\mathrm{Spin}(4,3)$, a split form of the Cayley form.

The Hodge decomposition of $\mathfrak{g}$ has dimensions 5, 11, 5. Indeed, the complex structure $J$ of $A$ fixes the class $\gamma_2$ of type $(2,2)$ and lies in the image of the Hodge group, which is connected and contained in the group of $\mathfrak{g}$; as $J^2 = -1$ is the nontrivial central element of $\mathrm{Spin}(4,3)$, the image of $J$ in $\mathrm{SO}(7,\mathbb{C})$ is an involution with $2k$ eigenvalues $-1$ and $k$ odd,

since a lift of such an involution squares to $(-1)^k$. The eigenvalue $-1$ of $\operatorname{Ad} J$ on $\mathfrak{g}\otimes\mathbb{C}$ then has multiplicity $2k(7-2k)$, which is 10 or 6, and it contains the $(-1)$-eigenspace of $\mathfrak{sl}(V_+)$, of dimension 8; so $k=1$, and $\mathfrak{g}^{-1,1}\oplus\mathfrak{g}^{1,-1}$ has dimension 10.

The Hodge locus of $\gamma$ near $A$ is the orbit of $A$ under the connected real group $G$ of $\mathfrak{g}$, since that orbit has dimension $\dim\mathfrak{g}^{-1,1}=5$ and lies in the locus, which near $A$ is smooth of dimension 5 (Proposition 17.11(i) and (iv)). The Hodge group of a very general member $T$ has Lie algebra an ideal of $\mathfrak{g}$, hence all of it, so the Hodge classes of $T$ are 1, $\gamma_2$ and the class of a point, and $\mathrm{NS}(T)=0$. The locus therefore contains tori that are not algebraic, and the proof of Theorem 17.5 can be started there. It breaks down at its last step: Proposition 17.4 needs a Kähler class on which the degree-four Chern character integrates to zero, and here

$$\int_A \gamma_2\,\kappa_h^2=\frac{|c_2|}{6}\Big(\operatorname{sign}(c_2)\,\sigma_2(h)+4\operatorname{Re}\big(\phi\det h_{01,23}\big)\Big)\int_A\theta^4,\qquad |\phi|=1,$$

for the Kähler class $\kappa_h=i\sum h_{jk}z_j\wedge\overline{z}_k$ of a positive hermitian $h$ in a unitary frame for $\theta$, and the right-hand side has the sign of $c_2$. Indeed, writing $h=\sum_j\lambda_j w_j w_j^*$ with $(w_j)$ orthonormal, Cauchy–Binet gives $\sigma_2(h)\pm4\operatorname{Re}(\phi\det h_{01,23})=\sum_{j<k}\lambda_j\lambda_k\big(1\pm4\operatorname{Re}(\phi\,\xi_{01}\overline{\xi}_{23})\big)$ with $\xi=w_j\wedge w_k$ of norm one. The Plücker relation and the inequality of the means give $|\xi_{01}\xi_{23}|\le\frac14$, so each term is nonnegative, and by Parseval the brackets sum to 6 over $j<k$, so the sum is at least $6\min_j\lambda_j^2>0$. Since $G$ fixes $\gamma_2$ and carries Kähler cones to Kähler cones, the same holds on every member of the locus. So at this ratio the argument excludes at most a torsion-free sheaf with $c_2>0$, which Bogomolov's inequality already forbids to be $\theta$-semistable, and it excludes no complex. With $R=c_2^2\int\theta^4/\int\omega^2$, the ratio that equals $\frac34$ here, Figure 23 draws the infimum of the ratio as a function of $R$. Since Markman's theorem settles every Weil fourfold, this remark concerns the shape of (P2′), not the conjecture.

*Remark* 17.13 (The obstructions for a polarised character). The obstruction theorems proved from Section 9 on for a Chern character on the Weil line divide as follows for a polarised character. The following hold unchanged, with $r(\gamma)$ in place of $2n(2n-1)$ where a number occurs: Theorem 9.20 and Corollary 18.21, whose only hypothesis is that $\operatorname{ch}(E)$ stay of Hodge type to first order along $\mathcal{D}_{n,n}$; Theorem 18.5 and Corollary 18.6, since a polarised character is a flat Hodge class in every degree; Lemma 16.38; and the first two parts of Theorem 16.37, which become Theorem 17.9(iv). The following hold only when $c_0=\cdots=c_{2n-2}=0$, so that $H^{0,2}$ still kills $\gamma$: Theorem 9.7, Proposition 9.10, Corollary 9.18, Theorem 9.19 and Theorem 16.31(iii). The following hold only when $\ker H$ contains a vector with $w\neq0$, which needs $\rho\le2$, the vector giving a compensated class of $P\wedge Q$ with a component in $T_+\otimes T_-$: Theorem 16.31(iv), Corollary 16.32 and Theorem 16.39. The following do not extend: Theorem 17.5, outside Theorem 17.9(v); the positivity of $\chi(E,E)$ in Proposition 9.11 and with it Theorem 16.40(i)–(iii), by Proposition 17.10(ii); and Theorem 20.32. So for a general shape, $\rho=3$, what these results exclude is a line bundle summand off the polarisation line and a Chern character effective in the exponential basis, and nothing else; Proposition 17.14 below extends the first to summands with a scalar Atiyah class. An object with no such summand, whose Chern character is not effective, and with $\dim\operatorname{Ext}^2(E,E)=7n^2-2n$, is not excluded by anything proved here, and none is known; by Lemma 17.16 below it may be taken indecomposable.

**Proposition 17.14** (Summands with a scalar Atiyah class)**.** *Let $A$ be of $(K,-1,n)$-Weil type with $n\ge2$, let $T=T_{[A]}\mathcal{D}_{n,n}$, and let $E$ be a perfect complex on $A$ with $v\lrcorner\operatorname{ch}(E)=0$*

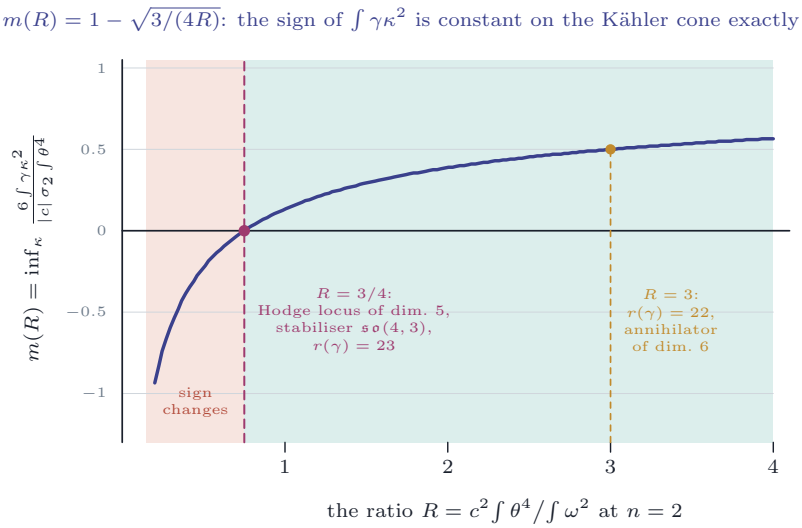



FIGURE 23. The exceptional ratio of Remark 17.12. By the identity displayed there, which is linear in $\gamma$, the infimum over the Kähler cone of $6\int\gamma\kappa^2/(|c|\,\sigma_2(h)\int\theta^4)$ is $m(R) = 1 - \sqrt{3/(4R)}$, approached at the boundary of the cone by a positive hermitian form of rank two whose Plücker coordinates satisfy $|\xi_{01}\xi_{23}| = \frac{1}{4}$. It is negative exactly for $R < 3/4$, and $R = 3/4$ is also where the Hodge locus of $\gamma$ jumps from dimension 4 to 5 and its stabiliser becomes $\mathfrak{so}(4,3)$.

*for every $v \in T$. Suppose $E$ has a direct summand $F[p]$ with $\mathrm{At}(F) = \delta\cdot\mathrm{id}_F$, where $\delta$ is a real class of type $(1,1)$ that is not a real multiple of $\theta$, and $\mathrm{rk}\,F \neq 0$. Then there is $v \in T$ with $\mathrm{ev}_E(v) \neq 0$ and $\sigma_E(\mathrm{ev}_E(v)) = 0$. In particular no finite direct sum of shifted bundles $\pi_*L$, $\pi$ an isogeny, among them the simple semi-homogeneous bundles, has a polarised character $\gamma$ with $N \neq 0$ and injective semiregularity map.*

*Proof.* As in Theorem 9.20, $\mathrm{ev}_E(v) = v\lrcorner\mathrm{At}(E)$ on $H^1(T_A)$ and the Atiyah class of a direct sum is block diagonal, so the component of $\mathrm{ev}_E(v)$ in $\mathrm{Ext}^2(F,F)$ is $(v\lrcorner\delta)\cdot\mathrm{id}_F$; the map $z \mapsto z\cdot\mathrm{id}_F$ on $H^{0,2}$ is injective because $\mathrm{tr}(z\cdot\mathrm{id}_F) = \mathrm{rk}(F)\,z$, and some $v \in T$ has $v\lrcorner\delta \neq 0$ by Proposition 9.15(i). On the other hand $\sigma_E(\mathrm{ev}_E(v)) = v\lrcorner\mathrm{ch}(E) = 0$. For the last sentence, the proof of Lemma 18.91(i) gives $\mathrm{At}(\pi_*L) = \delta\cdot\mathrm{id}$ with $\pi^*\delta = c_1(L)$ for any line bundle $L$, and $\mathrm{ch}(\pi_*L) = \mathrm{rk}(\pi_*L)\,e^{\delta}$; if every $\delta$ were a multiple of $\theta$ the Chern character of the sum would lie in $\mathbb{Q}[\theta]$, which does not contain $\omega$ because $\omega\theta = 0$ and $\theta^{n+1} \neq 0$. ∎

Theorem 16.29 asks that $\sigma_E$ be injective, but its proof uses only that the relative deformations of $E$ dominate the family, and Pridham's form of Theorem 16.28, in which $\sigma_E$ annihilates every obstruction, gives this under a weaker hypothesis. That hypothesis allows an object that is obstructed to first order, which is exactly the way in which Theorem 9.20, Corollary 18.21 and Proposition 17.14 exclude an object.

**Proposition 17.15** (A flatness form of the criterion)**.** *Let $n \geq 2$, $s_0 \in \mathcal{D}_{n,n}$, and let $E$ be a bounded complex of vector bundles on $A_{s_0}$ with $\mathrm{ch}(E) = N\omega_1 + \sum_k c_k\theta^k$ and $N \neq 0$. Let $K_E$ be the base of a miniversal deformation of $E$ on the fixed variety $A_{s_0}$, put $e = \dim\mathrm{Ext}^1(E,E)$ and $k = \dim\ker\sigma_E$. By Theorem 16.28 in Pridham's form, $\sigma_E$*

*annihilates every obstruction, both on $A_{s_0}$ and along the Artinian deformations of $A_{s_0}$ in $\mathcal{D}_{n,n}$.*

(i) $\dim K_E \geq e - k$.

(ii) *If* $\dim K_E = e - k$, *then* $\Sigma = \mathcal{D}_{n,n}$, *and the Hodge conjecture holds in every degree for every member of the family with* $\mathrm{Hg}(A) = \mathrm{SU}(V, H)$.

(iii) $\dim K_E = e - k$ *as soon as the cone* $\{\xi \in \mathrm{Ext}^1(E, E) : \xi \circ \xi = 0\}$ *has codimension* $k$.

*For* $k = 0$ *this is Theorem 16.29.*

*Proof.* Let $R$ be the completed local ring of $\mathcal{D}_{n,n}$ at $s_0$, regular of dimension $n^2$. By hypothesis $\ker \sigma_E$ is a complete obstruction space for the deformations of the pair $(A_s, E)$ over Artinian $R$-algebras, whose relative tangent space is $\mathrm{Ext}^1(E, E)$; so this functor has a hull $P/J$ with $P = R[[x_1, \ldots, x_e]]$ and $J$ generated by at most $k$ elements [FM98], and its fibre over the closed point of $R$ is a hull of the absolute functor, of dimension $\dim K_E$. Krull's height theorem gives $\dim P/J \geq n^2 + e - k$, and the same count on the fibre gives (i). If $\dim K_E = e - k$, the dimension inequality for a local homomorphism [Mat86, Theorem 15.1] gives $\dim P/J \leq n^2 + e - k$, so equality holds, $J$ has height $k$ and is generated by a regular sequence, $P/J$ is Cohen–Macaulay, and it is flat over $R$ by miracle flatness [Mat86, Theorem 23.1]: a local homomorphism from a regular local ring to a Cohen–Macaulay local ring is flat exactly when the dimension of the target is the dimension of the source plus that of the fibre, and here both sides equal $n^2 + e - k$. The analytic Kuranishi germ of pairs, as in the proof of Theorem 17.5, has $P/J$ as its completion, so it is flat over the germ of $\mathcal{D}_{n,n}$ at $s_0$ and therefore open onto a neighbourhood $U$ of $s_0$. Every $A_s$ with $s \in U$ then carries a bounded complex of vector bundles, algebraic by GAGA, whose Chern character is the parallel transport of $\mathrm{ch}(E)$, so $N\omega_{1,s}$ differs from an algebraic class by a polynomial in $\theta$ and is algebraic; Corollary 16.5 gives (ii). For (iii), the quadratic part of the Kuranishi map of $E$ is $\xi \mapsto \xi \circ \xi$, so the tangent cone of $K_E$ lies in the cone of the hypothesis, and $\dim K_E \leq e - k$. Every step uses only deformations along $\mathcal{D}_{n,n}$, which are projective, whereas the proof of Theorem 17.5 needs non-projective ones. ∎

No object is known that meets (iii) with $k > 0$. Among direct sums the flatness form fails for a large class (Remark 17.29 below); for indecomposable objects nothing in this paper excludes it, and the proposition says only that injectivity of $\sigma_E$ is more than propagation needs.

17.3. **Candidate objects for the polarised criterion.** The polarised criterion asks for one perfect complex. The results below say what such a complex can be made of, for $K$ imaginary quadratic: a direct sum meets the criterion only through one indecomposable summand (Lemma 17.16); no sum of objects whose Ext algebra is an exterior algebra meets it (Corollary 17.17); the Orlov products of secant objects meet its twisted form, with $\dim \mathrm{Ext}^2$ equal to the number of Theorem 17.9, at $n = 2$ and $n = 3$ (Theorem 17.19), and cannot meet it at $n = 4$ when the factors have no negative self-extensions (Proposition 17.20); line bundles on smooth divisors, which give the certificates, give no factor from $n = 4$ on, nor do bundles restricted from the abelian variety at $n = 5$ (Proposition 17.23); and at $n = 4$ a combination of natural objects needs at least eight of them, on one conic (Theorem 17.27). None of this gives a new case of the Hodge conjecture: the certificates are for twisted objects, in families where Markman's theorem already holds.

**Lemma 17.16** (The summands of an object meeting the criterion)**.** *Let $A$ be a member of a $(K, -1, n)$-Weil family, $K$ imaginary quadratic and $n \geq 2$, let $T = T_{[A]}\mathcal{D}_{n,n} \subseteq H^1(T_A)$, and let $E = \bigoplus_i F_i[k_i]$ be a finite direct sum of perfect complexes with $v \lrcorner \mathrm{ch}(E) = 0$ for*

*every $v \in T$, as holds for every polarised character $N\omega_1 + \sum_k c_k\theta^k$. Suppose that $\sigma_E$ is injective on $\mathrm{ev}_E(T)$; this holds if $\sigma_E$ is injective, and under the weakened criterion of* [Mar25b, Question 11.4].

(i) *For every $i$ and every $v \in T$, $\mathrm{ev}_{F_i}(v) = 0$ and $v \lrcorner \mathrm{ch}(F_i) = 0$, and every $\mathrm{ch}(F_i)$ lies in $\bigoplus_k \mathbb{Q}\theta^k \oplus W(A)$.*

(ii) *If $\mathrm{ch}(E) = N\omega_1 + \sum_k c_k\theta^k$ with $N \neq 0$, some $F_i$ has $\mathrm{ch}(F_i) = \omega' + \sum_k c'_k\theta^k$ with $\omega'$ a nonzero rational Weil class, and then $\dim \mathrm{Ext}^2(F_i, F_i) \geq r(\mathrm{ch}\, F_i) \geq 2n(2n-1)$. If $\sigma_E$ is injective, so is every $\sigma_{F_i}$, and that $F_i$ meets the polarised criterion on its own.*

*So the polarised criterion is a statement about indecomposable objects.*

*Proof.* For $v \in T$, $\sigma_E(\mathrm{ev}_E(v)) = v \lrcorner \mathrm{ch}(E) = 0$ by the commutative square of Setup 18.54, so $\mathrm{ev}_E(v) = 0$. The Atiyah class is natural, so $\mathrm{ev}_E(v) = v \lrcorner \mathrm{At}(E)$ is block diagonal with the blocks $\pm\,\mathrm{ev}_{F_i}(v)$, which therefore vanish, and $v \lrcorner \mathrm{ch}(F_i) = \sigma_{F_i}(\mathrm{ev}_{F_i}(v)) = 0$. By Proposition 17.11(i) every homogeneous component of $\mathrm{ch}(F_i)$, a rational class of type $(k,k)$, stays of Hodge type on the germ of $\mathcal{D}_{n,n}$ at $A$, which in the chart of that proposition is the linear space $T$; so it is a Hodge class at a member with Hodge group $\mathrm{SU}(V,H)$, and Theorem 4.14 puts it in $\mathbb{Q}\theta^k$, or in $\mathbb{Q}\theta^n \oplus W$ for $k = n$. That is (i). For (ii), the Weil parts of the $\pm\,\mathrm{ch}(F_i)$ add up to $N\omega_1 \neq 0$, so one of them is nonzero; Theorem 17.9 gives $\dim \mathrm{Ext}^2(F_i, F_i) \geq r(\mathrm{ch}\, F_i)$ with no hypothesis on $\sigma$, and $r \geq (4+0)n^2 - 2n = 2n(2n-1)$ for $n \geq 3$, while at $n = 2$ the seven values of Figure 22 are at least 12. Injectivity passes to the summands by Lemma 21.38. By the Krull–Schmidt property of $D^b(A)$ the $F_i$ may be taken indecomposable. ∎

The Weil part $\omega'$ of the summand need not be a multiple of $\omega_1$; propagating any nonzero rational Weil class propagates the whole Weil plane, through the algebraic action of $K$. The lemma is the Weil-family analogue of Corollary 21.41(ii).

**Corollary 17.17** (Sums of objects with exterior Ext algebras)**.** *In Lemma 17.16, let every $F_i$ have $\dim \mathrm{Ext}^2(F_i, F_i) < 2n(2n-1)$. This holds when $\mathrm{Ext}^*(F_i, F_i) \cong \bigwedge^* \mathbb{C}^{2n}$, and so for line bundles, for the sheaves $\mathcal{O}_B \otimes L$ with $B \subseteq A$ an abelian subvariety, for simple semi-homogeneous bundles, for skyscraper sheaves, and for the images of all of these under autoequivalences of $D^b(A)$. Then no such sum with $\mathrm{ch}(E) = N\omega_1 + \sum_k c_k\theta^k$ and $N \neq 0$ has $\sigma_E$ injective on $\mathrm{ev}_E(T)$: it fails the polarised criterion and the weakened criterion, some $v \in T$ obstructs it to first order, and $\dim \mathrm{Ext}^2(E, E) > r(\mathrm{ch}\, E)$.*

*Proof.* The first assertion is Lemma 17.16(ii), since $\binom{2n}{2} = n(2n-1) < 2n(2n-1)$, and $\dim \mathrm{Ext}^2(E, E) \geq r(\mathrm{ch}\, E)$ with equality only if $\sigma_E$ is injective (Theorem 17.9). For $\mathcal{O}_B$ choose a complementary abelian subvariety $B'$; the isogeny $\mu\colon B \times B' \to A$ is étale with group $G = B \cap B'$, and $\mu^*\mathcal{O}_B = \bigoplus_{g \in G} \mathcal{O}_{B \times \{g\}}$ with $G$ permuting the summands simply transitively, so $\mathrm{Ext}_A(\mathcal{O}_B, \mathcal{O}_B) = \mathrm{Ext}_{B \times B'}(\mathcal{O}_{B \times 0}, \mathcal{O}_{B \times 0}) = \bigwedge^*(\mathbb{C}^{\dim B} \oplus \mathbb{C}^{\dim B'})$. For a simple semi-homogeneous bundle $V$, $V \otimes V^\vee$ is a sum of line bundles in $\mathrm{Pic}^0$ of which $\mathcal{O}$ occurs once [Muk78], so $\mathrm{Ext}^*(V, V) = H^*(\mathcal{O}_A)$. Equivalences preserve Ext algebras. ∎

By Lemma 17.16 the objects that Remark 17.13 allows may be taken indecomposable. The corollary recovers, for polarised characters, the part of Theorem 20.32 that concerns objects with exterior Ext algebras, and it extends Proposition 17.14, whose summand has a scalar Atiyah class, to the subtori, the points and the Fourier–Mukai images. In the language of pure spinors, a summand whose character is a rational multiple of the pure spinor of a rational maximal isotropic subspace is excluded unless that character lies in

$\mathbb{Q}[\theta]$, because the rational maximal isotropic subspaces whose pure spinors are flat over $\mathcal{D}_{n,n}$ are those with pure spinor 1, the class of a point or $e^{c\theta}$.

The secant objects of Section 14 have their own Hochschild profile, which Theorem 18.56(i) and (ii) compute in degrees one and two.

**Proposition 17.18** (The Hochschild profile of a secant class)**.** *Let $X$ be a principally polarised abelian $n$-fold, $n \geq 2$, with polarisation $\theta$, let $d \geq 1$, and let $v = a\,u_\theta + b\,v_\theta$ with $(a,b) \in \mathbb{Q}^2 \setminus \{0\}$, in the notation of (30). The rank of contraction $HT^k(X) \to H^*(X,\mathbb{C})$, $\xi \mapsto \xi \lrcorner v$, is 1 for $k = 0$ and $k = n$, $2\binom{n}{k}$ for $1 \leq k \leq n-1$, and 0 for $k > n$. A perfect complex $F$ on $X$ with $\operatorname{ch}(F) = v$ has $\dim \operatorname{Ext}^k(F,F) \geq 2\binom{n}{k}$ for $k = 1, 2$, and*

$$\chi(F,F) = \begin{cases} (-1)^{n/2}\, 2^{n-1} d^{\,n/2-1}(a^2 d + b^2) & n \text{ even}, \\ 0 & n \text{ odd}. \end{cases}$$

*Proof.* Put $s = \sqrt{-d}$ and $w = \frac{1}{2}(a - bs/d)$. By Theorem 14.5(i), $v = w\,e^{s\theta} + \overline{w}\,e^{-s\theta}$, and $w \neq 0$. By Lemma 18.55(i), $\xi \lrcorner e^{\pm s\theta} = y_\pm e^{\pm s\theta}$, where $y_\pm \in H^{0,k}(X)$ is the component of $e^{\pm s\theta}\xi$ in $\bigwedge^k H^{0,1}$. The map $\xi \mapsto (y_+, y_-)$ is onto $H^{0,k} \oplus H^{0,k}$ for $k \geq 1$: $\xi = z \in \bigwedge^k H^{0,1}$ gives $(z,z)$, and $\xi = z' \otimes \partial$ with $z' \in \bigwedge^{k-1} H^{0,1}$ gives $\pm s\, z' \wedge (\partial \lrcorner \theta)$, where $\partial \lrcorner \theta$ runs over $H^{0,1}$. If $w\,y_+ e^{s\theta} + \overline{w}\,y_- e^{-s\theta} = 0$, the components of degrees $k$ and $k+2$ give $wy_+ + \overline{w}y_- = 0$ and $(wy_+ - \overline{w}y_-) \wedge \theta = 0$, and for $k < n$ wedge product with $\theta = \sum_a p_a \wedge q_a$ is injective on $\bigwedge^k H^{0,1}$, so $y_\pm = 0$. So the rank is $2\binom{n}{k}$ for $1 \leq k < n$. For $k = n$, $y \wedge \theta = 0$ for $y \in H^{0,n}$ and the image is the line spanned by $wy_+ + \overline{w}y_-$; for $k > n$, $H^{0,k} = 0$; and for $k = 0$ the rank is one. The bounds on $\operatorname{Ext}^1$ and $\operatorname{Ext}^2$ follow, as in Theorem 18.56(i) and (ii), from the compatibility of the characteristic morphism with contraction into the Chern character. Finally the Todd class is trivial, so $\chi(F,F) = \int_X \operatorname{ch}(F)^\vee \operatorname{ch}(F)$ with $\operatorname{ch}(F)^\vee = w\,e^{-s\theta} + \overline{w}\,e^{s\theta}$, and $\int_X (e^{2s\theta} + e^{-2s\theta}) = (2s)^n(1 + (-1)^n)$ because $\int_X \theta^n = n!$; with $|w|^2 = (a^2 d + b^2)/(4d)$ this is the formula. ∎

At $n = 4$ and $(a,b) = (1,3)$, the class of Proposition 18.52, this gives $\chi(\mathcal{F},\mathcal{F}) = 8d(d+9)$, the rank recorded there.

**Theorem 17.19** (Twisted certificates at $n = 2$ and $n = 3$)**.** *Let $d \geq 1$, $K = \mathbb{Q}(\sqrt{-d})$, let $\Phi$ be Orlov's equivalence of Proposition 18.52(v), and for a perfect complex $F$ on a principally polarised $X$ put $E = \Phi(F \boxtimes F^\vee)$ on $A = X \times \widehat{X}$, with the $K$-action of (29), the Poincaré class $\ell$ and the polarisation $\eta = d\beta + \widehat{\beta}$ of the split member (Notation 15.1).*

(i) *Let $X$ be the Jacobian of a curve $C$ of genus two, $C \subset X$ by an Abel–Jacobi map, and $F = \mathcal{O}_C(p)$ for a point $p \in C$. Then $\operatorname{ch}(F) = v_\theta$, $\dim \operatorname{Ext}^k(F,F) = 1, 4, 1$, and $\dim \operatorname{Ext}^k(E,E) = 1, 8, 18, 8, 1$ for $k = 0, \ldots, 4$, which are the ranks of contraction from $HT^k(A)$ into $\operatorname{ch}(E)$. So $\dim \operatorname{Ext}^2(E,E) = 18 = r(\operatorname{ch} E)$ and $\sigma_E$ is injective.*

(ii) *Let $X$ be the Jacobian of a non-hyperelliptic curve $C$ of genus three, $i\colon C^{(2)} \to X$ the embedding of the theta divisor, $x$ the class of $\{p+q : q \in C\}$ and $\theta$ the restriction of the polarisation, and let $d = k(k+1)$ with $k \geq 1$ and $F = i_*\mathcal{O}_{C^{(2)}}((2k+1)x - k\theta)$. Then $\operatorname{ch}(F) = v_\theta$, $\dim \operatorname{Ext}^k(F,F) = 1, 6, 6, 1$, and $\dim \operatorname{Ext}^k(E,E) = 1, 12, 48, 74, 48, 12, 1$. So $\dim \operatorname{Ext}^2(E,E) = 48 = r(\operatorname{ch} E)$ and $\sigma_E$ is injective.*

(iii) *In both cases $\operatorname{ch}(E)$ is not killed by $T = T_{[A]}\mathcal{D}_{n,n}$, and no line bundle twist of $E$ has a character killed by $T$; the twisted character $\kappa = \operatorname{ch}(E)e^{\ell/2}$ is killed by $T$, has the form $N'\omega' + \sum_k c_k \eta^k$ with $\omega'$ a nonzero rational Weil class, and has the same contraction ranks as $\operatorname{ch}(E)$. A class $B \in \operatorname{NS}(A)_\mathbb{Q}$ with $\operatorname{ch}(E)e^B$ killed by $T$ is congruent to $\ell/2$ modulo $\mathbb{Q}\eta$. At $n = 2$, $\kappa$ has $c_1 = c_3 = 0$ and lies at the exceptional*

*ratio of Remark 17.12, and its annihilator in $H^1(T_A)$ has dimension 5; at $n=3$ that annihilator has dimension $9=n^2$.*

*Proof.* (i) By Grothendieck–Riemann–Roch $\mathrm{ch}(i_*\mathcal{O}_C(p))=[C]+(1+1-2)[\mathrm{pt}]=\theta=v_\theta$, since $v_\theta=\theta$ when $n=2$. For the curve, $\mathrm{Ext}^1(\mathcal{O}_C,\mathcal{O}_C)=H^1(\mathcal{O}_C)\oplus H^0(N_C)$ with $N_C=K_C$ by adjunction, of dimension $2+2$, and $\mathrm{Ext}^0=\mathrm{Ext}^2=\mathbb{C}$ by Serre duality; twisting by $\mathcal{O}_C(p)$ changes nothing. The Ext algebra of $F^\vee$ is that of $F$, $\Phi$ is an equivalence, and by the Künneth formula $\mathrm{Ext}^*(E,E)=\mathrm{Ext}^*(F,F)\otimes\mathrm{Ext}^*(F^\vee,F^\vee)$, which gives $1,8,18,8,1$. By Proposition 17.18 and the Künneth decomposition $HT^2(X\times X)=HT^2\otimes 1\oplus HT^1\otimes HT^1\oplus 1\otimes HT^2$ the rank of contraction into $\mathrm{ch}(F)\boxtimes\mathrm{ch}(F^\vee)$ is $1+4\cdot 4+1=18$, and it is transported by $\Phi$ as in the proof of Theorem 18.56(iv). The other degrees are the same count. Equality in degree two forces $\sigma_E$ injective, as in Theorem 17.9. (ii) On $C^{(2)}$ one has $x^2=1$, $x\theta=3$, $\theta^2=6$ and $K_{C^{(2)}}=\theta$, $i_*x=\frac12\Theta^2$ and $i_*\theta=\Theta^2$, so Riemann–Roch gives $\mathrm{ch}_2(F)=0$ and $\mathrm{ch}_3(F)=-k(k+1)[\mathrm{pt}]$, that is $\mathrm{ch}(F)=\Theta-d[\mathrm{pt}]=v_\theta$. The local-to-global spectral sequence, with $h^p(\mathcal{O}_{C^{(2)}})=1,3,3$ and $h^p(K_{C^{(2)}})=3,3,1$, gives $\dim\mathrm{Ext}^1,\dim\mathrm{Ext}^2\le 6$, and Proposition 17.18 gives $\ge 6$; $\mathrm{Ext}^0=\mathbb{C}$ and Serre duality gives $\mathrm{Ext}^3=\mathbb{C}$. The rest is as in (i), with $r(\mathrm{ch}\,E)=6n^2-2n=48$ by Theorem 18.56(iv).

(iii) By Proposition 17.21 below, whose proof does not use this theorem, $\kappa$ lies in $\mathbb{Q}[\eta]\oplus W(A)$ and is killed by $T$; both summands are defined over $\mathbb{Q}$, so its Weil part $N'\omega'$ is rational, and $r(\kappa)=r(\mathrm{ch}\,E)$ is Lemma 18.55(i). The proof of Proposition 17.21 also gives $\kappa$ explicitly. At $n=2$, $\mathrm{ch}(F)=\theta=\frac{1}{2s}(e^{s\theta}-e^{-s\theta})$ and $\mathrm{ch}(F^\vee)=-\theta$, so, with $t=s/2d$,

$$\kappa=-(e^{t\eta}+e^{-t\eta})-\tfrac{1}{4d}(\gamma^2-d\ell^2)=-2+\tfrac{1}{4d}\eta^2-\tfrac{1}{192d^2}\eta^4-\tfrac{1}{4d}(\gamma^2-d\ell^2).$$

Hence $c_1=c_3=0$ and $c_2=1/4d$. Write $\gamma-s\ell=a_1+a_2$ and $\gamma+s\ell=b_1+b_2$ for the decompositions over the two factors of $X=E_1\times E_2$, as in the proof of Proposition 17.21. Then $a_i^2=b_i^2=0$, so the Weil part is $\omega=-\frac{1}{4d}(a_1a_2+b_1b_2)$, and $a_ib_i=\gamma_i^2+d\ell_i^2=-4d\,\beta_i\widehat{\beta}_i$; so $\int\omega^2=\frac{2}{16d^2}\int a_1b_1a_2b_2=2$, while $\int\eta^4=6\int\eta_1^2\eta_2^2=24d^2$. Thus $c_2^2\int\eta^4=\frac32=\frac34\int\omega^2$, the exceptional ratio of Proposition 17.11(iv), where the annihilator has dimension 5. At $n=3$, $\mathrm{ch}(F)=v_\theta=\frac{1}{2s}(e^{s\theta}-e^{-s\theta})$ and $\mathrm{ch}(F^\vee)=-v_\theta$, and the same computation gives the $\eta$-part $-2s(e^{t\eta}-e^{-t\eta})=2\eta+\cdots$; so $c_1=2\neq 0$ and the annihilator is $T_{[A]}\mathcal{D}_{3,3}$, of dimension 9, by Proposition 17.11(iii).

For $B$, put $B'=B-\frac12\ell$, so that $\mathrm{ch}(E)e^B=\kappa e^{B'}$. Contraction with $v\in T$ is a derivation and kills $\kappa$, so $v\lrcorner(\kappa e^{B'})=\kappa\,(v\lrcorner B')\,e^{B'}$. Multiplication by $\kappa$ is injective on $H^2(A,\mathbb{C})$: at $n=2$ the degree zero part of $\kappa$ is $-2$, and at $n=3$ its lowest part is $2\eta$, and multiplication by $\eta$ is injective from $H^2$ to $H^4$ on the sixfold $A$. So $T$ kills $\kappa e^{B'}$ exactly when $v\lrcorner B'=0$ for all $v\in T$. By Proposition 17.11(i) the real class $B'$ of type $(1,1)$ then stays of type $(1,1)$ on the germ of $\mathcal{D}_{n,n}$ at $A$, hence on all of $\mathcal{D}_{n,n}$, and at a very general member $\mathrm{NS}\otimes\mathbb{Q}=\mathbb{Q}\eta$ (Lemma 7.5); so $B'\in\mathbb{Q}\eta$. The classes $x_j\xi_j$ appear in $\frac12\ell+c\eta$ with coefficient $\frac12$ for every $c$, so no integral class is congruent to $\frac12\ell$ modulo $\mathbb{Q}\eta$, on any $X$: no line bundle twist of $E$, and in particular not $E$ itself, has a character killed by $T$. ∎

In both cases the numerical form of the criterion holds with equality, so the injectivity of $\sigma_E$ comes out of a dimension count and does not need [Mar25b, Question 11.4]. The certificates concern the twisted object only: the character $\kappa$ stays of Hodge type along $\mathcal{D}_{n,n}$, while $\mathrm{ch}(E)$ does so only on a locus of dimension 3 at $n=2$ and 6 at $n=3$, so propagation needs the extension of Theorem 16.28 to twisted objects, which Perry proves [Per26], and an argument extracting untwisted cycles, as in [Mar25b, Section 2]. The families concerned, of trivial discriminant with $n=2$ and $n=3$, are Markman's.

**Proposition 17.20** (No Orlov product at $n = 4$)**.** *Let $X$ be a principally polarised abelian fourfold, $d \geq 1$, and $F_1, F_2$ perfect complexes on $X$ with characters in the secant plane $P_\theta \setminus \{0\}$ and $\operatorname{Ext}^{<0}(F_i, F_i) = 0$, for instance sheaves. Then $E = \Phi(F_1 \boxtimes F_2^\vee)$ has $\dim \operatorname{Ext}^2(E, E) > 88 = r(\operatorname{ch} E)$.*

*Proof.* By the Künneth formula $\dim \operatorname{Ext}^2(E,E) = e_0e_2' + e_1e_1' + e_2e_0'$ with $e_k = \dim \operatorname{Ext}^k(F_1, F_1)$ and $e_k'$ likewise for $F_2$, and by Proposition 17.18 $e_0 \geq 1$, $e_1 \geq 8$, $e_2 \geq 12$; so the sum is at least $12 + 64 + 12 = 88$, which is $r(\operatorname{ch} E)$ by Theorem 18.56(iv). Equality would force $e_0, e_1, e_2 = 1, 8, 12$. Since $e_k = 0$ for $k < 0$ and Serre duality gives $e_k = e_{4-k}$, this would give $\chi(F_1, F_1) = 1 - 8 + 12 - 8 + 1 = -2$, while Proposition 17.18 gives $\chi(F_1, F_1) = 8d(a^2d + b^2) > 0$. ∎

Without $\operatorname{Ext}^{<0} = 0$ the count changes, because negative Ext groups of the $F_i$ contribute products $e_{-j}e'_{2+j}$ to $\operatorname{Ext}^2(E, E)$ and change the Euler characteristic. The next two results treat the template in every dimension: the twisted character is always flat, and the count says exactly which secant objects would give equality.

**Proposition 17.21** (Flatness in every dimension)**.** *Let $(X, \theta)$ be a principally polarised abelian $n$-fold, $n \geq 1$, let $d \geq 1$ and $s = \sqrt{-d}$, and put $\Psi(Z) = \operatorname{ch}(\Phi(Z))e^{\ell/2}$ for classes $Z$ on $X \times X$, extended linearly, with $\gamma = d\beta - \widehat{\beta}$ as in Theorem 15.2 and $\eta = d\beta + \widehat{\beta}$. For $\sigma = \pm 1$, $\Psi(e^{\sigma s\theta} \boxtimes e^{\sigma s\theta})$ is a nonzero multiple of $e^{\sigma(s/2d)\eta}$, and $\Psi(e^{\sigma s\theta} \boxtimes e^{-\sigma s\theta})$ is a nonzero multiple of the Weil class $(\gamma - \sigma s\ell)^n$. Consequently, for perfect complexes $F_1, F_2$ with $\operatorname{ch}(F_i) \in P_\theta$, the class $\kappa = \operatorname{ch}(\Phi(F_1 \boxtimes F_2^\vee))e^{\ell/2}$ lies in $\mathbb{Q}[\eta] \oplus W(A)$, is killed by $T = T_{[A]}\mathcal{D}_{n,n}$, and has Hankel rank $\rho \leq 2$; for $F_1 = F_2 \neq 0$ its Weil part is nonzero.*

*Proof.* A symplectic basis of $H^1(X, \mathbb{Z})$ for $\theta$ identifies $H^*(X, \mathbb{Q})$, with $\theta$, with the tensor product of $n$ copies of the cohomology of an elliptic curve with its class $\theta_i$, $\theta = \sum_i \theta_i$. The action of $\Phi$ on cohomology is the correspondence $(1 \times \Phi_P) \circ \mu_*$ with $\mu(a, b) = (a - b, b)$, and $\mu$, the Poincaré class $\ell = \sum_i \ell_i$, the classes $\eta$ and $\gamma$, and the exponentials $e^{\pm s\theta} = \bigotimes_i e^{\pm s\theta_i}$ all decompose as tensor products over $i$ of classes of even degree, so no signs arise. Each identity is therefore the tensor product of $n$ copies of the identity for $n = 1$, up to a nonzero scalar, where for the Weil class one uses $(\gamma - \sigma s\ell)^n = n! \prod_i (\gamma_i - \sigma s\ell_i)$, valid because the factors commute and $(\gamma_i - \sigma s\ell_i)^2 = 0$: by Theorem 15.2(i) at $n = 1$ the class $\gamma_i - \sigma s\ell_i$ spans the top exterior power of a two-dimensional eigenspace.

It remains to treat $n = 1$. Let $y_1, y_2$ be a symplectic basis of $H^1(X)$, $z_1, z_2$ its copy on the second factor of $X \times X$ and $\xi_1, \xi_2$ the dual basis of $H^1(\widehat{X})$, so that $\theta = \beta = y_1y_2$, $\widehat{\beta} = \xi_1\xi_2$, $\ell = y_1\xi_1 + y_2\xi_2$ and $\ell^2 = -2\beta\widehat{\beta}$. On cohomology $\Phi$ sends $w$ to $\int_z \mu^* w \cdot e^{z_1\xi_1 + z_2\xi_2}$, the integral being over the second factor; with the opposite sign of the Poincaré class, $\ell$ is replaced by $-\ell$ in this proposition and in Theorem 17.19. For $a, b \in \{\pm 1\}$,

$$\mu^*(e^{as\theta} \boxtimes e^{bs\theta}) = (1 + as(y_1 + z_1)(y_2 + z_2))(1 + bs\, z_1z_2),$$
$$e^{z_1\xi_1 + z_2\xi_2} = 1 + z_1\xi_1 + z_2\xi_2 - z_1z_2\xi_1\xi_2,$$

and collecting the coefficient of $z_1z_2$ in the product gives

$$\operatorname{ch}\Phi(e^{as\theta} \boxtimes e^{bs\theta}) = (a + b)s - ab\, d\, \beta - \widehat{\beta} - as\, \ell - as\, \beta\widehat{\beta}.$$

Multiplying by $e^{\ell/2} = 1 + \frac{1}{2}\ell - \frac{1}{4}\beta\widehat{\beta}$, and using $\beta\ell = \widehat{\beta}\ell = 0$, gives $\Psi = 2as - d\beta - \widehat{\beta} - \frac{as}{2}\beta\widehat{\beta} = 2as\, e^{(as/2d)\eta}$ when $a = b$, and $\Psi = d\beta - \widehat{\beta} - as\ell = \gamma - as\ell$ when $a = -b$. So for general $n$ the multiples are $(2\sigma s)^n$ and $1/n!$.

For the consequence, write $\mathrm{ch}(F_1) = w_1 e^{s\theta} + \overline{w}_1 e^{-s\theta}$ and $\mathrm{ch}(F_2^\vee) = w_2 e^{-s\theta} + \overline{w}_2 e^{s\theta}$ as in the proof of Proposition 17.18; the terms $e^{\pm s\theta} \boxtimes e^{\pm s\theta}$ give multiples of $e^{\pm(s/2d)\eta}$ and the mixed terms give the Weil part $w_1 w_2 c_+ \alpha_+ + \overline{w}_1 \overline{w}_2 c_- \alpha_-$, with $c_\pm \neq 0$. The classes of $\mathbb{Q}[\eta] \oplus W(A)$ are Hodge classes on every member of the family, so $T$ kills $\kappa$, and the $\eta$-part lies in the span of the two exponentials $e^{\pm(s/2d)\eta}$, which gives $\rho \leq 2$. ∎

**Proposition 17.22** (The equality criterion for the Orlov template)**.** *Let $(X, t)$ be a polarised abelian $n$-fold, $n \geq 2$, $d \geq 1$, and $F_1$, $F_2$ perfect complexes with characters in $P_t \setminus \{0\}$; put $e_k = \dim \mathrm{Ext}^k(F_1, F_1)$ and $e'_k = \dim \mathrm{Ext}^k(F_2, F_2)$ for all $k \in \mathbb{Z}$, and $E = \Phi(F_1 \boxtimes F_2^\vee)$.*

(i) $\dim \mathrm{Ext}^2(E, E) = \sum_{k \in \mathbb{Z}} e_k e'_{2-k} \geq R_n$, *where $R_n = 6n^2 - 2n$ for $n \geq 3$ and $R_2 = 18$, and $R_n = r(\mathrm{ch}\, E) = r(\kappa)$.*

(ii) *Equality holds if and only if $(e_0, e_1, e_2) = (e'_0, e'_1, e'_2) = (1, 2n, r_2)$, with $r_2 = n(n-1)$ for $n \geq 3$ and $r_2 = 1$ for $n = 2$, and $e_{-j} e'_{2+j} = e_{2+j} e'_{-j} = 0$ for every $j \geq 1$.*

(iii) *For $F_1 = F_2 = F$ and $n \geq 3$ the last condition says that the set $S = \{j \geq 1 : \mathrm{Ext}^{-j}(F, F) \neq 0\}$ lies in $[n-1, \infty)$ and that $j \in S$ implies $j - (n-2) \notin S$; for $n = 2$ it says $\mathrm{Ext}^{<0}(F, F) = 0$.*

(iv) *Equality implies that $\sigma_E$ is injective.*

*Proof.* (i) $\Phi$ is an equivalence and the Künneth formula holds in all degrees, negative ones included, and dualising is an anti-equivalence, so $\mathrm{Ext}^*(E, E) = \mathrm{Ext}^*(F_1, F_1) \otimes \mathrm{Ext}^*(F_2, F_2)$. By Proposition 17.18, $e_0 \geq 1$, $e_1 \geq 2n$ and $e_2 \geq r_2$, so the three terms with $k = 0, 1, 2$ already give $R_n$; the equality $R_n = r(\mathrm{ch}\, E)$ is the Künneth count of Theorem 18.56(iv) and Theorem 17.19, and $r(\kappa) = r(\mathrm{ch}\, E)$ by Lemma 18.55(i). (ii) Every term is nonnegative, and $e_0 e'_2 \geq e'_2 \geq r_2 > 0$ with equality only if $e_0 = 1$ and $e'_2 = r_2$, and likewise for the other two terms; every other product must vanish. (iii) Serre duality gives $e_k = e_{n-k}$. For $1 \leq j \leq n-2$, $e_{2+j}$ is at least the rank of contraction from $HT^{2+j}(X)$ into $\mathrm{ch}(F)$, which is positive by Proposition 17.18, because the characteristic morphism is compatible with contraction into the Chern character in every degree [CRV12]; so $e_{-j} = 0$. For $j \geq n-1$, $e_{2+j} = e_{-(j-(n-2))}$. (iv) The composite $\sigma_E \circ \mathrm{ev}_E$ is contraction into $\mathrm{ch}(E)$, of rank $R_n \leq \mathrm{rk}\, \sigma_E \leq \dim \mathrm{Ext}^2(E, E)$, so equality forces $\sigma_E$ to be injective. ∎

For $n = 2$ and $n = 3$ the certificates of Theorem 17.19 attain the equality; for $n = 4$ it is excluded by Proposition 17.20 when $\mathrm{Ext}^{<0} = 0$, with negative Ext groups in degrees at most $-(n-1)$ the Euler characteristic leaves room in every dimension, and from $n = 5$ on it leaves room without them (Proposition 17.23(i)). If a secant object $F$ attaining the equality existed for one $d$ and unboundedly many $n$, then, since $\kappa$ is flat in every dimension by Proposition 17.21, a twisted form of Theorem 16.29 together with Proposition 20.13 would make the Weil classes of every family of $\mathbb{Q}(\sqrt{-d})$ algebraic. That is the secant route of (C1) and (C2) in Section 20.5, with the question of [Mar25b] replaced by a dimension count, and it lies off every minimal route (Theorem 20.70(iv)).

Both certificates are sheaves on theta divisors, a curve of genus two and the surface $C^{(2)}$. The next result explains why this construction stops at $n = 3$ and what a continuation would need.

**Proposition 17.23** (Sheaves on divisors in the Orlov template)**.** *Let $(X, \theta)$ be a principally polarised abelian $n$-fold and $d \geq 1$; put $S(x) = \sum_{j \geq 0} (-d)^j x^{2j} / (2j+1)!$, so that $v_\theta = \theta S(\theta)$, and $T_b(x) = bx/(1 - e^{-bx})$ for an integer $b \geq 1$. Write $\beta_0 = \beta_n = 1$, $\beta_k = 2\binom{n}{k}$ for $0 < k < n$ and $\beta_k = 0$ otherwise; these are the Betti numbers of the connected sum of two real $n$-tori.*

(i) *Let $n \geq 5$ and let $F$ be a perfect complex with $\operatorname{ch}(F) \in P_\theta \setminus \{0\}$. There are integers $e_k$, zero for $k \notin [0,n]$, with $e_k = e_{n-k}$, $(e_0, e_1, e_2) = (1, 2n, n(n-1))$, $e_k \geq \beta_k$ for all $k$, and $\sum_k (-1)^k e_k = \chi(F,F)$. So Serre duality, the contraction bounds $e_k \geq \beta_k$ (Proposition 17.18 and [CRV12]) and the Euler characteristic allow equality in Proposition 17.22 with $\operatorname{Ext}^{<0}(F,F) = 0$. For odd $n$ one may take $e_k = \beta_k$, and at $n = 5$ this is the only choice; for even $n$ the sequence $(\beta_k)$ has Euler characteristic $-2 \neq \chi(F,F)$, so some $e_k$ with $3 \leq k \leq n-3$ must exceed $\beta_k$.*

(ii) *Let $n \geq 3$, let $i\colon D \hookrightarrow X$ be a smooth effective divisor, and let $G$ be a vector bundle of rank $r \geq 1$ on $D$ with $\operatorname{ch}(G) \in i^*H^*(X,\mathbb{Q})$. If $\operatorname{ch}(i_*G) \in P_\theta$, then $[D] = b\theta$ for an integer $b \geq 1$ and $\operatorname{ch}(G) = r\, i^*\big(S(\theta)\, T_b(\theta)\big)$; in particular $c_1(G) = \frac{rb}{2}\,\theta|_D$, $\operatorname{ch}_2(G) = r\big(\frac{b^2}{12} - \frac{d}{6}\big)\theta^2|_D$ and $r \geq 2$. If $n \geq 4$, every line bundle on $D$ satisfies the hypothesis, so no line bundle $L$ on a smooth divisor has $\operatorname{ch}(i_*L) \in P_\theta$, and $rb$ is even. If $n \geq 5$, $b = 1$ and $r = 2$, then $c_2(G) = \frac{d+1}{3}\,\theta^2|_D$, which is integral only for $d \equiv 2 \pmod 3$.*

(iii) *Let $n \geq 2$, let $i\colon D \hookrightarrow X$ be a smooth divisor with $[D] = b\theta$, $V$ a vector bundle on $X$, $G = V|_D$ and $F = i_*G$. Then*

$$\dim \operatorname{Ext}^k(F,F) = h^k(\operatorname{End} G) + h^{k-1}(\operatorname{End} G \otimes \mathcal{O}_D(D))$$

*for every $k$. For $b = 1$ this is $\beta_k + h^k(\operatorname{End}_0 G) + h^{k-1}(\operatorname{End}_0 G \otimes \mathcal{O}_D(D))$, so for a line bundle $V$ the sheaf $F$ has exactly the profile required in Proposition 17.22(ii). For $b \geq 2$, $\dim \operatorname{Ext}^1(F,F) \geq b^n + n - 1 > 2n$.*

(iv) *At $n = 5$, for no smooth divisor $D$ and no vector bundle $V$ on $X$ is $i_*(V|_D)$ a factor of an Orlov product attaining the equality of Proposition 17.22.*

*Proof.* (i) The rank $a$ of $F$ is an integer, and $c_1(F) = b\theta$ with $\theta$ primitive, so $b \in \mathbb{Z}$; for even $n$ the number $c = 2^{n-1}d^{n/2-1}(a^2 d + b^2)$ is therefore a positive integer divisible by $2^{n-1}$, and $\chi(F,F) = (-1)^{n/2}c$ by Proposition 17.18, while $\chi(F,F) = 0$ for odd $n$. The conditions fix $e_0, e_1, e_2$ and, by symmetry, $e_{n-2}, e_{n-1}, e_n$, with the values $\beta_k$; for $n = 5$ these are all the $e_k$. Since $\sum_{k=0}^{n}(-1)^k\binom{n}{k} = 0$, the sequence $(\beta_k)$ has Euler characteristic 0 for odd $n$ and $-2$ for even $n$. For odd $n$ take $e_k = \beta_k$. For even $n \geq 6$ take $e_k = \beta_k$ for $k \neq n/2$ and $e_{n/2} = \beta_{n/2} + c + 2(-1)^{n/2} \geq \beta_{n/2}$, which is possible because $c \geq 2^{n-1} > 2$ and $3 \leq n/2 \leq n-3$; its Euler characteristic is $-2 + (-1)^{n/2}(c + 2(-1)^{n/2}) = \chi(F,F)$.

(ii) The rank of $i_*G$ is zero, so $\operatorname{ch}(i_*G) = b' v_\theta$ for some $b' \in \mathbb{Q}$, and comparing degree two gives $r[D] = b'\theta$. As $[D]$ is integral and nonzero and $\theta$ is primitive, $[D] = b\theta$ with $b$ a positive integer, and $D$ is ample. The tangent bundle of $X$ is trivial and the normal bundle of $D$ is $\mathcal{O}_D(D)$, so $\operatorname{td}(D) = i^*T_b(\theta)^{-1}$; writing $\operatorname{ch}(G) = i^*y$, Grothendieck–Riemann–Roch and the projection formula give

$$\operatorname{ch}(i_*G) = i_*\big(i^*(y\, T_b(\theta)^{-1})\big) = y\, T_b(\theta)^{-1}\, b\theta = y\,(1 - e^{-b\theta}).$$

So $\operatorname{ch}(i_*G) = rb\,\theta S(\theta)$ says $b\theta\, z\, T_b(\theta)^{-1} = 0$ for $z = y - r\,S(\theta)T_b(\theta)$, that is $\theta z = 0$. Let $z_k$ be the component of $z$ in $H^{2k}(X,\mathbb{Q})$. If $2k \leq n-1$, multiplication by $\theta$ is injective on $H^{2k}(X,\mathbb{Q})$ by the hard Lefschetz theorem, so $z_k = 0$. If $2k \geq n$, the Gysin map $i_*$ is injective on $H^{2k}(D,\mathbb{Q})$, being dual to the restriction $H^{2n-2-2k}(X,\mathbb{Q}) \to H^{2n-2-2k}(D,\mathbb{Q})$, which is onto by the Lefschetz hyperplane theorem, and $i_*i^*z_k = b\theta z_k = 0$; so $i^*z_k = 0$. Hence $\operatorname{ch}(G) = r\, i^*(S(\theta)T_b(\theta))$, and the expansion $S(x)T_b(x) = 1 + \frac{b}{2}x + \big(\frac{b^2}{12} - \frac{d}{6}\big)x^2 + O(x^3)$ gives $c_1$ and $\operatorname{ch}_2$. If $r = 1$ then $\operatorname{ch}_2(G) = \frac{1}{2}c_1(G)^2$, so $\big(\frac{b^2}{24} + \frac{d}{6}\big)\theta^2|_D = 0$, which is false: $\int_D (\theta|_D)^{n-1} = b\,n! \neq 0$ and $n - 1 \geq 2$. If $n \geq 4$, the Lefschetz hyperplane theorem

gives $H^2(X,\mathbb{Z}) \cong H^2(D,\mathbb{Z})$, so every line bundle $L$ on $D$ has $\operatorname{ch}(L) = i^*e^\lambda$ for some $\lambda \in H^2(X,\mathbb{Z})$, and $\theta|_D$ is primitive, so the integrality of $c_1(G)$ makes $rb$ even. If $n \geq 5$, $b = 1$ and $r = 2$, then $c_2(G) = \frac{1}{2}c_1^2 - \operatorname{ch}_2 = \frac{d+1}{3}\theta^2|_D = \frac{2(d+1)}{3}\, i^*(\frac{1}{2}\theta^2)$. The class $\frac{1}{2}\theta^2$ is integral and primitive. The affine variety $X \setminus D$ has the homotopy type of a CW complex of dimension $n$, so by Lefschetz duality $H^4(X,D;\mathbb{Z}) \cong H_{2n-4}(X \setminus D;\mathbb{Z}) = 0$ and $H^5(X,D;\mathbb{Z}) \cong H_{2n-5}(X \setminus D;\mathbb{Z})$ is torsion free; hence $i^*\colon H^4(X,\mathbb{Z}) \to H^4(D,\mathbb{Z})$ is injective with torsion-free cokernel, so $i^*(\frac{1}{2}\theta^2)$ is primitive modulo torsion, and $c_2(G)$ is integral only if 3 divides $2(d+1)$.

(iii) Let $s$ be the section of $\mathcal{O}_X(D)$ vanishing on $D$. The sheaf $F$ has the locally free resolution $0 \to V(-D) \xrightarrow{s} V \to F \to 0$, and applying $\operatorname{Hom}(-,F)$ gives a long exact sequence with $\operatorname{Ext}^k(V,F) = H^k(\operatorname{End} G)$ and $\operatorname{Ext}^k(V(-D),F) = H^k(\operatorname{End} G \otimes \mathcal{O}_D(D))$, in which the map between these two groups is multiplication by $s|_D = 0$. So it splits into short exact sequences $0 \to H^{k-1}(\operatorname{End} G \otimes \mathcal{O}_D(D)) \to \operatorname{Ext}^k(F,F) \to H^k(\operatorname{End} G) \to 0$. Write $\operatorname{End} G = \mathcal{O}_D \oplus \operatorname{End}_0 G$. On an abelian variety $H^k(\mathcal{O}_X(D)) = 0$ for $k > 0$ and $H^k(\mathcal{O}_X(-D)) = 0$ for $k < n$, and $h^0(\mathcal{O}_X(D)) = b^n$ by Riemann–Roch. From $0 \to \mathcal{O}_X(-D) \to \mathcal{O}_X \to \mathcal{O}_D \to 0$ one gets $h^k(\mathcal{O}_D) = \binom{n}{k}$ for $k \leq n-2$, and also for $k = n-1$ when $b = 1$, because $H^n(\mathcal{O}_D) = 0$ makes the map $H^n(\mathcal{O}_X(-D)) \to H^n(\mathcal{O}_X)$ between lines onto. From $0 \to \mathcal{O}_X \to \mathcal{O}_X(D) \to \mathcal{O}_D(D) \to 0$ one gets $h^0(\mathcal{O}_D(D)) = b^n - 1 + n$ and $h^k(\mathcal{O}_D(D)) = \binom{n}{k+1}$ for $k \geq 1$. For $b = 1$ the two contributions of $\mathcal{O}_D$ add up to $\beta_k$ in every degree, which gives the formula; for a line bundle $V$, $\operatorname{End}_0 G = 0$, and $F$ is a sheaf, so $\operatorname{Ext}^{<0}(F,F) = 0$. For $b \geq 2$, $\dim \operatorname{Ext}^1(F,F) \geq h^0(\mathcal{O}_D(D)) = b^n + n - 1$, and $b^n > n+1$.

(iv) Such a product needs $\operatorname{ch}(F) \in P_\theta$ for $F = i_*(V|_D)$, so $[D] = b\theta$ by (ii), and its factors have $(e_0,e_1,e_2) = (1,10,20)$ by Proposition 17.22(ii); by (iii) $b = 1$. Then Serre duality gives $e_k = \beta_k$ for every $k$, so by (iii) $H^*(\operatorname{End}_0 G) = 0$ and $\chi(D,\operatorname{End}_0 G) = 0$. On the other hand (ii) gives $\operatorname{ch}(G) = r\, i^*(S(\theta)T_1(\theta))$, so $\operatorname{ch}(\operatorname{End} G) = \operatorname{ch}(G)\operatorname{ch}(G)^\vee = r^2 i^*\big(S(\theta)^2 T_1(\theta)T_1(-\theta)\big)$, and Hirzebruch–Riemann–Roch on $D$ gives

$$\chi(D,\operatorname{End} G) = r^2 \int_X S(\theta)^2\, \frac{\theta^2}{e^\theta - 1} = \frac{r^2(32d^2 - 20d - 1)}{6},$$

while $\chi(D,\mathcal{O}_D) = \int_X (1 - e^{-\theta}) = 1$. So $6\chi(D,\operatorname{End}_0 G) = r^2(32d^2 - 20d - 1) - 6 \geq 5$, because $32d^2 - 20d - 1 \geq 11$ for $d \geq 1$. ∎

So from $n = 5$ on the numbers allow a secant object attaining the equality, the line bundles on theta divisors have the Ext groups it requires in every dimension, and what fails is their character. The line bundle of Theorem 17.19(ii) has $c_1 = (2k+1)x - k\theta$, which by Proposition 17.23(ii), valid at $n = 3$, is not restricted from the Jacobian; from $n = 4$ on the Lefschetz theorem forbids such classes. At $n = 5$, by (ii) and (iv), a vector bundle $G$ on a smooth divisor with $i_*G$ a factor attaining the equality is neither a line bundle nor restricted from $X$; if $G$ lives on a theta divisor and has Chern character restricted from $X$, its rank is even, and rank two needs $d \equiv 2 \pmod 3$. Bundles of this kind that are not restricted from $X$, sheaves on singular divisors, sheaves that are not locally free on their support, and objects of positive rank remain; we know no object of these kinds with a secant character and a small Ext algebra.

The singular divisors closest to the two certificates are the theta divisors of Jacobians: both certificates are line bundles on the symmetric product $C^{(g-1)}$, pushed forward along the Abel–Jacobi map, for $g = 2, 3$. From $g = 4$ on the theta divisor is singular and the Lefschetz argument of Proposition 17.23(ii) no longer applies, since $C^{(g-1)}$ carries the class

$x$, which does not come from the Jacobian. The character excludes this continuation all the same.

**Proposition 17.24** (The Jacobian continuation stops at genus three)**.** *Let $C$ be a smooth projective curve of genus $g \geq 2$, $X = J(C)$ with its principal polarisation $\Theta$, and $u\colon C^{(g-1)} \to X$ the Abel–Jacobi map, whose image is a translate of the theta divisor. Let $x$ be the class of the divisor $p + C^{(g-2)}$ and $\theta = u^*\Theta$, let $L$ be a line bundle on $C^{(g-1)}$ with $c_1(L) = \alpha x + \beta\theta$, and put $\gamma = \beta - \frac{1}{2}$. Then the perfect complex $Ru_*L$ has*

$$\operatorname{ch}(Ru_*L) = \Theta + \Big(\frac{\alpha}{2} + \gamma\Big)\Theta^2 + c_3\Theta^3 + c_4\Theta^4 + \cdots,$$

*and when $\alpha = -2\gamma$, that is $\alpha + 2\beta = 1$,*

$$c_3 = -\frac{4\gamma^2 - 1}{24}, \qquad c_4 = -\frac{\gamma(4\gamma^2 - 1)}{72}.$$

*Consequently $\operatorname{ch}(Ru_*L) \in P_\Theta$ for some $d \geq 1$ exactly when $g = 2$ and $\alpha + 2\beta = 1$, or $g = 3$, $\alpha + 2\beta = 1$ and $d = \beta(\beta - 1)$; these are the sheaves of Theorem 17.19(i) and (ii). For $g \geq 4$ no line bundle on $C^{(g-1)}$ with $c_1$ in the span of $x$ and $\theta$ has a secant character.*

*Proof.* The Todd class of $X$ is 1, so by Grothendieck–Riemann–Roch $\operatorname{ch}(Ru_*L) = u_*(e^{\alpha x + \beta\theta}\operatorname{td}(C^{(g-1)}))$. By Macdonald [Mac62], [ACGH85, Chapter VII, Section 5], the total Chern class of $C^{(k)}$ is $(1+x)^{k-g+1}e^{-\theta/(1+x)}$; for $k = g-1$ the first factor is 1. In Macdonald's description of the cohomology ring $\theta = \sum_{i=1}^{g}\sigma_i$ with $\sigma_i^2 = 0$, so $e^{-\theta/(1+x)} = \prod_i (1 + x - \sigma_i)/(1+x)$ and, with $\operatorname{Td}(y) = y/(1-e^{-y})$ and $q = \operatorname{Td}'/\operatorname{Td}$,

$$\operatorname{td}(C^{(g-1)}) = \prod_i \frac{\operatorname{Td}(x - \sigma_i)}{\operatorname{Td}(x)} = \prod_i (1 - \sigma_i q(x)) = e^{-\theta q(x)}.$$

Since $\log \operatorname{Td}(y) = \frac{y}{2} - \frac{y^2}{24} + O(y^4)$, we have $-q(x) = -\frac{1}{2} + \frac{x}{12} + O(x^3)$. The class $x^a$ is that of the image of $C^{(g-1-a)}$ under $D \mapsto D + p_1 + \cdots + p_a$, which $u$ maps birationally onto a translate of $W_{g-1-a}$, of class $\Theta^{a+1}/(a+1)!$ by Poincaré's formula [ACGH85, Chapter I, Section 5]; with $\theta = u^*\Theta$ and the projection formula, $u_*(x^a\theta^b) = \Theta^{a+b+1}/(a+1)!$. So the coefficient $c_k$ of $\Theta^k$ is $\sum_{a+b=k-1} m_{ab}/(a+1)!$, where $m_{ab}$ is the coefficient of $x^a\theta^b$ in $e^{\alpha x}e^{\theta(\gamma + x/12)}$ for $k \leq 4$ (the term $O(x^3)$ first contributes to $\Theta^5$), namely $m_{ab} = \sum_j \frac{\alpha^{a-j}}{(a-j)!}\frac{\gamma^{b-j}}{(b-j)!}\frac{1}{12^j j!}$. This gives $c_1 = 1$, $c_2 = \frac{\alpha}{2} + \gamma$,

$$c_3 = \frac{\alpha^2}{12} + \frac{\alpha\gamma}{2} + \frac{1}{24} + \frac{\gamma^2}{2}, \qquad c_4 = \frac{\alpha^3}{144} + \frac{\alpha^2\gamma}{12} + \frac{\alpha}{72} + \frac{\alpha\gamma^2}{4} + \frac{\gamma}{24} + \frac{\gamma^3}{6},$$

and at $\alpha = -2\gamma$ these are the stated values.

The rank of $Ru_*L$ is 0, so $\operatorname{ch}(Ru_*L) \in P_\Theta$ means $\operatorname{ch}(Ru_*L) = v_\Theta = \Theta - \frac{d}{6}\Theta^3 + \frac{d^2}{120}\Theta^5 - \cdots$, the coefficient of $\Theta$ being 1. For $g = 2$ this is $c_2 = 0$. For $g \geq 3$ it also asks $c_3 = -d/6$, that is $d = \gamma^2 - \frac{1}{4} = \beta(\beta - 1)$, which settles $g = 3$. For $g \geq 4$ it asks in addition $c_4 = 0$; as $4\gamma^2 - 1 = 4d$, this is $\gamma d = 0$, so $d = 0$ or $\gamma = 0$, where $d = -\frac{1}{4}$. At $g = 3$, $L = \mathcal{O}((2k+1)x - k\theta)$ has $\beta = -k$ and $d = k(k+1)$, and at $g = 2$, $C^{(1)} = C$ and $L = \mathcal{O}_C(p)$ has $\alpha = 1$, $\beta = 0$. ∎

So the continuation of the two certificates must leave the line bundles on $C^{(g-1)}$: a bundle of higher rank, a sheaf on the theta divisor of a principally polarised abelian variety that is not a Jacobian, or a complex of positive rank.

*Remark* 17.25 (Untwisting). The template $E = \Phi(F_1 \boxtimes F_2^\vee)$ does not give an untwisted object meeting the polarised criterion by the obvious operations. No line bundle twist makes $\mathrm{ch}(E)$ flat (Theorem 17.19(iii)). At $n = 2$ the object $S^2(E[1]) \otimes P$, for $P$ a line bundle with $c_1(P) = \ell$, is untwisted with flat character, of rank 3 and at the exceptional ratio after a twist by a multiple of $\eta$, but its number is $r = 23$, which is odd, while Serre duality and the evenness of the Mukai pairing on an abelian variety make $\dim \mathrm{Ext}^2 \equiv \chi(E, E) \equiv 0 \pmod 2$; so equality is impossible. The object $\bigwedge^2(E[1]) \otimes P$ has no Weil part. For $E \otimes V$ with $V$ a simple semi-homogeneous bundle of slope $\ell/2$ [Muk78], the character $\mathrm{rk}(V)\kappa$ is flat, but $\sigma_{E\otimes V}$ kills the summands $\mathrm{Ext}^2(E, E \otimes P_\xi)$, $\xi \neq 0$, of $\mathrm{Ext}^2(E \otimes V, E \otimes V)$ coming from the decomposition of $V \otimes V^\vee$, and at $n = 2$ these are nonzero; this last point uses the decomposition of $\mathrm{End}\, V$ and the way $\Phi$ intertwines $\otimes P_\xi$ with translations, which we recall without proof. None of this shows that the polarised criterion must be met in twisted form; it shows that this template meets it only there.

At $n = 4$ the natural objects of the split member can be described completely, and the description turns the criterion into a statement about points on a quadric.

**Setup 17.26** (The quadric of Lagrangians)**.** Let $X$ be a very general principally polarised abelian fourfold, $U = H^1(X, \mathbb{Q})$, and $A = X \times \widehat{X}$ with the $K$-action of (29), so that $H^1(A, \mathbb{Q}) = U \otimes \mathbb{Q}^2$ and the rational Hodge ring $\mathcal{H}_A$ of $A$ is the ring of $\mathrm{Sp}(U)$-invariants in $\bigwedge^*(U \otimes \mathbb{Q}^2)$, of dimension 55, with $1, 3, 6, 10, 15, 10, 6, 3, 1$ in degrees $0, 2, \dots, 16$. The symmetric pairing on $H^1(A, \mathbb{Q}) \oplus H_1(A, \mathbb{Q}) = U \otimes \mathbb{Q}^4$ is the product of the symplectic form of $U$ and a symplectic form on $\mathbb{Q}^4$, so the maximal isotropic subspaces stable under $\mathrm{Sp}(U)$ are the $U \otimes L$ with $L$ a Lagrangian plane in $\mathbb{Q}^4$. Let $Q \subset \mathbb{P}^4$ be the Lagrangian Grassmannian $\mathrm{LG}(2, 4)$, a smooth quadric threefold in its Plücker embedding. Call a perfect complex on $A$ *natural* if it is a line bundle, a sheaf $\mathcal{O}_B \otimes L$ on an abelian subvariety, a point, or the image of one of these under an autoequivalence. Then the character of a natural $F$ is a multiple of the pure spinor of $U \otimes L_F$ for a rational point $L_F \in Q$, and there is a linear isomorphism $\Phi_4 \colon H^0(Q, \mathcal{O}(4))^\vee \to \mathcal{H}_A$ with $\mathrm{ch}(F) = \Phi_4(\nu_4(L_F))$ up to a rational factor, $\nu_4$ the fourth Veronese map. Under it the powers of $\eta$ span the image of a conic $C_\eta \subset Q$, and the Weil plane is the rational part of the line spanned by $\nu_4(L_+)$ and $\nu_4(L_-)$ for a conjugate pair of $K$-points $L_\pm$ of $Q$; the line $l_W = L_+L_-$ meets $Q$ only in $L_\pm$, and the plane of $C_\eta$ is its polar plane.

The pure spinor description of the characters of natural objects is classical, and we recall it without proof. The abelian subvarieties $B_{(p,q)} = \{(px, q\varphi x)\}$, with $\varphi \colon X \to \widehat{X}$ the polarisation, form a conic of natural objects, and on it the two points $L_\pm$ have coordinates $(p : q) = (1 : \mp\sqrt{-d})$: by Theorem 14.2(iv) the eigenspaces $W_\pm$ are the graphs of $\mp\sqrt{-d}\,\beta$. So the Weil classes are the values of the octic $(p, q) \mapsto [B_{(p,q)}]$ at those points.

**Theorem 17.27** (Minimal support at $n = 4$)**.** *In Setup 17.26, let $F_1, \dots, F_s$ be natural objects whose points $L_{F_i}$ are pairwise distinct and not on $C_\eta$, and suppose that $\sum_i m_i \,\mathrm{ch}(F_i) + P = N\omega_1 + \sum_k c_k \eta^k$ with $m_i \in \mathbb{Q}$, $P \in \mathbb{Q}[\eta]$ and $N \neq 0$. Then $s \geq 8$. If $s = 8$, the eight points lie on one smooth conic $\Pi \cap Q$ with $\Pi$ a plane containing $l_W$, the conic is disjoint from $C_\eta$, and $\sum_i m_i \,\mathrm{ch}(F_i) = N\omega_1$ exactly. Conversely any eight rational points of such a conic carry such a relation with every $m_i \neq 0$.*

*Proof.* Through $\Phi_4$ the relation reads $\sum_i m_i \nu_4(L_{F_i}) + P' = c_+\nu_4(L_+) + c_-\nu_4(L_-)$ in $H^0(Q, \mathcal{O}(4))^\vee$, with $P'$ in the span of $\nu_4(C_\eta)$ and $c_\pm$ conjugate and nonzero because $N \neq 0$. A quartic $f$ vanishing on $C_\eta$, at $L_-$ and at every $L_{F_i}$ gives $c_+ f(L_+) = 0$; so it suffices to produce such an $f$ with $f(L_+) \neq 0$ unless the configuration is as stated. Let $h_-$ be the

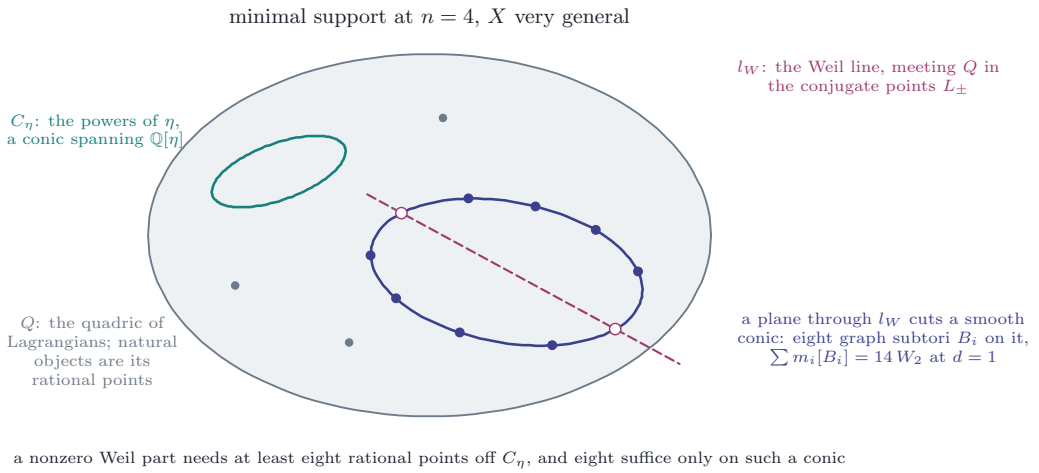



FIGURE 24. Theorem 17.27. The quadric $Q$ of Lagrangians; the rational points are the Chern characters of the natural objects on the split member at $n = 4$, up to the Veronese map $\nu_4$. The powers of the polarisation fill the conic $C_\eta$ (teal), and the Weil line is spanned by the conjugate points $L_\pm$ (open circles) in which the line $l_W$ (dashed) meets $Q$. A combination of natural characters with a nonzero Weil part needs at least eight rational points off $C_\eta$, and with eight they lie on one conic cut out by a plane through $l_W$ (indigo), such as the eight graph subvarieties $B_i$ of Example 17.28; the grey points are natural objects off any such conic, which no relation of this size can use.

linear form $B(L_-, \cdot)$, with $B$ the bilinear form of the quadric $Q$; it vanishes on the polar plane of $l_W$, hence on $C_\eta$, and at $L_-$, but not at $L_+$, since $l_W \not\subset Q$. If the points split into at most three groups $G_k$ with $L_+$ outside the span of each, linear forms $l_k$ vanishing on $G_k$ and not at $L_+$ give the quartic $h_- l_1 l_2 l_3$, which kills everything but $c_+$, so $c_+ = 0$. A rational line through $L_+$ also passes through $L_-$, so it is $l_W$, which has no rational point on $Q$; hence groups of one or two points never contain $L_+$ in their span, and a triple does only if it lies in a plane through $l_W$. For $s \le 6$ split into pairs. For $s = 7$, either some triple is not in a plane through $l_W$, or all seven lie in one such plane $\Pi$, hence on $C'' = \Pi \cap Q$. That conic is smooth, since a degenerate one is a pair of conjugate lines with at most one rational point, and it misses $C_\eta$, since $\Pi$ meets the plane of $C_\eta$ in one point $z$ and $z \in Q$ would put the lines $zL_\pm$ in $Q$. The quartics vanishing on $C_\eta$ restrict onto all binary octics on $C'' \cong \mathbb{P}^1$, and nine distinct points impose independent conditions on octics, so $L_\pm$ and the seven points are independent. For $s = 8$, if no plane through $l_W$ holds seven of the points, choose two triples not in such planes and a pair; if exactly seven lie in $\Pi$, modify the quartic by a multiple of one vanishing on $\Pi$ and at the eighth point; if all eight lie on $C''$, the ten points $L_\pm, L_{F_1}, \dots, L_{F_8}$ of the rational normal curve $\nu_4(C'')$ of degree eight satisfy exactly one relation, with every coefficient nonzero, and it is rational by Galois symmetry. Its part along the span of $\nu_4(C_\eta)$ vanishes, because quartics restrict onto the sum of the octics on the two disjoint conics, so the relation is pure. ∎

Figure 24 draws the quadric of Lagrangians, the conic $C_\eta$ and a conic through eight natural characters.

**Example 17.28** (Fourteen times a Weil class)**.** Let $d = 1$ and write $(\gamma - \sqrt{-1}\,\ell)^4 = W_1 + \sqrt{-1}\,W_2$ with $W_1, W_2$ rational, as in Theorem 15.2(iii). The eight subvarieties $B_i = B_{(p_i,q_i)}$ with

$$(p_i, q_i) = (1,-3),\ (1,-2),\ (1,2),\ (1,3),\ (2,-1),\ (2,1),\ (3,-1),\ (3,1)$$

and $m = (-1, 16, -16, 1, -16, 16, 1, -1)$ satisfy

$$\textstyle\sum_i m_i [B_i] = 14\, W_2,$$

with the sign of $\varphi$ for which $B_{(p,q)}$ has conormal forms $p\xi_j - q x_{n+j}$ and $p\xi_{n+j} + q x_j$; with the opposite sign the right side is $-14W_2$. Among the 12870 sets of eight subvarieties with $|p|, |q| \le 3$ this one has the least $\sum_i m_i^2$, namely 1028, and the least $\sum_i |m_i|$, namely 68. In general the coefficients of the relation of Theorem 17.27 for eight subvarieties are proportional to $\big((q_i^2 + dp_i^2)\prod_{j\neq i}(p_i q_j - p_j q_i)\big)^{-1}$. The sum $E = \bigoplus_i \mathcal{O}_{B_i}^{\oplus |m_i|}[k_i] \oplus \mathcal{O}(\eta) \oplus \mathcal{O}(2\eta) \oplus \mathcal{O}(3\eta)$, with $k_i = 0$ for $m_i > 0$ and 1 otherwise, has $\mathrm{ch}(E) = 14W_2 + \sum_k c_k \eta^k$ with $k!\, c_k = 1 + 2^k + 3^k$, so $\rho = 3$ and $r = 104$, while $\dim \mathrm{Ext}^2(E, E) = 28868$; by Corollary 17.17 it is far from the criterion, and every direct sum of natural objects with $N \neq 0$ has $\dim \mathrm{Ext}^2 \ge 8 \cdot 28 = 224$.

*Remark* 17.29 (The flatness form for direct sums). Lemma 17.16 does not apply to the flatness form of Proposition 17.15, but a direct sum fails that form too when its cross-extensions cannot interact. Let $E = \bigoplus_i (\mathcal{O}_{B_i} \otimes L_i)^{\oplus m_i}[k_i]$ with the $B_i$ pairwise distinct abelian subvarieties of dimension $n$ meeting pairwise in finite sets, such as the $B_{(p,q)}$. The extensions between different summands are concentrated in degree $n$, so an arrow of $\mathrm{Ext}^1$ between summands raises the shift by $n-1$, and a chain of $a \ge 1$ arrows lands in $\mathrm{Ext}^{2+a(n-1)} \neq \mathrm{Ext}^2$. Every term of the Maurer–Cartan equation that involves a cross-extension therefore lands in a zero group, the base $K_E$ is the product of the varieties of commuting $2n$-tuples of $m_i \times m_i$ matrices with the space of cross-extensions, and a count of dimensions, using the simultaneously diagonalisable component, gives $\dim K_E > e - k$; so (ii) and (iii) of Proposition 17.15 fail. We give this in outline only. Indecomposable objects, among them iterated extensions of at least eight natural blocks, and sums whose cross-extensions compose to nonzero classes are not covered.

## 18. Secant objects on the split member

18.1. **Sums of line bundles and the positivity obstruction.** Theorem 16.29 asks for one perfect complex at one base point. The cheapest candidate is a sum of line bundles on a split member, because there the Weil classes are already polynomials in $\beta$, $\widehat{\beta}$ and $\ell$ by Theorem 15.2, so the Chern character one wants is manifestly in reach. This subsection settles that candidate. It does not exist, and the obstruction is not semiregularity: it is the Chern character condition, and it fails because the norm form of $K$ is positive definite.

The norm rules out three different constructions in this paper, by three different mechanisms, and they should be kept apart. Theorem 5.6 uses the norm *character* of $K^\times$ and an equivariance hypothesis. Proposition 15.18 uses the fact that the part $R$ of $H^2$ on which $K^\times$ acts through $\tau^2$ and $\overline{\tau}^2$ has no Hodge classes at a general point, which is a statement about the Hodge group. Theorem 18.5 below uses neither: it uses that $\mathrm{N}_{K/\mathbb{Q}}$ is a positive definite quadratic form on $K$ and that the multiplicities of a filtration are positive integers. The three are independent, and only the third is available at a special point, which is exactly where the object of Proposition 16.27 has to live.

**Lemma 18.1** (The semiregularity map of a split object)**.** *Let $Y$ be a smooth projective complex variety and let*

$$E = \bigoplus_{i=1}^{s} L_i[p_i]^{\oplus m_i}, \qquad \lambda_i = c_1(L_i),$$

*with the $L_i$ line bundles, the $\lambda_i$ pairwise distinct and $p_i \in \mathbb{Z}$. Then for $\zeta = (\zeta_{ij}) \in \operatorname{Ext}^2(E,E)$,*

$$\sigma_E(\zeta)_q = \frac{1}{q!}\sum_{i=1}^{s}(-1)^{p_i}\operatorname{tr}(\zeta_{ii})\,\lambda_i^q, \tag{40}$$

*so $\sigma_E$ annihilates $\operatorname{Ext}^2(L_i[p_i], L_j[p_j])$ for $i \neq j$ and the traceless part of each diagonal block. Consequently $\sigma_E$ is injective if and only if all three of the following hold:*

(i) *$m_i = 1$ for every $i$;*
(ii) *$H^{\,2+p_i-p_j}(Y, L_jL_i^{-1}) = 0$ for all $i \neq j$;*
(iii) *the map*

$$\Phi\colon H^2(\mathcal{O}_Y)^{\oplus s} \longrightarrow H^*(Y,\mathbb{C}), \qquad (\zeta_i) \longmapsto \sum_i e^{\lambda_i}\zeta_i,$$

*is injective.*

*Proof.* The Atiyah class of a direct sum is the direct sum of the Atiyah classes, and that of a line bundle is its first Chern class, so $\operatorname{At}(E)$ is diagonal with entries $\lambda_i$ and $\operatorname{At}(E)^q$ is diagonal with entries $\lambda_i^q$. The matrix product $\zeta \circ \operatorname{At}(E)^q$ therefore has $(i,j)$ entry $\zeta_{ij}\lambda_i^q$, and the trace keeps only the diagonal; a shift by $p$ multiplies the trace by $(-1)^p$. That is (40), and it exhibits the two spaces the map kills.

For the equivalence, the off-diagonal part of $\operatorname{Ext}^2(E,E)$ is $\bigoplus_{i\neq j}\operatorname{Ext}^2(L_i[p_i],L_j[p_j]) = \bigoplus_{i\neq j} H^{2+p_i-p_j}(L_jL_i^{-1})$, which must vanish, giving (ii); the diagonal block at $i$ is $H^2(\mathcal{O}_Y)\otimes \operatorname{End}(\mathbb{C}^{m_i})$ and only its trace survives, so $m_i = 1$, giving (i). What is left is $\bigoplus_i H^2(\mathcal{O}_Y)$, and by (40) an element $(\zeta_i)$ of it dies exactly when $\sum_i \lambda_i^q\zeta_i = 0$ for every $q \geq 0$, which is (iii) after absorbing the signs $(-1)^{p_i}$ into the $\zeta_i$ and the factorials into the grading. ∎

Condition (ii) is cheap, because the shifts $p_i$ are at our disposal: only their parities are fixed by the Chern character, and a line bundle whose first Chern class is nondegenerate of index $k$ has cohomology in degree $k$ alone, so one may choose the $p_i$ to avoid every index that occurs. Condition (iii) is a rank condition, and condition (i) is free. What is not free is the Chern character itself, and that is the subject of the rest of this subsection.

*Notation* 18.2 (Coordinates on the divisor classes). Let $A_0 = X \times \widehat{X}$ be a split member, $n \geq 2$, and keep $\gamma = d\beta - \widehat{\beta}$ from Notation 15.1. Write

$$\theta^{\pm} = \gamma \mp \sqrt{-d}\,\ell \in \textstyle\bigwedge^2 V_{\pm}, \qquad R_0 = \mathbb{Q}\gamma \oplus \mathbb{Q}\ell \subset R.$$

Every $\theta \in R_0$ is $\theta = u\theta^+ + \overline{u}\,\theta^-$ for a unique $u \in K$, and in coordinates $u = g + h\sqrt{-d}$ corresponds to $\theta = 2g\gamma + 2dh\,\ell$. By Proposition 15.19(iv) the group $\mathrm{NS}(A_0)_{\mathbb{Q}}$ at a split member with $X$ general is $\mathbb{Q}\eta \oplus R_0$, so every divisor class $\lambda$ is $e\eta + \theta$ with $e \in \mathbb{Q}$ and $\theta \in R_0$, and we write $u(\lambda) \in K$ for the coordinate of its $R_0$-part.

**Lemma 18.3** (A basis in degree four)**.** *For $n \geq 2$ the six products $\beta^2, \widehat{\beta}^2, \ell^2, \beta\widehat{\beta}, \beta\ell, \widehat{\beta}\ell$ are linearly independent in $H^4(A_0,\mathbb{Q})$. Equivalently the six classes $\eta^2$, $\eta\theta^+$, $\eta\theta^-$, $(\theta^+)^2$, $(\theta^-)^2$, $\theta^+\theta^-$ are a basis of the degree four part of the subring generated by the divisor classes.*

*Proof.* Grade $H^*(A_0,\mathbb{Q}) = \bigwedge^* U \otimes \bigwedge^* U^\vee$ by bidegree. Then $\beta$ has bidegree $(2,0)$, $\widehat{\beta}$ has $(0,2)$ and $\ell$ has $(1,1)$, so the six products have bidegrees $(4,0)$, $(0,4)$, $(2,2)$, $(2,2)$, $(3,1)$, $(1,3)$. All are distinct except for the pair $\beta\widehat{\beta}$ and $\ell^2$, and those two are independent in $\bigwedge^2 U \otimes \bigwedge^2 U^\vee$ because $\beta\widehat{\beta} = \beta \otimes \widehat{\beta}$ is decomposable there while $\ell^2 = -2\sum_{i<j}(x_i x_j)\otimes(\xi_i\xi_j)$ is diagonal in the induced bases, of tensor rank $\binom{2n}{2} > 1$. The second description follows because the change of basis between the two sets of six classes is invertible over $K$: $\eta = d\beta + \widehat{\beta}$ and $\gamma = d\beta - \widehat{\beta}$ span the same plane as $\beta$ and $\widehat{\beta}$ since $\det\left(\begin{smallmatrix} d & 1 \\ d & -1 \end{smallmatrix}\right) = -2d \neq 0$, and $\theta^{\pm}$ span the same plane as $\gamma$ and $\ell$ over $\mathbb{C}$. ∎

The next lemma says the same thing at every abelian variety of Weil type, not only at a split member, and it says it without coordinates. It is the fact on which the positivity argument turns.

**Lemma 18.4** ($\theta^+\theta^-$ is never proportional to $\eta^2$)**.** *Let $(A,\iota,\eta)$ be of $(K,-1,n)$-Weil type with $n \geq 2$, and let $0 \neq \theta^{\pm} \in \bigwedge^2 V_{\pm}$. Then $\theta^+\theta^-$ is not a scalar multiple of $\eta^2$ in $H^4(A,\mathbb{C})$.*

*Proof.* Both classes lie in the summand $\bigwedge^2 V_+ \otimes \bigwedge^2 V_-$ of $H^4(A,\mathbb{C})$, the isotypic component for the character $\mathrm{N}^2$, so the question is one of tensor rank there. The polarisation class $\eta$ lies in $V_+ \otimes V_-$ and is the hermitian form of Proposition 3.7, hence nondegenerate: in dual bases $\eta = \sum_{a=1}^{2n} v_a \wedge w_a$ with $v_a$ a basis of $V_+$ and $w_a$ a basis of $V_-$. Squaring,

$$\eta^2 = \sum_{a,b}(v_a \wedge w_a)(v_b \wedge w_b) = -2\sum_{a<b}(v_a v_b)\otimes(w_a w_b),$$

which is diagonal in the induced bases of $\bigwedge^2 V_+$ and $\bigwedge^2 V_-$ and therefore has tensor rank exactly $\binom{2n}{2}$. The class $\theta^+\theta^-$ is, up to sign, $\theta^+ \otimes \theta^-$, of tensor rank one. For $n \geq 2$ one has $\binom{2n}{2} \geq 6 > 1$, so the two cannot be proportional. ∎

**Theorem 18.5** (The positivity obstruction)**.** *Let $(A,\iota,\eta)$ be of $(K,-1,n)$-Weil type with $n \geq 2$, and suppose that the only divisor classes on which $K^\times$ acts through the norm character are the multiples of $\eta$, that is*

$$\mathrm{NS}(A)_{\mathbb{Q}} \cap (V_+ \otimes V_-) = \mathbb{Q}\,\eta. \tag{41}$$

*Let $E$ be a perfect complex on $A$ whose Chern character is effective in the exponential basis, meaning*

$$\mathrm{ch}(E) = \sum_{i=1}^{s} m_i\, e^{\lambda_i}, \qquad m_i \in \mathbb{Q}_{>0}, \quad \lambda_i \in \mathrm{NS}(A)_{\mathbb{Q}}, \tag{42}$$

*and write $\lambda_i = e_i\eta + \theta_i$ with $\theta_i$ in the part $R$ of $H^2(A,\mathbb{Q})$ carrying the characters $\tau^2$ and $\overline{\tau}^2$, and $u_i \in K$ for the $\theta^+$-coordinate of $\theta_i$. If $\mathrm{ch}_2(E)$ is a flat Hodge class, then*

$$\sum_{i=1}^{s} m_i\, \mathrm{N}_{K/\mathbb{Q}}(u_i) = 0, \tag{43}$$

*and since $\mathrm{N}_{K/\mathbb{Q}}$ is positive definite and every $m_i$ is positive, $u_i = 0$ for every $i$. Hence every $\lambda_i$ is a rational multiple of $\eta$, and $\mathrm{ch}_k(E) \in \mathbb{Q}\,\eta^k$ for every $k$. In particular $\mathrm{ch}_n(E)$ has no component on the Weil line.*

*Proof.* From (42), $\mathrm{ch}_2(E) = \frac{1}{2}\sum_i m_i\lambda_i^2$, and hypothesis (41) is what makes the decomposition $\lambda_i = e_i\eta + \theta_i$ available with $e_i$ a scalar. Decompose $H^4(A,\mathbb{C})$ into $K^\times$-isotypic pieces

and keep the one for $\mathrm{N}^2$, namely $\bigwedge^2 V_+ \otimes \bigwedge^2 V_-$. Writing $\theta_i = u_i\theta^+ + \overline{u}_i\theta^-$ for a fixed choice of generators $\theta^\pm$ of $\bigwedge^2 V_\pm$, one has

$$\lambda_i^2 = e_i^2\eta^2 + 2e_i\eta\theta_i + u_i^2(\theta^+)^2 + \overline{u}_i^2(\theta^-)^2 + 2\mathrm{N}_{K/\mathbb{Q}}(u_i)\,\theta^+\theta^-,$$

and of these five terms only the first and the last are in the $\mathrm{N}^2$ piece: $\eta\theta_i$ carries $\mathrm{N}\tau^2$ and $\mathrm{N}\overline{\tau}^2$, and $(\theta^\pm)^2$ carry $\tau^4$ and $\overline{\tau}^4$, all four distinct from $\mathrm{N}^2$. So the $\mathrm{N}^2$-part of $\mathrm{ch}_2(E)$ is

$$\tfrac{1}{2}\Big(\sum_i m_i e_i^2\Big)\eta^2 + \Big(\sum_i m_i \mathrm{N}_{K/\mathbb{Q}}(u_i)\Big)\theta^+\theta^-.$$

Now compare with the flat Hodge classes of degree four. For $n \geq 3$ they are $\mathbb{Q}\eta^2$ by Theorem 4.14. For $n = 2$ they are $\mathbb{Q}\eta^2 \oplus W(A)$, and $W(A)$ spans the summands for $\tau^4$ and $\overline{\tau}^4$, which are not $\mathrm{N}^2$. In both cases the $\mathrm{N}^2$-part of a flat Hodge class of degree four is a multiple of $\eta^2$. By Lemma 18.4 the class $\theta^+\theta^-$ is not such a multiple, so its coefficient must vanish, which is (43).

Finally $\mathrm{N}_{K/\mathbb{Q}}(g + h\sqrt{-d}) = g^2 + dh^2$ with $d > 0$, so $\mathrm{N}_{K/\mathbb{Q}}(u) \geq 0$ with equality only at $u = 0$, and a sum of nonnegative terms with positive weights vanishes only when every term does. Then $\mathrm{ch}_k(E) = \frac{1}{k!}(\sum_i m_i e_i^k)\eta^k$, and $\eta^k$ is not on the Weil line by Theorem 4.14. ∎

**Corollary 18.6** (Objects filtered by line bundles)**.** *Let $A$ satisfy (41), which holds at every split member of Theorem 15.2 and at every quaternionic base point of Theorem 15.25. Then Theorem 16.29 cannot be applied at $A$ with any of the following:*

(a) *a coherent sheaf admitting a filtration by line bundles, in particular any direct sum of line bundles;*

(b) *a simple semi-homogeneous vector bundle in the sense of Mukai* [Muk78]*, since such a bundle of rank $r$ has $\mathrm{ch}(E) = r\,e^{c_1(E)/r}$;*

(c) *any iterated extension of the objects in (a) and (b).*

*In each case the failure is not of the semiregularity hypothesis but of the Chern character hypothesis of Proposition 16.27: no such $E$ has a Chern character which is a flat section of Hodge classes with a nonzero Weil component.*

*Proof.* First, (41) holds at the points named. At a split member $\mathrm{NS}(A)_\mathbb{Q}$ has rank three with basis $\beta, \widehat{\beta}, \ell$ by Proposition 15.19(iv), and of these $\eta$ spans the norm-character part while $\gamma$ and $\ell$ span $R$. At a quaternionic point $\mathrm{NS}(A)_\mathbb{Q}$ is the space of Rosati-symmetric elements of $B = (-d, b)_\mathbb{Q}$, which is spanned by 1, $j$ and $ij$; the first gives $\eta$ and the other two are semilinear over $K$, hence lie in $R$. In both cases the norm-character part of $\mathrm{NS}(A)_\mathbb{Q}$ is exactly $\mathbb{Q}\eta$.

Each of (a), (b) and (c) has a Chern character of the shape (42) with positive multiplicities, by additivity of ch along a filtration and by Mukai's computation of the Chern character of a simple semi-homogeneous bundle. The conclusion of Theorem 18.5 then contradicts the requirement that $\mathrm{ch}_n(E)$ have a nonzero component on the Weil line. ∎

*Remark* 18.7 (Scope of the theorem, and the objects of Markman). At a very general member of the family $\mathrm{NS}(A)_\mathbb{Q} = \mathbb{Q}\eta$, so Theorem 18.5 is true for the empty reason that there are no $\theta_i$ to begin with. Its content is at the special members, which is exactly where Theorem 16.29 needs its object to live and where the base points of Theorem 15.27 are. That is the opposite of the usual situation in this paper, where the special members are the easy ones.

The theorem does not touch the objects of Markman [Mar25a, Mar25b]. Those are sheaves whose class in $K_0$ is not a positive combination of exponentials of divisor classes, so (42) fails for them at the first step; and they live at very general points, where, as just

noted, the statement is vacuous. What the theorem does is remove the class of objects one would reach for first, and it removes it for a reason which is visible before any cohomology is computed.

*Remark* 18.8 (Signed sums of line bundles). Theorem 18.5 uses positivity of the multiplicities and nothing else, so it says exactly where to go next: the object must be a complex whose class in $K_0$ is not effective, that is one in which the line bundles appear with multiplicities of both signs. Such objects exist. Writing $E = F \oplus G[1]$ with $F$ and $G$ sums of line bundles, the class in $K_0$ is $[F]-[G]$ and (43) becomes $\sum_{L_i \subset F} \mathrm{N}(u_i) = \sum_{L_j \subset G} \mathrm{N}(u_j)$, which is satisfiable; and Lemma 18.1 still applies verbatim, with the shifts absorbing condition (ii).

The object of Theorem 16.29, if it exists at a split member, is not a sum of line bundles in any arrangement of signs and shifts: Corollary 18.21 below proves this in every dimension. It must have $\mathrm{Ext}^2$ that the trace does not see, that is, it must be indecomposable in the derived category. This description of the remaining construction is sharper than the one in Remark 17.7.

18.2. **An explicit object and the Prouhet–Tarry–Escott problem.** Proposition 16.27 produces the object of Theorem 16.29 by an existence argument: the Chern character is an isomorphism onto the Chow ring with rational coefficients, so a $K$-theory class with the right character exists. That is enough for the theorem and tells one nothing about the object. This subsection writes it down. The answer turns the question into a Diophantine one, and the Diophantine problem has a classical name.

**Setup 18.9** (Objects supported on $R$)**.** Let $A$ be of $(K,-1,n)$-Weil type, $n \geq 2$, satisfying (41), and keep $\theta^{\pm}$ from Notation 18.2. For $u \in K$ with $\theta(u) = u\theta^+ + \overline{u}\,\theta^- \in \mathrm{NS}(A)_{\mathbb{Q}}$, and for multiplicities $m_i \in \mathbb{Z}$, consider

$$v = \sum_{i=1}^{s} m_i\, e^{\theta(u_i)}, \qquad T_{q,r} = \sum_{i=1}^{s} m_i\, u_i^q\, \overline{u}_i^r \quad (q,r \geq 0).$$

**Proposition 18.10** (The Chern character in terms of the $u_i$)**.** *In Setup 18.9,*

$$v_k = \sum_{\substack{q+r=k \\ 0 \leq q,r \leq n}} \frac{T_{q,r}}{q!\,r!}\,(\theta^+)^q(\theta^-)^r, \tag{44}$$

*because $(\theta^{\pm})^q = 0$ for $q > n$. Consequently $v$ is a flat section of Hodge classes with a nonzero Weil component if*

$$T_{q,r} = 0 \ \text{ for all } 0 \leq q,r \leq n \text{ with } (q,r) \notin \{(0,0),(n,0),(0,n)\}, \qquad T_{n,0} \neq 0, \tag{45}$$

*and then $v_0 = T_{0,0} = \sum_i m_i$, $v_k = 0$ for $0 < k < 2n$ with $k \neq n$, and*

$$v_n = \frac{1}{n!}\Big(T_{n,0}(\theta^+)^n + \overline{T}_{n,0}(\theta^-)^n\Big),$$

*a nonzero rational multiple of a generator of the Weil line, by Theorem 15.2.*

*Proof.* Expand $\theta(u_i)^k$ by the binomial theorem, divide by $k!$ and collect; the vanishing of $(\theta^{\pm})^q$ for $q > n$ is because $\theta^{\pm} \in \bigwedge^2 V_{\pm}$ and $\dim_{\mathbb{C}} V_{\pm} = 2n$. Under (45) the only surviving terms of (44) are the ones listed, and $(\theta^+)^n$ spans $\bigwedge^{2n} V_+$, which is the Weil line over $\mathbb{C}$. ∎

So the search for the object of Theorem 16.29 is the search for a finitely supported integer-valued function on $\mathcal{O}_K$ that annihilates every monomial $u^q\overline{u}^r$ with $q,r\le n$ except three of them. Two things follow at once, and they pull in opposite directions.

**Lemma 18.11** (Equal norms force the Prouhet–Tarry–Escott condition)**.** *Suppose all the $u_i$ in Setup 18.9 have the same norm $N=\mathrm{N}_{K/\mathbb{Q}}(u_i)>0$. Then (45) holds if and only if*

$$\sum_i m_i u_i^k = 0 \quad \text{for } k=0,1,\dots,n-1, \qquad \sum_i m_i u_i^n \neq 0. \tag{46}$$

*With $m_i=\pm1$ this says that the $u_i$ split into two multisets on the circle $\mathrm{N}=N$ whose power sums agree through degree $n-1$ and differ at degree $n$: a Prouhet–Tarry–Escott pair of degree $n-1$, constrained to a norm circle of $\mathcal{O}_K$.*

*Proof.* For $q\ge r$ one has $u^q\overline{u}^r=N^r u^{q-r}$, so $T_{q,r}=N^r\sum_i m_i u_i^{q-r}$, and the pairs $(q,r)$ with $q,r\le n$ realise every difference $k=q-r$ from $0$ to $n$. The excluded pairs $(0,0)$ and $(n,0)$ are the ones with $r=0$ and $k=0$ or $k=n$; but $k=0$ also occurs at $(1,1),\dots,(n,n)$, which are not excluded, so $\sum_i m_i=0$ is still required. The remaining differences $k=1,\dots,n-1$ each occur at some non-excluded pair, and $k=n$ occurs only at $(n,0)$. ∎

**Theorem 18.12** (Equal norms cannot be semiregular)**.** *Let $E=\bigoplus_i L_i[p_i]$ realise (46) with all norms equal. Then $\mathrm{ch}_0(E)=0$, and the semiregularity map $\sigma_E$ has a kernel of dimension at least $n^2$.*

*Proof.* By Lemma 18.11 the case $k=0$ of (46) gives $\sum_i m_i=0$, which is $\mathrm{ch}_0(E)$. For the kernel, take $\zeta_i=m_i\zeta$ for a single $\zeta\in H^2(\mathcal{O}_A)$; then by Lemma 18.1(iii)

$$\Phi\big((m_i\zeta)_i\big)=\Big(\sum_i m_i e^{\lambda_i}\Big)\zeta=\mathrm{ch}(E)\,\zeta,$$

which by Proposition 18.10 is a multiple of $\omega\,\zeta$ with $\omega$ a generator of the Weil line. Now $W(A)_{\mathbb{C}}$ is spanned by $\bigwedge^{2n}V_+$ and $\bigwedge^{2n}V_-$, and the Weil type condition says $\dim V_\pm^{0,1}=n$; writing $H^2(\mathcal{O}_A)=\bigwedge^2(V_+^{0,1}\oplus V_-^{0,1})$, the product of a generator of $\bigwedge^{2n}V_+$ with $\zeta$ kills the components of $\zeta$ in $\bigwedge^2 V_+^{0,1}$ and in $V_+^{0,1}\otimes V_-^{0,1}$, and likewise with the roles exchanged. So $\omega\,\zeta=0$ for every $\zeta$ in $V_+^{0,1}\otimes V_-^{0,1}$, a space of dimension $n^2$, and all of those lie in the kernel of $\Phi$. ∎

**Corollary 18.13** (At least $n{+}1$ norms)**.** *If $E$ as in Setup 18.9 satisfies (45) and $\mathrm{ch}_0(E)\neq0$, then the $u_i$ take at least $n+1$ distinct norms.*

*Proof.* Let $N_1,\dots,N_\nu$ be the distinct norms and $M_j$ the sum of the $m_i$ over the level $\mathrm{N}(u_i)=N_j$. The conditions $T_{r,r}=0$ for $r=1,\dots,n$ read $\sum_j M_j N_j^r=0$ for $r=1,\dots,n$. The matrix $(N_j^r)_{1\le r\le n,\,1\le j\le\nu}$ has rank $\min(n,\nu)$ because the $N_j$ are distinct and nonzero, so $\nu\le n$ forces every $M_j=0$ and hence $\mathrm{ch}_0(E)=\sum_j M_j=0$. ∎

*The smallest solution.* At $n=3$ and $K=\mathbb{Q}(\sqrt{-3})$ the Prouhet–Tarry–Escott condition of Lemma 18.11 has a particularly simple solution: the norm circle $\mathrm{N}=1$ carries the six roots of unity of $K$, and the two cosets of the cube roots of $-1$ and of $+1$ are a Prouhet–Tarry–Escott pair of degree two. Explicitly, with $\zeta_6$ a primitive sixth root of unity, the cube roots of $-1$ are $\zeta_6,\zeta_6^3,\zeta_6^5$ and those of $+1$ are $1,\zeta_6^2,\zeta_6^4$; each triple sums to zero and has zero sum of squares, while the sums of cubes are $-3$ and $+3$.

**Proposition 18.14** (An explicit object at $n=3$)**.** *Let $K=\mathbb{Q}(\sqrt{-3})$, $n=3$, and let $A_0=X\times\widehat{X}$ be a split member. In the basis $\beta,\widehat{\beta},\ell$ of $\mathrm{NS}(A_0)_{\mathbb{Q}}$ put*

$$\lambda_1=(3,-1,3),\quad \lambda_2=(-6,2,0),\quad \lambda_3=(3,-1,-3),$$

*and $\lambda_{3+i} = -\lambda_i$ for $i = 1, 2, 3$, with $m_i = +1$ for $i \le 3$ and $m_i = -1$ for $i \ge 4$. Then the six classes are distinct and integral, the perfect complex*

$$E = (L_1 \oplus L_2 \oplus L_3) \ \oplus \ (L_4 \oplus L_5 \oplus L_6)[3]$$

*has*

$$\mathrm{ch}_k(E) = 0 \quad (k \neq 3), \qquad \mathrm{ch}_3(E) = 12\,\omega_1,$$

*and every difference $\lambda_j - \lambda_i$ with $i \neq j$ is nondegenerate of index exactly 3, so that conditions (i) and (ii) of Lemma 18.1 hold for this shift. Condition (iii) fails: the map $\Phi$ has rank 75 instead of 90, with kernel of dimension $15 = \dim H^2(\mathcal{O}_{A_0})$.*

*Proof.* With $d = 3$, the class $\theta(u) = 2g\gamma + 6h\ell$ of $u = g + h\sqrt{-3}$ has coordinates $(6g, -2g, 6h)$, so $\lambda_1, \lambda_2, \lambda_3$ are the images of the cube roots $\zeta_6, \zeta_6^3, \zeta_6^5$ of $-1$ and $\lambda_4, \lambda_5, \lambda_6$ those of the cube roots of $+1$; the six classes are distinct and integral. All six have norm 1, and $\sum_i m_i u_i^k$ is 0 for $k = 0, 1, 2$ and $-3 - 3 = -6$ for $k = 3$, so (46) holds at $n = 3$. By Lemma 18.11 and Proposition 18.10, $\mathrm{ch}_k(E) = 0$ for $k \neq 3$ and $\mathrm{ch}_3(E) = -((\theta^+)^3 + (\theta^-)^3)$, and by Example 15.3 at $n = 3$, $\omega_1 \pm \sqrt{-3}\,\omega_2 = -\frac{1}{6}(\theta^{\mp})^3$; so $\mathrm{ch}_3(E) = 12\,\omega_1$.

Every difference is $\lambda_j - \lambda_i = \theta(w)$ with $w = u_j - u_i \neq 0$, and $\theta(w)$ is nondegenerate of index $n$ for every $w \in K^\times$. Indeed $\theta(1) = 2\gamma$ and $\theta(\sqrt{-d}) = 2d\ell$ are of type $(1,1)$, hence so are $\theta^{\pm}$, so $\theta^+ \in V_+^{1,0} \otimes V_+^{0,1}$, and $V_+^{0,1}$ is the conjugate of $V_-^{1,0}$. In the splitting of the tangent space dual to $H^{1,0} = V_+^{1,0} \oplus V_-^{1,0}$, the hermitian form of $\theta(w)$ therefore has the block form $\left(\begin{smallmatrix} 0 & wP \\ \bar{w}P^* & 0 \end{smallmatrix}\right)$, with $P$ the block of $2\gamma$, which is invertible because $\gamma$ is nondegenerate. Its eigenvalues are plus and minus the singular values of $wP$, so it has signature $(n, n)$, and a line bundle with this class has cohomology only in degree $n$ [BL04, Chapter 3]. For the shift, condition (ii) asks that $2 + p_i - p_j \neq 3$ whenever $i \neq j$, that is $p_i - p_j \neq 1$; within each of the two blocks the difference is 0, and between them it is $\pm 3$.

For the rank, the $u_i$ are the six sixth roots of unity, each once, and $\overline{u}_i = u_i^{-1}$. Absorb the signs into the $\zeta_i$ as in Lemma 18.1(iii) and expand $e^{\lambda_i} = e^{u_i\theta^+} e^{\overline{u}_i\theta^-}$:

$$\Phi(\zeta) = \sum_{a,b=0}^{3} \frac{(\theta^+)^a (\theta^-)^b}{a!\,b!}\, Z_{a-b}, \qquad Z_k = \sum_i u_i^k \zeta_i,$$

with $k$ read modulo 6. The $Z_0, \dots, Z_5$ determine the $\zeta_i$, by a Vandermonde matrix. Split $H^{0,2}(A_0) = \bigwedge^2 V_+^{0,1} \oplus (V_+^{0,1} \otimes V_-^{0,1}) \oplus \bigwedge^2 V_-^{0,1}$, write $Z_k = Z_k^+ + Z_k^0 + Z_k^-$ accordingly, and grade $H^*(A_0) = \bigwedge^* V_+ \otimes \bigwedge^* V_-$ by the Hodge types of the two factors. The term $(\theta^+)^a(\theta^-)^b Z_k^0$ has types $(a, a+1)$ and $(b, b+1)$, the term with $Z_k^+$ has types $(a, a+2)$ and $(b, b)$, and the term with $Z_k^-$ has types $(a, a)$ and $(b, b+2)$. These are different for different $(a, b)$ and different pieces, so every term vanishes separately. Since $\theta^{\pm}$ pairs $V_{\pm}^{1,0}$ nondegenerately with $V_{\pm}^{0,1}$, both of dimension 3, multiplication by $(\theta^{\pm})^a$ is injective on $\bigwedge^j V_{\pm}^{0,1}$ when $a + j \le 3$ and zero otherwise; in bases with $\theta^+ = \sum_i e_i f_i$ it sends $f_J$ to a signed sum of the $e_I f_{I \cup J}$ with $I \cap J = \emptyset$, $|I| = a$, and different $J$ give disjoint supports. Hence $Z_k^0 = 0$ unless no $a, b \le 2$ have $a - b \equiv k$, which leaves $k = 3$; $Z_k^+ = 0$ unless no $a \le 1$, $b \le 3$ have $a - b \equiv k$, which leaves $k = 2$; and likewise $Z_k^- = 0$ unless $k = 4$. So $\ker \Phi$ is $V_+^{0,1} \otimes V_-^{0,1}$ at $k = 3$ together with $\bigwedge^2 V_+^{0,1}$ at $k = 2$ and $\bigwedge^2 V_-^{0,1}$ at $k = 4$, of dimension $9 + 3 + 3 = 15 = \dim H^2(\mathcal{O}_{A_0})$, and $\Phi$ has rank $90 - 15 = 75$. The first summand is the kernel of Theorem 18.12. ∎

*Remark* 18.15 (What the construction settles). Three things are settled. The object of Proposition 16.27 is not an abstraction: at $n = 3$ over $\mathbb{Q}(\sqrt{-3})$ it is six line bundles and a

shift, and its Chern character is exactly 12 times a Weil class with every other graded piece zero. The construction is completely general, and Lemma 18.11 identifies the constraint as a Prouhet–Tarry–Escott problem on a norm circle, a question of a kind that can be attacked directly. And Corollary 18.13 says what any semiregular solution must look like: the $u_i$ must take at least $n+1$ distinct norms, because otherwise the rank is zero and the argument of Theorem 18.12 produces a kernel of dimension $n^2$.

This leaves the existence of a solution with $n+1$ norms. The level sums are then forced up to scale (Proposition 18.20), and Corollary 18.21 shows that no solution could serve, because a line bundle summand off the polarisation line is obstructed along the family whatever the norms.

*The obstruction classes of the explicit object.* Corollary 9.18 and Remark 9.21 say that the kernel of $\sigma_E$ contains a copy of the tangent space of the Weil family, and that this is a statement about where the semiregularity map is blind, not a statement that the object is obstructed. For the explicit object the obstruction classes can be written down, and they are not zero. The computation is elementary and we record it in general.

Recall the conventions of Section 9.3: a first-order deformation of $A$ is $v \in \mathrm{Hom}(H^{1,0}, H^{0,1}) = H^1(A, T_A)$, extended by zero on $H^{0,1}$ and acting on $H^*(A,\mathbb{C})$ as a derivation $v\lrcorner$; a class $\xi$ of type $(p,p)$ stays of type $(p,p)$ to first order along $v$ exactly when $v\lrcorner\xi = 0$. For a line bundle $L$ with $\lambda = c_1(L)$ the Atiyah class is $\lambda \in H^1(\Omega^1_A)$, its contraction with $v$ is $v\lrcorner\lambda \in H^2(\mathcal{O}_A) = H^{0,2}(A)$, and this is the obstruction to deforming $L$ along $v$; a shift does not change it. For $E = \bigoplus_i L_i[p_i]$ the Atiyah class is diagonal with entries $\lambda_i$, as in the proof of Lemma 18.1, so the obstruction to deforming $E$ along $v$ is the diagonal element

$$\mathrm{ob}_E(v) = \big(v\lrcorner\lambda_i\big)_i \in \bigoplus_i H^{0,2}(A) \subset \mathrm{Ext}^2(E,E). \tag{47}$$

**Proposition 18.16** (The obstruction map of a split object)**.** *Let $A_0 = X \times \widehat{X}$ be a split member with $\beta$ principal and $X$ general, so that $\mathrm{NS}(A_0)_{\mathbb{Q}} = \langle \beta, \widehat{\beta}, \ell\rangle$ by Proposition 15.19(iv). Let $T = T_{[A_0]}\mathcal{D}_{n,n}$ be the space of polarised first-order deformations commuting with $K$, of dimension $n^2$ by Proposition 9.14, and put*

$$T_{\mathrm{spl}} = \big\{\, v \in \mathrm{Hom}(H^{1,0}, H^{0,1}) \;:\; v\lrcorner\beta = v\lrcorner\widehat{\beta} = v\lrcorner\ell = 0 \,\big\}.$$

*Let $E = \bigoplus_{i=1}^{s} L_i[p_i]$ with $\lambda_i = c_1(L_i)$ and let $\mathrm{ob}_E\colon T \to \bigoplus_i H^{0,2}(A_0)$ be the map (47) restricted to $T$.*

(i) *$T_{\mathrm{spl}}$ is the tangent space to the split locus: it has dimension $n(n+1)/2$, it is contained in $T$, and it consists of the deformations of $X \times \widehat{X}$ induced by the polarised deformations of $(X,\beta)$.*

(ii) *If the classes $\lambda_i$ together with $\eta$ span $\mathrm{NS}(A_0)_{\mathbb{Q}}$, then $\ker \mathrm{ob}_E = T_{\mathrm{spl}}$ and $\mathrm{rank}\,\mathrm{ob}_E = n(n-1)/2$, the codimension of the split locus in $\mathcal{D}_{n,n}$.*

(iii) *If $\mathrm{ch}_k(E) \in \mathbb{Q}\eta^k$ for $k \neq n$ and $\mathrm{ch}_n(E) \in \mathbb{Q}\eta^n \oplus W(A_0)$, then $\sigma_E \circ \mathrm{ob}_E = 0$ on $T$: for every $v \in T$,*

$$\sigma_E(\mathrm{ob}_E(v)) = v\lrcorner\,\mathrm{ch}(E) = 0. \tag{48}$$

(iv) *For the object of Proposition 18.14, at $n=3$ and $K = \mathbb{Q}(\sqrt{-3})$: $\ker \mathrm{ob}_E = T_{\mathrm{spl}}$ has dimension 6 and $\mathrm{ob}_E$ has rank 3; the image of $\mathrm{ob}_E$ lies in $\ker\sigma_E$; the subspace $\Delta = \{z\cdot \mathrm{id}_E : z \in \mathrm{Ann}_{H^{0,2}}(\omega_1)\}$ of Corollary 9.18 has dimension 9 and meets the image of $\mathrm{ob}_E$ only in zero, so that $\Delta \oplus \mathrm{im}\,\mathrm{ob}_E$ is a twelve-dimensional subspace of the fifteen-dimensional kernel of $\sigma_E$; and $\mathrm{ob}_E(v)$ has components $v\lrcorner\lambda_i$ on $L_i$ and*

*$-v\lrcorner\,\lambda_i$ on $L_i^{-1}[3]$, so it lies on the line of vectors of the shape $(z,\dots,z)$ only when it vanishes. In particular $E$ has a nonzero first-order obstruction along every $v\in T$ not tangent to the split locus, and no element $z\cdot\mathrm{id}_E$ of $\mathrm{Ext}^2(E,E)$ cancels it.*

*Proof.* (i) Let $P = H^{1,0}(X) \subset H^1(X,\mathbb{C}) = U_{\mathbb{C}}^\vee$ with basis $p_1,\dots,p_n$, so that $p_1,\dots,p_n,\overline{p}_1,\dots,\overline{p}_n$ is a basis of $U_{\mathbb{C}}^\vee$, and let $q_a,\overline{q}_a$ be the dual basis of $U_{\mathbb{C}} = H^1(\widehat{X},\mathbb{C})$; the dual of a conjugate basis is the conjugate of the dual basis, so the notation is consistent. The Poincaré class is the identity element of $U^\vee\otimes U$,

$$\ell = \sum_{a=1}^{n}(p_a\wedge q_a + \overline{p}_a\wedge\overline{q}_a),$$

and it is of type $(1,1)$, which forces $H^{1,0}(\widehat{X}) = \langle\overline{q}_a\rangle$ and $H^{0,1}(\widehat{X}) = \langle q_a\rangle$: the Hodge structure of $\widehat{X}$ is the dual one. Since $\beta$ is of type $(1,1)$ and nondegenerate, $\beta = \sum_{a,b} h_{ab}\,p_a\wedge\overline{p}_b$ with $h\in\mathrm{GL}_n(\mathbb{C})$; and since $\beta$ is principal with Gram matrix $B$ in a symplectic basis of $U$ and $\widehat{\beta}$ has Gram matrix $-B^{-1} = B$ in the dual basis of $U^\vee$, the Gram matrix of $\widehat{\beta}$ in the basis $p_a,\overline{p}_a$ is minus the inverse of the Gram matrix $\left(\begin{smallmatrix}0 & h\\ -h^\top & 0\end{smallmatrix}\right)$ of $\beta$ in the basis $q_a,\overline{q}_a$, which gives $\widehat{\beta} = -\sum_{a,b}(h^{-1})_{ab}\,\overline{q}_a\wedge q_b$.

Now $H^{1,0}(A_0) = \langle p_a\rangle\oplus\langle\overline{q}_a\rangle$ and $H^{0,1}(A_0) = \langle\overline{p}_a\rangle\oplus\langle q_a\rangle$, so a deformation $v$ is four $n\times n$ matrices,

$$v(p_a) = \sum_c(A_{ca}\,\overline{p}_c + C_{ca}\,q_c),\qquad v(\overline{q}_a) = \sum_c(B_{ca}\,\overline{p}_c + D_{ca}\,q_c),$$

and $v$ kills $\overline{p}_a$ and $q_a$. Contracting,

$$v\lrcorner\,\ell = \sum_{a,c}\Big(A_{ca}\,\overline{p}_c\wedge q_a + C_{ca}\,q_c\wedge q_a + B_{ca}\,\overline{p}_a\wedge\overline{p}_c + D_{ca}\,\overline{p}_a\wedge q_c\Big),$$

and reading off the three kinds of monomials, $v\lrcorner\,\ell = 0$ if and only if $D = -A^\top$ and $B$, $C$ are symmetric. Next

$$v\lrcorner\,\beta = \sum_{a,b,c} h_{ab}\Big(A_{ca}\,\overline{p}_c\wedge\overline{p}_b + C_{ca}\,q_c\wedge\overline{p}_b\Big),$$

so $v\lrcorner\,\beta = 0$ if and only if $Ch = 0$, that is $C = 0$, and $Ah$ is symmetric. Likewise $v\lrcorner\,\widehat{\beta} = 0$ if and only if $B = 0$ and $Dh^{-1}$ is symmetric. Given $D = -A^\top$, the last condition reads $A^\top h^{-1} = h^{-\top}A$, which after multiplication by $h^\top$ on the left and by $h$ on the right is $h^\top A^\top = Ah$, the symmetry of $Ah$ again. Hence

$$T_{\mathrm{spl}} = \{(A,B,C,D) = (A,0,0,-A^\top)\ :\ Ah\text{ symmetric}\},$$

of dimension $n(n+1)/2$. It is the tangent space to the split locus: the same computation on $X$ alone shows that the polarised first-order deformations of $(X,\beta)$ are the matrices $A$ with $Ah$ symmetric, the space $T_{[X]}\mathcal{A}_n$ of dimension $n(n+1)/2$; the deformation of $\widehat{X}$ induced by $A$ is the dual one, $-A^\top$; and the map $A\mapsto(A,0,0,-A^\top)$ is injective. Finally $T_{\mathrm{spl}}\subseteq T$: for $v\in T_{\mathrm{spl}}$, $v\lrcorner\,\eta = 0$ because $\eta\in\langle\beta,\widehat{\beta}\rangle$, so $v$ is polarised, and $v\lrcorner\,\omega_1 = v\lrcorner\,\omega_2 = 0$ because $\omega_1$ and $\omega_2$ are polynomials in $\beta,\widehat{\beta},\ell$ by Theorem 15.2 and $v\lrcorner$ is a derivation; by Proposition 9.14 such a $v$ commutes with $K$.

(ii) Every $v\in T$ has $v\lrcorner\,\eta = 0$. If the $\lambda_i$ and $\eta$ span $\mathrm{NS}(A_0)_{\mathbb{Q}} = \langle\beta,\widehat{\beta},\ell\rangle$, then $v\lrcorner\,\lambda_i = 0$ for all $i$ is equivalent, for $v\in T$, to $v\lrcorner\,\beta = v\lrcorner\,\widehat{\beta} = v\lrcorner\,\ell = 0$, so $\ker\mathrm{ob}_E = T\cap T_{\mathrm{spl}} = T_{\mathrm{spl}}$ by (i), and the rank is $n^2 - n(n+1)/2 = n(n-1)/2$, which is the codimension of Proposition 15.19(iii).

(iii) By (40), $\sigma_E(\zeta) = \sum_i (-1)^{p_i} \zeta_{ii}\, e^{\lambda_i}$ for a diagonal $\zeta$. Since $v \lrcorner$ is a derivation and $v \lrcorner \lambda_i$ has even degree, $v \lrcorner \lambda_i^q = q\, \lambda_i^{q-1} (v \lrcorner \lambda_i)$ and so $v \lrcorner e^{\lambda_i} = (v \lrcorner \lambda_i)\, e^{\lambda_i}$. Therefore

$$\sigma_E(\mathrm{ob}_E(v)) = \sum_i (-1)^{p_i} (v \lrcorner \lambda_i)\, e^{\lambda_i} = v \lrcorner \sum_i (-1)^{p_i} e^{\lambda_i} = v \lrcorner \mathrm{ch}(E).$$

For $v \in T$ one has $v \lrcorner \eta^k = k \eta^{k-1} (v \lrcorner \eta) = 0$, and $v \lrcorner \omega = 0$ for every $\omega \in W(A_0)$ by Proposition 9.14; under the hypothesis on $\mathrm{ch}(E)$ this gives (48).

(iv) The three classes $\lambda_1 = \gamma + 3\ell$, $\lambda_2 = -2\gamma$, $\lambda_3 = \gamma - 3\ell$ span $\langle \gamma, \ell \rangle$, and $\langle \eta, \gamma \rangle = \langle \beta, \widehat{\beta} \rangle$ by the determinant in the proof of Lemma 18.3, so (ii) applies and gives the kernel and the rank; Proposition 18.14 gives the hypothesis of (iii), so the image lies in $\ker \sigma_E$. The subspace $\Delta$ has dimension $\dim \mathrm{Ann}_{H^{0,2}}(\omega_1) = n^2 = 9$ by Lemma 9.6. The components of $\mathrm{ob}_E(v)$ on $L_i$ and on $L_i^{-1}[3]$ are $v \lrcorner \lambda_i$ and $v \lrcorner (-\lambda_i) = -v \lrcorner \lambda_i$, so an element of the image of the shape $(z, \dots, z)$ has $z = -z$, hence is zero; this is $\Delta \cap \mathrm{im}\, \mathrm{ob}_E = 0$, and with $\dim \ker \sigma_E = 15$ from Proposition 18.14 the dimension count follows. ∎

*Remark* 18.17 (Meaning of the obstruction classes). Proposition 18.16 sharpens Remark 9.21 in one respect and exposes a limit of the criterion in another. First, for the explicit object the obstruction classes along the family are nonzero, and they are nonzero exactly in the $n(n-1)/2$ directions normal to the split locus, which are the directions in which the Weil class has to be transported. Second, those classes lie in the kernel of $\sigma_E$ by (48), and (48) holds for every object whose Chern character is a flat section of Hodge classes along the family, so the semiregularity map cannot detect the obstruction of any such object. The obstruction is therefore nonzero, and the criterion cannot see it. For the explicit object the situation is the worst possible: the twelve dimensions $\Delta \oplus \mathrm{im}\, \mathrm{ob}_E$ of the kernel are accounted for by the two mechanisms of Corollary 9.18 and (47), the remaining three lie on the even side of Corollary 9.8 as well, and none of the fifteen is an artefact. The classes of the shape $z \cdot \mathrm{id}_E$ are the ones a change of the twisting class would contribute, one and the same $z$ to every rank one summand; the last assertion of (iv) says that no such change removes the obstruction. So the explicit object does not move off the split locus to first order, either as a complex or as a twisted complex, and the question of Question 15.13 for it is not one of verifying a hypothesis but of replacing the object.

The positive content is a recipe. By (ii), a split object whose classes span $\mathrm{NS}(A_0)_{\mathbb{Q}}$ with $\eta$ has obstruction map of rank $n(n-1)/2$ against the family, and the spanning hypothesis is more than is needed: a single class $\theta \in \mathrm{NS}(A_0)_{\mathbb{Q}}$ not proportional to $\eta$ already has $\{v \in T : v \lrcorner \theta = 0\} = T_{\mathrm{spl}}$. Indeed, every $v \in T$ kills $\eta$ and commutes with $K$, so it is a pair $(v_+, v_-)$ with $v_\pm \in \mathrm{Hom}(V_\pm^{1,0}, V_\pm^{0,1})$. Since $\theta^\pm$ is of type $(1,1)$ in $\bigwedge^2 V_\pm$, the $R_0$-part $u\theta^+ + \overline{u}\,\theta^-$ of $\theta$, with $u \neq 0$, gives $v \lrcorner \theta = u\, v_+ \lrcorner \theta^+ + \overline{u}\, v_- \lrcorner \theta^-$, whose two terms lie in the different summands $\bigwedge^2 V_+^{0,1}$ and $\bigwedge^2 V_-^{0,1}$ of $H^{0,2}(A_0)$. So the kernel of $v \mapsto v \lrcorner \theta$ on $T$ does not depend on $u \neq 0$; it is therefore the common kernel for $\gamma$ and $\ell$, the cases $u = 1$ and $u = \sqrt{-d}$, which is $T_{\mathrm{spl}}$ by (ii). So every split object built from line bundles other than powers of $\eta$ is obstructed in exactly the normal directions to the split locus, and Theorem 18.5 says that an object built from powers of $\eta$ carries no Weil class. What survives is the statement of Remark 9.9: the object must have positive rank.

*The evaluation map of the explicit object.* Theorem 9.19 places the copy $\Delta$ of the tangent space inside the image of the evaluation map. For a split object the whole of $\mathrm{ev}_E$ can be written down, and for the explicit object the entire kernel of $\sigma_E$ turns out to lie in its image. We identify $HH^2(A_0)$ with

$$HT^2(A_0) = H^{0,2}(A_0) \;\oplus\; \mathrm{Hom}(H^{1,0}, H^{0,1}) \;\oplus\; \textstyle\bigwedge^2 (H^{1,0})^\vee \tag{49}$$

by the Hochschild–Kostant–Rosenberg isomorphism, the three summands being $H^2(\mathcal{O})$, $H^1(T)$ and $H^0(\bigwedge^2 T)$ for the trivial tangent bundle. An element $x = (z, v, \pi)$ acts on $H^*(A_0, \mathbb{C})$ by $z\wedge$, by the derivation $v\lrcorner$ of Section 9.3, and by the double contraction $\pi\lrcorner$, which removes two $(1,0)$ slots; we write $x\lrcorner\xi$ for the sum of the three.

The double contraction is a second order operator. For a class $\lambda$ of type $(1,1)$, which has a single $(1,0)$ slot, $\pi\lrcorner\lambda = 0$ and $\pi\lrcorner\lambda^k = \binom{k}{2}\lambda^{k-2}\,(\pi\lrcorner\lambda^2)$, because $\pi\lrcorner$ is a composite of two interior products, each of which is an antiderivation, and the only nonzero terms are those in which the two interior products hit two different factors. Summing over $k$ with the factorials gives $\pi\lrcorner e^{\lambda} = \frac{1}{2}(\pi\lrcorner\lambda^2)\,e^{\lambda}$, exactly parallel to $v\lrcorner e^{\lambda} = (v\lrcorner\lambda)\,e^{\lambda}$ for the first order operator $v\lrcorner$. This is the whole content of the compatibility (51) below: the semiregularity map of a line bundle is multiplication by $e^{\lambda}$, and every summand of $HT^2$ acts on $e^{\lambda}$ as multiplication by a class in $H^{0,2}$ which is precisely the value of $\mathrm{ev}_L$. The same identity, for arbitrary $E$, is the commutative square of Buchweitz and Flenner [BF03, Corollary 4.3] and its Hochschild form in [Mar25b, Lemma 11.3]; on an abelian variety the Todd class is trivial and no correction to the HKR map is needed [Cal05, BF08].

**Proposition 18.18** (The evaluation map of a split object)**.** *Let $A_0$ be a split member with $X$ general and let $E = \bigoplus_{i=1}^{s} L_i[p_i]$ with $\lambda_i = c_1(L_i)$ and all $\mathrm{Ext}^2(L_i[p_i], L_j[p_j])$, $i \neq j$, zero, so that $\mathrm{Ext}^2(E,E) = \bigoplus_i H^{0,2}(A_0)$.*

(i) *Under (49), $\mathrm{ev}_E$ is diagonal:*

$$\mathrm{ev}_E(z,v,\pi)_i = z + v\lrcorner\lambda_i + \tfrac{1}{2}\,\pi\lrcorner\lambda_i^2 \;\in\; H^{0,2}(A_0) = \mathrm{Ext}^2\big(L_i[p_i], L_i[p_i]\big). \tag{50}$$

(ii) *For every $x \in HT^2(A_0)$,*

$$\sigma_E\big(\mathrm{ev}_E(x)\big) = x\lrcorner\,\mathrm{ch}(E), \tag{51}$$

*and consequently $\ker\sigma_E \cap \mathrm{im}\,\mathrm{ev}_E = \mathrm{ev}_E\big(\{x : x\lrcorner\,\mathrm{ch}(E) = 0\}\big)$. The weakened hypothesis of* [Mar25b, Question 11.4] *holds for $E$ if and only if $\mathrm{ev}_E$ kills every $x$ that preserves $\mathrm{ch}(E)$ to first order.*

(iii) *$\dim HT^2(A_0) = 2\binom{2n}{2} + 4n^2$ and $\dim\mathrm{Ext}^2(E,E) = s\binom{2n}{2}$; for $s \geq 5$ and every $n \geq 2$ the second exceeds the first, so $\mathrm{ev}_E$ is not surjective and* [Mar25b, Lemma 11.3] *cannot apply to a sum of five or more line bundles.*

(iv) *For the object of Proposition 18.14, with $\lambda_i = u_i\theta^+ + \overline{u}_i\theta^-$ as in Notation 18.2: the space $\{x : x\lrcorner\,\mathrm{ch}(E) = 0\}$ is $\mathrm{Ann}_{H^{0,2}}(\omega_1) \oplus \{v : v\lrcorner\omega_1 = 0\} \oplus \{\pi : \pi\lrcorner\omega_1 = 0\}$, of dimensions 9, 18 and 9; the first and the third summands are both carried by $\mathrm{ev}_E$ onto $\Delta$; the second is carried onto*

$$\Sigma = \big\{\,(u_i a + \overline{u}_i b)_{i=1}^{6} \;:\; a \in \textstyle\bigwedge^2 V_+^{0,1},\; b \in \bigwedge^2 V_-^{0,1}\,\big\}, \qquad \dim\Sigma = n(n-1) = 6,$$

*which contains the image of the obstruction map of Proposition 18.16; $\Delta \cap \Sigma = 0$; and*

$$\ker\sigma_E = \Delta \oplus \Sigma \subseteq \mathrm{im}\,\mathrm{ev}_E\,.$$

*So every direction in which $\sigma_E$ is blind is the value of $\mathrm{ev}_E$ on a Hochschild class that preserves $\mathrm{ch}(E)$ to first order, and the weakened hypothesis fails for $E$ by all fifteen dimensions. The rank of $\mathrm{ev}_E$ is 45.*

*Proof.* (i) The evaluation map is the action of $HH^*(A_0)$ on $\mathrm{Ext}^*(E,E)$, which under HKR is contraction of a polyvector in $H^q(\bigwedge^p T)$ with $\frac{1}{p!}\mathrm{At}(E)^p$ [BF08]; for $p = 0$ this is $z \cdot \mathrm{id}_E$ as in the proof of Theorem 9.19, for $p = 1$ it is $v\lrcorner\mathrm{At}(E)$, the map (47), and for $p = 2$ it is $\frac{1}{2}\pi\lrcorner\mathrm{At}(E)^2$. The Atiyah class of $E$ is diagonal with entries $\lambda_i$, so all three are diagonal with the entries displayed.

(ii) By (40), $\sigma_E(\zeta) = \sum_i (-1)^{p_i} \zeta_i e^{\lambda_i}$ for diagonal $\zeta$. The three terms of (50) are exactly the factors by which $z\wedge$, $v\lrcorner$ and $\pi\lrcorner$ multiply $e^{\lambda_i}$: the first trivially, the second because $v\lrcorner$ is a derivation and $v\lrcorner\lambda_i$ has even degree, the third by the formula for $\pi\lrcorner e^{\lambda}$ above. Hence $\sigma_E(\mathrm{ev}_E(x)) = \sum_i (-1)^{p_i}\, x\lrcorner e^{\lambda_i} = x\lrcorner \operatorname{ch}(E)$. The description of $\ker\sigma_E \cap \operatorname{im}\mathrm{ev}_E$ follows: $\mathrm{ev}_E(x)$ lies in $\ker\sigma_E$ if and only if $x\lrcorner \operatorname{ch}(E) = 0$.

(iii) The three summands of (49) have dimensions $\binom{2n}{2}$, $4n^2$ and $\binom{2n}{2}$. The inequality $s\binom{2n}{2} > 2\binom{2n}{2} + 4n^2$ is $(s-2)(2n-1) > 4n$, and $4n/(2n-1) \le 8/3$ for $n \ge 2$, so it holds for $s \ge 5$.

(iv) Since $\operatorname{ch}(E) = 12\omega_1$ is of type $(3,3)$, the three summands of $x\lrcorner \operatorname{ch}(E)$ have types $(3,5)$, $(2,4)$ and $(1,3)$, so the kernel is the direct sum of the three kernels. The first is $\mathrm{Ann}_{H^{0,2}}(\omega_1)$, of dimension $n^2 = 9$ by Lemma 9.6. The second is $\mathrm{Hom}(V_+^{1,0}, V_+^{0,1}) \oplus \mathrm{Hom}(V_-^{1,0}, V_-^{0,1})$, of dimension $2n^2 = 18$, by the proof of Proposition 9.14, whose argument for (i)$\Leftrightarrow$(iii) does not use the polarisation. For the third, decompose $\bigwedge^2(H^{1,0})^\vee$ as $\bigwedge^2(V_+^{1,0})^\vee \oplus (V_+^{1,0})^\vee \otimes (V_-^{1,0})^\vee \oplus \bigwedge^2(V_-^{1,0})^\vee$; here $\omega_1$ is a combination of $(\theta^+)^3$ and $(\theta^-)^3$ with both coefficients nonzero, $(\theta^\pm)^3$ spans $\bigwedge^3 V_\pm^{1,0} \otimes \bigwedge^3 V_\pm^{0,1}$, the outer blocks of $\pi$ act injectively on the respective factor $\bigwedge^3 V_\pm^{1,0}$ and land in different summands of $H^{1,3}$, and the middle block kills both; so the third kernel is the middle block, of dimension $n^2 = 9$.

At a split point $\theta^\pm$ are of type $(1,1)$, so $\theta^\pm \in V_\pm^{1,0} \otimes V_\pm^{0,1}$, and they are nondegenerate pairings because $(\theta^\pm)^3 \ne 0$. Write $\lambda_i^2 = u_i^2(\theta^+)^2 + \overline{u}_i^2(\theta^-)^2 + 2\mathrm{N}(u_i)\,\theta^+\theta^-$. A middle block $\pi$ kills $(\theta^\pm)^2$, which have no $V_\mp^{1,0}$ slot, so $\frac{1}{2}\pi\lrcorner\lambda_i^2 = \mathrm{N}(u_i)\,\pi\lrcorner(\theta^+\theta^-)$; the six $u_i$ of Proposition 18.14 are $\pm\zeta_6, \pm 1, \pm\zeta_6^{-1}$, all of norm one, so this value is independent of $i$ and $\mathrm{ev}_E(\pi) = z_\pi \cdot \mathrm{id}_E$ with $z_\pi = \pi\lrcorner(\theta^+\theta^-)$. The map $\pi \mapsto z_\pi$ from the middle block to $V_+^{0,1} \otimes V_-^{0,1}$ is the tensor product of the two nondegenerate pairings, hence an isomorphism onto the middle block of $H^{0,2}$, which is $\mathrm{Ann}_{H^{0,2}}(\omega_1)$ by Lemma 9.6. So $\mathrm{ev}_E$ carries the third kernel onto $\Delta$, as it carries the first by Theorem 9.19.

For $v = (v_+, v_-)$ in the second kernel, $v\lrcorner\lambda_i = u_i\, v_+\lrcorner\theta^+ + \overline{u}_i\, v_-\lrcorner\theta^-$ with $a = v_+\lrcorner\theta^+ \in \bigwedge^2 V_+^{0,1}$ and $b = v_-\lrcorner\theta^- \in \bigwedge^2 V_-^{0,1}$; as $v_\pm$ ranges over $\mathrm{Hom}(V_\pm^{1,0}, V_\pm^{0,1})$, $a$ and $b$ range over the whole of $\bigwedge^2 V_\pm^{0,1}$ because the pairings $\theta^\pm$ are nondegenerate, which is the surjectivity of $\mathrm{alt}(Vc)$ in (26). So the image is $\Sigma$. The map $(a,b) \mapsto (u_i a + \overline{u}_i b)_i$ is injective, because $(u_1, \overline{u}_1)$ and $(u_2, \overline{u}_2)$ are linearly independent in $\mathbb{C}^2$, $u_1/\overline{u}_1 = \zeta_6^2$ differing from $u_2/\overline{u}_2 = 1$; hence $\dim\Sigma = 2\binom{n}{2} = n(n-1) = 6$. The obstruction map of Proposition 18.16 is the restriction to the polarised $K$-commuting $v$, so its image lies in $\Sigma$. An element of $\Sigma$ of the shape $(z, \dots, z)$ has $u_i a + \overline{u}_i b = z = -z$ on comparing $i$ with $3+i$, so $z = 0$ and then $a = b = 0$: $\Delta \cap \Sigma = 0$. By (ii), $\Delta \oplus \Sigma \subseteq \ker\sigma_E \cap \operatorname{im}\mathrm{ev}_E$, and its dimension $9 + 6 = 15$ equals $\dim\ker\sigma_E$ by Proposition 18.14, so the three spaces coincide.

For the rank, each term of (50) is a scalar depending on $i$ times a class independent of $i$. Write $\mathbf{u}^k = (u_i^k)_i \in \mathbb{C}^6$ and $M = V_+^{0,1} \otimes V_-^{0,1}$. The class $z$ contributes $\mathbf{u}^0 \otimes z$. Split $v$ into its four blocks $v_{\varepsilon\varepsilon'} \colon V_\varepsilon^{1,0} \to V_{\varepsilon'}^{0,1}$; then $v\lrcorner\theta^+$ is the sum of $v_{++}\lrcorner\theta^+ \in \bigwedge^2 V_+^{0,1}$ and $v_{+-}\lrcorner\theta^+ \in M$, the first ranging over $\bigwedge^2 V_+^{0,1}$ as above and the second over $M$ because $\theta^+$ is a nondegenerate pairing, and symmetrically for $v\lrcorner\theta^-$ with the blocks $v_{--}$ and $v_{-+}$. As $\overline{u}_i = u_i^{-1}$, the values on the second summand of (49) are therefore $\mathbf{u}^1 \otimes (a+c) + \mathbf{u}^{-1} \otimes (b+c')$ with $a$, $b$, $c$, $c'$ ranging independently over $\bigwedge^2 V_+^{0,1}$, $\bigwedge^2 V_-^{0,1}$, $M$, $M$. On the third summand, the outer block $\pi_{++}$ kills $(\theta^-)^2$ and $\theta^+\theta^-$, and $\pi_{++} \mapsto \frac{1}{2}\pi_{++}\lrcorner(\theta^+)^2$ is an isomorphism onto $\bigwedge^2 V_+^{0,1}$ (in bases with $\theta^+ = \sum_j e_j f_j$ it sends $e_j^\vee \wedge e_k^\vee$ to $\pm f_j f_k$); likewise for $\pi_{--}$, and

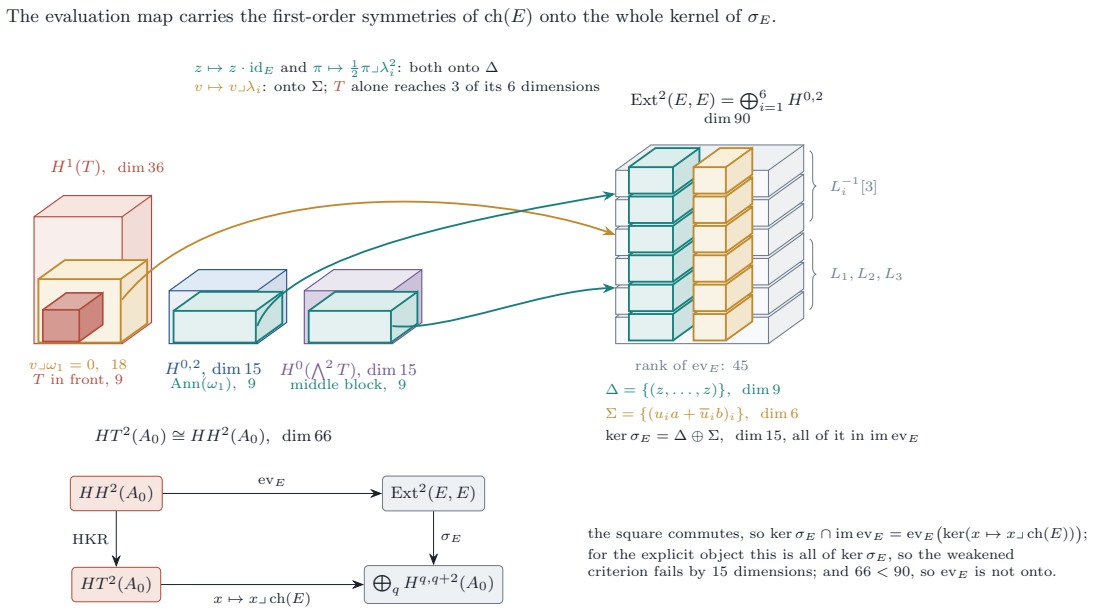



FIGURE 25. Proposition 18.18 for the explicit object at $n = 3$. Left: the three summands of $HT^2(A_0)$, drawn with heights proportional to their dimensions, and inside each the part that preserves $\mathrm{ch}(E)$ to first order: the annihilator of $\omega_1$, the deformations killing $\omega_1$ with the polarised $K$-commuting tangent space $T$ in front, and the middle block of bivectors. Right: $\mathrm{Ext}^2(E,E)$ as six stacked copies of $H^{0,2}$, one per summand of $E$, and inside it the kernel of $\sigma_E$: the column $\Delta$, one and the same class in every layer, and the column $\Sigma$, whose entries depend on the summand through $u_i$. The three arrows are the evaluation map on the three kernels; the two teal ones both land on $\Delta$, the ochre one on $\Sigma$, and together they cover the kernel. Below, the commutative square that makes this a statement about the weakened criterion of [Mar25b, Question 11.4].

the middle block contributes $\mathbf{u}^0 \otimes z_\pi$. So

$$\operatorname{im} \mathrm{ev}_E = \mathbf{u}^0 \otimes H^{0,2} + \mathbf{u}^{\pm 1} \otimes \left(\textstyle\bigwedge^2 V_\pm^{0,1} \oplus M\right) + \mathbf{u}^{\pm 2} \otimes \textstyle\bigwedge^2 V_\pm^{0,1}.$$

The $u_i$ are the six sixth roots of unity, so the vectors $\mathbf{u}^k$ for $k = 0, \pm 1, \pm 2$ are distinct characters of the cyclic group they form and are linearly independent. The sum is direct, and the rank is $15 + 2 \cdot 12 + 2 \cdot 3 = 45$. ∎

Figure 25 draws the three kernels and their images in $\mathrm{Ext}^2(E,E)$.

*Remark* 18.19 (Question 11.4 for split objects). For a split object the two criteria stand or fall together, and for the explicit object both fall completely. Theorem 9.19 makes the first $n^2$ dimensions of the failure uniform over every concentrated object in every dimension; Proposition 18.18(iv) accounts for the remaining $n(n-1)$ at $n = 3$ by the deformations of the two halves $V_\pm$ of the Hodge structure separately, and the equality $\ker \sigma_E = \Delta \oplus \Sigma$ says that the two mechanisms exhaust the kernel: nothing in it is accidental. None of this touches the case for which the question was asked. A sheaf of positive rank has $z \cdot \mathrm{ch}(E) \neq 0$ for every $z \neq 0$, so its $\Delta$ is zero, and whether $\mathrm{ev}_E$ kills the deformations that preserve $\kappa(E)$

is a statement about the sheaf and not about its Chern character. Markman's candidate in [Mar25b, Section 12], a rank one complex on a principally polarised abelian fourfold with Chern character on the secant $\langle e^{\pm\sqrt{-d}\Theta}\rangle$, is exactly such a sheaf, and he records that the verification remains to be done. Theorem 18.56 reduces that verification to one identity on the fourfold, and Theorem 18.61 shows that the identity fails: the candidate does not satisfy the weakened criterion either.

*Solutions of nonzero rank.* We return to the Diophantine problem of Setup 18.9. By Corollary 18.13 a solution with $\mathrm{ch}_0(E) \neq 0$ needs at least $n+1$ distinct norms, and multiplicity one bounds each level sum by the number of elements of $\mathcal{O}_K$ of that norm.

**Proposition 18.20** (The level sums)**.** *Keep Setup 18.9 and suppose $E$ satisfies (45) with $\mathrm{ch}_0(E) \neq 0$, so that the $u_i$ take $\nu \geq n+1$ distinct norms by Corollary 18.13. Write $N_1 < \cdots < N_\nu$ for those norms, $M_j$ for the sum of the $m_i$ at level $N_j$, and $r_K(N) = \#\{u \in \mathcal{O}_K : \mathrm{N}_{K/\mathbb{Q}}(u) = N\}$. If $\nu = n+1$ then, with $P = \prod_k N_k$, $S = \mathrm{ch}_0(E)$ and $D_j = N_j \prod_{k\neq j}(N_j - N_k)$,*

$$M_j = (-1)^n \frac{S\,P}{D_j}, \tag{52}$$

*the least $S > 0$ making every $M_j$ an integer is $S_{\min} = \mathrm{lcm}_j\big(|D_j|/\gcd(|D_j|, P)\big)$, and multiplicity one, Lemma 18.1(i), forces $|M_j| \leq r_K(N_j)$ for every $j$.*

*Proof.* The conditions $T_{r,r} = 0$ for $r = 1, \ldots, n$ say that the linear functional $f \mapsto \sum_j M_j f(N_j)$ kills $x, x^2, \ldots, x^n$, hence sends every polynomial $f$ of degree at most $n$ to $S\,f(0)$. Applying it to $f(x) = \prod_{k\neq j}(x - N_k)$, which has degree $n$, gives

$$M_j \prod_{k\neq j}(N_j - N_k) = S\,(-1)^n \prod_{k\neq j} N_k,$$

and multiplying by $N_j$ turns this into (52). The value of $S_{\min}$ is the definition of a least common multiple, and the bound on $|M_j|$ holds because each $u$ of norm $N_j$ contributes at most one term $m_i = \pm 1$ to $M_j$. ∎

So the level sums are determined by the norms up to the scale $S$, and they compete with the supply $r_K(N_j)$ of elements at each level. That competition is arithmetic. A second mechanism, in the cohomology of $A$, closes the split route for every $n$ and every set of norms at once.

**Corollary 18.21** (Sums of line bundles fail both criteria)**.** *Let $A$ be of $(K, -1, n)$-Weil type with $n \geq 2$, and let $E = \bigoplus_i L_i[p_i]^{\oplus m_i}$ be a sum of shifted line bundles, twisted by a common class or not, whose (twisted) Chern character remains of Hodge type to first order along $\mathcal{D}_{n,n}$ and has a nonzero Weil component. Then $\sigma_E$ is not injective on the image of $\mathrm{ev}_E$: $E$ is neither semiregular nor semiregular in the weakened sense of* [Mar25b, Question 11.4]*, and it is obstructed to first order along some direction of the Weil family. In particular every object of Setup 18.9 satisfying (45), for every $n \geq 2$, every field $K$, every rank and every set of norms, fails both criteria.*

*Proof.* Write $\lambda_i$ for the twisted first Chern classes of the summands. If every $\lambda_i$ were a real multiple of $\eta$, it would be a rational multiple, both classes being rational, and the twisted Chern character $\sum_i m_i e^{\lambda_i}$ would be a polynomial in $\eta$; its component in degree $2n$ would then lie in $\mathbb{Q}\eta^n$, which meets the Weil line in zero by Proposition 5.1, against the hypothesis. So some $\lambda_i$ is not a real multiple of $\eta$, and Theorem 9.20 applies to the summand $L_i[p_i]$. For an object of Setup 18.9 the classes are $\lambda_i = \theta(u_i) \in R_0$, and $R_0 \cap \mathbb{Q}\eta = 0$; no twist is

needed because $c_1(E) = \theta(T_{1,0}) = 0$ by (45), as $(1,0)$ is not an excluded pair for $n \geq 2$; and (45) gives a flat Chern character with a nonzero Weil component by Proposition 18.10. ■

*Remark* 18.22 (What the corollary asks of an object). By Corollary 18.21 there is no sum of line bundles, at any member of any Weil family in any dimension, that could run the semiregularity argument in either form. The reason is Proposition 9.15(i): the only divisor class that stays of type $(1,1)$ across the whole family is the polarisation, so every line bundle summand off that line is obstructed in some direction of the family, and the semiregularity map, whose composite with the obstruction is contraction with the flat class $\mathrm{ch}(E)$, cannot see it. The object the argument needs is one with no rank one summand off the polarisation line, of positive rank, indecomposable enough that its Atiyah class contracted with every direction of $T$ vanishes. Markman's reflexive sheaves at $n = 3$ are of that kind, and nothing here says that their analogues at $n \geq 4$ do not exist.

18.3. **Convolutions of line bundles.** After Corollary 18.21 the next objects built from line bundles are their convolutions, the iterated cones of shifted line bundles. Their Chern character is the same signed sum of exponentials as for a direct sum, but their Ext groups are cut down by the maps between the pieces, so the argument of Theorem 9.20 no longer applies as it stands. This subsection shows that fewer than $2n$ pieces with nondegenerate differences never serve (Theorem 18.26); writes down, at a CM member for $K = \mathbb{Q}(i)$, a signed sum of $2 \cdot 4^{n-1}$ line bundles that is exactly a real Weil class (Proposition 18.27); and determines, for the convolutions of those pieces with three multiples of the polarisation at $n = 3$, which products in the twisted differential can act on the 525 diagonal classes of degree two, of which at least 468 must disappear before the polarised criterion can hold (Theorem 18.31). Nothing enters the diagonal, cup products remove at most 276 of these classes, and everything else would have to be removed by fourfold products along a path through the three multiples and one of the Weil pieces: one family of them for some placements of the multiples, and two families that exclude each other for the others (Proposition 18.32); one order of the pieces gives a convolution that solves the Maurer–Cartan equation (Proposition 18.33). Off the diagonal, the classes between two pieces that differ in every coordinate are reached by no product (Lemma 18.34), and no convolution of these pieces meets the polarised criterion (Theorem 18.35). At $n = 4$ the same classes, now between pieces that differ in three coordinates, are reached by no product in any convolution, and they exclude every convolution whose 128 pieces $L_\zeta$ occupy two shifts (Theorem 18.37) or three consecutive ones (Theorem 18.38); with the groups between pieces whose shifts differ by two or three (Lemma 18.39) they exclude every convolution whose pieces have their shifts within five consecutive values (Corollary 18.44); the diagonal classes at a lonely shift (Lemma 18.45) and the interleaved shifts (Proposition 18.47) extend this to six (Corollary 18.48).

**Setup 18.23** (Convolutions)**.** Let $A$ be an abelian variety of dimension $g$ with a flat Kähler metric, and let $L_1, \dots, L_m$ be line bundles on $A$ with hermitian metrics of constant curvature, hermitian forms $H_1, \dots, H_m$ and pairwise distinct first Chern classes. For integers $d_1, \dots, d_m$ put $P_a = L_a[-d_a]$. A *convolution* of $P_1, \dots, P_m$ is an object $E$ given by a one-sided twisted complex[17]

$$\Big( \bigoplus_a \mathcal{A}^{0,*}(L_a)[-d_a],\ \bar\partial + \delta \Big), \qquad \delta = (\delta_{ab})_{a<b}, \quad \delta_{ab} \in \mathcal{A}^{0,1+d_a-d_b}\big(L_a^{-1} \otimes L_b\big),$$

[17]Every object obtained from $P_1, \dots, P_m$ by iterated cones in this order is of this form, because the Dolbeault complexes form a differential graded enhancement of the full subcategory of line bundles. All the objects of this subsection are given in this way.

with $\bar\partial\delta+\delta\circ\delta=0$, so that $\mathrm{ch}(E)=\sum_a(-1)^{d_a}e^{c_1(L_a)}$. Write $\mathcal{A}(a,b)=\mathcal{A}^{0,*}(L_a^{-1}\otimes L_b)$ and $\mathcal{H}(a,b)$ for its harmonic forms, so that $\mathcal{H}^q(a,b)\cong H^q(A,L_a^{-1}\otimes L_b)$, and write $\pi$, $G$ and $h=\bar\partial^*G$ for the harmonic projection, the Green operator and the homotopy, with $1-\pi=\bar\partial h+h\bar\partial$ and $\pi h=h\pi=h^2=0$. A form of degree $q$ in $\mathcal{A}(a,b)$ is a morphism $P_a\to P_b$ of degree $w=q-d_a+d_b$; we call $q$ its Ext degree and, when $w=1$, call $q-1$ its *excess*. When $H_b-H_a$ is nondegenerate, $\mathcal{H}(a,b)$ is concentrated in the single degree $n(H_b-H_a)$, the number of negative eigenvalues [BL04, Chapter 3].

**Lemma 18.24** (The harmonic model)**.** *In Setup 18.23:*

(i) *$E$ is isomorphic to a convolution of the same pieces with $h\delta=0$, and then $\delta=\delta^H-h(\delta\circ\delta)$ with $\delta^H=\pi\delta$ harmonic. Substituting repeatedly, every $\delta_{ab}$ is a finite sum of planar binary trees whose leaves are components $\delta^H_{cd}$ with $c<d$, whose internal vertices compose, and whose internal edges carry $-h$.*

(ii) *$\mathrm{Ext}^*(E,E)$ is the cohomology of $\mathcal{H}_E=\bigoplus_{a,b}\mathcal{H}(a,b)$, graded by $w$, with the differential $d_E=\sum_{k\ge0}\pi\,Q(-hQ)^k$, where $Q=[\delta,-]$.*

(iii) *For $x\in\mathcal{H}(a,b)$, the component of $d_E(x)$ in $\mathcal{H}(c,e)$ is a sum of terms $\pm\pi\,T$, one for each planar binary tree $T$ with $k\ge2$ leaves, internal vertices composing, $-h$ on the internal edges other than the root, whose leaves read in order are the components along a path $c=a_0\to a_1\to\cdots\to a_k=e$ in which one step is $x\colon a\to b$ and the others are components $\delta^H_{a_ja_{j+1}}$ with $a_j<a_{j+1}$. The* Ext *degree of such a term is the sum of the* Ext *degrees of its leaves minus $k-2$.*

The terms with $k$ leaves in (iii) are the Massey products of length $k$ of the pieces;[18] (iii) is all that is used of them below.

*Proof.* (i) Suppose $h\delta_{cd}=0$ for $d-c<r$. Conjugating by $1+g$, with $g$ of degree 0 supported at distance $r$, replaces $\delta_{ab}$ for $b-a=r$ by $X_{ab}+\bar\partial g_{ab}$, where $X_{ab}$ depends only on $\delta$, and leaves the components at smaller distance unchanged. With $g_{ab}=-hX_{ab}$ the new component is $X_{ab}-\bar\partial hX_{ab}$, and $h\bar\partial h=h(1-\pi-h\bar\partial)=h$ gives $h(X_{ab}-\bar\partial hX_{ab})=0$. Induction on $r$ gives $h\delta=0$, and then $\delta=\pi\delta+\bar\partial h\delta+h\bar\partial\delta=\delta^H-h(\delta\circ\delta)$ by the Maurer–Cartan equation. The component $(\delta\circ\delta)_{ab}$ involves only components at distance less than $b-a$, so the substitution ends. (ii) The endomorphism complex of $E$ is $\bigoplus_{a,b}\mathcal{A}(a,b)$ with the differential $\bar\partial+Q$. Blockwise, the inclusion of harmonic forms, $\pi$ and $-h$ form a strong deformation retraction of $(\bigoplus\mathcal{A}(a,b),\bar\partial)$ onto $(\mathcal{H}_E,0)$, and $Q$ raises $b-a$, so $hQ$ is nilpotent. The basic perturbation lemma gives a homotopy equivalence of $(\bigoplus\mathcal{A}(a,b),\bar\partial+Q)$ with $(\mathcal{H}_E,d_E)$, with the chain maps $\iota_E=\sum_k(-hQ)^k$ on $\mathcal{H}_E$ and $\pi_E=\pi\sum_k(-Qh)^k$. (iii) Each $Q$ in $d_E$ composes with one component of $\delta$ on the left or on the right; writing that component as in (i) gives the trees, the edges from (i) and those from $d_E$ being exactly the internal edges other than the root. Composition adds degrees, $h$ lowers them by one, and a binary tree with $k$ leaves has $k-2$ internal edges. ∎

**Lemma 18.25** (The index of a sum)**.** *For hermitian forms $X,Y$ on $\mathbb{C}^g$, $n(X+Y)\le n(X)+n(Y)$. If $H_1+\cdots+H_k=0$, then $\sum_j n(H_j)\ge g-\dim\ker H_i$ for every $i$; in particular $\sum_j n(H_j)\ge g$ when one $H_i$ is nondegenerate.*

*Proof.* A subspace on which $X+Y$ is negative definite meets the subspaces on which $X\ge0$ and $Y\ge0$, of codimensions $n(X)$ and $n(Y)$, only in zero, so its dimension is at most

[18] In the language of $A_\infty$-categories they are the higher products $m_k$ of the minimal model of the line bundles, evaluated on the Maurer–Cartan element $\delta^H$ of the twisted complex; for $k=2$ they are cup products.

$n(X)+n(Y)$. For the second statement, $-H_i=\sum_{j\neq i}H_j$ gives $g-n(H_i)-\dim\ker H_i=n(-H_i)\leq\sum_{j\neq i}n(H_j)$. ∎

**Theorem 18.26** (Fewer than $2n$ pieces)**.** *Let $A$ be of $(K,-1,n)$-Weil type with $n\geq 2$, and let $E$ be a convolution of $m<2n$ shifted line bundles whose differences $H_b-H_a$ are all nondegenerate. Suppose that $\operatorname{ch}(E)$ remains of Hodge type to first order along $\mathcal{D}_{n,n}$ and has a nonzero Weil component, as every polarised character $N\omega_1+p(\theta)$ with $N\neq 0$ does. Then there is $v\in T_{[A]}\mathcal{D}_{n,n}$ with*

$$\operatorname{ev}_E(v)\neq 0 \qquad \text{and} \qquad \sigma_E(\operatorname{ev}_E(v))=0,$$

*so $\sigma_E$ is not injective, and for a polarised character $\dim\operatorname{Ext}^2(E,E)>r(\operatorname{ch}E)$ by Theorem 17.9(iv).*

*Proof.* First, no element of degree one of $\mathcal{H}_E$ has a differential with a nonzero component in the diagonal classes $\mathcal{H}^2(c,c)=H^2(A,\mathcal{O}_A)$. Such a component is a sum of terms of Lemma 18.24(iii) whose $k$ leaves have degree one and run around a closed path $c\to\cdots\to c$; the pieces before the step $x$ increase from $c$ and those after it increase to $c$, so the path passes through $k$ distinct pieces and $k\leq m$. Its leaves have Ext degrees $q_j=n(H_{a_{j+1}}-H_{a_j})$, and the output has Ext degree $\sum_j q_j-(k-2)=2$, so $\sum_j q_j=k<2n$. The $k$ differences are nondegenerate and add up to zero, so Lemma 18.25 gives $\sum_j q_j\geq g=2n$, a contradiction. Hence taking diagonal components gives a well defined map $\pi_D\colon\operatorname{Ext}^2(E,E)\to\bigoplus_a H^2(A,\mathcal{O}_A)$.

Next, $\operatorname{ev}_E(v)=v\lrcorner\operatorname{At}(E)$, as in the proof of Theorem 9.20. In the twisted complex the Atiyah class is represented by the commutator of the Chern $(1,0)$-connection $\bigoplus_a\nabla_a$ with $\bar\partial+\delta$; its diagonal entries are the curvatures $F_a$, constant forms representing $c_1(L_a)$, and its other entries $\nabla\delta_{ab}$ lie above the diagonal. So $\operatorname{ev}_E(v)$ is represented by an upper triangular $z$ with $z_{aa}=v\lrcorner F_a$, and in $\mathcal{H}_E$ by $\pi_E(z)$. The operators $Q$ and $h$ preserve upper triangular forms and $Q$ makes them strictly upper triangular, so the diagonal of $\pi_E(z)$ is that of $\pi(z)$, and $\pi_D(\operatorname{ev}_E(v))=(v\lrcorner c_1(L_a))_a$.

If every $c_1(L_a)$ were a real multiple of the polarisation, it would be a rational one and $\operatorname{ch}(E)$ a polynomial in it, whose component in degree $2n$ meets the Weil line in zero by Proposition 5.1. So some $c_1(L_a)$ is not, and by Proposition 9.15(i) some $v\in T_{[A]}\mathcal{D}_{n,n}$ has $v\lrcorner c_1(L_a)\neq 0$; then $\pi_D(\operatorname{ev}_E(v))\neq 0$, while $\sigma_E(\operatorname{ev}_E(v))=v\lrcorner\operatorname{ch}(E)=0$. ∎

Theorem 9.20 is the case $\delta=0$, with any number of pieces. A convolution reaches its diagonal classes once it has $2n$ pieces, when a cycle through all of them can have total index $2n$, the equality case of Lemma 18.25. A convolution meeting the polarised criterion must therefore have at least $2n$ pieces, or pieces with degenerate differences. The next construction has both.

**Proposition 18.27** (A pure Weil class from line bundles)**.** *Let $E_0=\mathbb{C}/\mathbb{Z}[i]$, $A=E_0^{2n}$ with coordinates $z_1,\dots,z_{2n}$ and $n\geq 2$, let $K=\mathbb{Q}(i)$ act by $i$ on $z_1,\dots,z_n$ and by $-i$ on $z_{n+1},\dots,z_{2n}$, and let $\theta$ be the product principal polarisation; then $A$ is of $(\mathbb{Q}(i),-1,n)$-Weil type, a CM member as in Remark 19.6. Write $B_j$ for the surface with coordinates $(z_j,z_{n+j})$, so that $A=B_1\times\cdots\times B_n$. For $p\in\mathbb{Z}$ and $\zeta\in\mu_4^n$ let $L_\zeta$ be a line bundle whose hermitian form is $\left(\begin{smallmatrix}p & \zeta_j\\ \bar\zeta_j & p\end{smallmatrix}\right)$ on $B_j$ and zero between different surfaces, and put $\alpha=\prod_j\frac{i}{2}\,dz_j\wedge d\bar z_{n+j}$.*

(i) *$\alpha$ and $\overline{\alpha}$ span the Weil classes of $A$, and*

$$\sum_{\prod\zeta=1} e^{c_1(L_\zeta)}-\sum_{\prod\zeta=-1} e^{c_1(L_\zeta)}=2\cdot 4^{n-1}\,(\alpha+\overline{\alpha}).$$

(ii) *For distinct integers $t_1, t_2, t_3$ and signs $\sigma_i$, the character $\gamma = 2\cdot 4^{n-1}(\alpha+\overline{\alpha})+\sum_i \sigma_i e^{t_i\theta}$ has Hankel rank 3, so $r(\gamma) = 7n^2 - 2n$. A convolution with this character of the $2\cdot 4^{n-1}$ bundles $L_\zeta$, with $(-1)^d = \prod \zeta$, and of three line bundles $M_i$ with $c_1(M_i) = t_i\theta$ has $2\cdot 4^{n-1}+3$ pieces and $(2\cdot 4^{n-1}+3)\,n(2n-1)$ diagonal classes of degree two: 525 against $r = 57$ at $n = 3$, and 3668 against 104 at $n = 4$.*[19]

*Proof.* The hermitian forms have imaginary parts integral on $\mathbb{Z}[i]^{2n}$, so the line bundles exist. On $B_j$ write $c_1$ of the restriction as $p(a_j + b_j) + \zeta_j c_j + \overline{\zeta}_j c'_j$, where $a_j = \frac{i}{2} dz_j \wedge d\bar{z}_j$, $b_j = \frac{i}{2} dz_{n+j} \wedge d\bar{z}_{n+j}$, $c_j = \frac{i}{2} dz_j \wedge d\bar{z}_{n+j}$ and $c'_j = \frac{i}{2} dz_{n+j} \wedge d\bar{z}_j$. Since $c_j^2 = c_j'^2 = 0$,

$$e^{c_1(L_\zeta)} = \prod_j f_j(\zeta_j), \qquad f_j(z) = e^{p(a_j+b_j)}(1 + zc_j + \overline{z}c'_j + c_j c'_j).$$

For $u \in \mu_4$, $\frac{1}{2}(u + \overline{u})$ is $u$ when $u = \pm 1$ and 0 when $u = \pm i$, so the left side of (i) is $\frac{1}{2}(\prod_j \sum_z z f_j(z) + \prod_j \sum_z \overline{z} f_j(z))$. Now $\sum_{z\in\mu_4} z = \sum z^2 = 0$ and $\sum |z|^2 = 4$, so $\sum_z z f_j(z) = 4e^{p(a_j+b_j)} c'_j = 4c'_j$, because $a_j c'_j$ and $b_j c'_j$ repeat a differential, and likewise $\sum_z \overline{z} f_j(z) = 4c_j$. This gives $2\cdot 4^{n-1}(\prod c'_j + \prod c_j) = 2\cdot 4^{n-1}(\overline{\alpha} + \alpha)$. The differentials $dz_j$, $j \le n$, and $d\bar{z}_{n+j}$ span the $i$-eigenspace of $K$ on $H^1(A, \mathbb{C})$, and $\alpha$ spans its top exterior power, which is a Weil class, and $\overline{\alpha}$ the conjugate one. (ii) The moments $\mu_m = \sum_i \sigma_i t_i^m$ make the Hankel matrix of Setup 17.8 a product of Vandermonde matrices of rank 3, and Theorem 17.9(ii) gives $r$. The counts are immediate. ∎

The pieces $L_\zeta$ and $L_{\zeta'}$ have a degenerate difference as soon as $\zeta$ and $\zeta'$ share a coordinate, so Theorem 18.26 does not apply to them, and there are more than $2n$ of them. The rest of this subsection computes what the Massey products can do to the diagonal classes of such a convolution. The pieces are external products over the surfaces $B_j$, and the first tool is a bound on how far a product can lower the degree on each surface.

**Lemma 18.28** (Degree drop on a product)**.** *Let $A = B_1 \times \cdots \times B_n$ with a product metric, and let every $L_a$ be an external product with a product metric. Decompose the leaves of a term of Lemma 18.24(iii) with $k$ leaves into tensor products of harmonic forms on the factors. Then the output has degree $s_j - d_j$ on $B_j$, where $s_j$ is the sum of the degrees of the leaves on $B_j$ and $d_j \ge 0$, $\sum_j d_j = k - 2$, and*

$$d_j \le \max(0,\, u_j - 2),$$

*where $u_j$ is the number of leaves whose component on $B_j$ is not a multiple of the unit section of $\mathcal{O}_{B_j}$ (such a unit occurs only where source and target agree on $B_j$).*

*Proof.* The Laplacian is $\Delta = \sum_j \Delta_j$, so $h = \sum_j h_j$ with $h_j = \bar{\partial}_j^* G = G\bar{\partial}_j^*$, which lowers the degree on $B_j$ by one and keeps the others; expand every internal edge accordingly, and let $d_j$ be the number of edges carrying $h_j$. Let $V_j$, $U_j$ and $W_j$ be the forms whose component on $B_j$ is $\Delta_j$-harmonic, a multiple of the unit, or in the image of $\bar{\partial}_j^*$. The operators $G$ and $h_i$, $i \ne j$, preserve all three, $h_j$ kills $V_j$ and $W_j$, $\pi$ kills $W_j$, and multiplication by an element of $U_j$ preserves $V_j$, $U_j$ and $W_j$. For an edge $e$ carrying $h_j$ let $N(e)$ be the set of leaves below $e$ whose component on $B_j$ is not a unit. If $|N(e)| \le 1$ the value below $e$ lies in $V_j$ and the term vanishes. If $e$ lies below the next such edge $e'$ with $N(e) = N(e')$, everything composed in between lies in $U_j$, the value at $e'$ lies in $W_j$, and the term vanishes; if $N(e)$ is the set of all $u_j$ such leaves for the highest such edge, the value at the root lies in $W_j$ and $\pi$ kills it. So in a nonzero term the sets $N(e)$ are distinct, of size at least two,

[19] At $n = 2$ the same construction has $8 + 3 = 11$ pieces and 66 diagonal classes against $r = 24$.

different from the whole set, and pairwise nested or disjoint. Together with the whole set they form a family of that kind, which has at most $u_j - 1$ members, since its members are the internal vertices of a forest with $u_j$ leaves in which every internal vertex has at least two children. Hence $d_j \le u_j - 2$. ∎

**Proposition 18.29** (Nothing enters the diagonal)**.** *Keep Proposition 18.27 with $n \ge 3$, and let $t_1, t_2, t_3$ be distinct with $|t_i - p| \ge 2$. For every convolution $E$ of the pieces $L_\zeta$ and $M_i$, in any order, with any shifts of the given parities and with the pieces twisted by any line bundles of degree zero, no element of degree one of $\mathcal{H}_E$ has a differential with a nonzero diagonal component. Consequently*

$$\dim \operatorname{Ext}^2(E,E) \;\ge\; \dim \ker\big(d_E|_{D^2}\big), \qquad D^2 = \bigoplus_a H^2(A, \mathcal{O}_A).$$

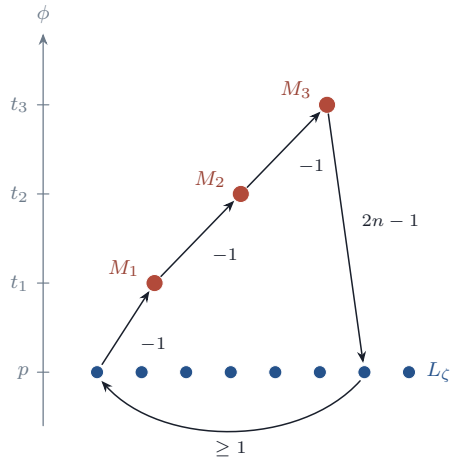
*Proof.* As in the proof of Theorem 18.26, a term that reaches the diagonal runs around a closed path of distinct pieces whose $k$ steps have degree one and total excess 0. The step from $L_\zeta$ to $L_{\zeta'}$ has Ext degree 1 on each surface where $\zeta_j \ne \zeta'_j$, the difference there having eigenvalues $\pm|\zeta'_j - \zeta_j|$, and at least 0 elsewhere. If $\prod \zeta = \prod \zeta'$ the two differ on at least two surfaces; otherwise the step has even Ext degree, the shifts having opposite parities. Either way its excess is at least 1. The difference between $L_\zeta$ and $M_i$ has eigenvalues $t_i - p \pm 1$ on every surface, so it is definite: the step has Ext degree 0 and excess $-1$ if it goes up the value $\phi$, where $\phi(L_\zeta) = p$ and $\phi(M_i) = t_i$, and Ext degree $2n$ and excess $2n - 1$ if it goes down; the same holds between the $M_i$. A maximal run of the cycle through the $M_i$ starts and ends at the value $p$, or is the whole cycle, so it has at least one step down and at most three steps up; its excess is at least $2n - 1 - 3 \ge 2$. So every cycle has excess at least 1. Twisting by line bundles of degree zero leaves the forms unchanged and can only make some $\mathcal{H}(a,b)$ vanish. For the inequality, the diagonal components of cocycles of degree two vanish on coboundaries, so $\dim \operatorname{Ext}^2(E,E)$ is at least the dimension of their image, which contains the kernel of $d_E$ on $D^2$. ∎

At $n = 2$ the bound on a run is 0 and a cycle through the three multiples can reach the diagonal; this is why the analysis below starts at $n = 3$.

**Proposition 18.30** (Cup products on the diagonal)**.** *Keep Proposition 18.29 with $n = 3$, so that $D^2$ has dimension $35 \cdot 15 = 525$. The terms of $d_E$ on $D^2$ with two leaves, the cup products, vanish on the 45 diagonal classes of the $M_i$, and on the 480 diagonal classes of the $L_\zeta$ they have a kernel of dimension at least 204. So their kernel on $D^2$ has dimension at least 249.*

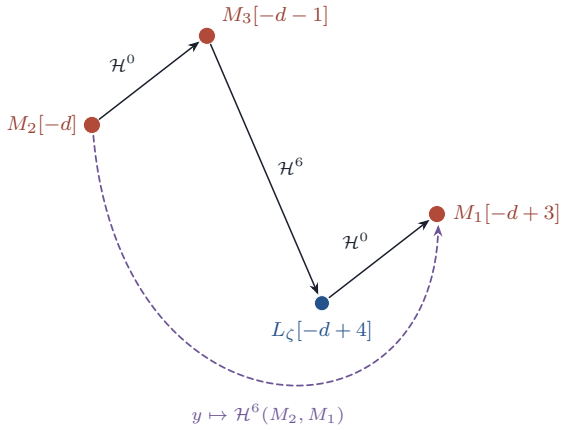
*Proof.* For $y = (y_a) \in D^2$ the cup products in $\mathcal{H}(a,b)$ are $(y_a - y_b) \wedge \delta^H_{ab}$, since $y$ has even degree. On a surface where the pieces differ, $\delta^H_{ab}$ has degree 1, the only degree of the cohomology there, and the product with a component of $y$ of positive degree lands in a degree where the cohomology vanishes. The pieces $M_i$ differ from every other piece on every surface, which gives the first statement. Write $y_a$ by its type $\tau \in \{0,1,2\}^3$, $|\tau| = 2$, the degrees on the three surfaces: three types $(2,0,0)$ of dimension one and three types $(1,1,0)$ of dimension four. The degree count on each surface and the parity of Ext degrees show that $L_\zeta$ and $L_{\zeta'}$ can carry a nonzero cup product on the type $\tau$ only if either $\zeta'$ differs from $\zeta$ by a sign in one coordinate outside the support of $\tau$, or $\zeta'$ is $\zeta$ multiplied by $(i,i)$ or $(-i,-i)$ in two coordinates and $\tau$ is 2 on the third. Let $\Gamma_\tau$ be the graph on the 32 pieces with these edges. If each component $y^\tau$ is constant on the connected components of $\Gamma_\tau$, every cup product vanishes. For $\tau = (2,0,0)$ the coordinate $\zeta_1$ is constant along the edges, and the eight pieces with a given $\zeta_1$ are connected, so $\Gamma_\tau$ has 4 components; for

(a) the excess of a cycle, $n \geq 3$




excess of a step of degree one: $-1$ up ($\mathrm{Ext}^0$), $2n-1$ down ($\mathrm{Ext}^{2n}$), at least 1 from $L_\zeta$ to $L_{\zeta'}$ this cycle: $3\cdot(-1)+(2n-1)+1 = 2n-3 > 0$; reaching $H^2(\mathcal{O}_A)$ needs a cycle of excess 0

(b) the one path that acts, $n = 3$




Ext degrees $0, 6, 0$, excesses $-1, 5, -1$: the product with $y \in H^2(\mathcal{O}_A)$ lands in degree 6, in a space of dimension $(t_2 - t_1)^6$; the shifts force $\sigma_1 = \sigma_3 = -\sigma_2$

FIGURE 26. The pieces of Proposition 18.27 at their values $\phi$, with $\phi(L_\zeta) = p$ and $\phi(M_i) = t_i$. (a) A step of degree one between pieces with a definite difference has Ext degree 0 going up and $2n$ going down, so excess $-1$ or $2n-1$; a step between two $L_\zeta$ has excess at least 1. The cycle drawn runs up through all three $M_i$ and down once, the worst case, and still has excess $2n-3 > 0$, so for $n \geq 3$ no term reaches the diagonal (Proposition 18.29). (b) The one path whose products act on the diagonal at $n = 3$ when the $t_i$ lie above $p$ (Theorem 18.31(i)), drawn at the same heights and in the order of the convolution, with the Ext degrees of its steps and the shifts they force; the dashed arrow is the value of the product with a diagonal class $y$.

$\tau = (1,1,0)$ the edges change the sign of $\zeta_3$ only, and $\Gamma_\tau$ has 16 components. The kernel therefore contains a space of dimension $3\cdot 4\cdot 1 + 3\cdot 16\cdot 4 = 12 + 192 = 204$. ■

**Theorem 18.31** (The diagonal on $E_0^6$)**.** *Let $n = 3$ and let $E$ be a convolution of the 32 bundles $L_\zeta$ and of $M_1, M_2, M_3$ as in Proposition 18.27, with $p+2 \leq t_1 < t_2 < t_3$ and signs $\sigma_i$, so that* $\mathrm{ch}(E) = 32(\alpha + \overline{\alpha}) + \sum_i \sigma_i e^{t_i\theta}$ *and the polarised criterion asks for* $\dim \mathrm{Ext}^2(E,E) = 57$.

(i) *The Massey products of length at least three vanish on $D^2$, except those of length four along a path $M_2 \to M_3 \to L_\zeta \to M_1$ taken in this order in the convolution, which occur only if $\sigma_1 = \sigma_3 = -\sigma_2$ and take values in $\mathcal{H}^6(M_2, M_1) \cong H^6(A, \mathcal{O}_A(-(t_2 - t_1)\theta))$, of dimension $(t_2 - t_1)^6$.*

(ii) $\dim \mathrm{Ext}^2(E,E) \geq 249 - \min(249, (t_2-t_1)^6)$, *and* $\dim \mathrm{Ext}^2(E,E) \geq 249$ *unless the order and the signs are as in (i). In particular $E$ does not meet the polarised criterion when $t_2 - t_1 \leq 2$, since $249 - 64 = 185 > 57$, nor when the signs or the order differ from (i).*

(iii) *If $E$ meets the polarised criterion, then the length four products of (i) have rank at least $249 - 57 = 192$ on the kernel of the cup products, and $t_2 - t_1 \geq 3$.*

*The same holds for $t_1 < t_2 < t_3 \leq p - 2$ with $E$ replaced by its dual.*

*Proof.* (i) By Lemma 18.24(iii) a term with $k \geq 3$ leaves acting on $y \in H^2(\mathcal{O}_A)$ at a piece $c$ runs along a chain $s = a_0 \to \cdots \to a_{k-1} = e$ of distinct pieces, increasing in the order of the convolution, whose $k - 1$ steps are components of degree one, with $y$ inserted at $c$. Its value lies in $\mathcal{H}(s, e)$, in Ext degree $2 + \sum_j q_j - (k - 2) = 3 + \varepsilon$, where $\varepsilon$ is the sum of the excesses of the steps; so $-3 \leq \varepsilon \leq 3$, and $\mathcal{H}(s, e)$ must be nonzero in degree $3 + \varepsilon$. By the proof of Proposition 18.29 a step between two $L_\zeta$ has excess at least 1, with equality only between pieces of opposite parities, and a step between pieces at different values of $\phi$ has excess $-1$ and Ext degree 0 going up, excess 5 and Ext degree 6 going down. Let $U$ and $D$ be the numbers of steps up and down and $\lambda$ the total excess of the steps between two $L_\zeta$, so that $\varepsilon = 5D - U + \lambda$. For two distinct $L_\zeta$, $L_{\zeta'}$ differing in $m \geq 1$ coordinates, $\mathcal{H}(L_\zeta, L_{\zeta'})$ vanishes above degree $6 - m$; for pieces at different values it sits in degree 0 if the target is higher and in degree 6 if it is lower.

Two observations remove chains. First, a chain of total excess $-2$ has no step down, and so consists of steps between $L_\zeta$ followed by at most three steps up. Second, let $a \to b \to c$ be two consecutive steps of Ext degree 0, sections $x_{ab}$ and $x_{bc}$ of line bundles. The Maurer–Cartan equation $\pi(\delta \circ \delta) = 0$ of the proof of Proposition 18.33(i) has, in $\mathcal{H}^0(a, c)$, one term for each chain from $a$ to $c$ of total excess $-2$. If $a \to b \to c$ is the only such chain, the equation reads $x_{ab}x_{bc} = 0$, a product of holomorphic sections being holomorphic and so harmonic; since $A$ is connected, $x_{ab}$ or $x_{bc}$ vanishes, and every term along a chain through both steps is zero.

Here every step up ends at one of $M_1, M_2, M_3$, so $U \leq 3$, and $D \leq 1$. If $D = 0$, the chain is $r \geq 0$ pieces $L_\zeta$ followed by multiples in increasing order. When it ends at an $L_\zeta$, $k \geq 3$ gives $\lambda \geq 2$, so $s$ and $e$ differ in one coordinate and $\lambda = 2$; then the two steps change the parity twice, whereas $s$ and $e$ have opposite parities. When it ends at a multiple, $e$ lies above $s$, so $\varepsilon = -3$, which forces $U = 3$, $\lambda = 0$ and the chain $L_\zeta \to M_1 \to M_2 \to M_3$; by the first observation $L_\zeta \to M_1 \to M_2$ is the only chain of excess $-2$ from $L_\zeta$ to $M_2$, and the second removes it. If $D = 1$, then $U \geq 2 + \lambda$, so $(U, \lambda)$ is $(2, 0)$, $(3, 0)$ or $(3, 1)$ and $\varepsilon$ is 3, 2 or 3. For $\varepsilon = 3$ the piece $s$ is a multiple and $e$ lies below it. With $(3, 1)$ the three steps up would end at three multiples other than $s$, which do not exist. With $(2, 0)$ the chain has three steps, and a step down lands at an $L_\zeta$, since a fourth multiple does not exist. It is $M_1 \to M_2 \to M_3 \to L_\zeta$, removed by the second observation at $M_1 \to M_2 \to M_3$; or $M_a \to M_b \to L_\zeta \to M_c$ with $t_a < t_b$ and $t_c < t_a$, that is $M_2 \to M_3 \to L_\zeta \to M_1$; or $M_a \to L_\zeta \to M_b \to M_c$ with $t_b < t_c < t_a$, that is $M_3 \to L_\zeta \to M_1 \to M_2$, removed at $L_\zeta \to M_1 \to M_2$. For $\varepsilon = 2$, with $(U, \lambda) = (3, 0)$, the pieces $s$ and $e$ are $L_\zeta$ differing in one coordinate; the last step is the step down, preceded by three steps up from $s$, so the chain is $L_\zeta \to M_1 \to M_2 \to M_3 \to L_{\zeta'}$, removed at $L_\zeta \to M_1 \to M_2$.

Only $M_2 \to M_3 \to L_\zeta \to M_1$ remains. Its steps have Ext degrees 0, 6 and 0, so degree one forces $d_{M_3} = d_{M_2} + 1$, $d_L = d_{M_3} - 5$ and $d_{M_1} = d_L + 1$; this gives the signs, and the value lies in $\mathcal{H}^{3+d_{M_2}-d_{M_1}}(M_2, M_1) = \mathcal{H}^6(M_2, M_1)$, whose dimension is $|\chi(\mathcal{O}_A((t_1 - t_2)\theta))| = (t_2 - t_1)^6$. (ii) By Proposition 18.29 it suffices to bound the kernel of $d_E$ on $D^2$. By Proposition 18.30 the cup products have a kernel $Y$ of dimension at least 249, and on $Y$ the differential is the sum of the products of (i), of rank at most $(t_2 - t_1)^6$. (iii) follows from (ii). For the dual, $E \mapsto E^\vee$ preserves $\mathrm{Ext}^2$ and sends the configuration with $p$, $t_i$ to the one with $-p$, $-t_i$, the set of the $L_\zeta$ being stable under $\zeta \mapsto -\zeta$. ∎

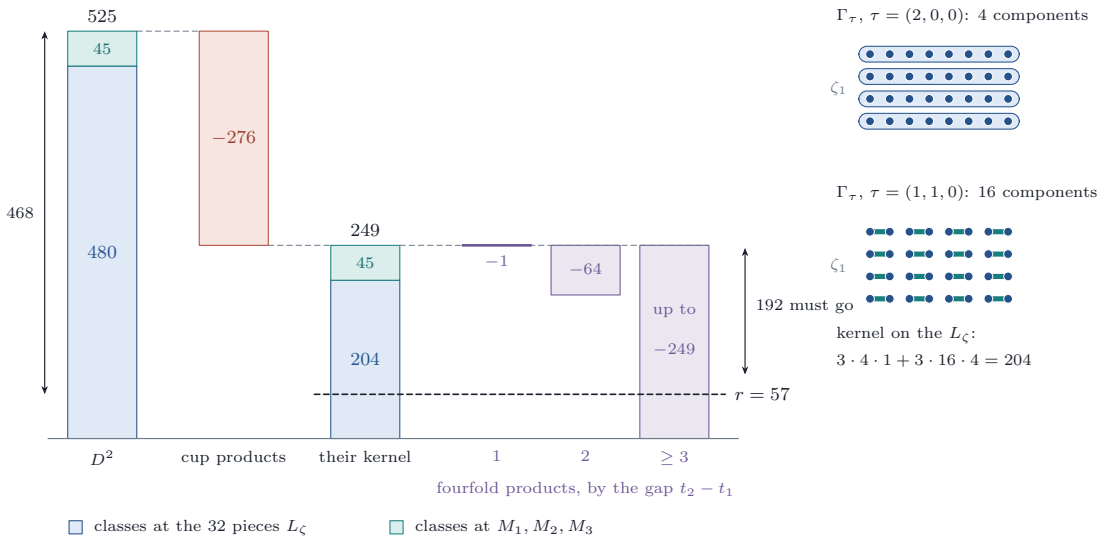



FIGURE 27. The 525 diagonal classes of degree two at $n = 3$ (Proposition 18.30 and Theorem 18.31). The cup products remove 276 of them for general components, and at most that many in any case, leaving 204 at the pieces $L_\zeta$ and the 45 at the $M_i$; when the $t_i$ lie on one side of $p$, the fourfold products of the surviving path remove at most $(t_2 - t_1)^6$ more, which is 1 or 64 for the gaps 1 and 2 and no constraint from the third gap on. The criterion asks for 57, so 468 classes must disappear in all and 192 after the cup products. On the right, the graphs $\Gamma_\tau$ on the 32 pieces, one row for each value of $\zeta_1$: for $\tau = (2, 0, 0)$ each row is a component, for $\tau = (1, 1, 0)$ the components are the pairs differing in the sign of $\zeta_3$; a class constant on the components of every $\Gamma_\tau$ has vanishing cup products.

The placements of the $t_i$ on both sides of $p$ are not covered by Theorem 18.31. There two families of fourfold products occur instead of one, and no convolution carries both.

**Proposition 18.32** (Placements on both sides of $p$)**.** *Keep Theorem 18.31, but let $t_1 \le p-2$ and $p+2 \le t_2 < t_3$.*

(i) *The Massey products of length at least three vanish on $D^2$ unless $\sigma_1 = \sigma_2 = -\sigma_3$. In that case the only ones that act have length four and run along one of the paths*

$$\text{(a)}\quad M_2 \to M_3 \to M_1 \to L_\zeta, \qquad \text{(b)}\quad M_3 \to M_1 \to L_\zeta \to M_2,$$

*taken in this order in the convolution, with $\prod \zeta = \sigma_3$. The products along (a) take values in $\mathcal{H}^6(M_2, L_\zeta)$, of dimension $\big((t_2 - p)^2 - 1\big)^3$, and those along (b) in $\mathcal{H}^6(M_3, M_2)$, of dimension $(t_3 - t_2)^6$. A convolution carries at most one of the two kinds, since (a) puts $M_2$ before $M_3$ and (b) puts $M_3$ before $M_2$.*

(ii) *If $E$ meets the polarised criterion, then $\sigma_1 = \sigma_2 = -\sigma_3$, and the products along (a) or those along (b) have rank at least 192 on the kernel of the cup products. For (b) this needs $t_3 - t_2 \ge 3$. The sixteen targets of (a) have dimension $16\big((t_2 - p)^2 - 1\big)^3 \ge 432$, so for (a) the count gives no condition.*

*For $t_1 < t_2 \le p-2$ and $t_3 \ge p+2$ the same holds for $E^\vee[1]$, and the condition on the signs becomes $\sigma_2 = \sigma_3 = -\sigma_1$.*

*Proof.* (i) We use the notation and the two observations of the proof of Theorem 18.31(i). Now $\phi(M_1) = t_1 < p < t_2 < t_3$; a step up ends at $M_2$, at $M_3$, or at an $L_\zeta$ coming from $M_1$, so again $U \le 3$ and $D \le 1$. If $D = 0$, the values do not decrease along the chain. When $e$ lies above $s$, $\varepsilon = -3$ forces $U = 3$, $\lambda = 0$ and the chain $M_1 \to L_\zeta \to M_2 \to M_3$, removed at $L_\zeta \to M_2 \to M_3$, the only chain of excess $-2$ from $L_\zeta$ to $M_3$. When $e$ and $s$ have the same value, they are two $L_\zeta$ joined by steps between $L_\zeta$, which is excluded as before. If $D = 1$, $(U, \lambda)$ is $(2, 0)$, $(3, 0)$ or $(3, 1)$. For $\varepsilon = 3$, $e$ lies below $s$. With $(3, 1)$ the steps up end at $M_2$, $M_3$ and an $L_\zeta$ following $M_1$; then $s$ is an $L_\zeta$, so $e = M_1$, which is followed by a step: impossible. With $(2, 0)$ the chain has three steps and no two consecutive $L_\zeta$. As $M_1$ is the lowest piece, it is never the end of a step up, and a step up into an $L_\zeta$ comes from $M_1$; so two consecutive steps up end at $M_2$ or $M_3$. The chain cannot start at $M_1$. If it starts at an $L_\zeta$, it ends at $M_1$ with the step down, after two steps up: $L_\zeta \to M_2 \to M_3 \to M_1$, removed at $L_\zeta \to M_2 \to M_3$. If it starts at $M_3$, the first step goes down and the last two go up, ending at $M_2$; the middle piece is then an $L_\zeta$ entered from $M_1$, and the chain is $M_3 \to M_1 \to L_\zeta \to M_2$, path (b). If it starts at $M_2$, then $e \in \{L_\zeta, M_1\}$, so the last two steps are not both up; the first step goes up to $M_3$, the second down, and the third up into $e = L_\zeta$ from $M_1$: the chain is $M_2 \to M_3 \to M_1 \to L_\zeta$, path (a). For $\varepsilon = 2$, $(U, \lambda) = (3, 0)$ and $s$, $e$ are $L_\zeta$ differing in one coordinate. The last step enters $e$ from $M_1$, since a step down into $e$ would follow three steps up from $s$, one more than the values allow; $M_1$ is entered by the step down, after two steps up from $s$: $L_\zeta \to M_2 \to M_3 \to M_1 \to L_{\zeta'}$, removed at $L_\zeta \to M_2 \to M_3$. Along (a) the steps have Ext degrees 0, 6 and 0, so degree one forces $d_{M_3} = d_{M_2} + 1$, $d_{M_1} = d_{M_3} - 5$ and $d_L = d_{M_1} + 1$. Along (b) they have Ext degrees 6, 0 and 0, which forces $d_{M_1} = d_{M_3} - 5$, $d_L = d_{M_1} + 1$ and $d_{M_2} = d_L + 1$. This gives the signs and puts the values in $\mathcal{H}^{3+d_{M_2}-d_L}(M_2, L_\zeta) = \mathcal{H}^6(M_2, L_\zeta)$ and in $\mathcal{H}^{3+d_{M_3}-d_{M_2}}(M_3, M_2) = \mathcal{H}^6(M_3, M_2)$. On $B_j$ the line bundle $L_\zeta \otimes M_2^{-1}$ has the hermitian form $\left(\begin{smallmatrix} p-t_2 & \zeta_j \\ \bar\zeta_j & p-t_2 \end{smallmatrix}\right)$, with eigenvalues $p - t_2 \pm 1 < 0$ and determinant $(t_2 - p)^2 - 1$, which gives the first dimension. The second is that of Theorem 18.31(i) with $t_3$, $t_2$ in place of $t_2$, $t_1$. (ii) Propositions 18.29 and 18.30 do not depend on the placement of the $t_i$, so as in Theorem 18.31(ii) $\dim \mathrm{Ext}^2(E, E) \ge 249 - \rho$, where $\rho$ is the rank on the kernel of the cup products of the products of (i) that $E$ carries, and $\rho = 0$ unless $\sigma_1 = \sigma_2 = -\sigma_3$. The criterion asks for 57, so $\rho \ge 192$, and $(t_3 - t_2)^6 \ge 192$ exactly when $t_3 - t_2 \ge 3$. For the last statement, $E \mapsto E^\vee[1]$ preserves $\mathrm{Ext}^2$, sends $p$ and the $t_i$ to $-p$ and the $-t_i$, reverses the order of the pieces and changes every sign. ∎

The Maurer–Cartan equation can be solved for one order of the pieces.

**Proposition 18.33** (An explicit convolution)**.** *Keep Theorem 18.31 with $p = 0$ and $t = (2, 5, 6)$, take the pieces to be external products $L_\zeta = N_{1,\zeta_1} \boxtimes N_{2,\zeta_2} \boxtimes N_{3,\zeta_3}$ and $M_i = M_{i,1} \boxtimes M_{i,2} \boxtimes M_{i,3}$ of line bundles on the $B_j$, and order them as $M_2$, $M_3$, the sixteen $L_\zeta$ with $\prod \zeta = 1$, $M_1$, the sixteen $L_\zeta$ with $\prod \zeta = -1$, with the shifts 0, 1, −4, −3, −5.*

(i) *The components that the degrees allow are $x_1 \in \mathcal{H}^0(M_2, M_3)$, $x_2^\zeta \in \mathcal{H}^6(M_3, L_\zeta)$, $x_3^\zeta \in \mathcal{H}^0(L_\zeta, M_1)$, components in $\mathcal{H}^2(L_\zeta, L_{\zeta'})$ for the $48 + 96$ pairs with $\prod \zeta = 1 = -\prod \zeta'$ that differ in one or two coordinates, and components in $\mathcal{H}^6(M_2, L_{\zeta'})$, $\prod \zeta' = -1$, which we take to be zero. For harmonic components the Maurer–Cartan equation is exactly*

$$x_1 x_2^\zeta = 0 \text{ in } H^6(A, M_2^{-1} \otimes L_\zeta) \text{ for every } \zeta, \qquad \sum_\zeta x_2^\zeta x_3^\zeta = 0 \text{ in } H^6(A, M_3^{-1} \otimes M_1),$$

*and every solution is the harmonic part of a convolution $E$ with* $\operatorname{ch}(E) = 32(\alpha+\overline{\alpha}) + e^{5\theta} - e^{6\theta} - e^{2\theta}$.

(ii) *There are solutions with every* $x_2^\zeta \neq 0$ *and* $x_1 x_2^\zeta = x_2^\zeta x_3^\zeta = 0$.

(iii) *For every solution* $\dim \operatorname{Ext}^2(E,E) \geq 2560$, *so $E$ does not meet the polarised criterion.*

*Proof.* (i) A component from $P_a$ to $P_b$ has Ext degree $1 + d_a - d_b$, and the forms give the cohomology of each pair: $\mathcal{H}(M_2, M_3)$ and $\mathcal{H}(L_\zeta, M_1)$ sit in degree 0, $\mathcal{H}(M_3, L_\zeta)$, $\mathcal{H}(M_2, L_{\zeta'})$ and $\mathcal{H}(M_2, M_1)$ in degree 6, and $\mathcal{H}(L_\zeta, L_{\zeta'})$ in degree 2 exactly for the pairs listed; no other pair has cohomology in the degree a component needs. Put $\delta = \delta^H - h(\delta \circ \delta)$ recursively, as in Lemma 18.24(i), and $\Phi = \bar\partial\delta + \delta\circ\delta$. Then $\Phi = \pi(\delta\circ\delta) + h\bar\partial(\delta\circ\delta)$ and $\bar\partial(\delta\circ\delta) = \Phi\circ\delta - \delta\circ\Phi$, so by induction on $b-a$ the twisted complex exists exactly when $\pi(\delta\circ\delta) = 0$. The components of $\pi(\delta\circ\delta)$ come from paths of two or more components; those that land in a nonzero group are $x_1 x_2^\zeta$ in $\mathcal{H}^6(M_2, L_\zeta)$ and $\sum_\zeta x_2^\zeta x_3^\zeta$ in $\mathcal{H}^6(M_3, M_1)$, while the product along $M_2 \to M_3 \to L_\zeta \to M_1$ would lie in $\mathcal{H}^5(M_2, M_1) = 0$ and $x_2^\zeta$ composed with a component in $\mathcal{H}^2(L_\zeta, L_{\zeta'})$ in degree eight. The signs $(-1)^d$ are $+,-,+,-,-$, which gives the Chern character by Proposition 18.27(i). (ii) Take $x_1$ and the $x_3^\zeta$ to be external products of nonzero sections on the $B_j$. On each $B_j$ the products with their factors map $H^2(B_j, M_3^{-1}L_\zeta)$, of dimension 35, to spaces of dimensions 24 and 16, so they have kernels of dimensions at least 11 and 19. A sum of external products whose factor on one surface lies in the first kernel and whose factor on another lies in the second is a nonzero $x_2^\zeta$ with $x_1 x_2^\zeta = x_2^\zeta x_3^\zeta = 0$, which solves (i).[20] (iii) The sixteen $L_\zeta$ with $\prod\zeta = 1$ have shift $-4$ and the others $-5$, so Lemma 18.34 below applies to every pair of them that differ in all three coordinates: there are 112 such pairs, and their groups $\mathcal{H}^3$ have dimensions adding up to $16 \cdot 160 = 2560$. ∎

The classes counted in (iii) are the ones no product can reach. Two pieces $L_\zeta$, $L_{\zeta'}$ that differ in every coordinate have a difference that is indefinite on every surface, so $\mathcal{H}(L_\zeta, L_{\zeta'})$ is $\bigotimes_j H^1(B_j, \cdot)$, concentrated in degree three, of dimension $D(\zeta,\zeta') = \prod_j |\zeta_j - \zeta'_j|^2$; no component can join them, and if their shifts differ by one their classes have degree two.

**Lemma 18.34** (Pieces that differ everywhere)**.** *Keep Proposition 18.29 with $n = 3$. Suppose that for some integer $r$ either $p+2 \leq t_1 < t_2 < t_3$ and $M_1$, $M_2$, $M_3$ have the shifts $r+1$, $r+4$, $r+5$, or $t_1 \leq p-2 < p+2 \leq t_2 < t_3$ and they have the shifts $r-1$, $r+3$, $r+4$ or $r-1$, $r+1$, $r+4$; these are the shifts forced by the paths of Theorem 18.31(i) and of Proposition 18.32(i), (a) and (b), with $r$ the shift of the $L_\zeta$ on the path. Let $L_\zeta$, $L_{\zeta'}$ differ in all three coordinates, with $d_{L_\zeta} = d_{L_{\zeta'}} + 1$. Then $d_E$ vanishes on $\mathcal{H}^3(L_\zeta, L_{\zeta'})$, and no element of degree one has a differential with a nonzero component there. Consequently*

$$\dim \operatorname{Ext}^2(E,E) \;\geq\; \dim\ker(d_E|_{D^2}) + \sum D(\zeta,\zeta'),$$

*the sum running over the pairs of this kind.*

*Proof.* Only the shifts are used, and the order of the pieces only removes components, so we ignore it. A component from $P_a$ to $P_b$ of Ext degree $q$ has $d_b = d_a + 1 - q$. Between distinct $L_\zeta$ the degree $q$ is at least 2: there is no $\mathcal{H}^0$, and $\mathcal{H}^1$ occurs only for pieces differing in one coordinate, which have opposite parities, while $q = 1$ would need equal shifts. So a component between two $L_\zeta$ lowers the shift by at least one, and by exactly one only in degree two. For a multiple $M_i$ let $\mu_i$ and $\nu_i$ be the shifts of an $L_\zeta$ that a component from

[20] Such solutions are far from unique: the equations of (i) are linear in each of $x_1$, $x_2^\zeta$, $x_3^\zeta$ separately, and the components on the 144 pairs do not enter them at all.

$M_i$ reaches and from which a component reaches $M_i$: $\mu_i = d_{M_i} - 5$ and $\nu_i = d_{M_i} - 1$ when $t_i > p$, and $\mu_i = d_{M_i} + 1$, $\nu_i = d_{M_i} + 5$ when $t_i < p$. Between the multiples the degrees allow only $M_2 \to M_3$ in the first case, $M_2 \to M_3 \to M_1$ in the second, and $M_3 \to M_1$ in the third. Let $\lambda_i$ be the largest $\mu_k$ and $\kappa_i$ the least $\nu_k$ over the multiples $M_k$ reachable from $M_i$, respectively reaching $M_i$, through these steps, $M_i$ included. The values, in the order $M_1$, $M_2$, $M_3$ and relative to $r$, are

| | $\mu$ | | | $\nu$ | | | $\lambda$ | | | $\kappa$ | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $r+1,\ r+4,\ r+5$ | $-4$ | $-1$ | $0$ | $0$ | $3$ | $4$ | $-4$ | $0$ | $0$ | $0$ | $3$ | $3$ |
| $r-1,\ r+3,\ r+4$ | $0$ | $-2$ | $-1$ | $4$ | $2$ | $3$ | $0$ | $0$ | $0$ | $2$ | $2$ | $2$ |
| $r-1,\ r+1,\ r+4$ | $0$ | $-4$ | $-1$ | $4$ | $0$ | $3$ | $0$ | $-4$ | $0$ | $3$ | $0$ | $3$ |

and in all three cases, for all $i$ and $k$,

$$\lambda_i \le r \le \kappa_k, \qquad \kappa_i \ge \lambda_i + 2, \qquad \lambda_i \le \mu_i + 2, \qquad \nu_i \le \kappa_i + 2, \tag{53}$$

while $\mu_i \le \lambda_i$ and $\nu_i \ge \kappa_i$ by definition. A run $L \to M_k \to \cdots \to M_i \to L'$ through multiples goes from the shift $\nu_k \ge \kappa_k$ to $\mu_i \le \lambda_k \le \kappa_k - 2$, so every chain of components from one $L$ to another lowers the shift, by at least two if it passes through a multiple and by exactly one only if it is a single component of degree two. In particular a chain from an $L$ of shift $x$ into $M_i$ has $x \ge \kappa_i$, and a chain from $M_i$ to an $L$ of shift $y$ has $y \le \lambda_i$.

By Lemma 18.24(iii), a term of $d_E$ on a class $u$ of Ext degree $q_u$ in $\mathcal{H}(a', b')$ runs along a chain $c \to \cdots \to a'$, the step $u$, and a chain $b' \to \cdots \to e$, and since the degrees of the components telescope, its value lies in $\mathcal{H}(c, e)$ in Ext degree

$$q_u + 1 + (d_c - d_{a'}) + (d_{b'} - d_e).$$

Write $a = L_\zeta$ and $b = L_{\zeta'}$, so that $d_a = d_b + 1$.

*Nothing reaches* $\mathcal{H}^3(a, b)$. Let $u$ have degree one, so $q_u = 1 + d_{a'} - d_{b'}$, with chains from $a$ to $a'$ and from $b'$ to $b$ and $(a', b') \ne (a, b)$. If $a'$ and $b'$ are both pieces $L$, then $d_{a'} \le d_a$ and $d_{b'} \ge d_b$, not both with equality, so $q_u \le 1$; $q_u = 0$ is impossible, and $q_u = 1$ needs either $a' = b'$, and then a chain from $a$ to $b$ lowering the shift by one, a single component in $\mathcal{H}^2(a, b) = 0$, or $a'$, $b'$ differing in one coordinate, of opposite parities, against $d_{a'} = d_{b'}$. If $a' = M_i$ and $b'$ is an $L$, then $u$ lies in the degree of a component, so $d_{b'} = \mu_i$, while $d_{b'} \ge d_b = d_a - 1 \ge \kappa_i - 1 \ge \lambda_i + 1 > \mu_i$. If $a'$ is an $L$ and $b' = M_k$, then $d_{a'} = \nu_k$ and $d_a - d_b \ge \nu_k - \lambda_k \ge \kappa_k - \lambda_k \ge 2$. If $a' = M_i$ and $b' = M_k$, then $d_a - d_b \ge \kappa_i - \lambda_k$; for $i = k$ this is at least 2, and for $i \ne k$ the class $u$ lies in the degree of a step $M_i \to M_k$, so $\lambda_k \le \lambda_i$ and again $d_a - d_b \ge 2$. Each case contradicts $d_a - d_b = 1$.

*$d_E$ vanishes on* $\mathcal{H}^3(a, b)$. Here $u$ is the class itself and a term lands in $\mathcal{H}(c, e)$, $(c, e) \ne (a, b)$, in degree $3 + d_c - d_e$, with chains from $c$ to $a$ and from $b$ to $e$. If $c$ and $e$ are both pieces $L$, then $d_c - d_e \ge 2$; $c = e$ is impossible, and otherwise a class of degree at least 5 needs $c$, $e$ differing in one coordinate, in degree 5, so $d_c - d_e = 2$ against their opposite parities. If $c = M_i$ and $e$ is an $L$, the degree is that of a component when $d_e = \mu_i + 2$, but $d_e \le d_b < d_a \le \lambda_i \le \mu_i + 2$. If $c$ is an $L$ and $e = M_k$, it is that of a component when $d_c = \nu_k - 2$, but $d_c \ge d_a > d_b \ge \kappa_k \ge \nu_k - 2$. If $c = M_i$ and $e = M_k$, then $d_a \le \lambda_i \le r \le \kappa_k \le d_b$, against $d_a > d_b$.

For the inequality: a coboundary has no component on $D^2$ (Proposition 18.29) and none on these groups, so the cocycles of $\ker(d_E|_{D^2})$ and of these groups span a space that meets the coboundaries only in zero. Twisting the pieces by line bundles of degree zero does not change $D(\zeta, \zeta')$, the forms being nondegenerate on every surface, and can only remove components. ∎

**Theorem 18.35** (No convolution of these pieces meets the criterion)**.** *Keep Proposition 18.29 with $n = 3$: let $E$ be any convolution of the 32 bundles $L_\zeta$ and of $M_1$, $M_2$, $M_3$, with $t_1 < t_2 < t_3$ and $|t_i - p| \geq 2$, in any order, with any shifts of the given parities and with the pieces twisted by any line bundles of degree zero. Then $\dim \mathrm{Ext}^2(E,E) \geq 69$. In particular no such convolution meets the polarised criterion, which asks for 57.*

*Proof.* Replacing $E$ by $E^\vee$ or $E^\vee[1]$, which preserves $\mathrm{Ext}^2$, we may assume that the $t_i$ are as in Theorem 18.31 or in Proposition 18.32. If no fourfold product of those results acts, $\dim \mathrm{Ext}^2(E,E) \geq 249$. Otherwise the path fixes the shifts of the $M_i$ as in Lemma 18.34, with $r$ the shift of every $L_\zeta$ on a path; let $P$ be the sixteen pieces of the parity of $r$, $Q$ the other sixteen, and $m$ the number of pieces on paths.

The fourfold products act on $D^2$ only through the classes at the $M_i$ and at the $m$ pieces on paths, so their rank is at most $45 + 15m$, and by Proposition 18.30 $\dim \ker(d_E|_{D^2}) \geq 249 - 45 - 15m = 204 - 15m$. If $m \leq 9$ this is at least 69.

Let $m \geq 10$, so that at least ten pieces of $P$ have the shift $r$. Every $W \in Q$ has 3, 6 and 7 partners in $P$ differing from it in one, two and three coordinates, and for the seven $D = 16$ six times and $D = 64$ once.[21] If $d_W = r \pm 1$, at most six pieces of $P$ lack the shift $r$, so one of the seven has it, and Lemma 18.34 gives at least 16 classes. If $d_W \neq r \pm 1$, consider for each type $\tau$ of the form $(1,1,0)$ the partner $W^\tau \in P$ obtained by changing the sign of the coordinate where $\tau$ vanishes. By the proof of Proposition 18.30, $W^\tau$ is the only piece with which $W$ can carry a nonzero cup product on the type $\tau$, and conversely; a cup product between them needs a component of degree two, hence shifts differing by one. If they do, the classes of type $\tau$ equal at $W$ and $W^\tau$ and zero elsewhere have vanishing cup products, and $W^\tau$, of shift $d_W \pm 1 \neq r$, lies on no path; if they do not, the classes of type $\tau$ at $W$ alone have vanishing cup products. Either way these 4 classes lie in $\ker(d_E|_{D^2})$, 12 for each such $W$, and those of different $W$ are independent, since $W \mapsto W^\tau$ is injective. Adding over $Q$, Lemma 18.34 gives $\dim \mathrm{Ext}^2(E,E) \geq 16 \cdot 12 = 192$. ∎

At $n = 4$ there are 128 pieces $L_\zeta$, and the classes that no product reaches are again those between pieces that differ in three coordinates and have adjacent shifts. The shifts along a chain are now controlled by counting alone.

**Lemma 18.36** (Shifts along chains at $n = 4$)**.** *Keep Proposition 18.27 with $n = 4$, take the $L_\zeta$ to be external products $N_{1,\zeta_1} \boxtimes \cdots \boxtimes N_{4,\zeta_4}$ of line bundles on the $B_j$, let $t_1, t_2, t_3$ be distinct with $|t_i - p| \geq 2$, and let $E$ be a convolution of the $L_\zeta$ and of $M_1, M_2, M_3$ with shifts of the given parities. Call a component of $\delta^H$, or an element of degree one of $\mathcal{H}_E$ in one group $\mathcal{H}(a,b)$, a step from $P_a$ to $P_b$; a step of* Ext *degree $q$ has $d_b = d_a + 1 - q$. Put $\phi(L_\zeta) = p$ and $\phi(M_i) = t_i$.*

(i) *A step between distinct $L_\zeta$ has* Ext *degree at least two, so it lowers the shift by at least one, and by exactly one only in* Ext *degree two. A step in $\mathcal{H}^1(a,a) = H^1(A, \mathcal{O}_A)$ keeps the shift.*

(ii) *A step between an $L_\zeta$ and a multiple, or between two multiples, has* Ext *degree* 0 *if it goes up the value $\phi$ and* 8 *if it goes down, so it raises the shift by one or lowers it by seven.*

(iii) *A sequence of steps from an $L_\zeta$ to an $L_{\zeta'}$ through multiples, distinct before one marked step and distinct after it, where the marked step may lie in $\mathcal{H}^1(a,a)$ at a multiple, lowers the shift by at least four. Consequently a sequence of steps from one*

[21] The ratios $\zeta'_j/\zeta_j$ of two pieces of opposite parities have product $-1$; when all three differ from 1 they are $(-1,-1,-1)$, with $D = 4^3$, or a permutation of $(i,-i,-1)$, with $D = 2 \cdot 2 \cdot 4$.

*$L_\zeta$ to another, at most one of them not a component, lowers the shift by at least one, and by exactly one only if it consists of steps between pieces $L_\zeta$, one of* Ext *degree two and the others in $\mathcal{H}^1(a,a)$.*

*Proof.* (i) For $\zeta \neq \zeta'$ the group $\mathcal{H}(L_\zeta, L_{\zeta'})$ is the tensor product over the surfaces of $H^1$ of a line bundle with a nondegenerate form, on the $k \geq 1$ surfaces where $\zeta$ and $\zeta'$ differ, and of $H^*(B_j, \mathcal{O}_{B_j})$ on the others, the factors of $L_\zeta$ and $L_{\zeta'}$ on those being equal. So its degrees are at least $k$, and degree one needs $k = 1$. Then the ratio $\zeta'_j/\zeta_j \neq 1$ in the one coordinate is the ratio of $\prod \zeta'$ and $\prod \zeta$, which is $-1$, and the shifts have opposite parities, while a step of Ext degree one keeps the shift. (ii) The forms of $M_i \otimes L_\zeta^{-1}$ have the eigenvalues $t_i - p \pm 1$ on every surface, which have the sign of $t_i - p$; the form of $M_k \otimes M_i^{-1}$ is $(t_k - t_i)\,\mathrm{Id}$. A definite form has cohomology in degree 0 if it is positive and $2n = 8$ if it is negative [BL04, Chapter 3].[22] (iii) Let $U$ and $D$ be the numbers of steps up and down outside $\mathcal{H}^1(a,a)$; by (ii) the shift changes by $U - 7D$. The values start and end at $p$ and pass through the values of multiples, so $D \geq 1$. If $D = 1$, the values increase strictly before the step down and strictly after it, so those before it lie above $p$ and those after it below $p$, each multiple occurs once, and $U \leq 3$: the drop $7 - U$ is at least four.[23] If $D \geq 2$, at most six multiples occur, counted with repetition, so there are at most seven steps outside $\mathcal{H}^1(a,a)$, $U \leq 7 - D$, and the drop $7D - U \geq 8D - 7$ is at least nine. For the consequence, split the sequence at the pieces $L_\zeta$ it visits: each part through multiples lowers the shift by at least four, and by (i) each other part lowers it by at least one, unless it lies in $\mathcal{H}^1(a,a)$. ∎

**Theorem 18.37** (Two shifts on $E_0^8$)**.** *Keep Lemma 18.36, and let $L_\zeta$ and $L_{\zeta'}$ differ in exactly three coordinates and have the shifts $d_{L_\zeta} = d_{L_{\zeta'}} + 1$. Then $\mathcal{H}(L_\zeta, L_{\zeta'})$ has its classes of degree two in $\mathcal{H}^3(L_\zeta, L_{\zeta'})$, of dimension $D(\zeta,\zeta') = \prod_{\zeta_j \neq \zeta'_j} |\zeta_j - \zeta'_j|^2$, which is* 16 *or* 64*, and:*

(i) *for every convolution, no element of degree one has a differential with a nonzero component in $\mathcal{H}^3(L_\zeta, L_{\zeta'})$;*

(ii) *if the $L_\zeta$ occupy two consecutive shifts, those with $\prod \zeta = \epsilon$ the shift $s$ and the others $s - 1$, for a sign $\epsilon$ and an integer $s$, then $d_E$ vanishes on these groups, and for every order of the pieces, every choice of the shifts of the $M_i$ and every placement of the $t_i$,*

$$\dim \mathrm{Ext}^2(E,E) \;\geq\; 64 \cdot 640 = 40960.$$

*In particular $E$ does not meet the polarised criterion, which asks for* $\dim \mathrm{Ext}^2(E,E) = 104$.

*Proof.* Write $W = L_\zeta$ and $V = L_{\zeta'}$. By the proof of Lemma 18.36(i), $\mathcal{H}(W,V)$ has degrees 3, 4 and 5, the last two from the surface where $\zeta$ and $\zeta'$ agree, and its classes of degree $w = 2$ have Ext degree $2 + d_W - d_V = 3$; the dimension of $\mathcal{H}^3(W,V)$ is the product over the three other surfaces of the determinants $|\zeta_j - \zeta'_j|^2$, as in Lemma 18.34.

[22] This is Mumford's index theorem: a line bundle on an abelian variety whose hermitian form is nondegenerate with $i$ negative eigenvalues has cohomology only in degree $i$. On $B_j = E_0 \times E_0$ the form of $M_i \otimes L_\zeta^{-1}$ is $\left(\begin{smallmatrix} t_i - p & -\zeta_j \\ -\bar\zeta_j & t_i - p \end{smallmatrix}\right)$, with eigenvalues $t_i - p \pm 1$, and $|t_i - p| \geq 2$ makes both of the sign of $t_i - p$; over the four surfaces the index is 0 or 8. The hypothesis $|t_i - p| \geq 2$ is what keeps the steps between a piece and a multiple out of the middle degrees.

[23] The bound is attained: with $p$ below the three values $t_i$, the run $L_\zeta \to M_1 \to M_2 \to M_3 \to L_{\zeta'}$, up three times and down once, changes the shift by $3 - 7 = -4$. With $p$ in the middle, $t_1 < p < t_2 < t_3$, the run up through $M_2$ and $M_3$, down to $M_1$ and up to $L_{\zeta'}$ changes it by $3 - 7 = -4$ as well.

(i) As in the proof of Lemma 18.34, a term of the differential of an element $u$ of degree one in $\mathcal{H}(a', b')$ with a component in $\mathcal{H}(W, V)$ runs along a chain of components $W \to \cdots \to a'$, the step $u$, and a chain $b' \to \cdots \to V$, with $(a', b') \neq (W, V)$. This is a sequence of steps from $W$ to $V$, one of them $u$, lowering the shift by one. By Lemma 18.36(iii) it consists of steps between pieces $L_\zeta$, one of Ext degree two and at most one, $u$, in $\mathcal{H}^1(a, a)$. If $u$ is not in $\mathcal{H}^1(a, a)$, it is the step of degree two and both chains are empty, against $(a', b') \neq (W, V)$. If $u$ is in $\mathcal{H}^1(a, a)$, the rest is a single component $W \to V$ of Ext degree two, in $\mathcal{H}^2(W, V) = 0$.

(ii) A term of $d_E$ on $x \in \mathcal{H}^3(W, V)$ runs along chains of components $c \to \cdots \to W$ and $V \to \cdots \to e$, not both empty, and lands in $\mathcal{H}(c, e)$ in Ext degree $3 + 1 + (d_c - d_W) + (d_V - d_e) = 3 + d_c - d_e$. No piece $L_{\zeta''}$ other than $W$ lies on the first chain: by Lemma 18.36(iii) it would have the shift at least $d_W + 1 = s + 1$. Likewise only multiples follow $V$ on the second chain, and each chain has at most three pieces besides $W$ or $V$. Let $U$ and $D$ be the numbers of steps up and down along both chains, so that $U + D \leq 6$; by Lemma 18.36(ii), $d_W = d_c + U_1 - 7D_1$ and $d_e = d_V + U_2 - 7D_2$, and the Ext degree is

$$3 + d_c - d_e = 4 - U + 7D.$$

If $c = e$, this degree is 3, so $U = 7D + 1$, and $D \geq 1$ because the values return to $\phi(c) \neq p$; then $U \geq 8$, which is impossible. If $c \neq e$, the group $\mathcal{H}(c, e)$ sits in degree 0 if $\phi(e) > \phi(c)$ and in degree 8 if $\phi(e) < \phi(c)$, by Lemma 18.36(ii). Degree 0 needs $U = 4 + 7D \geq 4$ with $D = 0$, so the values increase strictly, the multiples are distinct and $U \leq 3$. Degree 8 needs $7D = U + 4$, so $D = 1$ and $U = 3$; the values increase strictly before the step down, from $\phi(c)$, and after it, up to $\phi(e) < \phi(c)$, so the five values along the path, with $W$ and $V$ counted once, are distinct, and four of them are values of multiples, of which there are three. So no term reaches a nonzero group, and $d_E$ vanishes on $\mathcal{H}^3(W, V)$.

By (i) and (ii) the classes of these groups are cocycles of degree two, and no nonzero combination of them is a coboundary, since every coboundary has zero components there. Each piece of the shift $s$ has 28 partners of the other parity differing from it in exactly three coordinates,[24] all of the shift $s - 1$, and the 28 groups have dimensions adding up to 640; there are 64 pieces of the shift $s$. ∎

Figure 28 draws the shifts along a run. The hypothesis of (ii) is the arrangement of the explicit convolution of Proposition 18.33 at $n = 3$, where the sixteen pieces of each parity share a shift. The pieces must not be twisted by line bundles of degree zero here, unlike in Theorem 18.35: a twist that is nontrivial on the surface where $\zeta$ and $\zeta'$ agree removes $\mathcal{H}(L_\zeta, L_{\zeta'})$.

With a third shift, a component of Ext degree two can follow or precede a class of $\mathcal{H}^3(L_\zeta, L_{\zeta'})$, so the differential no longer vanishes on these classes. When the three shifts are consecutive, such terms reach only the groups between the highest and the lowest shift, which are too small to remove the classes.

**Theorem 18.38** (Three consecutive shifts on $E_0^8$)**.** *Keep Lemma 18.36, and suppose that the shifts of the $L_\zeta$ lie in $\{s, s-1, s-2\}$ for an integer $s$. By their parities, for a sign $\epsilon$, the pieces with $\prod \zeta = \epsilon$ have the shift $s - 1$ and the others the shift $s$ or $s - 2$; let $T$ and $B$ be the sets of pieces of the shifts $s$ and $s - 2$. Then, for every order of the pieces, every choice of the shifts of the $M_i$ and every placement of the $t_i$:*

[24] The ratios $\zeta'_j/\zeta_j$ in the three coordinates have product $-1$, so they are $(-1, -1, -1)$, with $D = 64$, or a permutation of $(i, -i, -1)$, with $D = 16$; with four places for the coordinate where they agree this gives $4 \cdot 7 = 28$ partners and $4 \cdot (64 + 6 \cdot 16) = 640$ classes.

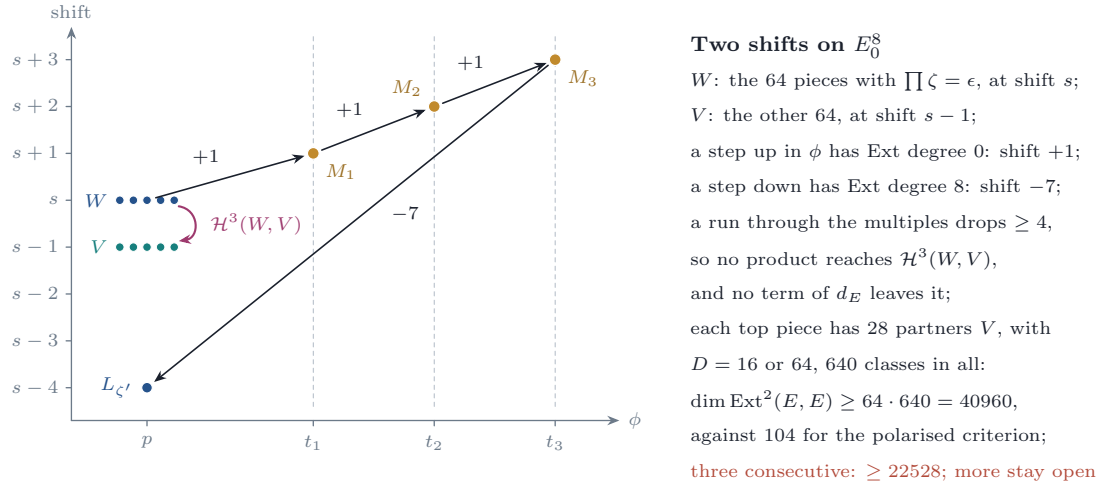



FIGURE 28. Lemma 18.36 and Theorem 18.37, in the placement $p<t_1<t_2<t_3$. Horizontal: the value $\phi$ of a piece; vertical: its shift. The pieces $L_\zeta$ with $\prod\zeta=\epsilon$ sit at the shift $s$ and the others at $s-1$. A run through the multiples, up three times with Ext degree 0 and down once with Ext degree 8, ends four shifts lower, the least drop of Lemma 18.36(iii), so it never joins a top piece $W$ to a partner $V$ one shift lower; the group $\mathcal{H}^3(W,V)$ survives in $\mathrm{Ext}^2(E,E)$.

(i) *every two pieces $L_\zeta$, $L_{\zeta'}$ that differ in exactly three coordinates have shifts differing by one, and the groups $\mathcal{H}^3(L_\zeta,L_{\zeta'})$ of Theorem 18.37, taken from the piece of the larger shift, have dimensions adding up to* 40960*;*

(ii) *$d_E$ maps the sum of these groups into $\bigoplus_{c\in T,\,e\in B}\mathcal{H}^5(c,e)$, and on them it consists of cup products with a single component of* Ext *degree two;*

(iii)
$$\dim \mathrm{Ext}^2(E,E)\ \geq\ 40960-\sum_{c\in T,\,e\in B}\dim\mathcal{H}^5(c,e)\ \geq\ 40960-18432=22528.$$

*In particular $E$ does not meet the polarised criterion. When $T$ or $B$ is empty this is Theorem 18.37(ii).*

*Proof.* (i) The ratios $\zeta'_j/\zeta_j$ in the three coordinates have product $-1$, so the two pieces have opposite parities: one has the shift $s-1$ and the other $s$ or $s-2$. Each of the 64 pieces of the shift $s-1$ has 28 such partners, whose groups have dimensions adding up to 640, as in the proof of Theorem 18.37; this gives $64\cdot 640=40960$.

(ii) Let $x\in\mathcal{H}^3(a,b)$ with $d_a=d_b+1$. As in the proof of Theorem 18.37(ii), a term of $d_E(x)$ runs along chains of components $c\to\cdots\to a$ and $b\to\cdots\to e$, not both empty, and lands in $\mathcal{H}(c,e)$ in Ext degree $3+d_c-d_e$. The shifts of the pieces $L_\zeta$ differ by at most two, so by Lemma 18.36(iii) no part of a chain between two pieces $L_\zeta$ passes through a multiple. Hence a chain $c\to\cdots\to a$ that starts at a multiple $M_i$ is a run $M_i=m_0\to\cdots\to m_r\to L_1$ of distinct multiples, $r\leq 2$, followed by pieces $L_\zeta$ only; let $U_1$ and $D_1$ be the numbers of its steps up and down the value $\phi$ before $L_1$, so that $d_{L_1}=d_{M_i}+U_1-7D_1$ by Lemma 18.36(ii) and $U_1+D_1=r+1\leq 3$. Likewise a chain $b\to\cdots\to e$ ending at a multiple $M_k$ ends with a run from a piece $L_2$ with $U_2$ steps up

and $D_2$ down, $U_2 + D_2 \le 3$ and $d_{M_k} = d_{L_2} + U_2 - 7D_2$. Put $L_1 = c$ if $c$ is a piece $L_\zeta$ and $L_2 = e$ if $e$ is one, and $\delta = d_{L_1} - d_{L_2}$; then $1 = d_a - d_b \le \delta \le 2$.

If $c$ and $e$ are both pieces $L_\zeta$, each nonempty chain lowers the shift by at least one, so $3 + d_c - d_e \ge 5$, while $\delta \le 2$ gives at most 5. So the degree is 5, $c \in T$, $e \in B$, and the two chains together consist of one component, of Ext degree two by Lemma 18.36(i): the term is a cup product of $x$ with that component.

If $c = M_i$ and $e$ is a piece $L_\zeta$, the degree is $3 + \delta - U_1 + 7D_1$, and $\mathcal{H}(c, e)$ sits in degree 0 or 8 by Lemma 18.36(ii). Degree 0 needs $U_1 \ge 3 + \delta \ge 4$, and degree 8 needs $7D_1 - U_1 = 5 - \delta \in \{3, 4\}$, so $D_1 \ge 1$ and $U_1 \ge 3$; both contradict $U_1 + D_1 \le 3$. The case of a piece $c$ and a multiple $e$ is the same with the second run.

If $c = M_i$ and $e = M_k$, the degree is $3 + \delta - U + 7D$ with $U = U_1 + U_2 \le 6$ and $D = D_1 + D_2$. For $i = k$ it is 3, so $U - 7D = \delta \ge 1$; $D = 0$ is impossible, since the values would increase from $t_i$ to $p$ along the first run and from $p$ to $t_i$ along the second, and $D \ge 1$ gives $U \ge 8$. For $i \ne k$, $\mathcal{H}(c, e)$ sits in degree 0 if $t_k > t_i$ and 8 if $t_k < t_i$. Degree 0 needs $U = 3 + \delta + 7D$, so $D = 0$ and $U \ge 4$; but then the multiples of the first run have values below $p$ and those of the second values above $p$, so they are distinct and $U \le 3$. Degree 8 needs $7D - U = 5 - \delta$, so $D = 1$ and $U \ge 3$: the runs have at least four steps, so some multiple lies on both. One run increases strictly. If it is the second, its values lie in $(p, t_k]$, while those of the first lie at or above $t_i > t_k$ before its step down and below $p$ after it. If it is the first, its values lie in $[t_i, p)$, while those of the second lie above $p$ before its step down and at or below $t_k < t_i$ after it. Either way no multiple lies on both runs, a contradiction.

(iii) By Theorem 18.37(i), which holds for every convolution, no coboundary has a component in these groups, so the kernel of $d_E$ on their sum injects into $\mathrm{Ext}^2(E, E)$, and by (ii) its dimension is at least 40960 minus the dimension of the targets. For pieces $c$, $e$ of the same parity, $\mathcal{H}^5(c, e)$ is nonzero only when they differ in two or three coordinates, and each piece has such partners with groups $\mathcal{H}^5$ of dimensions adding up to 960.[25] So the sum in (iii) is at most $\min(|T|, |B|) \cdot 960 \le 32 \cdot 960 = 30720$, which already gives $\dim \mathrm{Ext}^2(E, E) \ge 10240$. The dimension of $\mathcal{H}^5(c, e)$ depends only on the ratio $e/c$, so the sum is the weight of the cut $(T, B)$ of a weighted Cayley graph on the group of ratios of two pieces of one parity, of order 64 and degree 960. For the vector $x = \pm 1$ of the cut, the weight is $\frac{1}{4}(64 \cdot 960 - x^{\mathsf{T}} A x) \le 16\,(960 - \lambda)$, with $A$ the weighted adjacency matrix and $\lambda$ its least eigenvalue. Its eigenvalues are the character sums $\sum_g \dim \mathcal{H}^5(1, g)\, \chi(g)$. A character is $\chi(g) = \prod_j g_j^{k_j}$ with $k \in (\mathbb{Z}/4)^4$, up to adding $(1, 1, 1, 1)$; put $\omega_j = i^{k_j}$, $W = \sum_j \omega_j$, $\varrho = \sum_j \omega_j^2$ and $\Omega = \prod_j \omega_j$. By the list in the footnote, the ratios that move the pair of coordinates $\{j, l\}$ contribute $64\,\omega_j^2 \omega_l^2 + 32\,\Re(\omega_j \overline{\omega}_l)$, and those that fix only the coordinate $o$ contribute $32\,\Re(\Omega \overline{\omega}_o (W - \omega_o))$. Summing,

$$\chi\big(\textstyle\sum_g \dim \mathcal{H}^5(1, g)\, g\big) = 32\varrho^2 + 16|W|^2 - 192 + 32\,\Re(\Omega)\big(|W|^2 - 4\big).$$

If $\Re\Omega = 0$, the sum of the $k_j$ is odd, so an odd number of them are odd, $|\varrho| = 2$, and the value is at least $-64$. If $\Re\Omega = 1$, an even number are odd: with none or four, $|\varrho| = 4$ and the value is at least 192; with two, $\varrho = 0$, and $W = 0$ would force the $k_j$ to be $0, 1, 2, 3$, whose sum is not divisible by 4, so $|W|^2 \ge 4$ and the value is at least $-128$. If $\Re\Omega = -1$, the value is $32\varrho^2 - 16|W|^2 - 64$, and $|W|^2 \le 8 + 2\varrho^2$: when all $k_j$ have the same parity,

[25] The ratios $e_j/c_j \ne 1$ have product 1. In two coordinates they are $(-1, -1)$, with $D = 16$, or $(i, -i)$, $(-i, i)$, with $D = 4$; then $\mathcal{H}^5 = \mathcal{H}^1 \otimes \mathcal{H}^1 \otimes \mathcal{H}^3(B_j \times B_k, \mathcal{O})$ has dimension $4D$, and the six pairs of coordinates give $6 \cdot 64 + 12 \cdot 16 = 576$. In three coordinates they are a permutation of $(i, i, -1)$ or of $(-i, -i, -1)$, with $D = 16$, and $\mathcal{H}^5 = (\mathcal{H}^1)^{\otimes 3} \otimes H^2(B_l, \mathcal{O})$ has dimension 16; the four triples give $4 \cdot 6 \cdot 16 = 384$. In four coordinates $\mathcal{H}(c, e)$ sits in degree 4.

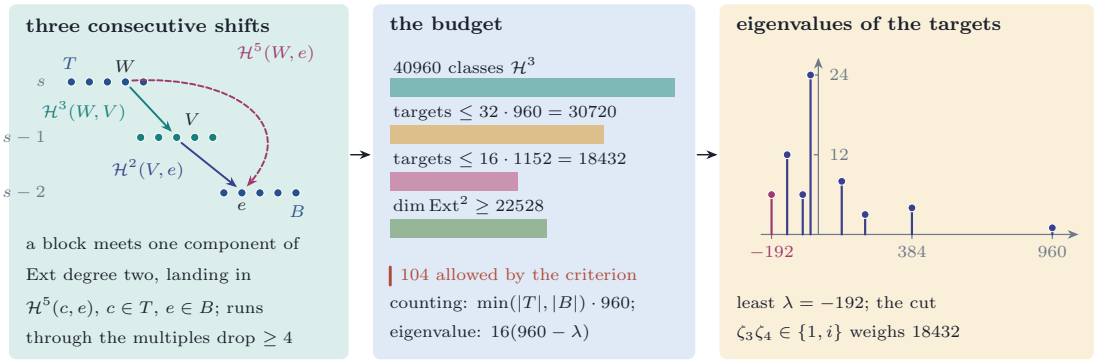



FIGURE 29. Theorem 18.38. Left: the pieces $L_\zeta$ at three consecutive shifts, those of one parity at $s-1$ and the others split between $T$ at $s$ and $B$ at $s-2$; a group $\mathcal{H}^3$ between two pieces that differ in three coordinates (teal) can meet a component of Ext degree two (indigo) only in a group $\mathcal{H}^5(c,e)$ with $c \in T$ and $e \in B$ (magenta). Middle: the 40960 classes against the targets, at most 30720 by counting and 18432 by the least eigenvalue, against the 104 that the criterion allows. Right: the eigenvalues of the Cayley graph of the targets.

$|W| \leq 4 = |\varrho|$, and when two are odd, $\varrho = 0$ and $W$ is a sum of two of $\pm 1$ and two of $\pm i$, so $|W|^2 \leq 8$. Hence $\lambda \geq -192$,[26] and the sum is at most $16 \cdot 1152 = 18432$. ∎

Figure 29 draws the three shifts and the budget.

The groups of Theorem 18.37 join two pieces whose shifts differ by one. All groups between pieces whose shifts differ by three, and thirteen kinds between pieces whose shifts differ by two, are also left alone by the differential, and with them the argument covers every arrangement of the shifts within five consecutive values. For pieces $a$ and $b$ call $d_a - d_b$ the *spread* of $\mathcal{H}(a,b)$; its classes of degree two have Ext degree $2 + d_a - d_b$. For $\zeta, \zeta' \in \mu_4^4$ call $\zeta'/\zeta = (\zeta'_j/\zeta_j)_j$ their *ratio*; it *moves* the coordinates where it differs from 1, and it changes the parity, $\prod \zeta' = -\prod \zeta$, exactly when the product of its entries is $-1$. As before, $D(\zeta,\zeta') = \prod_{\zeta_j \neq \zeta'_j} |\zeta_j - \zeta'_j|^2$.

**Lemma 18.39** (Groups of spread two and three)**.** *Keep Lemma 18.36, and let $E$ be any convolution of the $L_\zeta$ and the $M_i$, in any order, with any shifts of the given parities and any placement of the $t_i$.*

(i) *For a class $x$ of degree two in $\mathcal{H}(L_\zeta, L_{\zeta'})$ of positive spread, every term of $d_E(x)$ has all its leaves between pieces $L_{\zeta''}$.*

(ii) *Let $\prod \zeta = \prod \zeta'$, let $\zeta'/\zeta$ move all four coordinates, one of them by $-1$, and let $d_{L_\zeta} = d_{L_{\zeta'}} + 2$. The classes of degree two of $\mathcal{H}(L_\zeta, L_{\zeta'})$ form $\mathcal{H}^4(L_\zeta, L_{\zeta'})$, of dimension $D(\zeta,\zeta') \in \{64, 256\}$; no element of degree one has a differential with a nonzero component in it, and $d_E$ vanishes on it.*

(iii) *Let $\prod \zeta = -\prod \zeta'$ and $d_{L_\zeta} = d_{L_{\zeta'}} + 3$. The classes of degree two of $\mathcal{H}(L_\zeta, L_{\zeta'})$ lie in $\mathcal{H}^5(L_\zeta, L_{\zeta'})$, and $d_E$ vanishes on them.*

[26] The eight values are 960, 384, 192, 96, −32, −64, −128 and −192. The bound 18432 is attained by the cut $\zeta_3\zeta_4 \in \{1,i\}$ against $\zeta_3\zeta_4 \in \{-1,-i\}$, so 22528 is the best bound that this count gives.

(iv) *Let $\zeta$ and $\zeta'$ differ in exactly three coordinates, with $d_{L_\zeta} = d_{L_{\zeta'}} + 1$. Every term of $d_E$ on $\mathcal{H}^3(L_\zeta, L_{\zeta'})$ runs along chains of components $c \to \cdots \to L_\zeta$ and $L_{\zeta'} \to \cdots \to e$ between pieces $L_{\zeta''}$ that together lower the shift by one or two.*

*Proof.* Write $a = L_\zeta$, $b = L_{\zeta'}$ and $\delta = d_a - d_b$. As in the proof of Theorem 18.37(ii), a term of $d_E$ on a class of degree two in $\mathcal{H}(a, b)$ runs along chains of components $c \to \cdots \to a$ and $b \to \cdots \to e$, not both empty, and lands in $\mathcal{H}(c, e)$ in Ext degree $3 + d_c - d_e$, which is 3 if $c = e$.

(i) Let $L_1$ be the first piece $L_{\zeta''}$ of the first chain and $L_2$ the last of the second, and put $X = (d_{L_1} - d_a) + (d_b - d_{L_2})$. By Lemma 18.36(iii), $X \geq 0$, and $X \geq 4$ if a multiple lies between $L_1$ and $a$ or between $b$ and $L_2$. Before $L_1$ the first chain is a run of distinct multiples, with $U_1$ steps up and $D_1$ steps down the value $\phi$ and $U_1 + D_1 \leq 3$; after $L_2$ the second chain is a run with $U_2$ and $D_2$ steps, $U_2 + D_2 \leq 3$. By Lemma 18.36(ii) the term has Ext degree $3 + \delta + X - U + 7D$, where $U = U_1 + U_2$ and $D = D_1 + D_2$. If $c$ and $e$ are pieces, this is $3 + \delta + X$; a multiple on a chain makes it at least 8, but $\mathcal{H}^8(c, e) \neq 0$ only for $c = e$, where the degree is 3. If $c = M_i$ and $e$ is a piece, $\mathcal{H}(c, e)$ sits in degree 0 if $t_i < p$ and in degree 8 if $t_i > p$. Degree 0 needs $U_1 \geq 3 + \delta + X \geq 4$. Degree 8 needs a step down on the run from $t_i$ to $p$, so $D_1 \geq 1$, and then $U_1 = 7D_1 - 5 + \delta + X \geq 3$. Both contradict $U_1 + D_1 \leq 3$, and a piece $c$ with a multiple $e$ is the same. Let $c = M_i$ and $e = M_k$. For $i = k$ the degree is 3, so $U - 7D = \delta + X \geq 1$; $D \geq 1$ would give $U \geq 8$, and $D = 0$ would make the values increase from $t_i$ to $p$ and then from $p$ to $t_i$. For $i \neq k$, $\mathcal{H}(c, e)$ sits in degree 0 if $t_k > t_i$ and in degree 8 if $t_k < t_i$. Degree 0 needs $U = 3 + \delta + X + 7D$, so $D = 0$ and $U \geq 4$; but then the first run lies below $p$ and the second above, so the multiples are distinct and $U \leq 3$. Degree 8 needs $D \geq 1$, since with $D = 0$ the values would satisfy $t_i < p < t_k$; then $7D - U = 5 - \delta - X \leq 4$ gives $D = 1$ and $U \geq 3$, so the runs have at least four steps and share a multiple, which the last paragraph of the proof of Theorem 18.38(ii) excludes.

(ii) By the proof of Lemma 18.36(i), $\mathcal{H}(a, b)$ is the tensor product of four groups $H^1$ and sits in degree 4, of dimension $D(\zeta, \zeta')$; the ratio is $(-1, -1, -1, -1)$, with $D = 256$, or a permutation of $(i, -i, -1, -1)$, with $D = 64$.[27] A coboundary with a component in $\mathcal{H}(a, b)$ is a sum of terms of Lemma 18.24(iii) along a path $a \to \cdots \to b$ of morphisms of degree one, one of them an element $u$ and the others components, that lowers the shift by two. By Lemma 18.36(iii) no multiple lies on it, and by Lemma 18.36(i) each step between distinct pieces lowers the shift by at least one. So the path consists of a component of Ext degree three from $a$ to $b$, which lies in $\mathcal{H}^3(a, b) = 0$, and a class of $\mathcal{H}^1(a, a)$ or $\mathcal{H}^1(b, b)$; or of two steps $a \to c \to b$ of Ext degree two, possibly with a class of $H^1(A, \mathcal{O}_A)$ at one of the three pieces. The output has degree one on every surface. Without the class of $H^1$ the two steps have degrees adding up to one on every surface, so they move disjoint pairs of coordinates and have degree zero elsewhere; a step of Ext degree two changes the parity, so its two moved entries have product $-1$ and are $(i, i)$ or $(-i, -i)$, and $\zeta'/\zeta$ has no entry $-1$. With the class of $H^1$, Lemma 18.28 gives a drop of one on a surface where all three leaves have positive degree, and the output has degree at least two there.

Now let a term of $d_E$ on $\mathcal{H}^4(a, b)$ carry the components $y_1, \ldots, y_m$, $m \geq 1$, between pieces by (i), and land in $\mathcal{H}(c, e)$; let $K$ be the set of coordinates fixed by the ratio of $c$ and $e$, and $X \geq m$ the drop of the chains. The output has Ext degree $5 + X$. Let $n_j$

[27] A ratio that moves every coordinate has entries in $\{i, -1, -i\}$ and product 1. With a $-1$ among them it is $(-1, -1, -1, -1)$ or has exactly two entries $-1$ and the entries $i$ and $-i$, twelve ratios; together thirteen, and $256 + 12 \cdot 64 = 1024$. The other eight ratios have all four entries in $\{i, -i\}$ and $D = 16$; (ii) makes no statement about them.

be the number of the $y_i$ of positive degree on $B_j$; the class $x$ has degree one there, so Lemma 18.28 bounds the drop on $B_j$ by $\max(0, n_j - 1)$. For $j \notin K$ the output has degree 1 on $B_j$, so the degrees of the $y_i$ there add up to at most $\max(0, n_j - 1)$, which forces $n_j = 0$. For $j \in K$ the $y_i$ undo the entry $\zeta'_j/\zeta_j \neq 1$, so $n_j \geq 1$, and the output has degree at least $1 + n_j - (n_j - 1) = 2$ there, hence exactly 2. So $5 + X = (4 - |K|) + 2|K|$, that is $X = |K| - 1$. The components change the parity $X$ times modulo two, and together their ratio is $\zeta/\zeta'$ on $K$ and 1 elsewhere, so

$$\prod_{j \in K} \zeta'_j/\zeta_j = (-1)^{|K|-1}.$$

For the ratio $(-1, -1, -1, -1)$ the left side is $(-1)^{|K|}$. For a permutation of $(i, -i, -1, -1)$ the left side is real only if $K$ contains both or neither of the coordinates moved by $\pm i$, and then it is again $(-1)^{|K|}$. So no term is nonzero.

(iii) The classes of degree two have Ext degree 5. By (i) a term of $d_E$ on them lands in Ext degree $6 + X$, with $X \geq 1$ the drop of the chains between pieces. For $X \geq 2$ this is at least 8, which needs $c = e$ and degree 3. For $X = 1$ the pieces $c$, $e$ have shifts differing by four, hence the same parity, so their ratio moves at least two coordinates and $\mathcal{H}(c, e)$ has no degree above 6.

(iv) By (i) a term lands in Ext degree $4 + X$, with $X \geq 1$ the drop of the chains between pieces. For $X \geq 4$ the degree is at least 8, which needs $c = e$. For $X = 3$ the degree is 7, which needs a ratio moving one coordinate, hence opposite parities, while $d_c - d_e = 4$ is even. ∎

The groups of (ii) are reached by no coboundary and left by no term in every convolution, so they survive in $\mathrm{Ext}^2(E, E)$ as soon as the shifts give them the spread two. That happens along every edge of a Cayley graph that crosses between two shift values of one parity, and a cut of that graph is never light.

**Lemma 18.40** (Cuts of a Cayley graph)**.** *Let $G$ be a finite abelian group, $S \subset G \setminus \{0\}$ a generating set and $w\colon S \to \mathbb{R}_{>0}$ a weight, extended by 0 to $G$. For every subset $T \subset G$ with $T \neq \emptyset, G$,*

$$\sum_{c \in T,\ e \notin T} w(e - c) \ \geq \ \min_{H} |H| \cdot w(S \setminus H),$$

*the minimum over the proper subgroups $H$ of $G$. For the group $G_0$ of the 64 ratios of pieces of one parity, with $S$ the thirteen ratios of Lemma 18.39(ii) and $w = D$, the minimum is 1024, attained only at $H = 0$, so every such cut weighs at least 1024.*

*Proof.* The left side is $\sum_{s \in S} w(s)\,|(T + s) \setminus T|$. Let $H$ be the stabiliser of $T$, a proper subgroup since $T \neq \emptyset, G$. Then $T$ and $T + s$ are unions of cosets of $H$, and for $s \notin H$ the set $(T + s) \setminus T$ is nonempty, since $|T + s| = |T|$, so it has at least $|H|$ elements. This gives the inequality. In $G_0$, written additively as the quadruples in $(\mathbb{Z}/4)^4$ of sum 0, the ratios of type $(i, -i, -1, -1)$ generate, since the differences of two of them give every $e_j - e_k$. The weights are 256 once and 64 otherwise, and $|S \cap H| \leq |H| - 1$. So $|H|\,w(S \setminus H)$ is 1024 for $H = 0$, at least $|H|\,(1024 - 256 - 64(|H| - 2))$, which is 1536, 2560 and 3072, for $|H| = 2, 4, 8$, and at least $64|H| \geq 1024$ for $|H| \geq 16$, because some generator lies outside $H$.[28] ∎

[28] The orders of the subgroups of $G_0 \cong (\mathbb{Z}/4)^3$ are powers of two. A cut at one vertex weighs exactly 1024, so the bound is sharp.

**Theorem 18.41** (One parity at two or three values)**.** *Keep Lemma 18.36, and suppose that the shifts of the pieces $L_\zeta$ with $\prod\zeta=\epsilon$ take exactly two values $s$ and $s-2$, or exactly three values $s+2$, $s$ and $s-2$, for a sign $\epsilon$ and an integer $s$. Then, for every order of the pieces, all shifts of the other pieces, every choice of the shifts of the $M_i$ and every placement of the $t_i$,*

$$\dim \mathrm{Ext}^2(E,E) \;\geq\; 1024.$$

*In particular $E$ does not meet the polarised criterion.*

*Proof.* Let $T$ be the set of pieces of parity $\epsilon$ at the shift $s$; it is neither empty nor all of them. Two pieces of parity $\epsilon$, one in $T$ and the other not, have shifts differing by exactly two. When their ratio is one of the thirteen of Lemma 18.39(ii), the group $\mathcal{H}^4$ from the piece of the larger shift to the other is reached by no coboundary and killed by no term, so its classes inject into $\mathrm{Ext}^2(E,E)$, and those of different pairs lie in different summands $\mathcal{H}(a,b)$. Their dimensions add up to the weight of the cut $(T,\complement T)$ of the Cayley graph of Lemma 18.40, since the pieces of parity $\epsilon$ form a coset of $G_0$; so they add up to at least 1024. ∎

**Theorem 18.42** (Shifts three apart)**.** *Keep Lemma 18.36, and suppose that the pieces $L_\zeta$ with $\prod\zeta=\epsilon$ all have the shift $s$ and the others have shifts in $\{s-3,s+3\}$. Then, for every order, every choice of the shifts of the $M_i$ and every placement of the $t_i$,*

$$\dim \mathrm{Ext}^2(E,E) \;\geq\; 64\cdot 1072 - 128\cdot 8 = 67584.$$

*Proof.* Every piece $W$ of parity $-\epsilon$ and every piece $V$ of parity $\epsilon$ have shifts differing by three. By Lemma 18.39(iii) the classes of degree two of the group from the higher to the lower of the two lie in $\mathcal{H}^5$, and $d_E$ vanishes on them. For each $W$ these groups have dimensions adding up to 1072.[29] A coboundary with a component in such a group runs along a path of morphisms of degree one that lowers the shift by three; no multiple lies on it, by Lemma 18.36(iii), and a step between distinct pieces lowers the shift by at least one, while no piece has a shift strictly between those of $W$ and $V$. So the path is the component between $W$ and $V$ together with a class of $H^1(A,\mathcal{O}_A)$ at one of them, and the components of all coboundaries in these groups come from $\bigoplus_a \mathcal{H}^1(a,a)$ over the 128 pieces, of dimension $128\cdot 8=1024$. The $64\cdot 1072=68608$ classes are cocycles, and at most 1024 dimensions of them are coboundaries. ∎

When one parity sits at a single shift, the groups $\mathcal{H}^3$ of Theorem 18.37 survive next to it unless pieces of the other parity crowd it from both sides.

**Proposition 18.43** (A gap of four)**.** *Keep Lemma 18.36, let the pieces $L_\zeta$ with $\prod\zeta=\epsilon$ all have the shift $y$, and let $T$ be the set of pieces of parity $-\epsilon$ at the shift $y+1$. If $T$ is not empty and no piece has the shift $y-1$ or $y+3$, then for every order, every choice of the shifts of the $M_i$ and every placement of the $t_i$, the groups $\mathcal{H}^3(W,V)$ with $W\in T$ and $V$ differing from $W$ in exactly three coordinates are reached by no coboundary and killed by no term, and*

$$\dim \mathrm{Ext}^2(E,E) \;\geq\; 640\,|T| \;\geq\; 640.$$

[29] Let the ratio move $k$ coordinates. For $k=1$ it is $-1$ in one coordinate, $D=4$, and $\mathcal{H}^5$ is $H^1$ tensored with the part of degree four of $H^*(\mathcal{O})$ on the other three surfaces, of dimension $\binom{6}{4}=15$: four partners with 60 each. For $k=2$ it is $(i,i)$ or $(-i,-i)$, $D=4$, and the part of degree three on the other two surfaces has dimension $\binom{4}{3}=4$: twelve partners with 16 each. For $k=3$ the part of degree two on the last surface has dimension one, and the 28 partners give 640 as in Theorem 18.37. For $k=4$ the group sits in degree 4. So $240+192+640=1072$.

*The same holds with $y+1$, $y-1$, $y+3$ replaced by $y-1$, $y+1$, $y-3$ and the groups $\mathcal{H}^3(V,W)$.*

*Proof.* By Theorem 18.37(i), which holds for every convolution, no coboundary reaches these groups. By Lemma 18.39(iv) a term of $d_E$ on $\mathcal{H}^3(W,V)$ runs along chains of components between pieces into $W$ and out of $V$ that lower the shift by one or two in all; the first would pass through a piece at the shift $y+2$ or $y+3$, the second through one at $y-1$ or $y-2$. The shifts $y\pm 2$ have parity $\epsilon$, whose pieces are all at $y$, and $y-1$, $y+3$ are excluded. Each $W\in T$ has 28 partners $V$, all at the shift $y$, with groups of dimensions adding up to 640. The second statement has the same proof, with the chains into $V$ and out of $W$ in place of those into $W$ and out of $V$. ∎

**Corollary 18.44** (Five consecutive shifts)**.** *Keep Lemma 18.36. If the shifts of the* 128 *pieces $L_\zeta$ lie in five consecutive integers, then, for every order, every choice of the shifts of the $M_i$ and every placement of the $t_i$, $\dim \mathrm{Ext}^2(E,E)\geq 640$; no such convolution meets the polarised criterion.*

*Proof.* Let the values be $s,\dots,s-4$, with one parity at values in $\{s,s-2,s-4\}$ and the other at values in $\{s-1,s-3\}$. If the second parity takes both its values, or the first takes two values two apart or all three, Theorem 18.41 applies. Otherwise the second parity sits at one value $y$ and the first at one value or at $\{s,s-4\}$. One value each, at an odd distance at most three: Theorem 18.37 or Theorem 18.42. The first at $\{s,s-4\}$: for $y=s-1$ the pieces at $y+1=s$ are not empty and $y-1$, $y+3$ are empty, and for $y=s-3$ the pieces at $y-1=s-4$ are not empty and $y+1$, $y-3$ are empty, so Proposition 18.43 applies. The least of the bounds is 640. ∎

Within six consecutive values two arrangements are left, up to exchanging the parities: each parity at one shift, five apart, and the parities at $\{s,s-4\}$ and $\{s-1,s-5\}$. The first is settled by the diagonal, whose classes at a shift with no other piece nearby can be reached only through all three multiples, and the second by the groups of Theorem 18.37 once more.

**Lemma 18.45** (The diagonal at a lonely shift)**.** *Keep Lemma 18.36, and let the pieces $L_\zeta$ with $\prod\zeta=\epsilon$ all have the shift $s$ and the others the shift $s-g$, for a sign $\epsilon$ and an odd integer $g\geq 5$. Let $y\in\mathcal{H}^2(a,a)$ be a diagonal class at a piece $a$, and let a term of $d_E(y)$ in Lemma 18.24(iii) be nonzero. Then $a$ is the only piece on its path, the path passes through $M_1$, $M_2$ and $M_3$, and the shifts of $a$, $M_1$, $M_2$, $M_3$ are pairwise incongruent modulo* 4.

*Proof.* Write the path as $c\to\cdots\to a\to\cdots\to e$, with the chain of components into $a$ and the chain out of $a$ not both empty, so that $c\neq e$, since the components of a convolution go up the order of the pieces. Let $L_1$ and $L_2$ be the first and the last piece on it and $X=d_{L_1}-d_{L_2}$; before $L_1$ and after $L_2$ the path is a run of distinct multiples, with $U$ steps up and $D$ steps down the value $\phi$ in all. As in the proof of Lemma 18.39(i), now with $\delta=0$, the term has Ext degree $3+X-U+7D$ by Lemma 18.36(ii). A sequence of components from one piece to another lowers the shift by Lemma 18.36(iii), and the pieces have the shifts $s$ and $s-g$; so $X\in\{0,g\}$, and $X=0$ exactly when $a$ is the only piece on the path.

If $c$ and $e$ are pieces, then $U=D=0$ and $X=g$, and $\mathcal{H}^{3+g}(c,e)=0$ since $3+g\geq 8$ and $c\neq e$. If $c$ is a multiple and $e$ a piece, then $\mathcal{H}(c,e)$ sits in degree 0 or 8. Degree 0 needs $U=3+X+7D$, so $X=D=0$ and $U=3$. Degree 8 needs $7D-U=5-X$: for $X=0$ this gives $D=1$ and $U=2$; for $X=g$ it gives $D=0$ and $U=g-5$, which is at

least one because the run before $L_1$ is not empty and at most three, so $g = 7$; but then the path lowers the shift by seven from $L_1$ to $e$, by a single component, of Ext degree eight between distinct pieces, or by a run through multiples, and neither is possible.[30] A piece $c$ with a multiple $e$ is the same with the runs exchanged. If $c$ and $e$ are multiples, both runs are nonempty and together pass through at most three multiples, so $U + D \leq 3$ and $U + D \geq 2$. Degree 0 needs $U = 3 + X + 7D$, so $X = D = 0$ and $U = 3$. Degree 8 needs $7D - U = 5 - X$; $D = 0$ gives $U = X - 5 \geq 2$, so $X = g = 7$, which is excluded as before; $D = 1$ gives $U = 2 + X \leq 2$, so $X = 0$ and $U = 2$; $D \geq 2$ gives $U \geq 9$.

So in every case $X = 0$ and $U+D = 3$: the path consists of $a$ and the three multiples, with three steps. A step up raises the shift by one and a step down lowers it by seven, so every step adds one modulo 4, and the four nodes of the path have the shifts $d_c, d_c+1, d_c+2, d_c+3$ modulo 4 in the order of the path.[31] ■

**Proposition 18.46** (Single shifts)**.** *Keep Lemma 18.36, and suppose that the pieces $L_\zeta$ with $\prod \zeta = \epsilon$ all have the shift $s$ and the others all have the shift $s - g$, for a sign $\epsilon$ and an odd integer $g$. Then, for every order, every choice of the shifts of the $M_i$ and every placement of the $t_i$,*

$$\dim \mathrm{Ext}^2(E, E) \;\geq\; 64 \cdot 28 = 1792.$$

*Proof.* Exchanging the parities, we may take $g > 0$. For $g = 1$ and $g = 3$ this is Theorem 18.37 and Theorem 18.42. Let $g \geq 5$. If a term of $d_E$ were nonzero on a diagonal class at a piece of shift $s$ and another on a diagonal class at a piece of shift $s - g$, then by Lemma 18.45 the shifts of the three multiples would be pairwise incongruent modulo 4 and miss both the residue of $s$ and that of $s - g$, which differ, since $g$ is odd. So at one of the two shifts $d_E$ vanishes on the 28 diagonal classes of each of the 64 pieces there, and by Proposition 18.29, which holds for $n = 4$, these 1792 classes inject into $\mathrm{Ext}^2(E, E)$.[32] ■

**Proposition 18.47** (Interleaved shifts)**.** *Keep Lemma 18.36, and suppose that the shifts of the pieces $L_\zeta$ with $\prod \zeta = \epsilon$ take exactly the two values $s$ and $s - 4$, and those of the others exactly the two values $s - 1$ and $s - 5$. Then, for every order, every choice of the shifts of the $M_i$ and every placement of the $t_i$, $\dim \mathrm{Ext}^2(E, E) \geq 640$.*

*Proof.* Let $A$ and $A'$ be the pieces of parity $\epsilon$ at the shifts $s$ and $s - 4$, and $B$ and $B'$ those of parity $-\epsilon$ at $s - 1$ and $s - 5$; all four are nonempty. Let $W$ and $V$ differ in exactly three coordinates, with $d_W = d_V + 1$, that is $W \in A$ and $V \in B$, or $W \in A'$ and $V \in B'$. By Theorem 18.37(i) no coboundary reaches $\mathcal{H}^3(W, V)$, and by Lemma 18.39(iv) a term of $d_E$ on it passes through a piece at the shift $d_W + 1$ or $d_W + 2$ or at $d_V - 1$ or $d_V - 2$. These are $s + 1$, $s + 2$, $s - 2$, $s - 3$ for $W \in A$ and $s - 3$, $s - 2$, $s - 6$, $s - 7$ for $W \in A'$, and no piece has these shifts. So the groups inject into $\mathrm{Ext}^2(E, E)$.

Let $\Gamma$ be the group of the ratios of two pieces, of order 128, which acts simply transitively on the pieces, and $S_3 \subset \Gamma$ the 28 ratios that move exactly three coordinates and change

[30] Between distinct pieces $\mathcal{H}^8 = 0$. A run from a piece through $m \leq 3$ distinct multiples to a piece has $m + 1$ steps, at least one up and one down, and lowers the shift by $7D' - U'$ with $U' + D' = m + 1$: by 6, 5, 4 for $D' = 1$, by 13, 12 for $D' = 2$ and by 20 for $D' = 3$, never by seven.

[31] With the values $t_1 < t_2 < t_3$ and $p$, the path with $U = 3$ visits the four nodes in the order of their values; a path with $U = 2$ and $D = 1$ ending in degree eight visits the largest $j$ values in increasing order and then the others, $j = 1, 2, 3$. So each placement has exactly four such paths, and for each the shifts of the multiples are determined by $d_a$.

[32] The proof does not look at the classes between the two parities. For $g = 5$ these are $64 \cdot 16 = 1024$ classes of degree two in groups $\mathcal{H}^7$, on which $d_E$ vanishes by Lemma 18.39(i); but the coboundaries of the loops in $\mathcal{H}^1(a, a)$, and, when two multiples have suitable shifts, the products along runs $W \to M_i \to M_j \to V$, reach them, and counting alone gives nothing there.

the parity, with the weight $D$, which adds up to 640 (Theorem 18.37). With $X = A \cup B'$ the pairs above are exactly the edges of the Cayley graph of $(\Gamma, S_3)$ between $X$ and its complement $A' \cup B$; so the dimensions add up to the weight of that cut, and $X$ is neither empty nor everything. Written additively, as the quadruples in $(\mathbb{Z}/4)^4$ of even sum, the differences of two ratios of type $(i, -i, -1, 1)$ give every $e_j - e_k$, which generate the quadruples of sum 0 modulo 4, and the elements of $S_3$ have sum 2 modulo 4; so $S_3$ generates $\Gamma$. In Lemma 18.40, $H = 0$ gives 640; for $|H| = 2, 4, 8, 16$ at most $|H| - 1$ elements of $S_3$ lie in $H$, at most four of them of weight 64, so $|H|\, w(S_3 \setminus H)$ is at least $2 \cdot 576$, $4 \cdot 448$, $8 \cdot 336$ and $16 \cdot 208$; for $|H| = 32$ some $r \in S_3$ lies outside $H$, and so does $-r$, which gives weight at least 32, or 64 if $r = -r$; and for $|H| = 64$ the weight outside $H$ is at least 16. So every such cut weighs at least 640.[33] ■

**Corollary 18.48** (Six consecutive shifts)**.** *Keep Lemma 18.36. If the shifts of the* 128 *pieces $L_\zeta$ lie in six consecutive integers, then, for every order, every choice of the shifts of the $M_i$ and every placement of the $t_i$,* $\dim \mathrm{Ext}^2(E, E) \geq 640$*; no such convolution meets the polarised criterion.*

*Proof.* By Corollary 18.44 we may assume that the shifts take both values $s$ and $s - 5$ of the window $s, \dots, s - 5$; let the parity at $s$ have its values in $\{s, s - 2, s - 4\}$ and the other in $\{s - 1, s - 3, s - 5\}$. If one of them takes two values two apart, or three, Theorem 18.41 applies. Otherwise the first takes $\{s\}$ or $\{s, s - 4\}$ and the second $\{s - 5\}$ or $\{s - 1, s - 5\}$. For $\{s\}$ and $\{s - 5\}$ this is Proposition 18.46, and for $\{s, s - 4\}$ and $\{s - 1, s - 5\}$ it is Proposition 18.47. For $\{s\}$ and $\{s - 1, s - 5\}$, Proposition 18.43 applies in its second form with $y = s$: the shift $s - 1$ is taken and $s + 1$, $s - 3$ are not. For $\{s, s - 4\}$ and $\{s - 5\}$ it applies in its first form with $y = s - 5$: the shift $s - 4$ is taken and $s - 6$, $s - 2$ are not. The least of the bounds is 640. ■

For the arrangements outside these results the diagonal still carries a large part of the obstruction.

**Proposition 18.49** (Cup products on the diagonal at $n = 4$)**.** *Keep Lemma 18.36, so that $D^2$ has dimension* $131 \cdot 28 = 3668$*. For every convolution the cup products vanish on the* 84 *diagonal classes of the $M_i$, and on the* 3584 *diagonal classes of the $L_\zeta$ they have a kernel of dimension at least* 400*. So their kernel on $D^2$ has dimension at least* 484*, and a convolution meeting the polarised criterion needs products of length at least three that remove at least* 380 *diagonal classes.*

*Proof.* As in the proof of Proposition 18.30, the cup products in $\mathcal{H}(a, b)$ are $(y_a - y_b) \wedge \delta^H_{ab}$, they vanish on the classes of the $M_i$, and for $L_\zeta$, $L_{\zeta'}$ and a type $\tau$ of degree two on the four surfaces they vanish on the part of type $\tau$ unless $\zeta'/\zeta$ fixes the coordinates in the support of $\tau$. So a class whose part of type $\tau$ depends only on the coordinates of $\zeta$ in the support of $\tau$ has vanishing cup products. The four types $(2, 0, 0, 0)$ have dimension one and four values of the coordinate, the six types $(1, 1, 0, 0)$ have dimension four and sixteen values of the two coordinates: $4 \cdot 4 + 6 \cdot 16 \cdot 4 = 400$. By Proposition 18.29, which holds for $n = 4$, the kernel of $d_E$ on $D^2$ injects into $\mathrm{Ext}^2(E, E)$, and it is at most 104 only if the longer products remove 380 dimensions of these 484. ■

Figure 30 and Figure 31 draw the arrangements and the bounds.

[33] The subgroups of $\Gamma \cong (\mathbb{Z}/4)^3 \times \mathbb{Z}/2$ have orders that are powers of two. A cut at a single vertex weighs exactly 640, so the bound is sharp.

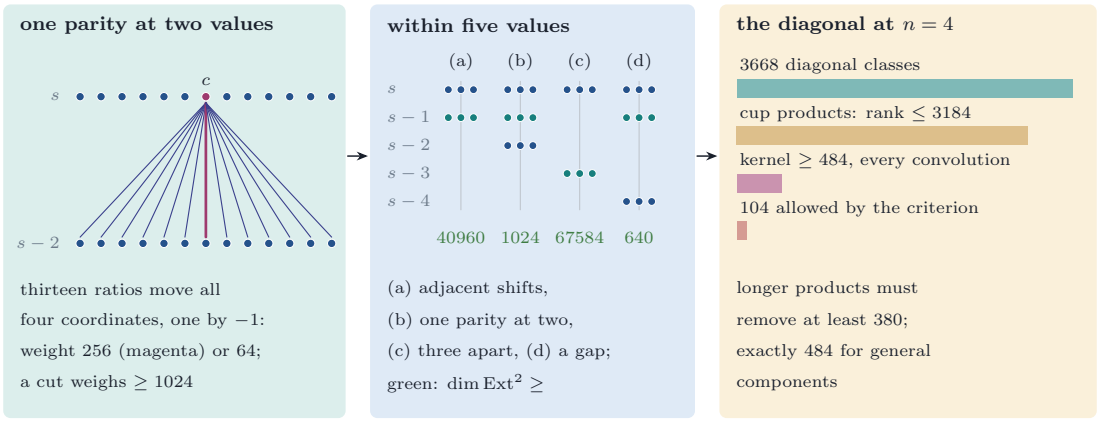



FIGURE 30. The shift arrangements of the pieces $L_\zeta$ on $E_0^8$. Left: one parity at two values two apart; a piece $c$ is joined to the thirteen pieces whose ratio moves every coordinate, one by $-1$, and the groups $\mathcal{H}^4$ of the edges crossing between the values survive, weighing at least 1024 (Lemma 18.39(ii), Lemma 18.40). Middle: the four kinds of arrangement within five consecutive values, with the result that excludes each and its bound on $\dim \mathrm{Ext}^2$. Right: the diagonal at $n = 4$; the cup products leave at least 484 of the 3668 classes, and the criterion allows 104 (Proposition 18.49).

*Remark* 18.50 (What the route gives). At $n = 3$ the route through convolutions of these pieces is decided. The diagonal obstruction of Theorem 18.31 leaves room only for the fourfold products along one path, and the explicit convolution of Proposition 18.33 solves the Maurer–Cartan equation. The obstruction that decides the route lies off the diagonal, in classes that no product of any length can reach: those between pieces that differ in every coordinate and have adjacent shifts (Lemma 18.34), and, at pieces whose shifts are not adjacent to the shift of the paths, the classes of type $(1, 1, 0)$. No order and no choice of shifts or twists avoids both, and Theorem 18.35 excludes every convolution of the 35 pieces; in the explicit convolution the first kind alone gives 2560 classes against the 57 allowed. In each placement exactly two of the eight sign patterns carry products, and the two are exchanged by a symmetry of the construction.[34] The member $E_0^6$ with this Weil structure has trivial discriminant, where the Weil classes are algebraic on every member by Markman's theorem [Mar25a, Mar25b], so a convolution meeting the criterion here would have given a second proof of a known case rather than a new one. At $n = 4$ the classes between pieces that differ in three coordinates and have adjacent shifts are again reached by nothing, in every convolution, and when the pieces $L_\zeta$ occupy two shifts, as in the explicit convolution, no term of $d_E$ leaves them, so that $\dim \mathrm{Ext}^2(E, E) \geq 40960$ against the 104 allowed (Theorem 18.37). With three consecutive shifts the differential reaches these classes, but only through cup products into the groups between the highest and the lowest shift, and at least 22528 of them survive (Theorem 18.38). The groups of spread three, and those of spread two whose ratio moves every coordinate, one of them by

[34] The automorphism of $A$ that is $-1$ on the coordinate $z_{n+1}$ and the identity on the others changes $\alpha$ into $-\alpha$, fixes $\theta$ and permutes the $L_\zeta$, changing $\zeta_1$ into $-\zeta_1$; composed with the shift $[1]$ it changes every $\sigma_i$ into $-\sigma_i$ and keeps everything else.

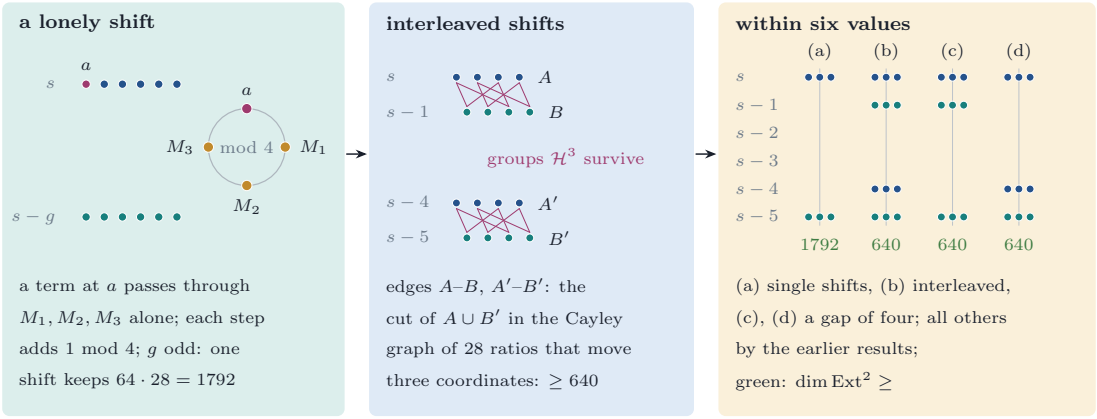



FIGURE 31. The arrangements within six consecutive values on $E_0^8$. Left: the parities at single shifts $s$ and $s-g$, $g$ odd and at least five; a term on a diagonal class at a piece $a$ runs through $M_1$, $M_2$, $M_3$ and no other piece, and since each step adds one to the shift modulo 4 the four nodes have distinct residues, so the multiples serve at most one of the two shifts and the 1792 diagonal classes at the other survive (Lemma 18.45, Proposition 18.46). Middle: the interleaved shifts; the groups $\mathcal{H}^3$ between $A$ and $B$ and between $A'$ and $B'$ survive and form a cut of the Cayley graph of the 28 ratios that move three coordinates, of weight at least 640 (Proposition 18.47). Right: the four arrangements left within six values by the results for five values, each with its bound on $\dim \mathrm{Ext}^2$ (Corollary 18.48).

$-1$, are untouched as well, in every convolution (Lemma 18.39): the latter cross every cut between two shift values of one parity, and weigh at least 1024 there (Theorem 18.41); with one parity at a single shift, those of spread three, or of spread one beside a gap, survive in their turn (Theorem 18.42, Proposition 18.43). Together they exclude every convolution whose pieces $L_\zeta$ have their shifts within five consecutive values (Corollary 18.44). The two arrangements left within six values fall as well: with each parity at a single shift, at any odd distance, the diagonal classes at one of the two shifts can be reached only through all three multiples, and the multiples serve at most one of them (Lemma 18.45, Proposition 18.46); the interleaved shifts leave the groups of Theorem 18.37 on a cut (Proposition 18.47). So every convolution whose shifts lie in six consecutive values fails (Corollary 18.48). For wider arrangements the cup products still leave 484 diagonal classes, so products of length at least three would have to remove 380 (Proposition 18.49), and products of length three and four among the $L_\zeta$ pass every test of Theorem 18.31. Whether some convolution of the 131 pieces whose shifts do not lie in six consecutive values, and whose parities do not each sit at one shift, meets the criterion at $n = 4$ is open. Markman's theorem does not reach the eightfolds of Weil type, so such a convolution would give a new case. Figure 26 shows the excess count and the surviving path, and Figure 27 the budget of diagonal classes.

*Markman's candidate in dimension eight.* Above $n = 3$ there is one concrete candidate of this kind. Markman proposes it in [Mar25b, Section 12] and notes that it remains to be checked against the weakened criterion. Its Chern character is a finite computation in the ring $\mathbb{Q}[\Theta]/(\Theta^5)$. We carry the computation out: it places the candidate at a rational

point of the secant plane of Definition 14.4 and reduces what has to be checked to a single condition, which Theorem 18.61 decides in the negative.

**Setup 18.51** (Markman's object)**.** Let $(X,\Theta)$ be a principally polarised abelian fourfold, $d\geq 3$ odd and squarefree, $K=\mathbb{Q}(\sqrt{-d})$ and $N=(d+9)/2$. Let $D_i$, $i\in\mathbb{Z}/N\mathbb{Z}$, be $N$ translates of $\Theta$ in general position: any two, three or four of them meet transversally, in a surface, a curve and $\Theta^4=24$ points, and any five have empty intersection. Put $Z_i=D_i\cap D_{i+1}$, $Z=\bigcup_i Z_i$, let $C_i=Z_i\cap Z_{i+1}=D_i\cap D_{i+1}\cap D_{i+2}$ be the curves along which consecutive components meet, and let $\nu\colon\widetilde{Z}\to Z$ be the partial normalisation of $Z$ at $m=(d+9)(2d-1)$ of its isolated points of self-intersection, the points of $Z_i\cap Z_j$ for $i,j$ at cyclic distance at least three. Put

$$\mathcal{F}=[\mathcal{O}_X\longrightarrow\nu_*\mathcal{O}_{\widetilde{Z}}]\otimes\mathcal{O}_X(3\Theta),$$

the complex placed in degrees 0 and 1.

**Proposition 18.52** (Markman's candidate on the secant plane)**.** *In Setup 18.51, with $t=\Theta$ and $u_t,v_t$ as in (30) for $n=4$:*

(i) *$\chi(\mathcal{O}_{Z_i})=14$, $\chi(\mathcal{O}_{C_i})=-36$, the isolated self-intersection points of $Z$ number $12N(N-5)=(d+9)(3d-3)$, and $\chi(\mathcal{O}_{\widetilde{Z}})=110N-12N^2+2N(2d-1)=(d+9)(27-d)$;*

(ii) *$\operatorname{ch}(\nu_*\mathcal{O}_{\widetilde{Z}})=N\Theta^2-2N\Theta^3+\chi(\mathcal{O}_{\widetilde{Z}})\,[\mathrm{pt}]$;*

(iii) *$\operatorname{ch}(\mathcal{F})=u_t+3v_t$, that is*

$$\operatorname{ch}(\mathcal{F})=1+3\Theta-\tfrac{d}{2}\Theta^2-\tfrac{d}{2}\Theta^3+\tfrac{d^2}{24}\Theta^4,$$

*the rational point $(a,b)=(1,3)$ of the secant plane $P_\Theta$; with the twist $\mathcal{O}_X(\Theta)$ in place of $\mathcal{O}_X(3\Theta)$ the degree two component would instead be $-\frac{d+8}{2}\Theta^2$, so the twist in* [Mar25b, Section 12] *must be read as $3\Theta$;*

(iv) *$\mathrm{N}_{K/\mathbb{Q}}(3+\sqrt{-d})=d+9=2N$, with $N$ the number of divisors, and by Theorem 14.7 the Bogomolov discriminant of $\mathcal{F}$ is $\Delta(\mathcal{F})=(d+9)\Theta^2$;*

(v) *the objects $\Phi(\mathcal{F}\boxtimes\mathcal{F})$ and $\Phi(\mathcal{F}\boxtimes\mathcal{F}^\vee)$ on $X\times\widehat{X}$ have ranks $\int_X\operatorname{ch}(\mathcal{F})^2=8d(d-9)$ and $\int_X\operatorname{ch}(\mathcal{F})\operatorname{ch}(\mathcal{F}^\vee)=8d(d+9)$, up to sign, both nonzero for squarefree $d$; the same formula gives $-8q$ for the pair of secant sheaves of* [Mar25b, Section 11] *on a threefold, which is his rank $8q$ up to the shift.*

*Consequently $\mathcal{F}\boxtimes\mathcal{F}$ satisfies conditions (1), (2b) and (2c) of Markman's strategy* [Mar25b, Section 4]*: the transform has nonzero rank, its normalised Chern character is invariant under the special Mumford–Tate group by* [Mar25b, Corollary 9.2]*, and its Weil component is nonzero by* [Mar25b, Proposition 10.2]*, the relevant coefficient being $\mathrm{N}(c)$ with $c=\frac{1}{2}(1+3/\sqrt{-d})\neq 0$. What remains is condition (2a) in its weakened form: for $E=\Phi(\mathcal{F}\boxtimes\mathcal{F})$, the map $\mathrm{ev}_E$ must kill every class in $H^1(T)\oplus H^0(\bigwedge^2 T)$ that preserves $\kappa(E)$ to first order.*

*Proof.* On a principally polarised abelian fourfold $\chi(\mathcal{O}_X(k\Theta))=k^4$ and $\Theta^4=24$. The Koszul complex of $r$ transversal translates of $\Theta$ gives ch of the structure sheaf of their intersection as $(1-e^{-\Theta})^r$, so $\operatorname{ch}(\mathcal{O}_{Z_i})=\Theta^2-\Theta^3+\frac{7}{12}\Theta^4$, $\operatorname{ch}(\mathcal{O}_{C_i})=\Theta^3-\frac{3}{2}\Theta^4$ and ch of four divisors is $\Theta^4$. Integrating gives $\chi(\mathcal{O}_{Z_i})=14$, $\chi(\mathcal{O}_{C_i})=-36$ and 24 points; the same numbers are the Koszul sums $0-2\cdot 1+16$, $0-3+48-81$ and $0-4+96-324+256$ of the values $\chi(\mathcal{O}_X(-k\Theta))=k^4$.

(i), (ii). At every point $Z$ is locally a union of coordinate subspaces of $\mathbb{C}^4$: along $C_i$ two surfaces $\{x_1=x_2=0\}\cup\{x_2=x_3=0\}$, at a point of $C_i\cap C_{i+1}$ three of them, at an isolated point $\{x_1=x_2=0\}\cup\{x_3=x_4=0\}$. The ideals of such an arrangement are

monomial and distribute over sums, so the Mayer–Vietoris complex

$$0 \to \mathcal{O}_Z \to \bigoplus_i \mathcal{O}_{Z_i} \to \bigoplus_{i<j} \mathcal{O}_{Z_i \cap Z_j} \to \bigoplus_{i<j<k} \mathcal{O}_{Z_i \cap Z_j \cap Z_k} \to \cdots$$

with scheme-theoretic intersections is exact. Pairs at cyclic distance one meet in $C_i$, $N$ of them; pairs at distance at least two meet in the 24 points of four divisors, $N(N-3)/2$ of them; the only triples with nonempty intersection are the consecutive ones $Z_i \cap Z_{i+1} \cap Z_{i+2} = C_i \cap C_{i+1}$, 24 points each, $N$ of them; and no four components meet, since that would involve at least five divisors. Hence

$$\operatorname{ch}(\mathcal{O}_Z) = N \operatorname{ch}(\mathcal{O}_{Z_i}) - N \operatorname{ch}(\mathcal{O}_{C_i}) - \tfrac{N(N-3)}{2}\Theta^4 + N\Theta^4,$$

whose degree two part is $N\Theta^2$, degree three part $-N\Theta^3 - N\Theta^3 = -2N\Theta^3$, and degree four part $\chi(\mathcal{O}_Z)[\mathrm{pt}]$ with $\chi(\mathcal{O}_Z) = 14N + 36N - 12N(N-3) + 24N = 110N - 12N^2$. The isolated singular points are those of pairs at distance at least three, of which there are $N(N-5)/2$, giving $12N(N-5) = 3(d+9)(d-1)$ points; the pairs at distance two lie on the double curves and are not isolated. At an isolated point the two branches meet transversally, and separating them gives $0 \to \mathcal{O}_Z \to \nu_*\mathcal{O}_{\widetilde{Z}} \to k(p) \to 0$, so each of the $m$ normalisations raises $\chi$ by one and leaves the lower degrees of ch unchanged. With $N = (d+9)/2$, $\chi(\mathcal{O}_{\widetilde{Z}}) = 110N - 12N^2 + 2N(2d-1) = (d+9)(55 - 3(d+9) + 2d - 1) = (d+9)(27-d)$.

(iii) $\operatorname{ch}(\mathcal{F}) = e^{3\Theta}(1 - \operatorname{ch}(\nu_*\mathcal{O}_{\widetilde{Z}}))$. The degree two part is $(\frac{9}{2} - N)\Theta^2 = -\frac{d}{2}\Theta^2$, which is where $N = (d+9)/2$ enters; with the twist $k\Theta$ it would be $(\frac{k^2}{2} - N)\Theta^2$, equal to $-\frac{d}{2}\Theta^2$ exactly for $k^2 = 9$, and $k = -3$ is excluded by the degree three part. The degree three part is $(\frac{27}{6} - 3N + 2N)\Theta^3 = (\frac{9}{2} - N)\Theta^3 = -\frac{d}{2}\Theta^3$, and the degree four part is $(\frac{81}{24} - \frac{9N}{2} + 6N - \frac{\chi(\mathcal{O}_{\widetilde{Z}})}{24})\Theta^4 = \frac{81 + 36N - \chi(\mathcal{O}_{\widetilde{Z}})}{24}\Theta^4 = \frac{d^2}{24}\Theta^4$ by (i). Comparing with (30) at $n = 4$, $u_t = 1 - \frac{d}{2}t^2 + \frac{d^2}{24}t^4$ and $v_t = t - \frac{d}{6}t^3$, gives $\operatorname{ch}(\mathcal{F}) = u_t + 3v_t$.

(iv) This follows from $\mathrm{N}(3 + \sqrt{-d}) = 9 + d$ and (31) with $(a, b) = (1, 3)$.

(v) Orlov's equivalence is $\Phi = (\mathrm{id} \times \Phi_{\mathcal{P}})^{-1} \circ \mu^*$ with $\mu(x, y) = (x + y, y)$, so the restriction of $\Phi(\mathcal{F}_1 \boxtimes \mathcal{F}_2)$ to $\{x\} \times \widehat{X}$ is the Fourier–Mukai transform of $\tau_x^*\mathcal{F}_1 \otimes \mathcal{F}_2$, whose derived fibre at $L \in \widehat{X}$ is $R\Gamma(X, \tau_x^*\mathcal{F}_1 \otimes \mathcal{F}_2 \otimes L^{\pm 1})$; the rank of a perfect complex is the Euler characteristic of its derived fibre, and $\chi(\tau_x^*\mathcal{F}_1 \otimes \mathcal{F}_2 \otimes L) = \int_X \operatorname{ch}(\mathcal{F}_1)\operatorname{ch}(\mathcal{F}_2)$ by Riemann–Roch, translations and $L \in \mathrm{Pic}^0$ acting trivially on cohomology. With $\operatorname{ch}(\mathcal{F}^\vee) = u_t - 3v_t$, the involution $\tau$ changing the sign of the odd degrees, $\int u_t^2 = 8d^2$, $\int u_t v_t = 0$ and $\int v_t^2 = -8d$ give the two values. On a threefold $\int (u_t + v_t)^2 = 2\int u_t v_t = -8q$.

For the final assertion, condition (2b) is the $\mathrm{Spin}(V_{\mathbb{Q}})_B$-invariance of $\kappa(E)$, which [Mar25b, Corollary 9.2] derives from $\operatorname{ch}(\mathcal{F}) \in B$ and the nonvanishing of the rank. Condition (2c) is [Mar25b, Proposition 10.2], which for $K$ imaginary quadratic reduces to the nonvanishing of the component of $\operatorname{ch}(\mathcal{F}) \otimes \operatorname{ch}(\mathcal{F})$ in $\ell_W \otimes \ell_{\overline{W}}$, and writing $\operatorname{ch}(\mathcal{F}) = c\,\ell_W + \bar{c}\,\ell_{\overline{W}}$ with $\ell_W = e^{\sqrt{-d}\Theta}$ gives that coefficient as $c\bar{c}$. ∎

*Remark* 18.53 (The remaining condition). Proposition 18.52 does not verify Markman's candidate. It verifies every statement about the candidate that concerns classes, and it shows that the one remaining statement concerns the object. For $E = \Phi(\mathcal{F} \boxtimes \mathcal{F})$ on the eightfold $X \times \widehat{X}$ of Weil type, with $\kappa(E) = \operatorname{ch}(E)e^{-c_1(E)/\operatorname{rk} E}$, that statement reads: every $x \in H^1(T) \oplus H^0(\bigwedge^2 T)$ with $x \lrcorner \kappa(E) = 0$ satisfies $\mathrm{ev}_E(x) = 0$ in $\mathrm{Ext}^2(E, E)$. The $H^2(\mathcal{O})$ summand imposes nothing, because $\operatorname{rk} E \neq 0$ makes $z \cdot \kappa(E) \neq 0$ for $z \neq 0$. By Theorem 9.20 the condition fails as soon as $E$ has a rank one direct summand whose class is not a multiple of the polarisation, so the first question about $E$ is whether it is

indecomposable. Markman's object at $n = 3$ is reflexive of rank $8q$ away from a divisor, and he proves the condition by descending to a quotient on which $\sigma_E$ is injective on an invariant subspace [Mar25b, Proposition 11.5]. Whether the same can be done for $\Phi(\mathcal{F} \boxtimes \mathcal{F})$ is a question about the Ext groups of one object on one eightfold, not about any Hodge class. Theorem 18.56 below moves the question from the eightfold to the fourfold and from $E$ to $\mathcal{F}$ itself: the condition is equivalent to the single identity (58) in $\mathrm{Ext}^2(\mathcal{F}, \mathcal{F} \otimes \Omega_X^2)$, the other half of it holding by construction. Theorem 18.61 shows that the identity fails, so the answer is negative.

*Reduction to the fourfold.* The condition of Remark 18.53 concerns an object on an eightfold. It is equivalent to a condition on the factor $\mathcal{F}$ on the fourfold, and there it splits into two conditions, one of which holds for Markman's object because the object is constructed uniformly over the moduli of principally polarised fourfolds. The reduction is not special to that object. It works for every secant object on every abelian $n$-fold with $n \geq 3$, and its ingredients are three computations in the Hodge algebra of a single abelian variety, which we carry out by hand. We first fix conventions.

**Setup 18.54** (The weakened criterion for one object)**.** Let $Y$ be an abelian variety and $F$ a nonzero perfect complex on $Y$. Write $HT^2(Y) = H^{0,2}(Y) \oplus H^1(T_Y) \oplus H^0(\bigwedge^2 T_Y)$ as in (49), with $x = (z, v, \pi)$ acting on $H^*(Y, \mathbb{C})$ by $x \lrcorner$, and likewise $HT^1(Y) = H^{0,1}(Y) \oplus H^0(T_Y)$, with $(z, \theta)$ acting by $z\wedge$ and by the interior product $\theta \lrcorner$. Normalise the Atiyah class $\mathrm{At}(F) \in \mathrm{Ext}^1(F, F \otimes \Omega_Y^1)$ so that $\mathrm{ch}(F) = \mathrm{tr} \exp \mathrm{At}(F)$; for a line bundle $L$ it is then $c_1(L)$, as in (47).[35] The evaluation map of Theorem 9.19 is, under HKR,

$$\mathrm{ev}_F(z, v, \pi) = z \cdot \mathrm{id}_F + v \lrcorner \mathrm{At}(F) + \tfrac{1}{2}\, \pi \lrcorner \mathrm{At}(F)^2 \ \in\ \mathrm{Ext}^2(F, F), \tag{54}$$

the contraction of the polyvector with the exponential Atiyah class [Hua21, Theorem A], [BF08]; its restriction to $H^1(T_Y)$ is the obstruction map, whose kernel is the space of first order deformations of $Y$ to which $F$ extends [Tod09], [Mar25a, Section 8.3]. Put

$$\mathrm{Ann}_{HT^2}(\mathrm{ch}\, F) = \{x \in HT^2(Y) : x \lrcorner\, \mathrm{ch}(F) = 0\},$$

the Hochschild classes preserving $\mathrm{ch}(F)$ to first order. We say that $F$ *satisfies the weakened criterion* if $\mathrm{ev}_F$ vanishes on $\mathrm{Ann}_{HT^2}(\mathrm{ch}\, F)$. By the commutative square $\sigma_F \circ \mathrm{ev}_F = (\,\cdot\,) \lrcorner\, \mathrm{ch}(F)$ of [BF08, Proposition 6.2.1 and Corollary 6.3.2], in the form quoted in [Mar25b, Lemma 11.3],

$$\mathrm{ev}_F\big(\mathrm{Ann}_{HT^2}(\mathrm{ch}\, F)\big) = \ker \sigma_F \cap \mathrm{im}\, \mathrm{ev}_F,$$

so the weakened criterion is exactly the hypothesis of [Mar25b, Question 11.4], that the restriction of $\sigma_F$ to the image of $\mathrm{ev}_F$ is injective. For a split object this is Proposition 18.18(ii).

**Lemma 18.55** (The $B$-field transform)**.** *Let $B \in H^{1,1}(Y)$. For $x = (z, v, \pi) \in HT^2(Y)$ put*

$$e^B x = (z + v \lrcorner B + \tfrac{1}{2}\, \pi \lrcorner B^2,\ v + \pi \lrcorner B,\ \pi), \tag{55}$$

*where $\pi \lrcorner B \in \mathrm{Hom}(H^{1,0}, H^{0,1}) = H^1(T_Y)$ is the contraction of one slot of $\pi$ with $B$: for $\pi = \partial_a \wedge \partial_b$ it sends $p_b$ to $\partial_a \lrcorner B$, $p_a$ to $-\partial_b \lrcorner B$ and the other basis vectors to zero. Then $e^B$ is a linear automorphism of $HT^2(Y)$ with inverse $e^{-B}$, and:*

(i) *$x \lrcorner (w\, e^B) = \big((e^B x) \lrcorner w\big)\, e^B$ for every even class $w \in H^{\mathrm{ev}}(Y, \mathbb{C})$;*

[35] This is the convention of [Tod09] and of [Mar25a, footnote 14]. Buchweitz and Flenner use the opposite sign, which replaces $\mathrm{At}(F)$ by $-\mathrm{At}(F)$; this leaves $\mathrm{At}(F)^2$ unchanged and changes only the sign of the middle term of $\mathrm{ev}_F$, which does not affect the vanishing statements below.

(ii) *for a line bundle $L$ with $c_1(L) = B$, $\mathrm{ev}_{F\otimes L}(x) = \mathrm{ev}_F(e^B x)$ and $\mathrm{Ann}_{HT^2}(\mathrm{ch}(F\otimes L)) = e^{-B}\,\mathrm{Ann}_{HT^2}(\mathrm{ch}\,F)$; hence $F$ satisfies the weakened criterion if and only if $F\otimes L$ does.*

*The computation in (ii) is formal: it uses only that the twisted object has Atiyah class $\mathrm{At}(F) + B\cdot\mathrm{id}_F$ and Chern character $\mathrm{ch}(F)e^B$. It therefore applies verbatim to the twisted sheaves of* [Mar25b, Section 2]*, with $B$ rational. In particular, for $F$ of nonzero rank $r$, the criterion for the twisted object $F\otimes L^{-1}$ with class $\kappa(F) = \mathrm{ch}(F)e^{-c_1(F)/r}$ is the same condition as the criterion for $F$ with class $\mathrm{ch}(F)$.*

*Proof.* The computation takes place in a graded algebra on which the interior products $\iota_a = \partial_a \lrcorner$ act as odd derivations. It uses only that $w$ is even; that $B$ is even and central with $\iota_a\iota_b B = 0$, which holds for $B$ of type $(1,1)$ and $\partial_a, \partial_b$ of type $(1,0)$; and that the odd scalar classes $\iota_a B$ anticommute with every odd element. In $H^*(Y,\mathbb{C})$ these properties are clear. They also hold in $\bigoplus_{p,q}\mathrm{Ext}^p(F, F\otimes\Omega_Y^q) = \mathrm{Ext}^*(F,F)\otimes\bigwedge^* H^{1,0}$, on which the polyvectors act through the second factor by odd derivations, because the classes coming from $H^*(Y,\mathbb{C})$ are graded central in the Yoneda algebra. First, $\iota_a(B^k) = k(\iota_a B)B^{k-1}$, since $\iota_a B$ is odd and $B$ is even, so $\iota_a e^B = (\iota_a B)e^B$. For $w$ even this gives $v\lrcorner(we^B) = (v\lrcorner w)e^B + w\sum_a v(p_a)(\iota_a B)e^B = (v\lrcorner w + (v\lrcorner B)w)e^B$, and $z(we^B) = (zw)e^B$. For the bivector, with $\pi\lrcorner = \iota_a\iota_b$,

$$\iota_b(we^B) = (\iota_b w)e^B + w(\iota_b B)e^B,$$

and applying $\iota_a$, using that $\iota_b w$ and $\iota_b B$ are odd,

$$\iota_a\iota_b(we^B) = \big[\iota_a\iota_b w + (\iota_a B)\iota_b w - (\iota_b B)\iota_a w + w\big((\iota_a\iota_b B) - (\iota_b B)(\iota_a B)\big)\big]e^B.$$

The middle two terms are $(\pi\lrcorner B)\lrcorner w$ for the element $\pi\lrcorner B$ of $\mathrm{Hom}(H^{1,0}, H^{0,1})$ displayed in the statement, acting as a derivation, and the last bracket is $\frac12\iota_a\iota_b(B^2) = \iota_a((\iota_b B)B)$. Summing the three actions gives (i). Applying (i) with $B$ replaced by $-B$ and then by $B$, and using $(\pi\lrcorner B)\lrcorner B = 2(\iota_a B)(\iota_b B) = \pi\lrcorner B^2$ when $\iota_a\iota_b B = 0$, shows $e^{-B}e^B = \mathrm{id}$.

(ii) The Atiyah class of $F\otimes L$ is $\mathrm{At}(F) + B\cdot\mathrm{id}_F$, the Atiyah class being additive on tensor products [BF03, Section 3], and the two summands commute, so $\exp\mathrm{At}(F\otimes L) = \exp\mathrm{At}(F)\cdot e^B$. By (i) in the algebra $\mathrm{Ext}^*(F,F)\otimes\bigwedge^* H^{1,0}$, with $w = \exp\mathrm{At}(F)$,

$$x\lrcorner\exp\mathrm{At}(F\otimes L) = \big((e^B x)\lrcorner\exp\mathrm{At}(F)\big)e^B,$$

and the right hand side has the same component in $\mathrm{Ext}^2(F,F)\otimes\bigwedge^0$ as $(e^B x)\lrcorner\exp\mathrm{At}(F)$, because $e^B - 1 = B + \frac12 B^2 + \cdots$ raises the exterior degree. This is $\mathrm{ev}_{F\otimes L}(x) = \mathrm{ev}_F(e^B x)$. By (i) in $H^*(Y,\mathbb{C})$ with $w = \mathrm{ch}(F)$, $x\lrcorner\mathrm{ch}(F\otimes L) = 0$ if and only if $(e^B x)\lrcorner\mathrm{ch}(F) = 0$, which is the statement on the annihilators. Hence $\mathrm{ev}_{F\otimes L}(\mathrm{Ann}_{HT^2}(\mathrm{ch}(F\otimes L))) = \mathrm{ev}_F(\mathrm{Ann}_{HT^2}(\mathrm{ch}\,F))$. ∎

**Theorem 18.56** (Reduction to the factor)**.** *Let $(X,t)$ be a polarised abelian variety of dimension $n\ge 3$, let $d\ge 1$, and let $F$ be a perfect complex on $X$ with $\mathrm{ch}(F) = a\,u_t + b\,v_t$ for some $(a,b)\in\mathbb{Q}^2\setminus\{0\}$, in the notation of* (30)*. Then:*

(i) *No nonzero class in $HT^1(X)$ preserves $\mathrm{ch}(F)$: the map $HT^1(X)\to H^*(X,\mathbb{C})$, $y\mapsto y\lrcorner\mathrm{ch}(F)$, is injective. Its two lowest components form the matrix $\left(\begin{smallmatrix} a & b\\ b & -ad\end{smallmatrix}\right)$ of determinant $-(a^2d+b^2) = -\mathrm{N}_{K/\mathbb{Q}}(b + a\sqrt{-d}) \neq 0$.*

(ii) *The classes in $HT^2(X)$ preserving $\mathrm{ch}(F)$ to first order are*

$$\mathrm{Ann}_{HT^2}(\mathrm{ch}\,F) = \{(0,v,0) : v\lrcorner t = 0\}\ \oplus\ \{(\tfrac{d}{2}\,\pi\lrcorner t^2,\ 0,\ \pi) : \pi\in H^0(\textstyle\bigwedge^2 T_X)\}, \qquad (56)$$

*the polarised first order deformations of $(X,t)$, of dimension $n(n+1)/2$, and the Poisson directions each compensated by a class in $H^{0,2}$, of dimension $n(n-1)/2$; the total is $n^2=\dim\mathcal{D}_{n,n}$, and the space does not depend on $(a,b)$.*

(iii) *$F$ satisfies the weakened criterion if and only if*
   (a) *$v\lrcorner\,\mathrm{At}(F)=0$ in $\mathrm{Ext}^2(F,F)$ for every $v\in H^1(T_X)$ with $v\lrcorner t=0$, that is, $F$ extends to first order along every polarised deformation of $(X,t)$; and*
   (b) *$\mathrm{At}(F)^2=-d\,t^2\cdot\mathrm{id}_F$ in $\mathrm{Ext}^2(F,F\otimes\Omega_X^2)$.*

   *The trace of (b) is automatic: $\operatorname{tr}\mathrm{At}(F)^2=2\,\mathrm{ch}_2(F)=-ad\,t^2=\operatorname{tr}(-d\,t^2\cdot\mathrm{id}_F)$, so (b) is the vanishing of the trace free part of the square of the Atiyah class.*

(iv) *Let $F_1,F_2$ be two such complexes, $G=F_1\boxtimes F_2$ on $X\times X$ and $E=\Phi(G)$ on $X\times\widehat{X}$, with $\Phi$ Orlov's equivalence as in Proposition 18.52(v). Then $\mathrm{Ann}_{HT^2}(\mathrm{ch}\,G)=\mathrm{Ann}_{HT^2}(\mathrm{ch}\,F_1)\otimes 1\oplus 1\otimes\mathrm{Ann}_{HT^2}(\mathrm{ch}\,F_2)$, of dimension $2n^2$, and*

$$E\text{ satisfies the weakened criterion}\iff G\text{ does}\iff F_1\text{ and }F_2\text{ do.}$$

*Proof.* Choose a basis $p_1,\dots,p_n$ of $H^{1,0}(X)$ and a basis $q_1,\dots,q_n$ of $H^{0,1}(X)$ with $t=\sum_a p_a\wedge q_a$, which is possible because $t$ is a positive class of type $(1,1)$; let $\partial_a$ be the dual basis of $T$, so that $\iota_a p_b=\delta_{ab}$ and $\iota_a q_b=0$. Write $\mathrm{ch}(F)=\sum_k c_k t^k$ with $c_k\,k!=a(-d)^{k/2}$ for $k$ even and $c_k\,k!=b(-d)^{(k-1)/2}$ for $k$ odd, so that

$$c_{k+1}\,(k+1)!=-d\,c_{k-1}\,(k-1)!\qquad(k\geq 1).\tag{57}$$

Two facts about the Hodge algebra are used throughout: for $v\in H^1(T)$ acting as a derivation, $v\lrcorner t^k=k\,(v\lrcorner t)\,t^{k-1}$; and for a bivector $\pi$ of type $(2,0)$ in the tangent directions, $\pi\lrcorner t=0$ because $t$ has a single $(1,0)$ slot in each term, so $\pi\lrcorner t^k=\binom{k}{2}(\pi\lrcorner t^2)\,t^{k-2}$ by the footnote to (49). Finally, multiplication by $t$ is injective on $H^{0,1}$ when $n\geq 2$ and on $H^{0,2}$ when $n\geq 3$: $q_a\wedge t$ contains the monomial $p_c q_c q_a$ for any $c\neq a$, and $q_a q_b\wedge t$ contains $p_c q_c q_a q_b$ for any $c\notin\{a,b\}$, and distinct monomials are independent.

(i) For $y=(z,\theta)$ the class $y\lrcorner\,\mathrm{ch}(F)$ has components of type $(k-1,k)$ equal to $c_{k-1}z\,t^{k-1}+c_k\,\theta\lrcorner t^k=(c_{k-1}z+k\,c_k\,(\theta\lrcorner t))t^{k-1}$. For $k=1$ this is $az+b\,(\theta\lrcorner t)$ and for $k=2$ it is $(bz-ad\,(\theta\lrcorner t))t$, by (57). Both $z$ and $\theta\lrcorner t=\sum_a\theta^a q_a$ lie in $H^{0,1}$, on which $t$ acts injectively, so $y\lrcorner\,\mathrm{ch}(F)=0$ forces $(z,\theta\lrcorner t)$ into the kernel of the displayed matrix, whose determinant is $-a^2d-b^2<0$. Hence $z=0$ and $\theta\lrcorner t=0$, and the latter gives $\theta=0$ because $t$ is nondegenerate.

(ii) For $x=(z,v,\pi)$ the component of $x\lrcorner\,\mathrm{ch}(F)$ of type $(k-1,k+1)$ is

$$c_{k-1}z\,t^{k-1}+c_k\,v\lrcorner t^k+c_{k+1}\,\pi\lrcorner t^{k+1}=\Big(c_{k-1}z+k\,c_k\,W+\tbinom{k+1}{2}c_{k+1}\,P\Big)t^{k-1},$$

with $W=v\lrcorner t$ and $P=\pi\lrcorner t^2$, both in $H^{0,2}$. By (57), $\binom{k+1}{2}c_{k+1}=-\frac{d}{2}c_{k-1}$, so the bracket is $c_{k-1}Z+k\,c_k W$ with $Z=z-\frac{d}{2}P$. For $k=1$ and $k=2$ this gives $aZ+bW$ and $(bZ-adW)\,t$, and $t$ is injective on $H^{0,2}$ because $n\geq 3$; the same determinant forces $Z=W=0$. Conversely $Z=W=0$ kills every component. So $\mathrm{Ann}_{HT^2}(\mathrm{ch}\,F)$ is the set of $(z,v,\pi)$ with $v\lrcorner t=0$ and $z=\frac{d}{2}\,\pi\lrcorner t^2$, which is (56); the two summands are independent because the first has $\pi=0$ and the second is the graph of a linear map from the $\pi$ component. A $v\in\mathrm{Hom}(H^{1,0},H^{0,1})$ has $v\lrcorner t=\sum_a v(p_a)\wedge q_a$, which vanishes exactly when the matrix of $v$ is symmetric; this is the tangent space of the polarised deformations, of dimension $n(n+1)/2$, as in Proposition 18.16(i). The bivectors contribute $\dim\bigwedge^2 T=n(n-1)/2$. Nothing in the description involves $(a,b)$.

(iii) By (ii) and (54), $\mathrm{ev}_F$ vanishes on $\mathrm{Ann}_{HT^2}(\mathrm{ch}\,F)$ if and only if it vanishes on both summands. On the first it is $v \mapsto v \lrcorner \mathrm{At}(F)$, which is (a). On the second it is

$$\tfrac{d}{2}(\pi \lrcorner t^2)\cdot \mathrm{id}_F + \tfrac{1}{2}\,\pi \lrcorner \mathrm{At}(F)^2 = \tfrac{1}{2}\,\pi \lrcorner (\mathrm{At}(F)^2 + d\,t^2 \cdot \mathrm{id}_F),$$

since $\pi \lrcorner (t^2 \cdot \mathrm{id}_F) = (\pi \lrcorner t^2)\cdot \mathrm{id}_F$. Now $\Omega_X^2 = \mathcal{O}_X \otimes \bigwedge^2 H^{1,0}$, so $\mathrm{Ext}^2(F, F\otimes\Omega_X^2) = \mathrm{Ext}^2(F,F)\otimes\bigwedge^2 H^{1,0}$, and the contractions with all $\pi \in \bigwedge^2 (H^{1,0})^\vee$ vanish on an element of this space if and only if the element is zero, the pairing being perfect. This is (b). For the trace, $\mathrm{ch}(F) = \mathrm{tr}\exp \mathrm{At}(F)$ gives $\mathrm{ch}_2(F) = \frac{1}{2}\,\mathrm{tr}\,\mathrm{At}(F)^2$, and $\mathrm{ch}_2(F) = c_2 t^2 = -\frac{ad}{2}t^2$, while $\mathrm{tr}(\mathrm{id}_F) = \mathrm{rk}\,F = c_0 = a$.

(iv) The Hochschild cohomology of a product is the tensor product, $HT^2(X\times X) = HT^2\otimes HT^0 \oplus HT^1\otimes HT^1 \oplus HT^0\otimes HT^2$, the Atiyah class of $G$ is $\mathrm{At}(F_1)\boxtimes 1 + 1\boxtimes \mathrm{At}(F_2)$ [BF03, Proposition 3.14], [Mar25a, (8.4.1)], so $\exp \mathrm{At}(G) = \exp \mathrm{At}(F_1)\boxtimes \exp \mathrm{At}(F_2)$, and $\mathrm{Ext}^2(G,G)$ is the corresponding Künneth sum of $\mathrm{Ext}^i(F_1,F_1)\otimes\mathrm{Ext}^j(F_2,F_2)$. A polyvector on a factor contracts only the forms of that factor, so for $x_1 \in HT^i(X)$ and $x_2 \in HT^j(X)$ with $i+j=2$,

$$(x_1\otimes x_2)\lrcorner \mathrm{ch}(G) = \pm(x_1\lrcorner \mathrm{ch}\,F_1)\otimes(x_2\lrcorner \mathrm{ch}\,F_2), \qquad \mathrm{ev}_G(x_1\otimes x_2) = \pm\,\mathrm{ev}_{F_1}(x_1)\otimes \mathrm{ev}_{F_2}(x_2),$$

with $\mathrm{ev}(1) = \mathrm{id}$ on $HT^0$ and Koszul signs that play no role; this is the decomposition [Mar25a, (8.4.2), (8.4.3)]. Let $x = x_{20}+x_{11}+x_{02}$ be a class in $\mathrm{Ann}_{HT^2}(\mathrm{ch}\,G)$ according to the three summands. Since $\mathrm{ch}(F_i)$ is even, $x_{11}\lrcorner \mathrm{ch}(G)$ lies in $H^{\mathrm{odd}}\otimes H^{\mathrm{odd}}$ while the other two terms lie in $H^{\mathrm{ev}}\otimes H^{\mathrm{ev}}$, so $x_{11}\lrcorner \mathrm{ch}(G) = 0$ separately; and $x_{11}\lrcorner \mathrm{ch}(G)$ is the image of $x_{11}$ under the tensor product of the two maps of (i), which is injective, so $x_{11}=0$. For the even part write $x_{20} = x_1\otimes 1$ and $x_{02} = 1\otimes x_2$; then $(x_1\lrcorner \mathrm{ch}\,F_1)\otimes\mathrm{ch}(F_2) + \mathrm{ch}(F_1)\otimes(x_2\lrcorner \mathrm{ch}\,F_2) = 0$. Applying a linear form on the second factor taking the value 1 on $\mathrm{ch}(F_2)$ shows $x_1\lrcorner \mathrm{ch}(F_1) = \lambda\,\mathrm{ch}(F_1)$ for a scalar $\lambda$, and then $x_2\lrcorner \mathrm{ch}(F_2) = -\lambda\,\mathrm{ch}(F_2)$. But every component of $x_1\lrcorner \mathrm{ch}(F_1)$ has Hodge type $(p,p+2)$ while $\mathrm{ch}(F_1)\neq 0$ has type $(p,p)$; so $\lambda = 0$ and $x_i \in \mathrm{Ann}_{HT^2}(\mathrm{ch}\,F_i)$. This proves the description of $\mathrm{Ann}_{HT^2}(\mathrm{ch}\,G)$ and its dimension $2n^2$ by (ii). On it $\mathrm{ev}_G(x_1\otimes 1 + 1\otimes x_2) = \pm\,\mathrm{ev}_{F_1}(x_1)\otimes \mathrm{id}_{F_2} \pm \mathrm{id}_{F_1}\otimes \mathrm{ev}_{F_2}(x_2)$, whose two terms lie in different Künneth summands and vanish exactly when $\mathrm{ev}_{F_i}(x_i)=0$, as $\mathrm{id}_{F_i}\neq 0$. So $G$ satisfies the criterion if and only if $F_1$ and $F_2$ do.

For the transport under $\Phi$ we use the commutative diagram of [Mar25a, Lemma 8.3.10]: an equivalence $\Phi\colon D^b(A)\to D^b(B)$ of derived categories of abelian varieties induces isomorphisms $\Phi^{HT}\colon HT^2(A)\to HT^2(B)$ and $\Phi^H\colon H^*(A,\mathbb{C})\to H^*(B,\mathbb{C})$ with $\Phi^H(x\lrcorner \mathrm{ch}\,G) = \Phi^{HT}(x)\lrcorner \mathrm{ch}\,\Phi(G)$ [CRV12], and $\Phi\circ \mathrm{ev}_G = \mathrm{ev}_{\Phi(G)}\circ\Phi^{HT}$ on $\mathrm{Hom}(G,G[2])\to\mathrm{Hom}(\Phi G,\Phi G[2])$ [Hua21, Theorem A]; on abelian varieties the Todd class is trivial, so there is no Duflo correction [Mar25a, Remark 8.3.11]. The first square gives $\Phi^{HT}(\mathrm{Ann}_{HT^2}(\mathrm{ch}\,G)) = \mathrm{Ann}_{HT^2}(\mathrm{ch}\,E)$ and the second $\mathrm{ev}_E(\mathrm{Ann}_{HT^2}(\mathrm{ch}\,E)) = \Phi(\mathrm{ev}_G(\mathrm{Ann}_{HT^2}(\mathrm{ch}\,G)))$, with $\Phi$ an isomorphism on $\mathrm{Ext}^2$. So $E$ satisfies the criterion if and only if $G$ does. ∎

Figure 32 draws the Künneth stack and the two conditions at $n=4$.

**Corollary 18.57** (The remaining condition on the fourfold)**.** *In Setup 18.51 let $E = \Phi(\mathcal{F}\boxtimes\mathcal{F})$. Then $E$ satisfies the weakened criterion of* [Mar25b, Question 11.4] *if and only if*

$$\mathrm{At}(\mathcal{F})^2 = -d\,\Theta^2\cdot \mathrm{id}_{\mathcal{F}} \qquad \text{in } \mathrm{Ext}^2(\mathcal{F},\mathcal{F}\otimes\Omega_X^2). \tag{58}$$

*Equivalently, writing $\mathrm{At}(\mathcal{F}) = \sum_a A_a\otimes p_a$ with $A_a\in\mathrm{Ext}^1(\mathcal{F},\mathcal{F})$ in a basis $p_a$ of $H^{1,0}(X)$, and $\Theta = \sum_a p_a\wedge q_a$ with $q_a\in H^{0,1}(X)$, all the commutators $A_aA_b - A_bA_a\in\mathrm{Ext}^2(\mathcal{F},\mathcal{F})$ are the scalars $\pm 2d\,(q_a\wedge q_b)\cdot\mathrm{id}_{\mathcal{F}}$ prescribed by $\Theta^2$, the sign being fixed by the Koszul rule.*

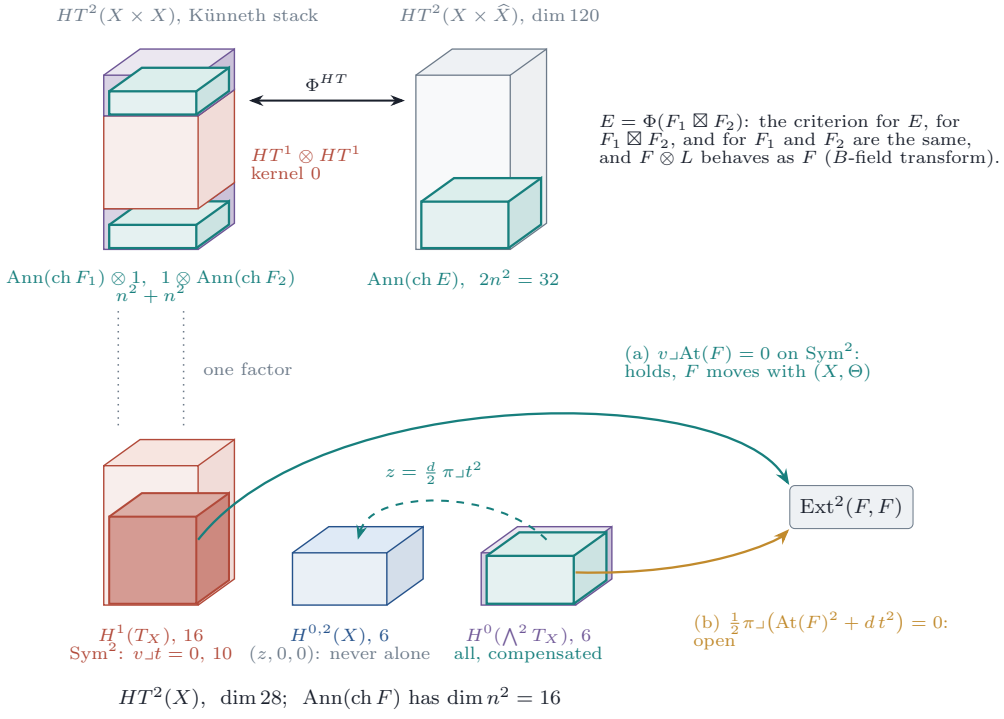



FIGURE 32. Theorem 18.56 for $n = 4$. Top: the Künneth stack of $HT^2(X \times X)$, whose middle summand $HT^1 \otimes HT^1$ meets the kernel in zero by (i), so that the classes preserving $\mathrm{ch}(F_1 \boxtimes F_2)$ are two copies of the kernel on one factor; across Orlov's isomorphism, $HT^2$ of the eightfold with the $2n^2$ classes preserving $\mathrm{ch}(E)$. Bottom: $HT^2(X)$ as three slabs with heights proportional to their dimensions, and inside them the $n^2$ classes of (56): the polarised deformations in front of $H^1(T_X)$, and the whole of $H^0(\bigwedge^2 T_X)$, each bivector tilted into $H^{0,2}$ by its compensation $\frac{d}{2}\, \pi \lrcorner t^2$; nothing in $H^{0,2}$ alone preserves $\mathrm{ch}(F)$. On the right, the two conditions that the evaluation map imposes on the two parts, (a) closed by the construction of $\mathcal{F}$ and (b) the identity (58).

*Proof.* By Proposition 18.52(iii), $\mathrm{ch}(\mathcal{F}) = u_\Theta + 3v_\Theta$, so Theorem 18.56 applies with $n = 4$, $(a, b) = (1, 3)$ and $F_1 = F_2 = \mathcal{F}$, and $E$ satisfies the criterion if and only if $\mathcal{F}$ satisfies (a) and (b). It remains to show that (a) holds. The data of Setup 18.51 are a principally polarised fourfold $(X, \Theta)$ and $N$ points $x_i \in X$ with $D_i = \tau_{x_i}^* \Theta$. The conditions imposed on the $D_i$, that the intersections of two, three and four of them be smooth of the expected codimension and that five of them be disjoint, are open in the parameter space of such data over the local universal deformation $S$ of $(X, \Theta)$, by the openness of smoothness for a proper flat family and the properness of the projection. The isolated points of $Z$, being transversal intersections of four divisors, form a finite étale cover of the open set where the conditions hold, so the $m$ points to be normalised can be followed locally; the partial normalisation separates two smooth branches at each of them and commutes with base change. The construction therefore produces a perfect complex $\mathcal{F}_S$ on the universal family over a neighbourhood of the base point in $S$, whose fibre at the base point is $\mathcal{F}$ and whose base has Kodaira–Spencer image the whole space of polarised deformations, of dimension $10 = n(n+1)/2$. The obstruction $v \lrcorner \mathrm{At}(\mathcal{F})$ to extending $\mathcal{F}$ along $v$ vanishes for every $v$ along which $\mathcal{F}$ extends [Tod09], and this is (a).

For the reformulation, $\Omega_X^1 = \mathcal{O}_X \otimes H^{1,0}$ gives $\mathrm{At}(\mathcal{F}) = \sum_a A_a \otimes p_a$, and $\mathrm{At}(\mathcal{F})^2 = -\sum_{a<b}(A_aA_b - A_bA_a) \otimes p_a \wedge p_b$ by the Koszul rule, while $\Theta^2 = -2\sum_{a<b}(p_a \wedge p_b)(q_a \wedge q_b)$ for $\Theta = \sum_a p_a \wedge q_a$. Comparing coefficients of $p_a \wedge p_b$ gives the commutator relations. ∎

*Remark* 18.58 (Scope of the reduction). The reduction accomplishes three things. First, it removes the eightfold: the condition of Remark 18.53, on an object of rank $8d(d-9)$ on $X \times \widehat{X}$ of Weil type, is equivalent to one identity (58) for a rank one complex on the fourfold, in the Yoneda algebra $\mathrm{Ext}^*(\mathcal{F}, \mathcal{F})$. The dimension of the space of classes that has to be killed drops from $2n^2$ to $n^2$ and, by (a), to $n(n-1)/2 = 6$. At $n = 3$ the count $n^2 = 9$ is the nine-dimensional kernel of [Mar25a, Corollaries 8.3.12 and 8.5.2], which (56) writes out for every secant object in every dimension. Second, the half of that space made of geometric deformations imposes nothing, because Markman's construction is uniform over the moduli of principally polarised fourfolds. Third, by Lemma 18.55 the two forms of the criterion, for $E$ with $\mathrm{ch}(E)$ and for the twisted object with $\kappa(E)$, coincide, so nothing is lost by working with the untwisted object.

What remains is the identity itself, which is a Heisenberg relation: the components of the Atiyah class in the $n$ tangent directions must commute up to the scalars prescribed by $\Theta^2$, and the Chern character forces only the trace of the relation. The identity says exactly that $\mathcal{F}$ deforms, as an object of the deformed category, along the Poisson directions of $X$ compensated by the gerbe classes $\frac{d}{2}\pi \lrcorner \Theta^2$; these are the noncommutative first order deformations of $X$ with a gerby correction, in the language of [Mar25a, Section 8.5]. It holds in the one case where the method is known to work. For Markman's sheaves $\mathcal{F}_1 = I_{\bigcup C_i}(\Theta)$ on the Jacobian threefold of [Mar25b, Section 11.1], with $\mathrm{ch} = u_\Theta + v_\Theta$ and $d = q$, the kernel of the obstruction map equals the annihilator of the Chern character [Mar25a, Corollary 8.3.12], which is stronger than the weakened criterion; so by Theorem 18.56(iii) these sheaves satisfy $\mathrm{At}(\mathcal{F}_1)^2 = -q\,\Theta^2 \cdot \mathrm{id}$ on the threefold. For the rank one complex of Setup 18.51 on the fourfold, the identity is a computation of Yoneda products in $\mathrm{Ext}^*(\mathcal{F}, \mathcal{F})$, and the computation can be made local. Below we carry it out at one point of the support and find that the identity fails.

*Failure at an isolated double point.* The identity (58) can be tested locally. The evaluation map is an algebra homomorphism [Mar25a, (8.3.3)], so on a bivector $\theta \wedge \theta'$ it is the Yoneda

product of the classes

$$A_\theta = \mathrm{ev}_{\mathcal{F}}(\theta) = \theta \lrcorner \mathrm{At}(\mathcal{F}) \in \mathrm{Ext}^1(\mathcal{F},\mathcal{F}), \qquad \theta \in H^0(T_X),$$

which are the first order deformations of $\mathcal{F}$ along the translations of $X$. The product of two global classes restricts to the product of their germs. At a point where the support of $\mathcal{F}$ has two smooth branches of codimension two meeting transversally, the germs multiply nontrivially; since the compensating classes $z \cdot \mathrm{id}_{\mathcal{F}}$ have zero germs, the identity fails. Markman's construction has such points and cannot avoid them.

**Lemma 18.59** (Localisation)**.** *Let $F$ be a perfect complex on a smooth projective variety $Y$, let $\mathcal{E}xt^m(F,F)$ be the sheaf associated with $U \mapsto \mathrm{Hom}_{D(U)}(F|_U, F|_U[m])$, and let $e\colon \mathrm{Ext}^m(F,F) \to H^0(Y, \mathcal{E}xt^m(F,F))$ send a morphism $F \to F[m]$ to its germs. Then $e$ is compatible with Yoneda products; $e(z \cdot \mathrm{id}_F) = 0$ for $z \in H^m(\mathcal{O}_Y)$, $m \geq 1$; and for $\theta \in H^0(T_Y)$ the germ of $A_\theta$ at $p$ is the class of the jet extension $0 \to F_p \to J_\theta(F_p) \to F_p \to 0$ of the germ $F_p$ along the derivation $\theta$ of $\mathcal{O}_{Y,p}$, which is $\mathcal{O}_{Y,p}$-linear in $\theta$.*

*Proof.* Composition of morphisms commutes with restriction to open sets; this is the first assertion. For the second, $z \cdot \mathrm{id}_F$ restricted to an affine open $U$ is $z|_U \cdot \mathrm{id}_{F|_U}$, and $z|_U \in H^m(U, \mathcal{O}_U) = 0$. For the third, the Atiyah class of $F$ is the class of the first jet sequence $0 \to F \otimes \Omega^1_Y \to J^1(F) \to F \to 0$ [BF03, Section 3], and its restriction to $U$ is the jet sequence of $F|_U$. Contracting with $\theta$ is pushing out along $\mathrm{id} \otimes \theta\colon F \otimes \Omega^1 \to F$, which produces the extension $J_\theta(F)$ whose underlying sheaf is $F \oplus F$ with $f \cdot (s, s') = (fs + \theta(f)s', fs')$. Linearity in $\theta$ is linearity of the contraction. ∎

**Lemma 18.60** (Two branches through a point)**.** *Let $\mathcal{O}$ be the local ring of $\mathbb{C}^4$ at the origin or its completion $\mathbb{C}[[x_1,\dots,x_4]]$; let $P_1 = \{x_1 = x_2 = 0\}$ and $P_2 = \{x_3 = x_4 = 0\}$, two smooth surfaces meeting transversally at the origin; put $N_1 = \langle \partial_1, \partial_2 \rangle$ and $N_2 = \langle \partial_3, \partial_4 \rangle$, the normal spaces of $P_1$ and of $P_2$ at the origin, so that $\bigwedge^2 T_0 = \bigwedge^2 N_1 \oplus N_1 \otimes N_2 \oplus \bigwedge^2 N_2$. For a germ of vector field $\theta$ and a perfect complex $F$ of $\mathcal{O}$-modules write $a_\theta \in \mathrm{Ext}^1_{\mathcal{O}}(F,F)$ for the class of the jet extension along $\theta$.*

(i) *For $F = I = I_{P_1} \cap I_{P_2}$, the ideal of the union, $\mathrm{Ext}^2_{\mathcal{O}}(I,I) \cong N_1 \otimes N_2$, a vector space of dimension four on which the maximal ideal acts trivially, and $a_\theta a_{\theta'}$ is, up to a fixed nonzero scalar, the component of $\theta(0) \wedge \theta'(0)$ in $N_1 \otimes N_2$. In particular $a_\theta a_{\theta'} \neq 0$ unless $\theta(0) \wedge \theta'(0) \in \bigwedge^2 N_1 \oplus \bigwedge^2 N_2$.*

(ii) *For $F = K = [\mathcal{O} \to \mathcal{O}_{P_1} \oplus \mathcal{O}_{P_2}]$, the complex in degrees $0$ and $1$ whose cohomology is $I$ in degree $0$ and the residue field $k$ in degree $1$, $a_\theta a_{\theta'} \neq 0$ unless $\theta(0) \wedge \theta'(0) \in N_1 \otimes N_2$.*

*In both cases the bilinear map $(\theta, \theta') \mapsto a_\theta a_{\theta'}$ is nonzero, and it depends only on $\theta(0) \wedge \theta'(0)$.*

*Proof.* Write $A = \mathbb{C}[x_1,x_2]$, $B = \mathbb{C}[x_3,x_4]$, $R = A \otimes_{\mathbb{C}} B$, with maximal ideals $\mathfrak{m}_A$, $\mathfrak{m}_B$ at the origin, and $k = \mathbb{C}$. The ideal of the union is $I = \mathfrak{m}_A R \cap \mathfrak{m}_B R = \mathfrak{m}_A \otimes_{\mathbb{C}} \mathfrak{m}_B$, because the two ideals are generated by disjoint sets of variables. Everything below is homogeneous, the Ext groups in degree two are of finite length, and finite length modules and jet classes are unchanged by the flat base change from $R$ to $\mathcal{O}$; so we compute over $R$. For finitely generated modules $M, M'$ over $A$ and $N, N'$ over $B$, the tensor product of free resolutions resolves $M \otimes_{\mathbb{C}} N$ and the Künneth formula gives

$$\mathrm{Ext}^n_R(M \otimes N, M' \otimes N') = \bigoplus_{i+j=n} \mathrm{Ext}^i_A(M,M') \otimes_{\mathbb{C}} \mathrm{Ext}^j_B(N,N'), \tag{59}$$

compatibly with Yoneda products up to the Koszul sign, since a product of chain maps on the tensor product of resolutions is the tensor product of the products.

The basic case is a single plane. The Koszul complex $0 \to A \xrightarrow{(-x_2,x_1)^t} A^2 \xrightarrow{(x_1,x_2)} \mathfrak{m}_A \to 0$ is a free resolution, so $\mathfrak{m}_A$ has projective dimension one and $\operatorname{Ext}^i_A(\mathfrak{m}_A,-)=0$ for $i\geq 2$; and $\operatorname{Ext}^1_A(\mathfrak{m}_A,\mathfrak{m}_A)=\operatorname{coker}\big(\operatorname{Hom}(A^2,\mathfrak{m}_A)\to\operatorname{Hom}(A,\mathfrak{m}_A)\big)=\mathfrak{m}_A/\mathfrak{m}_A^2$, a two-dimensional space killed by $\mathfrak{m}_A$. On a complex of free modules with differential $d$ the jet extension along a derivation $D$ is the complex with terms $P_i\oplus P_i$ and differential $\left(\begin{smallmatrix} d & -D(d)\\ 0 & d\end{smallmatrix}\right)$, $D(d)$ the entrywise derivative, because the twisted module $J_D(P_i)$ is free on the elements $(e,0)$ and $(0,e)$ and $d(0,e)=\sum_l d_{le}(0,e_l)-\sum_l D(d_{le})(e_l,0)$; so the jet class is represented by the degree $-1$ chain map $-D(d)$. For $D=\partial_i$ on the Koszul complex this is $e\mapsto \delta_{i2}f_1-\delta_{i1}f_2$ on the generator $e$ of the last term, whose image in $\mathfrak{m}_A/\mathfrak{m}_A^2$ is $\delta_{i2}x_1-\delta_{i1}x_2$:

$$\alpha_1 := a_{\partial_1} = -[x_2], \qquad \alpha_2 := a_{\partial_2} = [x_1],$$

a basis of $\operatorname{Ext}^1_A(\mathfrak{m}_A,\mathfrak{m}_A)$, which we identify with $N_1$ by $\partial_i\mapsto\alpha_i$. Similarly $\beta_3,\beta_4$ form a basis of $\operatorname{Ext}^1_B(\mathfrak{m}_B,\mathfrak{m}_B)$, identified with $N_2$.

(i) By (59) with $M=M'=\mathfrak{m}_A$, $N=N'=\mathfrak{m}_B$,

$$\operatorname{Ext}^2_R(I,I)=\operatorname{Ext}^1_A\otimes\operatorname{Ext}^1_B=N_1\otimes N_2,$$

the other two summands vanishing by the projective dimension, and it is killed by $\mathfrak{m}_A$ and $\mathfrak{m}_B$. The derivation $\partial_1$ acts on the factor $A$, so the jet extension of $I=\mathfrak{m}_A\otimes\mathfrak{m}_B$ along it is $J_{\partial_1}(\mathfrak{m}_A)\otimes\mathfrak{m}_B$ and $a_{\partial_1}=\alpha_1\otimes 1$; likewise $a_{\partial_2}=\alpha_2\otimes 1$, $a_{\partial_3}=1\otimes\beta_3$, $a_{\partial_4}=1\otimes\beta_4$. Hence $a_{\partial_i}a_{\partial_j}=\pm\alpha_i\otimes\beta_j$ for $i\leq 2<j$, the four of them a basis of $N_1\otimes N_2$, while $a_{\partial_1}a_{\partial_2}=\alpha_1\alpha_2\otimes 1=0$ and $a_{\partial_3}a_{\partial_4}=0$ because $\operatorname{Ext}^2_A(\mathfrak{m}_A,\mathfrak{m}_A)=0$. For germs $\theta=\sum_i c^i\partial_i$ and $\theta'=\sum_j c'^j\partial_j$ the linearity of Lemma 18.59 and the triviality of the module structure give $a_\theta a_{\theta'}=\sum_{i,j}c^i(0)c'^j(0)\,a_{\partial_i}a_{\partial_j}$, the image of $\theta(0)\wedge\theta'(0)$ under the projection of $\bigwedge^2 T_0$ onto $N_1\otimes N_2$ followed by the isomorphism $\partial_i\otimes\partial_j\mapsto\pm\alpha_i\otimes\beta_j$.

(ii) Let $Q=\mathcal{O}_{P_1}\oplus\mathcal{O}_{P_2}$. The map $\mathcal{O}\to Q$ has kernel $I$ and cokernel $\mathcal{O}/(I_{P_1}+I_{P_2})=k$, so $K$ sits in an exact triangle $I\xrightarrow{\iota}K\xrightarrow{\pi}k[-1]\xrightarrow{\varepsilon}I[1]$. The jet extension along $\theta$ is an exact functor, natural in its argument, so the jet classes commute with $\iota$ and $\pi$; and for the residue field $a^k_\theta$ depends only on $\theta(0)$, because $\operatorname{Ext}^1(k,k)$ is killed by $\mathfrak{m}$ and the class is linear in $\theta$. Therefore

$$\pi[2]\circ a^K_\theta a^K_{\theta'}=a^k_\theta a^k_{\theta'}\circ\pi=\pi^*(a^k_{\theta(0)}a^k_{\theta'(0)}),\qquad \pi^*\colon \operatorname{Ext}^2(k,k)\to\operatorname{Ext}^1(K,k),$$

and it suffices to show that the right hand side is nonzero. The long exact sequence of the triangle for $\operatorname{Hom}(-,k)$ reads

$$\operatorname{Hom}(I,k)\xrightarrow{\varepsilon^*}\operatorname{Ext}^2(k,k)\xrightarrow{\pi^*}\operatorname{Ext}^1(K,k),$$

so $\ker\pi^*=\operatorname{im}\varepsilon^*$. Now $k=k_A\otimes k_B$ and, by (59), $\operatorname{Ext}^*(k,k)$ is the tensor product of $\operatorname{Ext}^*_A(k,k)$ and $\operatorname{Ext}^*_B(k,k)$. On the Koszul resolution $A\to A^2\to A$ of $k_A$ the jet chain maps of $\partial_1,\partial_2$ are $-D(d)$ as above, and their composite $A\to A$ is the scalar $-1$, whose class in $\operatorname{Ext}^2_A(k,k)=\operatorname{Hom}(A,k)=k$ is nonzero; the degree one classes $\gamma_1,\gamma_2$ of $\partial_1,\partial_2$ are the basis $-e_1^*,-e_2^*$ of $\operatorname{Ext}^1_A(k,k)=\operatorname{Hom}(A^2,k)$, and their product spans $\operatorname{Ext}^2_A(k,k)$. With the analogous classes $\gamma_3,\gamma_4$ over $B$, the products $\gamma_1\gamma_2\otimes 1$, $\gamma_i\otimes\gamma_j$ $(i\leq 2<j)$ and $1\otimes\gamma_3\gamma_4$ form a basis of $\operatorname{Ext}^2(k,k)=k\oplus(k^2\otimes k^2)\oplus k$, so $\partial_i\wedge\partial_j\mapsto a^k_{\partial_i}a^k_{\partial_j}$ identifies $\bigwedge^2(N_1\oplus N_2)$ with $\operatorname{Ext}^2(k,k)$, and $a^k_{\theta(0)}a^k_{\theta'(0)}$ is the image of $\theta(0)\wedge\theta'(0)$. The class $\varepsilon$ is the Künneth product $\varepsilon_A\otimes\varepsilon_B$ of the classes of $0\to\mathfrak{m}_A\to A\to k\to 0$ and its $B$-analogue, because $K$ is the two-step truncation of the tensor product of the complexes $[A\to k_A]$ and $[B\to k_B]$ and $\varepsilon$ is the inclusion of the omitted term $k_A\otimes k_B$. Hence $\varepsilon^*$ maps $\operatorname{Hom}(I,k)=\operatorname{Hom}(\mathfrak{m}_A,k)\otimes\operatorname{Hom}(\mathfrak{m}_B,k)$ into $\operatorname{Ext}^1_A(k,k)\otimes\operatorname{Ext}^1_B(k,k)=N_1\otimes N_2$, and onto it, since $\operatorname{Hom}(\mathfrak{m}_A,k)\to\operatorname{Ext}^1_A(k,k)$ is an isomorphism: it is the connecting

homomorphism of the sequence, injective because $\mathrm{Hom}(A,k)\to\mathrm{Hom}(\mathfrak{m}_A,k)$ is zero and surjective because $\mathrm{Ext}^1_A(A,k)=0$. So $\ker\pi^*=N_1\otimes N_2$ and $\pi^*(\theta(0)\wedge\theta'(0))\neq 0$ exactly when $\theta(0)\wedge\theta'(0)\notin N_1\otimes N_2$. ∎

**Theorem 18.61** (Markman's candidate does not satisfy the weakened criterion)**.** *In Setup 18.51, for every odd squarefree $d\geq 3$, every choice of the translates $D_i$ in general position and every choice of the $m$ points to be normalised:*

(i) *the identity (58) fails: there is a constant bivector $\pi\in H^0(\bigwedge^2 T_X)$ with $\frac{1}{2}\pi\lrcorner(\mathrm{At}(\mathcal{F})^2+d\,\Theta^2\cdot\mathrm{id}_{\mathcal{F}})\neq 0$ in $\mathrm{Ext}^2(\mathcal{F},\mathcal{F})$;*

(ii) *$\mathcal{F}$ does not satisfy the weakened criterion: the class $x_{\mathcal{F}}=(\frac{d}{2}\,\pi\lrcorner\Theta^2,0,\pi)$ has $x_{\mathcal{F}}\lrcorner\,\mathrm{ch}(\mathcal{F})=0$ and $\mathrm{ev}_{\mathcal{F}}(x_{\mathcal{F}})\neq 0$;*

(iii) *for every perfect complex $F_2$ on $X$ with Chern character on the secant plane, in particular for $F_2=\mathcal{F}$ and $F_2=\mathcal{F}^\vee$, the object $E=\Phi(\mathcal{F}\boxtimes F_2)$ on the eightfold $X\times\widehat{X}$ of Weil type does not satisfy the weakened criterion of* [Mar25b, Question 11.4]*: the class $x=\Phi^{HT}(x_{\mathcal{F}}\otimes 1)$ satisfies $x\lrcorner\,\mathrm{ch}(E)=0$ and $\mathrm{ev}_E(x)\neq 0$, so $\sigma_E$ is not injective on the image of $\mathrm{ev}_E$.*

*Proof.* (i) The surfaces $Z_i$ and $Z_j$ at cyclic distance at least three meet in the 24 points of $D_i\cap D_{i+1}\cap D_j\cap D_{j+1}$, and there are $12N(N-5)=(d+9)(3d-3)$ such points, of which $m=(d+9)(2d-1)$ are normalised; the remaining $(d+9)(d-2)\geq 12$ are not. Let $p$ be one of the latter. The four divisors through $p$ meet transversally, so their local equations $x_1,\dots,x_4$ form a regular system of parameters of $\mathcal{O}_{X,p}$ and identify its completion with $\mathbb{C}[[x_1,\dots,x_4]]$, with $D_i=\{x_1=0\}$, $D_{i+1}=\{x_2=0\}$, $D_j=\{x_3=0\}$ and $D_{j+1}=\{x_4=0\}$; hence $Z_i=P_1$ and $Z_j=P_2$ in the notation of Lemma 18.60. The completion is flat, so a class whose completed germ is nonzero is nonzero. No other $D_k$ passes through $p$, since five divisors have empty intersection, so no other component of $Z$ does. Near $p$ the map $\nu$ is an isomorphism, $\nu_*\mathcal{O}_{\widetilde{Z}}=\mathcal{O}_Z$, and $\mathcal{F}$ is the ideal sheaf $I_Z=I_{P_1}\cap I_{P_2}$ twisted by the locally trivial $\mathcal{O}_X(3\Theta)$: the germ $\mathcal{F}_p$ is the module $I$ of Lemma 18.60(i).

Let $\theta,\theta'\in H^0(T_X)$ be constant vector fields whose values at $p$ are $\partial_1$ and $\partial_3$; they exist because the constant fields span every tangent space. Put $\pi=\theta\wedge\theta'$. Since $\mathrm{ev}_{\mathcal{F}}$ is an algebra homomorphism, $\mathrm{ev}_{\mathcal{F}}(\pi)=\frac{1}{2}\pi\lrcorner\,\mathrm{At}(\mathcal{F})^2=A_\theta A_{\theta'}$. By Lemma 18.59 its germ at $p$ is $a_\theta a_{\theta'}$, which is nonzero by Lemma 18.60(i), because $\partial_1\wedge\partial_3$ has nonzero component in $N_1\otimes N_2$. The germ of $\frac{d}{2}(\pi\lrcorner\Theta^2)\cdot\mathrm{id}_{\mathcal{F}}$ at $p$ is zero by Lemma 18.59. So the germ at $p$ of $\frac{1}{2}\pi\lrcorner(\mathrm{At}(\mathcal{F})^2+d\Theta^2\cdot\mathrm{id}_{\mathcal{F}})$ is $a_\theta a_{\theta'}\neq 0$, and the class itself is nonzero.

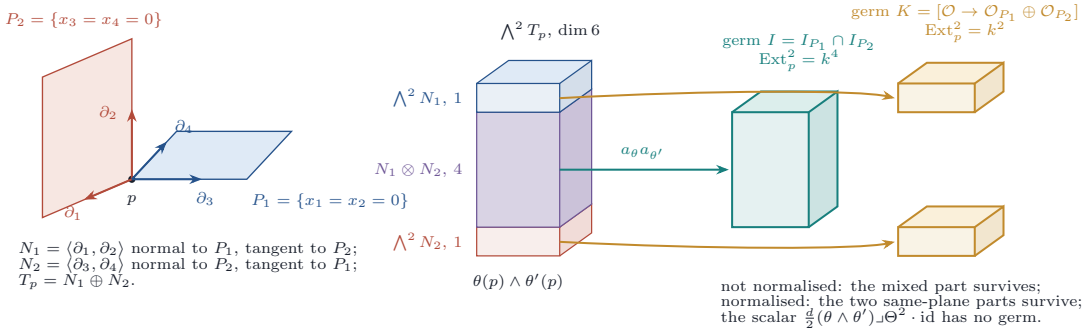
(ii) By Theorem 18.56(ii) the class $x_{\mathcal{F}}$ preserves $\mathrm{ch}(\mathcal{F})$, and by (54) $\mathrm{ev}_{\mathcal{F}}(x_{\mathcal{F}})$ is the class of (i).

(iii) Both $\mathcal{F}$ and $\mathcal{F}^\vee$ are admissible as $F_2$, since $\mathrm{ch}(\mathcal{F}^\vee)=u_\Theta-3v_\Theta$ lies on the secant plane. By Theorem 18.56(iv), $E$ satisfies the criterion only if both factors do, and $\mathcal{F}$ does not. The class $x$ of the statement is the transport of $x_{\mathcal{F}}\otimes 1\in\mathrm{Ann}_{HT^2}(\mathrm{ch}(\mathcal{F}\boxtimes F_2))$, on which ev is $\pm\,\mathrm{ev}_{\mathcal{F}}(x_{\mathcal{F}})\otimes\mathrm{id}_{F_2}\neq 0$; since $\Phi$ is an isomorphism on $\mathrm{Ext}^2$, $\mathrm{ev}_E(x)\neq 0$. The last assertion follows from the commutative square of Setup 18.54: $\sigma_E(\mathrm{ev}_E(x))=x\lrcorner\,\mathrm{ch}(E)=0$ with $\mathrm{ev}_E(x)\neq 0$. ∎

Figure 33 draws the local computation at the double point.

*Remark* 18.62 (Normalised points). At a normalised point the germ of $\mathcal{F}$ is the complex $K$ of Lemma 18.60(ii), and the same argument, with $\theta(p)=\partial_1$ and $\theta'(p)=\partial_2$, produces a nonzero germ. The failure therefore does not depend on which points are separated. Separating all of them would not help even if the construction allowed it: normalising every

At an isolated double point of the support the translation classes multiply to a nonzero germ.




At every isolated double point of $Z$, of either kind, some constant bivector $\pi$ has $\mathrm{ev}_{\mathcal{F}}(\pi) + \frac{d}{2}(\pi\lrcorner\Theta^2)\,\mathrm{id} \neq 0$; the class $\pi + \frac{d}{2}\,\pi\lrcorner\Theta^2$ preserves $\mathrm{ch}(\mathcal{F})$, so the weakened criterion fails for $\mathcal{F}$ and for $\Phi(\mathcal{F} \boxtimes \mathcal{F})$.

FIGURE 33. Lemma 18.60 and Theorem 18.61. Left: two smooth surfaces of a fourfold meeting transversally at $p$, with the normal spaces $N_1$ and $N_2$, each tangent to the other branch. Centre: the bivectors at $p$, split into $\bigwedge^2 N_1$, $N_1 \otimes N_2$ and $\bigwedge^2 N_2$, with heights proportional to the dimensions. Right: the local $\mathrm{Ext}^2$ of the germ of $\mathcal{F}$ at $p$ in the two cases, and the part of the bivectors that the products of the translation classes carry onto it: the mixed part when the point is not normalised, the two same-plane parts when it is. The compensating scalar has no germ, so in either case the identity (58) fails at $p$.

isolated point changes $\chi(\mathcal{O}_{\widetilde{Z}})$ to $110N - 12N^2 + 12N(N-5) = 50N$, while the secant plane requires $(d+9)(27-d) = 72N - 4N^2$, and the two agree only for $N = 11/2$. The two local computations are complementary: at a point of type (i) the obstruction lies in $N_1 \otimes N_2$, at a point of type (ii) in $\bigwedge^2 N_1 \oplus \bigwedge^2 N_2$, and together these exhaust $\bigwedge^2 T_p$.

**Corollary 18.63** (A local obstruction in every dimension)**.** *Let $(X, t)$ be a polarised abelian variety of dimension $n \geq 4$ and $F$ a perfect complex on $X$ with $\mathrm{ch}(F) = au_t + bv_t$, $(a, b) \neq (0, 0)$. Suppose that at some point $p$ the germ $F_p$ is isomorphic, in the derived category of $\mathcal{O}_{X,p}$-modules, to the ideal of the union of two smooth germs of codimension two meeting transversally along a germ of codimension four, or, when $n = 4$, to the complex $[\mathcal{O} \to \mathcal{O}_{P_1} \oplus \mathcal{O}_{P_2}]$ of two such germs. Then $F$ does not satisfy the weakened criterion, and neither does $\Phi(F \boxtimes F_2)$ on $X \times \widehat{X}$ for any secant $F_2$. In particular, in every dimension $n \geq 4$, the union $Z$ of the intersections $D_i \cap D_{i+1}$ of consecutive translates of a polarisation in general position contains, for every pair $(i, j)$ of nonconsecutive indices, a subvariety $Z_i \cap Z_j$ of codimension four along which $Z$ has two transversal branches; any secant object of the form $[\mathcal{O}_X \to \nu_* \mathcal{O}_{\widetilde{Z}}] \otimes L$ in which $\nu$ is an isomorphism at a general point of one such $Z_i \cap Z_j$ fails the criterion.*

*Proof.* In formal coordinates $x_1, \ldots, x_n$ at $p$ with the two branches $\{x_1 = x_2 = 0\}$ and $\{x_3 = x_4 = 0\}$, which exist because the four local equations of the branches form part of a regular system of parameters, the ideal is $I \otimes_{\mathbb{C}} \mathbb{C}[x_5, \ldots, x_n]$ with $I$ as in Lemma 18.60, and (59) with the third factor gives $\mathrm{Ext}^2(I_p, I_p) = N_1 \otimes N_2 \otimes \mathcal{O}_{W,p}$, where $W$ is the intersection

of the branches. The jet classes along $\partial_5, \dots, \partial_n$ vanish, since these derivations act on a free factor, and the products of the classes along $\partial_1, \dots, \partial_4$ are as in the lemma. Reducing modulo the maximal ideal of $\mathcal{O}_{W,p}$ shows that $a_\theta a_{\theta'} \neq 0$ whenever $\theta(p) \wedge \theta'(p)$ has a nonzero component in $N_1 \otimes N_2$. When $n = 4$ and $F_p$ is the complex $[\mathcal{O} \to \mathcal{O}_{P_1} \oplus \mathcal{O}_{P_2}]$, Lemma 18.60(ii) applies directly. The rest of the argument is that of Theorem 18.61, using Theorem 18.56 for $n \geq 3$. For the last assertion, $Z_i \cap Z_j = D_i \cap D_{i+1} \cap D_j \cap D_{j+1}$ is a transversal intersection of four divisors, of codimension four, and at a general point of it no fifth divisor passes, so $Z$ has exactly the two branches $Z_i$ and $Z_j$ there. ∎

*Remark* 18.64 (Scope of the local obstruction). Theorem 18.61 answers the question posed in [Mar25b, Section 12]: the secant objects proposed there do not satisfy the weakened criterion of [Mar25b, Question 11.4], for any discriminant. The reason is local and geometric. That question asks whether the semiregularity map may be required to be injective only on the image of the evaluation map. For a secant object this image contains, by Theorem 18.56, the products $A_\theta A_{\theta'}$ of the translation classes shifted by the scalars $\frac{d}{2}(\theta \wedge \theta') \lrcorner \Theta^2$, and the semiregularity map kills them all. For the criterion to hold these shifted products must vanish, and they cannot: at an isolated double point of the support, the translation classes of the two branches multiply to a nonzero local class, while a scalar has no local class at all. The fourth ingredient of Remark 14.16 can accordingly be stated more sharply. A secant object of positive rank on a fourfold that satisfies the weakened criterion must have, at every point, germs whose translation classes multiply to zero in the local $\mathrm{Ext}^2$. By Lemma 18.60 this excludes every support with two transversal branches of codimension two through a point, whether or not the branches are separated, and by Corollary 18.63 it excludes the chain of consecutive intersections of translates in every dimension. An object supported on a smooth surface, or on surfaces meeting only along curves, is not excluded by this argument, but no such object with Chern character on the secant plane is known.

18.4. **Smooth supports and four discriminants.** Theorem 18.61 and Corollary 18.63 exclude every secant object whose support has two transversal branches of codimension two through a point. This leaves supports that are smooth or that meet themselves only along curves, and Remark 18.64 ends with the observation that no such object is known. In this subsection we explain why the local argument cannot reach the smooth case, and then determine what a smooth support can be. The answer is narrow: on the fourfold the invariants of the support are forced, and the discriminant is confined to a finite list.

We begin with the fact that at a smooth point there is nothing local left to compute.

**Proposition 18.65** (No local obstruction at a smooth point)**.** *Let $Y$ be a smooth variety with $\omega_Y \cong \mathcal{O}_Y$ and let $S \subset Y$ be smooth of codimension two. Then*

$$\mathcal{E}xt^0(\mathcal{I}_S, \mathcal{I}_S) = \mathcal{O}_Y, \qquad \mathcal{E}xt^1(\mathcal{I}_S, \mathcal{I}_S) = N_{S/Y}, \qquad \mathcal{E}xt^q(\mathcal{I}_S, \mathcal{I}_S) = 0 \quad (q \geq 2),$$

*and the same holds for $\mathcal{I}_S \otimes L$ for any line bundle $L$. Consequently, for $F = \mathcal{I}_S \otimes L$ both sides of (58) have vanishing germs in $\mathcal{E}xt^2$, and Lemma 18.59 decides nothing.*

*Proof.* Twisting by $L$ changes neither side, so take $F = \mathcal{I}_S$. The sheaf $\mathcal{I}_S$ is torsion free of rank one, and its dual is $\mathcal{O}_Y$ because $S$ has codimension two; hence every local endomorphism of $\mathcal{I}_S$ is multiplication by a function, and $\mathcal{E}xt^0(\mathcal{I}_S, \mathcal{I}_S) = \mathcal{O}_Y$. Apply $\mathcal{H}om(-, \mathcal{I}_S)$ to

$$0 \to \mathcal{I}_S \to \mathcal{O}_Y \to \mathcal{O}_S \to 0. \tag{60}$$

Since $\mathcal{E}xt^q(\mathcal{O}_Y, \mathcal{I}_S) = 0$ for $q \geq 1$, the long exact sequence gives

$$\mathcal{E}xt^q(\mathcal{I}_S, \mathcal{I}_S) \cong \mathcal{E}xt^{q+1}(\mathcal{O}_S, \mathcal{I}_S) \qquad (q \geq 1). \tag{61}$$

Now apply $\mathcal{H}om(\mathcal{O}_S,-)$ to (60). Because $S$ is smooth of codimension two, hence a local complete intersection, the Koszul resolution gives $\mathcal{E}xt^k(\mathcal{O}_S,\mathcal{O}_Y)=0$ for $k\neq 2$ and $\mathcal{E}xt^2(\mathcal{O}_S,\mathcal{O}_Y)=\omega_S\otimes\omega_Y^{-1}=\omega_S$, while $\mathcal{E}xt^k(\mathcal{O}_S,\mathcal{O}_S)=\bigwedge^k N_{S/Y}$ for $0\le k\le 2$ and zero otherwise. Computing both from the same Koszul complex shows that the map

$$\rho\colon \mathcal{E}xt^2(\mathcal{O}_S,\mathcal{O}_Y)=\omega_S\longrightarrow \mathcal{E}xt^2(\mathcal{O}_S,\mathcal{O}_S)=\textstyle\bigwedge^2 N_{S/Y}=\omega_S$$

induced by $\mathcal{O}_Y\to\mathcal{O}_S$ is the identity: in local coordinates in which $S=\{x_1=x_2=0\}$, both groups are computed as $M/(x_1,x_2)M$ for $M=\mathcal{O}_Y$ and $M=\mathcal{O}_S$, and the map is reduction modulo $(x_1,x_2)$, which is the identity on $\mathcal{O}_S$.

Consider the long exact sequence of $\mathcal{H}om(\mathcal{O}_S,-)$ in degrees two and three. Since $\rho$ is surjective and $\mathcal{E}xt^3(\mathcal{O}_S,\mathcal{O}_Y)=0$, the segment

$$\mathcal{E}xt^2(\mathcal{O}_S,\mathcal{O}_Y)\xrightarrow{\rho}\mathcal{E}xt^2(\mathcal{O}_S,\mathcal{O}_S)\to\mathcal{E}xt^3(\mathcal{O}_S,\mathcal{I}_S)\to\mathcal{E}xt^3(\mathcal{O}_S,\mathcal{O}_Y)=0$$

gives $\mathcal{E}xt^3(\mathcal{O}_S,\mathcal{I}_S)=0$, so $\mathcal{E}xt^2(\mathcal{I}_S,\mathcal{I}_S)=0$ by (61). For $k\ge 3$ both $\mathcal{E}xt^k(\mathcal{O}_S,\mathcal{O}_Y)$ and $\mathcal{E}xt^k(\mathcal{O}_S,\mathcal{O}_S)$ vanish, so $\mathcal{E}xt^{k+1}(\mathcal{O}_S,\mathcal{I}_S)=0$ and $\mathcal{E}xt^k(\mathcal{I}_S,\mathcal{I}_S)=0$. In degrees one and two, since $\rho$ is injective, the segment

$$0=\mathcal{E}xt^1(\mathcal{O}_S,\mathcal{O}_Y)\to N_{S/Y}\to\mathcal{E}xt^2(\mathcal{O}_S,\mathcal{I}_S)\to\mathcal{E}xt^2(\mathcal{O}_S,\mathcal{O}_Y)\xrightarrow{\rho}\mathcal{E}xt^2(\mathcal{O}_S,\mathcal{O}_S),$$

in which $N_{S/Y}=\mathcal{E}xt^1(\mathcal{O}_S,\mathcal{O}_S)$, gives $\mathcal{E}xt^2(\mathcal{O}_S,\mathcal{I}_S)\cong N_{S/Y}$, and this is $\mathcal{E}xt^1(\mathcal{I}_S,\mathcal{I}_S)$ by (61).

For the last assertion, the right side of (58) has vanishing germs by Lemma 18.59, and the left side lies in $\mathrm{Ext}^2(F,F\otimes\Omega_Y^2)$, whose sheaf of germs in the $\mathcal{E}xt^2$ direction is $\mathcal{E}xt^2(F,F)\otimes\Omega_Y^2=0$. ∎

*Remark* 18.66 (The global form of the criterion). Proposition 18.65 is the precise reason why the argument of Theorem 18.61 stops at a smooth support, and it also shows what takes its place. Since $\mathcal{E}xt^2=0$, the local-to-global spectral sequence for $\mathrm{Ext}^2(\mathcal{I}_S,\mathcal{I}_S)$ has only two surviving rows, $H^1(S,N_{S/Y})$ and $H^2(Y,\mathcal{O}_Y)$, and they sit in different steps of the filtration: the first is the quotient that records the geometry of $S$, the second the subgroup of scalars. A product $A_\theta A_{\theta'}$ of translation classes therefore has a well defined image in $H^1(S,N_{S/Y})$, while the scalar on the right of (58) has image zero there. The criterion thus splits into two conditions of different kinds: a vanishing in $H^1(S,N_{S/Y})$, which is a statement about the normal bundle and not about any point, and an identity in $H^2(Y,\mathcal{O}_Y)$. At a double point the first condition already fails locally. At a smooth support it is a global question, and what is decided here is which surfaces $S$ are available at all.

The invariants of a smooth support are far from free. Since the Todd class of an abelian variety is trivial, Grothendieck–Riemann–Roch converts the Chern character of the object into the complete set of numerical invariants of the surface, and three classical inequalities then constrain them.

**Theorem 18.67** (The invariants of a smooth support)**.** *Let $(X,\Theta)$ be a principally polarised abelian fourfold, let $d>0$, let $b\in\mathbb{Z}_{>0}$, and let $\mathcal{F}=\mathcal{I}_S(b\Theta)$ be a coherent sheaf with*

$$\mathrm{ch}(\mathcal{F})=u_\Theta+b\,v_\Theta$$

*and $S\subset X$ smooth. Write $N=\frac{1}{2}\mathrm{N}_{K/\mathbb{Q}}(b+\sqrt{-d})=\frac{1}{2}(b^2+d)$, $\iota\colon S\hookrightarrow X$ and $\lambda=\iota^*\Theta$. Then $S$ is a surface and*

$$\begin{aligned}
&[S]=N\Theta^2, && \iota_*K_S=\tfrac{4bN}{3}\,\Theta^3, && \chi(\mathcal{O}_S)=4N(2b^2-N),\\
&K_S^2=12N(4b^2-N), && e(S)=12N(4b^2-3N), && K_S\cdot\lambda=32bN,
\end{aligned}\tag{62}$$

*with $\lambda^2 = 24N$ and $[S]^2 = 24N^2$. Moreover $K_S$ is globally generated, hence nef; $S$ is minimal of general type; and*

$$\tfrac{4}{9}b^2 \;\le\; N \;\le\; b^2, \qquad \textit{equivalently} \qquad -\tfrac{1}{9}b^2 \;\le\; d \;\le\; b^2. \tag{63}$$

*The lower bound is the Hodge index theorem and is implied by $d > 0$; the upper bound is the Bogomolov–Miyaoka–Yau inequality. In terms of $b$ and $d$ the five invariants read*

$$\chi(\mathcal{O}_S) = (b^2+d)(3b^2-d), \quad K_S^2 = 3(b^2+d)(7b^2-d), \quad e(S) = 3(b^2+d)(5b^2-3d), \tag{64}$$

*and the defect in the Bogomolov–Miyaoka–Yau inequality is*

$$3e(S) - K_S^2 = 24\,(b^4 - d^2), \tag{65}$$

*so that the inequality amounts exactly to $d \le b^2$. Finally, $K_S$ is never numerically proportional to $\lambda$.*

*Proof.* Since $\mathrm{ch}_0(\mathcal{F}) = 1$ and $c_1(\mathcal{F}) = b\Theta$, the sheaf $\mathcal{F}(-b\Theta) = \mathcal{I}_S$ has $\mathrm{ch}_0 = 1$, $\mathrm{ch}_1 = 0$, so $S$ has codimension at least two, and

$$\mathrm{ch}(\mathcal{O}_S) = 1 - \mathrm{ch}(\mathcal{I}_S) = 1 - \mathrm{ch}(\mathcal{F})\, e^{-b\Theta}.$$

Put $z = b + \sqrt{-d}$, so that $z\overline{z} = 2N$. Since $u_\Theta = \Re e^{\sqrt{-d}\Theta}$ and $\sqrt{d}\, v_\Theta = \Im e^{\sqrt{-d}\Theta}$, we have $\mathrm{ch}(\mathcal{F}) = \Im(z\, e^{\sqrt{-d}\Theta})/\sqrt{d}$, hence $\mathrm{ch}(\mathcal{F})e^{-b\Theta} = \Im(z\, e^{-\overline{z}\Theta})/\sqrt{d}$, whose coefficient of $\Theta^k$ is $(-1)^k\Im(z\overline{z}^k)/(\sqrt{d}\,k!) = (-1)^{k+1}\,2N\Im(z^{k-1})/(\sqrt{d}\,k!)$. With $\Im z = \sqrt{d}$, $\Im z^2 = 2b\sqrt{d}$ and $\Im z^3 = (3b^2-d)\sqrt{d}$ this gives

$$\mathrm{ch}_2(\mathcal{O}_S) = \tfrac{b^2+d}{2}\Theta^2 = N\Theta^2, \quad \mathrm{ch}_3(\mathcal{O}_S) = -\tfrac{2bN}{3}\Theta^3, \quad \mathrm{ch}_4(\mathcal{O}_S) = \tfrac{3b^4+2db^2-d^2}{24}\Theta^4.$$

The first identity says $S$ has codimension exactly two, so it is a surface, with class $N\Theta^2$. Because $T_X$ is trivial, $\mathrm{td}(T_X) = 1$, and Grothendieck–Riemann–Roch for $\iota$ reads $\mathrm{ch}(\iota_*\mathcal{O}_S) = \iota_*\,\mathrm{td}(T_S)$. Comparing degrees three and four,

$$\mathrm{ch}_3(\mathcal{O}_S) = \iota_*\tfrac{c_1(T_S)}{2} = -\tfrac{1}{2}\iota_*K_S, \qquad \mathrm{ch}_4(\mathcal{O}_S) = \iota_*\tfrac{c_1^2+c_2}{12} = \chi(\mathcal{O}_S)\,[\mathrm{pt}],$$

the second by Noether's formula. With $\Theta^4 = 24\,[\mathrm{pt}]$ these give $\iota_*K_S = \frac{4bN}{3}\Theta^3$ and, after substituting $d = 2N - b^2$, $\chi(\mathcal{O}_S) = 3b^4 + 2db^2 - d^2 = 4N(2b^2 - N)$. Then $K_S \cdot \lambda = \int_X \iota_*K_S \cdot \Theta = \frac{4bN}{3}\cdot 24 = 32bN$ and $\lambda^2 = \int_X [S]\cdot\Theta^2 = 24N$.

For the last two invariants, the normal bundle sequence and the triviality of $T_X$ give $c(T_S)c(N_{S/X}) = 1$, so $c_1(N_{S/X}) = K_S$ and $c_2(N_{S/X}) = K_S^2 - e(S)$; the self-intersection formula $[S]^2 = \int_S c_2(N_{S/X})$ then reads

$$24N^2 = K_S^2 - e(S),$$

while Noether's formula reads $12\chi(\mathcal{O}_S) = K_S^2 + e(S)$. Solving the two gives the stated $K_S^2$ and $e(S)$.

The conormal sequence presents $\Omega^1_S$ as a quotient of $\Omega^1_X|_S = \mathcal{O}_S^{\oplus 4}$, so $\Omega^1_S$ is globally generated, and therefore so is $K_S = \det\Omega^1_S$, a quotient of $\bigwedge^2\mathcal{O}_S^{\oplus 4}$. A globally generated line bundle is nef. A globally generated rank two bundle on a surface has $c_2 \ge 0$, the class $c_2$ being represented by the vanishing scheme of a general section, which is empty or finite; hence

$$e(S) = c_2(\Omega^1_S) = 12N(4b^2 - 3N) \;\ge\; 0,$$

so $N \le \frac{4}{3}b^2 < 4b^2$ and therefore $K_S^2 = 12N(4b^2 - N) > 0$. A nef divisor of positive self-intersection is big, so $\kappa(S) = 2$ and, $K_S$ being nef, $S$ is minimal of general type. The Bogomolov–Miyaoka–Yau inequality $c_1^2 \le 3c_2$ therefore applies and reads $K_S^2 \le 3e(S)$.

Substituting $N = \frac{1}{2}(b^2 + d)$ in (62) gives (64), hence $3e(S) - K_S^2 = 24(b^4 - d^2)$, and the inequality is $d \le b^2$, that is $N \le b^2$.

For the lower bound, $\lambda$ is ample on $S$, so the Hodge index theorem gives $(K_S \cdot \lambda)^2 \ge K_S^2 \, \lambda^2$, that is $(32bN)^2 \ge 12N(4b^2 - N) \cdot 24N$, which simplifies to $9N \ge 4b^2$. Since $N = \frac{1}{2}(b^2 + d)$ with $d > 0$, this holds automatically. Equality in the Hodge index theorem occurs exactly when $K_S$ is numerically proportional to $\lambda$, hence exactly when $9N = 4b^2$, and this never happens, because $9N = \frac{9}{2}(b^2 + d) > 4b^2$ for $d > 0$. ∎

Figure 34 draws the Bogomolov–Miyaoka–Yau defect over the quarter plane of $(b, d)$.

**Corollary 18.68** (Four discriminants on the fourfold)**.** *Let $(X, \Theta)$ be a principally polarised abelian fourfold and $d > 0$ squarefree. A sheaf $\mathcal{I}_S(3\Theta)$ with Chern character $u_\Theta + 3v_\Theta$, the value of Proposition 18.52(iii), and with smooth support $S$ exists only if $d \le 9$, that is $d \in \{1, 2, 3, 5, 6, 7\}$; if in addition $N = (9 + d)/2$ is an integer, as in Setup 18.51, only*

$$d \in \{1,\ 3,\ 5,\ 7\}, \qquad N \in \{5,\ 6,\ 7,\ 8\}.$$

*For these values the invariants are forced to be those of Table 3. In particular, for $d > 9$ no such sheaf has smooth support.*

*Proof.* This is immediate from (63) at $b = 3$ and the arithmetic of (62), and the table evaluates (62) at $b = 3$. The value $d = 9$ is the equality case of the Bogomolov–Miyaoka–Yau inequality, where $K_S^2 = 3e(S) = 2916$ and $S$ would have to be a quotient of the ball; it is excluded because 9 is not squarefree. ∎

| $d$ | $N$ | $\chi(\mathcal{O}_S)$ | $K_S^2$ | $e(S)$ | $K_S \cdot \lambda$ | $\lambda^2$ |
|---|---|---|---|---|---|---|
| 1 | 5 | 260 | 1860 | 1260 | 480 | 120 |
| 3 | 6 | 288 | 2160 | 1296 | 576 | 144 |
| 5 | 7 | 308 | 2436 | 1260 | 672 | 168 |
| 7 | 8 | 320 | 2688 | 1152 | 768 | 192 |

TABLE 3. The only invariants a smooth support can have on an abelian fourfold, by Theorem 18.67 and Corollary 18.68, at $b = 3$ and with $N$ an integer. Every entry is determined by $\mathrm{ch}(\mathcal{F})$ alone. The slope $K_S^2/\chi(\mathcal{O}_S)$ runs from 93/13 at $d = 1$ to 42/5 at $d = 7$, and reaches the Bogomolov–Miyaoka–Yau ceiling of 9 only at the excluded value $d = 9$.

*Remark* 18.69 (Status of the smooth case)*.* The local argument does not apply here, for a structural reason given by Proposition 18.65: the sheaf $\mathcal{E}xt^2$ of a smooth ideal vanishes, so both sides of (58) are locally zero, and the comparison of germs that decides Theorem 18.61 has nothing to compare.

The surface is rigidly constrained. At $b = 3$ its class, its canonical class, its Euler characteristic, its Euler number and its two intersection numbers with the polarisation are all determined by $d$, and they are compatible with existence only for $d \le 9$. This is the content of Corollary 18.68, and it restricts the construction: for all but finitely many imaginary quadratic fields the secant construction of Setup 18.51 with smooth support is not available, whatever the geometry of the surface.

What remains is a single existence question. For $d \in \{1, 3, 5, 7\}$, does a principally polarised abelian fourfold carry a smooth surface $S$ with the invariants of Table 3, and if

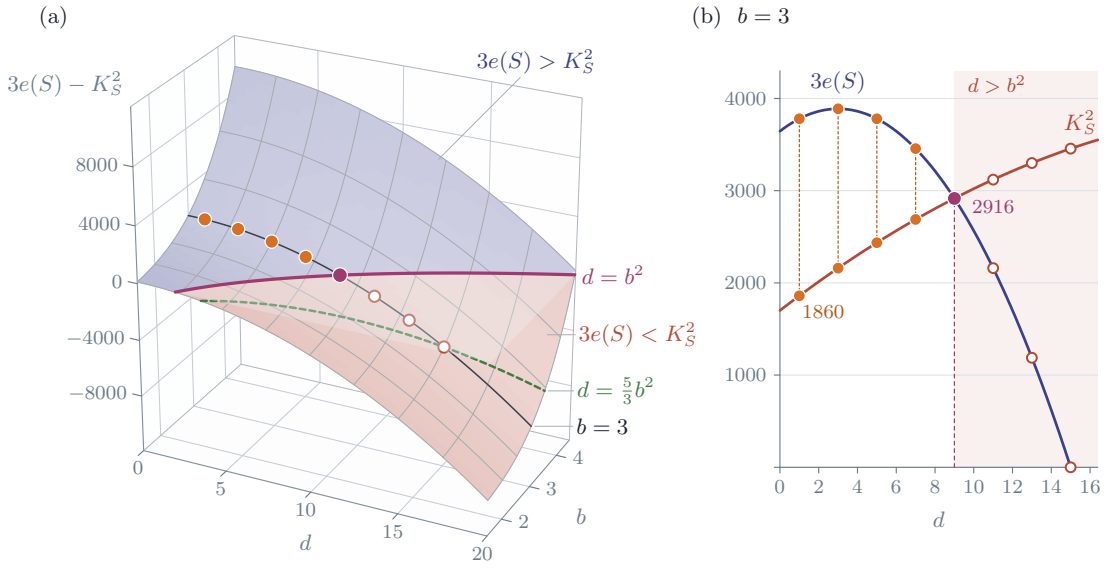



FIGURE 34. Theorem 18.67 over the quarter plane of $(b, d)$. The height of the surface is the defect $3e(S) - K_S^2 = 24(b^4 - d^2)$ in the Bogomolov–Miyaoka–Yau inequality for the surface the theorem attaches to the pair, compressed by a signed cube root so that four orders of magnitude fit one frame; the compression is odd and monotone, so the zero locus and the sign are the true ones. The surface is drawn indigo where the defect is positive and clay where it is negative, and it crosses zero along $d = b^2$, which is therefore the whole content of the inequality. The weaker wall $d = \frac{5}{3}b^2$, where $e(S)$ itself changes sign, is drawn dashed and lies above it. On the line $b = 3$, which is the value Proposition 18.52(iii) forces on the fourfold, the four squarefree discriminants $d = 1, 3, 5, 7$ with $N$ an integer sit above the wall and are marked with their drop lines; $d = 9$ sits exactly on it, the equality case, and $d = 11, 13, 15$ sit below and are excluded.

so, does the image of $A_\theta A_{\theta'}$ in $H^1(S, N_{S/X})$ vanish for all $\theta, \theta' \in H^0(T_X)$? The first half is a question about surfaces in abelian fourfolds with prescribed Chern numbers, of the kind that is answered by a construction or by a classification argument; the second is a question about one cohomology group of one normal bundle. Neither is settled here. The simplest surfaces of the right class are ruled out by one entry of the table: a complete intersection of divisors in $|a\Theta|$ and $|b'\Theta|$ has class $ab'\Theta^2$, so $ab' = N$, and canonical class $(a + b')\lambda$ by adjunction, so $K_S \cdot \lambda = 24(a + b')N$; since $ab' = N \geq 5$ forces $a + b' \geq 5$, this exceeds the required $96N$. The surface, if it exists, is not cut out by two theta divisors, nor by any two divisors algebraically equivalent to multiples of $\Theta$; nor, by Corollary 18.84 below, is its ideal sheaf presented by any two bundles whose classes lie in the subring generated by line bundles. It must also move with $X$ to first order in every polarised direction, in the exact sense of Proposition 18.73, while the injectivity that Lemma 18.88 asks of it holds automatically when $X$ is simple (Lemma 18.70).

A positive answer to both, for a surface that moves in this way, together with an affirmative answer to [Mar25b, Question 11.4], would produce Weil classes on abelian eightfolds for at most four discriminants; the other discriminants, every $n \geq 5$ and every

field of degree larger than two would still have to be reached through Proposition 20.13 and the propagation statement.

Two of the conditions on a smooth support are decided by the table, and the third, which Theorem 18.56(iii)(a) imposes on every candidate, has an exact form. The first is the injectivity of $H^2(\mathcal{O}_X) \to H^2(\mathcal{O}_S)$ that Lemma 18.88 below requires of every support.

**Lemma 18.70** (The injectivity condition for a smooth support)**.** *Let $X$ be a simple complex abelian fourfold, let $V = H^0(T_X)$, and let $S \subset X$ be a smooth surface. The following are equivalent:*

(i) *the restriction $H^2(\mathcal{O}_X) \to H^2(\mathcal{O}_S)$ is injective;*
(ii) *no nonzero translation-invariant holomorphic 2-form on $X$ restricts to zero on $S$;*
(iii) *$S$ is not Lagrangian for any translation-invariant holomorphic symplectic form on $X$;*
(iv) *the Gauss image of $S$ in $\mathrm{Gr}(2,V) \subset \mathbb{P}(\bigwedge^2 V)$ spans $\mathbb{P}(\bigwedge^2 V)$.*

*If $e(S) \neq [S]^2$, they hold. In particular they hold for every smooth support of Table 3, where $e(S) = 12N(36-3N)$ and $[S]^2 = 24N^2$ agree only for $5N = 36$.*

*Proof.* The restriction $\iota^*\colon H^2(X,\mathbb{C}) \to H^2(S,\mathbb{C})$ is a morphism of Hodge structures defined over $\mathbb{Q}$, so it commutes with complex conjugation, which interchanges the $(0,2)$ and $(2,0)$ parts; hence $\iota^*$ is injective on $H^{0,2}(X) = H^2(\mathcal{O}_X)$ if and only if it is injective on $H^{2,0}(X) = \bigwedge^2 V^\vee$, the translation-invariant 2-forms. A holomorphic 2-form on a compact surface is zero as a class in $H^{2,0}(S) = H^0(K_S)$ if and only if it is zero as a form; this is (i)⇔(ii). The restriction of $p \wedge q \in \bigwedge^2 V^\vee$ to $S$ is the pull-back of the Plücker coordinate $p \wedge q$ along the Gauss map $\gamma\colon S \to \mathrm{Gr}(2,V)$, $s \mapsto T_sS$, as a section of $K_S = \gamma^*\mathcal{O}(1)$, so a 2-form vanishes on $S$ if and only if the corresponding hyperplane contains $\gamma(S)$; this is (ii)⇔(iv).

A nonzero 2-form on $V \cong \mathbb{C}^4$ has rank two or four. If $\omega$ has rank four, $\omega|_S = 0$ says that every tangent plane of $S$ is $\omega$-isotropic, that is, $S$ is Lagrangian for $\omega$. If $\omega = \alpha \wedge \beta$ has rank two, with $\alpha, \beta \in V^\vee$ independent, then $\alpha|_S$ and $\beta|_S$ are independent holomorphic 1-forms on $S$, because $S$ generates the simple $X$ and a nonzero invariant 1-form restricts nontrivially to a generating subvariety, and $\alpha|_S \wedge \beta|_S = 0$. By the theorem of Castelnuovo and de Franchis [BHPV04] there is a fibration $f\colon S \to C$ onto a curve of genus at least two with $\alpha|_S$ and $\beta|_S$ pulled back from $C$. A fibre $F$ of $f$ is a curve in $X$ on which $\alpha$ and $\beta$ vanish, so all its tangent lines lie in the fixed plane $B = \ker\alpha \cap \ker\beta \subset V$. The abelian subvariety generated by $F - F$ is the image of a sum map from a power of $F$, whose differential at a general point is spanned by tangent vectors of $F$; so its tangent space lies in $B$, its dimension is at most two, and it is nonzero because $F$ is not a point. This contradicts the simplicity of $X$. Hence only the rank-four case occurs, which is (ii)⇔(iii).

Finally, if $S$ is Lagrangian for a symplectic $\omega$, then $\omega$ identifies $T_S$ with the annihilator of $T_S$ in $\Omega^1_X|_S$, which is the conormal bundle, so $N_{S/X} \cong \Omega^1_S$ and $c_2(N_{S/X}) = e(S)$; since $T_X$ is trivial, $c_2(N_{S/X}) = [S]^2$ by the self-intersection formula. So $e(S) \neq [S]^2$ excludes (iii). For the table, Theorem 18.67 gives $[S]^2 = 24N^2$ and $e(S) = 12N(36-3N)$ at $b = 3$, and $24N^2 = 12N(36-3N)$ reads $5N = 36$. For general $b$ the two agree only when $N = \frac{4}{5}b^2$, that is $d = \frac{3}{5}b^2$, whose first integral instance is $b = 5$, $d = 15$. ∎

*Remark* 18.71 (The class $K_S - 4\lambda$). Theorem 18.67 records $K_S \cdot \lambda = 96N$ and $\lambda^2 = 24N$ at $b = 3$, so $K_S \cdot \lambda = 4\lambda^2$, and that $K_S$ is never numerically proportional to $\lambda$. The sharper statement is that

$$\delta = K_S - 4\lambda \in \ker\bigl(\iota_*\colon H^2(S,\mathbb{Q}) \to H^6(X,\mathbb{Q})\bigr), \qquad \delta^2 = -12N(N-4) < 0,$$

because $\iota_*K_S = 4N\Theta^3$ and $\iota_*\lambda = \Theta \cup [S] = N\Theta^3$. So $\delta$ is a nonzero algebraic class of type $(1,1)$, orthogonal by the projection formula to the whole image of $H^2(X,\mathbb{Q})$ in $H^2(S,\mathbb{Q})$, and of negative square: the Picard number of $S$ is at least two, and the kernel of the Gysin map on $H^2(S,\mathbb{Q})$, a sub-Hodge structure, contains an algebraic class. For general $b$ the class is $K_S - \frac{4b}{3}\lambda$, with square $\frac{4N}{3}(4b^2 - 9N)$, and its vanishing is the equality case $9N = 4b^2$ of the Hodge index bound in Theorem 18.67. This excludes none of the four cases; it describes them.

The third condition is the one that Theorem 18.56(iii)(a) imposes on every candidate and that the construction of Setup 18.51 meets by its uniformity over the moduli of principally polarised fourfolds (Corollary 18.57): the object must extend to first order along every polarised deformation of $(X,\Theta)$. For $\mathcal{F} = \mathcal{I}_S(3\Theta)$ with $S$ smooth this is a condition on $S$. If $S$ extends along a deformation $X_v$ of $X$ that keeps $\Theta$ of type $(1,1)$, then so does $\mathcal{F}$. Conversely, the obstruction $v \lrcorner \mathrm{At}(\mathcal{F}) \in \mathrm{Ext}^2(\mathcal{F},\mathcal{F})$ has, under the edge map to $H^1(S, \mathcal{E}xt^1(\mathcal{F},\mathcal{F})) = H^1(S, N_{S/X})$ of Remark 18.66, the image of $v$ under $H^1(T_X) \to H^1(\iota^*T_X) \to H^1(N_{S/X})$, by Lemma 18.59 applied on the overlaps of a cover; and that image is the obstruction $\mathrm{ob}_S(v)$ to extending the subscheme $S$ along $X_v$. So condition (a) for $\mathcal{F}$ is the statement that $S$ extends to first order along every polarised deformation of $X$, and Bloch's semiregularity map [Blo72] puts it in a form that names the object it concerns. For $S \subset X$ smooth of codimension two with normal bundle $N = N_{S/X}$, let

$$\pi\colon H^1(S,N) \xrightarrow{\cong} H^1(S, N^\vee \otimes K_S) \xrightarrow{j} H^1(S, \iota^*\Omega^1_X \otimes K_S) \xrightarrow{\iota_*} H^3(X,\Omega^1_X)$$

be the composite of $N \cong N^\vee \otimes \bigwedge^2 N$, of the conormal inclusion $N^\vee \to \iota^*\Omega^1_X$ tensored with $K_S$, and of the Gysin map, with $K_X \cong \mathcal{O}_X$. This is the semiregularity map of Bloch for codimension two, and it satisfies $\pi(\mathrm{ob}_S(v)) = v \lrcorner [S]$ for every $v \in H^1(X,T_X)$.

**Lemma 18.72** (Duality)**.** *Under Serre duality $H^1(X,T_X) \times H^3(X,\Omega^1_X) \to H^4(\mathcal{O}_X) = \mathbb{C}$,*

$$\{v \in H^1(X,T_X) : S \text{ extends to first order along } v\} = \pi\big(H^1(S,N)\big)^\perp.$$

*Proof.* $S$ extends along $v$ if and only if $\mathrm{ob}_S(v) = 0$, and $\mathrm{ob}_S$ is the composite of the restriction $r\colon H^1(X,T_X) \to H^1(S,\iota^*T_X)$ with the map induced by $\iota^*T_X \to N$. The Serre dual of $r$ is the Gysin map $\iota_*\colon H^1(S,\iota^*\Omega^1_X \otimes K_S) \to H^3(X,\Omega^1_X \otimes K_X)$, and the Serre dual of $H^1(\iota^*T_X) \to H^1(N)$ is the map $H^1(N^\vee \otimes K_S) \to H^1(\iota^*\Omega^1_X \otimes K_S)$ induced by the conormal inclusion. So the dual of $\mathrm{ob}_S$ is $\iota_* \circ j$, which is $\pi$ after the identification $N \cong N^\vee \otimes K_S$, and the kernel of a linear map is the annihilator of the image of its dual. ∎

**Proposition 18.73** (The first-order condition is a rank condition)**.** *Let $S \subset X$ be a smooth surface with $[S] = N\Theta^2$, $N \neq 0$. Then $\pi(H^1(S,N)) \supseteq \Theta \cup H^{0,2}(X)$, a subspace of dimension six, and the following are equivalent:*

(i) *$S$ extends to first order along every polarised deformation of $(X,\Theta)$, that is, $\mathcal{I}_S(3\Theta)$ satisfies condition (a) of Theorem 18.56(iii);*
(ii) *$\pi(H^1(S,N)) = \Theta \cup H^{0,2}(X)$;*
(iii) *$\operatorname{rank}\pi = 6$, the least value possible.*

*In particular a smooth secant support is as far from semiregular in Bloch's sense as a surface with this class can be: $\dim\ker\pi = h^1(N) - 6$, with $h^1(N) = 2h^0(N) - 2\chi(\mathcal{O}_S) + 24N^2 \geq 8 - 2\chi(\mathcal{O}_S) + 24N^2$, which is at least 88, 296, 568 and 904 for $N = 5,6,7,8$.*

*Proof.* The polarised deformations form $P = \{v : v \lrcorner \Theta = 0\}$, of dimension ten in $H^1(T_X) = V \otimes H^{0,1}(X)$. For $z \in H^{0,2}(X)$ the pairing is $\langle v, \Theta \cup z\rangle = (v \lrcorner \Theta) \cup z$, so $\Theta \cup H^{0,2}(X) \subseteq P^\perp$;

the map $z \mapsto \Theta \cup z$ is injective by hard Lefschetz, so this subspace has dimension six, and $\dim P^{\perp} = 16 - 10 = 6$, so $P^{\perp} = \Theta \cup H^{0,2}(X)$. By Lemma 18.72, (i) says $P \subseteq \pi(H^1(N))^{\perp}$, that is $\pi(H^1(N)) \subseteq P^{\perp}$. On the other hand $\pi(\mathrm{ob}_S(v)) = v \lrcorner [S] = 2N\,\Theta \cup (v \lrcorner \Theta)$, and $v \mapsto v \lrcorner \Theta$ is surjective onto $H^{0,2}(X)$, because $\Theta$ identifies $V$ with $\overline{V}^{\vee}$ and the contraction becomes the exterior product $V \otimes \overline{V}^{\vee} \to \bigwedge^2 \overline{V}^{\vee}$. So $\pi(H^1(N)) \supseteq \Theta \cup H^{0,2}(X)$ always, and (i), (ii), (iii) are equivalent. For $h^1(N)$, Riemann–Roch with $c_1(N) = K_S$ and $c_2(N) = [S]^2 = 24N^2$ gives $\chi(N) = 2\chi(\mathcal{O}_S) - 24N^2$, Serre duality with $N^{\vee} \otimes K_S \cong N$ gives $h^2(N) = h^0(N)$, and $h^0(N) \geq 4$ because the four translation vector fields map injectively to $H^0(N)$: a translation vector field tangent to $S$ everywhere would restrict to a nowhere vanishing vector field on $S$, so that $e(S) = c_2(T_S) = 0$, while a secant support has $e(S) = 12N(36 - 3N) \neq 0$ for $N = 5, 6, 7, 8$ by Theorem 18.67. ∎

*Remark* 18.74 (What the rank condition asks). Proposition 18.73 does not decide the condition; it converts it into the statement that the kernel of Bloch's map has codimension exactly six, as it has for the complete intersections $\Theta \cap \Theta_x$, which do move with $X$. The kernel of $\pi$ is $j^{-1}(V^{\vee} \otimes \ker(H^1(K_S) \to H^3(\mathcal{O}_X)))$, and for $q(S) = 4$ it is the image in $H^1(N^{\vee} \otimes K_S)$ of $H^0(\Omega^1_S \otimes K_S)$ modulo $V^{\vee} \otimes H^0(K_S)$. A count with $\chi(\Omega^1_S \otimes K_S) = 2\chi(\mathcal{O}_S) + 24N^2$ and $h^1(T_S) \leq h^0(N) + 12$ shows the rank condition to be consistent with every inequality available, so no exclusion follows at this level; what would decide it is $\ker \pi$ for an actual surface, and no candidate surface is known.

18.5. **Depth and the local obstruction.** By Proposition 18.65, at a smooth point of the support there is nothing local to compare. This is a special case of a much more general statement, and the general statement shows exactly which supports Theorem 18.61 and Corollary 18.63 exclude and which lie beyond their reach. The dividing line is neither smoothness nor the local complete intersection property, but depth.

**Theorem 18.75** (Cohen–Macaulay supports carry no local obstruction)**.** *Let $Y$ be a smooth variety with $\omega_Y \cong \mathcal{O}_Y$, let $Z \subset Y$ be a closed subscheme of codimension two, and let $p \in Z$. If the local ring $\mathcal{O}_{Z,p}$ is Cohen–Macaulay, then*

$$\mathcal{E}xt^q(\mathcal{I}_Z, \mathcal{I}_Z)_p = 0 \qquad \text{for every } q \geq 2,$$

*and the same holds for $\mathcal{I}_Z \otimes L$ for every line bundle $L$. Consequently both sides of (58) have vanishing germ at $p$, and Lemma 18.59 decides nothing there.*

*Proof.* Twisting by $L$ changes no $\mathcal{E}xt$ sheaf, so take $F = \mathcal{I}_Z$. Write $A = \mathcal{O}_{Y,p}$, a regular local ring of dimension $m$, and $I = \mathcal{I}_{Z,p}$. By hypothesis $A/I$ is Cohen–Macaulay of dimension $m - 2$, so $\operatorname{depth} A/I = m - 2$, and the Auslander–Buchsbaum formula gives

$$\operatorname{pd}_A(A/I) = m - \operatorname{depth}_A(A/I) = 2.$$

From the exact sequence $0 \to I \to A \to A/I \to 0$ with $A$ free it follows that $\operatorname{pd}_A I = 1$. A module of projective dimension one has a free resolution of length one, so $\operatorname{Ext}^q_A(I, M) = 0$ for every $A$-module $M$ and every $q \geq 2$; taking $M = I$ gives the vanishing, since $\mathcal{E}xt^q(\mathcal{I}_Z, \mathcal{I}_Z)_p = \operatorname{Ext}^q_A(I, I)$. The last assertion follows from Lemma 18.59: the right side of (58) is $z \cdot \mathrm{id}_F$ with $z \in H^2(\mathcal{O}_Y)$ and has vanishing germs, and the left side lies in a sheaf whose $\mathcal{E}xt^2$ factor is zero. ∎

*Remark* 18.76 (Sharpness of the hypothesis). Two observations show that Theorem 18.75 is the right statement.

First, Cohen–Macaulayness is strictly weaker than the more familiar conditions that could be imposed instead. Among codimension two germs in $\mathbb{C}^4$, the smooth germ

$(x_1, x_2)$, two planes meeting along a line, $(x_1, x_2x_3)$, three planes meeting along a line, $(x_1, x_2x_3x_4)$, the double plane $(x_1^2, x_2)$ and the quadric cone $(x_1, x_2^2 - x_3x_4)$ are complete intersections. The fat plane $(x_1, x_2)^2$ and the cone over the twisted cubic are the ideals of $2 \times 2$ minors of $\left(\begin{smallmatrix} x_1 & x_2 & 0 \\ 0 & x_1 & x_2 \end{smallmatrix}\right)$ and of $\left(\begin{smallmatrix} x_1 & x_2 & x_3 \\ x_2 & x_3 & x_4 \end{smallmatrix}\right)$, of the expected codimension two, so they are Cohen–Macaulay by the Hilbert–Burch theorem, although neither is a local complete intersection, both needing three generators. By Theorem 18.75, $\mathcal{E}xt^2(\mathcal{I}_Z, \mathcal{I}_Z)$ vanishes at all seven. Reducedness, irreducibility and the complete intersection property are therefore all irrelevant; what matters is the projective dimension, which by Auslander–Buchsbaum is a matter of depth.

Second, the vanishing fails at the germs that the construction of Setup 18.51 produces. Two planes meeting at a point and three planes meeting pairwise at a point are reduced surface germs that become disconnected when the point is removed, so they have depth 1: at least 1 because they are reduced, and at most 1 by the connectedness theorem recalled below. A plane with an embedded point has depth 0. By Auslander–Buchsbaum the projective dimensions of $\mathcal{O}_{Z,p}$ are 3, 3 and 4, and the proof of Corollary 18.77(i) below gives $\mathcal{E}xt^q(\mathcal{I}_Z, \mathcal{I}_Z)_p \neq 0$ for $q = 2$, 2 and 3. By Hartshorne's connectedness theorem, a scheme whose local ring at $p$ is Cohen–Macaulay is connected in codimension one at $p$, so a support with two components through $p$ meeting only at $p$ is never Cohen–Macaulay there. This is precisely the configuration analysed in Lemma 18.60.

**Corollary 18.77** (The local dichotomy)**.** *Let $(X, t)$ be a polarised abelian variety of dimension $n \geq 4$ and $F$ a perfect complex with $\operatorname{ch}(F) = au_t + bv_t$, $(a, b) \neq (0, 0)$, whose germ at each point is the germ of $\mathcal{I}_Z \otimes L$ for a subscheme $Z$ of codimension two, and let $p \in Z$.*

(i) *If $Z$ is not Cohen–Macaulay at $p$, then $\mathcal{E}xt^q(\mathcal{I}_Z, \mathcal{I}_Z)_p \neq 0$ for $q = \operatorname{pd} \mathcal{I}_{Z,p} \geq 2$, so the comparison of germs of Lemma 18.59 has room at $p$. If moreover $Z$ has at $p$ two smooth branches of codimension two meeting transversally, as in Lemma 18.60 and Corollary 18.63, then $F$ does not satisfy the weakened criterion, nor does $\Phi(F \boxtimes F_2)$ for any secant $F_2$.*

(ii) *If $Z$ is Cohen–Macaulay, then there is no local obstruction at any point. Condition (b) of Theorem 18.56(iii) is then a single global condition: it holds if and only if the image of $A_\theta A_{\theta'}$ in $H^1(Z, \mathcal{E}xt^1(\mathcal{I}_Z, \mathcal{I}_Z))$ vanishes for all $\theta, \theta' \in H^0(T_X)$, the scalar on the right of (58) being forced by the trace. Condition (a), the extension of $F$ to first order along the polarised deformations, is not affected by the depth and remains to be checked.*

*In case (ii), $\mathcal{I}_Z$ has projective dimension one at every point, so it is locally the ideal of maximal minors of a Hilbert–Burch matrix; if $Z$ is moreover smooth, then $\mathcal{E}xt^1(\mathcal{I}_Z, \mathcal{I}_Z) = N_{Z/X}$ and the invariants of $Z$ are those of Theorem 18.67.*

*Proof.* (i) Put $A = \mathcal{O}_{X,p}$, of dimension $n$, and $I = \mathcal{I}_{Z,p}$. Since $A/I$ is not Cohen–Macaulay, $\operatorname{depth} A/I < \dim A/I \leq n - 2$, so $\operatorname{pd}_A(A/I) \geq 3$ by the Auslander–Buchsbaum formula and $q = \operatorname{pd}_A I \geq 2$. In a minimal free resolution $F_\bullet \to I$ the entries of the differentials lie in the maximal ideal $\mathfrak{m}$, so $\operatorname{Ext}^q_A(I, I)$, the cokernel of $\operatorname{Hom}(F_{q-1}, I) \to \operatorname{Hom}(F_q, I)$, surjects onto $\operatorname{Hom}(F_q, I) \otimes A/\mathfrak{m} \neq 0$ by Nakayama's lemma. At two transversal branches the conclusion is Theorem 18.61 and Corollary 18.63. (ii) Theorem 18.75 gives $\mathcal{E}xt^q = 0$ for $q \geq 2$ at every point, so the local-to-global spectral sequence for $\operatorname{Ext}^2(\mathcal{I}_Z, \mathcal{I}_Z)$ has only the two rows $H^1(\mathcal{E}xt^1)$ and $H^2(\mathcal{O}_X)$, and these lie in different steps of the filtration, the second being the subgroup of scalars. The right side of (58) lies in that subgroup, so its image in the quotient $H^1(\mathcal{E}xt^1)$ is zero, and (b) forces the image of the left side there to vanish.

Conversely, if that image vanishes, then $\mathrm{At}(F)^2 = z\cdot \mathrm{id}_F$ for some $z \in H^2(\mathcal{O}_X)\otimes \bigwedge^2 H^{1,0}$, the map $z \mapsto z\cdot \mathrm{id}_F$ being injective because its composite with the trace is multiplication by $\mathrm{rk}\, F = 1$; taking traces, $z = \mathrm{tr}\,\mathrm{At}(F)^2 = -d\,t^2$ by the last assertion of Theorem 18.56(iii), which is (b). The statement on projective dimension was established in the proof of Theorem 18.75, and the Hilbert–Burch theorem is the structure theorem for ideals of projective dimension one over a regular local ring; the last clause is Proposition 18.65 together with Theorem 18.67. ∎

*Remark* 18.78 (Normalisation and depth). Corollary 18.77 explains the tension recorded in Remark 18.62. The partial normalisation performed in Setup 18.51 separates the two branches at an isolated double point, and separating them is exactly what would make the local ring Cohen–Macaulay there; the normalisation is thus an attempt to reach case (ii). Remark 18.62 shows that the attempt cannot succeed: normalising every isolated point moves $\chi(\mathcal{O}_{\widetilde{Z}})$ to $50N$, while the secant plane demands $(d+9)(27-d) = 72N - 4N^2$, and the two agree only at $N = 11/2$. Repairing the depth destroys the Chern character, and keeping the Chern character keeps a point where the depth fails. This is the precise form of the obstruction, and it explains why a construction assembled from intersections of translates of a polarisation cannot be made to work: by Corollary 18.63 such a chain always has a point of the bad type, and by Remark 18.62 it cannot be separated without leaving the plane.

18.6. **Hilbert–Burch resolutions.** Corollary 18.77 leaves one class of supports and, beside the extension along the polarised deformations, one condition on them. The class consists of the Cohen–Macaulay subschemes of codimension two, and for these the proof of Theorem 18.75 gives more than the vanishing it was used for: the ideal sheaf has projective dimension one at every point. This is the Hilbert–Burch shape, whose invariants are polynomials in the twists.

**Setup 18.79** (A resolution of the support)**.** Let $(X,\Theta)$ be a principally polarised abelian fourfold with $\mathrm{NS}(X)_{\mathbb{Q}} = \mathbb{Q}\Theta$, let $Z \subset X$ be Cohen–Macaulay of codimension two, and suppose $\mathcal{I}_Z$ admits a global resolution

$$0 \longrightarrow E_1 \xrightarrow{\ \varphi\ } E_0 \longrightarrow \mathcal{I}_Z \longrightarrow 0, \qquad \mathrm{rk}\, E_0 = r+1, \quad \mathrm{rk}\, E_1 = r, \tag{66}$$

with $E_0 = \bigoplus_{i=1}^{r+1}\mathcal{O}(e_i\Theta)$ and $E_1 = \bigoplus_{j=1}^{r}\mathcal{O}(f_j\Theta)$ sums of line bundles. Put $p_i = e_i + b$ and $q_j = f_j + b$.

**Lemma 18.80** (The resolution in moments)**.** *In Setup 18.79,* $\mathrm{ch}(\mathcal{I}_Z(b\Theta)) = u_\Theta + b\,v_\Theta$ *holds if and only if*

$$\sum_{i=1}^{r+1} p_i^k - \sum_{j=1}^{r} q_j^k = c_k, \qquad k = 0,1,2,3,4, \qquad (c_k) = (1,\ b,\ -d,\ -db,\ d^2). \tag{67}$$

*The equation at $k=0$ is the rank count and is automatic. In addition, $\varphi$ can be injective only if $\max_i e_i \geq \max_j f_j$, and $\mathcal{O}_X$ embeds in $E_0^\vee$ only if $\min_i e_i \leq 0$.*

*Proof.* From (66), $\mathrm{ch}(\mathcal{I}_Z) = \mathrm{ch}(E_0) - \mathrm{ch}(E_1)$, so $\mathrm{ch}(\mathcal{I}_Z(b\Theta)) = \sum_i e^{p_i\Theta} - \sum_j e^{q_j\Theta}$. Comparing coefficients of $\Theta^k/k!$ with those of $u_\Theta + bv_\Theta$, which are the $c_k$ displayed, gives (67). For the last two statements, recall that on an abelian variety $\mathrm{Hom}(\mathcal{O}(f\Theta),\mathcal{O}(e\Theta)) = H^0(\mathcal{O}((e-f)\Theta))$ vanishes unless $e \geq f$. If $\varphi$ is injective, each summand of $E_1$ must map nonzero to $E_0$, which requires $\max_i e_i \geq f_j$ for every $j$. Dualising (66) and using $\mathcal{I}_Z^\vee = \mathcal{O}_X$ gives an injection $\mathcal{O}_X \hookrightarrow E_0^\vee$, which requires $-e_i \geq 0$ for some $i$. ∎

The case $r = 1$ consists of the complete intersections, and it is settled for every $b$ and every $d$ by an argument that needs no search.

**Theorem 18.81** (No complete intersection supports a secant object)**.** *Let $(X, \Theta)$ be a principally polarised abelian fourfold with $\mathrm{NS}(X)_{\mathbb{Q}} = \mathbb{Q}\Theta$, let $d > 0$ and $b \geq 1$. No complete intersection $Z = D_1 \cap D_2$ of two divisors satisfies $\mathrm{ch}(\mathcal{I}_Z(b\Theta)) = u_\Theta + b\, v_\Theta$.*

*Proof.* Such a $Z$ is Setup 18.79 with $r = 1$, the resolution being the Koszul complex of the two equations. Put $w(x) = x^2 + d$, which is strictly positive on $\mathbb{R}$ because $d > 0$, and read (67) in the combinations $c_{k+2} + d\, c_k$ for $k = 0, 1, 2$. Those three numbers are

$$c_2 + dc_0 = -d + d = 0, \qquad c_3 + dc_1 = -db + db = 0, \qquad c_4 + dc_2 = d^2 - d^2 = 0,$$

so that

$$\sum_i w(p_i)\, p_i^k = \sum_j w(q_j)\, q_j^k, \qquad k = 0, 1, 2.$$

The two sides are the moments of order 0, 1 and 2 of the positive measures $\nu_P = \sum_i w(p_i)\delta_{p_i}$ and $\nu_Q = \sum_j w(q_j)\delta_{q_j}$, so those measures have the same mass, the same mean and the same variance. At $r = 1$ the measure $\nu_Q$ has one atom, hence variance zero, so $\nu_P$ has variance zero as well, and a positive measure with two atoms has variance zero only if the atoms coincide. Hence $p_1 = p_2$, and equality of the means forces $q_1 = p_1$. The resolution (66) then cancels, and $\mathcal{I}_Z \cong \mathcal{O}(e_1\Theta)$ would be a line bundle, which the ideal sheaf of a subscheme of codimension two is not.

In terms of the divisors, the same computation shows that their degrees $a_1$ and $a_2$ would have to satisfy $a_1 + a_2 = \frac{4}{3}b$ and $(b - a_1)(b - a_2) = \frac{1}{2}(\frac{1}{3}b^2 + d)$; the quadratic with roots $b - a_1$ and $b - a_2$ then has discriminant $-\frac{2}{9}b^2 - 2d < 0$, so the degrees are not even real. ∎

For $r \geq 2$ the moment conditions alone no longer decide the question. The Chern characters of the bundles of Setup 18.79, and of every bundle built from line bundles, lie in one lattice, and the secant plane meets that lattice only along a congruence. Write a class $\xi \in \mathbb{Q}[\Theta]/(\Theta^5)$ as $\xi = \sum_{k=0}^{4} c_k(\xi)\, \Theta^k/k!$, as in Lemma 18.80, so that $c_k(e^{j\Theta}) = j^k$, and let $\mathcal{L}_\Theta$ be the subgroup generated by the classes $e^{j\Theta} = \mathrm{ch}\, \mathcal{O}_X(j\Theta)$, $j \in \mathbb{Z}$. It is a subring, since $e^{a\Theta}e^{b\Theta} = e^{(a+b)\Theta}$, and it contains ch of every bundle obtained from line bundles algebraically equivalent to multiples of $\Theta$ by direct sums, tensor products, duals, exterior and symmetric powers, kernels of surjections and cokernels of injections between such bundles, and extensions; in particular the kernel bundles $M_{m\Theta} = \ker\big(H^0(\mathcal{O}(m\Theta)) \otimes \mathcal{O}_X \to \mathcal{O}_X(m\Theta)\big)$, of class $m^4 - e^{m\Theta}$, and their duals and twists.

**Lemma 18.82** (Binomial moments)**.** *Define $m_k(\xi) \in \mathbb{Q}$, $0 \leq k \leq 4$, by*

$$c_i(\xi) = \sum_{k=0}^{i} S(i, k)\, k!\, m_k(\xi), \tag{68}$$

*where $S(i, k)$ are the Stirling numbers of the second kind; explicitly*

$$m_0 = c_0, \qquad m_1 = c_1, \qquad m_2 = \tfrac{1}{2}(c_2 - c_1),$$
$$m_3 = \tfrac{1}{6}(c_3 - 3c_2 + 2c_1), \qquad m_4 = \tfrac{1}{24}(c_4 - 6c_3 + 11c_2 - 6c_1).$$

*Then $\xi \in \mathcal{L}_\Theta$ if and only if $m_k(\xi) \in \mathbb{Z}$ for $k = 0, \ldots, 4$.*

*Proof.* For $\xi = \sum_j a_j e^{j\Theta}$ one has $c_i(\xi) = \sum_j a_j j^i$, and $j^i = \sum_k S(i, k)\, k! \binom{j}{k}$, so (68) holds with $m_k(\xi) = \sum_j a_j \binom{j}{k} \in \mathbb{Z}$. Conversely, the five vectors $\big(\binom{j}{0}, \ldots, \binom{j}{4}\big)$ for $j = 0, \ldots, 4$ form a unitriangular matrix, so every integer vector $(m_0, \ldots, m_4)$ is an integer combination of

them, and the corresponding $\sum_{j=0}^{4} a_j e^{j\Theta}$ lies in $\mathcal{L}_\Theta$ with the prescribed moments. Since (68) is invertible over $\mathbb{Q}$, the moments determine $\xi$. ∎

Equivalently, sending $e^{j\Theta}$ to $x^j$ identifies $\mathbb{Q}[\Theta]/(\Theta^5)$ with $\mathbb{Q}[y]/(y^5)$, $y = x-1$, the class $e^{j\Theta}$ going to $(1+y)^j$; then $m_k(\xi)$ is the coefficient of $y^k$ and $\mathcal{L}_\Theta$ is $\mathbb{Z}[y]/(y^5)$.

**Theorem 18.83** (The secant plane and the lattice of line bundles)**.** *For $(a,b) \in \mathbb{Z}^2$ and $d \geq 1$,*

$$m(a\,u_\Theta + b\,v_\Theta) = \Big(a,\ b,\ -\frac{ad+b}{2},\ \frac{3ad+2b-bd}{6},\ \frac{ad^2-11ad+6bd-6b}{24}\Big).$$

*Consequently $a\,u_\Theta + b\,v_\Theta \in \mathcal{L}_\Theta$ if and only if*

$$ad \equiv b \pmod 2, \qquad 3ad+2b-bd \equiv 0 \pmod 6,$$
$$ad^2-11ad+6bd-6b \equiv 0 \pmod{24}.$$

*For $a=1$ and $b=3$ the moments are $\big(1,3,-\frac{d+3}{2},1,\frac{(d+9)(d-2)}{24}\big)$, and $u_\Theta + 3v_\Theta \in \mathcal{L}_\Theta$ if and only if $d$ is odd and $24 \mid (d+9)(d-2)$, that is, if and only if*

$$d \equiv 15 \text{ or } 23 \pmod{24}.$$

*Proof.* In the variable $y = x-1$ put $L = \log(1+y) = y - \frac{1}{2}y^2 + \frac{1}{3}y^3 - \frac{1}{4}y^4$ modulo $y^5$. Under $e^{j\Theta} \mapsto x^j$ the class $\Theta$ goes to $L$, hence $u_\Theta \mapsto 1 - \frac{d}{2}L^2 + \frac{d^2}{24}L^4$ and $v_\Theta \mapsto L - \frac{d}{6}L^3$. With $L^2 = y^2 - y^3 + \frac{11}{12}y^4$, $L^3 = y^3 - \frac{3}{2}y^4$ and $L^4 = y^4$ modulo $y^5$, the coefficients of $1, y, \dots, y^4$ in $a\,u_\Theta + b\,v_\Theta$ are the five displayed expressions, and the congruences are the integrality of the last three. For $b=3$ the second reads $3d+6-3d = 6$, so $m_3 = 1$, and the third reads $d^2+7d-18 = (d+9)(d-2)$. Modulo 8 the condition is $8 \mid (d+1)(d-2)$ with $d$ odd, hence $d \equiv 7 \pmod 8$; modulo 3 it is $3 \mid d(d-2)$, hence $d \equiv 0, 2 \pmod 3$; together $d \equiv 15, 23 \pmod{24}$. ∎

**Corollary 18.84** (Split resolutions, for every rank and every twist)**.** *Let $Z \subset X$ be Cohen–Macaulay of codimension two with a resolution $0 \to E_1 \to E_0 \to \mathcal{I}_Z \to 0$ in which $\operatorname{ch} E_0$ and $\operatorname{ch} E_1$ both lie in $\mathcal{L}_\Theta$: for instance Setup 18.79 with any rank and any twists, or $E_0$ and $E_1$ built from such line bundles and from the bundles $M_{m\Theta}$ by the operations listed above. If $\operatorname{ch}(\mathcal{I}_Z(3\Theta)) = u_\Theta + 3v_\Theta$, then $d \equiv 15$ or $23 \pmod{24}$. In particular this never happens for the four discriminants of Corollary 18.68.*

*Proof.* $\operatorname{ch}(\mathcal{I}_Z(3\Theta)) = e^{3\Theta}(\operatorname{ch} E_0 - \operatorname{ch} E_1) \in \mathcal{L}_\Theta$, and Theorem 18.83 applies. ∎

*Remark* 18.85 (What the congruence measures). The fractional part of $m_4$ is the invariant that separates the secant plane from the line bundles. For $d \in \{1,3,5,7\}$ it is $-\frac{5}{12}, \frac{1}{2}, \frac{3}{4}, \frac{1}{3}$ modulo $\mathbb{Z}$, and any object with Chern character $u_\Theta + 3v_\Theta$ must carry a class outside $\mathcal{L}_\Theta$ with exactly that defect. Two checks. A point has $m = (0,0,0,0,1/24)$, and the partial normalisation of Setup 18.51 at $(d+9)(2d-1)$ isolated points shifts $m_4$ by $(d+9)(2d-1)/24$; since $(d+9)(d-2) + (d+9)(2d-1) = 3(d+9)(d-1) \equiv 0 \pmod{24}$ for odd $d$, the unnormalised chain, which lies in $\mathcal{L}_\Theta$, is moved onto the plane by exactly the right defect, as it must be. For a smooth support the class of the structure sheaf has $m(\mathcal{O}_S) = (0,0,N,-3N,N(83-2N)/12)$, and $N(83-2N)/12$ is not an integer for $N \in \{5,6,7,8\}$; so a smooth secant support is a surface whose structure sheaf is not in the line-bundle subring, unlike every complete intersection of translates of $\Theta$ and every degeneracy locus of a map between bundles in $\mathcal{L}_\Theta$. The Fourier–Mukai transforms of line bundles and the push-forwards of line bundles along isogenies do have classes outside $\mathcal{L}_\Theta$, which is why the semi-homogeneous bundles need the separate argument of Theorem 18.92.

The lattice constrains the classes of the two terms of a resolution; their ranks constrain the Chern classes, because a vector bundle of rank $s$ has $c_k = 0$ for $k > s$, and on a fourfold this bites for $s \leq 3$.

**Proposition 18.86** (Resolutions of rank one and two)**.** *Let $(X, \Theta)$ be a principally polarised abelian fourfold, let $b \geq 1$ and $d > 0$, and let $Z \subset X$ be of codimension two with a resolution*

$$0 \longrightarrow E_1 \longrightarrow E_0 \longrightarrow \mathcal{I}_Z \longrightarrow 0$$

*by vector bundles of ranks $r$ and $r+1$, and with $\operatorname{ch}(\mathcal{I}_Z(b\Theta)) = u_\Theta + b\, v_\Theta$.*

(i) *$r \geq 2$, whatever the bundles. Equivalently, $Z$ is not the zero locus of a section of a vector bundle of rank two.*

(ii) *If $r = 2$ and the Chern classes of $E_1$ are polynomials in $\Theta$, as they are when $\operatorname{Hdg}^2(X) = \mathbb{Q}\Theta$ and $\operatorname{Hdg}^4(X) = \mathbb{Q}\Theta^2$, then, writing $c_1(E_1(b\Theta)) = a\Theta$ with $a \in \mathbb{Z}$,*

$$3d - b^2 - 4ab \equiv 0 \pmod{12}.$$

*At $b = 3$ this says $d \equiv 3 \pmod 4$.*

*In particular, at the discriminants $d = 1$ and $d = 5$ of Corollary 18.68, a resolution whose Chern classes are polynomials in $\Theta$ has $r \geq 3$.*

*Proof.* Put $E = E_1(b\Theta)$, $G = E_0(b\Theta)$ and $F = \mathcal{I}_Z(b\Theta)$, so that $0 \to E \to G \to F \to 0$ is exact and $c(G) = c(E)\,c(F)$. Newton's identities applied to $\operatorname{ch}(F) = 1 + b\Theta - \frac{d}{2}\Theta^2 - \frac{db}{6}\Theta^3 + \frac{d^2}{24}\Theta^4$ give, with $N = \frac{1}{2}(b^2 + d)$,

$$c(F) = 1 + b\Theta + N\Theta^2 + \tfrac{bN}{3}\Theta^3 + \tfrac{N(b^2-3d)}{12}\Theta^4.$$

Since $E$ and $G$ are vector bundles, $c_k(E) = 0$ for $k > r$ and $c_k(G) = 0$ for $k > r+1$.

(i) Let $r = 1$ and $\ell = c_1(E)$, so that $c(G) = (1 + \ell)\,c(F)$. The rank two bundle $G$ has $c_3(G) = \frac{bN}{3}\Theta^3 + N\ell\,\Theta^2 = 0$, and cup product with $\Theta^2$ is an isomorphism $H^2(X, \mathbb{Q}) \to H^6(X, \mathbb{Q})$ by hard Lefschetz, so $\ell = -\frac{b}{3}\Theta$. Then

$$c_4(G) = c_4(F) + \ell\, c_3(F) = N\Big(\frac{b^2 - 3d}{12} - \frac{b^2}{9}\Big)\Theta^4 = -\frac{N(b^2 + 9d)}{36}\,\Theta^4,$$

which is not zero, since $N > 0$ and $d > 0$. A section of a rank two bundle $V$ vanishing in codimension two exactly on $Z$ gives the Koszul resolution $0 \to \det V^\vee \to V^\vee \to \mathcal{I}_Z \to 0$, which has $r = 1$; conversely, at $r = 1$ the map $E_1 \to E_0$ is a section of $E_0 \otimes E_1^\vee$ with zero locus $Z$.

(ii) Let $r = 2$, and write $c_1(E) = a\Theta$ and $c_2(E) = \frac{m}{2}\Theta^2$ with $a, m \in \mathbb{Z}$, since $\Theta$ and $\frac{1}{2}\Theta^2$ are primitive integral classes. Now $c_3(E) = c_4(E) = 0$, and the rank three bundle $G$ has

$$c_4(G) = c_4(F) + c_1(E)\,c_3(F) + c_2(E)\,c_2(F) = 0.$$

Integrating, with $\int_X \Theta^4 = 24$, and dividing by $2N$ gives $6m = 3d - b^2 - 4ab$. By Riemann–Roch with trivial Todd class, $\chi(E) = \int_X \operatorname{ch}_4(E) = a^4 - 2a^2m + \frac{1}{2}m^2$ is an integer, so $m$ is even, which is the congruence. At $b = 3$ it reads $3(d-3) \equiv 12a \equiv 0$ modulo 12, that is, $d \equiv 3 \pmod 4$. ∎

*Remark* 18.87 (What the ranks exclude). Part (i) extends Theorem 18.81 from two divisors to the zero locus of a section of any rank two bundle, stable or not, so the construction of Serre produces no secant support; it holds for every $X$, with no condition on $\mathrm{NS}(X)$. Part (ii) holds for arbitrary bundles whose Chern classes are polynomials in $\Theta$, not only for sums of line bundles, and at $b = 3$ it excludes $r = 2$ whenever $d \not\equiv 3 \pmod 4$. The integrality of the Chern classes of $E_0$ is automatic, since $c(E_0(b\Theta)) = c(E_1(b\Theta))\,c(F)$ and $c(F)$ is integral, so it is the vanishing above the rank, and not integrality, that constrains.

Neither condition restricts $r \geq 3$: a bundle of rank four has no vanishing Chern class on a fourfold, and at $r = 3$ the only condition, $c_4(E) = 0$, involves $E$ alone.

Corollary 18.84 says where a split resolution can have the right Chern character. A stronger statement holds: no object with a split resolution satisfies the criterion, at any rank and for any discriminant (Theorem 18.89). It rests on a necessary condition that we isolate, because it applies to every object and can be tested on any candidate before any global computation is attempted.

**Lemma 18.88** (Morphisms to a line bundle are killed by $H^2(\mathcal{O})$)**.** *Let $(X, t)$ be a polarised abelian variety of dimension $n \geq 3$, let $F$ be a perfect complex with $\operatorname{ch}(F) = au_t + bv_t$, $(a, b) \neq (0, 0)$, which satisfies the weakened criterion of Setup 18.54, and let $L$ be a line bundle with $c_1(L) \in \mathbb{Q}\,t$. Then for every $m \in \mathbb{Z}$, every morphism $p\colon F \to L[m]$, every morphism $q\colon L[m] \to F$, and every $z \in H^2(\mathcal{O}_X)$,*

$$z \cdot p = 0 \quad \text{in } \operatorname{Hom}(F, L[m+2]), \qquad z \cdot q = 0 \quad \text{in } \operatorname{Hom}(L[m], F[2]),$$

*where $z \cdot (\,\cdot\,)$ is the $H^*(\mathcal{O}_X)$-module structure on morphisms in $D^b(X)$. In particular, if $F = \mathcal{I}_Z \otimes L'$ for a closed subscheme $Z$ and a line bundle $L'$ with $c_1(L') \in \mathbb{Q}t$, then the restriction $H^{n-2}(\mathcal{O}_X) \to H^{n-2}(\mathcal{O}_Z)$ is injective.*

*Proof.* For a constant bivector $\pi$ the class $x_\pi = (\frac{d}{2}\,\pi \lrcorner t^2, 0, \pi)$ preserves $\operatorname{ch}(F)$ by Theorem 18.56(ii), so $\operatorname{ev}_F(x_\pi) = 0$ by hypothesis. The Hochschild action is by natural transformations $\mathrm{id} \to \mathrm{id}[2]$ of $D^b(X)$, so for $p\colon F \to L[m]$ we have $\operatorname{ev}_{L[m]}(x_\pi) \circ p = p \circ \operatorname{ev}_F(x_\pi) = 0$, where $\operatorname{ev}_{L[m]} = \operatorname{ev}_L[m]$. By (50), $\operatorname{ev}_L(x_\pi) = c_\pi \cdot \mathrm{id}_L$ with $c_\pi = \frac{1}{2}\,\pi \lrcorner (\lambda^2 + d\,t^2) \in H^{0,2}(X)$ and $\lambda = c_1(L) = e\,t$, that is, $c_\pi = \frac{e^2+d}{2}\,\pi \lrcorner t^2$. Since $e^2 + d > 0$ and $\pi \mapsto \pi \lrcorner t^2$ is an isomorphism from $H^0(\bigwedge^2 T_X)$ onto $H^{0,2}(X)$, as in the proof of Theorem 18.56(iii), $c_\pi$ runs over all of $H^2(\mathcal{O}_X)$ as $\pi$ varies; as $(c_\pi \cdot \mathrm{id}_L) \circ p = c_\pi \cdot p$, this proves the assertion for $p$. The assertion for $q$ follows in the same way from the square $\operatorname{ev}_F(x_\pi) \circ q = q \circ \operatorname{ev}_{L[m]}(x_\pi)$. For the last assertion, apply the first to the inclusion $p\colon \mathcal{I}_Z \otimes L' \to L'$. After twisting by $L'^{-1}$, the sequence $0 \to \mathcal{I}_Z \to \mathcal{O}_X \to \mathcal{O}_Z \to 0$ gives an exact sequence

$$\operatorname{Ext}^2(\mathcal{O}_Z, \mathcal{O}_X) \longrightarrow H^2(\mathcal{O}_X) \longrightarrow \operatorname{Ext}^2(\mathcal{I}_Z, \mathcal{O}_X),$$

in which the second map is $z \mapsto z \cdot p$. So $z \cdot p = 0$ for all $z$ means that the first map is surjective, and by Serre duality, $\omega_X$ being trivial, the first map is dual to the restriction $H^{n-2}(\mathcal{O}_X) \to H^{n-2}(\mathcal{O}_Z)$. ∎

**Theorem 18.89** (No two-term complex of powers of the polarisation satisfies the criterion)**.** *Let $(X, t)$ be a polarised abelian variety of dimension $n \geq 3$ and $M$ an ample line bundle with $c_1(M) \in \mathbb{Q}t$. Let $A$ and $B$ be direct sums of tensor powers $M^{\otimes e}$, $e \in \mathbb{Z}$, let $\varphi\colon A \to B$ be any morphism, and let $F$ be the cone of $\varphi$, or any shift of it. If $F \neq 0$ and $\operatorname{ch}(F) = au_t + bv_t$ with $(a, b) \neq (0, 0)$, then $F$ does not satisfy the weakened criterion. In particular, no object with a resolution $0 \to E_1 \to E_0 \to F \to 0$ by sums of powers of $M$ satisfies it, whatever the rank.*

*Proof.* A pair of summands $M^{\otimes e}$ of $A$ and $M^{\otimes e}$ of $B$ joined by a nonzero entry of $\varphi$, which is then a nonzero scalar, can be cancelled without changing the cone up to isomorphism; do this until no such pair remains. If $A = 0$ or $B = 0$ after cancelling, $F$ is a shifted sum of line bundles, $\operatorname{ev}_F(x_\pi)$ is diagonal with the nonzero entries $c_\pi \cdot \mathrm{id}$ of the proof of Lemma 18.88, and the criterion fails as in Corollary 18.21; so suppose both are nonzero and consider the triangle $A \xrightarrow{\varphi} B \xrightarrow{q} F \to A[1]$. Let $M^{\otimes b_j}$ be a summand of $B$, $\iota_j$ its

inclusion into $B$, and $q_j = q \circ \iota_j \colon M^{\otimes b_j} \to F$. Applying $\operatorname{Hom}(M^{\otimes b_j}, -)$ to the triangle gives the exact sequence

$$\operatorname{Hom}(M^{\otimes b_j}, A[2]) \xrightarrow{\varphi_*} \operatorname{Hom}(M^{\otimes b_j}, B[2]) \xrightarrow{q_*} \operatorname{Hom}(M^{\otimes b_j}, F[2]),$$

and $z \cdot q_j = q_*(z \cdot \iota_j)$ for $z \in H^2(\mathcal{O}_X)$. Now $\operatorname{Hom}(M^{\otimes b_j}, B[2]) = \bigoplus_k H^2(M^{\otimes(b_k - b_j)})$, and $H^2(M^{\otimes m}) = 0$ for $m \neq 0$: for $m > 0$ by Kodaira vanishing, $\omega_X$ being trivial, and for $m < 0$ because a negative line bundle on an abelian variety has cohomology only in degree $n \geq 3$. So $z \cdot \iota_j$ is the element with component $z$ in the summand $k = j$ and zero elsewhere. It lies in the image of $\varphi_*$ only if there is $\psi \in \operatorname{Hom}(M^{\otimes b_j}, A[2]) = \bigoplus_i H^2(M^{\otimes(a_i - b_j)})$ with $(\varphi \circ \psi)_j = \sum_i \varphi_{ji}\psi_i = z$; but $\psi_i \neq 0$ forces $a_i = b_j$, and for such $i$ the entry $\varphi_{ji}$ is a scalar, zero after the cancellation. So $z \cdot q_j \neq 0$ for every $z \neq 0$, which contradicts Lemma 18.88. The last assertion is the case of a resolution, where $F \cong \operatorname{coker} \varphi$ is the cone of $\varphi \colon E_1 \to E_0$. ∎

The powers of the polarisation are not the only bundles on which the compensated Poisson classes act by scalars. The same holds for every bundle whose Atiyah class is a multiple of the identity, and on an abelian variety these are the semi-homogeneous bundles of Mukai [Muk78]. The argument of Theorem 18.89 goes through for them, with the cohomology of the fibre product of two isogenies in place of the cohomology of a power of $\Theta$. We state the result in a self-contained form.

**Definition 18.90** (Isogeny blocks)**.** Let $(X, \Theta)$ be a polarised abelian variety. An *isogeny block* is a vector bundle $E = \pi_* L$, where $\pi \colon Y \to X$ is an isogeny and $L$ is a line bundle on $Y$ with $c_1(L) = \nu\, \pi^* \Theta$ in $H^2(Y, \mathbb{Q})$, $\nu \in \mathbb{Q}$; its *slope* is $\nu$, so that $c_1(E) = \nu \deg(\pi)\, \Theta$. The block is *simple* if $\tau_g^* L \not\cong L$ for every nonzero $g \in \ker \pi$. The powers $\mathcal{O}(e\Theta)$ are the blocks with $\pi = \mathrm{id}$.

**Lemma 18.91** (Three properties of a block)**.** *Let $E = \pi_* L$ be an isogeny block of slope $\nu$ on an abelian $n$-fold.*

(i) $\operatorname{At}(E) = \nu\Theta \cdot \mathrm{id}_E$*; hence for a constant bivector $\pi$ the compensated class $x_\pi$ of Theorem 18.56(ii) acts by* $\operatorname{ev}_E(x_\pi) = \frac{\nu^2 + d}{2}\, (\pi \lrcorner \Theta^2) \cdot \mathrm{id}_E$*, and Lemma 18.88 holds with $E$ in place of the line bundle $L$.*

(ii) *For a second block $E' = \pi'_* L'$ of slope $\nu'$, $\operatorname{Ext}^q(E', E) = H^q(Z, M)$ with $Z = Y' \times_X Y$, a disjoint union of abelian varieties isogenous to $X$, and $M$ a line bundle whose class on each component is $(\nu - \nu')$ times the pullback of $\Theta$. If $\nu \neq \nu'$ and $n \geq 3$ then $\operatorname{Ext}^2(E', E) = 0$. If $E$ is simple then $\operatorname{Ext}^2(E, E) = H^2(\mathcal{O}_X) \cdot \mathrm{id}_E$.*

(iii) *A simple block is $\mu$-stable with respect to $\Theta$; hence a nonzero morphism between simple blocks of the same slope is an isomorphism.*

*Proof.* (i) Since $\pi$ is étale, $\pi^* \Omega^1_X = \Omega^1_Y$ and $\pi_*$ is exact, so the first jet sequence of $\pi_* L$ is the direct image of that of $L$, and $\operatorname{At}(\pi_* L)$ is the image of $\operatorname{At}(L)$ under $\pi_* \colon \operatorname{Ext}^1_Y(L, L \otimes \pi^* \Omega^1_X) \to \operatorname{Ext}^1_X(\pi_* L, \pi_* L \otimes \Omega^1_X)$. As $\operatorname{At}(L) = c_1(L) \cdot \mathrm{id}_L = \nu\, \pi^* \Theta \cdot \mathrm{id}_L$, the projection formula gives $\nu\Theta \cdot \mathrm{id}_{\pi_* L}$. Then (54) with a scalar Atiyah class gives the value of $\operatorname{ev}_E(x_\pi)$, and the proof of Lemma 18.88 used nothing else about $L$.

(ii) Adjunction for the finite étale $\pi'$, flat base change along the cartesian square with vertices $Z$, $Y'$, $Y$, $X$, and the projection formula give $\operatorname{Ext}^q_X(\pi'_* L', \pi_* L) = H^q(Y', L'^{-1} \otimes \pi'^* \pi_* L) = H^q(Z, p'^* L'^{-1} \otimes p^* L)$, with $p, p'$ the two projections; the class of $M = p'^* L'^{-1} \otimes p^* L$ is $(\nu - \nu')$ times the pullback of $\Theta$ on every component, which is an abelian variety finite étale over $Y'$. A line bundle on an abelian $n$-fold whose class is a positive multiple of an ample class has cohomology only in degree 0, and one whose class is a negative multiple

only in degree $n$, by Kodaira vanishing and Serre duality with trivial canonical bundle; so $H^2(Z, M) = 0$ when $\nu \neq \nu'$ and $n \geq 3$. When $E' = E$, the components of $Y \times_X Y$ are the graphs of the translations by $g \in \ker \pi$, on which $M$ restricts to $L^{-1} \otimes \tau_g^* L \in \mathrm{Pic}^0(Y)$; a nontrivial element of $\mathrm{Pic}^0$ has no cohomology, so $\mathrm{Ext}^q(E, E) = \bigoplus_g H^q(\mathcal{O}_Y)$ over the $g$ with $\tau_g^* L \cong L$, which for a simple block is $g = 0$ alone, and $H^q(\mathcal{O}_Y) = H^q(\mathcal{O}_X)$ through $\pi$.

(iii) The pullback $\pi^* E = \bigoplus_{g \in \ker \pi} \tau_g^* L$ is a direct sum of line bundles of one slope, hence $\mu$-semistable, and semistability descends along the finite étale $\pi$; so $E$ is semistable. If $F \subset E$ is a saturated subsheaf of the same slope, then $\pi^* F \subset \pi^* E$ is a saturated subsheaf of the same slope of a direct sum of pairwise non-isomorphic line bundles, the $\tau_g^* L$ being distinct because the block is simple, hence a partial sum $\bigoplus_{g \in S} \tau_g^* L$; it is invariant under the translations by $\ker \pi$, which permute the summands transitively, so $S$ is empty or everything and $F = 0$ or $F = E$. Finally, a nonzero morphism $f$ between $\mu$-stable bundles of the same slope is injective and generically surjective, so the two bundles have the same rank; for blocks of the same slope $\nu$ they then have the same $c_1$, since $c_1(E) = \nu \operatorname{rk}(E)\, \Theta$, and $\det f$ is a nonzero section of a numerically trivial line bundle, hence nowhere zero. So $f$ is an isomorphism. ∎

**Theorem 18.92** (No two-term complex of simple blocks satisfies the criterion)**.** *Let $(X, \Theta)$ be a polarised abelian variety of dimension $n \geq 3$, let $A$ and $B$ be direct sums of simple isogeny blocks, let $\varphi\colon A \to B$ be any morphism and let $F$ be the cone of $\varphi$ or a shift of it. If $F \neq 0$ and $\mathrm{ch}(F) = a u_\Theta + b v_\Theta$ with $(a, b) \neq (0, 0)$, then $F$ does not satisfy the weakened criterion. In particular, no object with a resolution $0 \to E_1 \to E_0 \to F \to 0$ by sums of simple isogeny blocks satisfies it, whatever the rank.*

*Proof.* Cancel, by Gaussian elimination in the two-term complex, every pair of isomorphic blocks of $A$ and $B$ joined by an entry of $\varphi$ that is an isomorphism; this changes neither the cone nor the block structure of the remaining terms. By Lemma 18.91(iii), every remaining entry between blocks of the same slope is zero. If $A$ or $B$ is empty after cancelling, $F$ is a shifted sum of blocks, on each of which $\mathrm{ev}(x_\pi)$ is a nonzero scalar by (i). Otherwise the rest is the proof of Theorem 18.89 with Lemma 18.91 in place of the vanishing of $H^2(M^{\otimes m})$. For a block $E'$ of $B$ with inclusion $\iota$ and $q' = q \circ \iota\colon E' \to F$, the class $z \cdot \iota \in \mathrm{Hom}(E', B[2])$ has component $z \cdot \mathrm{id}_{E'} \neq 0$ in $\mathrm{Ext}^2(E', E') = H^2(\mathcal{O}_X) \cdot \mathrm{id}_{E'}$ by (ii). It lies in the image of $\varphi_*$ only if $\sum_i \varphi_{ji} \psi_i = z \cdot \mathrm{id}_{E'}$ for some $\psi_i \in \mathrm{Ext}^2(E', A_i)$, and $\psi_i \neq 0$ forces $A_i$ to have the slope of $E'$ by (ii), for which $\varphi_{ji} = 0$. So $z \cdot q' \neq 0$, contradicting Lemma 18.88 in the form given by (i). ∎

*Remark* 18.93 (Semi-homogeneous bundles). By Mukai [Muk78], the simple isogeny blocks are exactly the simple semi-homogeneous bundles, a bundle $E$ being semi-homogeneous if for every $x$ there is $P \in \mathrm{Pic}^0(X)$ with $\tau_x^* E \cong E \otimes P$; and every semi-homogeneous bundle on a variety with $\mathrm{NS}(X)_{\mathbb{Q}} = \mathbb{Q}\Theta$ is an iterated extension of such blocks and their twists by $\mathrm{Pic}^0$. We do not use this. In these terms, Theorem 18.92 says that the bundles in a Hilbert–Burch resolution of a candidate cannot all have scalar Atiyah class. The obstruction is always the same: the compensated Poisson classes act on a block by the scalar $\frac{\nu^2 + d}{2}\, \pi \lrcorner \Theta^2$, which would vanish only for $\nu^2 = -d$, and cones of maps between blocks inherit the scalar through the summand inclusions, because $\mathrm{Ext}^2$ between blocks of different slopes vanishes. A candidate must therefore be built from a bundle whose Atiyah class is not a scalar, that is, one which is not projectively flat.

**Corollary 18.94** (Split resolutions and the remaining shapes)**.** *Let $(X, \Theta)$ be a principally polarised abelian fourfold.*

(i) *No secant object of rank one whose support has a resolution* (66) *by sums of powers of* $\Theta$ *satisfies the weakened criterion, for any* $r$*, any twists and any* $d$*, whether or not its matrix degenerates in codimension two.*

(ii) *If a secant object* $\mathcal{I}_Z(b\Theta)$ *satisfies the criterion, then* $Z$ *has no point at which two smooth branches of codimension two meet transversally, no two-term resolution of* $\mathcal{I}_Z$ *splits into powers of* $\Theta$ *or into simple semi-homogeneous bundles, and the restriction* $H^2(\mathcal{O}_X) \to H^2(\mathcal{O}_Z)$ *is injective. If* $Z$ *is Cohen–Macaulay,* $\mathcal{I}_Z$ *has projective dimension one. If* $b = 3$ *and* $Z$ *is smooth, then* $d \leq 9$*, so* $d \in \{1, 3, 5, 7\}$ *when* $N = (9 + d)/2$ *is an integer as in Setup 18.51, the two terms of a resolution do not both have classes in* $\mathcal{L}_\Theta$*, and when* $X$ *is simple the injectivity is automatic.*

(iii) *At* $b = 3$*, smooth supports and split resolutions are disjoint, for every rank and every twist of the resolution: a smooth support forces* $d \leq 9$*, and a resolution by bundles with classes in* $\mathcal{L}_\Theta$ *forces* $d \equiv 15, 23 \pmod{24}$*, hence* $d \geq 15$*, so that, even without (i), no support could have both properties.*

*Proof.* Part (i) is Theorem 18.89 with $M = \mathcal{O}_X(\Theta)$, the resolution (66) presenting $\mathcal{I}_Z(b\Theta)$ as the cone of $\varphi \otimes \mathcal{O}(b\Theta)$. In (ii), the branches are excluded by Corollary 18.77(i), projective dimension one comes from Corollary 18.77(ii), the non-splitting from Theorem 18.89 and Theorem 18.92, the injectivity from the last assertion of Lemma 18.88, the bound on $d$ from Corollary 18.68, the statement on the two terms from Corollary 18.84, and the automatic injectivity from Lemma 18.70. Part (iii) compares Corollary 18.68 with Corollary 18.84: no $d \leq 9$ is congruent to 15 or 23 modulo 24. ∎

## Part 4. Weil classes of CM fields

### 19. Weil classes of every CM field

The Weil classes of an imaginary quadratic field are the case $m = 1$ of a construction that makes sense for every CM field $F$ of degree $2m$, and the methods of this paper apply to the question whether those classes are algebraic. Every theorem of Section 14 that turns the conjecture for a Weil family into a propagation statement is a theorem about a connected period domain, a flat class on it, a group whose rational points act with dense orbits, and a base point where the class is a polynomial in divisor classes, and none of these four ingredients depends on the field being quadratic. This section carries the four over to every CM field; the only one that needs a new argument is the base point, which Theorem 19.5 supplies. Nothing here is conditional on a question about semiregularity, because the base point is a CM point: the Weil classes there are written out as polynomials with rational coefficients in divisor classes, each divisor class is algebraic by the Lefschetz theorem on $(1, 1)$ classes, and products of algebraic classes are algebraic. Nothing is claimed about the whole Hodge ring at a CM point, which can contain classes that are not polynomials in divisor classes.

#### 19.1. The families.

**Setup 19.1** (Weil type for a CM field)**.** Let $F$ be a CM field of degree $2m$ with maximal totally real subfield $F_0$, and write $\alpha \mapsto \overline{\alpha}$ for the nontrivial automorphism of $F/F_0$, which is complex conjugation under every embedding. The real places of $F_0$ are $\tau_1, \dots, \tau_m$, and each extends to a conjugate pair $\sigma_\tau, \overline{\sigma}_\tau$ of embeddings of $F$ into $\mathbb{C}$. Let $V$ be an $F$-vector space of dimension $2n$, let $H \colon V \times V \to F$ be a nondegenerate hermitian form, $F$-linear in the first variable, and fix a totally imaginary $\xi \in F^\times$, that is $\overline{\xi} = -\xi$. Put

$$E(x, y) = \operatorname{Tr}_{F/\mathbb{Q}}(\xi\, H(x, y)),$$

a nondegenerate alternating $\mathbb{Q}$-bilinear form with $E(\alpha x, y) = E(x, \overline{\alpha} y)$. For each $\tau$ let $V_\tau = V \otimes_{F_0,\tau} \mathbb{R}$, a vector space of dimension $2n$ over $F \otimes_{F_0,\tau} \mathbb{R} \cong \mathbb{C}$, and let $(p_\tau, q_\tau)$ be the signature of the hermitian form $H_\tau$ induced by $H$ on it. We say that $(V, H)$ is of $(F, n)$-*Weil type* if $p_\tau = q_\tau = n$ for every $\tau$, and we call the class of $(-1)^n \det H$ in $F_0^\times / \mathrm{N}_{F/F_0}(F^\times)$ its discriminant $\delta$; it is totally positive, and for $F = K$ it is the discriminant of Definition 15.22. A polarised abelian variety $(A, \iota, E)$ of dimension $2mn$ with $\iota\colon F \to \mathrm{End}^0(A)$ and $H^1(A, \mathbb{Q}) \cong V$ as $F$-modules with polarisation $E$ is of $(F, n, \delta)$-Weil type when $(V, H)$ is. As in Lemma 3.2, we work throughout with $V = H^1$; a complex structure on $V_\mathbb{R}$ is the same datum as one on the tangent space. The Weil classes of $A$ are

$$W_F(A) = \textstyle\bigwedge_F^{2n} V \;\subseteq\; \bigwedge_\mathbb{Q}^{2n} V = H^{2n}(A, \mathbb{Q}),$$

the $\mathbb{Q}$-subspace whose complexification is $\bigoplus_\sigma \bigwedge^{2n} V_\sigma$, where $V_\sigma = V \otimes_{F,\sigma} \mathbb{C}$ runs over the $2m$ eigenspaces of $F$ on $V_\mathbb{C}$; it is a one-dimensional $F$-vector space, of dimension $2m$ over $\mathbb{Q}$. Write

> $\mathbf{W}(F, n, \delta)$: every abelian variety of $(F, n, \delta)$-Weil type has algebraic Weil classes.

The period domain $\mathcal{D}_F$ is the set of complex structures $J$ on $V_\mathbb{R}$ commuting with $F$ and polarised by $E$, and the group $G_F = \mathrm{Res}_{F_0/\mathbb{Q}} \mathrm{SU}(V, H)$ acts on it. We fix an arithmetic subgroup $\Gamma \subset G_F(\mathbb{Q})$ and write $\mathcal{S}_F = \Gamma \backslash \mathcal{D}_F$ for the quotient, $p\colon \mathcal{D}_F \to \mathcal{S}_F$ for the projection, $\pi\colon \mathcal{A} \to \mathcal{S}_F$ for the universal family and $\Sigma_F \subseteq \mathcal{D}_F$ for the set of $s$ at which $W_F(A_s)$ is algebraic. For $m = 1$ this is Setup 15.12 and Setup 16.1, with $\omega_s$ a generator of the line.

*Remark* 19.2 (Every such abelian variety is a member). If $A$ is any abelian variety with a CM field $F \subseteq \mathrm{End}^0(A)$ such that $\dim_F H^1(A, \mathbb{Q}) = 2n$ and $\dim V_\sigma^{1,0} = n$ for every embedding $\sigma$, then $A$ is of $(F, n, \delta)$-Weil type for some $\delta$: the Rosati involution of any polarisation restricts to a positive involution of $F$, hence to conjugation, and the argument of Lemma 4.5 writes the polarisation as $\mathrm{Tr}_{F/\mathbb{Q}}(\xi H)$ for a unique hermitian $H$, whose signatures are then forced by the Hodge numbers as in Proposition 19.3(i). These are exactly the abelian varieties whose Weil classes for $F$ are Hodge classes, in the sense of [MZ99], so $\mathbf{W}(F, n, \delta)$ over all $(F, n, \delta)$ is the statement that all Weil classes of all CM fields are algebraic.

**Proposition 19.3** (The Weil classes of a CM field)**.** *Keep Setup 19.1, with $(V, H)$ of $(F, n)$-Weil type.*

(i) *A polarised complex structure $J$ is the same as an $H_\tau$-orthogonal decomposition $V_\tau = V_\tau^+ \oplus V_\tau^-$ for every $\tau$, with $H_\tau$ definite of the sign of $\mathrm{Im}\, \sigma_\tau(\xi)$ on $V_\tau^+$ and of the opposite sign on $V_\tau^-$; then $\dim V_{\sigma_\tau}^{1,0} = \dim V_\tau^+ = n$ and $\dim V_{\overline{\sigma}_\tau}^{1,0} = \dim V_\tau^- = n$, so every class in $W_F(A_J)$ is of type $(n, n)$. The spaces $W_F(A_J)$ form a flat subsystem $W_F$ of $R^{2n}\pi_*\mathbb{Q}$ of rank $2m$ on which $G_F$ acts trivially, and $\mathcal{D}_F \cong \prod_\tau \mathcal{D}_{n,n}$ is connected of dimension $mn^2$.*

(ii) *If one nonzero element of $W_F(A)$ is algebraic, every element is.*

(iii) *If $\mathrm{Hg}(A_s) = G_F$, the Hodge ring of $A_s$ is generated by $\mathrm{NS}(A_s)_\mathbb{Q}$ and by $W_F$, so the Hodge conjecture for $A_s$ in every degree is equivalent to the algebraicity of $W_F$. For $n \geq 2$ the space $\mathrm{NS}(A_s)_\mathbb{Q}$ is the $m$-dimensional space of classes $\eta_t = [E(t\,\cdot, \cdot)]$, $t \in F_0$.*

(iv) *Two members of $(F, n, \delta)$-Weil type have isomorphic $(V, H)$, so every one of them is a fibre of $\pi$.*

*Proof.* (i) A complex structure commuting with $F$ preserves each $V_\tau$, since $F_0 \otimes \mathbb{R}$ acts diagonally, and is $\mathbb{C}$-linear on it for the structure $F \otimes_{F_0,\tau} \mathbb{R} \cong \mathbb{C}$, so it is $\pm\sqrt{-1}$ on two complementary $\mathbb{C}$-subspaces $V_\tau^{\pm}$. Write $\sigma_\tau(\xi) = \sqrt{-1}\, r_\tau$ with $r_\tau \in \mathbb{R}^\times$. For $x \in V_\tau^{\epsilon}$ and $y \in V_\tau^{\epsilon'}$ one has $H_\tau(Jx, Jy) = \epsilon\epsilon' H_\tau(x,y)$, so $E(Jx,Jy) = E(x,y)$ forces $E(x,y) = 0$ for $\epsilon \neq \epsilon'$, and replacing $x$ by $\sqrt{-1}\, x$ then forces $H_\tau(x,y) = 0$: the decomposition is orthogonal. For $x \in V_\tau^{\epsilon}$, $E(x,Jx) = 2\,\mathrm{Re}(\sqrt{-1}\, r_\tau\, \overline{\epsilon\sqrt{-1}}\, H_\tau(x,x)) = 2\epsilon r_\tau H_\tau(x,x)$, so positivity is the sign condition stated. Conversely such a decomposition defines a polarised $J$. Under $V_\tau \otimes_{\mathbb{R}} \mathbb{C} = V_{\sigma_\tau} \oplus V_{\overline{\sigma}_\tau}$ the space $V^{1,0}$, the $\sqrt{-1}$-eigenspace of $J$ on $V_{\mathbb{C}}$, meets $V_{\sigma_\tau}$ in the image of $V_\tau^+$ and $V_{\overline{\sigma}_\tau}$ in the conjugate of the image of $V_\tau^-$, which gives the two equalities of dimensions. Since $H_\tau$ has signature $(n,n)$ and is definite, of opposite signs, on the complementary orthogonal subspaces $V_\tau^{\pm}$, both have dimension $n$. So each $V_\sigma$ is the sum of its intersections with $V^{1,0}$ and $V^{0,1}$, both of dimension $n$, and a generator $\alpha_\sigma$ of $\bigwedge^{2n} V_\sigma$ is of type $(n,n)$. The subspace $\bigwedge_F^{2n} V$ of $H^{2n}(A_s, \mathbb{Q})$ is defined by the action of $F$ on $V$ alone and does not depend on $s$, so it is flat; $\mathrm{SU}(V,H)$ acts on $\bigwedge_F^{2n} V$ by the $F$-determinant, which is 1. Finally, the choice of $V_\tau^+$ is a point of the bounded symmetric domain of $\mathrm{SU}(V_\tau, H_\tau) \cong \mathrm{SU}(n,n)$, of dimension $n^2$, and $\mathcal{D}_F$ is the product over $\tau$.

(ii) For $\alpha \in \mathcal{O}_F$ the endomorphism $\iota(\alpha)$ acts on $\bigwedge_F^{2n} V$ by $\alpha^{2n}$, through an algebraic correspondence, so the algebraic classes in $W_F$ form a $\mathbb{Q}$-subspace stable under multiplication by every $\alpha^{2n}$. By the polarisation identity, in characteristic zero every product $x_1 \cdots x_{2n}$ of elements of a field is a rational combination of $2n$-th powers of sums of the $x_i$; taking $x_2 = \cdots = x_{2n} = 1$ shows that the $2n$-th powers span $F$ over $\mathbb{Q}$, and since every element of $F$ is a rational multiple of an element of $\mathcal{O}_F$, so do the $\alpha^{2n}$ with $\alpha \in \mathcal{O}_F$. Hence the $\mathbb{Q}$-span of $\{\alpha^{2n} w\}$ is $Fw = W_F$ for every $w \neq 0$.

(iii) Hodge classes are the invariants of the Hodge group, which here is $G_F$. Over $\mathbb{C}$ it is $\prod_\tau \mathrm{SL}(U_\tau)$ with $U_\tau = V_{\sigma_\tau}$, and $H$ identifies $V_{\overline{\sigma}_\tau}$ with $U_\tau^\vee$, so $\bigwedge^* V_{\mathbb{C}} = \bigotimes_\tau \bigwedge^*(U_\tau \oplus U_\tau^\vee)$ with the $\tau$-th factor of the group acting on the $\tau$-th tensor factor alone. The invariants of a product group in a tensor product of representations are the tensor product of the invariants, and the invariants of $\mathrm{SL}(U)$ in $\bigwedge^*(U \oplus U^\vee)$ are generated by the pairing $\psi \in U \otimes U^\vee$ and the two determinants, by the computation of Theorem 4.14. The classes $\eta_t$ are Hodge classes: $J$ commutes with $t$ and preserves $E$, hence preserves the $F_0$-twisted form $E_t = E(t\,\cdot, \cdot) = \mathrm{Tr}(\xi t H)$, which for totally positive $t$ is again a polarisation of $A_s$. The complexified span of the $\eta_t$ is $\bigoplus_\tau \mathbb{C}\psi_\tau$, and the complexification of $W_F$ contains all the determinants. So the $\mathbb{Q}$-algebra generated by $\mathrm{NS}_{\mathbb{Q}}$ and $W_F$ has complexification equal to the ring of invariants, hence equals $\mathrm{Hdg}^*(A_s)$. Since divisor classes are algebraic, the conjecture reduces to $W_F$, and by (ii) to one class. For the last sentence, the invariants in $\bigwedge^2 V_{\mathbb{C}}$ are $\bigoplus_\tau \mathbb{C}\psi_\tau$ when $n \geq 2$: a summand $\bigwedge^2 U_\tau$ or $\bigwedge^2 U_\tau^\vee$ has an $\mathrm{SL}(U_\tau)$-invariant only if $\dim U_\tau = 2n$ equals 2, and a summand mixing two different $\tau$ has none. So $\mathrm{NS}(A_s)_{\mathbb{Q}}$ has dimension $m$, and it contains the classes $\eta_t$, which form an $m$-dimensional space because $t \mapsto \eta_t$ is injective on $F_0$, $E$ being nondegenerate.

(iv) Two hermitian forms over $F/F_0$ of the same rank, the same discriminant and the same signature at every real place of $F_0$ are isomorphic [Sch85, Ch. 10, §6]. So the datum $(F, n, \delta)$ determines $(V,H)$ up to isomorphism, and the Hodge structure of any member is, by (i), a polarised complex structure on that $V_{\mathbb{R}}$, that is a point of $\mathcal{D}_F$. ∎

**Lemma 19.4** (The Hodge group of a very general member)**.** *Keep Setup 19.1. For every $s \in \mathcal{D}_F$ one has $\mathrm{Hg}(A_s) \subseteq G_F$, and there is a countable union $Z$ of proper closed analytic subsets of $\mathcal{D}_F$ such that $\mathrm{Hg}(A_s) = G_F$ for every $s \notin Z$.*

*Proof.* The Hodge group of $A_s$ commutes with $\mathrm{End}^0(A_s) \supseteq F$ and preserves $E$, so it lies in $\mathrm{Res}_{F_0/\mathbb{Q}}\,\mathrm{U}(V,H)$: an $F$-linear map preserving $E = \mathrm{Tr}_{F/\mathbb{Q}}(\xi H)$ preserves $H$, because $H$ is recovered from $E$ and the action of $F$ as in Lemma 4.5. It fixes every Hodge class, in particular the Weil classes, which are Hodge classes by Proposition 19.3(i); an element $g$ of $\mathrm{Res}_{F_0/\mathbb{Q}}\,\mathrm{U}(V,H)$ acts on $\bigwedge_F^{2n} V$ by $\det_F(g)$, so $\det_F(g) = 1$ and $g \in G_F$.

For the second statement let $T$ be the set of rational tensors $t \in \bigoplus_{a,b} V^{\otimes a} \otimes (V^\vee)^{\otimes b}$ that are not fixed by $G_F$, a countable set, and for $t \in T$ let $Z_t$ be the set of $s$ at which $t$ is a Hodge class, of type $(p,p)$ with $2p = a - b$, for the Hodge structure induced by $A_s$. The condition is that $t$ lies in the $p$-th step of the Hodge filtration, which varies holomorphically with $s$, so $Z_t$ is a closed analytic subset of $\mathcal{D}_F$. It is proper. Otherwise $t$ is fixed by $h_s(\mathrm{U}(1))$ for every $s \in \mathcal{D}_F$, where $h_s$ is the Hodge representation, which acts on the $(p,q)$ part through $z^{q-p}$, and hence by the subgroup $N$ of $G_F(\mathbb{R})$ that these tori generate. Since $\mathcal{D}_F$ is one orbit of $G_F(\mathbb{R})^+$ and $h_{gs} = g h_s g^{-1}$, the group $N$ is a connected normal subgroup of $G_F(\mathbb{R})^+ = \prod_\tau \mathrm{SU}(V_\tau, H_\tau)$, so its Lie algebra is an ideal of $\bigoplus_\tau \mathfrak{su}(V_\tau, H_\tau)$, a sum of some of these simple summands. It contains every summand, because $h_s(\sqrt{-1})$ acts on $V_\tau$ as the complex structure $J$, which is $\sqrt{-1}$ on $V_\tau^+$ and $-\sqrt{-1}$ on $V_\tau^-$ by Proposition 19.3(i), and is therefore not central in the factor $\mathrm{SU}(V_\tau, H_\tau)$. So $N = G_F(\mathbb{R})^+$, which is Zariski dense in the connected group $G_F$, and $t$ would be fixed by $G_F$, contrary to $t \in T$.

Put $Z = \bigcup_{t \in T} Z_t$. The Hodge group of $A_s$ is the subgroup of $\mathrm{GL}(V)$ fixing every rational Hodge class in the spaces $V^{\otimes a} \otimes (V^\vee)^{\otimes b}$ [Del82, §3]. For $s \notin Z$ each such class is fixed by $G_F$, so $G_F \subseteq \mathrm{Hg}(A_s)$, and equality follows from the first statement. ∎

## 19.2. **A base point in every family.**

**Theorem 19.5** (A base point in every family, for every CM field)**.** *Let $(V,H)$ be of $(F,n)$-Weil type with discriminant $\delta$. Then $\mathcal{D}_F$ contains a point $s_0$ with the following properties.*

(i) *$A_{s_0}$ is isogenous to $B_\Phi^n \times B_{\overline{\Phi}}^n$, where $B_\Phi$ is an abelian $m$-fold with complex multiplication by $F$ of type $\Phi = \{\sigma : \mathrm{Im}\,\sigma(\xi) > 0\}$ and $B_{\overline{\Phi}}$ its conjugate; there is an $H$-orthogonal basis $e_1, \dots, e_{2n}$ of $V$ with $H(e_i, e_i)$ totally positive for $i \le n$ and totally negative for $i > n$, and $Fe_i = H^1(B_\Phi, \mathbb{Q})$ for $i \le n$, $Fe_i = H^1(B_{\overline{\Phi}}, \mathbb{Q})$ for $i > n$.*

(ii) *For $i \le n$ put $i' = i+n$ and let $D_{ii'} \subseteq H^2(A_{s_0}, \mathbb{Q})$ be the $F$-balanced part of $Fe_i \wedge Fe_{i'}$, spanned by the classes*

$$\delta_i(f) = \sum_k (\omega_k f\, e_i) \wedge (\omega_k^*\, e_{i'}), \qquad f \in F,$$

*for dual $\mathbb{Q}$-bases $(\omega_k)$, $(\omega_k^*)$ of $F$ with respect to the trace form. Every $\delta_i(f)$ is a Hodge class of type $(1,1)$, hence a divisor class, and $D_{ii'}$ is a one-dimensional $F$-vector space with $\alpha \cdot \delta_i(f) = \delta_i(\alpha f)$.*

(iii) *The Weil classes are the products*

$$w(f) = \sum_{k_1, \dots, k_{n-1}} \delta_1(\omega_{k_1} f) \cdot \delta_2(\omega_{k_1}^* \omega_{k_2}) \cdots \delta_{n-1}(\omega_{k_{n-2}}^* \omega_{k_{n-1}}) \cdot \delta_n(\omega_{k_{n-1}}^*), \qquad f \in F,$$

*which lie in $W_F(A_{s_0})$, span it as $f$ runs over $F$, and are polynomials with rational coefficients in divisor classes. In particular $W_F(A_{s_0})$ is algebraic, and $\Sigma_F \neq \emptyset$ for every $F$, every $n \ge 1$ and every $\delta$.*

*Proof.* (i) Since $H_\tau$ has signature $(n,n)$ at every $\tau$, the determinant $\det H$ has sign $(-1)^n$ at every real place, so $b = (-1)^n \det H$, a representative of $\delta$, is totally positive, and the

diagonal form $H' = \operatorname{diag}(b, 1, \dots, 1, -1, \dots, -1)$ with $n$ positive and $n$ negative entries has the same dimension, the same discriminant and the same signatures as $H$. Hermitian forms over $F/F_0$ are classified by these invariants [Sch85, Ch. 10, §6], so $(V, H) \cong (F^{2n}, H')$ and the standard basis is the required orthogonal basis $e_i$, with $a_i = H(e_i, e_i)$. Write $\sigma_\tau(\xi) = \sqrt{-1}\, r_\tau$ and define $J$ on $V_\tau = \bigoplus_i \mathbb{C}e_i$ to be $\epsilon_{i,\tau}\sqrt{-1}$ on $\mathbb{C}e_i$, with $\epsilon_{i,\tau} = \operatorname{sign}(r_\tau)\operatorname{sign}(\tau(a_i))$. The decomposition $V_\tau^+ = \bigoplus_{\epsilon_{i,\tau}=+} \mathbb{C}e_i$, $V_\tau^- = \bigoplus_{\epsilon_{i,\tau}=-} \mathbb{C}e_i$ is $H_\tau$-orthogonal, with $H_\tau$ of sign $\operatorname{sign}(r_\tau)$ on $V_\tau^+$ and of the opposite sign on $V_\tau^-$, so Proposition 19.3(i) makes $J$ a point $s_0$ of $\mathcal{D}_F$. On $Fe_i \otimes \mathbb{R} = \prod_\tau \mathbb{C}e_i$ the structure $J$ is multiplication by the element $(\epsilon_{i,\tau}\sqrt{-1})_\tau$ of $F \otimes_{\mathbb{Q}} \mathbb{R}$, so each $Fe_i$ is a rational sub-Hodge structure of weight one with $F$-action, polarised by the restriction of $E$; that is, it is the $H^1$ of an abelian $m$-fold with complex multiplication by $F$, of CM type $\Phi_i = \{\sigma : (Fe_i)_\sigma \subseteq V^{1,0}\}$, which by the computation in Proposition 19.3(i) is $\{\sigma_\tau^{\epsilon_{i,\tau}}\}$, where $\sigma^+ = \sigma$ and $\sigma^- = \overline{\sigma}$. For $i \le n$ the $a_i$ are totally positive, so $\epsilon_{i,\tau} = \operatorname{sign}(r_\tau)$ and $\Phi_i = \{\sigma : \operatorname{Im}\sigma(\xi) > 0\} = \Phi$; for $i > n$ the signs are reversed and $\Phi_i = \overline{\Phi}$. So $V = \bigoplus_i Fe_i$ splits $A_{s_0}$ up to isogeny as stated.

(ii) Write $L_{i,\sigma} = (Fe_i)_\sigma$, a line, so that $V_{\mathbb{C}} = \bigoplus_{i,\sigma} L_{i,\sigma}$ and $L_{i,\sigma}$ has type $(1,0)$ exactly when $\sigma \in \Phi_i$. For dual bases, $\sum_k \sigma(\omega_k)\sigma'(\omega_k^*) = 1$ if $\sigma = \sigma'$ and 0 otherwise, because the matrices $(\sigma(\omega_k))$ and $(\sigma(\omega_k^*))$ are inverse transposes by the definition of the trace form. Let $\ell_{i,\sigma}$ be the component of $e_i$ in $L_{i,\sigma}$, a generator of that line, so that $\omega e_i = \sum_\sigma \sigma(\omega)\,\ell_{i,\sigma}$ for $\omega \in F$. Hence in $Fe_i \otimes Fe_{i'} \otimes \mathbb{C} = \bigoplus_{\sigma,\sigma'} L_{i,\sigma} \otimes L_{i',\sigma'}$ the class $\delta_i(f)$ has component $\sigma(f)\,\ell_{i,\sigma} \wedge \ell_{i',\sigma}$ on the diagonal $\sigma = \sigma'$ and zero off it. In particular it is $F$-balanced: $\alpha \in F$ acting through the first factor and through the second agree on it, both multiplying the component at $\sigma$ by $\sigma(\alpha)$, and this is the identity $\alpha \cdot \delta_i(f) = \delta_i(\alpha f)$. Since $\Phi_{i'} = \overline{\Phi}_i$, exactly one of $L_{i,\sigma}$, $L_{i',\sigma}$ has type $(1,0)$, so every component, hence $\delta_i(f)$, has type $(1,1)$; and $\delta_i(f)$ is rational, being a sum of wedges of rational vectors. By the Lefschetz theorem on $(1,1)$ classes it is a divisor class. As $f$ varies the components $\sigma(f)$ vary independently, so $D_{ii'} \otimes \mathbb{C} = \bigoplus_\sigma L_{i,\sigma} \otimes L_{i',\sigma}$ and $D_{ii'} \cong F$.

(iii) The complexification of $W_F$ is spanned by $\alpha_\sigma = \pm \prod_{i \le n} \ell_{i,\sigma} \wedge \ell_{i',\sigma}$, a generator of $\bigwedge^{2n} V_\sigma = \bigwedge^{2n}(\bigoplus_i L_{i,\sigma})$. In the product $w(f)$, the component indexed by $(\sigma_1, \dots, \sigma_n)$, with $\sigma_i$ the embedding chosen in the $i$-th factor, carries the coefficient $\sigma_1(f) \prod_{j=1}^{n-1} \sum_{k_j} \sigma_j(\omega_{k_j})\sigma_{j+1}(\omega_{k_j}^*)$, which vanishes unless $\sigma_1 = \cdots = \sigma_n$ and then equals $\sigma_1(f)$. So $w(f) = \sum_\sigma \sigma(f)\,\alpha_\sigma$ up to the sign of the reordering, an element of $W_F \otimes \mathbb{C}$ which is rational because it is a rational polynomial in rational classes; as $f$ runs over $F$ the vectors $(\sigma(f))_\sigma$ span $\mathbb{C}^{2m}$, so the $w(f)$ span $W_F$. Each $w(f)$ is a rational combination of $n$-fold products of the divisor classes $\delta_i(\cdot)$, and a product of divisor classes is algebraic. ∎

*Remark* 19.6 (The base point for a quadratic field). For $m = 1$ the point $s_0$ is a member isogenous to $E^{2n}$ for an elliptic curve $E$ with complex multiplication by $K$, at which the whole Hodge ring is generated by divisor classes, since the Hodge group is a one-dimensional torus. It is a second base point in every family of Section 14, beside the quaternionic one of Theorem 15.27, and it lies in the product locus of Theorem 15.10. The formula of Theorem 19.5(iii) then gives the class at that point explicitly.

### 19.3. **The reduction and the propagation.**

**Theorem 19.7** (The reduction and the propagation, for every CM field)**.** *Fix $(F, n, \delta)$ with $n \ge 2$, keep Setup 19.1, and let $\omega_s$ denote a generator of $W_F(A_s)$ chosen flatly.*

(i) $\mathbf{W}(F, n, \delta)$ *holds if and only if $\omega_s$ is algebraic for every $s \in \mathcal{D}_F$, and then the Hodge conjecture holds in every degree for every member of the family with* $\operatorname{Hg}(A_s) = G_F$.

(ii) *$\Sigma_F$ is stable under $G_F(\mathbb{Q})$, every $G_F(\mathbb{Q})$-orbit in $\mathcal{D}_F$ is dense, and $\Sigma_F$ is dense.*

(iii) $\Sigma_F = p^{-1}\big(\bigcup_{\underline{P},\underline{q}} \Sigma_{\underline{P},\underline{q}}\big)$ *with each* $\Sigma_{\underline{P},\underline{q}}$ *Zariski closed in* $\mathcal{S}_F$*, the union countable; either* $\Sigma_F = \mathcal{D}_F$*, or* $\Sigma_F$ *is meagre and the conjecture fails on a dense* $G_\delta$*.*

(iv) *If at every point* $s$ *of the orbit* $G_F(\mathbb{Q})\cdot s_0$ *the class* $\omega_s$ *is represented by a cycle whose Hilbert data comes from one finite set, then* $\Sigma_F = \mathcal{D}_F$*.*

(v) *If some perfect complex* $E$ *on* $A_{s_0}$ *with* $\mathrm{ch}(E) = N\omega_{s_0} + P$*, where* $N \neq 0$ *and* $P$ *is a polynomial with rational coefficients in the classes* $\eta_t$ *of Proposition 19.3(iii), has injective semiregularity map, then* $\Sigma_F = \mathcal{D}_F$*. Complexes with* $\mathrm{ch}(E) = N\omega_{s_0}$ *exist.*

*Proof.* (i) By Proposition 19.3(i) and (iv), every member has its Hodge structure at some point of $\mathcal{D}_F$, so $\mathbf{W}(F,n,\delta)$ is the algebraicity of $W_F$ at every point. By Proposition 19.3(ii) this is the algebraicity of one generator at every point, and by Proposition 19.3(iii) it gives the conjecture at the members with full Hodge group.

(ii) The stability is the argument of Lemma 16.6: $g \in G_F(\mathbb{Q})$ carries $J$ to $gJg^{-1}$ and induces, after clearing denominators, an isogeny $A_J \to A_{gJ}$ whose graph transports algebraic classes to algebraic classes, and $g$ fixes $W_F$ pointwise by Proposition 19.3(i).

For the density we show that the closure $L$ of $G_F(\mathbb{Q})$ in $G_F(\mathbb{R}) = \prod_\tau \mathrm{SU}(V_\tau, H_\tau)$ contains the identity component, which acts transitively on $\mathcal{D}_F$. First, the form $H$ is isotropic over $F_0$. It is indefinite at every real place, its signature being $(n,n)$. At every finite place $v$ its trace form is a quadratic form in $4n \geq 8$ variables over the local field $F_{0,v}$, hence isotropic, and a hermitian form is isotropic as soon as its trace form is [Sch85, Ch. 10, §1]. Since hermitian forms over the quadratic extension $F/F_0$ satisfy the Hasse principle [Sch85, Ch. 10, §6], $H$ is isotropic. Let $X = \{v \in V : H(v,v) = 0\}$ be the cone of isotropic vectors, a quadric cone in the $4n$-dimensional $F_0$-space $V$ with a nonzero $F_0$-point. Projection from that point shows that the projective quadric of its lines is $F_0$-rational, that is, $F_0$-birational to a projective space, so it satisfies weak approximation at the archimedean places, $F_0$ being dense in $F_0 \otimes \mathbb{R} = \prod_\tau \mathbb{R}$. Hence the $F_0$-points of $X$ are dense in its points over $F_0 \otimes \mathbb{R}$, which are the products $\prod_\tau X_\tau$ of the isotropic cones of the $H_\tau$.

For an isotropic vector $v \in V$ let $U_v \subseteq \mathrm{SU}(V,H)$ be the unipotent radical of the stabiliser of the line $Fv$, a unipotent $F_0$-group. The exponential identifies $U_v(F_0)$ with an $F_0$-vector space and $U_v(F_0 \otimes \mathbb{R}) = \prod_\tau U_{v,\tau}(\mathbb{R})$ with the corresponding real space, so $U_v(\mathbb{Q}) = U_v(F_0)$ is dense in $\prod_\tau U_{v,\tau}(\mathbb{R})$. Hence $L$ contains $\prod_\tau U_{v,\tau}(\mathbb{R})$ for every rational isotropic $v$, in particular the factor $U_{v,\tau}(\mathbb{R})$ for each $\tau$ separately, and its Lie algebra $\mathfrak{l}$ contains $\mathfrak{u}_{v,\tau}$ for every $\tau$ and every rational isotropic $v$. The map $v \mapsto \mathfrak{u}_{v,\tau}$ is continuous on $X_\tau \setminus 0$ and rational $v$ are dense in $\prod_\tau X_\tau$, so the closed subspace $\mathfrak{l} \cap \mathfrak{su}(V_\tau, H_\tau)$ contains $\mathfrak{u}_{v,\tau}$ for every isotropic $v \in V_\tau$. The span of these is stable under the adjoint action of $\mathrm{SU}(V_\tau, H_\tau)$, which permutes the isotropic lines, so it is a nonzero ideal of the simple Lie algebra $\mathfrak{su}(n,n)$, hence everything. Thus $\mathfrak{l} \supseteq \bigoplus_\tau \mathfrak{su}(V_\tau, H_\tau)$ and $L \supseteq G_F(\mathbb{R})^+$, so every $G_F(\mathbb{Q})$-orbit in $\mathcal{D}_F$ is dense. Since $\Sigma_F$ is nonempty by Theorem 19.5 and $G_F(\mathbb{Q})$-stable, it contains a dense orbit.

(iii) The proofs of Theorem 16.4 and Corollary 16.5 use only three facts: that $\mathcal{S}_F$ is a quasi-projective variety, by the theorem of Baily and Borel [BB66]; that the relative Hilbert schemes of $\mathcal{A}/\mathcal{S}_F$ are projective over it [Gro61]; and that $W_F$ is a flat subsystem consisting of Hodge classes, which is Proposition 19.3(i). They apply verbatim.

(iv) The argument is the proof of Theorem 16.9, which asks for a cycle chosen at each point (Remark 16.10). The orbit is dense by (ii), and a finite union of the closed sets of (iii) covering it forces one of them to be dense and closed, hence everything; then (i) applies. As in the quadratic case, the hypothesis is equivalent to the conclusion (Theorem 16.17), and the transported cycle does not satisfy it (Theorem 16.15).

(v) The existence of complexes with $\mathrm{ch}(E) = N\omega_{s_0}$ is the proof of Proposition 16.27, applied to the algebraic cycle of Theorem 19.5 with the isomorphism $K_0(A)_{\mathbb{Q}} \cong \mathrm{CH}^*(A)_{\mathbb{Q}}$ [Ful98]. The deformation argument is the proof of Theorem 16.29 for a polarised Chern character, as in Remark 17.6. Its only inputs are the theorem of Buchweitz and Flenner, (iii), and the flatness of $\mathrm{ch}(E)$ as a section of Hodge classes over $\mathcal{D}_F$; the last holds because $W_F$ is flat by Proposition 19.3(i) and the classes $\eta_t$ remain of Hodge type on the whole family, as in the proof of Proposition 19.3(iii). At a member $A_s$ reached by the deformation the component of degree $2n$ of the character is $N\omega_s + P_{2n}$, with $P_{2n}$ a polynomial in divisor classes, so $\omega_s$ is algebraic. ∎

**Corollary 19.8** (The Hodge conjecture for the very general member)**.** *Fix $(F, n, \delta)$ with $n \geq 2$. The following are equivalent.*

(a) *$\Sigma_F$ is closed in $\mathcal{D}_F$.*

(b) *$\Sigma_F$ has an interior point.*

(c) *$\Sigma_F = \mathcal{D}_F$, that is, the Weil classes are algebraic on every member of the family.*

(d) *The Hodge conjecture holds for the very general member of the family: there is a countable union of proper closed analytic subsets of $\mathcal{D}_F$ outside which the Hodge conjecture holds for $A_s$ in every degree.*

*Proof.* If (a) holds, $\Sigma_F$ is closed and dense by Theorem 19.7(ii), so (c) holds; and (c) implies (a) and (b). If (b) holds, $\Sigma_F$ is not meagre, since a nonempty open subset of the manifold $\mathcal{D}_F$ is not meagre by Baire's theorem, and Theorem 19.7(iii) gives (c). If (c) holds, the Hodge conjecture holds in every degree for every member with $\mathrm{Hg}(A_s) = G_F$ by Theorem 19.7(i), and these are the members outside the set $Z$ of Lemma 19.4, so (d) holds. Finally, if (d) holds with an exceptional set $Z'$, then at every $s \notin Z'$ the Weil classes, which are Hodge classes at every point by Proposition 19.3(i), are algebraic. So $\Sigma_F$ contains the complement of $Z'$, which is not meagre by Baire's theorem, since each proper closed analytic subset of the connected manifold $\mathcal{D}_F$ is nowhere dense; by Theorem 19.7(iii), $\Sigma_F = \mathcal{D}_F$. ∎

*Proof of Theorem 1.1.* Part (i) is Theorem 19.5, parts (ii) and (iii) are Theorem 19.7(ii) and (iii), and part (iv) is Corollary 19.8. ∎

*Remark* 19.9 (Scope of the reduction). The propagation statements (iv) and (v) are exactly the two forms (P1) and (P2) of Remark 17.7 with $F$ in place of $K$, (v) being (P2) in the polarised form of Remark 17.6. This section proves that they are the whole of the problem for every CM field, and that their hypothesis is never vacuous: the base point exists, its cycle is written down, and the orbit through it is dense. The theorem of Theorem 16.31 on what (v) forces transfers as well, with the annihilator of a class in $W_F$ computed over the $2m$ eigenspaces in place of two, and we do not repeat it. Nothing above depends on Question 11.4 of [Mar25b] or on any secant construction. The route through Weil restriction and CM points is independent of the route through secant sheaves and reaches every field at once; Markman has extended the route through secant sheaves to these fields [Mar25c] (see (C3) in Section 20.5, and Section 19.6 for the smallest case).

**Proposition 19.10** (Composite fields)**.** *Let $F = KF_0$ with $K$ imaginary quadratic, and let $A$ be of $(F, n, \delta)$-Weil type. Then $A$ is of $(K, -1, mn)$-Weil type for the restricted action, its Weil classes $W_K(A) \subseteq H^{2mn}(A, \mathbb{Q})$ lie in the $\mathbb{Q}$-span of the $m$-fold products of elements of $W_F(A)$, and $W_K(A)$ is algebraic whenever $W_F(A)$ is. In particular $\mathbf{W}(F, n, \delta)$ implies the algebraicity of the $K$-Weil classes on every member of the $F$-family, a closed subvariety*

*of dimension $mn^2$ of the $K$-family of dimension $m^2n^2$; the converse implication is not available, and Theorem 19.5 is not a consequence of Theorem 15.27.*

*Proof.* The eigenspace $V_{\tau_K}$ of $K$ on $V_{\mathbb{C}}$ for the embedding $\tau_K$ is $\bigoplus_{\sigma|\tau_K} V_\sigma$, of dimension $2mn$ with $\dim V_{\tau_K}^{1,0} = mn$ by Proposition 19.3(i), so $A$ is of $K$-Weil type with balanced signature, and $\bigwedge^{2mn} V_{\tau_K} = \bigotimes_{\sigma|\tau_K} \bigwedge^{2n} V_\sigma$ is spanned by $\prod_{\sigma|\tau_K} \alpha_\sigma$. So $W_K \otimes \mathbb{C}$ is contained in the complexification of the $\mathbb{Q}$-span $P$ of the $m$-fold products of elements of $W_F$, and a rational subspace whose complexification lies in that of $P$ lies in $P$. Since products of algebraic classes are algebraic, $W_K(A)$ is algebraic whenever $W_F(A)$ is. The last sentence of the statement is a dimension count: for $m \geq 2$ the $F$-family is a proper subvariety of the $K$-family, and the $K$-Weil classes on it are products that carry no information about their factors. ∎

19.4. **The special locus and an escaping sequence.** The properties of $\Sigma_F$ proved so far are shared by a set that is meagre, the locus of members whose Hodge group is smaller than $G_F$, and every construction of algebraic Weil classes available for $n \geq 4$ produces points of that locus only. In the other direction the bounded criterion (iv) of Theorem 19.7 needs far less than an orbit: by the theorem of Pila, Shankar and Tsimerman on the André–Oort conjecture, a bound on one sequence of CM points that leaves every proper special subvariety is enough (Figure 35).

**Definition 19.11** (Special subvarieties and the special locus)**.** Keep Setup 19.1, with $\Gamma$ a congruence subgroup. A *special subvariety* of $\mathcal{S}_F$ is an irreducible component of the image in $\mathcal{S}_F$ of a connected component of the locus of Hodge classes $\mathrm{Hdg}^p(\mathbb{V})$, where $\mathbb{V}$ runs over the variations $(R^1\pi_*\mathbb{Z})^{\otimes a} \otimes (R^1\pi_*\mathbb{Z})^{\vee \otimes b}$ of even weight, or $\mathcal{S}_F$ itself; these are the special subvarieties of $\mathcal{S}_F$ for $R^1\pi_*\mathbb{Z}$ in the sense of [KOU23].[36] A *CM point* is a point $p(s)$ at which $A_s$ is of CM type, that is, at which $\mathrm{Hg}(A_s)$ is a torus; the CM points are the special points of $\mathcal{S}_F$. The *special locus* is

$$\Sigma_F^{\mathrm{sp}} = \{\, s \in \mathcal{D}_F \;:\; \mathrm{Hg}(A_s) \neq G_F \,\}.$$

**Proposition 19.12** (The special locus)**.** *Fix $(F, n, \delta)$ with $n \geq 2$ and keep Definition 19.11.*

(i) *Special subvarieties are closed algebraic subvarieties of $\mathcal{S}_F$, and there are countably many of them. A special subvariety other than $\mathcal{S}_F$ is proper and nowhere dense.*

(ii) *$\Sigma_F^{\mathrm{sp}}$ is the set $Z$ of Lemma 19.4, and it is the preimage under $p$ of the union of the proper special subvarieties. In particular it is the preimage of a countable union of proper Zariski closed subsets of $\mathcal{S}_F$, and it is meagre.*

(iii) *If $\mathrm{End}^0(A_s) \neq F$, then $s \in \Sigma_F^{\mathrm{sp}}$. In particular $\Sigma_F^{\mathrm{sp}}$ contains every point at which $A_s$ is of CM type, the base point $s_0$ of Theorem 19.5, and, for $F$ imaginary quadratic, the split members of Setup 15.12, the quaternionic members of Theorem 15.25 and the products of Lemma 15.26.*

(iv) *$\Sigma_F^{\mathrm{sp}}$ is stable under $G_F(\mathbb{Q})$ and dense in $\mathcal{D}_F$.*

*Proof.* (i) Each image of a component of $\mathrm{Hdg}^p(\mathbb{V})$ is closed and algebraic by Theorem B.6, applied to the polarised variation $\mathbb{V}$, after a Tate twist when $a < b$, so its irreducible components are irreducible closed subvarieties. There are countably many pairs $(a, b)$, each $\mathrm{Hdg}^p(\mathbb{V})$ has countably many components by Lemma B.4, and an algebraic set has finitely

[36] By a theorem of Chevalley every algebraic subgroup of $\mathrm{GL}(V)$ is the stabiliser of finitely many tensors [Del82, §3], so the locus where the Hodge group lies in a given subgroup is a locus of Hodge tensors, and these subvarieties are the special subvarieties of $\mathcal{S}_F$ as a connected Shimura variety, the images of the domains of the reductive subgroups of $G_F$ that contain the Hodge group of one of their points.

many irreducible components. A proper closed subvariety of the irreducible variety $\mathcal{S}_F$ has empty interior.

(ii) Let $s \in Z$, so that some rational tensor $t$ not fixed by $G_F$ is a Hodge class of $A_s$, and let $C$ be the component of $\mathrm{Hdg}^p(\mathbb{V})$ through the point $(p(s), Nt)$, for $N$ with $Nt$ integral. Along $C$ the class is carried by parallel transport, so over $\mathcal{D}_F$ it is one of the flat sections $\gamma Nt$ with $\gamma \in \Gamma$ on each sheet. If the image of $C$ were $\mathcal{S}_F$, then by Lemma B.4 and Baire's theorem one such section $\gamma Nt$ would be a Hodge class on a nonempty open subset of $\mathcal{D}_F$, hence on all of it, since the locus $Z_{\gamma t}$ of Lemma 19.4 is closed analytic in the connected manifold $\mathcal{D}_F$; the proof of Lemma 19.4 then shows that $\gamma t$, and with it $t$, is fixed by $G_F$, a contradiction. So $p(s)$ lies in a proper special subvariety. Conversely, if $p(s)$ lies in a proper special subvariety, an irreducible component of the image of a component $C$ through a point $(p(s), v)$, then $v$ is a Hodge class of $A_s$. If $v$ were fixed by $G_F$ it would be fixed by $\Gamma$, hence a flat global section that is a Hodge class at every point by Lemma 19.4; its graph is open and closed in $\mathrm{Hdg}^p(\mathbb{V})$, so it would be $C$, with image $\mathcal{S}_F$, and the special subvariety would not be proper. Hence $s \in Z$. The rest follows from (i) and Lemma 19.4.

(iii) The endomorphism algebra $\mathrm{End}^0(A_s)$ is the algebra of Hodge classes in $\mathrm{End}(V)$, that is, the commutant of $\mathrm{Hg}(A_s)$ [Del82, §3]. Over $\mathbb{C}$ the group $G_F$ is $\prod_\tau \mathrm{SL}(V_{\sigma_\tau})$, acting on $V_{\sigma_\tau}$ by the standard representation and on $V_{\overline{\sigma}_\tau}$, which $H$ pairs perfectly with $V_{\sigma_\tau}$, by its dual. These $2m$ representations are irreducible and pairwise non-isomorphic, the standard representation of $\mathrm{SL}_{2n}$ and its dual being non-isomorphic for $2n \geq 3$, so the commutant of $G_F$ in $\mathrm{End}(V_{\mathbb{C}})$ is $\mathbb{C}^{2m}$, and its commutant in $\mathrm{End}_{\mathbb{Q}}(V)$ is a commutative algebra of dimension $2m$ containing $F$, hence equal to $F$. So $\mathrm{Hg}(A_s) = G_F$ forces $\mathrm{End}^0(A_s) = F$. A member of CM type has a commutative Hodge group, and $G_F$ is not commutative; at $s_0$ the Hodge group lies in that of $B_\Phi^n \times B_{\overline{\Phi}}^n$, a torus. The split members contain the projections to $X$ and to $\widehat{X}$ in $\mathrm{End}^0(A)$, the quaternionic members contain a quaternion algebra, and the products contain the projections to their factors; none of these lies in $K$.

(iv) An element $g \in G_F(\mathbb{Q})$ induces an isogeny $A_s \to A_{gs}$ acting on $V$ by $g$, so $\mathrm{Hg}(A_{gs}) = g\,\mathrm{Hg}(A_s)g^{-1}$, and $gG_Fg^{-1} = G_F$. By (iii) the set contains $s_0$, hence its orbit, which is dense by Theorem 19.7(ii). ∎

**Corollary 19.13** (What closing the locus requires)**.** *Fix $(F, n, \delta)$ with $n \geq 2$. The set $\Sigma_F^{\mathrm{sp}}$ has every property of $\Sigma_F$ used in Theorem 1.1: it contains the explicit CM point $s_0$, it is dense and stable under $G_F(\mathbb{Q})$, and it is the preimage of a countable union of Zariski closed subsets of $\mathcal{S}_F$; and it is meagre. Consequently $\Sigma_F = \mathcal{D}_F$ does not follow from these properties of $\Sigma_F$, and it holds only if the Weil classes are algebraic at some $s$ with $\mathrm{Hg}(A_s) = G_F$, where they span, together with the products of divisor classes, every Hodge class of $A_s$.*

*Proof.* The properties are Proposition 19.12(ii) to (iv), and meagreness is (ii). If $\Sigma_F = \mathcal{D}_F$, then $\Sigma_F$ contains the complement of $\Sigma_F^{\mathrm{sp}}$, which is nonempty by Baire's theorem. The last assertion is Proposition 19.3(iii). ∎

The base points of Theorem 19.5, Theorem 15.27 and Theorem 15.10, the split members of Theorem 15.14(c), their orbits under $G_F(\mathbb{Q})$, and the members on which the divisorial method of Remark 15.29 works all lie in $\Sigma_F^{\mathrm{sp}}$ by Proposition 19.12(iii). These are all the members at which this paper exhibits algebraic Weil classes, so the part of $\Sigma_F$ exhibited here fits inside the meagre alternative, and the two criteria of Theorem 19.7, a bound or a semiregular object, are what would leave it. The first can be weakened as follows.

**Theorem 19.14** (One escaping sequence suffices)**.** *Fix $(F, n, \delta)$ with $n \geq 2$ and keep Definition 19.11.*

(i) *There is a sequence $y_1, y_2, \dots$ of distinct CM points in $p(G_F(\mathbb{Q}) \cdot s_0)$ such that every proper special subvariety of $\mathcal{S}_F$ contains only finitely many of them.*

(ii) *Let $y_1, y_2, \dots$ be any sequence of distinct CM points of $\mathcal{S}_F$ with this property. Then $\Sigma_F = \mathcal{D}_F$ if and only if there is a finite set $\mathcal{T}$ of pairs $(\underline{P}, \underline{q})$ such that for every $k$ the class $\omega$ is represented at $y_k$ by a cycle whose Hilbert data lies in $\mathcal{T}$.*

*Proof.* (i) By Proposition 19.12(i) the proper special subvarieties form a countable family $Y_1, Y_2, \dots$ of nowhere dense closed subsets. The image $p(G_F(\mathbb{Q}) \cdot s_0)$ is dense in $\mathcal{S}_F$, since the orbit is dense in $\mathcal{D}_F$ by Theorem 19.7(ii) and $p$ is continuous and surjective, and it consists of CM points by Proposition 19.12(iii), (iv). For each $k$ the complement of $Y_1 \cup \dots \cup Y_k \cup \{y_1, \dots, y_{k-1}\}$ is a nonempty open set, so it contains a point $y_k$ of the image. Then $y_k \in Y_j$ forces $k < j$, and $Y_j$ contains at most $j - 1$ of the points.

(ii) If $\Sigma_F = \mathcal{D}_F$, then the countably many closed sets $\Sigma_{\underline{P},\underline{q}}$ of Theorem 19.7(iii) cover $\mathcal{S}_F$, so by Baire's theorem one of them has an interior point, and being Zariski closed in the irreducible $\mathcal{S}_F$ it is all of $\mathcal{S}_F$; take $\mathcal{T}$ to be that one pair. Conversely, let $W$ be the union of the finitely many $\Sigma_{\underline{P},\underline{q}}$ with $(\underline{P}, \underline{q}) \in \mathcal{T}$, a Zariski closed set that contains every $y_k$, and let $Y \subseteq W$ be the Zariski closure of the set of the $y_k$. The variety $\mathcal{S}_F$ is a connected Shimura variety, and by the theorem of Pila, Shankar and Tsimerman [PST21, Theorem 1.1] a subvariety of a connected Shimura variety contains only finitely many maximal special subvarieties.[37] Each $y_k$, being a special point of $Y$, lies in a maximal special subvariety contained in $Y$, so one of these finitely many, say $M$, contains infinitely many $y_k$. By hypothesis $M$ is not proper, so $M = \mathcal{S}_F$, and therefore $W = \mathcal{S}_F$. As $\mathcal{S}_F$ is irreducible, one of the finitely many closed sets making up $W$ is all of $\mathcal{S}_F$, and $\Sigma_F = \mathcal{D}_F$ by Theorem 19.7(iii). ∎

*Remark* 19.15 (What the sequence changes). Theorem 19.14 replaces the dense orbit of Theorem 19.7(iv) by a discrete sequence, which may be chosen inside the orbit or among CM points with complex multiplication by different fields; what it must do is leave each proper special subvariety. Without the theorem of Pila, Shankar and Tsimerman the Zariski closure of such a sequence could be a proper subvariety that is not special, and a bound on it would give nothing. Like Theorem 16.17, the statement is an equivalence, so it shrinks the set on which a bound is needed and not the problem. Read on the transported cycle the bound remains subject to Theorem 16.15; read on a representative chosen afresh at each $y_k$, it is not.

19.5. **Weil tori of a CM field and the pure form.** The argument of Theorem 17.5 uses the field only through the eigenspaces of its action, and it extends to every CM field. For $m \geq 2$ the Hodge ring of a very general Weil torus is larger than for an imaginary quadratic field, since products of Weil classes over a set of embeddings closed under conjugation are Hodge classes there, and a Weil torus may carry subvarieties of codimension $2n$; but it carries none of codimension $n$, and that is all the argument needs.

**Setup 19.16** (Weil tori of a CM field)**.** Keep Setup 19.1, let $n \geq 2$ and let $A$ be of $(F, n, \delta)$-Weil type, with $V_{\mathbb{C}} = \bigoplus_\sigma V_\sigma$ as there. Complex conjugation carries $V_{\overline{\sigma}}$ onto $V_\sigma$. A complex structure on $V_{\mathbb{R}}$ commuting with $F$, with $H^{1,0} \cap V_\sigma$ of dimension $n$ for every $\sigma$, is the same as a choice, for each real place $\tau$, of complementary $n$-dimensional subspaces

[37] This is the André–Oort conjecture, proved for $\mathcal{A}_g$ by Tsimerman [Tsi18] and in general in [PST21].

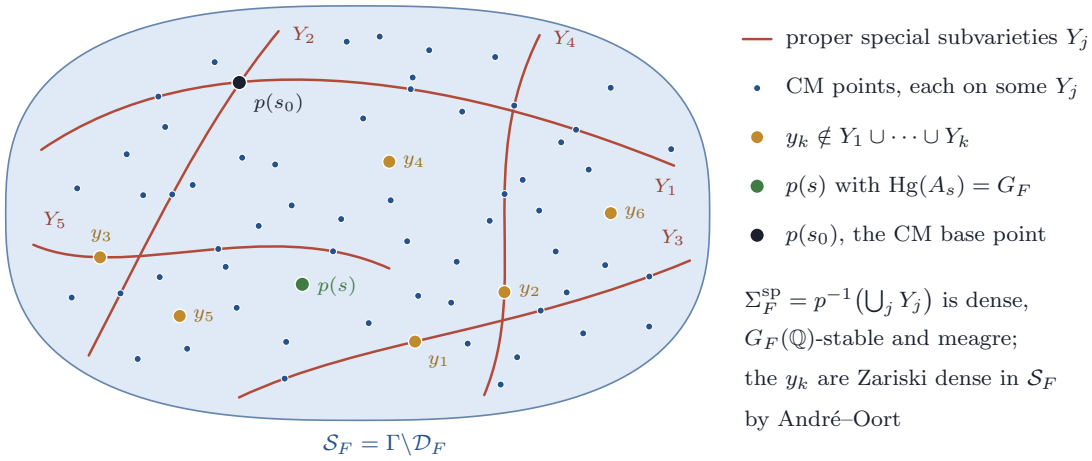



FIGURE 35. Proposition 19.12 and Theorem 19.14 on the quotient $\mathcal{S}_F$. The curves are five of the countably many proper special subvarieties $Y_j$, whose union is the image of the special locus; it contains $p(s_0)$ and is dense and meagre. The small dots are CM points, each lying on some $Y_j$, drawn or not. The points $y_1, \dots, y_6$ begin an escaping sequence chosen as in the proof of Theorem 19.14(i): $y_k$ lies on no $Y_j$ with $j \le k$, so $y_1$ may lie on $Y_3$ and $y_2$ on $Y_4$. A point $p(s)$ at which the Hodge group is all of $G_F$ lies on none of them.

$X_\tau, Y_\tau \subset V_{\sigma_\tau}$: then $H^{1,0} \cap V_{\sigma_\tau} = X_\tau$, $H^{0,1} \cap V_{\sigma_\tau} = Y_\tau$, and on $V_{\overline{\sigma}_\tau}$ the two parts are the conjugates of $Y_\tau$ and $X_\tau$. Call the complex torus obtained in this way a Weil torus of $F$, and let $S_F$ be the connected complex manifold of these choices, of dimension $2mn^2$. Its tangent space at $A$ is $\mathrm{Hom}_F(H^{1,0}, H^{0,1})$, the polarised family $\mathcal{D}_F$ is the submanifold on which $E$ is of type $(1,1)$, and $S_F$ contains non-algebraic tori. For $\sigma = \sigma_\tau$ the line $\bigwedge^{2n} V_\sigma = \bigwedge^n X_\tau \otimes \bigwedge^n Y_\tau$ is of type $(n,n)$, and likewise for $\overline{\sigma}_\tau$, so $W_F$ consists of Hodge classes on every member of $S_F$. For a set $\Sigma$ of embeddings write $\alpha_\Sigma = \prod_{\sigma\in\Sigma} \alpha_\sigma$, with $\alpha_\sigma$ spanning $\bigwedge^{2n} V_\sigma$.

**Lemma 19.17** (Hodge classes and subvarieties of a very general Weil torus)**.** *Let $T$ be a very general member of $S_F$ and $d = 2mn = \dim T$.*

(i) *Every Hodge class of $T$ lies in the span of the $\alpha_\Sigma$. In particular* $\mathrm{Hdg}^k(T) = 0$ *unless $n$ divides $k$, and* $\mathrm{Hdg}^n(T) = W_F(T)$.

(ii) *Let $\kappa_0$ be the Kähler class of a hermitian form on the tangent space that is block diagonal for the splitting of $H^{1,0}$ into the $H^{1,0} \cap V_\sigma$. Then $\int_T \alpha_\Sigma\, \kappa_0^{d-n|\Sigma|} = 0$ unless $\Sigma$ is closed under conjugation.*

(iii) *Every irreducible analytic subset of $T$ other than a point and $T$ itself has codimension $ns$ with $s$ even, hence at least $2n$.*

*Proof.* (i) As in the proof of Lemma 17.2: a rational class of Hodge type at a very general member is of Hodge type on all of $S_F$, and the Mumford–Tate group $M$ of a very general member lies in $G = \mathrm{Res}_{F/\mathbb{Q}} \mathrm{GL}_F(V)$, since $F$ acts by endomorphisms. Here $G(\mathbb{R}) = \prod_\tau \mathrm{GL}(V_{\sigma_\tau})$ acts transitively on $S_F$, and the $\mathrm{Ad}\, G(\mathbb{R})$-span of the complex structure of one member is an ideal of $\bigoplus_\tau \mathfrak{gl}_{2n}(\mathbb{C})$ whose projection to every factor contains a non-central element, $i$ on $X_\tau$ and $-i$ on $Y_\tau$. An ideal of a sum of simple real Lie algebras

is a sum of some of them, and each $\mathfrak{sl}_{2n}(\mathbb{C})$ is simple as a real Lie algebra, so the span contains $\bigoplus_\tau \mathfrak{sl}_{2n}(\mathbb{C})$ and $M$ contains $\mathrm{Res}_{F/\mathbb{Q}}\,\mathrm{SL}_F(V)$. Over $\mathbb{C}$ this group is $\prod_\sigma \mathrm{SL}(V_\sigma)$ acting on $\bigwedge^* V_{\mathbb{C}} = \bigotimes_\sigma \bigwedge^* V_\sigma$, and the invariants of $\mathrm{SL}(V_\sigma)$ in $\bigwedge^* V_\sigma$ are $\bigwedge^0$ and $\bigwedge^{2n}$. So the invariants are spanned by the $\alpha_\Sigma$, of degree $2n|\Sigma|$, and a rational class in the span of the $\alpha_\sigma$ lies in $W_F$.

(ii) Choose the hermitian form so that $\kappa_0 = \sqrt{-1}\sum_\sigma \sum_k \xi_{\sigma,k} \wedge \overline{\xi}_{\sigma,k}$ with $\xi_{\sigma,k}$ a basis of $H^{1,0} \cap V_\sigma$; each term lies in $V_\sigma \otimes V_{\overline{\sigma}}$. A product of classes has nonzero integral only if its degree in every $V_\rho$ is $2n$. The factor $\alpha_\Sigma$ has degree $2n$ in $V_\rho$ for $\rho \in \Sigma$ and 0 otherwise, and every term of a power of $\kappa_0$ has the same degree in $V_\rho$ as in $V_{\overline{\rho}}$. For $\rho \in \Sigma$ that degree must be 0, so $V_{\overline{\rho}}$ must be filled by $\alpha_\Sigma$, that is $\overline{\rho} \in \Sigma$.

(iii) Let $Z$ be irreducible of codimension $c$ with $0 < c < d$. For a Kähler class $\kappa$, $\int_T [Z]\,\kappa^{d-c} = \int_Z \kappa^{d-c} > 0$, so $[Z]$ is a nonzero Hodge class of degree $2c$, and by (i) $c = ns$ and $[Z]$ is a combination of the $\alpha_\Sigma$ with $|\Sigma| = s$. Complex conjugation permutes the embeddings without fixed points, so a set closed under it has even size; if $s$ were odd, (ii) would give $\int_T [Z]\,\kappa_0^{d-c} = 0$. ∎

**Proposition 19.18** (Coherent sheaves on a very general Weil torus of $F$)**.** *Let $T$ be a very general member of $S_F$. Every coherent analytic sheaf $\mathcal{F}$ on $T$ has $\mathrm{ch}_k(\mathcal{F}) = 0$ in $H^{2k}(T,\mathbb{Q})$ for $1 \le k \le 2n-1$.*

*Proof.* The Chern classes of [Gri10] are Hodge classes, additive, and compatible with restriction to open subsets, so by Lemma 19.17(i) only $k = n$ needs an argument. First let $\mathcal{F}$ be supported on a proper analytic subset $W$. Its restriction to $T \smallsetminus W$ is zero, so $\mathrm{ch}_n(\mathcal{F})$ comes from $H^{2n}_W(T,\mathbb{Q})$, which is isomorphic to the Borel–Moore homology $H_{2d-2n}(W,\mathbb{Q})$; by Lemma 19.17(iii) every component of $W$ has codimension at least $2n$, so $W$ has real dimension at most $2d-4n < 2d-2n$ and that group vanishes. In general, the torsion of $\mathcal{F}$ is of this kind, and so is the cokernel of the inclusion of a torsion-free sheaf in its reflexive hull. Every coherent sheaf has $c_1 = 0$, since $\mathrm{Hdg}^1(T) = 0$, so as in the proof of Proposition 17.4 a torsion-free sheaf has a Jordan–Hölder filtration whose factors $Q_j$ are stable for every Kähler class, and $\mathrm{ch}_n(Q_j) = \mathrm{ch}_n(Q_j^{**})$. By Bando and Siu [BS94], $\int_T c_2(Q_j^{**})\,\kappa_0^{d-2} \ge 0$, with equality exactly when $Q_j^{**}$ is a projectively flat bundle. Here $c_2(Q_j^{**})$ lies in $\mathrm{Hdg}^2(T)$, which is zero for $n \ge 3$ and is $W_F(T)$ for $n = 2$, where the integral vanishes by Lemma 19.17(ii) with $|\Sigma| = 1$. So $Q_j^{**}$ is projectively flat with $c_1 = 0$, hence flat, and $\mathrm{ch}_n(Q_j) = 0$. Summing over the factors and adding the torsion gives $\mathrm{ch}_n(\mathcal{F}) = 0$. ∎

**Theorem 19.19** (The pure form is never met, for any CM field)**.** *Let $n \ge 2$, let $A$ be of $(F,n,\delta)$-Weil type, and let $E$ be a perfect complex on $A$ whose Chern character stays of Hodge type on every member of $S_F$ and has $\mathrm{ch}_n(E) \ne 0$; for instance $\mathrm{ch}(E) = c + N\omega$ with $c \in \mathbb{Q}$, $N \ne 0$ and $\omega$ a nonzero rational Weil class. Then $\sigma_E$ is not injective, and $\dim \mathrm{Ext}^2(E,E) > r(\mathrm{ch}\,E)$. In particular no complex of the pure form $\mathrm{ch}(E) = N\omega_{s_0}$ in Theorem 19.7(v), with or without a rank, has injective semiregularity map, whatever the CM field.*

*Proof.* The proof of Theorem 17.5 applies with $S_F$ in place of $S$: an injective $\sigma_E$ would let $E$ deform, by the theorem of Buchweitz and Flenner in Pridham's form and Kuranishi's method, to a bounded complex of holomorphic vector bundles on every member of a neighbourhood of $A$ in $S_F$, with Chern character the parallel transport of $\mathrm{ch}(E)$; at a very general member $\mathrm{ch}_n$ of that complex vanishes by Proposition 19.18, while it is the transport of $\mathrm{ch}_n(E) \ne 0$. A class $c + N\omega$ is of Hodge type on all of $S_F$ by Setup 19.16. The inequality follows from the square $\sigma_E \circ \mathrm{ev}_E = (\,\cdot\,) \lrcorner \mathrm{ch}(E)$, since $\dim \mathrm{Ext}^2(E,E) \ge \mathrm{rk}\,\mathrm{ev}_E \ge r(\mathrm{ch}\,E)$

and equality in both forces $\mathrm{ev}_E$ onto with kernel the annihilator, and then $\sigma_E$ injective, as in Theorem 17.9(iv). ∎

For $m = 1$ this is Theorem 17.5, and for $m = 1$, $n = 2$ the vanishing of Proposition 19.18 is Voisin's theorem [Voi02b, Theorem 10]. So the criterion of Theorem 19.7(v) needs a polynomial $P$ in the classes $\eta_t$ that is not constant, for every CM field and every $n \geq 2$. At a quartic field and $n = 2$ this pushes the least possible value of $\dim \mathrm{Ext}^2(E, E)$ for a complex meeting it from 80 to 88 (Corollary 19.30).

19.6. **Quartic CM fields at $n = 2$.** By Theorem 20.70(ii) the only form of (P2) on a minimal route to the conjecture is the one for the split families of the CM fields of degree at least four, and by Proposition 20.13(i) the Weil classes of any family of such a field in a higher dimension give those of its families at $n = 2$. For a quartic field these are families of abelian eightfolds, the smallest families of a field of degree at least four. This subsection determines how far the available mechanisms reach there.

Divisor classes reach the Weil classes exactly on a Noether–Lefschetz locus, which has codimension two when $\delta$ is trivial (Proposition 19.21, Corollary 19.22). For biquadratic $F$, Markman's theorem reaches them on a further locus of dimension four (Proposition 19.23). For $F = \mathbb{Q}(\zeta_5), \mathbb{Q}(\zeta_8), \mathbb{Q}(\zeta_{12})$ and trivial $\delta$, Schoen's cycles on Prym varieties of cyclic covers reach them on a locus of dimension at most six, at whose very general points the Hodge group is all of $G_F$ (Remark 19.24). The Casimir class of the quadratic form on the weight two part $R_F$ below, the natural class of degree four attached to it, is algebraic exactly when the Weil classes are (Proposition 19.25), and correspondences with abelian varieties whose simple factors have dimension at most seven cannot help at a member with full Hodge group (Proposition 19.27). For a general shape the polarised criterion asks $\dim \mathrm{Ext}^2(E, E) = 112$ of one complex (Theorem 19.28). No Orlov product of sheaves on a principally polarised fourfold with real multiplication meets the criterion through a dimension count, whatever the quartic field (Theorem 19.37), and no Orlov product of complexes does so unless both factors have nonzero self-Ext groups in some degree at most $-3$ (Remark 19.39). Those products are in any case candidates only for a twisted form of the criterion, the form of Markman's reduction (Remark 19.38). Objects with pure spinor characters give no Weil class at all (Proposition 19.40). Together these results fix the loci that divisors, scalar extension and correspondences reach inside $\mathcal{D}_F$, and the numbers that one complex has to attain on the rest.

Throughout, $F$ is a quartic CM field and $F_0$ its real quadratic subfield, with real places $\tau_1, \tau_2$, and $(V, H)$ is of $(F, 2)$-Weil type with discriminant $\delta$; so $m = n = 2$, the members are abelian eightfolds, and $\mathcal{D}_F$ has dimension $mn^2 = 8$. By Proposition 19.3(i), $H^1(A, \mathbb{C}) = \bigoplus_\sigma V_\sigma$ over the four embeddings $\sigma$ of $F$, with $\dim V_\sigma^{1,0} = \dim V_\sigma^{0,1} = 2$. Put

$$R_F = \textstyle\bigwedge_F^2 V \;\subseteq\; \bigwedge_{\mathbb{Q}}^2 V = H^2(A, \mathbb{Q}),$$

the $\mathbb{Q}$-subspace with complexification $\bigoplus_\sigma \bigwedge^2 V_\sigma$; for $m = 1$ it is the space $R$ of Proposition 15.18. It is an $F$-vector space of dimension 6, of $\mathbb{Q}$-dimension 24, with Hodge numbers $(4, 16, 4)$. Fix a generator $\Omega$ of the $F$-line $W_F = \bigwedge_F^4 V$, so that $W_F = \{a\Omega : a \in F\}$, and define a symmetric $F$-bilinear form $q$ on $R_F$ by $x \wedge_F y = q(x, y)\,\Omega$. Let $h$ be the hermitian form on $R_F$ induced by $H$, $h(x_1 \wedge x_2, y_1 \wedge y_2) = \det\big(H(x_i, y_j)\big)$, and $h_4$ the one induced on $\bigwedge_F^4 V$; in a basis $e_1, \dots, e_4$ with $\Omega = e_1 \wedge e_2 \wedge e_3 \wedge e_4$, $h_4(\Omega, \Omega)$ is $\det H$, so its class in $F_0^\times / \mathrm{N}_{F/F_0}(F^\times)$ is $\delta$. With dual bases $(\omega_k)$, $(\omega_k^*)$ of $F$ for the trace form, as in Theorem 19.5, put $x \star y = \sum_k (\omega_k x) \cup (\omega_k^* y)$ for $x, y \in R_F$. Then

$$x \star y = q(x, y)\,\Omega, \tag{69}$$

because the complexification of $(\omega_k x)\cup(\omega_k^* y)$ is $\sum_{\sigma,\sigma'}\sigma(\omega_k)\sigma'(\omega_k^*)\,x_\sigma\wedge y_{\sigma'}$, the sum over $k$ of the coefficient is 1 for $\sigma=\sigma'$ and 0 otherwise, as in the proof of Theorem 19.5(ii), and $\sum_\sigma x_\sigma\wedge y_\sigma$ is the complexification of $q(x,y)\Omega$. Finally let $\theta_j\in H^{1,1}(A)$ be the component of the polarisation that pairs $V_\sigma$ with $V_{\overline\sigma}$ for the two embeddings $\sigma,\overline\sigma$ over $\tau_j$. Then $\eta_t=\sum_j\tau_j(t)\,\theta_j$ for the classes $\eta_t$ of Proposition 19.3(iii), and $\mathbb{C}[\theta_1,\theta_2]$ is the complexification of the $\mathbb{Q}$-algebra generated by the $\eta_t$.

**Proposition 19.20** (Hodge classes, and a quaternion algebra acting on $R_F$)**.** *Let $A$ be a member of an $(F,2,\delta)$-family.*

(i) *If $\mathrm{Hg}(A)=G_F$, the space of Hodge classes of $A$ has dimension 2 in degree two, spanned by the $\eta_t$, and 7 in degree four, spanned by the three products of two $\eta_t$ and by $W_F(A)$, which meets the span of those products in 0.*

(ii) *The map $\phi\colon R_F\to R_F$ defined by $q(x,y)=h(x,\phi y)$ satisfies $\phi(ax)=\overline{a}\,\phi(x)$ for $a\in F$ and $\phi^2=h_4(\Omega,\Omega)^{-1}$, and it commutes with $\mathrm{SU}(V,H)$. So $\phi$ and the action of $F$ are Hodge endomorphisms of $R_F$ at every member, and the Hodge classes in $R_F$ form a module over the algebra $\mathcal{Q}_\delta=F\oplus F\phi$.*

(iii) *$\mathcal{Q}_\delta$ is the cyclic quaternion algebra $(F/F_0,\delta)$ over $F_0$, which is split if and only if $\delta$ is trivial. If $\mathrm{Hg}(A)=G_F$, then $\mathcal{Q}_\delta$ is the whole algebra of Hodge endomorphisms of $R_F$.*

(iv) *If $\delta$ is trivial, then $R_F=W\otimes_{F_0}F$ for a rational sub-Hodge structure $W$, stable under $F_0$, of dimension 12 and with Hodge numbers $(2,8,2)$.*

*Proof.* (i) As in the proof of Proposition 19.3(iii), over $\mathbb{C}$ the group $G_F$ is $\mathrm{SL}(U_1)\times\mathrm{SL}(U_2)$ with $\dim U_j=4$, and $H^1(A,\mathbb{C})=\bigoplus_j(U_j\oplus U_j^\vee)$. The invariants of $\mathrm{SL}(U)$ in $\bigwedge^a U\otimes\bigwedge^b U^\vee$ form a line if $a=b$ or $\{a,b\}=\{0,4\}$, and are zero otherwise. In degree two this leaves the bidegree $(1,1)$ in one factor: two lines, spanned by $\theta_1$ and $\theta_2$. In degree four it leaves $(2,2)$ in one factor or $(1,1)$ in both, three lines spanned by $\theta_1^2$, $\theta_1\theta_2$ and $\theta_2^2$, which are nonzero because $\dim U_j=4$; and $(4,0)$ or $(0,4)$ in one factor, four lines spanning $W_F\otimes\mathbb{C}$. These lie in different summands, so $W_F(A)$ meets the span of the products in 0.

(ii) Choose an $H$-orthogonal basis $e_1,\dots,e_4$ of $V$ and put $a_i=H(e_i,e_i)\in F_0$. Replacing $\Omega$ by $\lambda\Omega$ replaces $\phi$ by $\overline{\lambda}^{-1}\phi$ and $h_4(\Omega,\Omega)$ by $\mathrm{N}(\lambda)h_4(\Omega,\Omega)$, so we may take $\Omega=e_1\wedge e_2\wedge e_3\wedge e_4$, with $h_4(\Omega,\Omega)=a_1a_2a_3a_4$. For $J=\{j<j'\}$ put $e_J=e_j\wedge e_{j'}$ and $a_J=a_ja_{j'}$, and let $J^c$ be the complement, with $e_{J^c}\wedge e_J=\epsilon_J\Omega$. The $e_J$ form an $h$-orthogonal $F$-basis with $h(e_J,e_J)=a_J$, and $q(e_K,e_J)$ is $\epsilon_J$ for $K=J^c$ and 0 otherwise, so $\phi(e_J)=\epsilon_Ja_{J^c}^{-1}e_{J^c}$. Two-forms commute, so $\epsilon_{J^c}=\epsilon_J$ and $\phi^2(e_J)=(a_Ja_{J^c})^{-1}e_J=h_4(\Omega,\Omega)^{-1}e_J$. Semilinearity holds because $q$ is linear and $h$ antilinear in the second variable. An element of $\mathrm{SU}(V,H)$ preserves $h$ and fixes $\Omega$, hence preserves $q$ and commutes with $\phi$, and it commutes with $F$. The Hodge group of every member lies in $G_F$, because the circle defining the Hodge structure preserves $E$, commutes with $F$ and acts on each $V_\sigma$ with determinant $z^2z^{-2}=1$; so $\phi$ and $F$ commute with it, and they are Hodge endomorphisms.

(iii) The relations $\phi a=\overline{a}\phi$ and $\phi^2=h_4(\Omega,\Omega)^{-1}$ present $F\oplus F\phi$ as the cyclic algebra $(F/F_0,h_4(\Omega,\Omega)^{-1})$, which is $(F/F_0,\delta)$ because a square in $F_0$ is a norm; and a cyclic algebra $(F/F_0,c)$ is split exactly when $c$ is a norm from $F$. It has $\mathbb{Q}$-dimension 8, since $\phi$ is not $F$-linear. If $\mathrm{Hg}(A)=G_F$, the Hodge endomorphisms of $R_F$ are the commutant of $G_F$. Over $\mathbb{C}$, $R_F\otimes\mathbb{C}=\bigoplus_j(\bigwedge^2U_j\oplus\bigwedge^2U_j^\vee)$, where $\bigwedge^2U_j\cong\bigwedge^2U_j^\vee$ is irreducible for $\mathrm{SL}(U_j)$ and trivial for the other factor, so the commutant is $M_2(\mathbb{C})\times M_2(\mathbb{C})$, of dimension 8.

(iv) If $\delta$ is trivial, $h_4(\Omega,\Omega)=\mathrm{N}(b)$ for some $b\in F$, and $\psi=b\phi$ satisfies $\psi^2=b\overline{b}\,\phi^2=1$. Its fixed space $W$ is an $F_0$-subspace and a sub-Hodge structure, $\psi$ being a Hodge endomorphism,

and an element $\beta \in F$ with $\overline{\beta} = -\beta \neq 0$ maps it isomorphically onto the $(-1)$-eigenspace, since $\psi\beta = -\beta\psi$. So $R_F = W \oplus \beta W = W \otimes_{F_0} F$, and $W$ has half the dimension and half the Hodge numbers of $R_F$. ∎

**Proposition 19.21** (What divisor classes reach)**.** *Let $A$ be a member of an $(F, 2, \delta)$-family.*

(i) *The following are equivalent:* (a) *$W_F(A)$ lies in the subalgebra of $H^*(A, \mathbb{Q})$ generated by $\mathrm{NS}(A)_{\mathbb{Q}}$;* (b) *$W_F(A)$ meets that subalgebra in a nonzero class;* (c) *$R_F$ contains a nonzero rational Hodge class. When they hold, $R_F$ contains a rational Hodge class $t$ with $q(t,t) \neq 0$, and $W_F(A) = \{(ft) \star t : f \in F\}$, a space of polynomials with rational coefficients in divisor classes.*

(ii) *For a nonzero $\mathcal{Q}_\delta$-submodule $M \subseteq R_F$ let $\mathcal{D}(M) \subseteq \mathcal{D}_F$ be the set of points at which $M$ consists of Hodge classes, and let $\mathrm{NL}(R_F) \subseteq \mathcal{D}_F$ be the set of points at which* (c) *holds. Then $\mathrm{NL}(R_F) = \bigcup_M \mathcal{D}(M)$, a countable union. Each nonempty $\mathcal{D}(M)$ is a closed complex submanifold, isomorphic to a product of two bounded symmetric domains of type IV and dimension $4 - d$ each, where $d = \dim_F M$, and $d = 1$ occurs only if $\delta$ is trivial. So if $\delta$ is trivial, $\mathrm{NL}(R_F)$ is the union of the nonempty $\mathcal{D}(M)$ with $d = 1$, each of dimension exactly 6; if $\delta$ is not trivial, every $\mathcal{D}(M)$ has dimension at most 4.*

(iii) *$\mathrm{NL}(R_F)$ contains the base point $s_0$ of Theorem 19.5, and for every $\delta$ it contains the locus of products $B_1 \times B_2$ of two abelian fourfolds of $(F,1)$-Weil type, which has dimension 4. The split locus, of members isogenous to $X \otimes_{\mathcal{O}_{F_0}} \mathcal{O}_F$ with $X$ an abelian fourfold with $F_0 \subseteq \mathrm{End}^0(X)$, has dimension 6, lies in the family of trivial discriminant, and is contained in $\mathrm{NL}(R_F)$.*

*Proof.* (i) Since $W_F(A) \neq 0$, (a) implies (b). Let $F^\times$ act on $H^*(A, \mathbb{Q})$ by $\alpha \mapsto \iota(\alpha)^*$, which on $\bigotimes_\sigma \bigwedge^{a_\sigma} V_\sigma$ is the character $\prod_\sigma \sigma^{a_\sigma}$. The isotypic parts for the characters $\sigma^2$ in degree two and $\sigma^4$ in degree four are $R_F$ and $W_F$. The projections $p_R$ and $p_W$ onto them are rational, the two sets of characters being Galois orbits, and they are morphisms of Hodge structures, being $\mathbb{Q}$-linear combinations of the $\iota(\alpha)^*$. The characters occurring in $H^2$ are the products $\rho\rho'$, and the only way to write $\sigma^4$ as a product of two of them is $\sigma^2 \cdot \sigma^2$, so $p_W(D \cup D') = q(p_R D, p_R D')\,\Omega$ for $D, D' \in H^2(A, \mathbb{Q})$. If a nonzero $w \in W_F$ equals $\sum_i D_i \cup D_i'$ with $D_i, D_i' \in \mathrm{NS}(A)_{\mathbb{Q}}$, then $w = p_W(w) = \sum_i q(p_R D_i, p_R D_i')\,\Omega$, so some $p_R D_i$ is a nonzero rational Hodge class in $R_F$, and (b) implies (c). Conversely, let $t \in R_F$ be a nonzero rational Hodge class. The part of type $(1,1)$ of $\bigwedge^2 V_\sigma$ is $V_\sigma^{1,0} \wedge V_\sigma^{0,1}$, and $H$ is definite of opposite signs on the two factors by Proposition 19.3(i), so $h$ is negative definite on it at each real place, and $h(t,t) \in F_0$ is totally negative. By Proposition 19.20(ii), $q(t, \phi t) = h(t, \phi^2 t) = h_4(\Omega, \Omega)^{-1} h(t,t) \neq 0$, so one of the rational Hodge classes $t$, $\phi t$, $t + \phi t$, call it $t'$, has $q(t', t') \neq 0$. Each $\omega_k f t'$ and each $\omega_k^* t'$ is a rational Hodge class in $H^2$, hence a divisor class, and by (69) $(ft') \star t' = f\, q(t', t')\,\Omega$, which runs over $W_F$ as $f$ runs over $F$. So (c) implies (a), with the stated formula.

(ii) The Hodge classes in $R_F$ form a $\mathcal{Q}_\delta$-module, so $\mathrm{NL}(R_F)$ is the union of the $\mathcal{D}(M)$, and there are countably many $M$. Fix a real place $\tau$, let $\sigma$ be an embedding over it, and put $R_\tau = R_F \otimes_{F_0,\tau} \mathbb{R}$, a complex vector space of dimension 6 through $F \otimes_{F_0,\tau} \mathbb{R} = \mathbb{C}$, identified with $\bigwedge^2 V_\sigma$. Since $\det H$ has sign $(-1)^n = 1$ at $\tau$, the map $\phi$ induces an antilinear $\phi_\tau$ on $R_\tau$ with $\phi_\tau^2 = \lambda^2$, $\lambda > 0$. Its eigenspace $W_\tau = \{x : \phi_\tau x = \lambda x\}$ is a real form of $R_\tau$ and a real sub-Hodge structure with $W_\tau \otimes \mathbb{C} \cong \bigwedge^2 V_\sigma$, of Hodge numbers $(1,4,1)$, on which $q = \lambda h$ is real of signature $(2,4)$, positive on the plane $\Pi_\tau$ of real classes of type $(2,0) + (0,2)$ and negative on those of type $(1,1)$. The group $\mathrm{SU}(V_\tau, H_\tau)$ acts on $W_\tau$ preserving $q$, through the isogeny $\mathrm{SU}(2,2) \to \mathrm{SO}(2,4)^0$, and the map from $\mathcal{D}_{2,2}$ to the space of oriented positive

planes in $W_\tau$ that sends a point to $\Pi_\tau$ is the equivariant isomorphism of the symmetric domains of this group onto one component, $\mathcal{D}^{\mathrm{IV}}(W_\tau)$, of dimension 4. For a quadratic space $P$ of signature $(2,k)$ write likewise $\mathcal{D}^{\mathrm{IV}}(P)$, of dimension $k$. Let $M$ have $F$-dimension $d$. Then $M_\tau = M \otimes_{F_0,\tau} \mathbb{R}$ is stable under $\phi$ and $F$, so $M_\tau = P_\tau \otimes \mathbb{C}$ with $P_\tau = M_\tau \cap W_\tau$ of real dimension $d$, and $M$ consists of Hodge classes exactly when every $P_\tau$ is of type $(1,1)$, that is, orthogonal to $\Pi_\tau$. So $\mathcal{D}(M) = \prod_\tau \mathcal{D}^{\mathrm{IV}}(P_\tau^\perp)$, which is empty unless every $P_\tau$ is negative definite and then has dimension $2(4-d)$. If $d=1$, then $M = Fx$ with $\phi x = bx$ for some $b \in F$, and $\phi^2 x = \bar{b}b\,x$ gives $h_4(\Omega,\Omega)^{-1} = \mathrm{N}(b)$, so $\delta$ is trivial. If $\delta$ is trivial, $\mathcal{Q}_\delta \cong M_2(F_0)$, every nonzero module contains one with $d=1$, and $\mathcal{D}(M) \subseteq \mathcal{D}(M')$ for $M' \subseteq M$.

(iii) The classes $\delta_i(f)$ of Theorem 19.5(ii) have complexification in $\bigoplus_\sigma \bigwedge^2 V_\sigma$, so they lie in $R_F$, and they are Hodge classes at $s_0$. For a product, $V = V_1 \oplus V_2$ with $H$-orthogonal $F$-planes of signature $(1,1)$ at each place, and $\bigwedge^2_F V_1 \subseteq R_F$ is $W_F(B_1)$, of type $(1,1)$ by Proposition 19.3(i). Every $\delta$ is the product of the discriminants of two such planes, for instance of $\mathrm{diag}(1,-\delta)$ and $\mathrm{diag}(1,-1)$, and each factor varies in a family of dimension $m=2$. On the split locus $H^1(A,\mathbb{Q}) = U \otimes_{F_0} F$, with $U = H^1(X,\mathbb{Q})$ and the complex structure of $X$ on the first factor. The polarisation class $e$ of $X$ lies in $\bigwedge^2_{F_0} U$, the hermitian form of $A$ is $c$ times the sesquilinear extension of the $F_0$-bilinear form $e$, with $c \in F$ totally imaginary, and so $\det H = c^4 \,\mathrm{Pf}(e)^2$ is a square in $F_0$ and $\delta$ is trivial. The class $e \otimes 1 \in (\bigwedge^2_{F_0} U) \otimes_{F_0} F = R_F$ is fixed by $\mathrm{Hg}(A) = \mathrm{Hg}(X) \subseteq \mathrm{Res}_{F_0/\mathbb{Q}} \mathrm{Sp}(U,e)$, so it is a nonzero Hodge class. The polarised abelian fourfolds with real multiplication by $F_0$ form families of dimension $[F_0:\mathbb{Q}]\cdot 3 = 6$. ∎

**Corollary 19.22** (A base locus of codimension two)**.** *In every $(F,2,\delta)$-family, $\Sigma_F \supseteq \mathrm{NL}(R_F)$, and at every point of $\mathrm{NL}(R_F)$ the Weil classes are the polynomials in divisor classes of Proposition 19.21(i). If $\delta$ is trivial, $\mathrm{NL}(R_F)$ is a countable union of closed complex submanifolds of $\mathcal{D}_F$ of codimension exactly two, and it contains the split locus; if $\delta$ is not trivial, they have codimension at least four, and the product loci have codimension four. In both cases $\mathrm{NL}(R_F)$ contains the base point $s_0$ of Theorem 19.5 and is stable under $G_F(\mathbb{Q})$. At a point outside $\mathrm{NL}(R_F)$ no nonzero Weil class is a polynomial in divisor classes.*

*Proof.* Everything except the stability is Proposition 19.21. An element $g \in G_F(\mathbb{Q})$ commutes with $F$ and with $\phi$, so it carries $\mathcal{Q}_\delta$-submodules to $\mathcal{Q}_\delta$-submodules, and $g\,\mathcal{D}(M) = \mathcal{D}(gM)$. ∎

Theorem 19.5 gives one CM point, whose $G_F(\mathbb{Q})$-orbit is dense but countable. When $\delta$ is trivial, Corollary 19.22 replaces it by a locus of codimension two on which the Weil classes are balanced products of divisor classes, as they are at $s_0$; nothing is claimed about the rest of the Hodge ring at its points. It does not decide $\mathbf{W}(F,2,\delta)$: $\mathrm{NL}(R_F)$ is a countable union of proper closed submanifolds, hence meagre, and Theorem 19.7(iii) allows $\Sigma_F$ to be meagre. The last sentence of the corollary is the analogue of Proposition 15.18: no argument through divisor classes reaches a member outside $\mathrm{NL}(R_F)$. The codimension is two rather than $h^{2,0}(R_F) = 4$ (compare Remark 15.21) because Hodge classes in $R_F$ come in $\mathcal{Q}_\delta$-modules, and a module of $F$-dimension one costs one condition at each real place. Figure 36 shows where the Weil classes of such a family are known to be algebraic.

**Proposition 19.23** (Scalar extension from an imaginary quadratic subfield)**.** *Let $F = KF_0$ with $K$ imaginary quadratic; these are the biquadratic quartic CM fields, and the only ones containing an imaginary quadratic field. Let $C$ be an abelian fourfold of $(K,2,\delta)$-Weil type*

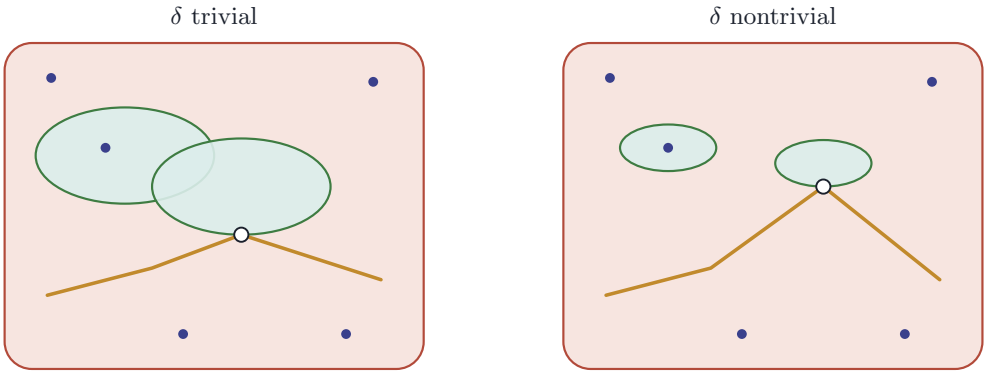



FIGURE 36. Where the Weil classes of an $(F, 2, \delta)$-family are known to be algebraic, $F$ a quartic CM field; each panel is the period domain $\mathcal{D}_F$, of dimension 8. The grass regions are components of the Noether–Lefschetz locus $\mathrm{NL}(R_F)$, on which $W_F$ consists of polynomials in divisor classes (Proposition 19.21, Corollary 19.22): of dimension exactly 6 when $\delta$ is trivial, where the quaternion algebra $\mathcal{Q}_\delta$ of Proposition 19.20 is split, and at most 4 otherwise. The ochre curve is the locus of scalar extensions of Weil fourfolds, of dimension 4 when $F$ is biquadratic, on which Markman's theorem applies (Proposition 19.23); it is not contained in $\mathrm{NL}(R_F)$ but meets it (ring) where the fourfold is a product of Weil surfaces. The indigo points are CM points; the base point $s_0$ of Theorem 19.5 lies on $\mathrm{NL}(R_F)$. For $F = \mathbb{Q}(\zeta_5), \mathbb{Q}(\zeta_8), \mathbb{Q}(\zeta_{12})$ and trivial $\delta$ the Prym locus of Remark 19.24, of dimension at most 6, is not drawn. On the rest of the domain, the very general member included, the algebraicity of the Weil classes is the open propagation statement.

*and $B_F = C \otimes_{\mathcal{O}_K} \mathcal{O}_F$, which is of $(F, 2, \iota(\delta))$-Weil type by Proposition 20.13(ii); write $(V_C, H_C)$ for the hermitian space of $C$. For $f \in F$ let $g_f \in \mathrm{Hom}(B_F, C) \otimes \mathbb{Q}$ be the element inducing $v \mapsto v \otimes f$ from $H^1(C, \mathbb{Q})$ to $H^1(B_F, \mathbb{Q}) = H^1(C, \mathbb{Q}) \otimes_K F$.*

(i) *$W_F(B_F)$ lies in the $\mathbb{Q}$-span of the classes $g_f^* w$, $f \in F$, $w \in W_K(C)$, which has dimension 10. Hence $W_F(B_F)$ is algebraic, by Markman's theorem $\mathbf{W}(K, 2, \delta)$ [Mar25a] (Proposition 20.12(ii)).*

(ii) *The members $B_F$ form a locus of dimension 4 in the $(F, 2, \iota(\delta))$-family. At those with $\mathrm{Hg}(C) = \mathrm{SU}(V_C, H_C)$ the space $R_F$ contains no nonzero Hodge class, so this locus is not contained in $\mathrm{NL}(R_F)$; it meets $\mathrm{NL}(R_F)$, for instance where $C = C_1 \times C_2$ with $C_1$, $C_2$ abelian surfaces of Weil type, and then $B_F$ is a product of two abelian fourfolds of $(F, 1)$-Weil type.*

*Proof.* (i) The map $v \mapsto v \otimes f$ commutes with the complex structures, that of $B_F$ being the one of $C$ tensored with $F$, so $g_f$ exists. Over $\mathbb{C}$, for each embedding $\kappa$ of $K$ let $V_\kappa$ be the $\kappa$-eigenspace of $K$ on $H^1(C, \mathbb{C})$ and $U_\kappa = F \otimes_{K,\kappa} \mathbb{C} \cong \mathbb{C}^2$, the sum of the lines $L_\sigma$, $\sigma|\kappa$, on which $F$ acts through $\sigma$. The $\kappa$-part of $H^1(B_F, \mathbb{C})$ is $V_\kappa \otimes U_\kappa$ and its $\sigma$-part is $V_\kappa \otimes L_\sigma$, so the $\sigma$-component $\bigwedge^4 V_\kappa \otimes L_\sigma^{\otimes 4}$ of $W_F(B_F) \otimes \mathbb{C}$ lies in $\bigwedge^4 V_\kappa \otimes \mathrm{Sym}^4 U_\kappa \subseteq \bigwedge^4 (V_\kappa \otimes U_\kappa)$. The map $g_f^*$ sends a generator $\alpha_\kappa$ of $\bigwedge^4 V_\kappa$ to $\alpha_\kappa \otimes f_\kappa^{\otimes 4}$, with $f_\kappa$ the image of $f$ in $U_\kappa$. The fourth powers of the elements of the Zariski dense subset $F$ of $U_\kappa$ span $\mathrm{Sym}^4 U_\kappa$, so the complexified span of the $g_f^* w$ is $\bigoplus_\kappa \bigwedge^4 V_\kappa \otimes \mathrm{Sym}^4 U_\kappa$, of dimension $2 \cdot 5$, and it contains $W_F(B_F) \otimes \mathbb{C}$. A rational subspace whose complexification lies in that of a rational subspace lies in it, and pull-backs of algebraic classes are algebraic.

(ii) The map $C \mapsto B_F$ embeds the four-dimensional domain of the $K$-family into $\mathcal{D}_F$. As rational Hodge structures $H^1(B_F, \mathbb{Q}) \cong H^1(C, \mathbb{Q})^2$, so $\mathrm{Hg}(B_F) = \mathrm{Hg}(C)$, and $R_F \otimes \mathbb{C} = \bigoplus_\sigma \bigwedge^2 V_\kappa \otimes L_\sigma^{\otimes 2}$, with $\bigwedge^2 V_\kappa$ irreducible and nontrivial for $\mathrm{SL}(V_\kappa)$; so when $\mathrm{Hg}(C) = \mathrm{SU}(V_C, H_C)$, $R_F$ has no invariants. For $C = C_1 \times C_2$ one has $B_F = (C_1 \otimes_{\mathcal{O}_K} \mathcal{O}_F) \times (C_2 \otimes_{\mathcal{O}_K} \mathcal{O}_F)$, and Proposition 19.21(iii) applies. ∎

Proposition 19.23 gives no new case of the Hodge conjecture: $B_F$ is isogenous to $C \times C$, the classes are Hodge classes there, and when $\mathrm{Hg}(C) = \mathrm{SU}(V_C, H_C)$ Proposition 20.24 already gives the conjecture for $C \times C$ from [Mar25a].

*Remark* 19.24 (Prym varieties of cyclic covers). For three cyclotomic fields published work gives a further locus, at whose very general points the Hodge group is the whole of $G_F$. Let $N \geq 3$, let $\widetilde{C} \to C$ be an étale cyclic cover of degree $N$ of a smooth projective curve of genus $g \geq 2$, and let $P \subseteq J(\widetilde{C})$ be the abelian subvariety on whose $H^1$ the covering group acts through characters of exact order $N$. Then $F = \mathbb{Q}(\zeta_N)$ acts on $P$, and every embedding of $F$ occurs in $H^{1,0}(P)$ with multiplicity $g - 1$ [Sch88, Lemma 1.6a], [PZ25, Lemma 5.1], so $P$ is of $(F, g-1)$-Weil type (Remark 19.2). The hermitian form is split, because the part of $H_1(\widetilde{C}, \mathbb{Z})$ on which the covering group acts through characters of exact order $N$ has a hyperbolic $\mathbb{Z}[\zeta_N]$-basis [Loo97, Proposition 2.2]; so $\delta = [(-1)^{g-1}]$, which is trivial for $g = 3$. Schoen proved that the Weil classes of $P$ are algebraic [Sch88, Theorem 2.0 with $r = 0$, Corollary 3.1], and Patel and Zhang give a proof by unramified geometric class field theory [PZ25, Theorems 1.1, 1.2 and 5.3, Remark 3.7]. By [Loo97, Theorem 2.4], for every $g \geq 2$, the monodromy group of the family of the $P$ over a finite cover of the moduli of such covers contains a subgroup of finite index of an arithmetic subgroup of $G_F$. Every factor $\mathrm{SU}(g-1, g-1)$ of $G_F(\mathbb{R})$ is noncompact, so by Borel's density theorem [Bor60] its Zariski closure contains $G_F$; the Hodge group of every member lies in $G_F$, so by André's theorem [And92, Theorem 1] a very general $P$ has $\mathrm{Hg}(P) = G_F$,

and by Proposition 19.3(iii) the Hodge conjecture holds for it in every degree. All of this is a consequence of [Sch88, Loo97, And92], and we make no claim of priority.

For $\varphi(N) = 4$, that is for $F = \mathbb{Q}(\zeta_5) = \mathbb{Q}(\zeta_{10})$, $\mathbb{Q}(\zeta_8)$ and $\mathbb{Q}(\zeta_{12})$, and $g = 3$, the $P$ are abelian eightfolds of $(F, 2, 1)$-Weil type. With their $G_F(\mathbb{Q})$-translates, which lie in $\Sigma_F$ by Theorem 19.7(ii), they fill a countable union of analytic subvarieties of $\mathcal{D}_F$, the *Prym locus* of $F$, of positive dimension, since the monodromy is infinite, and of dimension at most $3g - 3 = 6$ [Sch88, pp. 25–26]. At a very general point of it $\mathrm{Hg} = G_F$, so $R_F$ has no nonzero Hodge class there, each $\bigwedge^2 V_\sigma$ being a nontrivial irreducible representation of $\mathrm{SL}(V_\sigma)$, and the Hodge group is larger than that of a scalar extension of a Weil fourfold: such points lie outside $\mathrm{NL}(R_F)$, outside the locus of Proposition 19.23, and are not CM points. They are the only members with $\mathrm{Hg} = G_F$ at which we know the Weil classes of a quartic CM field to be algebraic. They do not decide $\mathbf{W}(F, 2, 1)$: the count $3g - 3 = 6 < 8 = \frac{1}{2}\varphi(N)(g-1)^2 = \dim \mathcal{D}_F$ of [Sch88, pp. 25–26] makes the union meagre, as $\mathrm{NL}(R_F)$ is, and Theorem 19.7(iii) allows $\Sigma_F$ to be meagre. Nor does Proposition 19.27 exclude them: the cycles come from $J(\widetilde{C})$, which has $P$ as a factor. For imaginary quadratic fields the count is $3n \geq n^2$ for $n \leq 3$, and there the Prym varieties do reach whole families: Schoen proved the algebraicity for the Weil fourfolds and for a family of Weil sixfolds of $\mathbb{Q}(\sqrt{-3})$ [Sch88, Theorem 3.2], [Sch98], and Koike for a family of Weil sixfolds of $\mathbb{Q}(\sqrt{-1})$ [Koi04, Theorem 2.1, Corollary 2.1].

**Proposition 19.25** (The Casimir class)**.** *Let $A$ be a member of an $(F, 2, \delta)$-family and $c \in F^\times$, and let*

$$C_q \in R_F \otimes_{\mathbb{Q}} R_F \subseteq H^2(A, \mathbb{Q}) \otimes H^2(A, \mathbb{Q}) \subseteq H^4(A \times A, \mathbb{Q})$$

*be the Casimir element $\sum_j u_j \otimes u_j^*$ of the nondegenerate symmetric form $\mathrm{Tr}_{F/\mathbb{Q}}(c^{-1}q)$ on $R_F$, for a basis $(u_j)$ and its dual basis $(u_j^*)$. Then $C_q$ is a Hodge class, $\Delta^* C_q = 6c\,\Omega$ for the diagonal $\Delta\colon A \to A \times A$, and $C_q$ lies in the $\mathbb{Q}$-span of the classes $f^*w$, $w \in W_F(A)$, $f(x, y) = \iota(a)x + \iota(b)y$ with $a, b \in \mathcal{O}_F$. Hence $W_F(A)$ is algebraic if and only if $C_q$ is.*

*Proof.* For an $F$-basis $(x_I)$ of $R_F$ and its $q$-dual basis $(x_I^q)$, the elements $\omega_k x_I$ and $c\,\omega_k^* x_I^q$ form dual bases for $\mathrm{Tr}_{F/\mathbb{Q}}(c^{-1}q)$. So by (69) $\Delta^* C_q = \sum_I x_I \star (c\,x_I^q) = \sum_I c\,\Omega = 6c\,\Omega$, and if $C_q$ is algebraic so is $W_F(A)$, by Proposition 19.3(ii). Conversely, $\mathrm{SU}(V, H)$ preserves $q$, so $C_q$ is invariant under the diagonal action of $G_F$, which contains $\mathrm{Hg}(A \times A)$, and $C_q$ is a Hodge class. Since $q$ pairs $\bigwedge^2 V_\sigma$ only with itself, the complexification of $C_q$ lies in $\bigoplus_\sigma \bigwedge^2 V_\sigma \otimes \bigwedge^2 V_\sigma$, inside the part $\bigoplus_\sigma \bigwedge^4(V_\sigma \otimes \mathbb{C}^2)$ of $H^4(A \times A, \mathbb{C})$ on which $F^\times$, acting diagonally, has the characters $\sigma^4$. The group $G_F$ acts on $V_\sigma$ through $\mathrm{SL}(V_\sigma)$, whose invariants in $\bigwedge^4(V_\sigma \otimes \mathbb{C}^2)$ are $\bigwedge^4 V_\sigma \otimes \mathrm{Sym}^4\mathbb{C}^2$ by Cauchy's formula, so the $G_F$-invariants in that part form the rational space $W_F(A) \otimes_F \mathrm{Sym}^4_F(F^2)$, of dimension 20. The map $f^*$ sends $w$ to $w \otimes (a, b)^{\otimes 4}$, and the fourth powers of the elements of $\mathcal{O}_F^2$ span $\mathrm{Sym}^4_F(F^2)$; so the classes $f^*w$ span that space, and $C_q$ is algebraic if $W_F(A)$ is. ∎

*Remark* 19.26 (What the exceptional isogeny gives). By the proof of Proposition 19.21(ii), at each real place $R_F$ becomes $W_\tau \otimes \mathbb{C}$ with $W_\tau$ of K3 type, $\mathrm{SU}(V_\tau, H_\tau)$ acting through $\mathrm{SU}(2, 2) \to \mathrm{SO}(2, 4)$, and when $\delta$ is trivial the $W_\tau$ are the real components of the rational structure $W$ of Proposition 19.20(iv). A route of Kuga–Satake type would start from this weight two structure and would have to return to degree four, on $A$ or on $A \times A$; by Proposition 19.25 the natural class it meets there, the Casimir class of $q$, is algebraic exactly when the Weil classes are. So this mechanism restates $\mathbf{W}(F, 2, \delta)$ at each member and does not reduce it.

**Proposition 19.27** (No correspondence from a small abelian variety)**.** *Let $A$ be a member with $\mathrm{Hg}(A) = G_F$, and let $X$ be an abelian variety such that $\mathrm{Hg}(X)$ has no normal subgroup isogenous to $G_F$; by the proof, this holds when every simple factor of $X$ has dimension at most 7, in particular when $\dim X \le 7$, and it holds when $X$ is of CM type. Then $\mathrm{Hg}(A \times X) = \mathrm{Hg}(A) \times \mathrm{Hg}(X)$, and $\mathrm{Hdg}^*(A^a \times X^b) = \mathrm{Hdg}^*(A^a) \otimes \mathrm{Hdg}^*(X^b)$ for all $a, b \ge 0$. So the class of an algebraic cycle $Z$ on $A^a \times X^b$ is $\sum_i \beta_i \otimes \xi_i$ with $\beta_i \in \mathrm{Hdg}^*(A^a)$ and linearly independent $\xi_i \in \mathrm{Hdg}^*(X^b)$; the correspondence $Z$ carries $H^*(X^b, \mathbb{Q})$ into the span of the $\beta_i$; and if the Hodge conjecture holds for $X^b$, every $\beta_i$ is algebraic.*

*Proof.* The group $G_F = \mathrm{Res}_{F_0/\mathbb{Q}} \mathrm{SU}(V, H)$ is almost simple over $\mathbb{Q}$. By Goursat's lemma the kernel $N$ of the projection $\mathrm{Hg}(A \times X) \to \mathrm{Hg}(X)$ maps onto a normal subgroup of $\mathrm{Hg}(A) = G_F$. If that subgroup is $G_F$, then $\mathrm{Hg}(A \times X)$ contains $G_F \times 1$ and is the product. Otherwise $N$ is finite, $\mathrm{Hg}(X)$ has a normal subgroup $H$ isogenous to $G_F$, contrary to the hypothesis, and the cocharacter $\mu$ defining the Hodge structure of $X$ projects through $H \to G_F$ to that of $A$. This situation also cannot occur when every simple factor of $X$ has dimension at most 7, or when $X$ is of CM type, as follows. Over $\mathbb{C}$, $H$ is $\mathrm{SL}_4 \times \mathrm{SL}_4$, and $\mu$ projects in each factor to $\mathrm{diag}(\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}, -\frac{1}{2})$ modulo the centre. Since $\mu$ acts on $H^1(X, \mathbb{C})$ with two weights differing by one, the weights of its projection spread by at most one on every irreducible constituent for $H$; for the representation of $\mathrm{SL}_4$ of highest weight $a\varpi_1 + b\varpi_2 + c\varpi_3$ the spread is $a + 2b + c$, and spreads add over the two factors. So every nontrivial constituent is the standard representation of one factor or its dual. These four are permuted transitively by $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$, as the embeddings of $F$ are; so a simple factor $Y$ of $X$ on whose $H^1$ the group $H$ acts nontrivially has $\dim_{\mathbb{Q}} H^1(Y, \mathbb{Q}) \ge 16$, that is $\dim Y \ge 8$. A torus has no such $H$. Next, $\mathrm{Hg}(A^a \times X^b)$ is $\mathrm{Hg}(A) \times \mathrm{Hg}(X)$ acting factorwise on $H^*(A^a) \otimes H^*(X^b)$, and the invariants of a product of groups in a tensor product of representations of the factors are the tensor product of the invariants. Finally, the intersection pairing is perfect on $\mathrm{Hdg}^*(X^b)$, the invariants being orthogonal to the other isotypic parts, so there are $\xi_j^* \in \mathrm{Hdg}^*(X^b)$ with $\int \xi_i \xi_j^* = \delta_{ij}$; they are algebraic when the conjecture holds for $X^b$, and then $\beta_j = pr_{A^a *}(Z \cdot pr_{X^b}^* \xi_j^*)$ is algebraic. ∎

So a correspondence with such an $X$ can produce a Weil class of $A$ only from an algebraic cycle on some $A^a \times X^b$ that already yields one on $A^a$. At a member with full Hodge group this excludes every abelian variety of dimension at most seven, among them the fourfolds and sixfolds of Weil type of the imaginary quadratic fields on which Markman's theorems are proved, and every CM abelian variety. The same argument, with real points in place of dimensions, excludes rational Hodge structures $Y$ of K3 type: by Zarhin's theorem [Zar83] the Hodge group of such a $Y$ is the restriction of scalars of an orthogonal or unitary group over the endomorphism field of its transcendental part, noncompact at exactly one real place, so its real Lie algebra has at most one simple factor isomorphic to $\mathfrak{su}(2,2)$, while that of $G_F$ has two; hence $\mathrm{Hg}(A \times Y) = \mathrm{Hg}(A) \times \mathrm{Hg}(Y)$. This last statement rests on Zarhin's theorem, which we quote without proof.

**Theorem 19.28** (The polarised criterion for a quartic field at $n = 2$)**.** *Let $A$ be a member of an $(F, 2, \delta)$-family, let $\omega = \sum_\sigma u_\sigma \alpha_\sigma$, with $\alpha_\sigma$ spanning $\bigwedge^4 V_\sigma$ and every $u_\sigma \ne 0$, as holds for every nonzero rational Weil class, let $p \in \mathbb{C}[\theta_1, \theta_2]$, put $\gamma = \omega + p$, and let $r(\gamma)$ be the rank of the contraction $HT^2(A) \to H^*(A, \mathbb{C})$, $\xi \mapsto \xi \lrcorner \gamma$. Write $T_\sigma = (V_\sigma^{1,0})^\vee \subseteq T_0 A$.*

(i) $\dim HT^2(A) = 28 + 64 + 28 = 120$, *and*

$$\mathrm{Ann}_{HT^2}(\omega) = \bigoplus_\sigma \mathrm{Hom}(V_\sigma^{1,0}, V_\sigma^{0,1}) \ \oplus \ \bigoplus_{\{\sigma,\sigma'\},\, \sigma \ne \sigma'} T_\sigma \otimes T_{\sigma'},$$

*of dimension* $0+16+24$*: the* $F$*-linear first order deformations of* $A$ *and the bivectors mixing two embeddings. So* $r(\omega)=80$*. Unlike the case* $m=1$ *of Theorem 16.31(i), no nonzero element of* $H^{0,2}(A)$ *kills* $\omega$*.*

(ii) $r(\gamma)\le 112$ *for every* $p$*. For a general* $p$*,* $r(\gamma)=112$*, and whenever* $r(\gamma)=112$ *the annihilator of* $\gamma$ *is the tangent space* $T_{[A]}\mathcal{D}_F$ *of the family, of dimension* 8*, inside* $H^1(T_A)$*. The value* 112 *is also taken at a general element, with nonzero component in* $W_F\otimes\mathbb{C}$*, of the complexified Hodge ring of a member with* $\mathrm{Hg}(A)=G_F$*.*

(iii) $r(\gamma)\ge 80$ *for every* $p$*, with equality exactly when* $p$ *is a constant.*

(iv) *The rank* $r(\gamma)$ *takes exactly the fifteen values* 80*,* 84*,* 87*,* 88*,* 91*,* 92*,* 94*,* 95*,* 96*,* 104*,* 107*,* 108*,* 110*,* 111*,* 112*; Theorem 19.29 computes it for every* $p$*. On the monomials* $\theta_1^i\theta_2^j$ *with a general coefficient it takes the six values* 80*,* 84*,* 88*,* 104*,* 108*,* 112*.*

(v) *Let* $E$ *be a perfect complex on* $A$ *with* $\mathrm{ch}(E)=N\omega_1+P$*, where* $N\neq 0$*,* $\omega_1$ *is a nonzero rational Weil class and* $P$ *is a polynomial with rational coefficients in the classes* $\eta_t$*. Then* $\dim\mathrm{Ext}^2(E,E)\ge r(\mathrm{ch}\,E)\ge 80$*. If* $\dim\mathrm{Ext}^2(E,E)=r(\mathrm{ch}\,E)$*, then* $\mathrm{ev}_E$ *is onto with kernel* $\mathrm{Ann}_{HT^2}(\mathrm{ch}\,E)$ *and* $\sigma_E$ *is injective, so that* $\mathbf{W}(F,2,\delta)$ *follows by Theorem 19.7(v).*

*Proof.* (i) $HT^2(A)=\bigwedge^2 H^{0,1}\oplus H^{0,1}\otimes T_0A\oplus\bigwedge^2 T_0A$, with $H^{0,1}$ and $T_0A$ of dimension 8. As in the proof of Theorem 16.31(i) the three summands map into different Hodge components, so the annihilator is the sum of the three annihilators; and the torus $(\mathbb{C}^\times)^4$ acting on each $V_\sigma$ by scalars separates the images further. A block $V_\rho^{0,1}\wedge V_{\rho'}^{0,1}$ of $\bigwedge^2 H^{0,1}$ kills $\alpha_\sigma$ exactly when $\sigma\in\{\rho,\rho'\}$, and otherwise maps injectively into a weight space that no other pair of a block and an embedding reaches; there are four embeddings, so some $\sigma$ lies outside $\{\rho,\rho'\}$, and no nonzero element of $\bigwedge^2 H^{0,1}$ kills $\omega$. A block $\mathrm{Hom}(V_\rho^{1,0},V_{\rho'}^{0,1})$ of $H^1(T_A)$ kills $\alpha_\sigma$ for $\sigma\neq\rho$; it kills $\alpha_\rho$ when $\rho'=\rho$, because $V_\rho^{0,1}$ is already exhausted in $\alpha_\rho$, and maps it injectively into its own weight space when $\rho'\neq\rho$. A bivector in $T_\rho\otimes T_{\rho'}$ with $\rho\neq\rho'$ kills every $\alpha_\sigma$, whose two factors of type $(1,0)$ both lie in $V_\sigma^{1,0}$, and $\bigwedge^2 T_\sigma$ maps $\alpha_\sigma$ to a nonzero multiple of a generator of $\bigwedge^2 V_\sigma^{0,1}$. The dimensions are $0$, $4\cdot 4$ and $6\cdot 4$.

(ii) The classes $\theta_j$ and the rational Weil classes stay of Hodge type along $\mathcal{D}_F$, so each is killed by $T_{[A]}\mathcal{D}_F\subseteq H^1(T_A)$, as in Proposition 17.11(i); hence $r(\gamma)\le 120-8$, and when equality holds the annihilator, of dimension 8, is $T_{[A]}\mathcal{D}_F$. Take a basis of $H^1(A,\mathbb{C})$ adapted to the $V_\sigma$ and to the Hodge decomposition, in which the polarisation pairs the basis of $V_\sigma^{1,0}$ with that of $V_{\overline{\sigma}}^{0,1}$. The torus acting diagonally on it contains a subtorus that multiplies each $\theta_j$ by a scalar and each $\alpha_\sigma$ by an arbitrary scalar, and contraction is equivariant; so $r(\omega+p)$ depends on $\omega$ only through a rescaling of the variables of $p$, and each statement of the theorem holds for every $\omega$ once it holds for one. The rank is lower semicontinuous in $p$, and it equals 112 at $p=\theta_1^2\theta_2^2$ by Theorem 19.29(iii) below, so it equals 112 for a general $p$. Every element of the complexified Hodge ring of a member with $\mathrm{Hg}(A)=G_F$ is invariant under $G_F$, so its components stay of Hodge type along $\mathcal{D}_F$ and it is killed by $T_{[A]}\mathcal{D}_F$; its rank is therefore at most 112. The ring contains $\omega+\theta_1^2\theta_2^2$, since it contains $W_F\otimes\mathbb{C}$ and the classes $\theta_1$, $\theta_2$, whose span is defined over $\mathbb{Q}$, and the rank is lower semicontinuous; so it is 112 at a general element of the ring, whose component in $W_F\otimes\mathbb{C}$ is not zero.

(iii) and (iv) are Theorem 19.29(iii) and (iv) below.

(v) By the commutative square $\sigma_E\circ\mathrm{ev}_E=(\,\cdot\,)\lrcorner\,\mathrm{ch}(E)$ of Setup 18.54, which holds on every abelian variety, $\dim\mathrm{Ext}^2(E,E)\ge\mathrm{rk}\,\mathrm{ev}_E\ge r(\mathrm{ch}\,E)$, and equality forces $\mathrm{ev}_E$ to be onto with kernel the annihilator and $\sigma_E$ to be injective, as in Theorem 17.9(iv); and

$r(\operatorname{ch} E) \geq 80$ by (iii), since the complexification of $\operatorname{ch}(E)$ is a $\gamma$ as above with $\omega = N\omega_1$. The last implication is Theorem 19.7(v), whose character is exactly of this form. ■

At $m = 1$ and $n = 2$ the corresponding numbers are 28, 12 and 24 (Theorem 17.9); in both cases the general value is the dimension of $HT^2$ minus that of the family. For the pure shape the argument of Theorem 16.31(ii) and (iv) transfers with the annihilator of Theorem 19.28(i): an injective $\sigma_E$ would force $A_\theta A_{\theta'} = 0$ for $\theta \in T_\rho$, $\theta' \in T_{\rho'}$, $\rho \neq \rho'$, and a smooth component of the support of codimension at least two along which $E$ is locally free of positive rank would be tangent to three of the four $T_\rho$. The pure shape is excluded, for every CM field, by Theorem 19.19.

The rank $r(\gamma)$ can be computed for every $p$. In the basis of the proof of Theorem 19.28(ii) write $x_{\sigma,1}, x_{\sigma,2}$ for the basis of $V_\sigma^{1,0}$, $y_{\sigma,1}, y_{\sigma,2}$ for that of $V_\sigma^{0,1}$, and $\partial_{\sigma,1}, \partial_{\sigma,2}$ for the dual basis of $T_\sigma$, so that $\theta_j = \sum_{\sigma|\tau_j} \sum_k x_{\sigma,k}\, y_{\overline{\sigma},k}$ and $\alpha_\sigma = x_{\sigma,1}x_{\sigma,2}y_{\sigma,1}y_{\sigma,2}$, and use the torus of that proof to make every $u_\sigma$ equal to 1. Expand $p = \sum_{i,j=0}^4 e_{ij}\, \theta_1^i \theta_2^j/(i!\, j!)$, so that $e_{ij}$ is a derivative of $p$ at 0, and put:

- $\mu(p) = 0$ if $\partial_1\partial_2 p = 0$ and $\mu(p) = 1$ otherwise, the derivatives taken in $\mathbb{C}[\theta_1, \theta_2]$;
- $\rho_1(p)$ the dimension of the span of $\partial_1 p$ and $\partial_1^2 p$ modulo $\theta_1^3$, which is the rank of the $2 \times 15$ matrix whose columns are the pairs $(e_{i,j}, e_{i+1,j})$, $1 \leq i \leq 3$, $0 \leq j \leq 4$; and $\rho_2(p)$ likewise, with the two indices exchanged;
- for $t = 1, 2$, with $\sigma, \overline{\sigma}$ the embeddings over $\tau_t$, $W_t \subseteq HT^2(A)$ the span of the eight elements $y_{\sigma,1} \wedge y_{\sigma,2}$, $y_{\overline{\sigma},1} \wedge y_{\overline{\sigma},2}$, $\partial_{\sigma,1} \wedge \partial_{\sigma,2}$, $\partial_{\overline{\sigma},1} \wedge \partial_{\overline{\sigma},2}$ and $y_{\sigma,k} \otimes \partial_{\overline{\sigma},l}$, $y_{\overline{\sigma},k} \otimes \partial_{\sigma,l}$ with $k \neq l$, and $R_t(p)$ the rank of $\xi \mapsto \xi \lrcorner \gamma$ on $W_t$;
- $h_1 = e_1e_3 - e_2^2$, $h_2 = e_2e_4 - e_3^2$, $h_3 = e_1e_4 - e_2e_3$ on $\mathbb{C}^4$, and $V_\pm \subset \mathbb{C}^4$ the set where $h_3 = \pm e_3$, $2h_2 = \pm e_4$ and $2h_1 + 1 = \pm e_2$.

**Theorem 19.29** (The rank of the criterion at a quartic field)**.** *In this notation:*

(i) $r(\gamma) = 64 + 16\mu(p) + 4\rho_1(p) + 4\rho_2(p) + R_1(p) + R_2(p)$.

(ii) $R_t(p) \in \{7, 8\}$. *If* $R_1(p) = 7$, *then* $(e_{10}, e_{20}, e_{30}, e_{40})$ *lies on* $V_+ \cup V_-$ *and* $\rho_1(p) = 2$; *conversely, if* $\mu(p) = 0$ *and* $(e_{10}, \dots, e_{40})$ *lies on* $V_+ \cup V_-$, *then* $R_1(p) = 7$, *as for* $p = \theta_1^2/2$. *The same holds for* $R_2$ *with* $(e_{01}, \dots, e_{04})$. *Each* $V_\pm$ *contains the surface of points* $(w\lambda \pm \frac{3}{2\lambda},\ w\lambda^2 \pm \frac{1}{2},\ w\lambda^3,\ w\lambda^4)$, $w \in \mathbb{C}$, $\lambda \in \mathbb{C}^\times$.

(iii) $r(\gamma) \geq 80$, *with equality exactly when* $p$ *is a constant, and* $r(\gamma)$ *takes exactly the fifteen values*

$$80,\ 84,\ 87,\ 88,\ 91,\ 92,\ 94,\ 95,\ 96,\ 104,\ 107,\ 108,\ 110,\ 111,\ 112,$$

*the values* 87, 91, 94, 95, 107, 110 *and* 111 *only when* $R_1$ *or* $R_2$ *is* 7. *The value* 100 *is not taken.*

(iv) *For* $\omega = \sum_\sigma u_\sigma \alpha_\sigma$ *with arbitrary nonzero* $u_\sigma$, *the same holds with* $e_{ij}$ *replaced by* $\kappa_1^i \kappa_2^j e_{ij}$, *where* $\kappa_t^4 = (u_\sigma u_{\overline{\sigma}})^{-1}$ *for the embeddings over* $\tau_t$. *The four choices of* $\kappa_t$ *fix* $V_+ \cup V_-$, *and* $\mu$, $\rho_1$, $\rho_2$ *do not depend on them.*

*Proof.* (i) Let $S$ be the torus of the basis that fixes $\theta_1$, $\theta_2$ and every $\alpha_\sigma$. If $x_{\sigma,k}$ has weight $a_{\sigma,k}$, then $y_{\sigma,k}$ has weight $-a_{\overline{\sigma},k}$ and $\partial_{\sigma,k}$ has weight $-a_{\sigma,k}$, and the only relations are $a_{\sigma,1} + a_{\sigma,2} = a_{\overline{\sigma},1} + a_{\overline{\sigma},2}$, one for each real place; so $S$ has dimension six. Attach to $y_{\sigma,k}$ the label $(\overline{\sigma}, k)$ and to $\partial_{\sigma,k}$ the label $(\sigma, k)$. A basis element of $HT^2$ is a product of two of these and has weight $-(a_P + a_Q)$ for its two labels $P, Q$, and two such weights agree exactly when the pairs of labels coincide or are $\{(\sigma,1),(\sigma,2)\}$ and $\{(\overline{\sigma},1),(\overline{\sigma},2)\}$. So $HT^2$ splits into 34 weight spaces: the eight lines spanned by $y_{\sigma,k} \otimes \partial_{\overline{\sigma},k}$; sixteen spaces of dimension four, one for each pair of labels over different real places; eight of dimension four, one for

each pair $\{(\sigma,k),(\overline{\sigma},l)\}$ over one place; and $W_1$, $W_2$. The class $\gamma$ is invariant under $S$, so contraction carries each weight space into the same weight of $H^*(A,\mathbb{C})$, and $r(\gamma)$ is the sum of the ranks on the 34 spaces.

Two rules compute these ranks. For a label $P=(\rho,k)$ with $\rho$ over $\tau_t$ put $y_P=y_{\overline{\rho},k}$, the element of $H^{0,1}$ that carries it. Contraction by $\partial_{\rho,k}$ is a derivation that maps $\theta_t$ to $\pm y_P$ and the other $\theta_{t'}$ to 0, so it maps $f(\theta_1,\theta_2)$ to $\pm y_P\,(\partial_t f)(\theta_1,\theta_2)$. Hence a basis element with labels $P,Q$ carries $p$ to $\pm y_P y_Q\,(Dp)(\theta_1,\theta_2)$, where $D$ is 1, $\partial_t$ or $\partial_t\partial_{t'}$ according as none, one or both labels are carried by elements of $T_0A$, and $t,t'$ are their places. Multiplication by $y_Py_Q$ removes from $\theta_1$ and $\theta_2$ the terms that contain $y_P$ or $y_Q$, and the products $\theta_1^a\theta_2^b$ of what remains are linearly independent as long as $a$ and $b$ do not exceed the numbers of remaining terms. Every monomial of $y_Py_Qf(\theta_1,\theta_2)$ contains, with each $x_{\sigma,i}$, its partner $y_{\overline{\sigma},i}$. The second rule is that the images of $\omega$ below contain some $x_{\sigma,i}$ without its partner, so they are independent of every image of $p$.

A line $y_{\sigma,k}\otimes\partial_{\overline{\sigma},k}$ carries $p$ to $\pm y_{\sigma,k}^2(\cdots)=0$ and $\alpha_{\overline{\sigma}}$ to $\pm y_{\sigma,k}x_{\overline{\sigma},k'}y_{\overline{\sigma},1}y_{\overline{\sigma},2}\neq 0$, $k'\neq k$: rank one. On a space of the second kind, with labels $P=(\rho,k)$ over $\tau_1$ and $Q=(\rho',l)$ over $\tau_2$, the four basis elements are $y_P\wedge y_Q$, $y_P\otimes\partial_{\rho',l}$, $y_Q\otimes\partial_{\rho,k}$ and $\partial_{\rho,k}\wedge\partial_{\rho',l}$. They carry $\omega$ to $y_Py_Q(\alpha_\rho+\alpha_{\rho'})$, $y_P\,(\partial_{\rho',l}\lrcorner\,\alpha_{\rho'})$, $y_Q\,(\partial_{\rho,k}\lrcorner\,\alpha_\rho)$ and 0, the factors $y_P$ and $y_Q$ killing $\alpha_{\overline{\rho}}$ and $\alpha_{\overline{\rho}'}$; these contain $x_{\rho,k'}$ or $x_{\rho',l'}$ with $k'\neq k$, $l'\neq l$, without its partner, and the first three are independent, having two, one and one such factors $x$ at different places. The bivector carries $\gamma$ to $\pm y_Py_Q(\partial_1\partial_2p)(\theta_1,\theta_2)$, with three terms left in each $\theta_t$, and $\partial_1\partial_2p$ has degree at most three in each variable; so this image is zero exactly when $\partial_1\partial_2p=0$: rank $3+\mu(p)$. On a space of the third kind, with labels $P=(\sigma,k)$ and $Q=(\overline{\sigma},l)$ over $\tau_1$, the element $y_P\wedge y_Q$ carries $\omega$ to $y_Py_Q(\alpha_{\sigma'}+\alpha_{\overline{\sigma}'})$, $\sigma'$ and $\overline{\sigma}'$ the embeddings over $\tau_2$, and the other three elements kill $\omega$: $y_P\otimes\partial_{\overline{\sigma},l}$ and $y_Q\otimes\partial_{\sigma,k}$ carry $\alpha_{\overline{\sigma}}$ and $\alpha_\sigma$ into multiples of $y_{\overline{\sigma},k}y_{\overline{\sigma},1}y_{\overline{\sigma},2}=0$ and $y_{\sigma,l}y_{\sigma,1}y_{\sigma,2}=0$, and the bivector meets no $\alpha$. They carry $p$ to $\pm y_Py_Q(\partial_1p)$, twice, and to $\pm y_Py_Q(\partial_1^2p)$, with two terms left in $\theta_1$ and four in $\theta_2$. So the rank is 1 plus the dimension of the span of $\partial_1p$ and $\partial_1^2p$ modulo $\theta_1^3$; in the basis $\theta_1^a\theta_2^j/(a!\,j!)$, $a\le 2$, their coefficient vectors are $(e_{a+1,j})$ and $(e_{a+2,j})$, and the rank is $1+\rho_1(p)$; over $\tau_2$ it is $1+\rho_2(p)$. The total is $8+16(3+\mu)+4(1+\rho_1)+4(1+\rho_2)+R_1+R_2$.

(ii) Over $\tau_1$ write $x_k,X_k,y_k,Y_k$ for $x_{\sigma,k},x_{\overline{\sigma},k},y_{\sigma,k},y_{\overline{\sigma},k}$, so that $\theta_1=\sum_k(x_kY_k+X_ky_k)$, $\alpha_\sigma=x_1x_2y_1y_2$ and $\alpha_{\overline{\sigma}}=X_1X_2Y_1Y_2$, and put $c=y_1y_2$, $C=Y_1Y_2$, $a_k=x_ky_1y_2Y_k$, $A_k=X_ky_kY_1Y_2$, $b=x_1x_2y_1y_2Y_1Y_2$ and $B=X_1X_2y_1y_2Y_1Y_2$. Let $\alpha'$ be the part of $\omega$ over $\tau_2$, so that $\alpha'\theta_2=0$, and put $\langle e_i\rangle=\sum_j e_{ij}\theta_2^j/j!$. Since $c$ kills the terms $X_ky_k$ of $\theta_1$, $c\,\theta_1^i/i!$ is $c$, $a_1+a_2$, $-b$ and 0 for $i=0$, 1, 2 and $i\ge 3$, so that $c\gamma=B+c(\alpha'+\langle e_0\rangle)+(a_1+a_2)\langle e_1\rangle-b\langle e_2\rangle$; in the same way, with $\partial$ acting as a left derivation, the other seven basis elements of $W_1$ carry $\gamma$ to

$$\begin{aligned}
C\gamma &= b+C(\alpha'+\langle e_0\rangle)+(A_1+A_2)\langle e_1\rangle-B\langle e_2\rangle,\\
\partial_{\sigma,1}\lrcorner\,\partial_{\sigma,2}\lrcorner\,\gamma &= -c+C\langle e_2\rangle+(A_1+A_2)\langle e_3\rangle-B\langle e_4\rangle,\\
\partial_{\overline{\sigma},1}\lrcorner\,\partial_{\overline{\sigma},2}\lrcorner\,\gamma &= -C+c\langle e_2\rangle+(a_1+a_2)\langle e_3\rangle-b\langle e_4\rangle,\\
\pm y_k\,\partial_{\overline{\sigma},k'}\lrcorner\,\gamma &= A_k+c\langle e_1\rangle+(a_1+a_2)\langle e_2\rangle-b\langle e_3\rangle,\\
\pm Y_k\,\partial_{\sigma,k'}\lrcorner\,\gamma &= a_k+C\langle e_1\rangle+(A_1+A_2)\langle e_2\rangle-B\langle e_3\rangle,
\end{aligned}$$

where $\{k,k'\}=\{1,2\}$ and the sign is + for $k=1$. The differences of the last two pairs are $A_1-A_2$ and $a_1-a_2$. In a basis containing $A_1-A_2$, $a_1-a_2$, $c\alpha'$ and $C\alpha'$, these four coordinates form the identity matrix on the two differences and on $c\gamma$, $C\gamma$, and vanish on the other four images. So $R_1$ is 4 plus the rank of the two images of the bivectors and of the halves of the sums of the last two pairs, $\frac12(A_1+A_2)+c\langle e_1\rangle+(a_1+a_2)\langle e_2\rangle-b\langle e_3\rangle$

and $\frac{1}{2}(a_1+a_2)+C\langle e_1\rangle+(A_1+A_2)\langle e_2\rangle-B\langle e_3\rangle$. In the coordinates along $c$, $a_1+a_2$, $b$, $C$, $-(A_1+A_2)$, $B$ times $\theta_2^j/j!$, and up to the signs of some columns, the matrix of these four vectors is $K=\left(K^0\,\middle|\,\operatorname{diag}(H,H)\right)$: the columns with $j=0$ form, with $e_k=e_{k0}$,

$$K^0=\begin{pmatrix} e_1 & e_2 & -e_3 & 0 & -\frac{1}{2} & 0\\ e_2 & e_3 & -e_4 & -1 & 0 & 0\\ 0 & \frac{1}{2} & 0 & e_1 & -e_2 & -e_3\\ -1 & 0 & 0 & e_2 & -e_3 & -e_4 \end{pmatrix},$$

and those with $j\ge 1$ form $\operatorname{diag}(H,H)$, with $H$ the $2\times 12$ matrix of the pairs $(e_{i,j},e_{i+1,j})$, $1\le i\le 3$, $1\le j\le 4$. So $R_1=4+\operatorname{rk}K$. Let $(a,b,c,d)$ annihilate the columns of $K^0$. The first two columns give $d=ae_1+be_2$ and $c=-2(ae_2+be_3)$, and the other four then say that $(a,b)$ is orthogonal to the rows of

$$Q=\begin{pmatrix} e_3 & e_4\\ -e_1e_2 & e_2^2-2e_1e_3-1\\ 2e_2^2-e_1e_3-\frac{1}{2} & e_2e_3\\ e_1e_4-2e_2e_3 & e_2e_4-2e_3^2 \end{pmatrix},$$

so $\operatorname{rk}K^0=2+\operatorname{rk}Q$. If $Q$ were zero, then $e_3=e_4=0$ by its first row, $e_2^2=\frac{1}{4}$ by its third, and the second row would have the entry $-\frac{3}{4}$. So $\operatorname{rk}K\ge 3$ and $R_1\ge 7$. If $R_1=7$, then $\operatorname{rk}Q=1$, and the $2\times 2$ minors of $Q$ vanish exactly on $V_+\cup V_-$. Indeed, with $P=2h_1+1$ and $m_{ij}$ the minor of rows $i$ and $j$, expanding gives

$$m_{12}=e_2h_3-e_3P,\quad 2m_{13}=e_4P-2e_2h_2,\quad m_{14}=2e_3h_2-e_4h_3,$$
$$2m_{23}=e_2^2-P^2,\quad m_{24}=Ph_3-e_2e_3,\quad 2m_{34}=e_3^2-h_2-4h_1h_2.$$

On $V_\pm$, $(P,h_3,2h_2)=\pm(e_2,e_3,e_4)$, which makes the first five vanish, and the last is $e_3^2-h_2-2(\pm e_2-1)h_2=e_3^2+h_2-e_2e_4=0$. Conversely, let all six vanish. If $e_2=e_3=e_4=0$ then $P=1$ and $m_{23}\ne 0$; so $(P,h_3,2h_2)=\lambda(e_2,e_3,e_4)$ for some $\lambda$, by the first three. If $e_2\ne 0$, then $m_{23}=0$ gives $\lambda=\pm1$. If $e_2=0$, then $P=0$, so $e_1e_3=-\frac{1}{2}$, and $h_3=e_1e_4=\lambda e_3$ and $2h_2=-2e_3^2=\lambda e_4$ give $e_4=-2\lambda e_3^2$ and then $\lambda^2=1$. In both cases the point lies on $V_\lambda$. If moreover $\rho_1(p)\le 1$, then $h_1=h_2=h_3=0$, and the minors of the second and third rows and of the first and third rows become $(e_2^2-1)/2$ and, once $e_2^2=1$, $e_4/2$; so $e_4=0$, $e_3^2=e_2e_4=0$ and $e_1e_3=0\ne e_2^2$, against $h_1=0$. Hence $\rho_1(p)=2$. If $\mu(p)=0$, then $e_{ij}=0$ for $i,j\ge 1$, so $H=0$ and $R_1=6+\operatorname{rk}Q$, which is 7 on $V_\pm$; the point $(0,1,0,0)$ of $p=\theta_1^2/2$ lies on $V_-$. The parametrised points satisfy $h_1=\pm\frac{1}{2}w\lambda^2-\frac{1}{4}$, $h_2=\pm\frac{1}{2}w\lambda^4$ and $h_3=\pm w\lambda^3$, which are the equations of $V_\pm$. The space $W_2$ is the same with the indices exchanged.

(iii) A nonzero $e_{ij}$ with $i,j\ge 1$ enters both Hankel matrices, so $\mu=1$ forces $\rho_1,\rho_2\ge 1$, and by (ii) $R_t=7$ forces $\rho_t=2$. The tuples $(\mu,\rho_1,\rho_2,R_1,R_2)$ allowed by these two conditions give exactly the fifteen values, and 80 only for $(0,0,0,8,8)$; $\rho_1=\rho_2=0$ means that $e_{ij}=0$ unless $i=j=0$, that is, $p$ is constant. Each value is taken. For $p=0$, $\theta_1$, $\theta_1^2$, $\theta_1^2+\theta_2$, $\theta_1^2+\theta_2^2$, $\theta_1\theta_2$, $\theta_1\theta_2^2$ and $\theta_1^2\theta_2^2$ the points $(e_{10},\dots,e_{40})$ and $(e_{01},\dots,e_{04})$ are 0, $(1,0,0,0)$ or $(0,2,0,0)$, which lie on neither $V_\pm$, as $2h_1+1$ is 1, 1 and $-7$ there; so $R_1=R_2=8$, and $(\mu,\rho_1,\rho_2)$ is $(0,0,0)$, $(0,1,0)$, $(0,2,0)$, $(0,2,1)$, $(0,2,2)$, $(1,1,1)$, $(1,1,2)$ and $(1,2,2)$, giving 80, 84, 88, 92, 96, 104, 108 and 112. For $p=\theta_1^2/2+q$ with $q$ one of 0, $\theta_2$, $\theta_2^2$, $\theta_2^2/2$ the point $(0,1,0,0)$ lies on $V_-$ and $\mu=0$, so $R_1=7$, and $R_2=7$ exactly for $q=\theta_2^2/2$; this gives 87, 91, 95 and 94. For $p=\theta_1^2/2+\theta_1\theta_2+q$ with $q$ one of 0, $\theta_2^2$, $\theta_2^2/2$, the only nonzero $e_{ij}$ with $i,j\ge 1$ is $e_{11}=1$, so the columns of $\operatorname{diag}(H,H)$ are multiples of $(1,0,0,0)$ and $(0,0,1,0)$; at $e=(0,1,0,0)$ the columns of $K^0$ span the space of $(1,0,0,0)$,

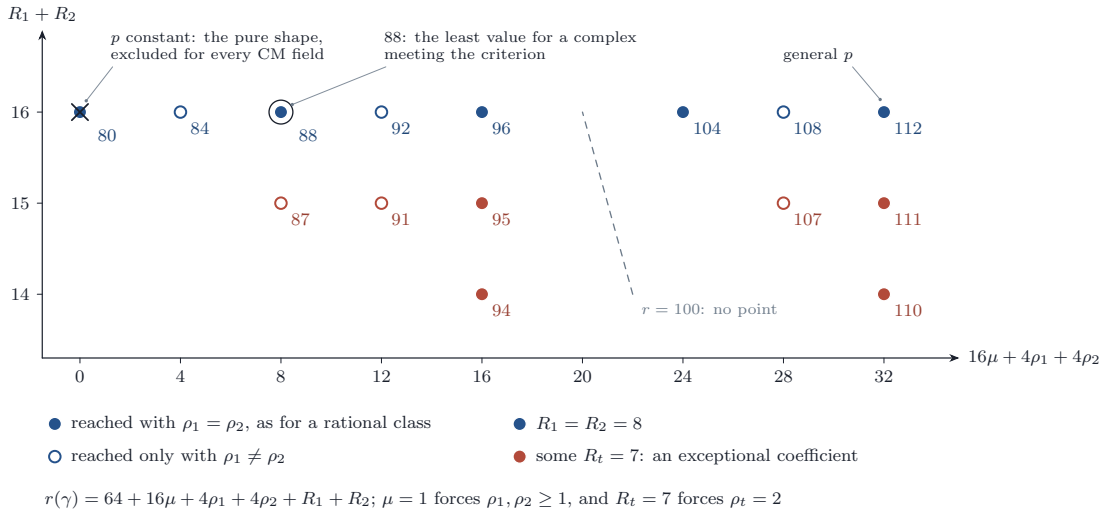



FIGURE 37. The rank $r(\gamma)$ of Theorem 19.29, over the plane of the Hankel part $16\mu + 4\rho_1 + 4\rho_2$ and of $R_1 + R_2$. The fifteen points are the tuples allowed by the two conditions of the theorem, and each carries its value. Filled points are reached with $\rho_1 = \rho_2$, as the character of a complex with rational coefficients is (Corollary 19.30(ii)); red points need an exceptional coefficient at one place at least. The value 80 is the pure shape, excluded for every CM field by Theorem 19.19; so 88 is the least rank a complex meeting the criterion can have. The level $r = 100$ meets no point.

$(0,0,1,0)$ and $(0,1,0,-1)$, so $\operatorname{rk} K = 3$ and $R_1 = 7$, and likewise $R_2 = 7$ for $q = \theta_2^2/2$. With $\mu = 1$ and $(\rho_1, \rho_2) = (2,1)$, $(2,2)$, $(2,2)$ this gives 107, 111 and 110.

(iv) The element of the torus of the proof of Theorem 19.28(ii) that multiplies $x_{\sigma,k}$ by $s_\sigma$ and $y_{\sigma,k}$ by $t_\sigma$, with $s_\sigma t_{\overline{\sigma}} = \kappa_t$ for $\sigma$ over $\tau_t$, multiplies $\theta_t$ by $\kappa_t$ and $\alpha_\sigma$ by $(s_\sigma t_\sigma)^2$, and these can be chosen to make every $u_\sigma (s_\sigma t_\sigma)^2$ equal to 1 exactly when $\kappa_t^4 = (u_\sigma u_{\overline{\sigma}})^{-1}$. It carries $\omega + p$ to the normalised class with coefficients $\kappa_1^i \kappa_2^j e_{ij}$, and contraction is equivariant. Replacing $\kappa_t$ by $-\kappa_t$ changes the signs of $e_1$ and $e_3$, which fixes $V_+$ and $V_-$, and replacing it by $\sqrt{-1}\,\kappa_t$ exchanges them; scaling the columns and the second row of the Hankel matrices does not change their ranks. ∎

The count of (i), drawn in Figure 37, is the analogue of the dimension counts of Theorem 17.9 for a field of degree four: sixty-four of the rank come from the Weil class alone, the Hankel terms measure how far $p$ is from an exponential in each variable, and the two eight-dimensional spaces $W_t$ carry the exceptional coefficients, the analogue of the exceptional ratio of Remark 17.12. Indeed $\rho_1(p) \leq 1$ exactly when $(e_{10}, \dots, e_{40})$ lies on the cone over the twisted cubic, that is, when the part of $p$ in $\theta_1$ alone is, up to a constant, a multiple of $e^{\lambda\theta_1} - 1$, of $\theta_1$ or of $\theta_1^4$.

**Corollary 19.30** (The least object the criterion asks for)**.** *Let $E$ be as in Theorem 19.28*(v).

(i) $\dim \operatorname{Ext}^2(E,E) \geq 81$.

(ii) *The coefficients of $P$ satisfy $e_{ji} = \iota(e_{ij})$ for the nontrivial automorphism $\iota$ of $F_0$, so $\rho_1 = \rho_2$, and $r(\operatorname{ch} E)$ is one of* 80, 88, 94, 95, 96, 104, 110, 111, 112.

(iii) *If* $\dim \mathrm{Ext}^2(E,E) = r(\mathrm{ch}\,E)$*, then* $P$ *is not constant and* $\dim \mathrm{Ext}^2(E,E)$ *is one of* 88, 94, 95, 96, 104, 110, 111, 112. *The value* 88 *requires* $\mu = 0$ *and* $\rho_1 = \rho_2 = 1$*, and it is the rank of* $N\omega_1 + \eta_s$ *for every* $s \in F_0^\times$.

*Proof.* (i) By Theorem 19.28(v), $\dim \mathrm{Ext}^2(E,E) \geq r(\mathrm{ch}\,E) \geq 80$. If $r(\mathrm{ch}\,E) = 80$, then $P$ is a constant by Theorem 19.29(iii), so $\mathrm{ch}(E) = c + N\omega_1$ and Theorem 19.19 makes the inequality strict. (ii) Write $\eta_t = \tau_1(t)\theta_1 + \tau_2(t)\theta_2$ as before Proposition 19.20. A polynomial with rational coefficients in the $\eta_t$ becomes, after exchanging $\theta_1$ and $\theta_2$, the polynomial whose coefficients are the images under $\iota$ of the old ones, so $e_{ji} = \iota(e_{ij})$; the Hankel matrix of $\rho_2$ is then the image under $\iota$ of that of $\rho_1$, which has the same rank. The values are those of Theorem 19.29(iii) with $\rho_1 = \rho_2$, the scaling of (iv) there changing neither $\mu$ nor the $\rho_t$. (iii) If $P$ were constant, Theorem 19.19 would give $\dim \mathrm{Ext}^2(E,E) > r(\mathrm{ch}\,E)$. For $P = \eta_s$ the pairs are $(\tau_t(s), 0)$, so $\rho_1 = \rho_2 = 1$, $\mu = 0$, and $R_t = 8$ by Theorem 19.29(ii). ∎

The Weil tori of Setup 19.16 exclude more than the pure shape. A polynomial in $\theta_1, \theta_2$ can stay of Hodge type on all of $S_F$, and then the argument of Theorem 19.19 applies to it.

**Proposition 19.31** (The characters that the Weil tori exclude)**.** *Let* $E$ *be as in Theorem 19.28*(v)*, and suppose that* $P$ *lies in the span of the* $\theta_1^i\theta_2^j$ *with* $i, j \in \{0, 4\}$*, as for* $\mathrm{ch}(E) = c + N\omega_1 + b\,\theta_1^4 + \iota(b)\,\theta_2^4$ *with* $b \in F_0$*. Then* $\sigma_E$ *is not injective and* $\dim \mathrm{Ext}^2(E,E) > r(\mathrm{ch}\,E)$*. Consequently, if* $\dim \mathrm{Ext}^2(E,E) = r(\mathrm{ch}\,E) = 88$*, then* $P = c + f(\theta_1) + f^\iota(\theta_2)$ *with* $c \in \mathbb{Q}$*, where* $f(\theta)$ *is a nonzero multiple either of* $\theta$ *or of* $e^{\lambda\theta} - 1$ *for some* $\lambda \neq 0$*, and* $f^\iota$ *is obtained by applying* $\iota$ *to the coefficients of* $f$.

*Proof.* In the basis before Theorem 19.29, $\theta_1$ is a sum of four products of two classes, one from $V_\sigma$ and one from $V_{\overline{\sigma}}$ for the two embeddings over $\tau_1$, in disjoint sets of basis vectors. So $\theta_1^4$ is 4! times their product, a nonzero element of $\bigwedge^4 V_\sigma \otimes \bigwedge^4 V_{\overline{\sigma}}$, that is a nonzero multiple of $\alpha_\sigma\alpha_{\overline{\sigma}}$, and likewise for $\theta_2^4$. Hence $\mathrm{ch}(E) = N\omega_1 + P$ lies in the span of the classes $\alpha_\Sigma$, which are of type $(p,p)$ on every member of $S_F$ (Setup 19.16); being rational, it is a Hodge class on every member. Its component of degree four is $N\omega_1 \neq 0$, since $P$ has none, and Theorem 19.19 applies. If $\dim \mathrm{Ext}^2(E,E) = r(\mathrm{ch}\,E) = 88$, then $\mu = 0$ and $\rho_1 = \rho_2 = 1$ by Corollary 19.30(iii), so $P$ has no mixed monomials, and by the paragraph after Theorem 19.29 its part in $\theta_1$ alone is a multiple of $e^{\lambda\theta_1} - 1$, of $\theta_1$ or of $\theta_1^4$; its part in $\theta_2$ alone is obtained by $\iota$, by Corollary 19.30(ii). In the third case $P$ lies in the span of 1, $\theta_1^4$ and $\theta_2^4$, which the first part excludes. ∎

The shape $c + b\,\theta_1^4 + \iota(b)\,\theta_2^4$ has rank 88, and with a multiple of $\theta_1^4\theta_2^4$ added it has rank 104, by Theorem 19.29: $\rho_1 = \rho_2 = 1$, $\mu$ is 0 or 1, and $R_1 = R_2 = 8$, since $(e_{10}, \dots, e_{40}) = (0, 0, 0, 24b)$ with $b \neq 0$ does not meet $V_+ \cup V_-$. Proposition 19.31 removes both. What remains at 88 is a Chern character whose polynomial part is exponential or linear in each $\theta_t$ separately, such as $c + \eta_s + N\omega_1$. For these the classes $\theta_t$ are not of Hodge type on the Weil tori, and the argument needs a family of complex tori along which the character stays of Hodge type and whose members are not algebraic. The next proposition, the analogue for a quartic field of Proposition 17.11(iii) and (iv), shows that there is none: the Hodge locus of such a character is the polarised family itself.

**Proposition 19.32** (The Hodge locus of a character at a quartic field)**.** *Keep the notation before Theorem 19.29, with* $\gamma = \omega + p$ *and every* $u_\sigma \neq 0$*, and for the embeddings* $\sigma, \overline{\sigma}$ *over* $\tau_t$ *put* $\omega_t = u_\sigma\alpha_\sigma + u_{\overline{\sigma}}\alpha_{\overline{\sigma}}$*. Call* $\tau_t$ *a* mid place *of* $p$ *if* $p$ *has a monomial* $\theta_1^i\theta_2^j$ *with nonzero coefficient whose exponent of* $\theta_t$ *is* 1, 2 *or* 3*, and call a mid place* exceptional *if the only such monomial is* $\theta_t^2$ *itself and its coefficient* $c_t$ *in* $p$ *satisfies* $c_t^2\theta_t^4 = \frac{3}{4}\omega_t^2$*, the ratio of Proposition 17.11(iv).*

(i) *The annihilator of $\gamma$ in $H^1(T_A)$ is the space of $F$-linear $\xi \in \bigoplus_\sigma \mathrm{Hom}(V_\sigma^{1,0}, V_\sigma^{0,1})$ with $\xi \lrcorner \theta_t = 0$ at every mid place, plus, at each exceptional place, a line spanned by an element of $\mathrm{Hom}(V_\sigma^{1,0}, V_{\overline\sigma}^{0,1}) \oplus \mathrm{Hom}(V_{\overline\sigma}^{1,0}, V_\sigma^{0,1})$ with both components nonzero. Its dimension is $8 + 4a + b$, where $a$ is the number of places that are not mid places and $b$ the number of exceptional places.*

(ii) *The connected component through $A$ of the Hodge locus of $\gamma$, in the space of complex structures on the real torus underlying $A$, is $S_F$ when neither place is a mid place, that is when $p$ lies in the span of the $\theta_1^i\theta_2^j$ with $i, j \in \{0, 4\}$; and it is $\mathcal{D}_F$ when both places are mid places and neither is exceptional. In the second case all its members are abelian varieties polarised by $\theta_1 + \theta_2$.*

(iii) *If both places are exceptional, then near $A$ the Hodge locus is the orbit of $A$ under $G_1 \times G_2$, where $G_t \cong \mathrm{Spin}(4, 3)$ is the connected stabiliser of $X_t = \omega_t + c_t\theta_t^2$ in the real linear group of $V_\sigma \oplus V_{\overline\sigma}$, and it has dimension 10. A very general member $T$ of it has $\mathrm{NS}(T) = 0$ and $\mathrm{Hdg}^2(T) \otimes \mathbb{C}$ contained in the span of $X_1$ and $X_2$, and on every member, for every Kähler class $\kappa$, $\int X_t\,\kappa^6$ is nonzero with the sign of $c_t$, in the normalisation in which $\theta_t$ is a real class with $\theta_t^2$ positive.*

(iv) *If $\gamma$ is the complexified character of Theorem 19.28(v), then both places are mid places or neither is, and both are exceptional or neither is; and an exceptional $\gamma$ has $r(\gamma) = 94$, or 110 when $p$ contains $\theta_1^4\theta_2^4$.*

*Proof.* (i) Grade $\bigwedge^* H^1$ by the degrees in the four $V_\sigma$, and write $\xi_{\sigma\sigma'}$ for the component of $\xi$ in $\mathrm{Hom}(V_{\sigma'}^{1,0}, V_\sigma^{0,1})$, which raises the degree in $V_\sigma$ by one and lowers that in $V_{\sigma'}$ by one. The monomial $\theta_1^i\theta_2^j$ has degree $i$ in the two embeddings over $\tau_1$ and $j$ in those over $\tau_2$, and $\alpha_\sigma$ has degree 4 in $V_\sigma$. By the proof of Theorem 19.28(i), $\xi_{\sigma\sigma}\lrcorner\omega = 0$, and for $\sigma \neq \sigma'$ the only term of $\xi_{\sigma\sigma'}\lrcorner\omega$ is $u_{\sigma'}\xi_{\sigma\sigma'}\lrcorner\alpha_{\sigma'}$, of degree 3 in $V_{\sigma'}$ and 1 in $V_\sigma$, which determines $\xi_{\sigma\sigma'}$. A term $\xi_{\rho\rho'}\lrcorner\theta_1^i\theta_2^j$ has that degree only if $\sigma, \sigma'$ lie over the same place $\tau_t$, $(i, j)$ is $(2, 0)$ or $(0, 2)$ with the 2 at $t$, and $(\rho, \rho') = (\sigma', \sigma)$. So every component $\xi_{\sigma\sigma'}$ with $\sigma, \sigma'$ over different places vanishes, and the equations split into the $F$-linear part and, at each place, the pair $\xi' = \xi_{\overline\sigma\sigma}$, $\xi'' = \xi_{\sigma\overline\sigma}$.

For $F$-linear $\xi$ put $D_t = \xi\lrcorner\theta_t \in V_\sigma^{0,1} \otimes V_{\overline\sigma}^{0,1}$; the map $\xi \mapsto (D_1, D_2)$ is onto, with kernel $T_{[A]}\mathcal{D}_F$. Then $\xi\lrcorner\theta_1^i\theta_2^j = i\,\theta_1^{i-1}\theta_2^j D_1 + j\,\theta_1^i\theta_2^{j-1}D_2$, the monomials lie in different degrees, and the two terms have different Hodge types at the two places, so each vanishes. Now $\theta_t^3 D_t = 0$ for every $D_t$, as a degree count shows, while multiplication by $\theta_t^k$ is injective on $V_\sigma^{0,1} \otimes V_{\overline\sigma}^{0,1}$ for $k \leq 2$: $\theta_t^2 y_{\sigma,k}y_{\overline\sigma,l}$ is 2 times the product of $y_{\sigma,k}y_{\overline\sigma,l}$ with the two terms of $\theta_t$ that avoid $y_{\sigma,k}$ and $y_{\overline\sigma,l}$, and different $(k, l)$ give different monomials. Multiplication by $\theta_{t'}^j$, $j \leq 4$, acts on the other factor of $\bigwedge^* H^1 = \bigwedge^*(V_\sigma \oplus V_{\overline\sigma}) \otimes \bigwedge^*(V_{\sigma'} \oplus V_{\overline{\sigma'}})$ and is nonzero there. So the conditions on the $F$-linear part are exactly $D_t = 0$ at the mid places, a space of dimension $16 - 4(2 - a) = 8 + 4a$.

At a place $\tau_t$ let $c_t$ be the coefficient of $\theta_t^2$ in $p$. The terms involving $\xi''$ are $u_{\overline\sigma}\xi''\lrcorner\alpha_{\overline\sigma} + c_t\,\xi'\lrcorner\theta_t^2$, of degree 3 in $V_{\overline\sigma}$ and 1 in $V_\sigma$, the same with $\sigma$ and $\overline\sigma$ and with $\xi'$ and $\xi''$ exchanged, and $i\,\theta_t^{i-1}\theta_{t'}^j(\xi'\lrcorner\theta_t)$ and its analogue for $\xi''$ for every other monomial with exponent $i \in \{1, 2, 3\}$ of $\theta_t$. Here $\xi'\lrcorner\theta_t$ lies in the line $\bigwedge^2 V_{\overline\sigma}^{0,1}$, on which multiplication by $\theta_t^{i-1}\theta_{t'}^j$ is injective for $i \leq 3$ by the same count. If such another monomial occurs, $\xi'\lrcorner\theta_t = \xi''\lrcorner\theta_t = 0$, and then $\xi' = \xi'' = 0$, the maps $\xi' \mapsto \xi'\lrcorner\alpha_\sigma$ and $\xi'' \mapsto \xi''\lrcorner\alpha_{\overline\sigma}$ being injective. Otherwise $\xi''\lrcorner\theta_t^2 = 2\theta_t\,(\xi''\lrcorner\theta_t)$ is a multiple of $\theta_t\,y_{\sigma,1}y_{\sigma,2}$, which is $\xi_0'\lrcorner\alpha_\sigma$ for one element $\xi_0'$, and likewise with $\sigma$ and $\overline\sigma$ exchanged; so $\xi'$ and $\xi''$ are multiples of $\xi_0'$ and $\xi_0''$. Explicitly, $\xi_0'(x_{\sigma,1}) = -y_{\overline\sigma,2}$ and $\xi_0'(x_{\sigma,2}) = y_{\overline\sigma,1}$ give $\xi_0'\lrcorner\alpha_\sigma = \theta_t\,y_{\sigma,1}y_{\sigma,2}$ and

$\xi_0' \lrcorner \theta_t = 2y_{\overline{\sigma},1}y_{\overline{\sigma},2}$, and symmetrically for $\xi_0''$. Writing $\xi' = a\xi_0'$ and $\xi'' = b\xi_0''$, the two remaining equations are $u_\sigma a + 4c_t b = 0$ and $4c_t a + u_{\overline{\sigma}} b = 0$, and their determinant vanishes exactly when $16c_t^2 = u_\sigma u_{\overline{\sigma}}$. Since $\theta_t^4 = 24\,\alpha_\sigma\alpha_{\overline{\sigma}}$ and $\omega_t^2 = 2u_\sigma u_{\overline{\sigma}}\alpha_\sigma\alpha_{\overline{\sigma}}$, that is the condition $c_t^2\theta_t^4 = \frac{3}{4}\omega_t^2$, and the solutions then form a line on which both components are nonzero.

(ii) By Proposition 17.11(i), applied to each homogeneous component, the Hodge locus of $\gamma$ in the chart at $A$ is the linear space $\mathrm{Ann}_{H^1(T_A)}(\gamma)$. If neither place is a mid place, that space is the space of all $F$-linear $\xi$, which is also the part of $S_F$ in the chart; $\gamma$ is of Hodge type on all of $S_F$, as in the proof of Proposition 19.31, and at every member of $S_F$ its component of degree four is $\omega$, whose annihilator in $H^1(T)$ is again the space of $F$-linear deformations (Theorem 19.28(i), whose proof uses only the decomposition into the $V_\sigma$). So $S_F$ is open in the Hodge locus; it is closed in the space of complex structures, since commuting with $F$ is a closed condition and the dimensions of the $H^{1,0} \cap V_\sigma$ are locally constant on it; and it is connected. If both places are mid places and neither is exceptional, the annihilator is $T_{[A]}\mathcal{D}_F$ by (i), at every member of $\mathcal{D}_F$, since the conditions of (i) are stated in terms of the flat classes $\theta_t$ and $\omega_t$; and the argument of Proposition 17.11(iii) shows that $\mathcal{D}_F$ is open, closed and connected in the Hodge locus.

(iii) At each place $X_t$ is, in the 8-dimensional space $V_\sigma \oplus V_{\overline{\sigma}}$ with the form $\theta_t$, the class of Remark 17.12, and the argument there applies word for word, with the factor $\mathfrak{su}(2,2)$ at $\tau_t$ of the Lie algebra of $G_F$ in place of the Hodge group: the stabiliser is $\mathfrak{so}(4,3)$, with an orbit of dimension 5 in the Grassmannian. The other components of $\gamma$ are constants, multiples of $\theta_t^4$, which is 24 times the volume form of $V_\sigma \oplus V_{\overline{\sigma}}$, and of $\theta_1^4\theta_2^4$, all fixed by $G_1 \times G_2$. So the orbit of $A$ lies in the Hodge locus, and it has dimension 10, which by (i) is that of the locus near $A$. A rational class of Hodge type at a very general member is of Hodge type on the whole orbit, hence killed by the complex structures of all its members, which lie in $\mathfrak{g}_1 \oplus \mathfrak{g}_2$ and generate an ideal projecting nontrivially to both simple factors, that is all of it. The invariants of $\mathfrak{so}(7,\mathbb{C})$ in the exterior algebra of its spin representation are 1, $X_t$ and the volume form (Remark 17.12), so in degree two there are none and in degree four only $X_1$ and $X_2$. For the sign, a Kähler class is $\kappa_h$ for a positive hermitian form $h$ on the tangent space; only the terms of $\kappa_h^6$ that contain the whole volume of the other place pair with $X_t$, and they give $\det(h_{t'})$ times $\kappa_{h'}^2$ for the Schur complement $h'$ of $h_{t'}$ in $h$, a positive hermitian form on the tangent space at $\tau_t$. So $\int X_t\kappa_h^6$ is a positive multiple of the integral of Remark 17.12 for $h'$, which has the sign of $c_t$; and $G_1 \times G_2$ fixes $X_t$ and carries Kähler cones to Kähler cones.

(iv) An automorphism of $\overline{\mathbb{Q}}$ inducing $\iota$ on $F_0$ fixes the rational class $\gamma$ and exchanges $\theta_1$ with $\theta_2$ and $\omega_1$ with $\omega_2$, so it carries the conditions at $\tau_1$ to those at $\tau_2$. An exceptional $p$ lies in the span of 1, $\theta_t^2$, $\theta_t^4$ and $\theta_1^4\theta_2^4$ with $c_t \neq 0$, so $\rho_1 = \rho_2 = 2$, $\mu = 1$ exactly when $\theta_1^4\theta_2^4$ occurs, and $(e_{10},\dots,e_{40})$ lies on $V_+ \cup V_-$, so $R_1 = R_2 = 7$ by Theorem 19.29(ii); the formula of Theorem 19.29(i) gives 94 or 110. ∎

**Corollary 19.33** (The Weil tori at rank 88)**.** *Let $E$ be as in Theorem 19.28*(v) *with* $\dim \mathrm{Ext}^2(E,E) = r(\mathrm{ch}\,E) = 88$*, so that* $\mathrm{ch}(E) = N\omega_1 + c + f(\theta_1) + f^\iota(\theta_2)$ *as in Proposition 19.31. Then the connected component through $A$ of the Hodge locus of* $\mathrm{ch}(E)$ *is $\mathcal{D}_F$, and every complex torus reached from $A$ by a deformation along which* $\mathrm{ch}(E)$ *stays of Hodge type is an abelian variety of the family. So no argument that deforms $E$ itself to a torus that is not algebraic can exclude it; the characters that such an argument excludes at* 88 *are exactly those of Proposition 19.31.*

*Proof.* The coefficient $e_{10}$ of $f$ is nonzero, both for a multiple of $\theta$ and for one of $e^{\lambda\theta}-1$, and $e_{01}=\iota(e_{10})$, so both places are mid places, with the monomials $\theta_1$ and $\theta_2$, and neither is exceptional. Proposition 19.32(ii) applies. ∎

As after Proposition 17.11, this concerns $E$ itself: an autoequivalence of $D^b(A)$ changes the character, and the proposition then applies to the new one. For the exceptional characters of Proposition 19.32(iii), of rank 94 or 110, the proof of Theorem 19.19 can be started, since the locus contains tori without divisors, but it breaks down as in Remark 17.12: the classes $X_t$ have integrals of constant sign against every Kähler class, so Bando and Siu give no vanishing: they bound the signs of the coefficients of $\mathrm{ch}_2$ of a torsion-free sheaf along $X_1$ and $X_2$, which a complex, an alternating sum, need not respect. So at a quartic field the Weil tori and their exceptional analogues exclude exactly the characters of Proposition 19.31, and what the criterion still asks for, at 88 and above, is the construction of an object.

The natural candidates are Markman's secant objects [Mar25c], in the twisted form in which he states his reduction. For them a $B$-field twist of $\mathrm{ch}(E)$, the class $\kappa$ of Theorem 19.37, is flat, and under a non-degeneracy condition its component in $H^4$ is a polynomial in the $\eta_t$ plus a nonzero Weil class [Mar25c, Proposition 10.2.1, Theorem 1.1.2]. But $\mathrm{ch}(E)$ itself is not flat, and $\kappa$ is not of the form $N\omega+P(\eta)$ in higher degrees (Remark 19.38), so they are candidates only for a twisted form of Theorem 19.28(v). The dimension count that works for imaginary quadratic fields (Theorem 17.19) can be tested on them exactly. Let $X$ be a principally polarised abelian fourfold with real multiplication by the ring of integers $\mathcal{O}_{F_0}$, write $\theta_1,\theta_2\in H^2(X,\mathbb{R})$ for the components of the polarisation at the two real places of $F_0$, so that $\theta=\theta_1+\theta_2$ and $\theta_j^3=0$, and let $\mathcal{R}$ be the $\mathbb{Q}$-subalgebra of $H^{\mathrm{ev}}(X,\mathbb{Q})$ spanned by the rational combinations of the $\theta_1^a\theta_2^b$, $0\le a,b\le 2$. A class $v=\sum_{a,b}V_{ab}\,\theta_1^a\theta_2^b/(a!\,b!)$ of $\mathcal{R}$ has a $3\times 3$ matrix $V$ with entries in $F_0\otimes\mathbb{R}$.

**Proposition 19.34** (The secant space of a quartic field)**.** *Let $F=F_0(\sqrt{-q})$ with $q\in F_0$ totally positive, let $t\in F_0$, $s_j=\sqrt{-1}\,\sqrt{\tau_j(q)}$, and for $\varepsilon\in\{\pm1\}^2$ put $e_\varepsilon=\exp\bigl(\sum_j(\tau_j(t)+\varepsilon_j s_j)\theta_j\bigr)$.*

(i) *The span $S(t,q)$ of the four $e_\varepsilon$ is a rational subspace of $\mathcal{R}$ of dimension $4=2^{[F_0:\mathbb{Q}]}$, secant to the spinor variety. For $t=0$ it is the secant space of* [Mar25c, §11.1, (11.1.2)] *for the polarisation $\theta$ and $q$; replacing $\theta$ by $f\theta$, $f\in F_0^\times$, replaces $q$ by $qf^2$, and a rational $F_0$-$B$-field $t$ transports Markman's $F$-structure by the rational automorphism of $X\times\widehat{X}$ acting on $H^*(X)$ as $e^{\theta_t}$, $\theta_t=\sum_j\tau_j(t)\theta_j$, and replaces $S(0,q)$ by $S(t,q)=e^{\theta_t}S(0,q)$.*

(ii) *For $v=\sum_\varepsilon w_\varepsilon e_\varepsilon\in S(t,q)$ let $\rho=\operatorname{rk}V$ and $N_w=\#\{\varepsilon : w_\varepsilon\neq 0\}$. The ranks of contraction from $HT^k(X)$ into $v$ are $1$, $8$ and $2\rho+4N_w$ for $k=0,1,2$, with $(\rho,N_w)\in\{(1,4),(2,4),(2,2)\}$, and $N_w=2$ only for biquadratic $F$ and $v$ in the secant plane of one of its two imaginary quadratic subfields* [Mar25c, §11.2.1].

(iii) *For $v=e^{\theta_t}(a\,u+v_f+c\,w)$ in the basis $u=(1-\frac12 q_1\theta_1^2)(1-\frac12 q_2\theta_2^2)$, $v_f=\theta_f-\frac12\theta_{\bar f q}\theta_1\theta_2$, $w=\theta_1\theta_2$ of $e^{-\theta_t}S(t,q)$, one has $\chi(v,v)=\int_X v^\vee v=4\bigl(\mathrm{N}(q)a^2+c^2+\mathrm{Tr}(\bar q f^2)\bigr)$, which is positive definite.*

*Proof.* (i) The Galois group permutes the four exponentials as it permutes the CM types of $F$, so their span is rational; since $\theta_j^3=0$, $e^{\pm s_j\theta_j}=(1-\frac12\tau_j(q)\theta_j^2)\pm s_j\theta_j$, and the four classes are independent. The conjugates in $\mathfrak{so}\cap\mathrm{End}^0(X\times\widehat{X})$ of the standard structure have eigenvalues $t\pm s$, and the pure spinors of their eigenspaces are the $e_\varepsilon$.

(ii) Over $\mathbb{C}$, $H^1(X) = U_1 \oplus U_2$ with $U_j$ the eigenspaces of $F_0$, and $HT^*(X)$ and $H^*(X)$ are tensor products over $j$, compatibly with contraction. By Lemma 18.55(i), $\xi \lrcorner e^{\lambda\theta_j} = y^\lambda(\xi) e^{\lambda\theta_j}$, and the blocks $HT_1^a \otimes HT_2^b$ have independent images because they shift the two Hodge bidegrees by $(a, b)$ while $v$ is balanced. The block $(1, 1)$ contributes $4N_w$, by the invertibility of the Hadamard matrix of the signs; the blocks $(2, 0)$ and $(0, 2)$ contribute $\rho$ each; the blocks $(1, 0)$ and $(0, 1)$ contribute 4 each in degree one. Reality gives $w_{-\varepsilon} = \overline{w_\varepsilon}$, and when $F$ is not biquadratic the Galois group acts transitively on the CM types, so $N_w = 4$.

(iii) One has $\int_X e^{c_1\theta_1 + c_2\theta_2} = c_1^2 c_2^2$, and $\chi$ is invariant under $e^{\theta_t}$; the form is positive definite because $q$ is totally positive. ∎

The integral classes of $\mathcal{R}$ are described by one congruence, and the description does not depend on the field.

**Lemma 19.35** (The integral classes of $\mathcal{R}$)**.** *Let $F_0$ be a real quadratic field with ring of integers $\mathcal{O}$ and different $\mathfrak{d}$, and let $X$ be a principally polarised abelian fourfold with real multiplication by $\mathcal{O}$. A class $v \in \mathcal{R}$ with matrix*

$$V = \begin{pmatrix} r & \bar{h} & \bar{g} \\ h & c & \bar{k} \\ g & k & m \end{pmatrix}, \qquad r, c, m \in \mathbb{Q}, \quad h, g, k \in F_0,$$

*lies in $H^{\mathrm{ev}}(X, \mathbb{Z})$ if and only if $r, c, m \in \mathbb{Z}$, $h, g, k \in \mathcal{O}$ and $c \equiv g$ modulo $\mathfrak{d}$.*

*Proof.* Integrality can be tested one prime $p$ at a time, and after a faithfully flat extension of $\mathbb{Z}_p$. Let $M = H_1(X, \mathbb{Z})$ and let $\langle\,,\rangle$ be the polarisation form on $M$, that is, $\theta$. The Rosati involution fixes $\mathcal{O}$, so $\langle\,,\rangle = \mathrm{Tr}_{F_0/\mathbb{Q}}\,\psi$ for a unique $\mathcal{O}$-bilinear alternating form $\psi$ with values in $\mathfrak{d}^{-1}$, and $\psi$ is perfect because $\theta$ is principal. Write $\mathcal{O} = \mathbb{Z}[\nu]$ and $\delta = \nu - \bar{\nu}$, a generator of $\mathfrak{d}$ with $\bar{\delta} = -\delta$, and put $\mathcal{O}_p = \mathcal{O} \otimes \mathbb{Z}_p$, a product of at most two discrete valuation rings. The form $\delta\psi$ on $M_p = M \otimes \mathbb{Z}_p$ is perfect with values in $\mathcal{O}_p$, so $M_p$ has a symplectic $\mathcal{O}_p$-basis $e, f, e^*, f^*$. Hence $M_p = \mathcal{O}_p \otimes_{\mathbb{Z}_p} W$, with $W$ the $\mathbb{Z}_p$-span of this basis, and $\langle\,,\rangle = B \otimes \omega_0$, where $\omega_0$ is the standard symplectic form of $W$ and $B(\alpha, \beta) = \mathrm{Tr}(\delta^{-1}\alpha\beta)$.

Let $\Lambda$ be the ring of integers of a finite extension of $\mathbb{Q}_p$ over which $F_0$ splits, and fix an embedding of $F_0$ in its field of fractions. The map $\alpha \otimes \lambda \mapsto (\alpha\lambda, \bar{\alpha}\lambda)$ identifies $\mathcal{O}_p \otimes_{\mathbb{Z}_p} \Lambda$ with the lattice of pairs $(x, y) \in \Lambda^2$ with $x \equiv y$ modulo $\delta$, whose basis is $(1, 1), (\delta, 0)$. Let $\varepsilon_1, \varepsilon_2$ be the coordinates and $u_1, \dots, u_4$ the basis of $W^\vee$ dual to $e, f, e^*, f^*$. The dual lattice has basis $\varepsilon_2$, $(\varepsilon_1 - \varepsilon_2)/\delta$, so $H^1(X, \Lambda)$ has the basis

$$Y_i = \varepsilon_2 \otimes u_i, \qquad Z_i = \delta^{-1}(X_i - Y_i), \qquad \text{where } X_i = \varepsilon_1 \otimes u_i,$$

and $H^*(X, \Lambda)$ is the exterior algebra on the $Y_i$ and $Z_i$. In these coordinates $B = \delta^{-1}(\varepsilon_1^2 - \varepsilon_2^2)$, and the components of $\theta = B \otimes \omega_0$ in the two eigenspaces of $\mathcal{O}$, the first being the one on which $\mathcal{O}$ acts through the fixed embedding, are

$$\theta_1 = \delta^{-1}(X_1X_3 + X_2X_4), \qquad \theta_2 = -\delta^{-1}(Y_1Y_3 + Y_2Y_4).$$

Substitute $X_i = Y_i + \delta Z_i$ in $v = \sum_{a,b} V_{ab}\,\theta_1^a\theta_2^b/(a!\,b!)$ and expand in the monomials in the $Y_i$ and $Z_i$. Up to sign, every coefficient is one of

$$r,\ m,\ h,\ k,\ c,\ g,\ g - c,\ \delta h,\ \delta k,\ \delta g,\ \delta^2 g,\ \frac{g - c}{\delta},\ \frac{h - \bar{h}}{\delta},\ \frac{k - \bar{k}}{\delta},\ \frac{2c - g - \bar{g}}{\delta^2},$$

and each of $r$, $m$, $h$, $k$, $c$, $g$ and $(g - c)/\delta$ occurs, so an integral class satisfies the conditions at $p$. Conversely, suppose that they hold. Every $x \in \mathcal{O}_p$ has $x - \bar{x} \in \delta\mathcal{O}_p$, which disposes

of $h-\bar h$ and $k-\bar k$; and $g=c+\delta g'$ with $g'\in\mathcal{O}_p$ gives $2c-g-\bar g=-\delta(g'-\bar g')\in\delta^2\Lambda$. So every coefficient lies in $\Lambda$. When $p$ is unramified in $F_0$, $\delta$ is a unit at $p$ and the congruence is empty. ∎

**Proposition 19.36** (Integral secant classes of small Euler characteristic)**.** *Let $F_0=\mathbb{Q}(\sqrt{D})$ with $D>1$ squarefree, let $X$ be a principally polarised abelian fourfold with real multiplication by $\mathcal{O}_{F_0}$, and let $v$ be an integral class in a space $S(t,q)$ of Proposition 19.34, with $\rho$ and $N_w$ as there.*

(i) *If $\rho=1$, then $\chi(v,v)\neq 4$.*

(ii) *If $\rho=2$ and $N_w=4$, then $\chi(v,v)\neq 6$.*

*Proof.* By Lemma 19.35 the entries of $V$ lie in $\mathcal{O}=\mathcal{O}_{F_0}$, the diagonal ones in $\mathbb{Z}$, and $c\equiv g$ modulo $\mathfrak{d}$; conjugation is trivial on $\mathcal{O}/\mathfrak{d}$.

(i) Write $V=\lambda\alpha\bar\alpha^T$ with $\alpha\in\mathcal{O}^3$, and let $\mathfrak{a}$ be the ideal generated by the entries of $\alpha$; replacing $\alpha$ by a multiple, we may take $\mathfrak{a}$ integral and prime to $\mathfrak{d}$. The entries $\lambda\alpha_i\bar\alpha_j$ lie in $\mathcal{O}$ and generate $\lambda\mathfrak{a}\bar{\mathfrak{a}}=\lambda\mathrm{N}(\mathfrak{a})\mathcal{O}$, so $\mu=\lambda\mathrm{N}(\mathfrak{a})\in\mathbb{Z}$. With $D'=\alpha_1^2-\alpha_0\alpha_2\in\mathfrak{a}^2$ one has $\chi=4\lambda^2\mathrm{N}(D')$, and $\mathrm{N}(D')\in\mathrm{N}(\mathfrak{a})^2\mathbb{Z}$, so $\chi=4$ forces $\mu=\pm1$ and $(D')=\mathfrak{a}^2$. Since conjugation is trivial modulo $\mathfrak{d}$, $\alpha_1\bar\alpha_1-\alpha_2\bar\alpha_0\equiv D'$ modulo $\mathfrak{a}\mathfrak{d}$, and the congruence $c\equiv g$ gives $\lambda D'\in\mathfrak{d}\,\bar{\mathfrak{a}}^{-1}$. The ideal of $\lambda D'$ is $\mathfrak{a}\bar{\mathfrak{a}}^{-1}$, so $\mathfrak{a}\subseteq\mathfrak{d}$, which is impossible for $\mathfrak{a}$ prime to $\mathfrak{d}$ and $\mathfrak{d}\neq(1)$.

(ii) The matrix $\operatorname{adj}V$ is Hermitian of rank one, so $\operatorname{adj}V=\mu\,\bar l\,l^T$ with $l\in\mathcal{O}^3$, $l^TV=0$ and $\mu\in\mathbb{Q}^\times$. Put $\Delta=l_1^2-4l_0l_2$ and $m=\mu^2\mathrm{N}(\Delta)$. Replacing $l$ by $\lambda l$, $\lambda\in F_0^\times$, replaces $\mu$ by $\mu/\mathrm{N}(\lambda)$ and $\Delta$ by $\lambda^2\Delta$, so $m$ depends only on $V$. As the entries of $\operatorname{adj}V$ are integral, $\mu$ is integral at every prime at which $l$ is primitive, and $l$ can be made primitive at the primes above any given $p$, because $\mathcal{O}\otimes\mathbb{Z}_{(p)}$ is a principal ideal domain; so $m\in\mathbb{Z}$. Let $J$ be the matrix with rows $(0,0,1)$, $(0,-2,0)$, $(1,0,0)$. Since $\int_X\theta_1^2\theta_2^2=4$, one has $\chi(v,v)=\operatorname{tr}(JVJ\bar V)$; and $\operatorname{adj}J$ has rows $(0,0,2)$, $(0,-1,0)$, $(2,0,0)$, so that $l^T(\operatorname{adj}J)\,l=-\Delta$ and

$$\sigma_2(JVJ\bar V)=\operatorname{tr}\operatorname{adj}(JVJ\bar V)=\mu^2\,\Delta\bar\Delta=m.$$

At a real place, $V=PWQ^T$ with $W=(w_\varepsilon)$ and $P$, $Q$ the $3\times2$ matrices with columns $(1,\pm s_1,s_1^2)$ and $(1,\pm s_2,s_2^2)$, when $t=0$; multiplication by $e^{\theta_t}$ replaces $V$ by $UV\bar U^T$ with $U$ unipotent and $U^TJU=J$, which conjugates $JVJ\bar V$, so we may take $t=0$. As $(1,z,z^2)\,J\,(1,z',z'^2)^T=(z-z')^2$ and $w_{-\varepsilon}=\overline{w_\varepsilon}$, the nonzero eigenvalues of $JVJ\bar V$ are $16\mathrm{N}(q)\big(|w_{++}|\pm|w_{+-}|\big)^2$, so

$$\chi^2-4m=\big(64\mathrm{N}(q)\,|w_{++}|\,|w_{+-}|\big)^2\geq0,$$

with equality exactly when $N_w=2$. Modulo $\mathfrak{d}$ one has $\bar V\equiv V$, $\operatorname{tr}(JV)=g-2c+\bar g\equiv0$ and $\sigma_2(JV)=\operatorname{tr}\big(\operatorname{adj}(V)\operatorname{adj}(J)\big)=\mu\,l^T(\operatorname{adj}J)\,\bar l\equiv-\mu\Delta$, with $l$ primitive at the primes dividing $\mathfrak{d}$. Hence

$$\chi\equiv\operatorname{tr}\big((JV)^2\big)=\operatorname{tr}(JV)^2-2\sigma_2(JV)\equiv2\mu\Delta\pmod{\mathfrak{d}}.$$

Now let $\chi=6$ and $N_w=4$. Then $1\leq m\leq8$, because $\Delta$ is totally negative, and $m$ satisfies three conditions.

(a) $4m\equiv36$ modulo $\mathfrak{d}\cap\mathbb{Z}$, which is $D\mathbb{Z}$ if $D\equiv1$ modulo 4 and $2D\mathbb{Z}$ otherwise. As $D$ is squarefree, this says that $m\equiv9$ modulo the odd part $D_o$ of $D$.

(b) $m\equiv0$ or 1 modulo 4. Take $l$ primitive at 2, so that $\mu\in\mathbb{Z}_{(2)}$. Since $\mathrm{N}(x+4y)\equiv\mathrm{N}(x)$ modulo 4 for $x,y\in\mathcal{O}\otimes\mathbb{Z}_{(2)}$, $\mathrm{N}(\Delta)\equiv\mathrm{N}(l_1^2)=\mathrm{N}(l_1)^2$, and $m\equiv\big(\mu\mathrm{N}(l_1)\big)^2$ modulo 4.

(c) If $D \equiv 2$ modulo 4, then $m$ is odd. The prime $\mathfrak{p}$ above 2 is ramified, $v_{\mathfrak{p}}(2) = 2$ and $v_{\mathfrak{p}}(\sqrt{D}) = 1$, so $v_{\mathfrak{p}}(\mathfrak{d}) = 3$ while $v_{\mathfrak{p}}(6) = 2$; the congruence $6 \equiv 2\mu\Delta$ modulo $\mathfrak{p}^3$ forces $v_{\mathfrak{p}}(\mu\Delta) = 0$, so $\mu$ and $\mathrm{N}(\Delta)$ are odd.

If $D \equiv 1$ modulo 4 and $D \neq 5$, then $D \geq 13$ and (a) has no solution in $[1, 8]$. If $D \equiv 3$ modulo 4, (a) gives $m \in \{3, 6\}$ for $D = 3$, $m = 2$ for $D = 7$ and nothing for $D \geq 11$, and (b) excludes these values. If $D = 2D'$, then (b) and (c) give $m \equiv 1$ modulo 4, which with (a) says $m \equiv 9$ modulo $4D'$, and this has no solution in $[1, 8]$ unless $D' = 1$. There remain $D = 5$, with $m = 4$, and $D = 2$, with $m \in \{1, 5\}$.

For $\mathbb{Q}(\sqrt{5})$, $m = 4$ means that $\Delta$ is $-2$ times a unit or minus the square of a unit, since 2 is inert; and $\Delta \equiv l_1^2$ modulo 4, while neither $-1$ nor twice a unit is a square modulo 4 in $\mathbb{Z}[\varphi]$. For $\mathbb{Q}(\sqrt{2})$, (c) makes $\mu$ odd, so $\mu = \pm 1$ and $\mathrm{N}(\Delta) \in \{1, 5\}$; the norms of the elements of $\mathbb{Z}[\sqrt{2}]$ prime to 2 are $\pm 1$ modulo 8, so $\mathrm{N}(\Delta) = 1$ and $\Delta = -(1 + \sqrt{2})^{2k}$, which is 3 or $1 + 2\sqrt{2}$ modulo 4, and neither is a square modulo 4 in $\mathbb{Z}[\sqrt{2}]$. ■

**Theorem 19.37** (No dimension-count certificate for quartic CM fields)**.** *Let $F = F_0(\sqrt{-q})$ be a quartic CM field with real quadratic subfield $F_0$, let $X$ be a principally polarised abelian fourfold with real multiplication by $\mathcal{O}_{F_0}$, and give $A = X \times \widehat{X}$ the action of $F$ of* [Mar25c, §11.1]*, transported by the $B$-field $e^{\theta_t}$ for a rational $t \in F_0$. Let $F_1, F_2$ be perfect complexes on $X$ with $\mathrm{Ext}^{<0}(F_i, F_i) = 0$, for instance sheaves, whose Chern characters lie in spaces $S(t', q')$ of Proposition 19.34, and let $E = \Phi(F_1 \boxtimes F_2^\vee)$ and $\kappa = \mathrm{ch}(E)e^{\ell/2}$. Put $v_1 = \mathrm{ch}(F_1)$ and $v_2 = \mathrm{ch}(F_2^\vee)$.*

(i) *$r(\kappa) = r(\mathrm{ch}\, E) = r^2(v_1) + 64 + r^2(v_2)$, where $r^2(v_i) \in \{12, 18, 20\}$ is given by Proposition 19.34(ii); so $r(\kappa) \leq 104$, and $r(\kappa) \in \{100, 102, 104\}$ unless $F$ is biquadratic and some $v_i$ has $N_w = 2$.*

(ii) *$\dim \mathrm{Ext}^2(E, E) > r(\kappa)$. So the injectivity of $\sigma_E$ cannot be obtained from a dimension count for any such $E$.*

(iii) *If $t = 0$ and both Chern characters lie in $S(0, q)$, then $\kappa$ is invariant under $G_F$, and so is a Hodge class on every member of the $(F, 2, 1)$-family. This is the case $[F : \mathbb{Q}] = 4$ of* [Mar25c, Proposition 10.2.1(1)] *and of the inclusion stated before* [Mar25c, Lemma 10.1.3]*, with other conventions for $\Phi$ and for the sign of the twist.*

*Proof.* (i) By Lemma 18.55(i), multiplication by $e^{\ell/2}$ conjugates contraction by an automorphism of $HT^2$, so $r(\kappa) = r(\mathrm{ch}\, E)$; and, as in the proof of Theorem 18.56(iv), $\Phi$ identifies $r(\mathrm{ch}\, E)$ with the rank of contraction from $HT^2(X \times X) = \bigoplus_{a+b=2} HT^a(X) \otimes HT^b(X)$ into $v_1 \boxtimes v_2$. Write $r^k$ for the rank of contraction from $HT^k(X)$. The block $HT^1 \otimes HT^1$ maps into $H^{\mathrm{odd}} \otimes H^{\mathrm{odd}}$ by the tensor product of the two contractions from $HT^1$, of rank $r^1(v_1)r^1(v_2) = 64$, and the other two blocks map into $H^{\mathrm{ev}} \otimes H^{\mathrm{ev}}$. If $(x_1 \lrcorner v_1) \otimes v_2 + v_1 \otimes (x_2 \lrcorner v_2) = 0$, then $x_1 \lrcorner v_1 = \lambda v_1$ for a scalar $\lambda$, and $\lambda = 0$ because contraction from $HT^2$ raises $q - p$ by two on classes of type $(p, q)$ while $v_1 \neq 0$ is of type $(p, p)$; so those two blocks contribute $r^2(v_1) + r^2(v_2)$, and $r^2(v_i) = 2\rho + 4N_w \in \{12, 18, 20\}$ by Proposition 19.34(ii), with 12 only for biquadratic $F$.

(ii) By the Künneth formula, as in Proposition 17.20, $\dim \mathrm{Ext}^2(E, E) = e_0e_2' + e_1e_1' + e_2e_0'$ with $e_k = \dim \mathrm{Ext}^k(F_1, F_1) \geq r^k(v_1)$, by the commutative square $\sigma_F \circ \mathrm{ev}_F = (\,\cdot\,) \lrcorner \mathrm{ch}(F)$, and likewise for $F_2$. So $\dim \mathrm{Ext}^2(E, E) \geq r(\kappa)$, with equality only if each $F_i$ is minimal, $\dim \mathrm{Ext}^k(F_i, F_i) = 1, 8, r^2(\mathrm{ch}\, F_i)$ for $k = 0, 1, 2$. For such $F_i$, Serre duality gives $\chi(F_i, F_i) = r^2 - 14$, that is 4, 6 or $-2$ for $(\rho, N_w) = (1, 4)$, $(2, 4)$, $(2, 2)$. The last is impossible because $\chi$ is positive on $S(t', q')$. The Chern character of a perfect complex on a complex torus is integral, since by the Künneth theorem the topological $K$-theory of a real torus is

generated by products of classes pulled back from its circle factors, whose Chern characters are integral. So the first two values are excluded by Proposition 19.36.

(iii) Over $\mathbb{R}$, $H^1(X,\mathbb{R}) = U_1 \oplus U_2$, with $U_j$ the eigenspace of $F_0$ for $\tau_j$, and the $U_j$ are orthogonal for $\theta$. As in the proof of Proposition 17.21, the correspondence $Z \mapsto \mathrm{ch}(\Phi(Z))e^{\ell/2}$, the Poincaré class $\ell = \ell_1 + \ell_2$, the classes $\eta$ and $\gamma$ and the exponentials $e_\varepsilon$ all decompose as tensor products over $j$ of classes of even degree, so that the image of $e_\varepsilon \boxtimes e^\vee_{\varepsilon'}$ is the tensor product over $j$ of the images of $e^{\varepsilon_j s_j \theta_j} \boxtimes e^{-\varepsilon'_j s_j \theta_j}$ in the $j$-th factor. On that factor $\sqrt{-q}$ acts as in the imaginary quadratic construction with $d$ replaced by $\tau_j(q)$, and the identities of Proposition 17.21 at $n = 2$, which are identities in the exterior algebra of a symplectic space with $d$ a symbol, show that each image is a multiple of $e^{\pm(s_j/2\tau_j(q))\eta_j}$ or of a Weil class $(\gamma_j \mp s_j\ell_j)^2$ of the $j$-th factor. These are invariant under the special unitary group $SU_j$ of the $j$-th factor, and $G_F(\mathbb{R}) = SU_1 \times SU_2$ acts factor by factor; so the rational class $\kappa$ is invariant under $G_F(\mathbb{R})$, which is Zariski dense in $G_F$, and the Hodge group of every member of the family lies in $G_F$. ∎

*Remark* 19.38 (The shape of the secant characters). Let $t = 0$ in Theorem 19.37(iii), and write $\eta = \eta_1 + \eta_2$, $\gamma = \gamma_1 + \gamma_2$ and $\ell = \ell_1 + \ell_2$ for the decompositions over the real places used in its proof: $\eta_j = \tau_j(q)\theta_j + \widehat{\theta}_j$ and $\gamma_j = \tau_j(q)\theta_j - \widehat{\theta}_j$, with $\theta_j$ and $\widehat{\theta}_j$ here the pull-backs to $A$ of the components of the polarisations of $X$ and $\widehat{X}$. The classes $\eta_t$ of $A$ are the combinations of $\eta_1$ and $\eta_2$, and the Weil classes at the two embeddings over $\tau_j$ are the multiples of $(\gamma_j \mp s_j\ell_j)^2$. Let $\kappa_{(a,b)}$ be the component of $\kappa$ of degree $a$ at $\tau_1$ and $b$ at $\tau_2$. Although $\kappa$ is flat, it does not lie in $\mathbb{C}[\eta_1,\eta_2] + W_F \otimes \mathbb{C}$ once $\kappa_4$ has a nonzero Weil part, unlike the classes of Proposition 17.21. Indeed, by the proof of Theorem 19.37(iii), $\kappa$ is a sum of products $X_1X_2$ in which each $X_j$ is a multiple of $e^{\pm c_j\eta_j}$, $c_j = s_j/(2\tau_j(q))$, or of $(\gamma_j \mp s_j\ell_j)^2$. The components $\kappa_{(4,0)}$ and $\kappa_{(4,8)}$ receive exactly the terms whose second factor is an exponential, through its parts of degree 0 and 8, and $c_2^4 = 1/(16\tau_2(q)^2)$ for both signs; so

$$\kappa_{(4,8)} = \frac{\kappa_{(4,0)}\,\eta_2^4}{384\,\tau_2(q)^2},$$

and symmetrically. The polynomials in $\eta_1,\eta_2$ of this bidegree are the multiples of $\eta_1^2\eta_2^4$ and $W_F$ has no component there, while $\kappa_{(4,0)}$ has a nonzero Weil part as soon as $\kappa_4$ has one, since a nonzero rational Weil class has nonzero components at all four embeddings. The class $\mathrm{ch}(E) = \kappa e^{-\ell/2}$ itself is not flat when $\mathrm{rk}(E) \neq 0$: for $Y$ in the Lie algebra of $G_F$, $D_Y\,\mathrm{ch}(E) = -\frac{1}{2}(D_Y\ell)\,\mathrm{ch}(E)$, whose part of degree two is $-\frac{1}{2}\,\mathrm{rk}(E)\,D_Y\ell$, and $\ell$ is not flat. So these objects, when their rank is not zero, are candidates only for a twisted form of Theorem 19.28(v), with hypothesis a flat twisted character $\kappa$ with $\kappa_4 = N\omega + P_4$, which needs the extension of Theorem 16.28 to twisted objects [Per26] and the extraction of untwisted cycles of [Mar25b, Section 2], as after Theorem 17.19. This is the form in which Markman states his reduction: he uses the normalised class $\mathrm{ch}(E)\exp(-c_1(E)/\,\mathrm{rk}\,E)$, which is $\kappa e^{-\kappa_1/\kappa_0}$ when $\mathrm{rk}\,E \neq 0$, asserts the shape of a polynomial plus a nonzero Weil class only for its component in $H^4$, and takes as propagation hypothesis that this component stays algebraic [Mar25c, Proposition 10.2.1, Corollary 10.2.3].

*Remark* 19.39 (Negative Ext groups). The hypothesis $\mathrm{Ext}^{<0}(F_i,F_i) = 0$ in Theorem 19.37(ii) can be replaced by a condition on the amplitude. Let $F_1, F_2$ be any perfect complexes with characters in spaces $S(t',q')$, let $e_k = \dim \mathrm{Ext}^k(F_1,F_1)$ and $e'_k = \dim \mathrm{Ext}^k(F_2,F_2)$, and put $\nu_i = \sum_{j\geq 3}(-1)^j \dim \mathrm{Ext}^{-j}(F_i,F_i)$. Then still $\dim \mathrm{Ext}^2(E,E) \geq r(\kappa)$, and equality forces, for each $i$, $\dim \mathrm{Ext}^k(F_i,F_i) = 1, 8, r^2(v_i)$ for

$k = 0, 1, 2$, $\operatorname{Ext}^{-1}(F_i, F_i) = \operatorname{Ext}^{-2}(F_i, F_i) = 0$ and $\nu_i = \big(\chi(v_i, v_i) - r^2(v_i) + 14\big)/2$. This gives $\nu_i \geq 2$ for $(\rho, N_w) = (1, 4)$ and $(2, 2)$, and $\nu_i \geq 1$ for $(2, 4)$ unless $F_0 = \mathbb{Q}(\sqrt{3})$ and $\chi(v_i, v_i) = 4$, where $\nu_i = -1$, a case that needs integral classes of that kind. So equality needs, for both $i$, some $\operatorname{Ext}^{-j}(F_i, F_i) \neq 0$ with $j \geq 4$ even, or $j \geq 3$ odd in the exceptional case, and it fails as soon as one $F_i$ has its cohomology sheaves in an interval of length at most 3, at most 2 in the exceptional case. These conditions are necessary, not sufficient. The strict inequality $\dim \operatorname{Ext}^2(E', E') > r(\operatorname{ch} E')$ passes to the images $E'$ of such an $E$ under derived equivalences, push-forward and pull-back along isogenies, and twists by line bundles or $B$-fields.

Indeed, $\dim \operatorname{Ext}^2(E, E) = \sum_{a \in \mathbb{Z}} e_a e'_{2-a}$ by the Künneth formula, and $e_a = e_{4-a}$ by Serre duality. The terms $a = 0, 1, 2$ give at least $r(\kappa)$, with equality only if $e_k$ and $e'_k$ are the ranks $r^k$ for $k \leq 2$, as in the proof of Theorem 19.37(ii); the terms $a = 3, 4, -1, -2$ are $e_1 e'_{-1}$, $e_0 e'_{-2}$, $e_{-1} e'_1$ and $e_{-2} e'_0$, which must vanish. Then $\chi(F_i, F_i) = 2e_0 - 2e_1 + e_2 + 2\sum_{j \geq 1} (-1)^j e_{-j}$ gives the value of $\nu_i$. The number $\chi(v, v)$ is a positive even integer: positive by Proposition 19.34(iii), and even because $\int_X x^2$ is even for $x \in H^4(X, \mathbb{Z}) = \bigwedge^4 H^1(X, \mathbb{Z})$. For $(1, 4)$, $\chi \in 4\mathbb{Z}$ by the proof of Proposition 19.36(i), and $\chi \neq 4$; for $(2, 2)$, $r^2 = 12$. For $(2, 4)$, $r^2 = 20$, and in the proof of Proposition 19.36(ii) $\chi^2 - 4m > 0$ with $m \geq 1$, so $\chi \geq 4$, and $\chi \neq 6$. If $\chi = 4$, then $m \in \{1, 2, 3\}$, and conditions (a) to (c) of that proof, with 36 replaced by 16, say that $m \equiv 4$ modulo the odd part of $D$, so that $D \in \{2, 3, 6\}$; that $m = 1$; and, for $D \equiv 2$ modulo 4, that $m$ is even, because $v_{\mathfrak{p}}(4) = 4 \geq 3$ here gives $v_{\mathfrak{p}}(\mu\Delta) \geq 1$. Only $D = 3$ remains. A complex with cohomology sheaves in an interval of length $l$ has $\operatorname{Ext}^{-j}(F, F) = 0$ for $j > l$. Equivalences preserve Ext algebras and, through the induced isomorphisms of $HT^*$ and $H^*$, ranks of contraction ([CRV12], as in the proof of Theorem 18.56(iv)); for an isogeny $g$, $g^* g_* E = \bigoplus_{x \in \ker g} t_x^* E$, so $\operatorname{Ext}^2(g_* E, g_* E)$ contains $\operatorname{Ext}^2(E, E)$ while $r(\operatorname{ch} g_* E) = r(\operatorname{ch} E)$, and likewise for $g^*$; twists are Lemma 18.55(i).

What Theorem 19.37 and Remark 19.39 leave for $\mathbf{W}(F, 2, \delta)$ is one of three things: an Orlov product both of whose factors have nonzero Ext groups in some even degree at most $-4$ (or odd degree at most $-3$ when $F_0 = \mathbb{Q}(\sqrt{3})$); a kernel on $X \times X$ that is not an external product; or a proof of the injectivity of $\sigma_E$ for some Orlov product that does not go through a dimension count, which is what Markman's question [Mar25c, Question 11.2.2] asks for. The next proposition shows that a kernel whose transform has a pure spinor as its character, for instance a line bundle on the graph of a homomorphism, gives no Weil class.

**Proposition 19.40** (Pure spinors give no Weil class)**.** *Keep the notation of Remark 19.38 on $A = X \times \widehat{X}$ with $t = 0$, let $V_j \subseteq H^1(A, \mathbb{R})$ be the part at $\tau_j$, and call a class flat if it is invariant under $G_F(\mathbb{R})$. Let $v = c e^{\beta} \Omega_N$ be a real pure spinor, with $c \in \mathbb{R}^\times$, $\beta$ a real 2-form, $N \subseteq H^1(A, \mathbb{R})$ and $\Omega_N$ a generator of $\det N$. If $v$ is flat, then $N$ is one of $0$, $V_1$, $V_2$, $H^1(A, \mathbb{R})$, and $v \in \mathbb{R}[\eta_1, \eta_2]$. This applies to the characters of line bundles on translates of abelian subvarieties, of points and of simple semi-homogeneous bundles, and of their images under derived equivalences and isogenies, twisted by any real $B$-field; for instance to $\Phi$ of a line bundle on the graph of a homomorphism $X \to X \times X$. The component in $W_F \otimes \mathbb{R}$ of such a class vanishes in every degree: no such object has a character, twisted as stated, of the form required by Theorem 19.28(v) or by its twisted form, whatever the quartic field.*

*Proof.* Over $\mathbb{R}$, $G_F(\mathbb{R}) = SU_1 \times SU_2$ with $SU_j$ acting on $V_j$ and trivially on the other part, and $V_j \otimes \mathbb{C}$ is the sum of the standard representation of $SU_j \otimes \mathbb{C} \cong \mathrm{SL}_4$ and its dual, which are not isomorphic (proof of Proposition 19.20(i)). So $V_1$ and $V_2$ are irreducible and not

isomorphic, and 0, $V_1$, $V_2$, $H^1(A,\mathbb{R})$ are the only stable subspaces. The flat classes are $\mathrm{Inv}_1 \otimes \mathrm{Inv}_2$, where by the same proof $\mathrm{Inv}_j \subseteq \bigwedge^* V_j$ is spanned by the powers of $\eta_j$ and the two Weil classes $(\gamma_j \mp s_j \ell_j)^2$; in degree two it is $\mathbb{R}\eta_j$, and $\eta_j^4$ spans $\bigwedge^8 V_j$.

If $v$ is real and pure over $\mathbb{C}$, then $N = \{\xi : \xi \wedge v = 0\}$ is defined over $\mathbb{R}$, and so are $c$ and the class of $\beta$ modulo $N \wedge H^1$. For $g \in G_F(\mathbb{R})$, $gv = v$ gives $gN = N$, and $g\Omega_N = \det(g|_N)\Omega_N = \Omega_N$ because the connected semisimple group $G_F(\mathbb{R})$ has no nontrivial character. Hence $(e^{g\beta-\beta} - 1)\Omega_N = 0$, whose part of lowest degree gives $g\beta - \beta \in N \wedge H^1$. So $N$ is one of the four stable subspaces, and the class of $\beta$ in $\bigwedge^2(H^1/N)$ is invariant; by complete reducibility it is the image of an invariant 2-form $a_1\eta_1 + a_2\eta_2$, and since $e^{n\wedge x}\Omega_N = \Omega_N$ for $n \in N$, $v = c\, e^{a_1\eta_1 + a_2\eta_2}\Omega_N$, with $\Omega_N$ a multiple of 1, $\eta_1^4$, $\eta_2^4$ or $\eta_1^4\eta_2^4$. The characters listed are pure spinors, and derived equivalences, isogenies and twists preserve purity; this is the classical description recalled in Setup 17.26 and after Corollary 17.17. ∎

*Markman's candidate for a quartic field.* Markman gives one explicit pair of sheaves for a biquadratic field [Mar25c, Example 11.2.7, Lemma 11.2.8] and asks whether the second can be chosen so that its semiregularity map is injective on the image of the evaluation map [Mar25c, Question 11.2.2]. For one object on the fourfold this is the weakened criterion of Setup 18.54, and the local method of Theorem 18.61 decides it for his pair: both sheaves fail it, at points of the curves that the construction glues in. Two local computations are needed, one for a sheaf that is locally free on a smooth support and one for the ideal of a curve in a smooth divisor. As in Lemma 18.60, for a germ of vector field $\theta$ and a perfect complex $F$ of modules over a local ring we write $a_\theta \in \mathrm{Ext}^1(F,F)$ for the class of the jet extension along $\theta$.

**Lemma 19.41** (Free germs on smooth supports)**.** *Let $\mathcal{O}$ be the local ring of a smooth complex variety $Y$ at a point $p$, or its completion, let $C \ni p$ be a smooth subvariety of codimension $c \geq 1$, put $N = T_pY/T_pC$, and let $F$ be a perfect complex of $\mathcal{O}$-modules quasi-isomorphic to a shift of $\mathcal{O}_{C,p}^{\oplus r}$, $r \geq 1$. Then $\mathrm{Ext}^2_{\mathcal{O}}(F,F)$ is a free $\mathcal{O}_{C,p}$-module isomorphic to $M_r(\mathcal{O}_{C,p}) \otimes \bigwedge^2 N$, and modulo the maximal ideal of $\mathcal{O}_{C,p}$ the product $a_\theta a_{\theta'}$ is, up to a sign that does not depend on $\theta, \theta'$, the element $\mathrm{id} \otimes (\bar\theta \wedge \bar\theta')$, where $\bar\theta, \bar\theta'$ are the images of $\theta(p), \theta'(p)$ in $N$. In particular $a_\theta a_{\theta'} \neq 0$ whenever $\theta(p)$, $\theta'(p)$ and $T_pC$ span a space of dimension $\dim C + 2$.*

*Proof.* Choose formal coordinates $x_1, \ldots, x_m$ at $p$ with $C = \{x_1 = \cdots = x_c = 0\}$. Ext groups of finitely generated modules and jet extensions commute with the flat base change from $R = \mathbb{C}[x_1, \ldots, x_m]$ to $\mathcal{O}$, and a shift changes neither the Ext algebra nor the jet classes, so we may take $F = (R/(x_1, \ldots, x_c))^{\oplus r}$ over $R$. Write $R = A_1 \otimes \cdots \otimes A_c \otimes B$ with $A_i = \mathbb{C}[x_i]$ and $B = \mathbb{C}[x_{c+1}, \ldots, x_m]$; then $F = k_1 \otimes \cdots \otimes k_c \otimes B^{\oplus r}$, with $k_i$ the residue field of $A_i$ at the origin. On the resolution $A_i \xrightarrow{x_i} A_i$ of $k_i$ the jet chain map of $\partial_i$ is $-\partial_i(x_i) = -1$, as in the proof of Lemma 18.60, so its class $\gamma_i$ spans $\mathrm{Ext}^1_{A_i}(k_i,k_i) = k_i$, and $\mathrm{Ext}^j_{A_i}(k_i,k_i) = 0$ for $j \geq 2$. The module $B^{\oplus r}$ is free, so its Ext algebra is $M_r(B)$ in degree zero and its jet classes vanish. By (59), applied factor by factor, $\mathrm{Ext}^*_R(F,F)$ is the graded tensor product of the exterior algebra on $\gamma_1, \ldots, \gamma_c$ with $M_r(B)$, and $a_{\partial_i} = \gamma_i \otimes \mathrm{id}$ for $i \leq c$, while $a_{\partial_i} = 0$ for $i > c$. Hence $\mathrm{Ext}^2_R(F,F) = \bigwedge^2\langle\gamma_1, \ldots, \gamma_c\rangle \otimes M_r(B)$, and $B = \mathcal{O}_C$ in these coordinates. For $\theta = \sum_i f^i\partial_i$ and $\theta' = \sum_i f'^i\partial_i$ the linearity of Lemma 18.59 gives

$$a_\theta a_{\theta'} = \sum_{i,j\leq c} f^i f'^j\, \gamma_i\gamma_j \otimes \mathrm{id} = \sum_{i<j\leq c} \left(f^i f'^j - f^j f'^i\right) \gamma_i\gamma_j \otimes \mathrm{id},$$

because the $\gamma_i$ anticommute and square to zero. The functions $x_1, \dots, x_c$ act by zero, so modulo the maximal ideal of $\mathcal{O}_C$ the coefficients are those of $\theta(p) \wedge \theta'(p)$ on $\partial_i \wedge \partial_j$, $i < j \leq c$, which are the coordinates of $\bar{\theta} \wedge \bar{\theta}'$ in $\bigwedge^2 N$. A section of a free module with nonzero reduction is nonzero. ∎

**Lemma 19.42** (A curve in a divisor)**.** *Let $\mathcal{O}$, $Y$ and $p$ be as in Lemma 19.41, let $D \ni p$ be a smooth divisor, let $C \subset D$ be a smooth subvariety of codimension two in $D$ through $p$, and let $F$ be quasi-isomorphic to a shift of the ideal $I_{C\subset D}$ of $C$ in $\mathcal{O}_D$, regarded as an $\mathcal{O}$-module. Put $\nu = T_pY/T_pD$ and $N' = T_pD/T_pC$. Then $\operatorname{Ext}^2_{\mathcal{O}}(F,F) \cong \mathcal{O}_{C,p} \otimes \nu \otimes N'$, and modulo the maximal ideal of $\mathcal{O}_{C,p}$ the product $a_\theta a_{\theta'}$ is, up to a fixed sign, the image of $\theta(p) \wedge \theta'(p)$ under*

$$\textstyle\bigwedge^2 T_pY \longrightarrow \bigwedge^2 (T_pY/T_pC) \longrightarrow \bigwedge^2 (T_pY/T_pC)/\bigwedge^2 N' \cong \nu \otimes N'.$$

*In particular $a_\theta a_{\theta'} \neq 0$ whenever $\theta(p) \wedge \theta'(p)$ does not lie in $\bigwedge^2 T_pD + T_pC \wedge T_pY$.*

*Proof.* With formal coordinates in which $D = \{x_1 = 0\}$ and $C = \{x_1 = x_2 = x_3 = 0\}$, and the same reduction to the polynomial ring, write $R = \mathbb{C}[x_1] \otimes \mathbb{C}[x_2, x_3] \otimes B$ with $B = \mathbb{C}[x_4, \dots, x_m]$. Then $I_{C\subset D} = k \otimes \mathfrak{m} \otimes B$, where $k$ is the residue field of $\mathbb{C}[x_1]$ and $\mathfrak{m} = (x_2, x_3)$. As in the proof of Lemma 18.60, $\mathfrak{m}$ has projective dimension one, $\operatorname{Ext}^1(\mathfrak{m},\mathfrak{m}) = \mathfrak{m}/\mathfrak{m}^2$ has the basis $\alpha_2 = a_{\partial_2}$, $\alpha_3 = a_{\partial_3}$ and is killed by $\mathfrak{m}$, and $\operatorname{Ext}^2(\mathfrak{m},\mathfrak{m}) = 0$; the factor $k$ contributes $\gamma_1$ in degree one, as in Lemma 19.41, and $B$ contributes nothing in positive degree. By (59)

$$\operatorname{Ext}^2_R(I_{C\subset D}, I_{C\subset D}) = \operatorname{Ext}^1(k,k) \otimes \operatorname{Ext}^1(\mathfrak{m},\mathfrak{m}) \otimes B = \langle \gamma_1 \rangle \otimes \langle \alpha_2, \alpha_3 \rangle \otimes B,$$

the other summands vanishing by the projective dimensions, and $a_{\partial_1} a_{\partial_j} = -a_{\partial_j} a_{\partial_1} = \pm \gamma_1 \otimes \alpha_j$ for $j = 2, 3$, while $a_{\partial_2} a_{\partial_3} = 1 \otimes \alpha_2\alpha_3 = 0$, $a^2_{\partial_1} = 0$ and $a_{\partial_i} = 0$ for $i \geq 4$. As before, $a_\theta a_{\theta'}$ reduces to the component of $\theta(p) \wedge \theta'(p)$ on $\partial_1 \wedge \langle \partial_2, \partial_3 \rangle$. In these coordinates $\partial_1$ spans $\nu$, the images of $\partial_2, \partial_3$ span $N'$, and the component is the image in $\bigwedge^2 (T_pY/T_pC)/\bigwedge^2 N'$. Its kernel on $\bigwedge^2 T_pY$ is spanned by $\partial_2 \wedge \partial_3$ and the $\partial_i \wedge \partial_j$ with $j \geq 4$, which is $\bigwedge^2 T_pD + T_pC \wedge T_pY$. ∎

Now keep the notation of Proposition 19.34, with $X$ an abelian fourfold with real multiplication by $F_0$ and $\theta = \theta_1 + \theta_2$ a polarisation in $\bigwedge^2_{F_0} H^1(X, \mathbb{Q})$. Over $\mathbb{C}$, $H^1(X) = U_1 \oplus U_2$ with $U_j$ the eigenspace of $F_0$ at $\tau_j$, and the tangent space splits accordingly as $T = T_1 \oplus T_2$, with $T_j$ of dimension two, dual to $U_j \cap H^{1,0}$; translation identifies every $T_pX$ with $T$. Let $\pi_j$ span $\bigwedge^2 T_j$ and put

$$x_j = \left(\tfrac{1}{2}\tau_j(q)\, \pi_j \lrcorner\, \theta_j^2,\ 0,\ \pi_j\right) \in HT^2(X), \qquad j = 1, 2. \tag{70}$$

The six classes $(0, v, 0)$ with $v \in \operatorname{Hom}(H^{1,0}, H^{0,1})$ preserving the decomposition by places and $v \lrcorner \theta_1 = v \lrcorner \theta_2 = 0$ are the first order deformations of $X$ with its real multiplication and polarisation; we call them the polarised $F_0$-linear deformations.

**Theorem 19.43** (The local obstruction for a quartic secant character)**.** *In this notation, let $q \in F_0$ be totally positive and $t \in F_0$.*

(i) *$x_j \lrcorner w = 0$ for every $w \in S(0, q)$, and $(e^{-\theta_t} x_j) \lrcorner w = 0$ for every $w \in S(t, q)$, $j = 1, 2$. If $w \in S(t, q)$ has $(\rho, N_w) = (2, 4)$, then $\operatorname{Ann}_{HT^2}(w)$ is spanned by $e^{-\theta_t} x_1$, $e^{-\theta_t} x_2$ and the six polarised $F_0$-linear deformations, and its bivector components span exactly $\langle \pi_1, \pi_2 \rangle$.*

(ii) *Let $F$ be a perfect complex on $X$ with $\mathrm{ch}(F) \in S(t,q)$, and let $\theta, \theta'$ be translation invariant vector fields with $\theta \wedge \theta' = \pi_j$. If $a_\theta a_{\theta'} \neq 0$ in $\mathrm{Ext}^2(F_p, F_p)$ at some point $p$, then $\mathrm{ev}_F(e^{-\theta_t} x_j) \neq 0$ and $F$ does not satisfy the weakened criterion. Neither do $\Phi(F \boxtimes G)$ and $\Phi(G \boxtimes F)$ on $X \times \widehat{X}$, for any nonzero perfect complex $G$ on $X$, with $\Phi$ as in Theorem 18.56(iv).*

(iii) *The hypothesis of* (ii) *holds at $p$, for $j = 1$ or for $j = 2$, whenever $F_p$ is a shift of a free module of positive rank over the local ring of a smooth curve through $p$, since $T_pC$ lies in at most one of $T_1, T_2$; it holds for both $j$ when $F_p$ is a shift of a free module over the residue field. It holds for $j$ when $F_p$ is a shift of the ideal of a smooth curve $C$ in a smooth divisor $D$ with $T_pC \not\subseteq T_j$ and $T_j \not\subseteq T_pD$.*

*Proof.* (i) Let $t = 0$. By the proof of Lemma 18.55(i), for a bivector $\pi$ of type $(2,0)$ in the tangent directions and a class $B$ of type $(1,1)$, $\pi \lrcorner e^B = \frac{1}{2}(\pi \lrcorner B^2)\, e^B$. The interior products by vectors of $T_j$ are odd derivations that vanish on $\bigwedge^* U_{j'}$ for $j' \neq j$, so $\pi_j \lrcorner (ab) = (\pi_j \lrcorner a)\, b$ for $a$ in the subalgebra generated by $U_j$ and $b$ an even class generated by $U_{j'}$. Applied to $e_\varepsilon = e^{\varepsilon_1 s_1 \theta_1} e^{\varepsilon_2 s_2 \theta_2}$, with $s_j^2 = -\tau_j(q)$, this gives $\pi_j \lrcorner e_\varepsilon = -\frac{1}{2}\tau_j(q)(\pi_j \lrcorner \theta_j^2)\, e_\varepsilon$ for all four signs, and so $x_j \lrcorner e_\varepsilon = 0$. A class $(0, v, 0)$ with $v$ preserving the places and killing $\theta_1$ and $\theta_2$ acts as a derivation that kills every polynomial in $\theta_1, \theta_2$, hence all of $S(0,q)$. For general $t$, Lemma 18.55(i) gives $y \lrcorner (w e^{\theta_t}) = ((e^{\theta_t} y) \lrcorner w) e^{\theta_t}$, so $e^{-\theta_t}$ carries $\mathrm{Ann}_{HT^2}(w)$ onto $\mathrm{Ann}_{HT^2}(w e^{\theta_t})$; it fixes the six classes $(0, v, 0)$, because $v \lrcorner \theta_t = 0$, and by (55) it does not change bivector components. The eight classes are independent: the six have no bivector component and are independent, and the other two have the independent components $\pi_1, \pi_2$. When $(\rho, N_w) = (2, 4)$ the rank of contraction from $HT^2(X)$, of dimension 28, is $2\rho + 4N_w = 20$ by Proposition 19.34(ii), so the annihilator has dimension 8 and the eight classes span it.

(ii) By (i), $y = e^{-\theta_t} x_j = (z, v, \pi_j)$ lies in $\mathrm{Ann}_{HT^2}(\mathrm{ch}\, F)$, and by (54) $\mathrm{ev}_F(y) = z \cdot \mathrm{id}_F + v \lrcorner \mathrm{At}(F) + \frac{1}{2}\pi_j \lrcorner \mathrm{At}(F)^2$. The last term is $A_\theta A_{\theta'}$, as in the proof of Theorem 18.61, and its germ at $p$ is $a_\theta a_{\theta'} \neq 0$ by Lemma 18.59. The germs of the first two terms vanish: $z \cdot \mathrm{id}_F$ by Lemma 18.59, and $v \lrcorner \mathrm{At}(F)$ because it is a composite $F \to F \otimes \Omega^1_X[1] \to F \otimes \Omega^1_X \otimes T_X[2] \to F[2]$ in which the middle arrow is $\mathrm{id} \otimes v$ with $v \in \mathrm{Ext}^1(\mathcal{O}_X, T_X) = H^1(T_X)$, and $v$ restricts to zero on every affine open set. So $\mathrm{ev}_F(y) \neq 0$ with $y \lrcorner \mathrm{ch}(F) = 0$, and $F$ fails the weakened criterion. For $G \boxtimes F$ and $F \boxtimes G$ the class $y \otimes 1$, resp. $1 \otimes y$, preserves the Chern character, and its evaluation $\pm \mathrm{ev}_F(y) \otimes \mathrm{id}_G$ is nonzero in its Künneth summand, as in the proof of Theorem 18.56(iv); the transport under $\Phi$ is the one given there.

(iii) Since $T_j$ is a plane, the image of $\pi_j$ in $\bigwedge^2(T/T_pC)$ is nonzero exactly when $T_j \cap T_pC = 0$. For a curve this says $T_pC \not\subseteq T_j$, which fails for at most one $j$ because $T_1 \cap T_2 = 0$; for a point it always holds. Lemma 19.41 then gives $a_\theta a_{\theta'} \neq 0$. In the last case the image of $\pi_j$ in $\bigwedge^2(T/T_pC)$ is the nonzero decomposable vector of the plane $(T_j + T_pC)/T_pC$, and it lies in $\bigwedge^2 N'$ only if that plane is $N'$, that is, only if $T_j + T_pC = T_pD$, which would put $T_j$ inside $T_pD$. So Lemma 19.42 applies. ∎

At a point where $F$ is locally free both sides of the local identity vanish, as at a smooth point of a surface in Proposition 18.65; the test sees only where the sheaf drops dimension or fails to be locally free on its support. Markman's construction produces exactly such points.

**Proposition 19.44** (The classes of Markman's candidate)**.** *Let $X$ be the Jacobian of a curve of genus four with its principal polarisation $\Theta$, with real multiplication by a real quadratic field $F_0$ and an automorphism $g$ with $g^* = \hat{\eta}(f)$ on $H^1(X, \mathbb{Q})$, where $f \in F_0$*

*has norm one and $f^2 \neq 1$; let $q > 0$ be rational, so that $F = F_0(\sqrt{-q})$ is biquadratic, and write $f_j = \tau_j(f)$. Let $\alpha_0 = \Theta - \frac{q}{6}\Theta^3$, the character of the sheaf $E_0 = F_q$ of* [Mar25a, Example 8.2.4]*, and $\beta' = g^*\Theta - \frac{q}{6}(g^{-1})^*\Theta^3$, a multiple of which is the character of the sheaf $E'$ of* [Mar25c, Example 11.2.7]*. With $A_j = 1 - \frac{q}{2}\theta_j^2$ and $\theta = \Theta$:*

(i) *$\alpha_0 = \theta_1 A_2 + \theta_2 A_1$ and $\beta' = f_1^2\theta_1 A_2 + f_2^2\theta_2 A_1$, both in $S(0,q)$. With $s = \sqrt{-q}$, the coefficients of $\beta'$ on the four $e_\varepsilon$ are $w_{++} = -w_{--} = (f_1^2 + f_2^2)/4s$ and $w_{+-} = -w_{-+} = (f_1^2 - f_2^2)/4s$, so $\beta'$ has $(\rho, N_w) = (2,4)$, while $\alpha_0$ lies on the secant plane of $\mathbb{Q}(\sqrt{-q})$ and has $(\rho, N_w) = (2,2)$.*

(ii) *The ranks of contraction from $HT^2(X)$ are 12 for $\alpha_0$ and 20 for $\beta'$. The annihilator of $\beta'$ is that of Theorem 19.43(i); that of $\alpha_0$ contains, besides it, the classes $(\frac{q}{2}\pi\lrcorner\Theta^2, 0, \pi)$ for every bivector $\pi$, by Theorem 18.56(ii) with $t = \Theta$ and $d = q$.*

(iii) *$\int_X \alpha_0\beta' = -4q\,\mathrm{Tr}_{F_0/\mathbb{Q}}(f^2) \neq 0$, which is Markman's formula for the rank of $\Phi(E_0 \boxtimes E')$* [Mar25c, Lemma 11.2.8]*; $\chi(\beta',\beta') = 4q\,\mathrm{Tr}_{F_0/\mathbb{Q}}(f^4)$ and $\chi(\alpha_0,\alpha_0) = 8q$; and the rank of contraction from $HT^2(X\times X)$ into $\alpha_0\boxtimes\beta'$ is $12+64+20 = 96$, as Theorem 19.37(i) predicts.*

*Proof.* (i) The automorphism $g$ acts on $U_j$ by $f_j$, hence on $\theta_j$ by $f_j^2$, and $f_1 f_2 = 1$. Since $\theta_j^3 = 0$, $\Theta^3 = 3\theta_1^2\theta_2 + 3\theta_1\theta_2^2$ and $(g^{-1})^*\Theta^3 = 3f_1^{-4}f_2^{-2}\theta_1^2\theta_2 + 3f_1^{-2}f_2^{-4}\theta_1\theta_2^2 = 3f_2^2\theta_1^2\theta_2 + 3f_1^2\theta_1\theta_2^2$, which gives the two expressions. With $t = 0$ and $e_\varepsilon = A_1A_2 - q\varepsilon_1\varepsilon_2\theta_1\theta_2 + s(\varepsilon_1\theta_1A_2 + \varepsilon_2\theta_2A_1)$, the class $\sum_\varepsilon w_\varepsilon e_\varepsilon$ equals $\beta'$ exactly when $\sum w_\varepsilon = \sum \varepsilon_1\varepsilon_2 w_\varepsilon = 0$, $s\sum\varepsilon_1 w_\varepsilon = f_1^2$ and $s\sum\varepsilon_2 w_\varepsilon = f_2^2$, whose solution is the one stated. All four coefficients are nonzero because $f_1^2 = f_2^2$ with $f_1f_2 = 1$ would force $f^2 = 1$. For $\alpha_0$ the same computation with $f_1 = f_2 = 1$ gives $w_{+-} = w_{-+} = 0$. The matrix $V$ of Proposition 19.34 has rank two in both cases.

(ii) The ranks are $2\rho + 4N_w$, by Proposition 19.34(ii); $\alpha_0 = v_\Theta$ in the notation of (30), with $d = q$.

(iii) By $\int_X \theta_1^2\theta_2^2 = 4$, $\int \alpha_0\beta' = -2q(f_1^2 + f_2^2 + f_1^{-2} + f_2^{-2}) = -4q\,\mathrm{Tr}(f^2)$, which is negative; the Euler pairings are Proposition 19.34(iii) in the basis given there. The last number is Theorem 19.37(i) with $r^2(\alpha_0) = 12$ and $r^2(\beta') = 20$. ∎

**Corollary 19.45** (Markman's candidate does not satisfy the weakened criterion)**.** *In Proposition 19.44, let $E'$ be the sheaf of* [Mar25c, Example 11.2.7]*, obtained by gluing $N$ general translates of $g^*F'$ to a line bundle on a curve $C'$ at the points where $C'$ meets them, and assume that $C'$ is reduced.*

(i) *$E'$ does not satisfy the weakened criterion, so it does not answer* [Mar25c, Question 11.2.2].

(ii) *$E_0$ does not satisfy the weakened criterion.*

(iii) *For every nonzero perfect complex $G$ on $X$, neither $\Phi(E_0 \boxtimes G)$ nor $\Phi(G \boxtimes E')$ nor $\Phi(G \boxtimes E'^\vee)$ satisfies it; in particular Markman's object $\Phi(E_0 \boxtimes E')$ does not.*

*Proof.* (i) Let $p$ be a smooth point of $C'$ that lies on none of the translates. Near $p$ the sheaf $E'$ is the line bundle on $C'$, so $E'_p \cong \mathcal{O}_{C',p}$, and $\mathrm{ch}(E')$ is a multiple of $\beta' \in S(0,q)$. Theorem 19.43(ii) and (iii) apply with the $j$ for which $T_pC' \not\subseteq T_j$.

(ii) In [Mar25a, Example 8.2.4] the sheaf $E_0 = F_q$ is the push-forward, from the theta divisor $\Theta \subset X$, of the ideal twisted by $\mathcal{O}(\Theta)$ of the union of a translate $W$ of the surface $W_2$ and $q$ translates $C_1, \dots, C_q$ of the curve $W_1$, each meeting $W$ in one point, and its character is $\alpha_0$. Let $p$ be a point of $C_1$ outside $W$, the other $C_i$ and the finite singular locus of $\Theta$. Near $p$, $E_0$ is the ideal of the smooth curve $C_1$ in the smooth divisor $\Theta$. Choose translation invariant fields with $\theta(p) \notin T_p\Theta$ and $\theta'(p) \in T_p\Theta \setminus T_pC_1$. Then $\theta(p) \wedge \theta'(p)$ has nonzero

image in $\nu \otimes N'$, and Lemma 19.42 gives $a_\theta a_{\theta'} \neq 0$. By Proposition 19.44(ii) the class $x = (\frac{q}{2}\,\pi \lrcorner \Theta^2, 0, \pi)$ with $\pi = \theta \wedge \theta'$ preserves $\alpha_0$, and the argument of Theorem 19.43(ii) gives $\mathrm{ev}_{E_0}(x) \neq 0$.

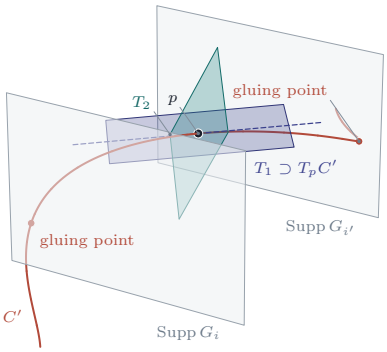
(iii) The first two cases are the last assertion of Theorem 19.43(ii), with the class of (ii) for $E_0$. For $E'^\vee$, the germ at $p$ is the dual of $\mathcal{O}_{C'}$, a shift of a line bundle on $C'$ because $C'$ is a local complete intersection of codimension three near $p$, and $\mathrm{ch}(E'^\vee) = -\mathrm{ch}(E')$ lies in $S(0,q)$. ∎

Figure 38 draws the support of Markman's sheaf near a point of the glued curve, with the two eigenplanes of the real multiplication.

*Remark* 19.46 (What is left of Markman's question). The question [Mar25c, Question 11.2.2] asks for a sheaf $F_2$ on $X$ whose character lies in $S(0,q)$ and forms with $\alpha_0$ a pair as in [Mar25c, Lemma 11.2.1], and whose semiregularity map is injective on the image of $HT^2(X)$ in $\mathrm{Ext}^2(F_2,F_2)$. Corollary 19.45 does not settle it. It decides it for the sheaf $E'$ of the example, and for every choice of $N$, of the translates and of the gluing, because the obstruction sits at any smooth point of the glued curve. By Theorem 19.43, a sheaf that answers the question, and likewise a first factor with character on the secant plane that meets the criterion, has no point at which it is locally free on a smooth curve or on a point, no point at which it is the ideal of a curve in a smooth divisor in the position of Theorem 19.43(iii), and at a point where it is locally free on a smooth surface $S$ the tangent plane $T_pS$ must meet both $T_1$ and $T_2$, by Lemma 19.41 with $c = 2$. Corrections of the character by sheaves on curves, the device of [Mar25c, Example 11.2.7], are therefore excluded. Sheaves that are locally free on smooth divisors, and the global part of the identity $\mathrm{ev}_F(e^{-\theta_t}x_j) = 0$, are not reached by this test. Even a positive answer would give an object with $\dim \mathrm{Ext}^2 \geq 97$, by Theorem 19.37(ii) and Proposition 19.44(iii), so its semiregularity could not come from a dimension count.

19.7. **Sextic CM fields.** The dimension count of Theorem 19.37 does not extend to CM fields of degree six, for a reason of parity (Remark 19.48). Let $F = F_0(\sqrt{-q})$ be a sextic CM field with totally real cubic subfield $F_0$, real places $\tau_1, \tau_2, \tau_3$ and $q \in F_0$ totally positive, and let $X$ be a principally polarised abelian sixfold with real multiplication by $\mathcal{O}_{F_0}$, so that $H^1(X,\mathbb{C}) = U_1 \oplus U_2 \oplus U_3$ with $U_j$ the eigenspace of $\tau_j$, of dimension four and Hodge type $(2,2)$. Let $\theta_j \in H^{1,1}(X)$ be the component of the polarisation on $U_j$, so that $\theta = \sum_j \theta_j$, $\theta_j^3 = 0$ and $\int_X \theta_1^2\theta_2^2\theta_3^2 = 8$. For $t \in F_0$ and $\varepsilon \in \{\pm 1\}^3$ put $s_j = \sqrt{-1}\,\sqrt{\tau_j(q)}$ and $e_\varepsilon = \exp(\sum_j (\tau_j(t) + \varepsilon_j s_j)\theta_j)$, and let $S(t,q)$ be the span of the eight $e_\varepsilon$. As in Proposition 19.34(i) it is a rational subspace of dimension $8 = 2^{[F_0:\mathbb{Q}]}$ of the algebra generated by the $\theta_j$, the Galois group permuting the $e_\varepsilon$ as it permutes the CM types of $F$; for $t = 0$ it is the secant space of Markman's construction for $F$ on $X \times \widehat{X}$ [Mar25c, §11.1], and $S(t,q) = e^{\sum_j \tau_j(t)\theta_j} S(0,q)$. A real class $v = \sum_\varepsilon w_\varepsilon e_\varepsilon$ has $w_{-\varepsilon} = \overline{w}_\varepsilon$. Write $N_w$ for the number of nonzero $w_\varepsilon$; $A_{jj'}$ for the number of sign pairs $(\sigma,\sigma')$ such that $w_\varepsilon \neq 0$ for some $\varepsilon$ with $\varepsilon_j = \sigma$ and $\varepsilon_{j'} = \sigma'$; $\rho_j$ for the rank of the $2 \times 4$ matrix $(w_\varepsilon)$ whose rows are indexed by $\varepsilon_j$ and whose columns by the two other signs; and, for $\{j,j',j''\} = \{1,2,3\}$ and $\sigma' = \pm 1$, $M^{(j;j')}_{\sigma'}$ for the $2 \times 2$ matrix $(w_\varepsilon)_{\varepsilon_j,\varepsilon_{j''}}$ with $\varepsilon_{j'} = \sigma'$ fixed. Let $r^k(v)$ be the rank of the contraction $HT^k(X) \to H^*(X,\mathbb{C})$, $\xi \mapsto \xi \lrcorner v$.

**Proposition 19.47** (The count in degree six)**.** *Let $v \in S(t,q)$ be a nonzero real class.*




$T_0X = T_1 \oplus T_2$: the eigenplanes of $F_0$.
$T_pC'$ lies in at most one of them; here in $T_1$.
$T_2 \cap T_pC' = 0$, so $\pi_2$ has nonzero image in $\bigwedge^2 N$, $N = T_pX/T_pC'$, and the germ of $\mathrm{ev}_{E'}(x_2)$ at $p$ is that image:
$x_2 \lrcorner \mathrm{ch}(E') = 0$ and $\mathrm{ev}_{E'}(x_2) \neq 0$.

pairs $\{k,l\}$ with $a_{\partial_k} a_{\partial_l} \neq 0$

| | | 12 | 13 | 14 | 23 | 24 | 34 | pd |
|---|---|---|---|---|---|---|---|---|
| (i) | $\mathcal{O}_C$, a line bundle on a smooth curve | ■ | ■ | | ■ | | | 3 |
| (ii) | $k(p)$, a skyscraper | ■ | ■ | ■ | ■ | ■ | ■ | 4 |
| (iii) | a curve ideal in a smooth divisor | ■ | ■ | | | | | 2 |
| (iv) | a point ideal in a smooth divisor | ■ | ■ | ■ | ■ | ■ | ■ | 3 |
| (v) | $\mathcal{O}_D$, a smooth divisor | | | | | | | 1 |
| (vi) | two planes meeting at a point | | ■ | ■ | ■ | ■ | | 2 |
| (vii) | maximal Cohen–Macaulay on a node | | | | | | | 1 |
| (viii) | the glued germ of Example 11.2.7 | | | | ■ | ■ | ■ | 3 |
| (ix) | reflexive, non-Cohen–Macaulay, node | ■ | | ■ | ■ | | ■ | 2 |
| (x) | $\mathcal{O}_C^2$, rank two on a smooth curve | ■ | ■ | | ■ | | | 3 |

coordinates adapted to the germ: a filled cell is a nonzero Yoneda square in the local $\mathrm{Ext}^2$; at a point of $C'$ the pairs of normal directions are filled, and one of $\pi_1, \pi_2$ projects onto them.

FIGURE 38. Theorem 19.43 and Corollary 19.45. Left: the support of Markman's sheaf $E'$ near a point $p$ of the glued curve $C'$, between two of the translated supports that the curve crosses at its gluing points. The tangent space at $p$ is the sum of the eigenplanes $T_1$ and $T_2$ of the real multiplication; the tangent line of $C'$ lies in at most one of them, here in $T_1$, so $\pi_2$ has nonzero image in $\bigwedge^2 N$, $N = T_pX/T_pC'$, which is the germ of $\mathrm{ev}_{E'}(x_2)$ at $p$. Right: for ten germs at the origin of $\mathbb{C}^4$, the pairs $\{k,l\}$ of coordinate directions whose jet classes have nonzero product in the local $\mathrm{Ext}^2$, and the projective dimension; rows (i), (ii) and (x) are Lemma 19.41, row (iii) is Lemma 19.42, row (vi) is Lemma 18.60.

(i) $r^1(v) = 12$,

$$r^2(v) = 4\,(A_{12}+A_{13}+A_{23}) + \rho_1 + \rho_2 + \rho_3, \qquad r^3(v) = 8N_w + 2\sum_{j\neq j'} \big(\mathrm{rk}\, M_+^{(j;j')} + \mathrm{rk}\, M_-^{(j;j')}\big).$$

*So $r^2(v) \leq 54$ and $r^3(v) \leq 112$, with equality when every $w_\varepsilon$ is nonzero and every $M_{\sigma'}^{(j;j')}$ is invertible, which holds outside a proper Zariski closed subset of $S(t,q)$. When $F$ contains an imaginary quadratic field $K$ and $v$ lies in the secant plane of $K$, so that $N_w = 2$, one has $(r^2, r^3) = (30, 40)$.*

(ii) $\chi(v,v) = \int_X v^\vee v = -64\,\mathrm{N}_{F_0/\mathbb{Q}}(q) \sum_\varepsilon |w_\varepsilon|^2 < 0$. *For a CM field of degree $2m$, with $X$ of dimension $2m$ and $2^m$ exponentials, the same computation gives $(-4)^m \mathrm{N}(q) \sum_\varepsilon |w_\varepsilon|^2$: the Euler form on the secant space is definite, of sign $(-1)^m$.*

(iii) *Let $F_1, F_2$ be perfect complexes on $X$ with $\mathrm{Ext}^{<0}(F_i, F_i) = 0$ and characters $v_i$ in spaces $S(t_i, q_i)$, let $A = X \times \widehat{X}$ carry the action of $F$ of* [Mar25c, §11.1], *transported by the exponential of a rational $F_0$-B-field, and put $E = \Phi(F_1 \boxtimes F_2^\vee)$ and $\kappa = \mathrm{ch}(E) e^{\ell/2}$. Then $r(\kappa) = r(\mathrm{ch}\, E) = r^2(v_1) + 144 + r^2(v_2) \leq 252$, while every class that stays of Hodge type over $\mathcal{D}_F$ has rank at most $\dim HT^2(A) - \dim \mathcal{D}_F = 276 - 12 = 264$. Moreover $\dim \mathrm{Ext}^2(E,E) \geq r(\kappa)$, with equality if and only if each $F_i$ has*

$$\dim \mathrm{Ext}^k(F_i, F_i) = 1,\ 12,\ r^2(v_i),\ e_3,\ r^2(v_i),\ 12,\ 1 \quad (k = 0, \ldots, 6),$$

*where* $e_3 = 2r^2(v_i) - 22 - \chi(v_i, v_i)$*, and this profile is compatible with the bound* $e_3 \geq r^3(v_i)$ *exactly when* $\chi(F_i, F_i) \leq -(r^3(v_i) - 2r^2(v_i) + 22)$*, that is* $\chi(F_i, F_i) \leq -26$ *for* $v_i$ *general in its secant space, and* $\chi(F_i, F_i) \leq -2$ *when* $N_w = 2$*.*

(iv) *If* $t_1 = t_2 = 0$ *and* $q_1 = q_2 = q$*, then* $\kappa$ *is invariant under* $G_F$*, and so is a Hodge class on every member of the* $(F, 2, 1)$*-family.*

*Proof.* (i) Choose a basis $x_{j1}, x_{j2}$ of $U_j^{1,0}$ and $y_{j1}, y_{j2}$ of $U_j^{0,1}$ with $\theta_j = x_{j1}y_{j1} + x_{j2}y_{j2}$, and let $\omega_j = y_{j1}y_{j2}$ span $\bigwedge^2 U_j^{0,1}$. Then $H^*(X, \mathbb{C})$ and $HT^*(X) = \bigwedge^*(H^{0,1} \oplus T)$ are the tensor products over $j$ of the corresponding algebras of the three factors, $HT^k(X) = \bigoplus_{a_1+a_2+a_3=k} HT_1^{a_1} \otimes HT_2^{a_2} \otimes HT_3^{a_3}$, and contraction is compatible with the factorisation; $v = \sum_\varepsilon w_\varepsilon \bigotimes_j e^{\lambda_j(\varepsilon_j)\theta_j}$ with $\lambda_j(\pm 1) = \tau_j(t) \pm s_j$, two distinct numbers. The blocks of $HT^k(X)$ shift the three Hodge bidegrees of $H^*(X)$ by different amounts, so their images are independent and $r^k(v)$ is the sum of their ranks. Inside a block the bookkeeping rests on two facts about one factor: the two classes $e^{\lambda_j(\pm 1)\theta_j}$ are linearly independent, and so are the subspaces $U_j^{0,1}e^{\lambda_j(\pm 1)\theta_j}$ of $H^*(U_j)$, since $u\, e^{\lambda\theta_j} = u + \lambda\, u\theta_j$ for $u \in U_j^{0,1}$; whereas $\omega_j e^{\lambda\theta_j} = \omega_j$ for every $\lambda$, so that the summands of a block through $HT_j^2$ share the factor $\omega_j$ and are combined, not juxtaposed. On one factor, contraction into $e^{\lambda\theta_j}$ is as follows. For $\xi \in HT_j^1$, $\xi \lrcorner e^{\lambda\theta_j} = y^\lambda(\xi)\, e^{\lambda\theta_j}$ with $y^\lambda(\xi) = \xi^{0,1} + \lambda\, \xi_T \lrcorner \theta_j \in U_j^{0,1}$, as in Lemma 18.55(i); $y^\lambda$ is onto, its kernel $K_j^\lambda$ has dimension two, and $HT_j^1 = K_j^{\lambda_j(+1)} \oplus K_j^{\lambda_j(-1)}$ because $\lambda_j(+1) \neq \lambda_j(-1)$ and $\xi_T \lrcorner \theta_j$ determines $\xi_T$. For $\xi \in HT_j^2$, since $\omega_j\theta_j = 0$, one finds $\omega_j \lrcorner e^{\lambda\theta_j} = \omega_j$, $(y_{ja} \otimes \partial_{x_{jb}}) \lrcorner e^{\lambda\theta_j} = \pm\lambda\, \omega_j$ for $a \neq b$ and 0 for $a = b$, and $(\partial_{x_{j1}} \wedge \partial_{x_{j2}}) \lrcorner e^{\lambda\theta_j} = \pm\lambda^2\omega_j$: the image of $HT_j^2$ is the line $\mathbb{C}\omega_j$, $\xi \lrcorner e^{\lambda\theta_j} = c(\xi, \lambda)\, \omega_j$, and the pairs $\big(c(\xi, \lambda_j(+1)), c(\xi, \lambda_j(-1))\big)$ fill $\mathbb{C}^2$. For $k \geq 3$, $HT_j^k$ kills $e^{\lambda\theta_j}$, because every element of it contains $\omega_j$ or $\partial_{x_{j1}} \wedge \partial_{x_{j2}}$ together with a further factor, and the further factor kills $\omega_j$.

Now count. The block $HT_j^1$ acts on $v$ by $\xi \mapsto \sum_\varepsilon w_\varepsilon\, y^{\lambda_j(\varepsilon_j)}(\xi) \bigotimes_{j'} e^{\lambda_{j'}(\varepsilon_{j'})\theta_{j'}}$; on the summand $K_j^{\lambda_j(-\sigma)}$ only the $\varepsilon$ with $\varepsilon_j = \sigma$ contribute, and they map it isomorphically onto $U_j^{0,1}$ tensored with a nonzero class as soon as one such $w_\varepsilon$ is nonzero. Both signs occur in the support of a nonzero real $w$, since $w_{-\varepsilon} = \overline{w}_\varepsilon$, so the block has rank 4 and $r^1(v) = 12$. The block $HT_j^1 \otimes HT_{j'}^1$ is the sum of the four summands $K_j^{\lambda_j(-\sigma)} \otimes K_{j'}^{\lambda_{j'}(-\sigma')}$, each of dimension four; the summand indexed by $(\sigma, \sigma')$ maps isomorphically onto $U_j^{0,1} \otimes U_{j'}^{0,1}$ tensored with the class $\sum_{\sigma''} w_\varepsilon e^{\lambda_{j''}(\sigma'')\theta_{j''}}$, which is nonzero exactly when the pair $(\sigma, \sigma')$ occurs in the support, and to zero otherwise; so the block has rank $4A_{jj'}$. The block $HT_j^2$ maps $\xi$ to $\sum_\sigma c(\xi, \lambda_j(\sigma))\, z_\sigma$ with $z_\sigma = \omega_j \otimes \sum_\mu w_{\sigma\mu} e_\mu$, where $\mu$ runs over the four sign patterns of the other two places and the $e_\mu$ are independent; as $(c(\xi, \lambda_j(+1)), c(\xi, \lambda_j(-1)))$ fills $\mathbb{C}^2$, the rank is the dimension of the span of $z_+, z_-$, which is $\rho_j$. This gives $r^2(v)$. In degree three the block $HT_1^1 \otimes HT_2^1 \otimes HT_3^1$ is the sum of eight summands $\bigotimes_j K_j^{\lambda_j(-\varepsilon_j)}$, of dimension eight, on each of which exactly one term of $v$ acts, isomorphically when $w_\varepsilon \neq 0$ and by zero otherwise: rank $8N_w$. The block $HT_j^2 \otimes HT_{j'}^1$ factors through $\mathbb{C}^2 \otimes HT_{j'}^1$ by $c \otimes \mathrm{id}$, which is onto, and on $\mathbb{C}^2 \otimes K_{j'}^{\lambda_{j'}(-\sigma')}$ the map is $M_{\sigma'}^{(j;j')} \otimes y^{\lambda_{j'}(\sigma')}$ followed by tensoring with independent classes, of rank $2 \operatorname{rk} M_{\sigma'}^{(j;j')}$; the two values of $\sigma'$ give independent images. The blocks $HT_j^3$ contribute nothing. This gives $r^3(v)$. The maximal values are $48 + 6 = 54$ and $64 + 6 \cdot 8 = 112$; the conditions for them are the nonvanishing of the $w_\varepsilon$ and of the

determinants of the $M_{\sigma'}^{(j;j')}$, which are polynomials in the coordinates of $v$, nonzero on $S(t,q)$. If $F \supseteq K$ and $v$ lies in the secant plane of $K$, the support of $w$ is a pair $\{\varepsilon_0, -\varepsilon_0\}$, the two CM types of $F$ induced from $K$; then $A_{jj'} = 2$, $\rho_j = 2$, $N_w = 2$ and every $M_{\sigma'}^{(j;j')}$ has one nonzero entry, so $(r^2, r^3) = (24+6, 16+24)$.

(ii) Since $\theta_j^3 = 0$, $\theta^{2m} = \frac{(2m)!}{2^m}\theta_1^2 \cdots \theta_m^2$, and $\int_X \theta^{2m}/(2m)! = 1$ gives $\int_X \theta_1^2 \cdots \theta_m^2 = 2^m$ and $\int_X \exp(\sum_j c_j\theta_j) = \prod_j c_j^2$. The dual of $e^{\lambda\theta_j}$ is $e^{-\lambda\theta_j}$, so $\int_X e_\varepsilon^\vee e_{\varepsilon'} = \prod_j (\lambda_j(\varepsilon_j') - \lambda_j(\varepsilon_j))^2$, which is 0 unless $\varepsilon' = -\varepsilon$, where it is $\prod_j (2s_j)^2 = (-4)^m \mathrm{N}(q)$. Hence $\chi(v,v) = (-4)^m \mathrm{N}(q) \sum_\varepsilon w_\varepsilon w_{-\varepsilon} = (-4)^m \mathrm{N}(q) \sum_\varepsilon |w_\varepsilon|^2$.

(iii) As in the proof of Theorem 19.37(i), $r(\kappa) = r(\mathrm{ch}\, E)$ by Lemma 18.55(i), and $\Phi$ identifies $r(\mathrm{ch}\, E)$ with the rank of contraction from $HT^2(X \times X) = \bigoplus_{a+b=2} HT^a(X) \otimes HT^b(X)$ into $v_1 \boxtimes v_2$; the block $HT^1 \otimes HT^1$ contributes $r^1(v_1) r^1(v_2) = 144$ and the other two contribute $r^2(v_1) + r^2(v_2)$, by the type argument given there. The tangent space $T_{[A]}\mathcal{D}_F \subseteq H^1(T_A)$, of dimension $mn^2 = 12$, kills every class that stays of Hodge type along $\mathcal{D}_F$, as in Proposition 17.11(i), and $\dim HT^2(A) = 66 + 144 + 66$. By the Künneth formula, with $\mathrm{Ext}^{<0} = 0$, $\dim \mathrm{Ext}^2(E,E) = e_0 e_2' + e_1 e_1' + e_2 e_0'$ where $e_k = \dim \mathrm{Ext}^k(F_1, F_1)$ and $e_k' = \dim \mathrm{Ext}^k(F_2, F_2)$; the commutative square $\sigma \circ \mathrm{ev} = (\,\cdot\,) \lrcorner \mathrm{ch}$ of Setup 18.54, valid in every degree, gives $e_k \geq r^k(v_1)$ and $e_k' \geq r^k(v_2)$, and $e_0, e_0' \geq 1$. So $\dim \mathrm{Ext}^2(E,E) \geq r^2(v_2) + 144 + r^2(v_1)$, with equality if and only if $e_0 = e_0' = 1$, $e_1 = e_1' = 12$, $e_2 = r^2(v_1)$ and $e_2' = r^2(v_2)$. Serre duality on the sixfold gives $e_k = e_{6-k}$, so $\chi(F_i, F_i) = 2(e_0 - e_1 + e_2) - e_3 = 2r^2(v_i) - 22 - e_3$, and by Riemann–Roch on an abelian variety $\chi(F_i, F_i) = \chi(v_i, v_i)$, which is (ii); this determines $e_3$, and $e_3 \geq r^3(v_i)$ is the stated inequality. With the maximal ranks it reads $\chi \leq -(112 - 108 + 22) = -26$, and with $N_w = 2$ it reads $\chi \leq -(40 - 60 + 22) = -2$.

(iv) The proof of Theorem 19.37(iii) applies with three factors in place of two: for $t = 0$ the classes $e_\varepsilon \boxtimes e_{\varepsilon'}^\vee$, the correspondence $Z \mapsto \mathrm{ch}(\Phi(Z)) e^{\ell/2}$ and the Poincaré class decompose as tensor products over $j$ of classes of even degree, on each factor $\sqrt{-q}$ acts as in the imaginary quadratic construction with $d$ replaced by $\tau_j(q)$, and the identities of Proposition 17.21 at $n = 2$ show that each image is a multiple of an exponential of $\eta_j$ or of a Weil class of the $j$-th factor, invariant under the special unitary group of that factor; $G_F(\mathbb{R})$ is the product of the three, and is Zariski dense in $G_F$. Statement (iv) is the case $[F : \mathbb{Q}] = 6$ of [Mar25c, Proposition 10.2.1(1)], with the inclusion $\widetilde{\phi}(B \otimes B) \subset A$ of [Mar25c, p. 33], which Markman proves for every CM field in his conventions for $\Phi$ and for the twist; we make no claim of priority. ∎

*Remark* 19.48 (The special role of degree four). In Theorem 19.37 the equality $\dim \mathrm{Ext}^2(E,E) = r(\kappa)$ determines the whole Ext profile of each factor, because on a fourfold $\mathrm{Ext}^2$ is the middle degree and Serre duality supplies the rest; the Euler characteristic of a minimal factor is then the small positive integer $r^2 - 14$, the Euler form on the secant space being positive definite, and the arithmetic of Proposition 19.36 excludes it. On a sixfold the same equality fixes $\mathrm{Ext}^0$, $\mathrm{Ext}^1$ and $\mathrm{Ext}^2$ and leaves the middle group $\mathrm{Ext}^3$ free, and the Euler form has the opposite sign: $\chi(F_i, F_i) = -64\mathrm{N}(q) \sum |w_\varepsilon|^2$ is a negative integer that $e_3$ absorbs. So no dimension count excludes a minimal Orlov product over a sextic field, and none does over any CM field of degree $2m \geq 6$, where $\mathrm{Ext}^3, \ldots, \mathrm{Ext}^m$ are free and the Euler form has sign $(-1)^m$. What Proposition 19.47(iii) leaves for such an object is the profile displayed there, with $\chi(F_i, F_i) \leq -26$ for a general secant character, and the flat twisted character of (iv), which stays of Hodge type over the whole twelve-dimensional period domain. Whether a sheaf with that profile exists on a

sixfold with real multiplication by a cubic field is not known; nothing in Theorem 19.37 bears on it. A certificate of the kind of Theorem 17.19 for a sextic field would be an Orlov product of such sheaves, with $\dim \mathrm{Ext}^2(E,E) = r(\kappa) \le 252$ on the twelvefold, whose twisted character has a nonzero component in the Weil classes $W_F(A)$; flatness does not give that component, and Lemma 19.50 computes it. In degree six as in degree four, the Orlov products never reach the rank of a class of general shape: $r(\kappa) \le 252$ against 264, and $r(\kappa) \le 104$ against 112.

The count of Proposition 19.47(iii) is a condition on the Euler characteristic of the character alone, so it can be tested on the lattice of integral characters, as Proposition 19.36 tests the quartic one. In degree four the test excludes every minimal character. In degree six it can exclude an integral character only if $-26 < \chi(v,v) < 0$ (Proposition 19.49(i),(ii)), and it excludes neither of the integral classes $\mathrm{Re}\, e^{\sqrt{-1}\,\theta}$ and $\mathrm{Im}\, e^{\sqrt{-1}\,\theta}$ (Proposition 19.49(iv)). For $t = 0$, the case in which Proposition 19.47(iv) makes $\kappa$ flat, put $B_j = 1 - \frac{1}{2}\tau_j(q)\,\theta_j^2$, so that $e^{\pm s_j\theta_j} = B_j \pm s_j\theta_j$, and for $c_0, c_3 \in \mathbb{Q}$ and $f, g \in F_0$ let

$$v(c_0, f, g, c_3) = c_0 B_1B_2B_3 + \sum_j \tau_j(f)\,\theta_j B_k B_l + \sum_j \tau_j(g)\,B_j\theta_k\theta_l + c_3\,\theta_1\theta_2\theta_3,$$

where $\{j,k,l\} = \{1,2,3\}$ in each sum.

**Proposition 19.49** (Integral flat characters in degree six)**.** *Keep the notation of Proposition 19.47.*

(i) *For every perfect complex $F$ on a complex torus of dimension six, $\chi(F,F)$ is even. So the condition of Proposition 19.47(iii) holds for every $F$ whose character has $N_w = 2$.*

(ii) *For every nonzero real $v \in S(t,q)$, $r^3(v) - 2r^2(v) \le 4$, with equality only if $(r^2(v), r^3(v)) = (54, 112)$. So for every shape the condition of Proposition 19.47(iii) is implied by $\chi(F_i,F_i) \le -26$.*

(iii) *The rational points of $S(0,q)$ are the classes $v(c_0,f,g,c_3)$, and*

$$\chi(v,v) = -8\big(\mathrm{N}(q)\,c_0^2 + \mathrm{Tr}(\mathrm{N}(q)\,q^{-1}f^2) + \mathrm{Tr}(q\,g^2) + c_3^2\big).$$

(iv) *For $q = 1$, the classes $\mathrm{Re}\, e^{\sqrt{-1}\,\theta} = v(1,0,-1,0)$ and $\mathrm{Im}\, e^{\sqrt{-1}\,\theta} = v(0,1,0,-1)$ are integral points of $S(0,1)$ with $\chi = -32$ and $N_w = 2$, so they satisfy the condition of Proposition 19.47(iii).*

*Proof.* (i) The Chern character $v = \mathrm{ch}(F)$ is integral, as in the proof of Theorem 19.37(ii), and $\chi(F,F) = \int_X v^\vee v$ by Riemann–Roch. Write $v = \sum_k v_{2k}$, so that $\int_X v^\vee v = \sum_k (-1)^k \int_X v_{2k} v_{12-2k}$. The terms $k$ and $6-k$ are equal, since $(-1)^k = (-1)^{6-k}$ and classes of even degree commute, so they contribute an even number when $k \ne 3$. For $k = 3$ write $v_6 = \sum_I a_I e_I$ in the basis of $\bigwedge^6 H^1(X,\mathbb{Z})$ given by the products $e_I$ of six elements of a basis of $H^1(X,\mathbb{Z})$; since $e_Ie_I = 0$ and $e_Ie_J = e_Je_I$, $v_6^2 = 2\sum_{I<J} a_Ia_Je_Ie_J$ is even. For $N_w = 2$ the condition of Proposition 19.47(iii) is $\chi(F,F) \le -2$, and $\chi(F,F)$ is a negative even integer by Proposition 19.47(ii).

(ii) For $j$ and $\sigma = \pm 1$ let $W_{j,\sigma}$ be the $2\times 2$ slice $(w_\varepsilon)_{\varepsilon_j = \sigma}$, with rows and columns indexed by the signs at the two other places. The matrices $M_\sigma^{(j';j)}$ are $W_{j,\sigma}$ and its transpose, so $\sum_{j\ne j'}\big(\mathrm{rk}\, M_+^{(j;j')} + \mathrm{rk}\, M_-^{(j;j')}\big) = 2\sum_{j,\sigma}\mathrm{rk}\, W_{j,\sigma}$. Since $w_{-\varepsilon} = \overline{w}_\varepsilon$, $W_{j,-}$ is the complex conjugate of $W_{j,+}$ with its rows and its columns reversed, so the two slices have the same

rank $k_j$. Let $S$ be the support of $w$ and $N = N_w = |S|$. By Proposition 19.47(i),

$$r^3(v) - 2r^2(v) = 8N - 8\,(A_{12} + A_{13} + A_{23}) + \sum_j (8k_j - 2\rho_j). \tag{71}$$

The set $S$ is stable under $\varepsilon \mapsto -\varepsilon$, which exchanges the two slices along each place; so $N$ is even, each slice has $N/2$ nonzero entries, $k_j \geq 1$, and $A_{jj'} \in \{2, 4\}$, the projection of $S$ to the signs at $j$ and $j'$ being stable under negation. Also $1 \leq \rho_j \leq 2$ and $k_j \leq 2$, and a slice whose nonzero entries lie in one row or one column has rank one. If $\rho_j = 1$, then $w_\varepsilon = a_{\varepsilon_j} W(\varepsilon')$, with $\varepsilon'$ the signs at the two other places, so every slice along a place $j' \neq j$ is $a \otimes$ (a row of $W$) and $k_{j'} \leq 1$. Now take the four values of $N$ in turn.

If $N = 8$, every $A_{jj'}$ is 4 and (71) is $-32 + \sum_j (8k_j - 2\rho_j)$. When every $\rho_j$ is 2 this is at most $-32 + 3 \cdot 12 = 4$, with equality exactly when every $k_j$ is 2, and then $r^2 = 48 + 6 = 54$ and $r^3 = 64 + 48 = 112$. When some $\rho_j$ is 1, the two other $k$ are at most 1 and (71) is at most $-32 + 14 + 6 + 6 = -6$.

If $N = 6$, $S$ misses one pair $\{\pm\varepsilon_0\}$. As $\varepsilon_0$ and $-\varepsilon_0$ differ at every place, they never have the same signs at two places, so every sign pair at two places occurs in $S$, every $A_{jj'}$ is 4, and (71) is at most $48 - 96 + 3 \cdot 14 = -6$.

If $N = 4$, $S = \{\pm\varepsilon, \pm\varepsilon'\}$, and after replacing $\varepsilon'$ by $-\varepsilon'$ if necessary, $\varepsilon$ and $\varepsilon'$ differ at exactly one place $j$. For the two other places $k$ and $l$ one has $A_{kl} = 2$ and $A_{jk} = A_{jl} = 4$. The two nonzero entries of a slice along $k$ are $w_{\pm\varepsilon}$ and $w_{\pm\varepsilon'}$ with the same sign, whose indices agree at $l$; they lie in one row or one column, so $k_k = 1$, and likewise $k_l = 1$. Hence (71) is at most $32 - 80 + 8\,(2 + 1 + 1) - 6 = -22$.

If $N = 2$, $S = \{\pm\varepsilon_0\}$, every $A_{jj'}$ is 2, every $k_j$ is 1, and every $\rho_j$ is 2, the two slices along $j$ having their nonzero entries at different positions; so (71) is $16 - 48 + 24 - 12 = -20$, the value $40 - 60$ of Proposition 19.47(i).

So $r^3 - 2r^2 \leq 4$, with equality only at $(54, 112)$, and the threshold $r^3 - 2r^2 + 22$ of Proposition 19.47(iii) is at most 26.

(iii) Since $\theta_j^3 = 0$ and $s_j^2 = -\tau_j(q)$, $e^{\pm s_j\theta_j} = B_j \pm s_j\theta_j$, so the eight exponentials and the eight products $\prod_j \beta_j$ with $\beta_j \in \{B_j, \theta_j\}$ span the same complex space, the change of basis being the tensor cube of an invertible $2 \times 2$ matrix. The Galois group permutes the $\tau_j$, the $\theta_j$ and the $B_j$ compatibly, so $\sum_a C_a \prod_j \beta_j$, indexed by the subsets $a$ of $\{1, 2, 3\}$ with $\beta_j = \theta_j$ for $j \in a$, is rational exactly when $C_{\pi(a)} = \sigma(C_a)$ for every $\sigma$ inducing $\pi$; the stabiliser of $\{j\}$, and of its complement, is the stabiliser of $\tau_j$, which gives the form $v(c_0, f, g, c_3)$. The involution $v \mapsto v^\vee$ and the integral are multiplicative over the three factors, with $\int_X \theta_1^2\theta_2^2\theta_3^2 = 8$; on one factor $B_j^\vee B_j = 1 - \tau_j(q)\theta_j^2$, $\theta_j^\vee\theta_j = -\theta_j^2$ and the mixed products are odd in $\theta_j$. So $\chi(v, v) = -8 \sum_a C_a^2 \prod_{j \notin a} \tau_j(q)$, which is the formula.

(iv) For $q = 1$, $s_j = \sqrt{-1}$ and $e_{(1,1,1)} = \prod_j e^{\sqrt{-1}\,\theta_j} = e^{\sqrt{-1}\,\theta}$. The polarisation is principal, so each $\theta^k/k!$ is integral, and so are the real and imaginary parts $\sum_k (-1)^k\theta^{2k}/(2k)!$ and $\sum_k (-1)^k\theta^{2k+1}/(2k+1)!$. They are $\frac{1}{2}(e_{(1,1,1)} + e_{(-1,-1,-1)})$ and $\frac{1}{2\sqrt{-1}}(e_{(1,1,1)} - e_{(-1,-1,-1)})$, so $N_w = 2$ and $\chi = -64\,(\frac{1}{4} + \frac{1}{4}) = -32$ by Proposition 19.47(ii); expanding $\prod_j (B_j + \sqrt{-1}\,\theta_j)$ gives the coordinates. ∎

The flatness of Proposition 19.47(iv) makes $\kappa$ a Hodge class on every member of the family, but in degree four the Hodge classes of a very general member are the products of divisor classes and the Weil classes $W_F(A)$ (Proposition 19.3(iii)), and only a nonzero component of $\kappa$ in $W_F(A)$ bears on $\mathbf{W}(F, 2, 1)$. We compute that component. Keep the notation of Proposition 19.47, with $t_1 = t_2 = 0$, $q_1 = q_2 = q$ and trivial $B$-field, put $d_j = \tau_j(q)$, and let $\Psi(Z) = \mathrm{ch}(\Phi(Z))e^{\ell/2}$ as in Proposition 17.21, so that $\kappa(v_1, v_2) = \Psi(v_1 \boxtimes v_2^\vee)$

for the characters $v_i$ of $F_i$. As in the proof of Proposition 19.47(iv), $H^1(A,\mathbb{C})$ is the sum of three factors $U_j \oplus U_j^\vee$, on each of which $\sqrt{-q}$ acts as in the imaginary quadratic construction with $\beta = \theta_j$ and $d = d_j$. On the $j$-th factor write $\widehat{\theta}_j$ and $\ell_j$ for the components of the dual polarisation and of the Poincaré class, and put $\gamma_j = d_j\theta_j - \widehat{\theta}_j$, as in (33), and $\eta_j = d_j\theta_j + \widehat{\theta}_j$, a positive multiple of the polarisation of the factor for which $\sqrt{-q}$ is anti-self-adjoint (Proposition 15.19(i)). For $\sigma = \pm 1$ let $\tau_{j,\sigma}$ be the embedding of $F$ over $\tau_j$ with $\tau_{j,\sigma}(\sqrt{-q}) = \sigma\sqrt{-1}\,\sqrt{\tau_j(q)} = \sigma s_j$, and $V_{j,\sigma} \subset H^1(A,\mathbb{C})$ its eigenspace, of dimension four. By Theorem 15.2(ii) with $n = 2$ and $d = d_j$, the line $\bigwedge^4 V_{j,\sigma}$ is spanned by

$$\omega_{j,\sigma} = -\frac{1}{2d_j^2}\bigl(\gamma_j - \sigma s_j \ell_j\bigr)^2, \tag{72}$$

so the six classes $\omega_{j,\sigma}$ span $W_F(A)\otimes\mathbb{C}$. Since $\eta_j$ pairs $V_{j,+}$ with $V_{j,-}$, the products $\eta_i\eta_k$ lie in summands of $H^4(A,\mathbb{C}) = \bigwedge^4\bigl(\bigoplus_{j,\sigma} V_{j,\sigma}\bigr)$ other than the six lines $\bigwedge^4 V_{j,\sigma}$. Let $\kappa_{j,\sigma}(v_1,v_2)$ be the coefficient of $\omega_{j,\sigma}$ in the component of $\kappa(v_1,v_2)$ in $\bigwedge^4 V_{j,\sigma}$; the *$F$-Weil part* of $\kappa$ is $\sum_{j,\sigma}\kappa_{j,\sigma}\,\omega_{j,\sigma}$, its component in $W_F(A)\otimes\mathbb{C}$. For $\varepsilon \in \{\pm1\}^3$ let $\varepsilon^{(j)}$ be the sign vector that agrees with $\varepsilon$ at $j$ and only there.

**Lemma 19.50** (The $F$-Weil part of an Orlov product)**.** *Let $v_i = \sum_\varepsilon w_\varepsilon^i e_\varepsilon \in S(0,q)$ for $i = 1,2$.*

(i) *The degree-four part of $\kappa(v_1,v_2)$ is a quadratic form in the $\eta_j$ plus $\sum_{j,\sigma}\kappa_{j,\sigma}\,\omega_{j,\sigma}$, with*

$$\kappa_{j,\sigma}(v_1,v_2) = -16\mathrm{N}(q)\,\tau_j(q)\sum_{\varepsilon_j=\sigma} w_\varepsilon^1\, w_{\varepsilon^{(j)}}^2.$$

(ii) *Let $v = v(c_0,f,g,c_3)$ be rational, and for $\{j,k,l\} = \{1,2,3\}$ write $f_j = \tau_j(f)$, $g_j = \tau_j(g)$ and $q_j = \tau_j(q)$. The numbers*

$$R_j = c_0^2 + \frac{f_k^2}{q_k} + \frac{f_l^2}{q_l} + \frac{g_j^2}{q_kq_l} - \frac{1}{q_j}\Bigl(f_j^2 + \frac{g_l^2}{q_k} + \frac{g_k^2}{q_l} + \frac{c_3^2}{q_kq_l}\Bigr),$$

$$I_j = c_0 f_j + \frac{f_k g_l}{q_k} + \frac{f_l g_k}{q_l} + \frac{c_3 g_j}{q_kq_l}$$

*are $\tau_j(R)$ and $\tau_j(I)$ for elements $R, I \in F_0$. Put $\Omega(v) = R + 2I/\sqrt{-q} \in F$. Then*

$$\kappa_{j,\sigma}(v,v) = -\mathrm{N}(q)\,\tau_j(q)\,\tau_{j,\sigma}\bigl(\Omega(v)\bigr).$$

*So the $F$-Weil part of $\kappa(v,v)$ is zero if $R = I = 0$, and otherwise all six $\kappa_{j,\sigma}(v,v)$ are nonzero.*

(iii) *If $v_1$ and $v_2$ are supported on one pair $\{\varepsilon_0,-\varepsilon_0\}$, the $F$-Weil part of $\kappa(v_1,v_2)$ is zero. For $q = 1$, $F$ contains $K = \mathbb{Q}(\sqrt{-1})$, the classes $\mathrm{Re}\,e^{\sqrt{-1}\,\theta}$ and $\mathrm{Im}\,e^{\sqrt{-1}\,\theta}$ span the plane $P_\theta$ of Definition 14.4 with $d = 1$, and for $v_i = a_i\,\mathrm{Re}\,e^{\sqrt{-1}\,\theta} + b_i\,\mathrm{Im}\,e^{\sqrt{-1}\,\theta}$ with $a_i, b_i \in \mathbb{Q}$ the class $\kappa(v_1,v_2)$ lies in $\mathbb{Q}[\eta]\oplus W_K(A)$, where $\eta = \theta + \widehat{\theta}$ and $W_K(A) \subseteq H^{12}(A,\mathbb{Q})$ are the Weil classes of $K$ on the twelvefold (Proposition 19.10); its part of degree at most four is*

$$-32(a_1a_2 + b_1b_2) + 16(a_1b_2 - a_2b_1)\,\eta + 4(a_1a_2 + b_1b_2)\,\eta^2.$$

*Proof.* (i) The classes $e_\varepsilon \boxtimes e_{\varepsilon'}^\vee = \bigotimes_j\bigl(e^{\varepsilon_j s_j\theta_j} \boxtimes e^{-\varepsilon'_j s_j\theta_j}\bigr)$ and $\Psi$ decompose over the three factors, as in the proof of Proposition 19.47(iv), and on the $j$-th factor the two identities of Proposition 17.21 at $n = 2$ read, with their constants,

$$\Psi_j\bigl(e^{\sigma s_j\theta_j}\boxtimes e^{\sigma s_j\theta_j}\bigr) = -4d_j\,e^{\sigma(s_j/2d_j)\eta_j}, \qquad \Psi_j\bigl(e^{\sigma s_j\theta_j}\boxtimes e^{-\sigma s_j\theta_j}\bigr) = \tfrac{1}{2}\bigl(\gamma_j - \sigma s_j\ell_j\bigr)^2 = -d_j^2\,\omega_{j,\sigma},$$

since $s_j^2 = -d_j$. So $\Psi(e_\varepsilon \boxtimes e_{\varepsilon'}^\vee)$ is the product over $j$ of the first class where $\varepsilon'_j = -\varepsilon_j$ and of the second where $\varepsilon'_j = \varepsilon_j$. In degree four a product with two or three factors of the second kind vanishes, one with none is a quadratic form in the $\eta_j$, and one with exactly one, at $j$, has $\varepsilon' = \varepsilon^{(j)}$ and equals $-d_j^2 \prod_{i\neq j}(-4d_i)\,\omega_{j,\varepsilon_j} = -16\mathrm{N}(q)\,d_j\,\omega_{j,\varepsilon_j}$. Summing $w^1_\varepsilon w^2_{\varepsilon'}\Psi(e_\varepsilon \boxtimes e^\vee_{\varepsilon'})$ over $\varepsilon$ and $\varepsilon'$ gives (i).

(ii) An element $h$ of the Galois group that fixes the place $j$ permutes the two others, so it fixes $R_j$ and $I_j$, which are symmetric in $k$ and $l$; hence $R_j, I_j \in \tau_j(F_0)$, and moving $j$ by the Galois group shows that they are the conjugates of two elements of $F_0$. Write $\varepsilon = (\sigma, \mu)$ with $\mu$ the signs at $k$ and $l$, so that $\varepsilon^{(j)} = (\sigma, -\mu)$. Since $e^{\pm s_i\theta_i} = B_i \pm s_i\theta_i$, the class $v = \sum_a C_a \prod_{i\in a}\theta_i \prod_{i\notin a} B_i$, with $C_\emptyset = c_0$, $C_{\{i\}} = f_i$, $C_{\{1,2,3\}\setminus\{i\}} = g_i$ and $C_{\{1,2,3\}} = c_3$, has $w_\varepsilon = \frac18 \sum_a C_a \prod_{i\in a}\varepsilon_i/s_i$. Grouping $a = b$ and $a = b\cup\{j\}$ for $b \subseteq \{k,l\}$ gives $w_{(\sigma,\mu)} = \frac18 \sum_b D_b \prod_{i\in b}\mu_i/s_i$ with $D_b = C_b + u\,C_{b\cup\{j\}}$ and $u = \sigma/s_j = \tau_{j,\sigma}(1/\sqrt{-q})$. Since $\sum_\mu \prod_{i\in b}\mu_i \prod_{i\in b'}(-\mu_i) = 4(-1)^{|b|}\delta_{bb'}$ and $s_i^2 = -q_i$,

$$\sum_{\varepsilon_j=\sigma} w_\varepsilon w_{\varepsilon^{(j)}} = \frac{1}{16}\Big((c_0+uf_j)^2 + \frac{(f_k+ug_l)^2}{q_k} + \frac{(f_l+ug_k)^2}{q_l} + \frac{(g_j+uc_3)^2}{q_kq_l}\Big),$$

and expanding with $u^2 = -1/q_j$ gives $\frac{1}{16}(R_j + 2uI_j) = \frac{1}{16}\tau_{j,\sigma}(\Omega(v))$. With (i) this is the formula. The embedding $\tau_{j,\sigma}$ is injective and $1$, $1/\sqrt{-q}$ are independent over $F_0$, so one $\kappa_{j,\sigma}(v,v)$ vanishes if and only if $R = I = 0$, and then all do.

(iii) If $w^1$ and $w^2$ vanish off $\{\varepsilon_0, -\varepsilon_0\}$, no term of (i) survives, since $\varepsilon^{(j)}$ is neither $\varepsilon$ nor $-\varepsilon$. For $q = 1$, $s_j = \sqrt{-1}$ and $e_{(1,1,1)} = e^{\sqrt{-1}\,\theta}$, whose real and imaginary parts are $u_\theta$ and $v_\theta$ of (30) with $d = 1$, supported on $\{\pm(1,1,1)\}$. On each factor $\sqrt{-1} = \sqrt{-q}$ acts as in the imaginary quadratic construction with $\beta = \theta_j$ and $d = 1$, so $K$ acts on $A$ by the action (29) determined by $\theta$ with $d = 1$, and Proposition 17.21 with $n = 6$ and $d = 1$ gives $\kappa \in \mathbb{Q}[\eta] \oplus W_K(A)$. For the constants write $v_1 = w_1 e^{\sqrt{-1}\,\theta} + \overline{w}_1 e^{-\sqrt{-1}\,\theta}$ and $v_2^\vee = a_2 \operatorname{Re} e^{\sqrt{-1}\,\theta} - b_2 \operatorname{Im} e^{\sqrt{-1}\,\theta} = w_2 e^{\sqrt{-1}\,\theta} + \overline{w}_2 e^{-\sqrt{-1}\,\theta}$, with $w_1 = \frac12(a_1 - \sqrt{-1}\,b_1)$ and $w_2 = \frac12(a_2 + \sqrt{-1}\,b_2)$. By the first identity in the proof of (i), $\Psi(e^{\pm\sqrt{-1}\,\theta} \boxtimes e^{\pm\sqrt{-1}\,\theta}) = -64\,e^{\pm(\sqrt{-1}/2)\eta}$, while the two mixed terms have degree twelve; so in degrees below twelve $\kappa$ is $-128\operatorname{Re}(w_1 w_2\, e^{(\sqrt{-1}/2)\eta})$, whose terms of degree 0, 2 and 4 are the stated ones. ∎

Markman states his reduction for the normalised class $\mathrm{ch}(E)\exp(-c_1(E)/\operatorname{rk} E)$, and it is the component of degree four of that class, a polynomial in the invariant divisor classes plus a Weil class, whose algebraicity on the deformations he needs [Mar25c, Proposition 10.2.1, Corollary 10.2.3]. When $\operatorname{rk} E \neq 0$ that class is $\kappa\exp(-\kappa_1/\operatorname{rk} E)$, with $\kappa_1$ the degree-two part of $\kappa$, which is a linear form in the $\eta_j$ by the proof of Lemma 19.50(i); so it has the same $F$-Weil part as $\kappa$.

For $q = 1$ the Orlov products of complexes with characters in $\mathbb{Q}\operatorname{Re} e^{\sqrt{-1}\,\theta} \oplus \mathbb{Q}\operatorname{Im} e^{\sqrt{-1}\,\theta}$ are therefore the template of Proposition 17.22 at $(n,d) = (6,1)$, seen on the twelvefold through the restriction of the action of $F$ to $K$. Their twisted characters are flat for the family of $K$ and have a nonzero $K$-Weil part in $H^{12}$ when $F_1 = F_2 \neq 0$ (Proposition 17.21), and they are not an input to $\mathrm{P2}_{\mathrm{split}}$. By Proposition 19.10, $\mathbf{W}(F,2,1)$ gives the $K$-Weil classes on the members of the $F$-family, and the converse is not available; a consequence for $K$ through Proposition 20.13 would still need a twisted form of the propagation of Theorem 16.29.

*Remark* 19.51 (Natural objects). Let $X$ be very general with real multiplication by $\mathcal{O}_{F_0}$, so that $\mathrm{NS}(X)_\mathbb{Q} = \{\theta_t = \sum_j \tau_j(t)\theta_j : t \in F_0\}$. Call *natural* the line bundles, the simple

semi-homogeneous bundles of [Muk78], the skyscraper sheaves of points, and the objects obtained from these by shifts, duals, twists by line bundles, pull-back by automorphisms commuting with $\mathcal{O}_{F_0}$, and $\Phi_{\mathcal{P}}$ followed by $\widehat{X} \cong X$. On the $j$-th factor write classes in $\langle 1, \theta_j, \theta_j^2/2\rangle$ as $(r, c, s)$, and put $\nu(t) = e^{t\theta_j} = (1, t, t^2)$ and $\nu(\infty) = (0, 0, 1)$. The character of a natural object is a multiple of $\bigotimes_j \nu(\tau_j(t))$ for one $t \in F_0 \cup \{\infty\}$, its slope: a simple semi-homogeneous bundle has character $r\, e^{\theta_t}$ [Muk78], a point has slope $\infty$, and the operations act on slopes by $t \mapsto -t$, $t + t'$, $tu^2$ with $u \in \mathcal{O}_{F_0}^{\times}$, and $-1/t$, the last by Theorem 6.5 applied factor by factor. Then

(a) no natural character lies in $S(0, q)$, for any totally positive $q$;

(b) if a real combination of natural characters of $k$ pairwise distinct slopes is a nonzero element of $S(0, q)$, then $k \geq 6$, and if $k = 6$ it has $N_w = 2$.

So no cone of a morphism between two natural objects has a nonzero character in $S(0, q)$, and by Lemma 19.50(iii) a class whose Orlov square has a nonzero $F$-Weil part needs at least seven slopes. Six are attained: $\operatorname{Re} e^{\sqrt{-1}\,\theta} = \frac{5}{2} - \frac{5}{6}(e^{\theta} + e^{-\theta}) + \frac{1}{12}(e^{2\theta} + e^{-2\theta}) - 10[\mathrm{pt}]$.

For (a): $S(0, q)$ is the tensor product of the planes $P_j = \langle B_j, \theta_j\rangle = \{s = -d_j r\}$, the form $c^2 - rs$, for which $\nu(t)$ and $\nu(\infty)$ are isotropic, is positive definite on $P_j$, and a decomposable tensor lies in $\bigotimes_j P_j$ only if each factor does. For (b), let $T = \sum_i a_i \bigotimes_j \nu(\tau_j(t_i))$ and $\lambda_1 = s + d_1 r$, which is positive on $\nu(t)$ and $\nu(\infty)$. Applying $\lambda_1$ to the first factor gives a relation with nonzero coefficients among the images, under the embedding of $\mathbb{P}^1 \times \mathbb{P}^1$ by forms of bidegree $(2, 2)$, of the $k$ points $(\tau_2(t_i), \tau_3(t_i))$, no two of which share a coordinate since each $\tau_j$ is injective. Any five such points are independent, since the product of the two vertical and the two horizontal lines through four of them misses the fifth; so $k \geq 6$. If $k = 6$, a form of bidegree $(2, 2)$ vanishing at five of the points vanishes at the sixth; applied to $CD$, with $C$ the curve of bidegree $(1, 1)$ through three of the points and $D$ running over the pencil of such curves through two others, whose base locus is those two points, this puts the sixth point on $C$. So the six points lie on the graph of a real Möbius transformation, and likewise for the other pairs of factors. Moving the second and third factors by these transformations, which act through $\mathrm{GL}_2$ on $\langle 1, \theta_j, \theta_j^2/2\rangle$, turns $T$ into a binary sextic, embedded in the tensor cube by $l^6 \mapsto (l^2)^{\otimes 3}$, that is annihilated by three definite binary quadrics, the transforms of the $\lambda_j$. Two non-proportional definite binary quadrics generate every binary form of degree at least three, and so would annihilate the sextic; as $T \neq 0$, the three quadrics are proportional to one quadric $\Lambda$, and the sextic lies in the kernel of contraction with $\Lambda$, spanned by $l_+^6$ and $l_-^6$ with $l_\pm$ the isotropic lines of $\Lambda$. Moving back, $T$ is a combination of two conjugate decomposable tensors in $S(0, q)$, which are $e_\varepsilon$ and $e_{-\varepsilon}$ for one $\varepsilon$; so $N_w = 2$.

## Part 5. Beyond the Weil classes

### 20. The reach of the obstructions

An obstruction theorem is useful only if its boundary is drawn exactly. This section draws the boundary of each of the main results, then states the statements through which the conjecture is reached and proves the implications of the literature that join them.

#### 20.1. Limits of the divisor obstruction.

*Remark* 20.1 (Markman's construction). The objects through which Markman proves the Weil classes of the sixfolds of trivial discriminant algebraic [Mar25a, Mar25b] are not obtained by restriction from an ambient projective space, and they are not $K$-equivariant.

Section 13 analyses exactly which hypotheses they escape and why, and Proposition 13.5 lists the three. No obstruction of this paper bears on those objects, and none needs to: their conclusion is a theorem. The candidates proposed for cases that remain open, in [Mar25b, Section 12] and [Mar25c, Example 11.2.7], are another matter, and Theorem 18.61 and Corollary 19.45 decide them.

*Remark* 20.2 (Twisted objects and gerbes). The hypothesis of Theorem 7.3 is that $\mathcal{F}$ is an object of $D^b(\mathbb{P}^{N-1})$, so a twisted sheaf on a gerbe over $\mathbb{P}^{N-1}$, or an object of a derived category of twisted sheaves, is not covered. What is covered, and what the proof uses, is that the Chern character of the restricted object lies in the subring of $H^*(A^\vee, \mathbb{Q})$ generated by divisor classes. Any construction whose input has that property is subject to the conclusion, whatever category it is phrased in.

*Remark* 20.3 (Other ambient varieties). Replacing $\mathbb{P}^{N-1}$ by another ambient variety $P$ changes the conclusion exactly when $H^*(P, \mathbb{Q})$ is not generated by divisor classes, since the proof only uses $\mathrm{ch}(\mathcal{F}) \in D^*(P)$ and the fact that $i^*$ carries $D^*(P)$ into $D^*(A^\vee)$. For $P$ a Grassmannian, a flag variety, or any variety whose cohomology is generated in degree two, nothing changes. For $P$ with exceptional Hodge classes of its own the argument does not apply, but then the problem has been moved rather than solved.

### 20.2. **Limits of the character obstruction.**

*Remark* 20.4 (Constructions that are not equivariant). Theorem 5.6 constrains eigenclasses. A construction whose output is not a $K^\times$-eigenclass is not constrained, and by Remark 5.8 such a construction must break the naturality of the $K$-action in an essential way: it must depend on a choice that is not $K$-invariant, and its output must have components in both extreme summands of the grading at once. These obstructions therefore point to constructions of that kind.

*Remark* 20.5 (Abelian varieties that are not very general). At special points of the moduli of Weil type the endomorphism algebra grows, the Néron–Severi group grows with it, and the Weil line can meet the divisor subring. This is the split case, where Weil classes are known to be algebraic by elementary means, and it is consistent with everything here: our statements about $W(A) \cap D^{2n}(A)$ assume $\mathrm{NS}(A) \otimes \mathbb{Q} = \mathbb{Q}\eta$, which fails there. Theorem 5.6 itself makes no such assumption and holds at every point, but at special points a class can be a sum of eigenclasses for several characters and the conclusion applies to each summand separately.

### 20.3. **Limits of the semiregularity statement.**

*Remark* 20.6 (Objects of positive rank). Theorem 9.7 and Corollary 9.18 both require the Chern character to lie on the Weil line, so that $z \mapsto z \cdot \mathrm{ch}(\mathcal{E})$ has no degree-two term. An object of nonzero rank has $\mathrm{rk}(\mathcal{E})\, z$ in that degree, the map is injective, and nothing here applies. Markman's sheaves are of this kind, and Remark 9.9 records why the distinction is not incidental to his argument.

*Remark* 20.7 (The weaker semiregularity hypothesis). What Corollary 9.18 obstructs is the injectivity of $\sigma_{\mathcal{E}}$ on the whole of $\mathrm{Ext}^2(\mathcal{E}, \mathcal{E})$. The variant proposed in [Mar25b, Question 11.4], injectivity on the image of an evaluation map, asks less. What we prove about it is confined to three groups of results. In Theorem 9.19, Proposition 18.18, Theorem 9.20 and Corollary 18.21 it fails. In Theorem 18.56, for a secant object of positive rank, it is shown to be equivalent to two conditions on the factor. In Theorem 18.61 and Corollary 18.63 it fails for Markman's candidate in dimension eight and for every secant object whose

support has two transversal branches of codimension two through a point. We do not decide the variant for any other object of positive rank, nor whether an affirmative answer to the question would follow from it.

*Remark* 20.8 (Deformation of the class itself). A kernel is not an obstruction class. Corollary 9.18 says that a sufficient criterion for the Weil class to stay algebraic is unavailable at the objects in question; it does not say the class fails to stay algebraic, and no statement in this paper does. Question 15.13 is asked in both directions.

20.4. **Limits of the hyperplane obstruction.**

*Remark* 20.9 (Families that are not linear systems on $X$). Theorem 10.3 is about smooth members of an ample linear system on $X$, and about the maps $j^*$ and $j_*$ they carry. It does not obstruct an argument that deforms $X$ itself, that passes to a different variety by a correspondence, or that uses a family in which $X$ appears as a fibre and the other fibres have the same dimension. In the last case the inductive parameter does not decrease, which is a different difficulty and the one recorded in Remark 10.10.

*Remark* 20.10 (Coniveau). A Hodge class of positive coniveau is supported on a proper closed subvariety, and the reduction to that subvariety is legitimate and lowers the dimension. Nothing here obstructs it. The obstruction concerns classes of coniveau zero, which is where the primitive middle cohomology of a general variety sits, and the generalised Hodge conjecture predicts that these are precisely the ones a coniveau argument cannot reach.

*Remark* 20.11 (The Kuga–Satake route). For a weight-two Hodge structure of K3 type the Kuga–Satake construction produces an abelian variety $\mathrm{KS}(T)$ and an injection of $T$, twisted once, into the endomorphisms of its first cohomology, carrying the Hodge class in question to a class of type $(0,0)$, that is to an endomorphism, whose graph is a cycle. The conclusion is transported back by a correspondence, not by a restriction, so Theorem 10.3 does not apply; this is Proposition 11.9. The route is limited by the requirement $h^{2,0}=1$, and Proposition 11.11 locates that limit exactly for abelian varieties of Weil type: no rational sub-Hodge structure of $H^2(A,\mathbb{Q})$ is of K3 type unless $n=1$. Section 11 treats the construction in full, including the algebraicity of the correspondence itself.

20.5. **The Weil statements and the closure map.** The results above close several routes and narrow one. The statements through which the conjecture is reached form a list, and we record each entry with its exact statement and with what this paper proves about it. Proposition 20.13 records the one way in which the entries on Weil classes are linked.

Fix an imaginary quadratic field $K=\mathbb{Q}(\sqrt{-d})$, an integer $n\geq 2$, and a class $\delta\in\mathbb{Q}^{\times}_{>0}/\mathrm{N}_{K/\mathbb{Q}}(K^{\times})$, the discriminant of the hermitian form. Write

> $\mathbf{W}(K,n,\delta)$: every abelian $2n$-fold of $(K,-1,n)$-Weil type with discriminant $\delta$ has algebraic Weil classes.

**Proposition 20.12** (The Weil statements, and how they are linked)**.** *For each triple $(K,n,\delta)$ the following hold.*

(i) $\mathbf{W}(K,n,\delta)$ *is equivalent to the algebraicity of the single class $\omega_s$ at every point $s$ of the period domain $\mathcal{D}_{n,n}$, and that domain is connected; this is Corollary 15.35.*

(ii) $\mathbf{W}(K,3,\delta_0)$ *holds for the trivial discriminant $\delta_0$* [Mar25a, Mar25b]*, and with it, by Proposition 20.13, $\mathbf{W}(K,2,\delta)$ for every $K$ and every $\delta$, which is how Markman obtains the fourfolds.*

(iii) *The period domains attached to distinct triples are disjoint, and Theorem 15.14 and Corollary 15.35 are statements about one triple at a time. The triples are*

*nevertheless not independent: by Proposition 20.13, $\mathbf{W}(K,N,\delta'')$ implies $\mathbf{W}(K,n,\delta)$ for every $n<N$ and every $\delta$, so the families of trivial discriminant in infinitely many dimensions give every imaginary quadratic family, and the families of a CM field of degree at least four give those of the imaginary quadratic fields it contains.*

*Proof.* (i) is Corollary 15.35 together with Proposition B.8. (ii) is the cited work together with Proposition 20.13(i); Remark 15.20 records that the construction of Section 15.1 reaches only $\delta_0$. For (iii), the discriminant is a discrete invariant of the hermitian form and is constant on $\mathcal{D}_{n,n}$ by Proposition 15.19(ii), while $n$ and $K$ are fixed by the endomorphism algebra, so the domains are disjoint; the links are Proposition 20.13, which passes between families through a product with a Weil surface or through a scalar extension, and not through a deformation. The product argument of Schoen that Markman uses to pass from sixfolds to fourfolds is an instance of such a link. ∎

**Proposition 20.13** (Descent and scalar extension)**.** *Let $F$ be a CM field of degree $2m$ and write $\mathbf{W}(F,n,\delta)$ for the statement that every abelian variety of $(F,n,\delta)$-Weil type has algebraic Weil classes, so that $\mathbf{W}(K,n,\delta)$ is the statement above when $m=1$.*

(i) Descent. *Let $B$ be of $(F,n,\delta)$-Weil type and $Y$ of $(F,1,\delta')$-Weil type. Then $X=B\times Y$, with the diagonal action of $F$ and the product polarisation, is of $(F,n+1,\delta\delta')$-Weil type, and the correspondence*

$$P(x)=pr_{B*}(x\cdot pr_Y^*(\eta_Y^{2m-2}\,y')),$$

*with $\eta_Y$ the polarisation of $Y$ and $y'$ a nonzero Weil class of $Y$, maps the Weil classes of $X$ isomorphically onto those of $B$. Since $y'$ lies in $H^2(Y)$ it is algebraic, so $P$ is algebraic, and every discriminant, a totally positive class in $F_0^\times/\mathrm{N}_{F/F_0}(F^\times)$, is that of some $Y$. Hence $\mathbf{W}(F,N,\delta'')$ for one $N$ and one $\delta''$ implies $\mathbf{W}(F,n,\delta)$ for every $n<N$ and every $\delta$.*

(ii) Scalar extension. *Let $K\subset F$ be CM fields and $B$ of $(K,n,\delta)$-Weil type. Then $B_F=B\otimes_{\mathcal{O}_K}\mathcal{O}_F$ is of $(F,n,\iota(\delta))$-Weil type, and for $j\colon B\to B_F$, $b\mapsto b\otimes 1$, one has $j^*\,\mathrm{Tr}_{F/\mathbb{Q}}(a\det_F)=\mathrm{Tr}_{K/\mathbb{Q}}(\mathrm{Tr}_{F/K}(a)\det_K)$, so $j^*$ maps the Weil classes of $B_F$ onto those of $B$. Hence $\mathbf{W}(F,n,\iota(\delta))$ implies $\mathbf{W}(K,n,\delta)$; every imaginary quadratic $K$ lies in the quartic CM field $K(\sqrt{2})$.*

*Proof.* (i) $H^1(X)=H^1(B)\oplus H^1(Y)$ as $F$-spaces and the hermitian form of $X$ is the orthogonal sum, of signature $(n+1,n+1)$ at every real place; its normalised discriminant is the product, and $H_Y=\mathrm{diag}(1,-t)$ with $t$ totally positive realises every class. The Weil classes of $X$ are $\bigoplus_\sigma\bigwedge^{2n}V_{B,\sigma}\otimes\bigwedge^2 V_{Y,\sigma}$, because the top exterior power of a direct sum is the tensor product of the top powers. On the $\sigma$-component, $P$ is multiplication by $\int_Y w_Y^\sigma\,\eta_Y^{2m-2}\,y'^{\bar\sigma}$, which is nonzero since $w_Y^\sigma\wedge y'^{\bar\sigma}$ fills the $(\sigma,\bar\sigma)$ block and $\eta_Y$ is nondegenerate on the others; the remaining pairings vanish for weight reasons under the norm-one torus of $F$. So $P$ is rational, diagonal over $\mathbb{C}$ with nonzero entries, and algebraic, and $\omega(B)$ is algebraic whenever $\omega(X)$ is; Lemma 16.3 passes from one Weil class to all of them.

(ii) $H^1(B_F)=H^1(B)\otimes_K F$, each eigenspace of $F$ is an eigenspace of $K$ of signature $(n,n)$, the hermitian form is $H_B\otimes F$ with determinant $\det H_B$, and a norm from $K$ is a norm from $F$. The identity for $j^*$ is the multiplicativity of the determinant under restriction of scalars, and $\mathrm{Tr}_{F/K}$ is onto. ∎

The case $K$ imaginary quadratic, $n=2$ and $N=3$ of (i) is Proposition 10 of Schoen's addendum [Sch98], the argument through which Markman obtains every Weil fourfold, at every discriminant, from the sixfolds of trivial discriminant [Mar25a]. Schoen multiplies a

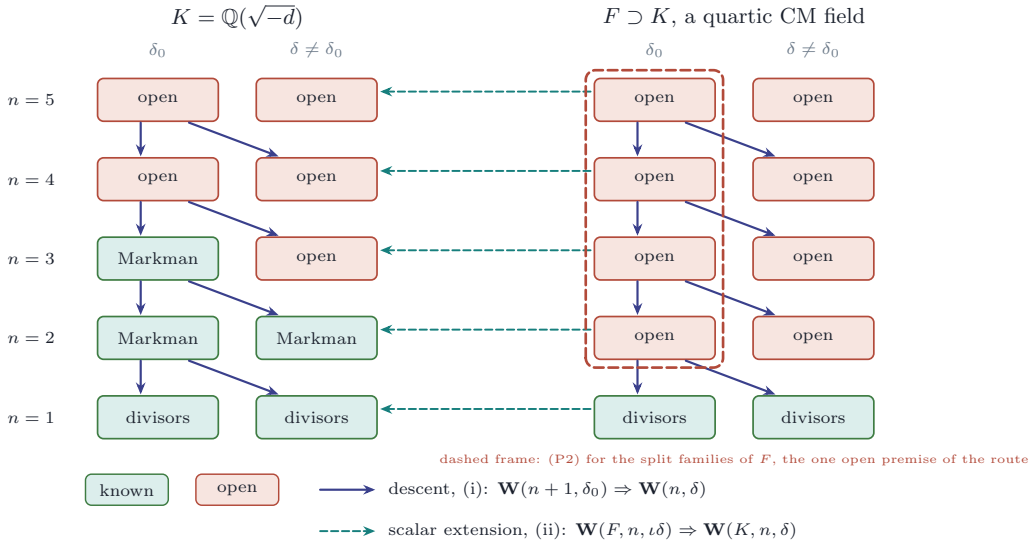



FIGURE 39. The Weil families of an imaginary quadratic field $K$ and of a quartic CM field $F \supset K$, one cell for each $n$ and each kind of discriminant, grass where the Weil classes are proved algebraic (divisor classes at $n = 1$; Markman [Mar25a, Mar25b] at $n = 2$ and at $n = 3$, $\delta = \delta_0$) and clay where they rest on the open propagation statement. The indigo arrows are the descent of Proposition 20.13(i), from the trivial discriminant in dimension $n + 1$ to every discriminant in dimension $n$; the dashed teal arrows are the scalar extension of Proposition 20.13(ii). The dashed frame is the propagation statement for the split families of the field of higher degree, which by Theorem 20.70(ii) is the only form of (P2) that any minimal route to the conjecture uses: every clay cell of the left panel lies below it.

Weil fourfold $A$ by a Weil surface $A'$ of the complementary discriminant, identifies $A \times A'$ up to isogeny with a member of a family of Weil sixfolds of fixed discriminant, and pushes forward the cycles $z \cdot (A \times D)$, for $D$ a divisor on $A'$ whose class spans its Weil classes; this is the correspondence $P$ of (i) with $m = 1$, and it involves no degeneration. What (i) adds is that the same argument works in every dimension and for every CM field. As a consequence the triples $(K, n, \delta)$ are not separate problems. The trivial discriminant families in infinitely many dimensions give every imaginary quadratic family, and by (ii) the split families of the CM fields of degree at least four give the imaginary quadratic fields as well. In particular the base points of nontrivial discriminant supplied by Theorem 15.27 are not needed on any route to the conjecture, since the split families already carry base points (34); what Theorem 15.27 shows is that propagation can be started in every family, not that it has to be. Figure 39 draws the web of families that Proposition 20.13 links. The list is as follows.

(C1) *One secant object on one fourfold.* There should exist a principally polarised abelian fourfold $(X, \Theta)$ and a perfect complex $\mathcal{F}$ on it with $\mathrm{ch}(\mathcal{F}) = u_\Theta + 3v_\Theta$ satisfying (58). By Theorem 18.56 and Corollary 18.57 the transform $\Phi(\mathcal{F} \boxtimes \mathcal{F})$ would then satisfy the weakened criterion, and by Proposition 18.52 the three conditions of Markman's strategy that concern only the Chern character hold at the point $(1, 3)$ for any such

$\mathcal{F}$, since they depend on $\mathrm{ch}(\mathcal{F})$ alone. Turning this into algebraicity needs one further input. The weakened criterion enters in place of the hypothesis Markman verifies, and the survival of the semiregularity theorem under the substitution is exactly [Mar25b, Question 11.4]. So (C1) is a demand on a construction, and an affirmative answer to that question is a separate prerequisite standing beside it; Theorem 20.70 places both. What this paper contributes is the exact form of the demand. The local part of the criterion is a condition on depth and nothing else: if the support is Cohen–Macaulay there is no local obstruction at any point (Theorem 18.75); if it is not, the local $\mathrm{Ext}^2$ is nonzero at the bad points, and where two smooth branches of codimension two meet transversally the criterion fails outright (Theorem 18.61, Corollary 18.63, Corollary 18.77). The candidate of [Mar25b, Section 12] is of the second kind, and it cannot be moved to the first: separating the branches is what would restore the depth, and Remark 18.78 shows that doing so moves the Chern character off the secant plane. What is needed is therefore a support without such points. For a Cohen–Macaulay support the identity (58) is the single vanishing of the image of $A_\theta A_{\theta'}$ in $H^1(Z, \mathcal{E}xt^1(\mathcal{I}_Z, \mathcal{I}_Z))$, and the sheaf must in addition extend to first order along the polarised deformations (Theorem 18.56(iii)). Such a support has one of two shapes (Corollary 18.94): a smooth one, where the invariants are forced outright and $d \leq b^2$ for the twist $b\Theta$, so $d \leq 9$ for the twist $3\Theta$ of Setup 18.51 (Theorem 18.67, Corollary 18.68), and a singular one. In neither may the Hilbert–Burch resolution split into powers of $\Theta$ or into simple semi-homogeneous bundles (Theorem 18.89, Theorem 18.92), and no complete intersection occurs (Theorem 18.81).

(C2) *The same for every $n$ and every discriminant.* A positive answer to (C1) reaches one triple, and the discriminant of that triple is the trivial one, because the object lives on $X \times \widehat{X}$ and every split member has trivial discriminant by Proposition 15.19(ii). The other triples are reached through Proposition 20.13: a positive answer in infinitely many dimensions gives $\mathbf{W}(K, n, \delta)$ for every $n$ and every $\delta$, so only uniformity in $n$ is asked for, and (C1) at $n = 4$, with Question 11.4, already gives every sixfold. No answer of this kind reaches the CM fields of degree at least four. What this paper contributes: three of the four ingredients of Question 13.7 are supplied for every $n$ and every $K$ (Theorem 14.15), including the existence of the secant sheaves themselves (Theorem 14.13); the fourth is isolated exactly (Remark 14.16).

(C3) *CM fields of degree larger than two.* For a CM field $F$ of degree $2m > 2$ acting on an abelian variety, the Weil classes span a space of dimension $2m$, and the secant framework of Section 14, which is built on a plane meeting the spinor variety in a conjugate pair, does not apply to them as it stands. Markman has extended it to these fields [Mar25c]: for an abelian variety $X$ with real multiplication by the totally real subfield $F_0$ of $F$, a polarisation $\Theta \in \bigwedge^2_{F_0} H^1(X, \mathbb{Q})$ and $q \in F_0$ with $F = F_0(\sqrt{-q})$, he makes $F$ act on $X \times \widehat{X}$ with split Weil type, replaces the plane by a rational subspace $B$ of dimension $2^{[F_0:\mathbb{Q}]}$ of the half-spin representation that is secant to the spinor variety, and shows that for sheaves $F_1, F_2$ on $X$ with characters in $B$ the class $\kappa(E) = \mathrm{ch}(E)\exp(-c_1(E)/\operatorname{rk} E)$ of $E = \Phi(F_1 \boxtimes F_2^\vee)$ stays of Hodge type on every member of the family, and that where it stays algebraic all Weil classes are algebraic [Mar25c, Theorem 1.1.2, Proposition 10.2.1]. He exhibits such a pair meeting his non-degeneracy criterion when $F$ is biquadratic and $X$ is the Jacobian of a curve of genus four with real multiplication by $F_0$ and

an automorphism acting through a unit of norm one [Mar25c, Example 11.2.7, Lemma 11.2.8]. That $\kappa(E)$ stays algebraic would follow from a semiregularity property of $E$, which is not addressed there [Mar25c, Question 11.2.2]; for the pair of that example the weaker property asked in the question fails at the points of the glued curves (Corollary 19.45). No result of this paper depends on a theorem of [Mar25c]; Section 19.6 and Section 19.7 use its construction as their setting. What this paper contributes: everything except the propagation. By Section 19 each such family has a connected period domain, a flat space of Weil classes, a dense set of rational orbits, a base point at a CM member where the classes are written as polynomials in divisor classes (Theorem 19.5), and the two propagation criteria (Theorem 19.7); so the higher CM fields are the same propagation problem as the quadratic ones, and not a separate entry. For a quartic field at $n = 2$, the smallest case of higher degree, Section 19.6 determines how far divisor classes, scalar extension and correspondences reach, and computes the number that the polarised criterion asks of one complex. For a sextic field, Section 19.7 shows that the dimension count, which excludes a minimal Orlov product over a quartic field, does not exclude one over a sextic field, and gives the profile such an object would have. Appendix C places the imaginary quadratic case inside Albert's classification.

(C4) *The non-split sixfolds.* $\mathbf{W}(K, 3, \delta)$ for $\delta \neq \delta_0$; the case $\delta = \delta_0$ is Markman's [Mar25a]. By Proposition 20.13(i) it would follow from $\mathbf{W}(K, 4, \delta_0)$, and by Proposition 20.13(ii) from $\mathbf{W}(K(\sqrt{2}), 3, \iota(\delta))$. What this paper contributes: a base point in every family, at a member with quaternionic multiplication (Theorem 15.27); the locus of such members is a Lagrangian Grassmannian of codimension $n(n-1)/2$ (Proposition 15.32), and the rest of the family can be reached from it only by propagation.

(C5) *Exceptional classes that are not of Weil type.* On an abelian variety the Hodge ring can contain exceptional classes that the Weil construction does not produce. Through dimension five it does not [MZ99], and in dimension eight it does: the self-product of an abelian fourfold of Mumford type carries two Hodge classes that are neither products of divisor classes nor Weil classes of any quotient (Proposition 20.74). So the Hodge conjecture for abelian varieties does not reduce to Weil classes, and the statement needed is the algebraicity of these classes themselves. What this paper contributes: the example, made exact, and the forms of their algebraicity studied in Section 21.

(C6) *Varieties that are not abelian.* A reduction of the Hodge conjecture for arbitrary smooth projective varieties to abelian varieties cannot be of Hodge-theoretic kind: the $H^3$ of a very general quintic threefold lies outside the tensor category generated by the cohomology of abelian varieties (Proposition 20.75), so its Hodge structure cannot be realised inside that of an abelian variety by any correspondence. Theorem 10.3 and Theorem 10.8 close the two routes through hyperplane sections that would be the natural candidates. What this paper contributes: those three obstructions and, on the positive side, special cases only (Proposition 20.26, Remark 20.68). Nor is this entry the complement of the others. The varieties that are not abelian include $A \times \mathbb{P}^1$ for every abelian variety $A$, and the conjecture for $A \times \mathbb{P}^1$ gives it for $A$, so the conjecture for the varieties that are not abelian is the whole conjecture (Proposition 20.33).

20.6. **Implications from the literature.** The theorems of this paper are not the only implications that end in the conjecture. The literature supplies others, and a frontier drawn without them overstates what is needed. This subsection proves those implications, or quotes them where they are theorems of the literature, so that Section 20.9 runs over both. Nothing in it is new except the assembly, but the assembly changes the answer.

**Definition 20.14** (Three statements about all varieties)**.** Here a variety is smooth, projective and defined over $\mathbb{C}$, except the base $S$ in (V).

(L) $B(X)$ holds for every variety $X$ (Definition 10.6).

(M) Every Hodge class on every variety $X$ is *motivated* in the sense of André [And96b, §2.1]: it equals $\mathrm{pr}_*(\alpha \cup *\beta)$ for some variety $Y$, algebraic classes $\alpha, \beta$ on $X \times Y$, the projection $\mathrm{pr}\colon X \times Y \to X$, and the Lefschetz involution $*$ of $H^*(X \times Y, \mathbb{Q})$ with respect to some ample class.

(V) For every smooth connected complex variety $S$, every smooth projective morphism $f\colon \mathcal{X} \to S$ and every global section $\xi$ of $R^{2p}f_*\mathbb{Q}$ over $S^{\mathrm{an}}$: if $\xi_s$ is algebraic at one point $s \in S$, then $\xi_t$ is algebraic at every point $t \in S$.

The Lefschetz involution of a variety $Z$ of dimension $e$ with ample class $L$ sends $L^r x$, for $x$ primitive of degree $j$ and $0 \le r \le e - j$, to $\pm L^{e-j-r}x$, with a sign that depends only on $j$ [And96b]; the sign plays no role below.

The statement (V) is the variational Hodge conjecture for algebraic classes; it asks nothing about Hodge classes, only that algebraicity, once present, survives deformation.

**Lemma 20.15** (Kleiman)**.** *Let $Z$ be a variety of dimension $e$ with ample class $L$, and suppose that $B(Z)$ holds. Then the Künneth projectors $\pi_k$ of $H^*(Z, \mathbb{Q})$, the projectors onto the summands $L^r P^j$ of the Lefschetz decomposition, and the Lefschetz involution $*$ are all induced by algebraic cycles on $Z \times Z$.*

*Proof.* Operators induced by algebraic cycles on $Z \times Z$ are closed under sums and rational multiples, and under composition, because the composite of two algebraic correspondences is again algebraic [Kle68]. The operator $L$ is induced by the image of a divisor of class $L$ under the diagonal, and $\Lambda$ by $B(Z)$. With $[L, \Lambda] = H$ and $H$ acting on $H^k$ by $k - e$ (the proof of Lemma 10.7), $H = L\Lambda - \Lambda L$ is induced by an algebraic cycle, and since the numbers $k - e$ for $0 \le k \le 2e$ are distinct,

$$\pi_k = \prod_{i \neq k} \frac{H - (i - e)}{k - i}$$

is as well. Put $\Omega = H^2 + 2H + 4\Lambda L$. For $x$ primitive of degree $j$ put $m = e - j \ge 0$. Then $\Lambda x = 0$ and $Hx = -mx$, and the relation

$$\Lambda L^s x = s(m - s + 1)\, L^{s-1}x \qquad (s \ge 1) \tag{73}$$

follows by induction from $\Lambda L = L\Lambda - H$: the case $s = 1$ is Lemma 10.7, and $\Lambda L^s x = L\Lambda L^{s-1}x - HL^{s-1}x = \big((s-1)(m-s+2) - (2s-2-m)\big)L^{s-1}x = s(m-s+1)L^{s-1}x$. So $\Omega x = (m^2 - 2m + 4m)x = m(m+2)x$, and $\Omega$ commutes with $L$, since $[L, H] = -2L$ and $[L, \Lambda L] = HL$ give $[L, \Omega] = -2(LH + HL) - 4L + 4HL = 0$; hence $\Omega$ acts on $L^r P^j$ by $m(m+2)$. Inside $H^k$ the summands are $L^r P^{k-2r}$, with $m = e - k + 2r$, and $m \mapsto m(m+2)$ is injective on $m \ge 0$, so Lagrange interpolation in $\Omega$ composed with $\pi_k$ gives the projector $p_{k,r}$ onto $L^r P^{k-2r}$ as an algebraic operator. Finally let $x \in P^j$ and $0 \le r \le m$. If $2r \le m$,

then $L^{m-2r}(L^r x) = L^{m-r}x$. If $2r > m$, put $t = 2r - m$, so $1 \le t \le r$; by (73),

$$\Lambda^t L^r x = \prod_{i=0}^{t-1} (r-i)(m-r+i+1)\; L^{m-r}x,$$

and every factor is at least 1, because $r-i \ge r-t+1 \ge 1$ and $m-r+i+1 \ge m-r+1 \ge 1$. So on $L^r P^j$ the involution is $\pm L^{m-2r}$ or a nonzero rational multiple of $\Lambda^{2r-m}$, and $* = \sum_{k,r} c_{k,r} T_{k,r}\, p_{k,r}$ with rational $c_{k,r}$ and $T_{k,r}$ a power of $L$ or of $\Lambda$. Every term is algebraic. ∎

**Proposition 20.16** (The conjecture is (L) and (M))**.** *The Hodge conjecture for every variety is equivalent to* (L) *and* (M) *together.*

*Proof.* Assume the conjecture. Then (L) holds by Theorem 12.2. An algebraic class $\xi$ on $X$ of dimension $d$ is motivated: take $Y$ a point, $\alpha = \xi$, and $\beta = \varepsilon h^d$ with $h$ the ample class and $\varepsilon = \pm 1$ the sign for which $*(h^d) = \varepsilon$, which exists because the unit $1 \in H^0$ is primitive of degree 0; then $\alpha \cup *\beta = \xi$. Every Hodge class is algebraic, so (M) holds.

Conversely assume (L) and (M), and let $\xi$ be a Hodge class on $X$. By (M), $\xi = \mathrm{pr}_*(\alpha \cup *\beta)$ with $\alpha, \beta$ algebraic on $X \times Y$. By (L), $B(X \times Y)$ holds, so $*$ is induced by an algebraic cycle (Lemma 20.15); hence $*\beta$, the product $\alpha \cup *\beta$ and its push-forward $\xi$ are algebraic. This is André's observation that motivated classes are algebraic as soon as the Lefschetz involutions are [And96b, §2.1]. ∎

**Proposition 20.17** (Two sources of (V))**.** *The Hodge conjecture implies* (V)*, and so does* (L)*.*

*Proof.* Let $f\colon \mathcal{X} \to S$, $\xi$ and $s$ be as in (V), with $\xi_s$ algebraic. As $S$ is smooth and connected it is irreducible, so two affine open subsets meet, and it suffices to treat $S$ affine, joining $s$ to an arbitrary $t$ through a point of the intersection of an affine neighbourhood of each. Then $\mathcal{X}$ is quasi-projective and, by resolution of singularities, has a smooth projective compactification $\overline{\mathcal{X}}$.

Assume the conjecture. Write $n = 2p$ and $\rho\colon H^n(\overline{\mathcal{X}}, \mathbb{Q}) \to H^0(S^{\mathrm{an}}, R^n f_* \mathbb{Q})$ for restriction; it is surjective by the theorem of the fixed part [Del71, 4.1.1(ii)]. For $t \in S$ the restriction $r_t\colon H^n(\overline{\mathcal{X}}, \mathbb{Q}) \to H^n(\mathcal{X}_t, \mathbb{Q})$ is $\rho$ followed by evaluation at $t$, and evaluation is injective because $S$ is connected; so $K = \ker r_t = \ker \rho$ does not depend on $t$. It is a sub-Hodge structure, and the Hodge structure $H^n(\overline{\mathcal{X}}, \mathbb{Q})$ is polarisable, hence semisimple, so $H^n(\overline{\mathcal{X}}, \mathbb{Q}) = K \oplus K'$ with $K'$ a sub-Hodge structure. Choose $\eta \in K'$ with $\rho(\eta) = \xi$. Then $r_s|_{K'}$ is an injective morphism of Hodge structures and $r_s(\eta) = \xi_s$ is of type $(p,p)$, so every component of $\eta$ of type other than $(p,p)$ maps to zero and is zero: $\eta$ is a Hodge class. So $\xi_t = r_t(\eta)$ is a Hodge class on $\mathcal{X}_t$ for every $t$, and algebraic by the conjecture.

Assume (L). The algebraic class $\xi_s$ is motivated (proof of Proposition 20.16), so $\xi_t$ is motivated for every $t$ by André's deformation theorem [And96b, Théorème 0.5], and a motivated class is algebraic under (L) (proof of Proposition 20.16). ∎

**Theorem 20.18** (Deligne, André; Abdulali, André)**.** (V) *implies the Hodge conjecture for every abelian variety, and so does* (L)*.*

*Proof.* Milne calls a class *accessible* if it lies in the least family of rational classes that contains the classes of algebraic cycles, is stable under pull-back by morphisms of varieties, and contains $t_t$ for every $t$ whenever $(t_s)_{s \in S}$ is a global section of $R^r f_* \mathbb{Q}$ for a smooth projective family over a smooth connected base and contains $t_s$ for one $s$; and he proves, following Deligne and André, that every Hodge class on an abelian variety is accessible

[Mil20, §5.1, Theorem 3]. The algebraic classes contain the classes of algebraic cycles and are stable under pull-back, and under (V) they have the third property. So they contain every accessible class, hence every Hodge class on an abelian variety; this is [Mil20, Remark 2]. The statement for (L) follows by Proposition 20.17, and is the theorem of Abdulali and André in the form of [Mil20, Theorem 4]. ∎

*Remark* 20.19 (The three in the literature). $B(X)$ is known for curves, where it is immediate, and for abelian varieties [Kle68, §2], and Corollary 20.22 below adds the total space of every Mumford family. (M) holds for the Hodge classes on abelian varieties [And96b, Théorème 0.6.2]. (V) holds in degree 2, where by the argument of Proposition 20.17 the class $\xi_t$ is a Hodge class and the Lefschetz theorem on $(1,1)$ classes applies; the one general mechanism for it in higher degree is the semiregularity theorem [Blo72, BF03, Pri24], which is the mechanism of Theorem 16.29. Theorem 20.18 says what is at stake for this paper: the single statement (V), which the conjecture itself implies, gives the Weil classes of every CM field and the classes beyond the Weil lines at once, by a route that uses none of the theorems proved here.

20.7. **The Lefschetz standard conjecture for abelian schemes over curves.** The statement $B(X)$ is known for abelian varieties $X$. This subsection proves it for the total space of an abelian scheme over a curve whose invariant cycles are algebraic, and so for the total space of every Mumford family. The proof assembles the operator $\Lambda$ from a relative part, which is a Pontryagin product and is algebraic on every abelian scheme, and a part along the base, which is algebraic exactly when the invariant cycles are.

**Proposition 20.20** ($\Lambda$ as a Pontryagin product)**.** *Let $A$ be an abelian variety of dimension $g$, $\ell \in H^2(A,\mathbb{Q})$ the class of an ample line bundle, $D = \int_A \ell^g/g!$ and $\gamma = \ell^{g-1}/(g-1)!$. Let $\mu\colon A\times A \to A$ be the addition and $x \star y = \mu_*(p_1^*x \cup p_2^*y)$ the Pontryagin product. Then the operator $\Lambda$ of Definition 10.6 for the class $\ell$ is*

$$\Lambda(x) = D^{-1}\, x \star \gamma.$$

*In particular $\Lambda$ is induced by the algebraic cycle $D^{-1}(p_1,\mu)_*(p_2^*Z)$ on $A\times A$, for any cycle $Z$ of class $\gamma$.*

*Proof.* Put $V = H_1(A,\mathbb{Q})$, so that $H^*(A,\mathbb{Q}) = \bigwedge^* V^\vee$ under cup product and $H_*(A,\mathbb{Q}) = \bigwedge^* V$ under the Pontryagin product of homology, and let $\mathrm{PD}(x) = x \cap [A]$. Cap product makes $H_*$ a module over $H^*$, and for a torus it is contraction; so $\mathrm{PD}(\ell \cup x) = \ell \cap \mathrm{PD}(x)$ is the contraction $\iota_\ell$ applied to $\mathrm{PD}(x)$. Poincaré duality commutes with $\mu_*$ and carries $p_1^*x \cup p_2^*y$ to the cross product of $\mathrm{PD}(x)$ and $\mathrm{PD}(y)$, without sign when $y$ has even degree, so $\mathrm{PD}(x\star\gamma) = \mathrm{PD}(x)\cdot\beta$ with $\beta = \mathrm{PD}(\gamma) \in \bigwedge^2 V$.

Choose a basis $x_1, y_1, \dots, x_g, y_g$ of $V^\vee$ with $\ell = \sum_i d_i\, x_i \wedge y_i$, $d_i > 0$, oriented so that $\int_A x_1\wedge y_1 \wedge \cdots \wedge x_g \wedge y_g = 1$; then $\int_A \ell^g = g!\, d_1 \cdots d_g$, so $D = d_1 \cdots d_g$. Let $e_i, f_i$ be the dual basis of $V$. Then $\gamma = \sum_j (\prod_{i\neq j} d_i) \prod_{i\neq j} x_i \wedge y_i$ and $\beta = \varepsilon D \sum_j d_j^{-1} e_j \wedge f_j$ with $\varepsilon = \pm 1$ independent of $j$, since the permutations of the pairs preserve the orientation and exchange the terms. The algebra $\bigwedge^* V$ is the graded tensor product of the algebras $\bigwedge^*\langle e_i, f_i\rangle$, and $\iota_\ell$ and multiplication by $D^{-1}\beta$ are sums over $i$ of even operators on the $i$-th factor, namely $d_i\, \iota_{x_i\wedge y_i}$ and $\varepsilon d_i^{-1} e_i \wedge f_i$. Operators on different factors commute, and on the $i$-th factor, with basis $1, e_i, f_i, e_i\wedge f_i$ and $\iota_{x_i\wedge y_i}(e_i\wedge f_i) = \sigma$ for a sign $\sigma$ fixed by the convention for contraction, the commutator $[\iota_\ell, D^{-1}\beta]$ acts by $\varepsilon\sigma$ on 1, by 0 on $e_i$ and $f_i$, and by $-\varepsilon\sigma$ on $e_i\wedge f_i$, that is by $\varepsilon\sigma(1-k_i)$ in degree $k_i$. Summing over $i$, $[\iota_\ell, D^{-1}\beta] = \varepsilon\sigma(g-k)$ on $\bigwedge^k V$. As PD carries $H^q$ to $\bigwedge^{2g-q} V$, the operator $\Lambda'(x) = D^{-1}x\star\gamma$ satisfies $[L,\Lambda'] = \varepsilon\sigma(q-g) = \varepsilon\sigma H$ on $H^q(A)$.

The sign is fixed by one value. For $x$ and $y$ of complementary degrees, $x \star y \in H^0(A)$ is $\int_{A\times A} p_1^*x \cup p_2^*y \cup \mu^*[\mathrm{pt}]$, and $\mu^*[\mathrm{pt}]$ is the class of the antidiagonal $\{(a,-a)\}$, so $x \star y = \int_A x \cup [-1]^*y$; for $y = \gamma$, of even degree, this is $\int_A x \cup \gamma$. Hence $\Lambda'(\ell) = D^{-1}\int_A \ell \cup \gamma = g$, while $\Lambda(\ell) = \Lambda(L\cdot 1) = g$ by Lemma 10.7. If $\varepsilon\sigma = -1$, then $-\Lambda'$ would be the operator of Definition 10.6 and $\Lambda'(\ell) = -g$. So $\varepsilon\sigma = 1$, and $\Lambda' = \Lambda$ by the uniqueness in Definition 10.6. Finally, the correspondence $(p_1,\mu)_*(p_2^*Z)$ acts by $x \mapsto p_{2*}\big(p_1^*x \cup (p_1,\mu)_*p_2^*[Z]\big) = \mu_*(p_1^*x \cup p_2^*\gamma)$, by the projection formula. ∎

**Theorem 20.21** (The Lefschetz operator of an abelian scheme over a curve; cf. Tankeev [Tan03])**.** *Let $C$ be a smooth projective connected curve and $\pi\colon W \to C$ an abelian scheme of relative dimension $g$ with $W$ projective. Suppose that every class in $H^*(W_c,\mathbb{Q})$ invariant under the monodromy of $R^*\pi_*\mathbb{Q}$ is the restriction of an algebraic class on $W$. Then there is an ample class $L_W$ on $W$ whose operator $\Lambda$ is induced by an algebraic cycle on $W\times W$: $B(W)$ holds.*

*Proof. Step 1: the decomposition.* Let $[n]\colon W \to W$ be multiplication by $n$ over $C$. It acts on $R^q\pi_*\mathbb{Q} = \bigwedge^q R^1\pi_*\mathbb{Q}$ by $n^q$. The Leray spectral sequence degenerates [Del71, 4.1.1(i)], and $[2]^*$ preserves the Leray filtration and acts on its graded pieces $H^p(C,R^q\pi_*\mathbb{Q})$ by $2^q$. So $H^k(W,\mathbb{Q}) = \bigoplus_q H^k_{(q)}$, where $H^k_{(q)}$ is the $2^q$-eigenspace of $[2]^*$ and maps isomorphically onto $H^{k-q}(C,R^q\pi_*\mathbb{Q})$; only $k-q \in \{0,1,2\}$ occurs. The projectors $P_q = \prod_{q'\neq q}([2]^* - 2^{q'})/(2^q - 2^{q'})$ are induced by algebraic cycles, polynomials in the graph of $[2]$. As $[2]^*$ is a ring homomorphism, $H_{(q)} \cup H_{(q')} \subset H_{(q+q')}$; as $\int_W [2]^*z = 2^{2g}\int_W z$, the integral vanishes on $H_{(q)}$ for $q \neq 2g$, so Poincaré duality pairs $H_{(q)}$ with $H_{(2g-q)}$ only; and $[2]_* = 2^{2g}([2]^*)^{-1}$ acts on $H_{(q)}$ by $2^{2g-q}$.

*Step 2: the ample class.* Let $\mathcal{L}$ be a relatively ample line bundle, replaced by $\mathcal{L}\otimes[-1]^*\mathcal{L}$ so that it is symmetric. Then $[-1]^*$ fixes $c_1(\mathcal{L})$ and acts on $H_{(q)}$ by $(-1)^q$, so $c_1(\mathcal{L})$ has no component in $H^2_{(1)}$, while $H^2_{(0)} = H^2(C,\mathbb{Q})$ is spanned by the class $f$ of a fibre. So $c_1(\mathcal{L}) = \ell + af$ with $\ell \in H^2_{(2)}$ algebraic and $a \in \mathbb{Q}$. Put $L_W = \ell + mf = c_1(\mathcal{L}) + (m-a)f$ with $m \in \mathbb{Q}$ so large that $L_W$ is a positive rational multiple of the class of an ample line bundle, which is possible because $\mathcal{L}$ is relatively ample. Write $L_{\mathrm{rel}}$ and $L_C$ for cup product with $\ell$ and with $mf$, so that $L_W = L_{\mathrm{rel}} + L_C$; on $H^p(C,R^q\pi_*\mathbb{Q})$ the first raises $q$ by two and the second raises $p$ by two.

*Step 3: the relative operator.* Put $\gamma = \ell^{g-1}/(g-1)!$, which is algebraic and lies in $H_{(2g-2)}$, and $D = \int_{W_c}\ell^g/g!$. Let $\mu, \mathrm{pr}_1, \mathrm{pr}_2\colon W \times_C W \to W$ be the addition and the projections, and $T(x) = \mu_*(\mathrm{pr}_1^*x \cup \mathrm{pr}_2^*\gamma)$. As in Proposition 20.20, $T$ is induced by the algebraic cycle $(\mathrm{pr}_1,\mu)_*(\mathrm{pr}_2^*Z)$ on $W\times W$. Since $\pi\circ\mathrm{pr}_1 = \pi\circ\mu$, the projection formula gives $T(x\cup\pi^*b) = T(x)\cup\pi^*b$, so $T$ commutes with $L_C$. Since $[2]\circ\mu = \mu\circ([2]\times_C[2])$, $[2]_*T(x) = \mu_*(\mathrm{pr}_1^*[2]_*x \cup \mathrm{pr}_2^*[2]_*\gamma)$, and for $x \in H_{(q)}$ this is $2^{2g-q}\,2^2\,T(x)$; so $T$ maps $H_{(q)}$ into $H_{(q-2)}$. A relative correspondence acts on the Leray spectral sequence through the maps it induces on the local systems, so on $H^p(C,R^q\pi_*\mathbb{Q})$ the operator $T$ is $H^p(C,T_\bullet)$, with $T_c$ the Pontryagin product with $\gamma|_{W_c}$ on the fibre. By Proposition 20.20, $D^{-1}T_c$ is the operator $\Lambda$ of the fibre, so $\Lambda_{\mathrm{rel}} = D^{-1}T$ satisfies $[L_{\mathrm{rel}},\Lambda_{\mathrm{rel}}] = H_{\mathrm{rel}}$, where $H_{\mathrm{rel}}$ acts on $H_{(q)}$ by $q-g$.

*Step 4: the invariant classes.* Restriction maps $H^q(W,\mathbb{Q})$ onto the monodromy invariants $I_q \subset H^q(W_c,\mathbb{Q})$ [Del71, 4.1.1(ii)], and it kills $H^q_{(q')}$ for $q' < q$, which lies in positive Leray degree; so it maps $H^q_{(q)}$ isomorphically onto $I_q$. Every element of $H^q_{(q)}$ is algebraic: if $v \in I_q$ is the restriction of an algebraic class $z$, then $P_qz$ is algebraic, lies in $H^q_{(q)}$ and restricts to $v$. The subspace $I_q$ is a sub-Hodge structure, as the image of a morphism of Hodge structures,

and it is stable under cup product with $\ell_c$ and under the operator $\Lambda$ of the fibre, which are the restrictions of $L_{\mathrm{rel}}$ and $\Lambda_{\mathrm{rel}}$. The polarisation of $H^q(W_c,\mathbb{Q})$ attached to $\ell_c$ has the form $(v,w)\mapsto \int_{W_c} v\cup S(w)$, with $S$ a polynomial in cup product with $\ell_c$, the operator $\Lambda$ and the projectors of the fibre, and its restriction to the sub-Hodge structure $I_q$ is again a polarisation [Voi02]. Since $S(I_q)\subset I_{2g-q}$, the pairing $I_q\times I_{2g-q}\to\mathbb{Q}$ is nondegenerate.

*Step 5: the operator along the base.* For $y\in H^q_{(q)}$ and $w\in H^{2g-q}_{(2g-q)}$ one has $\int_W (y\cup f)\cup w=\int_{W_c} y|_{W_c}\cup w|_{W_c}$. By Step 4 and Step 1 this shows that $L_C\colon H^q_{(q)}\to H^{q+2}_{(q)}$ is injective, and it is onto because $\dim H^{q+2}_{(q)}=\dim H^{2g-q}_{(2g-q)}=\dim I_{2g-q}=\dim I_q$, by Poincaré duality on $W$ and on the invariants. Choose bases $(a_j)$ of $H^q_{(q)}$ and $(b_k)$ of $H^{2g-q}_{(2g-q)}$, all algebraic by Step 4, let $M_{jk}=\int_W a_j\cup mf\cup b_k$, which is invertible, and put

$$\Lambda_C(x)=\sum_{j,k}(M^{-1})_{kj}\Big(\int_W x\cup b_k\Big)a_j\qquad (x\in H^{q+2}(W,\mathbb{Q})),$$

for every $q$. It is induced by the algebraic cycle $\sum_{j,k}(M^{-1})_{kj}\,b_k\times a_j$. By Step 1, $\int_W x\cup b_k$ vanishes for $x\in H^{q+2}_{(q')}$ with $q'\neq q$, so $\Lambda_C$ is zero on $H^p(C,R^q\pi_*\mathbb{Q})$ for $p\neq 2$, and $\Lambda_C(L_C a_i)=a_i$. Hence $[L_C,\Lambda_C]$ is $-1$ on $H^q_{(q)}$, $0$ on $H^{q+1}_{(q)}$ and $+1$ on $H^{q+2}_{(q)}$: it is the operator $H_C$ that acts on $H^p(C,R^q\pi_*\mathbb{Q})$ by $p-1$.

*Step 6: the cross relations and the conclusion.* $[L_C,\Lambda_{\mathrm{rel}}]=0$ by Step 3. Both $L_{\mathrm{rel}}\Lambda_C$ and $\Lambda_C L_{\mathrm{rel}}$ vanish off the summands with $p=2$, and on $L_C y$ with $y\in H^q_{(q)}$ both give $\ell\cup y$, because $\ell\cup y\in H^{q+2}_{(q+2)}$ and $L_C$ commutes with $L_{\mathrm{rel}}$; so $[L_{\mathrm{rel}},\Lambda_C]=0$. Put $\Lambda=\Lambda_{\mathrm{rel}}+\Lambda_C$, of degree $-2$. Then

$$[L_W,\Lambda]=[L_{\mathrm{rel}},\Lambda_{\mathrm{rel}}]+[L_C,\Lambda_C]=H_{\mathrm{rel}}+H_C,$$

which acts on $H^p(C,R^q\pi_*\mathbb{Q})\subset H^{p+q}(W)$ by $(q-g)+(p-1)=(p+q)-\dim W$. So $\Lambda$ is the operator of Definition 10.6 for $L_W$, by its uniqueness, and it is induced by an algebraic cycle by Steps 3 and 5. ∎

Figure 40 draws the summands of the proof in relative dimension four.

**Corollary 20.22** (Mumford families)**.** *Let $\pi\colon W\to C$ be a non-isotrivial abelian scheme of relative dimension four over a smooth projective connected curve, $W$ projective, whose very general fibre $X$ is a fourfold as in Proposition 20.74, with Hodge group the group $G$ of Mumford's construction* [Mum69]*, the restriction of scalars from a totally real cubic field $F$ of the norm one group of a quaternion algebra over $F$. Then $B(W)$ holds. Such families exist, over the compact Shimura curves of* [Mum69].

*Proof.* By André's theorem [And92, Theorem 1] the connected monodromy group is a normal subgroup of the derived Mumford–Tate group of $X$, which is $G$. It is not trivial, because the family is not isotrivial, and $G$ is $\mathbb{Q}$-simple, since over $\overline{\mathbb{Q}}$ its three factors are indexed by the embeddings of $F$ and permuted transitively by the Galois group; so it is $G$. The monodromy invariants in $H^*(X,\mathbb{Q})$ are therefore contained in the invariants of $G$, which are the Hodge classes of $X$, and these are the powers of the polarisation by Proposition 20.74(i). The powers of $\ell$ restrict to the powers of the polarisation, so every invariant class is the restriction of an algebraic class, and Theorem 20.21 applies. ∎

*Remark* 20.23 (Scope of the argument). The total space $W$ is a fivefold fibred in abelian fourfolds whose Hodge groups are not defined by endomorphisms; it is not a product and not an abelian variety. For abelian schemes over curves the mechanism of Theorem 20.21

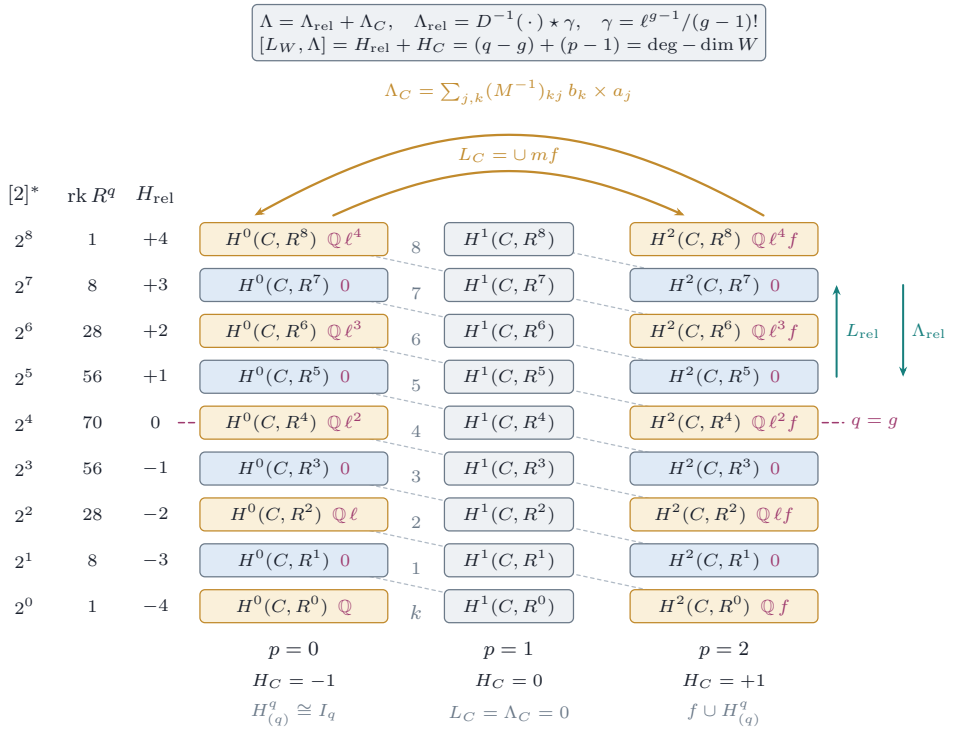



FIGURE 40. The proof of Theorem 20.21 for relative dimension four. The boxes are the summands $H^p(C, R^q\pi_*\mathbb{Q})$ of $H^*(W, \mathbb{Q})$, split off by the eigenvalue $2^q$ of $[2]^*$ shown on the left; the dotted diagonals join the summands of one $H^k(W, \mathbb{Q})$, $k = p + q$, and the dashed magenta line is the axis $q = g$ of hard Lefschetz on the fibres. The relative operators $L_{\mathrm{rel}}$ and $\Lambda_{\mathrm{rel}}$, in teal, move within a column; $\Lambda_{\mathrm{rel}}$ is a Pontryagin product with an algebraic class on every abelian scheme (Proposition 20.20). The base operators $L_C$ and $\Lambda_C$, in ochre, move along a row between the outer columns and kill the middle one; $\Lambda_C$ is the displayed cycle built from bases $(a_j)$, $(b_k)$ of the invariant summands, and is algebraic exactly when the invariant cycles are. Below each column is the weight of $H = [L_W, \Lambda]$ on it, and the panel at the top is Step 6 of the proof. The ochre boxes carry the invariant cycles and their images under $L_C$; for a Mumford family they are the lines listed in magenta on the right, spanned by $\ell^i f$ (and by $\ell^i$ in the left column), and the other boxes of the outer columns are zero (Corollary 20.22).

goes back to Tankeev [Tan03], who proves $B$ for the total space when certain canonically defined Hodge cycles are algebraic; the argument here does not use his hypotheses, and we make no claim of priority for Corollary 20.22. The difficulty lies in the hypothesis of Theorem 20.21. Applied to the fibre square $W \times_C W \to C$ the theorem gives $B(W \times_C W)$ as soon as every invariant class of $X \times X$ is the restriction of an algebraic class on $W \times_C W$; the invariant classes include the two exceptional classes of Proposition 20.74, so this is the smallest case of (F2) in a global form. Theorem 20.21 does not bear on (P2), on the Kuga–Satake class of Theorem 21.4, or on (L) for varieties that are not fibred in abelian varieties over a curve.

**Proposition 20.24** (Powers of a Weil variety; cf. [Mil22])**.** *Let $X$ be of $(K, -1, n)$-Weil type with $n \geq 2$ and $\mathrm{Hg}(X) = \mathrm{SU}(V, H)$. For every $k \geq 1$ the $\mathbb{Q}$-algebra $\mathrm{Hdg}^*(X^k)$ is generated by the divisor classes of $X^k$ and the pull-backs $f_a^*\omega$ of the Weil classes $\omega$ of $X$ along the homomorphisms $f_a = \sum_j a_j\, pr_j \colon X^k \to X$, $a \in \mathbb{Z}^k$. Consequently, if the Weil classes of $X$ are algebraic, the Hodge conjecture holds for every power of $X$. By [Mar25a] this is so for every such $X$ with $n = 2$, and with $n = 3$ and trivial discriminant.*

*Proof.* Write $H^1(X, \mathbb{C}) = V_+ \oplus V_-$ for the eigenspaces of $K$ and $W = V_+$. The polarisation makes $V_+$ and $V_-$ isotropic and pairs them perfectly, so $V_- = W^\vee$, and restriction to $V_+$ identifies $\mathrm{SU}(V, H)_{\mathbb{C}}$ with $\mathrm{SL}(W)$. As $\mathrm{Hg}(X^k)$ is $\mathrm{Hg}(X)$ acting diagonally on $H^1(X^k, \mathbb{C}) = (W \otimes U) \oplus (W^\vee \otimes U)$, $U = \mathbb{C}^k$, the Hodge classes of $X^k$, tensored with $\mathbb{C}$, are the $\mathrm{SL}(W)$-invariants in $\bigwedge^*\big((W \otimes U) \oplus (W^\vee \otimes U)\big)$. That algebra is an $\mathrm{SL}(W)$-equivariant quotient of a tensor algebra, with an equivariant section given by antisymmetrisation, so Weyl's first fundamental theorem for $\mathrm{SL}(W)$, in its tensor form and applied before antisymmetrisation, shows that its invariants are generated by the images of the contractions and the determinant tensors: the classes $D(u, u') = \sum_i (e_i \otimes u) \wedge (e_i^\vee \otimes u')$ and the polarised determinants, which by polarisation are combinations of $\det_u = \bigwedge_i (e_i \otimes u)$ and $\det_u^\vee = \bigwedge_i (e_i^\vee \otimes u)$, $u, u' \in U$. For $2n \geq 3$ the $D(u, u')$ span the invariants of degree two, which are the divisor classes tensored with $\mathbb{C}$. With $\alpha_\pm$ spanning $\bigwedge^{2n} V_\pm$ (Proposition 3.7) one has $f_a^*\alpha_+ = \det_a$ and $f_a^*\alpha_- = \det_a^\vee$, and since $u \mapsto \det_u$ is polynomial and $\mathbb{Z}^k$ is Zariski dense in $U$, these span the same space as all $\det_u$. So the $\mathbb{Q}$-algebra generated by $\mathrm{NS}(X^k)_{\mathbb{Q}}$ and the $f_a^*\omega$ is contained in $\mathrm{Hdg}^*(X^k)$ and has the same complexification. The generators are algebraic when the Weil classes of $X$ are, and they are for $n = 2$ and for $n = 3$ at trivial discriminant by [Mar25a]. ∎

The statement for powers is known (compare [Mil22], and [Li26] for the powers of CM fourfolds); we include the short proof because the next corollary uses it.

**Corollary 20.25** (Abelian schemes of small relative dimension)**.** *Let $\pi \colon W \to C$ be an abelian scheme of relative dimension $g$ over a smooth projective connected curve, with $W$ projective, and suppose that every monodromy invariant class in $H^*(W_c, \mathbb{Q})$ is a Hodge class, as happens when the identity component of the Zariski closure of the monodromy group is the Hodge group of a very general fibre.*

(i) *If $g \leq 4$, then $B(W)$ holds.*

(ii) *If the fibres are of $(K, -1, n)$-Weil type with connected monodromy group $\mathrm{SU}(V, H)$, and $n = 2$, or $n = 3$ with trivial discriminant, then $B$ holds for every fibre power $W \times_C \cdots \times_C W$. In particular $B(W_C)$ of Corollary 20.31 holds in these cases.*

*Proof.* By the theorem of the fixed part the invariant classes form a sub-Hodge structure of every fibre, with a Hodge structure independent of the point, so they are Hodge classes on every fibre.

(i) The Hodge conjecture holds for abelian varieties of dimension at most four, by the Lefschetz theorem on $(1,1)$ classes and hard Lefschetz up to dimension three and by [Mar25a] in dimension four, so each invariant class is algebraic on every fibre; Lemma 20.27 makes it the restriction of an algebraic class on $W$, and Theorem 20.21 applies.

(ii) The fibre power is an abelian scheme with projective total space and the monodromy acting diagonally. Its invariant classes are $\mathrm{SU}(V,H)$-invariants, hence Hodge classes at every member with $\mathrm{Hg} = \mathrm{SU}(V,H)$, an uncountable set of points, and there the Hodge conjecture holds for the fibre by Proposition 20.24; Lemma 20.27 and Theorem 20.21 conclude as before. For $W_C$ the connected monodromy group is $\mathrm{SU}(V,H)$ when it is Zariski dense, which is the hypothesis of Corollary 20.31(ii). ∎

The hypothesis on the monodromy in Corollary 20.25 can be checked with André's theorem [And92, Theorem 1]: the identity component of the Zariski closure of the monodromy group is always a normal subgroup of the derived Mumford–Tate group of a very general fibre, so the hypothesis holds whenever the family is not isotrivial and the Hodge group of a very general fibre is semisimple and $\mathbb{Q}$-simple, as for the Mumford families of Corollary 20.22.

**Proposition 20.26** (The total space of a Mumford family)**.** *Let $W \to C$ be as in Corollary 20.22, with $C$ the compact Shimura curve of* [Mum69] *of torsion-free level, or a finite étale cover of it. Then the Hodge conjecture holds for the fivefold $W$: its Hodge classes of degree $2p$ are spanned by $\ell^p$ and $f\ell^{p-1}$, with $f$ the class of a fibre.*

*Proof.* By Step 1 of the proof of Theorem 20.21, $H^k(W,\mathbb{Q}) = \bigoplus_q H^k_{(q)}$ with $H^k_{(q)} \cong H^{k-q}(C, R^q\pi_*\mathbb{Q})$, the summands being sub-Hodge structures cut out by algebraic projectors. The summand with $k = q$ is the space of invariants, which by the proof of Corollary 20.22 is spanned by the powers of the polarisation, restrictions of the powers of $\ell$. The summand with $k = q+2$ is $f \cup H^q_{(q)}$ by Step 5, spanned by the $f\ell^{q/2}$. It remains to see that $H^1(C, R^q\pi_*\mathbb{Q})$ carries no Hodge class. For $q$ even its weight $q+1$ is odd. For $q$ odd, over $\mathbb{C}$ the local system splits as a sum of $S^{k_1}V_1 \otimes U_\lambda \otimes M_\lambda$, where $V_1$ is the uniformising variation of the factor of $G$ split at the real place, $U_\lambda$ is a unitary variation of type $(0,0)$, $M_\lambda$ is constant of type $(a,a)$ with $2a + k_1 = q$, and $k_1 \equiv q \pmod 2$ because the central element $(-1,1,1)$ acts on $V$ by $-1$. When $C$ is étale over the Shimura curve the Kodaira–Spencer map of $V_1$ is an isomorphism, and then $H^1(C, S^{k_1}V_1 \otimes U_\lambda)$ has only the types $(k_1+1,0)$ and $(0,k_1+1)$ [Zuc79]; after the twist by $M_\lambda$ these are never balanced, since $k_1 \geq 1$. ∎

Statements of the kind of Proposition 20.26 are studied for Kuga fibre varieties, and we make no claim of priority. The étale hypothesis is used only at the end of the proof, and it cannot be dropped from the argument: over a ramified cover the Kodaira–Spencer map of $V_1$ vanishes at the branch points, the Higgs field of $S^{k_1}V_1$ is no longer an isomorphism, and the torsion of its cokernel contributes to $H^1(C, R^q\pi_*\mathbb{Q})$ in balanced types.

20.8. **The two targets as cases of (L).** The argument of Section 20.7 does more than add a case of (L). For an abelian scheme over a curve it shows that (L) for the total space is exactly the statement that algebraicity propagates along the curve. The base points of this paper then turn each of the two targets of Remark 20.73, one Weil family and the first classes beyond the Weil lines, into (L) for one explicit variety. The reductions are equivalences, so they locate the difficulty rather than remove it. Along the way the Hodge conjecture is proved for the self-product of a Mumford fourfold at every CM point (Theorem 20.29), and the two branches of Remark 16.36 are shown not to be realisable

by two objects (Theorem 20.32), an implication whose hypothesis Theorem 17.5 shows is never met.

**Lemma 20.27** (Spreading along a curve)**.** *Let $\pi\colon W \to C$ be a smooth projective morphism onto a smooth projective connected curve, $W$ projective, and $\xi$ a global section of $R^{2p}\pi_*\mathbb{Q}$. If $\xi_t$ is algebraic for uncountably many $t \in C$, there are an algebraic class $\tilde{\xi}$ on $W$ and an integer $d \geq 1$ with $\tilde{\xi}|_{W_t} = d\,\xi_t$ for every $t \in C$.*

*Proof.* The proof of Theorem 16.4 applies verbatim with $(\mathcal{A}, \mathcal{S}, \omega)$ replaced by $(W, C, \xi)$: the set of $t$ at which $\xi_t$ is algebraic is the union, over countably many tuples of Hilbert polynomials and rational coefficients, of the images of unions $T$ of connected components of fibre products of relative Hilbert schemes, projective over $C$, on which $\sum_i q_i[Z_i] = \xi$. Uncountably many $t$ force one image to be infinite, hence equal to $C$. Choose an irreducible component of that $T$ surjecting onto $C$, two of its points over distinct points of $C$, and an irreducible curve through them; its normalisation $C'$ maps finitely onto $C$, of some degree $d$. Pulling the universal subschemes back to $W' = W \times_C C'$ gives a cycle $z = \sum_i q_i[\mathcal{Z}_i]$ whose restriction to $W'_{c'} = W_{g(c')}$ is $\xi_{g(c')}$ for every $c' \in C'$. The projection $h\colon W' \to W$ is finite and flat of degree $d$, and by base change $h_*[z]$ restricts on $W_t$ to the sum of the restrictions over the points of $C'$ above $t$, counted with multiplicity, which is $d\,\xi_t$. ∎

**Theorem 20.28** ((L) for an abelian scheme over a curve is propagation)**.** *Let $\pi\colon W \to C$ be as in Theorem 20.21, but with no hypothesis on its invariant cycles, and keep the notation of its proof. Let $\Lambda_C$ be the operator that is zero on $H^p(C, R^q\pi_*\mathbb{Q})$ for $p \neq 2$ and inverts $L_C\colon H^q_{(q)} \to H^{q+2}_{(q)}$.*

(i) *The operator $\Lambda$ of Definition 10.6 for $L_W$ is $\Lambda_{\mathrm{rel}} + \Lambda_C$. So $B(W)$ holds for $L_W$ if and only if $\Lambda_C$ is induced by an algebraic cycle.*

(ii) *If $B(W)$ holds for $L_W$, then for every $c \in C$ and every algebraic class $y$ on $W_c$ the class $\Lambda_C(i_{c*}y)$ is algebraic; up to the factor $m$ it lies in $H^0(C, R^q\pi_*\mathbb{Q})$ and restricts on $W_c$ to the unique monodromy invariant class $v$ with $\int_{W_c} v \cup w = \int_{W_c} y \cup w$ for every invariant $w$ of complementary degree. In particular every global section of $R^{2p}\pi_*\mathbb{Q}$ that is algebraic at one point is the restriction of an algebraic class on $W$, and is algebraic at every point.*

(iii) *If at one point $c_0$ every monodromy invariant class of $H^*(W_{c_0}, \mathbb{Q})$ is algebraic, then $B(W)$ holds for $L_W$ if and only if every monodromy invariant class is the restriction of an algebraic class on $W$.*

*Proof.* (i) Steps 4 and 5 of the proof of Theorem 20.21 show that $L_C\colon H^q_{(q)} \to H^{q+2}_{(q)}$ is an isomorphism without using the hypothesis on invariant cycles: the nondegeneracy of $I_q \times I_{2g-q} \to \mathbb{Q}$ comes from the Hodge–Riemann relations alone. So $\Lambda_C$ is defined, and Steps 5 and 6, which use only that $\Lambda_C$ vanishes off $p = 2$ and inverts $L_C$, give $[L_W, \Lambda_{\mathrm{rel}} + \Lambda_C] = H$. By the uniqueness in Definition 10.6 this is $\Lambda$. As $\Lambda_{\mathrm{rel}}$ is induced by an algebraic cycle (Step 3), $\Lambda$ is if and only if $\Lambda_C$ is.

(ii) The Gysin map $i_{c*}$ commutes with $[2]_*$, which acts on $H^q(W_c)$ by $2^{2g-q}$, so $i_{c*}y \in H^{q+2}_{(q)}$, the summand with $p = 2$. If $y$ is algebraic so is $i_{c*}y$, and by (i) so is $\Lambda_C(i_{c*}y)$. Put $u = m\,\Lambda_C(i_{c*}y) \in H^q_{(q)}$, so $f \cup u = i_{c*}y$. For $w \in H^{2g-q}_{(2g-q)}$ the projection formula gives $\int_{W_c} u|_{W_c} \cup w|_{W_c} = \int_W (f \cup u) \cup w = \int_W i_{c*}y \cup w = \int_{W_c} y \cup w|_{W_c}$, and $w|_{W_c}$ runs over the invariants of complementary degree (Step 4); by the nondegeneracy of Step 4 this determines $u|_{W_c}$. If $y = \xi_c$ for a global section $\xi$, then $v = \xi_c$ has the property, so $u|_{W_c} = \xi_c$, and $u$, being a global section of $R^q\pi_*\mathbb{Q}$ equal to $\xi$ at $c$, restricts to $\xi_t$ at every $t$.

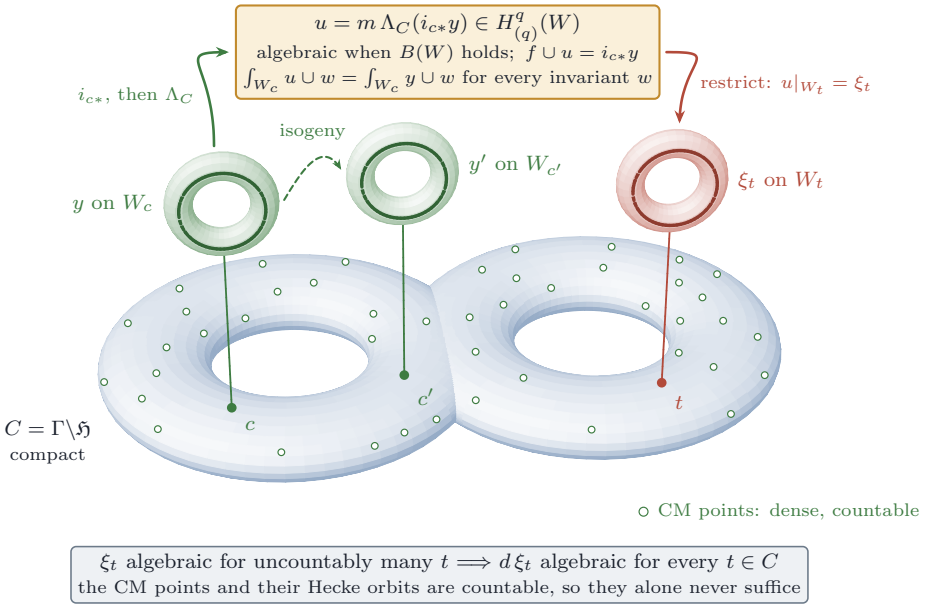



FIGURE 41. Theorem 20.28(ii) along a compact Shimura curve $C$. An algebraic class $y$ on one fibre, at a base point of a Weil family or at a CM point of a Mumford family (hollow dots: the CM points, countable and dense), is pushed into $W$ by the Gysin map and brought back to the invariant summand by the base operator $\Lambda_C$. When $B(W)$ holds the result $u$ is an algebraic class on $W$, pinned down by the displayed pairing against the invariant classes, and its restriction to a very general fibre is the invariant part of $y$. The dashed arcs are isogenies between CM members along Hecke orbits: they carry algebraic classes to algebraic classes but reach only countably many points, and Lemma 20.27, in the box, needs uncountably many. This is how Corollary 20.31 and Corollary 20.30 turn each target into one case of (L).

(iii) If every invariant class is a restriction of an algebraic class, $B(W)$ holds by Theorem 20.21. Conversely, under $B(W)$, (ii) applied at $c_0$ to each invariant class makes it the restriction of an algebraic class. ∎

Figure 41 draws part (ii) along a compact Shimura curve.

**Theorem 20.29** (The Mumford square at a CM point)**.** *Let $W \to C$ be a family as in Corollary 20.22 and $c \in C$ a CM point. Then every Hodge class on $X_c \times X_c$ is algebraic. In particular the two exceptional classes of Proposition 20.74 are algebraic on $X_c \times X_c$.*

*Proof.* The Hodge group $T$ of $X_c$ is a $\mathbb{Q}$-torus contained in $G$. It lies in a maximal $\mathbb{Q}$-torus of $G = \mathrm{Res}_{F/\mathbb{Q}}\mathrm{SL}_1(D)$, which is the restriction of scalars of the norm one torus of a quadratic extension of $F$ inside $D$, of rank three, one factor for each embedding of $F$. The cocharacter of the Hodge structure lies in the factor of the embedding at which $D$ splits, and the Galois group permutes the embeddings of $F$ transitively, so the smallest $\mathbb{Q}$-subtorus containing it has rank three: $T_{\mathbb{C}}$ is a maximal torus of $G_{\mathbb{C}} \cong \mathrm{SL}_2^3$. Its eight weights on $V = H^1(X_c, \mathbb{C}) = V_1 \otimes V_2 \otimes V_3$ are the vectors $(\pm 1, \pm 1, \pm 1)$, pairwise distinct. Hence $\mathrm{End}^0(X_c) \otimes \mathbb{C} = \mathrm{End}_T(V)$ is the algebra of diagonal matrices in a weight basis $(x_w)$.

For $n \geq 1$, $\mathrm{Hom}(X_c^n, X_c) \otimes \mathbb{C}$ is the space of $n$-tuples $(\varphi_1, \ldots, \varphi_n)$ of such matrices, acting by $x_w \mapsto x_w \otimes u_w$ with $u_w = (\varphi_{1,w}, \ldots, \varphi_{n,w}) \in \mathbb{C}^n$, so the vectors $u_w$ can be prescribed independently. The Hodge classes of $X_c^n$, tensored with $\mathbb{C}$, are spanned by the products of vectors $x_w \otimes u$, $u \in \mathbb{C}^n$, in $\bigwedge^*(V \otimes \mathbb{C}^n)$ in which the weights $w$, counted with multiplicity, sum to zero.

We show that for every $n \geq 1$ the ring $\mathrm{Hdg}^*(X_c^n)$ is generated by pull-backs of Hodge classes of $X_c$ along homomorphisms $X_c^n \to X_c$. Put $w_1 = (1,1,1)$, $w_2 = (1,-1,-1)$, $w_3 = (-1,1,-1)$ and $w_4 = (-1,-1,1)$; the weights are the $\pm w_i$, $w_1 + w_2 + w_3 + w_4 = 0$, and any three of the $w_i$ are linearly independent, so the integer relations among them are the multiples of $(1,1,1,1)$. Let $\alpha_i$ and $\beta_i$ be the multiplicities of $w_i$ and $-w_i$ in a nonempty multiset of weights with sum zero. Then $\alpha_i - \beta_i$ is independent of $i$; so either some $\alpha_i$ and $\beta_i$ are both positive, or all $\alpha_i$ are positive, or all $\beta_i$ are. In each case the multiset contains one of the sets $\{w_i, -w_i\}$, $\{w_1, w_2, w_3, w_4\}$, $\{-w_1, -w_2, -w_3, -w_4\}$, each of sum zero, and by induction every product above is, up to sign, a product of monomials $\bigwedge_{w \in P}(x_w \otimes u_w)$ over such sets $P$, whose weights are pairwise distinct. Choosing $f \in \mathrm{Hom}(X_c^n, X_c) \otimes \mathbb{C}$ with $f^* x_w = x_w \otimes u_w$ for $w \in P$ gives $\bigwedge_{w \in P}(x_w \otimes u_w) = f^*(\bigwedge_{w \in P} x_w)$, with $\bigwedge_{w \in P} x_w$ a Hodge class of $X_c$ tensored with $\mathbb{C}$. The map $f \mapsto f^*(t)$ is polynomial and the rational homomorphisms are Zariski dense in their complexification, so the same span is reached with $f$ defined over $\mathbb{Q}$ and $t$ rational. The Hodge classes of the abelian fourfold $X_c$ are algebraic because the Hodge conjecture holds for abelian fourfolds [Mar25a], and pull-backs and products of algebraic classes are algebraic. The case $n = 2$ is the theorem. ∎

**Corollary 20.30** (The first classes beyond the Weil lines are one case of (L))**.** *Let $W \to C$ be as in Corollary 20.22 and $W^{(2)} = W \times_C W \to C$, an abelian scheme of relative dimension eight with $W^{(2)}$ projective of dimension nine. The following are equivalent.*

(a) *$B(W^{(2)})$ holds for the class $L_{W^{(2)}}$.*

(b) *The two exceptional classes of $X_t \times X_t$ are algebraic for uncountably many $t \in C$.*

(c) *The Hodge conjecture holds for $X_t \times X_t$ for every $t \in C$.*

(d) *The Hodge conjecture holds for $W^{(2)}$; for this item $C$ is taken to be the compact Shimura curve of torsion-free level, or a finite étale cover of it.*

*Proof.* As in the proof of Corollary 20.22, the connected monodromy group is $G$, so the monodromy invariants in $H^*(X_t \times X_t, \mathbb{Q})$ are contained in the $G$-invariants, which are the Hodge classes of $X_t \times X_t$ for $t$ very general; conversely a $G$-invariant class is flat, being fixed by the monodromy. The Hodge group of every fibre lies in $G$, so the invariant classes are Hodge classes on every fibre. At a point $t$ that is not a CM point the Hodge group $H_t$ of $X_t$ is $G$. Indeed, if $H_t$ were smaller, the Hodge classes of the powers of $X_t$ that are not $G$-invariant would stay Hodge only on a discrete subset of the period domain, the upper half plane; every point of the orbit of the Hodge structure at $t$ under $H_t(\mathbb{R})^+$ carries them, so that orbit is a point, the cocharacter of the Hodge structure is central in $H_t$, and $H_t$, the smallest $\mathbb{Q}$-group containing it, is a torus: $t$ would be a CM point. So at such $t$ the Hodge classes of $X_t \times X_t$ are the invariant ones, and $\mathrm{End}^0(X_t) = \mathrm{End}_G(V) = \mathbb{Q}$, because $V \otimes \bar{\mathbb{Q}} = V_1 \otimes V_2 \otimes V_3$ is an irreducible representation of $G$; thus $X_t$ is as in Setup 21.1.

(c) implies (b) trivially. (b) implies (c): the exceptional classes are invariant, hence flat, so by Lemma 20.27 a multiple of each is the restriction of an algebraic class on $W^{(2)}$, and they are algebraic on every fibre. At a point that is not a CM point, Theorem 21.7 with $n = 2$ makes every Hodge class of $X_t \times X_t$ algebraic; at a CM point this is Theorem 20.29. (c) implies (a): by (c) every invariant class is algebraic on every fibre, so by Lemma 20.27 a multiple of it is the restriction of an algebraic class on $W^{(2)}$, and Theorem 20.21 gives

$B(W^{(2)})$. (a) implies (c): at a CM point $c_0$ every Hodge class of $X_{c_0} \times X_{c_0}$ is algebraic by Theorem 20.29, so by Theorem 20.28(iii) every invariant class is the restriction of an algebraic class, and is algebraic on every fibre. This is (c) at the points that are not CM points, where the Hodge classes are the invariant ones; at a CM point (c) is Theorem 20.29.

For (d), split $H^*(W^{(2)}, \mathbb{Q})$ into the summands $H^p(C, R^q)$, $p \in \{0, 1, 2\}$, as in the proof of Proposition 20.26. The summands with $p = 1$ carry no Hodge class, by the argument given there, applied to the fibre square. Those with $p = 2$ are $f \cup H^q_{(q)}$, and $f \cup u = i_{c*}(u|_{W_c^{(2)}})$ for any $c$; taking $c$ a CM point, where $u|_{W_c^{(2)}}$ is a Hodge class and so algebraic by Theorem 20.29, they are algebraic. The summands with $p = 0$ are the invariants, all of Hodge type. So (d) holds exactly when every invariant class is the restriction of an algebraic class, which by Theorem 20.28(iii), applied at a CM point, is (a). ∎

**Corollary 20.31** (One Weil family is one case of (L))**.** *Keep Setup 16.1, let $s_0 \in \mathcal{S}$ be the image of a base point of Theorem 15.27, and let $t_1 \in \mathcal{S}$ lie outside every proper Zariski closed set $\Sigma_{\underline{P},\underline{q}}$ of Theorem 16.4. Let $\kappa\colon C \to \mathcal{S}$ be a morphism from a smooth projective connected curve whose image contains $s_0$ and $t_1$, and $W_C = \mathcal{A} \times_{\mathcal{S}} C$.*

(i) *If $B(W_C)$ holds for $L_{W_C}$, then $\Sigma = \mathcal{D}_{n,n}$, and the Hodge conjecture holds in every degree for every member of the family with $\mathrm{Hg}(A) = \mathrm{SU}(V, H)$.*

(ii) *Conversely, if $\Sigma = \mathcal{D}_{n,n}$ and the monodromy of $W_C$ is Zariski dense in $\mathrm{SU}(V, H)$, then $B(W_C)$ holds for $L_{W_C}$.*

*Proof.* (i) The class $\kappa^*\omega$ is a global section of $R^{2n}\pi_*\mathbb{Q}$ on $C$, algebraic at a point over $s_0$ by Theorem 15.27. By Theorem 20.28(ii) it is algebraic at a point over $t_1$, so $t_1$ lies in some closed set $\Sigma_{\underline{P},\underline{q}}$, which is therefore all of $\mathcal{S}$. Theorem 16.4 gives $\Sigma = \mathcal{D}_{n,n}$, and Theorem 15.14(a) the rest. (ii) With dense monodromy the invariants are the $\mathrm{SU}(V, H)$-invariants, which are the Hodge classes of a member with $\mathrm{Hg}(A) = \mathrm{SU}(V, H)$: the powers of the polarisation and the Weil classes, by Theorem 4.14. All are algebraic on every fibre when $\Sigma = \mathcal{D}_{n,n}$, so they are restrictions of algebraic classes by Lemma 20.27, and Theorem 20.21 applies. ∎

Curves $\kappa$ as in Corollary 20.31 exist: the Baily–Borel compactification of $\mathcal{S}$ is projective [BB66], its boundary has codimension $2n - 1 \geq 3$, the rational boundary components of the domain of type $\mathrm{I}_{n,n}$ being of type $\mathrm{I}_{n-r,n-r}$ with $r \geq 1$, and a general complete intersection curve through $s_0$ and $t_1$ avoids it.

**Theorem 20.32** (Two branches cannot be two objects)**.** *Let $A$ be of $(K, -1, n)$-Weil type with $n \geq 2$, $\omega \neq 0$ a rational class on the Weil line, and $E$ a perfect complex with $\mathrm{ch}(E) = N\omega$, $N \neq 0$, and $\sigma_E$ injective. Let $E \cong \bigoplus_i F_i[k_i]$ be any decomposition. Then every summand $F_i$ with $\mathrm{ch}(F_i) \neq 0$ has a Yoneda product $\bigwedge^2 \mathrm{Ext}^1(F_i, F_i) \to \mathrm{Ext}^2(F_i, F_i)$ that is not injective. In particular $E$ is not a direct sum of shifts of line bundles, of skyscraper sheaves of points, or of their images under autoequivalences of $D^b(A)$, since the Ext algebra of each of these is the exterior algebra on its $\mathrm{Ext}^1$.*

*Proof.* By Theorem 16.37(ii) the kernel of $\mathrm{ev}_E$ in degree two is $P \wedge Q$. The characteristic morphism is compatible with the decomposition, so $\mathrm{ev}_{F_i}(p \wedge q) = \mathrm{ev}_{F_i}(p)\,\mathrm{ev}_{F_i}(q) = 0$ for all $p \in P$ and $q \in Q$. Suppose the Yoneda product of $F = F_i$ is injective on $\bigwedge^2 \mathrm{Ext}^1$. Then the images $\bar{P}$ and $\bar{Q}$ of $P$ and $Q$ in $\mathrm{Ext}^1(F, F)$ satisfy $\bar{p} \wedge \bar{q} = 0$ for all $\bar{p}, \bar{q}$, so either $\bar{P} = 0$, or $\bar{Q} = 0$, or $\bar{P} = \bar{Q}$ is a line. In the first case $\mathrm{ev}_F$ kills $P \wedge HH^1$, in the second $Q \wedge HH^1$, in the third all of $HH^2 = HH^1 \wedge HH^1$; by the compatibility $\sigma_F \circ \mathrm{ev}_F = (\,\cdot\,) \lrcorner \mathrm{ch}(F)$ of Setup 16.33 the same subspace annihilates $x = \mathrm{ch}(F)$.

These annihilators are computed in the free module of rank one that $H^*(A,\mathbb{C})$ is over $HH^*(A)=\bigwedge^*(P\oplus Q)$, with generator the holomorphic volume form $\Omega$: an element $y\cdot\Omega$ is killed by $P\wedge HH^1$ exactly when $p\wedge y$ has top degree for every $p\in P$, that is when $y\in\det P\wedge\bigwedge^* Q+\bigwedge^{\geq 4n-1}(P\oplus Q)$. Since $\det P\cdot\Omega=\pm\alpha_+$, the first summand gives the classes $\bigwedge^j Q\cdot\alpha_+$, of type $(n-b,n+a)$ with $a$ wedges by $V_-^{0,1}$ and $b$ contractions by $T_+$, and the second gives classes of type $(1,2n)$, $(0,2n-1)$ and $(0,2n)$. The components of $x$ are of type $(k,k)$, so $x\in\mathbb{C}\alpha_+$. Symmetrically the second case gives $x\in\mathbb{C}\alpha_-$ and the third $x=0$. Finally $x$ is rational, complex conjugation carries $V_+$ to $V_-$ and $H^{1,0}$ to $H^{0,1}$, so $\bar\alpha_+\in\mathbb{C}\alpha_-$, and a rational multiple of $\alpha_+$ or of $\alpha_-$ is zero. So $\operatorname{ch}(F)=0$. As $\sum_i(-1)^{k_i}\operatorname{ch}(F_i)=N\omega\neq 0$, some summand has nonzero Chern character, and the first statement follows; the second is the special case in which every summand has an exterior Ext algebra. ■

20.9. **Which hypotheses suffice.** The implications proved above, together with those of Section 20.6, determine exactly which combinations of statements yield the conjecture. The logical skeleton of the paper and of the literature it quotes is finite: a set of statements, each of them proved here, quoted from the literature, or taken as a hypothesis, and a set of inference rules, each of them one theorem. Consequences can then be taken.

The answer is only as strong as the rule set. A set of statements the rules do not carry to the conjecture is *insufficient* in the sense of the rules, and that means no more than that no rule derives the conjecture from it. An implication that is true and left out makes a set look necessary when it is not. One such implication is elementary, and it decides the shape of the answer, so we prove it first.

**Proposition 20.33** (The conjecture for the varieties that are not abelian is the conjecture)**.** *Let $A$ be a complex abelian variety of dimension $g\geq 1$ and let $X=A\times\mathbb{P}^1$. Then $X$ is a smooth complex projective variety that is not an abelian variety, and if every Hodge class of degree $2p$ on $X$ is algebraic, then so is every Hodge class of degree $2p$ on $A$. Consequently the statement*

> *the Hodge conjecture holds for every smooth complex projective variety that is not an abelian variety*

*implies the Hodge conjecture for every abelian variety, and it is equivalent to* **HC**.

*Proof.* $X$ is smooth and projective of dimension $g+1$. By the Künneth formula $h^{1,0}(X)=h^{1,0}(A)+h^{1,0}(\mathbb{P}^1)=g$, while an abelian variety of dimension $g+1$ has $h^{1,0}=g+1$; so $X$ is not an abelian variety. (Equally, $X$ contains the rational curves $\{a\}\times\mathbb{P}^1$, and every morphism from $\mathbb{P}^1$ to a complex torus is constant, since it lifts to the universal cover $\mathbb{C}^{g+1}$ [BL04, Chapter 1].)

Let $\mathrm{pr}\colon X\to A$ be the projection and $D=A\times\{0\}$, a smooth divisor that pr maps isomorphically onto $A$, so that $\mathrm{pr}_*[D]=1$ in $H^0(A,\mathbb{Q})$. Let $\alpha$ be a Hodge class of degree $2p$ on $A$. Pull-back along a morphism is a morphism of Hodge structures, so $\mathrm{pr}^*\alpha$ is a Hodge class of degree $2p$ on $X$, and by hypothesis $\mathrm{pr}^*\alpha=[Z]$ for an algebraic cycle $Z$ on $X$ with rational coefficients. The projection formula gives

$$\mathrm{pr}_*\big([Z]\cup[D]\big)=\mathrm{pr}_*\big(\mathrm{pr}^*\alpha\cup[D]\big)=\alpha\cup\mathrm{pr}_*[D]=\alpha.$$

The cycle class map is compatible with intersection products and with proper push-forward [Ful98, Chapter 19], so the left side is the class of the cycle $\mathrm{pr}_*(Z\cdot D)$ and $\alpha$ is algebraic. A point, the abelian variety of dimension zero, satisfies the conjecture trivially.

So the displayed statement gives the conjecture for every abelian variety, and with it the conjecture for every smooth complex projective variety. The converse is immediate: the displayed statement is a special case of **HC**. ∎

Since the conjecture for the varieties that are not abelian is the whole conjecture, the statement that belongs on a frontier is weaker: it asks for algebraicity only modulo what abelian varieties supply.

**Definition 20.34** (The conjecture modulo abelian varieties)**.** For a smooth complex projective variety $X$ and $p \geq 0$ let $\mathrm{Ab}^p(X) \subseteq H^{2p}(X, \mathbb{Q})$ be the span of the classes $\Gamma_*\gamma = \mathrm{pr}_{X*}([\Gamma] \cup \mathrm{pr}_B^*\gamma)$ of degree $2p$, where $B$ runs over the complex abelian varieties, the point included, $\gamma$ over the Hodge classes on $B$, and $\Gamma$ over the algebraic cycles on $B \times X$ with rational coefficients. The statement (F3′) is: for every $X$ and every $p$,

$$\mathrm{Hdg}^p(X) = \mathrm{Ab}^p(X).$$

Each $\Gamma_*\gamma$ is a Hodge class, since cup product with an algebraic class and push-forward preserve Hodge classes, so $\mathrm{Ab}^p(X) \subseteq \mathrm{Hdg}^p(X)$; and taking $B$ a point shows that $\mathrm{Ab}^p(X)$ contains every algebraic class. So (F3′) says that every Hodge class is algebraic modulo the images of Hodge classes of abelian varieties under algebraic correspondences, which is the form of (F3) that Remark 20.72 asks for.

**Proposition 20.35** (What is known of (F3′))**.**

(i) **HC** *implies* (F3′)*, and* (F3′) *together with the conjecture for abelian varieties implies* **HC***.*

(ii) (F3′) *holds on every abelian variety and on every product* $A \times \mathbb{P}^1$ *with* $A$ *abelian; the argument of Proposition 20.33 gives nothing for it.*

(iii) *Let* $\mathcal{A}$ *be the class of smooth complex projective varieties* $X$ *for which* $H^*(X, \mathbb{Q})$ *is spanned by classes* $\Gamma_* y$ *with* $B$ *abelian,* $\Gamma$ *an algebraic cycle on* $B \times X$ *and* $y \in H^*(B, \mathbb{Q})$*. Every* $X \in \mathcal{A}$ *satisfies* (F3′)*. The class* $\mathcal{A}$ *contains every abelian variety and every smooth projective curve, and it is closed under products and under images of surjective morphisms. So it contains every product of curves, every symmetric product of a curve, and every smooth projective variety onto which a product of curves or an abelian variety maps surjectively.*

(iv) *Every* $X$ *of dimension* $n$ *satisfies* (F3′) *in the degrees* $2p$ *with* $p \in \{0, 1, n-1, n\}$*, so every* $X$ *of dimension at most three satisfies it in every degree.*

*Proof.* (i) If **HC** holds, then $\mathrm{Hdg}^p(X)$ consists of algebraic classes, which lie in $\mathrm{Ab}^p(X)$. Conversely, if the conjecture holds for abelian varieties, every $\gamma$ in Definition 20.34 is algebraic, so every $\Gamma_*\gamma$ is, since composition with a correspondence carries algebraic classes to algebraic classes [Ful98, Chapter 16]; then (F3′) says that every Hodge class is algebraic.

(ii) On an abelian variety $A$ take $B = A$ and $\Gamma$ the diagonal. On $X = A \times \mathbb{P}^1$ the Künneth formula gives $H^*(X, \mathbb{Q}) = \mathrm{pr}^*H^*(A, \mathbb{Q}) \oplus \mathrm{pr}^*H^*(A, \mathbb{Q}) \cup \eta$, with $\eta$ the class of a point of $\mathbb{P}^1$ pulled back to $X$, which is algebraic. The first summand is $\Gamma_*H^*(A, \mathbb{Q})$ for $\Gamma$ the transpose of the graph of pr, and the second is $\Gamma'_*H^*(A, \mathbb{Q})$ for $\Gamma' = \Gamma \cdot \mathrm{pr}_X^*\eta$. So $X \in \mathcal{A}$, and (iii) applies.

(iii) Let $X \in \mathcal{A}$ and fix $p$. Since $H^{2p}(X, \mathbb{Q})$ is finite dimensional, finitely many classes $\Gamma_{j*}y_j$ span it, and we may take each $\Gamma_j$ of pure codimension $c_j$ on $B_j \times X$ and each $y_j$ of pure degree $k_j$, with $k_j + 2c_j - 2\dim B_j = 2p$. Then $\Gamma_{j*}\colon H^{k_j}(B_j, \mathbb{Q}) \to H^{2p}(X, \mathbb{Q})$ is a morphism of Hodge structures up to a Tate twist, and the sum of these maps is a surjective morphism of polarisable Hodge structures onto $H^{2p}(X, \mathbb{Q})$. Polarisable Hodge structures

are semisimple, so its kernel has a complement that is a sub-Hodge structure and maps isomorphically onto $H^{2p}(X,\mathbb{Q})$; hence every Hodge class of degree $2p$ is the image of a Hodge class, that is $\sum_j \Gamma_{j*}\gamma_j$ with each $\gamma_j$ a Hodge class on $B_j$. So $\mathrm{Hdg}^p(X) = \mathrm{Ab}^p(X)$.

An abelian variety lies in $\mathcal{A}$ through its diagonal. A curve $C$ of genus $g \geq 1$ lies in $\mathcal{A}$ through an Abel–Jacobi embedding $\iota\colon C \to J(C)$: the transpose of its graph induces $\iota^*$, which is an isomorphism on $H^0$ and $H^1$ and sends the class of the theta divisor to $g$ times the class of a point, so it is surjective; the projective line is the image of an elliptic curve under a double cover, and is covered by the last closure property. For products, if $H^*(X_i,\mathbb{Q})$ is spanned by the $\Gamma_{i*}y$, then by the Künneth formula $H^*(X_1\times X_2,\mathbb{Q})$ is spanned by the $(\Gamma_1\times\Gamma_2)_*(y_1\otimes y_2)$, with $\Gamma_1\times\Gamma_2$ an algebraic cycle on $(B_1\times B_2)\times(X_1\times X_2)$. For images, let $f\colon Y\to X$ be a surjective morphism with $Y\in\mathcal{A}$, let $r=\dim Y-\dim X$ and let $h$ be an ample class on $Y$. The correspondence $\Psi=[\Gamma_f]\cdot \mathrm{pr}_Y^* h^r$ on $Y\times X$ induces $\Psi_* y = f_*(y\cup h^r)$, and $\Psi_* f^* x = x\cup f_* h^r = c\,x$ with $c>0$ the degree of $h^r$ on a general fibre. So $\Psi_*$ is surjective, and $H^*(X,\mathbb{Q})$ is spanned by the classes $(\Psi\circ\Gamma)_* y$, with $\Psi\circ\Gamma$ algebraic [Ful98, Chapter 16]. A symmetric product of a curve is the image of a power of it.

(iv) Every class of degree 0 or $2n$ is algebraic, and every Hodge class of degree 2 is algebraic by the Lefschetz theorem on $(1,1)$ classes. For $n\geq 2$, the hard Lefschetz theorem makes cup product with $h^{n-2}$, for $h$ an ample class, an isomorphism $H^2(X,\mathbb{Q})\to H^{2n-2}(X,\mathbb{Q})$ of Hodge structures up to a Tate twist, so every Hodge class of degree $2n-2$ is $h^{n-2}$ times a Hodge class of degree 2, and is algebraic. Algebraic classes lie in $\mathrm{Ab}^p(X)$. When $n\leq 3$ every $p$ between 0 and $n$ is one of $0,1,n-1,n$. ∎

*Remark* 20.36 (The reach of algebraic correspondences). (F3′) is the statement that the Hodge conjecture reduces to abelian varieties along algebraic correspondences, and Proposition 20.35 records the cases of such a reduction that are theorems: the varieties of $\mathcal{A}$, whose whole cohomology is reached from abelian varieties by algebraic correspondences, and the degrees in which the conjecture itself is a theorem. The next degree is degree four on fourfolds outside $\mathcal{A}$ and outside the scope of Proposition 20.43 and Proposition 20.44. The square $S\times S$ of a K3 surface is the standard example when $\mathrm{End}_{\mathrm{Hdg}}(T(S))$ is a totally real field of degree at least two, that is, when $S$ has real multiplication. When that field is $\mathbb{Q}$ or a CM field, the Hodge conjecture holds on $S\times S$: by Proposition 20.38(i) in the first case, and in the second by [Bus19] and the fact that the elements of norm one span a CM field, as in [RM08] (Proposition 20.39). Its Hodge classes in $H^2(S)\otimes H^2(S)$ include the classes that induce the Hodge endomorphisms of the transcendental lattice $T(S)$, and $S\times S$ lies in $\mathcal{A}$ as soon as the Kuga–Satake embedding of $T(S)$ is induced by an algebraic cycle, the Kuga–Satake Hodge conjecture for $S$: the transpose of such a cycle maps the cohomology of the Kuga–Satake variety onto $T(S)$, and the rest of $H^*(S,\mathbb{Q})$ is algebraic. The rational Hodge isometries are algebraic [Bus19], but a real multiplication is not an isometry, and Remark 21.5 meets exactly this obstacle for the surfaces $S_\lambda$; Theorem 23.14 crosses it for two cyclic cubic fields by a dihedral family of correspondences, and Proposition 23.15 shows that this family reaches no other field.

For varieties whose cohomology is not of abelian type the position is starker. By Proposition 20.75 no correspondence moves the $H^3$ of a very general quintic threefold into the cohomology of an abelian variety, so such a variety does not lie in $\mathcal{A}$. That is no obstruction to (F3′) itself, since a Hodge class spans a Hodge structure of Tate type, which is of abelian type, and nothing in Hodge theory forbids it from lying in $\mathrm{Ab}^p(X)$. Nor is it a route: to put a Hodge class into $\mathrm{Ab}^p(X)$ one must construct an algebraic cycle on $B\times X$ whose class carries it, which is the kind of construction the Hodge conjecture

asks for, now on $B \times X$ instead of on $X$. The cases of (F3′) proved below outside $\mathcal{A}$ come from the Hodge conjecture itself, from hypotheses on zero-cycles, and, in degree four, from correspondences with members of $\mathcal{A}$ (Proposition 20.39 to Proposition 20.46); none of them reaches a variety in general position, and (F3′) enters Theorem 20.70 as a hypothesis.

Two further facts fix the position of (F3′). The first says how strong it is.

**Proposition 20.37** (The strength of (F3′))**.** (F3′) *implies* (M)*: every Hodge class on every smooth complex projective variety is motivated in the sense of André. In particular it implies that every such class is absolutely Hodge. Adding the implication* (F3′) ⇒ (M) *to the rule set of Setup 20.69 leaves the five minimal sufficient sets of Theorem 20.70(ii) unchanged.*

*Proof.* Let $\gamma \in \mathrm{Hdg}^p(X)$. By (F3′), $\gamma = \sum_j \Gamma_{j*}\beta_j$ with $\beta_j$ a Hodge class on an abelian variety $B_j$ and $\Gamma_j$ an algebraic cycle on $B_j \times X$. Every Hodge class on an abelian variety is motivated [And96b, Théorème 0.6.2], and the motivated classes form a graded $\mathbb{Q}$-algebra that contains the algebraic classes and is stable under pull-back and push-forward along morphisms [And96b, §2]. So $\mathrm{pr}_{B_j}^*\beta_j \cup [\Gamma_j]$ is motivated, and so is its push-forward $\Gamma_{j*}\beta_j$; hence $\gamma$ is motivated. Motivated classes are absolutely Hodge [And96b, §2]. For the last assertion, the new rule has the premise (F3′) and the conclusion (M). The sets $C_1$ and $C_2$ of the proof of Theorem 20.70 omit (F3′), and $C_3$ and $C_4$ contain both statements, so the four sets stay closed and that proof applies without change; the five sets remain sufficient, since adding a rule only enlarges closures. ∎

So on the route {L, F3′} the second member already contains (M), and that route is never easier than {L, M}. The second fact locates, on the powers of a variety of orthogonal type, the first class that (F3′) has to reach.

**Proposition 20.38** (Powers of a variety of orthogonal type)**.** *Let $Y$ be a smooth complex projective variety of even dimension $m$. Suppose that $H^k(Y,\mathbb{Q})$ is spanned by algebraic classes for $k \neq m$, that $H^m(Y,\mathbb{Q}) = N \oplus V$ with $N$ spanned by algebraic classes, the intersection form $\psi$ nondegenerate on $N$ and $V = N^\perp$, and that the Hodge group of $V$ is $\mathrm{SO}(V,\psi)$, with $t = \dim V \geq 3$. Let $\det(V) \in H^{mt}(Y^t,\mathbb{Q})$ be the image of a generator of $\bigwedge^t V \subseteq V^{\otimes t}$ under the Künneth embedding $V^{\otimes t} \subseteq H^{mt}(Y^t,\mathbb{Q})$.*

(i) *The Hodge conjecture holds for $Y^n$ for every $n < t$.*

(ii) $\det(V)$ *is a Hodge class, and* $\mathrm{Hdg}^*(Y^t)$ *is spanned by algebraic classes and* $\det(V)$.

(iii) $\det(V)$ *does not lie in the $\mathbb{Q}$-algebra generated by the pull-backs along the projections of the Hodge classes of the $Y^j$, $j < t$.*

(iv) *The Hodge conjecture holds for every power of $Y$ if and only if* $\det(V)$ *is algebraic, and* (F3′) *holds for every power of $Y$ if and only if* $\det(V) \in \mathrm{Ab}(Y^t)$.

*Proof.* Algebraic classes have even degree, so $H^k(Y,\mathbb{Q}) = 0$ for odd $k$. Let $[\Delta] = \sum_k \pi_k$ be the Künneth decomposition of the class of the diagonal, $\pi_k \in H^{2m-k}(Y,\mathbb{Q}) \otimes H^k(Y,\mathbb{Q})$. For $k \neq m$, $\pi_k = \sum_i e_i^\vee \otimes e_i$ for a basis $(e_i)$ of $H^k(Y,\mathbb{Q})$ and the Poincaré dual basis of $H^{2m-k}(Y,\mathbb{Q})$, both spanned by algebraic classes, so $\pi_k$ is algebraic; likewise the part $\pi_N$ of $\pi_m$ in $N \otimes N$, since $\psi$ is nondegenerate on $N$. So $\pi_V = [\Delta] - \sum_{k\neq m} \pi_k - \pi_N$, the class of $\psi^{-1}$ in $V \otimes V \subseteq H^{2m}(Y \times Y,\mathbb{Q})$, is algebraic.

The Hodge group of $Y^n$ is $\mathrm{SO}(V,\psi)$, acting on $H^*(Y^n,\mathbb{Q}) = \big(H^*(Y,\mathbb{Q})\big)^{\otimes n}$ through the factors $V$ and trivially on the rest, which is spanned by algebraic classes; so the Hodge classes are the $\mathrm{SO}(V)$-invariants. By the first fundamental theorem for the special

orthogonal group [GW09, Chapter 5], the invariants of $\mathrm{SO}(V)$ in $V^{\otimes j}$ are spanned by the complete contractions with $\psi^{-1}$ when $j < t$, and, when $j = t$, by these and the determinant tensor; in general they are spanned by products of contractions and determinant tensors on disjoint sets of $t$ factors. A contraction of the factors $a$ and $b$ of $Y^n$ is the pull-back of $\pi_V$ along the projection to those two factors, a determinant on a set $I$ of $t$ factors is the pull-back of $\det(V)$ along the projection to $Y^I$, and products of these with pull-backs of algebraic classes span the Hodge classes. This gives (i), (ii) and the first half of (iv). For (iii), the orthogonal group $\mathrm{O}(V, \psi)$ acts on $H^*(Y^j, \mathbb{Q})$ in the same way, and for $j < t$ every $\mathrm{SO}(V)$-invariant is $\mathrm{O}(V)$-invariant; pull-backs along projections and cup products commute with the diagonal action of $\mathrm{O}(V)$, so every element of the algebra in (iii) is $\mathrm{O}(V)$-invariant, while a reflection acts on $\det(V)$ by $-1$. For the second half of (iv), $\mathrm{Ab}^*(Y^n)$ contains the algebraic classes and is stable under pull-back along projections and under cup product: $\Gamma_*\gamma \cup \Gamma'_*\gamma'$ is the image of $\gamma \otimes \gamma'$ on $B \times B'$ under the correspondence $(\Gamma \times \Gamma')$ followed by the diagonal of $Y^n$. ■

For a K3 surface $S$ with $\mathrm{End}_{\mathrm{Hdg}}(T(S)) = \mathbb{Q}$ the Hodge group of $T(S)$ is $\mathrm{SO}(T(S))$ [Zar83]; for a very general one, $t = 21$. So the Hodge conjecture holds for $S^n$ with $n \leq 20$, and on $S^{21}$ it is equivalent to the algebraicity of the single class $\det(T(S))$. The argument is the classical one from invariant theory, and we make no claim of priority for it. What it adds here is the location of the first instance of (F3′) on the powers of such a variety that invariant theory does not reach: one class, in one degree, on one power.

The same invariant theory, with the motivic decompositions of the literature and the theorems of Buskin and Markman on Hodge isometries, gives the Hodge conjecture itself, and so (F3′), on the other varieties built from a K3 surface and on varieties of $\mathrm{K3}^{[n]}$ type, up to a real multiplication and one determinant class.

**Proposition 20.39** (Powers, Hilbert schemes and moduli spaces of a K3 surface)**.** *Let $S$ be a projective K3 surface, $T = T(S)_{\mathbb{Q}}$, $t = \dim T$ and $E = \mathrm{End}_{\mathrm{Hdg}}(T)$, a totally real field or a CM field* [Zar83]*.*

(i) *If $E$ is a CM field, the Hodge conjecture holds for every power $S^k$* [RM08, Theorem 5.4]*,* [Bus19]*, hence for every Hilbert scheme $S^{[n]}$* [dCM02]*, for every smooth projective moduli space of Gieseker stable (twisted) sheaves on $S$, or of $\sigma$-stable objects in $D^b(S, \alpha)$ for a generic stability condition $\sigma$ and a Brauer class $\alpha$* [Bül20, Theorem 0.1, Remark 2.1]*, and for the crepant resolutions of the ten-dimensional moduli spaces $M_\sigma(2v_0)$ of $\sigma$-semistable objects in $D^b(S, \alpha)$, with $v_0$ primitive, $v_0^2 = 2$ and $\sigma$ $v_0$-generic* [FFZ21, Theorem 1.3]*.*

(ii) *If $E = \mathbb{Q}$, the Hodge conjecture holds for $S^{[n]}$ for every $n < t(t+1)/2$, and for every moduli space $M$ as in* (i) *with $\dim M < t$. For $N = t(t+1)/2$ the following are equivalent: the Hodge conjecture for $S^{[N]}$; the algebraicity of the class $\det(T)$ on $S^t$ of Proposition 20.38; the Hodge conjecture for every power of $S$; and the Hodge conjecture for every $S^{[n]}$. For a very general $S$, the Hodge conjecture holds for $S^{[n]}$ with $n \leq 230$, and for $S^{[231]}$ if and only if $\det(T(S))$ on $S^{21}$ is algebraic.*

(iii) *If $E$ is totally real of degree at least two, the Hodge conjecture for $S \times S$, the Hodge conjecture for $S^{[2]}$, and the statement that every element of $E$ acts on $T$ through an algebraic cycle on $S \times S$ are equivalent.*

*Wherever the Hodge conjecture holds, so does* (F3′)*. This is a consequence of* [RM08, Bus19, dCM02, Bül20, FFZ21] *and Proposition 20.38; we make no claim of priority.*

*Proof.* We use three facts. First, $H^*(S,\mathbb{Q}) = A \oplus T$ with $A = H^0 \oplus \mathrm{NS}(S)_\mathbb{Q} \oplus H^4$ spanned by algebraic classes; the Künneth components of the diagonal of $S$ are algebraic [Mur90], and so is its part $\pi_T$ in $T \otimes T$, as in the proof of Proposition 20.38. The Hodge classes of $S^k$ are the invariants in $(A \oplus T)^{\otimes k}$ of the Hodge group of $T$, which acts trivially on $A$. Second, if $H^*(X,\mathbb{Q})$ is spanned by the images of the $H^*(Y_i,\mathbb{Q})$ under algebraic correspondences and the Hodge conjecture holds for every $Y_i$, it holds for $X$: by the semisimplicity argument of the proof of Proposition 20.35(iii) every Hodge class of $X$ is a sum of images of Hodge classes of the $Y_i$. Third, the correspondences $\Gamma_\nu$ of the proof of Proposition 20.42(ii) give an isomorphism of Chow motives between $h(S^{[n]})$ and $\bigoplus_\nu h(S^{(\nu)})(l(\nu)-n)$ [dCM02], so each $h(S^{(\nu)})(l(\nu)-n)$ is a direct summand of $h(S^{[n]})$, and the lift $\widetilde{\Gamma}_{\nu *}$ takes the same value on a class of $S^{l(\nu)}$ and on its average over $G_\nu$. Hence the Hodge conjecture holds for $S^{[n]}$ if and only if, for every partition $\nu$ of $n$, every $G_\nu$-invariant Hodge class of $S^{l(\nu)}$ is algebraic. By [Bül20, Theorem 0.1, Remark 2.1] the Chow motive of a moduli space $M$ as in (i) is a direct summand of a sum of Tate twists of motives $h(S^{k_i})$ with $1 \le k_i \le \dim M$, and by [FFZ21, Corollary 4.5] the Chow motive of a crepant resolution as in (i) lies in the tensor subcategory generated by $h(S)$, whose objects are direct summands of sums of Tate twists of the $h(S^k)$, the dual of $h(S)$ being $h(S)(2)$.

(i) Over $\mathbb{C}$ the Hodge group of $T$ is a product of groups $\mathrm{GL}(W_\sigma)$ acting on $T_\mathbb{C} = \bigoplus_\sigma (W_\sigma \oplus W_\sigma^\vee)$ [Zar83], and by the first fundamental theorem for the general linear group [GW09, Chapter 5] its invariants in $T_\mathbb{C}^{\otimes b}$ are spanned by products of contractions of one factor with another. Under the isomorphism $\mathrm{End}(T) \cong T \otimes T$, $\varphi \mapsto \tau_\varphi = \sum_i t_i^\vee \otimes \varphi(t_i)$, with $(t_i^\vee)$ the dual basis of a basis $(t_i)$ for the intersection form, the Hodge classes of $T \otimes T$ are the $\tau_\varphi$ with $\varphi \in E$, and these contractions lie in their complex span. The rational span of the products of the $\tau_\varphi$ on pairs of factors has the same complexification as the space of invariants, so the Hodge classes of $S^k$ are spanned by such products and pull-backs of algebraic classes. For $u \in E$ with $u\bar{u} = 1$ the map $\mathrm{id}_{\mathrm{NS}} \oplus u$ is a rational Hodge isometry of $H^2(S,\mathbb{Q})$, since the adjoint of $u$ is $\bar{u}$ [Zar83]; it is induced by an algebraic cycle [Bus19], and composing that cycle with $\pi_T$ on both sides gives $\tau_u$, which is therefore algebraic. The elements of norm one span $E$. Indeed, write $E = E_0(y_0)$ with $y_0^2 = -\delta$, $\delta \in E_0$, and let $y \in E$ with $\bar{y} = -y$; for $s \in \mathbb{Q}$ the element $u_s = (1+sy)(1-sy)^{-1}$ has norm one. A linear form on $E$ vanishing on all $u_s$ gives a rational function of $s$ vanishing on $\mathbb{Q}$, so it vanishes on the coefficients $1$, $2y$, $2y^2$ of the expansion of $u_s$ in powers of $s$. As $y$ runs over $E_0 y_0$, the $y$ span $E_0 y_0$ and the $y^2 = -a^2\delta$, $a \in E_0$, span $E_0$, since $2a = (a+1)^2 - a^2 - 1$. So every $\tau_u$, $u \in E$, is algebraic, and the Hodge conjecture holds for every $S^k$; this is the argument of [RM08, Lemma 5.3, Theorem 5.4], which takes the algebraicity of the isometries from Mukai's announcement; [Bus19] proves it in general. We use [RM08] only when $E$ is a CM field: for a totally real $E$ the invariants of the Hodge group are not generated by those of degree two, since they include determinants, as Proposition 20.38 shows for $E = \mathbb{Q}$, and the powers do not reduce to the square. The rest of (i) follows from the second and third facts.

(ii) Here the Hodge group of $T$ is $\mathrm{SO}(T)$ [Zar83], and $t \ge 3$, since a Hodge structure of K3 type of dimension two has complex multiplication. The Hodge conjecture holds for $S^k$, $k < t$, by Proposition 20.38(i), which gives the assertion on $M$ by the second fact. Let $\nu$ be a partition of $n$ with $a_j$ parts equal to $j$, so that $G_\nu = \prod_j \mathfrak{S}_{a_j}$ permutes the factors of $S^{l(\nu)}$ within blocks of sizes $a_j$, one block for each distinct part size. By the proof of Proposition 20.38, the Hodge classes of $S^{l(\nu)}$ are spanned by algebraic classes and by products $\beta = \det_I \cup \gamma$, where $\det_I$ is the pull-back of $\det(T)$ along the projection to

a set $I$ of $t$ factors and $\gamma$ is pulled back from the other factors. If two factors of $I$ lie in one block, the transposition in $G_\nu$ exchanging them fixes $\gamma$ and changes the sign of the alternating class $\det_I$, all classes involved being of even degree; so it takes $\beta$ to $-\beta$, and the average of $\beta$ over $G_\nu$ is zero. If $\nu$ has fewer than $t$ distinct part sizes, every set of $t$ factors meets some block twice, so a $G_\nu$-invariant Hodge class, being its own average, is the average of an algebraic class, and is algebraic. A partition with $t$ distinct part sizes has $n \geq 1+2+\cdots+t = t(t+1)/2$, so by the third fact the Hodge conjecture holds for $S^{[n]}$ when $n < t(t+1)/2$ (Figure 42). For $n = N$ the partition $(1, 2, \ldots, t)$ has $l(\nu) = t$ and $G_\nu = 1$, so the Hodge class $\det(T)$ of $S^t$ is algebraic if the Hodge conjecture holds for $S^{[N]}$. If $\det(T)$ is algebraic, the Hodge conjecture holds for every power of $S$ by Proposition 20.38(iv), and for every $S^{[n]}$ by the second fact; and the conjecture for $S^t$ contains the algebraicity of $\det(T)$. A very general $S$ has $E = \mathbb{Q}$ and $t = 21$, and $N = 231$.

(iii) The Hodge structure $T$ has no nonzero Hodge class, so the Hodge classes of $A \otimes T$ and $T \otimes A$ vanish, and those of $S \times S$ are the algebraic classes and the $\tau_\varphi$, $\varphi \in E$. A cycle inducing $\varphi$ on $T$, composed with $\pi_T$ on both sides, gives $\tau_\varphi$, and $\tau_\varphi$ induces $\varphi$ on $T$; this proves the equivalence for $S \times S$. By [dCM02], $h(S^{[2]}) = h(S^{(2)}) \oplus h(S)(-1)$, and the Hodge conjecture holds for $S$. The exchange of the two factors takes $\tau_\varphi$ to $\tau_{\varphi^*}$, and the adjoint involution of a totally real $E$ is trivial [Zar83], so every $\tau_\varphi$ is invariant under $\mathfrak{S}_2$ and the Hodge conjecture for $S^{[2]}$ asks for the same classes as that for $S \times S$. ■

**Proposition 20.40** (Varieties of $\mathrm{K3}^{[n]}$ type)**.** *Let $X$ be a smooth projective hyperkähler variety deformation equivalent to the Hilbert scheme of $n \geq 2$ points on a K3 surface, $q$ the Beauville–Bogomolov form on $H^2 = H^2(X, \mathbb{Q})$, $\mathrm{NS} = \mathrm{NS}(X)_\mathbb{Q}$, $T = \mathrm{NS}^\perp$ and $E = \mathrm{End}_{\mathrm{Hdg}}(T)$, a totally real field or a CM field whose adjoint involution for $q$ is complex conjugation* [Zar83]*. Let $\mathrm{SH}(X) \subseteq H^*(X, \mathbb{Q})$ be the subalgebra generated by $H^2$, and for $\varphi \in E$ with $\varphi^* = \varphi$ let $c_\varphi = \sum_i t_i^\vee \cup \varphi(t_i) \in H^4(X, \mathbb{Q})$, with $(t_i)$ a basis of $T$ and $(t_i^\vee)$ its dual basis for $q$.*

(i) *The Hodge classes of $\mathrm{SH}(X)$ are spanned by products of divisor classes and classes $c_\varphi$.*

(ii) *If $E$ is $\mathbb{Q}$ or a CM field, every Hodge class of $\mathrm{SH}(X)$ is algebraic.*

(iii) *Let $n = 2$. Then $\mathrm{SH}(X) = H^*(X, \mathbb{Q})$, with $b_4(X) = 276 = \dim \mathrm{Sym}^2 H^2$, and $c_2(X) = \frac{6}{5} q^\vee$, where $q^\vee \in \mathrm{Sym}^2 H^2$ is dual to $q$. So the Hodge conjecture holds for $X$ when $E$ is $\mathbb{Q}$ or a CM field. When $E$ is totally real of degree at least two, it holds for $X$ if and only if every element of $E$, acting on $T$ and by zero on $\mathrm{NS}$, is induced by an algebraic cycle on $X \times X$.*

*This is a consequence of* [Mar24, CM13]*; we make no claim of priority.*

*Proof.* (i) Multiplication $\mathrm{Sym}^k H^2 \to \mathrm{SH}^{2k}(X)$ is a surjective morphism of polarisable Hodge structures, so by semisimplicity every Hodge class of $\mathrm{SH}^{2k}(X)$ is the image of one of $\mathrm{Sym}^k H^2 = \bigoplus_{a+b=k} \mathrm{Sym}^a \mathrm{NS} \otimes \mathrm{Sym}^b T$, whose Hodge classes are $\mathrm{Sym}^a \mathrm{NS} \otimes (\mathrm{Sym}^b T)^{\mathrm{Hdg}}$. The isomorphism $\mathrm{End}(T) \cong T \otimes T$ of the proof of Proposition 20.39, now for $q$, carries the adjoint to the exchange of the factors, so $(\mathrm{Sym}^2 T)^{\mathrm{Hdg}}$ consists of the tensors of the self-adjoint $\varphi \in E$, whose images in $H^4$ are the $c_\varphi$. Over $\mathbb{C}$ the Hodge group of $T$ is a product of groups $\mathrm{SO}(T_\sigma)$ on $T_\mathbb{C} = \bigoplus_\sigma T_\sigma$, with $\dim T_\sigma = \dim_E T \geq 3$, when $E$ is totally real, and of groups $\mathrm{GL}(W_\sigma)$ on $T_\mathbb{C} = \bigoplus_\sigma (W_\sigma \oplus W_\sigma^\vee)$ when $E$ is a CM field [Zar83]. By the first fundamental theorems [GW09, Chapter 5], the invariant polynomials of one vector of $T_\sigma$, or of one vector of $W_\sigma$ and one of $W_\sigma^\vee$, are generated by the quadratic form, or by the pairing: a determinant needs $\dim T_\sigma$ distinct vectors. So $(\mathrm{Sym}^b T)^{\mathrm{Hdg}} \otimes \mathbb{C}$ is spanned by

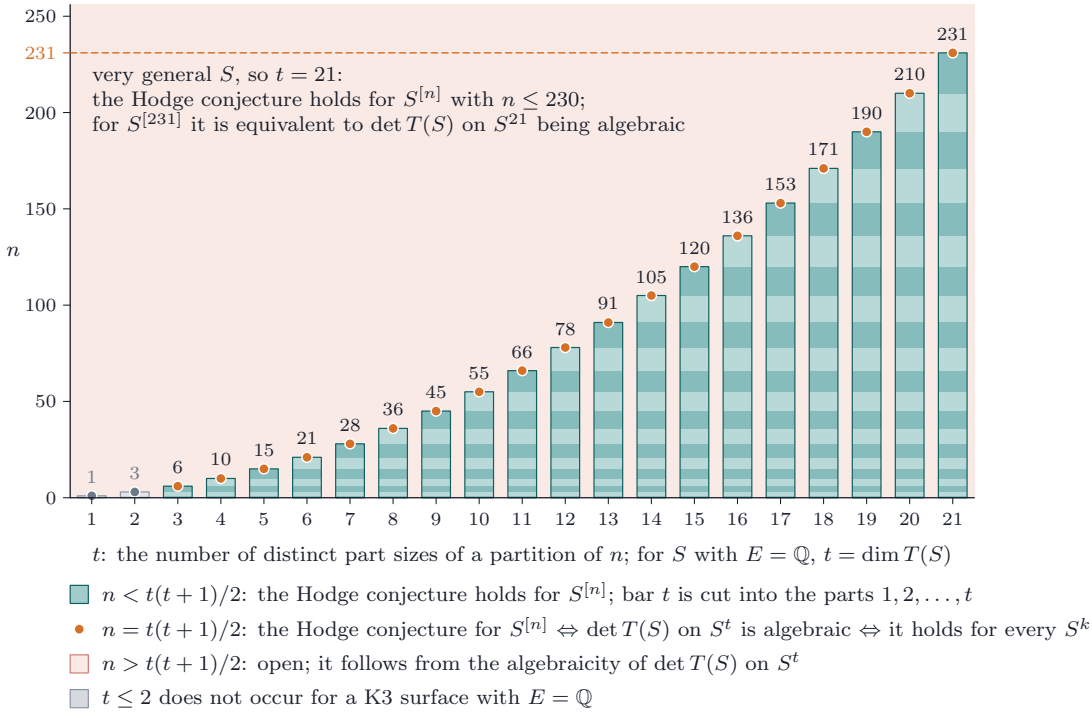



FIGURE 42. The threshold of Proposition 20.39(ii). For $t = 1, \dots, 21$ the bar rises to $t(t+1)/2$, the least $n$ that has a partition with $t$ distinct part sizes, and is cut into the parts $1, 2, \dots, t$ of the partition that attains it. For a K3 surface $S$ with $E = \mathrm{End}_{\mathrm{Hdg}}(T(S)) = \mathbb{Q}$ and $t = \dim T(S)$, which is at least 3, the Hodge conjecture holds for $S^{[n]}$ with $n$ below the bar; at its top it is equivalent to the algebraicity of the class $\det T(S)$ on $S^t$ of Proposition 20.38, and to the Hodge conjecture for every power of $S$; above it, it follows from that algebraicity. A very general $S$ has $t = 21$: the conjecture holds for $S^{[n]}$ with $n \le 230$, and for $S^{[231]}$ it is equivalent to the algebraicity of $\det T(S)$ on $S^{21}$, one of the frontier cases of (F3′) (Remark 20.68).

products of elements of $(\mathrm{Sym}^2 T)^{\mathrm{Hdg}} \otimes \mathbb{C}$, and the rational span of the products of classes $c_\varphi$, which has the same complexification, is $(\mathrm{Sym}^b T)^{\mathrm{Hdg}}$.

(ii) Divisor classes are algebraic, so by (i) it suffices that every $c_\varphi$ is. By [CM13, Theorem 1.1] the Lefschetz standard conjecture holds for $X$, so the Künneth projectors and the inverse of the isomorphism $L^{2n-2}\colon H^2 \to H^{4n-2}$, $L$ the cup product with an ample class $h$, are induced by algebraic cycles [Kle68, Kle94]. Let $u \in E$ with $u\bar{u} = 1$, and $u = 1$ if $E = \mathbb{Q}$. The map $f_u = \mathrm{id}_{\mathrm{NS}} \oplus u$ is a rational Hodge isometry of $H^2$, so by [Mar24, Theorem 1.1] there is an algebraic class $Z_u$ on $X \times X$ (the diagonal if $u = 1$) inducing on $H^2$ a nonzero rational multiple of $f_u$: the class of the correspondence $\tilde{f}$ of that theorem if it preserves degrees, and the class of $\tilde{f} \circ (c_2(X)^{n-1} \cup \cdot\,)$ if it reverses them. The class $Z_u$ may act in other degrees too; its Künneth component in $H^{4n-2}(X) \otimes H^2(X)$ is algebraic and induces a nonzero multiple of $f_u$ on $H^2$ and zero elsewhere. Applying $(L^{2n-2})^{-1} \otimes \mathrm{id}$ gives, up to a nonzero rational factor, the algebraic class $\sum_i b_i \otimes f_u(a_i)$, for a basis $(a_i)$ of $H^2$ and

the dual basis $(b_i)$ for $B(x,y)=\int_X xyh^{2n-2}$. Polarising the relation $\int_X \alpha^{2n}=c_X q(\alpha)^n$ of Fujiki and Beauville [Bea83, Fuj87] gives

$$B(x,y)=\frac{c_X}{2n-1}\,q(h)^{n-2}\big(q(h)q(x,y)+2(n-1)q(x,h)q(y,h)\big),$$

so NS and $T$ are orthogonal for $B$, and $B=\kappa q$ on $T$ with $\kappa=c_X q(h)^{n-1}/(2n-1)\neq 0$. With $(a_i)$ adapted to $\mathrm{NS}\oplus T$, the pull-back along the diagonal is $\sum_i b_i\cup f_u(a_i)=\nu+\kappa^{-1}c_{(u+\bar u)/2}$ with $\nu\in\mathrm{Sym}^2\mathrm{NS}$ algebraic, because cup product on $H^2$ is commutative and the exchange of factors turns $u$ into $\bar u$. So $c_{(u+\bar u)/2}$ is algebraic. When $E$ is a CM field its elements of norm one span it (proof of Proposition 20.39(i)), so the $(u+\bar u)/2$ span $E_0$, the self-adjoint part of $E$; when $E=\mathbb{Q}$, $u=1$ suffices. So every $c_\varphi$ is algebraic.

(iii) The variety $X$ has the Betti numbers of $S^{[2]}$, which by [dCM02] are those of $\mathrm{Sym}^2H^*(S)\oplus H^*(S)(-1)$: $H^{\mathrm{odd}}(X)=0$ and $b_4(X)=253+1+22=276=\dim\mathrm{Sym}^2H^2$. Here $c_X=3$, and the relation of Fujiki and Beauville gives $\int_X abcd=q(a,b)q(c,d)+q(a,c)q(b,d)+q(a,d)q(b,c)$ for $a,b,c,d\in H^2$. This form on $\mathrm{Sym}^2H^2$ is nondegenerate: in a $q$-orthogonal basis $(e_i)$ with $q(e_i,e_i)=d_i$, the $e_ie_j$ with $i<j$ are orthogonal to each other and to the squares, with $\int_X(e_ie_j)^2=d_id_j$, and on the squares the form has the matrix $D(J+2I)D$, with $D=\mathrm{diag}(d_i)$ and $J$ the matrix of ones, of determinant $2^{22}\cdot 25\prod_i d_i^2$. So $\mathrm{Sym}^2H^2\to H^4$ is injective, hence bijective, and with $H^6=h^2\cup H^2$ by the hard Lefschetz theorem, $\mathrm{SH}(X)=H^*(X,\mathbb{Q})$. By [Fuj87], $\int_X c_2(X)xy=C\,q(x,y)$ for a constant $C$, and the Riemann–Roch polynomial $\chi(L)=\binom{q(L)/2+3}{2}$ [EGL01] gives $C=30$; the Fujiki relation gives $\int_X q^\vee xy=25\,q(x,y)$, so $c_2(X)=\frac{6}{5}q^\vee$ by nondegeneracy. For $E=\mathbb{Q}$ this makes $c_1=q^\vee-q^\vee_{\mathrm{NS}}$ algebraic without [CM13], with $q^\vee_{\mathrm{NS}}\in\mathrm{Sym}^2\mathrm{NS}$ the class dual to $q|_{\mathrm{NS}}$. With (i) and (ii) this proves the Hodge conjecture for $X$ when $E$ is $\mathbb{Q}$ or a CM field. Let $E$ be totally real and $\varphi\in E$. If $\varphi\oplus 0$ is induced by an algebraic cycle, the computation of (ii) with $\varphi\oplus 0$ in place of $f_u$ makes $c_\varphi$ algebraic. Conversely, if $c_\varphi$ is algebraic, so is the correspondence $x\mapsto(L^2)^{-1}(c_\varphi\cup x)$. The Fujiki relation gives $\int_X c_\varphi xy=\mathrm{Tr}(\varphi)q(x,y)+2q(\varphi x,y)$ for $x\in T$ and $y\in H^2$, and $B(z,y)=q(h)q(z,y)$ for $z\in T$, so that correspondence acts on $T$ as $(\mathrm{Tr}(\varphi)+2\varphi)/q(h)$. The projector onto $T$, the Künneth projector $\pi_2$ minus $\sum_j n_j^\vee\otimes n_j$ over a basis $(n_j)$ of NS and the Hodge classes $n_j^\vee\in H^6$ dual to it, is algebraic by Proposition 20.35(iv); composing with it, $\varphi\oplus 0$ is induced by an algebraic cycle. By (i) the Hodge conjecture for $X$ is the algebraicity of the $c_\varphi$, which proves the equivalence. ∎

For varieties of generalised Kummer type every Hodge class in the subalgebra generated by $H^2$ is algebraic, whatever the field, and the Hodge conjecture holds in dimension four [FV25]. The preprint [O'G26] announces, with [Mar24], that rational Hodge isometries between varieties of the types $\mathrm{K3}^{[a_1^2+1]}$ and $\mathrm{K3}^{[a_2^2+1]}$ are induced by algebraic cycles; it is not used here.

**Corollary 20.41** (Cubic fourfolds and their Fano varieties of lines)**.** *Let $Y\subset\mathbb{P}^5$ be a smooth cubic fourfold, $T(Y)\subset H^4(Y,\mathbb{Q})$ the orthogonal complement of the Hodge classes, $E(Y)=\mathrm{End}_{\mathrm{Hdg}}(T(Y))$, and $F=F(Y)$ the Fano variety of lines on $Y$. The Hodge conjecture for $F$, the Hodge conjecture for $Y\times Y$, and the statement that every element of $E(Y)$ acts on $T(Y)$ through an algebraic cycle on $Y\times Y$ are equivalent. They hold when $E(Y)$ is $\mathbb{Q}$ or a CM field, for instance when $Y$ is very general.*

*Proof.* The Hodge conjecture holds for $Y$ [Zuc77], and also by Proposition 20.43, since $Y$ is covered by lines. So $H^*(Y,\mathbb{Q})$ is $T(Y)$ plus a space spanned by algebraic classes, on which

the intersection form is nondegenerate by the Hodge–Riemann relations, and as in the proof of Proposition 20.38 the part $\pi_T$ of the diagonal in $T(Y)\otimes T(Y)$ is algebraic. The Hodge structure $T(Y)$ is of K3 type up to a Tate twist and has no nonzero Hodge class, so $E(Y)$ is a field [Zar83], and as in Proposition 20.39(iii) the Hodge conjecture for $Y\times Y$ is the third statement. By [BD85], $F$ is a hyperkähler fourfold of $\mathrm{K3}^{[2]}$ type, and the universal line $P\subset F\times Y$ induces an isomorphism of Hodge structures $\alpha=P_*\colon H^4(Y,\mathbb{Q})\to H^2(F,\mathbb{Q})$ up to a Tate twist. Both $T(Y)$ and $T(F)$ are the sums of the isotypic components that are not of Tate type, so $\alpha(T(Y))=T(F)$ and $E(F)=\alpha E(Y)\alpha^{-1}$. Let $g$ be the Plücker class on $F$ and $e=P_*^t\circ(g^2\cup\,\cdot\,)\circ\alpha$, an algebraic correspondence that maps $T(Y)$ to itself, so that $e|_{T(Y)}\in E(Y)$. For $0\neq\sigma\in H^{3,1}(Y)$, $\int_Y e(\sigma)\bar\sigma=\int_F g^2\alpha(\sigma)\alpha(\bar\sigma)=q(g)\,q(\alpha(\sigma),\alpha(\bar\sigma))\neq 0$ by the Fujiki relation, since $q(g)>0$ and $\alpha(\sigma)$ spans $H^{2,0}(F)$. So $e|_{T(Y)}$ is invertible, its inverse is a polynomial in it and is algebraic, and the inverse of $\alpha$ on $T(F)$, which is $e^{-1}\circ P_*^t\circ(g^2\cup\,\cdot\,)$, is induced by an algebraic cycle. Hence $\varphi\in E(Y)$ acts through an algebraic cycle on $Y\times Y$ if and only if $\alpha\varphi\alpha^{-1}$ acts through one on $F\times F$, and Proposition 20.40(iii) proves the equivalences and the case where $E(Y)=E(F)$ is $\mathbb{Q}$ or a CM field. A very general $Y$ has $E(Y)=\mathbb{Q}$. ∎

The class $\mathcal{A}$ of Proposition 20.35(iii) is larger than the closure properties recorded there show. Four further operations preserve it, the last of them domination by correspondences, which brings in the varieties whose motives are known to be built from those of abelian varieties, and two classical families lie in it.

**Proposition 20.42** (More varieties of $\mathcal{A}$)**.** *Let $\mathcal{A}$ be as in Proposition 20.35(iii).*

(i) *If $Y\in\mathcal{A}$ and $Z\subset Y$ is a smooth closed subvariety with $Z\in\mathcal{A}$, then the blow-up $\mathrm{Bl}_Z Y$ lies in $\mathcal{A}$. If $Y\in\mathcal{A}$ and $V$ is a vector bundle on $Y$, then $\mathbb{P}(V)\in\mathcal{A}$.*

(ii) *If $S\in\mathcal{A}$ is a surface, then the Hilbert scheme $S^{[n]}$ lies in $\mathcal{A}$ for every $n\geq 1$.*

(iii) *Every Fermat hypersurface $X_m^r=\{x_0^m+\cdots+x_{r+1}^m=0\}\subset\mathbb{P}^{r+1}$, $r\geq 0$, $m\geq 1$, lies in $\mathcal{A}$.*

(iv) *A K3 surface $S$ lies in $\mathcal{A}$ as soon as its Kuga–Satake correspondence is induced by an algebraic cycle, that is, as soon as some algebraic cycle $\Gamma$ on $S\times B$, with $B$ an abelian variety, induces a nonzero map $T(S)_{\mathbb{Q}}\to H^2(B,\mathbb{Q})$.*

(v) *If $H^*(X,\mathbb{Q})$ is the sum of the images of the $H^*(Y_i,\mathbb{Q})$, with $Y_i\in\mathcal{A}$, under algebraic correspondences, then $X\in\mathcal{A}$; in particular $X\in\mathcal{A}$ when its Chow motive is a direct summand of a sum of Tate twists of motives of members of $\mathcal{A}$, for instance when it is of abelian type. So $\mathcal{A}$ contains the $2n$-dimensional generalised Kummer varieties $K_n(A)$ of abelian surfaces $A$* [Xu15, Theorem 2.7, Corollary 2.8]*, the smooth projective moduli spaces of Gieseker stable (twisted) sheaves, or of $\sigma$-stable objects for a generic stability condition $\sigma$, on abelian surfaces and on K3 surfaces in $\mathcal{A}$* [Bül20, Theorem 0.1, Remark 2.1]*; O'Grady's six-dimensional varieties $\widetilde{K}_v(A,H)$ obtained from moduli spaces of sheaves on an abelian surface $A$, with $v=2v_0$, $v_0$ primitive, effective and of square 2, and $H$ a $v$-generic polarisation* [Flo23, Theorem 1.1]*; the crepant resolutions of the ten-dimensional moduli spaces $M_\sigma(2v_0)$ of $\sigma$-semistable objects, with $v_0$ primitive, $v_0^2=2$ and $\sigma$ $v_0$-generic, on abelian surfaces and on K3 surfaces in $\mathcal{A}$* [FFZ21, Theorem 1.3, Corollary 4.5]*; and the cubic fourfolds whose lattice of algebraic 2-cycles has rank greater than 19, with their Fano varieties of lines, the eightfolds of Lehn, Lehn, Sorger and van Straten of those containing no plane, and the tenfolds of Laza, Saccà and Voisin of infinitely*

*many Pfaffian cubic fourfolds* [ABLP26, Theorem 3.5 and its proof, Proposition 3.9, Theorem 4.2].

*Consequently* (F3′) *holds on every smooth projective variety obtained from abelian varieties, curves, Fermat hypersurfaces and K3 surfaces as in* (iv) *by products, surjective images, blow-ups along smooth centres in* $\mathcal{A}$*, projective bundles and Hilbert schemes of points on surfaces, and on every variety dominated by these as in* (v).

*Proof.* (i) Let $c$ be the codimension of $Z$, $E=\mathbb{P}(N_{Z/Y})$ the exceptional divisor, $j\colon E\to\widetilde{Y}=\mathrm{Bl}_Z Y$ its inclusion, $\rho\colon E\to Z$ and $\pi\colon\widetilde{Y}\to Y$ the projections and $\xi=c_1(\mathcal{O}_E(1))$. By the blow-up formula [Voi02, §7.3.3],

$$H^*(\widetilde{Y},\mathbb{Q})=\pi^*H^*(Y,\mathbb{Q})\oplus\bigoplus_{i=0}^{c-2}j_*(\xi^i\cup\rho^*H^*(Z,\mathbb{Q})).$$

The first summand is the image of the transpose of the graph of $\pi$, and the $i$-th of the others is the image of the algebraic cycle $(\rho\times j)_*\xi^i$ on $Z\times\widetilde{Y}$. Composing with the correspondences that exhibit $Y$ and $Z$ as members of $\mathcal{A}$, which is allowed [Ful98, Chapter 16], shows that $H^*(\widetilde{Y},\mathbb{Q})$ is spanned by classes $\Gamma_*y$ with $B$ abelian. For $\mathbb{P}(V)$ the projective bundle formula $H^*(\mathbb{P}(V),\mathbb{Q})=\bigoplus_i\xi^i\cup\pi^*H^*(Y,\mathbb{Q})$ [Voi02, §7.3.3] does the same with the algebraic cycles $\Gamma_\pi^t\cdot\mathrm{pr}^*\xi^i$.

(ii) By de Cataldo and Migliorini [dCM02], for every partition $\nu$ of $n$ with $l(\nu)$ parts the incidence correspondence $\Gamma_\nu\subset S^{(\nu)}\times S^{[n]}$ between the product $S^{(\nu)}$ of symmetric products attached to $\nu$ and the Hilbert scheme induces, after a Tate twist by $n-l(\nu)$, an isomorphism $\bigoplus_\nu H^*(S^{(\nu)},\mathbb{Q})\to H^*(S^{[n]},\mathbb{Q})$. Let $q_\nu\colon S^{l(\nu)}\to S^{(\nu)}$ be the quotient by the finite group $G_\nu$. With rational coefficients the cycles and the cohomology of $S^{(\nu)}$ are the $G_\nu$-invariants of those of $S^{l(\nu)}$ [Ful98, Example 1.7.6], so $\Gamma_\nu$ pulls back to a cycle $\widetilde{\Gamma}_\nu$ with rational coefficients on the smooth variety $S^{l(\nu)}\times S^{[n]}$, whose action on $H^*(S^{l(\nu)},\mathbb{Q})$ is $\Gamma_{\nu*}\circ q_{\nu*}$ up to a nonzero factor. As $H^*(S^{(\nu)},\mathbb{Q})=q_{\nu*}H^*(S^{l(\nu)},\mathbb{Q})$, the $\nu$-th summand is the image of $H^*(S^{l(\nu)},\mathbb{Q})$ under $\widetilde{\Gamma}_\nu$. Products of members of $\mathcal{A}$ lie in $\mathcal{A}$, so $S^{[n]}\in\mathcal{A}$.

(iii) By induction on $r$, following the inductive structure of Shioda and Katsura [SK79]. For $r=0$, $X_m^0$ is a finite set of points, and $X_m^1$ is a curve. Let $r\geq 2$ and $r=a+b$ with $a,b\geq 1$; fix $\epsilon$ with $\epsilon^m=-1$ and put

$$\varphi\big((x),(y)\big)=\big(x_0y_{b+1}:\cdots:x_ay_{b+1}:\epsilon x_{a+1}y_0:\cdots:\epsilon x_{a+1}y_b\big),$$

a rational map $X_m^a\times X_m^b\dashrightarrow\mathbb{P}^{r+1}$; since $\sum_{i\leq a}x_i^m=-x_{a+1}^m$ and $\sum_{k\leq b}y_k^m=-y_{b+1}^m$, its image lies in $X_m^r$. It is defined by sections of $\mathcal{O}(1,1)$ whose common zero scheme is $\{x_{a+1}=0\}\times\{y_{b+1}=0\}$, the product $Z\cong X_m^{a-1}\times X_m^{b-1}$ of two smooth hyperplane sections, transversal because the Fermat hypersurfaces are smooth. Blowing up that reduced scheme resolves the map, so $\varphi$ extends to a morphism $\widetilde{\varphi}\colon\mathrm{Bl}_Z(X_m^a\times X_m^b)\to X_m^r$. It is dominant: at a point $z$ of $X_m^r$ with $\sum_{i\leq a}z_i^m\neq 0$ the preimages with $x_{a+1}=y_{b+1}=1$ are the points $x_i=\lambda z_i$, $y_k=\lambda z_{a+1+k}/\epsilon$ with $\lambda^m\sum_{i\leq a}z_i^m=-1$, and both equations reduce to this one because $\sum_{\mathrm{all}}z_i^m=0$. So $\widetilde{\varphi}$ is surjective, being proper. By induction $X_m^a$, $X_m^b$, $X_m^{a-1}$ and $X_m^{b-1}$ lie in $\mathcal{A}$, hence so do their products, the blow-up by (i) and $X_m^r$ as a surjective image; one may take $a=r-1$, $b=1$.

(iv) The transcendental part $T=T(S)_\mathbb{Q}$ is an irreducible Hodge structure, and $H^*(S,\mathbb{Q})=T\oplus T^\perp$ with $T^\perp$ spanned by algebraic classes. Let $h$ be an ample class on $B$, $d=\dim B$, and let $\phi=\Gamma_*^t\circ(\,\cdot\cup h^{d-2})\circ\Gamma_*$, an algebraic correspondence from $S$ to itself and a morphism of Hodge structures. It preserves $T$, because a morphism from the irreducible structure $T$ to the algebraic part, which is of Tate type, vanishes; so $\phi|_T$ lies in

the field $\mathrm{End}_{\mathrm{Hdg}}(T)$. For $0 \neq \sigma \in H^{2,0}(S)$, $\Gamma_*\sigma \neq 0$ because $T$ is irreducible and $\Gamma_*|_T \neq 0$; it is a class of type $(2,0)$ on $B$, hence primitive, and by the Hodge–Riemann relations $\int_S \phi(\sigma)\,\bar\sigma = \int_B \Gamma_*\sigma\,\overline{\Gamma_*\sigma}\,h^{d-2} \neq 0$. So $\phi|_T \neq 0$, it is invertible, and $T \subseteq \Gamma_*^t H^*(B,\mathbb{Q})$. With $T^\perp$ algebraic, $S \in \mathcal{A}$.

(v) If $H^*(X,\mathbb{Q}) = \sum_i \Psi_{i*}H^*(Y_i,\mathbb{Q})$ and each $H^*(Y_i,\mathbb{Q})$ is spanned by classes $\Gamma_{ij*}y$ with $B_{ij}$ abelian, then $H^*(X,\mathbb{Q})$ is spanned by the classes $(\Psi_i \circ \Gamma_{ij})_*y$, and composites of algebraic correspondences are algebraic [Ful98, Chapter 16]. If $h(X)$ is a direct summand of $\bigoplus_i h(Y_i)(n_i)$, through $\alpha$ and $\beta$ with $\beta\circ\alpha = \mathrm{id}$, then $\beta_*$ is surjective on cohomology, and its components are the $\Psi_i$; a motive of abelian type is such a summand with abelian varieties as the $Y_i$. The objects of the tensor subcategory generated by $h(Y)$ are direct summands of sums of Tate twists of the motives $h(Y^k)$, since the dual of $h(Y)$ is a Tate twist of $h(Y)$, and $Y^k \in \mathcal{A}$ when $Y \in \mathcal{A}$. The Chow motive of $K_n(A)$ lies in the tensor subcategory generated by $h(A)$ [Xu15, Theorem 2.7, Corollary 2.8], see also [FTV19, Theorem 1.5]; that of a moduli space as in (v) is a direct summand of a sum of Tate twists of motives of powers of the surface [Bül20, Theorem 0.1]; those of the varieties $\widetilde{K}_v(A,H)$ and of the ten-dimensional crepant resolutions lie in the tensor subcategory generated by the motive of the surface [Flo23, Theorem 1.1], [FFZ21, Corollary 4.5]; and the cubic fourfolds and hyperkähler varieties of [ABLP26] listed in (v) have Chow motives of abelian type [ABLP26, Theorem 3.5 and its proof, Proposition 3.9, Theorem 4.2].

The last assertion is Proposition 20.35(iii). ■

The known cases of (iv) are recalled in Remark 21.5; for the K3 surfaces among them whose powers are covered by [Var22] the Hodge conjecture itself holds on every power, which is stronger. What (iii) adds is a family on which (F3′) holds in every degree and dimension, while the Hodge conjecture for Fermat hypersurfaces is proved in a range of degrees and dimensions [Shi79]. Three recent preprints, if correct, extend that range to every Fermat variety of degree less than 65 other than 44, 51 and 52 [MMMV26], to the Fermat fourfolds of every odd degree at most 199, by a computer-assisted proof [Jum26], and to the Fermat fourfold of degree 33 [FK26]; (iii) is new for the degrees and dimensions outside these ranges. By (iii) and Proposition 20.35(i) the conjecture holds for $X_m^r$ as soon as it holds for the powers of the Jacobian of the Fermat curve of degree $m$. The inductive structure is Shioda and Katsura's, and the consequence for (F3′) is immediate; we claim no priority for it.

Item (v) collects consequences of published motivic results, and we claim no priority for them either. For a generalised Kummer variety $K_n(A)$ it is needed in general. The restriction from $A^{[n+1]}$, which lies in $\mathcal{A}$ by (ii), reaches exactly the $A[n+1]$-invariant part of $H^*(K_n(A),\mathbb{Q})$, since $A \times K_n(A) \to A^{[n+1]}$ is an étale Galois cover with group $A[n+1]$. When $n+1$ is prime the rest is spanned by the classes of the punctual Hilbert schemes at the points of $A[n+1]$, which are algebraic, and (ii) suffices; when $n+1$ is composite the rest contains transcendental classes, and [Xu15, FTV19] are needed. These results concern the varieties $K_n(A)$ themselves, not every variety of generalised Kummer type; for products of varieties $K_n(A)$ the Hodge conjecture itself is known [Xu15, Theorem 3.3], and for the varieties $\widetilde{K}_v(A,H)$ it holds on all self-products [Flo23, Corollary 1.2]. If the preprint [Ped26] is correct, (v) also puts in $\mathcal{A}$ the tenfolds of Laza, Saccà and Voisin of every cubic fourfold in $\mathcal{A}$, whose motives it makes direct summands of Tate twists of the motive of the fifth power of the cubic fourfold. In low dimension the algebraicity results of Bloch–Srinivas and Conte–Murre give more.

**Proposition 20.43** (Varieties whose zero-cycles are small)**.** *Let $X$ be a smooth complex projective variety of dimension $n$ whose group $\mathrm{CH}_0(X)_{\mathbb{Q}}$ is supported on a closed subvariety of dimension at most three. If $n \leq 5$, the Hodge conjecture holds for $X$, and so does* (F3′)*. This applies to every fourfold whose group of zero-cycles is supported on a proper closed subset, in particular every uniruled fourfold, and to every fivefold whose maximal rationally connected quotient has dimension at most three, in particular every rationally connected fivefold.*

*Proof.* By Bloch and Srinivas [BS83] every Hodge class of degree 4 on $X$ is algebraic. By Proposition 20.35(iv) the Hodge classes of degree 0, 2, $2n-2$ and $2n$ are algebraic, which settles $n \leq 4$. For $n = 5$ the remaining degree is $6 = 2n-4$, and cup product with $h$, for $h$ ample, is an isomorphism $H^4(X,\mathbb{Q}) \to H^6(X,\mathbb{Q})$ of Hodge structures up to a Tate twist by hard Lefschetz, so every Hodge class of degree 6 is $h$ times one of degree 4. A uniruled fourfold has $\mathrm{CH}_0$ supported on any ample divisor $D$, because a rational curve through a point meets $D$ and any two points of $\mathbb{P}^1$ are rationally equivalent; this case is the theorem of Conte and Murre [CM78]. A proper closed subset of a fourfold has dimension at most three.

Let $\phi\colon X \dashrightarrow R$ be the maximal rationally connected fibration of Campana and of Kollár, Miyaoka and Mori [Cam92, KMM92]: $R$, the maximal rationally connected quotient, is smooth and projective and unique up to birational equivalence, and $\phi$ is dominant with connected fibres, the general one rationally connected. Resolving its indeterminacy [Hir64] gives a smooth projective variety $\widetilde{X}$, a birational morphism $\pi\colon \widetilde{X} \to X$, an isomorphism over the domain of definition of $\phi$, and a surjective morphism $\widetilde{\phi}\colon \widetilde{X} \to R$ whose general fibre is smooth, by generic smoothness, and birational to a general fibre of $\phi$, hence rationally connected. Suppose $\dim R \leq 3$, and let $W \subset \widetilde{X}$ be a subvariety that $\widetilde{\phi}$ maps generically finitely onto $R$, of degree $e$. Any two points of a general fibre $F$ are joined by a rational curve, so every $x \in F$ satisfies $e[x] = [W \cap F]$ in $\mathrm{CH}_0(\widetilde{X})_{\mathbb{Q}}$. The general fibres cover a dense open subset of $\widetilde{X}$, and every zero-cycle is rationally equivalent to one supported on a given dense open subset, so $\mathrm{CH}_0(\widetilde{X})_{\mathbb{Q}}$ is supported on $W$ and $\mathrm{CH}_0(X)_{\mathbb{Q}} = \pi_*\mathrm{CH}_0(\widetilde{X})_{\mathbb{Q}}$ on $\pi(W)$, of dimension $\dim R \leq 3$. A rationally connected fivefold is the case where $R$ is a point. ∎

In degree four the hypothesis on zero-cycles can be relaxed to one on the fibres of a map.

**Proposition 20.44** (Degree four along rationally connected fibrations)**.** *Let $f\colon X \to Y$ be a surjective morphism of smooth complex projective varieties whose general fibre is rationally connected. Then $\mathrm{Hdg}^2(X) = \mathrm{Ab}^2(X)$ if and only if $\mathrm{Hdg}^2(Y) = \mathrm{Ab}^2(Y)$, and every Hodge class of degree four on $X$ is algebraic if and only if every one on $Y$ is. Both properties are therefore birational invariants of smooth projective varieties, and they hold for $X$ exactly when they hold for a smooth projective model of its maximal rationally connected quotient. Consequently* (F3′) *holds for every fourfold and every fivefold whose maximal rationally connected quotient is birational to a variety $R$ with $\mathrm{Hdg}^2(R) = \mathrm{Ab}^2(R)$, for instance a member of $\mathcal{A}$. With the Hodge conjecture for abelian fourfolds* [Mar25a, Mar25b]*, the Hodge conjecture holds for every fourfold birational to an abelian fourfold, and for every fivefold admitting a dominant rational map with rationally connected general fibre onto an abelian fourfold.*

*Proof.* Let $n = \dim X$ and $m = \dim Y$. Let $W$ be a component, dominating $Y$, of the intersection of $X$ with a general linear space of codimension $n-m$, so that $W \to Y$ is generically finite, of degree $e \geq 1$; let $i_W\colon \widetilde{W} \to X$ be a resolution of $W$ [Hir64] and

$g = f \circ i_W$. The correspondence $\Phi = \Gamma_f^t \circ \Gamma_g \circ \Gamma_{i_W}^t$ on $X \times X$ acts by $\Phi_* = f^* g_* i_W^*$ [Ful98, Chapter 16]. Let $\eta$ be the generic point of $X$ and $K$ its field of functions. The restriction of $\Phi^t$ to $\{\eta\} \times X$ is the zero-cycle $W_{f(\eta)}$, of degree $e$, that $W$ cuts on the fibre of $X_K$ over the point $f(\eta)$, and that of $[\Delta_X]$ is the point $\eta$ of the same fibre. Any two points of a general fibre of $f$ are rationally equivalent, and the argument of Bloch and Srinivas [BS83] (a countable algebraically closed field of definition, an embedding of an algebraic closure of $K$ into $\mathbb{C}$, specialisation and a norm) gives $e[\eta] = [W_{f(\eta)}]$ in $\mathrm{CH}_0(X_K)_{\mathbb{Q}}$. As $\mathrm{CH}_0(X_K)$ is the direct limit of the groups $\mathrm{CH}^n(U \times X)$ over the dense open subsets $U \subseteq X$, the cycle $e[\Delta_X] - \Phi^t$ vanishes on some $U \times X$, and by the localisation sequence [Ful98, Chapter 1] it is supported on $D \times X$ for a divisor $D \supseteq X \smallsetminus U$. The diagonal is symmetric, so $e[\Delta_X] - \Phi$ is supported on $X \times D$. Let $i\colon D' \to X$ be a resolution of $D$. Each component of a cycle supported on $X \times D$ is, up to a nonzero factor, the image of a subvariety of $X \times D'$ mapping generically finitely onto it, so $e[\Delta_X] - \Phi = (\mathrm{id} \times i)_* \Psi$ for a cycle $\Psi$ with rational coefficients on $X \times D'$, and the projection formula gives, for every $a \in H^k(X, \mathbb{Q})$,

$$e\, a = f^* g_* i_W^* a + i_* \Psi_* a, \qquad \Psi_* a \in H^{k-2}(D', \mathbb{Q}).$$

Let $a \in \mathrm{Hdg}^2(X)$. Then $g_* i_W^* a \in \mathrm{Hdg}^2(Y)$, and $\Psi_* a$ is a Hodge class of degree two on $D'$, algebraic by the Lefschetz theorem on $(1,1)$ classes. If $\mathrm{Hdg}^2(Y) = \mathrm{Ab}^2(Y)$, then $f^* g_* i_W^* a \in \mathrm{Ab}^2(X)$, because composites of algebraic correspondences are algebraic, and so $a \in \mathrm{Ab}^2(X)$; if every Hodge class of degree four on $Y$ is algebraic, $a$ is algebraic. Conversely, as in the proof of Proposition 20.35(iii), for $h$ ample on $X$ the correspondence $[\Gamma_f] \cdot \mathrm{pr}_X^* h^{n-m}$ sends $f^* b$ to $c\, b$, with $c > 0$ the degree of $h^{n-m}$ on a general fibre; so $b \in \mathrm{Hdg}^2(Y)$ lies in $\mathrm{Ab}^2(Y)$, or is algebraic, as soon as $f^* b$ does, or is.

A birational morphism is surjective with general fibre a point, and two birational smooth projective varieties are dominated through birational morphisms by a resolution of the closure of the graph [Hir64]; so both properties are birational invariants. The morphism $\widetilde{\phi}\colon \widetilde{X} \to R$ of the proof of Proposition 20.43, from a variety birational to $X$ onto its maximal rationally connected quotient, has rationally connected general fibre, which gives the statement on $R$. For $n \leq 5$ the degrees 0, 2, $2n-2$ and $2n$ are settled by Proposition 20.35(iv), and for $n = 5$ cup product with $h$ is an isomorphism $H^4(X, \mathbb{Q}) \to H^6(X, \mathbb{Q})$ of Hodge structures up to a Tate twist, so that $\mathrm{Hdg}^3(X) = h \cup \mathrm{Hdg}^2(X)$; members of $\mathcal{A}$ satisfy (F3′) by Proposition 20.35(iii). An abelian fourfold $B$ satisfies the Hodge conjecture [Mar25a, Mar25b]. A dominant rational map onto $B$ with rationally connected general fibre becomes, after its indeterminacy is resolved, a morphism whose general fibre is smooth and rationally connected, as in the proof of Proposition 20.43; so the fourfolds and fivefolds of the last assertion have every Hodge class of degree four algebraic, and the other degrees are settled as before. ∎

Proposition 20.44 reaches varieties that Proposition 20.42 does not: conic bundles with singular fibres over members of $\mathcal{A}$ of dimension three or four, and fourfolds and fivefolds birational to members of $\mathcal{A}$ without being known to lie in $\mathcal{A}$. A projective bundle over a member of $\mathcal{A}$ already lies in $\mathcal{A}$ by Proposition 20.42(i), and so, by Proposition 20.42(v) and the Leray–Hirsch theorem, does the source of a smooth morphism onto a member of $\mathcal{A}$ on which algebraic classes restrict to a basis of the cohomology of a fibre, such as a Severi–Brauer bundle. Proposition 20.44 is the relative form of the argument of Bloch and Srinivas, and we claim no priority for it.

*Remark* 20.45 (Zero-cycles alone go no further). Hypotheses on zero-cycles cannot carry (F3′) further without the fourfolds. If $\mathrm{CH}_0(X)_{\mathbb{Q}}$ is supported on a closed subset $Z$, with

resolution $j\colon Z' \to X$, the proof of Proposition 20.44, with the closure in $X \times Z$ of a zero-cycle on $Z_K$ rationally equivalent to the generic point in place of $\Phi^t$, gives a divisor with resolution $i\colon D' \to X$ and cycles $\Phi$ on $Z' \times X$ and $\Psi$ on $X \times D'$ with $a = \Phi_* j^* a + i_* \Psi_* a$ for every $a \in H^k(X, \mathbb{Q})$, where $\Psi_* a$ has degree $k-2$; this is the transpose of the decomposition of Bloch and Srinivas [BS83]. When $Z \neq X$ and $a$ has type $(n,0)$, $n = \dim X$, the first term vanishes, as $\dim Z' < n$, and the second has no component of type $(n,0)$; so $H^{n,0}(X) = 0$. A uniruled fivefold has its zero-cycles supported on an ample divisor, and for a fourfold $Y$ and $b \in \mathrm{Hdg}^2(Y)$ the class $\mathrm{pr}^* b$ is a Hodge class on $Y \times \mathbb{P}^1$ with $b = \mathrm{pr}_*(\mathrm{pr}^* b \cup \eta)$, for $\eta$ the class of $Y \times \{0\}$. So (F3′), and likewise the Hodge conjecture, holds for every uniruled fivefold if and only if it holds in degree four for every fourfold.

It also holds for every rational sixfold if and only if it holds for every fourfold. A rational sixfold has its zero-cycles supported on a point, so the proof of Proposition 20.43 settles every degree but six; and it is the image under a birational morphism of a variety obtained from $\mathbb{P}^6$ by blow-ups along smooth centres [Hir64], on which, by the blow-up formula [Voi02, §7.3.3], the Hodge classes of degree six are images under algebraic correspondences of Hodge classes of degree at most four on the centres, of dimension at most four, and lie in $\mathrm{Ab}^3$, or are algebraic, when the fourfolds satisfy (F3′), or the conjecture. Conversely, a fourfold $Y$ is birational to a hypersurface in $\mathbb{P}^5$; place it in a hyperplane of $\mathbb{P}^6$, and let $\widetilde{Y}$ be its smooth strict transform under an embedded resolution $M \to \mathbb{P}^6$ [Hir64]. On the rational sixfold $X = \mathrm{Bl}_{\widetilde{Y}} M$, with exceptional divisor $j\colon E \to X$ and projection $\rho\colon E \to \widetilde{Y}$, the normal bundle of $E$ is $\mathcal{O}_E(-1)$, so $a = -\rho_* j^* j_* \rho^* a$ for $a \in H^4(\widetilde{Y}, \mathbb{Q})$; this recovers $\mathrm{Hdg}^2(\widetilde{Y})$ from $\mathrm{Hdg}^3(X)$ through algebraic correspondences, and Proposition 20.44 passes from $\widetilde{Y}$ to $Y$.

Among smooth hypersurfaces, ordered by dimension and degree, the first family that these results do not cover is that of the sextic fourfolds: those of dimension at most three are covered by Proposition 20.35(iv), the fourfolds of degree at most five are Fano, hence rationally connected [Cam92, KMM92], and covered by Proposition 20.43, while a sextic fourfold has $h^{4,0} = 1$, so that its zero-cycles are not supported on a proper closed subset and it is its own maximal rationally connected quotient; the Fermat sextic lies in $\mathcal{A}$ by Proposition 20.42(iii). For a smooth sextic fourfold $Y$ in a hyperplane of $\mathbb{P}^6$ no resolution is needed, and the rational sixfold $\mathrm{Bl}_Y \mathbb{P}^6$, with $h^{6,0} = 0$, $h^{5,1} = 1$, $h^{4,2} = 426$ and $h^{3,3} = 1753$ by the blow-up formula [Voi02, §7.3.3] and the Hodge numbers 1, 426, 1752 of $Y$, satisfies (F3′) if and only if $Y$ does. For the Hodge conjecture these equivalences are folklore, and we claim no priority for them.

Surjective morphisms carry $\mathcal{A}$ to itself (Proposition 20.35(iii)). In degree four the morphism can be replaced by any dominant rational map, which reaches some of these sextic fourfolds.

**Proposition 20.46** (Degree four along dominant rational maps)**.** *Let $\varphi\colon Y \dashrightarrow X$ be a dominant rational map of smooth complex projective varieties.*

(i) *If $\mathrm{Hdg}^2(Y) = \mathrm{Ab}^2(Y)$, then $\mathrm{Hdg}^2(X) = \mathrm{Ab}^2(X)$; if every Hodge class of degree four on $Y$ is algebraic, so is every one on $X$.*

(ii) (F3′) *holds on every fourfold and every fivefold that admits a dominant rational map from a variety $Y$ with $\mathrm{Hdg}^2(Y) = \mathrm{Ab}^2(Y)$, for instance from a member of $\mathcal{A}$. The Hodge conjecture holds on every fourfold that admits a dominant rational map from an abelian fourfold, such as a smooth projective model of the quotient of an abelian fourfold by a finite group, and on every fivefold that admits one from the product of an abelian fourfold and a projective space.*

*Proof.* (i) Let $\Gamma \subset Y \times X$ be the closure of the graph of $\varphi$ and $\widetilde{Y} \to \Gamma$ a resolution of singularities [Hir64]. The projections give a birational morphism $\widetilde{Y} \to Y$ and a morphism $g\colon \widetilde{Y} \to X$, which is surjective, being proper and dominant. Both properties are birational invariants (Proposition 20.44), so they hold for $\widetilde{Y}$. Let $x \in \mathrm{Hdg}^2(X)$, let $h$ be ample on $\widetilde{Y}$ and $r = \dim \widetilde{Y} - \dim X$, and let $\Psi = [\Gamma_g] \cdot \mathrm{pr}_{\widetilde{Y}}^* h^r$, so that $\Psi_* g^* x = c\,x$ with $c > 0$, as in the proof of Proposition 20.35(iii). The class $g^*x$ is a Hodge class of degree four on $\widetilde{Y}$, hence a sum of classes $\Gamma_{j*}\gamma_j$ as in Definition 20.34, or algebraic. So $x = c^{-1}\sum_j (\Psi \circ \Gamma_j)_* \gamma_j$ lies in $\mathrm{Ab}^2(X)$, or $x$ is algebraic, since composites of algebraic correspondences are algebraic [Ful98, Chapter 16].

(ii) On a fourfold or a fivefold the degrees other than four and six are settled by Proposition 20.35(iv), and the same argument, the Lefschetz theorem on $(1,1)$ classes and hard Lefschetz, makes the Hodge classes of those degrees algebraic; on a fivefold cup product with an ample class is an isomorphism from $H^4$ onto $H^6$ of Hodge structures up to a Tate twist, so degree six follows from degree four. Members of $\mathcal{A}$ satisfy (F3′) (Proposition 20.35(iii)). An abelian fourfold $B$ satisfies the Hodge conjecture [Mar25a, Mar25b], and by the Künneth formula a Hodge class of degree four on $B \times \mathbb{P}^k$ is a sum of pull-backs of Hodge classes of degrees four, two and zero on $B$ multiplied by powers of the hyperplane class, so it is algebraic. A quotient $B/G$ by a finite group is dominated by $B$, and so is every smooth projective model of it. ∎

Lefschetz theory carries the two statements from a variety to its ample complete intersections, outside the middle degree always and in the middle degree for a very general member.

**Proposition 20.47** (Ample complete intersections)**.** *Let $X$ be a smooth complex projective variety of dimension $N$, $L$ an ample line bundle on $X$, and $Y = Y_c \subset Y_{c-1} \subset \cdots \subset Y_0 = X$ with each $Y_j$ a smooth divisor of $Y_{j-1}$ in the linear system of $L^{m_j}|_{Y_{j-1}}$, $m_j \geq 1$; put $r = \dim Y = N - c \geq 1$ and let $i\colon Y \to X$ be the inclusion.*

(i) *If $2p \neq r$, every Hodge class of degree $2p$ on $Y$ is $i^*\alpha$ for a Hodge class $\alpha$ of degree $2p$ on $X$. So if* (F3′)*, or the Hodge conjecture, holds for $X$ in degree $2p$, it holds for $Y$ in degree $2p$; if $r$ is odd and it holds for $X$ in every degree, it holds for $Y$ in every degree.*

(ii) *Let $r$ be even, $c = 1$ and $L^{m_1}$ very ample, and let $V \subset H^r(Y, \mathbb{Q})$ be the vanishing cohomology, the kernel of the Gysin map $i_*$. If $V$ is not of type $(r/2, r/2)$, then for a very general $Y$ in the linear system every Hodge class of degree $r$ on $Y$ is $i^*\alpha$ for a Hodge class $\alpha$ on $X$, and the conclusion of (i) holds in degree $r$ as well.*

*So the Hodge conjecture holds for every smooth ample divisor of odd dimension in $B \times \mathbb{P}^k$, for $B$ an abelian variety of dimension at most five, and for the very general member of every very ample linear system on an abelian fivefold; and* (F3′) *holds for every smooth ample divisor of odd dimension in a member of $\mathcal{A}$.*

*Proof.* (i) By the Lefschetz hyperplane theorem [Voi03, Chapter 1], applied to each $Y_j \subset Y_{j-1}$, the restriction $i^*\colon H^k(X, \mathbb{Q}) \to H^k(Y, \mathbb{Q})$ is an isomorphism for $k < r$ and injective for $k = r$. Let $2p < r$. Then $i^*$ is an isomorphism of Hodge structures in degree $2p$, so every Hodge class on $Y$ is $i^*$ of a Hodge class on $X$. Let $2p > r$. The Gysin map $i_*\colon H^{2p}(Y, \mathbb{Q}) \to H^{2p+2c}(X, \mathbb{Q})$ is Poincaré dual to $i^*$ in degree $2r - 2p < r$, hence an isomorphism, and $i_* i^*$ is cup product with $[Y] = m_1 \cdots m_c\, c_1(L)^c$. By the hard Lefschetz theorem [Voi02, Chapter 6], cup product with $c_1(L)^c$ maps $H^{2p}(X, \mathbb{Q})$ onto $H^{2p+2c}(X, \mathbb{Q})$

when $2p \geq N - c$, so $i^*$ is onto in degree $2p$. A surjective morphism of polarisable Hodge structures has a section that is a morphism of Hodge structures, so again every Hodge class on $Y$ is $i^*$ of a Hodge class on $X$. The restriction $i^*$ is induced by the transpose of the graph of $i$, so it carries $\mathrm{Ab}^p(X)$ into $\mathrm{Ab}^p(Y)$ and algebraic classes to algebraic classes, which gives the rest of (i).

(ii) The projection formula shows that $V$ is the orthogonal complement of $i^*H^r(X,\mathbb{Q})$ for the intersection form, and $H^r(Y,\mathbb{Q}) = i^*H^r(X,\mathbb{Q}) \oplus V$ as Hodge structures [Voi03, Chapter 2]; the Hodge classes in the first summand are the $i^*\alpha$ with $\alpha$ a Hodge class on $X$. Over the open set $U$ of the linear system parametrising smooth members, $V$ is a local system, and its monodromy group $\Gamma$ acts irreducibly [Voi03, Chapter 3]. For each class $v$ of the lattice of $V$ at a point of the universal cover of $U$, the set where the flat continuation of $v$ is of Hodge type is a closed analytic subset, either everything or nowhere dense; so at a very general point the Hodge classes of $V$ are those whose continuation is of Hodge type everywhere. They form a subspace $W$ stable under $\Gamma$, hence $W = 0$ or $W = V$, and $W = V$ would make $V$ of type $(r/2, r/2)$. So $W = 0$.

For the consequences, $B \times \mathbb{P}^k$ satisfies the Hodge conjecture by the Künneth formula, since $B$ does [Mar25a, Mar25b, MZ99]. For a very ample divisor $Y$ of an abelian fivefold $B$, $r = 4$, and $V$ is not of type $(2,2)$: $h^{4,0}(Y) = h^0(K_Y) = h^0(Y, \mathcal{O}_Y(Y)) = h^0(B, \mathcal{O}_B(Y)) + 4$, by the sequence $0 \to \mathcal{O}_B \to \mathcal{O}_B(Y) \to \mathcal{O}_Y(Y) \to 0$ and $h^1(\mathcal{O}_B(Y)) = 0$, exceeds $h^{4,0}(B) = 5$ because $h^0(\mathcal{O}_B(Y)) \geq 2$, so $V$ has a nonzero part of type $(4,0)$. Members of $\mathcal{A}$ satisfy (F3′) by Proposition 20.35(iii). ∎

For the Hodge conjecture these transfers are classical, and we claim no priority. What they add to (F3′) is a large class outside $\mathcal{A}$: the vanishing cohomology of an ample divisor of odd dimension is not reached from abelian varieties in general, yet it carries no Hodge class in even degree, since its only degree is the odd middle one.

A hypersurface of Delsarte type is defined by as many monomials as variables. Shioda observed that such a surface is dominated by a Fermat surface [Shi86]; the same monomial substitution works in every dimension, and with Proposition 20.46 it gives the following.

**Proposition 20.48** (Delsarte fourfolds)**.** *Let $A = (a_{ij})_{0 \leq i,j \leq r+1}$ be an invertible matrix of non-negative integers whose rows all sum to $m \geq 2$, let $F_A = \sum_i \prod_j x_j^{a_{ij}}$, suppose that $X_A = \{F_A = 0\} \subset \mathbb{P}^{r+1}$ is smooth, and let $d$ be a positive integer with $dA^{-1}$ integral, such as $|\det A|$.*

(i) *There is a dominant rational map from the Fermat hypersurface $X_d^r$ onto $X_A$, given by monomials.*

(ii) $\mathrm{Hdg}^2(X_A) = \mathrm{Ab}^2(X_A)$. *For $r = 4$, $X_A$ satisfies* (F3′)*, and it satisfies the Hodge conjecture if $X_d^4$ does.*

(iii) *If $m \geq 3$ and $F_A$ is a sum, in disjoint sets of variables, of terms $x^m$, chains $x_1^{m-1}x_2 + \cdots + x_{k-1}^{m-1}x_k + x_k^m$ and loops $x_1^{m-1}x_2 + \cdots + x_{k-1}^{m-1}x_k + x_k^{m-1}x_1$ with $k \geq 2$, then $A$ is invertible and $X_A$ is smooth. For $r = 4$ and $m = 6$ these polynomials have 29 shapes up to a renumbering of the variables, the Fermat sextic being one; each defines a smooth sextic fourfold, with $h^{4,0} = 1$, $h^{3,1} = 426$ and primitive $h^{2,2}$ equal to 1751. The least $d$ is 6 for the Fermat sextic, 24 or 30 for six others, those whose terms other than $x^m$ are all loops of length two or all chains of length two, at least 120 for the remaining twenty-two, and $5^6 - 1 = 15624$ for the loop $x_0^5x_1 + x_1^5x_2 + \cdots + x_5^5x_0$.*

*Proof.* (i) The matrix $dA^{-1}$ is integral for $d = |\det A|$, by the formula for the inverse through the adjugate. Let $B = dA^{-1}$. From $A\mathbf{1} = m\mathbf{1}$, with $\mathbf{1}$ the vector of ones,

$B\mathbf{1} = (d/m)\mathbf{1}$, so the rows of the integral matrix $B$ all sum to the integer $d/m$, and $AB = d \cdot I$. Choose $c \geq 0$ with every $b_{jk} + c \geq 0$ and let $\psi(y)_j = \prod_k y_k^{b_{jk}+c}$; the coordinates of $\psi$ are monomials of the same degree, so $\psi$ is a rational map $\mathbb{P}^{r+1} \dashrightarrow \mathbb{P}^{r+1}$, and

$$F_A(\psi(y)) = \sum_i \prod_k y_k^{\sum_j a_{ij}(b_{jk}+c)} = (y_0 \cdots y_{r+1})^{cm} \sum_i y_i^d.$$

On the torus $T$ of points with all coordinates nonzero, $\psi$ is induced by the homomorphism $y \mapsto (\prod_k y_k^{b_{jk}})_j$ of $\mathbb{G}_m^{r+2}$, which is surjective because $\det B \neq 0$. So every $z \in X_A \cap T$ is $\psi(y)$ for some $y \in T$, and the identity puts $y$ on $X_d^r$. A smooth hypersurface of degree at least two and positive dimension is irreducible, so it is not contained in a coordinate hyperplane, and $X_A \cap T$ and $X_d^r \cap T$ are dense. Hence $\psi$ restricts to a dominant rational map $X_d^r \dashrightarrow X_A$.

(ii) $X_d^r$ lies in $\mathcal{A}$ by Proposition 20.42(iii), so (i) and Proposition 20.46 give both assertions, the degrees other than four being settled as in the proof of Proposition 20.46(ii).

(iii) The exponent matrix is block diagonal, with blocks $m$, $(m-1)I + N + E$ for a chain, $N$ the nilpotent shift and $E$ the matrix whose only nonzero entry is a 1 in the last diagonal place, and $(m-1)I + P$ for a loop, $P$ the cyclic shift; their determinants are $m$, $m(m-1)^{k-1}$ and $(m-1)^k - (-1)^k$, which are nonzero. The partial derivatives of a sum in disjoint sets of variables are those of the summands, so it suffices that the partials of each summand vanish together only at 0. For $x^m$ this is clear. For a chain, $\partial F/\partial x_1 = (m-1)x_1^{m-2}x_2$ vanishes only if $x_1 = 0$ or $x_2 = 0$, and if $x_2 = 0$ then $\partial F/\partial x_2 = x_1^{m-1}$; so $x_1 = 0$, the same argument applies to $x_2, \ldots, x_{k-1}$, and then $\partial F/\partial x_k = m x_k^{m-1}$ gives $x_k = 0$. For a loop, $\partial F/\partial x_j = (m-1)x_j^{m-2}x_{j+1} + x_{j-1}^{m-1}$, indices modulo $k$. If one coordinate vanishes, so does the one before it, hence all of them; if none vanishes, the product of the relations $x_{j-1}^{m-1} = -(m-1)x_j^{m-2}x_{j+1}$ gives $\prod_j x_j^{m-1} = (1-m)^k \prod_j x_j^{m-1}$, which is impossible because $|1-m| \geq 2$. The number of shapes, drawn in Figure 43, is the coefficient of $t^6$ in $(1-t)^{-1}\prod_{k\geq 2}(1-t^k)^{-2}$, which is 29, and the least $d$ is the least common multiple of the orders of the cyclic groups attached to the terms in the proof of Corollary 20.53(iii): 6 for $x^6$, $6 \cdot 5^{k-1}$ for a chain and $5^k - (-1)^k$ for a loop of length $k$; so 24 or 30 when the terms other than $x^6$ are all loops or all chains of length two, and at least $\operatorname{lcm}(24, 30) = 120$, $5^3 + 1$ or $6 \cdot 5^2$ when both kinds of length two, or a loop or chain of length at least three, occur. The Hodge numbers are those of every smooth sextic fourfold, read off from the Jacobian ring [Voi03, Chapter 6]. ∎

A sextic fourfold has $h^{4,0} = 1$, so no hypothesis on zero-cycles or on rationally connected fibrations reaches it (Remark 20.45), and among the smooth sextic fourfolds only the Fermat one is covered by Proposition 20.42. The Delsarte sextics are finitely many up to isomorphism, since the torus rescales the coefficients of a Delsarte polynomial to 1, so they are isolated points in the 426-dimensional moduli of smooth sextic fourfolds: Proposition 20.48 adds points, not a family. By Proposition 20.48(ii) the Hodge conjecture for a Delsarte fourfold follows from that for one Fermat fourfold. For six of the sextic shapes that Fermat fourfold has degree 24 or 30, inside the range claimed in the preprint [MMMV26], so if it is correct the Hodge conjecture itself holds on these six; for the others the degree ranges from 120 to 18750, and from 120 to 3750 with the lattice degree of Corollary 20.53, beyond every known or claimed range [Shi79, MMMV26, Jum26]. The monomial cover is Shioda's, and we claim no priority for it.

Proposition 20.48 uses the covering map only in degree four, where algebraicity modulo abelian varieties descends along every dominant rational map. In higher degrees a birational

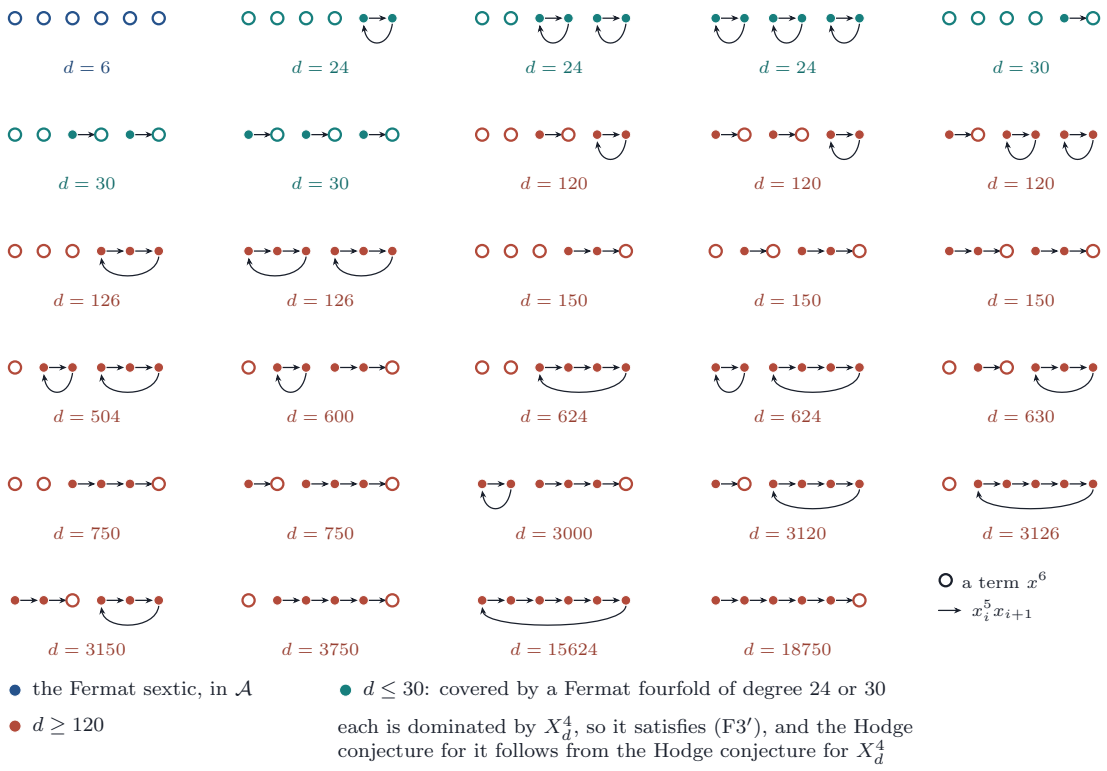



FIGURE 43. The 29 smooth sextic fourfolds of Proposition 20.48(iii), up to a renumbering of the variables. Each is drawn on its six variables: an arrow from $x_i$ to $x_j$ is a term $x_i^5 x_j$, and a ring is a term $x^6$, so a chain is a path of arrows ending at a ring and a loop is a cycle of arrows. Under each is the least $d$ with $dA^{-1}$ integral, the degree of the Fermat fourfold that covers it. All of them satisfy (F3′).

modification can add classes, so the boundary of a compactification has to be controlled. For a hypersurface of a torus defined by affinely independent monomials the boundary is again of the same kind, and the whole cohomology can be traced back to Fermat varieties of a single degree. We use toric varieties as in [CLS11]: a torus $T$ with character lattice $M$ and cocharacter lattice $N$, a fan $\Sigma$ in $N_{\mathbb{R}}$ and its toric variety $Y_\Sigma$, and for a cone $\tau \in \Sigma$ the orbit $O(\tau)$, whose character lattice is $\tau^\perp \cap M$, and its closure $V(\tau)$, the toric variety of the fan formed by the images of the cones containing $\tau$ in $N_{\mathbb{R}}/\mathbb{R}\tau$ [CLS11, Chapter 3]; for a ray $\rho$ we write $D_\rho = V(\rho)$. In what follows varieties need not be connected: a smooth projective variety lies in $\mathcal{A}$ when each of its connected components does, and the closure properties of Proposition 20.35(iii) and Proposition 20.42 hold component by component.

**Definition 20.49** (Simplex type)**.** Let $T$ be a torus of dimension $n$ with character lattice $M$. A Laurent polynomial $f = \sum_{i=0}^{k} c_i \chi^{u_i}$ on $T$ is of *simplex type* if every $c_i$ is nonzero and $u_0, \dots, u_k \in M$ are affinely independent. Let $L' \subseteq M$ be the lattice spanned by the $u_i - u_0$, $L$ its saturation, the set of $x \in M$ with a nonzero multiple in $L'$, and $e(f)$ the exponent of the finite group $L/L'$; put $V_f = \{f = 0\} \subseteq T$. A fan $\Sigma$ in $N_{\mathbb{R}}$ is *adapted* to $f$ if for every cone $\sigma \in \Sigma$ there is an index $i_\sigma$ with $\langle u_{i_\sigma}, v\rangle \leq \langle u_j, v\rangle$ for all $j$ and all $v \in \sigma$, that is, if every cone of $\Sigma$ lies in a cone of the normal fan of the simplex spanned by the

$u_i$. For a cone $\tau$ of an adapted fan let $I_\tau$ be the set of indices $i$ with $\langle u_i, v\rangle = \min_j \langle u_j, v\rangle$ for every $v \in \tau$, fix $i_\tau \in I_\tau$, and let $f_\tau = \sum_{i\in I_\tau} c_i \chi^{u_i - u_{i_\tau}}$, a Laurent polynomial of simplex type on the torus $O(\tau)$, since its exponents lie in $\tau^\perp \cap M$.

The compactification is Khovanskii's [Kho77]; for a simplex the proof is short, and we give it.

**Lemma 20.50** (The compactification)**.** *Let $f$ be of simplex type on $T$, let $\Sigma$ be a smooth fan adapted to $f$, and let $Z$ be the closure of $V_f$ in $Y_\Sigma$.*

(i) *$V_f$ is empty if $k = 0$. If $k \geq 1$, it is a finite étale cover, of degree $[L : L']$, of the product of a torus of dimension $n - k$ and the complement of $k + 1$ hyperplanes in general position in $\mathbb{P}^{k-1}$; in particular it is smooth.*

(ii) *$Z$ is smooth, and for every $\tau \in \Sigma$, $Z \cap O(\tau) = V_{f_\tau}$ and $Z$ meets $V(\tau)$ transversally: $Z \cap V(\tau)$ is empty or smooth of codimension $\dim \tau$ in $Z$, and it is the closure of $V_{f_\tau}$ in $V(\tau)$. So the divisors $Z \cap D_\rho$ of $Z$ are smooth and cross normally, and $Z \cap D_\rho$ is empty when $I_\rho$ has one element.*

(iii) *The fan of $V(\tau)$ is smooth and adapted to $f_\tau$.*

*Proof.* (i) If $k = 0$, $f$ is a monomial and has no zero on $T$. Let $k \geq 1$, choose a complement $C$ of the saturated lattice $L$ in $M$ and a basis $b_{k+1}, \dots, b_n$ of $C$. The characters $u_1 - u_0, \dots, u_k - u_0, b_{k+1}, \dots, b_n$ span a sublattice of index $[L : L']$ in $M$, so the map $T \to (\mathbb{C}^\times)^n$ that they define is an isogeny of that degree. The function $\chi^{-u_0} f$ is the pull-back of $c_0 + c_1 z_1 + \cdots + c_k z_k$, so $V_f$ is the full preimage of the product of $(\mathbb{C}^\times)^{n-k}$ and the set $H$ of points of $(\mathbb{C}^\times)^k$ with $c_0 + \sum_i c_i z_i = 0$. The projective closure of this affine hyperplane is a hyperplane $\mathbb{P}^{k-1}$ of $\mathbb{P}^k$, with homogeneous coordinates $w_0, \dots, w_k$, on which the $k + 1$ linear forms $w_i$ satisfy the single relation $\sum_i c_i w_i = 0$, with every $c_i \neq 0$; so any $k$ of them are independent, the hyperplanes $\{w_i = 0\}$ of $\mathbb{P}^{k-1}$ are in general position, and $H$ is their complement.

(ii) Let $\sigma \in \Sigma$, spanned by part $v_1, \dots, v_s$ of a basis $v_1, \dots, v_n$ of $N$, let $w_1, \dots, w_n$ be the dual basis of $M$, and let $t_j = \chi^{w_j}$, the coordinates of $U_\sigma \cong \mathbb{C}^s \times (\mathbb{C}^\times)^{n-s}$. Put $i = i_\sigma$. Every $u_j - u_i$ lies in the dual cone of $\sigma$, so $g = \chi^{-u_i} f = \sum_j c_j \chi^{u_j - u_i}$ is a regular function on $U_\sigma$, and $Z \cap U_\sigma \subseteq \{g = 0\}$. A face $\tau$ of $\sigma$ is spanned by the $v_j$ with $j$ in a subset $J$, and $O(\tau)$ is the locus where exactly the $t_j$ with $j \in J$ vanish. A character $\chi^x$ regular on $U_\sigma$ restricts to $O(\tau)$ as $\chi^x$ if $x \in \tau^\perp$ and as 0 otherwise [CLS11, Chapter 3]; since $\langle u_j - u_i, v\rangle \geq 0$ on $\sigma$, $u_j - u_i \in \tau^\perp$ exactly when $j \in I_\tau$. So $h = g|_{O(\tau)}$ is $\chi^{u_{i_\tau} - u_i} f_\tau$, a unit times $f_\tau$. Let $p \in O(\tau)$ with $h(p) = 0$ and put $a_j = c_j \chi^{u_j - u_i}(p)$ for $j \in I_\tau$. The derivative of $h$ at $p$ along the one-parameter subgroup of $O(\tau)$ given by $y$ in its cocharacter lattice is $\sum_j a_j \langle u_j - u_i, y\rangle$. If all of these vanished, then, with $\sum_j a_j = h(p) = 0$, the vector $\sum_j a_j (1, u_j - u_i)$ of $\mathbb{C} \oplus (\tau^\perp \cap M) \otimes \mathbb{C}$ would be zero, as the cocharacter lattice of $O(\tau)$ is dual to $\tau^\perp \cap M$; affine independence would give every $a_j = 0$, which is impossible at a point of the torus $O(\tau)$. So $dh(p) \neq 0$. Hence $\{g = 0\}$ is smooth at $p$ and transversal to $O(\tau)$ and to every $\{t_j = 0,\ j \in J'\}$ with $J' \subseteq J$, whose tangent space at $p$ contains that of $O(\tau)$. No component of $\{g = 0\}$ lies in a divisor $\{t_j = 0\}$, since $g$ restricted to the orbit of the ray of $v_j$ is a unit times a nonzero Laurent polynomial; so every component meets $T$, where $\{g = 0\}$ is $V_f$, and $\{g = 0\} = Z \cap U_\sigma$. This shows that $Z$ is smooth, that $Z \cap O(\tau) = V_{f_\tau}$ and that $Z \cap V(\tau)$ is smooth of codimension $\dim \tau$. As $Z \cap O(\tau')$ has codimension at least $\dim \tau'$ in $Z$ for every cone $\tau' \supsetneq \tau$, no component of $Z \cap V(\tau)$ lies in the smaller orbits, and $Z \cap V(\tau)$ is the closure of $V_{f_\tau}$. If $I_\rho$ has one element, $f_\rho$ is a nonzero constant.

(iii) The fan of $V(\tau)$ is smooth because $\Sigma$ is. Let $\sigma \supseteq \tau$. The index $i_\sigma$ attains the minimum on $\sigma$, hence on $\tau$, so $i_\sigma \in I_\tau$, and for $j \in I_\tau$ and $v \in \sigma$, $\langle u_{i_\sigma} - u_{i_\tau}, v\rangle \le \langle u_j - u_{i_\tau}, v\rangle$. Both sides depend only on the image of $v$ in $N_{\mathbb{R}}/\mathbb{R}\tau$, because the exponents of $f_\tau$ lie in $\tau^\perp$; so the image of $\sigma$ satisfies the condition of adaptedness for $f_\tau$. ∎

**Lemma 20.51** (Cohomology across a toric modification)**.** *Let $f$ be of simplex type on $T$, let $\Sigma_1$ and $\Sigma_0$ be smooth projective fans adapted to $f$, with $\Sigma_1$ refining $\Sigma_0$, let $\varphi\colon Y_1 = Y_{\Sigma_1} \to Y_0 = Y_{\Sigma_0}$ be the toric morphism, and let $Z_1$ and $Z_0$ be the closures of $V_f$. For each ray $\rho$ of $\Sigma_1$ that is not a ray of $\Sigma_0$ let $E_\rho = Z_1 \cap D_\rho$, with inclusion $i_\rho$. Then the kernel of $\varphi_*\colon H^k(Z_1,\mathbb{Q}) \to H^k(Z_0,\mathbb{Q})$ is contained in $\sum_\rho i_{\rho*}H^{k-2}(E_\rho,\mathbb{Q})$, and*

$$H^k(Z_1,\mathbb{Q}) = \varphi^*H^k(Z_0,\mathbb{Q}) + \sum_\rho i_{\rho*}H^{k-2}(E_\rho,\mathbb{Q}).$$

*Proof.* Let $W \subseteq Y_0$ be the union of the open sets $U_\tau$ with $\tau \in \Sigma_0 \cap \Sigma_1$. A cone of $\Sigma_1$ contained in such a $\tau$ is a face of it, since two cones of a fan meet in a common face; so $\varphi^{-1}(U_\tau) = U_\tau$, and $\varphi$ is an isomorphism over $W$. A cone of $\Sigma_1$ whose rays are all rays of $\Sigma_0$ lies in a cone $\tau_0$ of $\Sigma_0$, its rays are rays of $\tau_0$, and $\tau_0$ is simplicial, so it is a face of $\tau_0$ and lies in $\Sigma_0$. Hence $Y_1 \smallsetminus \varphi^{-1}(W)$ is the union of the orbits of the cones with a ray not in $\Sigma_0$, that is, of the divisors $D_\rho$ of those rays. So $U_1 = Z_1 \smallsetminus \bigcup_\rho E_\rho$ is the closure of $V_f$ in $\varphi^{-1}(W)$, and $\varphi$ maps it isomorphically onto $U_0 = Z_0 \cap W$, with $U_1 = Z_1 \cap \varphi^{-1}(U_0)$. The morphism $Z_1 \to Z_0$ is birational on each component, so $\varphi_*\varphi^* = \mathrm{id}$ by the projection formula, and every class is $\varphi^*\varphi_* a$ plus a class in the kernel of $\varphi_*$. Let $\varphi_* b = 0$. Push-forward along a proper map commutes with restriction to an open set and its preimage, so $b|_{U_1} = (\varphi|_{U_1})^*((\varphi_* b)|_{U_0}) = 0$. By Lemma 20.50(ii), $\bigcup_\rho E_\rho$ is a divisor with normal crossings and smooth components in the smooth projective variety $Z_1$, and by Deligne [Del71, §3.2] the weight spectral sequence of its complement $U_1$ degenerates at $E_2$; its term $E_2^{0,k}$, the cokernel of the Gysin map $\bigoplus_\rho H^{k-2}(E_\rho,\mathbb{Q}) \to H^k(Z_1,\mathbb{Q})$, is the image of $H^k(Z_1,\mathbb{Q})$ in $H^k(U_1,\mathbb{Q})$. So $b$ lies in the image of the Gysin map. ∎

**Theorem 20.52** (Hypersurfaces of simplex type)**.** *Let $f$ be a Laurent polynomial of simplex type on a torus $T$ of dimension $n$, with $e = e(f)$, let $Y$ be a projective toric variety with torus $T$, not necessarily smooth, and let $Z$ be the closure of $V_f$ in $Y$. Suppose that $Z$ is smooth.*

(i) *$Z \in \mathcal{A}$, so (F3′) holds on $Z$ in every degree.*

(ii) *If the Hodge conjecture holds for the Fermat varieties $X_e^j$ with $j \le \dim Z$, then it holds for $Z$.*

*Proof.* By induction on $n$. If $k = 0$, $Z$ is empty; let $k \ge 1$.

*The Fermat form.* With $C$ and $b_j$ as in the proof of Lemma 20.50(i), let $\widetilde{M}$ be the lattice with basis $w_i = (u_i - u_0)/e$ for $1 \le i \le k$ and $w_j = b_j$ for $k < j \le n$. It contains $M$, since $eL \subseteq L'$ gives $L \subseteq \frac{1}{e}L'$. Let $\widetilde{T}$ be the torus with character lattice $\widetilde{M}$, $\widetilde{N} \subseteq N$ its cocharacter lattice and $\pi\colon \widetilde{T} \to T$ the isogeny dual to $M \subseteq \widetilde{M}$. In the coordinates $z_i = \chi^{w_i}$ of $\widetilde{T}$, $\pi^*(\chi^{-u_0} f) = c_0 + \sum_{i\le k} c_i z_i^e$; call it $\widetilde{f}$. It is of simplex type, $V_{\widetilde{f}} = \pi^{-1}(V_f)$, and after the $z_i$ are rescaled it is $c_0(1 + \sum_{i\le k} z_i^e)$.

*The model.* Let $\Sigma_0$ be the fan in $\widetilde{N}_{\mathbb{R}}$ of $Y_0 = \mathbb{P}^k \times \mathbb{P}^{n-k}$, on whose torus $z_1, \dots, z_k$ are the affine coordinates of the first factor and $z_{k+1}, \dots, z_n$ those of the second. The closure of $V_{\widetilde{f}}$ in $Y_0$ is, after the rescaling, $Z_0 = X_e^{k-1} \times \mathbb{P}^{n-k}$, since the part of the Fermat variety in the torus is dense in it; $Z_0 \in \mathcal{A}$ by Proposition 20.42(iii) and Proposition 20.35(iii), $\mathbb{P}^{n-k}$

being the Fermat variety $X_1^{n-k}$. The fan $\Sigma_0$ is adapted to $\widetilde{f}$: $\min(0, e\langle w_1, v\rangle, \dots, e\langle w_k, v\rangle)$ depends only on the image of $v$ in the fan of $\mathbb{P}^k$, the normal fan of the standard simplex, on whose cones it is linear.

*The common refinement.* Let $\Sigma_Y$ be the fan of $Y$, the normal fan of a polytope $P$, as $Y$ is projective [CLS11, Chapter 2], and $\Sigma_0$ that of a polytope $Q$. The face of $P+Q$ on which a vector $v$ attains its minimum is the sum of those of $P$ and of $Q$, so the normal fan of $P+Q$ consists of the intersections of cones of $\Sigma_Y$ and $\Sigma_0$; it has a smooth projective refinement $\Sigma_1$ with respect to $\widetilde{N}$ [CLS11, Chapter 11]. It is adapted to $\widetilde{f}$, because $\Sigma_0$ is. Let $Y_1 = Y_{\Sigma_1,\widetilde{N}}$ and $Z_1$ the closure of $V_{\widetilde{f}}$ in it, smooth by Lemma 20.50(ii). The inclusion $\widetilde{N} \subseteq N$ maps each cone of $\Sigma_1$ into a cone of $\Sigma_Y$ and defines a proper toric morphism $\psi\colon Y_1 \to Y$ extending $\pi$ [CLS11, Chapter 3]; $\psi(Z_1)$ is closed, contains $\pi(V_{\widetilde{f}}) = V_f$ and lies in $Z$, so $\psi$ maps $Z_1$ onto $Z$. The identity of $\widetilde{N}$ defines the toric morphism $\varphi\colon Y_1 \to Y_0$, which maps $Z_1$ onto $Z_0$ and is an isomorphism over $V_{\widetilde{f}}$.

*The boundary.* By Lemma 20.50, each $E_\rho = Z_1 \cap D_\rho$ is the smooth closure of $V_{\widetilde{f}_\rho}$ in the smooth projective toric variety $D_\rho$ of dimension $n-1$. The exponents of $\widetilde{f}_\rho$ are the $e(w_i - w_{i_\rho})$ with $i \in I_\rho$, where $w_0 = 0$. The vectors $w_i - w_{i_\rho}$ with $i \le k$, $i \ne i_\rho$, together with $w_{k+1}, \dots, w_n$, form a basis of $\widetilde{M}$, so those with $i \in I_\rho$ span a saturated sublattice of $\widetilde{M}$, which lies in $\rho^\perp \cap \widetilde{M}$ and is saturated there. Hence $e(\widetilde{f}_\rho) = e$ if $I_\rho$ has at least two elements, and $E_\rho$ is empty otherwise. By induction every $E_\rho$ lies in $\mathcal{A}$, and it satisfies the Hodge conjecture if the $X_e^j$ with $j \le n-2$ do.

(i) By Lemma 20.51, $H^*(Z_1, \mathbb{Q})$ is spanned by the images of $H^*(Z_0, \mathbb{Q})$ and of the $H^*(E_\rho, \mathbb{Q})$ under $\varphi^*$ and the $i_{\rho *}$, which are induced by algebraic cycles, the transposed graph of $\varphi$ and the graphs of the $i_\rho$. So $Z_1 \in \mathcal{A}$ by Proposition 20.42(v), and $Z \in \mathcal{A}$ by Proposition 20.35(iii), as the image of $Z_1$ under $\psi$.

(ii) By the Künneth formula every Hodge class on $X_e^{k-1} \times \mathbb{P}^{n-k}$ is a sum of pull-backs of Hodge classes on $X_e^{k-1}$ multiplied by powers of the hyperplane class, so the conjecture holds on $Z_0$, as $k-1 \le n-1 = \dim Z$; it holds on the $E_\rho$ by induction. Let $a$ be a Hodge class on $Z_1$. Then $\varphi_* a$ is a Hodge class, hence algebraic, and so is $\varphi^*\varphi_* a$. The class $b = a - \varphi^*\varphi_* a$ is a Hodge class in the kernel of $\varphi_*$, hence in the image of the Gysin map of Lemma 20.51, a morphism of polarisable Hodge structures after a Tate twist; as in the proof of Proposition 20.35(iii) it is the image of a Hodge class, which is algebraic, so $b$ is algebraic. So the conjecture holds on $Z_1$, and it passes to $Z$ along $\psi$: for a Hodge class $x$ on $Z$, $\psi^* x$ is algebraic, and $\Psi_*\psi^* x = c\,x$ with $c > 0$ for the algebraic correspondence $\Psi$ of the proof of Proposition 20.35(iii). ∎

**Corollary 20.53** (Delsarte hypersurfaces of every dimension)**.** *Let $X_A \subset \mathbb{P}^{r+1}$, $r \ge 1$, be a smooth Delsarte hypersurface of degree $m$ as in Proposition 20.48, let $M = \{x \in \mathbb{Z}^{r+2} : \sum_j x_j = 0\}$, and let $e_A$ be the exponent of $M/M'$, where $M'$ is spanned by the differences $a_i - a_0$ of the rows of $A$.*

(i) *$X_A \in \mathcal{A}$, so (F3′) holds on $X_A$ in every degree, and the Hodge conjecture holds on $X_A$ if it holds on the Fermat varieties $X_{e_A}^j$ with $j \le r$. The number $e_A$ divides every $d$ of Proposition 20.48, and $[M : M'] = |\det A|/m$.*

(ii) *The same holds, with $j \le r+1$, for the cyclic cover $\{y^k = F_A(x)\}$ of $\mathbb{P}^{r+1}$ in the weighted projective space $\mathbb{P}(1, \dots, 1, m/k)$, for every $k$ dividing $m$, with $M = \{x \in \mathbb{Z}^{r+3} : \sum_{j \le r+1} x_j + (m/k)x_{r+2} = 0\}$ and $M'$ spanned by the differences of the exponents of $F_A - y^k$; it is smooth because $X_A$ is.*

(iii) *For the* 29 *sextic shapes of Proposition 20.48(iii),* $e_A$ *is the least* $d$ *there, except for the chain* $x_0^5x_1 + x_1^5x_2 + \cdots + x_4^5x_5 + x_5^6$ *and the loop, for which it is* 3125 *and* 2604 *instead of* 18750 *and* 15624. *For the Klein quartic* $x^3y + y^3z + z^3x$ *it is* 7.

*Proof.* (i) On the torus of $\mathbb{P}^{r+1}$, with character lattice $M$, $X_A$ is the closure of $V_f$ for $f = \chi^{-a_0}F_A$, as in the proof of Proposition 20.48(i). The rows of $A$ are linearly independent and lie on the affine hyperplane $\sum_j x_j = m$, which does not contain 0, so they are affinely independent, $f$ is of simplex type, $L' = M'$ has rank $r+1$ and $L = M$; so $e(f) = e_A$, and Theorem 20.52 gives the first two assertions. If $dA^{-1}$ is integral and $x \in M$, then $dx = \lambda A$ with $\lambda = dxA^{-1}$ integral and $\sum_i \lambda_i = d\sum_j x_j/m = 0$, as $A^{-1}\mathbf{1} = \mathbf{1}/m$; so $dx = \sum_i \lambda_i(a_i - a_0) \in M'$, and $e_A$ divides $d$. The lattice $M'$ is the intersection of $M$ with the lattice $R$ spanned by the rows, whose image under $x \mapsto \sum_j x_j$ is $m\mathbb{Z}$; the exact sequence $0 \to M/M' \to \mathbb{Z}^{r+2}/R \to \mathbb{Z}/m\mathbb{Z} \to 0$ gives $[M : M'] = |\det A|/m$.

(ii) The exponents of $F_A - y^k$ are linearly independent and lie on the hyperplane of weighted degree $m$, so the polynomial is of simplex type on the torus of $\mathbb{P}(1, \dots, 1, m/k)$, a projective toric variety whose character lattice is the given $M$. On the affine cone, the partial derivatives vanish together with $y^k - F_A$ only where $y = 0$ and $x$ is a singular point of the cone over $X_A$, that is $x = 0$; so the cover is quasi-smooth, and it misses the only singular point $(0 : \cdots : 0 : 1)$ of the weighted projective space, where $y^k = F_A(0)$ would force $y = 0$. So it is smooth. It is irreducible, $F_A$ being irreducible, and not contained in a coordinate hyperplane, so it is the closure of its part in the torus, and Theorem 20.52 applies.

(iii) The least $d$ with $dA^{-1}$ integral is the exponent of $\mathbb{Z}^{r+2}/R$, and the map $x \mapsto \sum_j x_j$ modulo $m$ is well defined on it because the rows sum to $m$. For a polynomial as in Proposition 20.48(iii) the group $\mathbb{Z}^{r+2}/R$ is the direct sum of one group for each term. The relations of a chain are $x_{j+1} = (1-m)x_j$ and $mx_k = 0$, those of a loop are $x_{j+1} = (1-m)x_j$, indices modulo $k$, and that of $x^m$ is $mx = 0$. So each summand is cyclic, generated by the class of its first variable, of order $N = m(m-1)^{k-1}$, $(m-1)^k - (-1)^k$ and $m$ respectively, and the sum map sends that generator to 1, since $1 - m \equiv 1$ modulo $m$; each $N$ is divisible by $m$. The least $d$ is the least common multiple of these orders, which gives the values of Proposition 20.48(iii). The group $M/M'$ is the kernel of the sum map. With one term only, the loop or the chain in six variables, the kernel is $m\mathbb{Z}/N\mathbb{Z}$, of exponent $N/m$, which is $15624/6 = 2604$ and $18750/6 = 3125$; for the Klein quartic, a loop with $m = 4$ and $k = 3$, it is $28/4 = 7$. With at least two terms, of orders $N_1, N_2, \dots$, the kernel contains the differences of the generators of the first and of the $b$-th summand, of order $\operatorname{lcm}(N_1, N_b)$; so its exponent is the least common multiple of all the $N_b$, which is the least $d$. ∎

For the Klein quartic the construction is classical: the isogeny has kernel of order $e_A^2/[M : M'] = 7$, and the Klein quartic is the quotient of the Fermat septic curve by a group of order seven; Figure 44 draws its Newton triangle and the smooth adapted fan of the proof. For $r = 4$, Corollary 20.53(i) recovers Proposition 20.48(ii), with the Fermat degree lowered to $e_A$. Its content lies in dimension five and more, where otherwise only the Fermat hypersurfaces are covered (Proposition 20.42(iii)), and in the cyclic covers, among them the Calabi–Yau varieties $\{y^2 = F_A(x)\}$ with $F_A$ of degree $2r + 4$ in $r + 2$ variables. Khovanskii's compactification and Shioda's cover are classical, and we claim no priority for the two lemmas. What Theorem 20.52 adds is that the boundary of the compactification is again of simplex type with the same exponent $e$, so that the whole cohomology, not only its part of top Hodge level, comes from Fermat varieties of one degree.

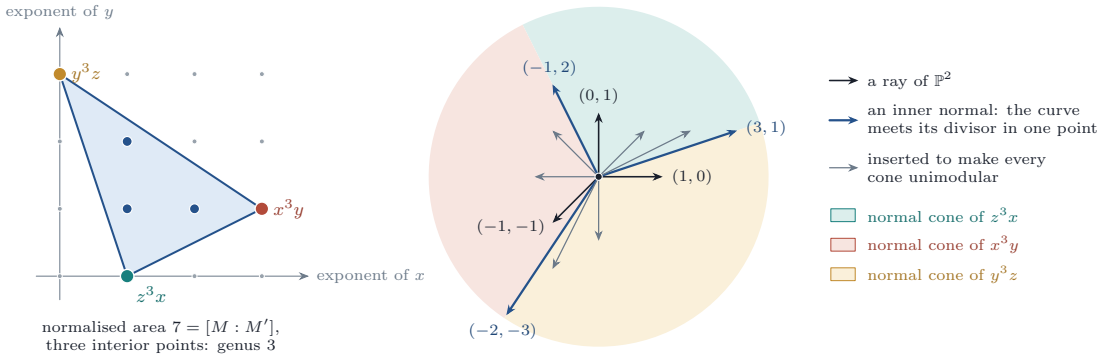



FIGURE 44. The Klein quartic $x^3y+y^3z+z^3x$ in the proof of Theorem 20.52. Left: its Newton triangle in the lattice $M$ of the torus of $\mathbb{P}^2$, in the chart $z = 1$, with normalised area $7 = [M : M']$ and three interior points, the genus. Right: the rays of a smooth fan adapted to it, the three rays of $\mathbb{P}^2$, the three inner normals of the triangle and the six rays inserted to make every cone unimodular. The curve meets only the three divisors of the inner normals, in one point each, the lattice length of the edge, so its Euler number is $-7 + 3 = -4$. With $e_A = 7$ the isogeny of the proof makes it the quotient of the Fermat septic by a group of order seven.

*Remark* 20.54 (Quotients by diagonal automorphisms). Let $G \subset T$ be a finite subgroup of the torus of $\mathbb{P}^{r+1}$ on which every character $\chi^{a_i-a_0}$ is trivial, so that $G$ acts on $X_A$; these $G$ are the subgroups of the group $G_A$ dual to $M/M'$. Then $f$ descends to a Laurent polynomial of simplex type on $T/G$, whose character lattice $M_G$, the characters trivial on $G$, contains $M'$; so $L' = M'$, $L = M_G$, and $e(f)$ becomes the exponent $e_G$ of $M_G/M'$, a divisor of $e_A$. As $\mathbb{P}^{r+1}/G$ is a projective toric variety with torus $T/G$ in which $X_A/G$ is the closure of $V_f/G$, Theorem 20.52 applies to every projective model $Z$ of $X_A/G$ that is the closure of $V_f/G$ in a projective toric variety with torus $T/G$ and is smooth, for instance in one whose fan is smooth and adapted to $f$ (Lemma 20.50): such a $Z$ lies in $\mathcal{A}$, and the Hodge conjecture holds on it if it holds on the $X^j_{e_G}$ with $j \le r$. For $G = G_A$, $M_G = M'$ and $e_G = 1$, so the conjecture holds on $Z$. For the Fermat hypersurface of degree $m = r + 2$ and the group of diagonal automorphisms of determinant one modulo scalars, the starting point of the mirror construction for $X_A$, $M' = mM$ and $M_G$ is the set of $x \in M$ whose coordinates are congruent modulo $m$, so $M_G/M' \cong \mathbb{Z}/m$ and $e_G = m = e_A$: the quotient needs the same Fermat varieties as $X_A$, and nothing is gained. For any finite group $G$ of automorphisms of $X_A$, every smooth projective model $X'$ of $X_A/G$ has $\mathrm{Hdg}^2(X') = \mathrm{Ab}^2(X')$, so (F3′) holds on it when it is a fourfold or a fivefold, by Proposition 20.46 and Corollary 20.53(i).

The Fermat hypersurfaces are the complete intersections of one diagonal hypersurface. Complete intersections of several diagonal hypersurfaces of the same degree lie in $\mathcal{A}$ as well when their coefficients are of Vandermonde type, because they are quotients of powers of one curve. Fix $d \ge 2$, $N \ge 2$ and $1 \le c \le N - 1$, put $r = N - c$, and call a $c \times (N + 1)$ matrix $(a_{ki})$ *of Vandermonde type* if there are distinct $\lambda_0, \dots, \lambda_N \in \mathbb{C}$, nonzero $\mu_0, \dots, \mu_N$ and $g \in \mathrm{GL}_c(\mathbb{C})$ with $a_{ki} = \mu_i \sum_{l=1}^{c} g_{kl}\lambda_i^{l-1}$; for $c \ge 2$ this says that the columns are distinct points of a rational normal curve of degree $c-1$ in $\mathbb{P}^{c-1}$. Put $w_i = \prod_{l\ne i}(\lambda_i - \lambda_l)^{-1}$

and

$$X_{d,r}(\lambda) = \Big\{ x \in \mathbb{P}^N : \sum_{i=0}^{N} w_i \lambda_i^k x_i^d = 0, \quad 0 \le k \le c-1 \Big\}.$$

For $r = 1$ this is the generalised Fermat curve of type $(d, N)$ of González-Diez, Hidalgo and Leyton [GDHL09], the Galois cover of $\mathbb{P}^1$ with group $H = \mu_d^{N+1}/\mu_d$, acting by $u_i \mapsto h_i u_i$, branched over the $\lambda_i$. The characters of $H$ are the $\chi_a(h) = \prod_i h_i^{a_i}$ with $a \in (\mathbb{Z}/d)^{N+1}$ and $\sum_i a_i = 0$; write $V_a \subset H^1(C, \mathbb{C})$ for the eigenspace on which $h^*$ acts by $\chi_a(h)$, $[a]$ for the orbit of $a$ under $(\mathbb{Z}/d)^\times$, and $B_{[a]}$ for the image of a multiple of the idempotent of $\mathbb{Q}[H]$ attached to $[a]$, an abelian subvariety of the Jacobian $J(C)$ whose first cohomology is the part $\bigoplus_{a' \in [a]} V_{a'}$ of $H^1(C, \mathbb{C})$.

**Theorem 20.55** (Diagonal complete intersections of Vandermonde type)**.** *Let $X \subset \mathbb{P}^N$ be the complete intersection of $c$ hypersurfaces $\sum_i a_{ki} x_i^d = 0$, $1 \le k \le c$, with $(a_{ki})$ of Vandermonde type, and let $C = X_{d,1}(\lambda)$ for the $\lambda_i$ of $(a_{ki})$.*

(i) *$X$ is smooth of dimension $r$ and isomorphic to $X_{d,r}(\lambda)$, and $C$ is a smooth curve of genus $1 + \frac{1}{2} d^{N-1}\big((N-1)(d-1) - 2\big)$.*

(ii) *The map $\Phi(u^{(1)}, \dots, u^{(r)}) = \big(\prod_j u_i^{(j)}\big)_i$ is a surjective morphism $C^r \to X_{d,r}(\lambda)$, and it induces an isomorphism $C^r/G \cong X_{d,r}(\lambda)$, where $G$ is the group of order $r!\, d^{N(r-1)}$ generated by the permutations of the factors and the $(h_1, \dots, h_r) \in H^r$ with $h_1 \cdots h_r = 1$.*

(iii) *$X \in \mathcal{A}$, so (F3′) holds on $X$ in every degree.*

(iv) *$\dim V_a = \#\{i : a_i \ne 0\} - 2$ for $a \ne 0$, and the Hodge conjecture holds on $X$ as soon as it holds on the abelian varieties $B_{[a]}$ with $\dim V_a \ge r$, which have dimension $\frac{1}{2}|[a]| \dim V_a$.*

*Proof.* Everything rests on the Lagrange identity: interpolating a polynomial $f$ of degree at most $N$ at the $\lambda_i$ and comparing the coefficients of $t^N$ gives $\sum_i w_i f(\lambda_i) = 0$ when $\deg f \le N - 1$.[38]

(i) Replacing the equations by their combinations under $g^{-1}$ and $x_i$ by $\nu_i x_i$ with $\nu_i^d \mu_i = w_i$ turns them into those of $X_{d,r}(\lambda)$. Every $c \times c$ minor of $(a_{ki})$ is $\det g \prod \mu_i$ times a Vandermonde determinant in distinct $\lambda_i$, so any $c$ columns are independent. At a point $x$ of $X$ the Jacobian matrix is $(d\, a_{ki} x_i^{d-1})$; if $S = \{i : x_i \ne 0\}$ had at most $c$ elements, $\sum_{i \in S} x_i^d a_i = 0$ would be a relation between at most $c$ columns, so $|S| > c$ and the Jacobian has rank $c$.[39] Hence $X$ is a smooth complete intersection of dimension $r$, connected because $r \ge 1$, and the same holds for $C$. The genus follows from the Riemann–Hurwitz formula for the cover $C \to \mathbb{P}^1$ of degree $d^N$ described below, which has $N + 1$ branch points of ramification index $d$.

[38] Lagrange interpolation writes $f(t) = \sum_i f(\lambda_i) \prod_{l \ne i} (t - \lambda_l)/(\lambda_i - \lambda_l)$ for $\deg f \le N$, and the coefficient of $t^N$ on the right is $\sum_i w_i f(\lambda_i)$. For $N = 2$ and $f = 1$ it says that $\sum_i \prod_{l \ne i} (\lambda_i - \lambda_l)^{-1} = 0$ for any three distinct numbers. The identity says that the vector $(w_i \lambda_i^k)_i$ is orthogonal to the values of every polynomial of degree at most $N - 1 - k$, which is how the equations of $X_{d,r}(\lambda)$ become the condition that the $d$-th powers of the coordinates are the values of one polynomial of degree at most $r$.

[39] The columns of the Jacobian matrix with $i \in S$ are the columns of $(a_{ki})$ multiplied by the nonzero numbers $d\, x_i^{d-1}$, and any $c$ of the columns of $(a_{ki})$ are independent; so $c$ of them already give rank $c$. Without the Vandermonde condition this fails: for two diagonal quadrics in $\mathbb{P}^3$ with proportional columns $i = 0, 1$, the points with $x_2 = x_3 = 0$ and $a_0 x_0^2 + a_1 x_1^2 = 0$ are singular.

(ii) For $1 \le k \le N-1$ let $\Lambda_k$ be the space of $y \in \mathbb{C}^{N+1}$ with $\sum_i w_i \lambda_i^l y_i = 0$ for $0 \le l \le N-k-1$. The vectors $(F(\lambda_i))_i$ with $\deg F \le k$ lie in $\Lambda_k$ by the identity, since $\deg t^l F \le N-1$; they form a space of dimension $k+1$, since $k < N+1$; and $\Lambda_k$ has dimension $k+1$, the $N-k$ equations having independent Vandermonde rows. So $\Lambda_k = \{(F(\lambda_i))_i : \deg F \le k\}$, and $C$ and $X_{d,r}(\lambda)$ are the sets of points whose $d$-th powers lie in $\Lambda_1$ and in $\Lambda_r$. The point $u \in C$ determines $(s:t) \in \mathbb{P}^1$ with $u_i^d = \kappa(t - \lambda_i s)$ for all $i$; this is the quotient $C \to C/H = \mathbb{P}^1$, whose fibres are the $H$-orbits. If $(u_i^{(j)})^d = t_j - \lambda_i s_j$, then $\Phi(u)_i^d = F(\lambda_i)$ for $F(t) = \prod_j (t_j - t s_j)$, a nonzero polynomial of degree at most $r < N+1$; so $\Phi(u) \in X_{d,r}(\lambda)$ and $\Phi(u) \ne 0$, and $\Phi$, given by forms of multidegree $(1,\dots,1)$ with no common zero, is a morphism. It is $G$-invariant. Conversely, if $x \in X_{d,r}(\lambda)$, then $(x_i^d)_i = (F(\lambda_i))_i$ for a unique nonzero $F$ of degree at most $r$, which factors as $\prod_j (t_j - t s_j)$, uniquely up to order and to rescaling the factors. Choosing $u^{(j)}$ over $(s_j : t_j)$ gives $\Phi(u)_i = \epsilon_i x_i$ with $\epsilon_i \in \mu_d$, and multiplying $u^{(1)}$ by $(\epsilon_i^{-1})$ gives a preimage. Two preimages have the same roots, so after a permutation $u'^{(j)} = h_j u^{(j)}$ with $h_j \in H$, and $\prod_j h_j$ acts trivially on the coordinates where $x_i \ne 0$; where $x_i = 0$ some $u_i^{(j)}$ vanishes and $h_{j,i}$ may be changed freely, so we may take $\prod_j h_j = 1$. Hence the fibres of $\Phi$ are the $G$-orbits, $\Phi$ induces a finite bijective morphism from the normal variety $C^r/G$ to the smooth variety $X_{d,r}(\lambda)$, and by Zariski's main theorem this is an isomorphism.[40] The order of $G$ is $r!\,|H|^{r-1}$.

(iii) $C$ is a curve, so $C^r \in \mathcal{A}$, and $X$ is the image of $C^r$ under a surjective morphism (Proposition 20.35(iii)).

(iv) For $a \ne 0$, the forms $t^j \prod_i (t-\lambda_i)^{-b_i/d}\,dt$ with $b_i \in \{0,\dots,d-1\}$, $b_i \equiv -a_i$, and $0 \le j \le \sum_i b_i/d - 2$, are holomorphic on $C$ and span the $\chi_a$-eigenspace of $H^{1,0}(C)$;[41] its dimension $\sum_i \langle -a_i/d\rangle - 1$ and that of the $\chi_{-a}$-eigenspace add up to $\#\{i : a_i \ne 0\} - 2$, and $H^1(C)^H = H^1(\mathbb{P}^1) = 0$. The group $H$ acts trivially on $H^0(C)$ and $H^2(C)$, so a tensor in $H^r(C^r) = \bigoplus \bigotimes_j H^{k_j}(C)$ invariant under the $(h_j)$ with $\prod_j h_j = 1$ has all its factors in $H^1$, with one character, or none: taking $h$ in one factor and $h^{-1}$ in another shows that two factors carry the same character, and a factor in $H^0$ or $H^2$ carries the trivial one. Taking the invariants of the permutations,

$$H^r(C^r,\mathbb{C})^G = \mathbb{C}\,\eta \;\oplus \bigoplus_{a \ne 0} \bigwedge^r V_a,$$

with $\eta$ the symmetrised product of $r/2$ classes of points when $r$ is even and $\eta = 0$ otherwise, and $W_{[a]} = \bigoplus_{a' \in [a]} \bigwedge^r V_{a'}$ is a sub-Hodge structure defined over $\mathbb{Q}$, zero unless $\dim V_a \ge r$. Let $\psi\colon C \to B_{[a]}$ be an Abel–Jacobi map followed by the projection, so that

[40] In characteristic zero a finite morphism that is bijective on closed points is generically étale of degree one, hence birational, and a finite birational morphism onto a normal variety is an isomorphism [Har77, Chapter III, Corollary 11.4]. The group $G$ acts faithfully on $C^r$, since $H$ is the group of deck transformations of $C \to \mathbb{P}^1$, so $\Phi$ has degree $|G|$.

[41] On the affine chart $s = 1$ put $u_i^d = t - \lambda_i$, so that the form is $t^j \prod_i u_i^{-b_i}\,dt$ and $h \in H$ multiplies it by $\prod_i h_i^{-b_i} = \chi_a(h)$; it is well defined on $H = \mu_d^{N+1}/\mu_d$ because $\sum_i b_i \equiv 0$. Over $\lambda_i$ the cover is ramified of index $d$, with a local parameter $z$ satisfying $z^d = t - \lambda_i$, and the form is a unit times $z^{-b_i}\,d(z^d) = d\,z^{d-1-b_i}\,dz$, holomorphic as $b_i \le d-1$. Over $t = \infty$, with $w = 1/t$ and $v^d = w$, it is a unit times $v^{d(\sum_i b_i/d - j - 2) + d - 1}\,dv$, holomorphic exactly when $j \le \sum_i b_i/d - 2$, the exponent of $v$ being an integer. Forms with different characters, or with different $j$, are independent, and their number is $\frac12 \sum_{a \ne 0}\big(\#\{i : a_i \ne 0\} - 2\big) = \frac12\big((N+1)(d^N - d^{N-1}) - 2(d^N - 1)\big)$, which is the genus of (i); so they form a basis of $H^{1,0}(C)$, and each eigenspace is spanned by the forms of its character. For $d = 2$ and $N = 4$ they are the five forms $dt/\sqrt{(t-\lambda_{i_1})\cdots(t-\lambda_{i_4})}$, one for each choice of four of the five indices, on a curve of genus five.

$\psi^*$ maps $H^1(B_{[a]},\mathbb{Q})$ isomorphically onto the $[a]$-part of $H^1(C,\mathbb{Q})$, and let $\sigma\colon C^r \to B_{[a]}$ be $\sum_j \psi\circ \mathrm{pr}_j$. Then $\sigma^*(\alpha_1\wedge\cdots\wedge\alpha_r) = \prod_k \sum_j \mathrm{pr}_j^*\psi^*\alpha_k$, whose component in $H^1(C)^{\otimes r}$ is the alternating sum of the $\psi^*\alpha_1\otimes\cdots\otimes\psi^*\alpha_r$. The Künneth projectors of a curve are algebraic,[42] and so is the average $e_G$ over $G$, so $e_G\circ\pi_1^{\otimes r}\circ\sigma^*$ is induced by an algebraic cycle on $B_{[a]}\times C^r$, and it maps $H^r(B_{[a]},\mathbb{Q})$ onto $W_{[a]}$: the alternating tensors in the $[a]$-part are the image of $\pi_1^{\otimes r}\circ\sigma^*$, and $e_G$ keeps those of one character. A surjective morphism of polarisable Hodge structures maps Hodge classes onto Hodge classes,[43] so the Hodge classes of $W_{[a]}$ are algebraic when those of $B_{[a]}$ are. Now let $x$ be a Hodge class of degree $r$ on $X = X_{d,r}(\lambda)$, the other degrees being multiples of powers of the hyperplane class by the Lefschetz hyperplane theorem. Then $\Phi^*x$ is a $G$-invariant Hodge class, its components in the $W_{[a]}$ and along $\eta$ are Hodge classes, being cut out by the algebraic projectors attached to the action of $H^r$, and they are algebraic; so $\Phi^*x$ is algebraic, and so is $x = \Phi_*\Phi^*x/|G|$. Finally $\dim B_{[a]}$ is half of $\sum_{a'\in[a]}\dim V_{a'} = |[a]|\dim V_a$. ∎

**Corollary 20.56** (Two diagonal hypersurfaces)**.**

(i) *Every smooth complete intersection of two diagonal hypersurfaces* $\sum_i a_i x_i^d = \sum_i b_i x_i^d = 0$ *of the same degree in* $\mathbb{P}^N$ *is of Vandermonde type, so it lies in* $\mathcal{A}$ *and satisfies* (F3′) *in every degree.*

(ii) *Let* $d\in\{3,4,6\}$ *and* $N\le 6$*, or* $d=2$ *and* $N\le 12$*. Then the Hodge conjecture holds on every complete intersection of Vandermonde type in* $\mathbb{P}^N$ *of* $N-r$ *diagonal hypersurfaces of degree* $d$*. In particular it holds on every smooth complete intersection of two diagonal hypersurfaces of degree* 3*,* 4 *or* 6 *in* $\mathbb{P}^6$*.*

(iii) *For* $d\in\{3,4,6\}$ *and* $N=6$*, each orbit* $[a]$ *of order at least three with six nonzero coordinates and* $\sum_i\langle a_i/d\rangle = 3$ *makes* $B_{[a]}$ *an abelian fourfold of Weil type for* $K=\mathbb{Q}(\zeta_d)$*, and* $W_{[a]}$ *is its space of Weil classes. On each of these fourfolds* $X$ *the Weil classes of the* 70*,* 490 *and* 6125 *such orbits, for* $d=3,4,6$*, give Hodge classes of degree four outside* $\mathbb{Q}h^2$*, so the Hodge classes of* $X$ *form a space of dimension at least* 141*,* 981 *and* 12251*, whereas the divisor classes of* $X$ *are the multiples of* $h$*.*

*Proof.* (i) If two columns $(a_i,b_i)$ and $(a_j,b_j)$ were proportional, or one of them zero, $X$ would contain a point supported on $\{i,j\}$, or the coordinate point $e_i$, at which the Jacobian has rank at most one; so the $N+1$ columns are distinct points of $\mathbb{P}^1$. Sending a point of $\mathbb{P}^1$ outside them to infinity writes them as $\mu_i(1,\lambda_i)$ with distinct $\lambda_i$, which is the Vandermonde form for $c=2$, and Theorem 20.55(iii) applies. (ii) For $d\in\{2,3,4,6\}$ every $a$ has order 1, 2, 3, 4 or 6, so $|[a]|\le 2$, with $|[a]|=1$ for the $a$ of order two, whose $V_a$ is real, so that $\dim B_{[a]}\le\dim V_a\le N-1$. When $d=2$ every $a\neq 0$ has order two and an even number of nonzero coordinates, at most 12 when $N\le 12$, so $\dim B_{[a]} = \frac12\dim V_a\le 5$. So every $B_{[a]}$ has dimension at most five,[44] and the Hodge conjecture holds on abelian varieties of dimension at most five [Mar25a, Mar25b, MZ99]; Theorem 20.55(iv) gives the conclusion, and (i) the last statement. (iii) For such $a$ the eigenspaces $V_a$ and $V_{-a}$ have dimension four

[42] For a curve $C$ with a point $o$ they are $\pi_0=[o\times C]$, $\pi_2=[C\times o]$ and $\pi_1=[\Delta_C]-\pi_0-\pi_2$, and $\pi_1^{\otimes r}$ is their exterior product on $C^r\times C^r$. The average $e_G = |G|^{-1}\sum_{g\in G}[\Gamma_g]$ of the graphs of the elements of $G$ acts on $H^*(C^r)$ as the projector onto the invariants, and $\Phi^*$ identifies $H^*(X_{d,r}(\lambda),\mathbb{Q})$ with $H^*(C^r,\mathbb{Q})^G$.

[43] The category of polarisable rational Hodge structures is semisimple, so the kernel of a surjective morphism $f\colon V\to W$ has a complement $V'$, a sub-Hodge structure that $f$ maps isomorphically onto $W$; a Hodge class of $W$ is then the image of a Hodge class of $V'$.

[44] Both bounds are sharp: for $d=3$ and $N=7$ a character with eight nonzero coordinates has $\dim V_a=6$ and $|[a]|=2$, so $\dim B_{[a]}=6$; for $d=2$ and $N=13$ a character with fourteen nonzero coordinates has $\dim B_{[a]}=6$.

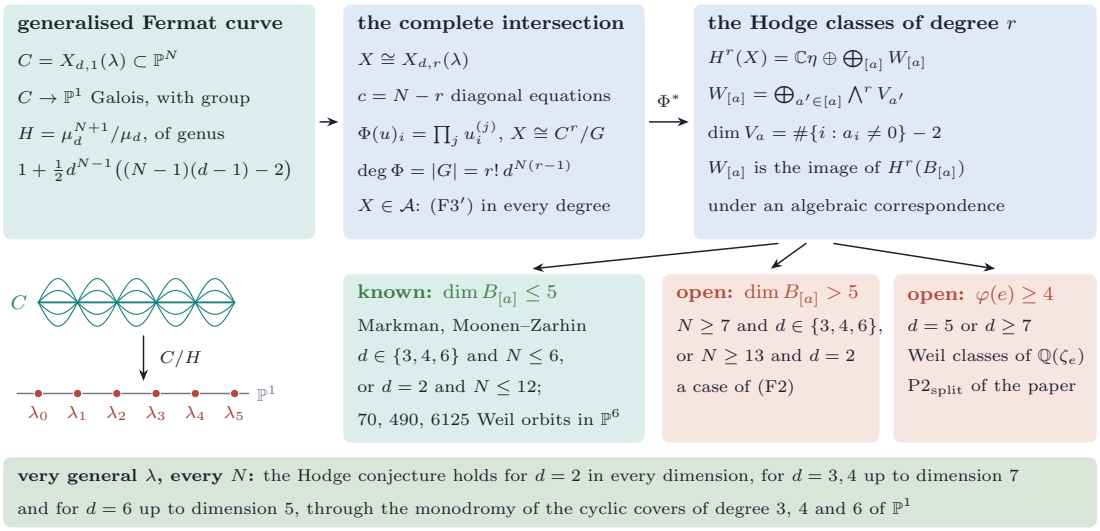



FIGURE 45. Theorem 20.55 and Corollary 20.56. Left: the generalised Fermat curve $C$ over $\mathbb{P}^1$, its sheets meeting over the branch points $\lambda_0, \dots, \lambda_N$ (five sheets stand for the $d^N$ of the cover). Top: $C$, the complete intersection $X \cong C^r/G$ with the map $\Phi$ of degree $|G|$, and the decomposition of $H^r(X)$ through $\Phi^*$. Bottom: where the Hodge conjecture for $X$ stands at an arbitrary configuration, according to the abelian varieties $B_{[a]}$ that carry its Hodge classes: known when they have dimension at most five, which happens for $d \in \{3,4,6\}$ and $N \leq 6$ and for $d = 2$ and $N \leq 12$, and open when they are larger, a case of (F2), or carry Weil classes of $\mathbb{Q}(\zeta_e)$ with $\varphi(e) \geq 4$, the case $\mathrm{P2}_{\mathrm{split}}$ of the paper. Foot: the very general configuration, Theorem 20.66.

and contain two-dimensional parts of type $(1,0)$, so $K$ acts on $H^{1,0}(B_{[a]})$ with multiplicities $(2,2)$ and $B_{[a]}$, of dimension four, is of Weil type; $\bigwedge^4 V_a$ and $\bigwedge^4 V_{-a}$ are its Weil lines, of type $(2,2)$ for every $\lambda$, and they lie in $H^4(C^4)^G = \Phi^* H^4(X)$. The zero coordinate of $a$ is one of seven, and the other six $a_i \in \{1, \dots, d-1\}$ add up to $3d$, which they do in 20, 141 and 1751 ways for $d = 3, 4, 6$, the coefficients of $x^{3d-6}$ in $(1 + x + \cdots + x^{d-2})^6$; removing the character of order two with every $a_i = d/2$ leaves 20, 140 and 1750, and $(\mathbb{Z}/d)^\times = \{\pm 1\}$ pairs them off, since $-a$ again has coordinate sum $3d$; this gives $7 \cdot 20/2 = 70$, $7 \cdot 140/2 = 490$ and $7 \cdot 1750/2 = 6125$ orbits. Distinct orbits give independent classes, and the hyperplane class lies in the part of the trivial character, which gives the bounds; the Picard group of a complete intersection of dimension at least three is generated by $h$. ■

Figure 45 draws the construction. For $c = 1$ the theorem is Proposition 20.42(iii) again, by a different route, and for $r = 1$ it restates the definition of $C$.

At a very general configuration the abelian varieties $B_{[a]}$ carry only the Hodge classes that the monodromy of the curves allows, and the conjecture can then be decided in every dimension for $d = 2$, up to dimension seven for $d = 3$ and $d = 4$, and up to dimension five for $d = 6$. Write $U \subset \mathbb{C}^{N+1}$ for the space of $(N+1)$-tuples of distinct numbers. The varieties $X_{d,r}(\lambda)$ with $\lambda \in U$ form a smooth projective family over $U$, and so do the curves $C$ and the maps $\Phi$, with $H$ acting on the family of curves.

**Lemma 20.57** (Hodge classes of a very general member)**.** *Let $\mathbb{V}$ be a polarised variation of $\mathbb{Z}$-Hodge structure of weight $2p$ on a smooth connected quasi-projective variety $S$, and let $\Pi = \pi_1(S, s)$ act on $\mathbb{V}_s$. If $s$ lies outside a countable union of proper Zariski closed subsets of $S$, the space $\mathrm{Hdg}(\mathbb{V}_s)$ of rational classes of type $(p,p)$ is stable under $\Pi$, and a subgroup of finite index of $\Pi$ fixes it pointwise.*

*Proof.* The locus of Hodge classes has countably many components (Definition B.3, Lemma B.4), and the image in $S$ of each is Zariski closed (Theorem B.6); let $s$ avoid the images that are not all of $S$. For $\alpha \in \mathrm{Hdg}(\mathbb{V}_s)$ the component through $(s,\alpha)$ maps onto $S$, locally biholomorphically onto its image, so the flat section $\tilde{\alpha}$ through $\alpha$ on the universal cover $\tilde{S}$ is of type $(p,p)$ on a nonempty open set. The set where it is of type $(p,p)$ is a closed analytic subset of the connected manifold $\tilde{S}$, so it is all of $\tilde{S}$, and every transport of $\alpha$ along a loop at $s$ is a Hodge class at $s$. The group $\Pi$ therefore preserves $\mathrm{Hdg}(\mathbb{V}_s)$, the lattice $\mathbb{V}_{s,\mathbb{Z}} \cap \mathrm{Hdg}(\mathbb{V}_s)$ and the flat form $\tilde{Q}$ of the proof of Lemma B.5, which is positive definite on the real classes of type $(p,p)$; so it acts on $\mathrm{Hdg}(\mathbb{V}_s)$ through a finite group, whose kernel is the subgroup of the statement. ∎

For the characters of order three, four and six the monodromy of the curves $D_a$ is determined by induction on the number of branch points. The step lets two branch points collide, and a lemma on two copies of SL of a hyperplane closes it. The base has four branch points, two of them with opposite exponents, whose full twist is a transvection (Figure 46).

**Setup 20.58** (Cyclic covers)**.** Let $m \in \{3,4,6\}$, $\zeta = e^{2\pi i/m}$ and $K = \mathbb{Q}(\zeta)$, and let $a = (a_1, \dots, a_k)$ with $a_i \in \mathbb{Z}/m$ nonzero, $\sum_i a_i = 0$ and $\gcd(m, a_1, \dots, a_k) = 1$, so that $a$ has order $m$. For $\lambda$ in the space $\mathrm{Conf}_k$ of $k$-tuples of distinct complex numbers let $D_a$ be the cyclic cover $y^m = \prod_i (t-\lambda_i)^{a_i}$ of $\mathbb{P}^1$, with representatives $0 < a_i < m$; it is unramified over $\infty$. Let $V_a \subset H^1(D_a, \mathbb{C})$ be the eigenspace of the deck group on which $H^{1,0}$ is spanned by the forms $t^j \prod_i (t-\lambda_i)^{-b_i/m}\, dt$, where $0 < b_i < m$, $b_i \equiv -a_i$ and $0 \le j \le \sum_i b_i/m - 2$ [DM86, §2], and put $n = \dim V_a$, $p = \dim V_a^{1,0} = \sum_i b_i/m - 1$ and $q = n - p$. Let $\Gamma_a \subset \mathrm{GL}(V_a)$ be the image of the pure braid group $P_k = \pi_1(\mathrm{Conf}_k)$ and $G_a^0$ the identity component of its Zariski closure.

**Lemma 20.59** (A model of the monodromy)**.** *Let $F_k$ be the free group on loops $x_1, \dots, x_k$ around $\lambda_1, \dots, \lambda_k$, put $t_i = \zeta^{a_i}$, let $\partial\colon K^k \to K$ map $e_i$ to $t_i - 1$, and let $E_a = \ker\partial / K\ell$, where $\ell = \sum_i t_1 \cdots t_{i-1} e_i$. A pure braid $\beta$ acts on $F_k$ by Artin's automorphism and on $K^k$ by $v \mapsto vJ_\beta$, where $J_\beta$ is the matrix of Fox derivatives $\partial\beta(x_i)/\partial x_j$ evaluated at $x_j \mapsto t_j$. This action preserves $\ker\partial$ and fixes $\ell$, and the representation of $P_k$ on $E_a$, of dimension $k-2$, is isomorphic to $V_a$, to its dual or to its complex conjugate. So $n = k-2$, and $G_a^0 \supseteq \mathrm{SL}(V_a)$ if and only if the identity component of the Zariski closure of the image of $P_k$ in $\mathrm{GL}(E_a)$ contains $\mathrm{SL}(E_a)$.*

*Proof.* Fox's formula $\sum_j (\partial w/\partial x_j)(x_j - 1) = w - 1$ [Fox53], evaluated at $x_j \mapsto t_j$, shows that the vector $\sum_j (\partial w/\partial x_j)(t)\, e_j$ of a word $w$ with $t(w) = 1$ lies in $\ker\partial$. It is the class of the lift of the loop $w$ in the summand of $H_1(D_a^\circ, K)$ on which the deck group acts through $x_i \mapsto t_i$, where $D_a^\circ$ is the unramified part of the cover, and that summand is $H_1(F_k, K_t) = \ker\partial$, computed from the free resolution of length one of the augmentation ideal of a free group.[45] Filling in the points over $\lambda_i$ adds no relation: the loop around such

[45] By Shapiro's lemma $H_1(D_a^\circ, K) = H_1(F_k, K[\mu_m])$, and its summand for the character $x_i \mapsto t_i$ is $H_1(F_k, K_t)$. The augmentation ideal of $\mathbb{Z}[F_k]$ is free on the $x_i - 1$, so $H_1(F_k, K_t)$ is the kernel of $\bigoplus_i K e_i \to K$, $e_i \mapsto t_i - 1$, the chain $e_i$ standing for the loop $x_i$ and a word $w$ for $\sum_j (\partial w/\partial x_j)(t) e_j$.

a point lifts $x_i^e$, $e$ the order of $t_i$, whose vector $(1+t_i+\cdots+t_i^{e-1})e_i$ is zero. Filling in the $m$ points over $\infty$, where the cover is unramified, kills the lift of $x_1\cdots x_k$, whose vector is $\ell\neq 0$. So $E_a$ is the summand of $H_1(D_a,K)$ for the character $x_i\mapsto t_i$, of dimension $k-2$. A pure braid fixes $x_1\cdots x_k$ and maps each $x_i$ to a conjugate of itself, so it preserves $t$, and the chain rule for Fox derivatives makes $\beta\mapsto J_\beta$ the action of $P_k$ on this summand.[46] Poincaré duality and the choice of the generator of the deck group identify it with $V_a$, its dual or its conjugate, and the automorphisms $g\mapsto {}^tg^{-1}$ and $g\mapsto\bar g$ of GL preserve SL. ■

**Lemma 20.60** (Collision of two branch points)**.** *In Lemma 20.59 suppose that $a_1+a_2\neq 0$, and let $a'=(a_1+a_2,a_3,\dots,a_k)$. Let $I=(1-t_2)e_1-(1-t_1)e_2$, the cycle of a disc around $\lambda_1$ and $\lambda_2$, and let $Y\subset E_a$ be the image of the cycles outside that disc, the vectors of $\ker\partial$ with $v_2=t_1v_1$. Then $E_a=Y\oplus KI$, and $\iota(v_1,v_3,\dots,v_k)=(v_1,t_1v_1,v_3,\dots,v_k)$ induces an isomorphism $E_{a'}\to Y$. If $\beta\in P_k$ is obtained from $\beta'\in P_{k-1}$ by doubling its first strand, then $\beta$ preserves $Y$ and $KI$, acts on $Y$ as $\beta'$ acts on $E_{a'}$, and multiplies $I$ by a power of $\zeta$.*

*Proof.* The vectors of $Y$ have $(v_1,v_2)$ proportional to $(1,t_1)$, and $(1-t_2,t_1-1)$ is proportional to $(1,t_1)$ only if $t_1t_2=1$; so $I\notin Y$. The map $\iota$ carries the kernel of $\partial'$, where $\partial'(e_1)=t_1t_2-1$, onto the cycles outside the disc, because $v_1(t_1t_2-1)=v_1(t_1-1)+t_1v_1(t_2-1)$, and it carries the vector $\ell'$ of $a'$ to $\ell$; as $\dim E_{a'}+1=\dim E_a$, both assertions about $Y$ follow. The doubled braid moves a disc containing $\lambda_1,\lambda_2$ rigidly. Its automorphism therefore maps $x_1\mapsto wx_1w^{-1}$, $x_2\mapsto wx_2w^{-1}$ and $x_j\mapsto u_j$ for $j\geq 3$, where $y\mapsto w'yw'^{-1}$, $x_j\mapsto u'_j$ is the automorphism of $\beta'$ on the free group on $y,x_3,\dots,x_k$, and $w,u_j$ are $w',u'_j$ with $y$ replaced by $x_1x_2$. For such a word $u$ the chain rule gives $\partial u/\partial x_1=\partial u/\partial y$ and $\partial u/\partial x_2=(\partial u/\partial y)\,x_1$, which at $t$ is the relation $v_2=t_1v_1$; together with $\partial(wx_iw^{-1})/\partial x_j=(1-t_i)\,\partial w/\partial x_j+t(w)\delta_{ij}$ at $t$, this shows that the rows of $J_\beta$ for $j\geq 3$, and the first row plus $t_1$ times the second, are the images under $\iota$ of the rows of $J_{\beta'}$; this is the equivariance of $\iota$. The same formula gives $IJ_\beta=(1-t_2)((1-t_1)\partial w+t(w)e_1)-(1-t_1)((1-t_2)\partial w+t(w)e_2)=t(w)I$, and $t(w)$ is a power of $\zeta$. ■

**Lemma 20.61** (Two hyperplanes)**.** *Let $E=Y\oplus L$ be a complex vector space with $\dim Y\geq 2$ and $\dim L=1$, let $S\subset\mathrm{SL}(E)$ be the subgroup acting on $Y$ through $\mathrm{SL}(Y)$ and trivially on $L$, and let $\gamma\in\mathrm{GL}(E)$ satisfy $\gamma L\neq L$ and $\gamma Y\neq Y$. Then the group $T$ generated by $S$ and $\gamma S\gamma^{-1}$ acts irreducibly on $E$. If $\dim Y\geq 3$, then $T=\mathrm{SL}(E)$. If $\dim Y=2$, every connected semisimple subgroup of $\mathrm{SL}(E)$ that contains $T$ is $\mathrm{SL}(E)$.*

*Proof.* The group $T$ is connected. As a representation of $S$, $E$ is the sum of the irreducible nontrivial $Y$ and the trivial $L$, so its $S$-stable subspaces are $0$, $L$, $Y$ and $E$, and those of $\gamma S\gamma^{-1}$ are $0$, $\gamma L$, $\gamma Y$ and $E$; by the hypotheses and the dimensions only $0$ and $E$ are in both lists, and $T$ is irreducible.

Let $\dim Y\geq 3$. As a representation of $\mathfrak{sl}(Y)$,

$$\mathfrak{sl}(E)=\mathfrak{sl}(Y)\oplus\mathrm{Hom}(L,Y)\oplus\mathrm{Hom}(Y,L)\oplus\mathbb{C}z,$$

$z$ acting by 1 on $Y$ and by $-\dim Y$ on $L$, and the four summands, the adjoint representation, $Y$, $Y^*$ and the trivial one, are pairwise non-isomorphic since $\dim Y\geq 3$. The Lie algebra $\mathfrak{t}$ of $T$ contains $\mathfrak{sl}(Y)$ and is stable under it, so it is $\mathfrak{sl}(Y)$ plus some of the other three summands. Without $\mathrm{Hom}(L,Y)$ it preserves $L$, and without $\mathrm{Hom}(Y,L)$ it preserves $Y$;

[46] Artin's automorphism of the half twist $\sigma_i$ of the $i$-th and $(i+1)$-st points is $x_i\mapsto x_ix_{i+1}x_i^{-1}$, $x_{i+1}\mapsto x_i$, the other generators being fixed; $P_k$ is generated by the full twists $A_{st}=\sigma_{t-1}\cdots\sigma_{s+1}\sigma_s^2\sigma_{s+1}^{-1}\cdots\sigma_{t-1}^{-1}$. For a pure braid the chain rule reads $J_{\beta\beta'}=J_{\beta'}J_\beta$ or $J_\beta J_{\beta'}$, according to the convention for composing braids, with no twist of the coefficients, since $t\circ\beta=t$.

both are excluded by irreducibility. The bracket of a nonzero map $L \to Y$ with a map $Y \to L$ not vanishing on its image has a nonzero component along $z$, so $\mathfrak{t} = \mathfrak{sl}(E)$ and $T = \mathrm{SL}(E)$.

Let $\dim Y = 2$ and let $G \supseteq T$ be connected and semisimple. Then $E$ is a faithful irreducible representation of dimension 3 of the Lie algebra $\mathfrak{g}$ of $G$. An irreducible representation of a product of simple Lie algebras is a tensor product of irreducible representations of the factors, and 3 is prime, so $\mathfrak{g}$ is simple. It lies in $\mathfrak{sl}(E)$, of dimension 8 and rank 2, and the simple Lie algebras of rank at most 2 are $\mathfrak{sl}_2$, $\mathfrak{sl}_3$, $\mathfrak{so}_5$ and $\mathfrak{g}_2$, of dimensions 3, 8, 10 and 14. If $\mathfrak{g} \cong \mathfrak{sl}_2$, then $\mathfrak{g}$ equals the Lie algebra of $S$, both being of dimension 3, so $G = S$, which preserves $L$, against the irreducibility of $T$. Hence $\mathfrak{g} = \mathfrak{sl}(E)$ and $G = \mathrm{SL}(E)$. ∎

**Lemma 20.62** (Merging two branch points)**.** *Let $m$ and $a$ be as in Setup 20.58, with $k \geq 6$ and $p, q \geq 1$. Then there are $i \neq j$ with $a_i + a_j \neq 0$ such that the vector $a'$ obtained by replacing $a_i$ and $a_j$ by $a_i + a_j$ has order $m$, and its numbers $p'$ and $q'$ are at least 1.*

*Proof.* Put $f_i = b_i/m \in [1/m, 1 - 1/m]$, so that $p + 1 = \sum_i f_i$ and $q + 1 = \sum_i (1 - f_i)$. Merging $a_i$ and $a_j$ keeps $p$ and lowers $q$ by one if $f_i + f_j < 1$, and lowers $p$ by one and keeps $q$ if $f_i + f_j > 1$. Since $a_i + a_j = -\sum_{l \neq i,j} a_l$, the order of $a'$ is that of the subgroup generated by the $a_l$ with $l \neq i, j$, the rest of $a$. Replacing $a$ by $-a$ exchanges $p$ and $q$, so we may assume $q \geq p$, hence $q \geq 2$ as $p + q = k - 2 \geq 4$; if $p \geq 2$, every merge keeps $p', q' \geq 1$ and only the order matters.

*The order.* For $m = 3$ two equal coordinates exist, their sum is not 0, and any nonzero rest generates $\mathbb{Z}/3$. For $m = 4$ the number of odd coordinates is even and positive; if it is at least 4, merge two equal odd ones, and if it is 2, merge an odd one with one of the at least four coordinates equal to 2. For $m = 6$ the rest generates $\mathbb{Z}/6$ exactly when it contains an odd coordinate and one prime to 3. The odd coordinates are even in number and the coordinates prime to 3 have sum divisible by 3, so each kind occurs at least twice. If exactly two coordinates are odd, the at least four others lie in $\{2, 4\}$; merge two equal ones among them. If exactly two are prime to 3, the others equal 3; merge a 3 with one of the two. Otherwise each kind occurs at least three times, and any two coordinates with nonzero sum can be merged; they exist, since pairwise sums 0 would force every coordinate to be 3.

*The case $p = 1$.* Then $\sum_i b_i = 2m$. Some $b_i$ equals 1, since otherwise $\sum_i b_i \geq 2k \geq 12 \geq 2m$, with equality only if $m = 6$ and every coordinate is 4, when $a$ has order three. Keep such a coordinate, $a_i = -1$, which generates $\mathbb{Z}/m$. The other $k - 1 \geq 5$ coordinates have $b$-sum $2m - 1$, so the two smallest values of $b$ among them add up to at most $2(2m - 1)/5 < m$; merge these two coordinates. Their sum is not 0, $p' = p = 1$, $q' = q - 1 = k - 4 \geq 2$, and the rest contains $a_i$, so $a'$ has order $m$. ∎

**Lemma 20.63** (Five branch points)**.** *Let $m$ and $a$ be as in Setup 20.58, with $k = 5$ and $p, q \geq 1$. Then $a_i + a_j = 0$ for some $i \neq j$, and the other three coordinates can be labelled $x, y, z$ so that $x + y \neq 0$ and $a' = (x + y, z, a_i, a_j)$ has order $m$ and $p' = q' = 1$.*

*Proof.* Replacing $a$ by $-a$ we may assume $p = 1$, so that $\sum_l b_l = 2m$ with five terms in $[1, m - 1]$; and $a_i + a_j = 0$ means $b_i + b_j = m$. For $m = 3$ the terms lie in $\{1, 2\}$ and exactly one is 2; take it and a 1. For $m = 4$, take a 3 and a 1 if both occur. If no term is 3, the terms lie in $\{1, 2\}$ and at least three are 2, which gives $2 + 2$; if no term is 1, they lie in $\{2, 3\}$ and add up to at least 10, which is impossible. For $m = 6$ suppose that no two terms add up to 6. Then at most one term is 3, and neither 1 and 5 nor 2 and 4 both occur. Terms in $\{1, 2, 3\}$ with at most one 3 add up to at most 11, and terms in $\{3, 4, 5\}$ to

at least 19. Terms in $\{1,3,4\}$, respectively $\{2,3,5\}$, cannot add up to 12, since the number $y$ of terms 4, respectively 5, would satisfy $3y \in \{7,5\}$, respectively $3y \in \{2,1\}$. This proves the first assertion.

The other three coordinates add up to 0 and have $b$-sum $m$, so any two of them, $x$ and $y$, have $x+y=-z\neq 0$ and $b$-sum less than $m$; merging them keeps $p'=1$, so $q'=1$, and $a'$ contains $c=a_i$ and $a_j=-c$. The order of $a'$ is that of the subgroup generated by $z$ and $c$. If $c$ generates $\mathbb{Z}/m$, any choice of $z$ will do. Otherwise $c$ is even, or $m=6$ and $c=3$. As $a$ has order $m$, one of $x,y,z$ is then odd, respectively prime to 3, and we take it as $z$; then $z$ and $c$ generate $\mathbb{Z}/m$. ∎

**Proposition 20.64** (Monodromy of cyclic covers)**.** *In Setup 20.58, if $p=0$ or $q=0$ then $\Gamma_a$ is finite, and if $n\geq 3$ and $p,q\geq 1$ then $G_a^0 \supseteq \mathrm{SL}(V_a)$.*

*Proof.* If $pq=0$, the polarisation is definite on $V_a$ up to sign, and $\Gamma_a$ preserves it and the $\mathbb{Z}[\zeta]$-lattice spanned by $H^1(D_a,\mathbb{Z})$; a discrete subgroup of a compact group is finite. Assume $p,q\geq 1$ and use the model of Lemma 20.59. Three facts are used throughout. If $u\in\Gamma_a$ is unipotent, the Zariski closure of its powers is the group $\exp(\tau\log u)$, so $\log u$ lies in the Lie algebra of $G_a^0$. The group $\Gamma_a$ normalises $G_a^0$. And $G_a^0$ is semisimple: by André's theorem [And92, Theorem 1] the identity component of the Zariski closure of the monodromy group of $H^1(D_a,\mathbb{Q})$ is a normal subgroup of the derived group of the Mumford–Tate group of a very general fibre, hence semisimple, and $G_a^0$ is its image in $\mathrm{GL}(V_a)$, the monodromy commuting with the deck group.

*Four points.* Let $k=4$, so $n=2$, and suppose that two coordinates of $a$ are opposite; relabel them $a_1$ and $a_2$, so that $t_1t_2=1$. The full twist $A_{12}$ maps $x_i\mapsto wx_iw^{-1}$ for $i=1,2$, with $w=x_1x_2$, and fixes $x_3,x_4$. As $t(w)=1$, the formula for $\partial(wx_iw^{-1})/\partial x_j$ in the proof of Lemma 20.60 gives $vJ_{A_{12}}=v+\varphi(v)\,\delta$, with $\delta=e_1+t_1e_2$ and $\varphi(v)=(1-t_1)v_1+(1-t_2)v_2$, which equals $\sum_{j\geq 3}(t_j-1)v_j$ on $\ker\partial$. Here $\delta\in\ker\partial$, $\varphi(\delta)=0$, $\varphi(\ell)=t_1\cdots t_k-t_1t_2=0$, the vector $\delta$ is not a multiple of $\ell$, whose third coordinate is $t_1t_2=1$, and $\varphi((t_3-1)e_1-(t_1-1)e_3)=-(t_1-1)(t_3-1)\neq 0$. So $A_{12}$ acts on $E_a$ as a unipotent $u\neq 1$, and $G_a^0$ is a connected semisimple subgroup of $\mathrm{SL}(E_a)\cong\mathrm{SL}_2$ containing $\exp(\tau\log u)$. The connected subgroups of $\mathrm{SL}_2$ of dimension one or two are solvable, so $G_a^0=\mathrm{SL}(E_a)$.

*The induction.* Let $n\geq 3$. If $n=3$, Lemma 20.63 gives, after a relabelling, a vector $a'=(a_1+a_2,a_3,a_4,a_5)$ of order $m$ with $a_1+a_2\neq 0$, $p'=q'=1$ and two opposite coordinates, so the identity component of the Zariski closure of the monodromy of $E_{a'}$ is $\mathrm{SL}(E_{a'})$ by the case of four points. If $n\geq 4$, Lemma 20.62 and a relabelling give $a_1+a_2\neq 0$ and $a'=(a_1+a_2,a_3,\ldots,a_k)$ of order $m$ with $p',q'\geq 1$, and by induction on $n$ that identity component contains $\mathrm{SL}(E_{a'})$. Let $H$ be the identity component of the Zariski closure of the doubled braids of Lemma 20.60. Then $H\subseteq G_a^0$, $H$ preserves $Y$ and $KI$ and acts trivially on $KI$, where the doubled braids act through a finite group, and its image in $\mathrm{GL}(Y)$ is the identity component of the Zariski closure of the monodromy of $E_{a'}$, which contains $\mathrm{SL}(Y)$. So $G_a^0$ contains the group $S$ of Lemma 20.61 for $E_a=Y\oplus KI$. Let $\gamma=A_{23}$, the full twist of the second and third points; its automorphism fixes $x_1$ and maps $x_2\mapsto x_2x_3x_2x_3^{-1}x_2^{-1}$ and $x_3\mapsto x_2x_3x_2^{-1}$. Then $\gamma(I)=I+(1-t_1)t_2I_{23}$ with $I_{23}=(1-t_3)e_2-(1-t_2)e_3$, and $\gamma(KI)\neq KI$, since $I$, $I_{23}$ and $\ell$ are independent, $\ell$ having a nonzero fourth coordinate. The functional $v\mapsto v_2-t_1v_1$ vanishes on $\ell$ and cuts out $Y$, and at $\gamma(v)$ for $v\in Y$ it equals $(t_3-1)(t_1t_2v_1-v_3)$, which is nonzero for the cycle $(1-t_4)e_3-(1-t_3)e_4$ of $Y$; so $\gamma Y\neq Y$. As $\gamma$ normalises $G_a^0$, this group contains $S$ and $\gamma S\gamma^{-1}$, and Lemma 20.61,

with $\dim Y = n-1$, gives $G_a^0 = \mathrm{SL}(E_a)$: for $n \geq 4$ directly, and for $n = 3$ because $G_a^0$ is connected and semisimple. ∎

For $m = 3$ the proposition also follows from the $\ell$-adic theorem of Achter and Pries [AP07, Corollary 3.10], which identifies the monodromy of trielliptic curves with a special unitary group.

**Lemma 20.65** (The discriminant of the new part)**.** *In Setup 20.58 let $n$ be even with $p = q = n/2$, let $L = H^1(D_a, \mathbb{Z})$ with the cup product $Q$, and let $L_a = L \cap (V_a \oplus \overline{V_a})$. Then $L_a$ is a free $\mathbb{Z}[\zeta]$-module of rank $n$, the $K$-hermitian form $H$ with $Q = \frac{1}{2}\operatorname{Tr}_{K/\mathbb{Q}}(\delta^{-1}H)$ on it, $\delta = \sqrt{-1}$ for $m = 4$ and $\delta = \sqrt{-3}$ otherwise (Lemma 4.5), has signature $(n/2, n/2)$, and the class of $(-1)^{n/2}\det H$ in $\mathbb{Q}^\times/\mathrm{N}(K^\times)$ is trivial for $m = 3$ and $m = 4$; for $m = 6$ it is trivial if and only if the number of $i$ with $a_i = 3$ is even.*

*Proof.* The deck transformation $g$ is an isometry of the unimodular lattice $L$, so its eigenspaces for eigenvalues $\mu, \nu$ are orthogonal unless $\mu\nu = 1$. Hence $L_a$, the part for the primitive $m$-th roots of unity, and the part $L^{\mathrm{old}}$ for the other eigenvalues are orthogonal primitive sublattices, each the orthogonal complement of the other, and $|\det(L_a)| = |\det(L^{\mathrm{old}})|$, which is 1 for $m = 3$, when $L^{\mathrm{old}} = 0$.[47] For a quotient map $\pi\colon D_a \to D'$ of prime degree $e$ that is ramified, $\pi^* H^1(D', \mathbb{Z})$ is primitive in $L$ and the cup product on it is $e$ times that of $D'$.[48] For $m = 4$, $L^{\mathrm{old}} = \pi^* H^1(D', \mathbb{Z})$ for the quotient $D'$ by $g^2$, ramified over the $\lambda_i$ with $a_i$ odd, so $|\det L^{\mathrm{old}}| = 2^{2g'}$, $g'$ the genus of $D'$. For $m = 6$, the quotients $D_2$ by $g^2$ and $D_3$ by $g^3$ are ramified over the $\lambda_i$ with $3 \nmid a_i$ and with $a_i$ odd, which exist since $a$ has order six, and $L^{\mathrm{old}}$ contains $A \oplus B$ with $A = \pi_2^* H^1(D_2, \mathbb{Z})$ and $B = \pi_3^* H^1(D_3, \mathbb{Z})$, of determinants $3^{2g_2}$ and $2^{2g_3}$; the quotient $L^{\mathrm{old}}/(A \oplus B)$ embeds in $A^\vee/A$ and in $B^\vee/B$, of coprime orders, so $L^{\mathrm{old}} = A \oplus B$ and $|\det L^{\mathrm{old}}| = 3^{2g_2}2^{2g_3}$.

On $L_a$ the deck transformation satisfies the cyclotomic equation of $\zeta$, so $L_a$ is a torsion-free module over the principal ideal domain $\mathbb{Z}[\zeta]$, free of rank $n$; the signature of $H$ is $(p, q)$ (Remark 4.6). The determinant of the trace form of a hermitian lattice gives $|\det(L_a)| = |d_K|^n \mathrm{N}(\frac{1}{2\delta})^n (\det H)^2$, that is $(\det H)^2$ for $K = \mathbb{Q}(i)$ and $(\det H)^2/4^n$ for $K = \mathbb{Q}(\sqrt{-3})$.[49] Hence $(-1)^{n/2}\det H$ equals $2^n$ for $m = 3$, $2^{g'}$ for $m = 4$ and $2^{n+g_3}3^{g_2}$ for $m = 6$, the sign being that of a form of signature $(n/2, n/2)$. For $m = 3$ this is $\mathrm{N}(2^{n/2})$, and for $m = 4$ a norm as $2 = \mathrm{N}(1+i)$. For $m = 6$, $3 = \mathrm{N}(\sqrt{-3})$ and $4 = \mathrm{N}(2)$ are norms and 2 is not, since $x^2 + xy + y^2$ is even only when $x$ and $y$ are, so that every norm has even 2-adic valuation; so the class is trivial exactly when $g_3$ is even. By the Riemann–Hurwitz formula $D_3$, the triple cover $y^3 = \prod_i (t - \lambda_i)^{a_i}$ branched over the $i$ with $3 \nmid a_i$, has genus $k - c_3 - 2 = n - c_3$, where $c_3$ is the number of $i$ with $a_i = 3$; as $n$ is even, this is the assertion. ∎

[47] If $M \subset L$ is primitive and $L$ is unimodular, restriction $L \to M^\vee$ is onto with kernel $M^\perp$, so $L/(M \oplus M^\perp) \cong M^\vee/M$, and likewise $\cong (M^\perp)^\vee/M^\perp$; the order of $M^\vee/M$ is $|\det M|$.

[48] A loop in $D'$ avoiding the branch points, composed with a loop around a branch point, whose monodromy generates the group of order $e$, lifts to a closed loop with the same image in $H_1(D', \mathbb{Z})$; so $\pi_*\colon H_1(D_a, \mathbb{Z}) \to H_1(D', \mathbb{Z})$ is onto, and its transpose $\pi^*$ is a split injection. The factor $e$ is $\pi_*\pi^* = e$.

[49] For a free $\mathcal{O}_K$-module with hermitian Gram matrix $M$ and $c \in K^\times$, the $\mathbb{Z}$-bilinear form $\operatorname{Tr}_{K/\mathbb{Q}}(c\, v^T M \bar{w})$ has determinant of absolute value $|d_K|^n \mathrm{N}(c)^n (\det M)^2$, where $d_K$ is the discriminant of $K$: for $n = 1$ this is the discriminant of the trace form of $\mathcal{O}_K$ scaled by $\mathrm{N}(cM)$. Changing the basis by $P \in \mathrm{GL}_n(K)$ multiplies both sides by $\mathrm{N}_{K/\mathbb{Q}}(\det P)^2$, and a basis in which $M$ is diagonal makes the form an orthogonal sum of forms of rank one. Here $d_K = -4$ and $\mathrm{N}(\frac{1}{2i}) = \frac{1}{4}$, or $d_K = -3$ and $\mathrm{N}(\frac{1}{2\sqrt{-3}}) = \frac{1}{12}$.

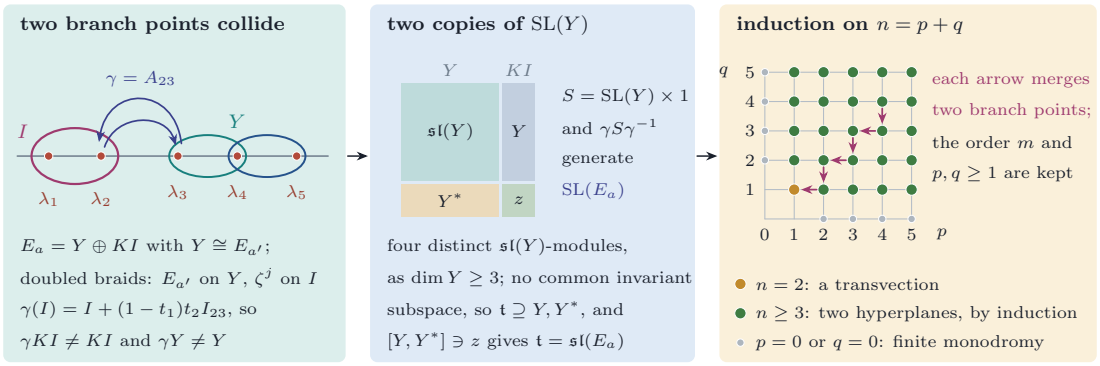



FIGURE 46. The proof of Proposition 20.64. Left: two branch points $\lambda_1, \lambda_2$ collide inside a disc (Lemma 20.60); the cycles outside the disc form the hyperplane $Y \cong E_{a'}$, the cycle $I$ of the disc spans the complementary line, and the twist $\gamma$ of the second and third points moves both. Middle: the two copies of $\mathrm{SL}(Y)$ of Lemma 20.61, which generate $\mathrm{SL}(E_a)$ when $\dim Y \geq 3$. Right: the induction on $n = k - 2$, each merge lowering $p$ or $q$ by one (Lemma 20.62, Lemma 20.63), down to $n = 2$, where the full twist of two points with opposite exponents is a transvection.

**Theorem 20.66** (Very general diagonal complete intersections)**.** *Keep the notation of Theorem 20.55, let $r \geq 2$ be even, and let $\lambda \in U$ lie outside a countable union of proper Zariski closed subsets of $U$. For a Hodge class $x$ of degree $r$ on $X = X_{d,r}(\lambda)$, write $x_{[a]}$ for the component of $\Phi^* x$ in $W_{[a]}$.*

(i) *If $a$ has order two, then $x_{[a]}$ is a rational multiple of the image of $\theta_a^{r/2}$, where $\theta_a$ is the class of a polarisation of $B_{[a]}$, under the correspondence of the proof of Theorem 20.55(iv); in particular $x_{[a]}$ is algebraic.*

(ii) *If $a$ has order $m \in \{3, 4, 6\}$ and $x_{[a]} \neq 0$, then $\dim V_a = r$ and $\dim V_a^{1,0} = r/2$. In that case $B_{[a]}$ is an abelian variety of dimension $r$ of Weil type for $\mathbb{Q}(\zeta_m)$, and $W_{[a]}$ is the image of its space of Weil classes. Its discriminant is trivial for $m = 3, 4$, and for $m = 6$ exactly when an even number of the coordinates of $a$ equal $d/2$.*

(iii) *The Hodge conjecture holds on $X_{d,r}(\lambda)$ for every $N$ when $d = 2$, when $d \in \{3, 4\}$ and $r \leq 6$, and when $d = 6$ and $r \leq 4$. For $d = 6$ and $r = 6$ the Hodge classes are algebraic modulo the spaces $W_{[a]}$ of the characters of order six of (ii) with an odd number of coordinates 3, which are spaces of Weil classes of abelian sixfolds of non-split Weil type for $\mathbb{Q}(\sqrt{-3})$ and occur for every $N$. For $d \in \{2, 3, 4, 6\}$ the Hodge classes of degree $r$ form a space of dimension*

$$1 + \binom{N+1}{r+2} T_d(r+2) + \epsilon_d \sum_{j \geq r/2+2} \binom{N+1}{2j},$$

*where $T_d(k)$ is the number of $(s_1, \ldots, s_k) \in \{1, \ldots, d-1\}^k$ with $\sum_i s_i = dk/2$, and $\epsilon_d$ is 1 for $d$ even and 0 for $d$ odd; thus $T_2(k) = 1$, $T_3(k) = \binom{k}{k/2}$, $T_4(6) = 141$ and $T_6(6) = 1751$.*

For odd $r$ there is nothing to prove, since every Hodge class of $X$ is then a multiple of a power of the hyperplane class. So the theorem gives the Hodge conjecture for the

very general complete intersection of Vandermonde type of diagonal quadrics in every dimension, of diagonal cubics or quartics up to dimension seven, and of diagonal sextics up to dimension five, in every $\mathbb{P}^N$.

*Proof.* Apply Lemma 20.57 to $H^r(X_{d,r}(\lambda),\mathbb{Z})$ over $U$, polarised by the hyperplane class, and let $\Pi'\subset\pi_1(U,\lambda)$ be the subgroup of finite index that fixes the Hodge classes. The monodromy commutes with $H$ and with $\Phi^*$, so $\Pi'$ fixes $x_{[a]}$ and each of its parts in the spaces $\bigwedge^r V_{a'}$. Let $m$ be the order of $a$, take representatives $0\le a_i<d$, and put $a_i'=ma_i/d$. The function $y=\prod_i u_i^{a_i}$ is invariant under the kernel of $\chi_a$ and satisfies $y^m=\prod_i(t-\lambda_i)^{a_i'}$, so the quotient of $C$ by that kernel is the cyclic cover $D_a$ of $\mathbb{P}^1$ of degree $m$ given by this equation, and pull-back identifies $H^1(D_a,\mathbb{Q})$ with the part $\bigoplus_{j\neq0}V_{ja}$ of $H^1(C,\mathbb{Q})$, compatibly with the monodromy. The image $B_{[a]}$ is isogenous to the abelian subvariety of $J(D_a)$ whose first cohomology is $\bigoplus_{j\in(\mathbb{Z}/m)^\times}V_{ja}$, all of $J(D_a)$ for $m\le3$, its polarisation being a rational multiple of the restriction of the principal one, since pull-back along a map of degree $e$ multiplies the cup product by $e$.

(i) For $m=2$, $D_a$ is the hyperelliptic curve $y^2=\prod_{i\in S}(t-\lambda_i)$ over the support $S$ of $a$, of genus $g=|S|/2-1$, and $H^1(D_a,\mathbb{Q})=V_a$. Move the base point so that the $\lambda_i$ with $i\in S$ are real and consecutive, $\lambda_{s_1}<\cdots<\lambda_{s_{2g+2}}$, the others lying away from the segment they span. The loop in which $\lambda_{s_k}$ and $\lambda_{s_{k+1}}$ turn once around each other, inside a disc that contains no other $\lambda_i$, makes $D_a$ acquire a node with vanishing cycle $c_k$, the lift of the segment $[\lambda_{s_k},\lambda_{s_{k+1}}]$, and acts by the square of the Picard–Lefschetz transvection, $T_k^2(z)=z+2\langle z,c_k\rangle c_k$ up to orientation [Voi03, Chapter 3].[50] The classes $c_1,\dots,c_{2g}$ satisfy $\langle c_k,c_{k+1}\rangle=\pm1$ and $\langle c_k,c_l\rangle=0$ for $|k-l|\ge2$, so their intersection matrix has determinant 1 and they form a basis of $H_1(D_a,\mathbb{Q})$; they are the distinguished basis of type $A$ of [ACa79]. Some power $T_k^{2e}$ with $e\ge1$ lies in the image of $\Pi'$; it is unipotent, so the Zariski closure of that image contains the one-parameter group $\exp(\tau N_k)$, $N_k(z)=\langle z,c_k\rangle c_k$, and its Lie algebra contains $N_1,\dots,N_{2g}$. These generate the Lie algebra $\mathfrak{sp}(V_a)$.[51] So the Zariski closure contains $\mathrm{Sp}(V_a)$, whose invariants in $\bigwedge^r V_a$ are the multiples of $\omega_a^{r/2}$, $\omega_a$ the polarisation form [FH91, Theorem 17.5].[52] The correspondence of the proof of Theorem 20.55(iv) maps $H^r(B_{[a]},\mathbb{Q})=\bigwedge^r H^1(B_{[a]},\mathbb{Q})$ isomorphically and equivariantly onto $\bigwedge^r V_a$; the class $\theta_a$ goes to an invariant nondegenerate form, a nonzero multiple of $\omega_a$, and $\theta_a^{r/2}$ to a nonzero multiple of $\omega_a^{r/2}$, which is algebraic.

(ii) For $m\in\{3,4,6\}$, $D_a$ is the cyclic cover of Setup 20.58 for the exponents $a_i'\in\{1,\dots,m-1\}$ on the support $S$ of $a$, and $V_a$ is its eigenspace, of dimension $n=|S|-2$ (Lemma 20.59). The map from $U$ to $\mathrm{Conf}_{|S|}$ that forgets the $\lambda_i$ outside $S$ is a fibration with connected fibres, so it is onto on fundamental groups, and $\Pi'$ maps onto a subgroup of finite index of $P_{|S|}$; the Zariski closure of its image in $\mathrm{GL}(V_a)$ therefore contains $G_a^0$. If $n\le2$, then $\bigwedge^r V_a$ is zero unless $r=n=2$. If $V_a^{1,0}$ or $V_a^{0,1}$ is zero, then $\bigwedge^r V_a$ has type $(r,0)$

[50] Near the collision the curve is $y^2=(t-\lambda_{s_k})(t-\lambda_{s_{k+1}})\,v$ with $v$ a unit, analytically $y^2=t^2-\epsilon$ with $\epsilon$ a multiple of $(\lambda_{s_{k+1}}-\lambda_{s_k})^2$; one turn of the two points around each other is two turns of $\epsilon$ around 0, and one turn of $\epsilon$ acts by the Dehn twist along the vanishing cycle, whose action on $H_1$ is the transvection $T_k$.

[51] Identify $\mathfrak{sp}(V)$ with $\mathrm{Sym}^2V$, the product $uv$ acting by $z\mapsto\langle z,u\rangle v+\langle z,v\rangle u$, so that $N_k=\frac12c_k^2$ and $[uv,wz]=\langle u,w\rangle vz+\langle u,z\rangle vw+\langle v,w\rangle uz+\langle v,z\rangle uw$ up to an overall sign. Then $[c_k^2,c_{k+1}^2]=\pm4c_kc_{k+1}$, and if all products $c_ic_j$ with $i,j\le k$ lie in the generated algebra, so do the $c_ic_{k+1}=\pm\frac12[c_ic_k,c_{k+1}^2]$ for $i<k$, because $c_{k+1}$ meets only $c_k$ among $c_1,\dots,c_k$. By induction every $c_ic_j$ lies in it, and these span $\mathrm{Sym}^2V$.

[52] The exterior algebra of $V$ is a representation of $\mathrm{Sp}(V)$ and of the $\mathfrak{sl}_2$ spanned by $\omega\wedge$ and its adjoint, and $\bigwedge^r V=\bigoplus_j\omega^j\wedge P^{r-2j}$, where $P^k$, the kernel of the contraction with $\omega$ in degree $k\le g$, is irreducible and nontrivial for $k\ge1$, so the invariants are the multiples of $\omega^{r/2}$, for $r$ even.

or $(0,r)$ and contains no class of type $(r/2,r/2)$. Otherwise $n\geq 3$, and Proposition 20.64 gives $G_a^0\supseteq \mathrm{SL}(V_a)$, whose invariants in $\bigwedge^r V_a$ vanish for $0<r<n$. In every case $x_{[a]}\neq 0$ forces $r=n$, and then $\bigwedge^r V_a$ is a line of type $(p_a,r-p_a)$ with $p_a=\dim V_a^{1,0}$, which contains a class of type $(r/2,r/2)$ only if $p_a=r/2$. The field $\mathbb{Q}(\zeta_m)$ then acts on $B_{[a]}$, of dimension $|[a]|\dim V_a/2=r$, with multiplicities $(r/2,r/2)$ on the tangent space and compatibly with the polarisation, the automorphism of $D_a$ preserving the cup product; so $B_{[a]}$ is of Weil type (Definition 3.1), and $W_{[a]}=\bigwedge^r V_a\oplus\bigwedge^r V_{-a}$ is the image of $\bigwedge^r_{\mathbb{Q}(\zeta_m)}H^1(B_{[a]},\mathbb{Q})$, its space of Weil classes. Its discriminant (Definition 15.22) is that of the lattice $L_a$ of Lemma 20.65: the isogeny with the abelian subvariety of $J(D_a)$ whose first cohomology is $L_a\otimes\mathbb{Q}$ scales the polarisation by a rational number and $\det H$ by its $r$-th power, a norm. The coordinates of $a$ equal to $d/2$ are those with $a_i'=3$ when $m=6$, and Lemma 20.65 gives the assertion.

(iii) Let $x$ be a Hodge class of degree $r$. The component of $\Phi^*x$ along $\eta$ is algebraic. For $d\in\{2,3,4,6\}$ every $a\neq 0$ has order 2, 3, 4 or 6, and by (i) and (ii) the components $x_{[a]}$ are images of powers of polarisations or of Weil classes of abelian $r$-folds of Weil type for $\mathbb{Q}(\sqrt{-3})$ or $\mathbb{Q}(i)$. These are algebraic for $r=2$ by the theorem of Lefschetz, for $r=4$ by Markman's theorem on abelian fourfolds of Weil type, and for $r=6$ by Markman's theorem on abelian sixfolds of split Weil type [Mar25a, Mar25b] when the discriminant is trivial: always for $d=3,4$, and for $d=6$ except along the characters of order six with an odd number of coordinates 3. So $\Phi^*x$ is algebraic in the cases stated, and so is $x=\Phi_*\Phi^*x/|G|$. For the dimension, the classes found are Hodge at every $\lambda$, and by (i) and (ii) there are no others. Each character of order two with $|S|\geq r+2$ gives one class, the image of $\theta_a^{r/2}$, nonzero since $r\leq\dim V_a$. A character of order at least three gives a class only if $|S|=r+2$ and $\dim V_a^{1,0}=\sum_{i\in S}(d-a_i)/d-1=r/2$, that is $\sum_i a_i=d(r+2)/2$ with representatives $0\leq a_i<d$, and then exactly one, in $\bigwedge^r V_a$. The characters with $|S|=r+2$ whose coordinates on $S$ average $d/2$ are counted by $T_d(r+2)$; for $d$ even they include the one of order two with all coordinates $d/2$, and the characters of order two with $|S|\geq r+4$ give the last sum. ∎

Figure 47 draws the steps of the proof. The dimensions of (iii) never exceed $h^{r/2,r/2}(X)$, and they equal it for a quadric, for the Fermat hypersurfaces (for cubics 7, 21 and 71 in dimensions 2, 4 and 6, for quartics 142 and 1108 in dimensions 4 and 6, for sextics 1752 in dimension 4) and for two quadrics in $\mathbb{P}^N$ with $N$ even $(N+2)$, where every class of type $(r/2,r/2)$ is algebraic. For two cubics in $\mathbb{P}^6$ the very general member has 141 Hodge classes of degree four against $h^{2,2}=267$, and for three quartics in $\mathbb{P}^7$ it has 3950 against $h^{2,2}=29872$ (Figure 48). Figure 49 shows, for each degree $d$ and dimension $r$, what is known for the very general member.

*Remark* 20.67 (What the diagonal complete intersections give). Theorem 20.55 adds to the domain of (F3′) a family of complete intersections of every dimension and every degree, parametrised by configurations of $N+1$ points on the line, among them all smooth complete intersections of two diagonal hypersurfaces; for $c\geq 3$ and $r\geq 2$ the condition on the columns is a real restriction. Corollary 20.56 shows on them how (F3′) and (F2) together give the conjecture: once the Hodge classes of $X$ are carried to the abelian varieties $B_{[a]}$, the conjecture for $X$ is the algebraicity of their Hodge classes, a question of type (F2), and for the fourfolds in $\mathbb{P}^6$ of degrees 3, 4 and 6 it is answered by Markman's theorem: their exceptional classes are Weil classes of abelian fourfolds of Weil type, 490 families of them for $d=4$. For $d=3$ these fourfolds are Fano and the conjecture also follows from the theorem of Conte and Murre on uniruled fourfolds [CM78]; for $d=4$ and $d=6$

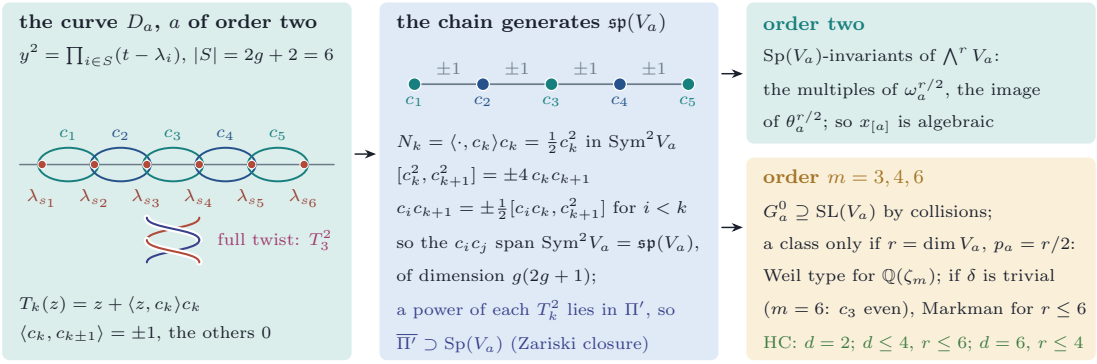



FIGURE 47. The proof of Theorem 20.66. Left: the hyperelliptic curve $D_a$ of a character of order two, drawn over the six points of its support for $g = 2$, with the chain of vanishing cycles $c_1, \dots, c_5$; a full twist of two neighbouring branch points, drawn as a braid, acts by the square of the transvection $T_k$. Middle: the chain as a graph, and the induction in $\mathrm{Sym}^2 V_a$ by which the logarithms $N_k$ generate $\mathfrak{sp}(V_a)$. Right: what the monodromy leaves of the Hodge classes, the powers of the polarisation for characters of order two, and for characters of order three, four and six the Weil classes of an abelian variety of Weil type for $\mathbb{Q}(\sqrt{-3})$ or $\mathbb{Q}(i)$.

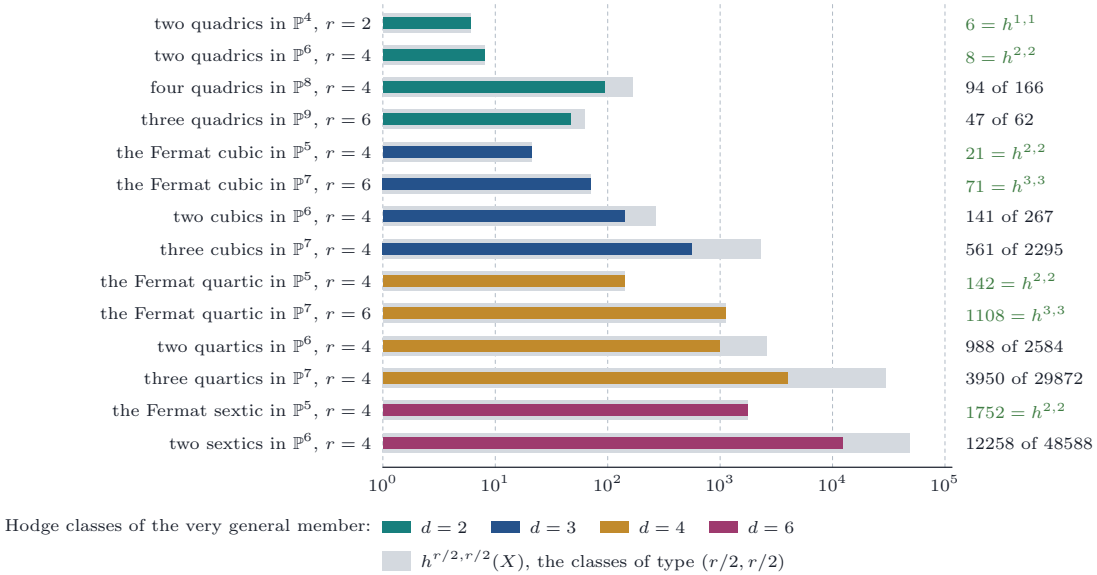



FIGURE 48. The dimension of the space of Hodge classes of degree $r$ on the very general member, from Theorem 20.66(iii), against the Hodge number $h^{r/2,r/2}$, on a logarithmic scale. Where the two agree, the space $H^{r/2,r/2}$ is spanned by Hodge classes at every configuration; elsewhere the very general member has fewer Hodge classes, and the members with more lie over a countable union of proper closed subsets of the space of configurations.

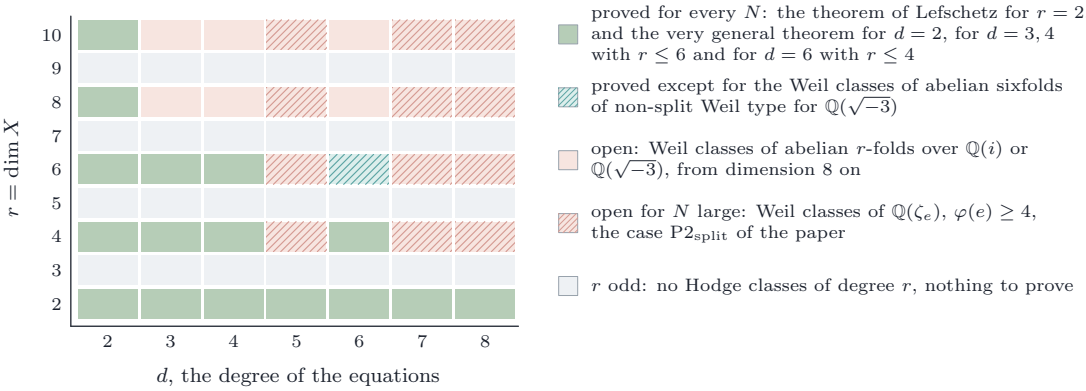



FIGURE 49. The Hodge conjecture for the very general complete intersection $X_{d,r}(\lambda)$ of Vandermonde type, for every $N$. Grass: proved, by the theorem of Lefschetz for $r = 2$ and by Theorem 20.66 for $d = 2$, for $d = 3, 4$ with $r \leq 6$ and for $d = 6$ with $r \leq 4$. Teal hatching: proved except for the Weil classes of abelian sixfolds of non-split Weil type for $\mathbb{Q}(\sqrt{-3})$. Clay: open, the Weil classes of abelian varieties of Weil type over $\mathbb{Q}(i)$ or $\mathbb{Q}(\sqrt{-3})$ in dimension at least eight, or, hatched, of a CM field of degree at least four (Remark 20.67). For odd $r$ there is nothing to prove.

they are of general type, with $h^{2,2} = 2584$ and 48588. Complete intersections of diagonal hypersurfaces, and of quadrics, are also the subject of Terasoma's paper [Ter88]. At a very general configuration Theorem 20.66 goes further, in every $\mathbb{P}^N$: for $d = 2$ in every dimension, for $d = 3$ and $d = 4$ up to dimension seven, and for $d = 6$ up to dimension five, through the monodromy of cyclic covers of degree three, four and six (Proposition 20.64). For $d = 3$ and $r \geq 8$ the conjecture for the very general member is the algebraicity of the Weil classes of the abelian $r$-folds $B_{[a]}$ of split Weil type for $\mathbb{Q}(\sqrt{-3})$, which is open from dimension eight on, and the same holds for $d = 4$ with $\mathbb{Q}(i)$. For $d = 6$ and $r = 6$ the open classes are those of the abelian sixfolds of non-split Weil type for $\mathbb{Q}(\sqrt{-3})$ of Theorem 20.66(iii), such as the one of the character $(1, 1, 2, 2, 3, 5, 5, 5)$ on eight points. Beyond these cases the same reduction meets (F2) itself: for $d = 5$ or $d \geq 7$ and $N$ large enough, some $B_{[a]}$ with $\dim V_a = r$ carries an action of $\mathbb{Q}(\zeta_e)$, $e \mid d$, with $\varphi(e) \geq 4$, and $W_{[a]}$ is the top exterior power of its first cohomology over that CM field, which consists of Hodge classes when the types are balanced: Weil classes for a CM field of degree at least four, of the kind treated in Theorem 1.1; and for larger $N$ the $B_{[a]}$ have dimension above five.

*Remark* 20.68 (Where (F3′) stops). With Proposition 20.35, Proposition 20.42 and Proposition 20.43, (F3′) holds on every variety of dimension at most three, on every fourfold whose zero-cycles are supported in dimension three (in particular every uniruled fourfold), on every fivefold whose zero-cycles are supported in dimension at most three, on every fourfold and fivefold whose maximal rationally connected quotient is birational to a variety satisfying (F3′) in degree four (Proposition 20.44), on every fourfold and fivefold dominated by a rational map from such a variety (Proposition 20.46), and on every member of $\mathcal{A}$, a class containing the Fermat hypersurfaces, the smooth Delsarte hypersurfaces with the cyclic covers branched along them (Corollary 20.53), and the complete intersections of

diagonal hypersurfaces of Vandermonde type, among them every smooth complete intersection of two diagonal hypersurfaces (Theorem 20.55, Corollary 20.56), of every degree and dimension, the Hilbert schemes of points on abelian surfaces, on products of curves and on the K3 surfaces of Proposition 20.42(iv), the generalised Kummer varieties $K_n(A)$ and the other varieties of Proposition 20.42(v), and every blow-up and projective bundle built from these. Through the Hodge conjecture itself it holds on the complete intersections of Corollary 20.56(ii), among them the fourfolds of general type cut out by two diagonal quartics or sextics in $\mathbb{P}^6$, on the very general complete intersections of diagonal quadrics, of diagonal cubics and quartics up to dimension seven and of diagonal sextics up to dimension five, of Theorem 20.66, and on the varieties of Proposition 20.39, Proposition 20.40 and Corollary 20.41: for a K3 surface $S$ with $E = \mathrm{End}_{\mathrm{Hdg}}(T(S))$ a CM field, on every power, Hilbert scheme and moduli space of $S$; for $E = \mathbb{Q}$ and $t = \dim T(S)$, on $S^k$ for $k < t$, on $S^{[n]}$ for $n < t(t+1)/2$ and on the moduli spaces of dimension less than $t$; on every fourfold of $\mathrm{K3}^{[2]}$ type whose field $E$ is $\mathbb{Q}$ or a CM field; on $F(Y)$ and $Y \times Y$ for every cubic fourfold $Y$ with such a field; and on every fourfold of generalised Kummer type [FV25]. Its frontier cases are these (Figure 50), the first two in degree four on fourfolds whose zero-cycles are not supported on a proper closed subset and which are not birational to a member of $\mathcal{A}$ by any construction above: $S \times S$ and $S^{[2]}$ for a K3 surface whose real multiplication is not induced by a cycle constructed so far, such as $S_\lambda$ (the known cases are recalled in Remark 21.5); the fourfolds of $\mathrm{K3}^{[2]}$ type with real multiplication, among them $F(Y)$ for a cubic fourfold $Y$ with real multiplication, and with it $Y \times Y$; and, for a very general K3 surface, the single class $\det T(S)$ on $S^{21}$ of Proposition 20.38, equivalently the Hodge conjecture for $S^{[231]}$. In the first two the rational Hodge isometries [Bus19, Mar24] act on $T$ by $\pm 1$, and a composite of Hodge similarities that returns to $T$ is an element $g \in E$ with $g^2 \in \mathbb{Q}$, the adjoint involution of $E$ being trivial; such elements span only the compositum of the quadratic subfields of $E$, which is $\mathbb{Q}$ for the cubic field of $S_\lambda$. Apart from proving the conjecture itself where zero-cycles are small, none of these results touches the step that Remark 20.36 identifies: putting a Hodge class into $\mathrm{Ab}^p(X)$ for a variety that no correspondence connects to an abelian variety. Theorem 20.70 is unchanged, since every statement proved here is a special case of one of the hypotheses of its rule set, not a new rule.

**Setup 20.69** (The rule set)**.** The rule set has forty-two statements and twenty-two rules. Ten statements are proved in this paper: the base point with an explicit cycle in every Weil family of an imaginary quadratic field (Theorem 15.27) and of a CM field of degree at least four (Theorem 19.5); for each of the two kinds of field, the reduction of the Weil classes of a family to one class at every point of one connected domain, and the stability of the algebraic locus under the rational points, with dense orbits (Section 12, Proposition 16.7, Theorem 19.7); and four statements on secant objects: the form of the weakened criterion for the transform (Theorem 18.56), the conditions of Markman's strategy that bear on the Chern character, the description of the local obstruction as a failure of Cohen–Macaulayness (Corollary 18.77), and three of the four ingredients of the construction in every dimension. Four are quoted: $\mathbf{W}(K, 2, \delta)$ for every $K$ and $\delta$, $\mathbf{W}(K, 3, \delta_0)$ [Mar25a, Mar25b], the deformation theorem for motivated classes [And96b, Théorème 0.5], and the accessibility of Hodge classes on abelian varieties [Mil20, §5.1, Theorem 3]. Sixteen are hypotheses: (F3), (F2) of Corollary 20.71 and (F3′) of Definition 20.34; (L), (M), (V) of Definition 20.14; (P2), in the polarised form of Remark 17.6, for four groups of families, the split and the other families of the imaginary quadratic fields and of the CM fields of degree at least

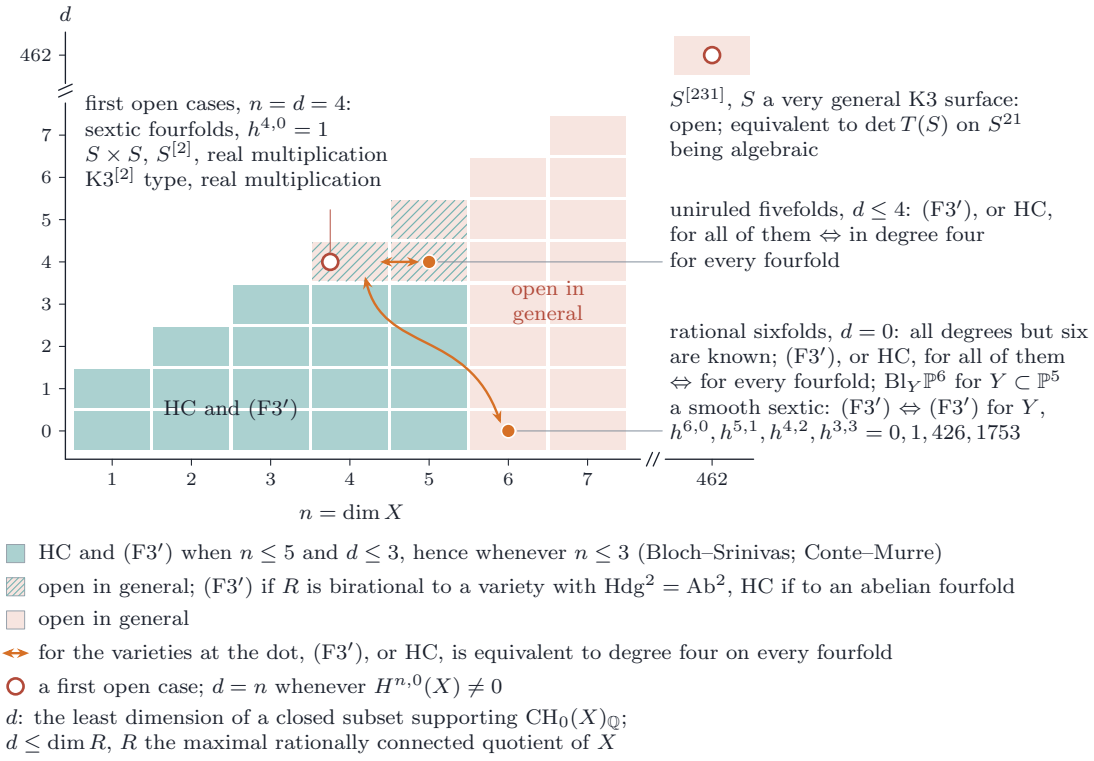



FIGURE 50. What arguments on zero-cycles and on rationally connected fibrations give for (F3′) and for the Hodge conjecture (HC in the figure) on a variety $X$, over the plane of $n = \dim X$ and the least dimension $d$ of a closed subset supporting $\mathrm{CH}_0(X)_{\mathbb{Q}}$; $d$ is at most the dimension of the maximal rationally connected quotient $R$ of $X$, and $d = n$ whenever $H^{n,0}(X) \neq 0$ (Remark 20.45). Both hold when $n \leq 5$ and $d \leq 3$ (Proposition 20.43), hence on every variety of dimension at most three. In the hatched cells (F3′) holds when $R$ is birational to a variety with $\mathrm{Hdg}^2 = \mathrm{Ab}^2$, for instance a member of $\mathcal{A}$, and the Hodge conjecture when $R$ is birational to an abelian fourfold (Proposition 20.44). The double arrows are the equivalences of Remark 20.45: (F3′), and the Hodge conjecture, hold for every uniruled fivefold, and for every rational sixfold, if and only if they hold in degree four for every fourfold; the sixfold $\mathrm{Bl}_Y\mathbb{P}^6$ of a smooth sextic fourfold $Y$ in a hyperplane carries the Hodge numbers listed. The rings are the frontier cases (Remark 20.68): the sextic fourfolds, the first family of smooth hypersurfaces beyond these results (the 29 Delsarte sextics lie in $\mathcal{A}$ by Corollary 20.53); $S \times S$ and $S^{[2]}$ for a K3 surface $S$ with real multiplication, such as $S_\lambda$; the fourfolds of $\mathrm{K3}^{[2]}$ type with real multiplication, among them $F(Y)$; and, beyond the breaks of the axes, $S^{[231]}$ for a very general K3 surface, which is equivalent to the algebraicity of $\det T(S)$ on $S^{21}$ (Proposition 20.39, Figure 42).

four, the third group giving $\mathrm{P2}_{\mathrm{split}}$; secant objects meeting the weakened criterion in every dimension, and an affirmative answer to [Mar25b, Question 11.4]; and, for a smooth and for a singular Cohen–Macaulay support on an abelian fourfold, its existence with the required extension property and the vanishing of the image of the product of translation classes. The other twelve are derived: **HC**, the conjecture for every abelian variety, written **AB**, and the Weil classes of every CM field, of the imaginary quadratic fields, of their families of trivial and of nontrivial discriminant, of the CM fields of degree at least four and of their split and their other families, together with one secant object on an abelian fourfold, $\mathbf{W}(K, 4, \delta_0)$, and the range settled in the literature. Each rule is a finite set of premises with one conclusion, and each is a theorem of this paper: Table 4 lists twenty-one of them, and the twenty-second derives the range settled in the literature from the two quoted Weil statements. For a set $T$ of statements write $\mathrm{Cn}(T)$ for the least set that contains $T$ and is closed under the rules, which exists because each rule has a single conclusion and the consequence operator is therefore monotone. Write $\mathbf{T}$ for the set of statements proved or quoted. Call a set $S$ of hypotheses *sufficient* if $\mathbf{HC} \in \mathrm{Cn}(\mathbf{T} \cup S)$, and *minimal* if no proper subset of $S$ is sufficient.

**Theorem 20.70** (The closure of what is proved here)**.** *With the notation of Setup 20.69:*

(i) $\mathbf{HC} \notin \mathrm{Cn}(\mathbf{T})$. *Nothing in this paper, combined with the theorems it quotes, implies the Hodge conjecture.*

(ii) *There are exactly five minimal sufficient sets:*

$$\{\mathrm{F3}\}, \quad \{\mathrm{L}, \mathrm{M}\}, \quad \{\mathrm{L}, \mathrm{F3}'\}, \quad \{\mathrm{V}, \mathrm{F3}'\}, \quad \{\mathrm{P2}_{\mathrm{split}}, \mathrm{F2}, \mathrm{F3}'\},$$

*with* (L), (M), (V) *as in Definition 20.14,* (F2), (F3) *as in Corollary 20.71, (F3′) as in Definition 20.34, and* $\mathrm{P2}_{\mathrm{split}}$ *the propagation statement* (P2)*, read in the polarised form of Remark 17.6, for the split families of the CM fields of degree at least four. No minimal set contains* (P2) *for an imaginary quadratic field or for a family of nontrivial discriminant: Proposition 20.13 derives those. The single hypothesis (F3) is sufficient, and it is equivalent to* **HC** *(Proposition 20.33); no other single hypothesis is. The sets* $\{\mathrm{L}, \mathrm{F3}\}$, $\{\mathrm{V}, \mathrm{F3}\}$ *and* $\{\mathrm{P2}_{\mathrm{split}}, \mathrm{F2}, \mathrm{F3}\}$ *are sufficient and not minimal. The last of the five is the only one with three elements and the only one whose derivation of* **HC** *uses a statement proved in this paper.*

(iii) *In each of the five, every element is necessary: removing any one leaves* **HC** *outside the closure.*

(iv) *The secant route lies in no minimal sufficient set. Granting both of its demands, a secant object satisfying the weakened criterion in every dimension and an affirmative answer to* [Mar25b, Question 11.4]*, puts* $\mathbf{W}(K, n, \delta_0)$ *in the closure, and with it, by Proposition 20.13,* $\mathbf{W}(K, n, \delta)$ *for every* $\delta$*; it leaves the Weil classes of the CM fields of degree at least four outside.*

(v) *No statement proved in this paper rests, at any depth, on a hypothesis, and the rule set is acyclic.*

(vi) *Every element of a minimal sufficient set other than* $\mathrm{P2}_{\mathrm{split}}$ *is a consequence of* **HC***, and* $\{\mathrm{F3}\}$ *and* $\{\mathrm{L}, \mathrm{M}\}$ *are each equivalent to it; no implication from* **HC** *to* (P2) *is known.*

*Proof.* (i) and (v). No rule has a proved or quoted statement as its conclusion, so each of them lies in $\mathrm{Cn}(\mathbf{T})$ by itself, on the strength of its proof in this paper or in the literature; this is the first assertion of (v). The twenty-second rule is the only one whose premises all lie in $\mathbf{T}$: every other rule has among its premises a hypothesis or a derived statement

| premises | conclusion | by |
|---|---|---|
| the conjecture for abelian varieties, (F3) | **HC** | Section 20.5 |
| (F3) | abelian varieties | Proposition 20.33 |
| the conjecture for abelian varieties, (F3′) | **HC** | Proposition 20.35 |
| the Weil classes of every CM field, (F2) | abelian varieties | Section 20.5 |
| $K$ imaginary quadratic, the fields of degree at least four | every CM field | Appendix C |
| $\delta = \delta_0$, $\delta \neq \delta_0$ | every discriminant | Proposition 20.12 |
| a base point, the reduction, the orbit, (P2) for the family | that family | Theorem 16.29 |
| the same three for a CM field of degree at least four, (P2) for the family | that family | Theorem 19.7 |
| its split families, its other families | the fields of degree at least four | Proposition 19.3 |
| $\delta = \delta_0$ | $\delta \neq \delta_0$ | Proposition 20.13 |
| the split families of the fields of degree at least four | their other families; $\delta = \delta_0$ | Proposition 20.13 |
| a secant object in every dimension, Question 11.4 | $\delta = \delta_0$ | Proposition 15.19 |
| a Cohen–Macaulay support, smooth or singular | one secant object | Corollary 18.77 |
| one secant object, Question 11.4 | $\mathbf{W}(K, 4, \delta_0)$ | Corollary 18.57 |
| (L), (M) | **HC** | Proposition 20.16 |
| (L), the deformation theorem for motivated classes | (V) | Proposition 20.17 |
| (V), the accessibility of Hodge classes on abelian varieties | abelian varieties | Theorem 20.18 |

TABLE 4. The rules of Setup 20.69 whose conclusion is derived, with the theorem that proves each; the rows for (P2) and for the Cohen–Macaulay support each stand for two rules, the row for descent over the fields of degree at least four for two, descent within a field and scalar extension down to the imaginary quadratic fields, the second row is Proposition 20.33, the third is Proposition 20.35(i) for the weaker statement (F3′) of Definition 20.34, the last three rows are the rules of Section 20.6, one of whose conclusions, (V), is itself a hypothesis of the rule set, and the one remaining rule records what the literature supplies on Weil classes. The bounded criterion (P1) is absent: by Theorem 16.17 it is equivalent to its own conclusion, so it is a restatement and not a premise.

other than the range settled in the literature. Hence $\mathrm{Cn}(\mathbf{T})$ is $\mathbf{T}$ together with that range, and it does not contain **HC**, which is (i). For acyclicity, list first the thirty statements that are proved, quoted or hypotheses other than (V); then one secant object, $\mathbf{W}(K, 4, \delta_0)$ and the range settled in the literature; the split and then the other families of the CM fields of degree at least four; the families of trivial and then of nontrivial discriminant of the imaginary quadratic fields; the imaginary quadratic fields, the CM fields of degree at least four and every CM field; and last (V), **AB** and **HC**. The premises of every rule come before its conclusion in this list.

(ii) and (iii). The five sets are sufficient: $\mathrm{F3} \Rightarrow$ **AB** (Proposition 20.33) and $\mathbf{AB} \wedge \mathrm{F3} \Rightarrow$ **HC**; $\mathrm{L} \wedge \mathrm{M} \Rightarrow$ **HC** (Proposition 20.16); (L) and the deformation theorem give (V) (Proposition 20.17), (V) and accessibility give **AB** (Theorem 20.18), and $\mathbf{AB} \wedge \mathrm{F3}' \Rightarrow$ **HC** (Proposition 20.35(i)), which settles $\{\mathrm{L}, \mathrm{F3}'\}$ and $\{\mathrm{V}, \mathrm{F3}'\}$; and $\mathrm{P2}_{\mathrm{split}}$ with the CM base point, the reduction and the orbit gives the split families of the CM fields of degree at least four (Theorem 19.7), descent gives their other families and the families of trivial discriminant of the imaginary quadratic fields (Proposition 20.13), hence the fields of degree at least four (Proposition 19.3), the nontrivial discriminant (Proposition 20.13), the imaginary quadratic fields (Proposition 20.12) and every CM field (Appendix C), and with (F2) this is **AB** and with (F3′) it is **HC**.

Let $C_1$, respectively $C_2$, be the set of all statements other than **HC**, (F3), (F3′) and (L), respectively (M). Let $C_3$ be the set of all statements other than **HC**, **AB**, the Weil classes of every CM field, of the fields of degree at least four and of their split families, (F3), (L), (V) and $\mathrm{P2}_{\mathrm{split}}$; and let $C_4$ be the set of all statements other than **HC**, **AB**, (F3), (L), (V) and (F2). Each $C_j$ contains $\mathbf{T}$ and not **HC**, and each is closed under the rules: every rule whose conclusion it omits has a premise it omits. For $C_1$ and $C_2$ the only omitted conclusion is **HC**, and the three rules with that conclusion have the premises (F3), (F3′), and (L) and (M). For $C_3$ and $C_4$ the rules with conclusion **HC** have the premises **AB** or (L); those with conclusion **AB** have the premises (F3), the Weil classes of every CM field together with (F2), and (V); the only rule with conclusion (V) has the premise (L); and for $C_3$ the rule for every CM field needs the fields of degree at least four, the rule for these needs their split families, and the rule for the split families needs $\mathrm{P2}_{\mathrm{split}}$. Since Cn is monotone, a set of hypotheses contained in some $C_j$ has its closure, together with $\mathbf{T}$, inside $C_j$, and is not sufficient. So a sufficient set meets each of

$$H_1 = \{\mathrm{F3}, \mathrm{F3}', \mathrm{L}\}, \qquad H_2 = \{\mathrm{F3}, \mathrm{F3}', \mathrm{M}\},$$
$$H_3 = \{\mathrm{F3}, \mathrm{L}, \mathrm{V}, \mathrm{P2}_{\mathrm{split}}\}, \qquad H_4 = \{\mathrm{F3}, \mathrm{L}, \mathrm{V}, \mathrm{F2}\},$$

the hypotheses omitted from $C_1, \ldots, C_4$. Let $S$ meet all four. If $\mathrm{F3} \in S$, then $S \supseteq \{\mathrm{F3}\}$. If $\mathrm{F3} \notin S$ and $\mathrm{L} \in S$, then $S$ meets $H_2$ in (F3′) or (M). If neither is in $S$, then $S$ contains (F3′), from $H_1$, and (V) or both $\mathrm{P2}_{\mathrm{split}}$ and (F2), from $H_3$ and $H_4$. So a set of hypotheses is sufficient exactly when it contains one of the five sets, and as none of the five contains another, they are the minimal sufficient sets. The other assertions of (ii) can be read off the list, and the derivations above show that only the last uses a statement proved here; (iii) is the minimality.

(iv) Granting the two demands of the secant route, the rule of Proposition 15.19 gives the families of trivial discriminant, descent those of nontrivial discriminant, and Proposition 20.12 the imaginary quadratic fields. The set formed by $\mathbf{T}$, the range settled in the literature, the two demands and these three statements is closed, since every other rule has a premise outside it: a hypothesis not granted, one secant object, or a Weil statement about the fields of degree at least four or about every CM field. Every base point the

secant route produces is a member $X \times \widehat{X}$, whose discriminant is trivial whatever $\beta$ is chosen (Proposition 15.19(ii)); the deformation argument stays inside the family of trivial discriminant, and it is Proposition 20.13(i), not the deformation, that carries the result to the other discriminants. The route has no base point over a field of degree at least four, so it reaches none of them, and none of its hypotheses lies in one of the five sets.

(vi) (L) follows from **HC** by Theorem 12.2, (M) by Proposition 20.16, (V) by Proposition 20.17, (F3′) by Proposition 20.35(i), and (F2) and (F3) are special cases of **HC**. The equivalence of {F3} with **HC** is Proposition 20.33, and {L, M} is equivalent to it by Proposition 20.16 and the first two implications. ■

**Corollary 20.71** (The three statements, one of which is the conjecture)**.** *The Hodge conjecture follows from the results of this paper together with the following three statements. No rule of Setup 20.69 derives it from (F1) and (F2) together, and it follows from (F3) alone, which is equivalent to it (Proposition 20.33); so the list is not a reduction of the conjecture to three smaller problems. With (F3) replaced by the weaker statement* (F3′) *of Definition 20.34, the conjecture modulo abelian varieties, the three statements (F1), (F2),* (F3′) *still suffice, and no rule of Setup 20.69 derives the conjecture from a proper subset of them. With the rule of Proposition 21.49, which the rule set does not contain, (F2) and* (F3′) *already suffice (Remark 21.50).*

(F1) One propagation statement, for every CM field. (P2), *in the polarised form of Remark 17.6: at one base point of each Weil family of each CM field some perfect complex whose Chern character is a nonzero multiple of a rational Weil class plus a polynomial in the polarisation has injective semiregularity map. Its first form, with Chern character exactly a multiple of the Weil class, is false for every imaginary quadratic field and every $n \geq 2$ (Theorem 17.5): such a complex would have to deform to non-algebraic Weil tori, whose coherent sheaves have no Chern character in degrees 1 to $2n-1$. The polarised form closes $\mathbf{W}(K, n, \delta)$ for every imaginary quadratic $K$, every $n$ and every $\delta$ at once, by Theorem 16.29, and $\mathbf{W}(F, n, \delta)$ for every CM field $F$ of higher degree, by Theorem 19.7, because those theorems put $\Sigma = \mathcal{D}$, so the Weil class is algebraic at every member of the family and not only at the general one, and because Theorem 15.27 and Theorem 19.5 supply the base point in every one of those families. By Proposition 20.13 it is enough to have it for the split families of the CM fields of degree at least four, in infinitely many dimensions. Theorem 16.31, Theorem 16.39, Theorem 20.32, Theorem 16.37 and Theorem 16.40 describe the object of the first form and are vacuous by Theorem 17.5; for the polarised form Theorem 17.9 computes the number that $\dim \mathrm{Ext}^2(E, E)$ has to reach, $7n^2 - 2n$ for a general shape. For one family its conclusion is equivalent to the Lefschetz standard conjecture for one variety (Corollary 20.31). The bounded criterion* (P1) *is not a second form of the same demand: by Theorem 16.17 it is equivalent to the conclusion it was to imply, and Theorem 16.15 shows that the one cycle it could have been tested on, the transport of the base cycle, does not satisfy it.*

(F2) The classes beyond the Weil lines. *Every Hodge class on an abelian variety that lies outside the subring generated by divisor classes and by Weil classes of quotients is algebraic. Through dimension five there is no such class* [MZ99]*; in dimension eight there are (Proposition 20.74), so this is not a reduction of the conjecture for abelian varieties to Weil classes, which is false as a statement about Hodge rings: it is the conjecture for those classes, and by Proposition 21.49 it is equivalent to the conjecture for every abelian variety. Its smallest case is brought down to one*

*Kuga–Satake class in Theorem 21.4; it holds at every CM point (Theorem 20.29), and on all members of a Mumford family it is equivalent to the Lefschetz standard conjecture for one ninefold (Corollary 20.30). Its classes are rigid along the base curve (Theorem 21.21), and one complex at one CM point with* $\mathrm{Ext}^2$ *of dimension* 119 *would close it (Theorem 21.27), as would a bound on the cycles that represent them at the closed points of the reductions of the curve modulo primes (Theorem 21.43).*

(F3) The varieties that are not abelian. *The Hodge conjecture holds for every smooth complex projective variety that is not an abelian variety. This is the Hodge conjecture itself. The varieties it covers include* $A \times \mathbb{P}^1$ *for every abelian variety* $A$*, and the conjecture for* $A \times \mathbb{P}^1$ *gives it for* $A$ *(Proposition 20.33); so (F3) implies the conjecture for abelian varieties, hence (F1) and (F2), and hence* **HC***. Nor is it a reduction to abelian varieties in the other direction: no realisation of the cohomology of an arbitrary variety inside that of abelian varieties exists (Proposition 20.75), and Theorem 10.3 and Theorem 10.8 close the two routes through hyperplane sections that would be the natural candidates for one.*

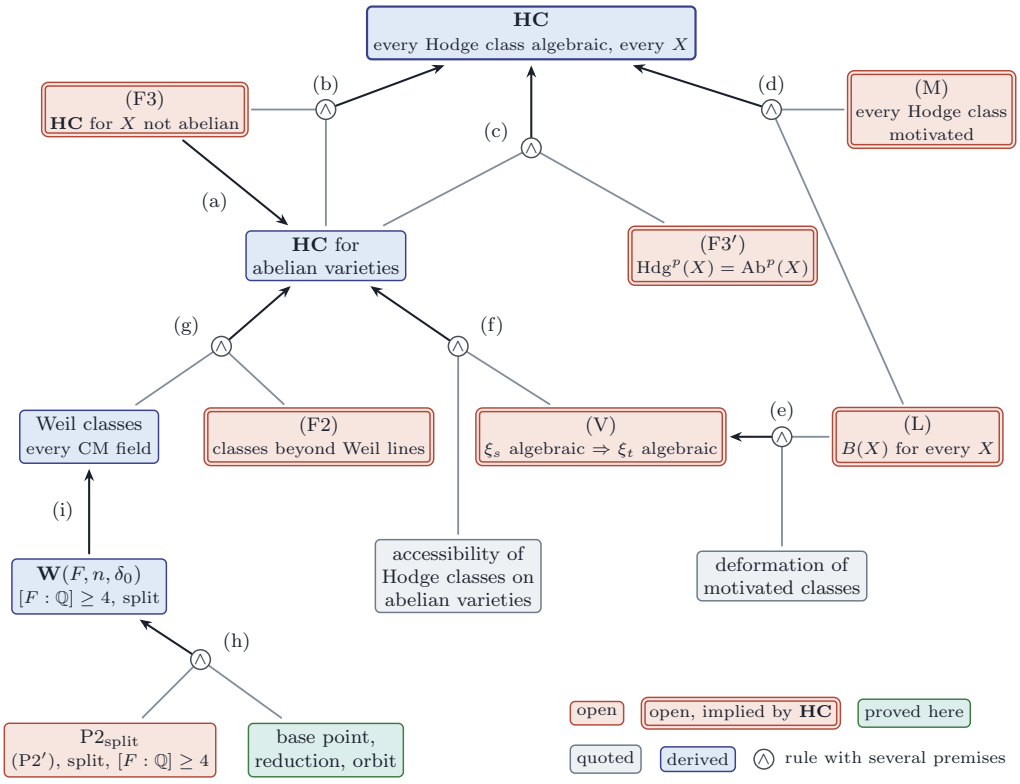
*Proof.* The three together are sufficient by the derivation of Theorem 20.70, and (F3) alone by Proposition 20.33. That (F1) and (F2) together are not sufficient in the rule set is Theorem 20.70(ii), since every minimal sufficient set contains (F3), (F3′) or (M); the statement about (F1), (F2), (F3′) is Theorem 20.70(ii) and (iii) for the last of the five minimal sets. The Weil classes of the CM fields of degree at least four are not a fourth statement: they are derived from (F1) by Theorem 19.7. Nor is the bounded criterion a premise, since by Theorem 16.17 it is a restatement of its own conclusion. ■

*Remark* 20.72 (Necessity relative to a rule set, and the wording of (F2)). A rule set can show that a set of statements suffices; it can show that a set is needed only relative to the implications written into it. A rule set that used (F3) only through (abelian varieties) ∧ (F3) ⇒ **HC**, without the elementary rule (F3) ⇒ (abelian varieties) of Proposition 20.33, would make the three statements of Corollary 20.71 look jointly necessary, although (F3) alone suffices.

The wording of (F2) has the same feature, one degree at a time. Let $R(A) \subseteq H^*(A, \mathbb{Q})$ be the subring generated by divisor classes and Weil classes of quotients. Suppose some Hodge class $\beta$ of degree $2p$ on $A$ lies outside $R(A)$, and let $\alpha$ be any Hodge class of degree $2p$. If $\alpha \notin R(A)$, (F2) makes it algebraic. If $\alpha \in R(A)$, then $\alpha + \beta \notin R(A)$, because $R(A)$ is closed under subtraction, so (F2) makes $\alpha + \beta$ and $\beta$ algebraic, and with them $\alpha$. So in every degree in which (F2) says anything it is the whole conjecture for that degree: on the square of a Mumford fourfold it gives every Hodge class of $H^4$, the six products of divisor classes included. Multiplying by an exceptional class of that square moves every Hodge class of every abelian variety outside the subring, and Proposition 21.49 deduces from this that (F2) is the whole conjecture for abelian varieties. The rule set does not contain that implication; Remark 21.50 says how Theorem 20.70(ii) changes when it is adjoined.

The form of (F3) that is not the conjecture itself asks only for algebraicity modulo classes coming from abelian varieties. It is (F3′) of Definition 20.34; Proposition 20.35 says what is known of it, and Remark 20.36 why it is not closed.

*Remark* 20.73 (The target, and what it would take). With the rules of Section 20.6, Proposition 20.33 and Proposition 20.35 in place, Theorem 20.70 changes the reading of Corollary 20.71 in three ways. First, the route through this paper is a route only with (F3) replaced by (F3′). As stated, its last statement (F3) is the conjecture (Proposition 20.33),




| minimal set | (F3) | (L) | (M) | (V) | (F3′) | (F2) | $\mathrm{P2}_{\mathrm{split}}$ | rules used |
|---|---|---|---|---|---|---|---|---|
| $S_1$ | ● | · | · | · | · | · | · | (a), (b) |
| $S_2$ | · | ● | ● | · | · | · | · | (d) |
| $S_3$ | · | ● | · | · | ● | · | · | (c), (e), (f) |
| $S_4$ | · | · | · | ● | ● | · | · | (c), (f) |
| $S_5$ | · | · | · | · | ● | ● | ● | (c), (g), (h), (i) |
| implied by **HC** | yes | yes | yes | yes | yes | yes | not known | |

FIGURE 51. The rule set of Setup 20.69 with the rules of Section 20.6, drawn as a hypergraph. A rule with several premises is a junction marked $\wedge$, joined to its premises and with one arrow to its conclusion. The letters name the results that prove the rules: (a) is Proposition 20.33; (b) and (g) are the closure map of Section 20.5; (c) is Proposition 20.35(i); (d) is Proposition 20.16; (e) is Proposition 20.17; (f) is Theorem 20.18; (h) is Theorem 19.7; and (i) is the chain of Proposition 20.13, Proposition 20.12, Proposition 19.3 and Appendix C. Clay marks a hypothesis, a double frame a hypothesis that the conjecture implies, grass what this paper proves, slate what it quotes, and indigo a derived statement. The table lists the five minimal sufficient sets of Theorem 20.70(ii), the rules each of them uses, and which of the hypotheses the conjecture implies; only $\mathrm{P2}_{\mathrm{split}}$, which asks for a particular complex, is drawn without that frame. The secant route, which reaches the imaginary quadratic families only, is drawn in Figure 52.

so (F1) and (F2), and everything this paper proves about Weil classes, drop out of every minimal derivation that uses (F3). With (F3′) in its place the route $\{\mathrm{P2}_{\mathrm{split}}, \mathrm{F2}, \mathrm{F3}'\}$ is minimal, and it is not the shortest: (F1) and (F2), which between them give the conjecture for abelian varieties along this route, are replaced on the routes of the literature by the single statement (V) (Theorem 20.18), or by (L), which implies it (Proposition 20.17). Second, every element of every minimal sufficient set except $\mathrm{P2}_{\mathrm{split}}$ is a consequence of the conjecture (Theorem 20.70(vi)), and two of the sets, $\{\mathrm{F3}\}$ and $\{\mathrm{L}, \mathrm{M}\}$, are each equivalent to it. (P2) is a statement about semiregularity maps, and a proof of the conjecture need not prove it; in its first form, with a Chern character that is exactly a Weil class, it is false (Theorem 17.5), and the route runs through the polarised form of Remark 17.6. Third, every minimal sufficient set contains (F3), (F3′) or (M): the varieties that are not abelian are reached only through the conjecture for them, which is the whole conjecture, through the conjecture modulo abelian varieties (Remark 20.36), or through motivated classes.

So the target at the centre is (L). It lies in two of the five minimal sets, it gives (V) and with it the whole abelian half, the conjecture implies it, and with (M) it is the conjecture. For each $X$ it asks for one explicit Hodge class on $X \times X$ to be algebraic, the class inducing $\Lambda$. This paper proves (L), (M), (V) and (F3′) on the classes of varieties named here: Corollary 20.22 adds the total spaces of Mumford families to the known cases of (L), Proposition 20.35 records the known cases of (F3′), and the routes of the literature use no other theorem of this paper.

*Two of the three are not reductions.* One could hope that (F2) and (F3) are reductions in the ordinary sense, statements to the effect that every Hodge class is carried, by some correspondence, to a class on a Weil type variety or on an abelian variety, so that the whole conjecture would follow from the Weil classes alone. Both hopes are refutable, and we refute them, because a frontier that contains a false statement is not a frontier.

**Proposition 20.74** (The Hodge ring is not generated by divisor and Weil classes)**.** *Let $X$ be a complex abelian fourfold whose Hodge group is a $\mathbb{Q}$-form of an almost direct product of three copies of $\mathrm{SL}_2$, acting on $V = H^1(X, \mathbb{Q})$ through the tensor product $V_1 \otimes V_2 \otimes V_3$ of the three standard representations, with $\mathrm{End}^0(X) = \mathbb{Q}$; such fourfolds exist and form families of dimension one, by Mumford* [Mum69]*, and by* [MZ99, 2.5(i)] *they are exactly the fourfolds with $\mathrm{End}^0(X) = \mathbb{Q}$ and Hodge group smaller than $\mathrm{Sp}_8$. Then:*

(i) *the Hodge ring of $X$ is generated by the polarisation:* $\dim_{\mathbb{Q}} \mathrm{Hdg}^1(X) = \dim_{\mathbb{Q}} \mathrm{Hdg}^2(X) = 1$*;*

(ii) $\dim_{\mathbb{Q}} \mathrm{Hdg}^1(X \times X) = 3$ *and* $\dim_{\mathbb{Q}} \mathrm{Hdg}^2(X \times X) = 8$*, while the products of divisor classes span a subspace of dimension* 6 *in $H^4(X \times X, \mathbb{Q})$;*

(iii) *the two remaining classes lie in $H^4(X \times X, \mathbb{Q})$ and are Weil classes of no CM field acting on $X \times X$, on a quotient of it, or on an abelian subvariety of it.*

*In particular the statement that every Hodge class on an abelian variety lies in the subring generated by divisor classes and Weil classes of quotients is false in dimension eight, though true through dimension five* [MZ99].

*Proof.* Hodge classes are the invariants of the Hodge group, so $\mathrm{Hdg}^k(Y) \otimes \mathbb{C}$ is the space of $\mathrm{Hg}(Y)_{\mathbb{C}}$-invariants in $\bigwedge^{2k} H^1(Y, \mathbb{C})$, and $\mathrm{Hg}(X \times X) = \mathrm{Hg}(X)$ acting diagonally on $V \oplus V$. Over $\mathbb{C}$ the group is $\mathrm{SL}_2^3$ and the weights of $V$ under its maximal torus are the eight sign vectors $(\pm 1, \pm 1, \pm 1)$. Write $V = V_1 \otimes W$ with $W = V_2 \otimes V_3$, on which $\mathrm{SL}_2^2 = \mathrm{Spin}(4)$ acts orthogonally. The Cauchy formula $\bigwedge^k (V_1 \otimes W) = \bigoplus_{|\lambda| = k} S_\lambda V_1 \otimes S_{\lambda'} W$ [FH91, §6.1], with $S^2 V_1 = S_{(3,1)} V_1 = V(2)$, $\bigwedge^2 V_1 = S_{(2,2)} V_1 = \mathbf{1}$ and $S^4 V_1 = V(4)$, gives

$\bigwedge^2 V = V(2)\otimes\bigwedge^2 W \oplus S^2W$ and $\bigwedge^4 V = V(4)\otimes\bigwedge^4 W \oplus V(2)\otimes S_{(2,1,1)}W \oplus S_{(2,2)}W$. Under $\mathrm{Spin}(4)$, $\bigwedge^2 W = V(2,0)\oplus V(0,2)$, $S^2W = V(2,2)\oplus \mathbf{1}$, $\bigwedge^4 W = \mathbf{1}$, $S_{(2,1,1)}W \cong \mathfrak{sl}(W) = \bigwedge^2 W \oplus S^2_0 W$, and $S_{(2,2)}W = V(4,0)\oplus V(0,4)\oplus V(2,2)\oplus \mathbf{1}$, the decomposition of the algebraic curvature tensors in dimension four into the self-dual and anti-self-dual Weyl parts, the traceless Ricci part and the scalar part. Hence

$$\bigwedge^2 V = V(2,2,0)\oplus V(2,0,2)\oplus V(0,2,2)\oplus \mathbf{1},$$
$$\bigwedge^4 V = V(2,2,2)\oplus \bigoplus\nolimits_{\mathrm{cyc}} V(4,0,0)\oplus \bigoplus\nolimits_{\mathrm{cyc}} V(2,2,0)\oplus \mathbf{1},$$

each with one trivial summand, which is (i). For (ii), $\bigwedge^4(V\oplus V) = \bigoplus_a \bigwedge^a V\otimes\bigwedge^{4-a}V$ and the invariants in $\bigwedge^a V\otimes\bigwedge^{4-a}V$ are $\mathrm{Hom}_{\mathrm{SL}^3_2}(\bigwedge^a V, \bigwedge^{4-a}V)$, since every summand is self-dual; the counts are $1,1,4,1,1$, total 8, the 4 coming from the four distinct irreducible summands of $\bigwedge^2 V$. Likewise the invariants in $\bigwedge^2(V\oplus V)$ number $1+1+1=3$. The divisor classes of $X\times X$ are the three symplectic forms $\psi_{11}$, $\psi_{12}$, $\psi_{22}$ on $V\oplus V$ built from the polarisation $\psi$ of $V$. Their six pairwise products are linearly independent: they lie in the summands $\bigwedge^a V\otimes\bigwedge^{4-a}V$ with $a = 4,3,2,2,1,0$, each is nonzero, and in the summand $a=2$ the classes $\psi_{11}\psi_{22}$ and $\psi_{12}^2$ correspond, as maps $\bigwedge^2 V\to\bigwedge^2 V$, to a multiple of the projection onto $\mathbb{C}\psi$ and to a multiple of the identity. The $\mathrm{Sp}_8$-invariants of $\bigwedge^4(V\oplus V)$ number $1+1+2+1+1=6$ by the same count, since $\bigwedge^1 V = V$, $\bigwedge^2 V = \bigwedge^2_0 V\oplus\mathbf{1}$ and $\bigwedge^3 V = \bigwedge^3_0 V\oplus V$ under $\mathrm{Sp}_8$; so the six products are all the classes a Hodge group equal to $\mathrm{Sp}_8$ would allow. For (iii), the Weil classes of an abelian variety $Y$ with respect to a CM field $F\subseteq \mathrm{End}^0(Y)$ are, by definition, the classes in $\bigwedge^{2k}_F H^1(Y,\mathbb{Q})\subseteq H^{2k}(Y,\mathbb{Q})$ with $2k = \dim_F H^1(Y,\mathbb{Q})$ (Setup 19.1). A CM field acting on $X\times X$ lies in $\mathrm{End}^0(X\times X) = M_2(\mathbb{Q})$, so it is imaginary quadratic, $2k=8$, and its Weil classes lie in $H^8$, while the two classes of (ii) have degree four. Every proper abelian subvariety and every proper quotient of $X\times X$ is isogenous to $X$, because $X$ is simple, and $\mathrm{End}^0(X)=\mathbb{Q}$ contains no CM field, so $X$ carries no Weil classes at all. This is the statement of [MZ99, 0.1(iii)] that the exceptional classes of $X\times X$ are not of Weil type. The last sentence is (ii) and (iii) together with the theorem of [MZ99] for dimension at most five. ∎

**Proposition 20.75** (No Hodge-theoretic reduction to abelian varieties)**.** *Let $Y$ be a very general smooth quintic threefold in $\mathbb{P}^4$, and let $V = H^3(Y,\mathbb{Q})$ with its polarised Hodge structure of weight three. Then $V$ is not isomorphic to a subquotient of any Hodge structure of the form $\bigoplus_i T_i$, where each $T_i$ is obtained from the $H^1$ of finitely many abelian varieties by tensor products, duals and Tate twists. In particular the Hodge structure $H^*(Y,\mathbb{Q})$ is not of abelian type, and no correspondence between $Y$ and an abelian variety, or a product of abelian varieties, induces an injection of $V$ into the cohomology of the latter.*

*Proof.* Write $\mathrm{MT}(W)$ for the Mumford–Tate group of a rational Hodge structure $W$ and $\mu_W\colon \mathbb{G}_m\to \mathrm{MT}(W)_{\mathbb{C}}$ for its Hodge cocharacter, acting on $W^{p,q}$ by $z^p$. The adjoint action of $\mu_W$ on $\mathrm{Lie}\,\mathrm{MT}(W)_{\mathbb{C}}\subseteq\mathrm{End}(W_{\mathbb{C}})$ has weights among the differences $p-p'$ of two Hodge indices of $W$.

If $W$ has weight one, these differences lie in $\{-1,0,1\}$. If $W'$ is a subquotient of a tensor construction $T$ on $W$, then $\mathrm{MT}(W')$ is the image of $\mathrm{MT}(W)$ acting on $W'$, and $\mu_{W'}$ is the image of $\mu_W$; so $\mathrm{Lie}\,\mathrm{MT}(W')$ is a quotient of $\mathrm{Lie}\,\mathrm{MT}(W)$ as a representation of $\mathbb{G}_m$ through $\mu$, and its weights are a subset of $\{-1,0,1\}$. The same holds with $W=\bigoplus H^1(A_i)$. Hence for every Hodge structure of abelian type the adjoint weights of the Hodge cocharacter lie in $\{-1,0,1\}$.

For $V = H^3(Y, \mathbb{Q})$ we show that the adjoint weight 3 occurs. The canonical bundle of a quintic threefold is trivial by adjunction, so $h^{3,0}(Y) = 1$; pick $0 \neq e \in H^{3,0}$. The intersection form $\psi$ on $V$ is symplectic and pairs $H^{3,0}$ with $H^{0,3}$ perfectly, so the rank one endomorphism $x \mapsto \psi(x, e)\, e$ of $V_{\mathbb{C}}$ is nonzero, sends $H^{0,3}$ to $H^{3,0}$ and kills the rest, and lies in $\mathfrak{sp}(V_{\mathbb{C}}, \psi)$ because $\psi(\psi(x, e)e, y) + \psi(x, \psi(y, e)e) = \psi(x, e)\big(\psi(e, y) + \psi(y, e)\big) = 0$; its adjoint weight is $3 - 0 = 3$. It remains to see that $\mathfrak{sp}(V_{\mathbb{C}}, \psi) \subseteq \operatorname{Lie} \mathrm{MT}(V)_{\mathbb{C}}$. Let $S$ be the parameter space of smooth quintic threefolds, connected and nonsingular, and $\mathbb{V}$ the polarisable variation of Hodge structure $R^3$ of the universal family. The connected algebraic monodromy group $G^0$ of $\mathbb{V}$ contains the connected monodromy group of any Lefschetz pencil of quintics, that is of hyperplane sections of the fifth Veronese image of $\mathbb{P}^4$; for a Lefschetz pencil of odd fibre dimension the image of the monodromy representation on the $\ell$-adic vanishing cohomology $E \otimes \mathbb{Q}_\ell$ is open in $\mathrm{Sp}(E \otimes \mathbb{Q}_\ell, \psi)$ [Del74, Theorem 5.10], so the Zariski closure of the monodromy group over $\mathbb{Q}$ is $\mathrm{Sp}(E, \psi)$; here $E = H^3(Y_t, \mathbb{Q})$, the vanishing cohomology being the orthogonal complement of the image of $H^3(\mathbb{P}^4, \mathbb{Q}) = 0$. Hence $G^0 \supseteq \mathrm{Sp}(V, \psi)$, and $G^0 \subseteq \mathrm{Sp}(V, \psi)$ since the monodromy preserves $\psi$. At a very general point $Y$ the group $G^0$ is a normal subgroup of the derived group of $\mathrm{MT}(V)$ [And92, Theorem 1], so $\mathrm{Sp}(V, \psi) \subseteq \mathrm{MT}(V)^{\mathrm{der}} \subseteq \mathrm{Sp}(V, \psi)$, and the adjoint weight 3 occurs in $\operatorname{Lie} \mathrm{MT}(V)$. So $V$ is not of abelian type, and neither is any Hodge structure containing it as a subquotient, in particular $H^*(Y, \mathbb{Q})$. A correspondence inducing an injection of $V$ into the cohomology of an abelian variety would exhibit $V$ as a sub-Hodge structure of a Tate twist of that cohomology, which is of abelian type. ■

20.10. **Routes from outside algebraic geometry.** By Proposition 21.49 and Proposition 20.35(i), the two statements (F2) and (F3′) are the two halves of the conjecture: the conjecture for abelian varieties, and the conjecture modulo abelian varieties. A route from another domain must therefore either produce algebraic cycles or arrive at one of the hypotheses of Setup 20.69. This subsection records what the routes most often proposed do. One of them, the analytic route through positive currents, can be made exact: the conjecture is the vanishing of an integrality gap in a mass minimisation problem whose real version is solved, and positivity, which is what analysis sees first, cannot distinguish an algebraic class from any other.

Throughout, $X$ is a smooth complex projective variety of dimension $n$, $h \in H^2(X, \mathbb{Z})$ is the class of an ample line bundle, $\omega_h$ is a Kähler form with class $h$, and for $c \in H^{2p}(X, \mathbb{Q})$ we put

$$L_h(c) = \int_X c \cup \frac{h^{n-p}}{(n-p)!} \in \mathbb{Q}.$$

A real $(p, p)$-form is *strongly positive* if at every point it is a nonnegative combination of forms $\mathrm{i}\beta_1 \wedge \overline{\beta}_1 \wedge \cdots \wedge \mathrm{i}\beta_p \wedge \overline{\beta}_p$ with $\beta_j$ of type $(1, 0)$; the closed positive currents of bidegree $(p, p)$ are the currents nonnegative on such forms of the complementary degree, and every smooth closed strongly positive form is one [Dem12, Chapter III].

**Proposition 20.76** (Positivity is no test)**.** *For every class $\alpha \in H^{p,p}(X, \mathbb{R})$, $0 \leq p \leq n$, there is an integer $N$ such that $\alpha + Nh^p$ is the class of a smooth closed strongly positive $(p, p)$-form. Consequently every real $(p, p)$-class becomes the class of a closed positive current after the addition of a multiple of $h^p$, and, since $h^p$ is algebraic, a Hodge class is algebraic if and only if the positive class $\alpha + Nh^p$ is: no property of a class implied by positivity, or by its being a limit of classes of effective cycles, separates the algebraic Hodge classes from the others.*

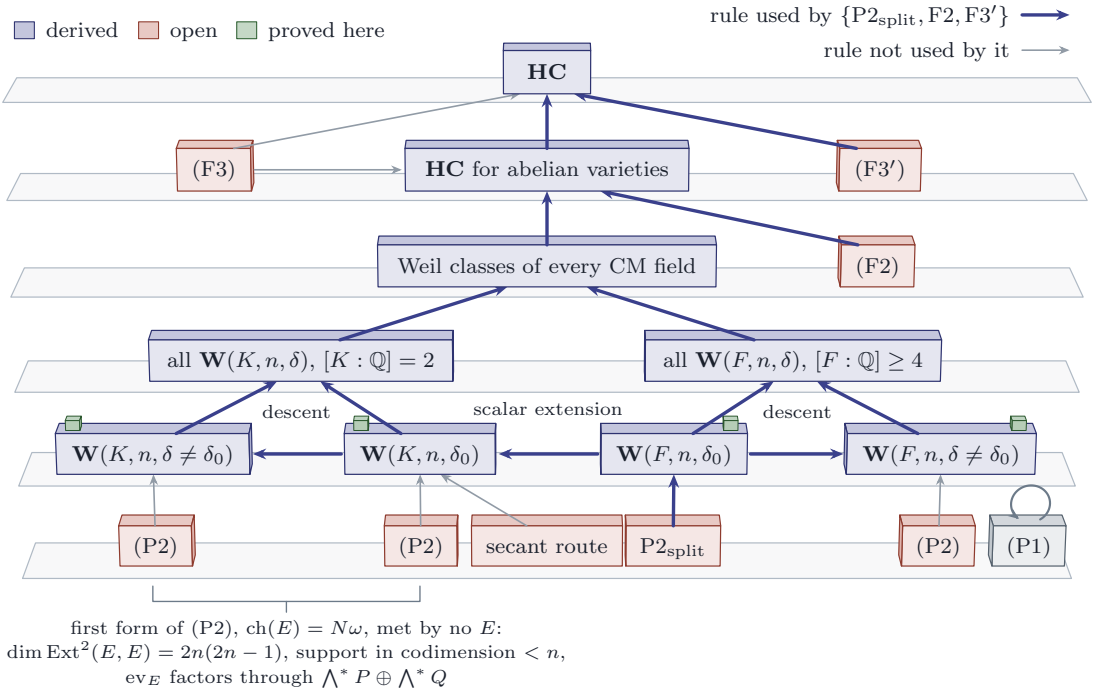



FIGURE 52. The route of this paper through the rule set of Theorem 20.70, drawn as a stack of plates, one per level of the derivation; the other routes are drawn in Figure 51. Heavy arrows are the rules used by the minimal set $\{\mathrm{P2}_{\mathrm{split}}, \mathrm{F2}, \mathrm{F3}'\}$, light arrows the rules it does not use. The bottom plate carries the propagation statement, read in the polarised form of Remark 17.6, once for each of the four kinds of Weil family, with the secant route beside it. The small grass blocks on the next plate are the base points of Theorem 15.27 and Theorem 19.5, one in every family. The three arrows inside that plate are Proposition 20.13: descent from the trivial discriminant to the others, and scalar extension from the fields of degree at least four to the quadratic fields. With them (P2) is needed only for the split families of the fields of higher degree, and the other three (P2) blocks are not used. (F2) is the conjecture for the Hodge classes that no Weil line reaches, and (F3) is the whole conjecture (Proposition 20.33), so the route is minimal only with (F3) replaced by (F3′) of Definition 20.34. The bracket under the two quadratic (P2) blocks records the first form of the criterion, in which the Chern character is exactly a Weil class, as Theorem 16.37, Theorem 16.39 and Corollary 16.35 describe it; by Theorem 17.5 that form is met by no complex. The secant route reaches the trivial discriminant, since the member it is built on is split, and through descent the rest of the quadratic case; it has no edge to the fields of higher degree. The bounded criterion (P1) stands apart, closed on itself: it is equivalent to its own conclusion.

*Proof.* Let $V = T_xX$ with the hermitian form $\omega_h(x)$, and let $\mathrm{SP}_p \subseteq \Lambda^{p,p}_{\mathbb{R}} V^\vee$ be the closed convex cone generated by the forms $\mathrm{i}^p \beta_1 \wedge \overline{\beta}_1 \wedge \cdots \wedge \beta_p \wedge \overline{\beta}_p$. These forms span $\Lambda^{p,p}_{\mathbb{R}} V^\vee$, so $\mathrm{SP}_p$ has nonempty interior, and $\omega_h(x)^p$ lies in that interior: in a unitary basis $dz_1, \ldots, dz_n$ of $V^\vee$,

$$\omega_h^p = p! \sum_{|J|=p} \Big(\frac{\mathrm{i}}{2}\Big)^p dz_J \wedge d\overline{z}_J,$$

and a generator $D = \mathrm{i}^p \beta_1 \wedge \overline{\beta}_1 \wedge \cdots \wedge \beta_p \wedge \overline{\beta}_p$ equals $|\det B|^2\, 2^p (\mathrm{i}/2)^p dz_1 \wedge d\overline{z}_1 \wedge \cdots \wedge dz_p \wedge d\overline{z}_p$ after a unitary change of coordinates taking the span of the $\beta_j$ to that of $dz_1, \ldots, dz_p$, with $B$ the matrix of the $\beta_j$ in that basis; so $\omega_h^p - \delta D \in \mathrm{SP}_p$ for $0 < \delta \le p!/(2^p |\det B|^2)$. As every real $(p,p)$-form is a finite real combination of generators, $\omega_h^p + \varepsilon\gamma \in \mathrm{SP}_p$ for every $\gamma$ and all small $\varepsilon$, which is interiority. The cone, the form $\omega_h(x)^p$ and the norm defined by $\omega_h(x)$ are carried into each other by unitary frames, so the radius $r > 0$ of a ball around $\omega_h(x)^p$ contained in $\mathrm{SP}_p$ does not depend on $x$. Represent $\alpha$ by a smooth closed real $(p,p)$-form $a$, for instance its harmonic representative, and let $N \ge \max_X |a|/r$. Then $a + N\omega_h^p = N(\omega_h^p + a/N)$ is strongly positive at every point, closed, and represents $\alpha + Nh^p$. The remaining assertions follow, the last because a class of the form $\alpha + Nh^p$ is positive for every $\alpha$. ∎

The proposition says what positivity cannot do. Demailly studied the cone of closed positive currents against the conjecture and constructed extremal closed strongly positive currents that are not integration currents on analytic sets [Dem82], so extremality does not single out the algebraic cycles either. What positivity can do is turn the conjecture into a variational statement, and the statement is exact. It is the Kähler case of a formulation that Harvey and Lawson give for every calibrated manifold [HL09, Section 6, Proposition 6.13 and Remarks 6.9 and 6.11]; we state it for the classes of this paper, with the effective representative that Lemma 20.77 supplies, and then apply it to the Mumford square. For a closed integral current $T$ of dimension $2(n-p)$ on $X$, Wirtinger's inequality [Fed69, 1.8.2], [HK74], [HL82, II.3] gives

$$\mathbf{M}(T) \;\ge\; T\Big(\frac{\omega_h^{n-p}}{(n-p)!}\Big) \;=\; L_h(c), \tag{74}$$

where $\mathbf{M}$ is the mass and $c \in H^{2p}(X, \mathbb{Z})$ is the Poincaré dual of the homology class of $T$, with equality if and only if the approximate tangent plane of $T$ is a complex subspace, positively oriented, at $\|T\|$-almost every point. The right side depends on $c$ alone.

**Lemma 20.77** (Complete intersection classes are big)**.** *For every algebraic class $\beta \in H^{2p}(X, \mathbb{Q})$ there is an integer $M$ such that $\beta + Mh^p$ is a positive rational combination of classes of irreducible subvarieties of codimension $p$; so some positive integer multiple of $\beta + Mh^p$ is the class of an effective algebraic cycle.*

*Proof.* Replacing $h$ by a multiple, which changes $M$, we may assume $h$ very ample. Let $Z \subseteq X$ be irreducible of codimension $p$, and let $d$ be such that the ideal sheaf $I_Z(d)$ is globally generated. The base locus of the linear system of divisors of degree $d$ containing $Z$ is exactly $Z$, so by Bertini $p$ general members $D_1, \ldots, D_p$ of it meet properly away from $Z$. Every component of $D_1 \cap \cdots \cap D_p$ has codimension at most $p$; a component not contained in $Z$ has codimension exactly $p$, because the $D_i$ meet properly off $Z$, and a component contained in $Z$ has codimension at least $p$ and so equals $Z$. The intersection is therefore the union of $Z$ and a closed set $W'$ of pure codimension $p$. Hence $d^p h^p = [D_1] \cdots [D_p] = m[Z] + [W']$ with $m \ge 1$ and $W'$ effective [Ful98, Chapter 7], and

$h^p - \varepsilon[Z]$ is an effective rational class for $0 < \varepsilon \le m/d^p$. Write $\beta = \sum_i n_i[Z_i]$ with $n_i \in \mathbb{Q}$ and choose $\varepsilon_i$ for each $Z_i$ with $n_i < 0$; then for $M \ge \sum_{n_i<0} |n_i|/\varepsilon_i$ the class $\beta + Mh^p$ is $\sum_{n_i>0} n_i[Z_i] + \sum_{n_i<0}(|n_i|/\varepsilon_i)(h^p - \varepsilon_i[Z_i]) + (M - \sum_{n_i<0}|n_i|/\varepsilon_i)h^p$, a positive rational combination of effective classes, $h^p$ itself being the class of a complete intersection. ∎

**Theorem 20.78** (The conjecture as a mass gap)**.** *Let $\alpha \in \mathrm{Hdg}^p(X)$ be a rational Hodge class.*

(i) *Let $N \ge 1$ be an integer with $N\alpha$ integral, let $M$ be an integer for which $\alpha + Mh^p$ is the class of a smooth closed strongly positive form, as every large $M$ is by Proposition 20.76, and put $c = N(\alpha + Mh^p)$. The minimum of the mass over closed real normal currents of dimension $2(n-p)$ in the class dual to $c$ is $L_h(c)$, and it is attained by a smooth strongly positive form. The minimum over closed integral currents in the same class exists and is at least $L_h(c)$.*

(ii) *The class $\alpha$ is algebraic if and only if there are integers $N \ge 1$ and $M \ge 0$, with $N\alpha$ integral, for which the minimum of the mass over closed integral currents in the class dual to $N(\alpha + Mh^p)$ equals $L_h(N(\alpha + Mh^p))$; and then every mass minimising integral current in that class is the integration current of an effective algebraic cycle.*

(iii) *Write $\|c\|_{\mathbb{Z}}$ for the minimum in* (i) *over integral currents. By a theorem of Federer, recorded in this form by Harvey and Lawson* [HL09, Section 6], [Fed69, Fed74], *$\lim_{k\to\infty} \|kc\|_{\mathbb{Z}}/k = L_h(c)$: the integrality gap closes asymptotically for every class. The class $\alpha$ is algebraic if and only if, for every sufficiently large $M$, the gap is closed at some finite $k$, that is, $\|kc\|_{\mathbb{Z}} = kL_h(c)$ for some $k \ge 1$.*

*In particular the conjecture for $X$ is the statement that for every rational Hodge class the integrality gap of this problem vanishes for some $N$ and $M$;* (F2) *is that statement for abelian varieties. Teh and Yang give a different sufficient condition in the same setting, through Lipschitz chains that are closed for $d$ and $d^c$ with positive $(k,k)$-part* [TY21, Theorem 4.2].

*Proof.* (i) The form $\omega_h^{n-p}/(n-p)!$ has comass one and calibrates exactly the complex $2(n-p)$-planes, which is Wirtinger's inequality [Fed69, 1.8.2]; so every closed real normal current $S$ in the class has $\mathbf{M}(S) \ge S(\omega_h^{n-p}/(n-p)!) = L_h(c)$. By Proposition 20.76 the class is represented by a smooth closed strongly positive form $a$, and for a strongly positive $(p,p)$-form the mass of the current it defines is $\int_X a \wedge \omega_h^{n-p}/(n-p)!$, because a strongly positive $2(n-p)$-vector is a nonnegative combination of complex unit planes, each of mass one and calibrated. So the minimum over real currents is $L_h(c)$ and $a$ attains it. Closed normal currents represent every real homology class and closed integral currents every integral one [FF60, 5.11], and by the compactness theorem of Federer and Fleming the mass attains its minimum on the closed integral currents in a given class [FF60, 9.6]; the lower bound is (74).

(ii) If $\alpha$ is algebraic, Lemma 20.77 gives $M$ and $N$ with $N(\alpha + Mh^p) = \sum_j n_j[V_j]$, $n_j \in \mathbb{Z}_{>0}$, $V_j$ irreducible of codimension $p$. The integration current $T = \sum_j n_j[V_j]$ is a closed locally integral current [Kin71, Theorem 3.1.1], its tangent planes are complex and positively oriented, so equality holds in (74) and $\mathbf{M}(T) = L_h(c)$ with $c = N(\alpha + Mh^p)$. Conversely let $T$ be a closed integral current in the class dual to $c$ with $\mathbf{M}(T) = L_h(c)$, for instance a mass minimiser when the minimum is $L_h(c)$. Equality in (74) makes $T$ a closed locally rectifiable current of bidimension $(n-p, n-p)$ which is positive, and by King's theorem [Kin71, Theorem 5.2.1] such a current is a holomorphic cycle $\sum_j n_j[V_j]$ with $n_j$ nonzero integers and $V_j$ irreducible analytic subvarieties of dimension $n-p$; positivity makes every $n_j$ positive. (Harvey and Shiffman characterised the holomorphic chains

among closed rectifiable currents of bidimension $(n-p, n-p)$ without positivity, under a hypothesis on the support that Alexander removed [HS74, Ale97].) By Chow's theorem [GH78, p. 167] the $V_j$ are algebraic, so $c$ is the class of an effective algebraic cycle and $\alpha$ is algebraic. This is the argument of [HL09, Proposition 6.13].

(iii) The limit is Federer's theorem on the real and integral masses of a class, as stated in the final note of [HL09, Section 6]; with (i) the limit is $L_h(c)$, since the real minimum is attained by a smooth form. If the gap is closed at $k$, then (ii), applied with $kN$ in place of $N$, makes $\alpha$ algebraic. Conversely, if $\alpha$ is algebraic, Lemma 20.77 gives $M_0$ such that $\alpha + M_0 h^p$ is a positive rational combination of classes of subvarieties; for $M \geq M_0$ so is $\alpha + M h^p$, since $h^p$ is the class of a complete intersection, and some multiple $kc$ is the class of an effective cycle, whose integration current attains the bound. For a fixed small $M$ the equivalence is not asserted, since it would need an effective multiple of every algebraic class represented by a strongly positive form, in codimension at least two. The last sentence of the theorem is the definition of the conjecture for $X$, and Proposition 21.49 for (F2). ■

**Example 20.79** (The Mumford square)**.** Let $Y = X \times X$ be the square of a Mumford fourfold at a very general point of its curve, with the product principal polarisation $\theta_Y$, and let $\omega$ be one of the two exceptional classes of Proposition 20.74, normalised in $H^2(X) \otimes H^2(X)$ with $a_0 = 0$ in the notation of Setup 21.18, so that it corresponds to an endomorphism of $H^2(X)$ killing $\theta$ with image in the primitive part. Then $L_{\theta_Y}(\omega) = 0$: the pairing of $\omega$ with $\theta_Y^6 = \sum_k \binom{6}{k} \theta^k \otimes \theta^{6-k}$ sees only the term $20\,\theta^3 \otimes \theta^3$, and $\int_X x \cup \theta^3 = 0$ for every primitive $x \in H^2(X)$. Also $L_{\theta_Y}(\theta_Y^2) = 8!/6! = 56$, and $4\theta_Y^2$ is the class of the complete intersection $D \cap D'$ of two general members of $|2\theta_Y|$, a base point free system, whose integration current has mass 224. By Theorem 20.78 and Corollary 21.45, the Hodge conjecture for every member of the Mumford family, and with it $B(W \times_C W)$, is equivalent to the following: for some integers $N \geq 1$ and $M \geq 1$ with $N\omega$ integral, the class dual to $N(M\theta_Y^2 + \omega)$ contains an integral current of mass exactly $56NM$; and any such current is an effective algebraic cycle representing the class. For $M$ divisible by four the class $NM\theta_Y^2$ contains one, $NM/4$ copies of $D \cap D'$, of mass $56NM$; so the question is whether the perturbation by $N\omega$, which changes the Wirtinger bound by nothing, can be absorbed without mass. By Theorem 20.29 the answer is yes at every CM point of the curve, with integers $N$ and $M$ depending on the point; and if the same $N$ and $M$ served at infinitely many CM points the answer would be yes everywhere, by Corollary 21.28, because the components of an effective cycle of degree $56NM$ have degree at most $56NM$ and the Hilbert polynomials of subvarieties of bounded degree form a finite set.

Figure 53 draws the theorem and the example.

20.11. **The routes at a glance.**

Table 5: The routes considered, and the boundary of each result.

| **Route** | **Result** | **Beyond the result** |
|---|---|---|
| Ambient object, restricted to $A^\vee$, transported by $\Phi_{\mathcal{P}}$ | closed by Theorem 7.3 | ambient varieties with exceptional classes |

*continued on the next page*

*Table 5, continued.*

| Route | Result | Beyond the result |
|---|---|---|
| Any $K$-equivariant construction with the norm character | closed by Theorem 5.6 | constructions that are not eigenclasses |
| Secant dimension count | no obstruction, Corollary 8.4 | none |
| Semiregularity dimension count | no obstruction, Corollary 9.3 | none |
| Semiregularity of an object whose Chern character sits on the Weil line | closed by Theorem 9.7 and Corollary 9.18 for direct sums, and for every object by Theorem 17.5, for every CM field by Theorem 19.19 | objects of positive rank; the weaker hypothesis of [Mar25b, Question 11.4] |
| The polarised criterion at a quartic CM field, $n = 2$ | computed for every shape, Theorem 19.29: a complex meeting it has $\dim \mathrm{Ext}^2(E, E) \geq 88$, Corollary 19.30 | existence of such a complex |
| Induction through hyperplane sections | closed by Theorem 10.3, Theorem 10.8 | correspondences, degenerations, coniveau |
| Kuga–Satake, from a weight two structure of K3 type | not obstructed, Proposition 11.9 | available only at $n = 1$, Corollary 11.12 |
| Divisors, on a member with quaternionic multiplication | succeeds, Theorem 15.25 | fails at full Hodge group, Proposition 15.18 |
| Propagation off the base locus | reduced to one bound or one rank, Theorem 16.9, Theorem 16.29 | the bound or the rank, Remark 17.7 |
| Objects with effective Chern character at a base point | closed by Theorem 18.5 | objects indecomposable in $D^b$, Remark 18.8 |

*continued on the next page*

*Table 5, continued.*

| Route | Result | Beyond the result |
|---|---|---|
| The weakened criterion for a secant object of positive rank | reduced to two conditions on the factor, Theorem 18.56, the first automatic for Markman's construction, Corollary 18.57; the identity fails for Markman's candidate, Theorem 18.61, and at every transversal double point, Corollary 18.63 | objects whose support has no such point, Remark 18.64 |
| A secant object $\mathcal{I}_S(b\Theta)$ with smooth support | invariants forced and $d \leq b^2$, so $d \leq 9$ at $b = 3$, Theorem 18.67, Corollary 18.68 | existence for $d \in \{1, 3, 5, 7\}$ at $b = 3$, one vanishing in $H^1(S, N_{S/X})$ and first-order extension, Remark 18.69 |
| The local part of the weakened criterion | it is a condition on depth: no obstruction at a Cohen–Macaulay point, Theorem 18.75, Corollary 18.77 | Cohen–Macaulay supports, where the identity becomes one global vanishing beside first-order extension |
| A support that is a complete intersection | closed for every $b$ and every $d$, Theorem 18.81 | supports of higher Hilbert–Burch rank, Proposition 18.86 |
| A support with a Hilbert–Burch resolution by line bundles or by simple semi-homogeneous bundles | closed for every rank, twist and $d$, Theorem 18.89, Theorem 18.92 | resolutions over bundles that are not projectively flat |
| The secant route, granted every demand it makes | reaches $\delta = \delta_0$, and every $\delta$ by Proposition 20.13, Theorem 20.70(iv) | the CM fields of degree at least four, which give the quadratic ones by Proposition 20.13(ii) |
| Propagation from a base point | closes every imaginary quadratic family at once, Theorem 16.9, Theorem 16.29 | (P2) for the family, Corollary 20.71 |

*continued on the next page*

*Table 5, continued.*

| Route | Result | Beyond the result |
|---|---|---|
| The bounded form (P1), read on the transported cycle | closed: the transport spreads each component by $c^{2n}/m_i$ and the data is unbounded whenever a component has finite stabiliser, Lemma 16.14, Theorem 16.15 | nothing; the other reading is closed too |
| The bounded form (P1), read on the minimal representative | closed: equivalent to the conjecture for the family, Theorem 16.17 | nothing; it is a restatement, so it is not a route |
| The pure form of (P2), for a given representative | forces the evaluation map to vanish on the $4n^2$-dimensional annihilator, and excludes every representative that is a sheaf on a smooth germ of codimension two of its support, Theorem 16.31, Corollary 16.32 | nothing: no representative exists, Theorem 17.5 |
| The same form, tested by a number | $\dim \operatorname{Ext}^2(E,E) = 2n(2n-1)$ at one base point would suffice, Theorem 16.37, and no representative attains it, Theorem 17.5 | nothing |
| The same form, for a representative in codimension $n$ | closed at every member of every family: the support must have a component of codimension less than $n$, and at a general member every component of codimension at least two has multiplicity zero, Lemma 16.38, Theorem 16.39 | nothing: no representative exists, Theorem 17.5 |

*continued on the next page*

*Table 5, continued.*

| Route | Result | Beyond the result |
|---|---|---|
| The Hochschild action on such a representative | the annihilator is the mixed part of a splitting of $HH^1$ and the action factors through a ring of dimension $2^{2n+1}-1$ in one of dimension $2^{4n}$, Theorem 16.34, Corollary 16.35 | nothing: no such complex exists, Theorem 17.5 |
| The shape of such a representative | an indecomposable summand carries the class; at $n=2$, and at $n=3$ granting the compatibility of Corollary 16.35(iii), its endomorphism algebra has dimension at least three, so no simple object serves, Theorem 16.40 | nothing: no object of any shape meets the pure form, Theorem 17.5 |
| The pure form of (P2), Chern character exactly a Weil class | false for every imaginary quadratic field and every $n \geq 2$: a semiregular such complex would deform to very general non-algebraic Weil tori, whose coherent sheaves have no Chern character below the top degree, Proposition 17.4, Theorem 17.5 | the polarised form (P2′), whose Chern character may carry powers of the polarisation, Remark 17.6 |
| The reduction to Weil classes, and to abelian varieties | false as a statement about Hodge rings in dimension eight, Proposition 20.74; impossible Hodge-theoretically, Proposition 20.75 | the algebraicity of the classes beyond the Weil lines, which is (F2), and (F3), which is the whole conjecture, Proposition 20.33 |
| The two classes on the self-product of a Mumford fourfold | a real multiplication by a cubic field on the primitive part of $H^2$, algebraic once the Kuga–Satake class of one K3 surface of Picard number thirteen is, Proposition 21.2, Theorem 21.4 | that Kuga–Satake class, Remark 21.5 |

*continued on the next page*

*Table 5, continued.*

| Route | Result | Beyond the result |
|---|---|---|
| Other routes to the same two classes | degeneration and larger families closed off by rigidity; semiregularity reduced to $\dim \mathrm{Ext}^2 = 119$ at one CM point; specialisation reduced to a degree bound, Theorem 21.21, Theorem 21.27, Corollary 21.28 | an object with $\dim \mathrm{Ext}^2 = 119$, or the degree bound, Remark 21.32 |
| The cycles at a CM point, for the same two classes | the fourfold is of Weil type for $k_c \subset \mathrm{End}^0(X_c)$ with the two exotic classes as its Weil line; pull-backs of that line along homomorphisms with components in $k_c$ span, with divisor products, 110 of the 132 Hodge classes of the square and no exceptional class, and an exceptional class is reached exactly by components generating the endomorphism algebra over $k_c$, Proposition 21.29 | a degree bound for cycles using the real multiplication, at infinitely many CM points, Remark 21.31 |
| Objects built from line bundles, for the same two classes | closed at every point of the curve: the Chern characters stay invariant under the Lefschetz group, Theorem 21.36 | steps that break that symmetry, Remark 21.37 |
| Twistor deformation of the square, for the same two classes | closed: the component of the Hodge locus among all complex tori is the Mumford curve, which contains no twistor line, Proposition 21.24 | the Kuga–Satake class, which moves in a seven-dimensional family |
| Five further routes, for the same two classes | one statement: three of them are equivalent to the target and two imply it, so any one of them suffices, Corollary 21.45 | the target itself, which is only the first case of (F2), Remark 21.46 |

*continued on the next page*

*Table 5, continued.*

| Route | Result | Beyond the result |
|---|---|---|
| Pull-back from a hyperkähler manifold, for the same two classes | the square is holomorphic symplectic and the classes are the twisted dual forms of the part of $H^2$ its symplectic class generates; a symplectic rational map to some $S_\mu^{[n]}$, $n \geq 5$, suffices, and no K3 surface or hyperkähler eightfold serves, Proposition 21.25 | such a map, Remark 21.26 |
| Reduction modulo $p$, for the same two classes | represented by cycles at every closed point of every reduction; the target is equivalent to a bound on their Hilbert data on a Zariski dense set, Proposition 21.42, Theorem 21.43 | the bound, Remark 21.44 |
| The Weil classes of every CM field | the same propagation problem, with a CM base point in every family, Theorem 19.5, Theorem 19.7 | (P2) for the split families |
| Routes from outside algebraic geometry | positivity of currents sees nothing; the conjecture is the vanishing of the integrality gap of a mass minimisation whose real version is solved, Proposition 20.76, Theorem 20.78 | a calibrated integral current in one class, Example 20.79 |

*continued on the next page*

*Table 5, continued.*

| **Route** | **Result** | **Beyond the result** |
|---|---|---|
| The closure, as a whole | five minimal sets of hypotheses: $\{\mathrm{F3}\}$ and $\{\mathrm{L}, \mathrm{M}\}$, each equivalent to the conjecture, and three through the conjecture modulo abelian varieties, $\{\mathrm{L}, \mathrm{F3}'\}$, $\{\mathrm{V}, \mathrm{F3}'\}$ and this route, $\{\mathrm{P2}_{\mathrm{split}}, \mathrm{F2}, \mathrm{F3}'\}$, Theorem 20.70, Proposition 20.33, Corollary 20.71, Remark 20.73 | the Lefschetz standard conjecture, (M), (V), (F2), (F3′) |
| Convolutions of the Weil pieces on $E_0^{2n}$ | at $n = 3$ every convolution has $\dim \mathrm{Ext}^2 \geq 69 > 57$, Theorem 18.35; at $n = 4$, with the pieces at two shifts, $\dim \mathrm{Ext}^2 \geq 40960 > 104$, Theorem 18.37, at three consecutive shifts $\dim \mathrm{Ext}^2 \geq 22528$, Theorem 18.38, with the shifts within six consecutive values $\dim \mathrm{Ext}^2 \geq 640$, Corollary 18.48, and with each parity at one shift, at any odd distance, $\dim \mathrm{Ext}^2 \geq 1792$, Proposition 18.46 | the other arrangements of the shifts at $n = 4$, Remark 18.50 |

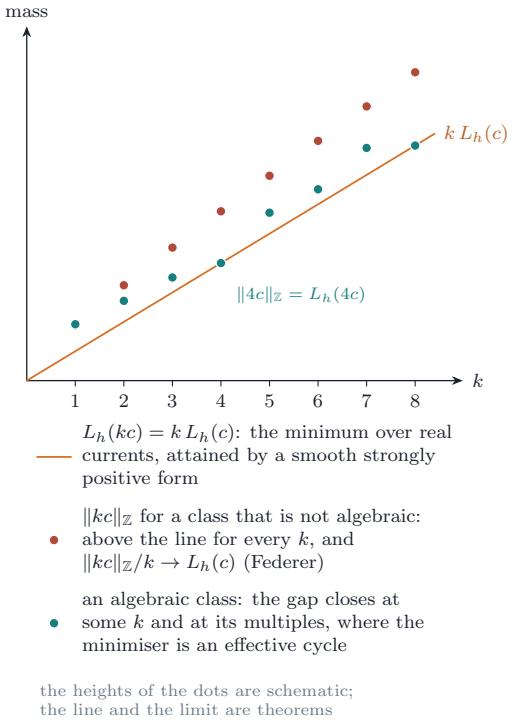
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*Table 5, continued.*

| Route | Result | Beyond the result |
|---|---|---|
| Diagonal complete intersections of Vandermonde type | quotients of powers of a generalised Fermat curve, so in $\mathcal{A}$; the conjecture holds when it holds on the abelian varieties $B_{[a]}$, and holds for two diagonal hypersurfaces of degree 3, 4, 6 in $\mathbb{P}^6$, Theorem 20.55, Corollary 20.56; it holds on the very general member for $d = 2$ in every dimension, for $d = 3, 4$ up to dimension seven and for $d = 6$ up to dimension five, Theorem 20.66 | $B_{[a]}$ of dimension above five, the Weil classes of $\mathbb{Q}(\zeta_e)$ with $\varphi(e) \geq 4$, and the very general member for $d = 3, 4, 6$ from dimension eight on and for $d = 6$ in dimension six, Remark 20.67 |
| The conjecture modulo abelian varieties, (F3′) | holds on the class $\mathcal{A}$ of varieties whose cohomology is reached from abelian varieties by algebraic correspondences (curves, abelian varieties, products, surjective images), in degrees 0, 2, $2n-2$, $2n$, and in dimension at most three, Proposition 20.35; on the Delsarte hypersurfaces and the diagonal complete intersections of Vandermonde type, Corollary 20.53, Theorem 20.55; on the fourfolds and fivefolds of Proposition 20.43, Proposition 20.44 and Proposition 20.46; through the Hodge conjecture on the varieties of Proposition 20.39 and Proposition 20.40 | degree four on fourfolds outside $\mathcal{A}$ not covered by these results, and beyond, for instance squares of K3 surfaces with real multiplication, such as $S_\lambda$, Remark 20.36, Remark 20.68 |

## 21. Hodge classes beyond the Weil lines

21.1. **The first classes beyond the Weil lines.** Statement (F2) is the Hodge conjecture for the Hodge classes on abelian varieties that no Weil line reaches, and Proposition 20.74




**The Mumford square** $Y = X \times X$, $\theta_Y = \theta \otimes 1 + 1 \otimes \theta$

$\theta_Y^6 = \sum_k \binom{6}{k} \theta^k \otimes \theta^{6-k}$

| $k$ | 0 | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|---|
| $\binom{6}{k}$ | 1 | 6 | 15 | 20 | 15 | 6 | 1 |
| $\int_Y \pi_0\, \theta^k \otimes \theta^{6-k}$ | 0 | 0 | 0 | 144 | 0 | 0 | 0 |
| $\int_Y \pi_{ij}\, \theta^k \otimes \theta^{6-k}$ | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

$\int_Y \pi_0\, \theta_Y^6 = 20 \cdot 144 = 2880$, so
$L_{\theta_Y}(\omega_a) = 2880\, a_0/6! = 4a_0$

$L_{\theta_Y}(\omega) = 0$ for an exceptional class with $a_0 = 0$

$L_{\theta_Y}(\theta_Y^2) = 8!/6! = 56$; $4\theta_Y^2 = [D \cap D']$ for $D, D' \in |2\theta_Y|$, of mass 224

open: an integral current of mass $56NM$ in the class dual to $N(M\theta_Y^2 + \omega)$; known at every CM point with $N, M$ depending on the point; the same $N, M$ at infinitely many CM points would give every member

FIGURE 53. Theorem 20.78 and Example 20.79. Left: the mass of a current in the class dual to $kc$ against $k$. The line $k\,L_h(c)$ is the minimum over closed real normal currents, attained by a smooth strongly positive form; the minima over closed integral currents lie on or above it, with $\|kc\|_{\mathbb{Z}}/k \to L_h(c)$ by Federer's theorem, and for $M$ large the class is algebraic exactly when one of them lies on the line, where the minimiser is an effective algebraic cycle. The heights of the dots are schematic. Right: the numbers of the Mumford square, in the split model. Of the seven terms $\binom{6}{k}\theta^k \otimes \theta^{6-k}$ of $\theta_Y^6$ only $k = 3$ pairs with $\pi_0 = \frac{1}{4}\theta \otimes \theta$, giving $20 \cdot 144 = 2880$, and none pairs with the primitive summands; so $L_{\theta_Y}(\omega_a) = 4a_0$, the Wirtinger bound of an exceptional class with $a_0 = 0$ is zero, and the perturbation of $NM\theta_Y^2$ by $N\omega$ changes the bound $56NM$ by nothing.

exhibits the first of them: two classes in $H^4(X \times X, \mathbb{Q})$ for an abelian fourfold $X$ of Mumford type. None of the constructions of the preceding sections acts on them. In this subsection we identify them as a real multiplication on the primitive part of $H^2(X)$, and we prove that they are algebraic as soon as the Kuga–Satake class of one K3 surface of Picard number thirteen attached to $X$ is algebraic. We do not prove that the Kuga–Satake class is algebraic, and we know of no proof. The reduction changes the kind of question: two classes on an eightfold are traded for one correspondence of a well-studied type, between a surface and its Kuga–Satake variety. The cubic field of the real multiplication enters through the Clifford relations, which force the blocks of the Kuga–Satake embedding to span it.

**Setup 21.1** (Mumford type)**.** Let $X$ be as in Proposition 20.74, with $V = H^1(X, \mathbb{Q})$, polarisation $\psi$ and polarisation class $\theta \in H^2(X, \mathbb{Q})$, and let $G = \mathrm{Hg}(X)$. Over $\bar{\mathbb{Q}}$ the Hodge

cocharacter is nontrivial on exactly one of the three factors of $G$: on $V_i$ it has weights $\pm a_i/2$ with $a_i \geq 0$, and $V$ has only the two weights $\pm\frac{1}{2}$, so exactly one $a_i$ is nonzero. Since $\mathrm{End}^0(X) = \mathbb{Q}$, it follows that $G$ is $\mathbb{Q}$-simple: the Lie algebra of a $\mathbb{Q}$-factor not containing that geometric factor would consist of Hodge classes in $\mathrm{End}(V)$, that is, of endomorphisms of $X$. So $G$ is the image in $\mathrm{GL}(V)$ of $\mathrm{Res}_{F/\mathbb{Q}} D^1$, where $F$ is a totally real cubic field, $D$ is a quaternion algebra over $F$ split at one real place $\sigma_3$ and definite at the other two, $\sigma_1$ and $\sigma_2$, with $\mathrm{Cor}_{F/\mathbb{Q}} D \cong M_8(\mathbb{Q})$, and $D^1$ is the group of elements of reduced norm one [Mum69, Gal00]; the kernel is the central subgroup of triples of signs with product one. Over $\bar{\mathbb{Q}}$, $V = V_1 \otimes V_2 \otimes V_3$, where $V_i$ is the standard representation of $D \otimes_{F,\sigma_i} \bar{\mathbb{Q}} = M_2(\bar{\mathbb{Q}})$; the Galois group permutes the three factors as it permutes the three embeddings, and $H^{1,0}(X) = V_1 \otimes V_2 \otimes V_3^{1,0}$. Put

$$T = \mathrm{Lie}\, G = D^0, \qquad q_\lambda(x,y) = \mathrm{Tr}_{F/\mathbb{Q}}(\lambda\, \mathrm{trd}(xy)) \quad (\lambda \in F^\times),$$

where $D^0$ is the space of quaternions of reduced trace zero, regarded as a $\mathbb{Q}$-vector space of dimension nine, and trd is the reduced trace. Over $\bar{\mathbb{Q}}$, $T = T_1 \oplus T_2 \oplus T_3$ with $T_i = \mathfrak{sl}(V_i)$, and $a \in F$ acts on $T_i$ by $\sigma_i(a)$. Write $m_a$ for multiplication by $a$. The space $T \subset \mathrm{End}(V)$ is a sub-Hodge structure of weight zero.

**Proposition 21.2** (The two classes are a real multiplication)**.** *In Setup 21.1:*

(i) $H^2(X,\mathbb{Q}) = \mathbb{Q}\theta \oplus U$, *where* $U$ *is the primitive part, of dimension* 27, *and over* $\bar{\mathbb{Q}}$

$$U = U_{12} \oplus U_{13} \oplus U_{23}, \qquad U_{ij} = S^2 V_i \otimes S^2 V_j \otimes \textstyle\bigwedge^2 V_k, \quad \{i,j,k\} = \{1,2,3\},$$

*each of dimension nine. The summand* $U_{12}$ *is of type* $(1,1)$, *and* $U_{13}$ *and* $U_{23}$ *have Hodge numbers* $(3,3,3)$.

(ii) *The algebra of Hodge endomorphisms of* $H^2(X,\mathbb{Q})$ *is* $\mathbb{Q} \times F$, *acting by scalars on* $\mathbb{Q}\theta$, *with* $a \in F$ *acting on* $U_{ij}$ *by* $\sigma_k(a)$. *Through Poincaré duality and the Lefschetz isomorphism* $\theta^2 \colon H^2 \to H^6$ *this algebra is the space of Hodge classes in the Künneth component* $H^2(X) \otimes H^2(X)$ *of* $H^4(X \times X, \mathbb{Q})$; *the subalgebra* $\mathbb{Q} \times \mathbb{Q}$ *is spanned by the products of divisor classes that lie there, and the two classes of Proposition 20.74(ii) may be taken to be the elements of* $F$ *of trace zero.*

(iii) $T(-1)$ *is a Hodge structure of K3 type with Hodge numbers* $(1,7,1)$, *and its Hodge endomorphisms are the* $m_a$, $a \in F$. *The* $G$*-invariant symmetric forms on* $T$ *are the* $q_\lambda$, *and* $q_\lambda$ *polarises* $T(-1)$ *if and only if* $\lambda$ *is totally positive.*

*Proof.* (i) is the decomposition of $\bigwedge^2(V_1 \otimes V_2 \otimes V_3)$ in the proof of Proposition 20.74, where $U_{ij}$ appears as the summand of highest weight 2 on the $i$-th and $j$-th factors. The Hodge cocharacter acts trivially on $V_1$ and $V_2$ and splits $V_3 = V_3^{1,0} \oplus V_3^{0,1}$, so $\bigwedge^2 V_3$ has type $(1,1)$ and $S^2 V_3$ has one dimension of each of the types $(2,0)$, $(1,1)$, $(0,2)$.

(ii) A Hodge endomorphism commutes with the Mumford–Tate group $G \cdot \mathbb{G}_m$, and $\mathbb{G}_m$ acts on $H^2$ by scalars, so the Hodge endomorphisms are the $G$-endomorphisms. Over $\bar{\mathbb{Q}}$ the four summands $\mathbb{Q}\theta$, $U_{12}$, $U_{13}$, $U_{23}$ are irreducible and pairwise non-isomorphic, so the $G$-endomorphisms are the combinations of the four projections. The Galois group fixes the projection onto $\mathbb{Q}\theta$ and permutes the other three as it permutes the indices $k$, that is, as it permutes the embeddings of $F$. For $a \in F$ an endomorphism acting on $U_{ij}$ by $\sigma_k(a)$ is therefore Galois invariant, hence rational, and with the scalars on $\mathbb{Q}\theta$ these exhaust the rational points, which form $\mathbb{Q} \times F$. Poincaré duality identifies $\mathrm{End}\, H^2(X,\mathbb{Q})$ with $H^6(X) \otimes H^2(X)$, and $\theta^2 \otimes 1$ carries $H^2 \otimes H^2$ onto it; both are given by algebraic classes, and on an abelian variety so is the inverse of the Lefschetz isomorphism [Kle68, §2]. The products of divisor classes in $H^2 \otimes H^2$ are $\psi_{11}\psi_{22}$ and $\psi_{12}^2$ in the notation of the proof of

Proposition 20.74. They are invariant under the whole of $\mathrm{Sp}(V,\psi)$, whose endomorphisms of $H^2 = \mathbb{Q}\theta \oplus U$ are spanned by the identity and the projection onto $\mathbb{Q}\theta$, and they are linearly independent, being a multiple of the projection and a multiple of the identity (proof of Proposition 20.74); so they span $\mathbb{Q}\times\mathbb{Q}$, and any complement of $\mathbb{Q}\times\mathbb{Q}$ in $\mathbb{Q}\times F$ may serve as the two remaining classes.

(iii) The adjoint action of the Hodge cocharacter is trivial on $T_1$ and $T_2$ and has weights $-1$, 0, 1 on $T_3$, so $T$ has Hodge numbers $(1,7,1)$ in the types $(-1,1)$, $(0,0)$, $(1,-1)$, and $T(-1)$ is of K3 type. Its Hodge endomorphisms are again the $G$-endomorphisms, and the argument of (ii), applied to the three pairwise non-isomorphic summands $T_i$, gives $F$ acting by multiplication. On each $T_i$ the invariant symmetric forms are the multiples of $\mathrm{tr}(xy)$, and $\mathrm{tr}(x_iy_i) = \sigma_i(\mathrm{trd}(xy))$ for $x,y\in D^0$, so the Galois-invariant ones are the $q_\lambda$. At $\sigma_1$ and $\sigma_2$ the algebra $D\otimes\mathbb{R}$ is Hamilton's quaternions, $T_i$ is of type $(1,1)$, and $\mathrm{trd}(x^2) = -2\,\mathrm{nrd}(x)$ is negative on nonzero pure quaternions. At $\sigma_3$ the form $\mathrm{tr}(xy)$ on $\mathfrak{sl}_2(\mathbb{R})$ is negative on the line of type $(1,1)$, the Lie algebra of the rotations fixed by the Hodge cocharacter, and positive on its orthogonal complement, the real points of the types $(2,0)$ and $(0,2)$. The Hodge–Riemann relations for a structure of K3 type ask for exactly these signs after multiplication by $\sigma_i(\lambda)$, so $q_\lambda$ polarises $T(-1)$ precisely when every $\sigma_i(\lambda)$ is positive. ∎

The two classes are thus the endomorphisms of $U$ given by the elements of $F$ of trace zero. A Hodge structure of K3 type with the same field sits beside them, and the next lemma connects the two.

**Lemma 21.3** (The product map)**.** *For $x,y\in T$ let $\mu(x\otimes y)\in H^2(X,\mathbb{Q})$ be the class that corresponds under $\psi$ to the alternating form $(v,w)\mapsto\psi(xv,yw)+\psi(yv,xw)$ on $V$. Then $\mu$ is a $G$-equivariant map $T\otimes T\to H^2(X,\mathbb{Q})$, and:*

(i) *for $i\neq j$ it restricts to an isomorphism $T_i\otimes T_j\to U_{ij}$, and it carries $T_i\otimes T_i$ onto $\mathbb{Q}\theta$;*

(ii) *for $e,\lambda\in F^\times$ put $\mu_e=\mu\circ(m_e\otimes m_e)$ and let $\mu_e^\dagger$ be its adjoint for the form $q_\lambda\otimes q_\lambda$ on $T\otimes T$ and the Lefschetz pairing $B(\alpha,\beta)=\int_X\alpha\beta\,\theta^2$ on $H^2(X,\mathbb{Q})$. Then $\mu_e\mu_e^\dagger$ preserves $\mathbb{Q}\theta$ and acts on $U$ as the element*

$$\nu\,\mathrm{N}_{F/\mathbb{Q}}(g)\,g^{-1}\in F,\qquad g=e^2/\lambda,$$

*where $\nu$ is a nonzero rational number independent of $e$ and $\lambda$.*

*Proof.* Equivariance is clear, since $\mu$ is built from $\psi$ and the action of $\mathrm{End}(V)$ on $V$. An element $x\in T_i$ acts on $V$ through the $i$-th factor and lies in the symplectic Lie algebra of $\psi=\epsilon_1\otimes\epsilon_2\otimes\epsilon_3$, so $\psi(xv,yw)=-\psi(v,xyw)$ and $\mu(x\otimes y)$ is the form $-\psi(v,(xy+yx)w)$. If $x\in T_i$ and $y\in T_j$ with $i\neq j$, then $xy=yx$ acts as $x\otimes y$ on $V_i\otimes V_j$, and the form is $-2$ times the product of $\epsilon_i(\cdot,x\,\cdot)$, $\epsilon_j(\cdot,y\,\cdot)$ and $\epsilon_k$; the first two are symmetric because $x$ and $y$ lie in $\mathfrak{sp}(\epsilon)=\mathfrak{sl}_2$, and $x\mapsto\epsilon(\cdot,x\,\cdot)$ is an isomorphism $\mathfrak{sl}_2\to S^2V^\vee$, so $\mu$ maps $T_i\otimes T_j$ isomorphically onto $U_{ij}$. If $i=j$, then $xy+yx=\mathrm{tr}(xy)$ on $V_i$, because $x$ and $y$ are trace-free $2\times2$ matrices, and $\mu(x\otimes y)=-\,\mathrm{tr}(xy)\,\theta$.

For (ii), $m_e$ is self-adjoint for $q_\lambda$, and the adjoint of a map out of $(T\otimes T,q_\lambda\otimes q_\lambda)$ is $(m_\lambda\otimes m_\lambda)^{-1}$ times its adjoint for $q_1\otimes q_1$, so $\mu_e\mu_e^\dagger=\mu\,(m_g\otimes m_g)\,\mu^{\dagger_1}$ with $\mu^{\dagger_1}$ the adjoint for $q_1\otimes q_1$. This map is $G$-equivariant, so it preserves $\mathbb{Q}\theta$ and each $U_{ij}$, and $\mu^{\dagger_1}$ carries $U_{ij}$ into $(T_i\otimes T_j)\oplus(T_j\otimes T_i)$, on which $m_g\otimes m_g$ is multiplication by $\sigma_i(g)\sigma_j(g)=\mathrm{N}(g)/\sigma_k(g)$. Hence $\mu_e\mu_e^\dagger$ is $\mathrm{N}(g)\sigma_k(g)^{-1}$ times $\mu\mu^{\dagger_1}$ on $U_{ij}$. The summands of $H^2(X,\bar{\mathbb{Q}})$ are pairwise non-isomorphic and self-dual, so $B$ is nondegenerate on $U_{ij}$, and by Schur's lemma the pull-back of $B$ along the isomorphism $\mu_0=\mu|_{T_i\otimes T_j}$ is $c_k\,q_1\otimes q_1$ with $c_k\neq0$; then $\mu_0\mu_0^{\dagger_1}=c_k$

on $U_{ij}$. As $\mu$ is symmetric in its two arguments and the exchange of the factors is an isometry of $q_1 \otimes q_1$, $\mu^{\dagger_1} u = \mu_0^{\dagger_1} u + s\,\mu_0^{\dagger_1} u$ with $s$ that exchange, and $\mu\mu^{\dagger_1} = 2c_k$ on $U_{ij}$. On the split model $V = \bar{\mathbb{Q}}^2 \otimes \bar{\mathbb{Q}}^2 \otimes \bar{\mathbb{Q}}^2$, $\psi = \epsilon \otimes \epsilon \otimes \epsilon$, the permutations of the three factors preserve $\psi$, hence $\theta$, $B$ and $\mu$, and also $q_1 = \sum_i \operatorname{tr}(x_i y_i)$, while they permute the $U_{ij}$; so $c_1 = c_2 = c_3$, and $\mu\mu^{\dagger_1}$ is one and the same nonzero number $\nu$ on the three summands $U_{ij}$, rational because $\mu\mu^{\dagger_1}$ is defined over $\mathbb{Q}$. By Proposition 21.2(ii) an endomorphism of $U$ acting on $U_{ij}$ by $\nu \mathrm{N}(g)/\sigma_k(g)$ is the element $\nu \mathrm{N}(g) g^{-1}$ of $F$. ■

Figure 54 draws Proposition 21.2 and Lemma 21.3 over the three factors of the Hodge group.

By the Kuga–Satake class of a projective K3 surface $S$ we mean the Hodge class on $S \times \mathrm{KS}(S) \times \mathrm{KS}(S)$ that induces the embedding of Proposition 11.7 for $T(S)_{\mathbb{Q}}$; its algebraicity is the Kuga–Satake Hodge conjecture for $S$ (Remark 11.8).

**Theorem 21.4** (One K3 surface for the two classes)**.** *In Setup 21.1, let $\lambda \in F$ be totally positive.*

(i) *There is a projective K3 surface $S_\lambda$ of Picard number thirteen and an isomorphism of Hodge structures $T(S_\lambda)_{\mathbb{Q}} \cong T(-1)$ carrying the intersection form to $q_\lambda$. Its Kuga–Satake variety is isogenous to $X^{32}$.*

(ii) *If the Kuga–Satake class of $S_\lambda$ is algebraic, then the real multiplication by $F$ on $T(S_\lambda)$ is induced by algebraic cycles on $S_\lambda \times S_\lambda$, and every Hodge class in $H^4(X \times X, \mathbb{Q})$ is algebraic; in particular the two classes of Proposition 20.74(ii) are algebraic.*

*Proof.* (i) The quadratic space $(T, q_\lambda)$ has dimension nine and signature $(2, 7)$ by Proposition 21.2(iii), and $H^2$ of a K3 surface with its intersection form has dimension 22 and signature $(3, 19)$. Over $\mathbb{Q}_p$ a nondegenerate quadratic space embeds isometrically in every nondegenerate space of dimension at least three larger, and over $\mathbb{R}$ the signatures allow it, so by the Hasse–Minkowski theorem for representations of forms by forms [OMe00, §§63, 66] there is an isometric embedding of $(T, q_\lambda)$ into $\Lambda_{\mathbb{Q}}$, where $\Lambda$ is the K3 lattice. The line $\ell = T(-1)^{2,0} \subset T_{\mathbb{C}} \subset \Lambda_{\mathbb{C}}$ satisfies $(\ell, \ell) = 0$ and $(\sigma, \bar{\sigma}) > 0$ for $0 \neq \sigma \in \ell$, so by the surjectivity of the period map there is a K3 surface $S_\lambda$ with a marking $H^2(S_\lambda, \mathbb{Z}) \cong \Lambda$ that carries $H^{2,0}(S_\lambda)$ to $\ell$ [Huy16, Chs. 6–7]. A rational class is of type $(1, 1)$ exactly when it is orthogonal to $\ell$, and its component in $T$ is then a rational class of type $(1, 1)$ in $T(-1)$, that is a $G$-invariant vector of $T = \operatorname{Lie} G$; there is none, since $G$ is semisimple. So $\mathrm{NS}(S_\lambda)_{\mathbb{Q}} = T^\perp$, of dimension 13 and signature $(1, 12)$, and $T(S_\lambda)_{\mathbb{Q}} = T$ with the form $q_\lambda$. A class of positive square in $\mathrm{NS}(S_\lambda)$ makes $S_\lambda$ projective [Huy16, Ch. 8].

For the Kuga–Satake variety, note first that over $\bar{\mathbb{Q}}$ the space $T$ is the orthogonal sum of the three ternary spaces $T_i$, so $C(T)$ is the graded tensor product of the $C(T_i)$. Scaling a ternary form does not change its even Clifford algebra, so $D^1$ is the spin group of $(D^0, \lambda \operatorname{trd})$ for every $\lambda$, and $\mathrm{Res}_{F/\mathbb{Q}} D^1$ maps to $\mathrm{Spin}(T, q_\lambda)$ by multiplication in the Clifford algebra, with the same kernel as its map to $\mathrm{GL}(V)$, so that $G$ is a subgroup of $\mathrm{Spin}(T, q_\lambda)$; as a representation of the $i$-th factor, $C(T_i)$ is four copies of $V_i$. Hence $C^+(T)$ is $V^{\oplus 32}$ as a representation of $G$ over $\bar{\mathbb{Q}}$, and over $\mathbb{Q}$ as well, because $V$ is absolutely irreducible. The Hodge structures on $C^+(T) = H^1(\mathrm{KS}(S_\lambda), \mathbb{Q})$ and on $V^{\oplus 32} = H^1(X^{32}, \mathbb{Q})$ are both given by lifts of weight one of the same homomorphism $\mathbb{S} \to \mathrm{MT}(T)_{\mathbb{R}}$ through $G \cdot \mathbb{G}_m$, and two such lifts coincide.[53] So $H^1(\mathrm{KS}(S_\lambda), \mathbb{Q}) \cong H^1(X^{32}, \mathbb{Q})$ as Hodge structures, and $\mathrm{KS}(S_\lambda)$ is

[53] They differ by a homomorphism from the anisotropic torus $\mathbb{S}/\mathbb{G}_m$ to the centre of $G \cdot \mathbb{G}_m$, whose identity component is $\mathbb{G}_m$, and such a homomorphism is trivial.

isogenous to $X^{32}$. For the form used by Galluzzi this is her theorem [Gal00]; van Geemen proves the corresponding splitting for every K3 surface with real multiplication [vG08].

(ii) Write $S = S_\lambda$ and $A = \mathrm{KS}(S)$, and let $Z_S$ be an algebraic cycle on $S \times A \times A$ inducing the Kuga–Satake embedding $T(S) \to \mathrm{End}\, H^1(A)$. Composing $Z_S$ with the graph of an isogeny $A \to X^{32}$ and with the graphs of the inclusions and projections of the factors of $X^{32}$ gives algebraic cycles on $S \times X \times X$ inducing the $32^2$ block components of the embedding, which are morphisms of Hodge structures $T(S) \to \mathrm{End}(V)$, up to a Tate twist; both lie in the tensor category generated by $H^1(X)$, so the blocks are equivariant for $\mathrm{MT}(X) = G \cdot \mathbb{G}_m$. Over $\bar{\mathbb{Q}}$, $\mathrm{End}(V) = S^2V \oplus \bigwedge^2 V$, and $T_i \cong S^2V_i$ occurs once in $S^2V$, as $S^2V_i \otimes \bigwedge^2 V_j \otimes \bigwedge^2 V_k$, and not at all in $\bigwedge^2 V$. So each block takes values in $T$, and through the identification of (i) it is a $G$-endomorphism of $T$, that is $m_{e_{ab}}$ for some $e_{ab} \in F$ by Proposition 21.2(iii). Over $\bar{\mathbb{Q}}$ the embedding is therefore $v \mapsto v \otimes N_i$ on $T_i$, acting on $H^1(A) = V \otimes \mathbb{Q}^{32}$, with $N_i = (\sigma_i(e_{ab}))_{a,b}$. The embedding $\Psi$ of Proposition 11.7 satisfies the Clifford relation $\Psi(v)\Psi(w) + \Psi(w)\Psi(v) = 2q_\lambda(v_0, v_0)\, q_\lambda(v, w)$, because $\Psi(v)\Psi(w)x = vwx\, v_0^2$. Take nonzero $v \in T_i$ and $w \in T_j$ with $i \neq j$. As endomorphisms of $V$ they commute and $vw \neq 0$, so the left side is $vw \otimes (N_iN_j + N_jN_i)$, while the right side is zero because $T_i$ and $T_j$ are orthogonal; hence $N_i$ and $N_j$ anticommute. For $v \in T_i$ with $q_\lambda(v, v) \neq 0$ the relation reads $v^2 \otimes N_i^2 = q_\lambda(v_0, v_0)\, q_\lambda(v, v)$, and $v^2$ is a nonzero scalar on $V_i$, so $N_i^2$ is a nonzero scalar. Three pairwise anticommuting invertible matrices are linearly independent: multiplying a relation $\sum_i c_iN_i = 0$ by $N_j$ on either side and adding gives $2c_jN_j^2 = 0$. If the $e_{ab}$ spanned a subspace of $F$ of dimension at most two, the three $N_i$ would lie in the span of at most two rational matrices, which is impossible; so the $e_{ab}$ span $F$, and two of them, $e$ and $e'$, are not rational multiples of each other. Then $(e/e')^2$ is irrational, since a cubic field contains no irrational element with rational square, so $e^2/\lambda$ and $e'^2/\lambda$ are not both rational. Fix a block $\kappa = m_e$ with $e^2/\lambda \notin \mathbb{Q}$, induced by a cycle $\Gamma$.

Consider first the real multiplication. The trace form $(a, b) \mapsto \mathrm{tr}(ab)$ on $\mathrm{End}(V)$ is induced by an algebraic class: for correspondences $a, b$ on $X$ the Lefschetz trace formula gives $\mathrm{tr}(a_*b_* \mid H^1) = -\deg((a \circ b \circ \pi_1) \cdot \Delta_X)$, where $\pi_1$ is the Künneth projector onto $H^1$, algebraic on an abelian variety [Kle68, §2]. So the adjoint $\kappa^\dagger$ of $\kappa$, for this form and the intersection form on $S$, is induced by a cycle built from the transpose of $\Gamma$, $\pi_1$ and $\Delta_X$, and $\kappa^\dagger\kappa$ is an algebraic endomorphism of $T(S)$. An element $x$ of $T_i$ acts on the eight-dimensional space $V$ through a two-dimensional factor, with multiplicity four, so $\mathrm{tr}_V(xy) = 4\,\mathrm{Tr}_{F/\mathbb{Q}}\,\mathrm{trd}(xy)$, and

$$q_\lambda(\kappa^\dagger\kappa x, y) = \mathrm{tr}_V(ex \cdot ey) = 4\,\mathrm{Tr}_{F/\mathbb{Q}}(e^2\,\mathrm{trd}(xy)) = q_\lambda(4e^2\lambda^{-1}x, y),$$

that is $\kappa^\dagger\kappa = m_{4e^2/\lambda}$. By the choice of $e$ this element is irrational, so it generates the cubic field $F$ and every $m_a$ is a polynomial in $\kappa^\dagger\kappa$. Composing with the projector of $H^2(S)$ onto $T(S)$, which is algebraic because it is the diagonal minus $[\mathrm{pt} \times S]$, $[S \times \mathrm{pt}]$ and $\sum_r d_r \times d_r^\vee$ over dual bases of $\mathrm{NS}(S)_\mathbb{Q}$, gives algebraic cycles on $S \times S$ inducing the real multiplication.

Consider next the classes on $X \times X$. The map $\mu_\kappa = \mu \circ (\kappa \otimes \kappa)$ from $T(S) \otimes T(S)$ to $H^2(X, \mathbb{Q})$ is induced by an algebraic cycle on $S \times S \times X$, obtained from $\Gamma \times \Gamma$, the divisor $m^*\theta - p_1^*\theta - p_2^*\theta$ on $X \times X$, whose class is $\psi$ in $H^1 \otimes H^1$, and the diagonal of $X$ by pull-back, push-forward and intersection, because $\mu$ uses nothing but $\psi$ and the composition of endomorphisms of $V$. The form $q_\lambda \otimes q_\lambda$ is the intersection form of $S \times S$ on $T(S) \otimes T(S)$ and $B$ is the intersection form of $X$ twisted by $\theta^2$, so the adjoint $\mu_\kappa^\dagger$ is induced by cup product with $\theta^2$ followed by the transposed cycle, and $E = \mu_\kappa\mu_\kappa^\dagger$ is an algebraic endomorphism of $H^2(X, \mathbb{Q})$. Since $\kappa = m_e$, Lemma 21.3(ii) says that $E$ preserves $\mathbb{Q}\theta$ and acts on $U$ as $\nu\mathrm{N}(g)g^{-1}$ with $g = e^2/\lambda$, and this element of $F$ is not rational, since $g$ is not.

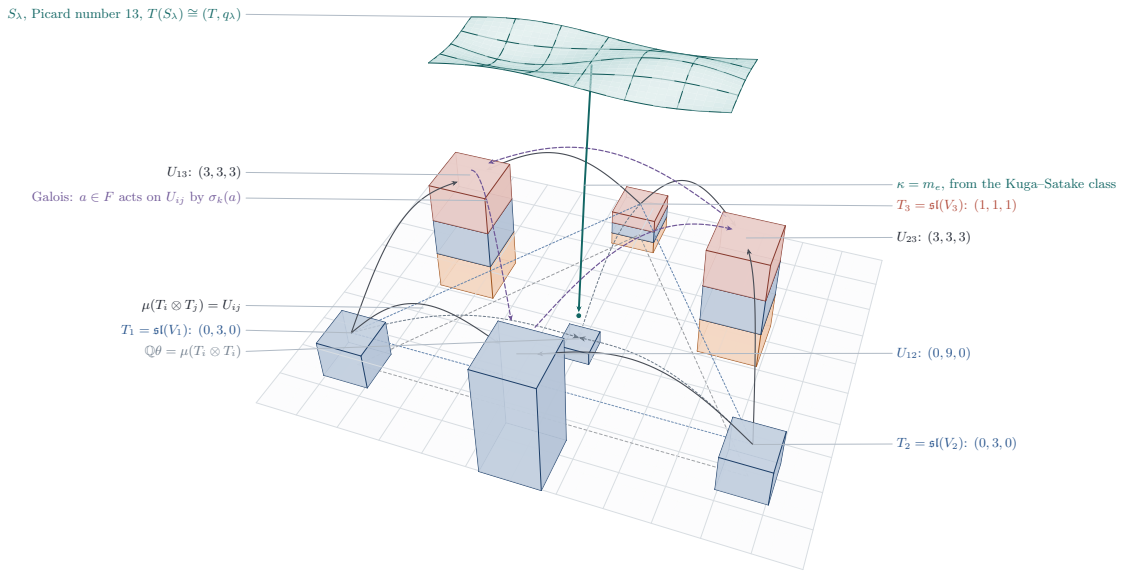



FIGURE 54. Proposition 21.2 and Lemma 21.3, drawn over the triangle of the three factors of the Hodge group; every height is a dimension. At the corners stand the summands $T_i = \mathfrak{sl}(V_i)$ of $T = \operatorname{Lie} G$, banded by Hodge type after the Tate twist $T(-1)$: blue for $(1,1)$, clay for $(2,0)$ and amber for $(0,2)$, so that only $T_3$ carries the K3 type. On the edges stand the three summands $U_{ij}$ of the primitive part of $H^2(X)$, of dimension nine, $U_{12}$ of pure type $(1,1)$ and the other two in three bands of three; in the middle is the line of $\theta$. The solid arcs are the product map $\mu$, which carries $T_i \otimes T_j$ onto $U_{ij}$, and the dashed ones carry $T_i \otimes T_i$ to $\mathbb{Q}\theta$. The violet arcs are the Galois group permuting the $U_{ij}$ as it permutes the embeddings of $F$, which is why the two exceptional classes may be taken to be the trace-zero part of a cubic field. Above, the K3 surface $S_\lambda$ of Theorem 21.4, whose Kuga–Satake class, if algebraic, comes down to $T$ as the map $\kappa = m_e$.

The projection $\pi_\theta$ onto $\mathbb{Q}\theta$ is algebraic, being given by the class $\theta \times \theta$ suitably normalised, so the algebraic Hodge endomorphisms of $H^2(X, \mathbb{Q})$ contain 1, $\pi_\theta$ and $E(1 - \pi_\theta)$, which generate $\mathbb{Q} \times F$, that is all of them by Proposition 21.2(ii). By the same part every Hodge class in $H^2(X) \otimes H^2(X)$ is then algebraic. The Hodge classes of $H^4(X \times X, \mathbb{Q})$ in the other Künneth components number one in each of $H^a \otimes H^{4-a}$, $a \neq 2$, by the count in the proof of Proposition 20.74, and are products of divisor classes. ■

*Remark* 21.5 (The Kuga–Satake class in the literature). Theorem 21.4 takes the algebraicity of the Kuga–Satake class of $S_\lambda$ as its input. The cases of the Kuga–Satake Hodge conjecture proved in the literature are these. They include the projective K3 surfaces with a Shioda–Inose structure, which relates the surface to an abelian surface; by Morrison these are the surfaces of Picard number 19 or 20, those of Picard number 18 with $T(S) \cong U \oplus T'$ and those of Picard number 17 with $T(S) \cong U^2 \oplus T'$ [Mor84, Theorem 6.3 and Corollary 6.4], and for these the proof uses in addition the Hodge conjecture for the powers of an abelian surface, whose Hodge classes are generated by divisor classes (see [MZ99]), and the fact that the Kuga–Satake variety of an abelian surface is isogenous to a power of it [Mor85].

With Buskin's theorem that rational Hodge isometries between K3 surfaces are algebraic [Bus19], this reaches the surfaces of Picard number at least 19, those of Picard number 18 with $T(S)_{\mathbb{Q}}$ isotropic, and those of Picard number 17 with $T(S)_{\mathbb{Q}}$ of Witt index two (Remark 21.17). It does not reach every surface of Picard number seventeen, as Morrison notes [Mor84, Remark 7.2]: for $T(S) = U \oplus \langle 6\rangle \oplus \langle -2\rangle \oplus \langle -2\rangle$, the transcendental lattice of the very general K3 surface of a family of dimension three [Mor84, Corollary 2.10], $T(S)_{\mathbb{Q}}$ has Witt index one, since $\langle 6, -2, -2\rangle$ is anisotropic over $\mathbb{Q}_3$: a primitive solution of $3x^2 = y^2 + z^2$ in $\mathbb{Z}_3$ would have $3 \mid y$ and $3 \mid z$, as $-1$ is not a square modulo 3, and then $3 \mid x$. The double planes branched along six lines, of Picard number at least sixteen, are settled by Paranjape's cycle; see [vG00]. Floccari proves the conjecture for the K3 surfaces attached to projective sixfolds of generalised Kummer type [Flo24], and then, for the K3 surfaces attached to all projective varieties of generalised Kummer type and for every projective K3 surface $S$ with $T(S)_{\mathbb{Q}}$ embedding isometrically in $(U^3 \oplus \langle -m\rangle)_{\mathbb{Q}}$ for some $m > 0$, both the conjecture and the Hodge conjecture for all powers [Flo25, Theorems 1.2 and 5.11]; their transcendental lattices embed, after scaling in the first case, in a rational quadratic space of rank seven and signature $(3, 4)$, so their transcendental rank is at most six. Ingalls, Logan and Patashnick prove the conjecture, with the Hodge conjecture for the square, for a general K3 surface in $\mathbb{P}^4$ with fifteen ordinary double points and for the K3 surfaces joined to it by chains of rational maps of finite degree [ILP22, Theorems 1.5, 4.8 and 6.4]. Bolognesi and Laterveer prove it for the triple covers of a quadric branched along a curve of genus four, which form two families of dimension nine; these carry a non-symplectic automorphism of order three, so their fields of Hodge endomorphisms are CM fields [BL24]. The surfaces $S_\lambda$, by contrast, have transcendental rank nine and real multiplication by a cubic field, and none of these cases contains them: a Shioda–Inose structure needs $T(S) \cong T(A)$ for an abelian surface $A$, hence a transcendental rank of at most five; the surfaces of Floccari, of Paranjape and of Ingalls, Logan and Patashnick have transcendental rank at most six; and the field of Hodge endomorphisms of $S_\lambda$ is totally real, not CM.

Schlickewei made the real multiplication algebraic on those double planes that have it, necessarily by a field of degree at most two, from Paranjape's cycle and van Geemen's description of their Kuga–Satake varieties [Sch10, vG08], and Varesco showed that when the Kuga–Satake Hodge conjecture holds for two hyperkähler varieties, every Hodge similarity between their transcendental lattices is algebraic up to the Lefschetz isomorphism [Var23], which reaches a quadratic field through its square root. More generally, if the Hodge endomorphisms of $T(S)$ form a totally real field $E$, then $E$ acts by self-adjoint maps [Zar83], so $g \in E$ is a similarity, with multiplier $\mu$, exactly when $g^2 = \mu \in \mathbb{Q}$; the Hodge similarities of $T(S)$ to itself, and with them all composites of Hodge similarities between transcendental lattices of K3 surfaces that start and end at $T(S)$, therefore span exactly the maximal multiquadratic subfield $E_{\mathrm{mq}}$ of $E$, the compositum of its quadratic subfields. For a cubic field $E_{\mathrm{mq}} = \mathbb{Q}$: a cubic field contains no irrational element with rational square, so no irrational element of it acts as a similarity. In the proof of Theorem 21.4 the element $\kappa^\dagger\kappa = m_{4e^2/\lambda}$ plays the role that the similarities play for quadratic fields, and the Clifford relations are what make it irrational. Buskin's theorem [Bus19] does not give the real multiplication either. A Hodge isometry of $T(S_\lambda)$ is a Hodge endomorphism, hence some $m_a$ by Proposition 21.2(iii), and it is an isometry only when $a^2 = 1$. For $b \in F^\times$ Buskin's theorem does give algebraic cycles between $S_\lambda$ and $S_{\lambda b^{-2}}$ that realise $m_b$, but every composite of them that returns to $S_\lambda$ is again a Hodge isometry of $T(S_\lambda)$, so it multiplies by an element of square one.

The surfaces $S_\lambda$ include those of Zhu, who computes the integral structure that the fourfold induces on $T$ and studies the K3 surfaces of Picard number thirteen that carry it [Zhu18]; the form of that structure is the corestriction of the reduced norm of $D^0$, which up to sign is $\frac{1}{2}q_1$. Theorem 21.4 applies to them as it stands, because the irrationality of $e^2/\lambda$ comes from the anticommuting blocks and not from $\lambda$. The explicit families closest to these are those of van Geemen and Schütt, with real multiplication by the cyclic cubic fields $\mathbb{Q}(\zeta_7+\zeta_7^{-1})$ and $\mathbb{Q}(\zeta_9+\zeta_9^{-1})$ and Picard numbers four and ten, on which the real multiplication is induced by explicit algebraic cycles [vGS25, Theorem 1.2 and Section 4.8]; their Picard numbers are not thirteen, and the Kuga–Satake class, which Theorem 21.4 needs and algebraic real multiplication does not supply, is not treated there.

By Theorem 21.4, then, the first case of (F2) is implied by one instance of the Kuga–Satake Hodge conjecture, for one K3 surface of Picard number thirteen; by Proposition 21.12 below that instance is equivalent to the first case of (F2) at $X$ together with a link, and it is not a reduction of it. Theorem 24.45 replaces the Kuga–Satake variety of dimension $2^7$ by an abelian sixfold and shows what the replacement does and does not give.

Theorem 21.4 trades the two exceptional classes for one Kuga–Satake class. The next results make this exchange precise. If the exceptional classes are algebraic, so is every Hodge class on every power of $X$ (Theorem 21.7); if the Kuga–Satake class is algebraic, so is every Hodge class on every product $S_\lambda^a \times X^b$ (Corollary 21.11). The Kuga–Satake class is algebraic exactly when the exceptional classes are and, in addition, some algebraic cycle links $S_\lambda$ to a power of $X$ (Proposition 21.12), and no such link can pass through an abelian variety of dimension at most five (Proposition 21.13). The theorem and the corollary are implications, and Remark 21.17 collects the forms they give to the first case of (F2).

**Proposition 21.6** (Where $T$ occurs)**.** *In Setup 21.1, write* $\mathrm{Hom}_{\mathrm{Hdg}}$ *for morphisms of rational Hodge structures.*

(i) *$T$ is isomorphic to no subquotient of $H^k(X,\mathbb{Q})(j)$, for any $k$ and $j$.*

(ii) *The $F$-vector space* $\mathrm{Hom}_{\mathrm{Hdg}}(T, H^2(X\times X,\mathbb{Q})(1))$ *is one-dimensional, and the copy of $T$ in $H^2(X\times X,\mathbb{Q})(1)$ lies in $H^1(X)\otimes H^1(X)$. For $n$ odd and for $n\in\{0,16\}$ the Hodge structure $T$ is isomorphic to no subquotient of any $H^n(X\times X,\mathbb{Q})(j)$.*

(iii) $\mathrm{Hom}_{\mathrm{Hdg}}(T, \mathrm{End}\, V)$ *is a one-dimensional $F$-vector space, spanned by the inclusion. For $\lambda\in F$ totally positive, after an isogeny* $\mathrm{KS}(S_\lambda)\to X^{32}$ *the Kuga–Satake embedding of $T(S_\lambda)$ is a $32\times 32$ matrix of maps $m_{e_{ab}}$ with $e_{ab}\in F$, and* $\mathrm{KS}(S_\lambda)$ *is isogenous to $B^4$, where $B$ is isogenous to $X^8$ and* $\mathrm{End}^0(B)=M_8(\mathbb{Q})$.

*Proof.* After the twists all the Hodge structures involved have weight zero and lie in the tensor category generated by $V$, whose Mumford–Tate group is $G\cdot\mathbb{G}_m$ with $\mathbb{G}_m$ acting trivially in weight zero; so their morphisms and subquotients are those of representations of $G$. Over $\bar{\mathbb{Q}}$, $T=T_1\oplus T_2\oplus T_3$ with $T_i\cong S^2V_i$, and the Galois group permutes the three summands transitively. Hence $T$ is irreducible, a representation of $G$ defined over $\mathbb{Q}$ contains $T_1$, $T_2$ and $T_3$ with one and the same multiplicity $m$, and its space of $G$-maps from $T$, of $\mathbb{Q}$-dimension $3m$, is a vector space over $\mathrm{End}_G(T)=F$ (Proposition 21.2(iii)) of dimension $m$.

For (i), write $V=V_1\otimes W$ with $W=V_2\otimes V_3$, on which $\mathrm{SL}(V_2)\times\mathrm{SL}(V_3)$ acts through $\mathrm{SO}(W,q)$, $q$ the product of the two symplectic forms. By the skew Cauchy formula $\bigwedge^k V=\bigoplus_\lambda S_\lambda V_1\otimes S_{\lambda'}W$, over the partitions $\lambda$ of $k$ with at most two rows and at most four columns, and $S_\lambda V_1\cong S^{\lambda_1-\lambda_2}V_1$ as a representation of $\mathrm{SL}(V_1)$. So $T_1$ occurs in $\bigwedge^k V$ only through $\lambda=(2,0)$, $(3,1)$ or $(4,2)$, and only if $S_{\lambda'}W$ has a nonzero invariant under $\mathrm{SL}(V_2)\times\mathrm{SL}(V_3)$. As representations of $\mathrm{SL}(W)$, $S_{(1,1)}W=\bigwedge^2 W$, $S_{(2,2,1,1)}W\cong\bigwedge^2 W$, and

$S_{(2,1,1)}W \cong \mathfrak{sl}(W)$, the kernel of the contraction $\bigwedge^3 W \otimes W \to \bigwedge^4 W$ with $\bigwedge^3 W \cong W^*$. Restricted to $\mathrm{SO}(W,q)$, $\bigwedge^2 W = \mathfrak{so}(W) = \mathfrak{sl}(V_2)\oplus\mathfrak{sl}(V_3)$ and $\mathfrak{sl}(W) = \mathfrak{so}(W)\oplus S^2V_2\otimes S^2V_3$, the second summand being the trace-free part of $S^2W$; none of these has a nonzero invariant. So $T_1$ occurs in no $\bigwedge^k V$, and by the same argument with the factors exchanged neither do $T_2$ and $T_3$. As $G$ is reductive, the subquotients of $\bigwedge^k V$ are its summands; that is (i).

For (ii), $H^n(X\times X) = \bigoplus_{a+b=n} \bigwedge^a V \otimes \bigwedge^b V$. The element of $G$ that is $-1$ on the first factor $\mathrm{SL}(V_1)$ and 1 on the other two acts on $V$ by $-1$, hence on $H^n(X\times X)$ by $(-1)^n$, and trivially on $T$; and $H^0$, $H^{16}$ are trivial representations. In $H^2(X\times X) = \bigwedge^2 V \oplus (V\otimes V)\oplus \bigwedge^2 V$ the summands $\bigwedge^2 V$ contain no $T_i$ by (i), and $V\otimes V = S^2V\oplus\bigwedge^2 V$ with

$$S^2(V_1\otimes V_2\otimes V_3) = S^2V_1\otimes S^2V_2\otimes S^2V_3 \oplus \bigoplus_{i=1}^{3} S^2V_i \otimes \bigwedge^2 V_j \otimes \bigwedge^2 V_k,$$

$\{i,j,k\} = \{1,2,3\}$, where $\bigwedge^2 V_j$ is trivial. So each $T_i$ occurs exactly once in $H^2(X\times X)$, inside $V\otimes V$, and $m = 1$. For (iii), $\operatorname{End} V = S^2V \oplus \bigwedge^2 V$ contains each $T_i$ once, and the proof of Theorem 21.4(ii) shows that the blocks of the Kuga–Satake embedding are maps $m_{e_{ab}}$. Finally $\mathrm{KS}(S_\lambda)$ is isogenous to $X^{32} = (X^8)^4$ by Theorem 21.4(i), and $\operatorname{End}^0(X^8) = M_8(\operatorname{End}^0 X) = M_8(\mathbb{Q})$. This is the splitting $\mathrm{KS} \sim B^{2^{d-1}}$, $d = [F:\mathbb{Q}]$, which van Geemen proves for K3 surfaces with real multiplication [vG08]; there $H^1(B,\mathbb{Q})$ is the corestriction to $\mathbb{Q}$ of the even Clifford algebra of $T$ over $F$, here $D$, and over $\bar{\mathbb{Q}}$ one has $\mathrm{Cor}_{F/\mathbb{Q}}(D) = \bigotimes_k D\otimes_{F,\sigma_k}\bar{\mathbb{Q}} = \bigotimes_k \operatorname{End}(V_k) = \operatorname{End}(V)$, which is $V^{\oplus 8}$ under left multiplication by $G$. ∎

**Theorem 21.7** (Every power of a Mumford fourfold)**.** *In Setup 21.1, suppose that the two exceptional classes of Proposition 20.74(ii) are algebraic on $X\times X$. Then every Hodge class on $X^n$ is algebraic, for every $n\geq 1$.*

*Proof.* Over $\bar{\mathbb{Q}}$ keep $V = V_1\otimes V_2\otimes V_3$ and $\psi = \epsilon_1\otimes\epsilon_2\otimes\epsilon_3$, as in the proof of Lemma 21.3. For $m\geq 1$ regard $V^{\otimes m} = H^1(X)^{\otimes m}$ as the Künneth component of $H^m(X^m,\mathbb{Q})$ of multidegree $(1,\dots,1)$, whose projector is algebraic [Kle68, §2]. Call an endomorphism of $V^{\otimes m}$ algebraic if it is the action on $V^{\otimes m}$ of an algebraic cycle on $X^m\times X^m$ that preserves $V^{\otimes m}$. Composing with the Künneth projector, the algebraic endomorphisms form a $\mathbb{Q}$-algebra, and they carry algebraic classes in $V^{\otimes m}$ to algebraic classes.

*Step 1: reduction to multilinear invariants.* Let $n\geq 1$ and $p\geq 0$. The map $V^{\otimes 2p}\otimes(\mathbb{Q}^n)^{\otimes 2p} \to \bigwedge^{2p}(V\otimes\mathbb{Q}^n) = H^{2p}(X^n,\mathbb{Q})$ sending $(v_1\otimes\cdots\otimes v_{2p})\otimes(a_1\otimes\cdots\otimes a_{2p})$ to $(v_1\otimes a_1)\wedge\cdots\wedge(v_{2p}\otimes a_{2p})$ is surjective and $G$-equivariant for the trivial action on $\mathbb{Q}^n$, so it is surjective on $G$-invariants, $G$ being reductive. For $a_1,\dots,a_{2p}\in\mathbb{Z}^n$ the image of $t\otimes(a_1\otimes\cdots\otimes a_{2p})$ is $f^*t$, where $f\colon X^n\to X^{2p}$ is the homomorphism with rows $a_1,\dots,a_{2p}$, and such tensors span $(\mathbb{Q}^n)^{\otimes 2p}$. So $\mathrm{Hdg}^p(X^n)$ is spanned by the classes $f^*t$ with $f$ a homomorphism and $t\in(V^{\otimes 2p})^G$, and it suffices to show that every element of $(V^{\otimes 2p})^G$ is algebraic.

*Step 2: degree four.* The space $(V^{\otimes 4})^G$ has dimension eight, the product of the dimensions of the three spaces of $\mathrm{SL}_2$-invariants in $V_k^{\otimes 4}$. Let $\mathcal{P}_0\subset(V^{\otimes 4})^G$ be the span of the multilinear components of the classes $f^*\xi$, for $\xi\in\mathrm{Hdg}^2(X\times X)$ and $f\colon X^4\to X^2$ a homomorphism, and of the products of two divisor classes of $X^4$. By Proposition 20.74(ii) $\mathrm{Hdg}^2(X\times X)$ is spanned by the six products of divisor classes and the two exceptional classes, so under the hypothesis every element of $\mathcal{P}_0$ is algebraic. For $t\in V^{\otimes 4}$, read as a correspondence from the first two factors of $X^4$ to the last two, let $\hat{t}$ be the endomorphism $v\mapsto t_*((\theta^3\times\theta^3)\cup v)$ of $V\otimes V$. The pairing $(v,w)\mapsto\int_X v\cup w\cup\theta^3$ on $V$ is nondegenerate by hard Lefschetz

and $G$-invariant, so $t \mapsto \hat{t}$ is an isomorphism $(V^{\otimes 4})^G \to \mathrm{End}_G(V \otimes V)$, and $\hat{t}$ is algebraic when $t$ is. We show that the $\hat{t}$ and the composites $\hat{u}\hat{t}$, for $t, u \in \mathcal{P}_0$, span $\mathrm{End}_G(V \otimes V)$. Over $\bar{\mathbb{Q}}$, $\mathrm{End}_G(V \otimes V) = \bigotimes_k \mathrm{End}_{\mathrm{SL}_2}(V_k \otimes V_k)$ is the algebra of functions on the eight sign patterns $\varepsilon \in \{\pm 1\}^3$, where $\varepsilon_k$ is the eigenvalue of the exchange $s_k$ of the two copies of $V_k$; the patterns with $\varepsilon_1\varepsilon_2\varepsilon_3 = -1$, the odd ones, make up $\bigwedge^2 V$.

(a) Let $f = (x_1 + x_2, x_3 + x_4)\colon X^4 \to X^2$. The multilinear component of $f^*\xi$ depends only on the Künneth component of $\xi$ in $H^2(X) \otimes H^2(X)$, and it is that component read as a tensor $t \in \bigwedge^2 V \otimes \bigwedge^2 V$, so $\hat{t}$ vanishes on $S^2V$ and takes values in $\bigwedge^2 V$. These components fill $(H^2 \otimes H^2)^G = \mathrm{End}_G(H^2)$, of dimension 4 by Proposition 21.2(ii); so the $\hat{t}$ span the functions supported on the four odd patterns.

(b) The map $g = (x_1 + x_3, x_2 + x_4)$ is $f$ composed with the exchange of the factors 2 and 3 of $X^4$, which permutes the factors of $V^{\otimes 4}$ up to sign. On $V_k^{\otimes 4}$ the $\mathrm{SL}_2$-invariants are spanned by $\epsilon_{13}\epsilon_{24}$, $\epsilon_{14}\epsilon_{23}$ and $\epsilon_{12}\epsilon_{34} = \epsilon_{13}\epsilon_{24} - \epsilon_{14}\epsilon_{23}$, where $\epsilon_{ab}$ is $\epsilon_k$ on the factors $a$ and $b$; up to a common sign their hats are 1, $s_k$ and $1 - s_k$. The exchange of the factors 2 and 3 swaps the first and the third and changes the sign of the second, so on functions of $\varepsilon_k$ it carries $\frac{1}{2}(1 - s_k)$ to $\frac{1}{2}$ and $\frac{1}{2}(1 + s_k)$ to $\frac{1}{2} - s_k$. Applied to the indicator of the odd pattern $(1, 1, -1)$ it gives $\frac{1}{2}(\frac{1}{2} - s_1)(\frac{1}{2} - s_2)$. The identity is $\hat{t}$ for the multilinear component of a product of two divisor classes, the pull-backs of the class $\psi$ on $X \times X$ along the projections of $X^4$ to the factors $(1, 3)$ and $(2, 4)$, so $h = (1 - 2s_1)(1 - 2s_2) - 1$ lies in the span of the $\hat{t}$, $t \in \mathcal{P}_0 \otimes \bar{\mathbb{Q}}$. As $(\varepsilon_1, \varepsilon_2)$ runs through $(1, 1)$, $(1, -1)$, $(-1, 1)$, $(-1, -1)$, $h$ takes the values $0, -4, -4, 8$, so $h(h + 4)/96$ is the indicator of $\varepsilon_1 = \varepsilon_2 = -1$; subtracting the indicator of $(-1, -1, -1)$, from (a), leaves that of $(-1, -1, 1)$. The same argument with the factors $V_k$ permuted gives the other two even patterns with two minus signs, and the indicator of $(1, 1, 1)$ is 1 minus the other seven.

So the $\hat{t}$ and $\hat{u}\hat{t}$ span $\mathrm{End}_G(V \otimes V) \otimes \bar{\mathbb{Q}}$. Since composition is bilinear and $\mathcal{P}_0 \otimes \bar{\mathbb{Q}}$ is spanned by the same pull-backs and products of a basis of Hodge classes, the $\mathbb{Q}$-algebra of algebraic $G$-endomorphisms of $V \otimes V$ has dimension 8: every $G$-endomorphism of $V \otimes V$ is algebraic.

*Step 3: the partial swaps.* Over $\bar{\mathbb{Q}}$, $\mathrm{End}_G(V \otimes V) = \bigotimes_k \mathrm{End}_{\mathrm{SL}_2}(V_k \otimes V_k)$, and $\mathrm{End}_{\mathrm{SL}_2}(V_k \otimes V_k)$ is spanned by 1 and the exchange $s_k$ of the two copies of $V_k$. In a basis of $V_k$ the matrix units $E_{ab}$ and $E_{ba}$ are dual for the trace form of $\mathfrak{gl}(V_k)$, and $\sum_{a,b} E_{ab} \otimes E_{ba} = s_k$; removing the centre, spanned by the identity 1 with $\mathrm{tr}(1 \cdot 1) = 2$, gives $s_k = \mathrm{Cas}_k + \frac{1}{2}$, with $\mathrm{Cas}_k = \sum_c x_c \otimes x^c$ for dual bases $(x_c)$, $(x^c)$ of $\mathfrak{sl}(V_k)$ under the trace form. So the $s_k$ come from the three summands $T_k$ of $T$. The Galois group permutes the $s_k$ as it permutes the embeddings $\sigma_k$, so for $a \in F$ the endomorphism $r_a = \sum_k \sigma_k(a) s_k$ is rational, hence algebraic by Step 2. For $m \geq 2$ and $1 \leq i < m$ let $r_a^{(i)}$ and $s_k^{(i)}$ act as $r_a$ and $s_k$ on the factors $i$ and $i + 1$ of $V^{\otimes m}$ and as the identity on the others; $r_a^{(i)}$ is algebraic, being induced by the product of a cycle inducing $r_a$ with the diagonals of the other factors. By Schur–Weyl duality $\mathrm{End}_{\mathrm{SL}_2}(V_k^{\otimes m})$ is the image of $\bar{\mathbb{Q}}[S_m]$, generated by the adjacent transpositions, and the commutant of $\mathrm{SL}_2^3$ in $\bigotimes_k V_k^{\otimes m}$ is the tensor product of the three commutants; so $\mathrm{End}_G(V^{\otimes m}) \otimes \bar{\mathbb{Q}}$ is generated by the $s_k^{(i)}$. For a basis $a_1, a_2, a_3$ of $F$ the matrix $(\sigma_k(a_j))$ is invertible, the square of its determinant being the discriminant of $F$, so each $s_k^{(i)}$ is a $\bar{\mathbb{Q}}$-linear combination of the $r_{a_j}^{(i)}$. The $\mathbb{Q}$-algebra generated by the $r_a^{(i)}$ therefore lies in $\mathrm{End}_G(V^{\otimes m})$ and spans it over $\bar{\mathbb{Q}}$, so it is $\mathrm{End}_G(V^{\otimes m})$: every $G$-endomorphism of $V^{\otimes m}$ is algebraic. Its dimension is the cube of $\dim \mathrm{End}_{\mathrm{SL}_2}(V_k^{\otimes m})$, the sum of the squares of the multiplicities in $V_k^{\otimes m}$: $8 = 2^3$ for $m = 2$ and $125 = 5^3$ for $m = 3$, where $V_k^{\otimes 3} = S^3 V_k \oplus V_k^{\oplus 2}$.

*Step 4: all multilinear invariants.* By the first fundamental theorem for $\mathrm{SL}_2$ the invariants in $V_k^{\otimes 2m}$ are spanned by the images of $\epsilon_k^{\otimes m}$ under the permutations of the $2m$ factors. So $(V^{\otimes 2m})^G \otimes \bar{\mathbb{Q}}$ is spanned by the images of $\psi^{\otimes m} = \bigotimes_k \epsilon_k^{\otimes m}$ under the operators that permute the factors of the three $V_k^{\otimes 2m}$ separately, which lie in $\mathrm{End}_G(V^{\otimes 2m}) \otimes \bar{\mathbb{Q}}$, and the map $\mathrm{End}_G(V^{\otimes 2m}) \to (V^{\otimes 2m})^G$, $e \mapsto e(\psi^{\otimes m})$, is surjective. The class $\psi$ in $V \otimes V$ is that of the divisor $m^*\theta - p_1^*\theta - p_2^*\theta$ on $X \times X$ (proof of Theorem 21.4(ii)), up to a nonzero factor, and $\psi^{\otimes m}$ is, up to sign, the product of its pull-backs along the projections of $X^{2m}$ to the pairs of factors $(2j-1, 2j)$; so it is algebraic, and by Step 3 so is every element of $(V^{\otimes 2m})^G$. Step 1 concludes. ∎

*Remark* 21.8 (Why a composite is needed). Step 2 cannot be carried out with pull-backs and products alone. The skew Cauchy decomposition $\bigwedge^4(V \otimes \mathbb{Q}^4) = \bigoplus_\lambda S_\lambda V \otimes S_{\lambda'}\mathbb{Q}^4$, over the partitions $\lambda$ of 4 with conjugates $\lambda'$, is functorial in $\mathbb{Q}^4$, and its summand $S^4V \otimes \bigwedge^4 \mathbb{Q}^4$ contains exactly one line of Hodge classes of $X^4$: $S_4$ acts on the two-dimensional space of $\mathrm{SL}_2$-invariants in $V_k^{\otimes 4}$ through its two-dimensional irreducible representation $\sigma$, and the trivial representation occurs once in $\sigma^{\otimes 3}$. That line is spanned by a class $\mathrm{Det} \in (V^{\otimes 4})^G$ on which the permutations of the factors of $X^4$ act by the sign, and whose restriction to the diagonal, on the split model $V = \bar{\mathbb{Q}}^2 \otimes \bar{\mathbb{Q}}^2 \otimes \bar{\mathbb{Q}}^2$, is an invariant quartic form of $\mathrm{SL}_2^3$, hence a nonzero multiple of Cayley's hyperdeterminant. Pull-backs along homomorphisms $X^4 \to X^2$ have no component in that summand, because $\bigwedge^4 \mathbb{Q}^2 = 0$, and products of divisor classes have none, because they are invariant under $\mathrm{Sp}(V, \psi)$ and $S^4V$ has no $\mathrm{Sp}(V, \psi)$-invariant. In the notation of Step 2 the hats of the elements of $\mathcal{P}_0$ therefore satisfy one linear condition, which is $v_{(1,-1,-1)} + v_{(-1,1,-1)} + v_{(-1,-1,1)} = 3v_{(1,1,1)}$ for the values $v$ on the even patterns, since the functions found in (a) and (b), those supported on the odd patterns, the constant, $h$ and its two images under the permutations of the factors $V_k$, satisfy it and span a space of dimension 7. So $\mathcal{P}_0$ has dimension 7 and is the space of Hodge classes of $X^4$ in the multilinear parts of the other summands, and Det is reached only through a composite of correspondences, which involves a push-forward. The composite $h^2$ of Step 2 is one: the indicator of $(-1, -1, 1)$ violates the condition. This is where Theorem 21.7 differs from Proposition 20.24, in which the first fundamental theorem for $\mathrm{SL}(W)$ lets pull-backs and products suffice. We have not found Theorem 21.7 in the literature; it is an implication between two cases of the Hodge conjecture.

At the CM points of a Mumford family the conclusion holds without any hypothesis.

**Proposition 21.9** (Powers at a CM point). *Let $W \to C$ be a family as in Corollary 20.22 and $c \in C$ a CM point. Then the Hodge conjecture holds for $X_c^n$ for every $n \geq 1$.*

*Proof.* The proof of Theorem 20.29 shows that for every $n \geq 1$ the ring $\mathrm{Hdg}^*(X_c^n)$ is generated by pull-backs of Hodge classes of the fourfold $X_c$ along homomorphisms $X_c^n \to X_c$, and these are algebraic [Mar25a]. ∎

The statement is also contained in the theorem of [Li26] on the powers of CM fourfolds.

**Corollary 21.10** (Powers and fibre powers of a Mumford family). *Let $W \to C$ be as in Corollary 20.22 and, for $n \geq 1$, let $W^{(n)} = W \times_C \cdots \times_C W$, with $n$ factors. The conditions of Corollary 20.30 are equivalent to each of the following.*

(e) *The Hodge conjecture holds for $X_t^n$ for every $t \in C$ and every $n \geq 1$.*

(f) *$B(W^{(n)})$ holds for $L_{W^{(n)}}$ for every $n \geq 1$.*

*Proof.* (b) implies (e): by Corollary 20.30, (b) gives (c), so the exceptional classes of $X_t \times X_t$ are algebraic for every $t$. At a point $t$ that is not a CM point the Hodge group of $X_t$ is

$G$, as shown in the proof of Corollary 20.30, and Theorem 21.7 applies; at a CM point Proposition 21.9 does. (e) implies (f): $W^{(n)}$ is an abelian scheme over $C$ with projective total space, and, as in the proof of Corollary 20.25, its monodromy invariant classes are Hodge classes on every fibre; by (e) they are algebraic on every fibre, hence restrictions of algebraic classes on $W^{(n)}$ by Lemma 20.27, and Theorem 20.21 gives $B(W^{(n)})$. (f) implies (a): take $n = 2$. ∎

**Corollary 21.11** (Products with the K3 surface)**.** *In Setup 21.1, let $\lambda \in F$ be totally positive and $S = S_\lambda$ as in Theorem 21.4. If the Kuga–Satake class of $S$ is algebraic, then the Hodge conjecture holds for $S^a \times X^b$ for all $a, b \geq 0$.*

*Proof.* By Theorem 21.4(ii) the exceptional classes are algebraic, so by Theorem 21.7 the Hodge conjecture holds for every power of $X$. In the proof of Theorem 21.4(ii) a block $\kappa = m_e$, $e \neq 0$, of the Kuga–Satake embedding is induced by a cycle on $S \times X \times X$; it is a map from $T(S)$ to $\operatorname{End} V$, which up to a twist is a Künneth component of $H^*(X \times X, \mathbb{Q})$. Its adjoint $\kappa^\dagger$ is algebraic, $\kappa^\dagger\kappa = m_{4e^2/\lambda}$, and every $m_a$, $a \in F$, is induced by a cycle on $S \times S$. So $L = m_{\lambda/(4e^2)}\kappa^\dagger$ is algebraic and $L\kappa$ is the identity of $T(S)$. The decomposition $H^*(S, \mathbb{Q}) = \mathbb{Q} \oplus \mathrm{NS}(S)_\mathbb{Q} \oplus T(S) \oplus H^4(S, \mathbb{Q})$ has algebraic projectors, as in that proof, so $H^*(S^a \times X^b, \mathbb{Q})$ is the sum of the summands $N \otimes T(S)^{\otimes r} \otimes H^*(X^b, \mathbb{Q})$, $N$ a tensor product of $a - r$ of the three other pieces, each cut out by an algebraic projector. Each such $N$ is spanned by products of algebraic classes and is of Tate type, so a Hodge class in the summand is a sum of products $p^*\eta \cup q^*\xi$ with $\eta$ algebraic on $S^{a-r}$ and $\xi$ a Hodge class in $T(S)^{\otimes r} \otimes H^*(X^b, \mathbb{Q}) \subset H^*(S^r \times X^b, \mathbb{Q})$. For such $\xi$ the class $(\kappa^{\otimes r} \otimes 1)\xi$ is a Hodge class on $X^{2r+b}$, hence algebraic, and $\xi = (L^{\otimes r} \otimes 1)(\kappa^{\otimes r} \otimes 1)\xi$ is algebraic. ∎

Corollary 21.11 strengthens Theorem 21.4(ii), which gives $X \times X$ only. Varesco proves an implication of the same kind, from the algebraicity of the Kuga–Satake correspondence to the Hodge conjecture for all powers, for families of K3 surfaces of generic Picard number sixteen [Var22]. The converse question, what the Kuga–Satake class asks for beyond the target, has a precise answer.

**Proposition 21.12** (What the Kuga–Satake route asks)**.** *In the situation of Corollary 21.11, the Kuga–Satake class of $S$ is algebraic if and only if*

(i) *the two exceptional classes of $X \times X$ are algebraic, and*
(ii) *(a link) for some $n \geq 1$ there is an algebraic cycle $Z$ on $S \times X^n$ whose action $Z_*$ is nonzero on $T(S)$.*

*Proof.* If the Kuga–Satake class is algebraic, (i) is Theorem 21.4(ii), and a nonzero block of the embedding, induced by a cycle on $S \times X \times X$, is a link with $n = 2$. Conversely assume (i) and (ii). By Theorem 21.7 the Hodge conjecture holds for every power of $X$. Since $T(S) \cong T(-1)$ is irreducible (proof of Proposition 21.6), $Z_*$ maps $T(S)$ isomorphically onto a sub-Hodge structure of some $H^{2k}(X^n, \mathbb{Q})(k-1)$, which is a direct summand, polarisable Hodge structures being semisimple. Projecting onto it and composing with an isomorphism onto the copy of $T(-1)$ in $H^2(X \times X, \mathbb{Q})$ (Proposition 21.6(ii)) gives a morphism of Hodge structures $H^{2k}(X^n, \mathbb{Q})(k-1) \to H^2(X \times X, \mathbb{Q})$, that is a Hodge class on $X^{n+2}$, which is algebraic. Its composite with $Z_*$ is an algebraic nonzero morphism $T(S) \to T(-1) \subset H^2(X \times X, \mathbb{Q})$, which is some $m_e$, $e \neq 0$, by Proposition 21.2(iii). The endomorphisms of $H^2(X \times X, \mathbb{Q})$ that act by $m_a$ on this copy of $T(-1)$ and by zero on a Hodge complement are Hodge classes on $X^4$, hence algebraic, so every $m_a \colon T(S) \to H^1(X) \otimes H^1(X)$, $a \in F$, is algebraic, and so is its composite with cup product with $\theta^3$ on one factor, which carries $H^1(X) \otimes H^1(X)$ onto the Künneth component of $H^8(X \times X, \mathbb{Q})$ that represents $\operatorname{End} V$.

By Proposition 21.6(iii), after an isogeny $\mathrm{KS}(S) \to X^{32}$ the Kuga–Satake embedding is a matrix of such maps, and the Kuga–Satake class is recovered from its blocks through the graphs of the isogeny and of the inclusions and projections of the factors of $X^{32}$; so it is algebraic. ∎

So the Kuga–Satake route is at least as strong as the first case of (F2) and is not a reduction of it: at the point $X$ it asks for (i), which is the first case of (F2) there, and in addition for a link. The Hodge conjecture implies both (i) and (ii). The route helps only if the Kuga–Satake class of $S_\lambda$ can be made algebraic by arguments proper to K3 surfaces. A link has to be a correspondence of a new kind: $T$ occurs in no $H^k(X)$ (Proposition 21.6(i)), and the next proposition shows that it occurs in the cohomology of no abelian variety of dimension below six.

**Proposition 21.13** (No small abelian carrier of $T$)**.** *In Setup 21.1, let $Y$ be an abelian variety such that $T(-1)$ is isomorphic to a subquotient of $H^k(Y,\mathbb{Q})(j)$ for some $k$ and $j$. Then $\dim Y \geq 6$. Consequently, if $\dim Y \leq 5$ and $S = S_\lambda$ is as in Theorem 21.4, every Hodge class on $S \times Y$, algebraic or not, acts by zero on $T(S)$, and no link as in Proposition 21.12(ii) is a composite $Z_2 \circ Z_1$ of algebraic cycles on $S \times Y$ and on $Y \times X^n$.*

*Proof.* Write $W = T(-1)$. As $W$ is a subquotient of a tensor construction on $H^1(Y)$, $\mathrm{MT}(Y)$ maps onto $\mathrm{MT}(W)$, whose derived group is the adjoint group $G^{\mathrm{ad}}$ of $G$, acting on $W$ through the adjoint representation. So the derived group of $\mathrm{Hg}(Y)$, an almost direct product of $\mathbb{Q}$-simple groups, maps onto $G^{\mathrm{ad}}$. The image of each factor is a normal subgroup of the $\mathbb{Q}$-simple group $G^{\mathrm{ad}}$, the images commute, and $G^{\mathrm{ad}}$ is not commutative, so exactly one factor $M$ maps onto $G^{\mathrm{ad}}$, with finite kernel. Over $\bar{\mathbb{Q}}$, $M$ is isogenous to $\mathrm{SL}_2^3$, with factors indexed by the embeddings $\sigma_1, \sigma_2, \sigma_3$ and permuted transitively by the Galois group, and the irreducible constituents of $H^1(Y,\bar{\mathbb{Q}})$ under $M$ are products $R_1 \otimes R_2 \otimes R_3$ of irreducible representations of the three factors.

Let $\mu_Y$ be the Hodge cocharacter of $Y$, with the two weights 0 and 1 on $H^1(Y)$, and write $\mu_Y = \mu'\mu''$ with fractional cocharacters $\mu'$ of $M$ and $\mu''$ commuting with $M$. The image of $\mu''$ in $\mathrm{MT}(W)$ commutes with $G^{\mathrm{ad}}$, so it is central, and the image of $\mu'$ in $G^{\mathrm{ad}}$ is the component there of the Hodge cocharacter of $W$, which by Proposition 21.2(iii) is trivial on the factors at $\sigma_1$ and $\sigma_2$ and has weights $-1, 0, 1$ on $T_3$. So $\mu'$ is trivial on the first two factors of $M$ and has weights $\pm\frac{1}{2}$ on the standard representation of the third. On $R_1 \otimes R_2 \otimes R_3$, tensored with an eigenvector of $\mu''$ in its multiplicity space, the weights of $\mu_Y$ therefore fill an interval of length $\dim R_3 - 1$, so $\dim R_3 \leq 2$. The constituents form a Galois-stable multiset and the Galois group carries the third factor to each of the others, so every constituent is $V_I = \bigotimes_{i\in I} V_i$ for a subset $I \subseteq \{1,2,3\}$, with a multiplicity $m_I$ that depends only on $|I|$, and $m_I \neq 0$ for some nonempty $I$, since $M$ acts nontrivially.

If $m_I \neq 0$ for some $I$ with $|I| = 2$, then $\dim H^1(Y) \geq 3 \cdot 4 = 12$. If $m_{\{1\}} \neq 0$, let $U = V_1 \otimes N$, $\dim N = m_{\{1\}}$, be the $V_1$-isotypic summand of $H^1(Y,\mathbb{C})$. The embedding $\sigma_1$ is real, so complex conjugation preserves the first factor of $M$ and $U$ is defined over $\mathbb{R}$. The connected group $\mathrm{MT}(Y)$ acts on $M$ preserving each factor, and $\mathrm{SL}_2$ has a single irreducible representation of dimension two, so $U$ is stable under $\mathrm{MT}(Y)(\mathbb{R})$ and is a real sub-Hodge structure of $H^1(Y,\mathbb{R})$, and $U^{1,0}$ and $U^{0,1}$ are complex conjugate and both nonzero. Since $\mu'$ acts trivially on $U$, $\mu_Y$ acts on $V_1 \otimes N$ through $N$, so $N$ carries both weights, $m_{\{1\}} \geq 2$, and $\dim H^1(Y) \geq 3 \cdot 2 \cdot 2 = 12$. Otherwise $H^1(Y,\bar{\mathbb{Q}})$ is $V^{\oplus m}$, $m \geq 1$, plus a trivial representation of $M$. If $m = 1$, every $H^k(Y)$ is, as a representation of $M$, a sum of copies of the $\bigwedge^a V$, none of which contains an $S^2V_i$ (Proposition 21.6(i)), while $W$ contains all three; so $m \geq 2$ and $\dim H^1(Y) \geq 16$. In every case $\dim Y = \frac{1}{2}\dim H^1(Y) \geq 6$.

For the rest, a Hodge class on $S \times Y$ acting nontrivially on the irreducible $T(S) \cong W$ carries it isomorphically onto a sub-Hodge structure of some $H^k(Y,\mathbb{Q})(j)$, and if $Z_2 \circ Z_1$ acts nontrivially on $T(S)$, so does $Z_1$. ■

Among the powers of $X$ the smallest carrier of $T$ is $X \times X$, of dimension eight, where $T(-1)$ sits in $H^2$ (Proposition 21.6(ii)). Whatever is known about abelian varieties of dimension at most five, in particular Markman's theorem in dimension four [Mar25a], no link passes through one.

*Remark* 21.14 (The bound six as a statement about Hodge structures). The bound of Proposition 21.13 is, as far as we know, the best that Hodge theory gives. Let $K \subset D$ be a CM quadratic extension of $F$, $\chi_i$ the character of the $i$-th factor of the torus $\mathrm{Res}_{F/\mathbb{Q}} K^1$ over $\bar{\mathbb{Q}}$, and $H = \bigoplus_i (V_i\chi_i \oplus V_i\chi_i^{-1})$, of dimension twelve, with types $(1,0)$ on $V_i\chi_i$ and $(0,1)$ on $V_i\chi_i^{-1}$ for $i = 1,2$, and with the Hodge cocharacter of $V_3$ on the two summands with $i = 3$: the signature $(2,0)$, $(2,0)$, $(1,1)$. The characters $\chi_i^{\pm 1}\chi_j^{\pm 1}$ with $i \neq j$ and $\chi_i^{\pm 2}$ are nontrivial, so the part of $\bigwedge^2 H$ on which the torus acts trivially is $\bigoplus_i V_i\chi_i \otimes V_i\chi_i^{-1} = \bigoplus_i (S^2 V_i \oplus \bigwedge^2 V_i)$, that is, $T_1 \oplus T_2 \oplus T_3$ plus three trivial summands. For $i = 1,2$ the summand $T_i$ is of type $(1,1)$, and $S^2V_3$ has one dimension in each of the types $(2,0)$, $(1,1)$, $(0,2)$, so this copy of $T$ has Hodge numbers $(1,7,1)$, those of $T(S_\lambda)$. We recall without proof that these are the Hodge structures on $H^1(Z,\mathbb{Q}) \cong D$ of the abelian sixfolds $Z$ with multiplication by $K$ over the Shimura curves of PEL type attached to $K$ and $D$, which go back to Shimura, and that at a suitable member the copy of $T$ is isomorphic to $T(S_\lambda)$, the Hodge structure on it depending only on the image of the Hodge cocharacter in $G^{\mathrm{ad}}$. Nothing in the paper uses this, and it would not give a link: that would still need an algebraic cycle on $S_\lambda \times Z$ acting nontrivially on $T(S_\lambda)$.

*Remark* 21.15 (The K3 surfaces at the CM points). Apply the construction of Theorem 21.4(i) at a CM point $c$ of a family as in Corollary 20.22, with the Hodge structure that $X_c$ induces on $T$, and call the surface $S_c$. By the proof of Theorem 20.29 the Hodge group of $X_c$ is $\mathrm{Res}_{F/\mathbb{Q}} L^1$ for a quadratic extension $L$ of $F$ contained in $D$; $L$ is a CM field, since the torus is compact at $\sigma_1$ and $\sigma_2$, where $D$ is definite, and at $\sigma_3$, where it contains the nontrivial image of the circle of the Hodge structure. The torus acts on $T = D^0$ by conjugation, with invariants $D^0 \cap L$, the three-dimensional space $L^0$ of elements of $L$ of reduced trace zero. So $\mathrm{NS}(S_c)_{\mathbb{Q}} = T^{\perp} \oplus L^0$ has rank sixteen. Write $D = L \oplus Lj$ with $jx = \bar{x}j$ for $x \in L$. Then $Lj$ is the $q_\lambda$-orthogonal complement of $L^0$ in $D^0$, because $\mathrm{trd}(yj) = 0$ for $y \in L$, so $T(S_c)_{\mathbb{Q}} = Lj$ has rank six; $t \in L^1$ acts on it by multiplication by $t^2$, so left multiplication by $L$ consists of Hodge endomorphisms and $T(S_c)$ has complex multiplication by the sextic CM field $L$. At the CM points the Kuga–Satake question therefore lies among K3 surfaces of Picard number sixteen, among which the known cases recalled in Remark 21.5 are particular families. Membership of some $S_c$ in one of them would not help: at $c$ the exceptional classes are already algebraic (Theorem 20.29), and what is missing is the propagation along $C$.

The last reduction replaces the eightfold $X \times X$ by a fourfold.

**Proposition 21.16** (A surface in $X$)**.** *In Setup 21.1, let $i\colon \Sigma \to X$ be a smooth surface that is the intersection of two ample divisors, and $L$ the cup product with $\theta$. Then $i_*i^* = cL^2$ on $H^2(X,\mathbb{Q})$ for a rational number $c > 0$. For $a \in F$ let $\rho_a$ be the Hodge endomorphism of $H^2(X,\mathbb{Q})$ that is zero on $\mathbb{Q}\theta$ and acts on $U$ as a (Proposition 21.2(ii)), and put $\rho'_a = c^{-1}i^*\rho_a L^{-2} i_*$, a Hodge endomorphism of $H^2(\Sigma,\mathbb{Q})$, where $L^{-2}$ inverts*

*$L^2\colon H^2(X)\to H^6(X)$. Then $\rho_a$ is induced by an algebraic cycle on $X\times X$ if and only if $\rho'_a$ is induced by an algebraic cycle on $\Sigma\times\Sigma$. So the two exceptional classes of $X\times X$ are algebraic if and only if the Hodge classes in $H^2(\Sigma)\otimes H^2(\Sigma)\subset H^4(\Sigma\times\Sigma,\mathbb{Q})$ that correspond to the $\rho'_a$, $a\in F$, are algebraic on the fourfold $\Sigma\times\Sigma$.*

*Proof.* By Proposition 20.74(i) the two divisors have classes $d_1\theta$ and $d_2\theta$ with $d_1,d_2>0$, so $i_*i^*\alpha=\alpha\cup[\Sigma]=d_1d_2\theta^2\alpha$ and $c=d_1d_2$. By hard Lefschetz $L^2\colon H^2(X)\to H^6(X)$ is an isomorphism, so $i^*$ is injective on $H^2(X)$, and $L^{-2}$ is induced by an algebraic cycle, $X$ being an abelian variety [Kle68, §2]. For $\alpha\in H^2(X)$, $\rho'_a(i^*\alpha)=c^{-1}i^*\rho_aL^{-2}(cL^2\alpha)=i^*\rho_a\alpha$. If $\beta\in H^2(\Sigma)$ is orthogonal to $i^*H^2(X)$, then $\int_X i_*\beta\cup\gamma=\int_\Sigma\beta\cup i^*\gamma=0$ for every $\gamma\in H^2(X)$, so $i_*\beta=0$ and $\rho'_a\beta=0$. Hence $\rho'_a$ is $i^*\rho_a(i^*)^{-1}$ on $i^*H^2(X)$ and zero on its orthogonal complement, and $\rho_a=c^{-1}L^{-2}i_*\rho'_ai^*$. Both formulas are composites of correspondences with the graph of $i$, its transpose and $L^{-2}$, so $\rho_a$ is algebraic if and only if $\rho'_a$ is. The exceptional classes are algebraic if and only if every $\rho_a$ is, by Proposition 21.2(ii), since 1 and the projection onto $\mathbb{Q}\theta$ are algebraic. The last statement uses Poincaré duality on $\Sigma$ and the algebraicity of the Künneth projectors of a surface [Mur90]. ∎

This is the usual reduction to a complete intersection surface, and it does not make the problem easier. By adjunction $K_\Sigma$ is the restriction of the sum of the two ample divisors, so $\Sigma$ is a surface of general type, and the $\rho'_a$ are Hodge classes of degree four on the self-product of a surface of general type.

*Remark* 21.17 (The forms of the first case of (F2)). Theorem 21.7, Corollary 21.11 and one direction of Proposition 21.12 are implications between cases of the Hodge conjecture; Proposition 21.9 follows from Markman's theorem and is also contained in [Li26]; the rest of the results from Proposition 21.6 to Proposition 21.16 are statements about Hodge structures and correspondences. Together they give the first case of (F2) many equivalent forms. By Corollary 20.30, Corollary 21.10 and Proposition 21.16 it is equivalent to each of the following: the algebraicity of the exceptional classes of $X_t\times X_t$ for uncountably many $t\in C$; the Hodge conjecture for every power of every fibre $X_t$; $B(W\times_C W)$; $B$ for every fibre power of $W$; the Hodge conjecture for $W\times_C W$, for the curves of Corollary 20.30(d); and, for uncountably many $t$, the algebraicity of the classes $\rho'_a$ on $\Sigma_t\times\Sigma_t$ for one complete intersection surface $\Sigma_t\subset X_t$. For a family defined over a number field it is also equivalent to the algebraicity, at one point $t\in C(\mathbb{C})\setminus C(\bar{\mathbb{Q}})$, of one cycle on one power of $X_t$ whose class is not fixed by the cyclic permutation of the three tensor factors of $V$ (Proposition 21.47). It is implied by the Kuga–Satake Hodge conjecture for the surfaces $S_\lambda$ attached to uncountably many fibres (Corollary 21.45(ii)), which by Proposition 21.12 is equivalent to it together with a link; the cases of the Kuga–Satake Hodge conjecture recalled in Remark 21.5 are of other shapes, and none has transcendental rank nine with real multiplication by a cubic field.

Two limits govern the constructions at hand. First, the Kuga–Satake class has a nonzero component in $T(S)\otimes H^*(A)$, $A$ a power of $X$, whereas a product $p^*\alpha\cup q^*\beta$ of pull-backs of algebraic classes of $S$ and of $A$ has none, the algebraic classes of $S$ lying in $H^0\oplus\mathrm{NS}(S)_\mathbb{Q}\oplus H^4$; so no product of pull-backs gives it, whatever is known on $A$, and a new correspondence between $S$ and an abelian variety is needed. Second, $\mathrm{Sp}(V,\psi)$ acts on $H^*(X^m,\mathbb{Q})=\bigwedge^*(V\otimes\mathbb{Q}^m)$ through $V$, compatibly with products, with pull-back and push-forward along homomorphisms and with integration, hence with the composition of correspondences. The divisor classes of $X^m$, the Hodge classes of $X$ and the Weil classes of every abelian variety isogenous to a power of $X$ are invariant under it: for a CM field $E\subset M_m(\mathbb{Q})=\mathrm{End}^0(X^m)$, each component over $\mathbb{C}$ of the top exterior power over $E$ of

$V \otimes \mathbb{Q}^m$ is the top exterior power of $V \otimes U$ for a complex vector space $U$ on which $V$ plays no part, that is $(\det V)^{\otimes \dim U} \otimes (\det U)^{\otimes 8}$, and $\mathrm{SL}(V)$ acts trivially on it. So every class obtained from these by the operations above is $\mathrm{Sp}(V,\psi)$-invariant, while the exceptional classes are not, the $\mathrm{Sp}(V,\psi)$-invariants of $H^4(X \times X, \mathbb{Q})$ being the six products of divisor classes (proof of Proposition 20.74). Proposition 21.48 extends this to every abelian variety, with the Hodge classes of the abelian varieties without isogeny factor $X$ and the Hodge classes of the powers of $X$ fixed by the cyclic permutation of the three tensor factors among the generators, and with a subgroup $G \cdot A_3$ of $\mathrm{Sp}(V,\psi)$ in place of $\mathrm{Sp}(V,\psi)$. What escapes it are classes on products $X^m \times Z$ that are not invariant under $G \cdot A_3$ acting through $V$, for instance, with $T(-1)$ in $H^2(Z)$ as in Remark 21.14, the Hodge classes of $X \times X \times Z$ that pair the two copies of $T(-1)$.

The statement needed can be taken in either of two forms. One is (V) along $C$ for the exceptional classes, from a CM point, where they are algebraic (Theorem 20.29), and Section 21.2 studies it. The other is (V) for the Kuga–Satake class along the seven-dimensional family of K3 surfaces whose Néron–Severi group contains $\mathrm{NS}(S_\lambda)$, on which it stays a Hodge class, from a member at which it is algebraic; it implies the Kuga–Satake Hodge conjecture for $S_\lambda$ and with it everything above.

The Kuga–Satake class is algebraic at some members of this family. Call the family $\mathcal{K}$, keep the embedding $T \subset \Lambda_\mathbb{Q}$ of the proof of Theorem 21.4(i), and for $S \in \mathcal{K}$ let $\kappa_S$ be the Kuga–Satake class of $T$ with the Hodge structure that $S$ induces on it, a Hodge class on $S \times A_S \times A_S$ with $H^1(A_S, \mathbb{Q}) = C^+(T)$; at $S_\lambda$ it is the class of Theorem 21.4. If $P = T(S)_\mathbb{Q}$ is smaller than $T$, then $N = P^\perp \cap T$ lies in $\mathrm{NS}(S)_\mathbb{Q}$. The complex structure on $C^+(T)$ is left multiplication by an element of $C^+(P \otimes \mathbb{R})$, which commutes with right multiplications; so $C^+(T) = C^+(P) \otimes C^+(N) \oplus C^-(P) \otimes C^-(N)$ is, as a Hodge structure, a sum of copies of $C^+(P)$, right multiplication by an anisotropic vector of $P$ carrying $C^-(P)$ onto $C^+(P)$, and the map $\Psi$ of Proposition 11.7, with $v_0 \in P$, carries $N$ to Hodge endomorphisms of $A_S$, left multiplication by $N$ commuting with that of $C^+(P \otimes \mathbb{R})$. So $\kappa_S$ is algebraic if and only if the Kuga–Satake class of $P$ is. By the argument recalled in Remark 21.5, $\kappa_S$ is therefore algebraic at the members of Picard number at least 19, which are dense in $\mathcal{K}$, at those of Picard number 18 with $T(S)_\mathbb{Q}$ isotropic and at those of Picard number 17 with $T(S)_\mathbb{Q}$ of Witt index two. In the last two cases $T(S)_\mathbb{Q} \cong L_\mathbb{Q}$ for an even lattice $L = U \oplus \langle 2a \rangle \oplus \langle -2b \rangle$ or $U^2 \oplus \langle -2b \rangle$, which embeds primitively in $U^3 \subset \Lambda$. The surjectivity of the period map gives a K3 surface $S'$ with $T(S') = L$, which has a Shioda–Inose structure [Mor84, Corollary 6.4], and a Hodge isometry $T(S)_\mathbb{Q} \cong T(S')_\mathbb{Q}$; by Witt's theorem it extends to a rational Hodge isometry $H^2(S, \mathbb{Q}) \cong H^2(S', \mathbb{Q})$, which is algebraic by [Bus19]. The case of Picard number 17 fills subfamilies of dimension three. The space $T$ has Witt index two: its signature $(2, 7)$ allows no totally isotropic subspace of dimension three, and an anisotropic kernel of dimension at least seven would be indefinite, hence isotropic by the Hasse–Minkowski theorem [OMe00]. So $T = H \oplus H \oplus K$ with $H$ a hyperbolic plane and $K$ negative definite, and each of the countably many rational subspaces $W \subset T$ of signature $(2, 3)$ and Witt index two, for instance $H \oplus H \oplus \mathbb{Q}k$ with $k \in K$, defines a subfamily $\mathcal{K}_W = \{S : T(S)_\mathbb{Q} \subseteq W\}$ of dimension three. At a very general point of a component of $\mathcal{K}_W$ one has $T(S)_\mathbb{Q} = W$, so $\kappa$ is algebraic there. After a finite cover with level structure the component carries a family of K3 surfaces and of Kuga–Satake abelian varieties, on which $\kappa$ is a flat section of Hodge classes, and, as in the proof of Lemma 20.27, the points at which it is algebraic form a countable union of images of components of relative Hilbert schemes, closed by properness; by the Baire category theorem one of them is the whole component, so $\kappa$ is algebraic at every point of $\mathcal{K}_W$. The

Mumford curve meets none of these loci: its surfaces have Picard number thirteen, and sixteen at the CM points (Theorem 21.4(i), Remark 21.15). What is missing is (V) for $\kappa$ along $\mathcal{K}$ from them, or a bound on the Hilbert data of cycles representing $\kappa$ over a Zariski dense union of the $\mathcal{K}_W$, which would give $\kappa$ on the whole of $\mathcal{K}$ by the same argument; neither is known.

21.2. **Rigidity and a numerical criterion for the Mumford classes.** The first case of (F2) is a propagation problem along one compact Shimura curve: the exceptional classes are algebraic at every CM point (Theorem 20.29), and they are algebraic on every member exactly when $B(W \times_C W)$ holds (Corollary 20.30). We ask here whether the propagation can be obtained without the Lefschetz standard conjecture, by one of the mechanisms that have proved other cases of the Hodge conjecture: degeneration, deformation into a larger family on which the class stays of Hodge type, specialisation to special points, or semiregularity. The first two cannot work; the third works exactly when the representing cycles have bounded degree; and the fourth reduces to one dimension count at one CM point. We do not prove that the classes are algebraic at a very general point.

**Setup 21.18** (The classes $\omega_a$)**.** Keep Setup 21.1 and let $W \to C$ be a family as in Corollary 20.22, with fibres $X_t$. Over $\bar{\mathbb{Q}}$ let $\pi_0, \pi_{12}, \pi_{13}, \pi_{23} \in H^2(X) \otimes H^2(X) \subset H^4(X \times X)$ be the classes that correspond, under the isomorphism $H^2 \otimes H^2 \cong \operatorname{End} H^2$ given by the form $\bigwedge^2 \psi$ on $H^2(X) = \bigwedge^2 V^\vee$, to the projections onto $\mathbb{Q}\theta$, $U_{12}$, $U_{13}$ and $U_{23}$ of Proposition 21.2(i), and for $a = (a_0, a_1, a_2, a_3)$ put

$$\omega_a = a_0\pi_0 + a_1\pi_{12} + a_2\pi_{13} + a_3\pi_{23}, \qquad L_0(a) = a_0 + a_1 + a_2 + a_3.$$

The four tensors are invariant under $G$ and under the exchange $\tau$ of the two factors of $X \times X$, because $\bigwedge^2 \psi$ is symmetric; so each $\omega_a$ is a flat section over $C$ of Hodge type at every point. The argument of Proposition 21.2(ii) shows that the rational ones are those with $a_0 \in \mathbb{Q}$ and $(a_1, a_2, a_3) = (\sigma_3(\alpha), \sigma_2(\alpha), \sigma_1(\alpha))$ for some $\alpha \in F$; we call such a class *exceptional* when $\alpha \notin \mathbb{Q}$. The products of divisor classes in $H^2 \otimes H^2$ are the rational $\omega_a$ with $\alpha \in \mathbb{Q}$, and $\pi_0$ is a multiple of $\mathrm{pr}_1^*\theta \cup \mathrm{pr}_2^*\theta$; so adding a rational multiple of $\pi_0$ to an exceptional class changes $L_0$ and not the question of algebraicity. For $c \in C$ write $Y = X_c \times X_c$, let $HT^2(Y) = H^{0,2}(Y) \oplus H^1(T_Y) \oplus H^0(\bigwedge^2 T_Y)$, of dimension $28 + 64 + 28 = 120$, act on $H^*(Y, \mathbb{C})$ by contraction as in Setup 18.54, and let $\kappa \in H^1(T_Y)$ be the Kodaira–Spencer class at $c$ of the family $W \times_C W \to C$. Put

$$e(\omega) = \dim HT^2(Y) - \dim \operatorname{Ann}_{HT^2(Y)}(\omega) = 120 - \dim \operatorname{Ann}_{HT^2(Y)}(\omega).$$

**Lemma 21.19** (Linear relations among conjugates)**.** *Let $F$ be a cubic number field with embeddings $\sigma_1, \sigma_2, \sigma_3$, let $\alpha \in F \setminus \mathbb{Q}$, and let $c_0, c_1, c_2, c_3, a_0 \in \mathbb{Q}$. The three conjugates $\sigma_i(\alpha)$ are pairwise distinct and nonzero, and if $c_0a_0 + \sum_i c_i\sigma_i(\alpha) = 0$ then $c_1 = c_2 = c_3$.*

*Proof.* The minimal polynomial of $\alpha$ has degree three, since $[F : \mathbb{Q}]$ is prime, and is separable, so the conjugates are distinct; they are nonzero because $\alpha \neq 0$. Let $\Gamma$ be the Galois group of a normal closure; it permutes the $\sigma_i$ transitively by $\gamma \circ \sigma_i = \sigma_{\gamma(i)}$. Put $\beta = \alpha - \operatorname{Tr}_{F/\mathbb{Q}}(\alpha)/3$, so that $\beta \neq 0$ has trace zero, and $r = -c_0a_0 - (c_1 + c_2 + c_3)\operatorname{Tr}_{F/\mathbb{Q}}(\alpha)/3 \in \mathbb{Q}$; the hypothesis reads $\sum_i c_i\sigma_i(\beta) = r$. Applying every $\gamma \in \Gamma$ and averaging, each $\sigma_j$ occurs $|\Gamma|/3$ times in the place of each $\sigma_i$, so $r = \frac{1}{3}(\sum_i c_i)\sum_j \sigma_j(\beta) = 0$. The relations $c \in \mathbb{Q}^3$ with $\sum_i c_i\sigma_i(\beta) = 0$ form a $\Gamma$-stable subspace $N$ containing $(1, 1, 1)$. The plane $\{\sum c_i = 0\}$ is an irreducible representation of $\Gamma$ over $\mathbb{Q}$: being transitive on three letters, $\Gamma$ is the alternating or the full symmetric group, so it contains the 3-cycles, which have no rational eigenvector in that plane. If $N$ were larger than the line of $(1, 1, 1)$, it would therefore contain the plane and be all of $\mathbb{Q}^3$, and $\sigma_1(\beta) = 0$. Hence $c \in \mathbb{Q}(1, 1, 1)$. ∎

**Lemma 21.20** (One exceptional class gives the others)**.** *Let $t \in C$. If one rational exceptional class $\omega$ is algebraic on $X_t \times X_t$, then every Hodge class in $H^2(X_t) \otimes H^2(X_t)$ is algebraic; in particular so are the two exceptional classes of Proposition 20.74.*

*Proof.* At a CM point every Hodge class of $X_t \times X_t$ is algebraic (Theorem 20.29). Let $t$ be any other point, so that the Hodge group of $X_t$ is $G$ (proof of Corollary 20.30), and write $X = X_t$. For $x \in H^2(X) \otimes H^2(X)$ let $E_x$ be the endomorphism $y \mapsto x_*(\theta^2 \cup y)$ of $H^2(X, \mathbb{Q})$, where $x_*$ is the action of $x$ as a correspondence. By Poincaré duality and the hard Lefschetz isomorphism $\theta^2 \colon H^2 \to H^6$, the map $x \mapsto E_x$ is a $G$-equivariant isomorphism onto $\operatorname{End} H^2(X, \mathbb{Q})$, so it carries the Hodge classes onto the Hodge endomorphisms, which are $\mathbb{Q} \times F$ by Proposition 21.2(ii). It carries the products of divisor classes, which are invariant under $\mathrm{Sp}(V, \psi)$, onto the $\mathrm{Sp}(V, \psi)$-equivariant endomorphisms, which are $\mathbb{Q} \times \mathbb{Q}$; hence it carries $\omega$ to an element $(r, \beta)$ with $\beta \notin \mathbb{Q}$. The algebraic Hodge classes form a subspace $A$ containing the products of divisor classes, and its image under $x \mapsto E_x$ is closed under composition: if $x_1, x_2$ are algebraic, then the composite correspondence $x_1 \circ \Delta_*(\theta^2) \circ x_2$ is algebraic, and so is its Künneth component in $H^2 \otimes H^2$, since the Künneth projectors of $X \times X$ are polynomials in the graphs of $[2] \times 1$ and $1 \times [2]$, which act on $H^i(X) \otimes H^j(X)$ by $2^i$ and $2^j$ (as in Step 1 of the proof of Theorem 20.21); that component $x$ has $E_x = E_{x_1} E_{x_2}$. So the image of $A$ is a subalgebra of $\mathbb{Q} \times F$ containing $\mathbb{Q} \times \mathbb{Q}$ and $(r, \beta)$, hence $\mathbb{Q} \times \mathbb{Q}[\beta] = \mathbb{Q} \times F$, because $\beta$ generates the cubic field $F$. Thus $A$ is the space of all Hodge classes in $H^2(X) \otimes H^2(X)$. ∎

**Theorem 21.21** (Rigidity of the exceptional classes)**.** *Let $c \in C$, $Y = X_c \times X_c$, and let $\omega = \omega_a$ be a rational exceptional class.*

(i) $\operatorname{Ann}_{H^1(T_Y)}(\omega) = \mathbb{C}\kappa$ *and* $\operatorname{Ann}_{H^{0,2}(Y)}(\omega) = 0$.

(ii) $\operatorname{Ann}_{H^0(\bigwedge^2 T_Y)}(\omega) = 0$ *if and only if* $L_0(a) = a_0 + \operatorname{Tr}_{F/\mathbb{Q}}(\alpha) \neq 0$. *In particular* $\operatorname{Ann}_{HT^2(Y)}(\omega) = \mathbb{C}\kappa$ *and* $e(\omega) = 119$ *whenever* $L_0(a) \neq 0$, *which can always be arranged by adding a rational multiple of* $\pi_0$.

(iii) *Let $\mathfrak{S}$ be the Siegel space of weight one Hodge structures on $H^1(Y, \mathbb{Q})$ polarised by $\psi \oplus \psi$, of dimension* 36. *The germ at $Y$ of the locus in $\mathfrak{S}$ where $\omega$ stays of Hodge type is the germ of the curve $\{X_t \times X_t\}$ through $Y$.*

*Proof. Notation.* Over $\mathbb{C}$ write $V = V_1 \otimes V_2 \otimes V_3$ and $\psi = \epsilon_1 \otimes \epsilon_2 \otimes \epsilon_3$, as in the proof of Lemma 21.3, and identify $V$ with $H^1(X_c, \mathbb{C})$ and with its dual through $\psi$. As at every point of $C$, the Hodge cocharacter lies in the third factor: $V_3^{1,0} = \mathbb{C}e$, $V_3^{0,1} = \mathbb{C}f$, $V^{1,0} = V_1 \otimes V_2 \otimes e$ and $V^{0,1} = V_1 \otimes V_2 \otimes f$. Let $N \in \mathfrak{sl}(V_3)$ be the nilpotent element with $Ne = f$, $Nf = 0$. Then $H^1(Y, \mathbb{C}) = V_{(1)} \oplus V_{(2)}$, two copies of $V$, with the form $\psi \oplus \psi$, and $\omega \in \bigwedge^2 V_{(1)} \otimes \bigwedge^2 V_{(2)}$.

*Step 1: operators.* For $J \subseteq \{1, 2, 3\}$ let $E_J \subset \operatorname{End} V$ be the tensor product of $\mathfrak{sl}(V_i)$ for $i \in J$ and of $\mathbb{C} \cdot 1$ for $i \notin J$. Then $\operatorname{End} V = \bigoplus_J E_J$, and the $E_J$ are irreducible and pairwise non-isomorphic under conjugation by $G$. The $\psi$-adjoint of an element of $\mathfrak{sl}(V_i)$ is its negative, so $E_J$ consists of $\psi$-skew operators when $|J|$ is odd and of $\psi$-symmetric ones when $|J|$ is even. The map $v \wedge w \mapsto \psi(v, \cdot)w - \psi(w, \cdot)v$ identifies $\bigwedge^2 V$ with the $\psi$-symmetric operators $E_\emptyset \oplus E_{12} \oplus E_{13} \oplus E_{23}$, which are $\mathbb{C}\theta$, $U_{12}$, $U_{13}$, $U_{23}$; a direct expansion shows that $\bigwedge^2 \psi$ becomes the form $\langle M, M' \rangle = \frac{1}{2} \operatorname{tr}(MM')$, and that the derivation action of $X \in \operatorname{End} V$ becomes $M \mapsto XM + MX^\psi$, that is $D_X M = [X, M]$ if $X$ is $\psi$-skew and $D_X M = XM + MX$ if $X$ is $\psi$-symmetric. In particular $D_X^\dagger = \mp D_X$ in the two cases, $\dagger$ denoting the adjoint for $\langle\,,\rangle$. By the Cayley–Hamilton theorem $XY = \frac{1}{2}[X, Y] + \frac{1}{2} \operatorname{tr}(XY)$

for $X, Y \in \mathfrak{sl}(V_i)$. Finally, by Setup 21.18, $\omega = \sum_j M_j \otimes P_a M^j$ for a basis $(M_j)$ of $\bigwedge^2 V$ with dual basis $(M^j)$, where $P_a$ acts on $E_\emptyset, E_{12}, E_{13}, E_{23}$ by $a_0, a_1, a_2, a_3$. As $P_a$ commutes with $G$, it preserves the Hodge types.

The space $\bigwedge^2 V^{0,1}$ consists of the $\psi$-symmetric operators that kill $V^{0,1}$ and take values in it, namely $\mathfrak{sl}(V_1) \otimes 1 \otimes N \oplus 1 \otimes \mathfrak{sl}(V_2) \otimes N \subset E_{13} \oplus E_{23}$, on which $P_a$ is $a_2$ and $a_3$. The map $v \otimes w \mapsto \psi(v, \cdot)w$ identifies $V^{0,1} \otimes V^{0,1}$ with $\operatorname{Hom}(V^{1,0}, V^{0,1}) = \operatorname{End} V_1 \otimes \operatorname{End} V_2 \otimes \mathbb{C}N = \bigoplus_{J \ni 3} E_J^-$, where $E_J^- = E_J \cap \operatorname{Hom}(V^{1,0}, V^{0,1})$ has dimension $1, 3, 3, 9$ for $J = \{3\}, \{1,3\}, \{2,3\}, \{1,2,3\}$.

*Step 2: the pieces.* The space $HT^2(Y)$ acts as follows: $H^{0,2} = \bigwedge^2 H^{0,1}$ by the wedge product, $H^1(T_Y) = \operatorname{Hom}(H^{1,0}, H^{0,1})$ by derivations, and $H^0(\bigwedge^2 T_Y) = \bigwedge^2 (H^{1,0})^\vee$, which $\psi \oplus \psi$ identifies with $\bigwedge^2 H^{0,1}$, by contraction. Write $\eta \in \bigwedge^2 H^{0,1}$ as $\eta_{11} + \eta_{12} + \eta_{22}$ with $\eta_{11} \in \bigwedge^2 V^{0,1}_{(1)}$, $\eta_{12} \in V^{0,1}_{(1)} \otimes V^{0,1}_{(2)}$, $\eta_{22} \in \bigwedge^2 V^{0,1}_{(2)}$, and $\xi \in H^1(T_Y)$ as a matrix $(\xi_{st})$ with $\xi_{st} \in \operatorname{Hom}(V^{1,0}_{(t)}, V^{0,1}_{(s)})$. The products $\eta_{11} \wedge \omega$, $\eta_{12} \wedge \omega$, $\eta_{22} \wedge \omega$ lie in different summands $\bigwedge^p V_{(1)} \otimes \bigwedge^q V_{(2)}$, and so do the three contractions, and $\xi_{11}\omega + \xi_{22}\omega$, $\xi_{12}\omega$, $\xi_{21}\omega$. So each annihilator is computed piece by piece, and $\tau$ exchanges the pieces 11 and 22, 12 and 21. For a rational exceptional class, Lemma 21.19 gives: $a_1, a_2, a_3$ are pairwise distinct and nonzero, and no linear form in $a_0, \dots, a_3$ with rational coefficients that are not all equal on $a_1, a_2, a_3$ vanishes. We use this for $a_i \pm a_j$ $(i \neq j)$, $a_2 + 3a_3$, $3a_2 + a_3$ and the three forms in Step 4.

*Step 3: $H^{0,2}$.* Let $(x_j)$ be a basis of $\bigwedge^2 V^{1,0}$ and $(x^j)$ the dual basis of $\bigwedge^2 V^{0,1}$. The component of $\omega$ in $\bigwedge^2 V^{1,0}_{(1)} \otimes \bigwedge^2 V^{0,1}_{(2)}$ is $\sum_j x_j \otimes K x^j$, where $K = P_a$ on $\bigwedge^2 V^{0,1}$ is invertible because $a_2 a_3 \neq 0$. For $\eta_{11} \neq 0$ the part of $\eta_{11} \wedge \omega$ in $H^{2,2}(X_c) \otimes H^{0,2}(X_c)$ is $\sum_j (\eta_{11} \wedge x_j) \otimes K x^j$, which is nonzero because $\bigwedge^2 V^{0,1} \otimes \bigwedge^2 V^{1,0} \to \bigwedge^4 V$ is injective; by $\tau$ the same holds for $\eta_{22}$. For $\eta_{12} = \sum_r v_r \otimes w_r$ the part of $\eta_{12} \wedge \omega$ in $H^{2,1} \otimes H^{0,3}$ is $\sum_{r,j} (v_r \wedge x_j) \otimes (w_r \wedge K x^j)$. If it vanishes, then, as $V^{0,1} \otimes \bigwedge^2 V^{1,0} \to \bigwedge^3 V$ is injective and the $K x^j$ span $\bigwedge^2 V^{0,1}$, $\sum_r \phi(v_r)\, w_r \wedge y = 0$ for every linear form $\phi$ on $V^{0,1}$ and every $y \in \bigwedge^2 V^{0,1}$; as $\dim V^{0,1} = 4$ this forces $\sum_r \phi(v_r) w_r = 0$ for every $\phi$, so $\eta_{12} = 0$. Hence $\operatorname{Ann}_{H^{0,2}(Y)}(\omega) = 0$.

*Step 4: $H^0(\bigwedge^2 T_Y)$.* Up to signs and one nonzero factor fixed by the normalisation of the contraction, $\iota_{\eta_{11}}\omega = \sum_j \langle \eta_{11}, M_j \rangle P_a M^j = P_a \eta_{11}$, which is nonzero for $\eta_{11} \neq 0$ because $a_2 a_3 \neq 0$; likewise for $\eta_{22}$. For $\eta_{12} = v \otimes w$ the contraction is $\sum_j M_j v \otimes (P_a M^j) w$, and through $v \otimes w \mapsto \psi(v, \cdot)w$ it becomes the operator $Q_a(R) = \sum_j (P_a M^j)\, R\, M_j$ with $R = \psi(v, \cdot)w$, because the $M_j$ are $\psi$-symmetric. For a basis $(X_k)$ of $\mathfrak{sl}(V_i)$ and the dual basis $(X^k)$ for the trace form, $\sum_k X^k R X_k$ is $\frac{3}{2}R$ for $R = 1$ and $-\frac{1}{2}R$ for $R \in \mathfrak{sl}(V_i)$ (take $X_k = h, e, f$ and $X^k = h/2, f, e$). The dual basis of $E_{ij}$ for $\langle\, , \rangle$ is the tensor product of these dual bases, and that of $E_\emptyset$ is $\frac{1}{4}$. Hence $Q_a$ acts on $E_J$ by the scalar

$$q_J(a) = \tfrac{1}{4} a_0 + \sum_{\{i,j\}} a_{\{i,j\}}\, \lambda_i \lambda_j, \qquad \lambda_i = -\tfrac{1}{2}\ (i \in J), \quad \lambda_i = \tfrac{3}{2}\ (i \notin J),$$

where $a_{\{1,2\}} = a_1$, $a_{\{1,3\}} = a_2$, $a_{\{2,3\}} = a_3$; on the four pieces $E_J^-$ this is

$$4q_3 = a_0 + 9a_1 - 3a_2 - 3a_3, \qquad 4q_{13} = a_0 - 3a_1 + a_2 - 3a_3,$$
$$4q_{23} = a_0 - 3a_1 - 3a_2 + a_3, \qquad 4q_{123} = L_0(a).$$

The first three do not vanish. So the annihilator in $H^0(\bigwedge^2 T_Y)$ is zero when $L_0(a) \neq 0$, and is the piece $E^-_{123}$ of $\eta_{12}$, of dimension nine, when $L_0(a) = 0$.

*Step 5: $H^1(T_Y)$, the diagonal pieces.* Put $\Phi(x \otimes y)(z) = \langle z, x \rangle y$, so that $\Phi(\omega) = P_a$, $\Phi((D \otimes 1)\omega) = P_a D^\dagger$ and $\Phi((1 \otimes D)\omega) = D P_a$. Thus $\xi_{11}\omega + \xi_{22}\omega = 0$ if and only if

$P_a D^{\dagger}_{\xi_{11}} + D_{\xi_{22}} P_a = 0$. This condition is $G$-equivariant in $(\xi_{11}, \xi_{22}) \in \operatorname{End} V \oplus \operatorname{End} V$, so its solutions are the pairs $(b_{11}X, b_{22}X)$ with $X \in E_J$ and $(b_{11}, b_{22})$ in a subspace $M_J \subseteq \mathbb{C}^2$; we need $J \ni 3$. For $X = \bigotimes X_i \in E_J$ and $M = \bigotimes M_i \in E_I$, the product rule above gives $XM \pm MX$ as $2^{1-|I\cap J|}$ times the sum, over $T \subseteq I \cap J$ with $|T| \equiv |J|$ modulo 2, of the tensor with factor $[X_i, M_i]$ for $i \in T$, $\operatorname{tr}(X_i M_i)$ for $i \in (I \cap J) \setminus T$, and the remaining factor of $X$ or $M$ elsewhere; this term lies in $E_{(I \triangle J) \cup T}$ and is nonzero for suitable $X$ and $M$. On the block from $E_I$ to $E_{I'}$ the condition reads $b_{22}a_I = b_{11}a_{I'}$ for $|J|$ odd and $b_{22}a_I = -b_{11}a_{I'}$ for $|J|$ even, with $a_\emptyset = a_0$, $a_{12} = a_1$, $a_{13} = a_2$, $a_{23} = a_3$.

- $J = \{3\}$: the blocks from $E_{13}$ to $E_{13}$ and from $E_{23}$ to $E_{23}$ give $(b_{22} - b_{11})a_2 = 0$, so $M_J = \mathbb{C}(1, 1)$.
- $J = \{1, 2, 3\}$: there is a block between any two distinct $E_{12}, E_{13}, E_{23}$, so $b_{22}a_1 = b_{11}a_2$ and $b_{22}a_2 = b_{11}a_1$, whence $(b_{11} + b_{22})(a_1 - a_2) = 0$ and $b_{11} = -b_{22}$; then $b_{22}(a_1 + a_2) = b_{22}(a_1 + a_3) = 0$, and $b_{22} \neq 0$ would give $a_2 = -a_1 = a_3$. So $M_J = 0$.
- $J = \{1, 3\}$: the blocks between $E_{12}$ and $E_{23}$ give $b_{22}a_1 + b_{11}a_3 = b_{22}a_3 + b_{11}a_1 = 0$, of determinant $a_1^2 - a_3^2 \neq 0$, so $M_J = 0$; and $J = \{2, 3\}$ likewise, with $a_1^2 - a_2^2 \neq 0$.

So on the diagonal pieces the annihilator is the line of $(N, N)$, with $N$ acting on both copies of $V$.

*Step 6: $H^1(T_Y)$, the mixed pieces.* The map $\xi_{12}$ kills $V^{0,1}_{(2)}$, so the part of $\xi_{12}\omega$ in $H^{0,3}(X_c) \otimes H^{1,0}(X_c)$ comes from the component $\sum_j K x^j \otimes x_j$ of $\omega$ in $\bigwedge^2 V^{0,1}_{(1)} \otimes \bigwedge^2 V^{1,0}_{(2)}$ only, and is $\sum_j K x^j \wedge \xi_{12}(x'_j) \otimes x''_j$, where $x \mapsto x' \otimes x''$ is the coproduct $\bigwedge^2 \to V \otimes V$. Identify $V^{1,0}$ and $V^{0,1}$ with $W = V_1 \otimes V_2$ through $e$ and $f$, with the symmetric form $q = \epsilon_1 \otimes \epsilon_2$ and a volume form; then $\bigwedge^2 W$ is the space of $q$-skew operators $\mathfrak{sl}(V_1) \otimes 1 \oplus 1 \otimes \mathfrak{sl}(V_2)$, the Hodge star is $+1$ on one summand and $-1$ on the other, and $K$ is $a_2$ on the first and $a_3$ on the second. Pair the part above with $b \in W$ in the first factor, through the volume form, and with $\beta \in W$ in the second, through $q$. Writing $\operatorname{vol}(y \wedge y') = q(*y, y')$ on $\bigwedge^2 W$ turns the pairing into the trace of $x \mapsto K * \big(\xi_{12}(\iota_\beta x) \wedge b\big)$ on $\bigwedge^2 W$, and summing over orthogonal bases of the two summands gives, up to a nonzero factor and a sign, $q\big(c_a(\xi_{12})\beta, b\big)$ with

$$c_a(\xi) = a_2 \sum_k (X^k \otimes 1)\, \xi\, (X_k \otimes 1) - a_3 \sum_k (1 \otimes X^k)\, \xi\, (1 \otimes X_k)$$

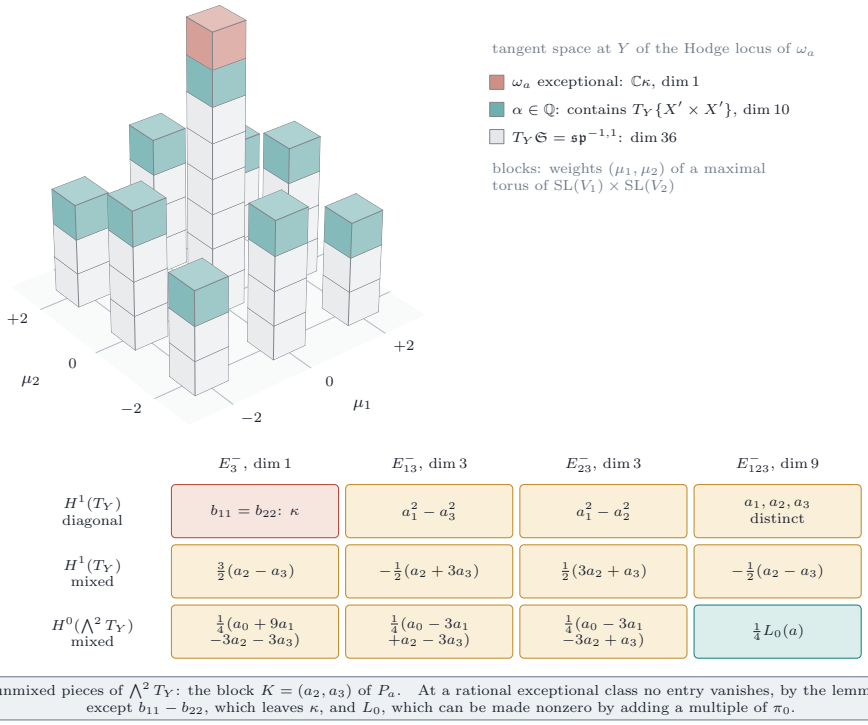
for $\xi \in \operatorname{End} W = \operatorname{End} V_1 \otimes \operatorname{End} V_2$. By the scalars of Step 4, $c_a$ acts on $1 \otimes 1$, $\mathfrak{sl}(V_1) \otimes 1$, $1 \otimes \mathfrak{sl}(V_2)$, $\mathfrak{sl}(V_1) \otimes \mathfrak{sl}(V_2)$, that is on $E_3^-$, $E_{13}^-$, $E_{23}^-$, $E_{123}^-$, by

$$\tfrac{3}{2}(a_2 - a_3), \qquad -\tfrac{1}{2}(a_2 + 3a_3), \qquad \tfrac{1}{2}(3a_2 + a_3), \qquad -\tfrac{1}{2}(a_2 - a_3),$$

none of which vanishes. So $\xi_{12}\omega \neq 0$ for $\xi_{12} \neq 0$, and by $\tau$ the same holds for $\xi_{21}$.

*Step 7: $\kappa$.* The family of Hodge structures along $C$ is the orbit of the Hodge structure at $c$ under the real points of the third factor of $G$, so the Kodaira–Spencer class $\kappa$ is the component of type $(-1, 1)$ of an element of $\mathfrak{sl}(V_3)$ acting on both copies of $V$, a nonzero multiple of $(N, N)$; it is nonzero because the Hodge structures along $C$ vary. By Steps 5 and 6, $\operatorname{Ann}_{H^1(T_Y)}(\omega) = \mathbb{C}\kappa$. With Steps 3 and 4 this proves (i) and (ii).

For (iii), a point of $\mathfrak{S}$ is a Hodge filtration $F^1$ on $H^1(Y, \mathbb{C})$, and $\omega$ is of Hodge type at it when $\omega \in F^2 H^4$, a holomorphic condition. For weight one every tangent direction of $\mathfrak{S}$ at $Y$ is an element $\xi$ of $\mathfrak{sp}(\psi \oplus \psi)^{-1,1} \subset H^1(T_Y)$, and it moves $F^2H^4$ by the derivation $\xi$ of $\bigwedge^* H^1$, which maps $H^{2,2}$ into $H^{1,3}$; so the Zariski tangent space of the Hodge locus is $\{\xi : \xi \lrcorner \omega = 0\}$, which is $\mathbb{C}\kappa$ by (i). This is the usual description of the tangent space of the Hodge locus of a class of type $(p, p)$ as the kernel of the contraction into $H^{p-1,p+1}$ [Voi03, Chapter 5], in its simplest case: a variation of weight one is subject to no horizontality




| | $E_3^-$, dim 1 | $E_{13}^-$, dim 3 | $E_{23}^-$, dim 3 | $E_{123}^-$, dim 9 |
|---|---|---|---|---|
| $H^1(T_Y)$ diagonal | $b_{11} = b_{22}$: $\kappa$ | $a_1^2 - a_3^2$ | $a_1^2 - a_2^2$ | $a_1, a_2, a_3$ distinct |
| $H^1(T_Y)$ mixed | $\frac{3}{2}(a_2 - a_3)$ | $-\frac{1}{2}(a_2 + 3a_3)$ | $\frac{1}{2}(3a_2 + a_3)$ | $-\frac{1}{2}(a_2 - a_3)$ |
| $H^0(\bigwedge^2 T_Y)$ mixed | $\frac{1}{4}(a_0 + 9a_1 - 3a_2 - 3a_3)$ | $\frac{1}{4}(a_0 - 3a_1 + a_2 - 3a_3)$ | $\frac{1}{4}(a_0 - 3a_1 - 3a_2 + a_3)$ | $\frac{1}{4}L_0(a)$ |

$H^{0,2}$ and the unmixed pieces of $\bigwedge^2 T_Y$: the block $K = (a_2, a_3)$ of $P_a$. At a rational exceptional class no entry vanishes, by the lemma on conjugates, except $b_{11} - b_{22}$, which leaves $\kappa$, and $L_0$, which can be made nonzero by adding a multiple of $\pi_0$.

FIGURE 55. Theorem 21.21. Top: the tangent space $\mathfrak{sp}^{-1,1}$ of the Siegel space $\mathfrak{S}$ at $Y = X_c \times X_c$, of dimension 36, split into nine blocks by the weights of a maximal torus of $\mathrm{SL}(V_1) \times \mathrm{SL}(V_2)$, one cube for each dimension. The tangent space of the Hodge locus of an exceptional class is the line $\mathbb{C}\kappa$ of the Mumford curve, in the block of weight $(0, 0)$; for $\alpha \in \mathbb{Q}$ it contains the tangent space of the diagonal Siegel space, of dimension 10 (Remark 21.22). Bottom: the scalars of Steps 4 to 6 of the proof on the pieces $E_J^-$ of $HT^2(Y)$; the pieces of $H^{0,2}$ and the unmixed pieces of $H^0(\bigwedge^2 T_Y)$ are controlled by the block $K$ of $P_a$ (Steps 3 and 4).

condition, so every direction of $\mathfrak{S}$ enters. The Hodge locus is an analytic germ whose Zariski tangent space is a line, so it lies in a smooth curve germ; it contains the smooth curve germ $\{X_t \times X_t\}$, whose tangent line is $\mathbb{C}\kappa$; so the two germs are equal. ∎

*Remark* 21.22 (Comparison with the divisor classes). For $\alpha \in \mathbb{Q}$ the class $\omega_a$ is a combination of products of divisor classes and is invariant under the whole of $\mathrm{Sp}(V, \psi)$ acting diagonally on $V \oplus V$, so its Hodge locus in $\mathfrak{S}$ contains the diagonal Siegel space $\{X' \times X'\}$, of dimension 10; in the proof above this shows in the scalars $a_2 - a_3$ of Step 6, which vanish when $a_1 = a_2 = a_3$. The products of divisor classes therefore move with $X' \times X'$ for every $X'$, whereas the exceptional classes move with $X_t \times X_t$ for $t \in C$ only and cease to be of Hodge type under every deformation of $Y$ off the curve, however small.

**Corollary 21.23** (No degeneration and no larger family)**.** *Let $\omega$ be a rational exceptional class on $X_c \times X_c$. Every connected complex analytic family of polarised abelian eightfolds through $X_c \times X_c$ on which $\omega$ stays of Hodge type lies, near each point of the curve, inside the curve $\{X_t \times X_t : t \in C\}$, which is compact. In particular no such family degenerates, and on no member of it other than the CM points of the curve is $\omega$ a combination of*

*products of divisor classes or of Weil classes of CM fields acting on the member or on its quotients.*

*Proof.* By Theorem 21.21(iii) the Hodge locus of $\omega$ coincides with the curve in a neighbourhood of every point of the curve, so a connected family through $Y$ on which $\omega$ stays of Hodge type maps into the curve near each of its points over the curve; by connectedness and the identity theorem its image lies in the curve. The curve is compact because $D$ is a division algebra, so that the arithmetic group of the Shimura curve is cocompact [Mum69]. Indeed, by Setup 21.1 the algebra $D$ is definite at $\sigma_1$ and $\sigma_2$, so $D \otimes_{F,\sigma_1} \mathbb{R}$ is Hamilton's quaternions and $D$ is not $M_2(F)$: it is a division algebra, split at the one real place $\sigma_3$. The norm one group of an order in $D$, embedded in $\mathrm{SL}_2(\mathbb{R})$ through $\sigma_3$, is a Fuchsian group of finite covolume with no parabolic element, since a parabolic $\gamma \neq \pm 1$ would make $\gamma \mp 1$ a nonzero nilpotent element of $D$; and a Fuchsian group of finite covolume without parabolic elements has compact quotient.

For the last assertion: a member of the curve other than a CM point has Hodge group $G$ (proof of Corollary 20.30), and on it $\omega$ lies outside the span of the products of divisor classes and of the Weil classes of quotients, by Proposition 20.74(ii) and (iii). At a CM point the class is already algebraic (Theorem 20.29), and the CM points, being countably many, do not carry algebraicity to the other members (Lemma 20.27 needs uncountably many points). ∎

The same annihilator controls deformations that are not algebraic at all, and so decides whether the twistor lines along which Markman moves sheaves in his proof for Weil classes can be used here.

**Proposition 21.24** (The Hodge locus among all complex tori, and twistor lines)**.** *Let $\omega$ be a rational exceptional class, $Y = X_c \times X_c$ with $c \in C$, and $\Lambda = H_1(Y, \mathbb{Z})$. Let $\mathcal{T}$ be the space of complex structures on the real torus $\Lambda_{\mathbb{R}}/\Lambda$, a complex manifold of dimension 64 with tangent space $H^1(T_Y)$ at $Y$, and $\mathcal{T}_\omega \subset \mathcal{T}$ the closed analytic subset on which $\omega$ is of type $(2,2)$.*

(i) *The connected component of $\mathcal{T}_\omega$ that contains $Y$ is the universal cover $\mathfrak{H}$ of the Mumford curve, the set of squares of members of the family, biholomorphic to the upper half plane. Every connected complex analytic family of complex tori, algebraic or not, through a member of the Mumford family on which $\omega$ stays of Hodge type lies in the family.*

(ii) *No twistor line through a member of the Mumford family, and no compact connected complex curve in $\mathcal{T}$ through it, keeps $\omega$ of Hodge type.*

(iii) *Nevertheless $\mathcal{T}_\omega$ contains twistor lines. The algebra $D \otimes_{F,\sigma_1} \mathbb{R}$ is Hamilton's quaternions and acts on $V_{\mathbb{R}}$ through the first factor of $G(\mathbb{R})$; for a unit quaternion $j$ with $j^2 = -1$, the complex structure $j \oplus j$ on $V_{\mathbb{R}} \oplus V_{\mathbb{R}}$ makes $\omega$ of type $(2,2)$, and these structures form a twistor line. On each factor $\psi(x, jy)$ has signature $(4,4)$, and a member whose Mumford–Tate group is $G$ is not an abelian variety. These twistor lines lie in components of $\mathcal{T}_\omega$ other than $\mathfrak{H}$.*

*Proof.* (i) The complex structures on $\Lambda_{\mathbb{R}}$ for which all the tensors invariant under $G$ are of Hodge type are those $J$ with $J^2 = -1$ in the real points of $G \cdot \mathbb{G}_m$, since a reductive group containing the homotheties is the stabiliser of its invariant tensors. Let $\mathfrak{H}$ be the connected component of this closed set that contains $Y$; it is closed in $\mathcal{T}$ and consists of the $J = 1 \otimes 1 \otimes j_3$ with $j_3 \in \mathrm{SL}_2(\mathbb{R})$, $j_3^2 = -1$, in the half plane that contains the structure of $Y$, that is, of the squares of the members of the Mumford family. At each of its points

the first order condition for $\omega$ to stay of type $(2,2)$ along $\xi \in H^1(T_Y)$ is $\xi \lrcorner \omega = 0$, as in the proof of Theorem 21.21(iii), where no polarisation was used, and by Theorem 21.21(i) the solutions form the line $\mathbb{C}\kappa$ tangent to $\mathfrak{H}$. So near each of its points $\mathcal{T}_\omega$ is the smooth curve germ of $\mathfrak{H}$, and $\mathfrak{H}$ is open in $\mathcal{T}_\omega$. Being also closed and connected, it is a connected component. A connected analytic family on which $\omega$ stays of Hodge type maps into $\mathcal{T}_\omega$, hence into $\mathfrak{H}$ once one of its points does.

(ii) A twistor line is a nonconstant holomorphic image of $\mathbb{P}^1$ in $\mathcal{T}$. If $\omega$ stayed of Hodge type along one through a point of $\mathfrak{H}$, or along any compact connected curve through it, the curve would lie in $\mathfrak{H}$ by (i), and a holomorphic map from a compact connected curve to the upper half plane is constant.

(iii) At $\sigma_1$ the quaternion algebra $D$ is definite (Setup 21.1), so the first factor of $G(\mathbb{R})$ is the group of unit quaternions, and $V_\mathbb{R}$ restricted to it is a module over $D \otimes_{F,\sigma_1} \mathbb{R} \cong \mathbb{H}$. The unit quaternions $j$ with $j^2 = -1$ form a two-sphere, and the complex structures given by left multiplication by them on a quaternionic vector space form a twistor line. Each $j$ lies in $G(\mathbb{R})$, and so does the circle $\cos\vartheta + j\sin\vartheta$, which fixes $\omega$; so $\omega$ is of type $(2,2)$. Inside $V_1 \otimes V_2 \otimes V_3$ the space $V_\mathbb{R}$ is the fixed space of a real structure that is quaternionic on the first two factors and real on the third, so $V_\mathbb{R} = U \otimes V_{3,\mathbb{R}}$ with $U$ the real form of $V_1 \otimes V_2$. On $U$ the form $g = \epsilon_1 \otimes \epsilon_2$ is symmetric and invariant under the compact group of pairs of unit quaternions, which acts irreducibly on $U \otimes \mathbb{C} = V_1 \otimes V_2$; so $g$ is definite. The complex structures of the Mumford family are $1 \otimes h$, with $h$ a complex structure on $V_{3,\mathbb{R}}$ for which $\epsilon_3(\cdot, h\cdot)$ is definite, and $\psi(x, (1\otimes h)y) = g \otimes \epsilon_3(\cdot, h\cdot)$ is definite. For $J = j \otimes 1 \otimes 1$ the form $g(\cdot, (j\otimes 1)\cdot)$ is alternating, $j \otimes 1$ being an isometry of $g$ with square $-1$, and $\psi(x, Jy) = g(\cdot, (j\otimes 1)\cdot) \otimes \epsilon_3$ is the product of nondegenerate alternating forms on spaces of dimensions 4 and 2; as the product of two alternating forms on $\mathbb{R}^2$ is the determinant on $\mathbb{R}^2 \otimes \mathbb{R}^2 = M_2(\mathbb{R})$, of signature $(2,2)$, it has signature $(4,4)$. At a member whose Mumford–Tate group is $G$, the Hodge classes of degree two on $V_\mathbb{R} \oplus V_\mathbb{R}$ are the forms $\psi \otimes s$ with $s$ a symmetric $2\times 2$ matrix, by Proposition 20.74(ii), and the signature of $\psi(\cdot, J\cdot) \otimes s$ is $(4(p+q), 4(p+q))$ when $s$ has signature $(p,q)$, so none is definite, none is a polarisation, and the member is not an abelian variety. The members of the Mumford family are not on these lines, since $\psi(x, Jy)$ is definite at the former and of signature $(4,4)$ at the latter, so by (i) the lines lie in other components. ∎

Markman's proof for Weil classes on abelian fourfolds moves a sheaf along twistor lines on which its Chern character stays of Hodge type. By Proposition 21.24 no such line passes through the square of a member of the Mumford family: twistor lines on which $\omega$ is of Hodge type exist, but they carry non-algebraic tori whose Hodge cocharacter lies in a definite factor of $D$, and they lie in components of the Hodge locus that the Mumford family never meets. The Kuga–Satake class of Theorem 21.4 behaves differently. The Kuga–Satake construction applies to every Hodge structure of K3 type on $(T, q_\lambda)$, so that class stays of Hodge type over the whole period domain of $(T, q_\lambda)$, of dimension seven, and not only along the curve; among the classes attached to the target, it is the only one for which deformation methods of this kind are not excluded.

The same space $T$ also makes the square a holomorphic symplectic variety, and this determines what the mechanism of van Geemen and Rapagnetta would have to supply here. They prove the Hodge conjecture for the general abelian fourfold of Weil type with trivial discriminant by pulling back the second Chern class of a hyperkähler sixfold along a map from the fourfold, which gives an algebraic class of codimension two that is not a product of divisor classes [vGR26]. For a projective hyperkähler manifold $M$ write $\sigma_M$ for

its symplectic form, $q_M$ for its Beauville–Bogomolov form and $T(M)$ for the orthogonal complement of $\mathrm{NS}(M)_{\mathbb{Q}}$ in $H^2(M,\mathbb{Q})$ for $q_M$.

**Proposition 21.25** (The square as a holomorphic symplectic variety)**.** *Let $c \in C$ be a point at which the Mumford–Tate group of $X = X_c$ is $G \cdot \mathbb{G}_m$, let $Y = X \times X$, and let $\Psi \in \bigwedge^2 V$ be the $G$-invariant bivector dual to $\psi$. For $x \in T$ put $\iota(x) = (x \otimes 1)\Psi$ in $V \otimes V = H^1(X,\mathbb{Q}) \otimes H^1(X,\mathbb{Q})$, a Künneth component of $H^2(Y,\mathbb{Q})$, and let $\iota_2 \colon S^2T \to H^4(Y,\mathbb{Q})$ be the map $xy \mapsto \iota(x) \cup \iota(y)$. Let $C_i \in S^2T_i$ be the Casimir element of the trace form of $\mathfrak{sl}(V_i)$, and for $\nu \in F$ let $c_\nu = \sum_i \sigma_i(\nu) C_i$, a rational element of $S^2T$; for $\nu \neq 0$ it is the element dual to the form $q_{1/\nu}$.*

(i) *$\iota$ is an embedding of Hodge structures $T(-1) \to H^2(Y,\mathbb{Q})$, its image is the only irreducible sub-Hodge structure of $H^2(Y,\mathbb{Q})$ with $h^{2,0} = 1$, and the holomorphic 2-form $\sigma_0$ spanning its part of type $(2,0)$ is symplectic.*

(ii) *$\iota_2(c_\nu) = \rho\,\omega_a$ with $a_0 = 3\,\mathrm{Tr}_{F/\mathbb{Q}}(\nu)$ and $\alpha = 4\nu - \mathrm{Tr}_{F/\mathbb{Q}}(\nu)$, in the notation of Setup 21.18, where $\rho$ is a nonzero rational number independent of $\nu$. So $\iota_2(c_\nu)$ is exceptional if and only if $\nu \notin \mathbb{Q}$, and every rational exceptional class is some $\iota_2(c_\nu)$ plus a rational multiple of $\pi_0$.*

(iii) *Let $M$ be a projective hyperkähler manifold and $f \colon Y \dashrightarrow M$ a rational map with $f^*\sigma_M \neq 0$. Then $f^*$ maps $T(M)$ isomorphically onto $\iota(T)$ and is compatible with products of classes in $T(M)$; $f$ is generically finite onto a subvariety of dimension eight on which $\sigma_M$ is generically nondegenerate; $b_2(M) \geq 10$; and $\dim M \geq 10$.*

(iv) *Suppose in (iii) that $M = S^{[n]}$ is the Hilbert scheme of $n$ points on a projective K3 surface $S$, and let $q_\lambda$ be the form on $T$ that corresponds to the intersection form of $S$ on $T(S) = T(M)$ under $\iota^{-1}f^*$ (Proposition 21.2(iii)). If $\lambda \notin \mathbb{Q}$, or if the real multiplication of $T(S)$ is induced by algebraic cycles on $S \times S$, then the Hodge conjecture holds for $Y$.*

*Proof.* (i) The map $\iota$ is $G$-equivariant because $\Psi$ is invariant, and injective. Its image lies in $S^2V$: for $x$ in the symplectic Lie algebra of $\psi$ one has $(x \otimes 1)\Psi = -(1 \otimes x)\Psi$, and the exchange of the two factors carries $(x \otimes 1)\Psi$ to $-(1 \otimes x)\Psi = (x \otimes 1)\Psi$, because $\Psi$ is alternating. An element $g$ of $G \cdot \mathbb{G}_m$ multiplies $\Psi$ by the multiplier $\chi(g)$ of $\psi$, so $(g \otimes g)\iota(x) = \chi(g)\,\iota(gxg^{-1})$: the map $\iota$ is equivariant for the adjoint action on $T$ twisted by $\chi$, which is the action defining $T(-1)$. Both Hodge structures come from the Hodge cocharacter through $G \cdot \mathbb{G}_m$, so $\iota$ is a morphism of Hodge structures from $T(-1)$. The sub-Hodge structures of $H^2(Y,\mathbb{Q})$ are its $G$-stable subspaces, since $\mathbb{G}_m$ acts by scalars. Over $\bar{\mathbb{Q}}$, $H^2(Y) = \bigwedge^2(V \otimes \bar{\mathbb{Q}}^2) = S^2V \oplus (\bigwedge^2 V)^{\oplus 3}$, and expanding $V \otimes V = \bigotimes_k (S^2V_k \oplus \bigwedge^2 V_k)$ gives $\bigwedge^2 V$ as the sum of the pieces of highest weights $(0,0,0)$, $(2,2,0)$, $(2,0,2)$, $(0,2,2)$ (an odd number of factors $\bigwedge^2 V_k$) and $S^2V$ as the sum of those of highest weights $(2,2,2)$, $(2,0,0)$, $(0,2,0)$, $(0,0,2)$. So the isotypic pieces of $H^2(Y)$ have these eight highest weights and dimensions $3, 27, 27, 27, 27, 3, 3, 3$. The Hodge cocharacter acts through $V_3$, on which it has one weight of each type, so $S^2V_3$ has one dimension of type $(2,0)$ and $\bigwedge^2 V_3$ none; the parts of type $(2,0)$ of the eight pieces have dimensions $0, 0, 9, 9, 9, 0, 0, 1$. The last three pieces occur once, in $S^2V$, and are $\iota(T_1)$, $\iota(T_2)$, $\iota(T_3)$. So an irreducible summand of highest weight $(2,0,2)$ or $(0,2,2)$ contributes 3 to $h^{2,0}$, one of weight $(2,2,2)$ contributes 9, one of weight $(0,0,2)$ contributes 1, and the others contribute nothing. A rational sub-Hodge structure $W$ is stable under the Galois group, which permutes the highest weights as it permutes the factors, transitively. So if $W \otimes \bar{\mathbb{Q}}$ has a summand of weight $(2,2,0)$, $(2,0,2)$ or $(0,2,2)$, it has summands of all three and $h^{2,0}(W) \geq 6$; if it has one of weight $(2,2,2)$, then $h^{2,0}(W) \geq 9$. If $h^{2,0}(W) = 1$, therefore, $W \otimes \bar{\mathbb{Q}}$ contains the piece $\iota(T_3)$

of weight $(0,0,2)$, which has multiplicity one, and with it $\iota(T_1)$ and $\iota(T_2)$; so $W \supseteq \iota(T)$, and $W = \iota(T)$ if $W$ is irreducible. The same argument shows that $\iota(T)$ is irreducible: a nonzero sub-Hodge structure of it contains one of the $\iota(T_i)$ over $\bar{\mathbb{Q}}$, hence all three. The part of type $(2,0)$ of $\iota(T)$ is spanned by $\sigma_0 = \iota(E_3)$, where $E_3 \in \mathfrak{sl}(V_3) \otimes \mathbb{C}$ maps $V_3^{0,1}$ onto $V_3^{1,0}$. Write $\Psi = \Psi_1 \otimes \Psi_2 \otimes \Psi_3$ with $\Psi_k \in \bigwedge^2 V_k$ dual to $\epsilon_k$, and $\Psi_3 = e_+ \otimes e_- - e_- \otimes e_+$ with $e_\pm$ spanning $V_3^{1,0}$ and $V_3^{0,1}$. Then $(E_3 \otimes 1)\Psi_3$ is a nonzero multiple of $e_+ \otimes e_+$, so $\sigma_0$ is a nonzero multiple of $\Psi_1 \otimes \Psi_2 \otimes e_+ \otimes e_+$: it lies in $H^{1,0}(X) \otimes H^{1,0}(X)$ and pairs $H^{1,0}$ of the first factor with $H^{1,0}$ of the second through the nondegenerate form dual to $\epsilon_1 \otimes \epsilon_2$ on $V_1 \otimes V_2$. So $\sigma_0$ has rank eight: it is a holomorphic symplectic form on $Y$.

(ii) For $x, y \in T$ the class $\iota(x) \cup \iota(y)$ lies in $H^2(X) \otimes H^2(X)$ and comes from the tensor $(x \otimes 1)\Psi \otimes (y \otimes 1)\Psi$ in $V^{\otimes 4}$ by antisymmetrising its two slots in the first factor of $Y$ and its two slots in the second. Read it, as in Setup 21.18, as a map from the first-factor slots to the second-factor slots. Contraction with $\epsilon_k$ turns $\Psi_k \otimes \Psi_k$ into $\gamma$ times the identity of $V_k \otimes V_k$ and, for $x, y \in T_k$, the tensor $(x \otimes 1)\Psi_k \otimes (y \otimes 1)\Psi_k$ into $\gamma\,(x \otimes y)$, with one and the same constant $\gamma \neq 0$, since $x$ and $y$ are antisymmetric for $\epsilon_k$ and the two signs cancel. So $\iota_2(C_1)$ corresponds, up to a nonzero constant $\gamma'$ independent of the factor, to the restriction to $\bigwedge^2 V$ of $\mathrm{Cas}_1 \otimes 1 \otimes 1$, where $\mathrm{Cas}_1 = \sum_c x_c \otimes x^c = s_1 - \frac{1}{2}$ is the operator of Step 3 of the proof of Theorem 21.7, $s_1$ the exchange of the two copies of $V_1$. On $\bigwedge^2 V$ the exchange $s_1$ is $-1$ on $\mathbb{Q}\theta$ and $U_{23}$ and $+1$ on $U_{12}$ and $U_{13}$, so $\iota_2(C_1) = \rho\,(3\pi_0 - \pi_{12} - \pi_{13} + 3\pi_{23})$ with $\rho = -2\gamma'$, and permuting the factors gives the two cyclic permutations of this identity. Summing with the coefficients $\sigma_i(\nu)$ gives $\iota_2(c_\nu) = \rho(3\,\mathrm{Tr}(\nu)\pi_0 + \sum \sigma_k(4\nu - \mathrm{Tr}(\nu))\pi_{ij})$, the sum over $\{i,j,k\} = \{1,2,3\}$ with $i < j$, which is $\rho\,\omega_a$ with the stated $a_0$ and $\alpha$. For rational $\nu \neq 0$ both $\iota_2(c_\nu)$ and $\omega_a$ are rational and nonzero, so $\rho \in \mathbb{Q}$. Since $\mathrm{Tr}(\nu) \in \mathbb{Q}$, the element $\alpha$ is rational exactly when $\nu$ is; and $\mathrm{Tr}(\alpha) = \mathrm{Tr}(\nu)$, so $\nu = (\alpha + \mathrm{Tr}(\alpha))/4$ recovers $\nu$ from any $\alpha \in F$. Two rational classes with the same $\alpha$ differ by a multiple of $\pi_0$, and $\iota_2(c_{\nu/\rho}) = \omega_a$ because $c_\nu$ is linear in $\nu$; so every rational exceptional class is some $\iota_2(c_\nu)$ plus a rational multiple of $\pi_0$. The Casimir element $C_i$ is dual to the trace form on $T_i$, so $c_\nu$ is dual to $\sum_i \sigma_i(\nu)^{-1}\,\mathrm{tr}(x_i y_i) = q_{1/\nu}(x,y)$.

(iii) Resolve the indeterminacy of $f$ by a sequence of blow-ups along smooth centres, $\pi\colon \tilde{Y} \to Y$ and $\tilde{f}\colon \tilde{Y} \to M$, and put $f^* = \pi_* \tilde{f}^*$, a morphism of Hodge structures. The kernel of $\pi_*$ on $H^2(\tilde{Y}, \mathbb{Q})$ is spanned by the classes of the exceptional divisors, so $\alpha \mapsto \tilde{f}^*\alpha - \pi^* f^* \alpha$ is a morphism of Hodge structures from $H^2(M, \mathbb{Q})$ to a space of Hodge classes. The Hodge structure $T(M)$ is irreducible: a sub-Hodge structure $W \subseteq T(M)$ with $W^{2,0} = 0$ consists of rational classes of type $(1,1)$, which lie in $\mathrm{NS}(M)_{\mathbb{Q}} \cap T(M) = 0$; if $W^{2,0} \neq 0$, then $\sigma_M \in W_{\mathbb{C}}$, the orthogonal complement of $W$ in $H^2(M, \mathbb{Q})$ is a sub-Hodge structure orthogonal to $\sigma_M$ and $\bar{\sigma}_M$, hence of type $(1,1)$ and contained in $\mathrm{NS}(M)_{\mathbb{Q}}$, and $W \supseteq T(M)$. Since $T(M)$ is not spanned by Hodge classes, that morphism vanishes on it, and $\tilde{f}^*\alpha = \pi^* f^* \alpha$ for $\alpha \in T(M)$. Because $\tilde{f}^*$ is a ring homomorphism and $\pi_*\pi^*$ is the identity, $f^*(\alpha\beta) = f^*\alpha \cup f^*\beta$ for $\alpha, \beta \in T(M)$. The kernel of $f^*$ on $T(M)$ is a sub-Hodge structure not containing $\sigma_M$, hence zero, and the image is irreducible with $h^{2,0} = 1$, hence $\iota(T)$ by (i). So $f^*\sigma_M$ is a nonzero multiple of $\sigma_0$ and has rank eight at every point of $Y$ where $f$ is defined; the rank of a pulled-back form is at most the dimension of the image, so $f$ is generically finite onto its image $Z$, of dimension eight, and $\sigma_M$ is generically nondegenerate on $Z$. As $M$ is projective, $b_2(M) = \dim T(M) + \dim \mathrm{NS}(M)_{\mathbb{Q}} \geq 9 + 1$; this excludes the generalised Kummer varieties, whose second Betti number is seven [Bea83]. Finally $\dim M \geq \dim Z = 8$; suppose that $\dim M = 8$. Then $f$ is dominant, and for $\alpha \in T(M)_{\mathbb{C}}$ the relation of Fujiki and Beauville

[Bea83, Fuj87] gives

$$\int_Y (f^*\alpha)^8 = \int_{\tilde{Y}} \tilde{f}^*(\alpha^8) = \deg(f)\int_M \alpha^8 = \deg(f)\, c_M\, q_M(\alpha)^4, \qquad c_M > 0.$$

Write $f^*\alpha = \iota(x)$ with $x = x_1 + x_2 + x_3 \in T_{\mathbb{C}}$ and $N_i = -\det x_i$. The form $x \mapsto q_M((f^*)^{-1}\iota(x))$ is invariant under $G$, so it is $q_\lambda(x,x) = \sum_i \sigma_i(\lambda)\,\mathrm{tr}(x_i^2) = 2\sum_i \sigma_i(\lambda) N_i$ for some $\lambda \in F^\times$ by Proposition 21.2(iii). On the other side, in a basis $(e_i)$ of $V$ one has $\iota(x) = \sum_{i,j} M_{ij}\, e_i^{(1)} \wedge e_j^{(2)}$, with $M$ the product of the matrix of $x$ and the constant invertible matrix of $\Psi$, and the 2-forms $e_i^{(1)} \wedge e_j^{(2)}$ commute, so $\iota(x)^8 = 8!\det(M)\prod_i e_i^{(1)} \wedge e_i^{(2)}$ is $8!\det(x)$ times a fixed volume form, and $\det(x_1 \otimes 1 \otimes 1 + 1 \otimes x_2 \otimes 1 + 1 \otimes 1 \otimes x_3) = \Delta(N_1, N_2, N_3)^2$ with $\Delta = N_1^2 + N_2^2 + N_3^2 - 2N_1N_2 - 2N_1N_3 - 2N_2N_3$; this is also the product of the eight numbers $\pm r_1 \pm r_2 \pm r_3$, where $\pm r_i$ are the eigenvalues of $x_i$ and $r_i^2 = N_i$. The $N_i$ are algebraically independent functions on $T_{\mathbb{C}}$, so the two expressions for $\int_Y (f^*\alpha)^8$ would make $\Delta^2$ a nonzero constant multiple of the fourth power of the linear form $\sum_i \sigma_i(\lambda) N_i$, and then, by unique factorisation, $\Delta$ a multiple of its square. But $\Delta$ is a nondegenerate ternary quadratic form, hence irreducible. So $\dim M \neq 8$, and since the dimension of a hyperkähler manifold is even, $\dim M \geq 10$.

(iv) By [Bea83] the map $H^2(S,\mathbb{Q}) \to H^2(S^{[n]},\mathbb{Q})$ induced by the universal subscheme $Z_n \subset S \times S^{[n]}$ is injective, carries the intersection form to the Beauville–Bogomolov form, and has $H^2(S^{[n]},\mathbb{Q}) = H^2(S,\mathbb{Q}) \oplus \mathbb{Q}\delta$ with $\delta$ algebraic, so $T(M) = T(S)$. Let $\Delta_T$ be the Künneth component in $T(S) \otimes T(S)$ of the class of the diagonal of $S$; it is algebraic, being the class of the diagonal minus $[\mathrm{pt} \times S]$, $[S \times \mathrm{pt}]$ and $\sum_r d_r \times d_r^\vee$ over dual bases of $\mathrm{NS}(S)_{\mathbb{Q}}$. Its image under the correspondence $Z_n \times Z_n$, pulled back along the diagonal of $M$, is the class $\sum_a e_a \cup e_a^\vee$ over dual bases of $T(S)$ for the intersection form, which is therefore algebraic on $M$, and by (iii) its image under $f^*$ is $\iota_2$ of the element of $S^2T$ dual to $q_\lambda$, that is $\iota_2(c_{1/\lambda})$, an algebraic class on $Y$. If $\lambda \notin \mathbb{Q}$ it is exceptional by (ii). If instead the real multiplication is algebraic, apply it to the first factor of $\Delta_T$: for $b \in F$ the class $\sum_a (m_b e_a) \cup e_a^\vee$ is algebraic, and its image is $\iota_2(c_{b/\lambda})$, which is exceptional for $b \notin \mathbb{Q}\lambda$. In both cases Lemma 21.20 makes every Hodge class in $H^2(X) \otimes H^2(X)$ algebraic; the Hodge classes in the other Künneth components of $H^4(Y,\mathbb{Q})$ are products of divisor classes, as at the end of the proof of Theorem 21.4; and the Hodge classes of $Y$ form a ring generated in degrees two and four (proof of Corollary 20.30). So every Hodge class of $Y$ is algebraic. ∎

*Remark* 21.26 (The mechanism of van Geemen and Rapagnetta). By Proposition 21.25(ii) the exceptional classes are, up to rational multiples of $\pi_0$, the classes $\iota_2(c_\nu)$ with $\nu \notin \mathbb{Q}$, the images of the elements dual to the forms $q_\lambda$, $\lambda = 1/\nu$, on the part $\iota(T)$ of $H^2(Y)$ generated by the symplectic class; the form with $\lambda$ rational gives a product of divisor classes. They are the analogues for $(Y,\sigma_0)$ of the dual Beauville–Bogomolov class, whose pull-back is what [vGR26] uses. By part (iii), a transfer of that argument needs a projective hyperkähler manifold of dimension at least ten with transcendental part $T(-1)$, such as $S_\mu^{[n]}$ with $n \geq 5$ for a surface $S_\mu$ of Theorem 21.4, and a rational map from $Y$ onto a symplectic subvariety of dimension eight. For $M = S_\mu^{[n]}$ the form of (iv) is $q_{\mu/e^2}$ for the $e \in F^\times$ with $\iota^{-1}f^* = m_e$ on $T(S_\mu) = T$, so $\lambda \notin \mathbb{Q}$ for every such map as soon as $\mu \notin \mathbb{Q}^\times F^{\times 2}$. No such map is known. It would give an algebraic cycle on $S_\mu \times X \times X$ inducing an isomorphism of $T(S_\mu)$ onto $\iota(T) \subset H^1(X) \otimes H^1(X)$, a correspondence of the kind that the Kuga–Satake class supplies in the proof of Theorem 21.4; so this route is the Kuga–Satake route in geometric form rather than an independent one. By (iii), maps to K3 surfaces and to hyperkähler eightfolds cannot serve.

**Theorem 21.27** (A numerical criterion for the Mumford classes)**.** *Let $c \in C$, $Y = X_c \times X_c$, $\omega = \omega_a$ a rational exceptional class with $L_0(a) \neq 0$, and $E$ a perfect complex on $Y$ with $\mathrm{ch}(E) = N\omega$, $N \neq 0$, and no other component.*

(i) $\dim \mathrm{Ext}^2(E,E) \geq 119$.

(ii) *If $\dim \mathrm{Ext}^2(E,E) = 119$, then $\mathrm{ev}_E \colon HT^2(Y) \to \mathrm{Ext}^2(E,E)$ is surjective with kernel $\mathbb{C}\kappa$, and $\sigma_E$ is injective. Conversely, if $\sigma_E$ is injective, then $\mathrm{ev}_E$ has rank 119 and kernel $\mathbb{C}\kappa$, and $E$ extends to first order along the curve.*

(iii) *If some such $E$ at one point $c$ has $\dim \mathrm{Ext}^2(E,E) = 119$, then $\omega_t$ is algebraic for every $t \in C$, the Hodge conjecture holds for $X_t \times X_t$ for every $t$, and $B(W \times_C W)$ holds.*

(iv) *At every CM point $c$ some perfect complex $E$ on $Y$ has $\mathrm{ch}(E) = N\omega$ for some $N \neq 0$ and no other component.*

*Proof.* (i) and (ii). By the square $\sigma_E \circ \mathrm{ev}_E = (\,\cdot\,) \lrcorner \mathrm{ch}(E)$ of Setup 18.54, $\ker \mathrm{ev}_E \subseteq \mathrm{Ann}_{HT^2}(\omega) = \mathbb{C}\kappa$ by Theorem 21.21(ii), so $\mathrm{ev}_E$ has rank at least 119, which is (i). If $\dim \mathrm{Ext}^2(E,E) = 119$, then $\mathrm{ev}_E$ is onto, its kernel has dimension one and lies in $\mathbb{C}\kappa$, so equals it; and $\sigma_E(\mathrm{ev}_E(x)) = N\, x \lrcorner \omega$ vanishes only for $x \in \mathbb{C}\kappa = \ker \mathrm{ev}_E$, so $\sigma_E$ is injective on $\mathrm{im}\, \mathrm{ev}_E = \mathrm{Ext}^2(E,E)$. Conversely, if $\sigma_E$ is injective, then $\sigma_E(\mathrm{ev}_E(\kappa)) = N\kappa \lrcorner \omega = 0$ forces $\mathrm{ev}_E(\kappa) = 0$, so the kernel is $\mathbb{C}\kappa$ and the rank is 119; and $\mathrm{ev}_E$ restricted to $H^1(T_Y)$ is the obstruction map [Tod09], which therefore vanishes at $\kappa$, so that $E$ extends to first order along the curve.

(iii) By (ii) $\sigma_E$ is injective. The Chern character of $E$ is $N\omega$, a flat section of Hodge type over the whole of $C$, so Theorem 16.28 applies to every infinitesimal deformation of $Y$ inside $W \times_C W \to C$. The proof of Theorem 16.29 then applies word for word, with the stack of perfect complexes on the fibres of $W \times_C W \to C$ [Lie06] in place of that of $\mathcal{A} \to \mathcal{S}$, or the analytic Kuranishi germ if $\mathrm{Ext}^{<0}(E,E) \neq 0$: it is smooth over $C$ at $[E]$, its image contains a nonempty open subset $U \subseteq C$, and for $t \in U$ a perfect complex $E_t$ on $X_t \times X_t$ in the same connected component has $\mathrm{ch}(E_t) = N\omega_t$, so $\omega_t$ is algebraic, and by Lemma 21.20 so are both exceptional classes. Since $U$ is uncountable, Corollary 20.30 gives the rest.

(iv) By Theorem 20.29 there is $\xi \in \mathrm{CH}^2(Y)_{\mathbb{Q}}$ with class $\omega$, and the argument of Proposition 16.27 gives $E$: the Chern character is an isomorphism $K_0(Y)_{\mathbb{Q}} \to \mathrm{CH}^*(Y)_{\mathbb{Q}}$ [Ful98, Ex. 15.2.16(b) and Cor. 18.3.2], so $\xi = \mathrm{ch}(u)$ for a unique $u$, and $Nu$ is the class of a perfect complex for a suitable $N$. ∎

Part (iii) is the Mumford analogue of Theorem 16.37(iii): one object, at one point, and one number, which is 119 here and $2n(2n-1)$ for a Weil family. For a Weil family that number is attained by no object (Theorem 17.5), because an object attaining it would have to deform to non-algebraic tori; the classes here are rigid, so that argument does not apply (Remark 17.6). The objects of (iv) exist, and the criterion asks that one of them, or another representative of the same class in $K_0$, attain the bound of (i). The rigidity of Theorem 21.21 is no obstacle: semiregularity moves an object along the Hodge locus of its Chern character, and the curve is exactly that locus, so a semiregular object would propagate along the curve and nowhere else, which is all that is needed.

**Corollary 21.28** (Bounded data at CM points)**.** *Fix a relatively ample line bundle on $W \times_C W \to C$, and let $\omega$ be a rational exceptional class. The following are equivalent.*

(a) *$\omega_t$ is algebraic for every $t \in C$; equivalently, by Lemma 21.20 and Corollary 20.30, the Hodge conjecture holds for $X_t \times X_t$ for every $t \in C$.*

(b) *There are a finite set $\Phi$ of pairs $(\underline{P},\underline{q})$, each a tuple of Hilbert polynomials with a tuple of rational numbers of the same length, and infinitely many points $c\in C$ at each of which $\omega_c=\sum_i q_i[Z_i]$ for subschemes $Z_i\subset X_c\times X_c$ with Hilbert polynomials $P_i$, for some $(\underline{P},\underline{q})\in\Phi$.*

*In particular the classes are algebraic on every member as soon as they are represented at infinitely many CM points by cycles of bounded degree.*

*Proof.* For a pair $(\underline{P},\underline{q})$ let $\Sigma_{\underline{P},\underline{q}}\subseteq C$ be the set of points at which $\omega$ is represented with these data. The proof of Theorem 16.4 applies verbatim to $(W\times_C W, C, \omega)$, as in Lemma 20.27, and shows that each $\Sigma_{\underline{P},\underline{q}}$ is Zariski closed. If (b) holds, then, $\Phi$ being finite, one pair of $\Phi$ serves at infinitely many points, so its $\Sigma$ is an infinite closed subset of the irreducible curve $C$, hence $C$, and (a) holds. If (a) holds, $C$ is the union of the countably many $\Sigma_{\underline{P},\underline{q}}$, and $C$ is uncountable, so one of them is infinite, hence equal to $C$; then (b) holds with $\Phi$ that one pair and every point of $C$. ∎

The proof of Theorem 20.29 represents the classes at every CM point, through Markman's theorem for abelian fourfolds [Mar25a] and the ring generated by divisor classes and pull-backs of Hodge classes of $X_c$ along homomorphisms $X_c\times X_c\to X_c$, but with no control of degree; isogenies between CM members, which move CM points along their Hecke orbits, do not bound the degree either. By Corollary 21.28 such a bound, at infinitely many CM points, is equivalent to the target; without it the CM points, being countably many, say nothing about a very general member (Lemma 20.27 needs uncountably many points).

The next proposition says which endomorphisms those pull-backs have to use. At a CM point the fourfold is of Weil type, its two Hodge classes of degree four that are not products of divisor classes form its Weil line, and the exceptional classes of the square are out of reach of everything built from that line with the Weil field alone: they need the real multiplication by the totally real quartic algebra of Rosati-fixed endomorphisms, and all of it.

**Proposition 21.29** (The Weil structure at a CM point, and what the cycles must use)**.** *Let $c\in C$ be a CM point, $X=X_c$, $Y=X\times X$, and let $e_w$, $w\in\{\pm1\}^3$, be a weight basis of $V=H^1(X,\mathbb{C})$ for the Hodge torus $T$ of $X$, as in the proof of Theorem 20.29, normalised so that $\psi(e_w,e_{-w})=\pm1$, with $e_w^{(1)}$, $e_w^{(2)}$ its two copies in $H^1(Y,\mathbb{C})$. Let $T_+$ be the set of the four weights with an even number of entries $-1$, let $T_-=-T_+$ be the set of the other four, let $t_\pm=\prod_{w\in T_\pm}e_w\in H^4(X,\mathbb{C})$, the factors in a fixed order, and let $\dagger$ be the Rosati involution of $\theta$ on $\mathrm{End}^0(X)$. For $\varphi\in\mathrm{End}^0(X)$ write $\varphi_w$ for its eigenvalue on $e_w$.*

(i) *$T_+$ and $T_-$ are the only sets of four weights with sum zero that contain no pair of opposite weights, and $\mathrm{Hdg}^2(X)\otimes\mathbb{C}$ is the direct sum of the span of the products of divisor classes and of $\mathbb{C}t_+\oplus\mathbb{C}t_-$. The elements of $\mathrm{End}^0(X)$ with $\varphi_w$ constant on $T_+$ and constant on $T_-$ form an imaginary quadratic field $k_c$; $(X,k_c,\theta)$ is of $(k_c,-1,2)$-Weil type, and its Weil line is $W(X)=(\mathbb{C}t_+\oplus\mathbb{C}t_-)\cap H^4(X,\mathbb{Q})$. The fixed algebra $K_c^+$ of $\dagger$ consists of the elements with $\varphi_w=\varphi_{-w}$ for every $w$; it is a product of totally real fields of total degree four, and $\mathrm{End}^0(X)=K_c^+\otimes_{\mathbb{Q}}k_c$.*

(ii) *For $\varphi,\psi\in\mathrm{End}^0(X)$ let $f^*_{\varphi,\psi}$ be the ring homomorphism $H^*(X,\mathbb{Q})\to H^*(Y,\mathbb{Q})$ that sends $e\in H^1(X,\mathbb{Q})$ to $\mathrm{pr}_1^*\varphi^*e+\mathrm{pr}_2^*\psi^*e$; for $\varphi,\psi\in\mathrm{End}(X)$ it is the pull-back along the homomorphism $\varphi\circ\mathrm{pr}_1+\psi\circ\mathrm{pr}_2\colon Y\to X$, and every homomorphism $Y\to X$ is of this form. For a $\mathbb{Q}$-subalgebra $R\subseteq\mathrm{End}^0(X)$ let $\Sigma_R\subseteq\mathrm{Hdg}^2(Y)$ be the $\mathbb{Q}$-span of the products of two divisor classes of $Y$ and of the classes $f^*_{\varphi,\psi}W(X)$ with $\varphi,\psi\in R$.*

*Then $\Sigma_{k_c} = \Sigma_{\mathbb{Q}}$, the span obtained from the maps $(x, y) \mapsto ax + by$ alone, has dimension* 110 *and meets the span of $\pi_0, \pi_{12}, \pi_{13}, \pi_{23}$ in the plane of the classes with $a_1 = a_2 = a_3$; it contains no exceptional class.*

(iii) *For a $\mathbb{Q}$-subalgebra $R \subseteq \mathrm{End}^0(X)$ the following are equivalent:* (a) *$\Sigma_R$ contains an exceptional class;* (b) *$\Sigma_R = \mathrm{Hdg}^2(Y)$, which has dimension* 132*;* (c) *$R$ separates the weights of $T_+$, that is, for any two distinct $w, w' \in T_+$ some $\varphi \in R$ has $\varphi_w \neq \varphi_{w'}$; equivalently, $R$ and $k_c$ generate $\mathrm{End}^0(X)$. In particular $\Sigma_{K_c^+} = \mathrm{Hdg}^2(Y)$, while $\Sigma_R$ contains no exceptional class if $R$ is a proper subalgebra of $K_c^+$ or a subalgebra of $k_c$.*

*Proof.* (i) The four weights $d_1 = (1,1,1)$, $d_2 = (1,-1,-1)$, $d_3 = (-1,1,-1)$, $d_4 = (-1,-1,1)$ of $T_+$ have sum zero, and any three of them are linearly independent, so every linear relation among them is a multiple of $d_1 + d_2 + d_3 + d_4 = 0$. A set of four weights containing no opposite pair contains exactly one of $d_i$, $-d_i$ for each $i$, say $\varepsilon_i d_i$, and has sum zero only if $\sum_i \varepsilon_i d_i = 0$, that is, if all $\varepsilon_i$ are equal: the set is $T_+$ or $T_-$. The Hodge classes of $X$ of degree four, tensored with $\mathbb{C}$, are spanned by the monomials $\prod_{w\in S} e_w$ over the sets $S$ of four weights with sum zero (proof of Theorem 20.29); the divisor classes are spanned by the four monomials $e_w e_{-w}$ of weight zero and degree two, their products are the six monomials of the sets that are unions of two opposite pairs, and the two remaining sets are $T_+$ and $T_-$.

The torus $T$ is defined over $\mathbb{Q}$, so the Galois group of $\bar{\mathbb{Q}}$ acts linearly on its character lattice and permutes the set of the eight weights of $V$. A linear map that permutes the weights carries opposite pairs to opposite pairs and sets with sum zero to sets with sum zero, hence carries $\{T_+, T_-\}$ to itself. Over $\bar{\mathbb{Q}}$ the algebra $\mathrm{End}^0(X) \otimes \bar{\mathbb{Q}} = \mathrm{End}_T(V_{\bar{\mathbb{Q}}})$ consists of the diagonal matrices $\sum_w x_w p_w$ in the weight basis, $p_w$ the projection to the weight space of $w$, and $\gamma \in \mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$ acts by $\gamma(\sum_w x_w p_w) = \sum_w \gamma(x_w) p_{\gamma w}$, because $\gamma$ carries the weight space of $w$ to that of $\gamma w$. As $\psi$ is $T$-invariant, $\psi(e_w, e_{w'}) = 0$ unless $w' = -w$, so the adjoint of $\sum_w x_w p_w$ for $\psi$, which is the Rosati involution, is $\sum_w x_{-w} p_w$. The subalgebra $A_W$ of the diagonal matrices that are constant on $T_+$ and on $T_-$, and the subalgebra $A^+$ of those with $x_w = x_{-w}$ for all $w$, are therefore stable under the Galois group and under $\dagger$; so $k_c = A_W \cap \mathrm{End}^0(X)$ and $K_c^+ = A^+ \cap \mathrm{End}^0(X)$ are $\mathbb{Q}$-forms of $A_W$ and $A^+$, of dimensions 2 and 4, and $K_c^+$ is the fixed algebra of $\dagger$. On $A_W$ the involution $\dagger$ exchanges the two scalars. The Rosati involution is positive: $\mathrm{Tr}(xx^\dagger) > 0$ for $x \neq 0$, with Tr the trace on $V$ [BL04, Section 5.1]. If $k_c$ were not a field it would contain the idempotent $p_+ = \sum_{w\in T_+} p_w$, and $\mathrm{Tr}(p_+ p_+^\dagger) = \mathrm{Tr}(p_+ p_-) = 0$; so $k_c$ is a quadratic field on which $\dagger$ is the nontrivial automorphism. If $k_c$ were real, $x = \sqrt{d} \in k_c$ would satisfy $x^\dagger = -x$ and $\mathrm{Tr}(xx^\dagger) = -\mathrm{Tr}(x^2) = -8d < 0$. So $k_c$ is imaginary quadratic and $\dagger$ is complex conjugation on it, which is condition (iii) of Definition 3.1. The eigenspaces of $k_c \otimes \mathbb{C}$ on $V$ are $V_\pm = \bigoplus_{w\in T_\pm} \mathbb{C} e_w$. The Hodge decomposition of $V$ is the eigenspace decomposition of the Hodge cocharacter, which takes values in the diagonal torus $T_{\mathbb{C}} \cdot \mathbb{G}_m$, so every $e_w$ is of pure type $(1,0)$ or $(0,1)$, and $t_+$ is of type $(p, 4-p)$ with $p = \dim(V_+ \cap H^{1,0}(X))$. Since $t_+$ is a Hodge class, $p = 2$: each eigenvalue of $\sqrt{-d}$ has multiplicity two on $H^{1,0}(X)$, which is condition (ii) of Definition 3.1, and condition (i) is $\dim X = 4$. The Weil line of Definition 3.6 is $(\bigwedge^4 V_+ \oplus \bigwedge^4 V_-) \cap H^4(X, \mathbb{Q}) = (\mathbb{C}t_+ \oplus \mathbb{C}t_-) \cap H^4(X, \mathbb{Q})$; it has dimension two by Proposition 3.7(i), so it is a rational complement of the products of divisor classes in $\mathrm{Hdg}^2(X)$. The algebra $K_c^+$ is commutative and semisimple, being a subalgebra of $\mathrm{End}^0(X)$, whose complexification is commutative, and its trace form $\mathrm{Tr}(xy) = \mathrm{Tr}(xy^\dagger)$ is positive definite; so every factor field $E$ of $K_c^+$ has $\mathrm{Tr}_{E/\mathbb{Q}}(x^2) > 0$ for $x \neq 0$ and is totally real. Finally the multiplication map $K_c^+ \otimes_{\mathbb{Q}} k_c \to \mathrm{End}^0(X)$ is an isomorphism, because over

$\bar{\mathbb{Q}}$ it is the isomorphism $\bar{\mathbb{Q}}^{\{\text{opposite pairs}\}} \otimes \bar{\mathbb{Q}}^{\{T_+,T_-\}} \to \bar{\mathbb{Q}}^{\{\text{weights}\}}$ induced by the bijection that sends a weight to the pair and the tetrahedron containing it, a bijection because the two weights of a pair lie in different tetrahedra, $T_-$ being $-T_+$.

(ii) Extend the notation: for diagonal $\varphi, \psi \in \mathrm{End}^0(X) \otimes \mathbb{C}$ with entries $\varphi_w$, $\psi_w$ let $f^*_{\varphi,\psi}$ be the ring homomorphism $H^*(X,\mathbb{C}) \to H^*(Y,\mathbb{C})$ with $e_w \mapsto \varphi_w e_w^{(1)} + \psi_w e_w^{(2)}$; for $\varphi, \psi \in \mathrm{End}^0(X)$ this is the map of the statement, since $\varphi^* e_w = \varphi_w e_w$, and every homomorphism $Y \to X$ is $\varphi \circ \mathrm{pr}_1 + \psi \circ \mathrm{pr}_2$ for its restrictions $\varphi, \psi$ to the two factors. Then

$$f^*_{\varphi,\psi} t_\pm = \prod_{w \in T_\pm} (\varphi_w e_w^{(1)} + \psi_w e_w^{(2)}) = \sum_{S \subseteq T_\pm} \pm \Big( \prod_{w \in S} \varphi_w \Big) \Big( \prod_{w \in T_\pm \setminus S} \psi_w \Big)\, e_S^{(1)}\, e_{T_\pm \setminus S}^{(2)}, \tag{75}$$

where $e_S^{(1)} e_{T\setminus S}^{(2)}$ is the product of the $e_w^{(1)}$, $w \in S$, and the $e_w^{(2)}$, $w \in T \setminus S$, taken in the order of $T$, and the sign is that of the shuffle. We call these 32 products the tetrahedron monomials. The Hodge classes of $Y$ of degree four, tensored with $\mathbb{C}$, are spanned by the monomials of weight zero in the sixteen classes $e_w^{(i)}$ (proof of Theorem 20.29). Such a monomial whose four weights contain an opposite pair is the product of two monomials of weight zero and degree two, that is, of two divisor classes: there are four of the form $e_w^{(1)} e_w^{(2)} e_{-w}^{(1)} e_{-w}^{(2)}$ and 96 with two distinct opposite pairs and a choice of copy for each of the four weights. A weight repeated twice, $2w + w' + w'' = 0$, forces $w' = w'' = -w$, so the remaining monomials have four distinct weights with no opposite pair, and by (i) they are the tetrahedron monomials. Hence $\dim \mathrm{Hdg}^2(Y) = 132$, the products of divisor classes span the 100 monomials of the first two kinds, and the tetrahedron monomials form a basis of the quotient $\Xi$ of $\mathrm{Hdg}^2(Y) \otimes \mathbb{C}$ by the span of the products of divisor classes. Two distinct weights of a tetrahedron $T$ differ in exactly two coordinates, and for $S = \{a, b\} \subseteq T$ the complement $T \setminus S$ has the same set of differing coordinates, because $a + b$ and the sum of the complement are opposite. Writing $D(a,b) \subseteq \{1,2,3\}$ for the set of coordinates in which two weights differ, the six tetrahedron monomials of $T$ with two factors from each copy fall into three pairs $\{e_S^{(1)} e_{T\setminus S}^{(2)}, e_{T\setminus S}^{(1)} e_S^{(2)}\}$, indexed by the two-element set $D(S) = D(T \setminus S)$.

For $\varphi, \psi \in k_c \otimes \mathbb{C} = A_W$, with values $\varphi_\pm$, $\psi_\pm$ on $T_\pm$, (75) reads $f^*_{\varphi,\psi} t_\pm = \sum_{k=0}^4 \varphi_\pm^k \psi_\pm^{4-k} s_{\pm,k}$, where $s_{\pm,k}$ is the sum, with the signs of (75), of the tetrahedron monomials of $T_\pm$ with $k$ factors from the first copy. Conversely the classes $(a\,\mathrm{pr}_1 + b\,\mathrm{pr}_2)^* t_\pm$ for five values of $a/b$ span the $s_{\pm,k}$, a Vandermonde matrix being invertible. Since $W(X) \otimes \mathbb{C} = \mathbb{C}t_+ \oplus \mathbb{C}t_-$, the complexifications of $\Sigma_{k_c}$ and of $\Sigma_{\mathbb{Q}}$ are both spanned by the products of divisor classes and the ten classes $s_{\pm,k}$, which are supported on disjoint nonempty sets of tetrahedron monomials; so $\Sigma_{k_c} = \Sigma_{\mathbb{Q}}$ has dimension $100 + 10 = 110$.

For the four $\pi$ we use the description of Setup 21.18. Fix a total order on the weights, and for an endomorphism $A$ of $H^2(X,\mathbb{C}) = \bigwedge^2 V$ let

$$c(A) = \sum_{a<b} A(e_a e_b) \otimes (e_a e_b)^\vee \in H^2(X) \otimes H^2(X)$$

be its class, where $(e_a e_b)^\vee$ is the dual basis for the form $\bigwedge^2 \psi$. As $\bigwedge^2 \psi(e_a e_b, e_c e_d) = \psi(e_a, e_c)\psi(e_b, e_d) - \psi(e_a, e_d)\psi(e_b, e_c)$ vanishes unless $\{c, d\} = \{-a, -b\}$, we have $(e_a e_b)^\vee = \chi(a)\chi(b)\, e_{-a} e_{-b}$ with $\chi(w) = \psi(e_w, e_{-w}) = \pm 1$, the two factors in the order matching $(a, b)$. Let $s_t$, $t = 1, 2, 3$, be the involution of $V \otimes V = \bigotimes_{u=1}^3 (V_u \otimes V_u)$ that exchanges the two factors $V_t$. The three $s_t$ commute, and $s_1 s_2 s_3$ is the exchange of the two slots of $V \otimes V$, which is $-1$ on $\bigwedge^2 V$; so each $s_t$ preserves $\bigwedge^2 V$, on which $s_1 s_2 s_3 = -1$, and $\bigwedge^2 V$ is the sum of the four joint eigenspaces with eigenvalues $(\varepsilon_1, \varepsilon_2, \varepsilon_3)$ of product $-1$: these are

$\mathbb{C}\theta = \bigwedge^2 V_1 \otimes \bigwedge^2 V_2 \otimes \bigwedge^2 V_3$ for $(-1,-1,-1)$ and $U_{ij} = S^2 V_i \otimes S^2 V_j \otimes \bigwedge^2 V_k$ for $\varepsilon_i = \varepsilon_j = 1$, $\varepsilon_k = -1$. The projection onto the eigenspace of $(\varepsilon_1, \varepsilon_2, \varepsilon_3)$ is $\prod_t (1 + \varepsilon_t s_t)/2$, which by the relation $s_1 s_2 s_3 = -1$ equals $(1 + \varepsilon_1 s_1 + \varepsilon_2 s_2 + \varepsilon_3 s_3)/4$. So

$$\pi_0 = c\Big(\frac{1 - s_1 - s_2 - s_3}{4}\Big), \qquad \pi_{12} = c\Big(\frac{1 + s_1 + s_2 - s_3}{4}\Big),$$

and $\pi_{13}$, $\pi_{23}$ likewise, with the plus signs at $s_1, s_3$ and at $s_2, s_3$. Let $a^t$ be the weight $a$ with its $t$-th coordinate changed; then $s_t(e_a e_b) = e_a e_b$ if $t \notin D(a,b)$ and $s_t(e_a e_b) = e_{a^t} e_{b^t}$ if $t \in D(a,b)$. Under the Künneth map to $H^4(Y, \mathbb{C})$ the term of the pair $\{a, b\}$ in $c(s_t)$ is, up to sign, the monomial $e^{(1)}_{a'} e^{(1)}_{b'} e^{(2)}_{-a} e^{(2)}_{-b}$ with $\{a', b'\} = \{a, b\}$ if $t \notin D(a,b)$ and $\{a', b'\} = \{a^t, b^t\}$ otherwise. If $t \notin D(a,b)$, or $D(a,b) = \{t\}$, in which case $a^t = b$, or $D(a,b) = \{1,2,3\}$, in which case $b = -a$ and the monomial is $e^{(1)}_{a^t} e^{(1)}_{-a^t} e^{(2)}_{-a} e^{(2)}_{a}$, the monomial contains an opposite pair and is a product of two divisor classes. In the remaining case $D(a,b) = \{t, t'\}$ with $t' \neq t$; then $b = a^{tt'}$, and the four weights $a^t$, $b^t = a^{t'}$, $-a$, $-b = a^{t''}$, where $\{t, t', t''\} = \{1,2,3\}$, are pairwise non-opposite, since any two of them differ in exactly two coordinates, and they have sum zero, because $a^t + b^t$ and $a + b$ agree outside the $t$-th coordinate, where both vanish. So

$$m_{ab} = \chi(a)\chi(b)\, e^{(1)}_{a^t} e^{(1)}_{b^t} e^{(2)}_{-a} e^{(2)}_{-b}, \qquad t = \min D(a,b),$$

is a tetrahedron monomial with $D(a^t, b^t) = D(-a, -b) = D(a,b)$, and since $s_{t'}(e_a e_b) = e_{a^{t'}} e_{b^{t'}} = e_{b^t} e_{a^t} = -e_{a^t} e_{b^t}$, the term of the pair in $c(s_{t'})$ is $-m_{ab}$, while its term in $c(s_{t''})$ is a product of divisor classes. For each two-element set $D$ put $u_D = \sum_{D(a,b)=D} m_{ab}$, a sum over the four pairs with difference set $D$: two of these pairs lie in each tetrahedron and $a^t$ lies in the other one, so $u_D$ has two monomials in each tetrahedron, all with coefficients $\pm 1$, and the three $u_D$ are supported on disjoint sets of monomials; the two monomials of $u_D$ in $T$ are those of the pair of $T$ indexed by $D$. In $\Xi$ this gives $c(1) \equiv 0$, $c(s_1) \equiv u_{12} + u_{13}$, $c(s_2) \equiv -u_{12} + u_{23}$, $c(s_3) \equiv -u_{13} - u_{23}$, hence

$$\pi_0 \equiv 0, \quad \pi_{12} \equiv \tfrac{1}{2}(u_{13} + u_{23}), \quad \pi_{13} \equiv \tfrac{1}{2}(u_{12} - u_{23}), \quad \pi_{23} \equiv -\tfrac{1}{2}(u_{12} + u_{13}), \tag{76}$$

and $\omega_a \equiv \frac{1}{2}((a_2 - a_3)u_{12} + (a_1 - a_3)u_{13} + (a_1 - a_2)u_{23})$ in $\Xi$: on the monomials of the pair of $T_\pm$ indexed by $\{1,2\}$, $\{1,3\}$, $\{2,3\}$ the class $\omega_a$ has coefficients of absolute values $|a_2 - a_3|/2$, $|a_1 - a_3|/2$, $|a_1 - a_2|/2$.

Now suppose that $\omega_a \in \Sigma_{\mathbb{Q}} \otimes \mathbb{C}$. Its image in $\Xi$ is a combination of the $s_{\pm,k}$, and since every monomial of $\omega_a$ has two factors from each copy, only $s_{+,2}$ and $s_{-,2}$ occur: $\omega_a \equiv x\, s_{+,2} + y\, s_{-,2}$. On the six monomials of $T_+$ the right side has coefficients of absolute value $|x|$. So the three differences $a_2 - a_3$, $a_1 - a_3$, $a_1 - a_2$ have a common absolute value $r$, and as $a_1 - a_3 = (a_1 - a_2) + (a_2 - a_3)$, $r \in \{0, 2r\}$; hence $r = 0$ and $a_1 = a_2 = a_3$. Conversely the classes with $a_1 = a_2 = a_3$ are the combinations of $\pi_0$ and $\pi_{12} + \pi_{13} + \pi_{23}$, which are invariant under $\mathrm{Sp}(V, \psi)$ and so are combinations of the products of divisor classes $\psi_{11}\psi_{22}$ and $\psi_{12}^2$ (proof of Proposition 21.2(ii)); they lie in $\Sigma_{\mathbb{Q}}$. An exceptional class has for $(a_1, a_2, a_3)$ the three distinct conjugates of an element of $F \setminus \mathbb{Q}$ (Lemma 21.19), so it does not lie in $\Sigma_{\mathbb{Q}}$.

(iii) We first show that (c) is equivalent to $R$ and $k_c$ generating $\mathrm{End}^0(X)$, and that $R$ separates the weights of $T_+$ if and only if it separates those of $T_-$. For $\varphi \in \mathrm{End}^0(X)$ the eigenvalue on $e_{-w}$ is $\varphi_{-w} = (\varphi^\dagger)_w$; since $\mathrm{End}^0(X)$ is a product of CM fields on which $\dagger$ is complex conjugation, and every embedding of a CM field into $\mathbb{C}$ commutes with complex conjugation, $\varphi_{-w} = \overline{\varphi_w}$. Hence $\varphi_w = \varphi_{w'}$ if and only if $\varphi_{-w} = \varphi_{-w'}$, and $R$ separates $T_+$ if and only if it separates $T_-$. The elements of $k_c$ are constant on each tetrahedron and

separate the two, and a subalgebra of $\bar{\mathbb{Q}}^{\{\text{weights}\}}$ containing the constants is the algebra of functions constant on the blocks of the partition it induces; so $R$ separates $T_+$ exactly when $R \cdot k_c \otimes \bar{\mathbb{Q}}$ separates all eight weights, that is, when $R \cdot k_c = \mathrm{End}^0(X)$.

(c) implies (b). If $R$ separates $T_+$, then $R \otimes \mathbb{C}$ maps onto $\mathbb{C}^{T_+}$ under $\varphi \mapsto (\varphi_w)_{w \in T_+}$, so the entries $\varphi_w$, $\psi_w$, $w \in T_+$, are eight independent coordinates on the image of $(R \otimes \mathbb{C})^2$. The sixteen coefficients in (75) are distinct monomials in these coordinates, hence linearly independent functions of $(\varphi, \psi) \in (R \otimes \mathbb{C})^2$; so no nonzero linear form on the span of the sixteen tetrahedron monomials of $T_+$ vanishes on all $f^*_{\varphi,\psi} t_+$, and these classes span that space. The same holds for $T_-$, and with the 100 products of divisor classes this is all of $\mathrm{Hdg}^2(Y) \otimes \mathbb{C}$. The map $(\varphi, \psi) \mapsto f^*_{\varphi,\psi} t_\pm$ is polynomial and $R \times R$ is Zariski dense in $(R \otimes \mathbb{C})^2$, so a linear form vanishing on its values at rational pairs vanishes on all values, and finitely many pairs $\varphi, \psi \in R$ give the same span. The $\mathbb{Q}$-span of the rational classes $f^*_{\varphi,\psi} W(X)$, $\varphi, \psi \in R$, and of the products of divisor classes is then a $\mathbb{Q}$-subspace of $\mathrm{Hdg}^2(Y)$ whose complexification is the whole space, hence is $\mathrm{Hdg}^2(Y)$.

(b) implies (a), as $\mathrm{Hdg}^2(Y)$ contains the exceptional classes.

(a) implies (c). Suppose that two distinct weights $u, u' \in T_+$ have $\varphi_u = \varphi_{u'}$ for every $\varphi \in R$, and let $v, v'$ be the other two weights of $T_+$. In (75) the sets $S = \{u, v\}$ and $S' = \{u', v\}$ then have the same coefficient for every $\varphi, \psi \in R$, up to the signs of the shuffles, which do not depend on $\varphi, \psi$. Hence every class in $\Sigma_R \otimes \mathbb{C}$ has coefficients of equal absolute value on the tetrahedron monomials $e^{(1)}_S e^{(2)}_{T_+ \setminus S}$ and $e^{(1)}_{S'} e^{(2)}_{T_+ \setminus S'}$, the products of divisor classes involving no tetrahedron monomial. The pairs $\{S, T_+ \setminus S\}$ and $\{S', T_+ \setminus S'\}$ are different partitions of $T_+$, so $D(S) \neq D(S')$, and for an exceptional class $\omega_a$ in $\Sigma_R$ two of the three numbers $|a_2 - a_3|$, $|a_1 - a_3|$, $|a_1 - a_2|$ would be equal. Each of the six resulting equations, $a_2 - a_3 = \pm(a_1 - a_3)$ and the four others, is a linear relation $c_1 a_1 + c_2 a_2 + c_3 a_3 = 0$ with rational coefficients not all equal, which Lemma 21.19 forbids for the conjugates of an element of $F \setminus \mathbb{Q}$. So (a) fails, and this proves the equivalence.

For the last assertions: $K_c^+ \otimes \bar{\mathbb{Q}}$ consists of all functions on the opposite pairs, and $T_+$ meets every pair once, so $K_c^+$ separates $T_+$; a proper subalgebra $R$ of $K_c^+$ has $R \otimes \bar{\mathbb{Q}}$ a proper subalgebra of the functions on the four pairs, which identifies two pairs and hence two weights of $T_+$; and a subalgebra of $k_c$ is constant on $T_+$. ∎

Figure 56 draws the weights of part (i), and Figure 57 the Hodge classes of parts (ii) and (iii).

The CM points fall into two kinds, according to whether the fourfold is simple; at the points of the second kind the real multiplication comes from the fixed cubic field $F$.

**Lemma 21.30** (Split CM points)**.** *Let $c \in C$ be a CM point, and let $L \subset D$ be the quadratic extension of $F$ whose norm one torus $\mathrm{Res}_{F/\mathbb{Q}} \ker(\mathrm{N}_{L/F})$ is the Hodge group of $X_c$ (proof of Theorem 20.29). Then $L$ is a CM field, and the following are equivalent:* (a) *$X_c$ is not simple;* (b) *$K_c^+$ is not a field;* (c) *$L = Fk$ for an imaginary quadratic field $k \subseteq L$. In that case $k_c \cong k$, $K_c^+ \cong \mathbb{Q} \times F$, and $X_c$ is isogenous to $E \times B$, where $E$ is an elliptic curve with complex multiplication by $k$ and $B$ is an abelian threefold of CM type on which the factor $F$ of $K_c^+$ acts, a real multiplication by $F$; the Weil line $W(X_c)$ lies in $H^1(E) \otimes H^3(B)$. At every other CM point $K_c^+$ is a totally real quartic field. Points of the second kind exist for infinitely many $k$: for every imaginary quadratic field $k$ such that $k \otimes_{\mathbb{Q}} F_v$ is a field at each finite place $v$ of $F$ at which $D$ is ramified.*

*Proof.* Write $T$ for the Hodge group. By Proposition 21.29(i), $\mathrm{End}^0(X_c) = K_c^+ \otimes k_c$ is a product of CM fields on which † is complex conjugation, and $T$ lies in the centraliser of

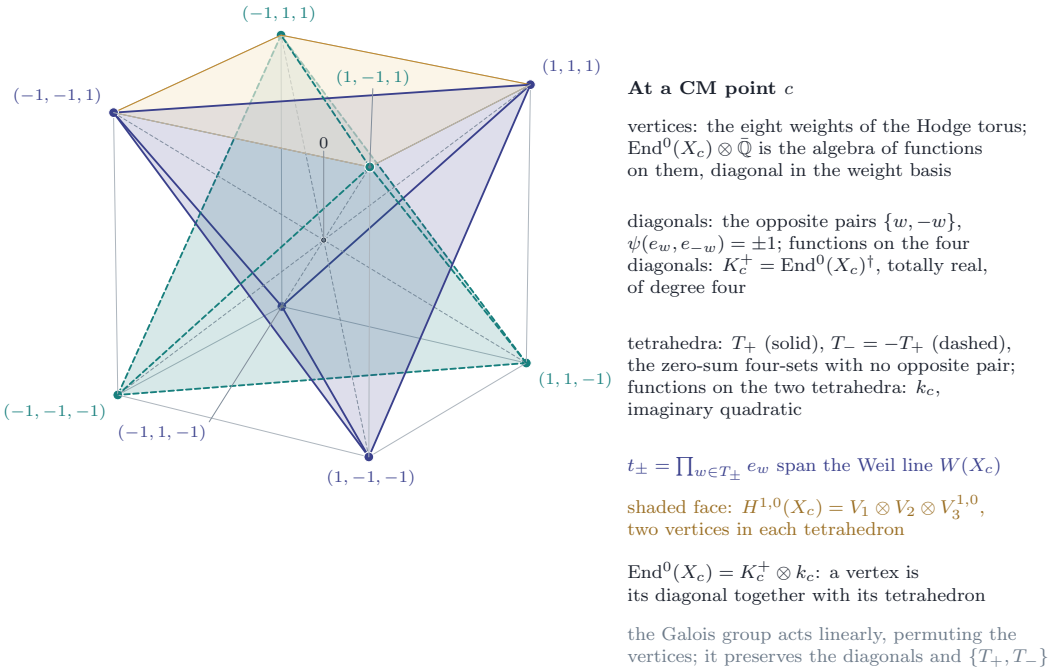
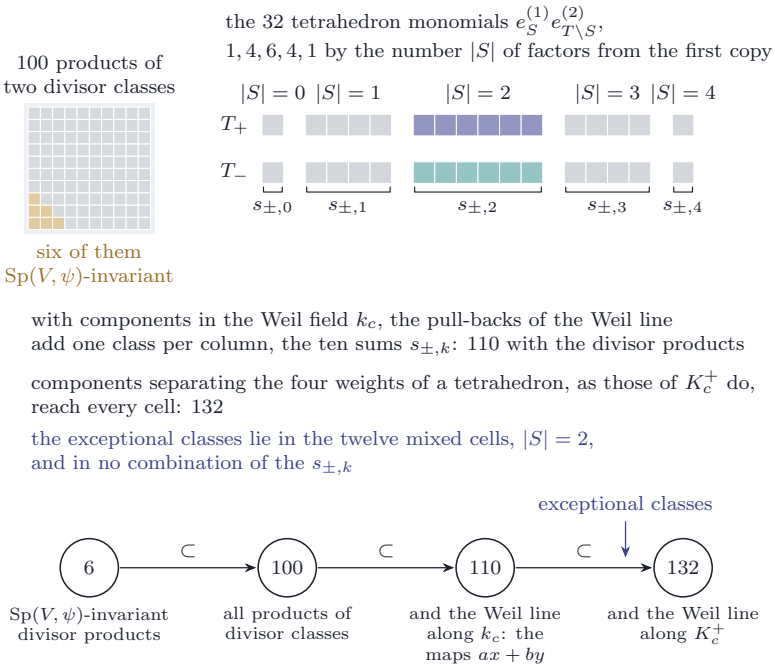



FIGURE 56. Proposition 21.29(i). The eight weights $(\pm1, \pm1, \pm1)$ of the Hodge torus at a CM point are the vertices of a cube. The four diagonals are the pairs of opposite weights, paired by the polarisation; the two inscribed tetrahedra $T_+$ (an even number of entries $-1$) and $T_- = -T_+$ are the only zero-sum four-sets without an opposite pair, and $t_\pm = \prod_{w\in T_\pm} e_w$ span the Weil line. The Galois group acts by linear maps permuting the vertices, so it preserves the diagonals and the pair of tetrahedra: the functions constant on the tetrahedra form the imaginary quadratic Weil field $k_c$, those constant on the diagonals the totally real quartic algebra $K_c^+$ of Rosati-fixed endomorphisms, and $\mathrm{End}^0(X_c) = K_c^+ \otimes k_c$ because a vertex is determined by its diagonal and its tetrahedron.

$\mathrm{End}^0(X_c)$, which over $\bar{\mathbb{Q}}$ is the diagonal torus, and in $\mathrm{Sp}(V, \psi)$; so $T(\mathbb{R})$ lies in the compact group of the $x \in (\mathrm{End}^0(X_c) \otimes \mathbb{R})^\times$ with $xx^\dagger = 1$. If $L \otimes_{F,\sigma} \mathbb{R}$ were $\mathbb{R} \times \mathbb{R}$ at a real place $\sigma$, the factor of $T(\mathbb{R})$ at $\sigma$ would be the norm one group of $\mathbb{R} \times \mathbb{R}$ over $\mathbb{R}$, which is not compact. Hence $L \otimes_{F,\sigma} \mathbb{R} = \mathbb{C}$ at each real place, and $L$ is a CM field.

Over $\bar{\mathbb{Q}}$, $T = \prod_\sigma T_\sigma$ with $T_\sigma = \ker(\mathrm{N}_{L/F}) \otimes_{F,\sigma} \bar{\mathbb{Q}}$ the diagonal torus of $\mathrm{SL}_2(\bar{\mathbb{Q}}) = (D \otimes_{F,\sigma} \bar{\mathbb{Q}})^1$ for the decomposition $L \otimes_{F,\sigma} \bar{\mathbb{Q}} = \bar{\mathbb{Q}} \times \bar{\mathbb{Q}}$ by the two embeddings of $L$ over $\sigma$, one of them chosen and called $\tau_\sigma$, the other $\bar\tau_\sigma$. The factor $V_\sigma$ of $V = \bigotimes_\sigma V_\sigma$ (Setup 21.1) has the weights $\tau_\sigma$ and $\bar\tau_\sigma = \tau_\sigma^{-1}$ on $T_\sigma$, with eigenvectors $v_{\sigma,\pm}$, and $e_w = \bigotimes_\sigma v_{\sigma,w_\sigma}$. An element $\gamma$ of the Galois group carries $V_\sigma$ to $V_{\gamma\sigma}$ and $v_{\sigma,+}$ to $v_{\gamma\sigma,\varepsilon_\gamma(\sigma)}$, where $\varepsilon_\gamma(\sigma) = 1$ if $\gamma\tau_\sigma = \tau_{\gamma\sigma}$ and $-1$ if $\gamma\tau_\sigma = \bar\tau_{\gamma\sigma}$; so it acts on the weights by $(\gamma w)_{\gamma\sigma} = \varepsilon_\gamma(\sigma) w_\sigma$.

If $L = Fk$, then $\mathrm{Hom}(L, \bar{\mathbb{Q}}) = \mathrm{Hom}(F, \bar{\mathbb{Q}}) \times \mathrm{Hom}(k, \bar{\mathbb{Q}})$ with the Galois group acting on both factors, and with $\tau_\sigma = \sigma \otimes \iota_0$ for a fixed embedding $\iota_0$ of $k$ the sign $\varepsilon_\gamma(\sigma) = \varepsilon_\gamma$ does not depend on $\sigma$: it is 1 if $\gamma$ fixes $k$ and $-1$ otherwise. Then $\gamma$ permutes the coordinates of a weight and changes all its signs when it moves $k$. It preserves each tetrahedron if it



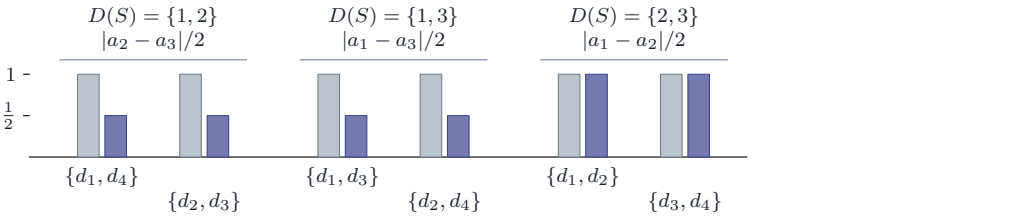




FIGURE 57. Proposition 21.29(ii) and (iii). Top: the 132 Hodge classes of degree four on the square at a CM point, one cell per monomial of weight zero. The 100 products of divisor classes are the cells with an opposite pair; the 32 tetrahedron monomials $e_S^{(1)} e_{T\setminus S}^{(2)}$ are arranged by tetrahedron and by the number $|S|$ of factors from the first copy. Pull-backs of the Weil line along homomorphisms with components in $k_c$ add only the ten symmetric sums $s_{\pm,k}$, one per column, giving 110; components separating the four weights of a tetrahedron, as those of $K_c^+$ do, separate the monomials and give all 132. Bottom: on the six mixed monomials of one tetrahedron, grouped in the three pairs by the coordinates in which the two weights of $S$ differ, the symmetric sum $s_{+,2}$ has coefficients $\pm 1$ throughout, while an exceptional class $\omega_a$ has coefficients of absolute value $|a_2 - a_3|/2$, $|a_1 - a_3|/2$, $|a_1 - a_2|/2$, drawn for $\pi_{12} - \pi_{13}$; two equal absolute values would be a linear relation among the conjugates of an irrational element of $F$.

fixes $k$ and exchanges the two otherwise, so $k_c \otimes \bar{\mathbb{Q}} = \bar{\mathbb{Q}}^{\{T_+,T_-\}}$ is the algebra of the Galois set $\mathrm{Hom}(k,\bar{\mathbb{Q}})$, and $k_c \cong k$. It carries the pair $\{\pm(1,1,1)\}$ to itself, so the projection $p_E$ onto $\mathbb{C}e_{(1,1,1)} \oplus \mathbb{C}e_{(-1,-1,-1)}$ lies in $\mathrm{End}^0(X_c)$, and by Poincaré reducibility $X_c$ is isogenous to $E \times B$ with $H^1(E,\mathbb{C})$ the image of $p_E$, of dimension two, so $E$ is an elliptic curve, on whose $H^1$ the field $k_c$ acts by two conjugate scalars: $E$ has complex multiplication by $k$. The threefold $B$ has $\mathrm{End}^0(B) \otimes \bar{\mathbb{Q}}$ containing the diagonal algebra of the six remaining weights, of dimension $6 = 2\dim B$, so $B$ is of CM type. Each of the six remaining weights has exactly one coordinate different from the other two, and $\gamma$ carries the opposite pair of the weight with distinguished coordinate $t$ to the pair with distinguished coordinate $\gamma t$; so the Galois set of the four opposite pairs is the fixed point $\{\pm(1,1,1)\}$ together with a copy of $\mathrm{Hom}(F,\bar{\mathbb{Q}})$, and $K_c^+ \cong \mathbb{Q} \times F$, the factor $\mathbb{Q}$ acting on $H^1(E)$ and the factor $F$ on $H^1(B)$. Finally $T_+$ consists of $(1,1,1)$ and three weights of $B$, and $T_-$ of $(-1,-1,-1)$ and three weights of $B$, so $t_\pm \in H^1(E) \otimes H^3(B)$ and $W(X_c)$ lies there. This proves that (c) implies (a) and (b), with the asserted structure.

$X_c$ is simple exactly when $\mathrm{End}^0(X_c) = K_c^+ \otimes k_c$ is a field, and since every factor field $E'$ of $K_c^+$ is totally real, $E' \otimes k_c$ is a field; so (a) and (b) are equivalent. Suppose (b). Then the Galois group does not act transitively on the four opposite pairs, whose algebra is $K_c^+$. It acts transitively on the three coordinates, as on the embeddings of $F$, so some $\gamma$ permutes them cyclically. For each coordinate $t$ the four pairs fall into two sets of two, according to the value of the coordinate $t$ on the representatives in $T_+$, and this assigns the three coordinates bijectively to the three partitions of the four pairs into two sets of two, compatibly with the Galois action, which preserves or exchanges $T_+$ and $T_-$. So the permutation of the pairs induced by $\gamma$ has order divisible by three, hence equal to three: it is a cycle of length three and fixes exactly one pair. The orbits of the Galois group on the pairs are therefore either one orbit or two, of sizes one and three, and by hypothesis some pair $\{w,-w\}$ is fixed: $\gamma w = \delta_\gamma w$ with $\delta_\gamma = \pm 1$ for every $\gamma$. Replace the choice of $\tau_\sigma$ by $\tau'_\sigma = \tau_\sigma$ if $w_\sigma = 1$ and $\tau'_\sigma = \bar{\tau}_\sigma$ if $w_\sigma = -1$; for the new choice $w$ becomes $(1,1,1)$, and the new signs are $\varepsilon'_\gamma(\sigma) = \varepsilon_\gamma(\sigma) w_\sigma w_{\gamma\sigma} = \delta_\gamma$ for all $\sigma$. So $\gamma$ carries the set $\Phi = \{\tau'_\sigma\}$ of three embeddings of $L$ to $\Phi$ if $\delta_\gamma = 1$ and to $\bar{\Phi}$ if $\delta_\gamma = -1$. The map $\gamma \mapsto \delta_\gamma$ is a character of the Galois group; it is nontrivial, for otherwise $\Phi$ would be a Galois stable set of three of the six embeddings and $L$ would not be a field; and complex conjugation carries every embedding $\tau$ of the CM field $L$ to $\bar{\tau}$, hence $\Phi$ to $\bar{\Phi}$, so the quadratic field $k$ cut out by the character is imaginary. Let $\iota_0$, $\bar{\iota}_0$ be the embeddings of $k$, with $\iota_0$ fixed by the kernel of the character. Then $\tau \mapsto (\tau|_F, \iota_0$ if $\tau \in \Phi$ and $\bar{\iota}_0$ otherwise) is a Galois equivariant bijection $\mathrm{Hom}(L,\bar{\mathbb{Q}}) \to \mathrm{Hom}(F,\bar{\mathbb{Q}}) \times \mathrm{Hom}(k,\bar{\mathbb{Q}})$, so $L \cong F \otimes_{\mathbb{Q}} k$, that is, $L = Fk$ for a copy of $k$ in $L$, which is (c). At a CM point where (c) fails, $K_c^+$ is therefore a field, quartic and totally real by Proposition 21.29(i).

Existence. Let $k$ be imaginary quadratic. By the theorem of Albert, Brauer, Hasse and Noether [PR94, §1.5], the CM field $Fk$ splits $D$, and so embeds in $D$ over $F$, exactly when $Fk \otimes_F F_v$ is a field at every place $v$ of $F$ at which $D$ is ramified, a quadratic extension of a local field splitting the local quaternion division algebra. At the two real places where $D$ is definite this holds because $k$ is imaginary, and at the finite ramified places it is the hypothesis. Given the finitely many residue characteristics $p$ of these places, choose for each a quadratic extension $k_p$ of $\mathbb{Q}_p$ not contained in $F_v$ for any $v$ above $p$: this is possible since the completions $F_v$ above $p$ have degrees over $\mathbb{Q}_p$ adding up to three, so at most one of them contains a quadratic extension of $\mathbb{Q}_p$, and then exactly one, while $\mathbb{Q}_p$ has at least three. Infinitely many imaginary quadratic fields $k$ have the completions $k_p$ at these $p$, and for them $F_v \otimes_{\mathbb{Q}_p} k_p$ is a field. For such $k$ let $T_k = \mathrm{Res}_{F/\mathbb{Q}} \ker(\mathrm{N}_{Fk/F}) \subset G$, a maximal

torus with $T_k(\mathbb{R})$ compact. A compact subgroup of $G(\mathbb{R})$ fixes a point $x$ of the symmetric space of $G(\mathbb{R})$, and the stabiliser of $x$ is the centraliser of $h_x(\mathbb{S})$; so $h_x(\mathbb{S})$ centralises $T_k(\mathbb{R})$, which is Zariski dense in $T_k$, and lies in $T_k(\mathbb{R})$. The Hodge group of the fibre $X_c$ over the image $c$ of $x$ is the smallest $\mathbb{Q}$-subgroup of $G$ containing $h_x(U(1))$, so it lies in $T_k$; it has rank three (proof of Theorem 20.29), so it equals $T_k$, and $c$ is a CM point with $L = Fk$. ■

*Remark* 21.31 (What the degree bound has to control).

(1) Markman's theorem [Mar25a], which the proof of Theorem 20.29 invokes, produces the cycles on $X_c$ through its Weil structure for $k_c$. Proposition 21.29 says that this structure is blind to the exceptional classes of the square: everything obtained from the Weil line and from divisors by pull-back along homomorphisms with components in $k_c$, in particular along the maps $(x, y) \mapsto ax + by$, and by products, spans 110 of the 132 Hodge classes and misses every exceptional class; and a representation of an exceptional class by pull-backs of the Weil line and products of divisors must use homomorphisms whose components, with $k_c$, generate the whole endomorphism algebra. At a CM point the exceptional classes are thus a statement about the real multiplication $K_c^+$, as they are at a general point about the real multiplication by $F$ (Proposition 21.2).

(2) For $\varphi, \psi \in \mathrm{End}(X_c)$ and a cycle $Z$ of codimension two on $X_c$, the degree $\int_Y f_{\varphi,\psi}^*[Z] \cup \theta_Y^6$ is, by (75) applied to the monomials of $[Z]$, a polynomial of degree four in the eigenvalues $\varphi_w$, $\psi_w$, which are algebraic integers. For cycles of this shape the bound of Corollary 21.28 therefore asks, at infinitely many CM points, for endomorphisms with bounded eigenvalues whose components generate $\mathrm{End}^0(X_c)$ over $k_c$. Only finitely many algebraic integers of degree at most eight have bounded conjugates, so the same finitely many minimal polynomials would have to occur at infinitely many points. At the split points of Lemma 21.30 the algebra $K_c^+ = \mathbb{Q} \times F$ does not depend on the point and the real multiplication is by $F$ on the threefold factor; the representation of the proof of Theorem 20.29 then uses, besides elements of $F$, the isogeny $X_c \to E \times B$, a cycle representing the Weil line of $E \times B$ in $H^1(E) \otimes H^3(B)$, and divisor classes, and these three are what varies with the point.

(3) At a split point the Weil line is a Hodge class in $H^1(E) \otimes H^3(B)$, a correspondence between an elliptic curve with complex multiplication by $k$ and a CM threefold with real multiplication by $F$, and its algebraicity is again Markman's theorem for the fourfold $E \times B$. A construction of that correspondence with a degree bounded independently of $k$, together with a bound on the isogeny and on the divisor classes, would give the target through Corollary 21.28. We do not have one.

*Remark* 21.32 (Routes to the Mumford target). We list the routes to the Mumford target that avoid the Lefschetz standard conjecture, with what is known about each; Figure 58 summarises them.

(1) *Degeneration.* Closed off: the base is compact and the Hodge locus is the base (Corollary 21.23), so no family on which $\omega$ stays of Hodge type approaches the boundary, and there is no limit mixed Hodge structure to specialise to.

(2) *Deformation into a larger Hodge locus*, on which $\omega$ would become a product of divisor classes or a Weil class, hence algebraic by a known theorem, and could then be carried back. Closed off by Theorem 21.21(iii), already to first order.

(3) *Hochschild deformations*: twisting $D^b(Y)$ by a $B$-field or a Poisson bivector, as in the gerby and noncommutative deformations used for Weil classes. Closed off: an

object with $\mathrm{ch}(E) = N\omega$ extends along no direction of $HH^2(Y)$ except that of the curve (Corollary 21.41(i)).

(4) *Constructions from line bundles*: cones, tensor products, duals, pull-backs and push-forwards along homomorphisms, and Fourier–Mukai functors whose kernels are built in the same way. Closed off at every point of the curve: everything so built has a Chern character invariant under the Lefschetz group, and the exceptional classes are not (Theorem 21.36).

(5) *Twistor and hyperholomorphic deformation*, the mechanism of Markman's proof for Weil classes: move a sheaf along twistor lines on which its Chern character stays of Hodge type. Closed off on the square: no twistor line through a member of the Mumford family keeps $\omega$ of Hodge type, and the twistor lines that do keep it of Hodge type carry non-algebraic tori and lie in other components of the Hodge locus (Proposition 21.24).

(6) *Specialisation.* The class is algebraic at every CM point (Theorem 20.29), on every power of $X_c$ [Li26, Theorem 1.3]; this suffices exactly when the degrees are bounded (Corollary 21.28), and no bound is known. The cycles that represent the class at a CM point must use the real multiplication by the Rosati-fixed algebra $K_c^+$, all of it; the Weil structure of the fourfold, with which Markman's theorem produces its cycles, reaches 110 of the 132 Hodge classes and none of the exceptional ones (Proposition 21.29, Remark 21.31).

(7) *Reduction modulo p.* At every closed point of every reduction of the curve the class is a $\mathbb{Q}_\ell$-combination of algebraic cycles (Proposition 21.42), by Li's theorem [Li26, Theorem 1.1]; this suffices, with no lifting step, exactly when the Hilbert data of the cycles are bounded on a Zariski dense set of closed points of the arithmetic model (Theorem 21.43), and no bound is known.

(8) *Semiregularity.* Reduced to one dimension count at one CM point (Theorem 21.27); an object attaining it has large odd self-extensions (Theorem 21.34), is not built from line bundles (Proposition 21.35) and splits only in a constrained way (Corollary 21.41); no such object is known.

(9) *Kuga–Satake.* Reduced to the Kuga–Satake class of one K3 surface of Picard number thirteen (Theorem 21.4), which the known cases of the Kuga–Satake Hodge conjecture do not reach (Remark 21.5). Its geometric form is the mechanism of van Geemen and Rapagnetta [vGR26], pull-back of the dual Beauville–Bogomolov class of a hyperkähler manifold: up to multiples of $\pi_0$, the exceptional classes are exactly such twisted dual forms of the part of $H^2$ of the square generated by its symplectic class, and a rational map onto a symplectic subvariety of some $S_\mu^{[n]}$, $n \geq 5$, would suffice; K3 surfaces and hyperkähler eightfolds cannot serve (Proposition 21.25, Remark 21.26).

(10) *Motivated classes.* Every Hodge class on an abelian variety is motivated [And96b, Théorème 0.6.2], so $\omega_t$ is motivated for every $t$; a motivated class is algebraic once the Lefschetz involutions of the auxiliary varieties are (Proposition 20.16). Along the family the auxiliary variety that suffices is $W \times_C W$ (Theorem 20.28), and its Lefschetz standard conjecture is equivalent to the target (Corollary 20.30): the route moves the problem and does not shorten it.

Five of the ten routes are closed off by theorems. Of the other five, two work at special points only, the CM points and the closed points of the reductions modulo $p$, and each suffices exactly when a degree bound holds; two are reduced to a single object, one Kuga–Satake class or one perfect complex with $\dim \mathrm{Ext}^2 = 119$ at one CM point; and one is

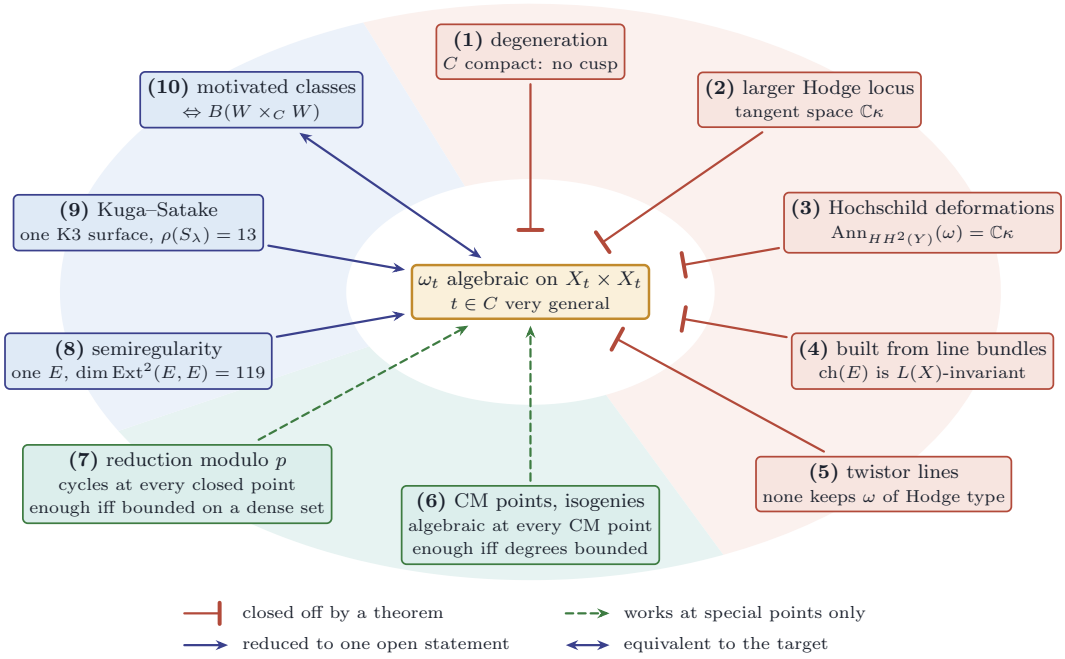



FIGURE 58. Remark 21.32: the ten routes to the Mumford target that avoid the Lefschetz standard conjecture. Clay, with a crossed dashed spoke: a route closed off by a theorem of Section 21.2 or Section 21.3. Grass: a route that works at special points only, at the CM points or at the closed points of the reductions modulo $p$, and suffices exactly when a degree bound holds (Corollary 21.28, Theorem 21.43). Indigo: a route reduced to one statement, of the same kind as the target or equivalent to it.

equivalent to the target. They are not five separate obligations: three of them are the target itself in other forms and the other two are stronger inputs that imply it, so any one of them would suffice (Corollary 21.45, Remark 21.46).

21.3. **Objects, reduction modulo $p$ and the motivic group.** Theorem 21.27 reduces the semiregularity route to one object at one CM point. We first derive what any object with the right Chern character must satisfy, whether or not it is semiregular: no sequence of standard operations produces it from line bundles; it extends along no first order deformation of $D^b(Y)$ other than the direction of the curve; if it is semiregular, it splits only into summands with no $\mathrm{Ext}^2$ between them; and its self-extensions are large in every degree, with an odd part that the Euler characteristic forces beyond what Hochschild cohomology supplies. We then turn to reduction modulo $p$. By Li's proof of the Tate conjecture for powers of abelian fourfolds over finite fields, the classes are represented by cycles at every closed point of every reduction of the curve, and the target is equivalent to a bound on those cycles on a Zariski dense set of closed points. Finally, the motivic group of $X$ is one of four groups, and the kinds of algebraic classes known on products of $X$ with other abelian varieties generate no exceptional class. None of this constructs such an object or proves the target.

**Lemma 21.33** (The contraction ranks are symmetric)**.** *Let $A$ be an abelian variety of dimension $g$ and $c \in H^*(A,\mathbb{C})$ a class each of whose components has Hodge type $(p,p)$ for some $p$. For $k \geq 0$ put $r_k(c) = \dim(HT^k(A) \lrcorner c)$, with the action of Setup 16.33. Then $r_k(c) = r_{g-k}(c)$ for $0 \leq k \leq g$, and $r_k(c) = 0$ for $k > g$.*

*Proof.* Write $H^{1,0} = H^{1,0}(A)$ and $A' = H^{0,1}(A)$, both of dimension $g$, so that $H^*(A,\mathbb{C}) = \bigwedge^* H^{1,0} \otimes \bigwedge^* A'$ and $HT^*(A) = \bigwedge^*(A' \oplus T)$ with $T = (H^{1,0})^\vee$. Fix a generator of $\bigwedge^g A'$ and let $\star\colon \bigwedge^q A' \to \bigwedge^{g-q} A'^\vee$ send $\eta$ to $\zeta \mapsto \eta \wedge \zeta$. Under $\Phi = \mathrm{id} \otimes \star$, wedging with $z \in A'$ becomes, up to a sign fixed on each bigraded piece, contraction with $z$ on the second factor, while $T$ still contracts the first. So $\Phi$ identifies the action of $HT^*(A)$ with the action of $\bigwedge^* U^\vee$ by interior products on $\bigwedge^* U$, where $U = H^{1,0} \oplus A'^\vee$ has dimension $2g$, up to signs that do not change ranks. A class of type $(p,p)$ goes to $\bigwedge^p H^{1,0} \otimes \bigwedge^{g-p} A'^\vee \subset \bigwedge^g U$, so $\eta = \Phi(c)$ lies in the middle degree whatever the $p$ of its components. Hence $r_k(c)$ is the rank of $\bigwedge^k U^\vee \to \bigwedge^{g-k} U$, $x \mapsto \iota_x \eta$, which is zero for $k > g$. Its transpose for the natural pairings is $\bigwedge^{g-k} U^\vee \to \bigwedge^k U$, $y \mapsto \pm \iota_y \eta$, because $\langle \iota_x \eta, y\rangle = \langle \eta, x \wedge y\rangle = \pm\langle \iota_y \eta, x\rangle$. A map and its transpose have the same rank. ∎

**Theorem 21.34** (The self-extensions of an object with an exceptional Chern character)**.** *Let $c \in C$ and $Y = X_c \times X_c$, and write $e_k = \dim \mathrm{Ext}^k(E,E)$ for a perfect complex $E$ on $Y$.*

(i) *For every $E$ and every $k$, $e_k \geq r_k(\mathrm{ch}\, E)$ and $e_k = e_{8-k}$.*

(ii) *For a rational exceptional class $\omega = \omega_a$ with $L_0(a) \neq 0$: $r_0 = r_8 = 1$, $r_1 = r_7 = 16$ and $r_2 = r_6 = 119$.*

(iii) *Let $\omega$ be a rational exceptional class and $\mathrm{ch}(E) = N\omega$ with $N \neq 0$. Then*

$$e_1 + e_3 = e_0 + e_2 + \tfrac{1}{2} e_4 + \sum_{j \geq 1} (-1)^j e_{-j}.$$

*If $\mathrm{Ext}^{<0}(E,E) = 0$, for instance if $E$ is a sheaf, then $\sum_{k \text{ odd}} e_k = \sum_{k \text{ even}} e_k$, and $e_1 + e_3 \geq 120 + \frac{1}{2} e_4$ when $L_0(a) \neq 0$.*

*Proof.* (i) The identity $\sigma_E \circ \mathrm{ev}_E = (\,\cdot\,) \lrcorner \mathrm{ch}(E)$ of Setup 16.33 holds in every degree, so the kernel of $\mathrm{ev}_E$ on $HT^k(Y)$ lies in the annihilator of $\mathrm{ch}(E)$, and the image of $HT^k(Y)$ in $\mathrm{Ext}^k(E,E)$ has dimension at least $r_k(\mathrm{ch}\, E)$. Serre duality with the trivial canonical bundle gives $e_k = e_{8-k}$.

(ii) $r_0 = 1$ because $\omega \neq 0$, and $r_2 = 119$ is Theorem 21.21(ii). If $w \in HT^1(Y)$ annihilates $\omega$, so does $w' \wedge w$ for every $w' \in HT^1(Y)$, since the annihilator is a left ideal; for $w \neq 0$ these products span a space of dimension 15, which cannot lie in $\mathrm{Ann}_{HT^2}(\omega) = \mathbb{C}\kappa$. So $r_1 = 16$, and Lemma 21.33 gives $r_8, r_7, r_6$.

(iii) Since $\mathrm{ch}(E)$ has only a component in $H^4$, $\mathrm{ch}(E^\vee)\,\mathrm{ch}(E)$ lies in $H^8(Y)$ and $\chi(E,E) = \int_Y \mathrm{ch}(E^\vee)\,\mathrm{ch}(E) = 0$, the Todd class being trivial. By (i) and Serre duality, which also pairs $\mathrm{Ext}^{-j}$ with $\mathrm{Ext}^{8+j}$,

$$0 = \chi(E,E) = 2(e_0 - e_1 + e_2 - e_3) + e_4 + 2\sum_{j \geq 1} (-1)^j e_{-j},$$

which is the displayed identity. With no negative extensions the odd and even parts are equal, and $e_0 \geq 1$, $e_2 \geq 119$ by (i) and (ii) when $L_0(a) \neq 0$. ∎

**Proposition 21.35** (Divisor classes do not reach the exceptional classes)**.** *Let $t \in C$ and let $\omega$ be a rational exceptional class. Then $\omega_t$ is not in the $\mathbb{Q}$-span of products of divisor classes of $X_t \times X_t$: that span meets the rational Hodge classes of $H^2 \otimes H^2$ exactly in the classes with $\alpha \in \mathbb{Q}$. Consequently no perfect complex whose Chern character lies in the*

*subring generated by divisor classes, in particular no bounded complex of direct sums of line bundles, has* $\mathrm{ch}(E) = N\omega$ *with* $N \neq 0$*. At a CM point* $c$ *the fourfold* $X_c$ *has eight Hodge classes in* $H^4$*, of which the products of divisor classes span six.*

*Proof.* At a point with Hodge group $G$, the divisor classes of $X_t \times X_t$ are the three classes of Proposition 20.74(ii), all invariant under $\mathrm{Sp}(V,\psi)$, and so are their products; the $\mathrm{Sp}(V,\psi)$-invariant part of $H^2 \otimes H^2$ is spanned by $\pi_0$ and $\pi_{12}+\pi_{13}+\pi_{23}$, the classes with $a_1 = a_2 = a_3$. At a CM point the Hodge group is a maximal torus of $G$ (proof of Theorem 20.29), so the Hodge classes are the monomials of weight zero. The proof of Proposition 21.29(ii) counts them: 16 divisor classes $e_w^{(i)}e_{-w}^{(j)}$, 132 Hodge classes in degree four, of which the products of divisor classes span 100, and modulo that span $\omega_a \equiv \frac{1}{2}((a_2-a_3)u_{12}+(a_1-a_3)u_{13}+(a_1-a_2)u_{23})$ with $u_{12},u_{13},u_{23}$ linearly independent. So the span meets that of the four $\pi$ in exactly the classes with $a_1 = a_2 = a_3$. For a rational class these coordinates are the conjugates of $\alpha$, which by Lemma 21.19 are equal only if $\alpha \in \mathbb{Q}$. The Chern character of a complex in the subring generated by divisor classes has its component in $H^4$ in the span of products of two divisor classes, which excludes $N\omega$. For $X_c$ itself the monomials of weight zero in $H^4$ are the six products $e_we_{-w}e_{w'}e_{-w'}$ over two opposite pairs, which are products of divisor classes, and the two tetrahedron monomials. ∎

At a CM point the Hodge classes of $X_c$ in $H^4$ that are not products of divisor classes are among the classes that the proof of Theorem 20.29 pulls back along homomorphisms $X_c \times X_c \to X_c$; they are algebraic by Markman's theorem [Mar25a]. Li proves, independently and more generally, that every Hodge class on every power of every complex CM abelian fourfold is algebraic [Li26, Theorem 1.3], which gives Theorem 20.29 for every power $X_c^n$ and not only for the square.

By Proposition 21.35, an object with $\mathrm{ch}(E) = N\omega$ cannot have its Chern character in the subring generated by divisor classes, so it cannot be assembled from line bundles alone. The next theorem shows that no sequence of the standard operations of the derived category, applied to line bundles on powers of $X_t$, gets any further.

**Theorem 21.36** (Objects generated by line bundles)**.** *Let* $t \in C$*,* $X = X_t$*, and let* $L(X) \subseteq \mathrm{Sp}(V,\psi)$ *be the Lefschetz group of* $X$*, the centraliser of* $\mathrm{End}^0(X)$ *in* $\mathrm{Sp}(V,\psi)$*. Let* $\mathcal{A}$ *be the class of abelian varieties isogenous to a power of* $X$*; through an isogeny* $X^n \to B$ *the group* $L(X)$ *acts diagonally on* $H^*(B,\mathbb{Q})$*, and the action does not depend on the isogeny. Let* $\mathcal{L}$ *be the smallest class of objects of the categories* $D^b(B)$*,* $B \in \mathcal{A}$*, that contains the line bundles and is closed under shifts, cones, derived tensor products, derived duals, translations, the functors* $Lf^*$ *and* $Rf_*$ *for homomorphisms* $f$ *between members of* $\mathcal{A}$*, and the Fourier–Mukai functors* $\Phi_K\colon D^b(B) \to D^b(B')$ *with kernel* $K \in \mathcal{L}$ *on* $B \times B'$*.*

(i) *Every object of* $\mathcal{L}$ *has an* $L(X)$*-invariant Chern character.*

(ii) $L(X) = \mathrm{Sp}(V,\psi)$ *if* $t$ *is not a CM point, and* $L(X)$ *is a maximal torus of* $\mathrm{Sp}(V,\psi)$ *if it is. In both cases the* $L(X)$*-invariant classes on* $X \times X$ *form the subring generated by divisor classes, of dimensions* $1,3,6,10,15,10,6,3,1$*, respectively* $1,16,100,304,454,304,100,16,1$*, in the degrees* $0,2,\dots,16$*.*

(iii) *No rational exceptional class* $\omega_t$ *is* $L(X)$*-invariant. Hence for no object* $E \in \mathcal{L}$ *on* $X \times X$ *and no* $N \neq 0$ *is* $\mathrm{ch}(E) - N\omega_t$ *invariant under* $L(X)$*; in particular* $\mathrm{ch}(E) \neq N\omega_t$*.*

(iv) *The* $\mathrm{Sp}(V,\psi)$*-module generated by* $\pi_{12}-\pi_{13}$ *is irreducible, of highest weight* $2\varpi_2$ *and dimension* 308*, and it meets the span of the four* $\pi$ *in the plane of the classes with* $a_0=0$ *and* $a_1+a_2+a_3=0$.

*Proof.* (i) Two isogenies $X^n\to B$ differ by an element of $\mathrm{End}^0(X^n)=M_n(\mathrm{End}^0(X))$, which commutes with $L(X)$ acting diagonally, so the action is well defined. For the same reason a homomorphism $f$ between members of $\mathcal{A}$ is $L(X)$-equivariant on $H^1$, so $f^*$ is equivariant on $H^*$, and so is $f_*$, which is $f^*$ conjugated by Poincaré duality, since $L(X)\subseteq\mathrm{SL}(V)$ fixes the orientation classes. Translations act trivially on cohomology. The Chern character is additive on distinguished triangles and multiplicative on derived tensor products, $\mathrm{ch}(E^\vee)=\sum_i(-1)^i\,\mathrm{ch}_i(E)$, $\mathrm{ch}(Lf^*E)=f^*\,\mathrm{ch}(E)$, and $\mathrm{ch}(Rf_*E)=f_*\,\mathrm{ch}(E)$ by the Grothendieck–Riemann–Roch theorem [Ful98, Theorem 15.2], the Todd class of an abelian variety being 1; and $\Phi_K=Rp_{2*}(p_1^*(\,\cdot\,)\otimes^L K)$. So the objects with $L(X)$-invariant Chern character form a class with all the closure properties, and it remains to see that it contains the line bundles, that is, that $c_1(\mathcal{O}(D))$ is $L(X)$-invariant. Pulled back along an isogeny, which is injective on rational cohomology, $c_1$ becomes a Hodge class of degree two on some $X^n$, an element of the invariants of the Hodge group in $\bigwedge^2(V^\vee\otimes\mathbb{Q}^n)=\bigwedge^2V^\vee\otimes S^2\mathbb{Q}^n\oplus S^2V^\vee\otimes\bigwedge^2\mathbb{Q}^n$. If $t$ is not a CM point, the Hodge group is $G$, whose invariants in $\bigwedge^2V$ are the multiples of $\psi$ and in $S^2V$ are zero, since every summand of $S^2(V_1\otimes V_2\otimes V_3)$ contains a factor $S^2V_i$; so the Hodge classes of degree two are $\mathrm{Sp}(V,\psi)$-invariant. If $t$ is a CM point, a monomial of degree two in a weight basis has weight zero for the Hodge torus exactly when its two weights are opposite, and then it has weight zero for $L(X)$, by the description in (ii).

(ii) If $t$ is not a CM point, $\mathrm{End}^0(X)=\mathbb{Q}$ and $L(X)=\mathrm{Sp}(V,\psi)$. At a CM point $\mathrm{End}^0(X)\otimes\mathbb{C}$ is the algebra of diagonal matrices in a weight basis (proof of Theorem 20.29), and $\psi$ pairs each weight with its opposite only, so $L(X)_\mathbb{C}$ is the torus of diagonal matrices $\mathrm{diag}(\lambda_w)$ with $\lambda_w\lambda_{-w}=1$, of rank four. For $t$ not CM, the first fundamental theorem for $\mathrm{Sp}(V,\psi)$ [GW09, Chapter 5] shows that its invariants in $\bigwedge^*(V\otimes\mathbb{Q}^2)$ are spanned by the products of the three quadratic invariants, which are the divisor classes. Their dimensions follow from the skew Cauchy formula $\bigwedge^{2m}(V\otimes\mathbb{Q}^2)=\bigoplus_\lambda S_\lambda V\otimes S_{\lambda'}\mathbb{Q}^2$, over the partitions $\lambda=(2^r,1^s)$ of $2m$ with at most two columns, and the rule that $S_\lambda V$ contains the trivial representation of $\mathrm{Sp}(V,\psi)$, then once, exactly when both columns of $\lambda$ have even length at most 8 [GW09]: the invariants of degree $2m$ have dimension $\sum(s+1)$, over the even $r,s\ge0$ with $2r+s=2m$ and $r+s\le8$, which is $1,3,6,10,15,10,6,3,1$ for $m=0,\dots,8$. At a CM point the invariants are spanned by the monomials of weight zero for $L(X)$; a set of weights $\pm e_j$ with sum zero is a union of opposite pairs, so each such monomial is, up to sign, a product of monomials of degree two and weight zero, which are divisor classes. For each opposite pair $\{w,-w\}$ the four vectors $e^{(1)}_{\pm w}$, $e^{(2)}_{\pm w}$ give one monomial of weight zero in degree 0, four in degree 2 and one in degree 4, so the numbers of these monomials are the coefficients of $(1+4y+y^2)^4$.

(iii) If $t$ is not a CM point, the $\mathrm{Sp}(V,\psi)$-invariant combinations of the $\pi$ are those with $a_1=a_2=a_3$ (Proposition 21.35), and the coefficients of $\omega_t$ are the distinct conjugates of $\alpha$ (Lemma 21.19). At a CM point use the map $x\mapsto E_x$ of the proof of Lemma 21.20, which is $\mathrm{Sp}(V,\psi)$-equivariant: it carries $\omega_t$ to an element $(r,\beta)$ of $\mathbb{Q}\times F$ with $\beta\notin\mathbb{Q}$, acting on $U_{ij}$ by the conjugate $\sigma_k(\beta)$ (Proposition 21.2(ii)), and $\omega_t$ is $L(X)$-invariant only if $E_{\omega_t}$ preserves the weight spaces of $L(X)$ in $\bigwedge^2V$. Take weight vectors $v=x\otimes y\otimes z$ and $w=x\otimes y'\otimes z'$ of weights $(1,1,1)$ and $(1,-1,-1)$ for the Hodge torus. They are not opposite, so $v\wedge w$ spans a weight space of $L(X)$, every weight of $L(X)$ on $V$ having

multiplicity one. But $v\wedge w=(x\cdot x)\otimes(y\cdot y'\otimes z\wedge z'+y\wedge y'\otimes z\cdot z')$ has nonzero components in $U_{12}$ and in $U_{13}$, on which $E_{\omega_t}$ acts by $\sigma_3(\beta)\neq\sigma_2(\beta)$. So the line is not preserved. The rest of (iii) follows from (i).

(iv) Write $H^2(X)=\mathbb{Q}\theta\oplus P$, with $P$ the primitive part, the irreducible $\mathrm{Sp}(V,\psi)$-module of highest weight $\varpi_2$ and dimension 27. Through the form $\bigwedge^2\psi$ the four $\pi$ are self-adjoint endomorphisms of $H^2(X)$, and $\pi_{12}-\pi_{13}$ kills $\theta$ and is a trace-free self-adjoint endomorphism of $P$, an element of $S^2P$ orthogonal to the invariant. Restricting $S^2(\bigwedge^2V)=S_{(2,2)}V\oplus\bigwedge^4V$ from $\mathrm{GL}(V)$ to $\mathrm{Sp}(V,\psi)$ gives, with dimensions $336=308+27+1$ and $70=42+27+1$, the decomposition $S^2P=V(2\varpi_2)\oplus V(\varpi_4)\oplus V(\varpi_2)\oplus\mathbb{Q}$. The element $\pi_{12}-\pi_{13}$ is $G$-invariant. The summand $V(\varpi_2)\cong P$ has no $G$-invariant, since $(\bigwedge^2V)^G=\mathbb{Q}\theta$, and neither has $V(\varpi_4)$, which lies in $\bigwedge^4V$, whose $G$-invariants are the line of $\theta^2$ (Proposition 20.74(i)), invariant under all of $\mathrm{Sp}(V,\psi)$. So $\pi_{12}-\pi_{13}$ is a nonzero vector of the irreducible module $V(2\varpi_2)$, which it generates; its dimension is 308 by Weyl's dimension formula. The same argument puts every trace-free combination of $\pi_{12},\pi_{13},\pi_{23}$ in $V(2\varpi_2)$, while $\pi_0$ and $\pi_{12}+\pi_{13}+\pi_{23}$ are $\mathrm{Sp}(V,\psi)$-invariant and independent modulo it; so adding the four $\pi$ raises the dimension by two. ∎

Figure 59 compares the Hodge classes of $X_t\times X_t$ with the subring generated by divisor classes, degree by degree.

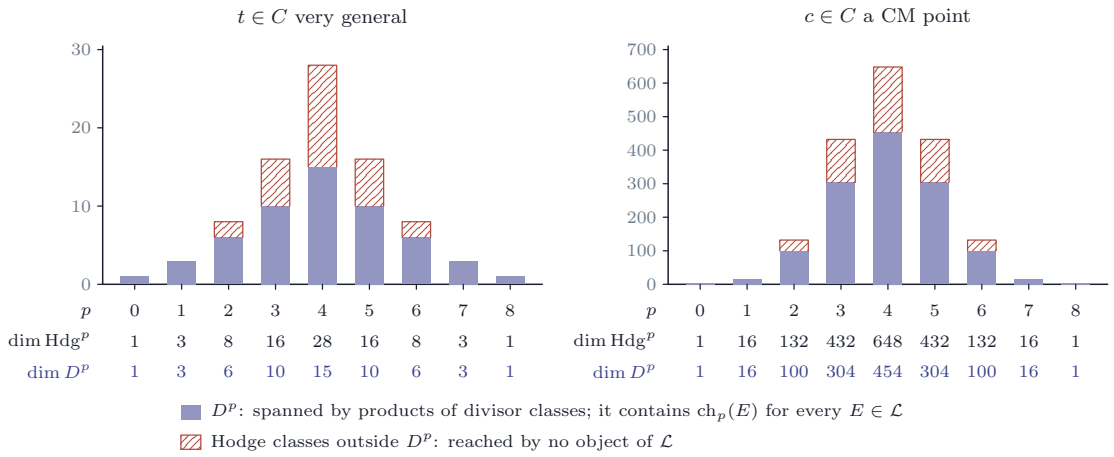



FIGURE 59. Theorem 21.36(ii). The Hodge classes of $X_t\times X_t$ degree by degree, against the subring $D^p$ generated by divisor classes, which contains the Chern character of every object of $\mathcal{L}$: at a very general point of the Mumford curve (left) and at a CM point (right). The hatched classes, the exceptional ones among them, are reached by no object of $\mathcal{L}$.

*Remark* 21.37 (Where a construction has to leave $\mathcal{L}$). Theorem 21.36 extends Proposition 21.35 from sums of line bundles to everything the standard operations make out of them, at every point of the curve, including the CM points, where the exceptional classes are algebraic. There the cycles that represent them are not Chern characters of objects of $\mathcal{L}$, and neither are the two Hodge classes of $X_c$ in degree four that are not products of divisor classes. At a point that is not CM, every Chern character of an object of $\mathcal{L}$ lies in the isotypic component of the trivial representation of $\mathrm{Sp}(V,\psi)$, while by Theorem 21.36(iv) an object with $\mathrm{ch}(E)=N\omega$ must have a nonzero component in the one

of highest weight $2\varpi_2$. Every construction of such an object therefore contains a step that does not commute with the Lefschetz group on cohomology: passing to a direct summand, a subobject or a Jordan–Hölder factor, choosing a point of a moduli space, or deforming from a member with more endomorphisms. The last is what semiregularity does along the curve (Theorem 21.27).

**Lemma 21.38** (The summands of a semiregular object)**.** *Let $Z$ be a smooth projective variety and $E_1, E_2$ perfect complexes on $Z$. On $\operatorname{Ext}^2(E_1\oplus E_2, E_1\oplus E_2) = \bigoplus_{i,j}\operatorname{Ext}^2(E_j,E_i)$ the semiregularity map is $\sigma_{E_1\oplus E_2}(\zeta) = \sigma_{E_1}(\zeta_{11}) + \sigma_{E_2}(\zeta_{22})$. So $\sigma_{E_1\oplus E_2}$ is injective if and only if $\sigma_{E_1}$ and $\sigma_{E_2}$ are injective, their images meet only in zero, and $\operatorname{Ext}^2(E_1,E_2) = \operatorname{Ext}^2(E_2,E_1) = 0$.*

*Proof.* The Atiyah class is natural, so $\operatorname{At}(E_1\oplus E_2)$ is the diagonal sum of $\operatorname{At}(E_1)$ and $\operatorname{At}(E_2)$. In $\sigma(\zeta) = \sum_q \frac{1}{q!}\operatorname{tr}(\zeta\circ\operatorname{At}^q)$ of Theorem 16.28 the composite $\zeta\circ\operatorname{At}^q$ has diagonal blocks $\zeta_{ii}\circ\operatorname{At}(E_i)^q$, and the trace of a block matrix of morphisms is the sum of the traces of its diagonal blocks. In particular the off-diagonal blocks $\zeta_{12}$ and $\zeta_{21}$ lie in the kernel, so injectivity forces them to vanish; on the diagonal blocks the map is $(\zeta_{11},\zeta_{22})\mapsto \sigma_{E_1}(\zeta_{11})+\sigma_{E_2}(\zeta_{22})$, which is injective exactly when both summands are injective and their images meet only in zero. ∎

The condition on the images cannot be dropped, and the objects that show it are the simplest ones: the structure sheaves of abelian subvarieties of half dimension, each of which is semiregular on its own.

**Lemma 21.39** (Abelian subvarieties of half dimension are semiregular)**.** *Let $A$ be an abelian variety of dimension $2n$ and $B\subseteq A$ an abelian subvariety of dimension $n$. Then $\operatorname{ev}_{\mathcal{O}_B}\colon HT^2(A)\to\operatorname{Ext}^2(\mathcal{O}_B,\mathcal{O}_B)$ is surjective, its kernel is $\operatorname{Ann}_{HT^2}([B])$, of dimension $6n^2-n$, and $\sigma_{\mathcal{O}_B}$ is injective.*

*Proof.* The normal bundle of $B$ is trivial, so $\operatorname{ch}(\mathcal{O}_B) = [B]$ by the Grothendieck–Riemann–Roch theorem for the embedding, and $\operatorname{Ext}^*(\mathcal{O}_B,\mathcal{O}_B)\cong\bigwedge^*\mathbb{C}^{2n}$ (proof of Corollary 17.17), so $\dim\operatorname{Ext}^2(\mathcal{O}_B,\mathcal{O}_B) = \binom{2n}{2} = 2n^2-n$, while $\dim HT^2(A) = 2\binom{2n}{2}+4n^2 = 8n^2-2n$. Choose a complementary abelian subvariety $B'$; the isogeny $B\times B'\to A$ identifies $H^1(A,\mathbb{C})$ with $H^1(B)\oplus H^1(B')$ compatibly with the Hodge decompositions, identifies $HT^2(A)$ with $HT^2(B\times B')$ compatibly with contraction, and carries $[B]$ to a nonzero multiple of $1\otimes[\mathrm{pt}_{B'}]$, where $[\mathrm{pt}_{B'}]$ spans $\bigwedge^{2n}H^1(B')$. Write $T = T_B\oplus T_{B'}$ for the tangent space and $H^{0,1}(A) = H^{0,1}(B)\oplus H^{0,1}(B')$. A class $\zeta\in H^{0,2}(A) = \bigwedge^2 H^{0,1}(A)$ has $\zeta\cup[B] = \zeta_{BB}\otimes[\mathrm{pt}_{B'}]$, with $\zeta_{BB}$ its component in $\bigwedge^2H^{0,1}(B)$, because every factor from $H^{0,1}(B')$ is killed by the top class of $B'$; so the annihilator in $H^{0,2}$ has dimension $\binom{2n}{2}-\binom{n}{2}$. For $a\otimes v\in H^{0,1}(A)\otimes T = H^1(T_A)$ the contraction is $a\wedge(v\lrcorner[B]) = a_B\otimes(v_{B'}\lrcorner[\mathrm{pt}_{B'}])$, since $v_B$ kills $[\mathrm{pt}_{B'}]$ and $a_{B'}\wedge(v_{B'}\lrcorner[\mathrm{pt}_{B'}])$ would have type $(n-1,n+1)$ on the $n$-fold $B'$; the map $v\mapsto v_{B'}\lrcorner[\mathrm{pt}_{B'}]$ has rank $n$, as does $a\mapsto a_B$, so the contraction has rank $n^2$ on the space $H^{0,1}(A)\otimes T$ of dimension $4n^2$ and the annihilator there has dimension $3n^2$. For $\vartheta\in\bigwedge^2T$ only the component in $\bigwedge^2T_{B'}$ contracts $[\mathrm{pt}_{B'}]$ nontrivially, and injectively; so the annihilator in $\bigwedge^2T$ has dimension $\binom{2n}{2}-\binom{n}{2}$. Altogether $\dim\operatorname{Ann}_{HT^2}([B]) = 2\binom{2n}{2}-2\binom{n}{2}+3n^2 = 6n^2-n$. By the square $\sigma_E\circ\operatorname{ev}_E = (\,\cdot\,)\lrcorner\operatorname{ch}(E)$ of Setup 18.54 the kernel of $\operatorname{ev}_{\mathcal{O}_B}$ lies in $\operatorname{Ann}_{HT^2}([B])$, so $\operatorname{ev}_{\mathcal{O}_B}$ has rank at least $(8n^2-2n)-(6n^2-n) = 2n^2-n = \dim\operatorname{Ext}^2(\mathcal{O}_B,\mathcal{O}_B)$: it is surjective and its kernel is the annihilator. If $\sigma_{\mathcal{O}_B}(y)=0$, write $y=\operatorname{ev}(x)$; then $x\lrcorner[B]=0$, so $x$ lies in the annihilator, which is the kernel, and $y=0$. ∎

*Remark* 21.40 (Sums of abelian subvarieties). Let $B_1, \dots, B_r \subseteq A$ be abelian subvarieties of dimension $n \geq 3$, translated so that any two of them meet in finitely many points or not at all, which generic translates achieve, and let $E = \bigoplus_j \mathcal{O}_{B_j}[k_j]$. The sheaves $\mathcal{E}xt^q(\mathcal{O}_{B_i}, \mathcal{O}_{B_j})$ for $i \neq j$ are supported on $B_i \cap B_j$ and vanish for $q \neq n$, by the Koszul resolution of $\mathcal{O}_{B_i}$ restricted to $B_j$, so $\operatorname{Ext}^2(\mathcal{O}_{B_i}[k_i], \mathcal{O}_{B_j}[k_j]) = H^{2+k_j-k_i-n}(B_i \cap B_j, \mathcal{E}xt^n)$ vanishes when all $k_j$ are equal and $n \geq 3$, and when $|k_i - k_j| \leq 1$ and $n \geq 4$; and each $\sigma_{\mathcal{O}_{B_j}}$ is injective by Lemma 21.39. In these cases every hypothesis of Lemma 21.38, applied to one summand and the sum of the others, holds except the one on the images. Now let $v \in HT^2(A)$ satisfy $v \lrcorner \operatorname{ch}(E) = \sum_j (-1)^{k_j} v \lrcorner [B_j] = 0$ with $v \lrcorner [B_1] \neq 0$. Then $\zeta = \bigoplus_j \operatorname{ev}_{\mathcal{O}_{B_j}}(v)$ is nonzero, because $v$ is not in the annihilator of $[B_1]$, and $\sigma_E(\zeta) = \sum_j (-1)^{k_j} v \lrcorner [B_j] = 0$: the images of the $\sigma_{\mathcal{O}_{B_j}}$ meet, and $E$ is not semiregular. This is exactly the situation at a member of a Weil family on which a Weil class is a combination of classes of abelian subvarieties of dimension $n$, as on a product of elliptic curves: a tangent direction $v$ of the family keeps $\operatorname{ch}(E)$ of Hodge type but moves any $[B_j]$ outside $\mathbb{Q}\theta^n \oplus W(A)$, and Corollary 17.17 excludes such sums for this reason. The sum is obstructed although its summands are not.

**Corollary 21.41** (Hochschild deformations and direct summands). *Let $c \in C$, $Y = X_c \times X_c$, $\omega = \omega_a$ a rational exceptional class with $L_0(a) \neq 0$, and $E$ a perfect complex on $Y$ with $\operatorname{ch}(E) = N\omega$, $N \neq 0$.*

(i) *$E$ extends to the first order deformation of $D^b(Y)$ given by $x \in HH^2(Y) = HT^2(Y)$ only if $x \in \mathbb{C}\kappa$. No B-field, no Poisson bivector and no commutative deformation off the curve carries it.*

(ii) *If $\sigma_E$ is injective and $E = E_1 \oplus E_2$ with both summands nonzero, then $\operatorname{Ext}^2(E_1, E_2) = \operatorname{Ext}^2(E_2, E_1) = 0$, each $\sigma_{E_i}$ is injective, each $E_i$ extends to first order along the curve, and $\kappa \lrcorner \operatorname{ch}(E_i) = 0$. If moreover $\dim \operatorname{Ext}^2(E, E) = 119$, then $\dim \operatorname{Ext}^2(E_i, E_i) = 120 - \dim \operatorname{Ann}_{HT^2}(\operatorname{ch} E_i)$, and*

$$\operatorname{Ann}_{HT^2}(\operatorname{ch} E_1) + \operatorname{Ann}_{HT^2}(\operatorname{ch} E_2) = HT^2(Y), \qquad \operatorname{Ann}_{HT^2}(\operatorname{ch} E_1) \cap \operatorname{Ann}_{HT^2}(\operatorname{ch} E_2) = \mathbb{C}\kappa.$$

*Proof.* (i) The kernel of $\operatorname{ev}_E$ on $HH^2(Y)$ is the space of deformations of $D^b(Y)$ along which $E$ deforms to first order [Tod09], [Mar25a, Section 8.3]. If $\operatorname{ev}_E(x) = 0$ then $N\, x \lrcorner \omega = \sigma_E(\operatorname{ev}_E(x)) = 0$, so $x \in \mathbb{C}\kappa$ by Theorem 21.21(ii). The three kinds of deformation named are the three summands of $HT^2(Y)$.

(ii) The first two assertions are Lemma 21.38. By Theorem 21.27(ii), $\operatorname{ev}_E(\kappa) = 0$, and $\operatorname{ev}_E$ is the diagonal sum of $\operatorname{ev}_{E_1}$ and $\operatorname{ev}_{E_2}$, so each vanishes at $\kappa$; then $\kappa \lrcorner \operatorname{ch}(E_i) = \sigma_{E_i}(\operatorname{ev}_{E_i}(\kappa)) = 0$. Put $e(x) = 120 - \dim \operatorname{Ann}_{HT^2}(x)$, the rank of contraction into $x$. It is subadditive in $x$, and by Theorem 21.34(i) $\dim \operatorname{Ext}^2(E_i, E_i) \geq e(\operatorname{ch} E_i)$. The cross terms vanish, so

$$119 = \dim \operatorname{Ext}^2(E, E) = \sum_i \dim \operatorname{Ext}^2(E_i, E_i) \geq e(\operatorname{ch} E_1) + e(\operatorname{ch} E_2) \geq e(N\omega) = 119,$$

and every inequality is an equality. Hence $\dim \operatorname{Ann}(\operatorname{ch} E_1) + \dim \operatorname{Ann}(\operatorname{ch} E_2) = 121$. The intersection of the two annihilators lies in $\operatorname{Ann}(N\omega) = \mathbb{C}\kappa$ and contains $\kappa$, so it is $\mathbb{C}\kappa$, and the sum has dimension 120. ∎

Part (ii) plays for the Mumford classes the role of Theorem 20.32: a semiregular object attaining the bound either does not split, or splits into two pieces whose annihilators fill $HT^2(Y)$ and meet only in the direction of the curve. The two annihilators are the analogue of the two branches $P$ and $Q$ of Remark 16.36.

**Proposition 21.42** (Tate classes at every closed point of every reduction)**.** *There are a finitely generated subring $R \subset \mathbb{C}$, a model $W_R \to C_R$ of $W \to C$ over $R$, with $C_R$ smooth and projective over $R$ with geometrically connected fibres and $W_R$ an abelian scheme, projective over $C_R$, and a finite surjective morphism $C'_R \to C_R$ from a second curve of the same kind, with the following properties. Put $Y' = (W_R \times_{C_R} W_R) \times_{C_R} C'_R$, with projection $\pi' \colon Y' \to C'_R$, and let $\omega$ be a rational exceptional class. For every prime $\ell$ there is a global section $\omega_\ell$ of the lisse sheaf $R^4\pi'_*\mathbb{Q}_\ell(2)$ on $C'_R[1/\ell]$ whose restriction to $C'_R \otimes_R \mathbb{C}$ is the pull-back of $\omega \otimes 1$; write $\omega_\ell(\bar{x})$ for its value at a geometric point over a point $x$.*

(i) *At a closed point $x$ of $C'_R[1/\ell]$ the residue field $\kappa(x)$ is finite, $Y'_x$ is the square of an abelian fourfold over it, and $\omega_\ell(\bar{x})$ is fixed by* $\mathrm{Gal}(\bar{\kappa}/\kappa(x))$.

(ii) *At such a point $\omega_\ell(\bar{x})$ is a $\mathbb{Q}_\ell$-linear combination of classes of algebraic cycles of codimension two on $Y'_x$, defined over $\kappa(x)$.*

(iii) *Let $s$ be a closed point of* $\operatorname{Spec} R[1/\ell]$ *and $\eta_s$ the generic point of the reduction $C'_s$. Then $\omega_\ell(\bar{\eta}_s)$ is fixed by the Galois group of the function field $\kappa(\eta_s)$, and it is not a $\mathbb{Q}_\ell$-linear combination of products of divisor classes of $Y'_{\bar{\eta}_s}$.*

*Every closed point $x$ of a reduction $C_s$ of $C$ is the image of a closed point $x'$ of $C'_s$, and $Y'_{x'} = Y_x \otimes_{\kappa(x)} \kappa(x')$; so (i) and (ii) hold at every closed point of every reduction of $C$, after a finite extension of its residue field.*

*Proof. The models.* The family $W \to C$ is defined over a subfield $k_1 \subset \mathbb{C}$ finitely generated over $\mathbb{Q}$, say as $W_1 \to C_1$. Let $K_1$ be the function field of $C_1$ and fix an embedding over $k_1$ of an algebraic closure $\bar{K}_1$ into $\mathbb{C}$; it sends the geometric generic point of $C_1$ to a point $t \in C$, and $Y_t = X_t \times X_t$ is the base change of $Y_{\bar{K}_1}$. The class $\omega_t$ is a Hodge class on an abelian variety, hence absolutely Hodge, so it comes from an absolutely Hodge class of $Y_{\bar{K}_1}$, and $\mathrm{Gal}(\bar{K}_1/K_1)$ acts on these through a finite quotient [Del82, Theorem 2.11 and Proposition 2.9]. Let $K' \subset \bar{K}_1$ be a finite extension of $K_1$ on which the action is trivial, $k'$ the algebraic closure of $k_1$ in $K'$, and $C'$ the smooth projective curve over $k'$ with function field $K'$; it is geometrically connected over $k'$ and maps finitely onto $C_1 \otimes_{k_1} k'$. Spreading out $W_1 \to C_1$, $C'$ and this map over a finitely generated subring of $k'$, and inverting one element of it, gives a normal $R$ and models with the stated properties.

*The section.* By smooth and proper base change $R^4\pi'_*\mathbb{Q}_\ell(2)$ is lisse on $C'_R[1/\ell]$. Its stalk at the geometric generic point $\operatorname{Spec} \bar{K}_1$ is $H^4(Y_{\bar{K}_1}, \mathbb{Q}_\ell(2)) = H^4(Y_t, \mathbb{Q}) \otimes \mathbb{Q}_\ell$, and $\omega_t \otimes 1$ is the $\ell$-adic component of an absolutely Hodge class, so it is fixed by $\mathrm{Gal}(\bar{K}_1/K')$. The scheme $C'_R[1/\ell]$ is normal and integral with function field $K'$, so $\mathrm{Gal}(\bar{K}_1/K')$ maps onto its fundamental group [Stacks, Tag 0BQI], and the vector extends to a global section $\omega_\ell$. On the connected curve $C'_R \otimes_R \mathbb{C}$ it agrees with the pull-back of $\omega \otimes 1$, the two being global sections of the same local system that agree at a point over $t$.

(i) and (ii). A scheme of finite type over $\mathbb{Z}$ has finite residue fields at its closed points. A global section of a lisse sheaf is fixed by the fundamental group of every point, which is (i); and (ii) is Li's theorem [Li26, Theorem 1.1], with $n = r = 2$, for the abelian fourfold that is the fibre of $W_R$ at the image of $x$, base changed to $\kappa(x)$.

(iii) The first assertion holds for the same reason as (i). For the second, let $\bar{\eta}$ be a geometric generic point of $C'_R$ and $\bar{s}$ a geometric point over $s$. The morphism $C'_R \to \operatorname{Spec} R$ is flat and proper with geometrically connected and geometrically reduced fibres, so the specialisation map $\pi_1(C'_{\bar{\eta}}) \to \pi_1(C'_{\bar{s}})$ is surjective [Stacks, Tag 0C0P], and it is compatible with the actions of the two groups on the first cohomology of the fibres of $W_R$, identified by cospecialisation. The image of $\pi_1(C'_{\bar{\eta}})$ contains that of $\pi_1(C'_{\mathbb{C}})$, the closure of the image of the topological fundamental group of a finite cover of $C$, whose Zariski closure contains

$G_{\mathbb{Q}_\ell}$ because the connected monodromy group of $W \to C$ is $G$ (proof of Corollary 20.22). So the Zariski closure of the monodromy of $W_R$ over $C'_{\bar{s}}$ contains $G_{\mathbb{Q}_\ell}$. A divisor class of $Y'_{\bar{\eta}_s}$ is defined over a finite extension of $\kappa(\eta_s)$, so it is fixed by an open subgroup of the Galois group, whose image in $\pi_1(C'_s)$ is open [Stacks, Tag 0BQI] and meets $\pi_1(C'_{\bar{s}})$ in an open subgroup; the Zariski closure of the image of an open subgroup has the same identity component, so the class is $G_{\mathbb{Q}_\ell}$-invariant. The $G$-invariants of $\bigwedge^2(V \oplus V)$ are the three divisor classes of Proposition 20.74(ii), which are $\mathrm{Sp}(V,\psi)$-invariant, so every product of divisor classes of $Y'_{\bar{\eta}_s}$ is $\mathrm{Sp}(V,\psi)$-invariant. The class $\omega_\ell(\bar{\eta}_s)$ corresponds to $\omega \otimes 1$, whose coefficients $a_1, a_2, a_3$ are the distinct conjugates of $\alpha$ (Lemma 21.19), and the $\mathrm{Sp}(V,\psi)$-invariant classes among the $\omega_a$ are those with $a_1 = a_2 = a_3$ (Proposition 21.35). The last sentence of the statement holds because $C'_R \to C_R$ is finite and surjective. ∎

At the generic point of each reduction the class is therefore a Tate class over a global function field that is not a combination of products of divisor classes, exactly as over $\mathbb{C}$, while at every closed point it is a combination of classes of cycles. The next theorem shows that a bound on those cycles is all that is missing, with no passage through $p$-adic Hodge theory.

**Theorem 21.43** (The Mumford target is a degree bound in positive characteristic)**.** *Keep Proposition 21.42, fix a prime $\ell$ and a relatively ample line bundle on $Y' \to C'_R$, and for a finite set $\Phi$ of tuples of Hilbert polynomials say that $\omega$ is* $\Phi$-represented *at a point $x$ of $C'_R[1/\ell]$ if $\omega_\ell(\bar{x}) = \sum_i q_i \,\mathrm{cl}(Z_i)$ for some $q_i \in \mathbb{Q}_\ell$ and closed subschemes $Z_i \subset Y'_{\bar{x}}$ of pure codimension two whose Hilbert polynomials form a tuple in $\Phi$. The following are equivalent.*

(a) *$\omega_t$ is algebraic for every $t \in C$.*

(b) *For some finite $\Phi$, the closed points of $C'_R[1/\ell]$ at which $\omega$ is $\Phi$-represented are Zariski dense in $C'_R$.*

(c) *For some finite $\Phi$ and some Zariski dense set $S$ of closed points of $\operatorname{Spec} R[1/\ell]$, the class $\omega$ is $\Phi$-represented at infinitely many closed points of $C'_s$ for every $s \in S$.*

*For a single closed point $s$ of $\operatorname{Spec} R[1/\ell]$, the class $\omega$ is $\Phi$-represented at infinitely many closed points of $C'_s$, for some finite $\Phi$, if and only if $\omega_\ell(\bar{\eta}_s)$ is a $\mathbb{Q}_\ell$-linear combination of classes of algebraic cycles on $Y'_{\bar{\eta}_s}$.*

*Proof.* One standard property of the $\ell$-adic cycle class is used beyond those over a field: if $\mathcal{O}$ is a discrete valuation ring in which $\ell$ is invertible, $\mathcal{Y} \to \operatorname{Spec}\mathcal{O}$ is smooth and projective, and $\mathcal{Z} \subset \mathcal{Y}$ is a closed subscheme flat over $\mathcal{O}$, then the specialisation isomorphism of $\ell$-adic cohomology carries the cycle class of the geometric generic fibre of $\mathcal{Z}$ to that of its geometric special fibre [SGA $4\frac{1}{2}$, Cycle]. The specialisation isomorphism also carries the values of a global section of a lisse sheaf to each other, by its construction.

*Step 1: the represented loci are closed.* For a tuple $\underline{P} = (P_1, \dots, P_m)$ let $\mathcal{H}_{\underline{P}}$ be the fibre product over $C'_R[1/\ell]$ of the relative Hilbert schemes $\mathrm{Hilb}^{P_i}(Y'/C'_R[1/\ell])$. It is projective over $C'_R[1/\ell]$ [Gro61], so it has finitely many connected components. On a component $T$, with universal subschemes $\mathcal{Z}_i$, put $A(h) = \{q \in \mathbb{Q}_\ell^m : \sum_i q_i \,\mathrm{cl}(\mathcal{Z}_{i,\bar{h}}) = \omega_\ell(\bar{h})\}$ for $h \in T$. If $h'$ is a specialisation of $h$, there is a discrete valuation ring mapping to $T$ with generic point to $h$ and closed point to $h'$ [Stacks, Tag 054F], and the property above, applied to the pull-backs of $Y'$ and of the $\mathcal{Z}_i$, gives $A(h) = A(h')$. Any two points of the connected noetherian scheme $T$ are joined by a chain of specialisations and generisations, through the generic points of its irreducible components, so $A$ is constant on $T$, say $A_T$. Let $\Sigma_{\underline{P}}$ be the union of the images in $C'_R[1/\ell]$ of the components $T$ with $A_T \neq \emptyset$. It is closed, as a finite union of images of proper morphisms, and it is the set of points at which $\omega$ is

$\{\underline{P}\}$-represented: at a point $x$ of the image of such a $T$, a closed point of the fibre of $T$ over $x$ has residue field finite over $\kappa(x)$ and gives subschemes of $Y'_{\bar{x}}$. Put $\Sigma_\Phi = \bigcup_{\underline{P}\in\Phi} \Sigma_{\underline{P}}$.

*Step 2: (b) implies (a).* $\Sigma_\Phi$ is closed and contains a dense set, so it is $C'_R[1/\ell]$ and contains the generic point $\xi$ of $C'_R$, whose residue field is $K'$. So $\omega_t \otimes 1 = \omega_\ell(\bar{\xi}) = \sum_i q_i \operatorname{cl}(Z_i)$ with $q \in \mathbb{Q}_\ell^m$ and subschemes $Z_i$ of $Y_{\bar{K}_1}$, and so of $Y_t$. The cycle class maps are compatible with the comparison isomorphism $H^4(Y_t, \mathbb{Q}) \otimes \mathbb{Q}_\ell \cong H^4(Y_t, \mathbb{Q}_\ell)$ [Del82, Section 1 and Example 2.1(a)], so the linear system $\omega_t = \sum_i q_i [Z_i]$, which has rational coefficients, is solvable over $\mathbb{Q}_\ell$, hence over $\mathbb{Q}$, and $\omega_t$ is algebraic. The $Z_i$ are defined over a finite extension of $K'$, so they spread out to subschemes, flat over a nonempty open subset [Stacks, Tag 052A], of the pull-back of $W \times_C W$ to a finite cover of $C$, and by the local constancy of the cycle class used in the proof of Theorem 16.4, $\omega$ is algebraic at every point of a nonempty Zariski open subset of $C$, which is uncountable. Corollary 20.30 gives (a).

*Step 3: (c) implies (b).* Let $D$ be the set of closed points at which $\omega$ is $\Phi$-represented and $\bar{D}$ its closure. For $s \in S$ the set $D \cap C'_s$ is infinite, so its closure is the irreducible curve $C'_s$, and $C'_s \subseteq \bar{D}$. The points $u$ of $\operatorname{Spec} R[1/\ell]$ over which the fibre of $\bar{D}$ has dimension at least one form a closed set, by the upper semicontinuity of the dimension of the fibres of a proper morphism [Stacks, Tag 0D4I]; it contains the dense set $S$, so it contains the generic point. The generic fibre of $\bar{D}$ is then a closed subset of dimension one of the irreducible curve $C'_R \otimes_R k'$, so it contains $\xi$, and $\bar{D} = C'_R[1/\ell]$.

*Step 4: (a) implies (c).* By (a) and Step 1, applied over $\bar{K}_1$ with the Betti class in place of the $\ell$-adic one, $\omega_t$ is the class of a rational combination of subschemes of $Y_{\bar{K}_1}$, defined over a finite extension $L$ of $K'$. Enlarge $R$ to a finitely generated subring of the algebraic closure of $k'$ in $L$, let $C''_R \to C'_R$ be the normalisation in $L$, and spread the subschemes out over $C''_R$; by generic flatness [Stacks, Tag 052A] they are flat over a dense open subset $U$ of $C''_R[1/\ell]$, with constant Hilbert polynomials. By Step 1 applied on $U$, $\omega$ is represented by their fibres at every point of $U$. The image of $U$ in $C'_R[1/\ell]$ contains a dense open subset $U'$, every closed point of $U'$ is the image of a closed point of $U$, and $U'$ meets $C'_s$ in a nonempty open, hence infinite, subset for every $s$ in a dense open subset of $\operatorname{Spec} R[1/\ell]$.

*The single reduction.* If $\omega$ is $\Phi$-represented at infinitely many closed points of $C'_s$, then $\Sigma_\Phi \cap C'_s$ is closed and infinite, so it is $C'_s$ and contains $\eta_s$. Conversely, cycles at $\bar{\eta}_s$ are defined over a finite extension of $\kappa(\eta_s)$, and spreading them out as in Step 4 over a finite cover of $C'_s$ represents $\omega$, with fixed Hilbert polynomials, at every closed point of a nonempty open subset of $C'_s$. ∎

*Remark* 21.44 (What is left in positive characteristic). By Proposition 21.42(ii) the class is represented at every closed point of $C'_R[1/\ell]$, so Theorem 21.43 turns the Mumford target into a statement of uniformity alone: the Hilbert data of some representation must stay in one finite set on a Zariski dense set of closed points, for instance at infinitely many closed points of infinitely many reductions. The passage to characteristic zero is then done by the specialisation of cycle classes along $C'_R$, and neither the $p$-adic variational Hodge conjecture nor a lifting of cycles is needed. Li's theorem gives no bound: where its proof carries cycles into a prescribed isogeny class, it does so by an isogeny whose degree it does not control [Li26, proof of Theorem 1.2, Step 3].

Nor do the natural representatives at the supersingular points supply the bound. At a closed point $x$ whose fibre is supersingular, the Frobenius eigenvalues on $H^2(Y'_{\bar{x}}, \mathbb{Q}_\ell(1))$ are roots of unity, so by Tate's theorem [Tat66] this space is spanned by divisor classes, and $\omega_\ell(\bar{x}) \in \bigwedge^4 H^1$ is a combination of products of two of them. But for a closed point $s$ of $\operatorname{Spec} R[1/\ell]$ and a finite set of Hilbert polynomials, only finitely many closed points of $C'_s$

carry such a combination $\sum_i q_i \operatorname{cl}(D_i)\operatorname{cl}(D_i')$ with effective divisors $D_i, D_i'$ whose Hilbert polynomials lie in that set: otherwise Step 1 of the proof of Theorem 21.43, run on relative Hilbert schemes of divisors, would give one at $\bar{\eta}_s$, against Proposition 21.42(iii). This holds whether or not the supersingular points are dense. They are dense in $C_R'$ when $R$ has dimension one. Indeed, for every prime $p \geq 5$ not dividing the discriminants of $F$ and $D$, which is all but finitely many, the supersingular points of the reduction at a prime over $p$ of Kisin's integral model of the Shimura curve, with hyperspecial level at $p$, are its whole Newton jumping locus and number $(p^r - 1)(g - 1)$ on each component of genus $g$, $r$ being the local degree of $F$ [SZ11, Theorem 1.2 and Corollary 5.17]. For all but finitely many $s$ the curve $C_s'$ maps finitely onto such a component, of genus at least two, so it has at least $p - 1$ supersingular points, while a closed subset of $C_R'$ of dimension one meets all but finitely many of the $C_s'$ in boundedly many points.

Within one reduction the bound is equivalent to the class being algebraic at the generic point of $C_s'$ (Theorem 21.43), a Tate class over a global function field that is not a combination of products of divisor classes (Proposition 21.42(iii)). This is the codimension two case of the variational Tate conjecture along $C_s'$, which Morrow proves for divisors in its crystalline form [Mor19]. From one reduction alone, the way back to characteristic zero would be the $p$-adic variational Hodge conjecture for $W \times_C W$ over the Witt vectors of $\kappa(s)$, with rational rather than $\ell$-adic coefficients and with the crystalline class of the cycle matching the de Rham class of $\omega$. Bloch, Esnault and Kerz lift such a class in $K_0 \otimes \mathbb{Q}$ of the special fibre to the formal completion for $p > 15$, the dimension being nine, and the algebraisation of the formal class is the remaining step of their programme [BEK14, Theorem 1.3]. The route therefore runs through the uniform bound or through this conjecture.

Five of the routes of Remark 21.32 survive the rigidity theorems: *specialisation*, *reduction modulo p*, *semiregularity*, *Kuga–Satake* and *motivated classes*. The next corollary shows that they are not five separate obligations.

**Corollary 21.45** (The five routes reduce to one statement)**.** *Let $W \to C$ be as in Corollary 20.22, let $\omega$ be a rational exceptional class, and let the* Mumford target *be the first case of* (F2)*: the Hodge conjecture for $X_t \times X_t$ for every $t \in C$.*

(i) *Each of the following is equivalent to the Mumford target: $\omega_t$ is algebraic for uncountably many $t \in C$; $B(W \times_C W)$ holds for $L_{W \times_C W}$; $\omega$ is represented with Hilbert data in one finite set at infinitely many points of $C$, for instance at infinitely many CM points; $\omega$ is represented with Hilbert data in one finite set on a Zariski dense set of closed points of the arithmetic model of Theorem 21.43.*

(ii) *Each of the following implies the Mumford target: a perfect complex $E$ at one point of $C$ with $\operatorname{ch}(E) = N\omega$, $N \neq 0$, no other component, and $\dim \operatorname{Ext}^2(E, E) = 119$; the algebraicity, for uncountably many $t$, of the Kuga–Satake class of the surface $S_\lambda$ of Theorem 21.4 attached to $X_t$; a rational map as in Proposition 21.25(iv), under its hypotheses, from $X_t \times X_t$ for uncountably many $t$.*

(iii) *There is a countable set $\Sigma_0 \subset C$ containing the CM points such that the Mumford target holds as soon as the Hodge conjecture holds for $X_c \times X_c$ at one point $c \notin \Sigma_0$.*

*Proof.* (i) The first two are Corollary 20.30, with Lemma 21.20 for one class in place of two, the third is Corollary 21.28 and the fourth is Theorem 21.43. (ii) The first is Theorem 21.27(iii). The second and the third give the Hodge conjecture for $X_t \times X_t$ at each of the uncountably many points, by Theorem 21.4(ii) and Proposition 21.25(iv), and then (i) applies. (iii) In the proof of Corollary 21.28 the set of points at which $\omega$ is algebraic is

the union of the countably many closed sets $\Sigma_{\underline{P},\underline{q}}$, each finite or equal to $C$. Let $\Sigma_0$ be the union of the finite ones and of the CM points. If $\omega_c$ is algebraic at a point $c \notin \Sigma_0$, then $c$ lies in some $\Sigma_{\underline{P},\underline{q}}$ that is not finite, which is $C$, and (i) applies. ■

*Remark* 21.46 (Which routes are needed). Of the five routes, specialisation, reduction modulo $p$ and motivated classes are the Mumford target itself in three other forms, by Corollary 21.45(i) and, for the motivated classes, Corollary 20.30; semiregularity and Kuga–Satake are stronger inputs that imply it. So any one route suffices. Each of the three forms is the whole of the problem, a bound on Hilbert data or a case of (L) that is equivalent to the target, and the two stronger inputs ask for an object and for a cycle.

The target can also be read at a single member, through the motivic group of $X$ rather than through cycles on the family; it then becomes the question of which of four groups this is.

**Proposition 21.47** (Four possible motivic groups)**.** *In Setup 21.1, let $M$ be the group of the Tannakian category of homological motives generated by $h(X)$, with the Betti realisation as fibre functor, let $M^1 = M \cap \mathrm{Sp}(V,\psi)$, and let $N$ be the normaliser of $G$ in $\mathrm{Sp}(V,\psi)$. Over $\bar{\mathbb{Q}}$ identify $V_1$, $V_2$ and $V_3$ with one plane, carrying $\epsilon_1$, $\epsilon_2$, $\epsilon_3$ to one form, and let $\zeta \in \mathrm{Sp}(V,\psi)(\bar{\mathbb{Q}})$ be the cyclic permutation $v_1 \otimes v_2 \otimes v_3 \mapsto v_3 \otimes v_1 \otimes v_2$ of the three factors. It normalises $G$, and other identifications change it by an element of $G(\bar{\mathbb{Q}})$; so it acts through $V$ on the Hodge classes of every power of $X$, independently of the choice.*

(i) *$N/G$ is a form of the symmetric group $S_3$, $M = \mathbb{G}_m \cdot M^1$, and $M^1$ is one of $G$, $G \cdot A_3$, $N$ and $\mathrm{Sp}(V,\psi)$, where $G \cdot A_3$ is the inverse image in $N$ of the subgroup of order three of $N/G$. The last three contain $\zeta$.*

(ii) *The following are equivalent: $M^1 = G$; the Hodge conjecture holds for $X^n$ for every $n \geq 1$; the two exceptional classes of $X \times X$ are algebraic; some algebraic class on some power of $X$ is not fixed by $\zeta$.*

(iii) *Let $\mathrm{Det} \in (V^{\otimes 4})^G \subset H^4(X^4,\mathbb{Q})$ be the Hodge class of Remark 21.8, and write $r_1$ also for the Hodge class $t \in (V^{\otimes 4})^G$ with $\hat{t} = r_1 = s_1 + s_2 + s_3$, in the notation of Steps 2 and 3 of the proof of Theorem 21.7. Both are fixed by $N$ and not by $\mathrm{Sp}(V,\psi)$, and each of them is algebraic if and only if $M^1 \neq \mathrm{Sp}(V,\psi)$. So the algebraicity of either excludes the last case only; by this argument it does not give the exceptional classes, which are not algebraic when $M^1$ is $G \cdot A_3$ or $N$.*

(iv) *Let $W \to C$ be as in Corollary 20.22, and suppose that it is defined over a number field; this holds for the restriction of the universal family of a fine moduli space of polarised abelian fourfolds with level structure to the normalisation of one of the Shimura curves of* [Mum69] *in it, and for its pull-backs to finite étale covers. Then the set $\Sigma_0$ constructed in the proof of Corollary 21.45(iii) lies in $C(\bar{\mathbb{Q}})$, and the Mumford target holds as soon as, at one point $t \in C(\mathbb{C}) \setminus C(\bar{\mathbb{Q}})$, some algebraic cycle on some power of $X_t$ has a class that is not fixed by $\zeta$.*

*Proof. The group.* On an abelian variety homological and numerical equivalence coincide [Lie68], so the homological motives generated by $h(X)$ and the Lefschetz motive form a semisimple abelian category [Jan92]. The Künneth projectors are algebraic [Kle68, §2], so after the usual change of sign in the commutativity constraint this is a Tannakian category over $\mathbb{Q}$, with the Betti realisation as a fibre functor, faithful by the definition of homological equivalence. Its group $M$ is reductive, the category being semisimple, and the $M$-invariants in the realisation of a motive $P$ are the image of $\mathrm{Hom}(\mathbf{1},P)$ [DM82, Theorem 2.11]; so for all $m$ and $p$ the $M$-invariants of $H^{2p}(X^m,\mathbb{Q})(p)$ are the classes of algebraic cycles. The

group $M$ acts faithfully on $V \oplus \mathbb{Q}(1)$ and fixes the polarisation class, so it lies in $\mathrm{GSp}(V,\psi)$ and acts on $\mathbb{Q}(1)$ through the multiplier. Through $\psi$ and the Künneth projectors every tensor space of $V \oplus \mathbb{Q}(1)$ is a sum of direct summands of spaces $H^k(X^k,\mathbb{Q})(j)$ cut out by algebraic projectors, so the $M$-invariants there are algebraic classes, hence Hodge classes, fixed by $\mathrm{MT}(X) = \mathbb{G}_m \cdot G$; as a reductive group is the subgroup fixing its invariants in the tensor spaces [Del82, Proposition 3.1], $\mathrm{MT}(X) \subseteq M$ and $G \subseteq M^1$. An element of $M$ with multiplier $\nu$ is a square root of $\nu$ times an element of $M^1$, so $M = \mathbb{G}_m \cdot M^1$. The homotheties act trivially on the spaces $H^{2p}(X^m,\mathbb{Q})(p)$, so there the algebraic classes are the $M^1$-invariants, as the Hodge classes are the $G$-invariants.

*The groups between $G$ and $\mathrm{Sp}(V,\psi)$.* Over $\bar{\mathbb{Q}}$ the Lie algebra $\mathfrak{sp}(V,\psi) \cong S^2V$ is the sum of $\mathrm{Lie}\, G = T_1 \oplus T_2 \oplus T_3$, where $T_i \cong S^2V_i \otimes \bigwedge^2 V_j \otimes \bigwedge^2 V_k$, and of $S^2V_1 \otimes S^2V_2 \otimes S^2V_3$, which is an irreducible representation of $G$ of dimension 27; so a $G$-stable subspace of $\mathfrak{sp}(V,\psi)$ containing $\mathrm{Lie}\, G$ is $\mathrm{Lie}\, G$ or $\mathfrak{sp}(V,\psi)$. If $H$ is a closed subgroup with $G \subseteq H \subseteq \mathrm{Sp}(V,\psi)$, its Lie algebra is such a subspace. If it is $\mathrm{Lie}\, G$, then $H^\circ = G$, which $H$ normalises, so $H \subseteq N$; if it is $\mathfrak{sp}(V,\psi)$, then $H^\circ = \mathrm{Sp}(V,\psi)$, which is connected, so $H = \mathrm{Sp}(V,\psi)$.

*The normaliser.* Since $V$ is absolutely irreducible, the centraliser of $G$ in $\mathrm{Sp}(V,\psi)$ is $\{\pm 1\}$, which lies in $G$ as the image of $(-1,1,1)$, and $G(\bar{\mathbb{Q}})$ maps onto the group of inner automorphisms of $G_{\bar{\mathbb{Q}}}$. So $N(\bar{\mathbb{Q}})/G(\bar{\mathbb{Q}})$ embeds in the outer automorphism group of $G_{\bar{\mathbb{Q}}}$, which permutes the three factors, $\mathrm{SL}_2$ having no outer automorphism. The six permutations of the factors, after identifications as above, preserve $\psi = \epsilon_1 \otimes \epsilon_2 \otimes \epsilon_3$ and normalise $G$, so $N(\bar{\mathbb{Q}})/G(\bar{\mathbb{Q}}) \cong S_3$. The Galois group acts on it by conjugation through its action on the three factors, which is transitive, and the subgroups of $S_3$ stable under conjugation by a transitive subgroup are 1, $A_3$ and $S_3$. So the algebraic subgroups of $N$ containing $G$ are $G$, $G \cdot A_3$ and $N$, and with the previous paragraph this proves (i), $\zeta$ mapping to an element of order three of $N/G$.

(ii) If $M^1 = G$, then $M = \mathrm{MT}(X)$, and every Hodge class on every power of $X$ is algebraic. The Hodge conjecture for $X \times X$ includes the exceptional classes. If these are algebraic, so is every rational exceptional class $\omega_a$ of Setup 21.18, and $\omega_a$ is not fixed by $\zeta$: $\zeta$ permutes the three factors cyclically, so it fixes $\theta$ and carries $U_{ij} = S^2V_i \otimes S^2V_j \otimes \bigwedge^2 V_k$ onto $U_{\zeta(i)\zeta(j)}$; labelling the factors so that $\zeta$ sends the $i$-th to the $(i+1)$-th, it carries $\pi_{12}$ to $\pi_{23}$, $\pi_{23}$ to $\pi_{13}$ and $\pi_{13}$ to $\pi_{12}$ and fixes $\pi_0$. So it fixes $\omega_a$ only if $a_1 = a_2 = a_3$, whereas these are the distinct conjugates of an irrational element of $F$ (Lemma 21.19). Finally, if $M^1 \neq G$, then $M^1$ contains $\zeta$ by (i), and every algebraic class, being $M^1$-invariant, is fixed by $\zeta$.

(iii) The class Det spans $(S^4V)^G \otimes \bigwedge^4 \mathbb{Q}^4$, where $(S^4V)^G$ is a line (Remark 21.8). As $N$ normalises $G$, it preserves this line, and it fixes it, because $N(\bar{\mathbb{Q}})$ is generated by $G(\bar{\mathbb{Q}})$ and the permutations of the factors, which fix Cayley's hyperdeterminant, a symmetric function of the three factors, and the restriction of Det to the diagonal is a nonzero multiple of it (Remark 21.8). $\mathrm{Sp}(V,\psi)$ fixes no nonzero element of $S^4V$, which would be a nonzero quartic form constant on the single orbit $V \setminus \{0\}$. The permutations of the factors conjugate $s_k$ into $s_{\sigma(k)}$, so they commute with $r_1$, as does $G$; and $t \mapsto \hat{t}$ is $\mathrm{Sp}(V,\psi)$-equivariant, since the pairing $\int_X v \cup w \cup \theta^3$ on $V$ is $G$-invariant and alternating, hence a multiple of the form $\psi$. So the class $r_1$ is fixed by $N$. On $S^2V \subset V \otimes V$ the endomorphism $r_1$ acts by 3 on $S^2V_1 \otimes S^2V_2 \otimes S^2V_3$ and by $-1$ on the $T_i$, since $s_k$ is 1 on $S^2V_k$ and $-1$ on $\bigwedge^2 V_k$; as $S^2V$ is irreducible under $\mathrm{Sp}(V,\psi)$, the class $r_1$ is not $\mathrm{Sp}(V,\psi)$-invariant. If $M^1 \neq \mathrm{Sp}(V,\psi)$, then $M^1 \subseteq N$ by (i), and the two classes are $M^1$-invariant, hence algebraic; if $M^1 = \mathrm{Sp}(V,\psi)$, they are not. When $M^1$ is $G \cdot A_3$ or $N$ it contains $\zeta$, which fixes no exceptional class.

(iv) A fine moduli space as in the statement is defined over a number field, and the curve contains infinitely many CM points, which are algebraic, an abelian variety with complex multiplication being defined over a number field; so the curve is the Zariski closure of a set of algebraic points and is defined over $\bar{\mathbb{Q}}$, and so are its normalisation, the restricted family and their finite étale covers. Now let $W \to C$ be defined over $\bar{\mathbb{Q}}$. The moduli map of the family is a finite morphism defined over $\bar{\mathbb{Q}}$, so the CM points of $C$, whose images are algebraic, lie in $C(\bar{\mathbb{Q}})$. The other points of $\Sigma_0$ lie in the finite sets $\Sigma_{\underline{P},\underline{q}}$ of the proof of Corollary 21.28, images of unions of connected components of fibre products of relative Hilbert schemes of $W \times_C W \to C$. These are base changes of schemes over $\bar{\mathbb{Q}}$, and each connected component of the base change is the base change of a connected component over $\bar{\mathbb{Q}}$, which is algebraically closed; so a finite $\Sigma_{\underline{P},\underline{q}}$ consists of algebraic points. If $t \notin C(\bar{\mathbb{Q}})$, then $t \notin \Sigma_0$, the Hodge group of $X_t$ is $G$, and (ii) for $X_t$ gives the Hodge conjecture for $X_t \times X_t$; Corollary 21.45(iii) gives the target. ∎

The four cases differ already on $X^4$ and $X^6$ (Figure 60). Over $\bar{\mathbb{Q}}$, $(V^{\otimes 2m})^G$ is the tensor product of three copies of the space of $\mathrm{SL}_2$-invariants of $(\bar{\mathbb{Q}}^2)^{\otimes 2m}$, of dimension $c = 2$ for $m = 2$ and $c = 5$ for $m = 3$, with a basis of non-crossing pairings that $\zeta$ and the transpositions of the factors permute. So the algebraic classes in the multilinear part $V^{\otimes 2m} \subset H^{2m}(X^{2m}, \mathbb{Q})$ have dimension $c^3$, $(c^3 + 2c)/3$, $(c^3 + 3c^2 + 2c)/6$ and $(2m-1)(2m-3)\cdots 1$ when $M^1$ is $G$, $G \cdot A_3$, $N$ and $\mathrm{Sp}(V, \psi)$, the last by the first fundamental theorem for $\mathrm{Sp}(V, \psi)$: 8, 4, 4, 3 on $X^4$ and 125, 45, 35, 15 on $X^6$. The middle two count the orbits of the basis under $\langle\zeta\rangle$ and under $S_3$: $\zeta$ fixes the $c$ triples of equal pairings and a transposition fixes $c^2$ of them. If $M$ were known to be connected, $M^1$ would be connected too, since $-1 \in G$ lies in its identity component; it would then be $G$ or $\mathrm{Sp}(V, \psi)$, and the algebraicity of Det alone would give the Hodge conjecture for every power of $X$. The argument does not decide whether $M$ is connected. It is the standard use of the motivic group as the stabiliser of the algebraic classes, and we claim no priority for Proposition 21.47. It proves no new case of the Hodge conjecture: it says what one new cycle on one power of one member would have to do.

The same invariance explains why the classes known or expected to be algebraic on abelian varieties stop short of the exceptional classes, on all abelian varieties at once.

**Proposition 21.48** (The system generated by the known classes)**.** *Keep Setup 21.1, and $\zeta$ and $G \cdot A_3$ as in Proposition 21.47. For an abelian variety $A$ let $A_X \subseteq A$ be the largest abelian subvariety isogenous to a power of $X$, and $A'$ the identity component of the intersection of the kernels of the homomorphisms $A \to X$. Then $A_X \times A' \to A$ is an isogeny,* $\mathrm{Hom}(X, A') = 0$*, and*

$$H^1(A, \mathbb{Q}) = \big(V \otimes \mathrm{Hom}^0(A_X, X)\big) \oplus H^1(A', \mathbb{Q}),$$

*the first summand embedded by $v \otimes f \mapsto f^*v$. Let $\mathrm{GL}(V)$ act on $H^*(A, \mathbb{Q}) = \bigwedge^* H^1(A, \mathbb{Q})$ through the factor $V$. Let $\mathcal{S}$ be the smallest family of subspaces $\mathcal{S}(A) \subseteq H^*(A, \mathbb{Q})$, over all complex abelian varieties $A$, that is closed under linear combinations, cup products, exterior products, and pull-back and push-forward along homomorphisms, and contains:*

(a) *the divisor classes;*
(b) *for every subfield $E \subseteq \mathrm{End}^0(A)$, the rational classes whose extension to $\bar{\mathbb{Q}}$ lies in $\bigoplus_\sigma \bigwedge^d H^1(A, \bar{\mathbb{Q}})_\sigma$, the sum over the embeddings $\sigma\colon E \to \bar{\mathbb{Q}}$ of the top exterior powers of the eigenspaces, $d = \dim_E H^1(A, \mathbb{Q})$; among them are the Weil classes;*
(c) *the Hodge classes of the abelian varieties without isogeny factor $X$;*
(d) *the Hodge classes of the powers of $X$ that are fixed by $\zeta$, among them* Det *and $r_1$.*

| $M = \mathbb{G}_m \cdot M^1$, $M^1 = M \cap \mathrm{Sp}(V,\psi)$ | dim of the algebraic classes in $V^{\otimes 2m}$: $X^4$, $m=2$ | $X^6$, $m=3$ | formula, $c=2$ or $5$ | $\zeta \in M^1$ | Det, $r_1$ algebraic | exceptional classes algebraic |
|---|---|---|---|---|---|---|
| $\mathrm{Sp}(V,\psi)$ | 3 | 15 | $(2m-1)!!$ | yes | no | no |
| dim: $9 < 36$ | | | | | | |
| $N$ | 4 | 35 | $(c^3+3c^2+2c)/6$ | yes | yes | no |
| index 2 | | | | | | |
| $G \cdot A_3$ | 4 | 45 | $(c^3+2c)/3$ | yes | yes | no |
| index 3 | | | | | | |
| $G = \mathrm{Hg}(X)$ | 8 | 125 | $c^3$ | no | yes | yes |

$M^1 = G \Leftrightarrow$ the Hodge conjecture for every $X^n \Leftrightarrow$ the two exceptional classes of $X \times X$ are algebraic $\Leftrightarrow$ some algebraic class on some power of $X$ is not fixed by $\zeta$

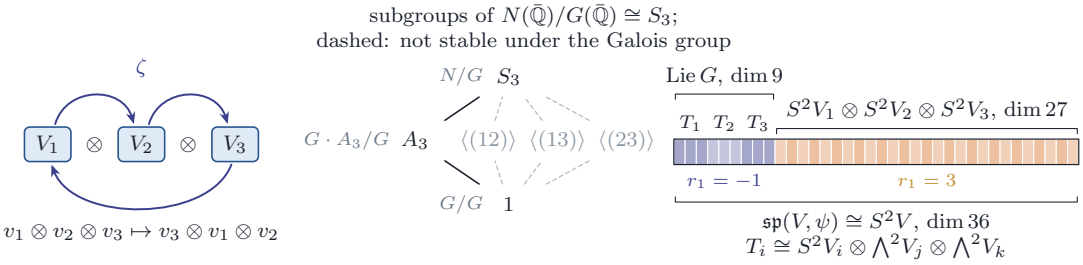



FIGURE 60. The four possible groups $M^1 = M \cap \mathrm{Sp}(V,\psi)$ of Proposition 21.47, where $M = \mathbb{G}_m \cdot M^1$, as the chain $G \subset G \cdot A_3 \subset N \subset \mathrm{Sp}(V,\psi)$: the first two inclusions have index 3 and 2, and $N$, whose identity component is $G$, has dimension 9 against 36. For each case the table gives the dimension of the algebraic classes in the multilinear part $V^{\otimes 2m}$ of $H^{2m}(X^{2m},\mathbb{Q})$ for $m=2$ and $m=3$, from the $c=2$, respectively $c=5$, non-crossing pairings; whether $M^1$ contains the cyclic permutation $\zeta$ of the three tensor factors; whether Det and $r_1$ are algebraic; and whether the two exceptional classes of $X \times X$ are. Only $M^1 = G$ makes the exceptional classes algebraic, and it is equivalent to the Hodge conjecture for every power of $X$ (Proposition 21.47(ii)); the proposition does not decide which of the four cases occurs. Below, from left to right: $\zeta$; the subgroups of $N(\bar{\mathbb{Q}})/G(\bar{\mathbb{Q}}) \cong S_3$, of which only 1, $A_3$ and $S_3$ are stable under the Galois group, which acts transitively on the three factors; and $\mathfrak{sp}(V,\psi) = \mathrm{Lie}\, G \oplus S^2V_1 \otimes S^2V_2 \otimes S^2V_3$, $36 = 9 + 27$, the second summand irreducible under $G$, with the eigenvalues $-1$ and 3 of $r_1$ on the two summands.

*Then every class of $\mathcal{S}$ is invariant under $G \cdot A_3$ acting through $V$, and the classes obtained from* (a), (b) *and* (c) *alone are invariant under* $\mathrm{Sp}(V,\psi)$. *The Hodge classes in $\mathcal{S}(X^m)$ are the Hodge classes of $X^m$ fixed by $\zeta$, and $\mathcal{S}$ contains no exceptional class of $X \times X$.*

*Proof.* Up to isogeny $A = A_X \times A'$ by Poincaré reducibility, with $A_X$ isogenous to some $X^k$ and no isogeny factor $X$ in $A'$, so $\mathrm{Hom}(X,A') = \mathrm{Hom}(A',X) = 0$, $X$ being simple. As $\mathrm{End}^0(X) = \mathbb{Q}$, $\mathrm{Hom}^0(A_X,X)$ has dimension $k$ and the displayed map is an isomorphism. A homomorphism $h\colon A \to B$ is, up to isogeny, $h_X \times h'$ with $h_X\colon A_X \to B_X$ and $h'\colon A' \to B'$, and $h^*$ acts on $H^1$ as $(1 \otimes (f \mapsto f \circ h_X)) \oplus h'^*$, which commutes with $\mathrm{GL}(V)$; so pull-backs are equivariant, and so are cup and exterior products, $\mathrm{GL}(V)$ acting on $H^*(A \times B) = H^*(A) \otimes H^*(B)$ diagonally. The orientation class of $A$ spans $\bigwedge^{\mathrm{top}} H^1(A) = (\det V)^{\otimes k} \otimes (\det \mathrm{Hom}^0(A_X,X))^{\otimes 8} \otimes \bigwedge^{\mathrm{top}} H^1(A')$, on which $\mathrm{SL}(V)$ acts trivially, so Poincaré duality and push-forwards are $\mathrm{SL}(V)$-equivariant. Since $G \cdot A_3 \subseteq \mathrm{Sp}(V,\psi) \subseteq \mathrm{SL}(V)$, the classes invariant under either group form a family closed under all the operations, and it remains to look at the generators.

(a) A divisor class of $A$ is the sum of its Künneth components in $\bigwedge^2 H^1(A_X)$, $H^1(A_X) \otimes H^1(A')$ and $\bigwedge^2 H^1(A')$, which are Hodge classes. The middle one corresponds to a homomorphism from $A'$ to the dual of $A_X$, which is zero. The first is a Hodge class of $A_X$, so a $G$-invariant of $\bigwedge^2(V \otimes \mathbb{Q}^k) = S^2V \otimes \bigwedge^2 \mathbb{Q}^k \oplus \bigwedge^2 V \otimes S^2\mathbb{Q}^k$; as $(S^2V)^G = 0$ and $(\bigwedge^2 V)^G = \mathbb{Q}\psi$ (proof of Theorem 21.36(i)), it lies in $\psi \otimes S^2\mathbb{Q}^k$, which $\mathrm{Sp}(V,\psi)$ fixes. On the third $\mathrm{GL}(V)$ acts trivially. (b) The field $E$ acts on $Q = \mathrm{Hom}^0(A_X,X)$ and on $H^1(A')$, and the $\sigma$-eigenspace of $H^1(A,\bar{\mathbb{Q}})$ is $V \otimes Q_\sigma \oplus H^1(A',\bar{\mathbb{Q}})_\sigma$. Its top exterior power is $(\det V)^{\otimes \dim Q_\sigma} \otimes (\det Q_\sigma)^{\otimes 8} \otimes \det H^1(A',\bar{\mathbb{Q}})_\sigma$, on which $\mathrm{SL}(V)$ acts trivially. (c) For such an abelian variety $A_X = 0$, and $\mathrm{GL}(V)$ acts trivially. (d) The Hodge classes of $X^m$ are the invariants of $G$, which acts through $V$, and those fixed by $\zeta$ are invariant under the group $G \cdot A_3$ that $G$ and $\zeta$ generate.

So every class of $\mathcal{S}$ is $G \cdot A_3$-invariant, and those obtained from (a), (b) and (c) are $\mathrm{Sp}(V,\psi)$-invariant. The Hodge classes of $X^m$ in $\mathcal{S}$ contain those of (d) and are fixed by $\zeta$, so they are exactly the Hodge classes fixed by $\zeta$; and an exceptional class of $X \times X$ is not fixed by $\zeta$ (proof of Proposition 21.47(ii)). ∎

So no argument that uses only these inputs and these operations reaches the exceptional classes. The inputs include all Weil classes, whether or not they are known to be algebraic, and with them Markman's cycles [Mar25a]; the whole Hodge conjecture for the abelian varieties without isogeny factor $X$; and the algebraicity of Det and $r_1$. A proof needs a cycle outside this system: on a product of a power of $X$ with an abelian variety, a class that is not invariant under $G \cdot A_3$ acting through $V$, or a cycle through a variety that is not an abelian variety, such as the Kuga–Satake class of $S_\lambda$ would provide (Proposition 21.12). Proposition 21.48 extends the second limit of Remark 21.17; it is an obstruction to a method, not to the target, and we claim no priority for it.

The same invariance, used the other way round, shows that the statement (F2) of Corollary 20.71 is not smaller than the conjecture for abelian varieties. For an abelian variety $Y$ let $R(Y) \subseteq H^*(Y,\mathbb{Q})$ be the subring generated by divisor classes and Weil classes of quotients, as in Remark 20.72.

**Proposition 21.49** (Statement (F2) is the conjecture for abelian varieties)**.** *Statement* (F2) *of Corollary 20.71 implies the Hodge conjecture for every abelian variety, and is therefore equivalent to it.*

*Proof.* The conjecture for abelian varieties contains (F2). For the converse, recall from Remark 20.72 that (F2) gives every Hodge class of degree $2q$ on $Y$ as soon as one Hodge

class of degree $2q$ on $Y$ lies outside $R(Y)$. Let $X$ be as in Setup 21.1, and let $\zeta$ act on the cohomology of every abelian variety through the factor $V$, as in Proposition 21.48. Every class of $R(Y)$ lies in $\mathcal{S}(Y)$, so it is fixed by $\zeta$, for every $Y$; and an exceptional class $e_0 \in H^4(X \times X, \mathbb{Q})$ is not (Proposition 21.48). Let $\theta$ be an ample class on $X \times X$, which $\zeta$ fixes. If $\zeta e_0 = \lambda e_0$, then $\lambda \neq 1$ and $e = e_0 + \theta^2$ has $\zeta e = \lambda e_0 + \theta^2$, which is not a multiple of $e$; otherwise put $e = e_0$. In either case $e$ is a Hodge class of degree 4 on $X \times X$ with $\zeta e \notin \bar{\mathbb{Q}}\, e$.

Let $A$ be an abelian variety and $\beta \neq 0$ a Hodge class of degree $2p$ on $A$, and put $Y = A \times X \times X$. As $\zeta$ acts on $H^*(Y) = H^*(A) \otimes H^*(X \times X)$ diagonally, a class $\beta \otimes e$ in $R(Y)$ would satisfy $\zeta\beta \otimes \zeta e = \beta \otimes e$, which forces $\zeta e \in \bar{\mathbb{Q}}\, e$. So the Hodge class $\beta \otimes e$ lies outside $R(Y)$, and (F2) makes it algebraic. In the same way $e \cup \theta^4 \notin R(X \times X)$, since otherwise $(\zeta e - e) \cup \theta^4 = 0$ and the hard Lefschetz theorem would give $\zeta e = e$; so (F2) makes every Hodge class of degree 12 on $X \times X$ algebraic. By the Hodge–Riemann bilinear relations the intersection pairing between the Hodge classes of degrees 4 and 12 on $X \times X$ is nondegenerate, so there is a Hodge class $e'$ of degree 12 with $\int_{X\times X} e \cup e' = 1$. Then

$$\beta = \mathrm{pr}_{A*}\big((\beta \otimes e) \cup \mathrm{pr}^*_{X\times X} e'\big),$$

with $\mathrm{pr}_A$ and $\mathrm{pr}_{X\times X}$ the projections of $Y$, and $\beta$ is algebraic. ∎

*Remark* 21.50 (The rule set with Proposition 21.49). Theorem 20.70 is a statement about the rule set of Setup 20.69, and Proposition 21.49 is a rule that the set does not contain. Adjoined to it, the rule changes the fifth minimal set of Theorem 20.70(ii): the minimal sufficient sets become

$$\{\mathrm{F3}\}, \quad \{\mathrm{L}, \mathrm{M}\}, \quad \{\mathrm{L}, \mathrm{F3}'\}, \quad \{\mathrm{V}, \mathrm{F3}'\}, \quad \{\mathrm{F2}, \mathrm{F3}'\},$$

and $\mathrm{P2}_{\mathrm{split}}$ lies in none of them. Indeed the new rule has the premise (F2) and the conclusion **AB**, the conjecture for abelian varieties, so the sets $C_1$, $C_2$ and $C_4$ of the proof of Theorem 20.70 stay closed, and a sufficient set meets $H_1$, $H_2$ and $H_4$. The sets meeting these three that are minimal with this property are the five above, by the argument of that proof without $H_3$, and each is sufficient: $\{\mathrm{F2}, \mathrm{F3}'\}$ gives **AB** by the new rule and then **HC** by Proposition 20.35(i). So the results of this paper on Weil classes do not shorten a route to the conjecture. What they do is reduce one case of it, the algebraicity of the Weil classes of every CM field, to the single statement $\mathrm{P2}_{\mathrm{split}}$, by a route that does not pass through (F2). Like (F3), which is the whole conjecture, (F2) is the whole conjecture for abelian varieties, carried by the classes that are hardest to reach; the exceptional classes of $X \times X$ are its smallest instance, and the proof above shows that the classes $\beta \otimes e$ built from them already carry all of it.

## 22. The main theorem in every family

Theorem 1.1 describes the algebraic locus of the Weil classes in one family: it contains an explicit CM point, it is dense, it is a countable union of closed algebraic strata, and it is the whole family as soon as it has an interior point. The first two properties use the Weil classes; the last two hold for every flat class on every smooth projective family (Lemma 22.2). This section carries the theorem to the families that carry (F2) and (F3′), and records what each construction of the paper contributes there.

For (F2) the families are abelian schemes with a member of CM type. Every Hodge class of every abelian variety lies in such a family (Proposition 22.4), and (F2) is equivalent to the statement that in every one of them the algebraic locus of a flat Hodge class that is algebraic at a CM member has an interior point (Theorem 22.5). The base points of

Theorem 19.5 are the CM seeds of the Weil families, and André's theorem makes the Weil classes of the split families seeds for every Hodge class of CM type. For (F3′) the cohomology of a variety splits into the part reached from abelian varieties by algebraic correspondences and a complement, and (F3′) is the statement that the complement carries no Hodge class (Proposition 22.7). On the first varieties where (F3′) is open, the squares of K3 surfaces with real multiplication, the algebraic locus of the real multiplication contains every CM point, is dense and stable under a group of rational Hodge isometries, and is either everything or meagre (Theorem 22.9). In both cases what is left is the interior point, and Theorem 22.10 lists what the constructions of the paper give towards it.

22.1. **Algebraic loci.**

**Definition 22.1** (Algebraic locus)**.** Let $S$ be a smooth connected complex quasi-projective variety, $f\colon \mathcal{X} \to S$ a smooth projective morphism, and $\xi$ a global section of $R^{2p}f_*\mathbb{Q}$ over $S^{\mathrm{an}}$.[54] The *algebraic locus* of $\xi$ is

$$\Sigma(\xi) = \{\, s \in S :\ \xi_s \text{ is the class of an algebraic cycle with rational coefficients on } \mathcal{X}_s \,\}.$$

**Lemma 22.2** (The dichotomy in every family)**.** *In Definition 22.1 the following hold.*

(i) $\Sigma(\xi)$ *is the union of countably many Zariski closed subsets of* $S$*.*

(ii) *Either* $\Sigma(\xi) = S$*, or* $\Sigma(\xi)$ *is meagre in* $S^{\mathrm{an}}$*.*

(iii) *The following are equivalent:* $\Sigma(\xi) = S$*;* $\Sigma(\xi)$ *has an interior point;* $\Sigma(\xi)$ *is not meagre;* $\xi_s$ *is algebraic for every* $s$ *outside a countable union of proper closed analytic subsets of* $S$*. If* $\Sigma(\xi)$ *is dense, they are also equivalent to* $\Sigma(\xi)$ *being closed.*

*Proof.* (i) The proof of Theorem 16.4 applies with $(\mathcal{A}, \mathcal{S}, \omega)$ replaced by $(\mathcal{X}, S, \xi)$, as in Lemma 20.27: for a tuple $\underline{P}$ of Hilbert polynomials the fibre product of the relative Hilbert schemes $\mathrm{Hilb}^{P_i}(\mathcal{X}/S)$ is projective over $S$ [Gro61], the class of a flat family of cycles is a flat section, two flat sections over a connected base that agree at one point agree everywhere, and the image of a proper morphism is closed. So the set of points at which $\xi_s = \sum_i q_i[Z_i]$ with $Z_i$ of Hilbert polynomial $P_i$ is Zariski closed, and $\Sigma(\xi)$ is the union of these strata over the countably many pairs $(\underline{P}, \underline{q})$.

(ii) and (iii). A Zariski closed subset of the irreducible variety $S$ that has an interior point in $S^{\mathrm{an}}$ is all of $S$, and a proper one is nowhere dense. If no stratum is $S$, then $\Sigma(\xi)$ is a countable union of nowhere dense closed sets, hence meagre, and has no interior point, since $S^{\mathrm{an}}$ is a Baire space. If one stratum is $S$, then $\Sigma(\xi) = S$. This gives (ii) and the equivalence of the first three conditions of (iii). The complement of a countable union of proper closed analytic subsets is not meagre, by the Baire category theorem, so the fourth condition implies the third, and it is implied by the first. Finally, a dense set that is closed is $S$. ∎

So the dichotomy of Theorem 1.1(iii) and the equivalences of Theorem 1.1(iv) hold for every flat class on every smooth projective family; what Theorem 1.1 adds for the Weil classes is a point of $\Sigma(\xi)$ and the density, and neither is automatic, since $\Sigma(\xi)$ may be empty.

[54] By Ehresmann's theorem $f$ is a locally trivial fibration of differentiable manifolds, so $R^{2p}f_*\mathbb{Q}$ is a local system with fibre $H^{2p}(\mathcal{X}_s, \mathbb{Q})$ at $s$. A global section is a class $\xi_s$ on every fibre that is locally constant under the identifications of nearby fibres, equivalently a class on one fibre fixed by the monodromy. The set of points $s$ at which $\xi_s$ is of type $(p,p)$ is in general a proper subset of $S$, and it is algebraic by Theorem B.6.

22.2. **(F2) is propagation from CM points.**

**Lemma 22.3** (Points of CM type are dense)**.** *Let $G$ be a connected reductive group over $\mathbb{Q}$, let $u_0\colon \mathrm{U}(1) \to G_{\mathbb{R}}$ be a homomorphism, and let $\mathcal{D}$ be the orbit of $u_0$ under conjugation by $G(\mathbb{R})^+$, with the topology of $G(\mathbb{R})^+/\mathrm{Stab}(u_0)$. Then the points $u \in \mathcal{D}$ for which $u(\mathrm{U}(1)) \subseteq T_{\mathbb{R}}$ for some maximal torus $T$ of $G$ defined over $\mathbb{Q}$ are dense in $\mathcal{D}$.*

*Proof.* Let $u_1 \in \mathcal{D}$ and let $W$ be a neighbourhood of $u_1$. The centraliser of the torus $u_1(\mathrm{U}(1))$ in $G_{\mathbb{R}}$ is connected and reductive, and a maximal torus of it defined over $\mathbb{R}$ is a maximal torus $T_1$ of $G_{\mathbb{R}}$ containing $u_1(\mathrm{U}(1))$. Write $\mathfrak{g}$ for the Lie algebra of $G$ and $\mathfrak{t}_1 \subseteq \mathfrak{g}_{\mathbb{R}}$ for that of $T_1$, and choose a regular element $Y_1 \in \mathfrak{t}_1$. The map $\varphi\colon G(\mathbb{R})^+ \times \mathfrak{t}_1 \to \mathfrak{g}_{\mathbb{R}}$, $(g, Y) \mapsto \mathrm{Ad}(g)Y$, has differential $(Z, Y') \mapsto [Z, Y_1] + Y'$ at $(1, Y_1)$, which is onto because $\mathfrak{g}_{\mathbb{R}} = \mathfrak{t}_1 \oplus [\mathfrak{g}_{\mathbb{R}}, Y_1]$ for a regular semisimple $Y_1$. So for neighbourhoods $N_1$ of 1 in $G(\mathbb{R})^+$ with $N_1 u_1 N_1^{-1} \subseteq W$, and $N_2$ of $Y_1$ in $\mathfrak{t}_1$ consisting of regular elements, the image $\varphi(N_1 \times N_2)$ contains a nonempty open subset of $\mathfrak{g}_{\mathbb{R}}$. The rational points $\mathfrak{g}(\mathbb{Q})$ are dense in $\mathfrak{g}_{\mathbb{R}}$, so that open subset contains some $Y \in \mathfrak{g}(\mathbb{Q})$, and $Y = \mathrm{Ad}(g)Y'$ with $g \in N_1$ and $Y' \in N_2$. As $Y$ is rational and regular semisimple, the identity component $T$ of its centraliser in $G$ is a maximal torus defined over $\mathbb{Q}$, and $T_{\mathbb{R}} = gT_1g^{-1}$ because the centraliser of $Y'$ has identity component $T_1$. Then $u = gu_1g^{-1}$ lies in $W$ and $u(\mathrm{U}(1)) \subseteq T_{\mathbb{R}}$. ∎

For an abelian variety the lemma is used with $G$ its Hodge group, the smallest algebraic $\mathbb{Q}$-subgroup of $\mathrm{GL}(H^1(A, \mathbb{Q}))$ whose real points contain the image of the homomorphism $\mathrm{U}(1) \to \mathrm{GL}(H^1(A, \mathbb{R}))$ that defines the Hodge structure. If $\mathrm{Hg}(A)$ lies in a $\mathbb{Q}$-torus $T$, then $A$ is of CM type.[55] Indeed $T$ lies in a maximal $\mathbb{Q}$-torus $T'$ of $\mathrm{GL}(H^1(A, \mathbb{Q}))$, which is the group of units of a commutative semisimple $\mathbb{Q}$-subalgebra $L \subseteq \mathrm{End}(H^1(A, \mathbb{Q}))$ of dimension $2 \dim A$; the algebra $L$ commutes with $T' \supseteq \mathrm{Hg}(A)$, so it lies in the commutant of $\mathrm{Hg}(A)$, which is $\mathrm{End}^0(A)$.

**Proposition 22.4** (A family with a CM member through every Hodge class)**.** *Let $A_0$ be a complex abelian variety and $\gamma$ a Hodge class of degree $2p$ on $A_0$. There are a smooth connected complex quasi-projective variety $S$, a polarised abelian scheme $\pi\colon \mathcal{A} \to S$, a global section $\xi$ of $R^{2p}\pi_*\mathbb{Q}$ that is a Hodge class at every point, and points $s_0$, $c \in S$, such that $\mathcal{A}_{s_0} \cong A_0$ with $\xi_{s_0} = \gamma$ under this isomorphism and $\mathcal{A}_c$ is of CM type.*

*Proof.* Let $\ell \geq 1$ be an integer with $\ell\gamma$ integral. We construct a section with value $\ell\gamma$ at $s_0$ and divide it by $\ell$ at the end; until then $\gamma$ stands for $\ell\gamma$. Fix a polarisation of $A_0$, put $V = H_1(A_0, \mathbb{Q})$, and fix a level $N \geq 3$. As in Setup 16.1, the quotient $\mathcal{S}_g$ of the Siegel space $\mathcal{H}$ of complex structures on $V_{\mathbb{R}}$ compatible with the polarisation, by the principal congruence subgroup of level $N$ of the stabiliser of $H_1(A_0, \mathbb{Z})$, is a smooth quasi-projective variety carrying a universal family $f\colon \mathcal{A}_g \to \mathcal{S}_g$, polarised by the fixed polarisation [BB66]; let $a_0 \in \mathcal{S}_g$ be the point of $A_0$ and $J_0 \in \mathcal{H}$ a lift, through which the fibre of $R^{2p}f_*\mathbb{Z}$ at every point of the image of $\mathcal{H}$ is identified with $H^{2p}(A_0, \mathbb{Z}) = \bigwedge^{2p} H_1(A_0, \mathbb{Z})^\vee$ by parallel transport in the simply connected space $\mathcal{H}$. Put $\mathbb{V} = R^{2p}f_*\mathbb{Z}$, a polarised variation of $\mathbb{Z}$-Hodge structure. By Theorem B.6 the connected component $\mathcal{C}$ of its locus of Hodge classes that contains $(a_0, \gamma)$ is a closed algebraic subvariety of the total space of $\mathbb{V}$; it has finitely many irreducible components.

[55] For an elliptic curve $G = \mathrm{SL}_2$ and $\mathcal{D}$ is the upper half-plane. The compact torus $u_\tau(\mathrm{U}(1))$ fixing $\tau$ lies in the real points of a maximal $\mathbb{Q}$-torus of $\mathrm{SL}_2$ exactly when $\tau$ is imaginary quadratic, and the torus is then the group of elements of norm one of $\mathbb{Q}(\tau)$. So the points of Lemma 22.3 are the curves with complex multiplication, and the lemma is the familiar density of the imaginary quadratic points in the upper half-plane.

Let $G = \mathrm{Hg}(A_0)$, which lies in the symplectic group of the polarisation and fixes $\gamma$ [Del82, §3], let $u_0\colon \mathrm{U}(1) \to G_{\mathbb{R}}$, $u_0(e^{i\theta}) = \cos\theta + J_0 \sin\theta$, be the homomorphism defining the Hodge structure, and let $\mathcal{D} = \{gJ_0g^{-1} : g \in G(\mathbb{R})^+\}$. For $g \in G(\mathbb{R})^+$ the complex structure $J = gJ_0g^{-1}$ is compatible with the polarisation and positive for it, since $g$ preserves the polarisation, so $\mathcal{D} \subseteq \mathcal{H}$; and $\mathcal{D}$ is the orbit of Lemma 22.3 when $J$ is identified with $u_J = gu_0g^{-1}$. For $J \in \mathcal{D}$ the group $u_J(\mathrm{U}(1))$ lies in $G(\mathbb{R})$, so $\mathrm{Hg}(A_J) \subseteq G$ fixes $\gamma$, which is therefore a Hodge class on $A_J$. The map sending $J \in \mathcal{D}$ to the image of $J$ in $\mathcal{S}_g$ together with $\gamma$ is thus a continuous map from the connected space $\mathcal{D}$ to the locus of Hodge classes of $\mathbb{V}$, so it takes values in $\mathcal{C}$. By Lemma 22.3 there are points $J_k \in \mathcal{D}$ converging to $J_0$ with $u_{J_k}(\mathrm{U}(1))$ in the real points of a $\mathbb{Q}$-torus of $G$; then $\mathrm{Hg}(A_{J_k})$ lies in that torus and $A_{J_k}$ is of CM type. One of the finitely many irreducible components of $\mathcal{C}$, say $\mathcal{C}_0$, contains the images of infinitely many $J_k$, and, being closed, also their limit $(a_0, \gamma)$.

Let $\rho\colon S \to \mathcal{C}_0$ be a resolution of singularities [Hir64]: $S$ is smooth, connected and quasi-projective, since $\mathcal{C}_0$ is irreducible and quasi-projective, and $\rho$ is projective and surjective. Let $\pi\colon \mathcal{A} \to S$ be the pull-back of $\mathcal{A}_g$ along $S \to \mathcal{C}_0 \to \mathcal{S}_g$, a polarised abelian scheme. A point of $\mathcal{C}_0$ is a pair $(a, v)$ with $v \in \mathbb{V}_a$ of Hodge type, and $(a, v) \mapsto v$ is a flat section of the pull-back of $\mathbb{V}$ to $\mathcal{C}_0$, because the total space of a local system is a covering space of the base; its pull-back to $S$, divided by $\ell$, is a flat section $\xi$ of $R^{2p}\pi_*\mathbb{Q}$ of Hodge type at every point. Take $s_0$ over $(a_0, \gamma)$ and $c$ over the image of some $J_k$ in $\mathcal{C}_0$. ∎

**Theorem 22.5** ((F2) is propagation from CM points)**.** *The following are equivalent.*

(i) (F2).

(ii) *The Hodge conjecture holds for every complex abelian variety.*

(iii) *For every polarised abelian scheme $\pi\colon \mathcal{A} \to S$ over a smooth connected complex quasi-projective variety and every global section $\xi$ of $R^{2p}\pi_*\mathbb{Q}$, if $\xi_s$ is algebraic at one point $s \in S$, then $\Sigma(\xi) = S$.*

(iv) *The same, for the points $s$ at which $\mathcal{A}_s$ is of CM type: if $\xi_c$ is algebraic at one point $c$ with $\mathcal{A}_c$ of CM type, then $\Sigma(\xi) = S$.*

(v) *For every $\pi$ and $\xi$ as in* (iii) *with $\xi_c$ algebraic at a point $c$ with $\mathcal{A}_c$ of CM type, $\Sigma(\xi)$ has an interior point.*

*Proof.* (i) and (ii) are equivalent by Proposition 21.49. If (ii) holds and $\xi_s$ is algebraic, then $\xi_t$ is a Hodge class at every $t$ by the first part of the proof of Proposition 20.17, which uses only the theorem of the fixed part and the semisimplicity of polarisable Hodge structures, and it is algebraic by (ii); so (ii) implies (iii). (iv) is the case of (iii) at points of CM type, and (iv) and (v) are equivalent by Lemma 22.2(iii).

Assume (iv). Let $F$ be a CM field and $n \geq 2$, and let $\delta_0$ be the trivial class, the discriminant of a hermitian space that is a sum of $n$ hyperbolic planes, each of determinant $-1$. The family of $(F, n, \delta_0)$-Weil type over its Shimura variety with a level structure, as in Setup 19.1 and Setup 16.1, is a polarised abelian scheme over a smooth connected quasi-projective base, on which the Weil classes are global sections (Proposition 19.3(i)). By Theorem 1.1(i) they are algebraic at a member isogenous to a product of abelian varieties with complex multiplication by $F$, which is of CM type, and by (iv) they are algebraic at every member. Let $A$ be an abelian variety of split Weil type for $F$ in the sense of [Mil20, §2.1]: $F$ acts on it with $\dim_F H^1(A, \mathbb{Q}) = 2n$ and $\dim V_\sigma^{1,0} = n$ for every embedding $\sigma$, and for some polarisation the hermitian form of Remark 19.2 has a totally isotropic subspace of dimension $n$, so that it is a sum of $n$ hyperbolic planes and $A$ is of $(F, n, \delta_0)$-Weil type. By Proposition 19.3(iv) the Hodge structure $H^1(A, \mathbb{Q})$ is isomorphic, compatibly with $F$, to that of a member $A_s$ of the family, so there is an isogeny $A \to A_s$ commuting with $F$; its

pull-back maps $W_F(A_s)$ onto $W_F(A)$ and algebraic classes to algebraic classes. So every abelian variety of split Weil type has algebraic Weil classes, for $n \geq 2$ by this argument and for $n = 1$ by the theorem of Lefschetz on $(1,1)$ classes. By André's theorem [Mil20, Theorem 1 and its proof], every Hodge class on an abelian variety $B$ of CM type is a sum $\sum f_\Delta^*(t_\Delta)$ of pull-backs along homomorphisms $f_\Delta\colon B \to A_\Delta$ of Weil classes $t_\Delta$ on abelian varieties $A_\Delta$ of split Weil type for CM fields $F_\Delta$; so the Hodge conjecture holds for every abelian variety of CM type. Now let $A_0$ be any abelian variety and $\gamma$ a Hodge class on it, and take $S$, $\pi$, $\xi$, $s_0$, $c$ from Proposition 22.4. Then $\xi_c$ is algebraic, so by (iv) $\xi_{s_0} = \gamma$ is algebraic. This is (ii). ∎

Figure 61 draws the theorem in one family.

The implication from (iii) to (ii) is the argument of Deligne and André as Milne records it [Mil20, §5, Theorem 3 and Remark 2].[56] In that argument the families can be taken to be polarised abelian schemes and the starting points to be of CM type, which is what (iv) records, and the converse holds. The seeds are explicit: for the Weil classes they are the base points of Theorem 19.5, and for every other Hodge class they are the CM members supplied by Proposition 22.4, at which the class is algebraic once the Weil classes of the split families are. For the Weil classes of the split family of a CM field, whose base point is of CM type, statement (v) is the condition of Theorem 1.1(iv). So the condition of Theorem 1.1(iv), required in every polarised abelian scheme with a member of CM type at which the class is algebraic, is exactly (F2); it is not a reduction of (F2) to something smaller, because the interior point it asks for is the conclusion of (iii).

### 22.3. **(F3′) and the part reached from abelian varieties.**

**Definition 22.6** (The abelian part)**.** For a smooth complex projective variety $X$ and $p \geq 0$ let $H^{2p}_{\mathrm{ab}}(X) \subseteq H^{2p}(X,\mathbb{Q})$ be the sum of the images of the maps $\Gamma_*\colon H^k(B,\mathbb{Q}) \to H^{2p}(X,\mathbb{Q})$, $y \mapsto \mathrm{pr}_{X*}([\Gamma] \cup \mathrm{pr}_B^* y)$, over all complex abelian varieties $B$, the point included, all $k$, and all algebraic cycles $\Gamma$ on $B \times X$ with rational coefficients and of the codimension that lands in degree $2p$.

**Proposition 22.7** ((F3′) is a statement about the complement of the abelian part)**.** *Let $X$ be a smooth complex projective variety and $p \geq 0$.*

(i) *$H^{2p}_{\mathrm{ab}}(X)$ is a sub-Hodge structure of $H^{2p}(X,\mathbb{Q})$, spanned by the images of finitely many of the maps $\Gamma_*$.*

(ii) $\mathrm{Ab}^p(X) = \mathrm{Hdg}^p(X) \cap H^{2p}_{\mathrm{ab}}(X)$.

(iii) *Let $T^{2p}_{\mathrm{ab}}(X)$ be a complement of $H^{2p}_{\mathrm{ab}}(X)$ in $H^{2p}(X,\mathbb{Q})$ as a Hodge structure. Then* (F3′) *holds for $X$ in degree $2p$ if and only if $T^{2p}_{\mathrm{ab}}(X)$ contains no nonzero Hodge class, in particular whenever the Hodge group of $T^{2p}_{\mathrm{ab}}(X)$ fixes no nonzero vector.*

(iv) *$X$ lies in the class $\mathcal{A}$ of Proposition 20.35(iii) if and only if $T^{2p}_{\mathrm{ab}}(X) = 0$ for every $p$ and the odd cohomology is reached in the same way.*

*Proof.* (i) An algebraic cycle $\Gamma$ of codimension $e$ on $B \times X$, with $\dim B = b$, induces a morphism of Hodge structures $H^k(B,\mathbb{Q}) \to H^{k+2e-2b}(X,\mathbb{Q})(e-b)$, so its image is a

[56] Deligne's proof that Hodge classes on abelian varieties are absolutely Hodge uses families of the same kind [Del82]. By his Principle B [Del82, §2] the property of being absolutely Hodge passes from one member of a family to every member, so a family through a given Hodge class with a member of CM type reduces the question to that member. Algebraicity is not known to pass from one member to all, and (iii) is precisely that property.

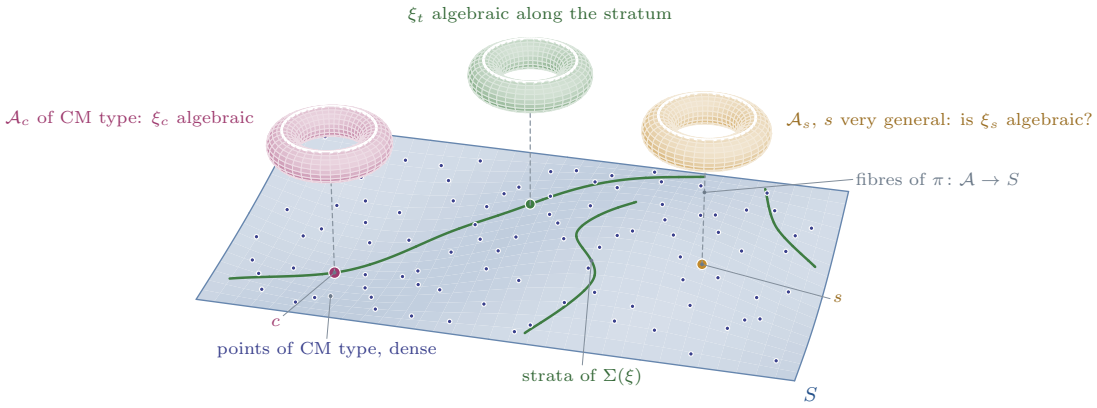



FIGURE 61. Theorem 22.5 in one family. The sheet is the base $S$ of a polarised abelian scheme $\pi\colon \mathcal{A} \to S$ through a given Hodge class, as in Proposition 22.4; the small dots are its points of CM type, dense by Lemma 22.3, and the green curves are strata of the algebraic locus $\Sigma(\xi)$ of the flat class $\xi$ (Lemma 22.2). Above three points the fibres are drawn as tori, and the loop on each stands for $\xi$, carried from fibre to fibre by parallel transport. At the CM member $\mathcal{A}_c$ the class is algebraic, and it stays algebraic along the stratum through $c$. At a very general member $\mathcal{A}_s$ it is algebraic exactly when $\Sigma(\xi)$ has an interior point, and (F2) is the statement that this happens in every such family.

sub-Hodge structure; a sum of sub-Hodge structures is one, and a finite-dimensional space is spanned by finitely many of them.

(ii) Every $\Gamma_*\beta$ with $\beta$ a Hodge class lies in both sets, so $\mathrm{Ab}^p(X)$ is contained in the intersection. Conversely take $(B_j, \Gamma_j, k_j)$, $1 \le j \le r$, as in (i), and let

$$\Phi = \sum_j \Gamma_{j*}\colon \bigoplus_j H^{k_j}(B_j, \mathbb{Q})(t_j) \longrightarrow H^{2p}_{\mathrm{ab}}(X),$$

with the Tate twists $t_j$ that make it a morphism of pure Hodge structures of weight $2p$; it is surjective. Polarisable Hodge structures form a semisimple category, so $\Phi$ has a section $\sigma$ that is a morphism of Hodge structures. For a Hodge class $\gamma \in H^{2p}_{\mathrm{ab}}(X)$ the class $\sigma(\gamma)$ is of type $(p,p)$, so each component $\beta_j \in H^{k_j}(B_j, \mathbb{Q})$ is a Hodge class, $k_j$ being even, and $\gamma = \sum_j \Gamma_{j*}\beta_j \in \mathrm{Ab}^p(X)$.

(iii) A complement exists by semisimplicity, and the Hodge classes of $H^{2p}(X, \mathbb{Q})$ are the sums of those of the two summands; by (ii) the Hodge classes of the first summand are $\mathrm{Ab}^p(X)$. So $\mathrm{Hdg}^p(X) = \mathrm{Ab}^p(X)$ exactly when the second has none, and a Hodge class of $T^{2p}_{\mathrm{ab}}(X)$ is a vector fixed by its Hodge group [Del82, §3]. (iv) is the definition of $\mathcal{A}$ read degree by degree. ∎

So (F3′) asks nothing of the abelian part, where by (ii) and Proposition 21.49 the Hodge conjecture reduces to (F2), and it asks of the complement that it carry no Hodge class. The abelian part contains every algebraic class, taking $B$ a point, so under the Hodge conjecture the complement carries none. To prove (F3′) for $X$ in degree $2p$ without it, one has to show that every Hodge class of $X$ is a sum of classes $\Gamma_*\beta$, that is, to construct algebraic cycles $\Gamma$ on products $B \times X$ that reach it, which is the step Remark 20.36

isolates. Proposition 20.35(iii), Proposition 20.42 and Theorem 20.52 are cases in which the complement is zero, and Proposition 20.39 and Proposition 20.40 cases in which the conjecture itself makes its Hodge classes algebraic.

22.4. **The first open case of (F3′) in families.** The frontier of (F3′) in Remark 20.68 begins with $S \times S$ and $S^{[2]}$ for a K3 surface $S$ with real multiplication. These surfaces come in families over Hermitian symmetric domains, and the argument of Theorem 1.1 applies to them with rational Hodge isometries in place of isogenies.

**Setup 22.8** (A family with real multiplication)**.** Let $E$ be a totally real field of degree $r \geq 2$ and $T$ a finite-dimensional $\mathbb{Q}$-vector space with an action of $E$ and a nondegenerate symmetric form $q$ for which $E$ acts by self-adjoint maps, so that $q = \mathrm{Tr}_{E/\mathbb{Q}}\, q_E$ for a unique $E$-bilinear form $q_E$. Put $m = \dim_E T \geq 3$, and suppose that $q_E$ has signature $(2, m-2)$ at one real place $\sigma_1$ of $E$ and is negative definite at the others, and that $(T, q)$ embeds isometrically in $\Lambda_{\mathbb{Q}}$, where $\Lambda$ is the K3 lattice; write $N = T^{\perp} \cap \Lambda$. Let $T_1 = T \otimes_{E,\sigma_1} \mathbb{R}$ and

$$\mathcal{D}_E = \{\, [\omega] \in \mathbb{P}(T_1 \otimes_{\mathbb{R}} \mathbb{C}) :\ q(\omega,\omega) = 0,\ q(\omega,\bar{\omega}) > 0 \,\}^{+},$$

one of its two components, a bounded symmetric domain of dimension $m-2$. For $\omega \in \mathcal{D}_E$ let $S_\omega$ be a K3 surface with a marking $H^2(S_\omega, \mathbb{Z}) \cong \Lambda$ carrying $H^{2,0}(S_\omega)$ to $\mathbb{C}\omega$, which exists by the surjectivity of the period map [Huy16, Chs. 6–7]; it is projective, since $N$ has signature $(1, 21 - rm)$ and so contains a class of positive square, which is of type $(1,1)$ [Huy16, Ch. 8]. Each $e \in E$ acts on $T_1 \otimes \mathbb{C}$ by the scalar $\sigma_1(e)$ and is self-adjoint, so it preserves the Hodge decomposition of $T$, and $T \subseteq H^2(S_\omega, \mathbb{Q})$ is a sub-Hodge structure containing $T(S_\omega)_{\mathbb{Q}}$. Write $\tilde{e}_\omega \in H^4(S_\omega \times S_\omega, \mathbb{Q})$ for the class of the endomorphism $e \oplus 0$ of $H^2(S_\omega, \mathbb{Q}) = T \oplus N_{\mathbb{Q}}$, a Hodge class, and let

$$\Sigma_E = \{\, \omega \in \mathcal{D}_E :\ \tilde{e}_\omega \text{ is algebraic for every } e \in E \,\}$$

be the algebraic locus of the real multiplication; as $e \mapsto \tilde{e}_\omega$ is $\mathbb{Q}$-linear, it suffices to ask this for $e$ in a $\mathbb{Q}$-basis of $E$. The set $\Sigma_E$ does not depend on the choice of the surfaces $S_\omega$. If $(S, \varphi)$ and $(S', \varphi')$ are two marked choices for $\omega$, then $\psi = \varphi'^{-1}\varphi$ is a rational Hodge isometry $H^2(S, \mathbb{Q}) \to H^2(S', \mathbb{Q})$, induced by an algebraic class $Z$ on $S \times S'$ by Buskin's theorem [Bus19], and $Z^t$ induces the adjoint $\psi^{-1}$; the composite $Z \circ \tilde{e} \circ Z^t$ acts on $H^2(S', \mathbb{Q})$ as $\psi(e \oplus 0)\psi^{-1}$, which is $e \oplus 0$ for the marking $\varphi'$, and by zero on $H^0$ and $H^4$, so it is the class $\tilde{e}'$ of $S'$, and $\tilde{e}$ is algebraic if and only if $\tilde{e}'$ is. Let $G = \mathrm{Res}_{E/\mathbb{Q}}\, \mathrm{SO}(T, q_E)$, so that $G(\mathbb{R}) = \mathrm{SO}(T_1) \times \prod_{\sigma \neq \sigma_1} \mathrm{SO}(T_\sigma)$; write $G(\mathbb{R})^+$ for the identity component of this Lie group, which acts transitively on $\mathcal{D}_E$ through its first factor, and $\Gamma_{\mathbb{Q}} = G(\mathbb{Q}) \cap G(\mathbb{R})^+$.

The surfaces with real multiplication of [vG08, vGS25] lie in families of this kind,[57] and so do the surfaces $S_\lambda$ of Theorem 21.4, for which $E$ is cubic, $m = 3$ and $\mathcal{D}_E$ is a disc.

**Theorem 22.9** (The algebraic locus of a real multiplication)**.** *Keep Setup 22.8.*

(i) *The points $\omega$ at which $S_\omega$ has complex multiplication, that is, at which the Hodge group of $T(S_\omega)_{\mathbb{Q}}$ is commutative, are dense in $\mathcal{D}_E$, and they lie in $\Sigma_E$.*

(ii) *$\Sigma_E$ is stable under $\Gamma_{\mathbb{Q}}$, and $\Gamma_{\mathbb{Q}}$ is dense in $G(\mathbb{R})^+$.*

(iii) *$\Sigma_E$ is dense in $\mathcal{D}_E$, and either $\Sigma_E = \mathcal{D}_E$ or $\Sigma_E$ is meagre.*

[57] For a projective K3 surface $S$ the transcendental part $T(S)_{\mathbb{Q}}$ is an irreducible Hodge structure whose algebra of Hodge endomorphisms is a field, totally real or CM [Zar83]; $S$ has real multiplication when this field is totally real and not $\mathbb{Q}$. The space $T(S)_{\mathbb{Q}}$ is then a vector space over the field, so its degree divides $\dim T(S)_{\mathbb{Q}}$, and a very general member of a family of Setup 22.8 has $T(S_\omega)_{\mathbb{Q}} = T$ and real multiplication by exactly $E$, as the proof of Theorem 22.9 shows.

(iv) *The following are equivalent:* $\Sigma_E$ *is closed;* $\Sigma_E$ *has an interior point;* $\Sigma_E$ *is not meagre;* $\Sigma_E = \mathcal{D}_E$*; the Hodge conjecture holds for* $S_\omega \times S_\omega$ *for every* $\omega$ *outside a countable union of proper closed analytic subsets of* $\mathcal{D}_E$*. When they hold, the Hodge conjecture and* (F3′) *hold for* $S_\omega \times S_\omega$ *and* $S_\omega^{[2]}$ *at every* $\omega$ *at which* $\mathrm{End}_{\mathrm{Hdg}}(T(S_\omega)_{\mathbb{Q}})$ *is* $E$ *or a CM field.*

*Proof.* We first record two facts about the very general point. The Hodge structure of $S_\omega$ on $T$ is given by $u_\omega\colon \mathrm{U}(1) \to \mathrm{SO}(T_{\mathbb{R}})$, acting by $z^2$ on $\mathbb{C}\omega$, by $z^{-2}$ on $\mathbb{C}\bar\omega$ and trivially on their orthogonal complement; it commutes with $E$, lies in $G(\mathbb{R})^+$, and $u_{g\omega} = g u_\omega g^{-1}$ for $g \in G(\mathbb{R})^+$.

(a) *No nonzero* $t \in T$ *is orthogonal to every* $\omega \in \mathcal{D}_E$. The lines $\mathbb{C}\omega$, $\omega \in \mathcal{D}_E$, span $T_1 \otimes \mathbb{C}$, since $\mathcal{D}_E$ is a nonempty open subset of the nondegenerate quadric $q(\omega,\omega) = 0$ in $\mathbb{P}(T_1 \otimes \mathbb{C})$, so such a $t$ would have zero image in $T_1$; but $T \to T_1$ is injective, since $T$ is a free $E$-module and $\sigma_1$ is injective.

(b) *An endomorphism* $f$ *of* $T$ *that is an endomorphism of Hodge structures at every* $\omega \in \mathcal{D}_E$ *is multiplication by an element of* $E$. Such an $f$ commutes with every $u_\omega$, hence with the elements $X_\omega$ of the Lie algebra of $G(\mathbb{R})$ that span the Lie algebras of the groups $u_\omega(\mathrm{U}(1))$. They lie in $\mathfrak{so}(T_1) \oplus 0$, which acts on $T' = \bigoplus_{\sigma \neq \sigma_1} T_\sigma$ by zero, and their span is stable under $G(\mathbb{R})^+$, since $u_{g\omega} = g u_\omega g^{-1}$, so it is an ideal of $\mathfrak{so}(T_1)$. That ideal has nonzero projection to every simple ideal, as the eigenvalues $2i, -2i, 0, \dots, 0$ of $X_\omega$ on $T_1 \otimes \mathbb{C}$ show also for $m = 4$, where $\mathfrak{so}(T_1) \otimes \mathbb{C} \cong \mathfrak{sl}_2 \oplus \mathfrak{sl}_2$ and $T_1 \otimes \mathbb{C}$ is the tensor product of the two standard representations; so it is all of $\mathfrak{so}(T_1)$, and $f$ commutes with $\mathfrak{so}(T_1)$. For $m \geq 3$ the representation of $\mathfrak{so}(T_1)$ on $T_1 \otimes \mathbb{C}$ is irreducible and nontrivial, so $f$ preserves $T_1 = \mathfrak{so}(T_1) \cdot T_{\mathbb{R}}$ and $T'$ and acts on $T_1 \otimes \mathbb{C} = T \otimes_{E,\sigma_1} \mathbb{C}$ by a scalar $\lambda$. An automorphism $\tau$ of $\mathbb{C}$, acting on $T \otimes_{\mathbb{Q}} \mathbb{C}$ through the second factor, commutes with $f$ and maps $T \otimes_{E,\sigma_1} \mathbb{C}$ onto $T \otimes_{E,\tau\sigma_1} \mathbb{C}$, so $f$ acts on the latter by $\tau(\lambda)$; and every embedding of $E$ is of the form $\tau\sigma_1$. So $f \otimes 1$ is multiplication by an element $e$ of $E \otimes_{\mathbb{Q}} \mathbb{C}$ on $T \otimes_{\mathbb{Q}} \mathbb{C} = T \otimes_E (E \otimes_{\mathbb{Q}} \mathbb{C})$, and $e$ is fixed by $\mathrm{Aut}(\mathbb{C})$ because $f$ is defined over $\mathbb{Q}$; hence $e \in E$.

For $t \in T$ the set of $\omega$ with $t \perp \omega$ is a closed analytic subset of $\mathcal{D}_E$, and for $f \in \mathrm{End}_{\mathbb{Q}}(T)$ so is the set where $f$ is an endomorphism of Hodge structures, namely where $f(\omega) \in \mathbb{C}\omega$ and $f^*(\omega) \in \mathbb{C}\omega$, $f^*$ the adjoint. By (a) and (b) these sets are proper for $t \neq 0$ and $f \notin E$. There are countably many of them, and outside their union $\mathrm{NS}(S_\omega)_{\mathbb{Q}} \cap T = 0$, so $T(S_\omega)_{\mathbb{Q}} = T$ and $\mathrm{End}_{\mathrm{Hdg}}(T(S_\omega)_{\mathbb{Q}}) = E$.

(i) The Hodge group of $T$ acts trivially on $T \cap \mathrm{NS}(S_\omega)_{\mathbb{Q}}$, whose orthogonal complement in $T$ is $P = T(S_\omega)_{\mathbb{Q}}$, so it is commutative exactly when the Hodge group $H$ of $P$ is. The group $G$ is connected and reductive, and $\omega \mapsto u_\omega$ identifies $\mathcal{D}_E$ with the orbit of Lemma 22.3, since $g \in G(\mathbb{R})^+$ fixes $u_\omega$ exactly when it preserves the line $\mathbb{C}\omega$, the eigenspace of $u_\omega(z)$ for $z^2$. By Lemma 22.3 the points at which $u_\omega(\mathrm{U}(1))$ lies in the real points of a maximal $\mathbb{Q}$-torus of $G$ are dense; at them the Hodge group of $T$, the smallest algebraic $\mathbb{Q}$-subgroup whose real points contain $u_\omega(\mathrm{U}(1))$, lies in that torus, so these points have complex multiplication. Now let $\omega$ be any point at which $H$ is commutative. The Hodge structure $P$ is irreducible [Zar83], so $P$ is an irreducible representation of the torus $H$, whose rational points are Zariski dense in it. The $\mathbb{Q}$-subalgebra $L$ of $\mathrm{End}(P)$ that $H(\mathbb{Q})$ generates is therefore commutative, and $P$ is a simple faithful $L$-module; so $L$ is a field with $\dim_L P = 1$, and its commutant $E' = \mathrm{End}_{\mathrm{Hdg}}(P)$ is $L$. By [Zar83] $E'$ is totally real or CM. If it were totally real, the embedding $\tau$ through which it acts on the line $H^{2,0}(S_\omega)$ would be real, and that line would lie in $P \otimes_{E',\tau} \mathbb{C}$, which is one-dimensional and stable under complex conjugation, so it would also contain the different line $H^{0,2}(S_\omega)$,

which is impossible. So $E'$ is a CM field, the Hodge conjecture holds for $S_\omega \times S_\omega$ by Proposition 20.39(i), and the Hodge classes $\tilde{e}_\omega$ are algebraic.

(ii) Let $g \in \Gamma_{\mathbb{Q}}$ and $\omega \in \Sigma_E$. Then $g\omega \in \mathcal{D}_E$, and $\phi = \mathrm{id}_N \oplus g$ is a rational isometry of $\Lambda_{\mathbb{Q}} = N_{\mathbb{Q}} \oplus T$ carrying $\mathbb{C}\omega$ to $\mathbb{C}g\omega$, so through the markings it is a rational Hodge isometry $H^2(S_\omega, \mathbb{Q}) \to H^2(S_{g\omega}, \mathbb{Q})$, induced by an algebraic class $Z \in H^4(S_\omega \times S_{g\omega}, \mathbb{Q})$ [Bus19], whose transpose induces $\phi^{-1}$. As $g$ is $E$-linear, $\phi \circ (e \oplus 0) \circ \phi^{-1} = e \oplus 0$, so, as in Setup 22.8, $\tilde{e}_{g\omega} = Z \circ \tilde{e}_\omega \circ Z^t$ is algebraic.

For the density of $\Gamma_{\mathbb{Q}}$, let $Y$ be an $E \otimes \mathbb{R}$-linear map of $T \otimes \mathbb{R}$ that is skew-adjoint for $q$ and has $1 - Y$ invertible. Its Cayley transform $c(Y) = (1 + Y)(1 - Y)^{-1}$ is $E \otimes \mathbb{R}$-linear, has $c(Y)^* c(Y) = 1$ because $1 - Y$ is the adjoint of $1 + Y$, and has determinant one for the same reason, so $c(Y) \in G(\mathbb{R})$; and $c$ is a homeomorphism onto the open neighbourhood $\{g \in G(\mathbb{R}) : 1 + g \text{ invertible}\}$ of 1, with inverse $g \mapsto (g - 1)(g + 1)^{-1}$. The $E$-linear skew-adjoint maps of $T$ are dense among these $Y$, and $c$ maps them into $G(\mathbb{Q})$. So $\Gamma_{\mathbb{Q}}$ is dense in a connected open neighbourhood of 1, its closure is an open subgroup of $G(\mathbb{R})^+$, and it is all of $G(\mathbb{R})^+$.

(iii) and (iv). Density follows from (i). If $\Sigma_E$ contains a nonempty open set $U_0$, then $\Sigma_E = \mathcal{D}_E$: for $\omega \in \mathcal{D}_E$ choose $\omega' \in U_0$ and $g_1 \in G(\mathbb{R})^+$ with $g_1\omega' = \omega$; the set of $g \in G(\mathbb{R})^+$ with $g^{-1}\omega \in U_0$ is open and contains $g_1$, so by (ii) it contains some $g \in \Gamma_{\mathbb{Q}}$, and $\omega = g(g^{-1}\omega) \in \Sigma_E$ by (ii).

It remains to show that $\Sigma_E$ has an interior point as soon as it is not meagre. Fix $h \in N$ with $h^2 > 0$ and $h \cdot \delta \neq 0$ for every $\delta \in N$ with $\delta^2 = -2$, which exists because the hyperplanes $\delta^\perp$ are locally finite in the positive cone of $N_{\mathbb{R}}$ [Huy16, Ch. 8]. Let $\mathcal{D}^\circ$ be the set of $\omega \in \mathcal{D}_E$ that are orthogonal to no $\delta \in \Lambda$ with $\delta^2 = -2$ and $h \cdot \delta = 0$. Writing such a $\delta$ as $n + t$ with $n \in N_{\mathbb{Q}}$ and $t \in T$, we have $t \neq 0$, because $N_{\mathbb{Q}} \cap \Lambda = N$, and $\delta \perp \omega$ exactly when $t \perp \omega$; so by (a) the complement of $\mathcal{D}^\circ$ lies in a countable union of proper closed analytic subsets and is meagre.

Let $\omega \in \mathcal{D}^\circ$. No class of $\mathrm{NS}(S_\omega)$ of square $-2$ is orthogonal to $h$, so after composing the marking with an isometry $\pm w$, $w$ a product of reflections in such classes, which fixes $\omega$ and so changes only the choice of $S_\omega$, the class $h$ is that of an ample line bundle $L$ on $S_\omega$ [Huy16, Ch. 8]. The deformations of the pair $(S_\omega, L)$ form a smooth family over a base that the local period map identifies with a neighbourhood of $\omega$ in the domain of periods orthogonal to $h$ [Huy16, Ch. 6], and that domain contains $\mathcal{D}_E$, since $h \in N$. Restricting to $\mathcal{D}_E$, we obtain an open neighbourhood $U$ of $\omega$ in $\mathcal{D}_E$ and a smooth family $\mathcal{X}_U \to U$ of K3 surfaces with markings, whose fibre over $u$ is a choice of $S_u$, carrying a line bundle that restricts to $L$ on $S_\omega$. On $S_\omega$ the bundle $L^{\otimes 3}$ is very ample and has no higher cohomology [Huy16, Ch. 2]; by semicontinuity and base change, after shrinking $U$, its direct image is free and its sections embed $\mathcal{X}_U$ in $\mathbb{P}^M \times U$ over $U$. The classes $\tilde{e}$ are flat sections of the fourth cohomology of $\mathcal{X}_U \times_U \mathcal{X}_U$, which is projective over $U$. As in the proof of Lemma 22.2(i), with the Hilbert schemes of $\mathbb{P}^M \times \mathbb{P}^M$ [Gro61], which are projective, for each $e$ in a $\mathbb{Q}$-basis of $E$ the set of $u \in U$ at which $\tilde{e}_u = \sum_i q_i[Z_i]$ with rational $q_i$ and $Z_i$ of Hilbert polynomial $P_i$ is the image, under a proper map, of a union of connected components of a fibre product of relative Hilbert spaces over $U$, and so it is closed in $U$. Taking the union over the countably many $(\underline{P}, \underline{q})$ and the intersection over the basis, $\Sigma_E \cap U$ is a countable union of subsets of $U$ that are closed in $U$.

The sets $U$ constructed for the points of $\mathcal{D}^\circ$ cover it, and countably many of them, $U_i$, suffice, since $\mathcal{D}_E$ has a countable basis. If $\Sigma_E$ is not meagre, neither is $\Sigma_E \cap \mathcal{D}^\circ$, so some $\Sigma_E \cap U_i$ is not meagre, and by the Baire category theorem in $U_i$ one of the countably many closed subsets of $U_i$ that make it up has an interior point. This proves (iii), and in (iv) the

equivalence of the first four conditions, since a dense closed set is everything. The fifth condition implies the third, because at each point where the Hodge conjecture holds for $S_\omega \times S_\omega$ the Hodge classes $\tilde{e}_\omega$ are algebraic, and the complement of a countable union of proper closed analytic subsets is not meagre.

If $\Sigma_E = \mathcal{D}_E$ and $\mathrm{End}_{\mathrm{Hdg}}(T(S_\omega)_\mathbb{Q}) = E$, then $E$ preserves $T(S_\omega)_\mathbb{Q}$, the smallest sub-Hodge structure of $T$ whose complexification contains $H^{2,0}(S_\omega)$, since each $e \neq 0$ maps it onto a sub-Hodge structure of the same dimension that contains $H^{2,0}(S_\omega)$; and each $e \in E$ acts on it through the algebraic class $\pi_T \circ \tilde{e}_\omega \circ \pi_T$, where $\pi_T$ is the algebraic projector onto $T(S_\omega)_\mathbb{Q}$ of the proof of Proposition 20.39; so Proposition 20.39(iii) gives the Hodge conjecture for $S_\omega \times S_\omega$ and $S_\omega^{[2]}$, and with it (F3′). If the field is CM, Proposition 20.39(i) gives the same. Outside the countable union found at the beginning of the proof the field is $E$, so the fourth condition implies the fifth. ∎

On the families of van Geemen and Schütt with real multiplication by $\mathbb{Q}(\zeta_7 + \zeta_7^{-1})$ and $\mathbb{Q}(\zeta_9 + \zeta_9^{-1})$ the real multiplication is induced by explicit cycles [vGS25, Theorem 1.2 and Section 4.8], and on the double planes branched along six lines that have real multiplication, necessarily by a quadratic field here, by Schlickewei's cycles [Sch10, vG08]. Whenever the members on which such cycles are known contain a nonempty open subset of $\mathcal{D}_E$, (iv) gives $\Sigma_E = \mathcal{D}_E$. For the family of the surfaces $S_\lambda$, Theorem 22.9 gives the real multiplication at every CM point and on a dense set stable under $\Gamma_\mathbb{Q}$, and Theorem 21.4(ii) gives it at $S_\lambda$ if the Kuga–Satake class of $S_\lambda$ is algebraic; whether $\Sigma_E$ is closed there is open. Since $\Sigma_E$ is stable under the group $\Gamma_\mathbb{Q}$, every $\Gamma_\mathbb{Q}$-orbit lies either in $\Sigma_E$ or in its complement: the Hodge isometries of (ii) move the real multiplication along an orbit and never bring it to an orbit where it is not yet algebraic, and by Remark 21.5 the composites of Hodge similarities that return to $T(S_\lambda)$ span only the maximal multiquadratic subfield of $E$, which is $\mathbb{Q}$ for a cubic field.

### 22.5. **What the constructions reach.**

**Theorem 22.10** (Where (F2) and (F3′) stand)**.**

(i) What carries over. *In every polarised abelian scheme of Theorem 22.5 the algebraic locus of a flat class is a countable union of closed algebraic strata and is either everything or meagre (Lemma 22.2); in the Weil families it is moreover dense and contains an explicit CM point (Theorem 1.1); in every family of Setup 22.8 the algebraic locus of the real multiplication is dense, contains every CM point, is stable under $\Gamma_\mathbb{Q}$ and is either everything or meagre (Theorem 22.9); and every Hodge class of every abelian variety lies in a polarised abelian scheme with a CM member (Proposition 22.4).*

(ii) (F2). *It holds if and only if the algebraic locus of every flat Hodge class algebraic at a CM member of a polarised abelian scheme has an interior point (Theorem 22.5). In the Weil families the constructions of the paper that supply an interior point are: an object with injective semiregularity map, or satisfying the flatness condition, at one point (Theorem 16.29, Theorem 19.7, Proposition 17.15); a bound on the Hilbert data of representatives along the Hecke orbit of a base point (Theorem 16.9), which is equivalent to its conclusion (Theorem 16.17); and the Lefschetz standard conjecture for the total space over a curve through a base point, which implies it and, when the monodromy is Zariski dense, follows from it (Corollary 20.31). For the squares of Mumford fourfolds the last two take the forms of Theorem 21.43 and Corollary 20.30. Markman's sheaves are objects of the first kind, and with them*

*he proves the algebraicity of the Weil classes on every abelian fourfold of Weil type and on the split sixfolds of every imaginary quadratic field* [Mar25a, Mar25b]*; with* [MZ99] *this gives the Hodge conjecture for abelian varieties of dimension at most five. The open cases named in Section 20 include the non-split Weil sixfolds of imaginary quadratic fields, the Weil eightfolds of quartic CM fields, where an object of the first kind needs* $\dim \mathrm{Ext}^2(E,E) \geq 88$ *(Corollary 19.30), and the squares of Mumford fourfolds, where the classes are algebraic at every CM point (Theorem 20.29) and on every member once one Kuga–Satake class is algebraic (Theorem 21.4), or once one complex at one CM point has* $\mathrm{Ext}^2$ *of dimension* 119 *(Theorem 21.27); on the square of a Mumford fourfold the two exceptional classes are algebraic exactly when one class on a product with an abelian sixfold is (Theorem 24.45).*

(iii) (F3′). *It holds if and only if, for every variety and every degree, the complement of the abelian part carries no Hodge class (Proposition 22.7). The complement is zero on the class* $\mathcal{A}$*, which contains the Fermat and Delsarte hypersurfaces of every dimension and the cyclic covers branched along them (Proposition 20.42, Corollary 20.53), and it carries no Hodge class on the varieties of Proposition 20.43, Proposition 20.44 and Proposition 20.46, where* (F3′) *is proved, and of Proposition 20.39 and Proposition 20.40, where the conjecture itself is. On the first open varieties,* $S \times S$ *and* $S^{[2]}$ *for a K3 surface with real multiplication, the algebraic locus of the real multiplication is dense, contains every CM point and is either the whole family or meagre (Theorem 22.9); the real multiplication is algebraic on the members of the families of* [vGS25] *and on the quadratic double planes of* [Sch10]*, and whether the locus is closed is open for the family of* $S_\lambda$*, except for the two cyclic cubic fields of conductor* 7 *and* 9 *with the algebra split at one real place, where Theorem 23.14 makes the locus the whole family. Beyond the varieties of abelian type, a Hodge class off the abelian part is removed only by a new algebraic cycle on a product* $B \times X$ *(Remark 20.36).*

*Neither (F2) nor* (F3′) *is proved here, and by Theorem 27.1 together they are the Hodge conjecture.*

*Proof.* Each statement is the result cited with it. For (ii), Markman's theorems are recalled in the introduction and their reach in Section 20, where the non-split sixfolds and the fields of degree at least four are among the open entries, and Theorem 19.29 and Corollary 19.30 give the number 88. For (iii), the known cases of the real multiplication are those recalled in Remark 21.5. ∎

## Part 6. Constructions at the frontier

### 23. Constructions at (F3′): real multiplication on K3 surfaces

23.1. **Where the first class is.** Let $S$ be a projective K3 surface, $T = T(S)_{\mathbb{Q}}$ and $E = \mathrm{End}_{\mathrm{Hdg}}\, T$. The Hodge classes of $S \times S$ are $H^0 \otimes H^4$, $H^4 \otimes H^0$ and $\mathrm{End}_{\mathrm{Hdg}}\, H^2(S) = \mathrm{End}(\mathrm{NS}_{\mathbb{Q}}) \oplus E$, since $T$ is irreducible with no class of type $(1,1)$ and $\mathrm{Hom}_{\mathrm{Hdg}}(T, \mathrm{NS}_{\mathbb{Q}}) = 0$. The first two pieces and $\mathrm{End}(\mathrm{NS}_{\mathbb{Q}})$ are spanned by products of divisor classes. So the Hodge conjecture for $S \times S$ is the algebraicity of $E$, and by Proposition 22.7 the same classes decide (F3′) for $S \times S$ once $T$ is not reached from abelian varieties. When $E$ is a CM field it is spanned by its elements of norm one, which are Hodge isometries of $T$ and algebraic by Buskin's theorem [Bus19]; see [Var23, Rem. 2.16]. When $E$ is totally

real, the Rosati involution is the identity on it [Zar83], every $e \in E$ is self-adjoint for the intersection form $q$, and $q(ex, ey) = q(e^2 x, y)$. The first class is a real multiplication.

23.2. **Chains of correspondences.**

**Lemma 23.1.** *Let $f\colon S \dashrightarrow S'$ be a dominant rational map of K3 surfaces of degree $d$ and $\Gamma_f$ the closure of its graph. Then $f_* = \Gamma_{f*}$ and $f^* = \Gamma_{f*}^t$ are morphisms of Hodge structures, $f^*$ maps $T(S')_{\mathbb{Q}}$ isomorphically onto $T(S)_{\mathbb{Q}}$ with $q_S(f^*a, f^*b) = d\, q_{S'}(a, b)$, and $f_* = d\,(f^*)^{-1}$ on $T(S)_{\mathbb{Q}}$.*

*Proof.* The graph is an algebraic cycle, so both maps are induced by algebraic classes and are morphisms of Hodge structures. Pull-back of a nowhere vanishing 2-form is a nonzero 2-form, so $f^*(T(S')_{\mathbb{Q}})$ is a sub-Hodge structure of $H^2(S)$ that meets $H^{2,0}(S)$; the smallest sub-Hodge structure of $H^2(S)$ containing $H^{2,0}$ is $T(S)_{\mathbb{Q}}$, so $f^*T(S')_{\mathbb{Q}} \supseteq T(S)_{\mathbb{Q}}$, and since $T(S')_{\mathbb{Q}}$ is irreducible and $f^*$ is injective on it ($f_* f^* = d$), equality of dimensions gives $f^*T(S')_{\mathbb{Q}} = T(S)_{\mathbb{Q}}$. The projection formula gives $q_S(f^*a, f^*b) = q_{S'}(f_* f^* a, b) = d\, q_{S'}(a, b)$ and $f_* f^* = d$. ∎

**Lemma 23.2.** *A Hodge isometry $\varphi\colon T(S)_{\mathbb{Q}} \to T(S')_{\mathbb{Q}}$ is induced by an algebraic class on $S \times S'$.*

*Proof.* The spaces $\mathrm{NS}(S)_{\mathbb{Q}}$ and $\mathrm{NS}(S')_{\mathbb{Q}}$ are the orthogonal complements of $T(S)_{\mathbb{Q}}$ and $T(S')_{\mathbb{Q}}$ in the two copies of $\Lambda_{\mathbb{Q}}$, isometric to each other through $\varphi$; by Witt's cancellation theorem the complements are isometric, so $\varphi$ extends to a Hodge isometry $H^2(S) \to H^2(S')$, which is algebraic by Buskin's theorem [Bus19] (see [Huy19, Thm. 0.1]). The projector of $H^2(S)$ onto $T(S)_{\mathbb{Q}}$ along $\mathrm{NS}(S)_{\mathbb{Q}}$ is algebraic, being 1 minus the projector onto NS, which is a sum $\sum_{i,j} c_{ij}\, D_i \otimes D_j$ over a basis of divisor classes with $(c_{ij})$ the inverse Gram matrix; composing gives $\varphi$ on $T$ and zero on NS. ∎

**Theorem 23.3** (Chains)**.** *Let $S$ be a projective K3 surface with $E$ totally real and $e \in E$ with $e^2 = n \in \mathbb{Q}_{>0}$. Suppose there are projective K3 surfaces $S = S_0, S_1, \dots, S_r$ and, for each $i$, either a dominant rational map $S_{i-1} \dashrightarrow S_i$ of degree $n_i$, a dominant rational map $S_i \dashrightarrow S_{i-1}$ of degree $n_i$, or a Hodge isometry $T(S_{i-1})_{\mathbb{Q}} \cong T(S_i)_{\mathbb{Q}}$ (and then $n_i = 1$), such that*

$$\prod_i n_i^{\pm 1} \equiv n \pmod{\mathbb{Q}^{\times 2}},$$

*the sign being $+$ for maps leaving $S_{i-1}$ and $-$ for maps arriving at it. Then $e$ is algebraic on $S \times S$.*

*Proof.* Composing along the chain the maps $f_*$ for maps leaving $S_{i-1}$, $f^*$ for maps arriving, and the classes of Lemma 23.2 for the isometries, gives an algebraic isomorphism $\phi\colon T(S_r)_{\mathbb{Q}} \to T(S)_{\mathbb{Q}}$ of Hodge structures with $q_S(\phi a, \phi b) = \mu\, q_{S_r}(a, b)$, where $\mu = \prod n_i^{\pm 1}$ by Lemma 23.1 (a map leaving contributes $f_* = n_i (f^*)^{-1}$, of multiplier $n_i$; a map arriving contributes $f^*$, of multiplier $n_i$ in the other direction, that is $n_i^{-1}$ for the composite in the direction of $\phi$). Write $\mu = nc^2$ with $c \in \mathbb{Q}^\times$. Since $e$ is self-adjoint with $e^2 = n$, it is a Hodge similarity of $T(S)_{\mathbb{Q}}$ of multiplier $n$. So $\psi = (c^{-1}\phi)^{-1} \circ e\colon T(S)_{\mathbb{Q}} \to T(S_r)_{\mathbb{Q}}$ is a Hodge isometry, algebraic by Lemma 23.2, and $e = (c^{-1}\phi) \circ \psi$ is a composite of algebraic correspondences, hence algebraic on $S \times S$. ∎

The case $r = 1$ with $S \dashrightarrow S_1$ the quotient by a symplectic automorphism of prime order $p$ is Varesco's theorem [Var23, Thm. 2.1], which gives $e = \sqrt{p}$ and is used for $p = 2, 3$. What a chain adds is a composite multiplier.

23.3. **Real multiplication by $\mathbb{Q}(\sqrt{6})$.** A symplectic automorphism of finite order on a K3 surface acts trivially on $T(S)$, its quotient has a K3 resolution $Y$ [Nik80b] (for involutions and their quotients see also [BN26]), and $T(Y)_{\mathbb{Q}} \cong T(S)_{\mathbb{Q}}(1/\mathrm{ord})$ by Lemma 23.1. The invariant lattice of a symplectic involution is $U^3 \oplus E_8(-2)$, and a K3 surface has a symplectic involution exactly when $E_8(-2)$ embeds primitively in its Néron–Severi lattice [Mor84, vGS07]. The invariant lattice of a symplectic automorphism of order three is $U \oplus U(3)^2 \oplus A_2^2$, and such an automorphism exists exactly when the Coxeter–Todd lattice $K_{12}(-2)$ embeds primitively in the Néron–Severi lattice [GS07, Thm. 4.1]. Put

$$V_2 = \left(U^3 \oplus E_8(-2)\right)_{\mathbb{Q}}, \qquad V_3 = \left(U \oplus U(3)^2 \oplus A_2^2\right)_{\mathbb{Q}},$$

of ranks 14 and 10 and signatures $(3,11)$ and $(3,7)$; over $\mathbb{Q}$, $U(3) \cong U$ and $E_8(-2) \cong \langle -1\rangle^8 \cong \langle -2\rangle^8$. Varesco [Var23, Props. 2.5 and 2.11] shows that a projective K3 surface whose transcendental space embeds isometrically in $V_2$ (resp. $V_3$) is Hodge isometric to one with a symplectic involution (resp. a symplectic automorphism of order three).

**Lemma 23.4** (Embedding criterion)**.** *Let $W$ and $V$ be rational quadratic spaces with $\operatorname{rk} V - \operatorname{rk} W \geq 3$. Then $W$ embeds isometrically in $V$ if and only if the signature of $V$ dominates that of $W$ in both signs. In that case a complement $W'$ exists with $V \cong W \oplus W'$.*

*Proof.* Necessity is clear. For sufficiency put $k = \operatorname{rk} V - \operatorname{rk} W \geq 3$ and let $(s_+, s_-)$ be the difference of signatures, $s_+ + s_- = k$. A quadratic space $W'$ of rank $k$ and signature $(s_+, s_-)$ with prescribed discriminant $d' = d(V)d(W)$ and prescribed Hasse invariants $\varepsilon'_p$ exists provided the $\varepsilon'_p$ equal 1 for almost all $p$, $\prod_{p\leq\infty} \varepsilon'_p = 1$, and $\varepsilon'_\infty$, $d'$ match the signature [Ser73, Chap. IV, Prop. 7], with no further condition when $k \geq 3$. Take $\varepsilon'_p = \varepsilon_p(V)\varepsilon_p(W)(d(W), d')_p$, the value forced by the formula $\varepsilon_p(W \oplus W') = \varepsilon_p(W)\varepsilon_p(W')(d(W), d(W'))_p$. The product formula for these holds because it holds for $V$, for $W$ and for the Hilbert symbol. Then $W \oplus W'$ and $V$ have the same rank, signature, discriminant and Hasse invariants at every place, so they are isometric by the Hasse–Minkowski theorem [Ser73, Chap. IV, Thm. 9], and $W$ embeds in $V$. ∎

**Theorem 23.5** (Two-dimensional families with algebraic real multiplication by $\mathbb{Q}(\sqrt{6})$)**.** *Let $S$ be a projective K3 surface with $\mathbb{Q}(\sqrt{6}) \subseteq E$ whose transcendental space $T = T(S)_{\mathbb{Q}}$ embeds isometrically in $V_3$.*

(i) *$\sqrt{6} \in E$ is algebraic on $S \times S$.*

(ii) *The Hodge conjecture, and* (F3′)*, hold for $S \times S$ and for the Hilbert square $S^{[2]}$.*

(iii) *Such surfaces exist in two-dimensional families, whose very general member has $\operatorname{rk} T = 8$ and $E = \mathbb{Q}(\sqrt{6})$ exactly. One such family is the one whose transcendental space is $E^4$ with the trace form of $q_E = \langle -2, -2, -2+\sqrt{6}, -1+\sqrt{6}\rangle$ over $E = \mathbb{Q}(\sqrt{6})$.*

*Proof.* (i) By [Var23, Prop. 2.11] $S$ is Hodge isometric to a K3 surface $S'$ with a symplectic automorphism $\sigma$ of order three, and by Lemma 23.2 the isometry is algebraic, so we may replace $S$ by $S'$. Let $Y$ be the minimal resolution of $S'/\sigma$ and $f_1 \colon S' \dashrightarrow Y$ the quotient map, of degree three, so that $T(Y)_{\mathbb{Q}} \cong T(1/3)$, of rank $\operatorname{rk} T \leq 8$ and signature $(2, \operatorname{rk} T - 2)$: the rank is at most 8 because $T$ embeds in $V_3$ of rank 10 and signature $(3,7)$ with a complement that must contain the remaining positive direction, so the complement has rank at least two. Now $T(Y)_{\mathbb{Q}}$ has rank at most 8 and signature dominated by $(3,11)$, and $14 - 8 \geq 3$, so by Lemma 23.4 it embeds in $V_2$. By [Var23, Prop. 2.5] $Y$ is Hodge isometric to a K3 surface $Y'$ with a symplectic involution $\iota$; let $Z$ be the resolution of $Y'/\iota$ and $f_2 \colon Y' \dashrightarrow Z$ the quotient, of degree two. The chain $S' \dashrightarrow Y \cong Y' \dashrightarrow Z$ has multiplier $3 \cdot 1 \cdot 2 = 6$, and Theorem 23.3 applies to $e = \sqrt{6}$ (Fig. 62).

TABLE 6. Rational invariants used in Theorem 23.5(iii). Hasse invariants are $+1$ at every prime not shown.

| space | rank | signature | $d$ | $\varepsilon_2$ | $\varepsilon_3$ | $\varepsilon_5$ |
|---|---|---|---|---|---|---|
| $T = \mathrm{Tr}_{E/\mathbb{Q}}\, q_E$ | 8 | $(2,6)$ | 10 | $+1$ | $-1$ | $+1$ |
| $\langle 1,-10\rangle$ | 2 | $(1,1)$ | $-10$ | $+1$ | $+1$ | $+1$ |
| $T \oplus \langle 1,-10\rangle$ | 10 | $(3,7)$ | $-1$ | $+1$ | $-1$ | $+1$ |
| $V_3$ | 10 | $(3,7)$ | $-1$ | $+1$ | $-1$ | $+1$ |
| $V_2$ | 14 | $(3,11)$ | $-1$ | $-1$ | $+1$ | $+1$ |

(ii) By the first paragraph of this section it remains to make $E$ algebraic. $E$ contains $\mathbb{Q}(\sqrt{6})$, and either $E = \mathbb{Q}(\sqrt{6})$ or $E$ is a CM field: a totally real $E$ has $\dim_E T \geq 3$, because the complex line $H^{2,0}$ lies in one eigenspace $T_\sigma \otimes \mathbb{C}$ and $\overline{H^{2,0}}$ in the same one, so that $T_\sigma \otimes \mathbb{R}$ has signature $(2,*)$ and $\dim_E T \geq 3$ follows from $q$ being negative definite on the other eigenspaces together with $T \otimes \mathbb{R}$ having exactly two positive directions; and $[E:\mathbb{Q}] \cdot \dim_E T = \mathrm{rk}\, T \leq 8$ then forces $[E:\mathbb{Q}] \leq 2$. If $E = \mathbb{Q}(\sqrt{6})$, (i) suffices. If $E$ is CM, it is spanned by its norm-one elements, Hodge isometries, algebraic by Buskin's theorem. So every Hodge class on $S \times S$ is algebraic, and (F3′) follows. For $S^{[2]}$ the Chow motive is a direct summand of twists of $h(S \times S)$, $h(S)$ and $\mathbf{1}$ [dCM02], so every Hodge class on $S^{[2]}$ is the image under an algebraic correspondence of a Hodge class on $S \times S$ or $S$, and both statements follow.

(iii) Let $E = \mathbb{Q}(\sqrt{6})$ with real embeddings $\sigma_1$, $\sigma_2$ ($\sqrt{6} \mapsto \pm\sqrt{6}$), and let $q_E = \langle a_1, a_2, a_3, a_4\rangle$ with $a_1, a_2$ negative at both places and $a_3, a_4$ positive at $\sigma_1$, negative at $\sigma_2$. The $\mathbb{Q}$-space $T = E^4$ with the form $\mathrm{Tr}_{E/\mathbb{Q}} \circ q_E$ has rank 8 and signature $(2,2)+(0,4) = (2,6)$, and multiplication by $\sqrt{6}$ is self-adjoint of square 6. For $q_E = \langle -2, -2, -2+\sqrt{6}, -1+\sqrt{6}\rangle$ one computes $d(T) = 10$, $\varepsilon_2 = 1$, $\varepsilon_3 = -1$, $\varepsilon_5 = 1$ and $\varepsilon_p = 1$ otherwise, while $V_3$ has $d = -1$, $\varepsilon_2 = 1$, $\varepsilon_3 = -1$, $\varepsilon_5 = 1$; and $T \oplus \langle 1,-10\rangle$ has the invariants of $V_3$ at every place, so $V_3 \cong T \oplus \langle 1,-10\rangle$ by Hasse–Minkowski, with a complement containing a positive vector (Table 6). Embed $V_3$ in $\Lambda_\mathbb{Q}$ as the invariant lattice of an order-three symplectic action, with orthogonal complement $K_{12}(-2)$, and let $N \subseteq \Lambda$ be the primitive sublattice with $N_\mathbb{Q} = T^\perp$; it contains $K_{12}(-2)$ primitively and a positive class. The Hodge structures of K3 type on $T$ for which $\sqrt{6}$ is a Hodge endomorphism are the periods $\omega \in \mathbb{P}(T_{\sigma_1} \otimes \mathbb{C})$ with $q(\omega) = 0$, $q(\omega, \bar\omega) > 0$, where $T_{\sigma_1}$ is the $\sigma_1$-eigenspace, of signature $(2,2)$; this is a domain of dimension two [Var23, Prop. 2.3]. For each such $\omega$ the surjectivity of the period map gives a K3 surface $S_\omega$ with $T(S_\omega)_\mathbb{Q} \subseteq T$, projective because $N$ contains a positive class, carrying an order-three symplectic automorphism because $K_{12}(-2) \subseteq N$ primitively. For $\omega$ outside a countable union of proper closed analytic subsets, $T(S_\omega)_\mathbb{Q} = T$ and $E = \mathbb{Q}(\sqrt{6})$, by the argument of Theorem 22.9, which needs only $\dim_E T \geq 3$. ∎

In Varesco's families $E$ is $\mathbb{Q}(\sqrt{2})$ or $\mathbb{Q}(\sqrt{3})$ [Var23, Thms. 2.9, 2.15]; van Geemen and Schütt obtain $\sqrt{2}$, $\sqrt{3}$, $\sqrt{7}$ from rational self-maps of elliptic K3 surfaces [vGS25, §§6–7]. The lattice condition of Theorem 23.5 is a genuine restriction: among the 165 sign-admissible diagonal forms $q_E$ with small coefficients, 98 embed in $V_3$ and 67 do not (a count made with the programs of the archive described in the statement on data availability).

23.4. **Isogenies of elliptic surfaces.** Varesco's quotients and the chains of Theorem 23.3 produce the square roots of the orders of symplectic automorphisms, 2 and 3, and their products. A different source of similarities is a rational self-map of a K3 surface, and the self-maps that exist in families come from isogenies of elliptic fibrations: van Geemen and

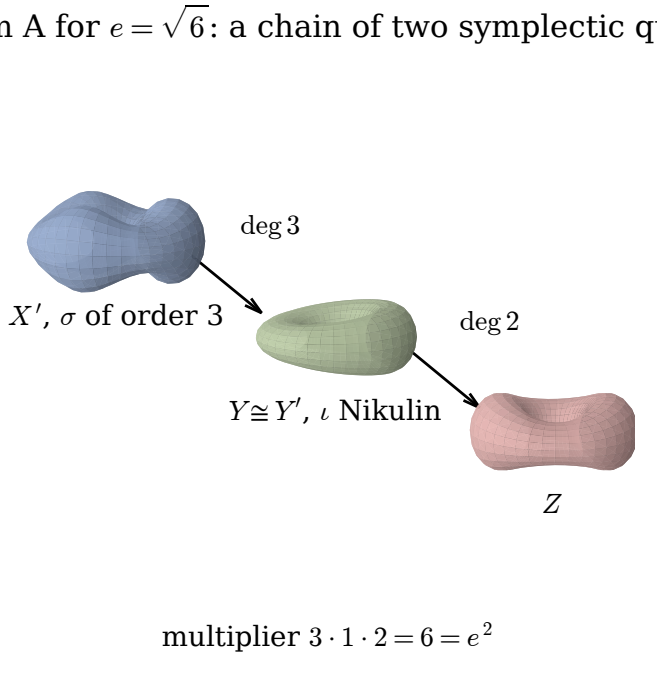



FIGURE 62. The chain of Theorem 23.5: the quotient of $S'$ by a symplectic automorphism of order three, a Hodge isometry (algebraic by Buskin), and the quotient of $Y'$ by a Nikulin involution. The multipliers compose to 6.

Schütt obtain $\sqrt{2}$ and $\sqrt{3}$ from isogenies of degree 2 and 3 in this way, and for degrees 5 and 7 they give one surface each [vGS25, §§6–7]. We show that the degrees 5 and 7 come in one-dimensional families, that in the families the isogeny is a complex multiplication by $\mathbb{Q}(\sqrt{-5})$ or $\mathbb{Q}(\sqrt{-7})$ and not a real one, the sign of its square being decided by how the twisting function behaves under the Fricke involution, and that the families satisfy the Hodge conjecture on all their powers.

Let $N \in \{5, 7\}$ and write $N^k$ for 125 when $N = 5$ and 49 when $N = 7$. The modular curve $X_0(N)$ is rational, with Fricke's coordinate $t$ in which the involution $(E, C) \mapsto (E/C, E[N]/C)$ is $w(t) = N^k/t$ [vGS25, §7.6].[58] The universal curve with a cyclic subgroup $C_t$ of order $N$ over $\mathbb{P}^1_t$ is

$$\mathcal{E}_t\colon\ y^2 = x^3 - 27c_4(t)\,x - 54c_6(t), \tag{77}$$

[58]The curve $X_0(N)$ parametrises elliptic curves with a cyclic subgroup of order $N$; it has genus zero exactly for $N \le 10$ and $N \in \{12, 13, 16, 18, 25\}$, and for $N = 5$ and $N = 7$ a coordinate is the Hauptmodul $t = \big(\eta(\tau)/\eta(N\tau)\big)^k$ with $k = 12/(N-1)$, so $k = 3$ for $N = 5$ and $k = 2$ for $N = 7$; this is why $N^k$ is 125 in one case and 49 in the other. The Fricke involution sends $\tau$ to $-1/N\tau$, which exchanges the two cusps 0 and $\infty$ of $X_0(N)$ and, on the moduli interpretation, replaces $(E, C)$ by the quotient $E/C$ with the image of $E[N]$ as its distinguished subgroup. Applying it twice gives $E/E[N] \cong E$, so it is an involution, and in the coordinate $t$ it reads $t \mapsto N^k/t$ because $\eta(-1/\tau) = \sqrt{-i\tau}\,\eta(\tau)$ turns the quotient of eta functions into its reciprocal up to the constant $N^{k/2}$, squared to $N^k$ by the exponent.

with, for $N = 7$,

$$c_4 = (t^2 + 13t + 49)(t^2 + 5t + 1),$$
$$c_6 = (t^2 + 13t + 49)(t^4 + 14t^3 + 63t^2 + 70t - 7),$$

and for $N = 5$,

$$c_4 = (t^2 + 10t + 5)(t^2 + 22t + 125),$$
$$c_6 = (t^2 + 4t - 1)(t^2 + 22t + 125)^2.$$

The discriminant is $1728\,t\,(t^2 + 13t + 49)^2$ for $N = 7$ and $1728\,t\,(t^2 + 22t + 125)^3$ for $N = 5$, of degree 5 and 7. So the relatively minimal model $R_N$ of (77) over $\mathbb{P}^1_t$ is a rational elliptic surface with fibres of type $\mathrm{I}_1$ at $t = 0$, $\mathrm{I}_N$ at $t = \infty$, and at the two roots $e_1, e_2$ of the quadratic factor, where $c_4$ vanishes, of type II for $N = 7$ and III for $N = 5$; the Euler numbers add up to $1 + N + 2 \cdot 2 = 12$ and $1 + N + 2 \cdot 3 = 12$. The roots satisfy $e_1 e_2 = N^k$, so $w$ exchanges them and exchanges 0 with $\infty$.

**Lemma 23.6** (The Fricke isogeny over the function field)**.** *$\mathcal{E}_t/C_t \cong \mathcal{E}_{w(t)}$ over $\mathbb{C}(t)$.*

*Proof.* The two curves have the same $j$-invariant, because $w$ is the Fricke involution of $X_0(N)$. Two elliptic curves over $\mathbb{C}(t)$ with the same non-constant $j$-invariant differ by a quadratic twist $\chi \in \mathbb{C}(t)^\times/\mathbb{C}(t)^{\times 2}$, and a twist ramified at a point of $\mathbb{P}^1$ changes the Kodaira type of the fibre there, exchanging $\mathrm{I}_n$ with $\mathrm{I}_n^*$, II with IV$^*$ and III with III$^*$. So it suffices to show that $\mathcal{E}_t/C_t$ has the same fibre types as $\mathcal{E}_{w(t)}$, which are $\mathrm{I}_N$ at 0, $\mathrm{I}_1$ at $\infty$ and II or III at $e_1, e_2$: then $\chi$ is unramified on $\mathbb{P}^1$, hence constant, hence trivial over $\mathbb{C}$.

At $t = 0$ the fibre of $\mathcal{E}$ is $\mathrm{I}_1$, the Tate curve $\mathbb{G}_m/q^{\mathbb{Z}}$ over $\mathbb{C}((t))$, whose only cyclic subgroup of order $N$ rational over $\mathbb{C}((t))$ is $\mu_N$; the quotient $\mathbb{G}_m/q^{N\mathbb{Z}}$ is $\mathrm{I}_N$. At $e_i$ the local monodromy $M \in \mathrm{SL}_2(\mathbb{Z})$ of $\mathcal{E}$ has order 6 or 4 and irreducible characteristic polynomial; the isogeny is given on the period lattices by a matrix $A \in M_2(\mathbb{Z})$ with $\det A = N$, and the monodromy of the quotient is $M' = AMA^{-1}$. The $\mathrm{SL}_2(\mathbb{Z})$-classes with this characteristic polynomial are those of $M$ and $M^{-1}$, which are the types II and II$^*$, or III and III$^*$. If $M' = gM^{-1}g^{-1}$ with $g \in \mathrm{SL}_2(\mathbb{Z})$, then $B = g^{-1}A$ conjugates $M$ to $M^{-1}$. All rational matrices doing so form the coset $B_0\,\mathbb{Q}(M)^\times$, where $B_0 = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ for $M = \left(\begin{smallmatrix} 0 & -1 \\ 1 & \mathrm{tr}M \end{smallmatrix}\right)$ has determinant $-1$ and the elements of the imaginary quadratic field $\mathbb{Q}(M)$ have positive norm; so $\det B < 0$, against $\det B = N$. Hence the quotient has the type of $M$ at $e_1, e_2$. At $\infty$ the $j$-invariant of the quotient has a simple pole, so the fibre is $\mathrm{I}_1$ or $\mathrm{I}_1^*$, and the Euler numbers of the other fibres add up to 11; since the total is 12 the fibre is $\mathrm{I}_1$. ∎

For $p \neq q$ in $\mathbb{P}^1_t$ let $u_{p,q} = (t - p)(t - q)$, and let $X_{p,q}$ be the elliptic K3 surface whose generic fibre is the quadratic twist $\mathcal{E}_t \otimes u_{p,q}$: the twist multiplies $c_4$ by $u^2$ and $c_6$ by $u^3$, giving Weierstrass coefficients of degrees 8 and 12, and adds two fibres of type $\mathrm{I}_0^*$ at $p$ and $q$ when these are not among $0, \infty, e_1, e_2$, by Tate's algorithm; the same bookkeeping of Weierstrass coefficients, fibre types and their resolutions in higher dimension is carried out in [Bha26b]. For a double cover $\psi\colon \mathbb{P}^1_s \to \mathbb{P}^1_t$ let $X_\psi$ be the relatively minimal model of $\psi^*\mathcal{E}$.

**Lemma 23.7.**

(i) *$X_{p,q}$ has Picard number at least 16: its Néron–Severi lattice contains $U \oplus A_6 \oplus D_4^2$ for $N = 7$ and $U \oplus A_4 \oplus A_1^2 \oplus D_4^2$ for $N = 5$. The same bound holds for $X_\psi$, whose fibres $2\,\mathrm{I}_1 + 2\,\mathrm{I}_N$, together with four fibres of type III when $N = 5$, and the pulled-back Mordell–Weil group of $R_N$, of rank 2, give rank at least 16 by the Shioda–Tate formula.*

(ii) *The fibrewise isogeny by $C_t$ is a dominant rational map of degree $N$*

$$f\colon X_{p,q} \dashrightarrow X_{w(p),w(q)}, \qquad f\colon X_\psi \dashrightarrow X_{w\circ\psi}.$$

(iii) *The deck involution of $\psi$ times $[-1]$ on the fibres is a symplectic involution $\iota$ of $X_\psi$, and the minimal resolution of $X_\psi/\iota$ is $X_{p,q}$ with $\{p,q\}$ the branch locus of $\psi$. Hence $T(X_\psi)_{\mathbb{Q}} \cong T(X_{p,q})_{\mathbb{Q}}(1/2)$ as in Lemma 23.1, and the two surfaces have the same Picard number.*

*Proof.* (i) Shioda–Tate with the fibres listed, and for $X_\psi$ the Mordell–Weil rank $8-6=2$ of $R_N$. (ii) By Lemma 23.6 the quotient of the generic fibre is $\mathcal{E}_{w(t)} \otimes u_{p,q}(t)$, and

$$u_{p,q}(w(t)) = \Big(\frac{N^k}{t} - p\Big)\Big(\frac{N^k}{t} - q\Big) = \frac{pq}{t^2}\, u_{w(p),w(q)}(t), \tag{78}$$

a constant times a square times $u_{w(p),w(q)}$. So $\mathcal{E}_{w(t)} \otimes u_{p,q}(t) \cong \mathcal{E}_{w(t)} \otimes u_{w(p),w(q)}(w(t))$, which is the pull-back along the automorphism $w$ of the base of the generic fibre of $X_{w(p),w(q)}$; pulling back along an automorphism of the base gives an isomorphic surface. For $X_\psi$, $(\psi^*\mathcal{E})/C = \psi^*(\mathcal{E}/C) \cong \psi^* w^* \mathcal{E}$. (iii) The quadratic twist by the function defining the double cover is the quotient of the base change by the deck involution composed with $[-1]$ on the fibres; the involution is symplectic with eight fixed points, the 2-torsion of the two fibres over the ramification points, and the quotient's resolution is the twist. ∎

**Lemma 23.8** (The sign of the Fricke isogeny)**.** *Let $\varphi_t\colon \mathcal{E}_t \to \mathcal{E}_{w(t)}$ be an isogeny over $\mathbb{C}(t)$ with kernel $C_t$, which exists by Lemma 23.6 and is unique up to sign, and let $\varphi_t^*(dx/y) = \lambda(t)\,dx/y$ in the models (77).*

(i) *$\varphi_{w(t)} \circ \varphi_t = [-N]$ on $\mathcal{E}_t$, and $\lambda(t)\lambda(w(t)) = -N$.*

(ii) *Let $u \in \mathbb{C}(t)$ satisfy $u(w(t)) = c(t)^2 u(t)$ with $c \in \mathbb{C}(t)$, let $X$ be the elliptic K3 surface with generic fibre $\mathcal{E}_t \otimes u$, and let $f\colon X \dashrightarrow X$ be the fibrewise isogeny by $C_t$ followed by the isomorphism $(x,y) \mapsto (c^2x, c^3y)$ from $\mathcal{E}_{w(t)} \otimes u(t)$ to $\mathcal{E}_{w(t)} \otimes u(w(t))$, the fibre of $X$ over $w(t)$. Then $f^*$ acts on $H^{2,0}(X)$ by a constant $\Lambda$ with*

$$\Lambda^2 = -\frac{N}{c(t)\,c(w(t))}.$$

*For $u = (t-p)(t-N^k/p)$ one has $c(t)c(w(t)) = 1$ and $\Lambda^2 = -N$; for $u = t^2 - N^k$ one has $c(t)c(w(t)) = -1$ and $\Lambda^2 = N$.*

*Proof.* (i) The composite is an endomorphism of $\mathcal{E}_t$ over $\mathbb{C}(t)$ of degree $N^2$. Since $j$ is not constant, $\mathrm{End}_{\mathbb{C}(t)}(\mathcal{E}_t) = \mathbb{Z}$, so the composite is $[N]$ or $[-N]$, and changing $\varphi$ to $-\varphi$ does not change it. We read off the sign on the upper half plane $\mathfrak{H}$. Let $\tau \mapsto t(\tau)$ be the quotient map $\mathfrak{H} \to X_0(N)$. By the moduli interpretation of $X_0(N)$ the pair $(\mathcal{E}_{t(\tau)}, C_{t(\tau)})$ is $(\mathbb{C}/(\mathbb{Z}\tau+\mathbb{Z}), \langle 1/N\rangle)$, analytically in $\tau$, and the Fricke involution lifts to $\tau \mapsto w(\tau) = -1/N\tau$. The map $f_\tau(z) = z/\tau$ carries $\mathbb{Z}\tau + \mathbb{Z}$ into $\mathbb{Z}w(\tau) + \mathbb{Z} = \mathbb{Z}/N\tau + \mathbb{Z}$, so it is an isogeny $\mathbb{C}/(\mathbb{Z}\tau+\mathbb{Z}) \to \mathbb{C}/(\mathbb{Z}w(\tau)+\mathbb{Z})$, and its kernel $\tau(\mathbb{Z}w(\tau)+\mathbb{Z})/(\mathbb{Z}\tau+\mathbb{Z}) = (\mathbb{Z}/N+\mathbb{Z}\tau)/(\mathbb{Z}\tau+\mathbb{Z})$ is $\langle 1/N\rangle$. Its composite with the map of the same form at $w(\tau)$ is $f_{w(\tau)}(f_\tau(z)) = z/\tau w(\tau) = -Nz$. On the set $\mathfrak{H}'$ of $\tau$ with $j(\tau) \notin \{0, 1728\}$ the automorphisms of $\mathcal{E}_\tau$ are $\pm 1$, so $\varphi_{t(\tau)} = \varepsilon(\tau) f_\tau$ with $\varepsilon(\tau) \in \{\pm 1\}$; $\varepsilon$ is continuous because $\varphi$ is, hence constant on $\mathfrak{H}'$, which is $\mathfrak{H}$ minus a discrete set and connected. So $\varphi_{w(t)} \circ \varphi_t = \varepsilon^2 f_{w(\tau)} \circ f_\tau = [-N]$ on $\mathfrak{H}'$, hence identically. Pulling back $dx/y$ along the composite gives $\lambda(t)\lambda(w(t))\,dx/y$ on one side and $-N\,dx/y$ on the other.

(ii) Over $\mathbb{C}(t)(\sqrt{u})$ the twist by $u$ is the same change of coordinates $(x,y) \mapsto (ux, u^{3/2}y)$ on $\mathcal{E}_t$ and on $\mathcal{E}_{w(t)}$, which multiplies $dx/y$ by $u^{-1/2}$ on both, so the twisted isogeny pulls

$dx/y$ back to $\lambda(t)\,dx/y$. The isomorphism $(x,y)\mapsto(c^2x,c^3y)$ pulls $dx/y$ back to $c(t)^{-1}dx/y$, and the holomorphic two-form of $X$, which is $dx/y\wedge dt$ on the Weierstrass model, restricts to the fibre over $w(t)$ as $dx/y\wedge d(w(t)) = w'(t)\,dx/y\wedge dt$. Hence $f^*(dx/y\wedge dt) = \Lambda(t)\,dx/y\wedge dt$ with $\Lambda(t)=w'(t)\lambda(t)/c(t)$. The pull-back of a holomorphic two-form along a dominant rational map of K3 surfaces is holomorphic off the finitely many points of indeterminacy, hence everywhere, so $f^*$ preserves the line $H^{2,0}(X)$ and $\Lambda$ is a constant. Then $\Lambda^2=\Lambda(t)\Lambda(w(t))=w'(t)w'(w(t))\,\lambda(t)\lambda(w(t))/c(t)c(w(t))$, where $w'(t)w'(w(t))=(w\circ w)'(t)=1$ and $\lambda(t)\lambda(w(t))=-N$ by (i). For $u=(t-p)(t-N^k/p)$ equation (78) with $q=N^k/p$ gives $u(w(t))=(N^k/t^2)\,u(t)$, so $c(t)=N^{k/2}/t$ for a fixed square root $N^{k/2}$, and $c(t)c(w(t))=N^k/t\,w(t)=1$. For $u=t^2-N^k$, $u(w(t))=N^{2k}/t^2-N^k=-(N^k/t^2)\,u(t)$, so $c(t)=iN^{k/2}/t$ and $c(t)c(w(t))=-1$. ∎

The programs of the archive described in the statement on data availability compute $\lambda$ itself: the kernel polynomial of $\varphi_t$ is the factor of degree $(N-1)/2$ of the $N$-th division polynomial over $\mathbb{Z}[t,x]$, Vélu's formulas give the quotient curve, and the isomorphism to (77) at $w(t)$ has $\lambda(t)^2=-t^2/25$ for $N=5$ and $-t^2/7$ for $N=7$, so that $\lambda(t)^2\lambda(w(t))^2=N^2$ and $\Lambda^2=-N$ on the pair twists directly; the program `code/fricke_sign.py` carries this out, and the Lean file checks the polynomial identity it reduces to.

**Theorem 23.9** (Fricke families)**.** *Let $N\in\{5,7\}$ and $K=\mathbb{Q}(\sqrt{-N})$. Let $\mathfrak{A}_N$ be the family of surfaces $X_p=X_{p,N^k/p}$ for $p\notin\{0,\infty,e_1,e_2,\pm\sqrt{N^k}\}$, and $\mathfrak{B}_N$ the family of surfaces $X_\psi$ with $\psi$ branched over a pair $\{b,N^k/b\}$.*

(i) *Every member $X$ has a dominant rational self-map $f$ of degree $N$ with $f^*\in E=\mathrm{End}_{\mathrm{Hdg}}\,T(X)_{\mathbb{Q}}$, $f^*f_*=N$, $(f^*)^2=-N$ and $f_*=-f^*$. So $K\subseteq E$, with $\sqrt{-N}=\pm f^*$ algebraic on $X\times X$, and $E$ is a CM field.*

(ii) *The families are not isotrivial, and every member has $\mathrm{rk}\,T(X)\in\{2,4,6\}$. At all but countably many members $E=K$ and the Hodge group of $T(X)$ is the unitary group of the $K$-hermitian form $h$ on $T(X)$ with $\mathrm{Tr}_{K/\mathbb{Q}}\,h=q_X$; at the others $T(X)$ is of CM type.*

(iii) *For every member $X$ and every $n\geq 1$ the Hodge conjecture holds for $X^n$ and for the Hilbert scheme $X^{[n]}$, and* (F3′) *holds there.*

Van Geemen and Schütt state that the Fricke parametrisations give one-dimensional families with complex multiplication by $\mathbb{Q}(\sqrt{-5})$ and $\mathbb{Q}(\sqrt{-7})$, and give the details for the two surfaces at the fixed points of $w$ only [vGS25, §7.6]; (i) and (ii) make their statement explicit, and (iii) is what the complex multiplication gives.

*Proof.* (i) $w$ exchanges $p$ and $N^k/p$, so Lemma 23.7(ii) gives $f\colon X_p\dashrightarrow X_p$, the map of Lemma 23.8(ii) for $u=(t-p)(t-N^k/p)$; for $\mathfrak{B}_N$, $w\circ\psi$ has the same branch locus as $\psi$, so $w\circ\psi=\psi\circ\sigma$ for a Möbius transformation $\sigma$ and $X_{w\circ\psi}\cong X_\psi$. The correspondence of a dominant rational map preserves NS and its transpose does too, so it preserves $T=\mathrm{NS}^\perp$ and acts there as a morphism of Hodge structures, nonzero since $f^*\omega=\Lambda\omega$ with $\Lambda\neq 0$; on a resolution $S$ of the graph with $p_1\colon S\to X$ birational and $p_2$ of degree $N$, both $p_1^*T$ and $p_2^*T$ are irreducible sub-Hodge structures of $H^2(S)$ containing its unique $(2,0)$-line, so they coincide, and $f_*f^*=p_{2*}p_1^*p_{1*}p_2^*=N$ on $T$. The field $E$ acts faithfully on the line $H^{2,0}(X)$, and $f^*$ acts there by $\Lambda$ with $\Lambda^2=-N$ by Lemma 23.8(ii); so $(f^*)^2=-N$ in $E$, and $f_*=N(f^*)^{-1}=-f^*$. For $X_\psi$ the generic fibre is $\mathcal{E}_{\psi(s)}$ over $\mathbb{C}(s)$, $f$ is the fibrewise isogeny over $\sigma$, and $\sigma$ is an involution: in the coordinate $s$ with $s^2=(t-b)/(t-N^k/b)$ one computes $(w(t)-b)/(w(t)-N^k/b)=b^2/N^ks^2$, so $\sigma(s)=\kappa/s$ with $\kappa^2=b^2/N^k$; the argument of

Lemma 23.8(ii) with $c = 1$ and $\sigma$ in place of $w$ gives $\Lambda^2 = \lambda(t)\lambda(w(t))\,\sigma'(s)\sigma'(\sigma(s)) = -N$. The adjoint of $f^*$ for $q_X$ is $f_* = -f^*$, so the involution of $E$ induced by $q_X$ is not the identity; by Zarhin's theorem $E$ is totally real or a CM field [Zar83, Thm. 1.5.1], so it is a CM field, and $K = \mathbb{Q} + \mathbb{Q}f^* \subseteq E$. The class of the graph of $f$ is algebraic, its action on $H^2(X)$ is $f^*$ on $T$ and an endomorphism of $\mathrm{NS}(X)_\mathbb{Q}$, which is a sum of algebraic classes of $\mathrm{NS} \otimes \mathrm{NS}$; so $f^*$, extended by zero on NS, is an algebraic class on $X \times X$.

(ii) The elliptic fibration of $X_p$ has labelled singular fibres $\mathrm{I}_1$ at 0, $\mathrm{I}_N$ at $\infty$, II or III at $e_1, e_2$ and $\mathrm{I}_0^*$ at $p$ and $N^k/p$. An isomorphism of elliptic fibrations induces a Möbius transformation preserving this configuration; the transformations preserving $\{0, \infty\}$ and $\{e_1, e_2\}$ are $t \mapsto \lambda t$ and $t \mapsto \lambda/t$ with $\lambda e_1 \in \{e_1, e_2\}$, and since $e_1 + e_2 \neq 0$ and $e_1^2 \neq e_2^2$ they are the identity and $w$, which preserve $\{p, N^k/p\}$. So $X_p$ and $X_{p'}$ have isomorphic fibrations only when $p' \in \{p, N^k/p\}$. A K3 surface has finitely many elliptic fibrations up to automorphisms [Ste85], so the map $p \mapsto [X_p]$ to the moduli space has finite fibres and the family is not isotrivial; by local Torelli its period map has uncountable image. By (i) $T(X)_\mathbb{Q}$ is a $K$-vector space, so its rank is even, and it is at most 6 by Lemma 23.7(i). It is also an $E$-vector space, so $[E : \mathbb{Q}]$ divides $\operatorname{rk} T(X)$; if $E \neq K$ then $[E : \mathbb{Q}] \in \{4, 6\}$, $\dim_E T(X) = 1$ and $T(X)$ is of CM type. A Hodge structure of CM type on a sublattice of $M^\perp$ is determined by its endomorphism field, a subfield of the endomorphism algebra of the sublattice, and by an eigenline of that field, so the periods of such members form a countable subset of the period domain of $M^\perp$; the period map of the family is a non-constant holomorphic map from a connected curve, its fibres over a countable set are countable, so all but countably many members have $E = K$. There the Hodge group of $T(X)$ is the unitary group of the $K$-hermitian form $h$ with $\mathrm{Tr}_{K/\mathbb{Q}}\, h = q_X$ by Zarhin's theorem for CM fields [Zar83, Thm. 2.3.1]. The same holds for $\mathfrak{B}_N$ by Lemma 23.7(iii).

(iii) By (i) $E$ is a CM field containing $K$. If $E = K$ then $E = \mathbb{Q} + \mathbb{Q}f^*$ is algebraic. Otherwise every element of $E$ of norm one to its maximal real subfield is a Hodge isometry of $T$, extended by the identity on NS to a Hodge isometry of $H^2(X, \mathbb{Q})$, algebraic by Buskin's theorem [Bus19]. The algebraic classes form a $\mathbb{Q}$-subalgebra $E' \subseteq E$ containing these elements; for $x \in E$ with $\bar{x} \neq x$ and $t_1 \neq t_2$ in $\mathbb{Q}$ it contains $(t_i + x)/(t_i + \bar{x})$, hence their quotient $1 + (t_2 - t_1)/(t_1 + \bar{x})$, hence $\bar{x}$; so $E' = E$. In both cases $\mathrm{End}_{\mathrm{Hdg}}\, H^2(X) = \mathrm{End}(\mathrm{NS}) \oplus E$ is algebraic, and $\mathrm{Hdg}^2(X \times X) = \mathrm{End}_{\mathrm{Hdg}}\, H^2(X) \oplus H^0 \otimes H^4 \oplus H^4 \otimes H^0$ since $H^1 = H^3 = 0$, so the Hodge conjecture holds for $X \times X$. For a K3 surface whose endomorphism field is a CM field the Hodge classes in the tensor algebra of $T$ are generated by those of $T \otimes T$, and the Hodge conjecture holds for every power [Var25, Thm. 2.2, Cor. 2.3], [RM08, Prop. 5.2, Thm. 5.4]. It passes to the Hilbert schemes $X^{[n]}$, whose motives are direct summands of motives of powers of $X$ [dCM02], and (F3′) on a variety follows from the Hodge conjecture on it. ∎

The surface of [vGS25, §7.6] with real multiplication by $\sqrt{7}$ is the twist at $\{7, -7\}$, the two fixed points of $w$, and the surface of [vGS25, Prop. 7.7] is the twist at the fixed points $\pm 5\sqrt{5}$ for $N = 5$, which has Picard number 18 and complex multiplication as well: there $u = t^2 - N^k$, $c(t)c(w(t)) = -1$ and $\Lambda^2 = N$ by Lemma 23.8(ii), so $f^*$ is the real multiplication by $\sqrt{N}$. The families $\mathfrak{A}_N$ are the twists at the pairs exchanged by $w$, where $c(t)c(w(t)) = 1$ and $\Lambda^2 = -N$; the two signs are the two values of the constant by which the twisting function changes under $w$ once the square factor $N^k/t^2$ is removed. The chains of Theorem 23.3 apply to every member, and a member whose field $E$ contained $\sqrt{2}$, $\sqrt{3}$ or $\sqrt{6}$ would have that element algebraic as well; by Theorem 23.9(ii) such a member is of CM type, and we know none.

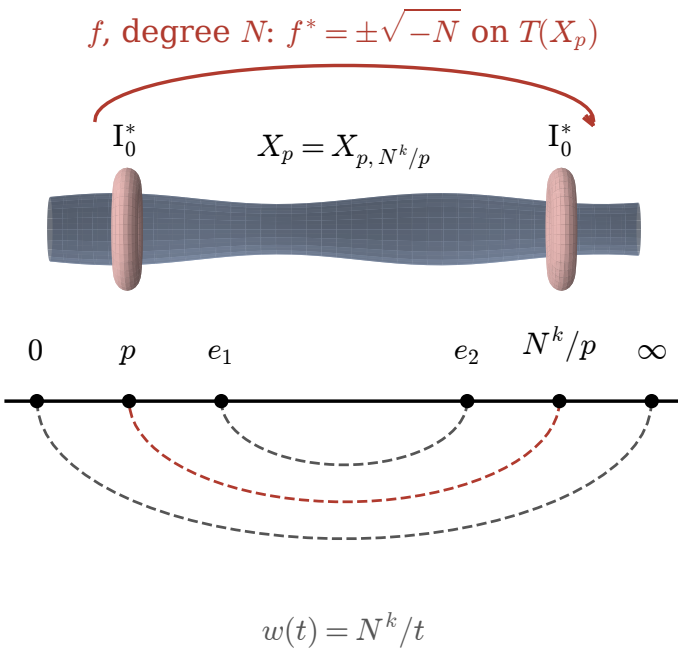



FIGURE 63. The family $\mathfrak{A}_N$ of Theorem 23.9. Over the Fricke coordinate $t$ of $X_0(N)$ the involution $w(t) = N^k/t$ exchanges the cusps $0$, $\infty$ and the elliptic points $e_1$, $e_2$; the twist $X_p$ at a pair $\{p, N^k/p\}$ exchanged by $w$ is carried to itself by the fibrewise isogeny $f$ of degree $N$, and $f^*$ is the complex multiplication by $\sqrt{-N}$ on the transcendental lattice.

*Remark* 23.10 (The Picard number). Theorem 23.9 does not say which of the three ranks the very general member has. The archive decides it by counting points. The surface $X_1$, the twist at $\{1, N^k\}$, has a Weierstrass model over $\mathbb{Z}$ with good reduction at $\ell = 11$ for $N = 7$ and at $\ell = 7$ for $N = 5$, primes split in $K$; the program `code/fricke_picard.py` counts its points over $\mathbb{F}_{\ell^i}$, $i \le 3$, fibre by fibre, subtracts the trace of Frobenius on the trivial lattice, and completes the characteristic polynomial of Frobenius on the rank-six complement of the trivial lattice by the functional equation, whose sign is fixed by the nonzero middle coefficient. The polynomials are $x^6 - 726x^3 + 11^6$ for $N = 7$ and $x^6 + 6x^5 + 56x^4 + 392x^3 + 2744x^2 + 14406x + 7^6$ for $N = 5$; in the variable $x/\ell$ neither has a root of unity among its roots, each is irreducible over $\mathbb{Q}$ and splits over $K$ into a cubic and its conjugate, as the action of $f^*$ requires. By the Tate conjecture for K3 surfaces over finite fields, used in the same way in [vGS25, §7], $\rho(X_1) = 16$, so the very general member of $\mathfrak{A}_N$ and of $\mathfrak{B}_N$ has $\rho = 16$, $\operatorname{rk} T = 6$, $E = K$ with $\dim_K T = 3$, and its Hodge group is the unitary group of a hermitian form of signature $(1, 2)$ on $K^3$. At the primes inert in $K$, $\ell = 5$ for $N = 7$ and $\ell = 11$ for $N = 5$, the odd traces vanish, as they must when Frobenius exchanges the two eigenspaces of $f^*$. No statement of this section rests on this computation.

23.5. **The reach of similarities.** Every correspondence used so far acts on transcendental spaces as a Hodge similarity. The next proposition says that this is not an accident of the examples but a property of every correspondence whose components are small, and it measures what a correspondence must look like to act otherwise.

**Proposition 23.11** (Structure of the transcendental action)**.** *Let $X$, $X'$ be projective K3 surfaces, $\Gamma \in \mathrm{CH}^2(X \times X')_{\mathbb{Q}}$, and $\gamma_T \colon T(X)_{\mathbb{Q}} \to T(X')_{\mathbb{Q}}$ the map induced on transcendental spaces by composing the action of $\Gamma$ with the orthogonal projections.*

(i) *$\gamma_T$ is zero or an isomorphism of Hodge structures.*

(ii) *If $\gamma_T \neq 0$ then $a = \gamma_T^{\dagger}\gamma_T$ lies in the maximal totally real subfield of $E = \mathrm{End}_{\mathrm{Hdg}}\, T(X)_{\mathbb{Q}}$ and is totally positive, and $\gamma_T$ is a similarity exactly when $a \in \mathbb{Q}$.*

(iii) *Let $\Gamma$ be irreducible, $S$ a resolution of its normalisation with projections $p_1, p_2$ of degrees $d_1, d_2$ to $X$, $X'$. If $p_1^* T(X)_{\mathbb{Q}} = p_2^* T(X')_{\mathbb{Q}}$ inside $H^2(S, \mathbb{Q})$, then $\gamma_T$ is a similarity of multiplier $d_1 d_2$. This holds whenever $p_g(S) \leq 1$.*

(iv) *If $\Gamma$ is irreducible and $\gamma_T$ is not a similarity, then $p_1^* T(X)_{\mathbb{Q}} \cap p_2^* T(X')_{\mathbb{Q}} = 0$, so $p_g(S) \geq 2$ and the transcendental rank of $S$ is at least twice that of $X$.*

(v) *The $\mathbb{Q}$-algebra generated by the actions on $T(X)_{\mathbb{Q}}$ of all correspondences from $X$ to itself whose irreducible components either fail to dominate a factor or have $p_g \leq 1$ is the multiquadratic subfield $E_2 \subseteq E$ generated by the square roots of the products $d_1 d_2$ that occur. When $[E : \mathbb{Q}]$ is odd it is $\mathbb{Q}$.*

*Proof.* (i) $T(X)_{\mathbb{Q}}$ is an irreducible Hodge structure [Zar83], and the image of $\gamma_T$ is a sub-Hodge structure of $H^2(X')$; if nonzero it contains the $(2,0)$-part, hence $T(X')_{\mathbb{Q}}$, and $\gamma_T$ is injective by irreducibility. (ii) $a$ is a Hodge endomorphism of $T(X)$, so $a \in E$, and it is self-adjoint, so it lies in the fixed field of the adjoint involution, the maximal totally real subfield. The form $q'(\gamma x, \gamma y) = q(x, ay)$ is a polarisation of the Hodge structure $T(X)$ up to sign on each real eigenspace of $E$, and comparison with $q$ gives $\sigma(a) > 0$ at every real place. A similarity of multiplier $\mu$ has $a = \mu$. (iii) Both $p_i^* T$ are irreducible sub-Hodge structures of $H^2(S)$ with a $(2,0)$-part; if $p_g(S) \leq 1$ they share the unique $(2,0)$-line, so they meet and coincide. When they coincide, $\gamma_T = p_{2*} p_1^* = d_2 (p_2^*)^{-1} p_1^*$ on $T$, so $p_2^* \gamma x = d_2\, p_1^* x$; since $q_S(p_1^* x, p_1^* y) = d_1 q(x,y)$ and $q_S(p_2^* x', p_2^* y') = d_2 q'(x', y')$, this gives $d_2\, q'(\gamma x, \gamma y) = q_S(p_2^* \gamma x, p_2^* \gamma y) = d_2^2\, q_S(p_1^* x, p_1^* y) = d_1 d_2^2\, q(x,y)$, that is $q'(\gamma x, \gamma y) = d_1 d_2\, q(x,y)$. (iv) If the two subspaces met they would coincide by irreducibility and (iii) would apply; so they are independent, their $(2,0)$-lines are distinct, and their sum has rank $2 \operatorname{rk} T(X)$ inside the transcendental part of $H^2(S)$. (v) A component that does not dominate a factor acts through NS or through $H^0$ and $H^4$, hence by zero on $T$. The others act by similarities by (iii), a composite of similarities returning to $X$ is an element $\Phi \in E$ with $\Phi^{\dagger}\Phi \in \mathbb{Q}$, and for totally real $E$ this is $\Phi^2 \in \mathbb{Q}$; such elements generate $E_2$. For a CM field $E$ the elements with $\Phi^{\dagger}\Phi \in \mathbb{Q}$ are the $\mathbb{Q}^{\times}$-multiples of the norm-one elements, which generate $E$, so the statement is about the totally real case, which is the one that matters for (F3′). An odd-degree field has $E_2 = \mathbb{Q}$. ∎

The correspondences of Lemmas 23.1 and 23.2, the Fourier–Mukai kernels between a K3 surface and its moduli spaces of sheaves, the rational Hodge isometries of hyper-Kähler varieties of $\mathrm{K3}^{[n]}$ type [Mar24], the Fricke isogenies of Theorem 23.9 and all their composites fall under (iii) or (v). A correspondence that realises an element of a cubic field must therefore have a component whose resolution has $p_g \geq 2$ and transcendental rank at least twice that of the surface. Such correspondences exist, and the only ones we know come from finite groups.

**Proposition 23.12** (Hecke correspondences)**.** *Let $Z$ be a smooth projective surface with an action of a finite group $G$, $H \leq G$ a subgroup, and $X$ a smooth projective surface with $p_g(X) = 1$ birational to $Z/H$; write $\pi\colon Z \dashrightarrow X$.*

(i) *For $g \in G$ the correspondence $\Gamma_g = (\pi\times\pi)_*[\text{graph of } g]$ acts on $H^2(X,\mathbb{Q})$ as the Hecke operator of the double coset $HgH$, and these operators span a $\mathbb{Q}$-algebra isomorphic to a quotient of $\operatorname{End}_G \mathbb{Q}[G/H]$ acting by Hodge endomorphisms.*

(ii) *Exactly one simple factor of this algebra acts nontrivially on $H^{2,0}(X)$; it is a field, it preserves $T(X)_\mathbb{Q}$, and it embeds into $E$ by algebraic correspondences. If $H^{2,0}(Z)^G = 0$ this factor is not the one on which $G$ acts trivially.*

(iii) *Every field obtained in this way is abelian over $\mathbb{Q}$. In particular a totally real cubic field $F$ whose Galois closure has group $S_3$ is realised by no correspondence built from a finite group acting on a cover of $X$.*

*Proof.* (i) Let $\tilde{Z} \to Z$ be an equivariant resolution and $\pi\colon \tilde{Z} \to X$ the quotient by $H$ followed by the resolution; then $\pi_*\pi^* = |H|$ and $\pi_* h^* = \pi_*$ for $h \in H$, so $\Gamma_g$ acts as $\pi_* g^* \pi^* = (\pi^*)^{-1} \circ \sum_{h\in HgH/H} h^* \circ \pi^*$ on $H^2(X,\mathbb{Q}) = H^2(\tilde{Z},\mathbb{Q})^H$, the Hecke operator of $HgH$. The Hecke operators of $(G,H)$ span $\operatorname{End}_G \mathbb{Q}[G/H]$, a semisimple algebra, and the action on $H^2(\tilde{Z},\mathbb{Q})^H$ factors through it. (ii) The idempotents of the simple factors are rational combinations of Hecke operators, so they are Hodge endomorphisms of $H^2(X)$; they sum to 1 and act on the line $H^{2,0}(X)$ by 0 or 1, so exactly one of them, $e_V$, acts as 1. A simple factor with nonzero image in the commutative ring $\operatorname{End}_{\mathrm{Hdg}} H^{2,0}(X) = \mathbb{C}$ through a ring homomorphism to a field is commutative, hence a field $\mathcal{H}_V$. For $f \in \mathcal{H}_V$, $f(T(X)_\mathbb{Q})$ is a sub-Hodge structure of the transcendental part of $H^2(X)$; if $f\omega \neq 0$ it contains $\omega$, hence $T(X)_\mathbb{Q}$, and if $f\omega = 0$ it has no $(2,0)$-part and lies in $\mathrm{NS} \cap T = 0$. So $\mathcal{H}_V \to \operatorname{End}_{\mathrm{Hdg}} T(X)_\mathbb{Q} = E$ is a homomorphism from a field, injective. The factor on which $G$ acts trivially fixes $H^{2,0}(\tilde{Z})^G = H^{2,0}(Z/G)$, which is zero by hypothesis, so it is not $\mathcal{H}_V$. (iii) The centres of the simple factors of $\operatorname{End}_G \mathbb{Q}[G/H]$ are the character fields $\mathbb{Q}(\chi_i)$ of the irreducible constituents of $\mathbb{Q}[G/H]$, generated by sums of roots of unity, so each lies in $\mathbb{Q}(\zeta_{|G|})$ and is abelian; a commutative simple factor is its centre. A non-abelian cubic field is not a subfield of an abelian field. ∎

For the dihedral group $D_n$ with $n$ odd, $H = \langle\tau\rangle$ of order two and $Z$ the elliptic surfaces $\tilde{\mathcal{E}}_a$ of [vGS25, §4.7], the algebra $\operatorname{End}_{D_n} \mathbb{Q}[D_n/H]$ is $\mathbb{Q} \times \prod_{d|n,\, d>1} \mathbb{Q}(\zeta_d)^+$, and the factor acting on $\omega$ is $\mathbb{Q}(\zeta_n)^+$; this is the content of [vGS25, §4.8], where the cycle $\Gamma_1 + \Gamma_{-1}$ induces $\zeta_n + \zeta_n^{-1}$. The components of these correspondences are birational to $\tilde{\mathcal{E}}_a$, which has $p_g \geq 2$, as Proposition 23.11(iv) requires.

**Corollary 23.13** (Squares of the dihedral families)**.** *Let $X$ be an elliptic K3 surface of one of the families of* [vGS25, Thm. 1.1]*, with real multiplication by $F = \mathbb{Q}(\zeta_n)^+$ for $n \in \{5,7,9,11\}$, and suppose $E = F$ or $E$ is a CM field; the very general member has $E = F$. Then the Hodge conjecture holds for $X \times X$, and* (F3′) *holds for $X \times X$ and $X^{[2]}$. This covers the cubic fields $\mathbb{Q}(\zeta_7)^+$ and $\mathbb{Q}(\zeta_9)^+$ on families of dimensions* 3 *and* 2*.*

*Proof.* If $E = F$, $E$ is algebraic by Proposition 23.12(ii) and [vGS25, §4.8]; if $E$ is CM, by Buskin's theorem as in Theorem 23.9(iii). The rest is as there. ∎

23.6. **The cyclic cubic fields $\mathbb{Q}(\zeta_7)^+$ and $\mathbb{Q}(\zeta_9)^+$.** The dihedral families reach the cyclic cubic fields only on their own members. For each of the two cubic fields they contain, $F = \mathbb{Q}(\zeta_n)^+$ with $n \in \{7,9\}$, there is a two-dimensional family on a transcendental lattice

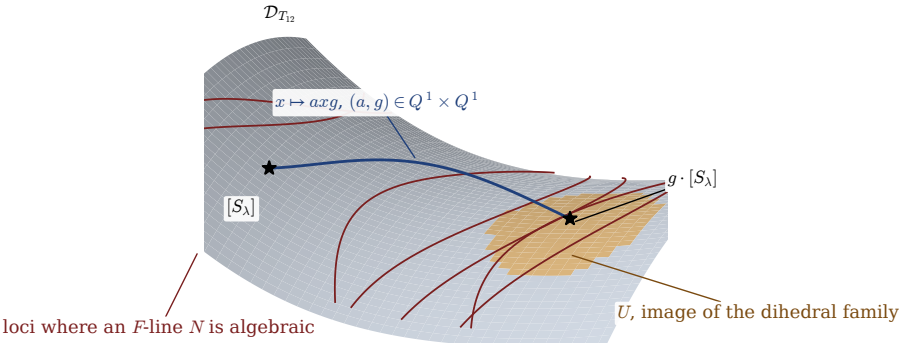



FIGURE 64. The proof of Theorem 23.14. The surface is the two-dimensional period domain of the twelve-dimensional space $T_{12}$ with real multiplication by $F = \mathbb{Q}(\zeta_7)^+$ or $\mathbb{Q}(\zeta_9)^+$; the shaded patch is the image of the two-dimensional dihedral family, on which the multiplication is algebraic. The curves are the loci on which an $F$-line $N$ becomes algebraic, and the surface of Mumford type is a point on one of them. The rational isometries $x \mapsto axg$ of the quaternion algebra are dense in the real group, so one of them moves the point into the patch; Buskin's theorem turns that isometry into a cycle.

of rank twelve: for $n = 9$ the family of [vGS25, §5.6], and for $n = 7$ the Dickson deformation of the one-parameter family of Artebani, Sarti and Taki with Picard lattice $U \oplus E_8$ [AST11, Ex. 6.1], which [vGS25] do not use, their family for $n = 7$ being the three-dimensional one of Picard number four. Two is the largest dimension a family with this real multiplication on a transcendental space of rank twelve can have, and this maximality lets the cycle travel: every K3 surface of Mumford type for $F$ and for the quaternion algebra ramified only at infinity is a Hodge isometric image of a member of the family, and Buskin's theorem carries the cycle along the isometry. The result settles the frontier case of Section 23.8 for these two fields and this algebra.

Write $\theta = \zeta_n + \zeta_n^{-1}$, so that $F = \mathbb{Q}(\theta)$, $K = \mathbb{Q}(\zeta_n) = F(\sqrt{-\delta})$ with $-\delta = (\zeta_n - \zeta_n^{-1})^2 = \theta^2 - 4$, and $\delta = 4 - \theta^2$ is totally positive. For $n = 7$ one has $\theta^3 + \theta^2 - 2\theta - 1 = 0$ and the three real embeddings send $\theta$ to $-1.802$, $-0.445$ and $1.247$; for $n = 9$ one has $\theta^3 - 3\theta + 1 = 0$ and the embeddings send $\theta$ to $-1.879$, $0.347$ and $1.532$. Let $p_n$ be the prime dividing $n$; it is totally ramified in $K$, so it lies under a single place of $F$, and $K/\mathbb{Q}$ is unramified outside $p_n$. The prime 2 has order three in $(\mathbb{Z}/n)^\times/\{\pm 1\} = \mathrm{Gal}(F/\mathbb{Q})$, so it is inert in $F$. Let $D_\infty$ be the quaternion algebra over $F$ ramified exactly at two real places and at no finite place. Put $\varepsilon = \theta$ for $n = 7$ and $\varepsilon = -\theta$ for $n = 9$; then $\varepsilon$ is a unit positive at exactly one real place, and $D_\infty = (-1, \varepsilon)_F$: the symbol is ramified at the two real places where $\varepsilon < 0$, split at every odd finite place because both entries are units there, and split at the one place above 2 by the parity of the number of ramified places. Its corestriction to $\mathbb{Q}$ is unramified everywhere and split at infinity, hence is $M_8(\mathbb{Q})$, so $D_\infty$ is one of Mumford's algebras and the fourfolds of [Mum69] exist for it. For $\mu \in F$ totally positive let $q_\mu$ be the quadratic form $\mathrm{Tr}_{F/\mathbb{Q}}(\mu\,\mathrm{trd}(y^2))$ on the nine-dimensional $\mathbb{Q}$-space $D_\infty^0$ of elements of reduced trace

zero, of signature $(2,7)$; the surfaces $S_\lambda$ of Theorem 21.4 for the fourfolds of $(F, D_\infty)$ have $T(S_\lambda)_\mathbb{Q} \cong (D_\infty^0, q_\lambda)$.

**Theorem 23.14** (Real multiplication by $\mathbb{Q}(\zeta_7)^+$ and $\mathbb{Q}(\zeta_9)^+$ on the surfaces of Mumford type)**.** *Let $n \in \{7,9\}$, $F = \mathbb{Q}(\zeta_n)^+$ and $D_\infty$ as above. Let $S$ be a projective K3 surface such that $T(S)_\mathbb{Q}$ with its intersection form is isometric to $(D_\infty^0, q_\mu)$ for some totally positive $\mu \in F$, by an isometry under which the multiplication by $F$ on $D_\infty^0$ acts on $T(S)_\mathbb{Q}$ by Hodge endomorphisms. Then $E = F$, the real multiplication by $F$ is induced by algebraic cycles on $S \times S$, and the Hodge conjecture and* (F3′) *hold for $S \times S$ and $S^{[2]}$. This applies to every surface $S_\lambda$ of a Mumford fourfold of $(F, D_\infty)$, at every point of its Shimura curve.*

*Proof.* Step 1, the family for $n = 9$. Let $\mathfrak{X} \to B$ be the two-dimensional family of [vGS25, §5.6]: elliptic K3 surfaces with very general Picard lattice $U \oplus A_2^4$,[59] deformations of the surfaces $y^2 = x^3 + bx + c_1 t^9 + c_0$ with a non-symplectic automorphism $\sigma$ of order nine, obtained by replacing $t^9$ by the Dickson polynomial $p_{9,a}(t)$. The transcendental lattice $T_0$ of a very general member has rank twelve and discriminant group $(\mathbb{Z}/3)^4$, is a $\mathbb{Z}[\zeta_9]$-module of rank two through $\sigma^*$, and carries the flat action of $F$ through $\sigma^* + \sigma^{*-1}$, which on every member of the family is induced by the cycle $\Gamma_1 + \Gamma_{-1}$ of [vGS25, §4.8] on a dihedral cover, as in Proposition 23.12. The period map of $B$ has two-dimensional image [vGS25, §5.6].

Step 1, the family for $n = 7$. The surfaces $X_b\colon y^2 = x^3 + bx + c_1 t^7 + c_0$ are the K3 surfaces of [AST11, Ex. 6.1] with $a = 0$ there: $\sigma(x,y,t) = (x, y, \zeta_7 t)$ is a non-symplectic automorphism of order seven, the fibration has a fibre of type $\mathrm{II}^*$ at $t = \infty$ and fourteen fibres of type $\mathrm{I}_1$, and for very general $b$ the Picard lattice is the invariant lattice $U \oplus E_8$ of $\sigma$, with $T_0 = U^2 \oplus E_8(-1)$ [AST11, Table 3]. Replacing $t^7$ by the Dickson polynomial $p_{7,a}(t) = t^7 - 7at^5 + 14a^2t^3 - 7a^3t$ gives the surfaces[60]

$$\mathfrak{X}_a\colon\ \ y^2 = x^3 + bx + c_1 p_{7,a}(t) + c_0, \tag{79}$$

where $p_{7,a}$ is the Dickson polynomial with $p_{7,a}(v + av^{-1}) = v^7 + a^7 v^{-7}$;[61] these are the deformations (4.4) of [vGS25, Prop. 4.6] with $n = 7$, $\alpha = b$ and $\beta(u) = c_1 u + c_0$; by that proposition each $\mathfrak{X}_a$ is a K3 surface on whose transcendental space $F$ acts by Hodge endomorphisms, induced by the cycle $\Gamma_1 + \Gamma_{-1}$ on the $D_7$-cover as before. The polynomial $c_1 p_{7,a} + c_0$ has degree seven, so at $t = \infty$ the coefficients of the Weierstrass equation vanish to orders 8 and 5 and the fibre of type $\mathrm{II}^*$ survives; hence $\mathrm{Pic}(\mathfrak{X}_a)$ contains the unimodular lattice $U \oplus E_8$ of rank ten, and since the Picard number of a very general member of a family is at most that of a very general member of any subfamily, here the locus $a = 0$, it equals $U \oplus E_8(-1)$, with $T_0 = U^2 \oplus E_8(-1)$, unimodular of rank twelve. The parameters $(b, c_1, c_0, a)$ modulo the scalings $(x,y,t) \mapsto (\lambda^2 x, \lambda^3 y, \mu t)$, which send $a$ to $a/\mu^2$, leave two, and the image of the period map is two-dimensional: otherwise it is the closure of the one-dimensional image of the locus $a = 0$, and a very general $\mathfrak{X}_a$ is isomorphic by the Torelli theorem to some $X_{b'}$, so carries a non-symplectic automorphism $\sigma'$ of order seven.

[59] Families of K3 surfaces polarised by a fixed lattice also occur as the fibres of Calabi–Yau threefolds, the quartic and quintic ones being studied in [Bha22]; the chains of this section act on the fibres only, and the Hodge classes of such a total space are not considered here.

[60] The Dickson polynomial $p_{n,a}$ is the unique polynomial of degree $n$ with $p_{n,a}(v + a/v) = v^n + (a/v)^n$; for $a \neq 0$ the map $t \mapsto p_{n,a}(t)$ from $\mathbb{P}^1$ to itself is a cover of degree $n$ with monodromy group the dihedral group $D_n$, totally ramified over $\infty$ [vGS25, Lem. 4.3], since it is the quotient of $v \mapsto v^n + (a/v)^n$, on which $D_n$ acts by $v \mapsto \zeta_n v$ and $v \mapsto a/v$, by the involution. For $a = 0$ it is the cyclic cover $t \mapsto t^n$. The identity $p_{n,a}(\mu t) = \mu^n p_{n,a/\mu^2}(t)$ follows from the defining equation with $v$ replaced by $\mu v$.

[61] Explicitly $p_{7,a}(t) = t^7 - 7at^5 + 14a^2t^3 - 7a^3t$, the case $n = 7$ of $p_{n,a}(t) = \sum_{k \le n/2} \frac{n}{n-k}\binom{n-k}{k}(-a)^k t^{n-2k}$; for $a = 0$ it is $t^7$, so the family $\mathfrak{X}_a$ contains the surfaces $X_b$.

Such a $\sigma'$ has no invariant vector in $T_0 \otimes \mathbb{Q}$, so its invariant lattice is $\mathrm{Pic}(\mathfrak{X}_a)$; it fixes the class of a fibre and the class of the zero section $O$, hence preserves the fibration and $O$, the only effective curve in its class, and it acts on the base with order seven, since an automorphism over the identity of the base that fixes $O$ is a group automorphism of the fibres, of order dividing six. It fixes $t = \infty$, the only reducible fibre, so it acts by $t \mapsto \zeta t + \beta$ with $\zeta$ a primitive seventh root of unity, and since the Weierstrass equation of a fibration with a chosen section is unique up to $(x, y) \mapsto (\lambda^2 x, \lambda^3 y)$, the polynomial $c = c_1 p_{7,a} + c_0$ satisfies $c(\zeta t + \beta) = \lambda^6 c(t)$. The coefficients of $t^7$, $t^6$ and $t^5$ give $\lambda^6 = 1$, $\beta = 0$ and $7ac_1(\zeta^5 - 1) = 0$, so $a = 0$. Thus the direction $a$ is not isotrivial and the image is two-dimensional. If $E \supsetneq F$ then $E$ has degree six or twelve and $\dim_E T_0 \otimes \mathbb{Q} \leq 2$, which allows at most one modulus by the count of [vGS25, §5.4]; so $E = F$ on the very general member.

In both cases the family is maximal: $\dim_F T_{12} = 4$ for $T_{12} = T_0 \otimes \mathbb{Q}$, the period domain of Hodge structures of K3 type on $T_{12}$ with $F$ acting by Hodge endomorphisms has dimension $4 - 2 = 2$, and the period map has two-dimensional image in its quotient by the arithmetic group, so by the open mapping theorem the image contains a nonempty open set $U$ of one of the two components of that domain.

Step 2, the quadratic structure. Since the intersection form $q$ of $T_{12}$ is $\sigma$-invariant there is a unique hermitian form $h$ on the two-dimensional $K$-space $T_{12}$ with $q = \mathrm{Tr}_{K/\mathbb{Q}} h$, and a unique $F$-bilinear form $\Phi = \mathrm{Tr}_{K/F} h$ with $q = \mathrm{Tr}_{F/\mathbb{Q}} \Phi$. Diagonalise $h = \langle h_1, h_2 \rangle$ with $h_i \in F^\times$. For $x = u + v\sqrt{-\delta}$ one has $\mathrm{Tr}_{K/F}(h_1 x \bar{x}) = 2h_1(u^2 + \delta v^2)$, so

$$\Phi \cong 2h_1 \langle 1, \delta, h, h\delta \rangle, \qquad h = h_2/h_1, \tag{80}$$

and $\langle 1, \delta, h, h\delta \rangle$ is the reduced norm form of the quaternion algebra $Q = (-\delta, -h_1 h_2)_F$; thus $(T_{12}, \Phi) \cong (Q, 2h_1 \mathrm{nrd})$ as $F$-quadratic spaces. The signature of $q$ is $(2, 10)$ and splits over the three real places of $F$ as $(2, 2)$ at the place $\sigma_1$ whose eigenspace carries $H^{2,0}$ and $(0, 4)$ at the two others. From (80), signature $(2, 2)$ at $\sigma_1$ forces $h_1 h_2 < 0$ there, and $(0, 4)$ at $\sigma_2, \sigma_3$ forces $h_1, h_2 < 0$ there; so $Q$ is split at $\sigma_1$ and ramified at $\sigma_2$ and $\sigma_3$, and after exchanging $h_1$ and $h_2$ we may take $h_1 > 0$ at $\sigma_1$ and $h_1 < 0$ at the other two places.

Step 3, $Q = D_\infty$. The lattice $T_0$ is unimodular at every prime $p \neq p_n$, and $K/\mathbb{Q}$ is ramified only at $p_n$, so for a place $v$ of $F$ not above $p_n$ the ring $\mathcal{O}_K \otimes \mathbb{Z}_p$ is a product of unramified extensions whose trace forms are perfect, and the hermitian $\mathcal{O}_{K,v}$-lattice $T_0 \otimes \mathbb{Z}_p$ is unimodular for $h$; its determinant $h_1 h_2$ is therefore a unit of $F_v$ times a norm from $K_v$. Units are norms from an unramified extension, so $(-\delta, -h_1 h_2)_v = 1$ at every finite $v \nmid p_n$, including the place above 2. The ramification set of $Q$ has even cardinality, contains $\sigma_2, \sigma_3$ and is otherwise contained in the single place above $p_n$; so that place is unramified as well, and $Q \cong D_\infty$. As a check independent of this argument, the rational invariants of $T_0 \otimes \mathbb{Q}$, computed from the Picard lattice by Witt cancellation in the K3 lattice, and those of $\mathrm{Tr}_{F/\mathbb{Q}}(\varepsilon\, \mathrm{nrd}_{D_\infty})$ agree for both $n$: signature $(2, 10)$, discriminant 1, Hasse invariant $-1$ at 2 and $+1$ at every odd prime, while the trace forms of the algebras over $F$ with the same real ramification and ramification at finite places differ from them at 7 or 43 for $n = 7$ and at 3 or 19 for $n = 9$.

Step 4, the surfaces of Mumford type inside $T_{12}$. For $n_0 \in Q^\times$ the $F$-line $N = F n_0$ has $\Phi(n_0, n_0) = 2h_1 \mathrm{nrd}(n_0)$, and its orthogonal complement is $n_0 Q^0$, with $\Phi(n_0 y, n_0 y') = -h_1 \mathrm{nrd}(n_0)\, \mathrm{trd}(y y')$ for $y, y' \in Q^0$; so $(N^\perp, \Phi) \cong (D^0_\infty, \mu\, \mathrm{trd})$ with $\mu = -h_1 \mathrm{nrd}(n_0)$, and $N$ is negative definite exactly when $\mu$ is totally positive. By the Hasse–Schilling norm theorem [Rei03, Thm. 33.15] the reduced norms of $Q^\times$ are the elements of $F^\times$ positive at $\sigma_2$ and $\sigma_3$, and $-h_1$ is negative at $\sigma_1$ and positive at the other two places, so every totally

positive $\mu$ occurs. Given $S$ as in the statement, choose $n_0$ with $\mu = -h_1\mathrm{nrd}(n_0)$ and an $F$-linear isometry $T(S)_{\mathbb{Q}} \cong n_0^{\perp}$; the period $\omega_0$ of $S$ becomes a point of the period domain of $T_{12}$ at which $Fn_0$ is algebraic and $F$ acts by Hodge endomorphisms, because $\omega_0$ lies in the $\sigma_1$-eigenspace. Replacing $S$ by its complex conjugate, whose period is $\bar{\omega}_0$ and for which the conclusion is equivalent, we may take $\omega_0$ on the component containing $U$.

Step 5, moving the point. The group $\mathrm{SO}(\Phi)(\mathbb{R})$ at the place $\sigma_1$ is $(\mathrm{SL}_2(\mathbb{R})\times\mathrm{SL}_2(\mathbb{R}))/\{\pm 1\}$ acting on $Q \otimes_{F,\sigma_1} \mathbb{R} \cong M_2(\mathbb{R})$ by $x \mapsto axg$, and it acts transitively on each component of the two-dimensional period domain. The group $Q^1 \times Q^1$ of pairs of elements of reduced norm one is simply connected, so by weak approximation [PR94, Thm. 7.8] its rational points are dense in its real points; hence there are $a, g \in Q^1(\mathbb{Q})$ with $x \mapsto axg$ carrying $\omega_0$ into $U$. This map is an $F$-linear isometry of $(T_{12}, \Phi)$, it carries $n_0^{\perp}$ onto $(an_0g)^{\perp}$ with $\mathrm{nrd}(an_0g) = \mathrm{nrd}(n_0)$, and it carries the Hodge structure of $T(S)$ to the Hodge structure on $(an_0g)^{\perp}$ determined by the image point, which is the period of a member $X_b$ of the family. The rational classes of type $(1,1)$ in $T_{12}$ at that period are exactly $F \cdot an_0g$, since a rational class orthogonal to the period decomposes into a component in the line and a component in $(an_0g)^{\perp}$ that is a rational class of type $(1,1)$ in a copy of $T(S)$, which has none. So $T(X_b)_{\mathbb{Q}} = (an_0g)^{\perp}$ with the restricted action of $F$, and $x \mapsto axg$ is a Hodge isometry $\varphi\colon T(S)_{\mathbb{Q}} \to T(X_b)_{\mathbb{Q}}$ that commutes with $F$.

Step 6, the cycle. $H^2(S,\mathbb{Q})$ and $H^2(X_b,\mathbb{Q})$ are both isometric to $\Lambda_{\mathbb{Q}}$, so by Witt cancellation $\mathrm{NS}(S)_{\mathbb{Q}} \cong \mathrm{NS}(X_b)_{\mathbb{Q}}$, and $\varphi$ extends to a rational Hodge isometry $H^2(S,\mathbb{Q}) \to H^2(X_b,\mathbb{Q})$, which is algebraic by Buskin's theorem [Bus19]. The action of $e \in F$ on $T(X_b)$ is induced by the cycle of Step 1, so $m_e = \varphi^{-1} \circ m_e \circ \varphi$ on $T(S)$ is induced by a cycle on $S \times S$. Since $T(S)$ has rank nine and $F \subseteq E$, $E$ is $F$ or a totally real field of degree nine with $\dim_E T = 1$, which [vG08] excludes; so $E = F$ is algebraic, and the Hodge conjecture for $S \times S$ and the rest follow as in Theorem 23.9(iii). ∎

The next proposition says exactly which pairs $(F, D)$ the mechanism reaches and where it stops.

**Proposition 23.15** (What the dihedral mechanism reaches)**.** *Let $n \in \{7, 9\}$ and $F = \mathbb{Q}(\zeta_n)^+$.*

(i) *Let $X$ be a K3 surface with a purely non-symplectic automorphism of order $n \in \{7,9\}$ whose transcendental lattice has rank twelve and is unimodular at every prime other than $p_n$. Then $(T(X)_{\mathbb{Q}}, \Phi)$ is isometric to $(D_\infty, c\,\mathrm{nrd})$ for some $c \in F^\times$, and every surface of Mumford type obtained from $X$ or from a rational isometric image of $X$ as in Step 4 has the algebra $D_\infty$. The Mumford algebras over $F$ ramified at two finite places are reached by no family of rank twelve with such an automorphism.*

(ii) *Let $F = \mathbb{Q}(\zeta_7)^+$ and let $T_{18} = U^2 \oplus A_6 \oplus E_8(-1)$ be the transcendental lattice of the three-dimensional family of* [vGS25, §5.4]*, with its $F$-bilinear form $\Phi_6$ of dimension six. For every Mumford algebra $D$ over $F$ and every totally positive $\lambda$ the space $(D^0, \lambda\,\mathrm{trd})$ embeds $F$-isometrically into $(T_{18} \otimes \mathbb{Q}, \Phi_6)$, so the four-dimensional period domain of this family contains the Shimura curve of the surfaces $S_\lambda$ of every pair $(F, D)$. The image of the family is a threefold in it, and Step 5 would need a rational isometry carrying the period of $S_\lambda$ into that threefold.*

(iii) *A finite group acting on the base $\mathbb{P}^1$ of an elliptic surface is cyclic, dihedral, tetrahedral, octahedral or icosahedral, and the centres of the simple factors of its rational group algebra are the fields $\mathbb{Q}(\zeta_m)$, $\mathbb{Q}(\zeta_m)^+$, $\mathbb{Q}(\sqrt{-3})$, $\mathbb{Q}$ and $\mathbb{Q}(\sqrt{5})$. The only cubic fields among them are $\mathbb{Q}(\zeta_7)^+$ and $\mathbb{Q}(\zeta_9)^+$, so the base change of elliptic surfaces along covers of $\mathbb{P}^1$ reaches no cyclic cubic field of conductor at least 13.*

*Proof.* (i) Steps 2 and 3 of the theorem use only the order of the automorphism, the rank, the signature and the unimodularity away from $p_n$, so they apply to $X$ and give $\Phi \cong 2h_1\mathrm{nrd}_{D_\infty}$. A rational isometry preserves $(T,\Phi)$ up to the $F$-similarity class, which determines the algebra, and Step 4 exhibits the orthogonal complement of every negative definite $F$-line as $(D^0_\infty, \mu\,\mathrm{trd})$; an algebra $D \neq D_\infty$ has $(D^0, \lambda\,\mathrm{trd}) \ncong (D^0_\infty, \mu\,\mathrm{trd})$ for all $\lambda,\mu$, since the even Clifford algebra of the ternary form $\lambda\,\mathrm{trd}$ is $D$.

(ii) $T_{18}\otimes\mathbb{Q}$ is a three-dimensional hermitian $K$-space and $\Phi_6 = 2\langle h_1, h_1\delta, h_2, h_2\delta, h_3, h_3\delta\rangle$ has signature $(2,4)$ at $\sigma_1$ and $(0,6)$ at $\sigma_2,\sigma_3$, by the same count as in Step 2. The form $q_3 = \lambda\,\mathrm{trd}$ on $D^0$ has signature $(2,1)$ at $\sigma_1$, where $D$ is split, and $(0,3)$ at the two ramified places. The Witt class of $\Phi_6 - q_3$ is represented at every real place by a negative definite ternary form and at every finite place by a ternary form, because a class of odd dimension has an anisotropic representative of dimension one or three; these local ternary forms have the global discriminant $d(\Phi_6)d(q_3)$, Hasse invariants $+1$ at almost every place and satisfying the product formula, since they are products of the invariants of $\Phi_6$ and $q_3$ and of Hilbert symbols of their discriminants; so by [OMe00, Thm. 72:1] there is a global ternary form $q_3'$ with these localisations, and $\Phi_6 \cong q_3 \perp q_3'$ by the Hasse–Minkowski theorem. The image $N$ of $q_3'$ is a negative definite $F$-subspace of dimension three with $N^\perp \cong (D^0, \lambda\,\mathrm{trd})$, and the Hodge structures of K3 type on $N^\perp$ with $F$ acting form the Shimura curve of the $S_\lambda$ of $(F,D)$. The family has three moduli [vGS25, §5.4] in a domain of dimension $6-2=4$.

(iii) The list of finite subgroups of $\mathrm{PGL}_2(\mathbb{C})$ is Klein's. The simple factors of $\mathbb{Q}[G]$ have centres the fields $\mathbb{Q}(\chi)$ generated by the values of the irreducible characters: $\mathbb{Q}(\zeta_m)$ for the cyclic group of order $m$, $\mathbb{Q}(\zeta_m)^+$ for the dihedral group of order $2m$ (its two-dimensional characters take the values $\zeta_m^k + \zeta_m^{-k}$), $\mathbb{Q}$ and $\mathbb{Q}(\sqrt{-3})$ for $A_4$, $\mathbb{Q}$ for $S_4$, and $\mathbb{Q}$ and $\mathbb{Q}(\sqrt{5})$ for $A_5$. By Proposition 23.12(iii) the correspondences cut out of the action act through these fields; $\mathbb{Q}(\zeta_m)$ is never cubic, and $\mathbb{Q}(\zeta_m)^+$ is cubic exactly for $m \in \{7,9,14,18\}$, where it is $\mathbb{Q}(\zeta_7)^+$ or $\mathbb{Q}(\zeta_9)^+$. ∎

For the fourfolds $X$ of $(F, D_\infty)$ the theorem leaves (F2) where it was: by Theorem 24.45(v) their exceptional classes are algebraic as soon as one algebraic cycle on $S_\lambda \times X^r$ acts nontrivially on $T(S_\lambda)$, and the real multiplication, now algebraic, supplies no such cycle.

23.7. **Powers of K3 surfaces and Kummer type.** The theorems of this section decide $S\times S$ and $S^{[2]}$. On the higher powers the Hodge classes are described by Varesco [Var25, Thm. 2.16]. Let $E$ be totally real, $m = \dim_E T$, and for each real embedding $\sigma$ of $E$ let $V_\sigma = T\otimes_{E,\sigma}\mathbb{C}$, so that $T_\mathbb{C} = \bigoplus_\sigma V_\sigma$. The line $\bigwedge^m V_\sigma \subset T_\mathbb{C}^{\otimes m}$ is spanned by a class $\det V_\sigma$ of type $(m,m)$, and the span of the $\det V_\sigma$ over all $\sigma$ is defined over $\mathbb{Q}$: it is $\bigwedge^m_E T \subset \bigwedge^m T \subset T^{\otimes m}$, a space of Hodge classes of dimension $[E:\mathbb{Q}]$ in $H^{2m}(S^m)$, the exceptional classes. Varesco's theorem is that every Hodge class in the tensor algebra of $T$ is obtained from the Hodge classes of $T\otimes T$, which are the elements of $E$, and from the exceptional classes by tensor products, permutations of the factors, linear combinations and contractions with $q_S$. Each of these operations carries algebraic classes on powers of $S$ to algebraic classes: a tensor product is an exterior product, a permutation is the pull-back along an automorphism of $S^k$, and the contraction of two factors is the push-forward along the projection $S^k \to S^{k-2}$ after cupping with the pull-back of the class of the diagonal of $S\times S$, since $q_S(a,b) = \int_{S\times S}(a\otimes b)\cup[\Delta_S]$.

**Proposition 23.16** (Hodge classes on the powers)**.** *Let $S$ be a projective K3 surface with $E$ totally real and $m = \dim_E T$, so that $m \geq 3$ as in the proof of Theorem 23.5(ii).*

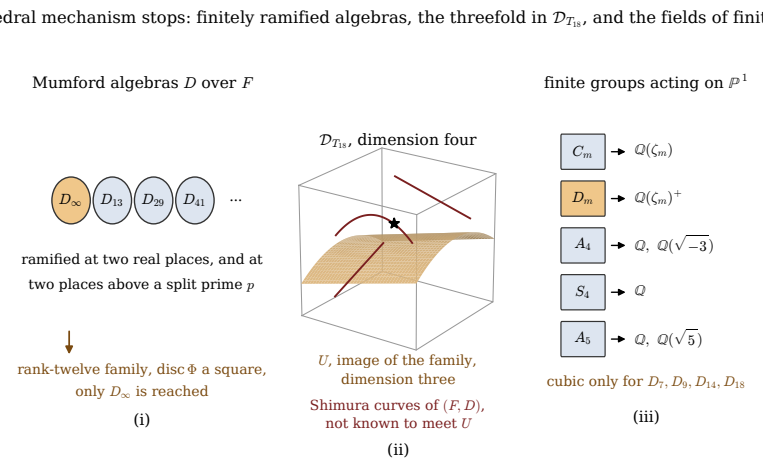



FIGURE 65. Where the dihedral mechanism stops, Proposition 23.15. (i) Over $F = \mathbb{Q}(\zeta_7)^+$ or $\mathbb{Q}(\zeta_9)^+$ the Mumford algebras are $D_\infty$ and the algebras $D_p$ ramified at two real places and at two places above a split prime $p$; a rank-twelve family unimodular away from $p_n$ has square discriminant and carries only $D_\infty$. (ii) The period domain of the rank-eighteen family is four-dimensional, the image of the family is a threefold, and the Shimura curves of the surfaces of Mumford type for the other algebras lie in the domain; whether the threefold meets them is the open transport condition. (iii) The finite subgroups of $\mathrm{PGL}_2$ acting on the base produce only $\mathbb{Q}(\zeta_m)$, $\mathbb{Q}(\zeta_m)^+$, $\mathbb{Q}$, $\mathbb{Q}(\sqrt{-3})$ and $\mathbb{Q}(\sqrt{5})$; a cubic field arises only from the dihedral groups $D_7$, $D_9$, $D_{14}$ and $D_{18}$.

(i) *The Hodge conjecture holds for every power $S^n$ if and only if the elements of $E$ are algebraic on $S \times S$ and the exceptional classes $\bigwedge^m_E T \subset H^{2m}(S^m)$ are algebraic. When $E = \mathbb{Q}$, $m = r = \operatorname{rk} T$ and the exceptional class is the determinant $\det T$, which spans $\bigwedge^r T \subset H^{2r}(S^r)$.*

(ii) *For a surface of Theorems 23.5, 23.14 or Corollary 23.13 with $E$ totally real, $E$ is algebraic, and the Hodge conjecture for all powers of $S$ is equivalent to the algebraicity of $\bigwedge^m_E T$, $m = \operatorname{rk} T/[E : \mathbb{Q}]$: $m = 3$ for the surfaces $S_\lambda$ of Theorem 23.14, with $\operatorname{rk} T = 9$; $m = 4$ for the very general members of the two-dimensional family of Theorem 23.14, with $\operatorname{rk} T = 12$.*

*Proof.* (i) If the Hodge conjecture holds on all powers, both kinds of classes are algebraic, being Hodge classes on $S \times S$ and on $S^m$. Conversely, $H^2(S) = \mathrm{NS}(S)_\mathbb{Q} \oplus T$, and $H^0(S)$, $\mathrm{NS}(S)_\mathbb{Q}$ and $H^4(S)$ are spanned by algebraic classes and are Tate twists of trivial Hodge structures. So a Hodge class of $H^*(S^n) = \bigotimes^n (H^0 \oplus \mathrm{NS}_\mathbb{Q} \oplus T \oplus H^4)$ is a sum of exterior products of algebraic classes with Hodge classes of the tensor powers $T^{\otimes k}$, $k \le n$, pulled back along the projections of $S^n$ onto $k$ of its factors. By Varesco's theorem and the remark preceding the proposition, the Hodge classes of $T^{\otimes k}$ are algebraic as soon as $E$ and the exceptional classes are. When $E = \mathbb{Q}$ there is one embedding, $V_\sigma = T_\mathbb{C}$ and $\bigwedge^m_E T = \bigwedge^r T$.

(ii) The theorems cited prove that $E$ is algebraic; $m = \operatorname{rk} T/[E : \mathbb{Q}]$ since $T$ is an $E$-vector space. The ranks are those of the nine-dimensional space $D^0_\infty$ and of the lattice $T_0$ in Step 1 of the proof of Theorem 23.14. ∎

**Proposition 23.17** (K3 surfaces of Kummer type over $\mathbb{Q}$)**.** *Let $S$ be a projective K3 surface, $r = \operatorname{rk} T$, let $\langle a_1, \dots, a_r\rangle$ be a diagonalisation of $q_S$ on $T$, $d(T) = \prod_i a_i \in \mathbb{Q}^\times/\mathbb{Q}^{\times 2}$ its discriminant and $\varepsilon_p(T) = \prod_{i<j}(a_i, a_j)_p$ its Hasse invariant at the prime $p$. The following are equivalent.*

(a) *$T$ embeds isometrically in $U^3_{\mathbb{Q}}$, the quadratic space $H^2(A, \mathbb{Q})$ of an abelian surface.*

(b) *There is a dominant rational map from $S$ to a Kummer surface.*

(c) *$r \le 3$; or $r = 4$ and $\varepsilon_p(T) = (-1,-1)_p$ at every prime $p$ at which $d(T)$ is a square in $\mathbb{Q}_p$; or $r = 5$ and $\varepsilon_p(T) = (-1,-1)_p$ at every prime $p$.*

*Here $(-1,-1)_p = 1$ for $p$ odd and $(-1,-1)_2 = -1$, so (c) asks for $\varepsilon_p(T) = 1$ at the odd primes concerned and $\varepsilon_2(T) = -1$. If (a)–(c) hold, then $S$ lies in the class $\mathcal{A}$ of Proposition 20.35, (F3′) holds for every power $S^n$, and the Hodge conjecture holds for every $S^n$ and every Hilbert scheme $S^{[n]}$.*

The equivalence of (a) and (b) is Ma's theorem [Ma12, Prop. 3.2], where (a) is also shown equivalent to an embedding of lattices $T(S) \hookrightarrow U^3$, and for a Kummer surface itself the Hodge conjecture on all powers follows from that on the powers of the abelian surface [Var25, §0.2]. What is added here is the form (c) of the condition, which reads it off the invariants of $T$, and the passage from Ma's rational map to the class $\mathcal{A}$.

*Proof.* (a)⇔(c). The space $V = U^3_{\mathbb{Q}} \cong \langle 1,-1\rangle^3$ has signature $(3,3)$, discriminant $-1$ and Hasse invariants $\varepsilon_p(V) = (-1,-1)_p^3 = (-1,-1)_p$. By Witt's theorem $T$ embeds in $V$ if and only if there is a quadratic space $W$ over $\mathbb{Q}$ with $T \oplus W \cong V$, and by the Hasse–Minkowski theorem [Ser73, Ch. IV, Thm. 9] this is a condition on invariants: $W$ must have dimension $6 - r$, signature $(1, 5-r)$, discriminant $d(W) = d(V)d(T)^{-1} = -d(T)$ and, from $\varepsilon_p(T \oplus W) = \varepsilon_p(T)\varepsilon_p(W)(d(T), d(W))_p$ with $(d(T), -d(T))_p = 1$, Hasse invariants $\varepsilon_p(W) = (-1,-1)_p\,\varepsilon_p(T)$. The signature forces $r - 2 \le 3$. A quadratic space over $\mathbb{Q}$ with prescribed dimension $k$, signature, discriminant and Hasse invariants exists if and only if the invariants satisfy the product formula, the real invariants are those of the signature, and at every prime the local existence condition holds [Ser73, Ch. IV, §2.3, Prop. 6, and §3.3, Prop. 7]; the first two conditions hold here because they hold for $T$ and for $V$ and $T \otimes \mathbb{R} \oplus W \otimes \mathbb{R} \cong V \otimes \mathbb{R}$, and the local condition is empty for $k \ge 3$, is $\varepsilon_p(W) = 1$ at every $p$ at which $d(W) = -1$ in $\mathbb{Q}_p^\times/\mathbb{Q}_p^{\times 2}$ for $k = 2$, and is $\varepsilon_p(W) = 1$ at every $p$ for $k = 1$. With $k = 6 - r$, $d(W) = -d(T)$ and $\varepsilon_p(W) = (-1,-1)_p\varepsilon_p(T)$ these are the three cases of (c).

(b) and the class $\mathcal{A}$. Let $f\colon S \dashrightarrow \mathrm{Km}(A)$ be dominant. By Lemma 23.1, $f^* = \Gamma_{f*}^t$ maps $T(\mathrm{Km}(A))_{\mathbb{Q}}$ onto $T$. Let $\beta\colon \tilde{A} \to A$ be the blow-up of the sixteen points of order two and $\pi\colon \tilde{A} \to \mathrm{Km}(A)$ the quotient by the involution; $\Gamma_A = (\beta \times \pi)_*[\tilde{A}]$ is an algebraic cycle on $A \times \mathrm{Km}(A)$ with $\Gamma_{A*} = \pi_*\beta^*$, and $H^2(\mathrm{Km}(A), \mathbb{Q}) = \pi_*\beta^* H^2(A, \mathbb{Q}) \oplus \bigoplus_{i=1}^{16} \mathbb{Q}[E_i]$ with $E_i$ the exceptional curves. Hence $T \subseteq (\Gamma_f^t \circ \Gamma_A)_* H^2(A, \mathbb{Q})$, and $H^2(S) = \mathrm{NS}(S)_{\mathbb{Q}} \oplus T$ is spanned by this image and by divisor classes, while $H^0(S)$ and $H^4(S)$ are spanned by the class of $S$ and the class of a point. So $S \in \mathcal{A}$, the point being the abelian variety of dimension zero, $S^n \in \mathcal{A}$ and (F3′) holds on $S^n$ by Proposition 20.35. For the Hodge conjecture, the proof of Proposition 20.35(i) writes a Hodge class of $S^n$ as $\sum_j \Gamma_{j*} y_j$ with $y_j$ a Hodge class on one of the abelian varieties spanning $H^*(S^n)$, here the powers $A^k$ with $k \le n$. The Hodge conjecture holds for every product of surfaces with $p_g = 1$ and $q = 2$ [RM08], in particular for $A^k$, so every $y_j$, and with it every Hodge class of $S^n$, is algebraic. The Chow motive of $S^{[n]}$ is a direct summand of a sum of Tate twists of the motives $h(S^k)$, $k \le n$ [dCM02], so every Hodge class of $S^{[n]}$ is the image under an algebraic correspondence of a Hodge class of some $S^k$. ∎

*Remark* 23.18 (The powers left open). (i) At $\rho(S) \geq 19$ the condition (c) is empty and $S$ has a Shioda–Inose structure [Mor84]. At $\rho(S) = 17$ or 18 it is a condition on the Hasse invariants, met by $U_{\mathbb{Q}}^2 \oplus \langle -2 \rangle$, by $U_{\mathbb{Q}} \oplus \langle 2, -2, -2 \rangle$ and by $\langle 2, 2, -2, -2 \rangle$, and failed at $p = 2$ and $p = 3$ by $\langle 2, 2, -6, -6 \rangle$, by $\langle 2, 2, -2, -6, -6 \rangle$ and by $\langle 2, 6, -2, -2, -2 \rangle$, the primes of failure coming in pairs by the product formula; all six are transcendental lattices of K3 surfaces of Picard number 18 or 17 by [Nik80] and the surjectivity of the period map. For $r \leq 5$ a totally real $E$ is $\mathbb{Q}$, since $m \geq 3$; so $E$ is $\mathbb{Q}$ or a CM field, and in the CM case the Hodge conjecture holds on all powers of $S$ with or without (c) [RM08]. At $\rho \geq 17$ the surfaces for which the Hodge conjecture on all powers is not decided by Propositions 23.16 and 23.17 are therefore those with $E = \mathbb{Q}$ and $T$ failing (c), and for them the one class left is $\det T \in H^{2r}(S^r)$, $r \in \{4, 5\}$. For $\rho \leq 16$ no surface is of Kummer type.

(ii) The Fricke surfaces have complex multiplication by $K = \mathbb{Q}(\sqrt{-N})$ at every member, and Theorem 23.9(iii) settles the Hodge conjecture on all their powers. Their transcendental spaces $U_{\mathbb{Q}}^2 \oplus \langle -1 \rangle \oplus \langle -N \rangle$ (Proposition 23.19) have discriminant $N$, so they are neither of Kummer type nor in the two families of Picard number 16 for which Varesco proves the Hodge conjecture on all powers from the algebraicity of the Kuga–Satake correspondence [Var25, Thm. 4.3, Exs. 4.4, 4.5], whose transcendental spaces $U_{\mathbb{Q}}^2 \oplus \langle -2 \rangle^2$ and $U_{\mathbb{Q}}^2 \oplus \langle -6 \rangle \oplus \langle -2 \rangle$ have discriminants 1 and 3 modulo squares; they are settled by the complex multiplication instead. For each surface of Proposition 23.16(ii) the exceptional classes are Hodge classes of degree $2m$ on $S^m$ reached by no correspondence of this section, all of which act on $T \otimes T$; whether they are algebraic, and with it whether (F3′) and the Hodge conjecture hold on $S^m$, is open. The one construction known to produce such classes is the Kuga–Satake correspondence, through which Varesco proves [Var25, Thm. 4.3].

**Proposition 23.19** (The transcendental lattice and the Kuga–Satake variety of the Fricke surfaces)**.** *Let $N \in \{5, 7\}$, $K = \mathbb{Q}(\sqrt{-N})$, let $M$ be the trivial lattice $U \oplus A_4 \oplus A_1^2 \oplus D_4^2$ or $U \oplus A_6 \oplus D_4^2$ of the elliptic fibration of the members of $\mathfrak{A}_N$, and let $T = M_{\mathbb{Q}}^{\perp} \subset H^2(X, \mathbb{Q})$, which is $T(X)_{\mathbb{Q}}$ when $\rho(X) = 16$.*

(i) *$T \cong U_{\mathbb{Q}}^2 \oplus \langle -1 \rangle \oplus \langle -N \rangle$ as a quadratic space over $\mathbb{Q}$, so $d(T) = N$ and the centre of the even Clifford algebra $C^+(T)$ is $K$.*

(ii) *The Kuga–Satake variety of $T$ is isogenous to $B^4$ for an abelian fourfold $B$ of $K$-Weil type with discriminant one. If $\rho(X) = 16$ and $\mathrm{End}_{\mathrm{Hdg}}\, T = K$, then $B$ is isogenous to $E_K \times B_3$, where $E_K$ is an elliptic curve with complex multiplication by $K$ and $B_3$ is a simple abelian threefold with $\mathrm{End}(B_3)_{\mathbb{Q}} = K$; in particular $B$ is not simple.*

*Proof.* (i) The root lattices are rationally diagonal: $D_n = \{x \in \mathbb{Z}^n : \sum x_i \in 2\mathbb{Z}\}$ has index two in $\mathbb{Z}^n$, so $D_n \otimes \mathbb{Q} \cong \langle 1 \rangle^n$, and $E_8 \supset D_8$ with index two gives $E_8 \otimes \mathbb{Q} \cong \langle 1 \rangle^8$; $A_n = \{x \in \mathbb{Z}^{n+1} : \sum x_i = 0\}$ is the orthogonal complement of the vector $(1, \dots, 1)$ of norm $n + 1$, so $(A_n \oplus \langle n + 1 \rangle) \otimes \mathbb{Q} \cong \langle 1 \rangle^{n+1}$; and $A_1^2 = \langle 2, 2 \rangle \cong \langle 1, 1 \rangle$ by the vectors $(1, 1)$ and $(1, -1)$ of $\mathbb{Z}^2$. Hence, with all forms taken positive definite, $A_4 \oplus A_1^2 \oplus D_4^2 \oplus \langle 1, 5 \rangle \cong \langle 1 \rangle^5 \oplus \langle 1 \rangle^2 \oplus \langle 1 \rangle^8 \oplus \langle 1 \rangle = \langle 1 \rangle^{16}$ and $A_6 \oplus D_4^2 \oplus \langle 1, 7 \rangle \cong \langle 1 \rangle^7 \oplus \langle 1 \rangle^8 \oplus \langle 1 \rangle = \langle 1 \rangle^{16}$ over $\mathbb{Q}$. The K3 lattice is $\Lambda = U^3 \oplus E_8(-1)^2$, so $\Lambda_{\mathbb{Q}} \cong U_{\mathbb{Q}}^3 \oplus \langle -1 \rangle^{16}$, while $M_{\mathbb{Q}} \oplus U_{\mathbb{Q}}^2 \oplus \langle -1, -N \rangle \cong U_{\mathbb{Q}}^3 \oplus \langle -1 \rangle^{16}$ by the two isometries just written. Thus $M_{\mathbb{Q}} \oplus T \cong \Lambda_{\mathbb{Q}} \cong M_{\mathbb{Q}} \oplus (U_{\mathbb{Q}}^2 \oplus \langle -1, -N \rangle)$, and Witt's cancellation theorem gives $T \cong U_{\mathbb{Q}}^2 \oplus \langle -1 \rangle \oplus \langle -N \rangle$. Its discriminant is $(-1)^2(-1)(-N) = N$, and the centre of $C^+(T)$ is $\mathbb{Q}(\sqrt{(-1)^3 d(T)}) = \mathbb{Q}(\sqrt{-N})$. The program `code/fricke_lattice.py` recomputes these isometries by the invariants of Hasse–Minkowski.

(ii) By Lombardo's theorem [Lom01, Thms. 6.2, 6.4], in the form of [Var25, Thm. 4.1], the Kuga–Satake variety of a polarised Hodge structure of K3 type on $U_{\mathbb{Q}}^2 \oplus \langle a\rangle \oplus \langle b\rangle$ with $a, b < 0$ is isogenous to $B^4$ for an abelian fourfold $B$ of $\mathbb{Q}(\sqrt{-ab})$-Weil type with discriminant one; here $(a, b) = (-1, -N)$, $\mathbb{Q}(\sqrt{-ab}) = K$ and $C^+(T) \cong M_4(K)$. Suppose $\rho(X) = 16$ and $\mathrm{End}_{\mathrm{Hdg}}\, T = K$. By Zarhin's theorem [Zar83, Thm. 2.3.1] the Hodge group of $T$ is $G = \mathrm{U}(W, h)$, the unitary group of the hermitian form $h$ on the three-dimensional $K$-space $W = T$ with $\mathrm{Tr}_{K/\mathbb{Q}}\, h = q$. Over $\mathbb{C}$ the two eigenspaces $W_\sigma$, $W_{\bar\sigma}$ of $K$ on $T_{\mathbb{C}}$ have dimension three and are isotropic for $q$, since $q(ax, y) = q(x, \bar a y)$, so $T_{\mathbb{C}} = W_\sigma \oplus W_\sigma^*$ and $G_{\mathbb{C}} = \mathrm{GL}(W_\sigma)$ acting on $W_\sigma^*$ by the dual representation. The Kuga–Satake Hodge structure on $H^1(B, \mathbb{Q})$ is the lift of that of $T$ along $\mathrm{Spin}(T) \to \mathrm{SO}(T)$ [vG08, §5], so the Hodge group of $B$ is the preimage $\widetilde{G}$ of $G$ in $\mathrm{Spin}(T)$, and $H^1(B, \mathbb{C}) = S^+ \oplus S^-$ is the sum of the two half-spin representations of $\mathrm{Spin}(T_{\mathbb{C}}) \cong \mathrm{SL}_4$, on which the two eigenspaces of the centre $K$ of $C^+(T)$ are $S^+$ and $S^-$. For the isotropic decomposition $T_{\mathbb{C}} = W_\sigma \oplus W_\sigma^*$ the spin representation is $\bigwedge^\bullet W_\sigma \otimes (\det W_\sigma)^{-1/2}$, with $S^+$ the even and $S^-$ the odd part. Restricted to $\widetilde{G}_{\mathbb{C}}$, $S^+ = (\bigwedge^0 W_\sigma \oplus \bigwedge^2 W_\sigma) \otimes \det^{-1/2}$ is the sum of the character $\chi = \det^{-1/2}$ and of the irreducible representation $W_\sigma^* \otimes \det^{1/2}$ of dimension three, and $S^- = (W_\sigma \oplus \bigwedge^3 W_\sigma) \otimes \det^{-1/2}$ is the sum of $W_\sigma \otimes \det^{-1/2}$ and of $\chi^{-1}$. The four summands are pairwise non-isomorphic, the two characters being distinct and the two three-dimensional representations having different weights, so they are the isotypic components of $H^1(B, \mathbb{C})$ under the connected group $\widetilde{G}_{\mathbb{C}}$. An automorphism of $\mathbb{C}$ over $\mathbb{Q}$ carries an isotypic component to one of the same dimension, since the representation is defined over $\mathbb{Q}$; so $U_{\mathbb{C}} = \chi \oplus \chi^{-1}$ and $V_{\mathbb{C}} = S^+ \oplus S^-$ minus it are stable under all of them and are the complexifications of $\mathbb{Q}$-subspaces $U$, $V$ with $H^1(B, \mathbb{Q}) = U \oplus V$. Both are stable under $\widetilde{G}$, hence sub-Hodge structures, and under $K$, which acts by scalars on $S^+$ and on $S^-$. By Poincaré reducibility $B$ is isogenous to $E_K \times B_3$ with $H^1(E_K, \mathbb{Q}) = U$ and $H^1(B_3, \mathbb{Q}) = V$. The algebra $\mathrm{End}_{\mathrm{Hdg}}(U) \otimes \mathbb{C} = \mathrm{End}(\chi) \oplus \mathrm{End}(\chi^{-1})$ has dimension two and contains $K$, so $\mathrm{End}(E_K)_{\mathbb{Q}} = K$; likewise $\mathrm{End}_{\mathrm{Hdg}}(V) \otimes \mathbb{C}$ has dimension two, so $\mathrm{End}(B_3)_{\mathbb{Q}} = K$, a field, and $B_3$ is simple; and $\mathrm{Hom}(E_K, B_3) = 0$ since $U$ and $V$ share no isotypic component. ∎

*Remark* 23.20 (Real multiplication at the fixed points). Theorem 23.9(iii) decides every power of every Fricke surface, so the Kuga–Satake variety of Proposition 23.19(ii) is not needed for them; the proposition records what the splitting of $T$ into the two eigenspaces of $K$ does to the Kuga–Satake construction. For a very general Hodge structure of K3 type on $U_{\mathbb{Q}}^2 \oplus \langle -1\rangle \oplus \langle -N\rangle$, with endomorphism field $\mathbb{Q}$, the fourfold $B$ is simple with a definite quaternion algebra as endomorphism algebra [vG08, Prop. 5.7]; the Fricke families lie in the locus of the four-dimensional period domain where the endomorphism field is $K$, and there $B$ splits. Real multiplication by $\sqrt{N}$ appears on the two twists at the fixed points of $w$, the surfaces of [vGS25, §7.6, Prop. 7.7], where $\Lambda^2 = N$ by Lemma 23.8(ii). A family of K3 surfaces of transcendental rank six with real multiplication by $\sqrt{5}$ or $\sqrt{7}$ on which the multiplication is algebraic is not known to us: the fixed points of $w$ are two, so the fixed-point twists are two surfaces, and Remark 23.22 says why the chains of Theorem 23.3 do not supply one.

23.8. **Where this stops.** The frontier case of (F3′) in Theorem 21.4 and Remark 21.5 is a K3 surface $S_\lambda$ of Picard number 13 with $T(S_\lambda)_{\mathbb{Q}} \cong T$, the nine-dimensional representation of a Mumford fourfold, and $E = F$ a totally real cubic field, which may have Galois group $S_3$. Theorem 23.14 settles it for the two cyclic fields $F = \mathbb{Q}(\zeta_7)^+$ and $\mathbb{Q}(\zeta_9)^+$ with the algebra $D_\infty$; the frontier is now $S_\lambda$ for every other pair $(F, D)$: for these two fields with

the algebras ramified at two finite places, for the cyclic cubic fields of conductor at least 13, and for every field that is not cyclic. There, by Proposition 23.11(v), no composite of similarities reaches $F$,[62] so Theorem 23.3, the Fricke isogenies and every correspondence of the kinds listed are blind to it; by Proposition 23.11(iv) a correspondence that does reach it has an irreducible component whose resolution has $p_g \geq 2$, transcendental rank at least 18 and $b_2 \geq 31$; and by Proposition 23.12(iii) that component is not the graph of an element of a finite group acting on a cover of $S_\lambda$ when $F$ is not cyclic. When $F$ is cyclic the dihedral families are the natural source, as van Geemen observed for $\mathbb{Q}(\zeta_9)^+$ [vG08, Rem. 7.9]; Theorem 23.14 is the statement that for the two fields of conductor 7 and 9 and the algebra $D_\infty$ the source actually reaches $S_\lambda$, and Proposition 23.15 says that it reaches nothing else. The only correspondences of this size that we know to act on a transcendental space of K3 type are the dihedral ones, which give abelian fields, and the Kuga–Satake correspondence into the cohomology of an abelian variety of dimension $2^7$, whose algebraicity is open [Del72, vG00]; in Section 24 we replace its abelian variety by a sixfold and show what the replacement does and does not give. Within quadratic fields the method reaches $\sqrt{d}$ for $d \in \{2, 3, 6\}$ in families, and the Fricke isogenies give complex multiplication by $\mathbb{Q}(\sqrt{-5})$ and $\mathbb{Q}(\sqrt{-7})$ on one-dimensional families, where Theorem 23.9 settles the Hodge conjecture on every power; real multiplication by $\sqrt{5}$ and $\sqrt{7}$ is algebraic on the two surfaces of [vGS25] and on no family known to us (Remark 23.20); for $\mathbb{Q}(\sqrt{10})$, $\mathbb{Q}(\sqrt{14})$ and the other composite discriminants no family is known on which the chain of Theorem 23.3 can be started. On the powers beyond the square, Proposition 23.16 reduces the Hodge conjecture for every surface of this section with totally real endomorphism field to its exceptional classes, Proposition 23.17 settles it with (F3′) for the surfaces of Kummer type over $\mathbb{Q}$, Theorem 23.9 settles it for every member of the Fricke families, and Remark 23.18 records what is left.

**Proposition 23.21** (Chains of symplectic quotients reach exactly $\sqrt{2}$, $\sqrt{3}$, $\sqrt{6}$)**.** *Let $S$ be a projective K3 surface whose endomorphism field $E$ is totally real and different from $\mathbb{Q}$, and let $S = S_0, \dots, S_r$ be a chain as in Theorem 23.3 in which every rational map is the quotient by a finite group of symplectic automorphisms. Then no degree $n_i$ is divisible by 5 or by 7, every $n_i$ divides a power of 24, and the multiplier $\prod n_i^{\pm 1}$ lies in $\{1, 2, 3, 6\}$ modulo squares. In particular no such chain makes $\sqrt{10}$, $\sqrt{14}$, $\sqrt{15}$ or $\sqrt{21}$ algebraic, and the surfaces with real multiplication by these fields are not reached by Theorem 23.3 in this form.*

*Proof.* Along the chain the transcendental spaces are isomorphic up to scaling of the form, by Lemma 23.1 and Lemma 23.2, so every $S_i$ has the same transcendental rank $t$ and the same endomorphism field $E$ as $S$. By Zarhin [Zar83, Thms. 2.2.1 and 2.3.1] the Hodge group of $T(S)_\mathbb{Q}$ is the group of $E$-linear special isometries of the $E$-quadratic space $W$ with $T(S)_\mathbb{Q} = \mathrm{Tr}_{E/\mathbb{Q}} W$, of dimension $m = t/[E : \mathbb{Q}]$ over $E$. If $m \leq 2$ this group is commutative

[62] Reaching $F$ means producing, by algebraic correspondences from $S_\lambda$ to itself, Hodge endomorphisms of $T(S_\lambda)_\mathbb{Q}$ that generate the cubic field $E = F$, which is what (F3′) asks of this surface, since the class of a generator of $F$ in $H^4(S_\lambda \times S_\lambda)$ is a Hodge class reached from no abelian variety as long as it is not algebraic. The correspondences built in this section, and every correspondence whose components have geometric genus at most one, act on $T(S_\lambda)$ as similarities, and a composite of similarities is a similarity; by Proposition 23.11(ii) a similarity lies in $\mathbb{Q}$ when $[E : \mathbb{Q}]$ is odd, so it generates nothing. The correspondences coming from finite group actions act through abelian fields by Proposition 23.12(iii), and a cubic field whose Galois closure has group $S_3$ is not abelian. The two propositions together say what a correspondence realising $F$ must look like: an irreducible component with $p_g \geq 2$ and transcendental rank at least eighteen, not arising from a finite group; no construction in the literature produces one.

(trivial for $m = 1$, a torus of dimension $[E : \mathbb{Q}]$ for $m = 2$). Then every $g \in \mathrm{Hdg}(\mathbb{Q})$ commutes with the Hodge group and is an endomorphism of the Hodge structure, so $g \in E$. The elements of the totally real field $E$ are self-adjoint for the form on $T(S)_{\mathbb{Q}}$ and $g$ is an isometry, so $g = g^* = g^{-1}$ and $g^2 = 1$. As $\mathbb{Q}$-points are Zariski dense in a connected linear algebraic group over $\mathbb{Q}$, the Hodge group would be trivial, which is impossible because $h^{2,0}(S) = 1$. So $m \geq 3$ and $t \geq 3[E : \mathbb{Q}] \geq 6$.

Let $G$ be a finite group of symplectic automorphisms of a K3 surface $X$ and $\Omega_G \subset \mathrm{NS}(X)$ its coinvariant lattice, negative definite because $G$ fixes the transcendental lattice and an ample class. If 5 divides $|G|$, then $G$ contains an element of order five, whose coinvariant lattice has rank 16 [Nik80b, § 5] and lies in $\Omega_G$, so $\operatorname{rk}\Omega_G \geq 16$; if 7 divides $|G|$ the same argument with the rank 18 of an element of order seven gives $\operatorname{rk}\Omega_G \geq 18$. The transcendental lattice $T(X)$ lies in $\Omega_G^{\perp} \subset H^2(X, \mathbb{Z})$, which has rank $22 - \operatorname{rk}\Omega_G \leq 6$ and signature $(3, 19 - \operatorname{rk}\Omega_G)$, and $T(X)$ has signature $(2, \operatorname{rk}T(X) - 2)$; so $\operatorname{rk}T(X) \leq 5$ when 5 divides $|G|$ and $\operatorname{rk}T(X) \leq 3$ when 7 does. Every symplectic automorphism of finite order has order at most eight [Nik80b], so if some $n_i = |G_i|$ is not of the form $2^a3^b$ then 5 or 7 divides it, the source of that quotient has transcendental rank at most five, and so has $S$; this contradicts $t \geq 6$. Hence every $n_i$ is of the form $2^a3^b$, and the multiplier is $2^a3^b$ modulo squares, that is 1, 2, 3 or 6. A multiplier $n$ makes only $e$ with $e^2 \equiv n$ algebraic, which gives the list. ∎

*Remark* 23.22. The proposition settles the lattice question of whether a chain $S \to Y \leftarrow S'$ through a symplectic involution and a symplectic automorphism of order five, of multiplier $2/5 \equiv 10$, can start on a surface with real multiplication by $\mathbb{Q}(\sqrt{10})$: it cannot, since $S'$ would have transcendental rank at most five and real multiplication by a real quadratic field needs rank at least six. The same bound shows that the surfaces of transcendental rank at most five with a symplectic automorphism of order five or seven have endomorphism field $\mathbb{Q}$ or a CM field; in the CM case the Hodge group is a torus and the chain closes only through the complex multiplication itself. Real multiplication by $\sqrt{10}$, $\sqrt{14}$ or any field outside $\{\mathbb{Q}(\sqrt{2}), \mathbb{Q}(\sqrt{3}), \mathbb{Q}(\sqrt{5}), \mathbb{Q}(\sqrt{6}), \mathbb{Q}(\sqrt{7})\}$, and by $\sqrt{5}$ or $\sqrt{7}$ beyond the two surfaces of [vGS25], therefore needs a correspondence that is not a composite of symplectic quotients, Hodge isometries and fibrewise isogenies, and none is known to us.

## 24. Constructions at (F2)

The first classes (F2) has to reach are the Weil classes of a very general abelian $2n$-fold of Weil type with $n \geq 3$, and the exceptional classes of a Mumford fourfold. We build an object at each.

24.1. **The polarised semiregularity criterion.** Let $A$ be a very general member of the polarised family $\mathcal{D}_{n,n}$ of abelian $2n$-folds of Weil type for an imaginary quadratic field $K$, with polarisation class $\theta$ and Weil plane $W(A)$. Then $\mathrm{Hdg}^*(A) = \mathbb{Q}[\theta] \oplus W(A)$ and $W(A) \cdot \theta = 0$ (Theorem 4.14). A class $\gamma \in H^{\mathrm{ev}}(A)$ has the polarised shape if

$$\gamma = N\omega + \sum_{k=0}^{2n} c_k \theta^k, \qquad N \neq 0,\ 0 \neq \omega \in W(A),\ c_k \in \mathbb{Q}.$$

Its Hankel rank $\rho \in \{0,1,2,3\}$ is the rank of the Hankel matrix of the sequence $(k!\,c_k)_k$ capped at three; it is the number of distinct exponentials $e^{a\theta}$ needed to write the polynomial part when that number is at most two, and 3 otherwise.[63]

**Theorem 24.1** (The polarised criterion, Theorem 17.9 restated)**.** *Let $\mathcal{E}$ be a perfect complex on $A$ with $\operatorname{ch}(\mathcal{E}) = \gamma$ of polarised shape and Hankel rank $\rho$. Then*

$$\dim \operatorname{Ext}^2(\mathcal{E},\mathcal{E}) \;\geq\; r(\gamma) := (4+\rho)\,n^2 - 2n,$$

*and if equality holds the semiregularity map[64] of $\mathcal{E}$ is injective, $\mathcal{E}$ deforms with $A$ along the whole of $\mathcal{D}_{n,n}$, and the Weil classes are algebraic on every member of the family with full Hodge group. If some $c_k \neq 0$ with $1 \leq k \leq 2n-1$, the first-order Hodge locus of $\gamma$ in $H^1(A,T_A)$ is the tangent space of $\mathcal{D}_{n,n}$, of dimension $n^2$. A perfect complex whose Chern character is exactly a multiple of a Weil class does not exist on a very general member, so the polynomial part is forced.*

The number $r(\gamma)$ is the rank of the contraction $HT^2(A) \to H^*(A)$, $\xi \mapsto \xi \lrcorner \gamma$, on the Hochschild cohomology $HT^2 = \bigoplus_{p+q=2} H^q(A, \bigwedge^p T_A)$, of dimension $\binom{4n}{2}$; its kernel is the annihilator $\operatorname{Ann}(\gamma)$, which meets $H^1(A,T_A)$ in the first-order Hodge locus. The semiregularity map $\sigma_{\mathcal{E}}$ of a perfect complex composes with the characteristic map $\mathrm{ev}_{\mathcal{E}} : HT^2 \to \operatorname{Ext}^2(\mathcal{E},\mathcal{E})$, $\xi \mapsto \xi \lrcorner \operatorname{At}(\mathcal{E})$, to the contraction into $\operatorname{ch}(\mathcal{E})$; this is how the lower bound arises, and it is the form in which the criterion is used below. Figure 66 shows $r(\gamma)$; the two values that matter below are $r = 57$ at $n=3$, $\rho=3$ and $r=104$ at $n=4$, $\rho=3$. On the product $E_0^6$ of six copies of the elliptic curve $\mathbb{C}/\mathbb{Z}[i]$ every twisted complex of line bundles has $\dim \operatorname{Ext}^2 \geq 69 > 57$ (Theorem 18.35); the criterion is not met there by any such object.

### 24.2. **The Weil plane of a product.**

**Lemma 24.2.** *Let $A$, $A'$ be of Weil type for the same field $K$, of dimensions $2m$, $2m'$, with the diagonal $K$-action on $A \times A'$. Then $W(A \times A')$ is a two-dimensional rational subspace of the four-dimensional space $W(A) \otimes W(A') \subseteq H^{2n}(A \times A')$, $n = m + m'$, and every rational $w \in W(A \times A')$ is $\sum_{i,j} a_{ij}\, \omega_i \otimes \omega'_j$ with $a_{ij} \in \mathbb{Q}$ for rational bases $(\omega_i)$, $(\omega'_j)$ of $W(A)$, $W(A')$.*

*Proof.* Over $\mathbb{C}$, $H^1(A \times A') = V \oplus V'$ splits into eigenspaces $V_\pm \oplus V'_\pm$ of $K$, and $\bigwedge_K^{2n}$ of the sum is spanned by $\det V_+ \otimes \det V'_+$ and $\det V_- \otimes \det V'_-$; these lie in the span of the four

[63] The Hankel matrix of a sequence $(a_k)_{k\geq 0}$ has entries $a_{i+j}$. If $a_k$ is a sum $\sum_j \lambda_j b_j^k$ of $m$ terms with distinct $b_j$ and nonzero $\lambda_j$, the matrix factors as $V^{\mathsf{T}} \operatorname{diag}(\lambda_j) V$ with $V$ the Vandermonde matrix of the $b_j$, so its rank is $m$; conversely a sequence whose Hankel matrix has finite rank $m$ satisfies a linear recurrence of order $m$ and is a sum of $m$ exponentials with polynomial coefficients. For the polynomial part $\sum_k c_k \theta^k$ of a Chern character, writing it as $\sum_j \lambda_j e^{b_j\theta}$ means $k!\,c_k = \sum_j \lambda_j b_j^k$, so the number of exponentials needed is the Hankel rank of $(k!\,c_k)$, provided the degree $2n$ allows the recurrence to be read off, which it does for $m \leq 2$; the cap at three records that the criterion of Theorem 24.1 does not distinguish further.

[64] For a perfect complex $\mathcal{E}$ on a smooth projective variety the semiregularity map of Buchweitz and Flenner [BF03] is the linear map $\sigma_{\mathcal{E}} \colon \operatorname{Ext}^2(\mathcal{E},\mathcal{E}) \to \prod_q H^{q+1}(X, \Omega_X^{q-1})$ sending $\xi$ to the trace of $\xi \circ \exp\bigl(-\operatorname{At}(\mathcal{E})\bigr)$, where $\operatorname{At}(\mathcal{E}) \in \operatorname{Ext}^1(\mathcal{E}, \mathcal{E} \otimes \Omega^1_X)$ is the Atiyah class; it generalises Bloch's map for a subvariety [Blo72], which is the case of the structure sheaf of the subvariety. Its value is that the obstruction to deforming $\mathcal{E}$ along a first-order deformation of $X$ lies in $\operatorname{Ext}^2(\mathcal{E},\mathcal{E})$ and is sent by $\sigma_{\mathcal{E}}$ to the obstruction to keeping $\operatorname{ch}(\mathcal{E})$ of Hodge type; so when $\sigma_{\mathcal{E}}$ is injective, $\mathcal{E}$ deforms wherever its Chern character stays a Hodge class, and the Chern character of the deformed complex is algebraic there. The dimension count of the theorem is the arithmetic of this statement: the rank of $\sigma_{\mathcal{E}}$ is at least the rank of the contraction into $\operatorname{ch}(\mathcal{E})$, so $\dim \operatorname{Ext}^2$ is at least $r(\gamma)$, and equality forces injectivity.

$r(\gamma) = (4+\rho)n^2 - 2n$: the rank the criterion demands

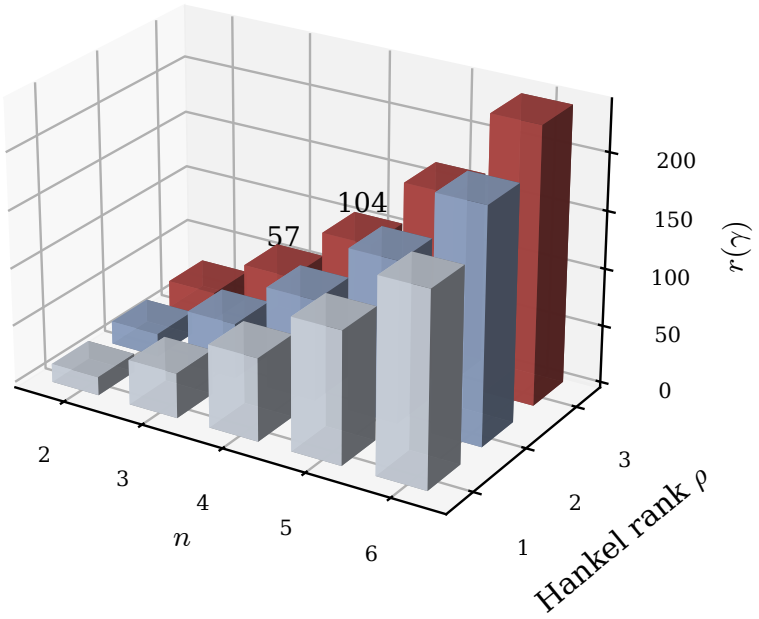



FIGURE 66. The lower bound $r(\gamma) = (4+\rho)n^2 - 2n$ of Theorem 24.1 for $2 \le n \le 6$ and Hankel rank $\rho = 1, 2, 3$. The two labelled bars are the product cases of Theorem 24.3.

products $\omega_\pm \otimes \omega'_\pm$ of the eigenvectors of $W(A) \otimes \mathbb{C}$ and $W(A') \otimes \mathbb{C}$. Both $W(A \times A')$ and $W(A) \otimes W(A')$ are defined over $\mathbb{Q}$, so the inclusion holds over $\mathbb{Q}$. The explicit formula is equation (36): $\omega_1(A \times A') = \omega_1 \otimes \omega'_1 - d\,\omega_2 \otimes \omega'_2$ for $K = \mathbb{Q}(\sqrt{-d})$. ∎

24.3. **An object with the polarised Chern character.**

**Theorem 24.3** (The product object)**.** *Let $A$, $A'$ be very general in their polarised Weil families, with ample classes $\theta_A$, $\theta_{A'}$, $\theta = \theta_A + \theta_{A'}$, and suppose that $W(A)$ and $W(A')$ are algebraic. Then there is a perfect complex $\mathcal{E}$ on $A \times A'$, a finite direct sum with shifts of exterior products*

$$F_i \boxtimes G'_j, \quad F_i \boxtimes L'^k, \quad L^l \boxtimes G'_j, \quad L^l \boxtimes L'^k,$$

*with $F_i$, $G'_j$ coherent sheaves on $A$, $A'$ and $L$, $L'$ the ample bundles, such that*

$$\operatorname{ch}(\mathcal{E}) = N\,w + \sum_{k=0}^{2n} c_k \theta^k, \qquad N \neq 0,\ 0 \neq w \in W(A \times A'),$$

*has the polarised shape with Hankel rank $\rho = 3$, so that $r(\operatorname{ch}\mathcal{E}) = 7n^2 - 2n$, and the first-order Hodge locus of $\operatorname{ch}(\mathcal{E})$ in $H^1(A \times A', T)$ is the tangent space of $\mathcal{D}_{n,n}$.*

The hypotheses hold in the two open cases nearest to the known ones: $A$, $A'$ very general Weil fourfolds of trivial discriminant, where the Weil classes are algebraic by Markman's theorem for discriminant one [Mar23] and the specialisation of algebraic classes, so that $A \times A'$ is an eightfold with $n = 4$; and $A$ such a fourfold and $A' = B$ a Weil surface of

any discriminant $d$, whose Weil plane is of type $(1,1)$ and algebraic by Lefschetz, so that $A\times B$ is a sixfold of discriminant $d$ with $n=3$.

*Proof.* Since $\mathrm{Hdg}^*(A)=\mathbb{Q}[\theta_A]\oplus W(A)$, algebraic classes on $A$ are rational combinations of Chern characters of coherent sheaves, and the Chern character of a coherent sheaf is a Hodge class, the algebraicity of $W(A)$ gives coherent sheaves $F_1$, $F_2$ with $\mathrm{ch}(F_i)=N_i\omega_i+p_i(\theta_A)$, $N_i\neq 0$, for a basis $(\omega_1,\omega_2)$ of $W(A)$; likewise $G'_1$, $G'_2$ on $A'$ with $\mathrm{ch}(G'_j)=N'_j\omega'_j+p'_j(\theta_{A'})$. Fix $0\neq w\in W(A\times A')$ rational, $w=\sum a_{ij}\omega_i\otimes\omega'_j$ by Lemma 24.2, and a common denominator $M$ of the numbers $Ma_{ij}/(N_iN'_j)$, so that these are integers. Negative multiplicities are realised by the shift $[1]$, which changes the sign of ch.

Step 1. $\mathcal{E}_0=\bigoplus_{i,j}(F_i\boxtimes G'_j)^{\oplus Ma_{ij}/(N_iN'_j)}$ has, by multiplicativity,

$$\mathrm{ch}(\mathcal{E}_0)=Mw+\sum_i\omega_i\otimes q'_i(\theta_{A'})+\sum_j q_j(\theta_A)\otimes\omega'_j+s, \tag{81}$$

with $q'_i$, $q_j$ polynomials and $s\in\mathbb{Q}[\theta_A]\otimes\mathbb{Q}[\theta_{A'}]$.

Step 2. The classes $e^{k\theta_{A'}}=\mathrm{ch}(L'^k)$, $k=0,\dots,2m'$, form a basis of $\mathbb{Q}[\theta_{A'}]/(\theta_{A'}^{2m'+1})$: the matrix $(k^j/j!)_{j,k}$ has determinant $\prod_j(1/j!)$ times the Vandermonde determinant of $0,1,\dots,2m'$, which is nonzero. Write $q'_i=\sum_k b_{ik}e^{k\theta_{A'}}$ and add $(F_i\boxtimes L'^k)^{\oplus -Mb_{ik}/N_i}$, after enlarging $M$ to clear denominators; this removes $\omega_i\otimes q'_i$ and adds only terms $p_i(\theta_A)\otimes e^{k\theta_{A'}}$, which lie in $\mathbb{Q}[\theta_A]\otimes\mathbb{Q}[\theta_{A'}]$. Symmetrically remove $q_j\otimes\omega'_j$ with summands $L^l\boxtimes G'_j$.

Step 3. The classes $e^{l\theta_A+k\theta_{A'}}=\mathrm{ch}(L^l\boxtimes L'^k)$ form a basis of $\mathbb{Q}[\theta_A]/(\theta_A^{2m+1})\otimes\mathbb{Q}[\theta_{A'}]/(\theta_{A'}^{2m'+1})$ by the same determinant, so the remaining part in $\mathbb{Q}[\theta_A]\otimes\mathbb{Q}[\theta_{A'}]$ is cancelled by line bundles with integer multiplicities and shifts. Finally add $(L\boxtimes L')^{\otimes k}$, $k=0,1,2$, with generic nonzero multiplicities $\mu_k$. The result is

$$\mathrm{ch}(\mathcal{E})=Mw+\sum_{k=0}^{2}\mu_k e^{k\theta},$$

a polarised shape whose polynomial part is a sum of three distinct exponentials with nonzero coefficients. Three exponentials are not one of the shapes of Hankel rank at most two, so $\rho=3$; and some $c_k$ with $1\le k\le 2n-1$ is nonzero, so the first-order Hodge locus is the tangent space of $\mathcal{D}_{n,n}$ by Theorem 24.1. ∎

The construction was carried out in the exterior-algebra model of $H^*(A\times A')$ for $(m,m')=(2,1)$ and $(2,2)$: the cross terms of $\mathrm{ch}(\mathcal{E}_0)$ vanish, nothing is left outside $W(A\times A')\oplus\mathbb{Q}[\theta]$ after Steps 2 and 3, the Hankel rank is 3, and the first-order Hodge locus has dimension 9, resp. 16, equal to $n^2$ (a count made with the programs of the archive described in the statement on data availability).

24.4. **What a direct sum cannot do.** The tangent space of $\mathcal{D}_{n,n}$ at a product point contains the tangent space of the product locus $\mathcal{D}_{m,m}\times\mathcal{D}_{m',m'}$, of dimension $m^2+m'^2$, and $n^2-m^2-m'^2=2mm'$ mixed directions. In the same model, each exterior product $F\boxtimes G'$ of sheaves of polarised shape has first-order Hodge locus inside $T\mathcal{D}_{n,n}$ equal to the tangent space of the product locus: all $2mm'$ mixed directions obstruct it, four for $A\times B$ and eight for $A\times A'$ (Fig. 67). The object $\mathcal{E}$ of Theorem 24.3 has the whole tangent space as its first-order locus because its Chern character is a single polarised class and not a product. Semiregularity is a different matter.

**Proposition 24.4** (Direct sums are never semiregular)**.**

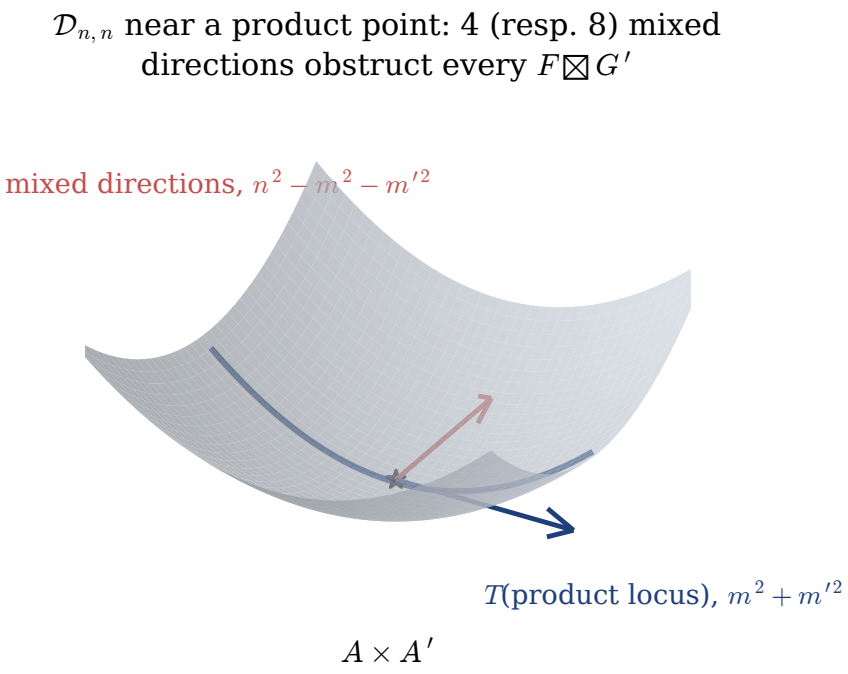



FIGURE 67. The domain $\mathcal{D}_{n,n}$ near a product point $A \times A'$. The curve is the product locus, whose tangent space has dimension $m^2 + m'^2$; the $2mm'$ mixed directions obstruct every exterior product $F \boxtimes G'$, and only the combined object of Theorem 24.3 has the full tangent space as first-order Hodge locus.

(i) *Let $\mathcal{E} = \bigoplus_k \mathcal{E}_k$ be a direct sum of at least two nonzero complexes. Then the semiregularity map $\sigma_{\mathcal{E}}$ vanishes on $\bigoplus_{k \neq l} \operatorname{Ext}^2(\mathcal{E}_k, \mathcal{E}_l)$. So $\sigma_{\mathcal{E}}$ is injective only if $\operatorname{Ext}^2(\mathcal{E}_k, \mathcal{E}_l) = 0$ for all $k \neq l$.*

(ii) *For every direct-sum realisation $\mathcal{E}$ of Theorem 24.3, $\sigma_{\mathcal{E}}$ is not injective and $\dim \operatorname{Ext}^2(\mathcal{E}, \mathcal{E}) > r(\operatorname{ch} \mathcal{E})$.*

*Proof.* (i) The semiregularity map of Buchweitz and Flenner is built from the traces $\xi \mapsto \operatorname{tr}(\xi \circ \operatorname{At}(\mathcal{E})^p)$ of compositions with powers of the Atiyah class [BF03, Sect. 4]. The Atiyah class of a direct sum is block diagonal, so for $\xi$ in an off-diagonal block $\operatorname{Ext}^2(\mathcal{E}_k, \mathcal{E}_l)$ the composition $\xi \circ \operatorname{At}(\mathcal{E})^p$ is off-diagonal, and the trace of an off-diagonal block is zero.

(ii) The summands of $\mathcal{E}$ include $F_1 \boxtimes L'^k$ and $F_1 \boxtimes L'^l$ for two exponents $k < l$, because Step 2 of the proof uses every power $L'^k$ with $0 \leq k \leq 2m'$ (the exponentials form a basis and $q_1'$ is a nonzero polynomial of positive degree for a sheaf of polarised shape, which has $p_1$ of positive degree by Theorem 24.1). By the Künneth formula

$$\operatorname{Ext}^2(F_1 \boxtimes L'^k, F_1 \boxtimes L'^l) \supseteq \operatorname{Ext}^2(F_1, F_1) \otimes H^0(A', L'^{l-k}), \tag{82}$$

and $\dim \operatorname{Ext}^2(F_1, F_1) \geq r(\operatorname{ch} F_1) \geq 12$ by Theorem 24.1 on the fourfold $A$ ($n = 2$, $\rho \geq 0$), while $H^0$ of an ample bundle is nonzero. So an off-diagonal block is nonzero, $\sigma_{\mathcal{E}}$ has a kernel by (i), and $\dim \operatorname{Ext}^2(\mathcal{E}, \mathcal{E}) > r$ because equality would force injectivity. ∎

**Proposition 24.5** (Convolutions)**.** *Let $\mathcal{E}_\delta$ be a one-sided twisted complex on pieces $P_1, \dots, P_N$ with shifts $s_a$, ordered so that the differential $\delta = (\delta_{ab})_{a<b}$, $\delta_{ab} \in \operatorname{Ext}^{1+s_a-s_b}(P_a, P_b)$, is strictly upper triangular; let $\gamma = \operatorname{ch}(\mathcal{E}_\delta) = \sum_a (-1)^{s_a} \operatorname{ch}(P_a)$.*

(i) *The semiregularity map $\sigma_{\mathcal{E}_\delta}$ vanishes on every class of $\operatorname{Ext}^2(\mathcal{E}_\delta, \mathcal{E}_\delta)$ represented by a strictly upper triangular cocycle. So the criterion holds for $\mathcal{E}_\delta$ only if every strictly upper triangular cocycle of degree two is a coboundary.*

(ii) *Suppose no term of the twisted differential carries a class of degree one into a diagonal block $\operatorname{Ext}^*(P_c, P_c)$. Let $v \in T\mathcal{D}_{n,n}$ have a nonzero mixed component. If some piece $P = X \boxtimes Y$ has $\operatorname{rk} Y\, c_1(X)/\theta_A \neq \operatorname{rk} X\, c_1(Y)/\theta_{A'}$, then $\operatorname{ev}_{\mathcal{E}_\delta}(v) \neq 0$ and $\sigma_{\mathcal{E}_\delta}(\operatorname{ev}_{\mathcal{E}_\delta}(v)) = 0$: the criterion fails.*

(iii) *Suppose the pieces split into two nonempty groups with $\operatorname{Ext}^*(P_a, P_b) = 0$ in both directions for $a$, $b$ in different groups. Then $\mathcal{E}_\delta = \mathcal{E}' \oplus \mathcal{E}''$ is the direct sum of the convolutions of the groups, and the criterion holds for $\mathcal{E}_\delta$ only if every $v \in T\mathcal{D}_{n,n}$ annihilates $\operatorname{ch}(\mathcal{E}')$ and $\operatorname{ch}(\mathcal{E}'')$ separately.*

*Proof.* (i) The components of $\sigma_{\mathcal{E}_\delta}(\xi)$ are the traces $\operatorname{tr}(\xi \circ \operatorname{At}(\mathcal{E}_\delta)^p)/p!$ [BF03, Sect. 4]. In the harmonic model of a twisted complex the Atiyah class is represented by $[\nabla, \bar\partial + \delta]$ with $\nabla$ the direct sum of the Chern connections of the pieces; it is block upper triangular with diagonal blocks $\operatorname{At}(P_a)$ and off-diagonal blocks $\nabla\delta_{ab}$. The homotopy transfer $\xi \mapsto \sum_k (-h[\delta, \cdot\,])^k \xi$ that produces the chain-level representative of $\xi$ raises or preserves the block position $b-a$, so a strictly upper triangular cocycle has a strictly upper triangular representative. The product of an upper triangular block matrix with a strictly upper triangular one is strictly upper triangular, and its trace, the sum of the traces of its diagonal blocks, is zero. The trace is a chain map for $\bar\partial + [\delta, \cdot\,]$ by cyclicity, so the vanishing holds in cohomology. Since $\sigma_{\mathcal{E}_\delta} \circ \operatorname{ev}_{\mathcal{E}_\delta}$ is contraction into $\gamma$, of rank $r(\gamma)$, the kernel of $\sigma_{\mathcal{E}_\delta}$ contains the image of the strictly upper triangular classes and $\dim \operatorname{Ext}^2 \geq r(\gamma) + \dim(\text{that image})$.

(ii) The Atiyah class of $X \boxtimes Y$ is $\operatorname{At}(X) \otimes 1 + 1 \otimes \operatorname{At}(Y)$, so the diagonal component of $\operatorname{ev}_{\mathcal{E}_\delta}(v)$ at $P$ is $v \lrcorner \operatorname{At}(P)$, with trace $v \lrcorner \operatorname{ch}_1(P)$. Write $v = v_A + v_{A'} + z \otimes d' + z' \otimes d$ with $v_A \in T\mathcal{D}_{m,m}$, $v_{A'} \in T\mathcal{D}_{m',m'}$, $z \in H^{0,1}(A)$, $d' \in H^0(T_{A'})$, $z' \in H^{0,1}(A')$, $d \in H^0(T_A)$. Since $v$ kills $\theta = \theta_A + \theta_{A'}$ and $v_A \lrcorner \theta_A = v_{A'} \lrcorner \theta_{A'} = 0$, we have $v \lrcorner \theta_{A'} = -v \lrcorner \theta_A = -z' \wedge (d \lrcorner \theta_A)$, nonzero when $d \neq 0$ because $\theta_A$ is nondegenerate and $z' \wedge$ is injective on $H^{0,1}(A)$; the case $d = 0$, $d' \neq 0$ is symmetric. On a very general member $c_1(X)$ and $c_1(Y)$ are multiples of the polarisations, so $v \lrcorner \operatorname{ch}_1(P) = (\operatorname{rk} Y\, c_1(X)/\theta_A - \operatorname{rk} X\, c_1(Y)/\theta_{A'})\, v \lrcorner \theta_A \neq 0$ under the hypothesis on $P$. Under the assumption on the differential, the diagonal components of a class in $\operatorname{Ext}^2(\mathcal{E}_\delta, \mathcal{E}_\delta)$ are well defined, so $\operatorname{ev}_{\mathcal{E}_\delta}(v) \neq 0$, while $\sigma_{\mathcal{E}_\delta}(\operatorname{ev}_{\mathcal{E}_\delta}(v)) = v \lrcorner \gamma = 0$ because $v$ is tangent to the Hodge locus of $\gamma$.

(iii) With no Ext groups between the groups there are no components of $\delta$ between them, so the twisted complex is block diagonal,

$$\begin{gathered} \operatorname{Ext}^2(\mathcal{E}_\delta, \mathcal{E}_\delta) = \operatorname{Ext}^2(\mathcal{E}', \mathcal{E}') \oplus \operatorname{Ext}^2(\mathcal{E}'', \mathcal{E}''), \\ \dim \operatorname{Ext}^2(\mathcal{E}_\delta, \mathcal{E}_\delta) \geq \dim HT^2 - \dim\big(\operatorname{Ann}(\operatorname{ch} \mathcal{E}') \cap \operatorname{Ann}(\operatorname{ch} \mathcal{E}'')\big), \end{gathered} \tag{83}$$

with $\sigma_{\mathcal{E}_\delta} = \sigma_{\mathcal{E}'} \oplus \sigma_{\mathcal{E}''}$; the inequality holds because for each summand the contraction into its Chern character factors through ev and $\sigma$, so that $\dim \operatorname{Ext}^2(\mathcal{E}', \mathcal{E}')$ is at least the rank of $\xi \mapsto \xi \lrcorner \operatorname{ch} \mathcal{E}'$, and likewise for $\mathcal{E}''$. Here $r(\gamma) = \dim HT^2 - \dim \operatorname{Ann}(\gamma)$ with $\operatorname{Ann}(\gamma) = T\mathcal{D}_{n,n}$. Equality $\dim \operatorname{Ext}^2 = r(\gamma)$ forces the common annihilator to be all of $T\mathcal{D}_{n,n}$. ∎

For the pieces of Theorem 24.3 the splitting in (iii) that one would like to use separates the pieces $F_i \boxtimes Y$ carrying a secant sheaf on $A$ from the pieces $L^l \boxtimes Y'$. Two statements settle what that splitting gives. The first computes the annihilator of the secant pieces by hand; the second shows that the splitting is never available.

**Theorem 24.6** (The annihilator of the secant pieces)**.** *Let $\mathcal{E}'$ be the sum, with the multiplicities and shifts of Theorem 24.3, of the pieces $F_i \boxtimes G'_j$ and $F_i \boxtimes L'^k$. Then*

$$\operatorname{ch}(\mathcal{E}') = Mw + \sum_i \frac{M}{N_i}\, p_i(\theta_A) \otimes R_i, \qquad R_i = \sum_j a_{ij}\omega'_j, \tag{84}$$

*where $(a_{ij})$ is the matrix of $w$ in the bases $(\omega_i)$, $(\omega'_j)$ and is invertible, and the annihilator of $\operatorname{ch}(\mathcal{E}')$ in $T\mathcal{D}_{n,n}$ is the tangent space $T\mathcal{D}_{m,m} \oplus T\mathcal{D}_{m',m'}$ of the product locus, of dimension $m^2 + m'^2$. If $\operatorname{Ext}^*(F_i, L^l) = 0$ for every power $L^l$ among the pieces, every convolution $\mathcal{E}_\delta$ of the pieces has $\dim \operatorname{Ext}^2(\mathcal{E}_\delta, \mathcal{E}_\delta) \geq r(\gamma) + 2mm'$, which is 61 at $n = 3$ and 120 at $n = 4$.*

*Proof.* Steps 1 and 2 of Theorem 24.3 add and subtract, for each $i$, the classes

$$\sum_j \frac{Ma_{ij}}{N_i N'_j} \operatorname{ch}(F_i) \operatorname{ch}(G'_j) \quad \text{and} \quad \frac{M}{N_i} \operatorname{ch}(F_i) \otimes q'_i,$$

where $q'_i = \sum_j (a_{ij}/N'_j) p'_j$; the difference is $\operatorname{ch}(F_i) \otimes R_i$, and summing over $i$ with $\operatorname{ch}(F_i) = N_i\omega_i + p_i(\theta_A)$ gives (84), the terms $\omega_i \otimes R_i$ adding up to $Mw$ by the definition of $a$. By Lemma 24.2 $w = x\,\omega_+ \otimes \omega'_+ + \bar{x}\,\omega_- \otimes \omega'_-$ with $x \neq 0$, a tensor of rank two, so $a$ is invertible and $R_1$, $R_2$ form a basis of the Weil plane of $A'$.

Write $V = H^1(A, \mathbb{C}) = V_+ \oplus V_-$ for the eigenspaces of $K$, $\omega_\pm = \det V_\pm$, and $V'_\pm$, $\omega'_\pm$ for their analogues on $A'$. A tangent vector $v \in H^1(T_{A\times A'})$ acts on cohomology as a derivation and decomposes as $v = v_A + v_{A'} + \mu + \mu'$ with $v_A \in H^{0,1}(A) \otimes H^0(T_A)$, $v_{A'}$ its analogue on $A'$, and the mixed parts $\mu \in H^{0,1}(A) \otimes H^0(T_{A'})$, $\mu' \in H^{0,1}(A') \otimes H^0(T_A)$. Let $v \in T\mathcal{D}_{n,n}$. The three Künneth components of $v \lrcorner \theta$ vanish separately: $v_A \lrcorner \theta_A = 0$, $v_{A'} \lrcorner \theta' = 0$ and $\mu \lrcorner \theta' + \mu' \lrcorner \theta_A = 0$. Contraction into a nondegenerate form is an isomorphism from $H^0(T)$ onto $H^{0,1}$, so $\mu \mapsto \mu \lrcorner \theta'$ and $\mu' \mapsto \mu' \lrcorner \theta_A$ are injective into $H^{0,1}(A) \otimes H^{0,1}(A')$, and $\mu = 0$ exactly when $\mu' = 0$. Next, $v \lrcorner (\omega_+ \otimes \omega'_+) = 0$. The term $(v_A \lrcorner \omega_+) \otimes \omega'_+$ has no component in $\Lambda^{2m} V_+ \otimes \Lambda^{2m'} V'_+$, because that component is the trace of $v_A$ on $V_+$ and $v_A$ sends $V_+^{1,0}$ into $V_+^{0,1}$ and kills $V_+^{0,1}$; it lies in $\Lambda^{2m-1} V_+ \otimes V_- \otimes \Lambda^{2m'} V'_+$, while the other three terms lie in $\Lambda^{2m} V_+ \otimes \Lambda^{2m'-1} V'_+ \otimes V'_-$, in $\Lambda^{2m-1} V_+ \otimes \Lambda^{2m'} V'_+ \otimes V'_-$ and in $\Lambda^{2m} V_+ \otimes V_- \otimes \Lambda^{2m'-1} V'_+$, four distinct weight spaces for $GL(V_+) \times GL(V_-) \times GL(V'_+) \times GL(V'_-)$. Hence $v_A \lrcorner \omega_+ = 0$, and in the same way $v_A \lrcorner \omega_- = 0$ and $v_{A'} \lrcorner \omega'_\pm = 0$. A vector of $T\mathcal{D}_{n,n}$ with vanishing mixed part therefore lies in $T\mathcal{D}_{m,m} \oplus T\mathcal{D}_{m',m'}$, and conversely every such sum kills $\theta$ and the Weil plane; the dimension $m^2 + m'^2$ is that of the two Weil families [Wei77].

Now let $v \in T\mathcal{D}_{n,n}$ have $\mu \neq 0$. It kills $Mw$; $v_A$ kills $p_i(\theta_A)$ and $v_{A'}$ kills $R_i$; so $v \lrcorner \operatorname{ch}(\mathcal{E}') = (\mu + \mu') \lrcorner \sum_k \theta_A^k \otimes R^{(k)}$, where $p_i = \sum_k c_{ik} \theta_A^k$ and $R^{(k)} = \sum_i (M c_{ik}/N_i) R_i$. Write $\mu = \sum_s z_s \otimes d'_s$ and $\mu' = \sum_s z'_s \otimes d_s$. The contraction of $\theta_A^k \otimes R^{(k)}$ against $\mu'$ is $k \sum_s \theta_A^{k-1} (d_s \lrcorner \theta_A) \otimes (z'_s \wedge R^{(k)})$, of Künneth bidegree $(2k-1, 2m'+1)$, and against $\mu$ it is $\sum_s (z_s \wedge \theta_A^k) \otimes (d'_s \lrcorner R^{(k)})$, of bidegree $(2k+1, 2m'-1)$. These bidegrees are pairwise distinct over all $k$ and both types, so every term vanishes on its own. Some $R^{(k)}$ is nonzero, since $p_1$, $p_2$ are not both zero and $R_1$, $R_2$ are a basis. For $R = \sigma\omega'_+ + \bar{\sigma}\omega'_-$ with $\sigma \neq 0$ the map $z' \mapsto z' \wedge R$ is injective on $H^{0,1}(A')$, because $z' \wedge \omega'_+ = z'_- \wedge \omega'_+$ and $z' \wedge \omega'_- = z'_+ \wedge \omega'_-$ lie in independent weight spaces and vanish only with $z'_\mp$; dually $d' \mapsto d' \lrcorner R$ is injective on $H^0(T_{A'})$. If the nonzero $R^{(k)}$ has $k \geq 1$, hard Lefschetz makes $\theta_A^{k-1}$ injective on $H^1(A)$,

since $k-1 \le 2m-1$, so the $\mu'$-term vanishes only if $\sum_s (d_s \lrcorner \theta_A) \otimes z'_s = 0$, that is $\mu' = 0$, and then $\mu = 0$. If $k = 0$ the $\mu$-term is $(\mathrm{id} \otimes (\,\cdot \lrcorner R^{(0)}))(\mu)$, which vanishes only if $\mu = 0$. Either way the mixed part is zero, against the assumption.

The bound is Proposition 24.5(iii): the common annihilator of $\mathrm{ch}\,\mathcal{E}'$ and $\mathrm{ch}\,\mathcal{E}''$ in $T\mathcal{D}_{n,n}$ lies in that of $\mathrm{ch}\,\mathcal{E}'$, so $\dim \mathrm{Ext}^2 \ge \dim HT^2 - m^2 - m'^2 = r(\gamma) + n^2 - m^2 - m'^2$, and $n = m + m'$. ∎

**Proposition 24.7** (Secant and line-bundle pieces are never orthogonal)**.** *For every realisation of Theorem 24.3 there are an index $i$ and a power $L^l$ among the pieces with $\chi(F_i, L^l) \neq 0$. Hence $\mathrm{Ext}^*(F_i, L^l) \neq 0$ and $\mathrm{Ext}^*(L^l, F_i) \neq 0$, and the hypothesis of the last sentence of Theorem 24.6 fails.*

*Proof.* The Todd class of $A$ is 1, $\omega_i \theta_A = 0$ and $\int \theta_A^4 = 24$, so Riemann–Roch gives $\chi(F_i, L^l) = \int p_i(-\theta_A) e^{l\theta_A} = P_i(l)$ with

$$P_i(l) = c_{i0} l^4 - 4c_{i1} l^3 + 12 c_{i2} l^2 - 24 c_{i3} l + 24 c_{i4}, \tag{85}$$

a nonzero polynomial because $p_i \neq 0$. Expand $p_i = \sum_l b_{il} e^{l\theta_A}$, $0 \le l \le 4$, modulo $\theta_A^5$, which is possible since the five exponentials span $\mathbb{Q}[\theta_A]/(\theta_A^5)$, and call the set $S_i$ of $l$ with $b_{il} \neq 0$ the support of $p_i$. Then $P_i(s) = \sum_{l \in S_i} b_{il} (s-l)^4$. If $P_i$ vanished on all of $S_i$ with $|S_i| \ge 2$, the vector $(b_{il})_{l \in S_i}$ would lie in the kernel of the symmetric matrix $\big((s-l)^4\big)_{s,l \in S_i}$. That matrix is nonsingular: for two elements its determinant is $-(s-l)^8$, for three it is $2\prod (s-l)^4$, for the five four-element subsets of $\{0,\dots,4\}$ it is 29952, 1032192, 6054912, 1032192, 29952, and for the whole set it is 7962624. So a support with at least two elements contains some $l$ with $P_i(l) \neq 0$.

Step 2 of Theorem 24.3 uses the pieces $L^l \boxtimes G'_j$ for every $l$ in the support of $q_j = \sum_i (M a_{ij}/N_i) p_i$, and the last step uses $L^k \boxtimes L'^k$ for three values of $k$ whose net multiplicities are nonzero. If $p_1$ and $p_2$ are independent, $q_1$ and $q_2$ span the same plane because $a$ is invertible, so the union of their supports is $S_1 \cup S_2$; if $p_2$ is a multiple of $p_1$, both $q_j$ are multiples of $p_1$, not both zero, with the same support. In either case every $l \in S_1 \cup S_2$ gives a piece $L^l \boxtimes G'_j$. If $|S_1| \ge 2$ or $|S_2| \ge 2$ we are done. Otherwise $p_i = b_i e^{l_i \theta_A}$ with $b_i \neq 0$. If $l_1 \neq l_2$ then $L^{l_2} \boxtimes G'_j$ is a piece and $P_1(l_2) = b_1 (l_2 - l_1)^4 \neq 0$. If $l_1 = l_2 = l$ the residue of Steps 1 to 3 lies in $e^{l\theta_A} \otimes \mathbb{Q}[\theta']$, so at least two of the three final exponents $k$ differ from $l$, and $P_i(k) = b_i (k-l)^4 \neq 0$. The vanishing of $\mathrm{Ext}^*(F_i, L^l)$ would force $\chi(F_i, L^l) = 0$; Serre duality gives $\mathrm{Ext}^*(L^l, F_i) \neq 0$ as well. ∎

Theorem 24.6 is therefore a conditional statement whose condition never holds: Proposition 24.5(iii) cannot be applied to the splitting between secant pieces and line-bundle pieces, and no bound on $\dim \mathrm{Ext}^2$ of a convolution follows from the Chern characters alone. The hypothesis of part (ii) can fail with line bundles only. The cone of a section of $L^l \boxtimes L'^{k'-k}$, viewed as a map $L^l \boxtimes L'^k \to L^l \boxtimes L'^{k'}$ with $k' > k$, is a twisted complex of two pieces in which the Serre dual of the section, a class of degree one, enters the diagonal block of the piece $L^l \boxtimes L'^k$; the only terms that can enter a diagonal block of a line-bundle piece are of this form, with the same power on $A$. Part (ii) applies to every piece $L^l \boxtimes L'^k$ with $l \neq k$ whose block receives no such term, which is how Step 3 of Theorem 24.3 uses it.

24.5. **Where the product object stops.** Every test that depends on $\mathrm{ch}(\mathcal{E})$ alone is passed: polarised shape, $N \neq 0$, Hankel rank 3, first-order Hodge locus equal to $T\mathcal{D}_{n,n}$, lower bound 57 or 104. The criterion then asks for $\dim \mathrm{Ext}^2(\mathcal{E}, \mathcal{E}) = r$, Proposition 24.4 shows that no direct sum achieves it, and Proposition 24.5 says what a convolution of

the same pieces must do: kill every strictly upper triangular class of degree two, carry classes of degree one from the lower triangle into the diagonal blocks of every piece with mismatched slopes, and, since by Proposition 24.7 the secant pieces and the line-bundle pieces are never orthogonal, carry classes between the two groups in a way that cancels the mixed directions, which by Theorem 24.6 the secant pieces alone leave uncancelled. All three demands are statements about the groups $\mathrm{Ext}^*(F_1, F_2)$, $\mathrm{Ext}^*(F_i, L^l)$, $\mathrm{Ext}^*(L^l, F_i)$ and their analogues on $A'$. Those groups are not computed in [Mar25a, Mar25b, Mar25c], where the only object with known Ext groups on a very general Weil fourfold of trivial discriminant is the twisted transform $\Phi(F \boxtimes F^\vee)$ with $\dim \mathrm{Ext}^k = 1, 8, 18, 8, 1$, whose Brauer class is half the Poincaré class and which therefore cannot be mixed with untwisted line bundles; and we have not computed them. Riemann–Roch fixes $\chi(F_i, L^l) = P_i(l)$ and the Künneth formula fixes everything on the line-bundle side, so the unknown is exactly the thirteen graded groups $\mathrm{Ext}^*(F_i, F_j)$ and $\mathrm{Ext}^*(F_i, L^l)$. This is the exact stopping point: the polarised criterion at a product point is a finite question about the Ext algebra of the secant sheaves, and Proposition 24.5 lists what the answer must contain for the route to stay open. The analogous question on $E_0^6$, for line bundles, closed negatively with $\dim \mathrm{Ext}^2 \geq 69$ (Theorem 18.35); that is an analogy and not a proof of the answer here.

At a point of complex multiplication the thirteen groups are known. We record them, since they turn the stopping point into a statement about one explicit variety.

**Lemma 24.8** (The Ext groups at a point of complex multiplication)**.** *Let $K = \mathbb{Q}(i)$, $E_0 = \mathbb{C}/\mathbb{Z}[i]$, and let $A_0 = E_0^6$ carry the $K$-action by $i$ on three factors and by $-i$ on the other three; it is a member of $\mathcal{D}_{3,3}$ for every discriminant, since the $K$-hermitian forms $\sum_{j\leq 3} a_j z_j \bar{w}_j - \sum_{j>3} a_j z_j \bar{w}_j$ with $a_j > 0$ are polarisations of every discriminant class. Let $L$ be ample on $A_0$ with class $\theta$, let $Z, Z' \subseteq A_0$ be abelian subvarieties of dimension three with $Z \cap Z'$ finite, and put $\chi_Z = (\theta|_Z)^3/6$, a positive integer.*

(i) *Every Hodge class on $A_0$ is a polynomial in divisor classes, and every product of three divisor classes is a rational combination of classes $[Z]$ of abelian threefolds. The Weil classes are such combinations, and no Weil class is a multiple of a single $[Z]$. For every $Z$, $\mathrm{ch}(\mathcal{O}_Z) = [Z]$ and $\mathrm{ch}(\mathcal{O}_Z \otimes L^a) = [Z]\, e^{a\theta}$.*

(ii) *$\mathrm{Ext}^k(\mathcal{O}_Z \otimes L^a, \mathcal{O}_{Z'} \otimes L^b) = 0$ for $k \neq 3$, and $\dim \mathrm{Ext}^3 = |Z \cap Z'| = (Z \cdot Z')$, for all $a, b$.*

(iii) *The groups $\mathrm{Ext}^k(\mathcal{O}_Z \otimes L^a, \mathcal{O}_Z \otimes L^b)$ are given by (86) below, with $N_Z \cong \mathcal{O}_Z^3$. For $a = b$ it has dimension $\binom{6}{k}$ and the characteristic map $HT^2(A_0) \to \mathrm{Ext}^2$ is onto; for $b > a$ it lives in degrees 0 to 3, of dimension $\binom{3}{k}(b-a)^3\chi_Z$; for $b < a$ in degrees 3 to 6, of dimension $\binom{3}{k-3}(a-b)^3\chi_Z$.*

(iv) *For $b > a$, $\mathrm{Ext}^k(\mathcal{O}_Z \otimes L^a, L^b)$ is concentrated in degree 3 and $\mathrm{Ext}^k(L^a, \mathcal{O}_Z \otimes L^b)$ in degree 0, both of dimension $(b-a)^3\chi_Z$; for $b < a$ the first is concentrated in degree 6 and the second in degree 3, both of dimension $(a-b)^3\chi_Z$; for $a = b$ the dimensions are $\binom{3}{k-3}$ and $\binom{3}{k}$.*

(v) *For $b > a$, $\mathrm{Ext}^k(L^a, L^b) = H^k(A_0, L^{b-a})$ is concentrated in degree 0, of dimension $(b-a)^6\theta^6/720$; for $b < a$ in degree 6; for $a = b$ it is $H^k(\mathcal{O}_{A_0})$, of dimension $\binom{6}{k}$.*

*Proof.* (i) The Hodge group of $A_0$ is the norm-one torus $K^1$ acting by $z$ on $V_+$ and $\bar{z}$ on $V_-$, so over $\mathbb{C}$ the Hodge classes of degree $2k$ are $\bigwedge^k V_+ \otimes \bigwedge^k V_-$, which is spanned by the products of $k$ elements of $V_+ \otimes V_- = \mathrm{Hdg}^2(A_0)_{\mathbb{C}}$; since $\mathrm{Hdg}^2 = \mathrm{NS}_{\mathbb{Q}}$ by Lefschetz, the $\mathbb{Q}$-span of the products of $k$ divisor classes has full dimension $\binom{6}{k}^2$. The classes

$f^*[0]$, $f \in \mathrm{Hom}(E_0^6, E_0) = \mathbb{Z}[i]^6$, are the rank-one hermitian forms $v\bar{v}^{\mathsf{T}}$ and span $\mathrm{NS}_{\mathbb{Q}}$. For $f_1, f_2, f_3$ independent over $K$ the map $f = (f_1, f_2, f_3)\colon A_0 \to E_0^3$ is a surjective homomorphism, hence flat, and $f_1^*[0]\, f_2^*[0]\, f_3^*[0] = f^*[0] = [f^{-1}(0)]$ is a positive multiple of the class of the abelian threefold $\ker(f)^\circ$; for dependent $f_i$ the product is zero. A Weil class $w$ satisfies $w\theta = 0$, while $[Z]\theta = i_*(\theta|_Z) \neq 0$ because $(\theta|_Z)^3 > 0$. Since $Z$ is a translate of a subgroup, $N_Z$ is trivial and $\mathrm{td}(N_Z) = 1$, so Grothendieck–Riemann–Roch gives $\mathrm{ch}(i_*\mathcal{O}_Z) = i_*(1) = [Z]$ and $\mathrm{ch}(\mathcal{O}_Z \otimes L^a) = [Z]e^{a\theta}$.

(ii) Since $\dim Z \cap Z' = 0$ and the tangent spaces are translation invariant, $T_Z \oplus T_{Z'} = T_{A_0}$ at every point of $Z \cap Z'$, so $Z$ and $Z'$ are locally transverse complete intersections and $Z \cap Z'$ is reduced. In local coordinates with $Z = \{x_1 = x_2 = x_3 = 0\}$ and $Z' = \{y_1 = y_2 = y_3 = 0\}$, the Koszul resolution of $\mathcal{O}_Z$ gives $\mathcal{E}xt^k(\mathcal{O}_Z, \mathcal{O}_{Z'})$ as the cohomology of the Koszul complex of $x_1, x_2, x_3$ on $\mathcal{O}_{Z'}$, a regular sequence there; so $\mathcal{E}xt^k = 0$ for $k \neq 3$ and $\mathcal{E}xt^3 = \mathcal{O}_{Z\cap Z'}$. The local-to-global spectral sequence has one row, $H^{k-3}(Z \cap Z', \mathcal{O})$, and twisting by line bundles does not change a sheaf on a finite reduced scheme; the intersection number equals $|Z \cap Z'|$ by transversality.

(iii) Let $\pi\colon A_0 \to A_0/Z$ be the quotient. The normal bundle $N_Z$ is the restriction of the trivial bundle $\pi^* T_{A_0/Z}$, so it extends to all of $A_0$, and by the theorem of Arinkin and Căldăraru [AC12] the derived restriction $i^*i_*\mathcal{O}_Z$ is formal, $i^*i_*\mathcal{O}_Z \cong \bigoplus_p \bigwedge^p N_Z^\vee[p]$. Hence

$$\mathrm{Ext}^k(\mathcal{O}_Z \otimes L^a, \mathcal{O}_Z \otimes L^b) = \mathrm{Ext}^k_Z(i^*i_*\mathcal{O}_Z, L^{b-a}|_Z) = \bigoplus_{p+q=k} H^q(Z, L^{b-a}|_Z) \otimes \textstyle\bigwedge^p N_Z. \quad (86)$$

For $a = b$ the dimension is $\sum_{p+q=k} \binom{3}{q}\binom{3}{p} = \binom{6}{k}$; for $b \neq a$ the cohomology of $L^{b-a}|_Z$ on the abelian threefold $Z$ sits in degree 0 or 3 according to the sign of $b - a$, with dimension $|b-a|^3\chi_Z$ by Riemann–Roch, which gives the two ranges of degrees and their dimensions. The characteristic map sends $H^2(\mathcal{O}_{A_0})$ onto $H^2(\mathcal{O}_Z)$ by restriction, $H^1(T_{A_0}) = H^1(\mathcal{O}) \otimes T_0$ onto $H^1(\mathcal{O}_Z) \otimes N_{Z,0}$ through the quotient $T_0 \to N_{Z,0}$, and $H^0(\bigwedge^2 T_{A_0}) = \bigwedge^2 T_0$ onto $\bigwedge^2 N_{Z,0}$; all three are surjective, the first because $H^*(A_0) \to H^*(Z)$ is onto for an abelian subvariety.

(iv) By Grothendieck duality for the regular embedding, whose relative dualising sheaf $\det N_Z$ is trivial, and by adjunction,

$$\mathrm{Ext}^k(\mathcal{O}_Z \otimes L^a, L^b) = H^{k-3}(Z, L^{b-a}|_Z),$$
$$\mathrm{Ext}^k(L^a, \mathcal{O}_Z \otimes L^b) = H^k(Z, L^{b-a}|_Z).$$

On $Z$ the cohomology of an ample bundle is concentrated in degree 0 and that of an anti-ample one in degree 3, of dimension $\chi = c_1^3/6$ by Riemann–Roch, and $c_1(L^m|_Z)^3 = m^3(\theta|_Z)^3$; for $m = 0$ one gets $H^*(\mathcal{O}_Z)$.

(v) is the same statement on $A_0$, with $\chi(L^m) = m^6\theta^6/720$. ∎

**Lemma 24.9** (Threefolds that meet along a positive-dimensional set)**.** *Let $Z, Z' \subseteq A_0$ be distinct abelian threefolds, let $W = Z \cap Z'$, of dimension $d \in \{0, 1, 2\}$, and put $c = 3 - d$ and $m = b - a$.*

(i) *$\mathcal{H}om(\mathcal{O}_Z, \mathcal{O}_{Z'}) = 0$, and $\mathcal{E}xt^q(\mathcal{O}_Z, \mathcal{O}_{Z'}) \cong \mathcal{O}_W^{\binom{d}{q-c}}$ for $c \leq q \leq 3$, zero otherwise.*

(ii) *$\mathrm{Ext}^k(\mathcal{O}_Z \otimes L^a, \mathcal{O}_{Z'} \otimes L^b) = 0$ unless $c \leq k \leq 3$ when $m > 0$, $3 \leq k \leq 3 + d$ when $m < 0$, and $c \leq k \leq 3 + d$ when $m = 0$. In particular the group vanishes in degrees $k \leq 2$ when $m < 0$, and $\mathrm{Ext}^1 \neq 0$ only when $d = 2$ and $m \geq 0$, where $\mathrm{Ext}^1 = H^0(W, L^m|_W)$.*

(iii) *For the threefolds $Z_\zeta$, $Z_{\zeta'}$ of Appendix D, $d$ is the number of indices $j$ with $\zeta_j = \zeta'_j$, $W$ is the product of the shared curves with $\prod_{\zeta_j \neq \zeta'_j} |\zeta_j - \zeta'_j|^2$ points, and for $d = 2$ and $m = 0$, $\dim \mathrm{Ext}^1 = |\zeta_l - \zeta'_l|^2 \in \{2, 4\}$, $l$ being the index at which they differ.*

*Proof.* (i) $W$ is a closed subgroup scheme of $A_0$, reduced in characteristic zero, so a disjoint union of translates of an abelian variety of dimension $d$, and it is nonempty since $Z$ and $Z'$ contain the origin. At a point of $W$ choose translation-invariant linear coordinates in which $Z = \{x_1 = x_2 = x_3 = 0\}$. The restrictions of $x_1, x_2, x_3$ to $Z'$ are linear forms cutting out $W$, of codimension $c$ in $Z'$; after a linear change $x_1, \dots, x_c$ restrict to a regular sequence on $Z'$ with zero locus $W$ and $x_{c+1}, \dots, x_3$ restrict to zero. The Koszul resolution of $\mathcal{O}_Z$ computes $\mathcal{E}xt^q(\mathcal{O}_Z, \mathcal{O}_{Z'})$ as the cohomology of the Koszul complex of these restrictions on $\mathcal{O}_{Z'}$, the tensor product of the Koszul complex of the regular sequence, with cohomology $\mathcal{O}_W$ in degree $c$, and of $d$ complexes $\mathcal{O}_{Z'} \xrightarrow{0} \mathcal{O}_{Z'}$; so $\mathcal{E}xt^q = \bigwedge^{q-c}(\mathcal{O}_W^d)$. Since $c \geq 1$, $\mathcal{H}om = 0$.

(ii) The local-to-global spectral sequence has $E_2^{p,q} = H^p(W, \mathcal{E}xt^q \otimes L^m|_W)$ with $0 \leq p \leq d$ and $c \leq q \leq 3$. On each component of $W$ the bundle $L^m|_W$ is ample for $m > 0$, with cohomology in degree 0 only, anti-ample for $m < 0$, with cohomology in degree $d$ only, and trivial for $m = 0$. This gives the three ranges. For $k = 1$ the only possible term is $E_2^{0,1}$, nonzero only when $c = 1$ and $m \geq 0$, and it survives to $E_\infty$ because $E_2^{1,0} = E_2^{2,0} = 0$.

(iii) In the coordinates of Appendix D, $Z_\zeta = \{u_j + \zeta_j v_j = 0,\ j = 1, 2, 3\}$; a shared index gives a form vanishing on $Z_{\zeta'}$, and an index with $\zeta_j \neq \zeta'_j$ gives the form $(\zeta_j - \zeta'_j)v_j$ on $Z_{\zeta'}$, whose zero locus in $C_{\zeta'_j}$ has $|\zeta_j - \zeta'_j|^2$ points by Lemma D.1(ii). So $W$ is as stated, and $\mathrm{Ext}^1 = H^0(W, \mathcal{O}_W)$ has dimension the number of components of $W$. ∎

Theorem 24.1 applies at $A_0$ with the same number $r = 57$: the rank of the contraction of $HT^2$ into $\mathrm{ch}(\mathcal{E})$ is invariant under $\mathrm{SU}(V, H)(\mathbb{R})$, which fixes the Weil classes and $\theta$ and acts transitively on $\mathcal{D}_{3,3}$, so it is the rank computed at a very general point, and the argument of Buchweitz and Flenner is local on the base. An object $\mathcal{E}$ with $\dim \mathrm{Ext}^2(\mathcal{E}, \mathcal{E}) = 57$ at $A_0$ would therefore deform along all of $\mathcal{D}_{3,3}$ and prove the Weil classes algebraic on every member with full Hodge group. By (i) every Chern character of polarised shape is realised at $A_0$ by a convolution of pieces $\mathcal{O}_{Z_a} \otimes L^{k_a}[s_a]$ and $L^k[t]$, with at least two subvariety pieces, since $w$ is not a multiple of one $[Z]$, and at least three powers of $L$, since the Hankel rank is three; by (ii) to (v) every group between two pieces is known. Appendix D writes the classes, the intersection numbers and every dimension of the lemma in coordinates for the threefolds that are products of elliptic curves of the three Weil surfaces, and exhibits the signed sum of thirty-two of them equal to $32(\alpha + \bar{\alpha})$. What the lemma decides is how the diagonal classes can disappear.

**Proposition 24.10** (No cancellation in two steps)**.** *Let $\mathcal{E}$ be a convolution of pieces $F_a[s_a]$ of the types of Lemma 24.8, whose threefolds meet pairwise in finite sets, and no two of which are isomorphic sheaves placed at shifts differing by at most one. A class of $\mathrm{Ext}^2(F_a, F_a)$ is cancelled in the spectral sequence of the convolution only through a closed chain $a = a_0, a_1, \dots, a_m = a$ of pieces with $\mathrm{Ext}^{k_i}(F_{a_{i-1}}, F_{a_i}) \neq 0$ and $\sum_i k_i = m$, and no such chain has $m = 2$. The chains with $m = 3$ are exactly those through three twists $k < k' < k''$ of one threefold, $\mathcal{O}_Z \otimes L^k[s] \to \mathcal{O}_Z \otimes L^{k'}[s+1] \to \mathcal{O}_Z \otimes L^{k''}[s+2] \to \mathcal{O}_Z \otimes L^k[s]$, with degrees $0, 0, 3$. Hence every cancellation among the diagonal classes of degree two is a Massey product of length at least three.*

*Proof.* The Maurer–Cartan components of the twisted complex and the class $\eta$ whose differential is computed all have total degree one, that is, a component in $\mathrm{Ext}^k(F_a, F_b)$ has $k + s_b - s_a = 1$. A product of $m$ such components along a closed chain lands in $\mathrm{Ext}^{\sum k_i - m + 2}(F_a, F_a)$, since the shifts cancel around the chain and the $m$-fold product lowers the degree by $m - 2$; so it reaches degree two exactly when $\sum k_i = m$. For $m = 2$ both components have degree at most two. Between two pieces that are not isomorphic sheaves, Lemma 24.8 gives the degrees: 3 and 3 between $\mathcal{O}_Z$ and $\mathcal{O}_{Z'}$; 0 to 3 one way and 3 to 6 the other between two twists of one $\mathcal{O}_Z$; 3 and 3 between $\mathcal{O}_Z \otimes L^a$ and $L^b$ with $a < b$, 6 and 0 with $a > b$, and 3 to 6 against 0 to 3 for $a = b$; 0 and 6 between two powers of $L$. In every case the two degrees add up to at least three. Between two copies of one sheaf at shifts $s$, $s'$ the two degrees are $1 - s' + s$ and $1 - s + s'$, which are both at most two only when $|s - s'| \leq 1$, the case excluded. For $m = 3$ the edges of degree at most two are, by the same list, the edges that raise the twist and the edges $L^a \to \mathcal{O}_Z \otimes L^a$; a closed chain cannot consist of them alone, so one edge has degree at least three and the other two have degree zero. Two edges of degree zero from a line bundle lead to a power of $L$ or to a subvariety piece of strictly larger twist, from which every edge back to the line bundle has degree six; from $\mathcal{O}_Z \otimes L^k$ they lead only to $\mathcal{O}_Z \otimes L^{k''}$ with $k'' > k$ through an intermediate twist $k'$, and the edge back has degree three, through $H^3(Z, L^{k-k''}|_Z) \otimes \bigwedge^0 N_Z$. These are the chains displayed, and their three groups are nonzero. ∎

For line-bundle pieces alone the same bookkeeping holds on $E_0^6$, and there the Massey products of length four are the ones that act, with the result $\dim \mathrm{Ext}^2 \geq 69$ (Theorem 18.35). With subvariety pieces and three powers of $L$ no closed chain of any length exists, and the criterion fails for a reason that involves no product at all.

**Proposition 24.11** (Convolutions with three line bundles fail)**.** *Let $\mathcal{E}$ be a convolution at $A_0$ of pieces $\mathcal{O}_{Z_a} \otimes L^{k_a}[s_a]$ and $L^{l_b}[t_b]$ in which no sheaf occurs twice, the threefolds $Z_a$ are pairwise distinct, there are at most three line bundles, and $\mathrm{Ext}^1(\mathcal{O}_{Z_a} \otimes L^{k_a}, \mathcal{O}_{Z_b} \otimes L^{k_b}) = 0$ whenever $s_a = s_b$. Suppose $\mathrm{ch}(\mathcal{E})$ has the polarised shape. Then no term of the twisted differential carries a class of total degree one into a diagonal block, and*

$$\dim \mathrm{Ext}^2(\mathcal{E}, \mathcal{E}) \;>\; r(\mathrm{ch}\,\mathcal{E}) :$$

*the criterion fails for $\mathcal{E}$. The hypothesis on $\mathrm{Ext}^1$ holds when the threefolds meet pairwise in finite sets, and for the thirty-two threefolds $Z_\zeta$ of Appendix D placed at one twist with shifts of the parity of $\zeta_1\zeta_2\zeta_3$.*

*Proof.* Step 1, no closed chain. As in the proof of Proposition 24.10, a term of the differential landing in a diagonal block runs along a closed chain $a_0, a_1, \ldots, a_m = a_0$ of pieces with a nonzero component in $\mathrm{Ext}^{k_i}(F_{a_{i-1}}, F_{a_i})$, $k_i = 1 + s_{a_{i-1}} - s_{a_i}$, so that $\sum_i k_i = m$; the pieces of the chain are distinct, because the differential is strictly upper triangular in the one-sided order. We show that no such chain exists. Between two distinct threefold pieces $\mathrm{Hom} = 0$ by Lemma 24.9(i), and a component of degree one needs $s_a = s_b$, which the hypothesis excludes; so every edge between two threefold pieces has degree at least two. By Lemma 24.8(iv) and (v), an edge $L^l \to \mathcal{O}_Z \otimes L^k$ has degree 0 when $k > l$, 3 when $k < l$ and 0 to 3 when $k = l$, so a degree at most one forces $k \geq l$; an edge $\mathcal{O}_Z \otimes L^k \to L^l$ has degree 3 when $l > k$, 6 when $l < k$ and 3 to 6 when $l = k$, so a degree at most four forces $l \geq k$; and an edge between two line bundles has degree 0 when it raises the twist and 6 when it lowers it. A chain of threefold pieces alone has $\sum_i k_i \geq 2m > m$. Otherwise the chain visits $\ell \leq 3$ line-bundle pieces; between consecutive visits it either passes directly, by

an edge that raises the twist ($u$ such edges in all) or lowers it ($w$ in all), or runs through $r \geq 1$ threefold pieces, entered with degree $e$ and left with degree $x$; let $R$ be the number of runs. Then $R + u + w = \ell$, $m = \ell + \sum_{\text{runs}} r$, a run contributes at least $e + x + 2(r-1)$ to $\sum_i k_i$, and so

$$\sum_i k_i - m \;\geq\; \sum_{\text{runs}} (e + x + r - 3) + 5w - u,$$

where each bracket is at least $0 + 3 + 1 - 3 = 1$. If $w \geq 1$ then $u \leq \ell - R - w \leq 2 - R$ and the right side is at least $R + 5 - (2 - R) > 0$. If $w = 0$ then $u = \ell - R \leq 3 - R$ and the right side is at least $2R - 3$, so $R \leq 1$; $R = 0$ is a cycle of edges that all raise the twist, which cannot close; and for $R = 1$ the single bracket must equal $u$, so $r \leq 2$ and $e + x \leq 5 - r$, that is $(e, x) \in \{(0,3), (0,4), (1,3)\}$ for $r = 1$ and $(e, x) = (0, 3)$ for $r = 2$. In both cases the run enters the first threefold, of twist $k$, from $L^l$ with $k \geq l$, leaves the last one, of twist $k'$, for $L^{l'}$ with $l' \geq k'$, and returns from $L^{l'}$ to $L^l$ by $u \geq 1$ edges that raise the twist, so $l' < l$ and $k' < k$. For $r = 1$ this reads $k < k$. For $r = 2$ the internal edge goes from the twist $k$ to the strictly smaller twist $k'$ and has degree two, which Lemma 24.9(ii) forbids. So there is no closed chain.

Step 2, the class $\mathrm{ev}_{\mathcal{E}}(v)$. By Step 1 the diagonal components of a class in $\mathrm{Ext}^2(\mathcal{E}, \mathcal{E})$ are well defined, as in Proposition 24.5(ii), and for $v \in H^1(A_0, T_{A_0})$ the diagonal components of $\mathrm{ev}_{\mathcal{E}}(v)$ are the classes $\mathrm{ev}_{F_a}(v) = v \lrcorner \mathrm{At}(F_a)$ of the pieces, the Atiyah class of the twisted complex being block upper triangular with diagonal blocks $\mathrm{At}(F_a)$. For a piece $F = \mathcal{O}_Z \otimes L^k$ the semiregularity map sends $\mathrm{ev}_F(v)$ to $v \lrcorner \mathrm{ch}(F) = v \lrcorner [Z] + k\, v \lrcorner ([Z]\theta) + \cdots$, whose component of lowest degree is $v \lrcorner [Z]$. There is at least one threefold piece, since the Weil component of $\mathrm{ch}(\mathcal{E})$ is nonzero and powers of $L$ contribute none. Step 3 gives $v \in T\mathcal{D}_{3,3}$ with $v \lrcorner [Z_a] \neq 0$ for such a piece, so $\mathrm{ev}_{\mathcal{E}}(v) \neq 0$; and $\sigma_{\mathcal{E}}(\mathrm{ev}_{\mathcal{E}}(v)) = v \lrcorner \mathrm{ch}(\mathcal{E}) = 0$, because $T\mathcal{D}_{3,3}$ annihilates $\theta$ and the Weil plane. So $\sigma_{\mathcal{E}}$ has a nonzero kernel, while its rank is at least the rank $r(\mathrm{ch}\,\mathcal{E})$ of $\sigma_{\mathcal{E}} \circ \mathrm{ev}_{\mathcal{E}}$; hence $\dim \mathrm{Ext}^2(\mathcal{E}, \mathcal{E}) \geq r(\mathrm{ch}\,\mathcal{E}) + 1$.

Step 3, no $[Z]$ is annihilated by $T\mathcal{D}_{3,3}$. The tangent space $T\mathcal{D}_{3,3} \subseteq H^1(A_0, T_{A_0}) = \mathrm{Hom}(V^{1,0}, V^{0,1})$ consists of the $K$-linear maps that are skew for the polarisation; it is the part $\mathfrak{g}^{-1,1}$ of the Lie algebra $\mathfrak{g}$ of $\mathrm{SU}(V, H)$ in the Hodge structure that $A_0$ induces on it, and the contraction $v \lrcorner$ on $H^*(A_0) = \bigwedge^* V_{\mathbb{C}}$ is the derivation by which $\mathfrak{g}_{\mathbb{C}}$ acts. Suppose $v \lrcorner [Z] = 0$ for all $v \in \mathfrak{g}^{-1,1}$. Since $[Z]$ is real it is annihilated by the conjugate space $\mathfrak{g}^{1,-1}$ as well, and $\mathfrak{g}^{-1,1} \oplus \mathfrak{g}^{1,-1}$, the off-diagonal blocks of $\mathfrak{sl}(V_+ \oplus V_-)$, generates $\mathfrak{g}_{\mathbb{C}} = \mathfrak{sl}_6$; so $[Z]$ is invariant under the connected group $\mathrm{SU}(V, H)$, in particular under its rational points. Every $g \in \mathrm{SU}(V, H)(\mathbb{Q}) \subseteq \mathrm{GL}_K(V) = \mathrm{End}^0(A_0)^\times$ is a quasi-isogeny of $A_0$, because $E_0$ has complex multiplication by $K$; for an integer $n$ with $\varphi = ng$ an isogeny, $\varphi^*[Z] = [Z]$ gives $\varphi_*[Z] = \varphi_*\varphi^*[Z] = \deg(\varphi)[Z]$, so the class of the abelian threefold $\varphi(Z)$, whose tangent space is $g\,T_Z$, is a positive multiple of $[Z]$. The group $\mathrm{SU}(V, H)(\mathbb{C}) \cong \mathrm{SL}_6(\mathbb{C})$ acts transitively on the three-dimensional subspaces of $V_+$, which determine the $K$-subspaces of $V$, the condition $g\,T_Z \cap T_Z = 0$ is Zariski open and nonempty in $g$, and $\mathrm{SU}(V, H)(\mathbb{Q})$ is Zariski dense in the connected group $\mathrm{SU}(V, H)$ [Bor91, Cor. 18.3]. So some $g$ makes $Z \cap \varphi(Z)$ finite; it contains the origin, and $(Z \cdot \varphi(Z)) = |Z \cap \varphi(Z)| \geq 1$ by Lemma 24.8(ii). But $(Z \cdot \varphi(Z))$ is a positive multiple of $[Z]^2$, and $[Z]^2$ is a multiple of $f^*([0]^2) = 0$ for the map $f\colon A_0 \to E_0^3$ of Lemma 24.8(i), since $[0]^2 = 0$ on $E_0^3$. This contradiction proves the claim, and the proposition.

For the last sentence of the statement: when the threefolds meet in finite sets every group between two of them sits in degree three by Lemma 24.8(ii); and two of the $Z_\zeta$ at the same shift have the same value of $\zeta_1\zeta_2\zeta_3$, hence differ in at least two coordinates, so $d \leq 1$ and $\mathrm{Ext}^1 = 0$ by Lemma 24.9(ii). ∎

Proposition 24.11 settles the configuration that Appendix D writes down, the thirty-two threefolds $Z_\zeta$ at one twist with three powers of $L$: no convolution of these thirty-five pieces meets the criterion. By Lemma 24.8(i) every object meeting the criterion at $A_0$ is a convolution of pieces $\mathcal{O}_Z \otimes L^k[s]$ and $L^l[t]$, so what remains are the convolutions with four or more line bundles, with a sheaf at two shifts, with a threefold at several twists, or with two threefold pieces at the same shift meeting along an abelian surface, as two of the $Z_\zeta$ with two equal coordinates do. Closed chains exist among them. With the threefolds at twist 0 and the line bundles of positive twist, every edge from a line bundle into a threefold piece and every edge out of one has degree three, so the shortest closed chain through a threefold piece runs through five line bundles at five consecutive shifts, $L^{l_1}[\sigma-2] \to L^{l_2}[\sigma-1] \to L^{l_3}[\sigma] \to L^{l_4}[\sigma+1] \to L^{l_5}[\sigma+2] \to \mathcal{O}_Z[\sigma] \to L^{l_1}[\sigma-2]$ with $0 < l_1 \le \cdots \le l_5$ and degrees $0,0,0,0,3,3$; one threefold at three twists carries the chain of Proposition 24.10; and two pieces $\mathcal{O}_{Z_\zeta} \otimes L^k[s]$, $\mathcal{O}_{Z_{\zeta'}} \otimes L^k[s]$ with two equal coordinates carry the chain of length two with degrees $1,1$, through the groups $H^0(W,\mathcal{O}_W)$ of Lemma 24.9(iii). The question at the stopping point is whether the diagonal components $\left(\mathrm{ev}_{F_a}(v)\right)_a$, $v \in T\mathcal{D}_{3,3}$, lie in the image of the products along such chains; the Hodge locus of the obstruction, the only other way the criterion could fail, is settled by Theorem 24.1 itself. The next subsection answers the question twice without computing a product: by counting the blocks that no chain reaches, and by observing that one threefold piece outside every closed chain decides it.

24.6. **Counting the remaining configurations at $E_0^6$.** The four kinds of convolution left open by Proposition 24.11 can be tested by a bound that uses no product at all. Fix pieces $F_a$ with shifts $s_a$ and a minimal $A_\infty$-model of their full subcategory, so that $\mathrm{Hom}^\bullet(\mathcal{E},\mathcal{E})$ for a convolution $\mathcal{E}$ of the $F_a$ is the direct sum of the blocks $\mathrm{Ext}^k(F_a,F_b)$ placed in total degree $k+s_b-s_a$, with the differential $x \mapsto \sum_{p,q} m_{p+q+1}(\delta,\dots,\delta,x,\delta,\dots,\delta)$ built from the twisting cochain $\delta$, whose components lie in the blocks of total degree one. Call a block $\mathrm{Ext}^k(F_a,F_b)$ of total degree two *isolated* when no block of total degree one maps to it and it maps to no block of total degree three along such a term: that is, there is no chain $a \to \cdots \to a'$ and $b' \to \cdots \to b$ of pieces with every step a nonzero block of total degree one and $(a',b') \neq (a,b)$ carrying a block of total degree one, and no such chain $a' \to \cdots \to a$, $b \to \cdots \to b'$ with $(a',b')$ carrying a block of total degree three.

**Lemma 24.12** (Isolated blocks)**.** *For every twisting cochain $\delta$ on the pieces $F_a$, the isolated blocks of total degree two inject into $\mathrm{Ext}^2(\mathcal{E},\mathcal{E})$; so $\dim \mathrm{Ext}^2(\mathcal{E},\mathcal{E})$ is at least the sum of their dimensions.*

*Proof.* An element of an isolated block is a cycle, since every term of its differential lands in a block of total degree three reached from it along a chain of components of $\delta$, and there is none; and it is not a boundary, since every term of the differential of a class of total degree one lands in a block reached from that class along such a chain, and none reaches an isolated block. The blocks are direct summands, so the cycles of the isolated blocks meet the boundaries only in zero. ∎

The bound depends only on the Ext groups of Lemmas 24.8 and 24.9 and on the shifts, not on $\delta$, so it can be computed over all configurations of a family at once. We did this for the configurations of the four kinds within a window of twists 1 to 4 and shifts $-3$ to 3 around the thirty-five pieces, keeping those whose Chern character has the polarised shape, and recorded for each the sums $C^1, C^2, C^3$ of the block dimensions in total degrees one

TABLE 7. The isolated-block bound of Lemma 24.12 over the configurations at $A_0 = E_0^6$. "Polarised" counts the configurations of the family in the window whose Chern character has the polarised shape; "least" is the smallest bound among them, with the configuration attaining it. The criterion needs $\dim \mathrm{Ext}^2 = 57$.

| family (least bound attained at) | polarised | least |
|---|---|---|
| thirty-five pieces, $L^1, L^2, L^3$ at shifts $0, 0, 0$ | 1 | 7213 |
| the same at shifts $2, 3, 4$ or $-3, -2, -1$ | 2 | 2605 |
| (a) four line bundles ($L^1[-2]$, $L^2[-1]$, $L^3[0]$, $L^4[1]$) | 2401 | 0 |
| (b) one threefold at a second shift | 0 | |
| (c) one threefold at three twists ($Z_{(1,1,1)} \otimes L^1$ at $[-1]$, $[0]$ and $\otimes L^2$ at $[1]$, $[2]$) | 432 | 0 |
| (d) a second threefold meeting $Z_\zeta$ along a surface, at the shift of $Z_\zeta$ and a neighbouring one ($Z_{(1,1,-1)}$ at $[0]$, $[1]$) | 6 | 5325 |

to three, the Euler characteristic $C^2 - C^1 - C^3$ and the isolated blocks (Table 7). The programs are in the archive of the Data availability section.

**Proposition 24.13** (What counting decides)**.** *At $A_0 = E_0^6$, with the pieces of Appendix D:*

(i) *For the thirty-five pieces of Proposition 24.11 with the three line bundles at any shifts in the window,* $\dim \mathrm{Ext}^2(\mathcal{E}, \mathcal{E}) \geq 2605$ *for every twisting cochain, by isolated blocks alone.*

(ii) *Adding to them a threefold $Z_{\zeta'}$ meeting $Z_\zeta$ along an abelian surface, at the shift of $Z_\zeta$ and at a neighbouring shift so that the Chern character keeps the polarised shape, gives* $\dim \mathrm{Ext}^2 \geq 5325$*: the configuration of two threefold pieces at the same shift meeting along a surface fails for every twisting cochain.*

(iii) *Adding one further copy of a threefold at a second shift within the window never gives the polarised shape.*

(iv) *For four line bundles and for one threefold at three twists the isolated blocks are* 0 *at the configurations of Table 7 and the Euler characteristic is negative throughout the window: the bound decides nothing there.*

*Proof.* (i), (ii) and (iv) are Lemma 24.12 applied to the block dimensions of Lemmas 24.8 and 24.9, enumerated over the window; (iii) is the enumeration of Chern characters, since a second copy of $\mathcal{O}_{Z_\zeta}$ at a shift of the other parity changes the Weil component by $\pm 2[Z_\zeta]$, which no choice of line bundles compensates. ∎

Proposition 24.13 closes the configuration (d) of two threefolds at one shift and every placement of the thirty-five pieces. For the configurations (a) and (c) the bound is silent, and a different argument decides them: it suffices that one threefold piece lies outside every closed chain. The configuration (a) with more line bundles is decided in Section 24.7, by hand up to five and by the enumeration of the archive up to seven.

Let $\Gamma$ be the directed graph whose vertices are the pieces $F_a$ and which has an edge $a \to b$, for $a \neq b$, when the block of total degree one, $\mathrm{Ext}^{1+s_a-s_b}(F_a, F_b)$, is nonzero. The chains of Lemma 24.12 are its paths, a closed chain in the sense of Proposition 24.10 is one of its directed cycles, and the components of every twisting cochain $\delta$ sit on its edges. Call a piece *chain-free* when it lies on no directed cycle of $\Gamma$; this is a property of the pieces and their shifts alone.

**Proposition 24.14** (A piece outside every closed chain)**.** *Let $\mathcal{E}$ be a convolution at $A_0$ of pieces $\mathcal{O}_{Z_a} \otimes L^{k_a}[s_a]$ and $L^{l_b}[t_b]$ whose Chern character has the polarised shape, and*

*suppose that some threefold piece $F_a = \mathcal{O}_{Z_a} \otimes L^{k_a}[s_a]$ is chain-free. Then, for every twisting cochain $\delta$,*

$$\dim \mathrm{Ext}^2(\mathcal{E},\mathcal{E}) \;\geq\; r(\mathrm{ch}\,\mathcal{E}) + 1,$$

*and the criterion fails for $\mathcal{E}$.*

*Proof.* Let $\eta \in \mathrm{Hom}^1(\mathcal{E},\mathcal{E})$ have components $\eta_{bc}$ in the blocks of total degree one, and consider the component of its differential in the block $\mathrm{Ext}^2(F_a,F_a)$. It is a sum of terms $m_{p+q+1}(\delta,\dots,\delta,\eta_{bc},\delta,\dots,\delta)$ with $p+q \geq 1$, and such a term is nonzero only if its arguments are composable and run from $F_a$ back to $F_a$: the $p$ components of $\delta$ on one side of $\eta_{bc}$ along edges of $\Gamma$ from $a$ to $b$, the $q$ components on the other side along edges from $c$ to $a$, and $\eta_{bc} \neq 0$. If $b = c$, the $p+q \geq 1$ components of $\delta$ form a closed walk of positive length, which is impossible because $\delta$ is strictly upper triangular in the one-sided order. If $b \neq c$, then $b \to c$ is an edge of $\Gamma$, and $a \to \cdots \to b \to c \to \cdots \to a$ is a closed walk through $a$ with at least two edges, whose consecutive vertices are distinct; its first return to $a$, with repeated vertices shortcut, is a directed cycle of $\Gamma$ through $a$. Since $F_a$ is chain-free, no term survives: the differential of every element of total degree one has zero component in $\mathrm{Ext}^2(F_a,F_a)$. Hence the map $\pi_a\colon \mathrm{Ext}^2(\mathcal{E},\mathcal{E}) \to \mathrm{Ext}^2(F_a,F_a)$ that reads off the component of a cocycle in that block is well defined on cohomology.

As in the proof of Proposition 24.5(ii), the Atiyah class of the twisted complex is block upper triangular with diagonal blocks $\mathrm{At}(F_b)$, and the homotopy transfer to the minimal model only raises the block position, so it preserves the diagonal blocks of an upper triangular element; hence for $v \in H^1(A_0,T_{A_0})$ the class $\mathrm{ev}_{\mathcal{E}}(v)$ satisfies $\pi_a(\mathrm{ev}_{\mathcal{E}}(v)) = \mathrm{ev}_{F_a}(v) = v \lrcorner \mathrm{At}(F_a)$. By Step 3 of the proof of Proposition 24.11 there is $v \in T\mathcal{D}_{3,3}$ with $v \lrcorner [Z_a] \neq 0$, and then $\mathrm{ev}_{F_a}(v) \neq 0$, because the semiregularity map of $F_a$ sends it to $v \lrcorner \mathrm{ch}(F_a)$, whose component of lowest degree is $v \lrcorner [Z_a]$. So $\mathrm{ev}_{\mathcal{E}}(v) \neq 0$, while $\sigma_{\mathcal{E}}(\mathrm{ev}_{\mathcal{E}}(v)) = v \lrcorner \mathrm{ch}(\mathcal{E}) = 0$, since $T\mathcal{D}_{3,3}$ annihilates $\theta$ and the Weil plane. The kernel of $\sigma_{\mathcal{E}}$ is nonzero and its rank is at least $r(\mathrm{ch}\,\mathcal{E})$, the rank of $\sigma_{\mathcal{E}} \circ \mathrm{ev}_{\mathcal{E}}$; the bound follows, and equality in Theorem 24.1 is excluded. ∎

**Corollary 24.15** (Configurations with a chain-free piece)**.** *Let $\mathcal{E}$ be a convolution at $A_0$ whose Chern character has the polarised shape and whose pieces are the thirty-two threefolds $\mathcal{O}_{Z_\zeta}$ of Appendix D at twist 0, at the shifts $s$ and $s+1$ of the parity of $\zeta_1\zeta_2\zeta_3$, together with further pieces, line bundles $L^l[t]$ and threefolds $\mathcal{O}_Z \otimes L^k[t]$, all of positive twist. The criterion fails for $\mathcal{E}$, for every twisting cochain, in each of the following cases.*

(i) *The further pieces are at most four line bundles, of pairwise distinct twists: then $\Gamma$ has no directed cycle.*

(ii) *Every further piece sits at a shift at most $s+2$: then no edge of $\Gamma$ ends at a piece of shift $s+1$, and the sixteen pieces there are chain-free.*

(iii) *The further pieces are at most five line bundles, at any shifts: then a cycle through a piece at shift $\sigma$ needs five line bundles at the shifts $\sigma-2,\dots,\sigma+2$, and the sixteen pieces at one of the two shifts are chain-free.*

*Both bounds are sharp: $L^l[s+3]$ enters every piece at shift $s+1$, and six line bundles $L^{l_1}[s-2],\dots,L^{l_6}[s+3]$ with $0 < l_1 \leq \cdots \leq l_6$ put each of the thirty-two pieces on a cycle of length six.*

*Proof.* Along a directed cycle $a_0, a_1, \dots, a_m = a_0$ of $\Gamma$ write $k_i = 1 + s_{a_{i-1}} - s_{a_i}$ for the degree of the block on the edge $a_{i-1} \to a_i$; the shifts cancel around the cycle, so $\sum_i k_i = m$. The degrees available on the edges are read off from Lemmas 24.8 and 24.9. Two of the thirty-two pieces at one shift have the same value of $\zeta_1\zeta_2\zeta_3$, hence differ in at least two

coordinates, so $d \leq 1$ and the degrees between them are 2 and 3; as Hom = 0 between distinct threefolds and a degree one would need equal shifts, every edge between two of the thirty-two has degree at least 2. An edge from a line bundle of positive twist into a piece of twist 0 has degree 3, and so has an edge from a piece of twist 0 into a line bundle of positive twist; an edge into a piece of twist 0 from a threefold piece of positive twist has degree at least 3; an edge between two line bundles has degree 0 when it raises the twist, 6 when it lowers it, and $1 + t - t' \geq 0$ between two copies of one line bundle at shifts $t$ and $t'$. Parts (i) and (iii) use of the thirty-two pieces only that they have one twist, pairwise distinct threefolds and no $\mathrm{Ext}^1$ at equal shifts, as in Proposition 24.11.

(i) A cycle among pieces of twist 0 alone has $\sum_i k_i \geq 2m$, so a cycle visits $\ell' \geq 1$ line-bundle pieces, $\ell' \leq 4$. Between two consecutive visits it either runs through $r \geq 1$ pieces of twist 0, entered and left with degree 3 and with $r - 1$ internal edges of degree at least 2, which contributes at least $2r + 4$ to $\sum_i k_i$ and $r$ to $m$, or it passes directly along an edge that raises the twist, $u$ such edges in all, or lowers it, $w$ in all. With $R$ runs, $R + u + w = \ell'$ and

$$0 = \sum_i k_i - m \;\geq\; \sum_{\text{runs}} (r + 4) + 6w - \ell' \;\geq\; 4R + 5w - u.$$

If $R \geq 1$ this gives $u \geq 4$ and $\ell' \geq 5$. If $R = 0$ it gives $w = 0$, and a cycle whose edges all raise the twist does not close. So $\Gamma$ has no cycle; every threefold piece is chain-free, there is one, and Proposition 24.14 applies.

(ii) An edge $u \to v$ into a piece $v$ at shift $s + 1$ has degree $1 + s_u - (s + 1) = s_u - s$, at most 2 when $s_u \leq s + 2$. No line bundle and no threefold piece of positive twist has an edge of degree at most 2 into a piece of twist 0, and an edge into $v$ from one of the thirty-two would have degree 0 or 1, which do not occur. So no edge ends at $v$.

(iii) Let a cycle pass through a piece $v$ of twist 0 at shift $\sigma$, and let $q$ count its direct edges between two copies of one line bundle, so that $R + u + w + q = \ell' \leq 5$ and, the $q$ edges contributing at least 0, $0 \geq 4R + 5w - u - q$ as in (i). Since $R \geq 1$, this forces $R = 1$, $w = 0$, $u + q = 4$, $\ell' = 5$ and equality throughout: the only run is $v$ itself, and the four direct edges have degree 0, so each raises the shift by one, while the two edges of degree 3 at $v$ lower it by two. The five line-bundle pieces therefore sit at the shifts $\sigma - 2, \ldots, \sigma + 2$, one at each, with twists that do not decrease along the shifts. Five line bundles occupy at most five shifts, and the five shifts around $s$ differ from the five around $s + 1$; so the pieces at one of the two shifts are chain-free.

For the sharpness, $\mathrm{Ext}^3(L^l, \mathcal{O}_{Z_\zeta}) \neq 0$ is the edge $L^l[s + 3] \to \mathcal{O}_{Z_\zeta}[s + 1]$; and the six line bundles carry the cycle $L^{l_1}[s - 2] \to \cdots \to L^{l_5}[s + 2] \to \mathcal{O}_{Z_\zeta}[s] \to L^{l_1}[s - 2]$ through every piece at shift $s$ and the cycle $L^{l_2}[s - 1] \to \cdots \to L^{l_6}[s + 3] \to \mathcal{O}_{Z_\zeta}[s + 1] \to L^{l_2}[s - 1]$ through every piece at shift $s + 1$, each edge being one of the nonzero groups listed. ■

*Remark* 24.16 (The configurations of Table 7). Corollary 24.15(i) covers the configuration (a) of the table, (ii) covers (c), whose further twists sit at shifts at most $2 = s + 2$, and (iii) covers every configuration of the thirty-two pieces with at most five line bundles. The programs of the archive rebuild $\Gamma$ from the dimensions of Lemmas 24.8 and 24.9 and find, in the window of the table, no cycle in any of the 2401 configurations of (a); at least fifteen chain-free threefold pieces in each of the 3024 configurations of (c) with further twists $1 \leq k < k' \leq 3$, twenty-two at the configuration of the table; no edge into the sixteen pieces at shift 1 from the thirty-two or from the 1560 pieces of positive twist at shifts $-3$ to 2, while $L^l[3]$ and $\mathcal{O}_{Z_{(1,1,1)}} \otimes L^1[3]$ do enter; in each of the 28 812 configurations of four or five line bundles of twists 1 to 5 at shifts $-3$ to 3 one class of sixteen pieces

chain-free, the other class, when it lies on cycles, having line bundles at its five shifts; and with $L^1[-2],\dots,L^6[3]$ every piece on a cycle. The Lean file checks the counts 32 and 22 of chain-free pieces at the configurations (a) and (c) of the table, the absence of entering edges from the line bundles $L^1$ to $L^4$ and the twists 1 to 3 of the sixty-four threefolds $Z_\zeta$ at shifts $-3$ to 2, the entering edges at shift 3 and the six-cycles. No statement above rests on these computations.

24.7. **Chains through line bundles.** Corollary 24.15 needs the threefolds at one twist and the further pieces of positive twist, and it stops at five line bundles because six of them put every threefold piece on a cycle. This subsection removes both restrictions. It allows the threefolds any twists and any shifts, it works for every polarisation of $A_0$ and hence at every discriminant, and in place of the absence of cycles it uses a property of the cycles that the Maurer–Cartan equation of the twisted complex forces. By hand this excludes every convolution of distinct threefolds and at most five line bundles; the enumeration of Remark 24.25, a computation of the archive, extends the exclusion to six and seven line bundles and to all but thirty-one arrangements of eight. The next subsection turns from counting pieces to the support of any object meeting the criterion.

**Setup 24.17** (Threefolds and line bundles at $A_0$)**.** Throughout this subsection $\mathcal{E}=\mathcal{E}_\delta$ is a convolution at $A_0=E_0^6$, in the sense of Proposition 24.5, of threefold pieces $\mathcal{O}_{Z_\zeta}\otimes L^{k_\zeta}[s_\zeta]$, $\zeta\in\mu_4^3$, and line-bundle pieces $L^{l_i}[t_i]$ such that (S1) each threefold $Z_\zeta$ occurs in at most one piece, (S2) the exponents $l_i$ are pairwise distinct, and (S3) two threefold pieces at one shift differ in at least two coordinates of $\zeta$, so that $\mathrm{Ext}^1=0$ between them by Lemma 24.9(ii) and (iii). The twists $k_\zeta$, $l_i$ and the shifts are arbitrary integers. For $h\in\mathbb{Z}^6_{>0}$ let $\theta_h=\sum_j(h_ja_j+h_{3+j}b_j)$ be the class of the hermitian form of Lemma 24.8 with coefficients $h$, so that $\theta_{(1,\dots,1)}=\theta$ is the class of $L$, let $\mathcal{D}_h$ be the family $\mathcal{D}_{3,3}$ of the discriminant of $\theta_h$, which contains $(A_0,\theta_h)$, and suppose that $\mathrm{ch}(\mathcal{E})$ has the polarised shape with respect to $\theta_h$: $\mathrm{ch}(\mathcal{E})=N\omega+\sum_k c_k\theta_h^k$ with $N\neq0$ and $0\neq\omega\in W$. The Weil plane $W$ of (94) does not depend on $h$.

Theorem 24.1 applies at $(A_0,\theta_h)$ with $r=57$ for every $h$, by the argument given before Proposition 24.10 with $\mathrm{SU}(V,H_h)$ in place of $\mathrm{SU}(V,H)$: the group fixes $W$ and $\theta_h$ and acts transitively on $\mathcal{D}_h$. The tangent space $T\mathcal{D}_h\subseteq H^1(A_0,T_{A_0})$ consists of the $K$-linear maps skew for $\theta_h$; it annihilates $\theta_h$ and $W$, and Step 3 of the proof of Proposition 24.11 holds for it word for word, $\mathrm{SU}(V,H_h)(\mathbb{Q})$ being again a group of quasi-isogenies of $A_0$ Zariski dense in $\mathrm{SU}(V,H_h)$: for every abelian threefold $Z$ there is $v\in T\mathcal{D}_h$ with $v\lrcorner[Z]\neq0$. Suppose now that the criterion holds for $\mathcal{E}$, that is $\dim\mathrm{Ext}^2(\mathcal{E},\mathcal{E})=57$. Then $\sigma_\mathcal{E}$ is injective, and for every $v\in T\mathcal{D}_h$ the class $\mathrm{ev}_\mathcal{E}(v)\in\mathrm{Ext}^2(\mathcal{E},\mathcal{E})$, which $\sigma_\mathcal{E}$ sends to $v\lrcorner\mathrm{ch}(\mathcal{E})=0$, vanishes. Everything below draws consequences from this vanishing.

Let $\Gamma$ be the graph of Section 24.6 on the pieces $F_a$ of $\mathcal{E}$, and let $\Gamma_\delta\subseteq\Gamma$ be the subgraph of the edges $a\to b$ on which the component $\delta_{ab}$ is nonzero. Since $\delta$ is strictly upper triangular in the one-sided order, $\Gamma_\delta$ has no directed cycle, and its walks are paths. The twisted complex lives over the minimal $A_\infty$-model of the pieces, whose underlying graded space is $\bigoplus_{a,b}\mathrm{Ext}^*(F_a,F_b)$ with the block $\mathrm{Ext}^k(F_a,F_b)$ in total degree $k+s_b-s_a$. The cochain $\delta$ satisfies the Maurer–Cartan equation $\sum_{k\geq2}m_k(\delta,\dots,\delta)=0$, and the differential of the twisted complex is $D(\eta)=\sum_{p,q\geq0}m_{p+q+1}(\delta,\dots,\delta,\eta,\delta,\dots,\delta)$ with $p$ components of $\delta$ before $\eta$ and $q$ after it. A term of either sum is nonzero only if its arguments are composable: along a path of $\Gamma_\delta$ for the Maurer–Cartan equation, and, for a term of $D(\eta)_{ad}$ with the argument $\eta_{bc}$, along a path of $\Gamma_\delta$ from $a$ to $b$, the edge $b\to c$ of $\Gamma$, and a path of $\Gamma_\delta$ from $c$ to $d$.

**Lemma 24.18** (The cocycle of a tangent vector)**.** *For $v \in H^1(A_0, T_{A_0})$ the class $\mathrm{ev}_{\mathcal{E}}(v)$ is represented in the minimal model by a cocycle $c(v)$ whose component in the diagonal block $\mathrm{Ext}^2(F_a, F_a)$ is $\mathrm{ev}_{F_a}(v) = v \lrcorner \mathrm{At}(F_a)$ and whose component in a block $(a, c)$ with $a \neq c$ is a sum of terms each of which is a composite along a path of $\Gamma_\delta$ from $a$ to $c$. In particular $c(v)_{ac} = 0$ when $\Gamma_\delta$ has no path from $a$ to $c$.*

*Proof.* As in the proof of Proposition 24.5(i), the Atiyah class of the twisted complex is represented in the harmonic model by $[\nabla, \bar\partial + \delta]$, block upper triangular with diagonal blocks $\mathrm{At}(F_a)$ and off-diagonal blocks $\nabla\delta_{ab}$, which vanish off the edges of $\Gamma_\delta$; so $v \lrcorner [\nabla, \bar\partial + \delta]$ has diagonal blocks $v \lrcorner \mathrm{At}(F_a)$ and off-diagonal blocks on the edges of $\Gamma_\delta$ only. The homotopy transfer $\xi \mapsto \sum_k (-h[\delta, \cdot])^k \xi$ to the minimal model composes each block of $\xi$ with components of $\delta$ on either side, the homotopy $h$ acting block by block; so every term of the component $(a, c)$ of the transferred cocycle is a composite of one block of $v \lrcorner [\nabla, \bar\partial + \delta]$, sitting on a vertex or an edge of $\Gamma_\delta$, with components of $\delta$ along edges of $\Gamma_\delta$ leading from $a$ to it and from it to $c$. For $a = c$ only the term with $k = 0$ occurs, since a composite of $k \geq 1$ components of $\delta$ from $a$ back to $a$ would be a cycle of $\Gamma_\delta$. ∎

Call a directed cycle of $\Gamma$ *effective* for $\delta$ when exactly one of its edges lies outside $\Gamma_\delta$.

**Proposition 24.19** (Every threefold piece lies on an effective cycle)**.** *Suppose $\mathcal{E} = \mathcal{E}_\delta$ meets the criterion. Then for every $v \in T\mathcal{D}_h$ there is $\eta$ of total degree one with $D(\eta) = c(v)$, and for every piece $F_a$*

$$\mathrm{ev}_{F_a}(v) = \sum m_{p+q+1}(\delta, \dots, \delta, \eta_{bc}, \delta, \dots, \delta),$$

*the sum running over the effective cycles $a \to \cdots \to b \to c \to \cdots \to a$ of $\Gamma$ through $a$, with $b \to c$ the edge outside $\Gamma_\delta$ and $p$, $q$ the numbers of edges before and after it. In particular every threefold piece lies on an effective cycle whose edge outside $\Gamma_\delta$ carries a nonzero component of $\eta$.*

*Proof.* The class $\mathrm{ev}_{\mathcal{E}}(v)$ vanishes, so its representative $c(v)$ of Lemma 24.18 is a coboundary $D(\eta)$. The diagonal component $D(\eta)_{aa}$ is the sum of the terms $m_{p+q+1}(\delta, \dots, \eta_{bc}, \dots, \delta)$ whose arguments are composable from $a$ to $a$: a path of $\Gamma_\delta$ from $a$ to $b$, the edge $b \to c$ with $\eta_{bc} \neq 0$, and a path of $\Gamma_\delta$ from $c$ to $a$. If $b = c$ the term is $m_1(\eta_{aa}) = 0$ when $p = q = 0$, and otherwise its components of $\delta$ form a cycle of $\Gamma_\delta$; so $b \neq c$. The two paths are increasing in the one-sided order, so they meet only in $a$; the edge $b \to c$ is not in $\Gamma_\delta$, as otherwise $\Gamma_\delta$ would contain a cycle; and the walk is a directed cycle of $\Gamma$ with exactly one edge outside $\Gamma_\delta$. By Lemma 24.18 the left side is $c(v)_{aa} = \mathrm{ev}_{F_a}(v)$. For a threefold piece choose $v \in T\mathcal{D}_h$ with $v \lrcorner [Z_a] \neq 0$; then $\mathrm{ev}_{F_a}(v) \neq 0$, as in the proof of Proposition 24.14, so some term of the sum is nonzero. ∎

**Lemma 24.20** (Two rules)**.** *Let $a = L^{l_a}[t]$, $b = L^{l_b}[t+1]$ and $c = L^{l_c}[t+2]$ be line-bundle pieces of $\mathcal{E}$ with $l_a < l_b < l_c$, so that $a \to b \to c$ is a path of $\Gamma$ whose edges carry sections, $\delta_{ab} \in H^0(L^{l_b - l_a})$ and $\delta_{bc} \in H^0(L^{l_c - l_b})$.*

(i) *The path $a \to b \to c$ is not the only path of $\Gamma_\delta$ from $a$ to $c$: either $\delta$ vanishes on one of its two edges, or $\Gamma_\delta$ has another path from $a$ to $c$.*

(ii) *Suppose $a \to b \to c$ is the only path of $\Gamma$ from $a$ to $c$, let $v \in T\mathcal{D}_h$ and let $\eta$ satisfy $D(\eta) = c(v)$. If $\delta_{bc} = 0 \neq \delta_{ab}$ then $\eta_{bc} = 0$, and if $\delta_{ab} = 0 \neq \delta_{bc}$ then $\eta_{ab} = 0$.*

*Proof.* (i) The component $(a, c)$ of the Maurer–Cartan equation is the sum, over the paths of $\Gamma_\delta$ from $a$ to $c$, of the products $m_k(\delta, \dots, \delta)$ along them. If $a \to b \to c$ were the only

such path it would read $m_2(\delta_{ab},\delta_{bc})=0$, and $m_2$ on sections is their product, which is nonzero for two nonzero sections of line bundles on the integral variety $A_0$.

(ii) A term of $D(\eta)_{ac}$ runs along a walk of $\Gamma$ from $a$ to $c$ made of a path of $\Gamma_\delta$, one edge carrying $\eta$ and a path of $\Gamma_\delta$. If the walk repeats a vertex, the repetition is a vertex common to the two paths of $\Gamma_\delta$, and shortcutting it leaves a path of $\Gamma_\delta$ from $a$ to $c$; under the hypothesis this path is $a\to b\to c$ with both components nonzero, which is excluded in either case. So every term runs along a path of $\Gamma$ from $a$ to $c$, that is along $a\to b\to c$, and $D(\eta)_{ac}=\pm m_2(\delta_{ab},\eta_{bc})\pm m_2(\eta_{ab},\delta_{bc})$. The right side $c(v)_{ac}$ vanishes by Lemma 24.18, since $\Gamma_\delta$ has no path from $a$ to $c$. When $\delta_{bc}=0\neq\delta_{ab}$ this gives $\delta_{ab}\eta_{bc}=0$, the product of the nonzero section $\delta_{ab}$ with $\eta_{bc}\in\operatorname{Ext}^0(L^{l_b},L^{l_c})=H^0(L^{l_c-l_b})$, so $\eta_{bc}=0$; the other case is the same with the roles exchanged. ■

**Lemma 24.21** (Both parities occur)**.** *Grade $H^*(A_0,\mathbb{C})$ by $\mathbb{Z}^3$, giving $dz_j$ and $d\bar z_{3+j}$ the weight $e_j$ and $d\bar z_j$, $dz_{3+j}$ the weight $-e_j$. Then $a_j$, $b_j$, $\theta$ and every $\theta_h$ have weight $0$, $c_j$ has weight $2e_j$, $\bar c_j$ has weight $-2e_j$, $W$ lies in the weights $\pm2(e_1+e_2+e_3)$, and the weight-zero part of $[Z_\zeta]$ is $X_0=\prod_j(a_j+b_j)$ for every $\zeta$. Consequently $\sum_\zeta(-1)^{s_\zeta}=0$ over the threefold pieces of $\mathcal{E}$: there are threefold pieces at even shifts and at odd shifts.*

*Proof.* The weights are read off from (91), (92) and (94), and in $[Z_\zeta]=\prod_j(a_j+b_j+\bar\zeta_j c_j+\zeta_j\bar c_j)$ a product of one term from each factor has weight zero only if it takes $a_j$ or $b_j$ from every factor, the $e_j$ being independent. The part of $\operatorname{ch}(\mathcal{E})$ in $H^6$ is $\sum_\zeta(-1)^{s_\zeta}[Z_\zeta]+P\theta^3$ with $P=\sum_i(-1)^{t_i}l_i^3/6$, by Lemma 24.8(i), and it equals $N\omega+c_3\theta_h^3$. Taking the parts of weight zero, with $Q=\sum_\zeta(-1)^{s_\zeta}$,

$$QX_0+P\theta^3=c_3\theta_h^3.$$

The squarefree cubic monomials in $a_1,b_1,\dots,a_3,b_3$ are linearly independent. The coefficient of $a_jb_ja_k$, $j\neq k$, is $0$ in $X_0$, $6$ in $\theta^3$ and $6h_jh_{3+j}h_k$ in $\theta_h^3$, so $P=c_3h_jh_{3+j}h_k$, and likewise $P=c_3h_jh_{3+j}h_{3+k}$. If $c_3\neq0$ these equations, over all $j\neq k$, give $h_k=h_{3+k}$ and $h_1h_4=h_2h_5=h_3h_6$, so all six $h_k$ equal one value $h$ and $P=c_3h^3$; if $c_3=0$ then $P=0$. The coefficient of $a_1a_2a_3$ is $1$ in $X_0$, $6$ in $\theta^3$ and $6h_1h_2h_3$ in $\theta_h^3$, so $Q+6P=6c_3h_1h_2h_3$, which equals $6P$ in both cases. Hence $Q=0$. There is a threefold piece, since $N\omega\neq0$ and the line bundles contribute nothing to the Weil component; so there are pieces of both parities. ■

The edges of $\Gamma$ between line-bundle pieces are of two kinds, by Lemma 24.8(v): an *up* edge $L^l[s]\to L^{l'}[s+1]$ with $l<l'$, of degree 0, and a *down* edge $L^l[s]\to L^{l'}[s-5]$ with $l>l'$, of degree 6; in general an edge of degree $k$ joins the shifts $s$ and $s-(k-1)$. By Lemma 24.8(iv) an edge $L^l\to\mathcal{O}_Z\otimes L^k$ has degree 0 when $k>l$, 3 when $k<l$ and 0 to 3 when $k=l$, an edge $\mathcal{O}_Z\otimes L^k\to L^l$ has degree 3 when $l>k$, 6 when $l<k$ and 3 to 6 when $l=k$, and by Lemma 24.9(ii) and (S3) an edge between two threefold pieces has degree at least 2, and at least 3 when it lowers the twist. A *run* is a maximal path of $\Gamma$ through threefold pieces from a line-bundle piece $L^l[s]$ to a line-bundle piece $L^{l'}[s-\beta]$; its *excess* is $\beta$, and it is *decreasing* when $l'<l$.

**Lemma 24.22** (The shape of a cycle)**.** *Let $C$ be a directed cycle of $\Gamma$ through a threefold piece, with $\ell$ line-bundle pieces, $R\geq1$ runs, $u$ up edges and $w$ down edges, so that $\ell=R+u+w$.*

(i) *A run through $r$ threefold pieces has excess at least $r$, with equality only when the twists do not decrease along it. A decreasing run has excess at least 3, with equality*

*only for* $L^l \to \mathcal{O}_Z \otimes L^k \to \mathcal{O}_{Z'} \otimes L^{k'} \to L^{l'}$ *with* $k' \le l' < l \le k$ *and degrees* $0, 3, 3$*, whose two threefold pieces have the parity of shift of* $L^{l'}$*.*

(ii) $u = \sum_{\text{runs}} \beta + 5w$*. If* $\ell \le 7$ *then* $w = 0$*, exactly one run is decreasing, and* $\ell \ge 2R+2$*.*

(iii) *If* $\ell \le 5$ *then* $R = 1$*: the line-bundle pieces of* $C$ *sit at* $\ell$ *consecutive shifts, joined by up edges in the order of their exponents, and the run leads from the highest back to the lowest. If* $\ell = 4$ *the run is the one of (i) with excess* 3*, and both its threefold pieces have the parity of the lowest shift.*

*Proof.* (i) An edge of degree $k$ lowers the shift by $k-1$. Along a run the entering edge has degree $e \ge 0$, the $r-1$ internal edges have degree at least 2 and the leaving edge has degree $x \ge 3$, so the excess is at least $-1 + (r-1) + 2 = r$, with equality only for $e = 0$, internal degrees 2 and $x = 3$, that is $l \le k_1 \le \cdots \le k_r \le l'$ for the twists. Let $l' < l$. With $r = 1$ the excess $(e-1) + (x-1)$ is at least 4: $e = 0$ forces $k \ge l > l'$, hence $x = 6$; $e = 3$ gives $2 + (x-1) \ge 4$; and $e \in \{1, 2\}$ forces $k = l > l'$, hence $x = 6$. With $r = 2$ and degrees $e, d, x$ the excess $e + d + x - 3$ equals 3 only for $(e, d, x) = (0, 3, 3)$: $e = 0$ forces $k_1 \ge l > l'$, and then either $k_2 \ge k_1 > l'$, so $x = 6$, or $k_2 < k_1$, so $d \ge 3$ and $x \ge 3$, with $x = 3$ exactly when $k_2 \le l'$; $e = 3$ gives at least $3 + 2 + 3$; $e = 1$ forces $k_1 = l$, and then $k_2 \ge l > l'$ gives $x = 6$ while $k_2 < k_1$ gives $d \ge 3$, each at least 7; $e = 2$ gives at least $2 + 2 + 3$. With $r \ge 3$ the excess is at least $r + 1 \ge 4$. Along the run with $(e, d, x) = (0, 3, 3)$ from $L^l$ at shift $s$ the shifts are $s + 1$, $s - 1$ and $s - 3$.

(ii) The shifts return to their value around $C$: each up edge adds 1, each down edge subtracts 5 and each run subtracts its excess. Each run has excess at least 1, so $\ell = R + u + w = R + \sum \beta + 6w \ge 2R + 6w$, and $w \ge 1$ forces $\ell \ge 8$. Let $w = 0$. The exponents also return to their value, and the up edges raise them, so some run is decreasing. With $D \ge 1$ decreasing runs, of excess at least 3, $\ell = R + \sum \beta \ge R + 3D + (R - D) = 2R + 2D$; so $\ell \le 7$ forces $D = 1$, and $\ell \ge 2R + 2$.

(iii) $\ell \le 5$ gives $R = 1$ by (ii), and $u = \ell - 1$: the line-bundle pieces form a chain of $\ell - 1$ up edges, each raising the shift by one and the exponent, closed by one run from the last to the first, which is decreasing. For $\ell = 4$ the run has excess $u = 3$, so it is the run of (i), and the parity of its threefold pieces is that of its exit, the lowest line bundle. ∎

**Theorem 24.23** (Convolutions with at most five line bundles fail)**.** *In Setup 24.17, if* $\mathcal{E}$ *has at most five line-bundle pieces then* $\dim \mathrm{Ext}^2(\mathcal{E}, \mathcal{E}) \ge 58$*: the criterion fails for* $\mathcal{E}$*, for every twisting cochain, for every twist of the threefolds and for every polarisation* $\theta_h$*.*

*Proof.* Suppose the criterion holds. By Lemma 24.21 there are threefold pieces of both parities of shift, and by Proposition 24.19 each lies on an effective cycle carrying a nonzero component of $\eta$, in particular on a directed cycle of $\Gamma$. A cycle through a threefold piece has at least four line-bundle pieces by Lemma 24.22(ii), so with at most three there is none. Let there be four or five, $L_1, \ldots, L_n$ in the order of their exponents.

With four, every cycle through a threefold piece uses all four, at consecutive shifts in the order of the exponents, and its threefold pieces have the parity of the lowest shift, by Lemma 24.22(iii); so the threefold pieces on cycles have one parity, and a threefold piece of the other parity lies on no cycle.

With five, a cycle through a threefold piece has four or five line bundles, at consecutive shifts in the order of the exponents. If the five are not at five consecutive shifts in that order, no cycle uses all five, and two cycles of four, omitting $L_m$ and $L_{m'}$ with $m \ne m'$, have the same lowest shift: the three line bundles common to both occupy three of four consecutive shifts in each; if these are consecutive, $x, x+1, x+2$, the omitted bundles sit at $x - 1$ or at $x + 3$, and at the same one of the two, since $L_m$ at $x - 1$ and $L_{m'}$ at $x + 3$

would put the five at five consecutive shifts in the order of the exponents; and if they leave a gap, both omitted bundles fill it. So again the threefold pieces on cycles have one parity, and a threefold piece of the other parity lies on no cycle.

It remains to treat $L_i$ at the shift $\sigma - 3 + i$, $i = 1, \dots, 5$. The cycles through threefold pieces are $C_5$ through all five, $C_{1234}$ and $C_{2345}$, each a chain of up edges followed by one run. The only path of $\Gamma$ from $L_i$ to $L_{i+2}$ is $L_i \to L_{i+1} \to L_{i+2}$: a path leaving the chain of up edges enters a run, which ends at least three shifts below its start, at $L_1$ or $L_2$, and from there the up edges lead only through line bundles already visited. By Lemma 24.20(i) applied to the three triples, $\delta$ vanishes on one of the edges $L_1 \to L_2$, $L_2 \to L_3$, on one of $L_2 \to L_3$, $L_3 \to L_4$, and on one of $L_3 \to L_4$, $L_4 \to L_5$. An effective cycle has exactly one edge outside $\Gamma_\delta$. So $C_5$, with two such edges, is not effective; $C_{1234}$ is effective only when its edge outside $\Gamma_\delta$ is $L_2 \to L_3$, and then $\delta_{12} \neq 0$, so that $\eta_{23} = 0$ by Lemma 24.20(ii); and $C_{2345}$ is effective only when that edge is $L_3 \to L_4$, with $\delta_{23} \neq 0$, so that $\eta_{34} = 0$. No effective cycle carries a nonzero component of $\eta$ on its edge outside $\Gamma_\delta$.

Each case contradicts Proposition 24.19. So the criterion fails, and since Theorem 24.1 gives $\dim \mathrm{Ext}^2 \geq 57$ with equality excluded, $\dim \mathrm{Ext}^2 \geq 58$. ∎

**Lemma 24.24** (The shifts along a cycle form an interval)**.** *The shifts of the line-bundle pieces along a directed cycle of* $\Gamma$*, or along a path of* $\Gamma$ *between two line-bundle pieces the second of which is at the higher shift, take every value between their least and their greatest. Hence, if the shifts of the line-bundle pieces of* $\mathcal{E}$ *do not form an interval of integers, the line bundles split along a gap into two groups such that every cycle of* $\Gamma$ *through a threefold piece, and every path* $a \to b \to c$ *of Lemma 24.20, has its line bundles in one group.*

*Proof.* Between consecutive line-bundle pieces of a cycle or a path the shift rises by exactly 1, along an up edge, or falls, along a down edge or a run; from the least shift to the greatest it therefore passes through every intermediate value. The three line bundles of Lemma 24.20 are at three consecutive shifts. ∎

*Remark* 24.25 (Six, seven and eight line bundles). The rules used in the proof of Theorem 24.23 depend on the shifts of the line bundles and the order of their exponents alone. The program of the archive described in the statement on data availability runs through every arrangement of $n \leq 8$ line bundles whose shifts form an interval, builds the graph of up edges, down edges and runs, admitting every excess and every parity of threefold shift that the degrees of Lemmas 24.8 and 24.9 allow for a run, and, for every set of up edges on which $\delta$ is taken to vanish that Lemma 24.20(i) admits, records the parities of the threefold pieces on the cycles with at most one edge in that set whose edge in the set is not forced to carry $\eta = 0$ by Lemma 24.20(ii). Runs and down edges are taken to carry nonzero components, and the condition that the edge outside $\Gamma_\delta$ point downwards in the one-sided order is dropped; both choices enlarge the set of cycles counted, so an arrangement for which no choice reaches both parities is one for which Proposition 24.19 and Lemma 24.21 exclude every twisting cochain. The counts are: 75 arrangements of four line bundles and 541 of five, none reaching a parity; 4683 of six and 47 293 of seven, of which 4 and 140 reach one parity and none both; and 545 835 of eight, of which 31 reach both. By Lemma 24.24 an arrangement whose shifts do not form an interval splits into arrangements of fewer line bundles, each contributing the parities it reaches, and with at most eight line bundles at most one part has six or more, so at most one parity is reached. Hence, in the Setup 24.17, every convolution with at most seven line-bundle pieces fails the criterion for every polarisation $\theta_h$, and so does every convolution with eight line bundles outside the thirty-one arrangements; those have their eight line bundles

at four to seven consecutive shifts with at most three at one shift, the simplest being $L^{l_1}[0], L^{l_2}[1], L^{l_3}[2], L^{l_4}[3], L^{l_5}[3], L^{l_6}[4], L^{l_7}[5], L^{l_8}[6]$ with $l_1 < \cdots < l_8$. These two statements are the output of the enumeration and rest on it; Theorem 24.23 rests on nothing computed.

24.8. **The support of an object meeting the criterion.** The results of this subsection do not count pieces. They start from one consequence of the criterion and read off where an object meeting it can be supported. Let $\mathcal{E}$ be a perfect complex on $A_0$ whose Chern character has the polarised shape with respect to $\theta_h$ and which meets the criterion, $\dim \mathrm{Ext}^2(\mathcal{E},\mathcal{E}) = 57$, so that $\sigma_{\mathcal{E}}$ is injective. For $v \in T\mathcal{D}_h$ the obstruction $\mathrm{ob}_{\mathcal{E}}(v) = v \lrcorner \mathrm{At}(\mathcal{E}) \in \mathrm{Ext}^2(\mathcal{E},\mathcal{E})$ to deforming $\mathcal{E}$ along $v$ is sent by $\sigma_{\mathcal{E}}$ to $v \lrcorner \mathrm{ch}(\mathcal{E})$ [BF03], which vanishes because $T\mathcal{D}_h$ annihilates $\theta_h$ and $W$; so

$$\mathrm{ob}_{\mathcal{E}}(v) = 0 \qquad \text{for every } v \in T\mathcal{D}_h. \tag{87}$$

Everything below is drawn from (87). The open subsets $U \subseteq A_0$ are Zariski open; inside the proofs we pass to the analytic spaces, which changes no Ext group and no Atiyah class on the projective $A_0$ [Ser56], and the cohomology with supports and the Chern characters with supports are those of the analytic spaces. For a closed subset $Z$ of an open $U$ we write $H^q_Z(U,F)$ and $\mathcal{H}^q_Z(F)$ for the cohomology and the local cohomology sheaves with supports in $Z$ [Har67]. Two facts are used throughout: if $Z$ is a submanifold of codimension $c$ and $F$ is locally free then $\mathcal{H}^q_Z(F) = 0$ for $q \neq c$ (depth below $c$, the Čech complex of $c$ local equations above $c$), so that $H^{c+p}_Z(U,F) = H^p(U, \mathcal{H}^c_Z(F))$; and if $C \subseteq Z$ is closed of codimension at least $c+1$ then $\mathcal{H}^0_C(\mathcal{H}^c_Z(F)) = 0$, because this sheaf embeds into $\mathcal{H}^c_C(F)$, which vanishes. So a section of $\mathcal{H}^c_Z(F)$ that vanishes outside $C$ vanishes.

For the threefolds we use the coordinates of Appendix D. Fix $\zeta \in \mu_4^3$ and $T = Z_\zeta$. The linear forms $n_m = z_m + \zeta_m z_{3+m}$ on the universal cover cut out the lift of $T$; their differentials $dn_m$, and the $dz_m$ for $m \leq 3$, are translation-invariant forms on $A_0$ and together a frame of $\Omega^1_{A_0}$. On a tubular neighbourhood $N$ of $T$ the $n_m$ themselves are single-valued, and $\pi(z) = z - \sum_m \bar{\zeta}_m n_m(z) \partial_{3+m}$ retracts $N$ onto $T$ with $z_m \circ \pi = z_m$; the pair $(n,\pi)$ identifies $N$ with $T \times D^3$, and a holomorphic function on $V \times (D^*)^3$, for $V \subseteq T$ open and $D^*$ the punctured disc, is a Laurent series in $n_1, n_2, n_3$ whose coefficients are holomorphic on $V$. The local cohomology $\mathcal{H}^3_T(\mathcal{O})$ is computed by the Čech complex of the cover of $N \setminus T$ by the three open sets $\{n_m \neq 0\}$, all of whose intersections are Stein; so on $T$ the sheaf $\mathcal{H}^3_T(\mathcal{O})$ is the sheaf of Laurent tails $\sum_{a,b,c \geq 1} g_{abc}\, n_1^{-a} n_2^{-b} n_3^{-c}$ with $g_{abc} \in \mathcal{O}_T$, and $\mathcal{H}^3_T(F) = \mathcal{H}^3_T(\mathcal{O}) \otimes F$ for $F$ locally free. The part of $\mathcal{H}^3_T(\mathcal{O})$ annihilated by the ideal $I_T$ is $\mathcal{O}_T \cdot [1/n_1 n_2 n_3]$. We write $\rho_T \colon \mathcal{H}^3_T(\mathcal{O}) \to \mathcal{O}_T$ for the map sending the class of $\sum g_{abc}\, n_1^{-a} n_2^{-b} n_3^{-c}$ to $g_{111}$, a morphism of sheaves of vector spaces on $T$ with $\rho_T(g\,[1/n_1 n_2 n_3]) = g|_T$ for $g$ holomorphic near $T$, extended coefficientwise to $\mathcal{H}^3_T(\Omega^q)$ in the frame above. For a coherent sheaf $G$ on $U$ supported in $T$ its multiplicity $\mu_T(G)$ is the length of $G$ at the generic point of $T$, the sum over $k \geq 0$ of the ranks of the $\mathcal{O}_T$-modules $I_T^k G / I_T^{k+1} G$; and $\mu_T(\mathcal{E}|_U) = \sum_q (-1)^q \mu_T(H^q(\mathcal{E}|_U))$ when every $H^q(\mathcal{E}|_U)$ is supported in $T$.

**Lemma 24.26** (The Chern character with supports)**.** *Let $U$ be open, $Z \subseteq U$ closed, and $\mathcal{F}$ a perfect complex on $U$ whose cohomology sheaves are supported in $Z$.*

*(i) The trace map $\mathrm{R}\mathcal{H}om(\mathcal{F},\mathcal{F}) \to \mathcal{O}_U$ factors in exactly one way through a morphism $\mathrm{tr}^Z \colon \mathrm{R}\mathcal{H}om(\mathcal{F},\mathcal{F}) \to \mathrm{R}\Gamma_Z \mathcal{O}_U$. Setting $\mathrm{ch}^Z_k(\mathcal{F}) = \mathrm{tr}^Z(\mathrm{At}(\mathcal{F})^k)/k! \in H^k_Z(U,\Omega^k)$ defines classes that forget to $\mathrm{ch}_k(\mathcal{F})$, are compatible with restriction to open subsets*

*and with enlarging $Z$, are additive in exact triangles, and satisfy*

$$v\lrcorner\,\mathrm{ch}^Z_{k+1}(\mathcal{F}) = \frac{1}{k!}\,\mathrm{tr}^Z\big(\mathrm{ob}_{\mathcal{F}}(v)\circ \mathrm{At}(\mathcal{F})^k\big) \in H^{k+2}_Z(U,\Omega^k)$$

*for $v \in H^1(U,T_U)$, where $\mathrm{ob}_{\mathcal{F}}(v) = v\lrcorner\mathrm{At}(\mathcal{F})$. In particular $\mathrm{ob}_{\mathcal{F}}(v) = 0$ implies $v\lrcorner\,\mathrm{ch}^Z_{k+1}(\mathcal{F}) = 0$ for every $k$.*

*(ii) For $T = Z_\zeta$ and every open $U$ meeting $T$ the class $[T] := \mathrm{ch}^T_3(\mathcal{O}_{T\cap U})$ in $H^0(U,\mathcal{H}^3_T(\Omega^3))$ is*

$$[T] = c\Big[\frac{dn_1\wedge dn_2\wedge dn_3}{n_1n_2n_3}\Big],$$

*with one constant $c\neq 0$, the same for all sixty-four threefolds.*

*(iii) If the cohomology sheaves of $\mathcal{F}$ are supported in $T\cap U$ then $\mathrm{ch}^T_3(\mathcal{F}) = \mu_T(\mathcal{F})\,[T]$. If they are supported in a union $Z$ of threefolds $T_i$ of the sixty-four, any two meeting in dimension at most two, then $\mathrm{ch}^Z_3(\mathcal{F}) = \sum_i \mu_{T_i}(\mathcal{F})[T_i]_Z$, where $[T_i]_Z$ is the image of $[T_i]$ in $H^3_Z(U,\Omega^3)$.*

*Proof.* (i) Let $j\colon U\setminus Z\to U$. In the triangle $\mathrm{R}\Gamma_Z\mathcal{O}\to\mathcal{O}\to\mathrm{R}j_*\mathcal{O}_{U\setminus Z}$ the third term receives no nonzero morphism from a complex $K$ whose cohomology sheaves are supported in $Z$, since $\mathrm{Hom}(K,\mathrm{R}j_*\mathcal{O}_{U\setminus Z}) = \mathrm{Hom}(K|_{U\setminus Z},\mathcal{O}_{U\setminus Z}) = 0$; so $\mathrm{Hom}(K,\mathrm{R}\Gamma_Z\mathcal{O})\to\mathrm{Hom}(K,\mathcal{O})$ is bijective. The complex $\mathrm{R}\mathcal{H}om(\mathcal{F},\mathcal{F})$ has cohomology supported in $Z$, which gives $\mathrm{tr}^Z$ and its uniqueness; the same bijection applied to the sources $\mathcal{F}^\vee\otimes\mathcal{F}\otimes\Omega^k$ transports every identity of the trace to $\mathrm{tr}^Z$: its compatibility with restriction to open subsets, with the morphism $\mathrm{R}\Gamma_Z\to\mathrm{R}\Gamma_{Z'}$ for $Z\subseteq Z'$, its additivity in triangles, and its graded cyclicity. The Atiyah class $\mathrm{At}(\mathcal{F})\in\mathrm{Ext}^1(\mathcal{F},\mathcal{F}\otimes\Omega^1)$ is compatible with restriction and with triangles, so the classes $\mathrm{ch}^Z_k$ have the listed properties, and cyclicity applied to the $k+1$ factors of $\mathrm{tr}^Z(\mathrm{At}^{k+1})$ gives $v\lrcorner\mathrm{tr}^Z(\mathrm{At}^{k+1}) = (k+1)\,\mathrm{tr}^Z((v\lrcorner\mathrm{At})\circ\mathrm{At}^k)$, which is the displayed identity.

(ii) The trace is $\mathcal{O}_U$-linear and $\mathrm{At}(\mathcal{O}_T)^3$ is a morphism between $\mathcal{O}_T$-modules, so $[T]$ is annihilated by $I_T$ on every open subset and is a section of $\mathcal{O}_{T\cap U}\,[1/n_1n_2n_3]\otimes\Omega^3$. By the compatibility with restriction it suffices to take $U = A_0$, where the coefficients are holomorphic functions on the compact connected $T$, hence constants: $[T] = \sum_a g_a[1/n_1n_2n_3]\,\omega_a$ with $g_a\in\mathbb{C}$ and $\omega_a$ running through the wedge products of three elements of the frame $dn_1,dn_2,dn_3,dz_1,dz_2,dz_3$. For $0<\lambda<1$ the map $\varphi_\lambda(z) = z-(1-\lambda)\sum_m\bar\zeta_m n_m(z)\partial_{3+m}$ is a biholomorphism of $N$ onto a smaller tubular neighbourhood of $T$, fixes $T$ pointwise, and satisfies $n_m\circ\varphi_\lambda = \lambda n_m$, $z_m\circ\varphi_\lambda = z_m$ and $\varphi_\lambda^*\mathcal{O}_T = \mathcal{O}_T$. The Atiyah class, the trace and the local cohomology are natural under biholomorphisms onto open subsets, and a section of $\mathcal{H}^3_T(\Omega^3)$ is determined by its restriction to any neighbourhood of $T$; so $\varphi_\lambda^*[T] = [T]$. If $\omega_a$ contains $a$ of the $dn_m$, then $\varphi_\lambda^*$ multiplies $g_a[1/n_1n_2n_3]\,\omega_a$ by $\lambda^{a-3}$; so $g_a = 0$ unless $a = 3$, and $[T] = c\,[dn_1\wedge dn_2\wedge dn_3/n_1n_2n_3]$. The automorphisms $z_{3+m}\mapsto i^{e_m}z_{3+m}$ of $A_0$ permute the sixty-four threefolds transitively and carry the forms $n_m$, $dn_m$ of one to those of another, so $c$ does not depend on $\zeta$. Forgetting supports sends $[T]$ to $\mathrm{ch}_3(\mathcal{O}_T)\in H^3(A_0,\Omega^3)$, which is the class of $T$ by the Riemann–Roch theorem for the embedding of $T$ [Ful98, Ch. 15] (the Chern character defined by the Atiyah class is the usual one on the projective $A_0$ [BF03]); and $[Z_\zeta]\theta^3 = 48\neq 0$ by Lemma D.1, so $c\neq 0$.

(iii) Both identities are between sections of a sheaf, so they may be checked locally on $U$ and, by the second fact recalled above, off a closed subset of codimension at least four. For the first, $\mathrm{ch}^T_3$ is additive and $\mathcal{F}$ is an iterated extension of its shifted cohomology sheaves, so let $G$ be a coherent sheaf supported in $T$; near a point of $T$ some $I_T^kG$ vanishes, and $G$ is an iterated extension of the $\mathcal{O}_T$-modules $G_k = I_T^kG/I_T^{k+1}G$, of ranks $r_k$ with $\sum_k r_k = \mu_T(G)$. Each $G_k$ is locally free of rank $r_k$ off a proper closed analytic subset of

$T$, where it is locally $\mathcal{O}_T^{r_k}$ and $\mathrm{ch}_3^T(G_k) = r_k[T]$ by additivity and (ii). For the second, let $C$ be the union of the pairwise intersections of the $T_i$, of codimension at least four. On $U \setminus C$ the $T_i$ are disjoint, $\mathrm{R}\Gamma_Z = \bigoplus_i \mathrm{R}\Gamma_{T_i}$ by the Mayer–Vietoris triangle for supports, and $\mathcal{F} \cong \mathrm{R}\Gamma_Z\mathcal{F} = \bigoplus_i \mathrm{R}\Gamma_{T_i}\mathcal{F}$, a sum of perfect complexes with cohomology supported in the $T_i$, whose multiplicities along $T_i$ are those of $\mathcal{F}$; so there $\mathrm{ch}_3^Z(\mathcal{F}) = \sum_i \mu_{T_i}(\mathcal{F})[T_i]_Z$ by additivity and the first identity. ∎

**Theorem 24.27** (Locality)**.** *Let $\mathcal{E}$ be as above, let $T = Z_\zeta$, and let $U \subseteq A_0$ be open with $T \cap U \neq \emptyset$, every cohomology sheaf of $\mathcal{E}|_U$ supported in $T$, and $\mu_T(\mathcal{E}|_U) \neq 0$. Then the union $S$ of the components of $T \setminus U$ of codimension one in $T$ is a nonempty ample divisor on $T$. In particular, if $T$ is an irreducible component of the support of $\mathcal{E}$ along which $\mathcal{E}$ has nonzero multiplicity, then the other components of the support cut $T$ in a set whose divisorial part is a nonempty ample divisor.*

The case of a shifted line bundle on $T \cap U$ is the case $\mu_T = \pm 1$ with one cohomology sheaf.

*Proof.* Let $v \in T\mathcal{D}_h$. Since $T$ is a subgroup, its normal bundle is $\mathcal{O}_T \otimes N_0$ with $N_0 = T_0A_0/T_0T$, and we write $\kappa_T(v) \in H^1(T, \mathcal{O}_T) \otimes N_0$ for the image of $v$ under $H^1(A_0, T_{A_0}) \to H^1(T, T_{A_0}|_T) \to H^1(T, N_T)$, and $\kappa_T(v)_m \in H^1(T, \mathcal{O}_T)$ for its image under $dn_m \colon N_0 \to \mathbb{C}$; on a cover of $A_0$ by open sets $U_a$ with $v$ represented by vector fields $v_{ab}$, the class $\kappa_T(v)_m$ is represented by the functions $dn_m(v_{ab})|_T$. By (87) and Lemma 24.26(i), (iii),

$$0 = v \lrcorner \mathrm{ch}_3^T(\mathcal{E}|_U) = \mu_T(\mathcal{E}|_U)\, v \lrcorner [T]$$

in $H^4_T(U, \Omega^2) = H^1(T \cap U, \mathcal{H}^3_T(\Omega^2))$. The right-hand side is represented by the cocycle $v_{ab} \lrcorner [T] = c\,[(\nu_1\, dn_2 \wedge dn_3 - \nu_2\, dn_1 \wedge dn_3 + \nu_3\, dn_1 \wedge dn_2)/n_1n_2n_3]$ with $\nu_m = dn_m(v_{ab})$, and $\rho_T$ sends it to $c\,(\kappa_T(v)_1\, dn_2 \wedge dn_3 - \kappa_T(v)_2\, dn_1 \wedge dn_3 + \kappa_T(v)_3\, dn_1 \wedge dn_2)$ restricted to $T \cap U$. As $\mu_T(\mathcal{E}|_U)c \neq 0$, every $\kappa_T(v)_m$ restricts to zero on $T \cap U$; the $dn_m$ are a basis of $N_0^\vee$, so $\kappa_T(v)$ lies in the kernel of the restriction $H^1(T, \mathcal{O}_T) \otimes N_0 \to H^1(T \cap U, \mathcal{O}_T) \otimes N_0$, for every $v \in T\mathcal{D}_h$.

We compute that kernel. Let $C$ be the union of the components of $T \setminus U$ of codimension at least two in $T$, so that $T \cap U = (T \setminus S) \setminus C$. Since $T$ is smooth, $\mathcal{O}_T$ has depth at least two along $C$, the local cohomology $H^1_C(T \setminus S, \mathcal{O})$ vanishes, and $H^1(T \setminus S, \mathcal{O}) \to H^1(T \cap U, \mathcal{O})$ is injective; so the kernel is that of $H^1(T, \mathcal{O}) \to H^1(T \setminus S, \mathcal{O})$, which is zero when $S = \emptyset$. Let $S \neq \emptyset$, let $H \subseteq T$ be the connected component of the stabiliser $\{x \in T : S + x = S\}$, an abelian subvariety, and let $q \colon T \to T' = T/H$. Each component of $S$ is $H$-invariant, $H$ being connected, so $S = q^{-1}(S')$ for the divisor $S' = q(S)$ of $T'$, whose stabiliser is the finite group of components of the stabiliser of $S$; hence $S'$ is ample [Mum70, §6, Application 1] and $T' \setminus S'$ is affine. By Poincaré reducibility there is an abelian subvariety $H' \subseteq T$ with $H \times H' \to T$ an isogeny, and the composite $H \times H' \to T \to T'$ is the projection to $H'$ followed by the isogeny $H' \to T'$. So the preimage of $T \setminus S$ in $H \times H'$ is $H \times U'$ with $U' \subseteq H'$ the preimage of the affine $T' \setminus S'$ under a finite map, which is affine. Pullback along the finite étale map $H \times U' \to T \setminus S$ is injective on $H^1(\mathcal{O})$, since composed with the trace it is multiplication by the degree, and by the Künneth formula and $H^1(U', \mathcal{O}) = 0$,

$$H^1(H \times U', \mathcal{O}) = H^1(H, \mathcal{O}) \otimes H^0(U', \mathcal{O}).$$

The composite $H^1(T, \mathcal{O}) \to H^1(H \times U', \mathcal{O})$ factors through $H^1(H \times H', \mathcal{O}) = H^1(H, \mathcal{O}) \oplus H^1(H', \mathcal{O})$, an isomorphism for the isogeny, followed by the restriction to $H \times U'$, which is injective on the first summand and zero on the second. So the kernel is the second summand, which is $q^*H^1(T', \mathcal{O})$: the classes of the invariant $(0,1)$-forms on $T$ vanishing

on the tangent space of $H$. Hence $\kappa_T(v)(\bar u)=0$ for every $v\in T\mathcal{D}_h$ and every $u\in T_0H$, where $H=T$ when $S=\emptyset$.

Finally we compute $\kappa_T(v)$ in the coordinates of Appendix D. Write $v=\sum_{k,l} v_{lk}\,d\bar z_k\otimes\partial_l$, so that $v$ replaces $dz_l$ by $\sum_k v_{lk}d\bar z_k$. The $K$-linear maps have $v_{lk}\neq0$ only when one of $k,l$ is at most 3 and the other at least 4, and

$$v\lrcorner\theta_h=\tfrac{i}{2}\sum_{k<l}(h_lv_{lk}-h_kv_{kl})\,d\bar z_k\wedge d\bar z_l$$

vanishes exactly when $h_lv_{lk}=h_kv_{kl}$ for all $k<l$; so $T\mathcal{D}_h$ is the nine-dimensional space of the $v$ with $v_{3+m,j}=Y_{mj}$ and $v_{j,3+m}=(h_{3+m}/h_j)Y_{mj}$ for $j,m\le3$, $Y$ any complex $3\times3$ matrix, and the other entries zero. The threefold $T=Z_\zeta$ is $\{z_j+\zeta_jz_{3+j}=0,\ j=1,2,3\}$, with tangent space spanned by $w_j=\zeta_j\partial_j-\partial_{3+j}$ and normal space $N_0$ with basis $f_m=[\partial_{3+m}]$, so that $[\partial_m]=\bar\zeta_mf_m$. The class $\kappa_T(v)$ is that of the invariant form $\sum_{k,l}v_{lk}\,d\bar z_k|_T\otimes[\partial_l]$, and on $\bar w_j=\bar\zeta_j\bar\partial_j-\bar\partial_{3+j}$, where $d\bar z_j(\bar w_j)=\bar\zeta_j$ and $d\bar z_{3+j}(\bar w_j)=-1$,

$$\kappa_T(v)(\bar w_j)=\sum_{m=1}^{3}\Big(\bar\zeta_jY_{mj}-\frac{h_{3+j}}{h_m}\,\bar\zeta_mY_{jm}\Big)f_m.$$

Let $0\neq u=\sum_j u_jw_j\in T_0T$ with $u_a\neq0$, choose $b\neq a$, and let $Y$ have the entry $Y_{ab}=1$ and all others 0. Then

$$\kappa_T(v)(\bar u)=\bar\zeta_b\Big(\bar u_bf_a-\frac{h_{3+a}}{h_b}\,\bar u_af_b\Big)\neq0.$$

So $T_0H=0$: $S$ is nonempty, $H$ is trivial, $S=S'$ has finite stabiliser, and $S$ is ample. ∎

Theorem 24.27 says where the rest of the support must cut a threefold of nonzero multiplicity; the next results say where it can lie. Fix $\zeta$ with $\zeta_1\zeta_2\zeta_3=\pm1$, let $T=Z_\zeta$, fix $j\in\{1,2,3\}$ with $\{j,k,l\}=\{1,2,3\}$, and let $\zeta^{(j)}$ be $\zeta$ with the sign of the $j$-th coordinate changed, $T'=Z_{\zeta^{(j)}}$. On the $j$-th factor $B_j$ the curves $C_{\zeta_j}$ and $C_{-\zeta_j}$ meet in the set $P_j$ of the four points of order two of $C_{\zeta_j}$ ($|2\zeta_j|^2=4$ by Lemma D.1), so

$$S:=T\cap T'=P_j\times C_{\zeta_k}\times C_{\zeta_l}$$

is a disjoint union of four surfaces, and $Z=T\cup T'$ is cut out near $S$ by $n_jn_j'=n_k=n_l=0$, where $n_j'=z_j-\zeta_jz_{3+j}$ is the $j$-th linear form of $T'$. For $y\in C_{\zeta_k}\times C_{\zeta_l}$ let $\Gamma_y=C_{\zeta_j}\times\{y\}\subseteq T$ and $\Gamma_y'=C_{-\zeta_j}\times\{y\}\subseteq T'$; each meets $S$ in the four points $P_j\times\{y\}$. On $C_{\zeta_j}$ one has $z_{3+j}=-\bar\zeta_jz_j$, so $n_j'|_{\Gamma_y}=2z_j$ is a coordinate vanishing at the four points and $dn_j'|_{\Gamma_y}=2\,dz_j$ is a nonzero invariant form; on $C_{-\zeta_j}$ likewise $n_j|_{\Gamma_y'}=2z_j$ and $dn_j|_{\Gamma_y'}=2\,dz_j$.

**Lemma 24.28** (The curves through the four points)**.** *Let $y=(y_k,y_l)\in C_{\zeta_k}\times C_{\zeta_l}$ with $y_k\notin C_\xi$ for $\xi\neq\zeta_k$ and $y_l\notin C_\xi$ for $\xi\neq\zeta_l$, which excludes finitely many values of each coordinate. The threefolds $Z_\xi$, $\xi\in\mu_4^3$, meeting $\Gamma_y$ or $\Gamma_y'$ are those with $\xi_k=\zeta_k$ and $\xi_l=\zeta_l$; among the thirty-two with $\xi_1\xi_2\xi_3=\pm1$ they are $T$ and $T'$ alone.*

*Proof.* $Z_\xi\cap\Gamma_y=(C_{\xi_j}\cap C_{\zeta_j})\times(\{y_k\}\cap C_{\xi_k})\times(\{y_l\}\cap C_{\xi_l})$; the first factor contains the origin, and $y_k\in C_{\xi_k}$ for $\xi_k\neq\zeta_k$ only for the finitely many $y_k\in C_{\xi_k}\cap C_{\zeta_k}$. The same holds for $\Gamma_y'$. The condition $\xi_k=\zeta_k$, $\xi_l=\zeta_l$, $\xi_1\xi_2\xi_3=\pm1$ leaves $\xi_j=\pm\zeta_j$. ∎

**Theorem 24.29** (Two threefolds meeting along a surface)**.** *Let $\mathcal{E}$ be as above and let $U\subseteq A_0$ be open such that every cohomology sheaf of $\mathcal{E}|_U$ is supported in $Z=T\cup T'$ and $\Gamma_y\cup\Gamma_y'\subseteq U$ for general $y\in C_{\zeta_k}\times C_{\zeta_l}$. Then $\mu_T(\mathcal{E}|_U)=\mu_{T'}(\mathcal{E}|_U)=0$.*

*Proof.* Let $\mu = \mu_T(\mathcal{E}|_U)$ and $\mu' = \mu_{T'}(\mathcal{E}|_U)$, so that by Lemma 24.26(iii)

$$\mathrm{ch}_3^Z(\mathcal{E}|_U) = \mu[T]_Z + \mu'[T']_Z.$$

Let $v \in T\mathcal{D}_h$ be the vector whose matrix $Y$, in the notation of the proof of Theorem 24.27, is $E_{jk}$, the matrix with the single entry $Y_{jk} = 1$. By (87) and Lemma 24.26(i) the class $v \lrcorner \mathrm{ch}_3^Z(\mathcal{E}|_U)$ vanishes in $H^4_Z(U, \Omega^2)$. The Mayer–Vietoris sequence for supports

$$H^4_S(U,\Omega^2) \to H^4_T(U,\Omega^2) \oplus H^4_{T'}(U,\Omega^2) \to H^4_Z(U,\Omega^2)$$

is exact, so some $s \in H^4_S(U,\Omega^2)$ maps to the pair

$$(\mu\, v\lrcorner[T],\ \pm\mu'\, v\lrcorner[T']).$$

Near $S$ the forms $dn_j, dn'_j, dn_k, dn_l, dz_k, dz_l$ are a frame of $\Omega^1$, the functions $n_j, n'_j, n_k, n_l$ are transverse coordinates to $S$, and the coordinates $z_k, z_l$ of $T$ restrict to coordinates on each component of $S$; as $S$ is smooth of codimension four, $H^4_S(U,\Omega^2) = H^0(S \cap U, \mathcal{H}^4_S(\Omega^2))$, and $s$ is a Laurent tail

$$s = \sum_{\alpha,\beta,\gamma,\delta \geq 1} c_{\alpha\beta\gamma\delta}\, n_j^{-\alpha}\, n_j'^{-\beta}\, n_k^{-\gamma}\, n_l^{-\delta}$$

in each coefficient of the frame of $\Omega^2$, the $c_{\alpha\beta\gamma\delta}$ holomorphic functions on $S \cap U$. The map $H^4_S(U,\Omega^2) \to H^4_T(U,\Omega^2) = H^1(T \cap U, \mathcal{H}^3_T(\Omega^2))$ forgets the support: since $\mathrm{R}\Gamma_S = \mathrm{R}\Gamma_S \mathrm{R}\Gamma_T$ it is the connecting map of $0 \to \mathcal{H}^3_T(\Omega^2) \to j_*\mathcal{H}^3_T(\Omega^2)|_{T\setminus S} \to \mathcal{H}^1_S(\mathcal{H}^3_T(\Omega^2)) \to 0$, which sends the tail $s$, read as a section over $(T \cap U) \setminus S$ of $\mathcal{H}^3_T(\Omega^2)$ whose coefficients have poles along $S$, to its class. Applying $\rho_T$ to that class gives the class in $H^1(T \cap U, \mathcal{O}_T)$ of the principal parts $\sum_\beta c_{1\beta11}\, n_j'^{-\beta}$ along $S$, and symmetrically for $T'$ with $\sum_\alpha c_{\alpha111}\, n_j^{-\alpha}$.

We compare the coefficients of $dn_j \wedge dn_l$. In $v\lrcorner[T']$ this coefficient is zero, because $[T'] = c\,[dn'_j \wedge dn_k \wedge dn_l / n'_j n_k n_l]$ contains no $dn_j$; in $v\lrcorner[T]$ it is $\pm c\,[dn_k(v)/n_j n_k n_l]$, and $\rho_T$ sends $\mu\, v\lrcorner[T]$ to $\pm\mu c\, \kappa_T(v)_k$. Let $c_{\alpha\beta\gamma\delta}$ be the coefficients of $dn_j \wedge dn_l$ in $s$. Restricting to the curve $\Gamma'_y \subseteq U$, $y$ general, the $T'$ side gives that the class of the principal parts $\sum_\alpha c_{\alpha111}(p)\, n_j^{-\alpha}$ at the four points $p \in P_j \times \{y\}$ vanishes in $H^1(\Gamma'_y, \mathcal{O})$; by Serre duality its pairing with the holomorphic form $dn_j|_{\Gamma'_y}$, the sum of the residues $\sum_p c_{1111}(p)$, is zero. Restricting to $\Gamma_y \subseteq U$, the $T$ side gives that the class of the principal parts $\sum_\beta c_{1\beta11}(p)\, n_j'^{-\beta}$ equals $\pm\mu c\, \kappa_T(v)_k|_{\Gamma_y}$ in $H^1(\Gamma_y, \mathcal{O})$. Pair both with $dn'_j|_{\Gamma_y}$. With the normalisation in which a class represented by a Dolbeault form $\phi$ pairs with $\omega$ to $(2\pi i)^{-1}\int \phi \wedge \omega$, a class given by principal parts $g_p$ pairs to $\sum_p \mathrm{Res}_p(g_p\omega)$ up to a sign independent of the class; so the left side pairs to $\pm\sum_p c_{1111}(p) = 0$. The class $\kappa_T(v)_k$ is represented by the invariant form $\sum_b (v_{kb} + \zeta_k v_{3+k,b})\, d\bar z_b|_T$, since $dn_k(\partial_k) = 1$, $dn_k(\partial_{3+k}) = \zeta_k$ and $dn_k$ kills the other $\partial_m$; on $\Gamma_y$ only $d\bar z_j$ and $d\bar z_{3+j} = -\zeta_j\, d\bar z_j$ survive, and the entries of $v$ for $Y = E_{jk}$ are $v_{k,3+j} = h_{3+j}/h_k$ and $v_{3+k,j} = 0$ among those that occur, so

$$\kappa_T(v)_k|_{\Gamma_y} = -\zeta_j \frac{h_{3+j}}{h_k}\, d\bar z_j|_{\Gamma_y},$$

$$\frac{1}{2\pi i}\int_{\Gamma_y} \kappa_T(v)_k \wedge dn'_j = -\frac{\zeta_j h_{3+j}}{\pi i\, h_k}\int_{C_{\zeta_j}} d\bar z_j \wedge dz_j \neq 0.$$

Hence $\mu = 0$. The coefficients of $dn'_j \wedge dn_l$ give $\mu' = 0$ in the same way, with the roles of $T$ and $T'$ exchanged: there $v\lrcorner[T]$ has coefficient zero, the restriction to $\Gamma_y$ paired with $dn'_j$ gives $\sum_p c'_{1111}(p) = 0$ for the coefficients $c'$ of that component, and on $\Gamma'_y$, where $d\bar z_{3+j} = \zeta_j\, d\bar z_j$, the class $\kappa_{T'}(v)_k|_{\Gamma'_y} = \zeta_j(h_{3+j}/h_k)\, d\bar z_j$ pairs with $dn_j$ to a nonzero number. ∎

**Proposition 24.30** (Where the rest of the support lies)**.** *Let $\mathcal{E}$ be as above, let $\zeta_1\zeta_2\zeta_3 = \pm 1$, and suppose that $T = Z_\zeta$ lies in the support of $\mathcal{E}$ with $\mu_T(\mathcal{E}|_{U_T}) \neq 0$, where $U_T$ is the complement of the components of the support other than $T$. Then for each $j \in \{1,2,3\}$ there is an irreducible component $R_j$ of the support of $\mathcal{E}$ of dimension at least three, not one of the thirty-two threefolds $Z_\xi$ with $\xi_1\xi_2\xi_3 = \pm 1$, which meets $\Gamma_y \cup \Gamma'_y$ for every $y \in C_{\zeta_k} \times C_{\zeta_l}$; and $R_j \cap (T \cup T')$ has an irreducible component that maps onto $C_{\zeta_k} \times C_{\zeta_l}$.*

*Proof.* Let $R$ be the union of the components of the support not contained in $T \cup T'$ and $U = A_0 \setminus R$. Then every cohomology sheaf of $\mathcal{E}|_U$ is supported in $T \cup T'$ and $\mu_T(\mathcal{E}|_U) = \mu_T(\mathcal{E}|_{U_T}) \neq 0$, so by Theorem 24.29 the curves $\Gamma_y \cup \Gamma'_y$ are not contained in $U$ for general $y$: the closed set $R \cap (T \cup T')$ projects onto a dense, hence the whole, subset of $C_{\zeta_k} \times C_{\zeta_l}$ under the proper projection $(z_k, z_l)$. Some irreducible component $R_j$ of $R$ then has $R_j \cap (T \cup T')$ projecting onto $C_{\zeta_k} \times C_{\zeta_l}$, so one of its components does, and $R_j$ meets every $\Gamma_y \cup \Gamma'_y$. If $\dim R_j \leq 2$ then $R_j = R_j \cap (T \cup T') \subseteq T \cup T'$, which is excluded; so $\dim R_j \geq 3$. By Lemma 24.28 no $Z_\xi$ with $\xi_1\xi_2\xi_3 = \pm 1$ other than $T$ and $T'$ meets $\Gamma_y$ or $\Gamma'_y$ for general $y$. ∎

**Corollary 24.31** (Objects supported on the thirty-two threefolds)**.** *No perfect complex $\mathcal{E}$ on $A_0$ whose support lies in the union of the thirty-two threefolds $Z_\xi$, $\xi_1\xi_2\xi_3 = \pm 1$, has a Chern character of polarised shape with respect to some $\theta_h$ and meets the criterion. In particular no convolution of pieces $\mathcal{O}_{Z_\xi} \otimes L^k[s]$ over these threefolds, at any twists and shifts, does so; the same holds for the thirty-two threefolds with $\xi_1\xi_2\xi_3 = \pm i$.*

*Proof.* Suppose $\mathcal{E}$ has polarised shape and meets the criterion. For each $\xi$ with $\xi_1\xi_2\xi_3 = \pm 1$ and each $j$, let $U$ be the complement of the thirty threefolds other than $Z_\xi$ and $Z_{\xi^{(j)}}$. Every cohomology sheaf of $\mathcal{E}|_U$ is supported in $Z_\xi \cup Z_{\xi^{(j)}}$, and $\Gamma_y \cup \Gamma'_y \subseteq U$ for general $y$ by Lemma 24.28; so $\mu_{Z_\xi}(\mathcal{E}) = 0$ by Theorem 24.29. Two distinct threefolds meet in dimension at most two, so Lemma 24.26(iii) with $U = A_0$ and $Z$ the union of the thirty-two gives $\mathrm{ch}_3^Z(\mathcal{E}) = \sum_\xi \mu_{Z_\xi}(\mathcal{E})[Z_\xi]_Z = 0$, and forgetting supports, $\mathrm{ch}_3(\mathcal{E}) = 0$. But $\mathrm{ch}_3(\mathcal{E}) = N\omega + c_3\theta_h^3$ with $N\omega \neq 0$, and $\omega \in W$ and $\theta_h^3$ are linearly independent, $W$ having the weights $\pm 2(e_1 + e_2 + e_3)$ and $\theta_h^3$ weight zero (Lemma 24.21). A convolution of pieces over the threefolds is supported in their union. The automorphism $z_4 \mapsto iz_4$ of $A_0$ preserves $\theta$, every $\theta_h$ and $W$, and exchanges the two families of thirty-two threefolds. ∎

Corollary 24.31 treats one family at a time; the threefolds of both families together need one more relation, read off from a line of four of them. Fix $j \in \{1,2,3\}$ with $\{j,k,l\} = \{1,2,3\}$ and $\xi_k, \xi_l \in \mu_4$. For $a \in \mu_4$ let $T_a = Z_x$ with $x_j = a$, $x_k = \xi_k$, $x_l = \xi_l$; two of the four lie in each family. Let $C_a = \{z_j + az_{3+j} = 0\} \subseteq B_j$, let $Q = \bigcup_a C_a$, a divisor on $B_j$, and let $\Pi = C_{\xi_k} \times C_{\xi_l} \subseteq B' = B_k \times B_l$, a surface, so that $A_0 = B_j \times B'$ and

$$Z := \bigcup_{a \in \mu_4} T_a = Q \times \Pi.$$

Two of the $T_a$ meet in $(C_a \cap C_b) \times \Pi$, a union of surfaces, so Lemma 24.26(iii) applies to $Z$. The divisor $Q$ is ample: the connected component $H$ of its stabiliser $\{x \in B_j : Q + x = Q\}$ is an abelian subvariety which, being connected, leaves each $C_a$ invariant, so $H \subseteq C_1 \cap C_i$, a finite set, and $Q$ has finite stabiliser [Mum70, §6, Application 1]. Hence $B_j \setminus Q$ is affine.

**Lemma 24.32** (The four threefolds through a surface)**.** *Let $\mathcal{E}$ be as above and let $U \subseteq A_0$ be open such that every cohomology sheaf of $\mathcal{E}|_U$ is supported in $Z$ and $Q \times \{y\} \subseteq U$ for*

*general $y \in \Pi$. Then the multiplicities $\mu_a = \mu_{T_a}(\mathcal{E}|_U)$ satisfy*

$$\sum_{a\in\mu_4} \mu_a = 0, \qquad \sum_{a\in\mu_4} a\,\mu_a = 0,$$

*that is, $\mu_a = m\,a^2$ for one integer $m$: $\mu_1 = \mu_{-1} = -\mu_i = -\mu_{-i}$. In particular, if one of the four threefolds carries no cohomology sheaf of $\mathcal{E}|_U$, none of them has nonzero multiplicity.*

*Proof.* Let $n_a = z_j + a z_{3+j}$, $n_k = z_k + \xi_k z_{3+k}$, $n_l = z_l + \xi_l z_{3+l}$ and $f = \prod_a n_a$, so that near $Z$ the set $Z$ is cut out by $f = n_k = n_l = 0$, a regular sequence, $Q = \{f = 0\}$ in $B_j$ and $\Pi = \{n_k = n_l = 0\}$ in $B'$. As for one threefold, the pair $(n_k, n_l)$ and the retraction along $\partial_{3+k}, \partial_{3+l}$ identify a tubular neighbourhood of $\Pi$ in $B'$ with $\Pi \times D^2$, and the local cohomology $\mathcal{H}^3_Z(F)$ of a locally free sheaf $F$, computed by the Čech complex of the cover of the complement of $Z$ by $\{f \neq 0\}$, $\{n_k \neq 0\}$, $\{n_l \neq 0\}$, is the sheaf on $Z$ of Laurent tails $\sum_{\beta,\gamma\geq 1} g_{\beta\gamma}\, n_k^{-\beta} n_l^{-\gamma}$ whose coefficients $g_{\beta\gamma}$ are sections of $\mathcal{H}^1_Q(F)$: over $W \times V$, with $W \subseteq B_j$ and $V \subseteq \Pi$ open and Stein, a coefficient is a section of $F$ on $(W \setminus Q) \times V$ modulo the sections on $W \times V$. In particular $\mathcal{H}^1_{C_a}(F) \subseteq \mathcal{H}^1_Q(F)$, and $[T_a] = c\,[dn_a \wedge dn_k \wedge dn_l / n_a n_k n_l]_Z$ has the single tail $n_k^{-1} n_l^{-1}$ with coefficient $c\,[dn_a/n_a] \wedge dn_k \wedge dn_l$, where $[dn_a/n_a] \in H^0(C_a, \mathcal{H}^1_{C_a}(\Omega^1_{B_j}))$ is the class of the form $dn_a/n_a$ with a pole along $C_a$.

Let $v \in T\mathcal{D}_h$ have the matrix $Y = E_{jk}$, as in the proof of Theorem 24.29: then $v$ replaces $dz_{3+j}$ by $d\bar z_k$ and $dz_k$ by $(h_{3+j}/h_k)\, d\bar z_{3+j}$ and kills the other $dz_m$, so $dn_a(v) = a\, d\bar z_k$, $dn_k(v) = (h_{3+j}/h_k)\, d\bar z_{3+j}$ and $dn_l(v) = 0$. By (87) and Lemma 24.26(i), (iii),

$$\begin{aligned} 0 = v \lrcorner \operatorname{ch}^Z_3(\mathcal{E}|_U) &= \sum_a \mu_a\, v \lrcorner [T_a]_Z \\ &= c \sum_a \mu_a \Big[ \frac{a\, d\bar z_k\, dn_k \wedge dn_l - (h_{3+j}/h_k)\, d\bar z_{3+j}\, dn_a \wedge dn_l}{n_a n_k n_l} \Big]_Z \end{aligned}$$

in $H^4_Z(U, \Omega^2) = H^1(Z \cap U, \mathcal{H}^3_Z(\Omega^2))$, the classes of the invariant $(0,1)$-forms $d\bar z_k$, $d\bar z_{3+j}$ in $H^1(U, \mathcal{O})$ acting by cup product. The decomposition $\Omega^2_{A_0} = \Omega^2_{B_j} \oplus (\Omega^1_{B_j} \boxtimes \Omega^1_{B'}) \oplus \Omega^2_{B'}$ into locally free summands splits the local cohomology; $dn_k \wedge dn_l$ lies in the third summand and $dn_a \wedge dn_l$ in the second, so the second summand of the identity reads

$$[d\bar z_{3+j}] \cup \sum_a \mu_a \Big[ \frac{dn_a \wedge dn_l}{n_a n_k n_l} \Big]_Z = 0$$

in $H^1(Z \cap U, \mathcal{H}^3_Z(\Omega^1_{B_j} \boxtimes \Omega^1_{B'}))$. Reading off the coefficient of $dn_l\, n_k^{-1} n_l^{-1}$ in the frame $dn_k, dn_l, dz_k, dz_l$ of $\Omega^1_{B'}$ is a morphism of sheaves of vector spaces $\mathcal{H}^3_Z(\Omega^1_{B_j} \boxtimes \Omega^1_{B'}) \to \mathcal{H}^1_Q(\Omega^1_{B_j}) \boxtimes \mathcal{O}_\Pi$ on $Z$, linear over the functions pulled back from $B_j$; restricting to $Q \times \{y\} \subseteq Z \cap U$ for general $y$, where the $(0,1)$-form $d\bar z_{3+j}$ restricts to the same form on $B_j$, gives

$$[d\bar z_{3+j}] \cup \sum_a \mu_a\, [dn_a/n_a] = 0$$

in $H^1(Q, \mathcal{H}^1_Q(\Omega^1_{B_j})) = H^2_Q(B_j, \Omega^1_{B_j})$. Since $B_j \setminus Q$ is affine, $H^1(B_j \setminus Q, \Omega^1) = 0$, and the exact sequence $H^1(B_j \setminus Q, \Omega^1) \to H^2_Q(B_j, \Omega^1) \to H^2(B_j, \Omega^1)$ shows that forgetting the support is injective on $H^2_Q(B_j, \Omega^1)$; it is compatible with the cup product by $H^1(B_j, \mathcal{O})$. The argument of Lemma 24.26(ii) in codimension one, with the scaling $z \mapsto z - (1-\lambda)\bar a\, n_a(z) \partial_{3+j}$ of $B_j$, which fixes $C_a$, multiplies $n_a$ by $\lambda$ and preserves $z_j$, shows that $\operatorname{ch}^{C_a}_1(\mathcal{O}_{C_a}) = c_1 [dn_a/n_a]$ in $H^0(C_a, \mathcal{H}^1_{C_a}(\Omega^1))$ with a constant $c_1$, the same for the four curves since $z_{3+j} \mapsto i z_{3+j}$

permutes them and carries $n_a$ to $n_{ia}$, and $c_1 \neq 0$ since forgetting the support gives $\mathrm{ch}_1(\mathcal{O}_{C_a}) = [C_a] \neq 0$. So the image of $[dn_a/n_a]$ in $H^1(B_j, \Omega^1)$ is $c_1^{-1}[C_a]$, and

$$[d\bar{z}_{3+j}] \cup \sum_a \mu_a [C_a] = 0 \quad \text{in } H^2(B_j, \Omega^1) = H^{1,2}(B_j).$$

By Lemma D.1(i) with $(u, v) = (z_j, z_{3+j})$, $[C_a] = \frac{i}{2}(dz_j + a\, dz_{3+j}) \wedge (d\bar{z}_j + \bar{a}\, d\bar{z}_{3+j})$, so

$$[C_a] \wedge d\bar{z}_{3+j} = \tfrac{i}{2}(dz_j + a\, dz_{3+j}) \wedge d\bar{z}_j \wedge d\bar{z}_{3+j},$$

and the invariant forms $dz_j \wedge d\bar{z}_j \wedge d\bar{z}_{3+j}$ and $dz_{3+j} \wedge d\bar{z}_j \wedge d\bar{z}_{3+j}$ are linearly independent in $H^{1,2}(B_j)$. Their coefficients $\sum_a \mu_a$ and $\sum_a a\mu_a$ vanish. Separating real and imaginary parts of the second relation gives $\mu_1 = \mu_{-1}$ and $\mu_i = \mu_{-i}$, and the first then gives $\mu_i = -\mu_1$; so $\mu_a = \mu_1 a^2$, and if some $\mu_a$ vanishes all do. ∎

**Theorem 24.33** (Objects supported on the sixty-four threefolds)**.** *No perfect complex $\mathcal{E}$ on $A_0$ whose support lies in the union of the sixty-four threefolds $Z_x$, $x \in \mu_4^3$, has a Chern character of polarised shape with respect to some $\theta_h$ and meets the criterion. In particular no convolution of pieces $\mathcal{O}_{Z_x} \otimes L^k[s]$ over threefolds of both families, at any twists and shifts, does so.*

*Proof.* Suppose $\mathcal{E}$ has polarised shape and meets the criterion, and write $\mu_x = \mu_{Z_x}(\mathcal{E})$. Fix $j$ and $(\xi_k, \xi_l)$ as above and let $U$ be the complement of the sixty threefolds $Z_{x'}$ with $(x'_k, x'_l) \neq (\xi_k, \xi_l)$. Every cohomology sheaf of $\mathcal{E}|_U$ is supported in $Z = \bigcup_a T_a$, and $\mu_{T_a}(\mathcal{E}|_U) = \mu_x$ for $x_j = a$, the generic point of $T_a$ lying in $U$. For $y = (y_k, y_l) \in \Pi$ with $y_k \notin C_\xi$ for $\xi \neq \xi_k$ and $y_l \notin C_\xi$ for $\xi \neq \xi_l$, which excludes finitely many values of each coordinate, $Z_{x'} \cap (Q \times \{y\}) = (C_{x'_j} \cap Q) \times (\{y_k\} \cap C_{x'_k}) \times (\{y_l\} \cap C_{x'_l})$ is empty unless $(x'_k, x'_l) = (\xi_k, \xi_l)$; so $Q \times \{y\} \subseteq U$ for general $y$, and Lemma 24.32 gives that $\mu_x/x_j^2$ does not depend on $x_j$ when $x_k, x_l$ are fixed. This holds for $j = 1, 2, 3$, so $\mu_x/(x_1^2 x_2^2 x_3^2) = \mu_x/\sigma(x)^2$ is independent of each coordinate of $x$ separately, hence constant: $\mu_x = m\,\sigma(x)^2$ for one integer $m$. Two distinct threefolds meet in dimension at most two, so Lemma 24.26(iii) with $U = A_0$ and $Z$ the union of the sixty-four, followed by forgetting supports and (95) with $m = 2$ there, gives

$$\mathrm{ch}_3(\mathcal{E}) = \sum_x \mu_x [Z_x] = m \sum_{\omega \in \mu_4} \omega^2 S(\omega) = m \prod_{j=1}^{3} T_{2,j} = 0.$$

But $\mathrm{ch}_3(\mathcal{E}) = N\omega + c_3\theta_h^3 \neq 0$, as in the proof of Corollary 24.31. A convolution of pieces over the threefolds is supported in their union. ∎

The theorem contains Corollary 24.31, since a complex supported on one family has $\mu_x = 0$ on the other, hence $m = 0$; the corollary is kept for its proof, which localises to two threefolds and is what Proposition 24.30 uses. The relations of Lemma 24.32 hold along every line of four threefolds in the support, whatever else the support contains, as long as the rest of the support misses $Q \times \{y\}$ for general $y$.

Theorem 24.33 removes every convolution built from threefold pieces alone, of one family or of both, at any twists and shifts; it leaves the convolutions whose support reaches outside the sixty-four threefolds. For these the next two results say what the rest of the support can be. By (94) the Weil class is $\omega = (p + qi)\alpha + (p - qi)\bar{\alpha}$ with $p, q \in \mathbb{Q}$; for $x \in \mu_4^3$ let $\sigma(x) = x_1 x_2 x_3$ and $c(x) = (x_1^2, x_2^2, x_3^2) \in \{\pm 1\}^3$. The fibres of $c$ are the eight orbits of the group $\{\pm 1\}^3$ acting on $\mu_4^3$ by sign changes; $\sigma(x)^2 = \prod_j c(x)_j$, so the four orbits with $\prod_j c_j = 1$ make up the family $\sigma = \pm 1$ and the other four the family $\sigma = \pm i$.

**Lemma 24.34** (The four orbits)**.** *Let $\nu\colon \mu_4^3 \to \mathbb{Q}$ satisfy $\sum_x \nu_x[Z_x] \equiv N\omega$ modulo $\theta_h H^4(A_0,\mathbb{C})$. Then for every $c \in \{\pm1\}^3$*

$$\sum_{c(x)=c} \nu_x \sigma(x) = \begin{cases} Np/4, & \prod_j c_j = 1, \\ -iNq/4, & \prod_j c_j = -1. \end{cases}$$

*So if $Np \neq 0$ each orbit of the family $\sigma = \pm1$ carries an $x$ with $\nu_x \neq 0$, and if $Nq \neq 0$ each orbit of the family $\sigma = \pm i$ does.*

*Proof.* For $\varepsilon \in \{\pm1\}^3$ let $X_\varepsilon = \prod_j (c_j \text{ if } \varepsilon_j = 1,\ \bar{c}_j \text{ if } \varepsilon_j = -1)$, of weight $2\varepsilon$; $X_\varepsilon$ spans the part of $H^6(A_0,\mathbb{C})$ of that weight, $H^4(A_0,\mathbb{C})$ has no class of weight $\pm 2\varepsilon$, and $\theta_h$ has weight zero, so the component of weight $2\varepsilon$ of $\theta_h H^4$ is zero and $\alpha = X_{(1,1,1)}$, $\bar{\alpha} = X_{(-1,-1,-1)}$. In $[Z_x] = \prod_j (a_j + b_j + \bar{x}_j c_j + x_j \bar{c}_j)$ the coefficient of $X_\varepsilon$ is $\prod_{\varepsilon_j=1} \bar{x}_j \prod_{\varepsilon_j=-1} x_j = \sigma(x) \prod_{\varepsilon_j=1} \bar{x}_j^2 = \sigma(x) \prod_{\varepsilon_j=1} c(x)_j$, as $x_j^4 = 1$. Writing $W_c = \sum_{c(x)=c} \nu_x \sigma(x)$, the coefficient of $X_\varepsilon$ in $\sum_x \nu_x [Z_x]$ is $\sum_c W_c \chi_\varepsilon(c)$ with $\chi_\varepsilon(c) = \prod_{\varepsilon_j=1} c_j$, and the $\chi_\varepsilon$ are the eight characters of $\{\pm1\}^3$. The hypothesis says that this coefficient is $N(p+qi)$ for $\varepsilon = (1,1,1)$, $N(p-qi)$ for $\varepsilon = (-1,-1,-1)$ and 0 otherwise. Fourier inversion gives $8W_c = N(p+qi)c_1c_2c_3 + N(p-qi)$, which is the claim. ∎

**Proposition 24.35** (A smooth complete intersection cannot carry the rest)**.** *Let $\mathcal{E}$ be as above and suppose that its support lies in $Y \cup \bigcup_{\sigma(x)=\pm1} Z_x$, where $Y \subseteq A_0$ is a smooth complete intersection of dimension $d \leq 5$ of divisors in the linear systems $|L_h^{m_i}|$ of powers of the polarisation $L_h$. Then:*

- *(i) $Np \neq 0$, and every orbit of the family $\sigma = \pm1$ contains a threefold $Z_x \not\subseteq Y$ along which $\mathcal{E}$ has nonzero multiplicity.*
- *(ii) If $Y$ meets every $Z_x$ in dimension at most one, then $\mathcal{E}$ does not exist: no such complex has polarised shape and meets the criterion.*

*Proof.* (i) Let $Y_1, \dots, Y_c$ be the divisors, $c = 6 - d \geq 1$. Each $A_0 \setminus Y_i$ is affine and smooth, as is every intersection of them, so has vanishing homology above degree six [AF59]; the Mayer–Vietoris spectral sequence of the cover of $A_0 \setminus Y$ by these $c$ open sets gives $H_q(A_0 \setminus Y) = 0$ for $q > 5 + c$, and Lefschetz duality $H^q(A_0, Y) \cong H_{12-q}(A_0 \setminus Y)$ gives $H^q(A_0, Y) = 0$ for $q \leq d$. So $H^{2d-6}(A_0,\mathbb{Q}) \to H^{2d-6}(Y,\mathbb{Q})$ is onto, as $2d - 6 < d$. For $\eta \in H^{2d-6}(A_0)$ the projection formula gives $i_* i^* \eta = \eta \cup [Y] = (\prod_i m_i)\theta_h^c \eta$, so $i_* H^{2d-6}(Y) \subseteq \theta_h H^4(A_0)$. Now let $\Sigma = Y \cup \bigcup_{x \in I} Z_x$, $I$ the set of $x$ with $\sigma(x) = \pm1$ and $Z_x \subseteq \operatorname{Supp} \mathcal{E}$, $Z_x \not\subseteq Y$. By the Riemann–Roch theorem for singular varieties [Ful98, Ch. 18], applied to the cohomology sheaves of $\mathcal{E}$, which are coherent sheaves on infinitesimal neighbourhoods of $\Sigma$, the class $\operatorname{ch}_3(\mathcal{E}) \in H^6(A_0,\mathbb{Q})$ is the image of a class $\tau \in A_3(\Sigma)_\mathbb{Q}$ (the Todd class of $A_0$ being 1). The localisation sequence $A_3(Y) \to A_3(\Sigma) \to A_3(\Sigma \setminus Y) \to 0$ is exact, $A_3(\Sigma \setminus Y)$ is free on the classes of the three-dimensional components $Z_x \setminus Y$, $x \in I$, and the component of $\tau$ there is $\sum_{x \in I} \mu_{Z_x}(\mathcal{E})[Z_x \setminus Y]$, the top-dimensional part of the Riemann–Roch class of a sheaf being its cycle. Hence

$$\operatorname{ch}_3(\mathcal{E}) - \sum_{x \in I} \mu_{Z_x}(\mathcal{E})[Z_x] \in i_* H^{2d-6}(Y) \subseteq \theta_h H^4(A_0),$$

and $\operatorname{ch}_3(\mathcal{E}) = N\omega + c_3\theta_h^3$ with $\theta_h^3 \in \theta_h H^4$. Lemma 24.34 applies to $\nu_x = \mu_{Z_x}(\mathcal{E})$ for $x \in I$ and $\nu_x = 0$ otherwise: since $\nu$ vanishes on the family $\sigma = \pm i$, $Nq = 0$, so $Np \neq 0$ as $N\omega \neq 0$, and each orbit of the family $\sigma = \pm1$ contains $x \in I$ with $\mu_{Z_x}(\mathcal{E}) \neq 0$.

(ii) Take $x$ from (i), $T = Z_x$. The multiplicity of $\mathcal{E}$ along $T$ on the complement of the other components equals $\mu_{Z_x}(\mathcal{E}) \neq 0$, so Proposition 24.30 gives a component $R_1$ of

the support, of dimension at least three and not among the thirty-two threefolds, hence contained in $Y$, such that $R_1 \cap (Z_x \cup Z_{x^{(1)}})$ contains a surface. But $Y \cap (Z_x \cup Z_{x^{(1)}})$ has dimension at most one. ∎

The hypotheses of Proposition 24.35(ii) hold, for instance, when the rest of the support is a smooth curve or a smooth surface cut out by divisors of the polarisation and meeting the threefolds in finitely many points; the smooth divisors $Y$ meet every $Z_x$ in a surface and are only constrained by (i) and by Theorem 24.27. When the support is large the localised obstruction says nothing at all:

**Proposition 24.36** (Where the localised obstruction stops)**.** *Let $\mathcal{E}$ be as above and suppose that its support contains an ample effective divisor $Y$. Then $v \lrcorner \operatorname{ch}^Z_{k+1}(\mathcal{E}|_U) = 0$ for every $v \in T\mathcal{D}_h$, every $k \geq 0$, every open $U \subseteq A_0$ and every closed $Z \subseteq U$ containing the support of $\mathcal{E}|_U$. In particular a convolution of line-bundle pieces $L^{l_i}[t_i]$ and threefold pieces with $\sum_i (-1)^{t_i} \neq 0$ has support $A_0$, and the criterion for it is decided by its products alone.*

*Proof.* Let $\Sigma$ be the support of $\mathcal{E}$ and $X = A_0 \setminus Y$, an affine variety, so that $H^1(X, T_X) = 0$ and $H^{k+1}(X, \Omega^k) = 0$ for $k \geq 0$. The restriction of $v$ to $X$ vanishes, so the image of $v \lrcorner \operatorname{ch}^\Sigma_{k+1}(\mathcal{E})$ under $H^{k+2}_\Sigma(A_0, \Omega^k) \to H^{k+2}_{\Sigma\setminus Y}(X, \Omega^k)$, which is $v|_X \lrcorner \operatorname{ch}^{\Sigma\setminus Y}_{k+1}(\mathcal{E}|_X)$, vanishes; by the exact sequence

$$H^{k+2}_Y(A_0, \Omega^k) \to H^{k+2}_\Sigma(A_0, \Omega^k) \to H^{k+2}_{\Sigma\setminus Y}(X, \Omega^k)$$

the class $v \lrcorner \operatorname{ch}^\Sigma_{k+1}(\mathcal{E})$ comes from a class $b \in H^{k+2}_Y(A_0, \Omega^k)$. The sequence

$$H^{k+1}(X, \Omega^k) \to H^{k+2}_Y(A_0, \Omega^k) \to H^{k+2}(A_0, \Omega^k)$$

shows that $b$ is determined by its image in $H^{k+2}(A_0, \Omega^k)$, which is $v \lrcorner \operatorname{ch}_{k+1}(\mathcal{E}) = 0$. So $v \lrcorner \operatorname{ch}^\Sigma_{k+1}(\mathcal{E}) = 0$, and its restriction to $U$, followed by the enlargement of supports from $\Sigma \cap U$ to $Z$, is $v \lrcorner \operatorname{ch}^Z_{k+1}(\mathcal{E}|_U)$. A complex of rank $\sum_i (-1)^{t_i} \neq 0$ has a cohomology sheaf of positive rank, whose support is $A_0$. ∎

What remains at $E_0^6$. By Theorem 24.33 every convolution of threefold pieces fails, of one family or of both, at any twists and shifts, which covers the threefolds at several twists of Setup 24.17; Theorem 24.23 and Remark 24.25 cover the convolutions with at most seven line bundles, and eight line bundles outside thirty-one arrangements. Left are the convolutions with eight line bundles in those thirty-one arrangements or with nine or more line bundles, with or without threefold pieces at any twists. For a convolution of line bundles the support is $A_0$ when $\sum_i (-1)^{t_i} \neq 0$, which happens for eight of the thirty-one arrangements, where the rank is $\pm 2$; there Proposition 24.36 says that only the products decide. For the other twenty-three the rank is zero and the support is a proper closed subset containing every threefold piece; Theorem 24.27 and Propositions 24.30 and 24.35 constrain it, and the products decide the rest. The products along the chains of Proposition 24.10 and along the cycles through the line bundles are computable in the theta-function model of Section 18, where the fourfold products of line-bundle pieces at distance at least two were found to give $\dim \operatorname{Ext}^2 \geq 69$; we have not computed them, and (F2) at $E_0^6$ stands or falls with them.

24.9. **Seven consecutive shifts at $E_0^8$.** At $n = 4$ the same route was taken in Section 18 on $E_0^8$, $E_0 = \mathbb{C}/\mathbb{Z}[i]$, with the 128 line bundles $L_\zeta$, $\zeta \in \mu_4^4$, $\prod \zeta = \pm 1$, whose signed Chern characters add up to a multiple of the Weil class, and three multiples $M_1, M_2, M_3$ of the polarisation; the criterion asks for $\dim \operatorname{Ext}^2(E, E) = 104$. In the setting of Lemma 18.36

the pieces of one parity, $\prod \zeta = \epsilon$, have shifts of one parity and the pieces of the other parity have shifts of the other, and Corollary 18.48 excludes every convolution whose 128 shifts lie in six consecutive integers. We extend this to seven. The groups used below are the groups $\mathcal{H}^k(W, V)$ between two pieces in the harmonic model of the convolution Lemma 18.24, as in Theorem 18.37 and Lemma 18.39: for pieces $W$, $V$ that differ in exactly three coordinates and have $d_W = d_V + 1$, the groups $\mathcal{H}^3(W, V)$, of total dimension 640 over the 28 ratios $W/V$ of this kind, are reached by no coboundary, and a term of $d_E$ on them passes through a piece at the shift $d_W + 1$, $d_W + 2$, $d_V - 1$ or $d_V - 2$.

**Proposition 24.37** (Two separated adjacent pairs)**.** *Let $E$ be a convolution as in Lemma 18.36, let $g \geq 4$, and suppose that the shifts of the pieces $L_\zeta$ take exactly the four values $s$, $s-1$, $s-g$ and $s-g-1$. Then for every order, every choice of the shifts of the $M_i$ and every placement of the $t_i$, $\dim \mathrm{Ext}^2(E, E) \geq 640$, and $E$ does not meet the polarised criterion.*

*Proof.* Let $P$, $Q$, $R$, $T$ be the sets of pieces at the shifts $s$, $s-1$, $s-g$, $s-g-1$; all four are nonempty. Two shifts that differ by one belong to pieces of opposite parity, so $P$ and $Q$ have opposite parities, and so do $R$ and $T$; $P$ and $R$ have the same parity when $g$ is even and opposite parities when $g$ is odd. Let $W$ and $V$ differ in exactly three coordinates with $d_W = d_V + 1$. Then $W \in P$ and $V \in Q$, or $W \in R$ and $V \in T$, since these are the only pairs of occupied shifts at distance one. A term of $d_E$ on $\mathcal{H}^3(W, V)$ passes through a piece at one of the shifts $s+1$, $s+2$, $s-2$, $s-3$ when $W \in P$, and $s-g+1$, $s-g+2$, $s-g-2$, $s-g-3$ when $W \in R$. Since $g \geq 4$ none of these is among $s$, $s-1$, $s-g$, $s-g-1$, so no term acts and the groups $\mathcal{H}^3(W, V)$ inject into $\mathrm{Ext}^2(E, E)$.

Let $\Gamma$ be the group of ratios of two pieces, of order 128, acting simply transitively on the pieces, and let $S_3 \subset \Gamma$ be the 28 ratios that move exactly three coordinates, with the weight given by the dimensions of the groups $\mathcal{H}^3$; every ratio in $S_3$ changes the parity. Put $X = P \cup T$ if $g$ is even and $X = P \cup R$ if $g$ is odd. In both cases $X$ is a union of one set of each parity, and its complement is the union of the other two. The edges of the Cayley graph of $(\Gamma, S_3)$ join pieces of opposite parity, so the edges between $X$ and its complement join $P$ to $Q$ and $R$ to $T$, and these are exactly the pairs $(W, V)$ above. Hence the dimensions of the surviving groups add up to the weight of the cut of $X$. The set $X$ is neither empty nor everything, and the proof of Proposition 18.47 shows, through Lemma 18.40, that every such cut weighs at least 640. ∎

For $g = 4$ this is Proposition 18.47. Together with the results of Section 18 it settles most arrangements in seven consecutive integers.

**Corollary 24.38** (Seven consecutive shifts)**.** *Let $E$ be a convolution as in Lemma 18.36 whose 128 pieces $L_\zeta$ have their shifts in $\{s, s-1, \dots, s-6\}$. Let $U \subseteq \{s, s-2, s-4, s-6\}$ and $U' \subseteq \{s-1, s-3, s-5\}$ be the values taken by the two parities. Then $\dim \mathrm{Ext}^2(E, E) \geq 640$ for every order, every choice of the shifts of the $M_i$ and every placement of the $t_i$, except possibly when $U$ is one of*

$$\{s, s-2, s-6\}, \quad \{s, s-4, s-6\}, \quad \{s, s-2, s-4, s-6\},$$

*$U'$ is one of*

$$\{s-1\}, \quad \{s-3\}, \quad \{s-5\}, \quad \{s-1, s-5\},$$

*and $(U, U')$ is neither $(\{s, s-2, s-6\}, \{s-5\})$ nor $(\{s, s-4, s-6\}, \{s-1\})$.*

*Proof.* If the shifts do not take both values $s$ and $s-6$ they lie in six consecutive integers and Corollary 18.48 applies; so $s, s-6 \in U$. If $U'$ contains two values two apart it is

$\{s-1,s-3\}$, $\{s-3,s-5\}$ or $\{s-1,s-3,s-5\}$, and Theorem 18.41 applies to the parity at $U'$ for every $U$. Otherwise $U'$ is $\{s-1\}$, $\{s-3\}$, $\{s-5\}$ or $\{s-1,s-5\}$. Let $U=\{s,s-6\}$. For $U'=\{s-3\}$ the two parities sit at $s-3$ and at $s-3\pm 3$, and Theorem 18.42 gives $\dim \mathrm{Ext}^2 \geq 67584$. For $U'=\{s-1\}$, Proposition 18.43 with $y=s-1$ applies, since the pieces at $y+1=s$ exist and no piece has the shift $s-2$ or $s+2$; for $U'=\{s-5\}$ its second form with $y=s-5$ applies, since the pieces at $s-6$ exist and no piece has the shift $s-4$ or $s-8$. For $U'=\{s-1,s-5\}$ the shifts take exactly the values $s$, $s-1$, $s-5$, $s-6$, and Proposition 24.37 applies with $g=5$. The same two applications of Proposition 18.43 settle $U=\{s,s-4,s-6\}$ with $U'=\{s-1\}$, where no piece has the shift $s-2$ or $s+2$, and $U=\{s,s-2,s-6\}$ with $U'=\{s-5\}$, where no piece has the shift $s-4$ or $s-8$. ∎

*Remark* 24.39 (Nine of the ten remaining arrangements). Nine of the ten arrangements left by Corollary 24.38 were decided by computation, which no statement of this paper uses. For each arrangement and each distribution of the 64 pieces of each parity over its values, a lower bound for $\dim \mathrm{Ext}^2(E,E)$ was obtained as the sum of three families of surviving groups: the groups $\mathcal{H}^4$ at spread two between pieces of one parity on the 13 ratios of Lemma 18.39(ii), bounded below by the cut bound of Lemma 18.40; the groups $\mathcal{H}^3$ between adjacent shifts, less what the terms into $\mathcal{H}^5$ and $\mathcal{H}^6$ can remove; and the groups $\mathcal{H}^5$ at spread three, on which $d_E$ vanishes (Lemma 18.39(iii)), less what the coboundaries can reach. The dimension carried by each set of edges between the sets of pieces at two shifts was bounded by the per-piece degrees, by the cut bounds and by the expander mixing lemma for the weighted Cayley graphs of $\Gamma$, with the largest nontrivial eigenvalue of each weighting, and the sum was minimised by linear programming, solved in double precision, for every distribution. Over eleven of the twelve arrangements of the statement, with the 1953 or 39711 distributions of the parity with three or four values and the 1 or 63 of the other, the least bound obtained is 125.8, against 104 required. For $U=\{s,s-2,s-4,s-6\}$ and $U'=\{s-1,s-5\}$ the bound is below 104 for 585 753 of the 2 501 793 distributions and vanishes for some of them, so the method decides nothing there and that arrangement remains open. The program is in the archive described in the statement on data availability. So every convolution whose 128 shifts lie in seven consecutive integers fails, by hand in eighteen of the twenty-eight arrangements and by the computation in nine others, except possibly in the arrangement just named; Proposition 24.37 settles the separated pairs at every distance.

24.10. **Weil sixfolds from cyclic triple covers of curves of genus four.** Abelian covers of curves give explicit Weil families. For an étale cyclic cover of degree $\ell \in \{3,4,6\}$ of a curve $B$ of genus four, defined by a line bundle $\eta$ of order $\ell$, the pairs $(B,\eta)$ have nine moduli, as many as the dimension of the domain $\mathcal{D}_F$, $F=\mathbb{Q}(\zeta_\ell)$, $n=3$, that receives the period map, and the image of this map is open near every point at which the multiplication map

$$\mu_\eta \colon H^0(K_B\otimes\eta)\otimes H^0(K_B\otimes\eta^{-1}) \longrightarrow H^0(K_B^{\otimes 2}) \tag{88}$$

is surjective (proof of Proposition 24.40(iv)). For $\ell=3$ the Prym map of these covers is generically finite, by Faber [Fab88], and surjectivity of $\mu_\eta$ at a general point follows. We exhibit an explicit point at which $\mu_\eta$ is an isomorphism.

**Proposition 24.40** (An explicit cover)**.** *For $\lambda\in\mathbb{C}\setminus\{0,1\}$ let $B_\lambda\subset\mathbb{P}^1\times\mathbb{P}^1$ be the curve $x^3+y^3+1+\lambda x^3y^3=0$, in affine coordinates $x$, $y$, and $\eta=\mathcal{O}(1,-1)|_{B_\lambda}$.*

(i) *$B_\lambda$ is a smooth curve of genus four with $K_{B_\lambda}=\mathcal{O}(1,1)|_{B_\lambda}$.*

(ii) *$\eta$ has order exactly three.*

(iii) *$H^0(K\otimes\eta)$ and $H^0(K\otimes\eta^{-1})$ are spanned by $1,x,x^2$ and by $1,y,y^2$, and $\mu_\eta$ is an isomorphism.*

(iv) *The period map of the étale cyclic triple covers of curves of genus four, with values in the domain $\mathcal{D}_F$ of $(\mathbb{Q}(\sqrt{-3}),3)$-Weil type that contains the $\chi$-part of $H^1$ of the cover $C_\lambda\to B_\lambda$ defined by $\eta$, is a local isomorphism at $(B_\lambda,\eta)$. Its image contains an open subset of $\mathcal{D}_F$, and the Weil classes are algebraic on every member of $\mathcal{D}_F$ as soon as they are algebraic on the abelian sixfolds $H_\chi(C_t)$ for the covers $C_t$ of a non-meagre subset of a neighbourhood of $(B_\lambda,\eta)$.*

*Proof.* (i) In bihomogeneous coordinates the curve is $F=x_1^3y_0^3+x_0^3y_1^3+x_0^3y_0^3+\lambda x_1^3y_1^3$, of bidegree $(3,3)$. In the chart $x_0=y_0=1$ the partial derivatives are $3x^2(1+\lambda y^3)$ and $3y^2(1+\lambda x^3)$. At a singular point with $x=0$ the equation gives $y^3=-1$ and then $\partial_yF=3y^2\neq0$; similarly $y\neq0$. So $x^3=y^3=-1/\lambda$, and the equation becomes $1-1/\lambda=0$, excluded. The equation is symmetric in $x$ and $y$; the substitution $(x,y)=(1/x',\lambda^{-1/3}y')$, after multiplication by $x'^3$, carries it to the same form in $x',y'$ with $\lambda$ replaced by $1/\lambda$; and the substitution $(x,y)=(\lambda^{-1/3}/x',\lambda^{-1/3}/y')$, after multiplication by $\lambda x'^3y'^3$, carries it to itself. So the other three charts reduce to the first. By adjunction $K_B=(K_{\mathbb{P}^1\times\mathbb{P}^1}+B)|_B=\mathcal{O}(1,1)|_B$, of degree $(1,1)\cdot(3,3)=6=2g-2$, so $g=4$.

(ii) Let $D_x$ and $D_y$ be the intersections of $B$ with the lines $x=\infty$ and $y=\infty$. At $x=\infty$ the equation reads $1+\lambda y^3=0$, so $D_x$ consists of three points $(\infty,c)$ with $c^3=-1/\lambda$, and at each of them $u=1/x$ is a local parameter, since the derivative of $1+u^3y^3+u^3+\lambda y^3$ in $y$ is $3\lambda c^2\neq0$; along $B$, $1+\lambda y^3=-u^3(1+y^3)$, so $\varphi=y^3+1/\lambda$ vanishes to order three there. Its other zeros would have $y^3=-1/\lambda$ and $x\neq\infty$, hence $x^3(1+\lambda y^3)=0=1+y^3$, which forces $\lambda=1$. Its poles are at $D_y$, three points at each of which $1/y$ is a local parameter by the same computation, of order three. So $\operatorname{div}\varphi=3D_x-3D_y$, and since $D_x\in|\mathcal{O}(1,0)|_B|$ and $D_y\in|\mathcal{O}(0,1)|_B|$, $\eta^{\otimes3}\cong\mathcal{O}_B$. If $\eta$ were trivial, $D_y$ would lie in $|\mathcal{O}_B(1,0)|$. From $0\to\mathcal{O}(-2,-3)\to\mathcal{O}(1,0)\to\mathcal{O}_B(1,0)\to0$ and $H^0(\mathcal{O}(-2,-3))=H^1(\mathcal{O}(-2,-3))=0$ by the Künneth formula, every section of $\mathcal{O}_B(1,0)$ is $a+bx$, so every divisor in that system is a fibre $x=c$ of $B$. The three points of $D_y$ have $x^3=-1/\lambda$, three distinct values of $x$, so $D_y$ is not such a fibre.

(iii) $K\otimes\eta=\mathcal{O}(2,0)|_B$ and $K\otimes\eta^{-1}=\mathcal{O}(0,2)|_B$. The kernels $\mathcal{O}(-1,-3)$ and $\mathcal{O}(-3,-1)$ of the restrictions have no cohomology in degrees 0 and 1, again by the Künneth formula, so restriction gives $H^0(K\otimes\eta)=H^0(\mathbb{P}^1\times\mathbb{P}^1,\mathcal{O}(2,0))$, spanned by $1,x,x^2$, and $H^0(K\otimes\eta^{-1})=H^0(\mathcal{O}(0,2))$, spanned by $1,y,y^2$. The product of the two is $H^0(\mathcal{O}(2,2))$, of dimension nine, and $\mu_\eta$ is its restriction to $B$, with values in $H^0(\mathcal{O}_B(2,2))=H^0(K^{\otimes2})$, of dimension $3g-3=9$. The kernel of the restriction is $H^0(\mathcal{O}(-1,-1))=0$, so $\mu_\eta$ is injective between spaces of dimension nine, hence an isomorphism.

(iv) Write $\pi\colon C_\lambda\to B=B_\lambda$ for the cover defined by $\eta$, so that $\pi_*\mathcal{O}_{C_\lambda}=\bigoplus_{i=0}^2\eta^i$ and $\mu_3$ acts on $\eta^i$ through the character $\zeta\mapsto\zeta^i$. The eigenspace of $H^1(C_\lambda,\mathbb{C})$ for the character $\chi$ of exponent one is $H^1(B,\mathbb{L})$ for the unitary local system $\mathbb{L}$ of rank one underlying $\eta$, with Hodge pieces $H^0(K\otimes\eta)$ of type $(1,0)$ and $H^1(\eta)\cong H^0(K\otimes\eta^{-1})^\vee$ of type $(0,1)$. Since $\deg\eta=0$ and $\eta\neq\mathcal{O}_B$, $h^0(\eta)=h^0(\eta^{-1})=0$, and Riemann–Roch gives $h^0(K\otimes\eta)=h^0(K\otimes\eta^{-1})=g-1=3$. The sum $H_\chi$ of the $\chi$- and $\bar\chi$-eigenspaces is defined over $\mathbb{Q}$, carries the polarisation induced by the intersection form and an action of $F=\mathbb{Q}(\sqrt{-3})$, and the Hermitian form on the $\chi$-eigenspace has signature $(3,3)$; so $H_\chi$ is of $(F,3)$-Weil type and the abelian sixfold $H_\chi(C_\lambda)$ lies in a domain $\mathcal{D}_F\cong U(3,3)/(U(3)\times U(3))$ of dimension nine. A line bundle of finite order deforms uniquely with the curve, so the

pairs $(B,\eta)$ are locally parametrised by $H^1(T_B)$, of dimension $3g-3=9$. By Griffiths' description of the differential of the period map [Voi02, Ch. 10], the differential at $(B,\eta)$ is the map $H^1(T_B)\to \mathrm{Hom}(H^0(K\otimes\eta),H^1(\eta))$ given by cup product, and after Serre duality, $H^1(T_B)^\vee=H^0(K^{\otimes 2})$ and $H^1(\eta)^\vee=H^0(K\otimes\eta^{-1})$, its transpose is $\mu_\eta$. By (iii) the differential is an isomorphism and the period map is a local isomorphism at $(B_\lambda,\eta)$; its image contains an open set $\Omega\subseteq\mathcal{D}_F$. A local biholomorphism sends a non-meagre subset of a neighbourhood to a non-meagre subset of $\Omega$, so under the hypothesis of the last assertion the algebraic locus of the Weil classes in $\mathcal{D}_F$ is not meagre, and by Lemma 22.2 it is all of $\mathcal{D}_F$. ∎

The curve $B_\lambda$ is a cyclic cover of $\mathbb{P}^1\times\mathbb{P}^1$ in disguise: it is invariant under $\mu_3^2$ acting on $x$ and $y$, and its quotient is the line $X+Y+1+\lambda XY=0$ in the coordinates $X=x^3$, $Y=y^3$. For $\ell=4$ and $\ell=6$ the same argument needs a curve of genus four with a line bundle of order four or six for which $\mu_\eta$ is surjective, and we have not found one.

24.11. **Integral characters at sextic CM fields.** For a sextic CM field $F=F_0(\sqrt{-q})$, with $F_0$ totally real cubic and $q\in F_0$ totally positive, Proposition 19.47 gives a count that would exclude a minimal object for the polarised criterion whose character has a component $v$ in the flat secant space $S(0,q)$ with $\chi(v,v)$ above an explicit threshold, and Proposition 19.49 shows that the condition of the count holds whenever $\chi(v,v)\le -26$, computes $\chi(v,v)=-8(\mathrm{Nm}(q)c_0^2+\mathrm{Tr}(\mathrm{Nm}(q)q^{-1}f^2)+\mathrm{Tr}(qg^2)+c_3^2)$ on the rational points $v(c_0,f,g,c_3)$, and exhibits integral points with $\chi=-32$ for $q=1$.

*Remark* 24.41 (No exclusion over twelve cubic fields). For the twelve totally real cubic fields of discriminant 49, 81, 148, 169, 229, 257, 316, 321, 361, 404, 469, 473, whose rings of integers are generated by one element, and for $q=1$ and $q=k+\alpha$ with $\alpha$ the generator and $k$ the least integer making $q$ totally positive, the lattice of integral points of $S(0,q)$ was computed exactly, in the model of Proposition 19.49, with the Gram matrix of $-\chi/8$, which is integral. No nonzero integral point has $-\chi\le 24$, and since $\chi\in 8\mathbb{Z}$ on the lattice, every nonzero integral point has $\chi\le -32$ and meets the condition of Proposition 19.47(iii). For $q=1$ the least value $-\chi=32$ is attained by four points of shape $N_w=2$, $r^2=30$, $r^3=40$; for $q=k+\alpha$ it is 32 for the discriminants 49, 81 and 288, 384, 912, 352, 480, 600, 960, 544, 2400, 1208 for the other ten. So over these fields the count of Proposition 19.47 excludes no integral character of a minimal object, and the route through it is closed there. This is a computation of the archive described in the statement on data availability, and nothing in the paper depends on it.

24.12. **The exceptional classes of a Mumford fourfold.** Keep Setup 21.1 and write $\psi$ for the polarisation form on $V$, $T=D^0$, $q_\lambda(x,y)=\mathrm{Tr}_{F/\mathbb{Q}}(\lambda\,\mathrm{trd}(xy))$ for $\lambda\in F$, and $m_a$ for multiplication by $a\in F$ on $T$. The copy $T_X(-1)\subseteq V\otimes V$, $x\mapsto((v,w)\mapsto\psi(xv,w))$, is symmetric and carries the form induced by $\psi\otimes\psi$, equal to $-4\,\mathrm{trd}(xy)$ summed over the three real places, that is $q_1$ up to the scalar $-4$. The exceptional classes of $X\times X$ are the elements $m_a$, $a\in F\setminus\mathbb{Q}$, of $\mathrm{End}_{\mathrm{Hdg}}T_X(-1)=F$ Proposition 21.2, and in Theorem 21.4 their algebraicity is derived from the Kuga–Satake class of a K3 surface $S_\lambda$ with $T(S_\lambda)_\mathbb{Q}\cong(T,q_\lambda)$. We replace $S_\lambda$ by an abelian variety.

**Setup 24.42** (The sixfold $Z_K$)**.** Let $K=F(\sqrt{-\delta})\subseteq D$ be a CM quadratic extension of $F$ inside $D$, $\delta\in F$ totally positive; every maximal commutative CM subfield of $D$ is of this form since $D$ is definite at two real places. Let $D$ carry the left action of $G=D^1$ and the right action of the torus $K^1=\ker(N_{K/F})$. Over $\bar{\mathbb{Q}}$, $D=D_1\oplus D_2\oplus D_3$ with $D_i=V_i\otimes W_i$, $V_i$ the standard representation of the $i$-th factor of $G$ and $W_i=\chi_i\oplus\chi_i^{-1}$ that of the $i$-th

factor of $K^1$. Fix a point $t$ of the Shimura curve $C$ of $X$, with Hodge cocharacter $\mu_X$ nontrivial only at the split place $\sigma_3$, and let $\mu_Z = \mu'\mu''$, with $\mu'$ the action of $\mu_X$ by left multiplication and $\mu''$ the cocharacter of $K^1$ of weight $\pm\frac{1}{2}$ on $\chi_1^{\pm1}$, $\chi_2^{\pm1}$ and 0 on $\chi_3^{\pm1}$. Then $D_{\mathbb{C}}$ has type $(1,0)$ on $V_1\chi_1$, $V_2\chi_2$, $V_3^{1,0}W_3$ and $(0,1)$ on the complements, so $D$ carries a rational Hodge structure of weight one with $h^{1,0}=6$ and Hodge group inside $G\cdot K^1$. For $c\in K$ with $\bar c=-c$ put $\psi_c(x,y)=\operatorname{Tr}_{F/\mathbb{Q}}\operatorname{trd}(\bar x\,y\,c)$. The abelian sixfold $Z_K=Z_{K,t}$ is the one with $H^1(Z_K)=D$ and polarisation $\psi_c$ for $c$ in the cone where $\psi_c$ is positive; over a finite cover of $C$ these form an abelian scheme [Shi63], and we use only the fibre at $t$.

**Lemma 24.43.** *$\psi_c$ is alternating, invariant under left multiplication by $D^1$ and right multiplication by $K^1$, and the $\psi_c$, $c\in K$ with $\bar c=-c$, fill the three-dimensional space of all $G\times K^1$-invariant alternating forms on $D$. Hence $\mathrm{NS}(Z_K)_{\mathbb{Q}}$ has dimension at least three at every point $t$ of $C$, and exactly three at every $t$ outside a countable subset of $C$.*

*Proof.* The reduced trace is invariant under conjugation and cyclic, so $\psi_c(y,x)=\operatorname{Tr}\operatorname{trd}(\bar y\,x\,c)=\operatorname{Tr}\operatorname{trd}(\bar c\,\bar x\,y)=\operatorname{Tr}\operatorname{trd}(\bar x\,y\,\bar c)$, and $\psi_c$ is alternating exactly when $\bar c=-c$. For $g$ with $\operatorname{nrd}(g)=1$, $\operatorname{trd}(\overline{gx}\,gy\,c)=\operatorname{trd}(\bar x\,\bar g g\,y\,c)=\operatorname{trd}(\bar x\,y\,c)$. For $k\in K^1$, $\operatorname{trd}(\overline{xk}\,yk\,c)=\operatorname{trd}(\bar k\,\bar x\,y\,k\,c)=\operatorname{trd}(\bar x\,y\,k\,c\,\bar k)=\operatorname{trd}(\bar x\,y\,c)$ because $K$ is commutative and $k\bar k=1$. Over $\bar{\mathbb{Q}}$ the invariant alternating forms live in the three summands $\bigwedge^2(V_i\otimes W_i)^{G\times K^1}=\bigwedge^2 V_i\otimes(\chi_i\chi_i^{-1})$, one dimension each, and the cross terms $D_i\otimes D_j$ carry the nontrivial characters $\chi_i^{\pm1}\chi_j^{\pm1}$; so the space has dimension $3=\dim_{\mathbb{Q}}K^-$, which the $\psi_c$ fill since $c\mapsto\psi_c$ is injective. The Hodge group $H_t$ of $Z_{K,t}$ lies in $G\cdot K^1$, so these three forms are Hodge classes and $\dim\mathrm{NS}(Z_{K,t})_{\mathbb{Q}}\geq 3$; a smaller group can have more invariants, so this is only a lower bound.[65] For the upper bound at a very general $t$, recall that the Hodge group at a very general point of a family is the generic Hodge group $H$, and that by Deligne and André [And92] the connected monodromy group is a normal subgroup of its derived group. The monodromy of the family over $C$ acts through left multiplication by an arithmetic subgroup of $G$, Zariski dense in $G$ because $G$ is $\mathbb{Q}$-simple and isotropic at the split place, so $G\subseteq H\subseteq G\cdot K^1$. The image of $H$ in the torus $G\cdot K^1/G$, a quotient of the three-dimensional torus $\operatorname{Res}_{F/\mathbb{Q}}K^1$ by a finite group, is the smallest $\mathbb{Q}$-subtorus through which the image of $\mu''$ factors. The cocharacter group of $\operatorname{Res}_{F/\mathbb{Q}}K^1$, tensored with $\mathbb{Q}$, is $\mathbb{Q}^3$, one coordinate for each real place of $F$, and the Galois group permutes the coordinates transitively up to sign; the orbit of $\mu''=(\frac12,\frac12,0)$ contains its three cyclic shifts, which are linearly independent. So $H=G\cdot K^1$ at a very general $t$, and its invariant alternating forms are exactly the three forms $\psi_c$, all of type $(1,1)$. ∎

**Lemma 24.44** (Where $T$ sits in $Z_K$)**.**

(i) *The $K^1$-invariants of $\bigwedge^2 D_i$ form a four-dimensional space: $S^2V_i\otimes\bigwedge^2 W_i$, the adjoint representation of $\mathfrak{sl}(V_i)$, and $\bigwedge^2 V_i\otimes\mathbf{1}$. The sum $T_Z$ of the three adjoint pieces is defined over $\mathbb{Q}$, is a sub-Hodge structure of $H^2(Z_K)$, and is isomorphic, as a Hodge structure with $G$-action, to $T_X(-1)$.*

(ii) *The form induced on $\bigwedge^2 D$ by $\psi_c$ restricts on $T_Z$ to $q_{c^2}$, up to a rational scalar independent of the place; here $c^2=-N_{K/F}(c)=-f^2\delta\in F$ for $c=f\sqrt{-\delta}$.*

*Proof.* (i) $\bigwedge^2(V_i\otimes W_i)=S^2V_i\otimes\bigwedge^2 W_i\oplus\bigwedge^2 V_i\otimes S^2W_i$; the torus acts trivially on $\bigwedge^2 W_i=\chi_i\chi_i^{-1}$ and on one summand of $S^2W_i=\chi_i^2\oplus\mathbf{1}\oplus\chi_i^{-2}$. The cross terms carry nontrivial

[65] The direction matters: if $H\subseteq H'$ then every $H'$-invariant is $H$-invariant, so containment of the Hodge group in a known group bounds the Hodge classes from below, never from above.

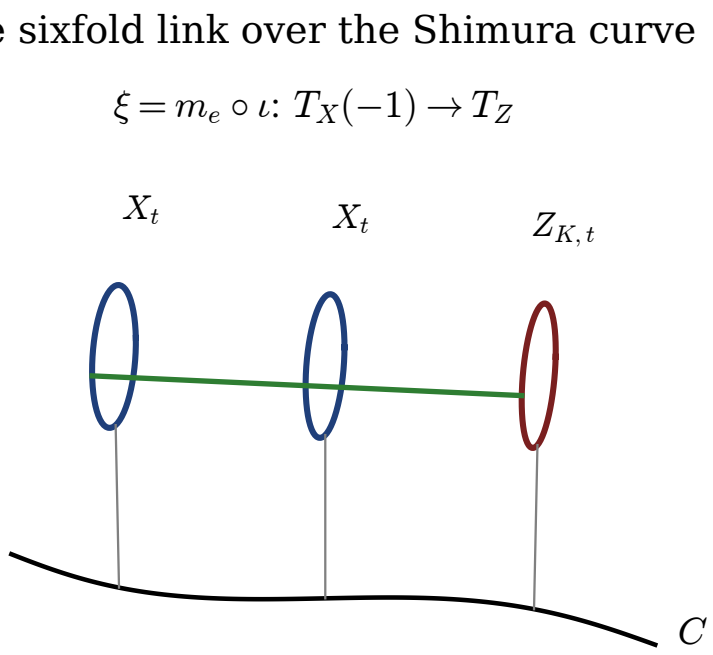



FIGURE 68. The sixfold link. Two copies of the Mumford fourfold $X_t$ and the sixfold $Z_{K,t}$ sit over the same point $t$ of the Shimura curve $C$; the class $\xi_t$ in the mixed Künneth component identifies the two copies of $T$ and is flat along $C$.

characters. The Galois group permutes the places, so $T_Z$ is defined over $\mathbb{Q}$; as a $G$-representation it is the adjoint representation $T$. The cocharacter $\mu''$ acts trivially on $K^1$-invariants and the centre acts on $\bigwedge^2 D$ with weight two, so the Hodge structure on $T_Z$ is the one induced by the adjoint action of $\mu'$, twisted to weight two; this is the recipe that gives $T_X(-1)$ at the same point $t$.

(ii) By Schur's lemma the induced form is, on each irreducible $T_i$, a multiple of $\mathrm{trd}(xy)$. In the split model at the place $i$ the element $c$ acts on $D_i$ by $\sigma_i(c)\,\mathrm{diag}(1,-1)$, whose square is $\sigma_i(c^2)$; the multiple is $\sigma_i(c)^2$ times a rational constant that is the same at every place, since the computation in the weight basis is identical at the three places. The sum over the places is $q_{c^2}$ up to that constant. ∎

**Theorem 24.45** (The sixfold link)**.** *Let $t \in C$ be a point at which $X = X_t$ has Hodge group $G$, and let $\xi$ be a nonzero element of* $\mathrm{Hom}_{\mathrm{Hdg}}(T_X(-1), T_Z) \subseteq H^4(X \times X \times Z_K)$.

(i) $\mathrm{Hom}_{\mathrm{Hdg}}(T_X(-1), T_Z) = F \cdot \iota$ *for the identification $\iota$ of Lemma 24.44, so $\xi = m_e \circ \iota$ for a unique $e \in F^\times$. The $G \cdot K^1$-invariant classes of degree four on $X \times X \times Z_K$ have dimension* 26*; they are Hodge classes at every $t$, at a very general $t$ they are all*

*the Hodge classes of degree four, and there $\xi$ and $\psi \otimes \mathrm{NS}(Z_K)$ are the only ones in the Künneth component $H^1(X) \otimes H^1(X) \otimes H^2(Z_K)$.*

(ii) *If $\xi$ is algebraic, then $\xi^\dagger \xi$, the composite with the adjoint for the Lefschetz pairing on $X \times X$ and the intersection pairing on $Z_K$, is an algebraic endomorphism of $T_X(-1)$ equal to $m_{a(e)}$ with $a(e) = \kappa\, e^2 c^2 \in F$, $\kappa \in \mathbb{Q}^\times$.*

(iii) *If $a(e) \notin \mathbb{Q}$, the exceptional classes of $X \times X$ are algebraic at $t$, and so is every Hodge class on every power of $X$.*

(iv) *$a(e) \notin \mathbb{Q}$ for every $e \in F^\times$ if and only if $K$ contains no imaginary quadratic field. If $K = F \cdot k$ with $k = \mathbb{Q}(\sqrt{-d})$ and $c = f\sqrt{-d}$, then $a(e) \in \mathbb{Q}$ exactly for $e \in \mathbb{Q} f^{-1}$.*

(v) *Conversely, if the exceptional classes are algebraic and some algebraic cycle on $X^r \times Z_K$ acts nonzero on $T_Z$, then every $\xi$ is algebraic.*

*Proof.* (i) The Hodge group of $X \times X \times Z_K$ lies in $G \cdot K^1$ and maps onto $\mathrm{Hg}(X) = G$; it acts on $T_X(-1)$ and on $T_Z$ through that image, since $K^1$ acts trivially on both. Both are the irreducible $G$-representation $T$ with $\mathrm{End}_G T = F$ (Proposition 21.2), so $\mathrm{Hom}_{\mathrm{Hdg}} = F\iota$. The count is the dimension of the kernel of the six raising and lowering operators of $\mathfrak{sl}_2^3$ on the torus-weight-zero part of $\bigwedge^4(V \oplus V \oplus D)$, computed by Künneth type: $(1,1,2)$ gives $3+3$, the line $F\iota$ and $\psi \otimes \mathrm{NS}(Z_K)$; $(2,2,0)$ gives $4 = \mathbb{Q} \oplus F$, the Hodge classes of $X \times X$ in $H^2 \otimes H^2$; $(0,0,4)$ gives $6 = S^2\mathrm{NS}(Z_K)$; $(2,0,2)$ and $(0,2,2)$ give 3 each; and $(4,0,0)$, $(0,4,0)$, $(3,1,0)$, $(1,3,0)$ one each, products of divisor classes. The sum is 26. The Hodge group of the product lies in $G \cdot K^1$, so these 26 classes are Hodge classes at every $t$, and 26 is a lower bound for the Hodge classes of degree four. At a very general $t$ the argument of Lemma 24.43, applied to the family $X \times X \times Z_K$ over $C$, whose monodromy is again Zariski dense in $G$, gives Hodge group exactly $G \cdot K^1$, and then the 26 classes are all of them; the description of the Künneth component $(1,1,2)$ follows.

(ii) The adjoint of a correspondence for the Lefschetz pairing on $H^2(X \times X)$ and the polarised intersection pairing on $Z_K$ is the transposed cycle composed with cup products with powers of the polarisations and the inverse Lefschetz isomorphism, which is algebraic on an abelian variety [Kle68, §2]; so $\xi^\dagger\xi$ is algebraic. By Lemma 24.44, $T_X(-1)$ carries $-4q_1$ and $T_Z$ carries a rational multiple of $q_{c^2}$, and for $\xi = m_e$, $q_{c^2}(m_e x, m_e y) = \mathrm{Tr}(c^2e^2\mathrm{trd}(xy)) = q_1(c^2e^2x, y)$, so $\xi^\dagger\xi = \kappa\, m_{e^2c^2}$.

(iii) An irrational $a \in F$ generates the cubic field, so every $m_b$, $b \in F$, is a polynomial in $\xi^\dagger\xi$, algebraic on $T_X(-1)$. Compose with the product map $\mu\colon H^2(X \times X) \supseteq T_X(-1) \to H^2(X)$ and its adjoint, both built from $\psi$, the diagonal and $\theta$ and hence algebraic (Lemma 21.3): the composite $\mu \circ (m_b \otimes 1) \circ \mu^\dagger$ acts on the primitive part $U$ of $H^2(X)$ by the element $\nu(\mathrm{Tr}\, b - b)/2$ of $F$, irrational when $b$ is. With the identity and the projector onto $\mathbb{Q}\theta$ this generates $\mathbb{Q} \times F = \mathrm{End}_{\mathrm{Hdg}} H^2(X)$, so by Proposition 21.2 the exceptional classes of $X_t \times X_t$ are algebraic, and then every Hodge class on every power of $X_t$ is algebraic (Theorem 21.7). The classes are flat along $C$, but by Lemma 22.2 algebraicity at $t$ gives algebraicity on the stratum through $t$ and, through the Hecke correspondences of $C$, on a countable dense set; whether some stratum is all of $C$ is not decided by this argument, so the conclusion is stated at $t$.

(iv) $a(e) = \kappa e^2 f^2 \delta$ is rational iff $e^2 f^2 \delta \in \mathbb{Q}$. If $-\delta \notin \mathbb{Q}^\times (F^\times)^2$ then $\delta \notin \mathbb{Q}^\times (F^\times)^2$ and $e^2 f^2 \delta \in \mathbb{Q}$ is impossible. If $-\delta = -dg^2$ with $d \in \mathbb{Q}^\times$, $g \in F^\times$, then with $c = f\sqrt{-d}$ one gets $a(e) = \kappa e^2 f^2 d$, rational iff $(ef)^2 \in \mathbb{Q}$ iff $ef \in \mathbb{Q}$, a cubic field having no irrational element with rational square. And $-\delta \in \mathbb{Q}^\times (F^\times)^2$ exactly when $K \supseteq \mathbb{Q}(\sqrt{-d})$ for some $d$.

(v) By Theorem 21.7 the Hodge conjecture holds for all powers of $X$ once the exceptional classes are algebraic. A cycle on $X^r \times Z_K$ acting nonzero on the irreducible $T_Z$ carries it

isomorphically onto a summand of $H^{2k}(X^r)(k-1)$; projecting onto that summand and composing with an isomorphism to $T_X(-1) \subseteq H^2(X \times X)$, a Hodge class on $X^{r+2}$ and algebraic, gives an algebraic nonzero element of $\mathrm{Hom}_{\mathrm{Hdg}}(T_Z, T_X(-1)) = F$; composing with the algebraic $m_a$ gives every element of that line, and transposing gives every $\xi$. ∎

**Proposition 24.46** (Hyper-Kähler carriers give $\pm 1$)**.** *Let $Y_1, \dots, Y_m$ be projective hyper-Kähler varieties of $K3^{[n_j]}$ type whose transcendental spaces are Hodge isometric to $(T, r_j q_{\lambda_j})$, $r_j \in \mathbb{Q}^\times$, $\lambda_j \in F$ totally positive. Every composite of rational Hodge isometries between them that returns to $Y_j$ acts on $T$ by $\pm 1$, and every composite of Hodge similarities by an element of $\mathbb{Q}$.*

*Proof.* A Hodge isometry $(T, r_j q_{\lambda_j}) \to (T, r_k q_{\lambda_k})$ is a Hodge endomorphism of $T$, hence $m_a$ for some $a \in F$, and it is an isometry iff $r_j \lambda_j = r_k \lambda_k a^2$. A composite returning to $j$ is $m_a$ with $a^2 = 1$, and a composite of similarities is $m_a$ with $a^2 \in \mathbb{Q}$, which in a cubic field forces $a \in \mathbb{Q}$. The isometries are algebraic by [Bus19, Mar24], which is why the statement matters: they realise $m_b$ between $S_\lambda$ and $S_{\lambda b^{-2}}$ and between their moduli spaces of sheaves, but every return is trivial. ∎

24.13. **Where the link stops.** Theorem 24.45 replaces the Kuga–Satake variety of Theorem 21.4, of dimension $2^7$, by an abelian sixfold over the same curve, and it replaces the irrationality that had to be extracted from anticommuting Clifford blocks by the arithmetic of $K$: for every $K$ that is not a compositum $F \cdot k$ every link closes the target. Its logical status is the same as the Kuga–Satake link: by (v) it is equivalent to the target together with one algebraic link, not a reduction to something smaller.

The missing step is the algebraicity of one class $\xi \in H^4(X \times X \times Z_K)$. It is exceptional: its component in $V \otimes V$ is the nine-dimensional $T$, not a multiple of $\psi$, so it is not invariant under $\mathrm{Sp}(V)$ and not in the Lefschetz algebra of $X \times X \times Z_K$ (Proposition 21.48); the theorems that prove the conjecture when the Hodge ring is generated by divisors do not apply, and no Weil class of $Z_K$ or of $X \times X$ is involved. At the points of complex multiplication of $C$ the class $\xi_c$ is a Hodge class on a CM abelian variety in a mixed Künneth component between $X_c \times X_c$ and $Z_{K,c}$ with $\mathrm{Hom}(X_c, Z_{K,c}) = 0$, so it is not a pull-back from one factor nor a product of lower-degree Hodge classes, and the generation theorems for CM abelian varieties do not reach it either. Unlike the Kuga–Satake class, the link is therefore not known even at the countably many special points. This is the price of staying inside abelian varieties, and it is the exact statement that (F2) needs at its smallest instance.

## 25. Devices that move the open statements without closing them

Three devices are often proposed for passing from the members where a class is known to be algebraic to the others, and two analytic descriptions of algebraic classes are exact. This section records what each one proves.

25.1. **Specialisation, degeneration and blow-ups.**

**Proposition 25.1.** *Let $\pi : \mathcal{X} \to B$ be a smooth projective family over an irreducible base and $\xi$ a flat Hodge class on its fibres.*

(i) *If $\xi_t$ is algebraic at the generic point of a curve $C \subseteq B$ then it is algebraic at every point of $C$. The converse implication, from one point of $C$ to its generic point, holds when $\xi_{t_0}$ is the Chern character of a perfect complex whose semiregularity map is injective, and in no greater generality is it known.*

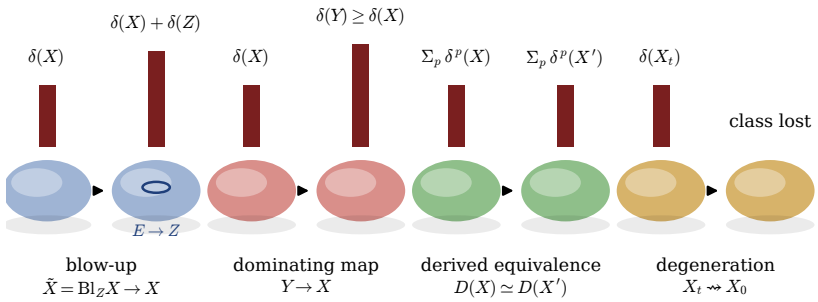



FIGURE 69. Where the defect goes under surgery. A blow-up along $Z$ carries the Hodge classes of $X$ to $\tilde{X}$ and adds those of $Z$, so the quotient Hdg/Alg grows by the defect of the centre; a dominating map $Y \to X$ carries the defect of $X$ into $Y$; a Fourier–Mukai equivalence permutes degrees and preserves the total defect; a degeneration to $X_0$ loses the class only in the special fibre, from which no cycle lifts.

(ii) *Let $\tilde{X}$ be the blow-up of a smooth projective variety $X$ along a smooth closed subvariety $Z$ of codimension $c$. Then* $\mathrm{Hdg}^k(\tilde{X})/\mathrm{Alg}^k(\tilde{X}) \cong \mathrm{Hdg}^k(X)/\mathrm{Alg}^k(X) \oplus \bigoplus_{i=1}^{c-1} \mathrm{Hdg}^{k-i}(Z)/\mathrm{Alg}^{k-i}(Z)$. *In particular the Hodge conjecture for $\tilde{X}$ is equivalent to the Hodge conjecture for $X$ and for $Z$, and a blow-up never reduces the defect of a variety.*

(iii) *For the Weil classes of $\mathcal{D}_{n,n}$ and the exceptional class of a Mumford fourfold, every degeneration to a fibre with a semistable model on which the limit class is algebraic has its lift to the generic fibre as its missing step, and for the Mumford fourfold no degeneration keeping the class of Hodge type exists, since the Hodge locus of the class is a compact Shimura curve.*

*Proof.* (i) Specialisation of cycles along a curve [Ful98, Ch. 20] sends an algebraic cycle on the generic fibre to one on each special fibre with the same flat cohomology class; this is the direction used by Schoen's degeneration and by Proposition 1.1. In the other direction the algebraic locus is a countable union of closed sets by Lemma 22.2, and a point in one stratum says nothing about the stratum being the whole base; the one proved lift is the semiregularity theorem of Bloch, Buchweitz and Flenner [Blo72, BF03], which is the content of Theorem 24.1. (ii) Manin's decomposition of the motive of a blow-up [Man68], $h(\tilde{X}) = h(X) \oplus \bigoplus_{i=1}^{c-1} h(Z)(-i)$, is realised by the algebraic correspondences $\pi^*$ and $j_* \circ p^* \circ$ (powers of the relative hyperplane class), which are compatible with Hodge classes and with algebraic classes; so both the Hodge classes and the algebraic classes of $\tilde{X}$ decompose accordingly, and the quotients add. Since every summand is non-negative, the defect of $\tilde{X}$ is at least that of $X$, and a blow-down along a smooth centre $Z$ lowers the defect by exactly that of $Z$ in the shifted degrees. (iii) A flat class has a limit in the limit mixed Hodge structure of a semistable fibre, and on a totally degenerate fibre the graded pieces are combinatorial and every class is algebraic on the components; the return to the generic fibre is the lift of (i), with the same unproved step. For a Mumford fourfold the Hodge locus of the exceptional class is the compact Shimura curve of Theorem 21.7, which has no boundary, so no fibre at which the class is still of Hodge type degenerates. ∎

25.2. **Dominating maps, birational models and derived equivalences.** Write $\delta^p(X) = \dim \mathrm{Hdg}^p(X) - \dim \mathrm{Alg}^p(X)$ for the defect of the conjecture in degree $2p$, $\delta^p_{\mathrm{ab}}(X) = \dim \mathrm{Hdg}^p(X) - \dim \mathrm{Ab}^p(X)$ for the defect of (F3′), and $\delta(X) = \sum_p \delta^p(X)$. The conjecture for $X$ is $\delta(X) = 0$. The following proposition records how the surgeries of birational geometry move these numbers; Figure 69 draws it.

**Proposition 25.2** (Defects under surgery)**.** *Let $X$, $Y$ be smooth complex projective varieties.*

(i) *If $f\colon Y \to X$ is surjective and generically finite, then $\delta^p(Y) \geq \delta^p(X)$ and $\delta^p_{\mathrm{ab}}(Y) \geq \delta^p_{\mathrm{ab}}(X)$ for every $p$. In particular no blow-up along a singular centre, weighted blow-up, resolution of a modification or finite cover of $X$ has a smaller defect than $X$.*

(ii) *If $X$ and $Y$ are birational, then $\delta^2(X) = \delta^2(Y)$ and $\delta^{d-1}(X) = \delta^{d-1}(Y)$, $d = \dim X$; for $3 \leq p \leq d-2$ the difference $\delta^p(X) - \delta^p(Y)$ is a signed sum of defects $\delta^{p-j}(Z)$, $j \geq 1$, of smooth projective varieties $Z$ of dimension less than $d$.*

(iii) *If $\Gamma \in \mathrm{CH}^*(X \times Y)_{\mathbb{Q}}$ induces an isomorphism from $H^*(X,\mathbb{Q})$ onto $H^*(Y,\mathbb{Q})$ whose inverse is induced by a cycle, for instance the kernel of an equivalence between the derived categories of $X$ and $Y$, then $\delta(X) = \delta(Y)$ and $\delta_{\mathrm{ab}}(X) = \delta_{\mathrm{ab}}(Y)$.*

(iv) *Let $\mathcal{X} \to \Delta$ be a projective family, smooth over the punctured disc, with a flat Hodge class $\xi$ on the smooth fibres that is algebraic on a crepant resolution $\hat{X}$ of the special fibre $X_0$, or on the components of a semistable model of $X_0$. Then $\xi$ is algebraic on $X_t$ for $t \neq 0$ if and only if the specialisation $H^{2p}(X_0) \to H^{2p}(X_t)$ sends an algebraic class to an algebraic class, which is the statement (V) for the family; the monodromy plays no role, since it fixes $\xi$.*

*Proof.* (i) $f_*f^*$ is multiplication by the degree, so $f^*$ is an injective morphism of Hodge structures and $f^*\mathrm{Hdg}^p(X) \subseteq \mathrm{Hdg}^p(Y)$. If $f^*\alpha$ is algebraic then so is $\deg(f)\,\alpha = f_*f^*\alpha$; hence $\mathrm{Alg}^p(Y) \cap f^*\mathrm{Hdg}^p(X) = f^*\mathrm{Alg}^p(X)$ and $f^*$ embeds $\mathrm{Hdg}^p(X)/\mathrm{Alg}^p(X)$ into $\mathrm{Hdg}^p(Y)/\mathrm{Alg}^p(Y)$. For Ab the same argument applies because $f_*$ of a class $\Gamma_*\gamma$ reached from an abelian variety is $(\Gamma_f \circ \Gamma)_*\gamma$, reached from the same one. The coarse space of a weighted blow-up has quotient singularities and $f_*f^* = \mathrm{id}$ with rational coefficients, so the argument is unchanged there.

(ii) By weak factorisation [AKMW02] a birational map between smooth projective varieties is a composite of blow-ups and blow-downs along smooth centres with smooth projective intermediate varieties, and Proposition 25.1(ii) gives the change of $\delta^p$ at each step as $\pm\sum_{j=1}^{c-1}\delta^{p-j}(Z)$. For $p = 2$ the terms are $\delta^1(Z)$ and $\delta^0(Z)$, which vanish by the Lefschetz $(1,1)$ theorem; for $p = d-1$ they are $\delta^{d-1-j}(Z)$ with $\dim Z \leq d-2$, hence $\delta^q(Z)$ with $q \geq \dim Z - 1$ after $j \geq 1$, and these vanish by hard Lefschetz and Lefschetz $(1,1)$ on $Z$.

(iii) By the projection formula $\Gamma_*$ sends $\bigoplus_p \mathrm{Hdg}^p(X)$ into $\bigoplus_p \mathrm{Hdg}^p(Y)$, since $[\Gamma]$ is a sum of classes of type $(i,i)$ and cup product and push-forward are morphisms of Hodge structures up to Tate twists; it sends algebraic classes to algebraic classes because $[\Gamma] \cup \mathrm{pr}_X^*[Z]$ is the class of a cycle and the push-forward of a cycle is a cycle; and it sends Ab to Ab by composing correspondences. The inverse does the same in the other direction, so the two total quotients are isomorphic. The kernel of a Fourier–Mukai equivalence and the kernel of its inverse are cycles by Mukai's and Orlov's description of the inverse functor [Orl97], which is the equivalence with kernel the dual of the original one shifted by the dimension.

(iv) $\xi$ is monodromy invariant, so by the local invariant cycle theorem it is the specialisation of a class in $H^{2p}(X_0)$, which pulls back to $\hat{X}$ or to the components of the semistable

model; the hypothesis says that the pulled back class is algebraic there, and push-forward keeps it algebraic on $X_0$. Cycles specialise from the generic fibre to the special fibre [Ful98, Ch. 20] and not the other way, so the remaining step is exactly the one in the statement. If the limit cycle on $X_0$ lifts to the family, its restriction to $X_t$ has class $\xi_t$; conversely if $\xi_t$ is algebraic the specialisation of a representing cycle is a cycle on $X_0$ with class the limit of $\xi$, and the specialisation map sends that class to $\xi_t$. The vanishing cycles, on which the monodromy acts, are orthogonal to the invariant part, and $\xi$ is invariant, so no condition on them enters. ∎

Items (i) to (iii) say that every surgery available in birational and derived geometry carries the defect of $X$ into its output as long as the open class survives there, and loses the class only in outputs of lower dimension, the centres, where it has become a defect of those varieties. Both open statements of this paper sit in degree four, (F2) for the Weil classes and the exceptional classes, (F3′) for the classes of $H^4(S \times S)$ on a K3 surface, and in degree four the defect is a strict birational invariant by (ii) and a derived invariant by (iii); so the open classes are present, and not known to be algebraic, on every birational model and every Fourier–Mukai partner of the varieties that carry them. Item (iv) is the reason that degenerations to products of elliptic curves, where every Hodge class is algebraic [MZ99], do not close (F2): the class returns to the smooth fibre only through (V). Flows of Kähler metrics change no complex structure between their surgeries, and the surgeries themselves are the contractions and flips of the minimal model programme, covered by (i) and (ii); the surgery of Perelman has no counterpart for Hodge structures, since it cuts a manifold along spheres and a Hodge structure has no local pieces to cut.

The devices therefore relocate the two open statements without weakening them. The one true reduction in shape is Theorem 22.5: the whole of (F2) is equivalent to a lift from CM points along polarised abelian schemes, and the lift along each scheme is a question about one semiregular object at one point.

25.3. **Residues and mass-minimising currents.** Two analytic descriptions of algebraic classes are exact, and we record what each reduces the conjecture to. The first is Griffiths' residue calculus on hypersurfaces [Gri69]. For a smooth hypersurface $X = \{F = 0\} \subset \mathbb{P}^{n+1}$ of even dimension $n = 2m$ and degree $d$, with Jacobian ring $R(F)$ and $\Omega = \sum_i (-1)^i x_i \, dx_0 \wedge \ldots \widehat{dx_i} \cdots \wedge dx_{n+1}$, the residues of the forms $P\Omega / F^{q+1}$ identify $H^{n-q,q}_{\mathrm{prim}}(X)$ with $R(F)_{(q+1)d-n-2}$.

**Proposition 25.3** (Residues of complete intersections)**.** *Let $X = \{F = 0\}$ be as above, and suppose that $F = \sum_{i=1}^{m+1} f_i g_i$ with $Z = \{f_1 = \cdots = f_{m+1} = 0\}$ a complete intersection of codimension $m+1$ in $\mathbb{P}^{n+1}$ whose ideal is generated by the $f_i$.*

(i) *Let $\theta$ be the hyperplane class and $J_H$ the Jacobian determinant of the $2m+2$ polynomials $f_1, g_1, \ldots, f_{m+1}, g_{m+1}$, which has degree $(m+1)(d-2)$. Then*

$$[Z] = \frac{\deg Z}{d}\, \theta^m - \frac{m!}{d}\, \operatorname{res}\Big( \frac{J_H \, \Omega}{F^{m+1}} \Big)^{m,m}$$

[VL22, Thm. 1.1].

(ii) *Let $J(F) \subseteq R(F)_{(m+1)(d-2)}$ be the span of the classes of the determinants $J_H$ over all such decompositions of $F$. If the rational span of the residues of $J(F)$ and $\theta^m$ contains $\mathrm{Hdg}^m(X)$, the Hodge conjecture holds for $X$.*

*Proof.* (i) is the cited theorem; the degree of the determinant is $\sum_i (\deg f_i + \deg g_i) - (n+2) = (m+1)d - 2m - 2$. (ii) By (i) every residue of an element of $J(F)$ is a rational combination

of $\theta^m$ and of classes of complete intersections in $X$, so it is algebraic. The Hodge conjecture for a hypersurface holds outside the middle degree by the Lefschetz hyperplane theorem, and in the middle degree it is the statement that $\mathrm{Hdg}^m(X)$ consists of algebraic classes. ■

The proposition replaces a cycle by a polynomial, and the open step becomes the statement that the polynomials of $J(F)$ have enough rational residues. Deciding whether a given element of $R(F)$ has a rational residue requires its periods, so the residue calculus moves the conjecture from cycles to transcendental numbers without closing it. On a Fermat hypersurface $\sum_i x_i^d = 0$ the linear subspaces cut out by $x_{2i} = \zeta_i x_{2i+1}$, $\zeta_i^d = -1$, are of this form, since $x_{2i}^d + x_{2i+1}^d$ factors into linear forms; whether their residues and those of the other decompositions exhaust $\mathrm{Hdg}^m(X)$ is again a question about periods.

## 26. Where the proof stops

This paper does not prove the Hodge conjecture. It proves that the conjecture is the pair of statements (F2) and (F3′) (Theorem 27.1), proves each of the two on the families and varieties listed there, and carries each to a first open instance at which a single object, a single class or a single correspondence decides it. This section says, without qualification, what those instances are and what is known about each.

The two open statements are not independent problems that happen to add up to the conjecture; they are two halves of one phenomenon. Both say that the algebraic locus of a flat Hodge class, a countable union of closed strata through a dense set of points of complex multiplication, is not meagre (Lemma 22.2, Theorem 22.5, Theorem 22.9). For (F2) the dense set is supplied by explicit cycles at CM points (Theorem 1.1, Theorem 20.29) and the first instance is a single class on an abelian fourteenfold or a single Ext group on an eightfold; for (F3′) the dense set is supplied by the K3 surfaces with complex multiplication and by the chains of Section 23, and the first instance is a cubic real multiplication on one K3 surface of Picard number 13, which Theorem 23.14 supplies for the fields $\mathbb{Q}(\zeta_7)^+$ and $\mathbb{Q}(\zeta_9)^+$ with the algebra ramified at two real places, and which stays open for every other pair of a cubic field and an algebra.

Each construction of this paper ends at a question with a definite answer. Whether some convolution of the pieces of Theorem 24.3 has $\dim \mathrm{Ext}^2 = 57$ is a finite computation in the Ext algebra of two sheaves on a fourfold; at the point $E_0^6$ of complex multiplication every group entering it is written down in Lemma 24.8, Proposition 24.11 answers it in the negative for every convolution with three powers of $L$, Proposition 24.13 answers it in the negative for two threefolds at one shift and for every placement of the thirty-five pieces, Proposition 24.14 with Corollary 24.15 for every further piece of positive twist within one shift of the thirty-two, Theorem 24.23 for at most five line bundles at any twists and every polarisation, the enumeration of Remark 24.25 for six and seven and for eight outside thirty-one arrangements, Corollary 24.31 and Theorem 24.33 for every convolution of threefold pieces, of one family or of both, at any twists and shifts, and for the configurations left, eight or more line bundles with or without threefolds, it is the rank of the products along the chains of Proposition 24.10 and along the cycles through the line bundles, the support of the object being constrained by Theorem 24.27 and Propositions 24.30 and 24.35. On $E_0^8$, where the criterion asks for $\dim \mathrm{Ext}^2 = 104$, every convolution of line bundles whose shifts lie in seven consecutive values fails outside ten arrangements (Corollary 24.38), and single shifts at any odd distance fail (Proposition 24.37). Whether the link $\xi$ is algebraic is a question about one Hodge class on one abelian variety, equivalent by Theorem 24.45(v) to the exceptional classes together with one more cycle. Whether a correspondence acting

on $T(S_\lambda)$ otherwise than by a similarity exists is a question about one K3 surface; for $F = \mathbb{Q}(\zeta_7)^+$ or $\mathbb{Q}(\zeta_9)^+$ and $D = D_\infty$ Theorem 23.14 answers it, and for every other pair $(F, D)$ it is open, with Proposition 23.15 marking the three places where the present mechanism fails. A negative answer to the first closes that route, as the analogous question on $E_0^6$ closed; a positive answer to any of them proves the corresponding statement at its first instance.

What a proof of the conjecture along the lines of this paper still needs is therefore one of the following, and nothing else.

(a) For (F2): a perfect complex on one abelian variety of Weil type at one CM point, with Chern character of polarised shape and $\dim \mathrm{Ext}^2$ equal to the number of Theorem 17.9, or the interior point of Theorem 22.5 supplied by some other mechanism; the constructions of Section 24 and Section 18 exclude every convolution of the pieces now known at $E_0^6$ except those named above.

(b) For (F2) on the squares of Mumford fourfolds: the algebraicity of one class $\xi$ on $X \times X \times Z_K$ (Theorem 24.45), of one Kuga–Satake class on one K3 surface of Picard number thirteen (Theorem 21.4), or the Lefschetz standard conjecture for one ninefold (Corollary 20.30).

(c) For (F3′) on the varieties of abelian type: a correspondence on $S_\lambda$ with an irreducible component of geometric genus at least two and transcendental rank at least eighteen, not the graph of an element of a finite group, for each cubic field other than $\mathbb{Q}(\zeta_7)^+$ and $\mathbb{Q}(\zeta_9)^+$ with $D_\infty$ (Proposition 23.11, Proposition 23.15), and families on which the chain of Theorem 23.3 can be started for the real quadratic fields other than $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{3})$ and $\mathbb{Q}(\sqrt{6})$.

(d) For (F3′) beyond abelian type: a Hodge class off the abelian part of a variety that no correspondence connects to an abelian variety, which is removed only by a new algebraic cycle on a product $B \times X$ (Remark 20.36); no such class is known, and none is excluded.

The devices of this section do not shorten this list: specialisation, degeneration, blow-ups, dominant maps, derived equivalences and residues each carry the open statements to another variety of the same or larger defect. The proved part of the paper is Theorem 27.1; the list above is the whole of what separates it from the conjecture.

## 27. The closure theorem

We collect the results of the paper in one theorem. Its first clause is Theorem 1.1; the second and third give the base of that theorem in its sharpest form; the fourth and fifth say what closing the algebraic locus requires and which constructions cannot supply it; and the last places the Weil classes inside the Hodge conjecture and states what Theorem 1.1 becomes when it is carried to every family. The theorems we quote from the literature are used as theorems: the semiregularity theorem of Buchweitz and Flenner [BF03], in the form for perfect complexes that Pridham proves [Pri24]; the theorem of Cattani, Deligne and Kaplan [CDK95]; the theorem of Baily and Borel [BB66]; the theorem of Pila, Shankar and Tsimerman on the André–Oort conjecture [PST21]; the description of the Hodge rings of abelian varieties of dimension at most five by Moonen and Zarhin [MZ99]; Markman's theorems on abelian varieties of Weil type [Mar25a, Mar25b]; the theorem of Bando and Siu on stable reflexive sheaves [BS94], Kuranishi's theorem in the form of Goldman and Millson [Kur65, GM88], Grivaux's Chern classes of coherent sheaves [Gri10] and Voisin's

theorem on Weil tori [Voi02b]; and the theorems of Deligne, André and Milne recorded in Section 20.6 [Del71, And96b, Mil20].

**Theorem 27.1** (Closure)**.** *Let $F$ be a CM field of degree $2m$, let $n \geq 2$, and let $\delta$ be a discriminant. Write $K = \mathbb{Q}(\sqrt{-d})$ when $m = 1$.*

(i) The algebraic locus. *The algebraic locus $\Sigma_F$ of the Weil classes contains an explicit CM point at which they are polynomials with rational coefficients in divisor classes, it is dense, it is a countable union of closed algebraic strata, and it is either the whole domain or meagre. It is closed if and only if the Hodge conjecture holds for the very general member of the family, and then the Weil classes are algebraic on every member (Theorem 1.1).*

(ii) One class on one connected domain. *For an abelian $2n$-fold $A$ of $(K, -1, n)$-Weil type with $\mathrm{Hg}(A) = \mathrm{SU}(V, H)$ one has $\dim_{\mathbb{Q}} \mathrm{Hdg}^k(A) = 1$ for $k \neq n$ and $3$ for $k = n$, and the Hodge conjecture for $A$ in every degree is the algebraicity of the single class $\omega_1$ of (11), written out with $2^{2n-1}$ integer coefficients. For the family of the triple $(K, n, \delta)$ the conjecture is equivalent to the algebraicity of the absolutely Hodge class $\omega_s$ at every point $s$ of the connected period domain $\mathcal{D}_{n,n}$ of dimension $n^2$ (Theorem 4.14, Corollary 4.15, Theorem 15.14, Corollary 15.35).*

(iii) A base point in every family. *Every family of every CM field contains a member isogenous to a product of CM abelian $m$-folds at which the Weil classes are polynomials in divisor classes (Theorem 19.5). For an imaginary quadratic field every family also contains a member with quaternionic multiplication extending $K$, at which the Weil classes are the real and imaginary parts of the $n$-th power of one explicit $(1, 1)$ class, and every product of $n$ abelian surfaces of Weil type is a member whose Weil classes are an explicit integral polynomial in $2n$ divisor classes (Theorem 15.25, Theorem 15.27, Theorem 15.10).*

(iv) Closing the locus. *If at every point of the orbit $G_F(\mathbb{Q}) \cdot s_0$ the Weil class is represented by a cycle whose Hilbert data comes from one finite set, then $\Sigma_F = \mathcal{D}_F$; this hypothesis is equivalent to its conclusion, and the transported cycle does not satisfy it whenever a component of the base cycle has finite stabiliser (Theorem 16.9, Theorem 16.17, Theorem 16.15). The bound is needed only along one sequence of CM points that leaves every proper special subvariety, and such a sequence exists inside the orbit (Theorem 19.14). The locus of members with smaller Hodge group, which contains every member at which this paper exhibits algebraic Weil classes, contains the base point, is dense, stable under $G_F(\mathbb{Q})$ and a countable union of closed algebraic strata, and is meagre, so closing the locus requires the algebraicity of the Weil classes at a member with full Hodge group (Proposition 19.12, Corollary 19.13). If at a base point some perfect complex whose Chern character is a nonzero multiple of a rational Weil class plus a polynomial in the divisor classes of the family has injective semiregularity map, or satisfies the flatness condition of Proposition 17.15, then $\Sigma_F = \mathcal{D}_F$ (Theorem 16.29, Theorem 19.7). When the Chern character is exactly a multiple of a Weil class, no complex at any member of any family, for any CM field, has injective semiregularity map (Theorem 17.5, Theorem 19.19). In the polarised form, for an imaginary quadratic field and $n \geq 3$, every object has $\dim \mathrm{Ext}^2(E, E) \geq (4 + \rho)n^2 - 2n$, with $\rho \leq 3$ the rank of the Hankel matrix of the coefficients, and an object attaining the bound meets the criterion (Theorem 17.9); at a quartic CM field and $n = 2$ the number takes fifteen values, an object meeting the criterion has $\dim \mathrm{Ext}^2(E, E) \geq 88$, and the characters of rank $88$ whose Hodge*

*locus leaves the polarised family are exactly those of $\theta^4$ shape, which the Weil tori exclude (Theorem 19.29, Corollary 19.30, Proposition 19.32, Corollary 19.33). The statement is needed only for the split families of the CM fields of degree at least four, which carry the Weil classes of every CM field by descent and scalar extension (Proposition 20.13), and for one family it is equivalent to the Lefschetz standard conjecture for the total space of the family over a curve through a base point (Corollary 20.31).*

(v) Constructions that do not reach a Weil class. *Each of the following fails to produce a Weil class on a general member, in the sense and under the hypotheses of the theorem cited, which Section 20 delimits:*
   (a) *an object restricted from a projective space to the dual and transported by the Poincaré kernel (Theorem 7.3);*
   (b) *equivariant for the field with the norm character (Theorem 5.6);*
   (c) *an object whose Chern character lies on the Weil line and on which $H^2(\mathcal{O}_A)$ acts faithfully, whether through semiregularity or through the weakened criterion of* [Mar25b, Question 11.4] *(Theorem 9.7, Corollary 9.18, Theorem 9.19);*
   (d) *an object with a line bundle summand off the polarisation line, in particular a sum of line bundles (Theorem 9.20, Corollary 18.21);*
   (e) *an object whose Chern character is effective at a base point (Theorem 18.5);*
   (f) *induction through hyperplane sections, in either direction (Theorem 10.3, Theorem 10.8);*
   (g) *Kuga–Satake from a weight two sub-structure of $H^2(A, \mathbb{Q})$ (Proposition 11.11, Corollary 11.12);*
   (h) *a divisorial argument at a general member of a family of nontrivial discriminant (Proposition 15.18);*
   (i) *a secant object whose support has two smooth branches of codimension two meeting transversally at a point, which includes the candidate of* [Mar25b, Section 12] *(Theorem 18.61, Corollary 18.63, Corollary 18.77);*
   (j) *a secant object of rank one whose support is a complete intersection of two divisors (Theorem 18.81), or a secant object quasi-isomorphic to a two-term complex of sums of powers of the polarisation or of simple semi-homogeneous bundles (Theorem 18.89, Theorem 18.92); on the fourfold, a secant sheaf $\mathcal{I}_S(3\Theta)$ with smooth support unless $d \leq 9$, that is $d \in \{1, 3, 5, 7\}$ in Setup 18.51, and one whose ideal sheaf is presented by two bundles with classes in the subring generated by line bundles unless $d \equiv 15, 23 \pmod{24}$ (Corollary 18.68, Corollary 18.84);*
   (k) *a representative of the Weil $K$-theory class that is a locally free sheaf on a smooth germ of codimension at least two of its support, or whose support has codimension at least $n$ (Theorem 16.31, Theorem 16.39);*
   (l) *at a quartic CM field, a complex with secant character that is locally free on a smooth curve at some point, in particular the explicit pair of* [Mar25c, Example 11.2.7] *(Theorem 19.43, Corollary 19.45);*
   (m) *a convolution of fewer than $2n$ line bundles with nondegenerate differences (Theorem 18.26); on $E_0^6$, every convolution of the Weil pieces with three multiples of the polarisation, and on $E_0^8$ every one whose pieces have their shifts within six consecutive values (Theorem 18.35, Corollary 18.48).*

(vi) The Weil classes inside the conjecture. *The Hodge conjecture is equivalent to the conjunction of* (F2)*, the algebraicity of the Hodge classes of abelian varieties outside the subring generated by divisor classes and Weil classes of quotients, and* (F3′)*, the algebraicity of every Hodge class modulo the images of Hodge classes of abelian varieties under algebraic correspondences; and* (F2) *alone is equivalent to the Hodge conjecture for abelian varieties (Proposition 21.49, Proposition 20.35). The statement* (F3′) *holds on every variety whose cohomology is reached from abelian varieties by correspondences, on every variety of dimension at most three, on every fourfold and fivefold whose zero-cycles are supported in dimension at most three or that is dominated by a variety of the first kind, on every smooth ample divisor of odd dimension in a variety of the first kind, and on every smooth Delsarte hypersurface of every dimension and every cyclic cover branched along one, and on every complete intersection of diagonal hypersurfaces of Vandermonde type, where the Hodge conjecture follows for two diagonal hypersurfaces of degree* 3*,* 4 *or* 6 *in* $\mathbb{P}^6$ *and for the very general member of degree* 2 *in every dimension, of degree* 3 *or* 4 *up to dimension seven and of degree* 6 *up to dimension five (Proposition 20.35, Proposition 20.42, Proposition 20.43, Proposition 20.44, Proposition 20.46, Proposition 20.47, Corollary 20.53, Theorem 20.55, Corollary 20.56, Theorem 20.66). The first instance of* (F2)*, the two exceptional classes on the square of a Mumford fourfold, is algebraic at every CM point of the family, is algebraic on every member once one Kuga–Satake class is algebraic, and is algebraic on every member exactly when the Lefschetz standard conjecture holds for one ninefold (Theorem 20.29, Theorem 21.4, Corollary 20.30). Over the rule set formed by the implications of the paper and of the literature, together with Proposition 21.49, the minimal sets of statements from which the conjecture follows are* $\{\mathrm{F3}\}$*,* $\{\mathrm{L},\mathrm{M}\}$*,* $\{\mathrm{L},\mathrm{F3}'\}$*,* $\{\mathrm{V},\mathrm{F3}'\}$ *and* $\{\mathrm{F2},\mathrm{F3}'\}$*, the first two being each equivalent to it (Theorem 20.70, Remark 21.50). Carried to every family, Theorem 1.1 gives exactly these two statements back:* (F2) *is equivalent to the statement that in every polarised abelian scheme with a member of CM type the algebraic locus of a flat Hodge class that is algebraic at such a member has an interior point (Theorem 22.5);* (F3′) *is equivalent to the absence of Hodge classes off the part of the cohomology reached from abelian varieties by algebraic correspondences (Proposition 22.7); and on the squares of K3 surfaces with real multiplication, the first varieties on which* (F3′) *is open, the algebraic locus of the real multiplication is dense, contains every CM point and is either the whole family or meagre (Theorem 22.9), and it is the whole family for real multiplication by* $\sqrt{2}$*,* $\sqrt{3}$ *and* $\sqrt{6}$ *in families, for complex multiplication by* $\mathbb{Q}(\sqrt{-5})$ *and* $\mathbb{Q}(\sqrt{-7})$ *on the Fricke families, and for the cubic fields* $\mathbb{Q}(\zeta_7)^+$ *and* $\mathbb{Q}(\zeta_9)^+$ *on the surfaces of the Mumford fourfolds of the algebra split at one real place (Theorem 23.5, Theorem 23.9, Theorem 23.14).*[66]

*Proof.* Each clause is the theorem or theorems cited with it, and no clause is used in the proof of another. For (i), Theorem 1.1 is proved at the end of Section 19.3. For (ii), Theorem 4.14 counts the Hodge classes, Corollary 4.15 reduces every degree to $\omega_1$, Theorem 4.9(i) shows that $\omega_1$ has integer coefficients and exactly $2^{2n-1}$ nonzero terms,

[66]These are the statements (L), (M) and (V) of Definition 20.14: $B(X)$ of Definition 10.6 for every smooth projective $X$; every Hodge class motivated in the sense of André [And96b, §2.1]; and, for every smooth projective morphism $f$ to a smooth connected base, a flat section of $R^{2p}f_*\mathbb{Q}$ that is algebraic at one point is algebraic at every point. The statement (F3) is the Hodge conjecture for the varieties that are not abelian, which is equivalent to the whole conjecture (Proposition 20.33).

Theorem 15.14(a) and (b) are the two implications for the family, and Proposition B.8 gives the connectedness and the dimension. For (iii), Theorem 19.5 gives the CM point, Theorem 15.25(v) the cycle at a quaternionic point, Theorem 15.27 places such a point in every family by the discriminant computation of Theorem 15.25(iv) and Lemma 15.26, and Theorem 15.10 treats products.

For (iv), the bounded criterion is Theorem 16.9 with Theorem 16.17 and Theorem 16.15, and its form for a CM field is Theorem 19.7(iv); the escaping sequence is Theorem 19.14, and the special locus is Proposition 19.12 with Corollary 19.13. The semiregularity criterion is Theorem 16.29 with Remark 17.6, carried to every CM field by Theorem 19.7(v); the flatness form is Proposition 17.15. The pure form is excluded by Theorem 17.5, which rests on Lemma 17.2, Lemma 17.3 and Proposition 17.4, and for every CM field by Theorem 19.19. The number of the polarised form is Theorem 17.9(i), (iv) and (v); the quartic rank and the Hodge loci are Theorem 19.29, Corollary 19.30, Proposition 19.32 and Corollary 19.33. The reduction to the split families is Proposition 20.13, and the equivalence with the Lefschetz standard conjecture is Corollary 20.31. For (v), each item is the statement of the theorem cited, under its own hypotheses, which Section 20 delimits item by item.

For (vi), Proposition 20.35(i) gives the Hodge conjecture from (F3′) and the conjecture for abelian varieties, and Proposition 21.49 gives the latter from (F2); the converse implications hold because (F2) is a case of the conjecture and (F3′) is weaker than it. The cases of (F3′) are Proposition 20.35(iii) and (iv), Proposition 20.42, Proposition 20.43, Proposition 20.44, Proposition 20.46, Corollary 20.53, Theorem 20.55, Corollary 20.56 and Theorem 20.66. The statements on the Mumford classes are Theorem 20.29, Theorem 21.4(ii) and Corollary 20.30. The statements on the families are Theorem 22.5, Proposition 22.7 and Theorem 22.9; the families on which the real multiplication is algebraic are Theorem 23.5, Theorem 23.9 and Theorem 23.14. The minimal sets are Theorem 20.70(ii) with the rule of Proposition 21.49 adjoined, as computed in Remark 21.50, and Theorem 20.70(vi) shows that {F3} and {L, M} are each equivalent to the conjecture. ■

The theorem arranges the paper around Theorem 1.1. Clause (ii) is the problem in its smallest form, one explicitly written class on one connected domain, and clause (iii) shows that the base case is never the difficulty: every family has a base point, for every field, every dimension and every discriminant, with a cycle written down rather than an existence statement. Clause (iv) reduces the closedness of the locus to one object at one point and computes the number that object must attain, and clause (v) shows that the constructions that fail do so for identifiable structural reasons: a divisor-generated Chern character cannot leave the divisor subring, a norm eigenclass sits in the wrong summand of the character grading, a Chern character on the Weil line has an annihilator exactly as large as the family, a support with two transversal branches of codimension two through a point carries a local obstruction that the compensating scalar cannot meet, and a curve along which an object is locally free is tangent to at most one of the two eigenplanes of the real multiplication. Clause (vi) shows where the Weil classes stand in the whole conjecture: the conjecture for abelian varieties is carried by the classes beyond the Weil lines, and the rest is algebraicity modulo abelian varieties, which holds on every variety whose cohomology comes from abelian varieties, on the Delsarte hypersurfaces of every dimension and on the diagonal complete intersections of Vandermonde type (Figure 70). Section 22 shows that the structure Theorem 1.1 finds in the Weil families, a CM seed, density and the dichotomy, is present in every family that carries (F2) or (F3′), and that in each of them the one missing property is the interior point that closes the locus (Theorem 22.10).

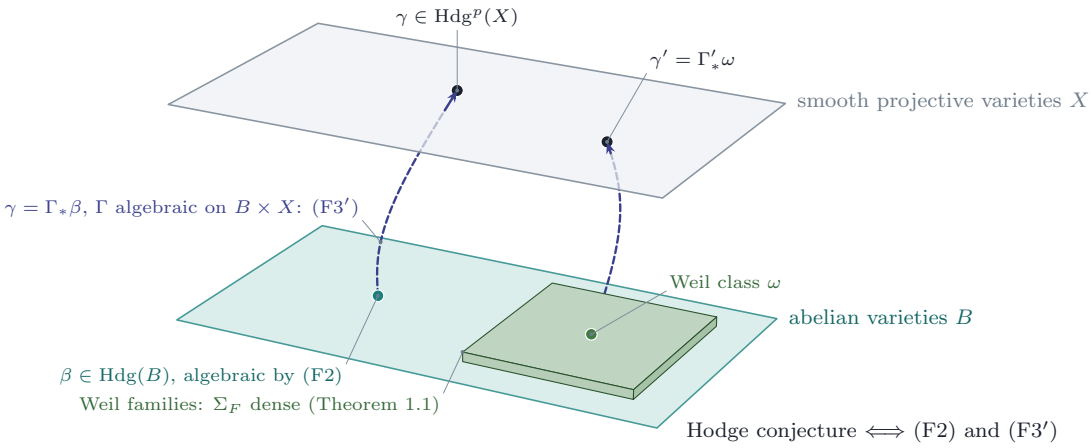



FIGURE 70. The two halves of the conjecture in Theorem 27.1(vi). A Hodge class $\gamma$ on a smooth projective variety $X$ is the image $\Gamma_*\beta$ of a Hodge class $\beta$ on an abelian variety $B$ under an algebraic correspondence $\Gamma$ on $B \times X$, which is (F3′); the class $\beta$ is algebraic, which is (F2), equivalently the conjecture for abelian varieties (Proposition 21.49). The raised plate on the lower sheet stands for the families of Weil type, on which Theorem 1.1 makes the algebraic locus of the Weil class $\omega$ dense; the dashed part of each correspondence is the part behind the upper sheet.

## APPENDIX A. THE EXTERIOR ALGEBRA MODEL IN COORDINATES

We record in coordinates the computation behind Theorem 6.5, both to pin down the sign in (20) and because the programs of the archive [BMB26] use this model.

Let $g \geq 1$ and let $W$ be the $\mathbb{Q}$-vector space with basis $e_1, \dots, e_{2g}, f_1, \dots, f_{2g}$. We identify $V = \langle e_1, \dots, e_{2g} \rangle$ with $H^1(A, \mathbb{Q})$ and $V^\vee = \langle f_1, \dots, f_{2g} \rangle$ with $H^1(A^\vee, \mathbb{Q})$, so that $\bigwedge^* W = H^*(A \times A^\vee, \mathbb{Q})$.

Number the generators so that the generator numbered $i$ is $e_i$ for $i \leq 2g$ and $f_{i-2g}$ for $i > 2g$. A basis of $\bigwedge^* W$ is then indexed by the subsets $S$ of $\{1, \dots, 4g\}$, the basis element $w_S$ being the wedge of the generators in $S$ in increasing order. The product is $w_S w_T = \varepsilon(S,T)\, w_{S \cup T}$ when $S \cap T = \emptyset$ and zero otherwise, where

$$\varepsilon(S,T) = (-1)^{\#\{(s,t) \in S \times T \,:\, s > t\}}. \tag{89}$$

The three distinguished elements are

$$\ell = \sum_{i=1}^{2g} e_i \wedge f_i, \qquad \theta = \sum_{j=1}^{g} e_j \wedge e_{g+j}, \qquad \theta' = \sum_{j=1}^{g} f_j \wedge f_{g+j},$$

and $p_*$ is the linear map $\bigwedge^* W \to \bigwedge^* V$ that sends $w_S$ to $\varepsilon(S \cap [1,2g],\, S \setminus [1,2g])\, w_{S \cap [1,2g]}$ when $S \supseteq \{2g+1, \dots, 4g\}$ and to zero otherwise.

**Lemma A.1.** *Let $0 \leq k \leq g$, let $T \subseteq \{1, \dots, g\}$ with $|T| = k$, let $U = \{1, \dots, g\} \setminus T$ and $S = U \cup (g + U)$. In the product $\theta'^k \cdot e^\ell$ the unique term contributing $w_S$ after $p_*$ is*

$$k! \Big( \prod_{j \in T} f_j f_{g+j} \Big) \cdot \Big( \prod_{i \in S} e_i f_i \Big),$$

*and its image under $p_*$ is $k!\,\sigma(g,k)\prod_{j\in U} e_j\wedge e_{g+j}$ with $\sigma(g,k)=(-1)^{g(g+1)/2+k}$, independent of $T$.*

*Proof.* The uniqueness of the term and the identification of $S$ are established in the proof of Theorem 6.5. The sign is the product of the two shuffle signs (89) relating the displayed word to the target word $\big(\prod_{j\in U} e_j e_{g+j}\big)(f_1\cdots f_{2g})$. Relabelling the symplectic pairs by a permutation $\pi$ of $\{1,\dots,g\}$ carries the word for $T$ to the word for $\pi(T)$ and multiplies both the displayed and the target word by the sign of the induced permutation of generators, so the quotient is unchanged. Since the permutations of $\{1,\dots,g\}$ act transitively on its $k$-subsets, $\sigma$ depends only on $(g,k)$. Evaluating at $T=\{1,\dots,k\}$, as in the proof of Theorem 6.5, gives the stated value. ∎

*Remark* A.2. At $g=2$ the computation can be done by hand. There $\ell=e_1f_1+e_2f_2+e_3f_3+e_4f_4$, $\theta=e_1e_3+e_2e_4$ and $\theta'=f_1f_3+f_2f_4$, and one finds $\Phi_{\mathcal{P}}(1)=-\frac{1}{2}\theta^2$, $\Phi_{\mathcal{P}}(\theta')=\theta$ and $\Phi_{\mathcal{P}}(\theta'^2)=-2$, in agreement with (20) for $g=2$, where $(-1)^{g(g+1)/2}=(-1)^3=-1$.

## Appendix B. The locus of Hodge classes

Deformation arguments about Hodge classes rest on one theorem, that of Cattani, Deligne and Kaplan [CDK95]: the locus in a family where a given class stays a Hodge class is an algebraic subvariety, not merely an analytic one. This appendix states the theorem, proves its elementary part, isolates the two analytic inputs that the rest requires, computes the example that matters here, and explains why the theorem is not a route to the Hodge conjecture. The last point is the one that bears on Section 14, and it is why the material is set out here rather than cited.

### B.1. **Variations of Hodge structure.**

**Definition B.1** (Variation of Hodge structure)**.** Let $S$ be a smooth connected complex quasi-projective variety. A *polarised variation of $\mathbb{Q}$-Hodge structure* of weight $w$ on $S$ consists of

(a) a local system $\mathbb{V}$ of finite dimensional $\mathbb{Q}$-vector spaces on $S$, with associated holomorphic bundle $\mathcal{V}=\mathbb{V}\otimes_{\mathbb{Q}}\mathcal{O}_S$ carrying the flat connection $\nabla$ whose flat sections are $\mathbb{V}$;
(b) a decreasing filtration $\mathcal{V}=F^0\supseteq F^1\supseteq\cdots$ by holomorphic subbundles, inducing on each fibre $\mathbb{V}_s$ a $\mathbb{Q}$-Hodge structure of weight $w$;
(c) a flat nondegenerate $(-1)^w$-symmetric pairing $Q\colon\mathbb{V}\otimes\mathbb{V}\to\mathbb{Q}_S$ polarising each fibre;

subject to Griffiths transversality

$$\nabla F^p\subseteq F^{p-1}\otimes\Omega^1_S. \tag{90}$$

The example to keep in mind is the geometric one: for $f\colon\mathcal{X}\to S$ smooth projective, $\mathbb{V}=R^wf_*\mathbb{Q}$ with its Gauss–Manin connection is such a variation, and (90) is Griffiths' theorem [Voi02, Ch. 10]. The other example we use is the Weil family of Proposition B.8 below, which is defined in the first instance by a period domain rather than by a family of varieties, and which carries a variation all the same.

*Notation* B.2. Write $\mathcal{V}^{\mathrm{an}}$ for the total space of $\mathcal{V}$, a complex manifold fibred over $S$ with fibre $\mathbb{V}_s\otimes\mathbb{C}$, and $\mathbb{V}^{\mathrm{tot}}\subset\mathcal{V}^{\mathrm{an}}$ for the subset of pairs $(s,\alpha)$ with $\alpha\in\mathbb{V}_s$, a covering space of $S$ with countable fibres.

B.2. **Hodge loci.**

**Definition B.3.** Let $\mathbb{V}$ be a polarised variation of weight $2p$ on $S$. Its *locus of Hodge classes* is

$$\mathrm{Hdg}^p(\mathbb{V}) = \{(s,\alpha) \in \mathbb{V}^{\mathrm{tot}} \ : \ \alpha \in F^p\mathcal{V}_s\}.$$

For $N \in \mathbb{Q}$ write $\mathrm{Hdg}^p_N$ for the part with $Q(\alpha,\alpha) = N$.

By conjugation symmetry, a rational $\alpha$ of weight $2p$ lies in $F^p$ exactly when it is of type $(p,p)$. The following two lemmas are elementary, and they give everything about $\mathrm{Hdg}^p$ except algebraicity.

**Lemma B.4** (Analyticity)**.** $\mathrm{Hdg}^p(\mathbb{V})$ *is a closed analytic subset of* $\mathbb{V}^{\mathrm{tot}}$*, and the projection* $\pi\colon \mathrm{Hdg}^p(\mathbb{V}) \to S$ *is locally biholomorphic onto its image on each connected component. In particular each component is a closed analytic subvariety of an open subset of* $S$*, and there are countably many components.*

*Proof.* Work on a simply connected open $U \subseteq S$ over which $\mathbb{V}$ is trivialised, $\mathbb{V}|_U \cong V_0 \times U$ for a fixed $\mathbb{Q}$-vector space $V_0$. Then $\mathbb{V}^{\mathrm{tot}}|_U$ is the disjoint union over $\alpha \in V_0$ of copies of $U$. Fix $\alpha$. The quotient $\mathcal{V}/F^p$ is a holomorphic vector bundle on $U$, and the image $\bar\alpha(s)$ of the flat section $\alpha$ in $(\mathcal{V}/F^p)_s$ is a holomorphic section of it. The set of $s \in U$ with $\alpha \in F^p\mathcal{V}_s$ is the zero locus of $\bar\alpha$, a closed analytic subset of $U$. Taking the union over the countable set $V_0$ gives the first statement, and shows that on the sheet indexed by $\alpha$ the projection $\pi$ is the inclusion of that zero locus into $U$, hence injective. There are countably many components because $S$ is covered by countably many such $U$, the set $V_0$ is countable, and each zero locus has countably many components. ∎

**Lemma B.5** (Finiteness at bounded norm)**.** *Fix* $s \in S$ *and* $N \in \mathbb{Q}$*. The set of* $\alpha \in \mathbb{V}_s$ *with* $\alpha$ *of type* $(p,p)$*,* $Q(\alpha,\alpha) = N$*, and* $\alpha$ *in a fixed lattice* $\mathbb{V}_{s,\mathbb{Z}}$*, is finite. The number of such* $\alpha$ *is locally constant on each component of* $\mathrm{Hdg}^p_N$*.*

*Proof.* Decompose $\alpha$ into its primitive parts for the Lefschetz decomposition attached to the polarisation. On the primitive part of type $(p,p)$ in weight $2p$ the Hodge–Riemann bilinear relations read $(-1)^p\,Q(\alpha,\overline{\alpha}) > 0$ for $\alpha \neq 0$, so the form $(-1)^pQ$ is positive definite on the real span of the primitive real $(p,p)$-classes; the same holds on each Lefschetz summand with an alternating sign, and the resulting form $\tilde Q$ obtained by flipping signs summand by summand is positive definite on the whole space of real $(p,p)$-classes. The conditions $Q(\alpha,\alpha) = N$ and $\alpha$ of type $(p,p)$ therefore confine $\alpha$ to a compact set for $\tilde Q$, and a compact set meets a lattice in finitely many points. The count is locally constant because the lattice and $Q$ are flat and the Hodge filtration varies continuously, so no Hodge class can appear or disappear without its $Q$-norm changing. ∎

**Theorem B.6** (Cattani, Deligne and Kaplan [CDK95])**.** [67] *Let* $S$ *be a smooth complex quasi-projective variety and* $\mathbb{V}$ *a polarised variation of* $\mathbb{Z}$*-Hodge structure of weight* $2p$ *on* $S$*. Then each connected component of* $\mathrm{Hdg}^p(\mathbb{V})$ *is a closed algebraic subvariety of* $\mathbb{V}^{\mathrm{tot}}$*, and its image in* $S$ *is a closed algebraic subvariety of* $S$*.*

We do not reproduce the proof, which draws on the whole asymptotic theory of degenerating Hodge structures developed in the two preceding decades. For families of abelian varieties the algebraicity of these loci also follows, by a different route, from Deligne's theorem that Hodge classes on abelian varieties are absolutely Hodge; Theorem B.6 is

[67] The paper is short and the proof is not: the two inputs isolated below, the nilpotent orbit theorem and the $\mathrm{SL}_2$-orbit theorem, carry the weight.

what carries it beyond abelian varieties. We record where the difficulty lies and which two theorems remove it, since the shape of the argument is what one needs in order to judge whether a deformation argument in this area is legitimate.

By Lemma B.4 a component $Z$ of $\mathrm{Hdg}^p$ is already a closed analytic subvariety of an open subset of $S$, and by Lemma B.5 it carries a locally constant $Q$-norm and is one sheet of a finite cover of its image. Choose a smooth projective compactification $\bar{S} \supseteq S$ with $\bar{S} \setminus S$ a normal crossing divisor. By Chow's theorem it is enough to show that $Z$ extends to an analytic subvariety of $\bar{S}$, that is, that $Z$ does not oscillate wildly as one approaches the boundary. This is an asymptotic question about the Hodge filtration, and it is answered by two theorems.

(A) *The nilpotent orbit theorem* of Schmid [Sch73]: near a boundary point the period map is approximated, to exponentially small error in the natural coordinates, by the orbit of a nilpotent Lie algebra element built from the local monodromy logarithms.

(B) *The* $\mathrm{SL}_2$*-orbit theorem* in several variables, of Cattani, Kaplan and Schmid [CKS86]: that nilpotent orbit is in turn approximated by an orbit of a fixed real reductive group, with uniform estimates in the several-variable setting.

Together these give a two-sided estimate for the Hodge norm of a flat section near the boundary, of the shape $\|\alpha\|_s \sim \prod_j (\log |t_j|^{-1})^{m_j}$ with integers $m_j$ determined by the weight filtration. A flat section that is Hodge on $Z$ has its norm for the positive definite form $\tilde{Q}$ of the proof of Lemma B.5 bounded by that lemma, and the estimate converts that bound into a statement that the possible boundary behaviours form a finite list. Algebraicity then follows from the analytic extension across the boundary together with Chow's theorem in $\bar{S}$.

*Remark* B.7. Both (A) and (B) are theorems about degenerating Hodge structures, and neither mentions algebraic cycles. The algebraicity in Theorem B.6 is algebraicity of a locus of periods, obtained by analysis, and it carries no information about algebraic cycles; Appendix B.4 develops the point.

B.3. **The Weil family.** The example relevant to this paper is the family of complex structures on the model of Setup 4.1.

**Proposition B.8** (The Weil period domain)**.** *Fix $K$, $n$, the $K$-space $V = K^{2n}$, and the hermitian form $H$ of signature $(n,n)$ from Lemma 4.5. The set of Hodge structures of weight one on $V$ that are polarised by $Q$, commute with the $K$-action, and are of $(K,-1,n)$-Weil type is in bijection with*

$$\mathcal{D}_{n,n} = \{ P \subset V_+ \ : \ \dim_{\mathbb{C}} P = n, \ H|_P \text{ negative definite} \},$$

*an open subset of the Grassmannian $\mathrm{Gr}(n,2n)$ and a bounded symmetric domain isomorphic to $\mathrm{U}(n,n)/(\mathrm{U}(n)\times\mathrm{U}(n))$. In particular*

$$\dim_{\mathbb{C}} \mathcal{D}_{n,n} = n^2,$$

*compared with $\dim \mathcal{A}_{2n} = n(2n+1)$ for the moduli of principally polarised abelian $2n$-folds.*

*Proof.* A Hodge structure of weight one on $V$ is a subspace $V^{1,0} \subset V_{\mathbb{C}}$ with $V_{\mathbb{C}} = V^{1,0} \oplus \overline{V^{1,0}}$. Commuting with the $K$-action means that $V^{1,0}$ is stable under the operator $S$ of Setup 4.1, so that it splits as $V^{1,0} = P \oplus P'$ with $P = V^{1,0} \cap V_+$ and $P' = V^{1,0} \cap V_-$. Conjugation interchanges $V_+$ and $V_-$, so $\overline{V^{1,0}} = \overline{P} \oplus \overline{P'}$ with $\overline{P} \subset V_-$, and the direct sum condition becomes $V_- = P' \oplus \overline{P}$, which determines $P'$ up to the choice of a complement; the Weil

condition $\dim P = \dim P' = n$ of Lemma 3.5 together with the polarisation condition pins it down as $P' = \overline{P}^{\perp_H}$. The structure is therefore determined by $P$.

For the positivity, extend $H$ to the $\mathbb{C}$-bilinear pairing between $V_+$ and $V_-$. By the computation (9) in the proof of Theorem 4.4, the second Riemann relation holds exactly when $iQ$ is positive on $V^{1,0}$, and on the $V_+$ part this is the condition that $H$ be negative definite on $P$, the sign being fixed by the convention of Lemma 4.5. The set of $n$-dimensional negative definite subspaces of a hermitian space of signature $(n,n)$ is the standard realisation of the symmetric domain of $\mathrm{U}(n,n)$: the unitary group acts transitively on it and the stabiliser of a point is $\mathrm{U}(n)\times\mathrm{U}(n)$, the unitary groups of $P$ and of $P^\perp$. It is open in $\mathrm{Gr}(n,2n)$ because negative definiteness is an open condition, and $\dim \mathrm{Gr}(n,2n) = n(2n-n) = n^2$.

For the comparison, $\dim \mathcal{A}_g = g(g+1)/2$, which is $n(2n+1)$ for $g = 2n$. ■

**Corollary B.9** (The Weil locus is algebraic)**.** *Inside $\mathcal{A}_{2n}$ the image of $\mathcal{D}_{n,n}$, that is the locus of principally polarised abelian $2n$-folds admitting an action of an order of $K$ of $(K,-1,n)$-Weil type, is a closed algebraic subvariety of dimension $n^2$.*

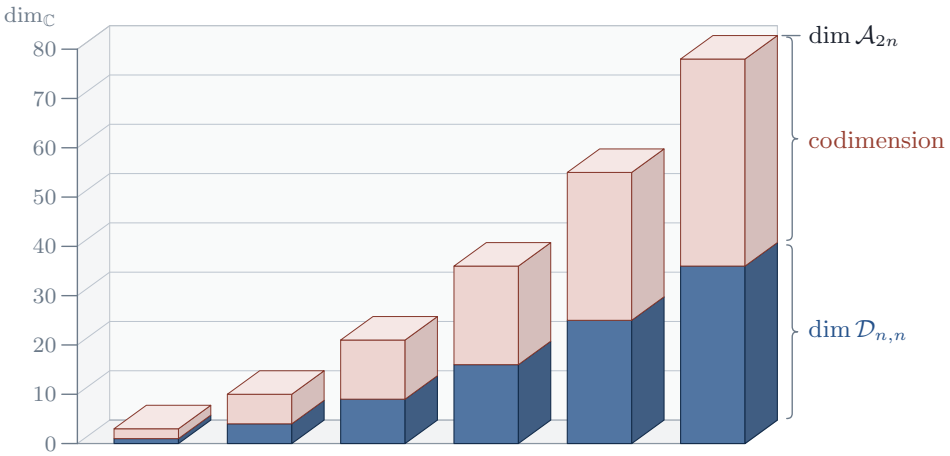
*Proof.* The condition of admitting such an action is the condition that a certain class in $\mathrm{End}(H^1)$, namely the graph of $\iota(\sqrt{-d})$, be a Hodge class, together with the balanced signature, which is locally constant. The locus is therefore a component of a locus of Hodge classes for the variation $\mathrm{End}(R^1 f_*\mathbb{Q})$ over $\mathcal{A}_{2n}$ with its level structure, and Theorem B.6 applies. The dimension is Proposition B.8. ■

## B.4. **Limits of the theorem.**

*Remark* B.10 (Algebraic locus, not algebraic class). Theorem B.6 says that the set of parameters where $\alpha$ is Hodge is cut out by polynomial equations. It says nothing about whether $\alpha$ is the class of a cycle at any one of those parameters. The two statements are independent in both directions: the Hodge locus of a class that happens to be algebraic everywhere is still known to be algebraic only through Theorem B.6, and conversely the algebraicity of the locus supplies no cycle. The distinction is easily blurred, and Corollary B.9 is exactly the kind of statement that invites it: the Weil locus is an algebraic subvariety, even a Shimura subvariety, and none of that produces the class $\omega_1$ of (11). Figure 71 compares the dimension of the Weil locus with that of $\mathcal{A}_{2n}$.

*Remark* B.11 (The variational Hodge conjecture). The property that would turn Theorem B.6 into a tool is the variational Hodge conjecture of Grothendieck: if $\mathcal{X} \to S$ is smooth projective over a connected base, $\alpha$ is a flat section of $R^{2p}f_*\mathbb{Q}$ that is Hodge at every point, and $\alpha_{s_0}$ is algebraic at one point $s_0$, then $\alpha_s$ is algebraic at every point. Its infinitesimal form is a theorem under a semiregularity hypothesis: if the cycle at $s_0$ is semiregular in the sense of Bloch [Blo72], then it deforms to first order, and hence formally, along the locus where its class stays Hodge. That theorem is the subject of Section 9, and Corollary 9.3 shows that its numerical hypothesis is not what obstructs the present problem.

*Remark* B.12 (The construction of Section 14). The construction of Section 14 produces, for every $n$, every $d$ and every nondegenerate $\beta$, an abelian variety of Weil type together with secant sheaves whose Chern classes are computed in closed form, and its output is algebraic on the split locus. The missing ingredient, described in Remark 14.16, is a statement along the algebraic locus of Corollary B.9: the Hodge type of the deformed object has to be controlled as the parameter moves off the split locus. Theorem B.6 guarantees that the locus along which one deforms is an algebraic variety, which is what makes the question well posed, but it does not answer the question. Question 13.7 states it precisely.




| $n$ | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| $n^2$ | 1 | 4 | 9 | 16 | 25 | 36 |
| $n(n+1)$ | 2 | 6 | 12 | 20 | 30 | 42 |
| $n(2n+1)$ | 3 | 10 | 21 | 36 | 55 | 78 |

FIGURE 71. The Weil locus inside the moduli of principally polarised abelian $2n$-folds, for $n = 1, \dots, 6$. Each prism has the height $n(2n+1)$ of the moduli space $\mathcal{A}_{2n}$; its lower part, of height $n^2$, is the dimension of the period domain $\mathcal{D}_{n,n}$ of Proposition B.8, and its upper part is the codimension $n(n+1)$. The codimension grows quadratically, so the Weil locus is thin, and by Corollary B.9 it is an algebraic subvariety. Both facts matter for propagation: the locus along which one would have to deform is small and is a variety, but neither fact produces a cycle on it.

## APPENDIX C. ENDOMORPHISM ALGEBRAS AND WEIL TYPE

Definition 3.1 asks for three things: an even dimension, an imaginary quadratic field acting with balanced eigenvalue multiplicities, and compatibility with the polarisation. This appendix places these conditions inside Albert's classification of endomorphism algebras. Two statements used in the body come out of it. The first is that the compatibility condition is automatic once the field is stable under the Rosati involution, so that it reduces to that stability (Lemma C.4). The second is that the balanced condition is not a normalisation but exactly the condition under which the Weil line carries a Hodge class (Proposition C.7); this is why the subject concerns balanced signatures alone.

### C.1. **Positive involutions.**

**Definition C.1.** Let $D$ be a finite dimensional semisimple $\mathbb{Q}$-algebra. An involution $\dagger$ of $D$, that is a $\mathbb{Q}$-linear anti-automorphism with $\dagger^2 = \mathrm{id}$, is *positive* if

$$\mathrm{Tr}_{D/\mathbb{Q}}(x\, x^{\dagger}) > 0 \qquad \text{for every } x \in D,\ x \neq 0,$$

where $\mathrm{Tr}_{D/\mathbb{Q}}$ is the trace of left multiplication on $D$ as a $\mathbb{Q}$-vector space.

For an abelian variety $A$ with a polarisation $\eta$, the algebra $D = \mathrm{End}^0(A) = \mathrm{End}(A) \otimes \mathbb{Q}$ is semisimple and the Rosati involution $\dagger$ attached to $\eta$ is positive [BL04, Ch. 5]. Everything below is a statement about the pair $(D, \dagger)$.

**Lemma C.2** (Totally real centre)**.** *Let $(D,\dagger)$ be as above with $D$ a division algebra, let $F$ be the centre of $D$ and $F_0=\{z\in F : z^\dagger = z\}$. Then $F_0$ is a totally real number field.*

*Proof.* For $z\in F_0$ the positivity condition reads $\mathrm{Tr}_{D/\mathbb{Q}}(z^2)>0$, and $\mathrm{Tr}_{D/\mathbb{Q}}$ restricted to $F_0$ is $[D:F_0]\cdot \mathrm{Tr}_{F_0/\mathbb{Q}}$, so $\mathrm{Tr}_{F_0/\mathbb{Q}}(z^2)>0$ for every nonzero $z\in F_0$. Suppose $F_0$ had a complex place $\sigma$. The embedding of $F_0$ into the product of its archimedean completions has dense image, so there is $z\in F_0$ with $\sigma(z)$ within $\frac{1}{10}$ of $i$ and $|\rho(z)|<\varepsilon$ for every other archimedean place $\rho$. Then $\mathrm{Tr}_{F_0/\mathbb{Q}}(z^2)=\sum_\rho \rho(z)^2$, where a complex place contributes twice the real part of $\rho(z)^2$. The real part of $\sigma(z)^2$ is less than $-1+\frac{2}{10}+\frac{1}{100}<-\frac{1}{2}$, so the term from $\sigma$ is less than $-1$, and the remaining terms are bounded in absolute value by $2[F_0:\mathbb{Q}]\varepsilon^2$, so the total is negative for $\varepsilon$ small. This contradicts positivity, so every place of $F_0$ is real. ∎

**Theorem C.3** (Albert [Alb34, Alb35a, Alb35b])**.** [68] *Let $(D,\dagger)$ be a division algebra over $\mathbb{Q}$ with a positive involution, $F$ its centre, $F_0$ the subfield fixed by $\dagger$, $e_0=[F_0:\mathbb{Q}]$ and $[D:F]=d^2$. Then $F_0$ is totally real, $[F:F_0]\in\{1,2\}$, and exactly one of the following holds.*

Type I. *$D=F=F_0$ is a totally real field and $\dagger$ is the identity.*

Type II. *$F=F_0$ is totally real and $D$ is a quaternion algebra over $F$ with $D\otimes_{F,\rho}\mathbb{R}\cong M_2(\mathbb{R})$ at every real place $\rho$; $\dagger$ is of orthogonal type.*

Type III. *$F=F_0$ is totally real and $D$ is a quaternion algebra over $F$ with $D\otimes_{F,\rho}\mathbb{R}\cong\mathbb{H}$ at every real place $\rho$; $\dagger$ is the canonical involution.*

Type IV. *$F$ is a CM field with maximal totally real subfield $F_0$, and $\dagger$ induces complex conjugation on $F$.*

*If moreover $D=\mathrm{End}^0(A)$ for an abelian variety $A$ of dimension $g$ with $\dagger$ the Rosati involution of a polarisation, then*

$$\text{I: } e_0 \mid g, \qquad \text{II: } 2e_0 \mid g, \qquad \text{III: } 2e_0 \mid g, \qquad \text{IV: } e_0 d^2 \mid g.$$

We quote Theorem C.3 as a classification. Lemma C.2 proves the part of it that we use directly, and the four types with their divisibility conditions are in [BL04, Ch. 9]. Figure 72 draws the signatures of an imaginary quadratic action and the dimensions of their families.

## C.2. **Compatibility with the polarisation.**

**Lemma C.4** (Rosati on an imaginary quadratic subfield)**.** *Let $A$ be an abelian variety with a polarisation $\eta$ and Rosati involution $\dagger$, and let $\iota\colon K\hookrightarrow\mathrm{End}^0(A)$ be an embedding of an imaginary quadratic field such that $\iota(K)$ is stable under $\dagger$. Then*

$$\iota(\tau)^\dagger=\iota(\overline{\tau}) \qquad \text{for every } \tau\in K.$$

*That is, condition* (iii) *of Definition 3.1 holds automatically, and Weil type is the conjunction of* (i), (ii) *and the $\dagger$-stability of $\iota(K)$.*

*Proof.* Put $D=\mathrm{End}^0(A)$. By stability, $\dagger$ restricts to an involutive anti-automorphism of the field $\iota(K)\cong K$. An anti-automorphism of a commutative ring is an automorphism, and

[68] The numbering of the four types is Albert's and has stuck, although it follows no natural order: types I to III have a totally real centre and type IV a CM one, so the natural division is two against two rather than three against one. Shimura's normalisation of the same classification, in terms of the associated symmetric domains, is the one that makes the moduli side transparent.

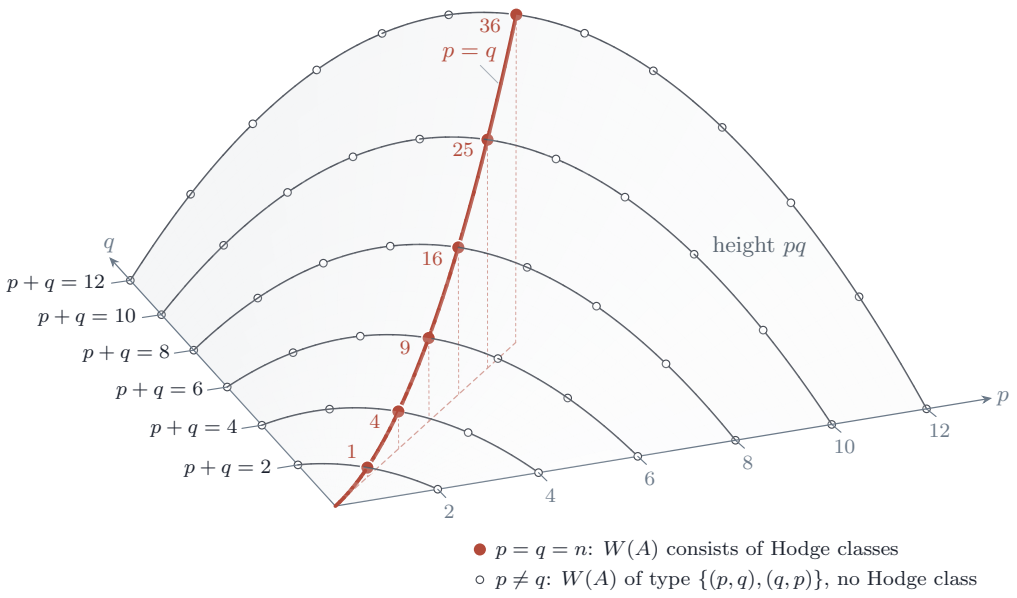



FIGURE 72. The signatures $(p, q)$ of an imaginary quadratic action, with $p + q = \dim A$, drawn on the surface of height $pq$, which is the dimension of the family of abelian varieties carrying that action, the unitary Shimura variety of signature $(p, q)$. Each curve on the surface collects the signatures available in one dimension, and on it the diagonal point is the highest. The divisibility conditions of Theorem C.3 put no constraint on which of them occur when $\mathrm{End}^0(A) = K$, by Proposition C.9(i). By Proposition C.7 the Weil line consists of Hodge classes exactly at the filled points on the ridge $p = q$, and at every open point it is a rational sub-Hodge structure of type $\{(p, 2n-p), (2n-p, p)\}$ carrying no Hodge class at all. The diagonal is the locus that Corollary B.9 shows to be an algebraic subvariety of the moduli.

$\mathrm{Gal}(K/\mathbb{Q})$ has order two, so the restriction is either the identity or complex conjugation. Suppose it were the identity, and put $x = \iota(\sqrt{-d})$. Then $x\,x^\dagger = x^2 = \iota(-d) = -d \cdot 1$, so

$$\mathrm{Tr}_{D/\mathbb{Q}}(x\,x^\dagger) = -d \cdot \dim_{\mathbb{Q}} D < 0,$$

contradicting the positivity of †. Hence † restricts to complex conjugation. ∎

*Remark* C.5. Stability is a real hypothesis: a subfield of $\mathrm{End}^0(A)$ need not be †-stable. What Lemma C.4 removes is the second half of the condition, the identification of $\dagger|_K$ with a specific automorphism. In particular, in the model of Setup 4.1 the polarisation need not be chosen to match the field, since any †-stable embedding does the same work.

## C.3. **The balanced signature.**

**Definition C.6** (Signature)**.** Let $\iota\colon K \hookrightarrow \mathrm{End}^0(A)$ be as above, $g = \dim A$. The operator $\iota(\sqrt{-d})$ acts on the tangent space $T_0A \cong \mathbb{C}^g$ with the two eigenvalues $\pm i\sqrt{d}$; write $p$ and $q = g - p$ for their multiplicities. The pair $(p, q)$ is the *signature* of $(A, \iota)$.

The signature is an isogeny invariant, since an isogeny induces an isomorphism on tangent spaces commuting with the action. It is locally constant in families: the two eigenspaces are the kernels of $\iota(\sqrt{-d}) \mp i\sqrt{d}$ on the relative tangent bundle, so their dimensions are

upper semicontinuous, and they sum to $g$. By duality $\iota(\tau)^*$ acts on $H^{1,0}(A)$ with the same multiplicities, so $\dim V_+^{1,0} = p$ and $\dim V_-^{1,0} = q$ in the notation of Section 3.

**Proposition C.7** (The balanced signature is forced)**.** *Let $A$ be an abelian variety of dimension $2n$ with a $\dagger$-stable embedding $\iota\colon K \hookrightarrow \mathrm{End}^0(A)$ of signature $(p,q)$, $p+q=2n$, and let*

$$W(A) = \left(\textstyle\bigwedge_{\mathbb{C}}^{2n} V_+ \oplus \bigwedge_{\mathbb{C}}^{2n} V_-\right) \cap H^{2n}(A,\mathbb{Q})$$

*as in Definition 3.6. Then $W(A)$ is a rational sub-Hodge structure of $H^{2n}(A,\mathbb{Q})$ of dimension two and of Hodge type $\{(p,2n-p),\,(2n-p,p)\}$, and the following are equivalent:*

(i) *$p=q=n$, that is $(A,\iota,\eta)$ is of $(K,-1,n)$-Weil type;*
(ii) *$W(A)$ contains a nonzero Hodge class;*
(iii) *$W(A)$ consists of Hodge classes.*

*Proof.* The space $\bigwedge^{2n} V_+$ is a line, and by (3) it equals $\bigwedge^p V_+^{1,0} \otimes \bigwedge^{2n-p} V_+^{0,1}$, which is of pure Hodge bidegree $(p,2n-p)$; the conjugate line $\bigwedge^{2n} V_-$ is of bidegree $(2n-p,p)$. Their sum is stable under conjugation and defined over $\mathbb{Q}$ by the argument of Proposition 3.7(i), which does not use the balanced condition, so $W(A)$ is a two-dimensional rational sub-Hodge structure of the stated type.

A Hodge class is an element of bidegree $(n,n)$. A nonzero element of $W(A)\otimes\mathbb{C}$ has a nonzero component in one of the two lines, so it is of bidegree $(n,n)$ only if $(p,2n-p)=(n,n)$ or $(2n-p,p)=(n,n)$, and either equality says $p=n$. This proves that (ii) implies (i). If $p=n$ then both lines are of bidegree $(n,n)$ and every element of $W(A)$ is a Hodge class, so (i) implies (iii); and (iii) implies (ii) because $\dim W(A) = 2 > 0$. ∎

*Remark* C.8. Proposition C.7 is the reason the literature speaks of abelian varieties of Weil type rather than of imaginary quadratic multiplication in general. For an unbalanced signature the space $W(A)$ is still a canonical two-dimensional rational sub-Hodge structure, and it is still not in the divisor subring, but it contains no Hodge class and therefore predicts nothing. The exceptional Hodge classes appear exactly at balance, and the locus of balance is the algebraic subvariety of Corollary B.9.

## C.4. **Endomorphism algebras of Weil type.**

**Proposition C.9** (Weil type in the classification)**.** *Let $A$ be an abelian variety of dimension $2n$ admitting a $\dagger$-stable embedding of an imaginary quadratic field $K$ with balanced signature, and suppose $\mathrm{End}^0(A)$ is a division algebra $D$ with centre $F$ as in Theorem C.3. Then:*

(i) *if $D=K$ then $D$ is of type IV with $e_0=1$ and $d=1$, and the divisibility condition of Theorem C.3 is vacuous, so every even dimension occurs;*
(ii) *the classification does not exclude type III with $F_0=\mathbb{Q}$: a totally definite quaternion algebra over $\mathbb{Q}$ contains imaginary quadratic subfields on which the canonical involution is conjugation;*
(iii) *$D$ is not of type I, because a totally real field contains no imaginary quadratic subfield.*

*Proof.* (i) If $D=K$ then $F=K$ is a CM field with $F_0=\mathbb{Q}$, so $e_0=1$; and $[D:F]=1$ gives $d=1$. The involution is conjugation by Lemma C.4, which is type IV. The condition $e_0 d^2 \mid g$ reads $1 \mid 2n$.

(ii) Let $D$ be a quaternion algebra over $\mathbb{Q}$ ramified at the infinite place, with standard generators $u,v$ satisfying $u^2=a<0$, $v^2=b<0$, $uv=-vu$, and canonical involution $x\mapsto\overline{x}$ fixing $\mathbb{Q}$ and negating $u,v,uv$. Then $\mathbb{Q}(u)\cong\mathbb{Q}(\sqrt{a})$ is imaginary quadratic, stable under the involution, and the involution restricts to conjugation on it. Whether such a $D$

TABLE 8. The Ext groups between two pieces at $A_0 = E_0^6$, from Lemma 24.8; $m = b - a$, $\chi_Z = 8$ for the threefolds $Z_\zeta$.

| from | to | degrees $k$ | $\dim \mathrm{Ext}^k$ |
|---|---|---|---|
| $\mathcal{O}_Z L^a$ | $\mathcal{O}_Z L^a$ | $0, \dots, 6$ | $\binom{6}{k}$: $1, 6, 15, 20, 15, 6, 1$ |
| $\mathcal{O}_Z L^a$ | $\mathcal{O}_Z L^b$, $m > 0$ | $0, \dots, 3$ | $\binom{3}{k}\, 8m^3$ |
| $\mathcal{O}_Z L^a$ | $\mathcal{O}_Z L^b$, $m < 0$ | $3, \dots, 6$ | $\binom{3}{k-3}\, 8\lvert m\rvert^3$ |
| $\mathcal{O}_Z L^a$ | $\mathcal{O}_{Z'} L^b$ | $3$ | $(Z \cdot Z') \in \{8, 16, 32, 64\}$ |
| $\mathcal{O}_Z L^a$ | $L^b$, $m > 0$ | $3$ | $8m^3$ |
| $\mathcal{O}_Z L^a$ | $L^b$, $m < 0$ | $6$ | $8\lvert m\rvert^3$ |
| $\mathcal{O}_Z L^a$ | $L^a$ | $3, \dots, 6$ | $\binom{3}{k-3}$ |
| $L^a$ | $\mathcal{O}_Z L^b$, $m > 0$ | $0$ | $8m^3$ |
| $L^a$ | $\mathcal{O}_Z L^b$, $m < 0$ | $3$ | $8\lvert m\rvert^3$ |
| $L^a$ | $\mathcal{O}_Z L^a$ | $0, \dots, 3$ | $\binom{3}{k}$ |
| $L^a$ | $L^b$, $m > 0$ | $0$ | $m^6$ |
| $L^a$ | $L^b$, $m < 0$ | $6$ | $\lvert m\rvert^6$ |
| $L^a$ | $L^a$ | $0, \dots, 6$ | $\binom{6}{k}$ |

occurs as $\mathrm{End}^0(A)$ together with a balanced signature is a separate question about the moduli, but nothing in the classification excludes it.

(iii) is immediate, since a totally real field has no element of negative square. ∎

*Remark* C.10 (The generic case). Case (ii) of Proposition C.9 is a special position: there $\mathrm{End}^0(A)$ is larger than $K$, the Hodge group is correspondingly smaller, and $\mathrm{Hdg}^n(A)$ can be larger than the three dimensions of Theorem 4.14. It is also a case in which the Weil classes can become accessible: Remark 20.5 records that at such points the Weil line can meet the divisor subring, which is the split situation. The hard case, and the one addressed throughout Sections 5, 7 and 10, is case (i) with $\mathrm{Hg}(A) = \mathrm{SU}(V, H)$.

*Remark* C.11 (Dimension four). For $n = 1$ the Weil line lies in $H^2(A, \mathbb{Q})$ and consists of divisor classes, algebraic by the Lefschetz theorem on $(1,1)$ classes, so it predicts nothing new; this is the same boundary as in Proposition 11.11. Moonen and Zarhin determined the Hodge classes on abelian varieties of dimension at most five [MZ99], and on simple abelian fourfolds in [MZ95]: through dimension three the Hodge ring is generated by divisor classes, and the first Hodge classes not of that form occur in dimension four, on abelian fourfolds of Weil type. Thus $n = 2$ is at once the first case of Definition 3.1 with exceptional Hodge classes, the first case where Theorem 4.14 gives three rather than one, and the first case of the Hodge conjecture for abelian varieties that needed more than divisor classes; Markman settled it [Mar25a, Corollary 1.6.1]. The next case is $n = 3$ with nontrivial discriminant, by Proposition 20.12(ii).

## APPENDIX D. THE POINT OF COMPLEX MULTIPLICATION IN COORDINATES

This appendix works out, in coordinates, the objects of Lemma 24.8 and Proposition 24.10 on $A_0 = E_0^6$: the Hodge ring, the Weil plane, an explicit family of abelian threefolds whose classes combine to the Weil class, their intersection numbers, and the dimensions of every Ext group between the pieces built from them.

D.1. **The forms and the Hodge ring.** Write $z_1, \dots, z_6$ for the coordinates of $A_0 = E_0^6$, $E_0 = \mathbb{C}/\mathbb{Z}[i]$, group the factors into the three surfaces $B_j = E_0^2$ with coordinates $(z_j, z_{3+j})$,

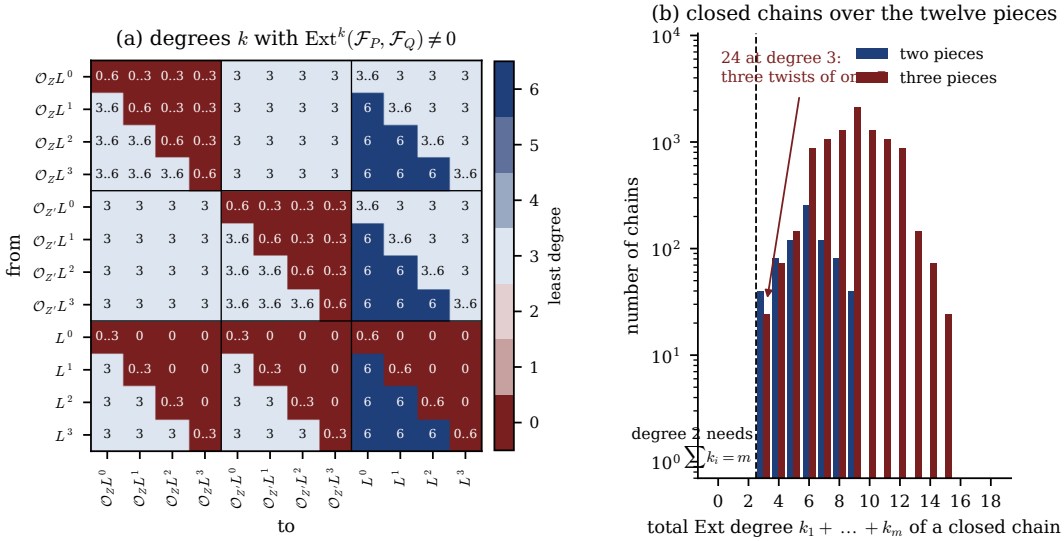



FIGURE 73. The degrees of the Ext groups between the twelve pieces $\mathcal{O}_Z L^k$, $\mathcal{O}_{Z'} L^k$, $L^k$, $k = 0, \ldots, 3$, of Table 8. Left: the degrees in which $\mathrm{Ext}^k(\mathcal{F}_P, \mathcal{F}_Q)$ is nonzero, coloured by the least one; degree zero occurs only from a lower twist to a higher one and from $L^a$ to $\mathcal{O}_Z L^a$. Right: the number of closed chains of two and of three distinct pieces by total degree, on a logarithmic scale; no two-piece chain has total degree below three, and the three-piece chains of total degree three are those of Proposition 24.10.

$j = 1, 2, 3$, and let $K = \mathbb{Q}(i)$ act by $i$ on $z_1, z_2, z_3$ and by $-i$ on $z_4, z_5, z_6$. Put

$$\begin{aligned} a_j &= \tfrac{i}{2}\, dz_j \wedge d\bar z_j, \qquad b_j = \tfrac{i}{2}\, dz_{3+j} \wedge d\bar z_{3+j}, \\ c_j &= \tfrac{i}{2}\, dz_j \wedge d\bar z_{3+j}, \quad \bar c_j = \tfrac{i}{2}\, dz_{3+j} \wedge d\bar z_j. \end{aligned} \tag{91}$$

In the real coordinates $z_j = x_j + iy_j$ one has $a_j = dx_j \wedge dy_j$, the volume form of the $j$-th factor, of integral one over $\mathbb{C}/\mathbb{Z}[i]$,[69] and $c_j = \frac{1}{2}(dx_j \wedge dy_{3+j} - dy_j \wedge dx_{3+j}) + \frac{i}{2}(dx_j \wedge dx_{3+j} + dy_j \wedge dy_{3+j})$, so that the real and imaginary parts of $c_j$ are rational classes and $\bar c_j$ is the complex conjugate of $c_j$. All four forms have type $(1,1)$. The polarisation is

$$\theta = \sum_{j=1}^{3} (a_j + b_j), \qquad \theta^6 = 6!\, a_1 b_1 a_2 b_2 a_3 b_3, \qquad \int_{A_0} \frac{\theta^6}{6!} = 1, \tag{92}$$

the principal polarisation of the product; the hermitian forms $\sum_{j\le 3} a'_j z_j \bar w_j - \sum_{j>3} a'_j z_j \bar w_j$ of Lemma 24.8 with all $a'_j = 1$ give $\theta$ itself.

The Hodge group of $A_0$ is the circle acting by $z$ on $H^{1,0}$ and by $\bar z$ on $H^{0,1}$, so the Hodge classes of degree $2k$ are exactly the rational classes of type $(k,k)$, of dimension $\binom{6}{k}^2$:

$$\dim \mathrm{Hdg}^{2k}(A_0) = 1,\ 36,\ 225,\ 400,\ 225,\ 36,\ 1 \quad (k = 0, \ldots, 6). \tag{93}$$

In degree two these are the 36 hermitian forms on $\mathbb{Z}[i]^6$, the Picard number of $E_0^6$. The $K$-eigenspaces of $H^1(A_0, \mathbb{C})$ are spanned by $dz_1, dz_2, dz_3, d\bar z_4, d\bar z_5, d\bar z_6$ (eigenvalue $i$) and by

[69] The fundamental domain of $\mathbb{Z}[i]$ is the unit square, so $\int_{E_0} dx \wedge dy = 1$; this normalisation makes every form in (91) integral on the lattice $\mathbb{Z}[i]^6$, and all intersection numbers below are integers computed as integrals of top forms.

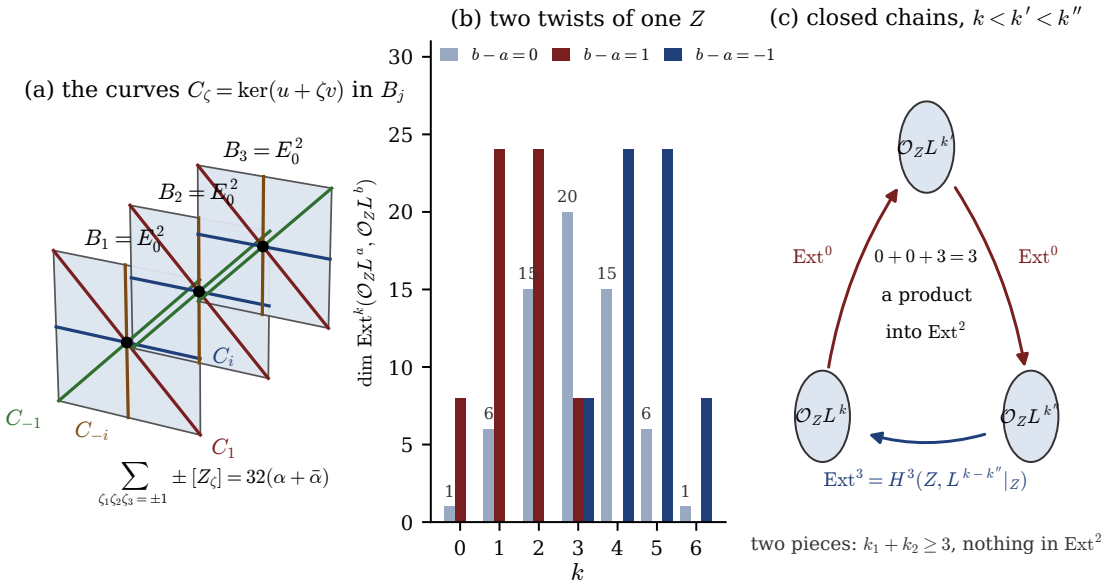



FIGURE 74. The point $E_0^6$ of complex multiplication. Left: the three Weil surfaces $B_j = E_0^2$ and the four elliptic curves $C_\zeta$ in each, kernels of $u+\zeta v$; a threefold $Z_\zeta$ picks one curve in each surface, and the 32 threefolds with $\zeta_1\zeta_2\zeta_3 = \pm 1$, signed by that product, add up to $32(\alpha+\bar\alpha)$. Middle: the dimensions of $\mathrm{Ext}^k(\mathcal{O}_Z\otimes L^a, \mathcal{O}_Z\otimes L^b)$ by degree for $b-a=0$ (the $\binom{6}{k}$), $b-a=1$ (degrees 0 to 3) and $b-a=-1$ (degrees 3 to 6). Right: the closed chains of Proposition 24.10; every two-step chain has total degree at least three, and the shortest chains reaching degree two run through three twists of one threefold with degrees $0,0,3$.

their conjugates, so the Weil plane $W = \bigwedge^6_K H^1(A_0,K) \subseteq H^6(A_0,\mathbb{Q})$ is spanned over $\mathbb{C}$ by

$$\alpha = c_1c_2c_3 \propto dz_1dz_2dz_3\, d\bar z_4 d\bar z_5 d\bar z_6 \qquad \text{and} \qquad \bar\alpha = \bar c_1\bar c_2\bar c_3, \tag{94}$$

and over $\mathbb{Q}$ by $\alpha+\bar\alpha$ and $i(\alpha-\bar\alpha)$. The relation $W\theta = 0$ is visible term by term: $c_ja_j$ contains $dz_j$ twice and $c_jb_j$ contains $d\bar z_{3+j}$ twice, so $\alpha\,(a_k+b_k) = 0$ for each $k$, since $\alpha$ already contains $dz_k$ and $d\bar z_{3+k}$.

D.2. **The elliptic curves $C_\zeta$ and the threefolds $Z_\zeta$.** For $\zeta\in\mu_4 = \{1,i,-1,-i\}$ let $C_\zeta \subseteq B_j$ be the kernel of $(u,v)\mapsto u+\zeta v$, an elliptic curve isomorphic to $E_0$, and for $\zeta = (\zeta_1,\zeta_2,\zeta_3)\in\mu_4^3$ let $Z_\zeta = C_{\zeta_1}\times C_{\zeta_2}\times C_{\zeta_3}\subseteq A_0$, an abelian threefold.

**Lemma D.1.** (i) *$[C_\zeta] = a_j + b_j + \bar\zeta\, c_j + \zeta\,\bar c_j$ in $H^2(B_j)$, and $[Z_\zeta] = \prod_j [C_{\zeta_j}]$.*

(ii) *$(C_\zeta\cdot C_{\zeta'}) = |\zeta-\zeta'|^2 \in \{0,2,4\}$ on $B_j$; $Z_\zeta\cap Z_{\zeta'}$ is finite exactly when $\zeta_j\neq\zeta'_j$ for all $j$, and then $|Z_\zeta\cap Z_{\zeta'}| = \prod_j|\zeta_j-\zeta'_j|^2 \in \{8,16,32,64\}$.*

(iii) *$\theta|_{Z_\zeta}$ has degree two on each factor, $(\theta|_{Z_\zeta})^3 = 48$ and $\chi_{Z_\zeta} = 8$; in particular $[Z_\zeta]\theta\neq 0$.*

(iv) *Writing* $S(\omega)=\sum_{\zeta_1\zeta_2\zeta_3=\omega}[Z_\zeta]$ *for* $\omega\in\mu_4$,

$$\sum_{\omega\in\mu_4}\omega^m S(\omega)=\prod_{j=1}^{3}T_{m,j},\qquad T_{m,j}=\sum_{\zeta\in\mu_4}\zeta^m[C_\zeta],$$
$$T_{m,j}=4(a_j+b_j),\ 4c_j,\ 0,\ 4\bar c_j\quad \text{for } m\equiv 0,1,2,3\pmod 4, \tag{95}$$

*and therefore*

$$S(1)-S(-1)=\sum_{\zeta_1\zeta_2\zeta_3=1}[Z_\zeta]-\sum_{\zeta_1\zeta_2\zeta_3=-1}[Z_\zeta]=32\,(\alpha+\bar\alpha). \tag{96}$$

*Proof.* (i) $C_\zeta$ is the fibre over 0 of the surjective homomorphism $f_\zeta\colon B_j\to E_0$, $w=u+\zeta v$, so $[C_\zeta]=f_\zeta^*[0]$ with $[0]=\frac{i}{2}dw\wedge d\bar w$, and

$$\begin{aligned}dw\wedge d\bar w&=(du+\zeta\,dv)\wedge(d\bar u+\bar\zeta\,d\bar v)\\&=du\wedge d\bar u+dv\wedge d\bar v+\bar\zeta\,du\wedge d\bar v+\zeta\,dv\wedge d\bar u,\end{aligned}$$

which is the formula. A product of subvarieties of the factors has the product class.

(ii) On $B_j$ the only nonzero products of two of the four forms are $a_jb_j$ and $c_j\bar c_j=\bar c_jc_j=-a_jb_j$ (reorder $du\,d\bar v\,dv\,d\bar u$ to $-du\,d\bar u\,dv\,d\bar v$), and $\int_{B_j}a_jb_j=1$. Hence $[C_\zeta][C_{\zeta'}]=2-\bar\zeta\zeta'-\zeta\bar\zeta'=|\zeta-\zeta'|^2$. Directly: a point of $C_\zeta\cap C_{\zeta'}$ has $(\zeta-\zeta')v=0$, so $v$ runs over the kernel of $\zeta-\zeta'$ on $E_0$, of order $|\zeta-\zeta'|^2$, and $u=-\zeta v$ is then determined. The statement for $Z_\zeta$ follows factor by factor.

(iii) $\int_{C_\zeta}(a_j+b_j)=\int_{B_j}(a_j+b_j)[C_\zeta]=2$, since $a_jc_j$, $a_j\bar c_j$, $b_jc_j$, $b_j\bar c_j$ all repeat a differential. So $\theta|_{Z_\zeta}=2(p_1+p_2+p_3)$ with $p_j$ the point class of the $j$-th factor, $(\theta|_{Z_\zeta})^3=3!\cdot 8=48$, and $\chi_{Z_\zeta}=48/6=8$.

(iv) The left side of (95) is $\sum_\zeta(\zeta_1\zeta_2\zeta_3)^m\prod_j[C_{\zeta_j}]$, which factors as the product of the three sums $T_{m,j}$; these are read off from (i), using $\sum_{\zeta\in\mu_4}\zeta^m=4$ or 0 according as $4\mid m$ or not. Then $S(1)-S(-1)=\frac12\sum_\omega(\omega+\omega^3)S(\omega)=\frac12(\prod_jT_{1,j}+\prod_jT_{3,j})=\frac12(64\alpha+64\bar\alpha)$. ∎

Equation (96) is the analogue, with abelian threefolds in place of line bundles, of the signed sum of the 32 line bundles on $E_0^6$ in Proposition 18.27: the 32 threefolds $Z_\zeta$ with $\zeta_1\zeta_2\zeta_3=\pm1$, taken with the sign $\zeta_1\zeta_2\zeta_3$, have classes adding up to 32 times the real Weil class. Together with three powers of $L$ this is a Chern character of polarised shape and Hankel rank three at $A_0$, and by Lemma 24.8(i) it is one of the shapes a convolution must realise. Two of these threefolds with $\zeta_j=\zeta_j'$ for one or two indices $j$ meet along a curve or a surface, so Lemma 24.8(ii) does not apply to them and Lemma 24.9 gives their groups; Proposition 24.11 shows that no convolution of these thirty-five pieces meets the criterion. Figure 74 draws the curves, the dimensions of the Ext groups and the chains.

D.3. **The dimensions in Lemma 24.8.** For the threefolds $Z=Z_\zeta$ the numbers of Lemma 24.8 are explicit. Table 8 lists, for every pair of piece types, the degrees $k$ in which $\mathrm{Ext}^k$ is nonzero and its dimension; here $m=b-a$ is the difference of the twists, $\chi_Z=8$, $Z'=Z_{\zeta'}$ with $\zeta_j\neq\zeta_j'$ for all $j$, and $(Z\cdot Z')=\prod_j|\zeta_j-\zeta_j'|^2$.

Each row of the table restates the corresponding item of Lemma 24.8, with $\chi(L^m|_Z)=m^3\chi_Z=8m^3$ from (iii) above and $\chi(L^m)=m^6\theta^6/720=m^6$ from (92). Two features of the table drive Proposition 24.10. Every group between two pieces of different twist is concentrated in degree 0, in degree 3 or 6, or in the range 0 to 3 for two twists of one threefold; and a group in degree at most two between distinct pieces occurs only from a lower twist to a higher one, or from $L^a$ to $\mathcal{O}_ZL^a$. The bookkeeping of the proposition

is then the following count. A closed chain of $m$ pieces whose edges have Ext degrees $k_1, \dots, k_m$ carries an $m$-fold product into $\mathrm{Ext}^{\sum k_i - m + 2}$ of the first piece. For $m = 2$ the table gives, for each unordered pair of distinct piece types, the two degrees and their sum:

$$\begin{array}{lll} (\mathcal{O}_Z L^a, \mathcal{O}_{Z'} L^b) & 3+3 & = 6, \\ (\mathcal{O}_Z L^a, \mathcal{O}_Z L^b),\ b > a & (0 \text{ to } 3) + (3 \text{ to } 6) & \geq 3, \\ (\mathcal{O}_Z L^a, L^b),\ b > a & 3+3 & = 6, \\ (\mathcal{O}_Z L^a, L^b),\ b < a & 6+0 & = 6, \\ (\mathcal{O}_Z L^a, L^a) & (3 \text{ to } 6) + (0 \text{ to } 3) & \geq 3, \\ (L^a, L^b) & 0+6 & = 6, \end{array} \tag{97}$$

so no two-step product reaches degree two. For $m = 3$ the sum must be 3 with two edges of degree zero, and the table leaves exactly the ascending chains $\mathcal{O}_Z L^k \to \mathcal{O}_Z L^{k'} \to \mathcal{O}_Z L^{k''}$, $k < k' < k''$, closed by the degree-three edge $H^3(Z, L^{k-k''}|_Z)$, of dimension $8(k'' - k)^3$. The program `code/cm_point_ext.py` enumerates the chains of two and three pieces over four twists and two threefolds and confirms both statements, together with the identities of Lemma D.1. Figure 73 shows the table as the matrix of the twelve pieces, and the distribution of the total degree over every closed chain of two and of three distinct pieces: the two-piece chains begin at total degree three, and the twenty-four ordered three-piece chains of total degree three are the eight ascending triples of twists of one threefold, each read from its three starting pieces.

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## Declarations

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**Competing interests.** The authors have no competing interests to declare.

**Ethics approval and consent.** This article reports no research involving human participants, human data or animals, so ethics approval and consent to participate or to publish were not required. The results are original and have not been published elsewhere except in the two earlier papers named on the title page, which this one replaces, and the work of others is credited by citation throughout.

**Data availability.** Two archives accompany the paper. The archive [BMB26], on Zenodo under the DOI `10.5281/zenodo.23225483`, holds programs that repeat much of the finite arithmetic of Parts 1 to 5 as an independent check, a Lean 4 certificate of part of that arithmetic, the Macaulay2 computations of local Ext algebras with their output, and the generators and TikZ sources of the figures of those parts; its programs perform 2218 checks, all of which pass, and the Lean kernel checks four hundred and nine theorems, each depending on no axiom or on propositional extensionality alone. The archive [BB26code], on Zenodo under the DOI `10.5281/zenodo.23790334`, holds the programs that repeat the

finite computations of Part 6 and Appendix D: the Hilbert symbols and lattice embeddings of Theorem 23.5, the kernel polynomials and quotient curves of the Fricke isogenies, the Ext dimensions and chain enumerations at $E_0^6$ and $E_0^8$, and the sixfold link, together with the sources of the figures of that part. Deep Bhattacharjee computed all the calculations, in C, Python, Julia, Macaulay2, Lean 4 and shell scripts, with the assistance of Claude Code, an AI model developed by Anthropic. Every result the programs certify can be reproduced from the archives with the programs and the Lean kernel, and the authors are responsible for the mathematical statements and their proofs. No other data were generated or analysed.

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